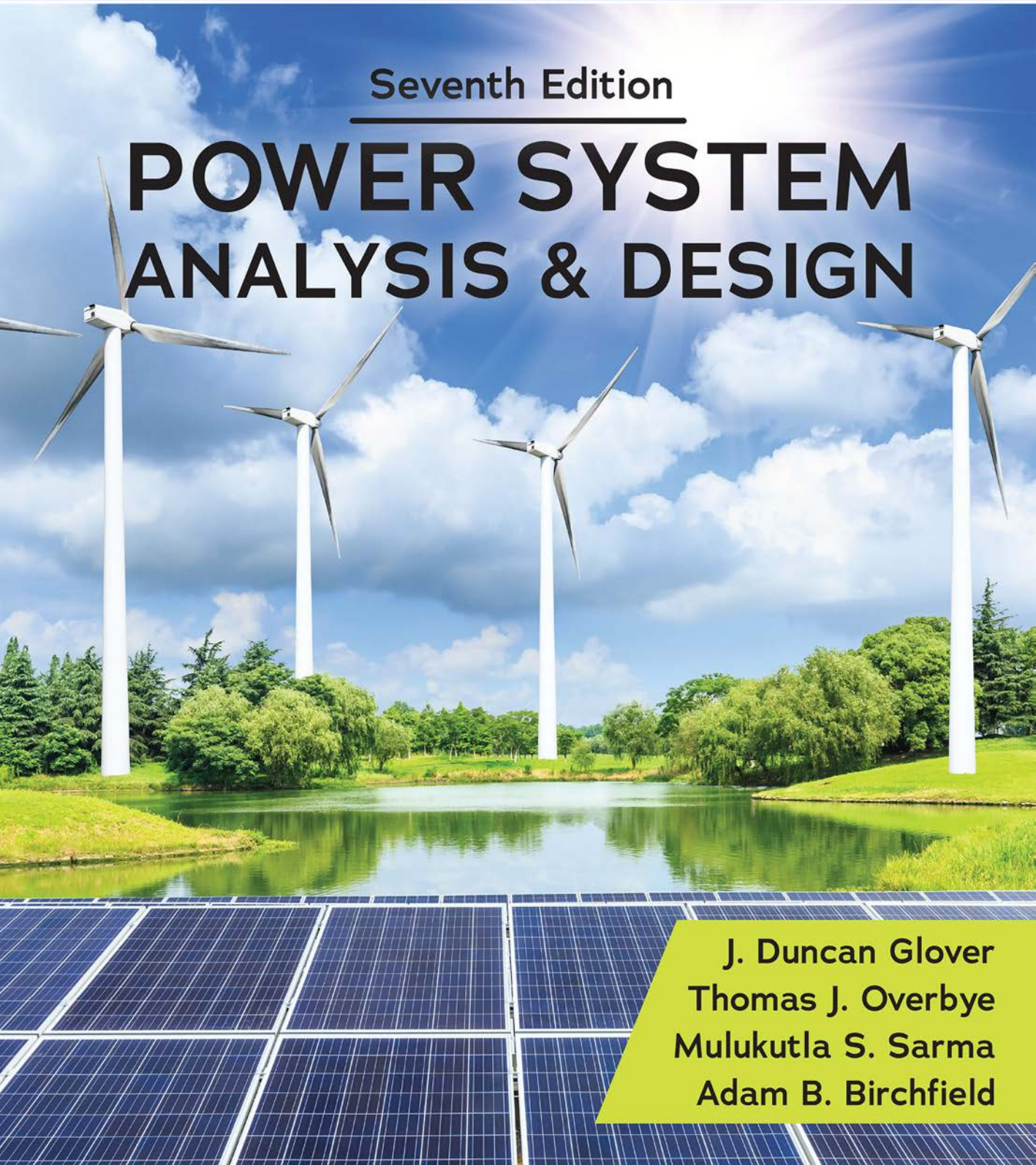
SI EDITION



Seventh Edition

POWER SYSTEM ANALYSIS & DESIGN

J. Duncan Glover
Thomas J. Overbye
Mulukutla S. Sarma
Adam B. Birchfield

CONVERSIONS BETWEEN U.S. CUSTOMARY UNITS AND SI UNITS

U.S. Customary unit		Times conversion factor		Equals SI unit	
		Accurate	Practical		
Acceleration (linear)					
foot per second squared	ft/s^2	0.3048*	0.305	meter per second squared	m/s^2
inch per second squared	$in./s^2$	0.0254*	0.0254	meter per second squared	m/s^2
Area					
circular mil	cmil	0.0005067	0.0005	square millimeter	mm^2
square foot	ft^2	0.09290304*	0.0929	square meter	m^2
square inch	$in.^2$	645.16*	645	square millimeter	mm^2
Density (mass)					
slug per cubic foot	$slug/ft^3$	515.379	515	kilogram per cubic meter	kg/m^3
Density (weight)					
pound per cubic foot	lb/ft^3	157.087	157	newton per cubic meter	N/m^3
pound per cubic inch	$lb/in.^3$	271.447	271	kilonewton per cubic meter	kN/m^3
Energy; work					
foot-pound	ft-lb	1.35582	1.36	joule (N·m)	J
inch-pound	in.-lb	0.112985	0.113	joule	J
kilowatt-hour	kWh	3.6*	3.6	megajoule	MJ
British thermal unit	Btu	1055.06	1055	joule	J
Force					
pound	lb	4.44822	4.45	newton ($kg·m/s^2$)	N
kip (1000 pounds)	k	4.44822	4.45	kilonewton	kN
Force per unit length					
pound per foot	lb/ft	14.5939	14.6	newton per meter	N/m
pound per inch	lb/in.	175.127	175	newton per meter	N/m
kip per foot	k/ft	14.5939	14.6	kilonewton per meter	kN/m
kip per inch	k/in.	175.127	175	kilonewton per meter	kN/m
Length					
foot	ft	0.3048*	0.305	meter	m
inch	in.	25.4*	25.4	millimeter	mm
mile	mi	1.609344*	1.61	kilometer	km
Mass					
slug	$lb-s^2/ft$	14.5939	14.6	kilogram	kg
Moment of a force; torque					
pound-foot	lb-ft	1.35582	1.36	newton meter	N·m
pound-inch	lb-in.	0.112985	0.113	newton meter	N·m
kip-foot	k-ft	1.35582	1.36	kilonewton meter	kN·m
kip-inch	k-in.	0.112985	0.113	kilonewton meter	kN·m

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CONVERSIONS BETWEEN U.S. CUSTOMARY UNITS AND SI UNITS (Continued)

U.S. Customary unit		Times conversion factor		Equals SI unit	
		Accurate	Practical		
Moment of inertia (area)					
inch to fourth power	$in.^4$	416,231	416,000	millimeter to fourth power	mm^4
inch to fourth power	$in.^4$	0.416231×10^{-6}	0.416×10^{-6}	meter to fourth power	m^4
Moment of inertia (mass)					
slug foot squared	$slug\text{-}ft^2$	1.35582	1.36	kilogram meter squared	$kg{\cdot}m^2$
Power					
foot-pound per second	ft-lb/s	1.35582	1.36	watt (J/s or N·m/s)	W
foot-pound per minute	ft-lb/min	0.0225970	0.0226	watt	W
horsepower (550 ft-lb/s)	hp	745.701	746	watt	W
Pressure; stress					
pound per square foot	psf	47.8803	47.9	pascal (N/m^2)	Pa
pound per square inch	psi	6894.76	6890	pascal	Pa
kip per square foot	ksf	47.8803	47.9	kilopascal	kPa
kip per square inch	ksi	6.89476	6.89	megapascal	MPa
Section modulus					
inch to third power	$in.^3$	16,387.1	16,400	millimeter to third power	mm^3
inch to third power	$in.^3$	16.3871×10^{-6}	16.4×10^{-6}	meter to third power	m^3
Velocity (linear)					
foot per second	ft/s	0.3048*	0.305	meter per second	m/s
inch per second	in./s	0.0254*	0.0254	meter per second	m/s
mile per hour	mph	0.44704*	0.447	meter per second	m/s
mile per hour	mph	1.609344*	1.61	kilometer per hour	km/h
Volume					
cubic foot	ft^3	0.0283168	0.0283	cubic meter	m^3
cubic inch	$in.^3$	16.3871×10^{-6}	16.4×10^{-6}	cubic meter	m^3
cubic inch	$in.^3$	16.3871	16.4	cubic centimeter (cc)	cm^3
gallon ($231\ in.^3$)	gal.	3.78541	3.79	liter	L
gallon ($231\ in.^3$)	gal.	0.00378541	0.00379	cubic meter	m^3

*An asterisk denotes an *exact* conversion factor

Note: **To convert from SI units to USCS units, *divide* by the conversion factor**

Temperature Conversion Formulas

$$T(°C) = \frac{5}{9}[T(°F) - 32] = T(K) - 273.15$$

$$T(K) = \frac{5}{9}[T(°F) - 32] + 273.15 = T(°C) + 273.15$$

$$T(°F) = \frac{9}{5}T(°C) + 32 = \frac{9}{5}T(K) - 459.67$$

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POWER SYSTEM ANALYSIS & DESIGN

SEVENTH EDITION, SI

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***Power System Analysis and Design,* Seventh Edition, SI**

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Production Service: RPK Editorial Services, Inc.

Compositor: MPS Limited

Designer: Nadine Ballard

Cover Image Source: ABCDstock/ Shutterstock.com

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Library of Congress Control Number: 2022934910

ISBN: 978-0-357-67619-6

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Printed in the United States of America
Print Number: 01 Print Year: 2022

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From J. Duncan Glover:

To my grandchildren: Natalie, John, Brigid, Emily, Anna, and Owen

From Thomas J. Overbye:

To Jo, Tim, Hannah, and Amanda

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Preface

The objective of this book is to present methods of power system analysis and design, particularly with the aid of a personal computer, in sufficient depth to give the student the basic theory at the undergraduate level. The approach is designed to develop students' thinking processes, enabling them to reach a sound understanding of a broad range of topics related to power system engineering, while motivating their interest in the electrical power industry. Because we believe that fundamental physical concepts underlie creative engineering and form the most valuable and permanent part of an engineering education, we highlight physical concepts while giving due attention to mathematical techniques. Both theory and modeling are developed from simple beginnings so that they can be readily extended to new and complex situations.

NEW TO THIS EDITION

We welcome Adam B. Birchfield as a co-author for this edition. Dr. Birchfield insures that the text will continue to be updated with the latest technological advances in power systems.

New chapter-opening case studies bring principles to life for students by providing practical, real-world engineering applications for the material discussed in each chapter.

Comprehensively revised problem sets ensure students have the practice they need to master critical skills.

KEY FEATURES

The text presents present-day, practical applications and new technologies along with ample coverage of the ongoing restructuring of the electric utility industry. It is supported by an ample number of worked examples, including illustrations, covering most of the theoretical points raised. It also includes PowerWorld Simulator version 22 to extend fully worked examples into computer implementations of the solutions. Version 22 includes power flow, optimal power flow, contingency analysis, short circuit, and transient stability.

The text includes a chapter on Power Distribution with content on Smart Grids.

It also includes discussions on modeling of wind turbines in power flow and transient stability.

Four design projects are included, all of which meet ABET requirements.

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POWERWORLD SIMULATOR

One of the most challenging aspects of engineering education is giving students an intuitive feel for the systems they are studying. Engineering systems are, for the most part, complex. While paper-and-pencil exercises can be quite useful for highlighting the fundamentals, they often fall short in imparting the desired intuitive insight. To help provide this insight, the book uses PowerWorld Simulator version 22 to integrate computer-based examples, problems, and design projects throughout the text.

PowerWorld Simulator was originally developed at the University of Illinois at Urbana-Champaign to teach the basics of power systems to nontechnical people involved in the electricity industry, with version 1.0 introduced in June 1994. The program's interactive and graphical design made it an immediate hit as an educational tool, but a funny thing happened—its interactive and graphical design also appealed to engineers doing analysis of real power systems. To meet the needs of a growing group of users, PowerWorld Simulator was commercialized in 1996 by the formation of PowerWorld Corporation. Thus while retaining its appeal for education, over the years PowerWorld Simulator has evolved into a top-notch analysis package, able to handle power systems of any size. PowerWorld Simulator is now used throughout the power industry, with a range of users encompassing universities, utilities of all sizes, government regulators, power marketers, and consulting firms.

In integrating PowerWorld Simulator with the text, our design philosophy has been to use the software to extend, rather than replace, the fully worked examples provided in previous editions. Therefore, except when the problem size makes it impractical, each PowerWorld Simulator example includes a fully worked hand solution of the problem along with a PowerWorld Simulator case. This format allows students to simultaneously see the details of how a problem is solved and a computer implementation of the solution. The added benefit from PowerWorld Simulator is its ability to easily extend the example. Through its interactive design, students can quickly vary example parameters and immediately see the impact such changes have on the solution. By reworking the examples with the new parameters, students get immediate feedback on whether they understand the solution process. The interactive and visual design of PowerWorld Simulator also makes it an excellent tool for instructors to use for in-class demonstrations. With numerous examples utilizing PowerWorld Simulator instructors can easily demonstrate many of the text topics. Additional PowerWorld Simulator functionality is introduced in the text problems and design projects.

The latest version of the valuable PowerWorld Simulator (version 22) is included and integrated throughout the text.

PREREQUISITES

As background for this course, it is assumed that students have had courses in electric network theory (including transient analysis) and ordinary differential

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equations and have been exposed to linear systems, matrix algebra, and computer programming. In addition, it would be helpful, but not necessary, to have had an electric machines course.

ORGANIZATION

The text is intended to be fully covered in a two-semester or three-quarter course offered to seniors and first-year graduate students. The organization of chapters and individual sections is flexible enough to give the instructor sufficient latitude in choosing topics to cover, especially in a one-semester course. The text is supported by an ample number of worked examples covering most of the theoretical points raised. The many problems to be worked with a calculator as well as problems to be worked using a personal computer have been revised in this edition.

After an introduction to the history of electric power systems along with present and future trends, *Chapter 2* orients the students to the terminology and serves as a brief review of fundamentals. The chapter reviews phasor concepts, power, network equations, single-phase as well as balanced three-phase circuits and a brief discussion of energy conversion.

Chapters 3 through 5 examine power transformers including the per-unit system, transmission-line parameters, and steady-state operation of transmission lines. *Chapter 6* examines power flows including the Newton-Raphson method, control of power flow, sparsity techniques, and power-flow modeling of wind and solar generation. Chapter 7 covers economic dispatch and optimal power flow, including coverage of unit commitment and markets. These chapters provide a basic understanding of power systems under balanced three-phase, steady-state, normal operating conditions.

Chapters 8 through 11, which cover symmetrical faults, symmetrical components, unsymmetrical faults, and system protection, come under the general heading of power system short-circuit protection. *Chapter 12* examines transient stability, which includes the swing equation, the equal-area criterion, and multimachine stability with modeling of wind turbine and solar PV systems. *Chapter 13* covers power system controls, including generator-voltage control, turbine-governor control, load frequency control, and power system stabilizer control. *Chapter 14* examines transient operation of transmission lines including power system overvoltages, insulation coordinationand surge protection.

Chapter 15 introduces the basic features of primary and secondary distribution systems as well as basic distribution components including distribution substation transformers, distribution transformers, and shunt capacitors. We list some of the major distribution software vendors followed by an introduction to distribution reliability, distribution automation, and smart grids.

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Preface to the SI Edition

This edition of *POWER SYSTEM ANALYSIS & DESIGN* has been adapted to incorporate the International System of Units (*Le Système International d'Unités* or SI) throughout the book.

Le Système International d'Unités

The United States Customary System (USCS) of units uses FPS (foot–pound–second) units (also called English or Imperial units). SI units are primarily the units of the MKS (meter–kilogram–second) system. However, CGS (centimeter–gram–second) units are often accepted as SI units, especially in textbooks.

USING SI UNITS IN THIS BOOK

In this book, we have used both MKS and CGS units. USCS (U.S. Customary Units) or FPS (foot-pound-second) units used in the US Edition of the book have been converted to SI units throughout the text and problems. However, in case of data sourced from handbooks, government standards, and product manuals, it is not only extremely difficult to convert all values to SI, it also encroaches upon the intellectual property of the source. Some data in figures, tables, and references, therefore, remains in FPS units. For readers unfamiliar with the relationship between the USCS and the SI systems, a conversion table has been provided inside the front cover.

To solve problems that require the use of sourced data, the sourced values can be converted from FPS units to SI units just before they are to be used in a calculation. To obtain standardized quantities and manufacturers' data in SI units, readers may contact the appropriate government agencies or authorities in their regions.

INSTRUCTOR RESOURCES

The Instructors' Solution Manual in SI units is available through your Sales Representative or online through the book website at http://login.cengage.com. A digital version of the ISM, Lecture Note PowerPoint slides for the SI text, as well as other resources are available for instructors registering on the book website.

Feedback from users of this SI Edition will be greatly appreciated and will help us improve subsequent editions.

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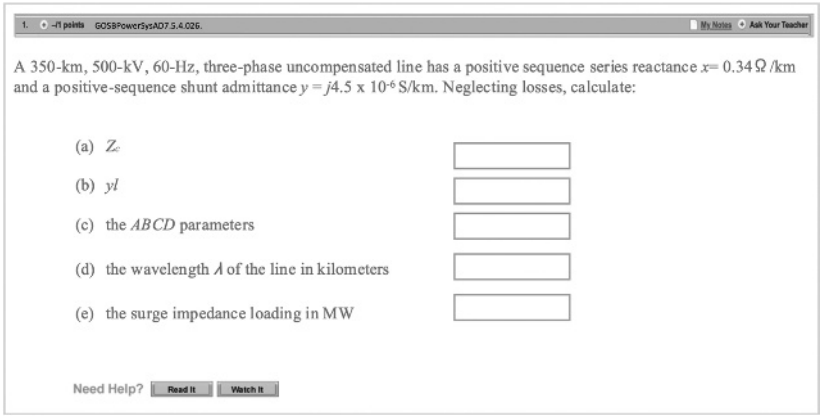

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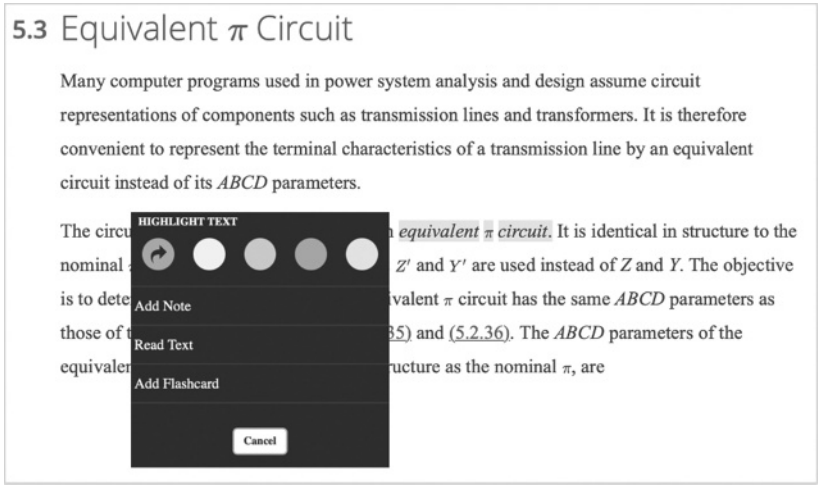

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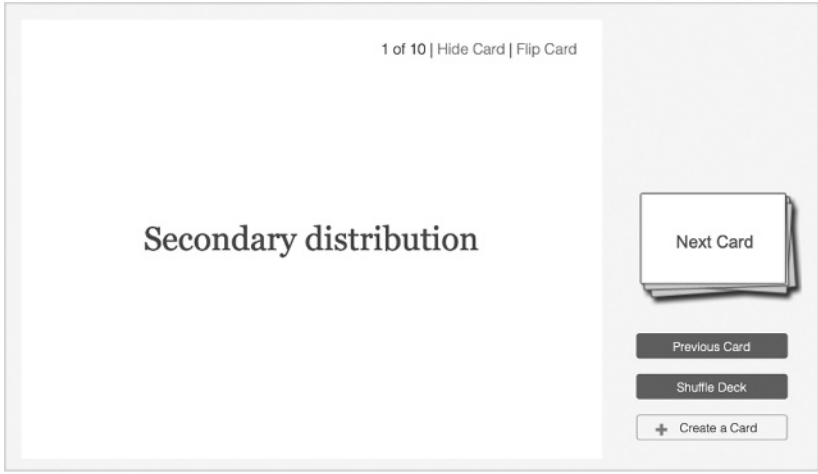

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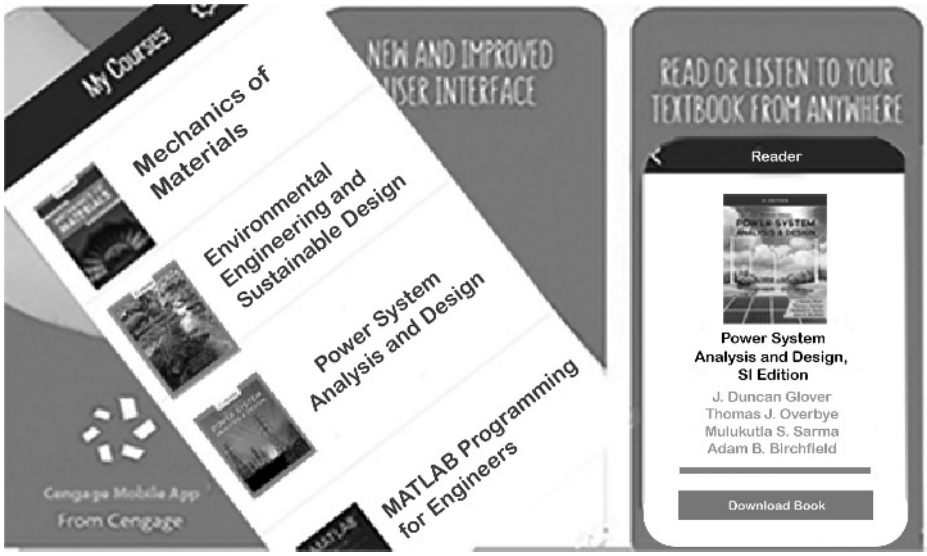

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The **Student Companion Site** includes a link to download the free student version of PowerWorld, Case Studies, and Student PowerPoint Notes.

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ACKNOWLEDGMENTS

The material in this text was gradually developed to meet the needs of classes taught at universities in the United States and abroad over the past 40 years. The original 13 chapters were written by the first author, J. Duncan Glover, *Failure Electrical LLC*, who is indebted to many people who helped during the planning and writing of this book. The profound influence of earlier texts written on power systems, particularly by W. D. Stevenson, Jr., and the developments made by various outstanding engineers are gratefully acknowledged. Details of sources can only be made through references at the end of each chapter, as they are otherwise too numerous to mention.

Chapter 7 (*Economic Dispatch and Optimal Power Flow*), included in Chapter 6 for prior editions, is now a separate chapter for the 7th edition.

Chapter 15 *(Power Distribution)* was a collaborative effort between Dr. Glover (Sections 15.1-15.7) and Co-author Thomas J. Overbye (Sections 15.8 & 15.9). Professor Overbye, *Texas A&M University also authored Sections 1.5, 2.8, 6.10, 6.11, 6.12, 7.4, 7.5, 12.6, 12.7, 12.8 and 13.4.* He further contributed to Section 13.1, provided PowerWorld Simulator examples and problems throughout the text and three design projects. Co-author Mulukutla Sarma, *Northeastern University (Emeritus), and Adam Birchfield, Texas A&M University,* contributed to end-of-chapter problems.

We commend the following Global Engineering team members at Cengage Learning: Timothy Anderson, Senior Product Manager; Sharon Chambliss, Senior Content Manager; and MariCarmen Constable, Learning Designer; as well as Rose Kernan of RPK Editorial Services, Inc. for their broad knowledge, skills, and ingenuity in publishing this edition. We also thank Matt Wood, Program Manager, Taylor Shenberger, Portfolio Marketing Manager, Simeon Lloyd-Wingard, Product Assistant.

Reviewers

Substantial contributions to prior editions of this text were made by a number of invaluable reviewers:

Max D. Anderson, *University of Missouri-Rolla*
Sohrab Asgarpoor, *University of Nebraska-Lincoln*
Kaveh Ashenayi, *University of Tulsa*
Ross Baldick, *University of Texas at Austin*
Thomas L. Baldwin, *Florida State University*
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Mahyar Zarghami, *California State University at Sacramento*

In conclusion, the objective in writing this text and the accompanying software package will have been fulfilled if the book is considered to be student-oriented, comprehensive, and up to date, with consistent notation and necessary detailed explanation at the level for which it is intended.

J. Duncan Glover
Thomas J. Overbye
Adam B. Birchfield
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List of Symbols, Units, and Notation

Symbol	Description	Symbol	Description
a	operator 1/120°	l	length
a_t	transformer turns ratio	l	length
A	area	L	inductance
A	transmission line parameter	**L**	inductance matrix
$\boldsymbol{A}$	symmetrical components transformation matrix	N	number (of buses, lines, turns, etc.)
		Q	reactive power
B	Susceptance	p.f.	power factor
B	Susceptance Matrix	$p(t)$	instantaneous power
B	loss coefficient	P	real power
B	frequency bias constant	q	Charge
B	phasor magnetic flux density	r	radius
B	transmission line parameter	R	resistance
C	capacitance	R	turbine-governor regulation constant
C	transmission line parameter		
D	damping	**R**	resistance matrix
D	distance	s	Laplace operator
D	transmission line parameter	S	apparent power
E	phasor source voltage	S	complex power
E	phasor electric field strengths	t	time
$\boldsymbol{f}$	frequency	T	period
G	conductance	T	temperature
G	conductance matrix	T	torque
H	normalized inertia constant	$v(t)$	instantaneous voltage
H	phasor magnetic field intensity	V	voltage magnitude (rms unless otherwise indicated)
$i(t)$	instantaneous current		
I	current magnitude (rms unless otherwise indicated)	V	phasor voltage
		$\boldsymbol{V}$	vector of phasor voltages
I	phasor current	X	reactance
$\boldsymbol{I}$	vector of phasor currents	**X**	reactance matrix
j	operator 1/90°	Y	phasor admittance
J	moment of inertia		

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Symbol	Description	Symbol	Description
$\boldsymbol{Y}$	admittance matrix	λ	Penalty factor
Z	phasor impedance	Γ	reflection or refraction λ coefficient
$\boldsymbol{Z}$	impedance matrix		
α	angular acceleration	θ	impedance angle
α	transformer phase shift angle	θ	angular position
β	current angle	μ	permeability
β	area frequency response characteristic	υ	velocity of propagation
		ω	radian frequency
δ	voltage angle	ρ	resistivity
δ	torque angle	τ	time in cycles
ϵ	permittivity	τ	transmission line transit time
Φ	magnetic flux		

	SI Units	English Units	
A	ampere	BTU	British thermal unit
C	coulomb	Cmil	circular mil
F	farad	ft	foot
H	henry	hp	horsepower
Hz	hertz	in	inch
J	joule	mi	mile
kg	kilogram		
m	meter		
N	newton		
rad	radian		
s	second		
S	siemen		
VA	volt-ampere		
var	volt-ampere reactive		
W	watt		
Wb	weber		
Ω	ohm		

Notation

Lowercase letters such as v(t) and i(t) indicate instantaneous values.
Uppercase letters such as V and I indicate rms values.
Uppercase letters in italic such as V and I indicate rms phasors.
Matrices and vectors with real components such as **R** and **I** are indicated by boldface type.
Matrices and vectors with complex components such as $\boldsymbol{Z}$ and $\boldsymbol{I}$ are indicated by boldface italic type.
Superscript T denotes vector or matrix transpose.
Asterisk (*) denotes complex conjugate.
PW highlights problems that utilize PowerWorld Simulator.

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1 Introduction

Blundell geothermal power plant near Milford, UT, USA. This 38-MW plant consists of two generating units powered by geothermal steam. Steam is created from water heated by magma at depths up to 6100 meters below Earth's surface. (Courtesy of PacifiCorp.)

LEARNING OBJECTIVES

At the conclusion of this chapter, you should be able to:

1. Briefly explain the history of the electric utility industry;
2. Discuss present and future trends in electric power systems;
3. Describe the restructuring of the electric utility industry;
4. Get up and running with PowerWorld Simulator, a power system analysis and simulation software package.

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Electrical engineers are concerned with every step in the process of generation, transmission, distribution, and utilization of electrical energy. The electric utility industry is probably the largest and most complex industry in the world. The electrical engineer who works in that industry encounters challenging problems in designing future power systems to deliver increasing amounts of electrical energy in a safe, clean, and economical manner.

CASE STUDY

The following article describes the impacts that Distributed Energy Resources (DERs) are having on bulk power systems in four of the United States: Hawaii, California, New York, and North Carolina; as well as in South Australia. As DERs continue to grow around the world, the aggregate amount of them is having impacts on bulk power system planning and operation. The power system industry is learning from prior experience and taking actions in anticipation of future DER penetration levels. The article also describes the ongoing efforts of the North American Reliability Corporation, in coordination with its stakeholders, to study the effect of increasing DER penetration on bulk power systems [8].

Transformation of the Grid*

Ryan Quint, Lisa Dangelmaier, Irina Green,
David Edelson, Vijaya Ganugula, Robert Kaneshiro,
James Pigeon, Bill Quaintance, Jenny Riesz,
and Naomi Stringer

DISTRIBUTED ENERGY RESOURCES (DERs) are unlocking new opportunities, and the grid is undergoing a dramatic transformation with unprecedented change. Yet as DERs continue to grow in North America and around the world, it is apparent that the aggregate amount of them is having an impact on bulk power system (BPS) planning and operation. The effects of DERs can be attributed to the uncertainty, variability, and lack of visibility of these resources at the BPS level. From a BPS perspective, the key grid planning impacts generally include the following:

- transmission-distribution coordination and data exchange
- visibility, dispatchability, and controllability
- DER ride-through capability

* Transformation of the Grid, by Ryan Quint et al, © 2019 IEEE. Reprinted, with permission, from *Power & Energy Magazine* (November/December 2019), pp. 35–45.

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- impacts to load-shedding programs
- aggregate DER modeling and changing reliability study approaches.

The industry is learning from prior experience and taking actions in anticipation of future DER penetration levels. Waiting for the effect of DERs to manifest before developing solutions may be extremely costly and even risk BPS reliability. However, the proactive development and coordination of requirements can ensure reliable operation of the BPS moving forward. The electric industry needs to address these challenges with innovative solutions earlier rather than later.

Distributed Energy Resource (DER) Impacts on Hawai'i's Grid Operation

The islands of Hawai'i, Oahu, and Maui have among the highest penetration levels of DERs in the United States, in terms of installed capacity relative to system size. Under favorable policies and the right economic circumstances, the rapid deployment of DERs can occur in an extremely short period of time. In only six years, the average distributed solar photovoltaic (D-PVs) contribution to meeting the gross peak daytime demand (net peak plus D-PVs) increased from 12 to 37% on Hawai'i island (see Figure 1). The instantaneous D-PVs penetration can exceed 71% of the daytime demand today, and it provides approximately 11% of the annual energy supply. With increasing DER penetration

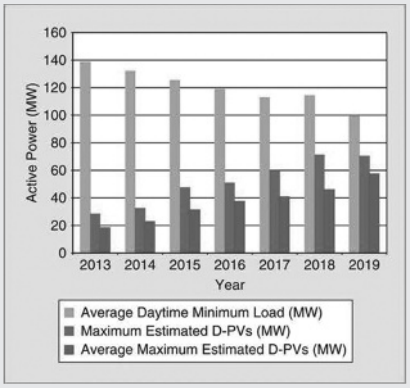

Figure 1 The rapid deployment of D-PVs for the island of Hawai'i.

levels come rapidly rising levels of variability and uncertainty. Hawai'i is now seeing more D-PVs systems with battery energy storage systems (BESSs), and it is anticipating the effect that these combined systems will have on the island grid. Needless to say, the long-term planning studies performed 10 years ago never anticipated such a major change.

The performance requirements needed for high DER penetration levels on the island grids were not initially supported by the requirements in IEEE Standards 1547-2003 and 1547a-2014. Hawai'i's DER interconnection standard (Rule 14 H) eventually deviated from IEEE Standard 1547 to support the continued integration of DERs. However, the new IEEE Standard 1547-2018 includes requirements that are beyond Rule 14 H and under consideration for inclusion in a revised Hawai'i standard. Based on Hawai'i's experience, robust interconnection requirements need to be in place well in advance

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of high penetration levels to support BPS reliability. It is extremely challenging and expensive to meet changing BPS needs by retroactively enhancing equipment that was installed with minimal performance requirements and capabilities.

For the Hawai'i grid, the following aspects of DER integration are top priorities:

- balancing solar PV variability with flexible energy resources that are capable of fast ramping and cycling and ensuring that regulation resources do not fall below minimum allowable dispatch levels
- managing frequency stability with sufficient frequency-responsive reserves and checking that newly installed DERs have active power-frequency controls enabled to assist the primary frequency response
- improving DER estimation techniques since nearly all DERs are behind-the-meter (BTM), with no visibility or control by the system operator; D-PVs output is estimated using field irradiance measurements provided to the operator via supervisory control and data acquisition
- ensuring effective system restoration with variable DER auto-reconnection and possibly adjusting reconnection criteria (although this will not address situations of high solar PV periods where DER control is necessary)
- studying the adverse impacts of DERs tripping during BPS faults and working with existing DER installations to modify trip settings and improve ride-through capabi-1ity, where possible.

There are also two issues related to frequency stability that are worth highlighting: the use of fast-responding battery energy storage to mitigate legacy DER tripping and the development of an adaptive UFLS program to ensure system security during severe resource-loss events.

Storage to Mitigate DER Loss

Some legacy DERs do not have a robust ride-through capability and, therefore, are subject to aggregate loss during over/under frequency and voltage excursions. The behavior of D-PVs is determined by the standard requirements at the time of interconnection, with different possible aggregate losses during these frequency and voltage excursions. The largest contingency concern for Hawai'i island is the loss of legacy DERs tripping at 60.5 Hz, as a large majority of the existing D-PVs is subject to trip at or near that frequency. The second-largest vulnerability is the loss of DERs during large-voltage excursions. On an island system like Hawai'i's, 60.5 Hz is possible after delayed fault clearing (due to transient swings following fault clearing) and major loss-of-load events, which can occur with transmission outages (both N-1 and N-1-1). The potential aggregate loss of D-PVs during high solar-production periods is roughly double the largest single-generator contingency, resulting in severe underfrequency conditions and the possible risk of system failure.

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The loss of D-PVs also exacerbates other loss-of-generation events, with some additional D-PVs loss during the underfrequency situation. To address this concern, a battery energy storage study was conducted to analyze the effect of legacy DER tripping for low voltage during transmission faults, low frequency following generating-unit trips, and high frequency following a transmission line fault. The study found that retrofitting legacy DERs to full ride-through capability was the most effective solution, but that it was too costly and impractical. The study also found the following:

- Storage can replace DER energy lost during transient voltage and frequency conditions, preventing excessive underfrequency protection and reducing the risk of system failure for disturbances during high solar PV production.
- The size of the storage necessary to mitigate reliability issues depends on the number of legacy DERs installed.
- Increased numbers of spinning reserves could reduce the size of the necessary storage but not eliminate the need for it. Increasing reserves also exacerbates excess-energy concerns.
- D-PVs protection systems are not consistently implemented, and performing sensitivity studies around the uncertainty of this DER behavior is critical.

Based on the studies, a BESS with 18 MW of capacity and a 30-minute duration was able to arrest frequency excursions and provide sufficient time to bring standby generation online. Increasing the duration to 1 hour enabled the BESS to respond to over-frequency conditions. Studies showed that two BESSs geographically and electrically separated from one another provided the best performance and grid resilience.

Underfrequency Load Shedding

The reliable operation of the Hawai'i island systems heavily relies on underfrequency load shedding (UFLS) that trips distribution circuits at preestablished frequency thresholds. However, as the penetration of DERs rapidly increases, variability in net loading on any given feeder poses significant challenges to the conventional UFLS design, which means that static UFLS arming is no longer effective. An adaptive UFLS program has been implemented to address this variability. Figure 2 is a dashboard of the adaptive program, including six instantaneous stages of UFLS arming between 59.1 and

UFLS STAGE DATA | Change to Monitor | Reset and Calculate | Mode: Active | System Load: 151.629 | Total Target: 120.559 | Total Available: 127.133

Stage	Frequency	Percent	Target Megawatts	Available Megawatts	Tolerance %	Tolerance	Delta Megawatts
Stage 1	59.1	5	7.53492	7.68113	5	0.377	−0.146
Stage 2	58.8	10	15.06987	14.93933	5	0.753	0.131
Stage 3	58.5	10	15.06987	15.45847	5	0.753	−0.1389
Stage 4	58.2	15	22.60476	22.63691	8	1.808	−0.032
Stage 5	57.9	10	15.06987	14.63551	8	1.206	0.434
Stage 6	57.6	20	30.13968	30.76797	25	7.535	−0.628
Kicker 1	59.3	5	7.53492	7.90598	5	0.377	−0.371
Kicker 2	59.5	5	7.53492	7.72132	5	0.377	−0.186

Figure 2 The adaptive UFLS-arming dashboard.

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57.6 Hz and two kicker stages that operate with time delays at 59.3 and 59.5 Hz. The target megawatt value is calculated based on system net loading at the time, and available distribution circuits are automatically armed every 15 minutes to achieve the target net load level. An arming tolerance percentage and value are depicted as well as the difference between the available megawatt arming and the target megawatt value. Stages 1–2 sum to 15% and stages 1–4 sum to 40% of the system net load based on the maximum allowable load shedding for N-1 and N-1-l unit trips, respectively. Each stage operates with an eight-cycle relay time plus breaker operation time. A rate-of-change-of-frequency (ROCOF) load shedding element was also included in stages 1 and 2, with a 0.5-Hz/s set point and nine-cycle relay time plus breaker operation time.

More than 40 substations were modified, including relay and communications upgrades, and 78% of customer circuits (up to 70% of peak load) are participating. The adaptive UFLS program has performed well during N-1 and N-1-1 events at various times of day and is an effective reliability improvement. Figure 3 provides an example of a severe N-1-1 generator-loss disturbance where stages 1–4 operated successfully to arrest the frequency and mitigate a potential large-scale outage.

Rooftop Solar PV Deployment

California has a state mandate that 33% of its annual energy must come from renewable sources by 2020, with a target of 60% by 2030, established by a bill, SB100, that took effect in September 2018. However, on some days, instantaneous output from the renewable sources already exceeds 50% of the total generation output. A large portion of that penetration consists of BTM D-PVs, and the rapid deployment of D-PVs is forcing changes in transmission planning, forecasting, and grid operations at the California Independent System Operator (CAISO).

Figure 4 provides an example of the effect that D-PVs are having

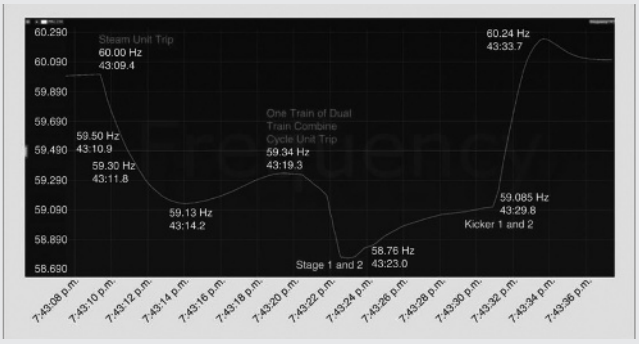

Figure 3 The operation of adaptive UFLS for a multiple-contingency event.

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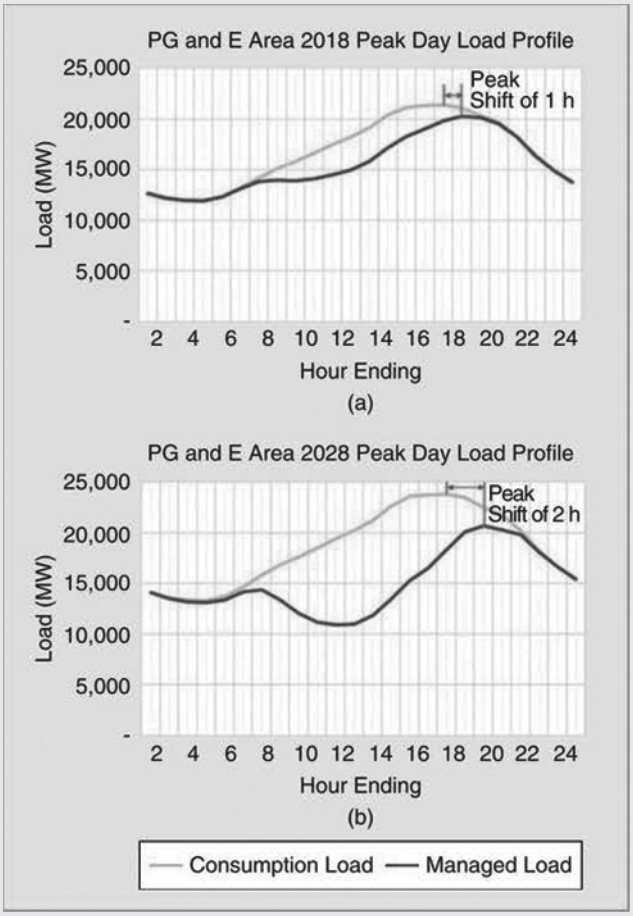

Figure 4 The PG&E net load impacts from DERs.

on the Pacific Gas & Electric (PG&E) net load profile, driving the need for different study times and net load levels from the CAISO's 2018–2019 transmission-planning process. The blue curves show the net load after subtracting the D-PVs' output, which is called the managed load. The figure clearly displays a shift in the peak net load time, which, in many parts of the state, is shifted outside the times when solar PV is available. D-PVs' penetration levels are also causing a drastic change in minimum net load levels, which need to be carefully studied. Sensitivity analysis is also needed to operate through and plan for ramping periods. All these issues are compounded by load modifiers, such as energy efficiency, demand response, time-of-use rates, and electric vehicle charging.

The uncertainty in net load forecasting, including D-PVs, is a concern

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for BPS reliability studies. The CAISO coordinates its load forecasts with and relies upon long-term forecasts produced by the California Energy Commission (CEC). The CEC's energy-demand forecast includes an hourly estimate of the consumption load and load modifiers to develop the managed outlook by which the transmission system is planned.

Since D-PVs are shifting net peak load periods, the selection of critical study conditions needs to change. Planning standards require an assessment of peak and off-peak conditions; however, with large numbers of DERs, critical system conditions may not occur during either of those periods. This requires even greater numbers of sensitivity studies of varying dispatch and load scenarios, including peak gross load, peak net load (at the transmission-distribution interface), and minimum net load conditions (for example, a spring weekend during off-peak load conditions). Tables 1 and 2 show how DERs were incorporated into the CAISO study scenarios during the 2018–2019 transmission planning process. In addition to these sensitivities, consideration for BPS-connected

	Day/Time			BTM-PV			AAEE			
Scenario	2020	2023	2028	2020	2023	2028	2020	2023	2028	Driver
Summer peak	10 Aug. HE 18	14 Aug. HE 19	14 Aug. HE 19	18%	3%	3%	90%	81%	76%	Hour of maximum managed load
Spring off peak	5 Apr. HE 12	6 Apr. HE 13	16 Apr. HE 13	79%	84%	85%	58%	56%	49%	Hour of minimum managed load
Winter off peak			13 Feb. HE 4			0%			28%	Hour of minimum consumption
Winter peak	15 Oct. HE 19	3 Oct. HE 18	3 Oct. HE 19	0%	1%	0%	77%	76%	72%	Hour of maximum managed load during winter

AAEE: additional achievable energy efficiency; HE: hour ending.

TABLE 1

The baseline scenarios and DER contributions.

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Scenario	Starting Baseline Case	BTM-PV		AAEE	
		Baseline	Sensitivity	Baseline	Sensitivity
Summer peak with high CEC forecasted load	2023 summer peak	3%	3%	81%	0%
Off peak with heavy renewable output and minimum gas generation commitment	2023 off peak	84%	99%	56%	56%
Summer peak with heavy renewable output and minimum gas generation commitment	2020 summer peak	18%	99%	81%	81%

TABLE 2

The sensitivity scenarios and DER contributions.

generation dispatch and assumptions for neighboring planning areas is critical as they impact local or wide-area reliability issues. This leads to the need for wider coordination between planning administrators.

With the increasing penetration of DERs, CAISO has explicitly incorporated the forecast resources in power flow and stability studies for the past three planning cycles to account for their unique operational characteristics that are different from the end-use load behavior. This is especially important for stability studies, since the results between DERs netted with load versus modeled ones are significantly different. With a large penetration of DERs in California, study results would be inaccurate if DERs were only netted with load and would lead to erroneous conclusions. Retail-scale BTM DERs are modeled in the power flow and dynamic stability studies as part of a composite load model, and utility-scale DERs are modeled as aggregated generators on the transmission buses. Figure 5 illustrates these impacts on transient voltage response following a simulated BPS fault event. The results are different because modeling the DER characteristics explicitly (including DER tripping of legacy equipment) allows the simulation to differentiate between DERs' response and gross load response. This is critical as penetration levels of DERs increase, although DER prevalence has a study impact in all cases, even at low penetration levels.

CAISO currently uses parameter values for the DER models based on the Western Electricity Coordinating Council and North American Electric Reliability Corporation (NERC) guidelines and performs multiple sensitivity studies

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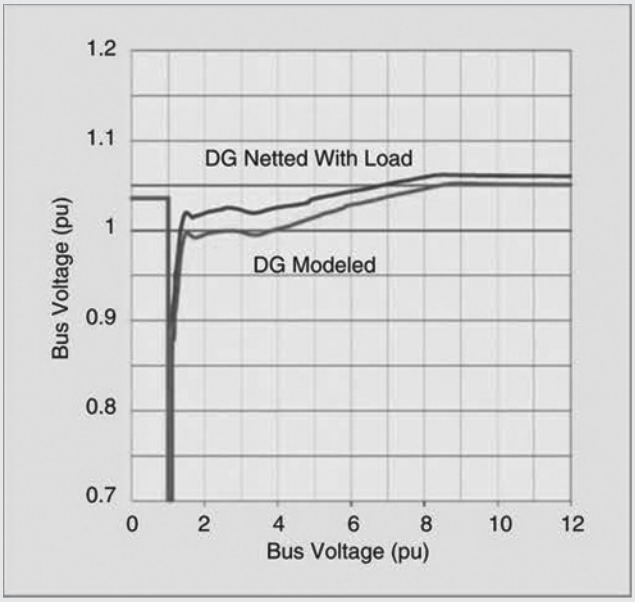

Figure 5 The 230-kV bus voltage close to a three-phase fault, with and without DER modeled. pu: per unit; DG: distributed generation.

to determine the impact that variations in the DER parameters have on BPS performance: The most sensitive parameters DER undervoltage trip settings, the fraction of DERs that recover when voltage returns, and whether DERs provide voltage regulation (and the characteristics of that regulation). Figure 6 illustrates the net load and DER response on a 230-kV bus and the BPS voltage for a three-phase fault close to the BPS bus. The plots show the DERs netted with load as well as two cases of DERs modeled explicitly with different voltage trip settings. One set uses the IEEE Standard 1547-2018 settings, and the other uses California Rule 21 settings. The DERs are on the different feeders and aggregated to the 230-kV bus. System performance differences are dominated by the different responses of the gross load and DERs. Although the difference is not that significant for one bus, the aggregate effect across a large portion of the system has a significant impact on the study results.

New York's DER Management

New York State is on the cusp of a transformation as it expects a significant penetration of DERs. New York's Reforming the Energy Vision and Clean Energy Standard initiatives, combined with efforts to expand transfer capability on the BPS, offer a comprehensive energy strategy to achieve its goal of 50% of its electricity delivered by renewable

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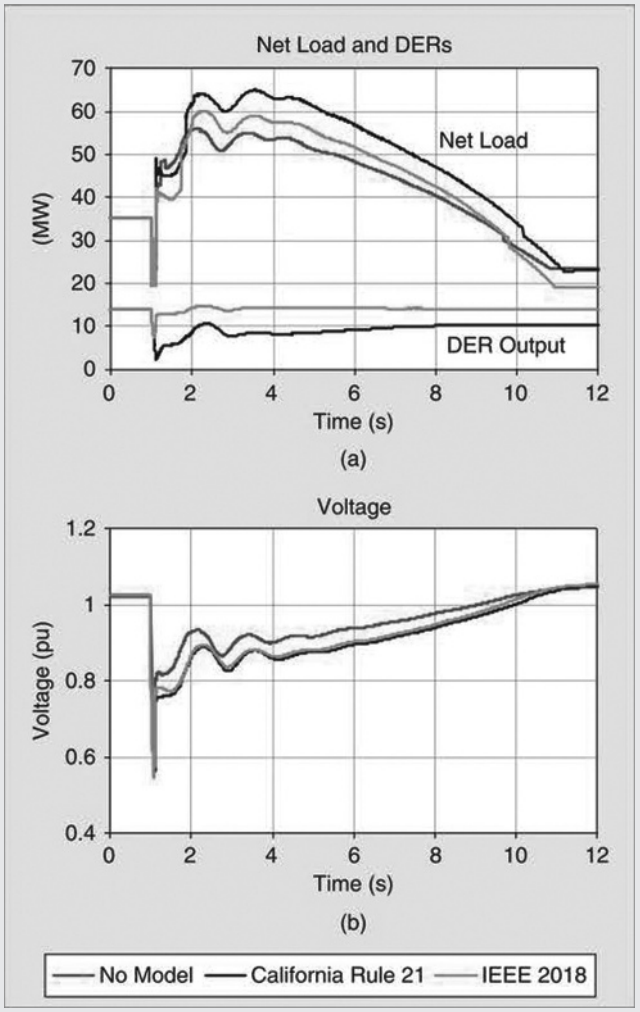

Figure 6 The sensitivities of net load, DER output, and BPS bus voltage to DER modeling settings.

energy sources by 2030. New York has also established a policy to achieve 1,500 MW of energy storage by 2025 and 3,000 MW by 2030.

The New York Independent System Operator (NYISO) expects DERs to play a significant role in achieving the state's goals. Due to the expected growth of DERs in the near future, NYISO is developing market-based mechanisms to integrate DERs into the

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wholesale electricity markets that it administers. NYISO's DER market design is expected to be implemented in 2021. The proposed design recognizes the unique characteristics of DERs. For example, the market design acknowledges that individual DERs may be small in size yet aggregate to a large capacity and that they can be fast-acting but with a limited capability to deliver energy for long durations. The proposed market design considers DERs that may have the capability to provide services to wholesale markets, retail markets, and end-use customers (see Figure 7).

NYISO's DER integration proposal focuses on enhancing operational coordination among BPS and distribution system operators and on developing market rules that allow DERs to aggregate to provide wholesale market services. Individual DERs that do not have sufficient capability to participate in the wholesale markets can aggregate with other DERs as a larger virtual resource. NYISO is lowering its minimum participation threshold from 1 MW to 100 kW for DERs. Aggregators can enroll many DERs of different technologies at different physical locations (as long as they are behind the same BPS transmission node) and choose to use DERs in the aggregation, or a subset of those DERs, to respond to NYISO market dispatch instructions. DER aggregators can submit offers to provide energy, regulation, operating reserves, and capacity. To determine the least-cost resource dispatch, NYISO will evaluate these offers with those from other BPS supply resources in its multi-interval, security-constrained economic dispatch to co-optimize energy, regulation, and operating reserves. Dispatch instructions from NYISO will be directed to the entire aggregation. DER aggregators will be responsible for

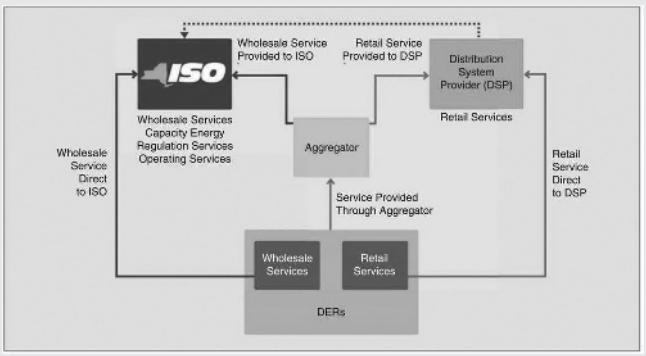

Figure 7 The concept of the NYISO wholesale and retail market designs incorporating DERs.

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1. receiving the aggregation-level dispatch instructions from NYISO in real time
2. identifying the dispatch of individual DERs within the aggregation
3. transmitting dispatch instructions to individual DERs
4. collecting responses from individual DERs
5. consolidating the responses to an aggregation level
6. communicating the aggregation-level response back to NYISO with real-time telemetry.

Performance will be measured for the DER aggregation rather than individual DERs. As NYISO encounters transmission constraints, its market dispatch signals can effectively dispatch the aggregation of DERs to address reliability issues at the lowest cost. This approach allows aggregators the flexibility to meet NYISO dispatches using DERs within the aggregation, based on availability, costs, and other factors.

Enhanced operational coordination among the wholesale system operator, distribution system platform (DSP) operator, aggregator, and individual DERs is necessary to ensure the reliability of the transmission and distribution systems. DERs may provide services to the wholesale and retail markets and to end-use customers (see Figure 7). Individual DERs that make up an aggregation may be located on different distribution feeders and have different impacts on each of them. Enhanced coordination ensures that the DSPs have situational awareness of the impacts of NYISO dispatch signals across the transmission-distribution system interface. In the event of conflicts between the transmission and distribution systems' reliability needs, under NYISO's proposed rules, priority will be given to maintaining the transmission system's reliability. DER aggregators must work with the DSPs to meet local reliability criteria and coordinate with NYISO to deliver the expected services.

Enhanced operational coordination starts two days prior to the operating day, when the DSP communicates any potential or expected outages to the DER aggregator. One day prior to the operating day, aggregators use this information to assess the availability of the overall aggregation and submit DER aggregation offers to the NYISO day-ahead market. NYISO will evaluate these offers with those from other BPS suppliers and schedule day-ahead market resources. Aggregators report to the DSP their intended dispatch of individual DERs to meet the NYISO day-ahead market schedule. DSPs can use this information to identify any potential reliability impacts on their systems and communicate any restrictions on individual DER dispatches to the aggregators.

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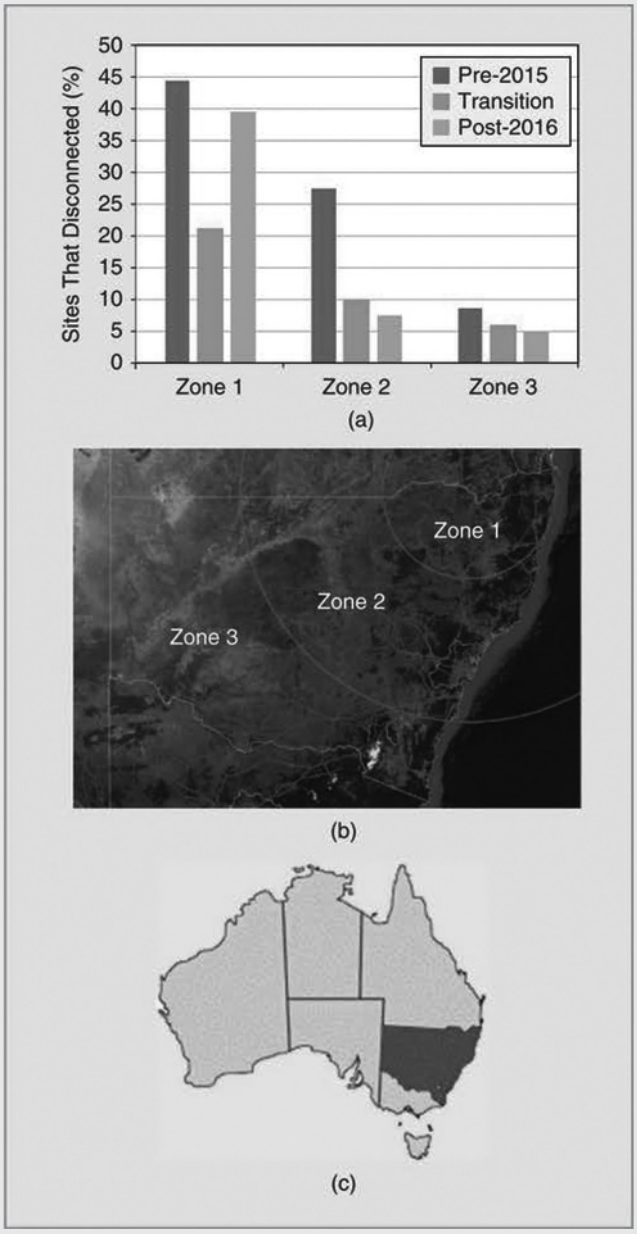

Figure 8 The geographic distribution of the monitored D-PVs sites disconnecting in New South Wales during the 25 August 2018 event.

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Ongoing communication will be necessary among the DSPs, DERs, and aggregators after the NYISO day-ahead market is posted, and it will continue in real time to identify and address changes in the distribution system conditions and potential impacts on individual DERs. Aggregators will be required to communicate aggregation-level changes to NYISO so that they are reflected in the real-time dispatch. The DSPs can also request NYISO operations to dispatch DER aggregations to address reliability issues. This enhanced operational coordination framework recognizes the transmission-distribution system interface and allows DERs to effectively provide services to both the BPS and DSP.

Managing DERs in Australia

The installed capacity of DERs, particularly D-PVs, is rapidly growing in Australia's National Electricity Market (NEM), with more than 6,000 MW installed and a projected increase to 10–17 GW by 2030 (with peak demand of roughly 35 GW). The South Australia region of the NEM has been a focus for analyzing D-PVs integration. This region is geographically the approximate size of Texas, yet with a demand of 600–3,000 MW, it has already experienced periods where more than half of its requirement has been supplied by D-PVs. By 2026–2027, the Australian Energy Market Operator (AEMO) predicts that the D-PVs capacity may be high enough to periodically supply the region's entire demand. The region is connected to the rest of the NEM via a single double-circuit ac interconnector, with the expectation that South Australia will continue to operate as an island if it is separated from the rest of the network. This raises challenges and questions about how to successfully operate when the majority of the load is supplied by D-PVs. To explore these challenges, AEMO launched a program to identify and implement the actions required to maintain the security and operability of the system.

One of the key focus areas in the program is understanding and managing the behavior of DERs during grid disturbances. If a large DER capacity disconnects during a disturbance, the loss exacerbates the disturbance and could lead to instability. AEMO has been gathering data on DER responses during grid disturbances and identified the following:

- D-PVs are more prone to tripping during grid disturbances than end-use loads.
- Net demand post-fault is higher than predisturbance levels, indicative of D-PVs disconnection.
- Data from actual grid disturbances in Victoria, South Australia, and New South Wales estimated that as much

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as 30–40% of the aggregate D-PYs may disconnect during a disturbance.

- Laboratory inverter testing showed multiple reasons for inverter tripping, including voltage phase jumps greater than 30°, ROCOF levels exceeding 0.4 Hz/s, and an inability to meet the ride-through requirements specified in standard AS/NZ4777.2-2015.
- Laboratory testing also showed sluggish responses for control modes (frequency–watt and volt–watt) that are compliant with the standard but provide little support to the grid.

Those findings highlight a number of potential security risks that the AEMO is trying to address. They also suggest a need to review Australian standards to more explicitly specify required responses.

Collecting suitable data related to D-PVs has been one of the main challenges that AEMO has faced in this program. Feeder-level monitoring captures aggregate DERs and load behavior but does not separate the two. Furthermore, disturbances during high D-PVs output periods are rare (for example, voltage disturbances are often associated with lighting strikes where cloud cover impacts D-PVs' output). AEMO has been working with Solar Analytics, Redfern, Australia, and the University of New South Wales, Sydney, Australia, to develop new techniques to analyze DER responses.

Solar Analytics provided AEMO with 1-minute-resolution data for 5,000 D-PVs systems during a disturbance on 25 August 2018. A fault occurred at the Queensland-New South Wales interconnector, causing a voltage dip across the northern part of New South Wales. Figure 8 shows the D-PVs' responses for systems installed prior to October 2015 (AS/NZ4777.3-2005 compliant), after October 2016 (AS/NZ4777.2-2015 compliant), and during the transition period (which could meet either standard). Nearly 45% of the pre-2015 D-PVs in zone 1 (a 540-km-diameter area) disconnected. Compliance to the newer standard appeared to have improved voltage disturbance ride-through capabilities, especially in zone 2 where the voltage dip was less severe. Future versions of DER standards can build on this improvement in capabilities, minimizing risks to power system security.

D-PVs can also provide valuable grid support. AS/NZ4777.2-2015 specifies that D-PVs should provide over-frequency droop response when frequency exceeds 50.25 Hz, with a linear ramp to zero generation by 52 Hz. This was observed during the August 2018 event, as illustrated in Figure 9. The fault led to the separation of Queensland and South Australia from the rest of the NEM, causing over-frequency in those regions.

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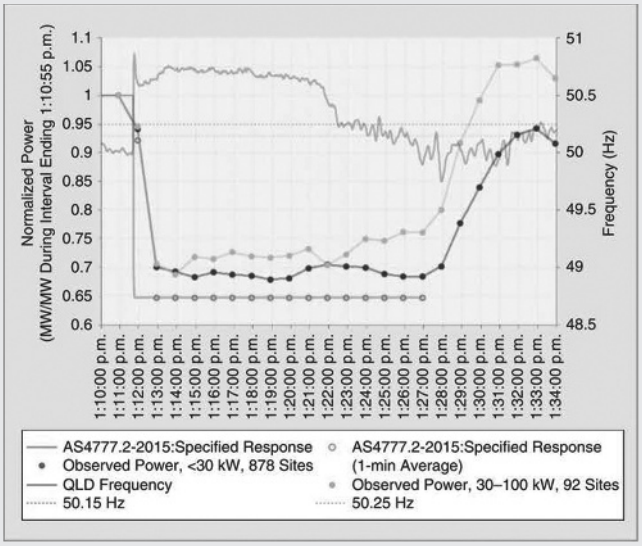

Figure 9 The behavior of post-2016 D-PVs' inverters in Queensland on 25 August 2018.

Many D-PVs systems behaved according to the standard and reduced output to assist with managing frequency. However, more than 15% of the D-PVs in Queensland and more than 30% in South Australia did not behave according to the standard, suggesting issues with compliance.

Those findings demonstrate a range of areas in which the DER standards in Australia need to be reviewed to ensure the reliability of the BPS. The review and implementation of advanced standards should be pursued years in advance of significant DER penetration due to the extensive stakeholder consultation that is required and the need for transition periods for manufacturers to implement new capabilities. Further, dynamic models that represent DER behavior during grid disturbances need to adapt quickly to those changes since these studies are of paramount importance for grid reliability.

The Duke Energy Progress Experience With DERs

Solar PV has grown significantly in North Carolina during the past five years due to government policies, the avoided cost of energy, available land, and falling prices for solar technology. Approximately 3,100 MW of solar PV capacity were connected in North Carolina, as of January 2019, across the Duke Energy North Carolina balancing areas. Roughly 2,300 MW are located in the Duke Energy

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Progress (DEP) region, with a winter peak demand of approximately 15,000 MW. About two-thirds of the solar PV capacity is in utility-scale DERs, with the most common DER capacity being 5 MW, due to past incentives. The rest is transmission connected, with sizes ranging from 20 to 80 MW. Very little solar PV is colocated with customer loads in North Carolina. As such, D-PVs in the state are modeled as a stand-alone generating resource at the distribution level. Location is critically important, and some areas of the transmission grid are saturated with generation and need upgrades to accommodate additional generation of any type. DER back-feeding onto the transmission grid is common, but even without it, DERs are having an impact.

Traditionally, utilities have understood and been able to accurately predict the daily and seasonal fluctuations of customer demand. Unplanned generation or transmission outages have long been incorporated into planning and operating practices. The significant growth of solar PV generation has added a new and independent dimension to planning and operating the BPS. Solar output does not follow load and is generally not dispatchable. Figure 10 presents an example of a daily load curve for generation and customer load in a DEP on a mild winter day. The top orange line represents a typical winter customer load shape, with dual peaks in the morning and evening. Generation resources are separated into baseload nuclear, regulating resources such as gas and coal generation, and solar PV. Solar is further split into distribution- and transmission-connected categories. As the figure shows, the customer load is at a minimum when solar generation is at its maximum on a winter day. Regulating resources that need to be online for the peaks have a minimum output represented by the red line and are potentially forced off at midday due to solar PV generation. However, many regulating resources have start-up and shut-down times of hours or days. The only solutions on some days are to sell energy to neighboring balancing areas at a low price, buy peak energy from neighbors at a high cost, and curtail solar PV generation.

Determining the planning conditions for the BPS requires understanding how the grid operates, which, as discussed, has become more complicated. Transmission planners examine the worst realistic conditions; in North Carolina, that has been the summer and winter peak loads and the valley load. With the addition of DERs, new operating hours are becoming the most limiting or worst case. As shown in Figure 10, generation output can vary significantly throughout the day. While the annual minimum customer load may occur at night, light load conditions on a mild Sunday afternoon during spring can closely correlate with the maximum solar PV output, resulting in the minimum net customer load. BPS voltages can be high, and power-flow patterns may occur that have never been observed in operating practice. During summer, while

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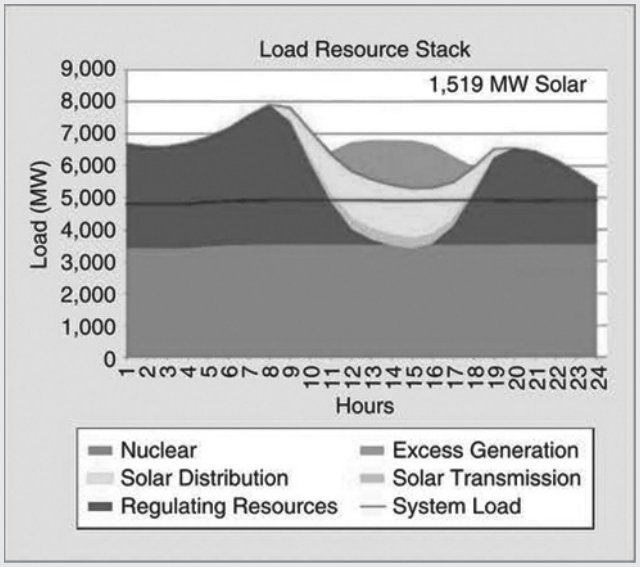

Figure 10 An example of a mild-winter load curve and resource stack in DEP.

the customer load may peak at 5 p.m., high solar PV output earlier in the day may increase flows on the transmission system.

Worst-case conditions can occur during nontraditional planning hours, and DEP is frequently reviewing the actual operating conditions to ensure that planners are focusing on the worst realistic conditions for the BPS. DEP is currently planning for various combinations of customer load and solar output, focusing on the following situations of stressed system conditions:

1. 100% summer peak load, 50% solar output (summer peak hour)
2. 90% summer peak load, 100% solar output (1 p.m. on a summer peak day)
3. 90% summer peak load, 0% solar output (sunset on a summer peak day)
4. 100% winter peak load, 0% solar output (winter peak hour)
5. 35% load, 0% solar output (a mild night during spring)
6. 40% load, 100% solar output (noontime on a mild spring Sunday).

NERC System Planning Impacts of DER Working Group

NERC, in coordination with its stakeholders, has been studying the effect of increasing DER penetration on the BPS. The focus of these efforts is ensuring that the BPS has sufficient essential reliability services: inertial response and stability,

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frequency control and balancing, and reactive power capability and voltage support. While modern inverter-based resources can be equipped with these capabilities and controls, changes in planning and operating paradigms must occur to utilize those capabilities. A BPS with increasing numbers of inverter-based resources, including DERs, presents new challenges that must be addressed to ensure a reliable grid.

NERC has strengthened its focus on DER impacts on the BPS and formed the System Planning Impacts of DER Working Group (SPIDERWG). This group is developing guidance to address the planning and modeling challenges that come with the increasing penetration of DERs across North America. NERC SPIDERWG membership includes distribution and transmission entities across North America, and the body is focusing on four primary topics: advancing aggregate DER modeling capabilities, verifying DER models and performance, developing study techniques with increasing penetration of DERs, and coordination between industry stakeholders.

Summary

While some may argue that the BPS is becoming obsolete in the face of DERs, others believe that it has never been more critical for the reliable delivery of power to end-use customers. However, we can all agree that the breadth of the DER impact on BPS reliability is immense. For now, we are focused on ride-through capability, evolving modeling and study techniques, changing control strategies and requirements, adjusting market designs, and the impending growth of energy storage. Interconnection standards are rapidly evolving to keep up with the transition from centralized control to truly distributed and decentralized architectures. Requirements for grid connection are more important now than ever before, and engagement from transmission providers and grid operators is essential for maintaining BPS reliability. Proactive improvements to interconnection requirements are imperative since retrofitting legacy DERs is cost prohibitive.

DERs pose one of the biggest evolutions in BPS planning and operation since the inception of polyphase ac power systems more than 130 years ago. Anyone who generates, transfers, controls, delivers, or utilizes electric energy should commit to these reliability efforts as we move toward an integrated energy system. Historically, these resources acted passively and were even required to trip offline during disturbances, but they are becoming a major component of the grid and must actively support it as a whole, including the mitigation of variability impacts and maintaining adequate levels of reliability at a reasonable cost. The successful path forward involves collaboration, communication, and coordination among these groups, breaking down barriers that have existed for many years.

For Further Reading

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1.1 HISTORY OF ELECTRIC POWER SYSTEMS

In 1878, Thomas A. Edison began work on the electric light and formulated the concept of a centrally located power station with distributed lighting serving a surrounding area. He perfected his light by October 1879, and the opening of his historic Pearl Street Station in New York City on September 4, 1882 marked the beginning of the electric utility industry (see Figure 1.1). At Pearl Street, dc generators, then called dynamos, were driven by steam engines to supply an initial load of 30 kW for 110-V incandescent lighting to 59 customers in a one-square-mile area. From this beginning in 1882 through 1972, the electric utility industry grew at a remarkable pace—a growth based on continuous reductions in the price of electricity due primarily to technological accomplishment and creative engineering [1].

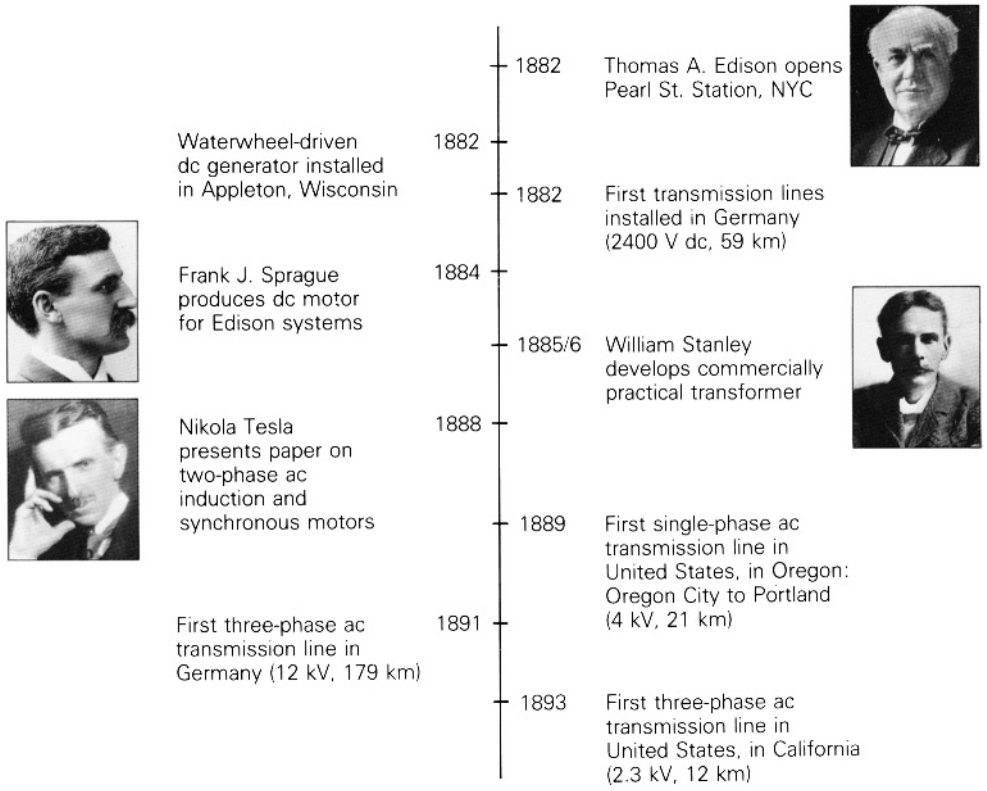

FIGURE 1.1

Milestones of the early electric utility industry [1]. (H.M. Rustebakke et al., *Electric Utility Systems Practice*, 4th Ed. (New York: Wiley, 1983). Reprinted with permission of John Wiley & Sons, Inc. Photos courtesy of Westinghouse Historical Collection.) (Photos courtesy of Westinghouse Historical Collection.) (Based on H.M. Rustebakke et al., Electric Utility Systems Practice, 4th. Ed. (New York: Wiley, 1983).)

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The introduction of the practical dc motor by Sprague Electric, as well as the growth of incandescent lighting, promoted the expansion of Edison's dc systems. The development of three-wire, 220-V dc systems allowed load to increase somewhat, but as transmission distances and loads continued to increase, voltage problems were encountered. These limitations of maximum distance and load were overcome in 1885 by William Stanley's development of a commercially practical transformer. Stanley installed an ac distribution system in Great Barrington, Massachusetts, to supply 150 lamps. With the transformer, the ability to transmit power at high voltage with corresponding lower current and lower line-voltage drops made ac more attractive than dc. The first single-phase ac line in the United States operated in 1889 in Oregon, between Oregon City and Portland—21 km at 4 kV.

The growth of ac systems, further encouraged in 1888 when Nikola Tesla presented a paper at a meeting of the American Institute of Electrical Engineers describing two-phase induction and synchronous motors, made evident the advantages of polyphase versus single-phase systems. The first three-phase line in Germany became operational in 1891, transmitting power 179 km at 12 kV. The first three-phase line in the United States (in California) became operational in 1893, transmitting power 12 km at 2.3 kV. The three-phase induction motor conceived by Tesla went on to become the workhorse of the industry.

In the same year that Edison's steam-driven generators were inaugurated, a waterwheel-driven generator was installed in Appleton, Wisconsin. Since then, most electric energy has been generated in steam-powered and in water-powered (called hydro) turbine plants. Today, steam turbines account for more than 80% of U.S. electric energy generation, whereas hydro turbines account for about 7%. Gas turbines are used in some cases to meet peak loads. Also, the addition of wind turbines and solar cells into the bulk power system is expected to grow considerably in the near future.

Steam plants are fueled primarily by gas, coal, oil, and uranium. From 2012 to 2019, gas and coal exchanged places as the two leading fuels used for electricity production in the United States. Gas increased from 30% in 2012 to 38% in 2019, due to the shale gas revolution in the USA, reduced carbon emissions, and the clean burning feature of gas. Coal, however, decreased from 37% in 2012 to 24% in 2019, due to the retirement of uneconomical coal-fired units and environmental concerns [2].

Starting in the 1990s, the choice of fuel for new power plants in the United States has been natural gas due to its availability and low cost as well as the higher efficiency, lower emissions, shorter construction lead times, safety, and lack of controversy associated with power plants that use natural gas. Natural gas is used to generate electricity by the following processes:

1. Gas combustion turbines use natural gas directly to fire the turbine.
2. Steam turbines burn natural gas to create steam in a boiler, which is then run through the steam turbine.
3. Combined cycle units use a gas combustion turbine by burning natural gas, and the hot exhaust gases from the combustion turbine are used to boil water that operates a steam turbine.

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4. Fuel cells powered by natural gas generate electricity using electrochemical reactions by passing streams of natural gas and oxidants over electrodes that are separated by an electrolyte. In 2019, approximately 38% of electricity in the United States was generated from natural gas [2, 3, 7].

In 1957, nuclear units with 90-MW steam-turbine capacity, fueled by uranium, were installed. As of April 2020, there were 440 nuclear reactors in operation in thirty countries with up to 1600-MW steam-turbine capacity. In 2019, approximately 19% of electricity in the United States was generated from uranium via 96 operating nuclear reactors at 58 power plants. However, the growth of nuclear capacity in the United States has been delayed by rising construction costs, licensing delays, and public opinion. Although there are no emissions associated with nuclear power generation, there are safety issues and environmental issues, such as the disposal of used nuclear fuel and the impact of heated cooling-tower water on aquatic habitats. In 2016, Tennessee Valley Authority's (TVA) Unit-2 reactor at the Watts Bar Nuclear Generating Station became the first U.S. reactor to enter commercial operation since 1996. Also in 2013, construction began on the Vogtle Electric Generating Plant Units 3 and 4 in the state of Georgia, with a November 2021 target in-service date for Unit 3. Future technologies for nuclear power are concentrated on safety and environmental issues [2, 3, 7].

Renewable energy sources have the advantage of providing a more sustainable use of finite energy sources with little or no contribution to polluting emissions. Renewable sources include conventional hydroelectric (water power), geothermal, wood, wood waste, all municipal waste, landfill gas, other biomass, solar, and wind power. In 2019, in the United States, approximately 18% of electricity was generated by renewable sources [2, 3]. Germany, which has an official governmental goal to generate 65% of electricity from renewable sources by 2030, achieved 46% in 2019 [13].

Substantial research efforts have shown nuclear fusion energy to be a promising technology for producing safe, pollution-free, and economical electric energy later in the 21st century and beyond. The fuel consumed in a nuclear fusion reaction is deuterium, of which a virtually inexhaustible supply is present in seawater.

The early ac systems operated at various frequencies including 25, 50, 60, and 133 Hz. In 1891, it was proposed that 60 Hz be the standard frequency in the United States. In 1893, 25-Hz systems were introduced with the synchronous converter. However, these systems were used primarily for railroad electrification (and many are now retired) because they had the disadvantage of causing incandescent lights to flicker. In California, the Los Angeles Department of Power and Water operated at 50 Hz but converted to 60 Hz when power from the Hoover Dam became operational in 1937. In 1949, Southern California Edison also converted from 50 to 60 Hz. Today, the two standard frequencies for generation, transmission, and distribution of electric power in

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the world are 60 Hz (in the United States, Canada, Mexico, and Brazil) and 50 Hz (in Europe, the former Soviet republics, China, South America, except Brazil, and India). In Japan, the western part of the country, including Kyoto, uses 60 Hz, while the eastern part, including Tokyo, uses 50 Hz. The advantage of 60-Hz systems is that generators, motors, and transformers in these systems are generally smaller than 50-Hz equipment with the same ratings. The advantage of 50-Hz systems is that transmission lines and transformers have smaller reactances at 50 Hz than at 60 Hz.

As shown in Figure 1.2, the rate of growth of electric energy generation in the United States was approximately 7% per year from 1902 to 1972. This corresponds to a doubling of electric energy generation every 10 years over the 70-year period. In other words, every 10 years, the industry installed a new electric system equal in energy-producing capacity to the total of what it had built since the industry began. The annual growth rate slowed after the oil embargo of 1973 to 1974. Kilowatt-hour generation in the United States increased by 3.4% per year from 1972 to 1980, and by only 0.7% per year from 2000 to 2011. In 2019, total net U.S. electricity generation was approximately 4.16×10^{12} kilowatt-hours (kWh), about the same as in 2010.

Along with increases in load growth, there have been continuing increases in the size of generating units (Table 1.1). The principal incentive to build larger units has been economy of scale—that is, a reduction in installed cost per kilowatt of capacity for larger units. However, there have also been steady improvements in generation efficiency. For example, in 1934, the average heat rate for steam generation in the U.S. electric industry was 17,950 BTU/kWh, which corresponds to 19% efficiency. By 1991, the average heat rate was 10,367 BTU/kWh, which corresponds to 33% efficiency. These improvements in thermal efficiency due to increases in unit size and in steam temperature and pressure, as well as to the use of steam reheat, have resulted in savings in fuel costs and overall operating costs.

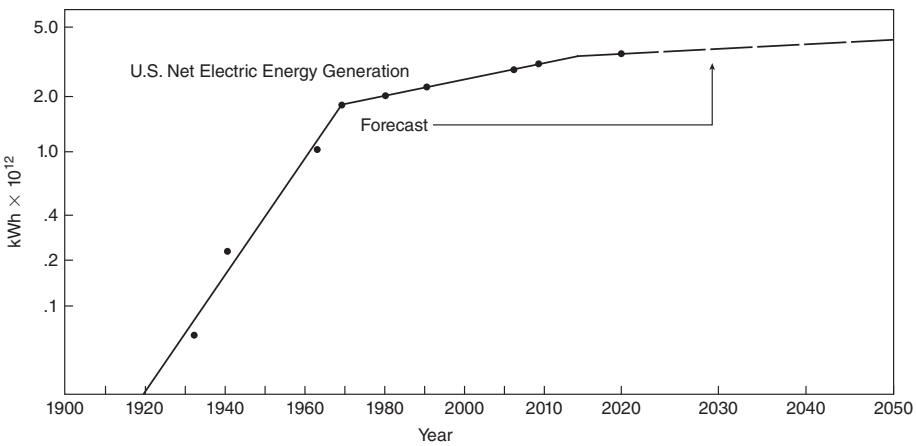

FIGURE 1.2

Growth of U.S. net electric energy generation [3]. (www.eia.gov, U.S. Energy Information Administration, Annual Energy Outlook 2019)

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Hydroelectric Generators		Generators Driven by Single-Shaft, 3600 r/min Fossil-Fueled Steam Turbines	
Size (MVA)	Year of Installation	Size (MVA)	Year of Installation
4	1895	5	1914
108	1941	50	1937
158	1966	216	1953
232	1973	506	1963
615	1975	907	1969
718	1978	1120	1974

TABLE 1.1

Growth of generator sizes in the United States [1].

(*Source:* Based on H.M. Rustebakke et al., *Electric Utility Systems Practice*, 4th. Ed. (New York: Wiley, 1983.)

There have been continuing increases, too, in transmission voltages (Table 1.2). From Edison's 220-V three-wire dc grid to 4-kV single-phase and 2.3-kV three-phase transmission, ac transmission voltages in the United States have risen progressively to 150, 230, 345, 500, and now 765 kV. And in 2009, in China, the first 1000-kV ultra-high voltage (UHV) ac transmission line, a 650-km line from Shaanxi Province to Hubei Province, began commercial operation [14]. The incentives for increasing transmission voltages have been: (1) increases in transmission distance and transmission capacity; (2) smaller line-voltage drops; (3) reduced line losses; (4) reduced right-of-way requirements per MW transfer; and (5) lower capital and operating costs of transmission. Today, one 765-kV three-phase line can transmit thousands of megawatts over hundreds of kilometers.

Voltage (kV)	Year of Installation
2.3	1893
44	1897
150	1913
165	1922
230	1923
287	1935
345	1953
500	1965
765	1969

TABLE 1.2

History of increases in three-phase transmission voltages in the United States [1].

(*Source:* Based on H.M. Rustebakke et al., *Electric Utility Systems Practice*, 4th. Ed. (New York: Wiley, 1983.)

The technological developments that have occurred in conjunction with ac transmission, including developments in insulation, protection, and control, are in themselves important. The following examples are noteworthy:

1. The suspension insulator;
2. The high-speed relay system, currently capable of detecting short-circuit currents within one cycle (0.017 s);
3. High-speed, extra-high-voltage (EHV) circuit breakers, capable of interrupting up to 63-kA three-phase, short-circuit currents within two cycles (0.033 s);
4. High-speed reclosure of EHV lines, which enables automatic return to service within a fraction of a second after a fault has been cleared;
5. The EHV surge arrester, which provides protection against transient overvoltages due to lightning strikes and line-switching operations;
6. Power-line carrier, microwave, and fiber optics as communication mechanisms for protecting, controlling, and metering transmission lines;

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7. The principle of insulation coordination applied to the design of an entire transmission system;
8. Energy control centers with supervisory control and data acquisition (SCADA) and with automatic generation control (AGC) for centralized computer monitoring and control of generation, transmission, and distribution;
9. Automated distribution features, including advanced metering infrastructure (AMI), reclosers, and remotely controlled sectionalizing switches with fault-indicating capability, along with automated mapping/facilities management (AM/FM) and geographic information systems (GIS) for quick isolation and identification of outages and for rapid restoration of customer services; and
10. Digital relays capable of circuit breaker control, data logging, fault locating, self-checking, fault analysis, remote query, and relay event monitoring/recording.

In 1954, the first modern high-voltage dc (HVDC) transmission line was put into commercial operation in Sweden between Vastervik and the island of Gotland in the Baltic Sea; it operated at 100 kV for a distance of 100 km. The first HVDC line in the United States was the ±400-kV (now ±500-kV), 1360-km Pacific Intertie line, installed between Oregon and California in 1970. As of 2019, 12 HVDC lines up to ±500 kV and 16 back-to-back ac-dc links had been installed in the United States, and a total of 146 HVDC lines up to ±800 kV had been installed worldwide [4]. And in 2019, in China, the first ±1100-kV UHVDC transmission line, a 3293-km line from Changji to Guquan, went into operation [15].

For an HVDC line embedded in an ac system, solid-state converters at both ends of the dc line operate as rectifiers and inverters. Since the cost of an HVDC transmission line is less than that of an ac line with the same capacity, the additional cost of converters for dc transmission is offset when the line is long enough. Studies have shown that overhead HVDC transmission is economical in the United States for transmission distances longer than about 600 km. However, HVDC also has the advantage that it may be the only feasible method to:

1. interconnect two asynchronous networks;
2. utilize long underground or underwater cable circuits;
3. bypass network congestion;
4. reduce fault currents;
5. share utility rights-of-way without degrading reliability; and
6. mitigate environmental concerns [5].

In the United States, electric utilities grew first as isolated systems, with new ones continuously starting up throughout the country. Gradually, however, neighboring electric utilities began to interconnect, to operate in parallel. This improved both reliability and economy. Figure 1.3 shows major 115-kV and higher-voltage, interconnected transmission in the United States. An interconnected system has many

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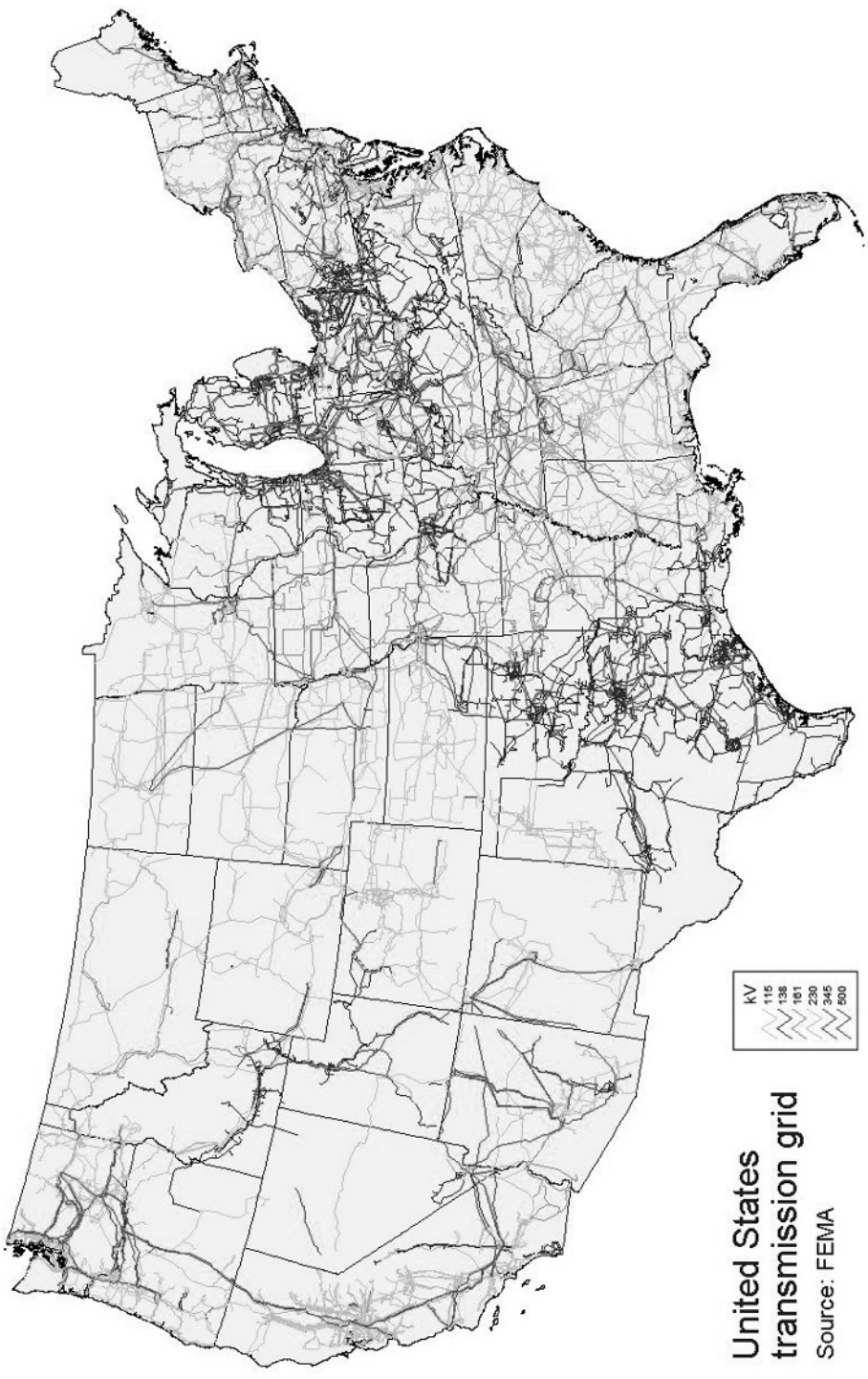

FIGURE 1.3

Major transmission in the United States (FEMA).

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advantages. An interconnected utility can draw upon another's rotating generator reserves during a time of need (such as a sudden generator outage or load increase), thereby maintaining continuity of service, increasing reliability, and reducing the total number of generators that need to be kept running under no-load conditions. Also, interconnected utilities can schedule power transfers during normal periods to take advantage of energy-cost differences in respective areas, load diversity, time zone differences, and seasonal conditions. For example, utilities whose generation is primarily hydro can supply low-cost power during high-water periods in spring/summer and can receive power from the interconnection during low-water periods in fall/winter. Interconnections also allow shared ownership of larger, more efficient generating units.

While sharing the benefits of interconnected operation, each utility is obligated to help neighbors who are in trouble, to maintain scheduled intertie transfers during normal periods, and to participate in system frequency regulation.

In addition to the benefits and obligations of interconnected operation, there are disadvantages. Interconnections, for example, have increased fault currents that occur during short circuits, thus requiring the use of circuit breakers with higher interrupting capability. Furthermore, although overall system reliability and economy have improved dramatically through interconnection, there is a remote possibility that an initial disturbance may lead to a regional blackout, such as the one that occurred in August 2003 in the northeastern United States and Canada.

1.2 PRESENT AND FUTURE TRENDS

The electric power industry is making long-term investment and planning decisions and is transforming the energy grid to be responsive to new resources, new technology options, and changing customer expectations. The industry advances economic growth, promotes business development and expansion, provides solid employment opportunities, enhances the quality of life for its users, and powers the world. Today, the United States electric power industry is a robust, $880-billion-plus industry with nearly 500,000 employees. In the United States economy, the industry represents approximately 5% of real gross domestic product (GDP) [6].

As shown in Figure 1.2, the growth rate in the use of electricity in the United States is projected to increase by about 0.7% per year from 2019 to 2030 [3]. Although electricity forecasts for the next eleven years are based on economic and social factors that are subject to change, 0.7% annual growth rate is considered necessary to generate the GDP anticipated over that period. Variations in forecasts of 0.5 to 1.0% annual growth from 2019 to 2030 are based on low-to-high ranges in economic growth. Average end-use price of electricity (2019 cents/kWh) is forecasted to remain flat at 10.4 cents per kilowatt-hour from 2019 to 2030 [2, 3].

Figure 1.4 shows the percentages of various fuels used to meet U.S. electric energy requirements for 2019 and those forecasted for 2030 and 2050. Several trends are apparent in the chart. One trend is that renewables are the fastest-growing source of electricity generation in the United States through 2050 because of

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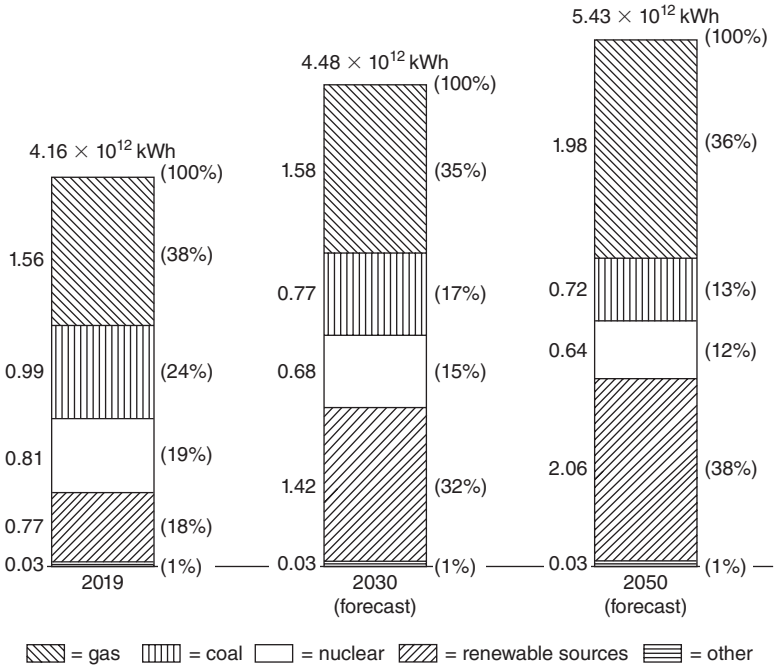

FIGURE 1.4

Net electric energy generation in the United States, by principal fuel types [2, 3]. Renewable sources include conventional hydroelectric, geothermal, wood, wood waste, all municipal waste, landfill gas, other biomass, solar, and wind power. (U.S. Energy Information Administration, Electric Power Annual 2019, Oct 2020, U.S. Energy Information Administration, Annual Energy Outlook 2020, www.eia.gov.)

continuing declines in the capital costs for solar and wind that are supported by federal tax credits and higher state-level renewables targets. A second trend is the continuing generation of electricity with gas via gas-fired turbines that are safe, clean, and more efficient than competing technologies. A third trend, with slow load growth and increasing electricity production from renewables, is that both coal-fired generation and nuclear electricity generation will decline through 2050 in the United States [2, 3].

Figure 1.5 shows the 2019 and forecasted 2030 U.S. net summer generating capability by principal fuel type. As shown, total U.S. net summer generating capability is forecasted to reach 1229 GW (1 GW = 1000 MW) by the year 2030, which represents a 1.5% annual forecasted growth in generating capacity from 2019 to 2030. The forecast assumes over 330 GW of new capacity additions from solar and wind generation and 147 GW of cumulative capacity retirements from 2019 to 2030 [2, 3, 7].

As of 2012, there were 476,247 circuit miles of existing transmission (above 200 kV) in North America, which includes the continental United States, Canada, and the northern portion of Baja California, Mexico. In North America from 2013 to 2018, there were plans for 18,236 circuit-miles of transmission additions; and from 2019 to 2029, there are plans for over 18,000 circuit-miles of transmission additions. According to the North American Electric Reliability Corporation (NERC), new transmission projects are being driven primarily to enhance reliability. Other reasons include congestion alleviation and integration of renewable energy sources [7].

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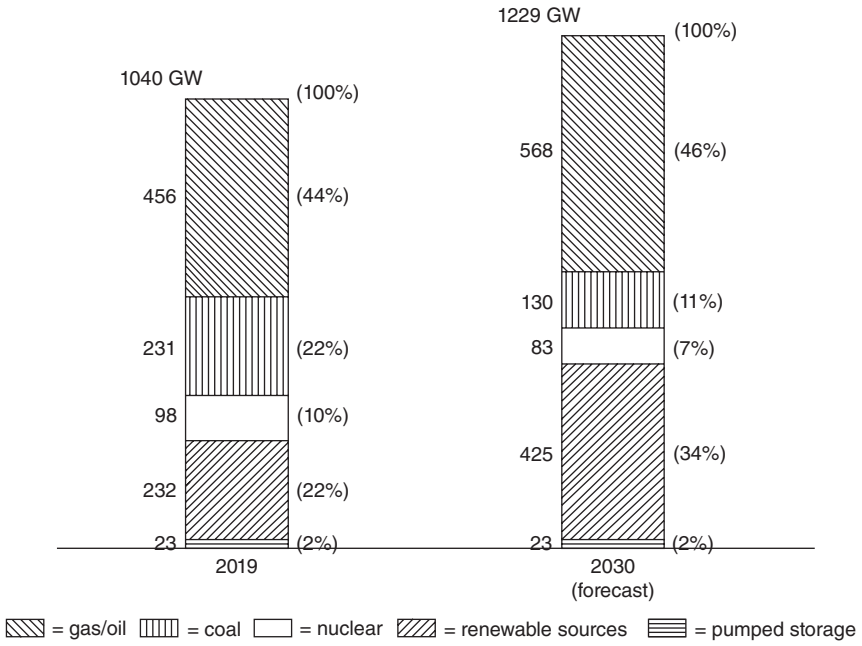

FIGURE 1.5

Installed net summer generating capability in the United States by principal fuel types [2, 3]. Renewable sources include conventional hydroelectric, geothermal, wood, wood waste, all municipal waste, landfill gas, other biomass, solar, and wind power. (U.S. Energy Information Administration, Electric Power Annual 2019, Oct 2020, U.S. Energy Information Administration, Annual Energy Outlook 2020, www.eia.gov.)

According to NERC, building new transmission is an ongoing challenge for the electricity industry. Transmission right-of-way issues required in the siting of new transmission lines that span multiple states and provinces are highly visible and require increased coordination among multiple regulating agencies and authorities. A lack of coordination regarding cost allocation, coupled with public opposition due to land use and property valuation concerns, has often resulted in extended delays in transmission construction. Further, ongoing state and provincial policies require continued integration of renewable resources, including wind and solar. These variable resources are most commonly built in remote areas where wind power densities and solar development are favorable. In many cases, the existing transmission network needs to be expanded to integrate these renewable resources and meet other state-wide goals [7].

Growth in distribution construction roughly correlates with growth in electric energy consumption. During the last three decades, many U.S. utilities converted older 2.4-, 4.1-, and 5-kV primary distribution systems to 12 or 15 kV. The 15-kV voltage class is widely preferred by U.S. utilities for new installations; higher primary distribution voltages, including 25 kV and 34.5 kV, are also utilized. Secondary distribution reduces the voltage for utilization by commercial and residential customers. Common secondary distribution voltages in the United States are 240/120 V, single-phase, three-wire; 208 Y/120 V, three-phase, four-wire; and 480 Y/277 V, three-phase, four-wire.

Transmission and distribution grids in the United States as well as other industrialized countries are aging and stressed by operational uncertainties and challenges never envisioned when they were developed many decades ago. There

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is a growing consensus in the power industry and among many governments that smart grid technology is the answer to the uncertainties and challenges. A smart grid is characterized by the following attributes:

1. Self-healing from power system disturbances;
2. Enables active participation by consumers in demand response;
3. Operates resiliency against both physical and cyberattacks;
4. Provides quality power that meets 21st century needs;
5. Accommodates all generation and energy storage technologies;
6. Enables new products, services, and markets; and
7. Optimizes asset utilization and operating efficiency.

The objective of a smart grid is to provide reliable, high-quality electric power to digital societies in an environmentally friendly and sustainable manner [10].

1.3 ELECTRIC UTILITY INDUSTRY STRUCTURE

The structure of the electric utility industry in the United States and other countries continues to undergo change. The previous structure of large, vertically integrated monopolies that existed until the last decade of the twentieth century is being replaced by a horizontal structure with generating companies, transmission companies, and distribution companies as separate business facilities.

In 1992, the United States Congress passed the Energy Policy Act, which has shifted and continues to further shift regulatory power from the state level to the federal level. The 1992 Energy Policy Act mandates the Federal Energy Regulatory Commission (FERC) to ensure that adequate transmission and distribution access is available to exempt wholesale generators (EWGs) and nonutility generation (NUG). In 1996, FERC issued the "MegaRule," which regulates Transmission Open Access (TOA).

TOA was mandated in order to facilitate competition in wholesale generation. As a result, a broad range of independent power producers (IPPs) and cogenerators now submit bids and compete in energy markets to match electric energy supply and demand. The retail structure of power distribution has become similar to the existing structure of the telephone industry; that is, consumers choose which electric energy supplier to buy power from. Also, with demand-side metering, consumers know the retail price of electric energy at any given time and may choose when to purchase it.

Overall system reliability has become a major concern as the electric utility industry adapts to the new horizontal structure. The North American Electric Reliability Corporation (NERC) was created after the 1965 Northeast blackout and is responsible for maintaining system standards and reliability. NERC coordinates its efforts with FERC and governmental organizations in Canada [11].

As shown in Figure 1.3, the transmission system in North America is interconnected in a large power grid known as the North American Power Systems

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Interconnection. NERC currently divides this grid into six regional entities including NPCC—the Northeast Power Coordinating Council, MRO—the Midwest Reliability Organization, and SERC—the Southeastern Electric Reliability Council. Members of the regional entities come from all segments of the electric industry: investor-owned utilities; federal power agencies; rural electric cooperatives; state, municipal, and provincial utilities; independent power producers; power marketers; and end-use customers. These members jointly perform regional planning studies and operate jointly to schedule generation.

The basic premise of TOA is that transmission owners treat all transmission users on a nondiscriminatory and comparable basis. In December 1999, FERC issued Order 2000, which calls for companies owning transmission systems to put transmission systems under the control of Regional Transmission Organizations (RTOs). Several of the NERC regions have either established independent system operators (ISOs) or planned for ISOs to operate the transmission system and facilitate transmission services. Maintenance of the transmission system remains the responsibility of the transmission owners.

At the time of the August 14, 2003, blackout in the northeastern United States and Canada, NERC reliability standards were voluntary. In August 2005, the U.S. Federal government passed the Energy Policy Act of 2005, which authorizes the creation of an electric reliability organization (ERO) with the statutory authority to enforce compliance with reliability standards among all market participants. As of June 18, 2007, FERC granted NERC the legal authority to enforce reliability standards with all users, owners, and operators of the bulk power system in the United States, and made compliance with those standards mandatory and enforceable. The National Energy Board of Canada formalized its recognition of NERC as the international Electric Reliability Organization by signing a memorandum of understanding in September 2006. Formal recognitions followed from the provinces of British Columbia, Manitoba, Nova Scotia, Ontario, Quebec and all other interconnected provinces between 2006 and 2009. Ontario was the first province to formally adopt NERC's mandatory Reliability Standards [11].

The objectives of electric utility restructuring are to increase competition, decrease regulation, and in the long run, lower consumer prices. There is a concern that the benefits from breaking up the old, vertically integrated utilities are unrealized if the new unbundled generation and transmission companies are able to exert market power. Market power refers to the ability of one seller or group of sellers to maintain prices above competitive levels for a significant period of time, which could be done via collusion or by taking advantage of operational anomalies that create and exploit transmission congestion. Market power can be eliminated by independent supervision of generation and transmission companies, by ensuring that there are an ample number of generation companies, by eliminating transmission congestion, and by creating a truly competitive market, where the spot price at each node (bus) in the transmission system equals the marginal cost of providing energy at that node, where the energy provider is any generator bidding into the system [9,12].

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1.4 COMPUTERS IN POWER SYSTEM ENGINEERING

As electric utilities have grown in size and the number of interconnections has increased, planning for future expansion has become increasingly complex. The increasing cost of additions and modifications has made it imperative that utilities consider a range of design options and perform detailed studies of the effects on the system of each option, based on a number of assumptions: normal and abnormal operating conditions, peak and off-peak loadings, and present and future years of operation. A large volume of network data must also be collected and accurately handled. To assist the engineer in this power system planning, digital computers and highly developed computer programs are used. Such programs include power-flow, stability, short-circuit, and transients programs.

Power-flow programs compute the voltage magnitudes, phase angles, and transmission-line power flows for a network under steady-state operating conditions. Other results, including transformer tap settings and generator reactive power outputs, are also computed. Today's computers have sufficient storage and speed to efficiently compute power-flow solutions for networks with 100,000 buses and 150,000 transmission lines. Results are usually viewed interactively on computer screens in the form of either tabular displays or single-line diagrams; the engineer uses these to modify the network with a mouse or from a keyboard and can readily visualize the results. The computer's large storage and high-speed capabilities allow the engineer to run the many different cases necessary to analyze and design transmission and generation-expansion options.

Stability programs are used to study power systems under disturbance conditions to determine whether synchronous generators and motors remain in synchronization. System disturbances can be caused by the sudden loss of a generator or transmission line, by sudden load increases or decreases, and by short circuits and switching operations. The stability program combines power-flow equations and machine-dynamic equations to compute the angular swings of machines during disturbances. The program also computes critical clearing times for network faults, and allows the engineer to investigate the effects of various machine parameters, network modifications, disturbance types, and control schemes.

Short-circuit programs are used to compute three-phase and line-to-ground faults in power system networks in order to select circuit breakers for fault interruption, select relays that detect faults and control circuit breakers, and determine relay settings. Short-circuit currents are computed for each relay and circuit-breaker location and for various system-operating conditions, such as lines or generating units out of service in order to determine minimum and maximum fault currents.

Transients programs compute the magnitudes and shapes of transient overvoltages and currents that result from lightning strikes and line-switching operations. The planning engineer uses the results of a transients program to determine insulation requirements for lines, transformers, and other equipment, and to select surge arresters that protect equipment against transient overvoltages.

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Other computer programs for power system planning include relay-coordination programs and distribution-circuits programs. Computer programs for generation-expansion planning include reliability analysis and loss-of-load probability (LOLP) programs, production cost programs, and investment cost programs.

1.5 POWERWORLD SIMULATOR

PowerWorld Simulator (PowerWorld) version 22 is a commercial-grade power system analysis and simulation package that accompanies this text. The purposes of integrating PowerWorld with the text are to provide computer solutions to examples in the text, to extend the examples, to demonstrate topics covered in the text, to provide a software tool for more realistic design projects, and to provide the readers with experience using a commercial grade power system analysis package. To use this software package, you must first install PowerWorld, along with all of the necessary case files onto your computer. The PowerWorld software and case files can be downloaded by going to the www.powerworld.com /gloveroverbyesarma Web page, and clicking on the **DownLoad PowerWorld Software and Cases for the 7th Edition** button. The remainder of this section provides the necessary details to get up and running with PowerWorld.

EXAMPLE 1.1

Introduction to PowerWorld Simulator

After installing PowerWorld, double-click on the PW icon to start the program. Power system analysis requires, of course, that the user provide the program with a model of the power system. With PowerWorld, you can either build a new case (model) from scratch or start from an existing case. Most of the examples in the book, as in this case, initially start from an existing case. PowerWorld uses the common ribbon user interface in which frequent commands, such as opening or saving a case, are available by clicking on the items on the top of the screen. So to open a case, click **File** and then select **Open Case**. This displays the Open Dialog. Select the Example 1_1 case in the Chapter 1 directory, and then click Open. The display should look similar to Figure 1.6. Note, the ribbon can be collapsed by clicking on the caret (^) symbol shown on the far right side of the ribbon; if collapsed, the ribbon can be restored by selecting the Window menu item and clicking on the small pin symbol that appears in the same location where the caret had been.

For users familiar with electric circuit schematics, it is readily apparent that Figure 1.6 does *not* look like a traditional schematic. This is because the system is drawn in what is called a oneline diagram form. A brief explanation is in order. Electric power systems range in size from small dc systems with peak power

(*Continued*)

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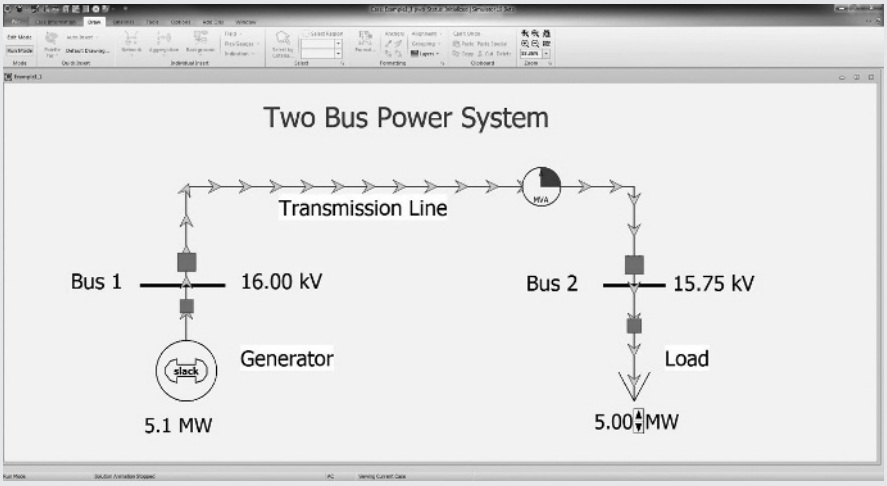
FIGURE 1.6
Example power system



demands of perhaps a few milliwatts (mW) to large continent-spanning interconnected ac systems with peak demands of hundreds of Giga-watts (GW) of demand ($1\ \text{GW} = 1 \times 10^9$ watts). The subject of this book and also PowerWorld are the high-voltage, high-power, interconnected ac systems. Almost without exception, these systems operate using three-phase ac power at either 50 or 60 Hz. As discussed in Chapter 2, a full analysis of an arbitrary three-phase system requires consideration of each of the three phases. Drawing such systems in full schematic form quickly gets excessively complicated. Thankfully, during normal operation, three-phase systems are usually balanced. This permits the system to be accurately modeled as an equivalent single-phase system (the details are discussed in Chapter 9, "Symmetrical Components"). Most power system analysis packages, including PowerWorld, use this approach. Then connections between devices are drawn with a single line joining the system devices, hence the term *oneline diagram*. However, do keep in mind that the actual systems are three-phase.

Figure 1.6 illustrates how the major power system components are represented in PowerWorld. By default, generators are shown as a circle with a "dog-bone" rotor, large arrows represent loads, and transmission lines are simply drawn as lines. In power system terminology, the nodes at which two or more devices join are called *buses*. In PowerWorld, thicker lines usually represent buses; the bus voltages can be shown in kilovolts (kV) in the fields close to the bus (e.g., to the right of the buses in Figure 1.6). In addition to voltages, power engineers are also concerned with how power flows through the system (the solution of the power flow problem is covered in Chapter 6, "Power Flows"). In PowerWorld, power flows can be visualized with arrows superimposed on the generators, loads, and transmission lines. The size and speed of the arrows indicate the direction of flow. One of the unique aspects of PowerWorld is its ability to animate power systems. To start the animation, select the **Tools** tab in the ribbon and then click on the green and black arrow button above **Solve**

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(i.e., the "Play" button). The oneline should come to life! While the oneline is being animated, you can interact with the system. Figure 1.6 represents a simple power system in which a generator is supplying power to a load through a 16-kV distribution system feeder. The solid red blocks on the line and load represent circuit breakers. To open a circuit breaker, simply click on it. Since the load is series-connected to the generator, clicking on any of the circuit breakers isolates the load from the generator resulting in a blackout. To restore the system, click again on the circuit breaker to close it and then again select the **Solve** button on the **Tools** ribbon. To vary the load, click on the up or down arrows between the load value and its "MW" field. Note that because of the impedance of the line, the load's bus voltage drops as its value is increased.

You can view additional information about most of the elements on the oneline by right-clicking on them. For example, right-clicking on the generator symbol brings up a local menu of additional information about the generator, while right-clicking on the transmission line brings up the local menu of information about the line. The meaning of many of these fields become clearer as you progress through the book. To modify the display itself, simply right-click on a blank area of the oneline. This displays the Onelines local menu. Select **Oneline Display Options** to display the Oneline Display Options dialog. From this dialog, you can customize many of the display features. For example, to change the animated flow arrow color, select Animated Flows from the options shown on the left side of the dialog. Then click on the green-colored box next to the Actual MW field (toward the bottom of the dialog) to change its color.

There are several techniques for panning and/or zooming on the oneline. One method to pan is to first click in an empty portion of the display and then press the keyboard arrow keys in the direction you would like to move. To zoom, just hold down the Ctrl key while pressing the up arrow to zoom in or the down arrow to zoom out. Alternatively, you can drag the oneline by clicking and holding the left mouse button down and then moving the mouse—the oneline should follow. To go to a favorite view from the Onelines local menu, select the **Go To View** to view a list of saved views.

If you would like to retain your changes after you exit PowerWorld, you need to save the results. To do this, select **File** in the upper-left portion of the display and then **Save Case As;** enter a different file name so as to not overwrite the initial case. One important note: PowerWorld actually saves the information associated with the power system model itself in a different file from the information associated with the oneline. The power system model is stored in *.pwb files (PowerWorld Binary file) while the Oneline Display information is stored in *.pwd files (PowerWorld Display file). For all the cases discussed in this book, the names of both files should be the same (except the different extensions). The reason for the dual file system is to provide flexibility. With large system models, it is quite common for a system to be displayed using multiple oneline diagrams. Furthermore, a single oneline diagram might be used at different times to display information about different cases.

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EXAMPLE 1.2

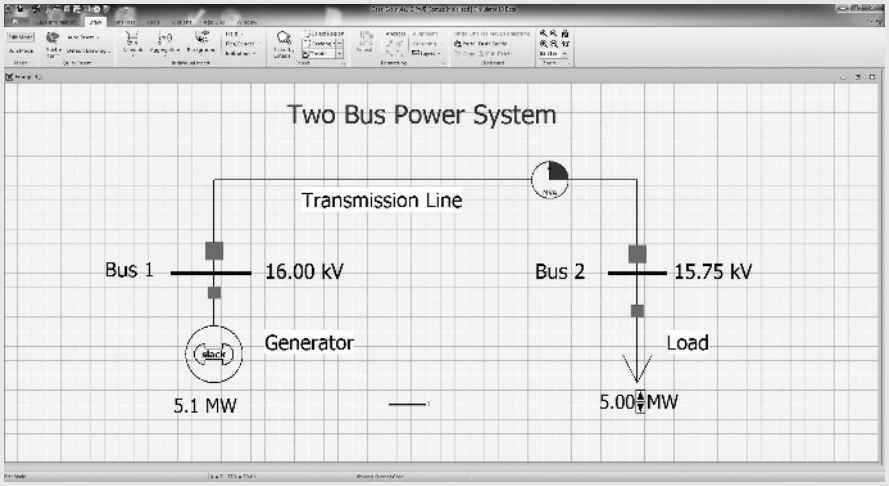
PowerWorld Simulator—Edit Mode

PowerWorld has two major modes of operations. The Run Mode, which was just introduced, is used for running simulations and performing analysis. The Edit Mode, which is used for modifying existing cases and building new cases, is introduced in this example. To switch to the Edit Mode, click on the **Edit Mode** button, which is located in the ribbon on the far left immediately below the File menu item. Use the Edit Mode to add an additional bus and load, as well as two new lines, to the Example 1_1 system.

When switching to the Edit Mode, notice that the ribbon changes slightly, with several of the existing buttons and icons disabled and others enabled. Also, the oneline now has a superimposed grid to help with alignment (the grid can be customized using the Grid/Highlight Unlinked options category on the Oneline Display Options dialog). In the Edit Mode, first add a new bus to the system. This can be done graphically by first selecting the **Draw** tab, then clicking on the **Network** button and selecting **Bus**. Once this is done, move the mouse to the desired oneline location and click (note the **Draw** tab is only available in the Edit Mode). The Bus Options dialog then appears. This dialog is used to set the bus parameters. For now, leave all the bus fields at their default values, except set Bus Name to "Bus 3" and set the nominal voltage to 16.0; note that the number for this new bus was automatically set to the one greater than the highest bus number in the case. Select **OK** to save the new bus and close the dialog. The oneline should look similar to Figure 1.7. You may wish to save your case now to avoid losing your changes.

FIGURE 1.7

Example 1.2—Edit Mode view with new bus



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By default, when a new bus is inserted, a bus field is also inserted. *Bus fields* are used to show information about buses on the onelines. In this case, the new field shows the bus name, although initially in rather small fonts. To change the field's font size, click on the field to select it, and then select the **Format** button (in the Draw ribbon) to display the Format dialog. Click on the **Font** tab and change the font's size to a larger value to make it easier to see. You can also change the size of the bus itself using the Format dialog, Display/Size tab. To see the bus voltage magnitude, add an additional bus field. On the Draw ribbon select **Field, Bus Field**, and then click near bus 3. This displays the Bus Field Options dialog. Make sure the bus number is set to 3, and that the "Type of Field" is Bus Voltage. Again, resize with the **Format, Font** dialog.

Next, insert some load at bus 3. This can be done graphically by selecting **Network, Load** and then clicking on bus 3. The Load Options dialog appears, allowing you to set the load parameters. Note that the load was automatically assigned to bus 3. Leave all the fields at their default values, except set the orientation to "Down," and enter 10.0 in the Constant Power column MW Value field. As the name implies, a constant power load treats the load power as being independent of bus voltage; constant power load models are commonly used in power system analysis (more detail on load models is given in Chapter 6). By default, PowerWorld "anchors" each load symbol to its bus. This is a handy feature when changing a drawing, since when you drag the bus, the load and all associated fields move as well. Note that two fields showing the load's real (MW) and reactive (Mvar) power were also auto-inserted with the load. Select the reactive power field and then choose **Delete** (located toward the right side of the **Draw** ribbon) to remove it. You should also resize the MW field using the **Format, Font** command.

Now join the bus 3 load to the rest of the system, which can be done by adding a line from bus 2 to bus 3. Select **Network, Transmission Line** and then click on bus 2. This begins the line drawing. During line drawing, PowerWorld adds a new line segment for each mouse click. After adding several segments place the cursor on bus 3 and double-click. The Transmission Line/Transformer Options dialog appears, allowing you to set the line's parameters. Note that PowerWorld should have automatically set the "from" and "to" bus numbers based upon the starting and ending buses (buses 2 and 3). If these values have not been set automatically, then you probably did not click exactly on bus 2 or bus 3; manually enter the values. Next, set the line's Series Resistance (R) field to 0.3, the Series Reactance (X) field to 0.6, and the MVA Limits Limit (A) field to 20 (the details of transformer and transmission line modeling is covered in Chapters 3 through 5). Select **OK** to close the dialog. Note that PowerWorld also auto-inserted two circuit breakers and a round "pie chart" symbol. The pie charts are used to show the percentage loading of the line. You can change the display size for these objects by right-clicking on them to display their option dialogs.

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EXAMPLE 1.3

PowerWorld Simulator—Run Mode

Next, switch back to Run Mode to solve and animate the new system developed in Example 1.2. Click on the **Run Mode** button (immediately below the **Edit Mode** button), select the **Tools** on the ribbon, and then click the green-and-black button above **Solve** to start the simulation. You should see the arrows flow from bus 1 to bus 2 to bus 3. Note that the total generation is now about 15.4 MW, with 15 MW flowing to the two loads and 0.4 MW lost to the transmission line resistance. To add the load variation arrows to the bus 3 load, right click on the Load MW field (not the load arrow itself) to display the field's local menu. Select **Load Field Information Dialog** to view the Load Field Options dialog. Set the Delta per Mouse Click field to 1.0, which changes the load by one MW per click on the up/down arrows. You may also wish to set the Digits to Right of Decimal to 2 to see more digits in the load field. Be sure to save your case. The new system now has one generator and two loads. The system is still radial, meaning that a break anywhere on the wire joining bus 1 to bus 2 would result in a blackout of all the loads. Radial power systems are quite common in the lower voltage distribution systems. At higher voltage levels, networked systems are typically used. In a networked system, each load has at least two possible sources of power. Convert the system to a networked system simply by adding a new line from bus 1 to bus 3. To do this, switch back to Edit Mode, and then repeat the previous line insertion process except you should start at bus 1 and end at bus 3; use the same line parameters as for the bus 2 to 3 line. Also before returning to Run Mode, right click on the blue Two Bus Power System title and change it to Three Bus Power System. Return to Run Mode and again solve. Your final system should look similar to the system shown in Figure 1.8. Note that now you can open any single line and still supply both loads—a nice increase in reliability!

With this introduction, you now have the skills necessary to begin using PowerWorld to interactively learn about power systems. If you'd like to take a look at some of the larger systems you'll be studying, in the Chapter 6 directory, open PowerWorld case Example 6_13. This case models a power system with 37 buses. Notice that when you open any line in the system, the flow of power immediately redistributes to continue to meet the total load demand. While the education version of PowerWorld is limited to 42 buses, the commercial version can handle cases with the tens of thousands of buses common in industry studies.

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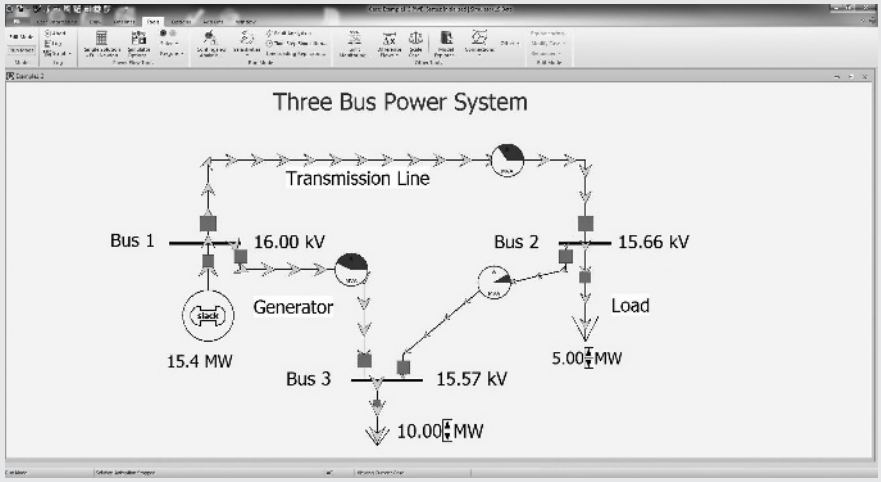
FIGURE 1.8

Example 1.3—New three-bus system



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2 Fundamentals

Operations Center at PJM. PJM is a regional transmission organization (RTO) that coordinates the movement of wholesale electricity and ensures the reliability of the high-voltage electric power system serving 61 million people in all or parts of Delaware, Illinois, Indiana, Kentucky, Maryland, Michigan, New Jersey, North Carolina, Ohio, Pennsylvania, Tennessee, Virginia, West Virginia and the District of Columbia, USA. (Courtesy of PJM Interconnection, LLC.)

LEARNING OBJECTIVES

At the conclusion of this chapter, you should be able to:

1. Explain the basic concepts of phasors, instantaneous power, complex power, network equations, and three-phase circuits.
2. Understand the notation and terminology used in this text.
3. Discuss the method of power factor correction, whereby a capacitor is connected in parallel with an inductive load.
4. Describe advantages of balanced three-phase versus single-phase power systems.
5. Explain the different forms of energy conversion in power systems.

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In this chapter, we review the basic concepts of phasors, instantaneous power, complex power, network equations, and elementary aspects of balanced three-phase circuits. Notation and terminology used throughout the text are also established. A brief introduction to energy conversion including the use of electronics in power systems, such as rectifiers and inverters, is also presented. Students who have already had courses in electric network theory and basic electric machines should find most of this chapter to be primarily refresher material.

CASE STUDY

Investing for the Future*

Young Ngo, Herschel Arant, Chris Bilby, and Amy Grice

The following paragraphs summarize the above article, which describes how small utilities are finding success with Advanced Distribution Management Systems (ADMS). Sign in at www.cengage.com to download this product and to access its online resources.

Real-life case studies are reviewed from three U.S. electric cooperatives: (1) Central Georgia Electric Membership Cooperative (Jackson, GA); (2) Holy Cross Energy (Glenwood Springs, CO); and (3) Peninsula Light Company (Western Pierce County, WA). These cooperatives are facing challenging issues, as are many other electric power distribution companies, including an escalating number of severe weather events; increasing penetrations of Distributed Energy Resources (DERs); and the need for grid modernization to achieve appropriate operational performance and financial returns.

Central Georgia Electric Membership Cooperative has completed its implementation of ADMS and has gained experience running Supervisory and Data Acquisition (SCADA), Fault Location, Isolation, and Service Restoration (FLISR), and other ADMS applications for many years. The utility has greatly improved its resiliency and reduced the time it takes to restore service to customers. In particular, it has achieved the following successful results: a decrease in its System Average Interruption Frequency Index of 32%, a nearly 50% decrease in System Average Interruption Duration Index, and 20% less time spent by field crews on outage repairs.

Holy Cross Energy has been preparing for grid modernization in small stages for years, beginning with the use of Graphic Information Systems (GIS), the subsequent installation of smart devices on reclosers and voltage regulators, and

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is now deploying line sensors and automated switching equipment. The utility is in the early stages of ADMS implementation. To date, it has used SCADA to monitor power flows and voltages at substations and reclosers, examine real-time data, and increase its control over assets. Ultimately, it aims to increase grid reliability, improve power quality, increase its use of renewable energy sources, and become more resilient to natural disasters.

Peninsula Light Company is halfway through its implementation of ADMS. Due to terrain, 20% of its distribution lines are overhead with exposure to vegetation, which causes outages that can trip both overhead and underground portions of lines. The utility first upgraded its SCADA to an integrated ADMS platform and applied FLISR to gain reliability improvements through automation. The most recent storm, which brought more snow to the service territory in 70 years, caused an outage of all transmission lines connected to the utility, and was the first system-wide outage since ADMS was commissioned. Previously, operators lost communication with field devices and were required to perform emergency maintenance on the legacy SCADA, which interfered with the ability to effectively manage restoration activities. With the ADMS in place, operators were able to focus solely on the outage and power restoration. The robustness of the ADMS has freed up time for system planning, Advanced Metering Infrastructure (AMI) deployment, and mobile workforce management tools, all of which added to productivity and improved efficiency.

The above three real-life cases illustrate how small utilities overcome challenges to meet their objectives: to focus on increasing resiliency in the face of escalating severe weather events; to address the requirements for greater interoperability between systems and devices due to the increasing penetration of DERs; and to achieve appropriate operational performance and financial returns via grid modernization [5].

2.1 PHASORS

A sinusoidal voltage or current at constant frequency is characterized by two parameters: a maximum value and a phase angle. A voltage

$$v(t) = \mathrm{V}_{\max} \cos(\omega t + \delta) \tag{2.1.1}$$

has a maximum value $\mathrm{V}_{\max}$ and a phase angle δ when referenced to $\cos(\omega t)$. The root-mean-square (rms) value, also called *effective value*, of the sinusoidal voltage is

$$\mathrm{V} = \frac{\mathrm{V}_{\max}}{\sqrt{2}} \tag{2.1.2}$$

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Euler's identity, $e^{j\phi} = \cos\phi + j\sin\phi$, can be used to express a sinusoid in terms of a phasor. For the above voltage,

$$\begin{aligned} v(t) &= \text{Re}\left[\text{V}_{\max} e^{j(\omega t+\delta)}\right] \\ &= \text{Re}\left[\sqrt{2}(\text{V}e^{j\delta})e^{j\omega t}\right] \end{aligned} \tag{2.1.3}$$

where $j = \sqrt{-1}$ and Re denotes "real part of." The rms phasor representation of the voltage is given in three forms—exponential, polar, and rectangular:

$$V = \underbrace{\text{V}e^{j\delta}}_{\text{exponential}} = \underbrace{\text{V}\angle\delta}_{\text{polar}} = \underbrace{\text{V}\cos\delta + j\text{V}\sin\delta}_{\text{rectangular}} \tag{2.1.4}$$

A phasor can be easily converted from one form to another. Conversion from polar to rectangular is shown in the phasor diagram of Figure 2.1. Euler's identity can be used to convert from exponential to rectangular form. As an example, the voltage

$$v(t) = 169.7\cos(\omega t + 60°) \text{ volts} \tag{2.1.5}$$

has a maximum value $\text{V}_{\max} = 169.7$ volts, a phase angle $\delta = 60°$ when referenced to $\cos(\omega t)$, and an rms phasor representation in polar form of

$$V = 120\angle 60° \quad \text{volts} \tag{2.1.6}$$

Also, the current

$$i(t) = 100\cos(\omega t + 45°) \text{ A} \tag{2.1.7}$$

has a maximum value $\text{I}_{\max} = 100$ A, an rms value $\text{I} = 100/\sqrt{2} = 70.7$ A, a phase angle of 45°, and a phasor representation

$$I = 70.7\angle 45° = 70.7e^{j45} = 50 + j50 \quad \text{A} \tag{2.1.8}$$

The relationships between the voltage and current phasors for the three passive elements—resistor, inductor, and capacitor—are summarized in Figure 2.2, where sinusoidal-steady-state excitation and constant values of R, L, and C are assumed.

When voltages and currents are discussed in this text, lowercase letters such as $v(t)$ and $i(t)$ indicate instantaneous values, uppercase letters such as V and I indicate rms values, and uppercase letters in italics such as V and I indicate rms phasors. When voltage or current values are specified, they shall be rms values unless otherwise indicated.

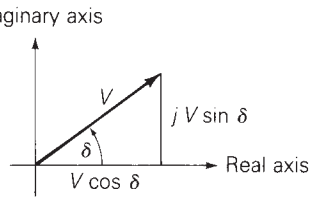

FIGURE 2.1
Phasor diagram for converting from polar to rectangular form

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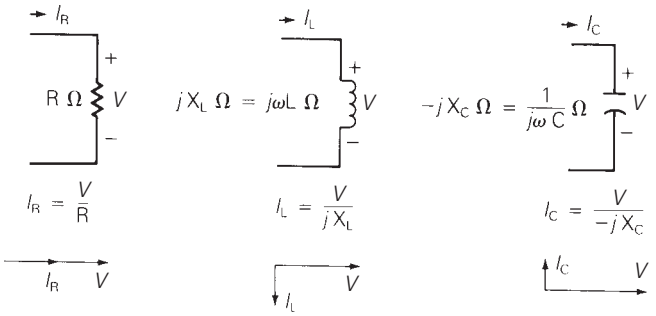

FIGURE 2.2

Summary of relationships between phasors *V* and *I* for constant R, L, and C elements with sinusoidal-steady-state excitation

2.2 INSTANTANEOUS POWER IN SINGLE-PHASE AC CIRCUITS

Power is the rate of change of energy with respect to time. The unit of power is a watt, which is a joule per second. Instead of saying that a load absorbs energy at a rate given by the power, it is common practice to say that a load absorbs power. The instantaneous power in watts absorbed by an electrical load is the product of the instantaneous voltage across the load in volts and the instantaneous current into the load in amperes. Assume that the load voltage is

$$v(t) = \mathrm{V}_{\max} \cos(\omega t + \delta) \text{ volts} \tag{2.2.1}$$

We now investigate the instantaneous power absorbed by purely resistive, purely inductive, purely capacitive, and general RLC loads. We also introduce the concepts of real power, power factor, and reactive power. The physical significance of real and reactive power is also discussed.

PURELY RESISTIVE LOAD

For a purely resistive load, the current into the load is in phase with the load voltage, $I = V/R$, and the current into the resistive load is

$$i_R(t) = \mathrm{I}_{\mathrm{Rmax}} \cos(\omega t + \delta) \tag{2.2.2}$$

where $\mathrm{I}_{\mathrm{Rmax}} = \mathrm{V}_{\max}/\mathrm{R}$. The instantaneous power absorbed by the resistor is

$$\begin{aligned} p_{\mathrm{R}}(t) &= v(t) i_{\mathrm{R}}(t) = \mathrm{V}_{\max} \mathrm{I}_{\mathrm{Rmax}} \cos^2(\omega t + \delta) \\ &= \tfrac{1}{2} \mathrm{V}_{\max} \mathrm{I}_{\mathrm{Rmax}} \{1 + \cos[2(\omega t + \delta)]\} \\ &= \mathrm{V} \mathrm{I}_{\mathrm{R}} \{1 + \cos[2(\omega t + \delta)]\} \ \ \mathrm{W} \end{aligned} \tag{2.2.3}$$

As indicated by (2.2.3), the instantaneous power absorbed by the resistor has an average value

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$$P_R = VI_R = \frac{V^2}{R} = I_R^2 R \quad W \tag{2.2.4}$$

plus a double-frequency term $VI_R \cos[2(\omega t + \delta)]$.

PURELY INDUCTIVE LOAD

For a purely inductive load, the current lags the voltage by 90°, $I_L = V/(jX_L)$, and

$$i_L(t) = I_{Lmax} \cos(\omega t + \delta - 90°) \quad A \tag{2.2.5}$$

where $I_{Lmax} = V_{max}/X_L$, and $X_L = \omega L$ is the inductive reactance. The instantaneous power absorbed by the inductor is*

$$\begin{aligned} p_L(t) &= v(t) i_L(t) = V_{max} I_{Lmax} \cos(\omega t + \delta) \cos(\omega t + \delta - 90°) \\ &= \tfrac{1}{2} V_{max} I_{Lmax} \cos[2(\omega t + \delta) - 90°] \\ &= VI_L \sin[2(\omega t + \delta)] \quad W \end{aligned} \tag{2.2.6}$$

As indicated by (2.2.6), the instantaneous power absorbed by the inductor is a double-frequency sinusoid with *zero* average value.

PURELY CAPACITIVE LOAD

For a purely capacitive load, the current leads the voltage by 90°, $I_c = V/(-jX_C)$, and

$$i_C(t) = I_{Cmax} \cos(\omega t + \delta + 90°) \quad A \tag{2.2.7}$$

where $I_{Cmax} = V_{max}/X_C$, and $X_C = 1/(\omega C)$ is the capacitive reactance. The instantaneous power absorbed by the capacitor is

$$\begin{aligned} p_C(t) &= v(t) i_C(t) = V_{max} I_{Cmax} \cos(\omega t + \delta) \cos(\omega t + \delta + 90°) \\ &= \tfrac{1}{2} V_{max} I_{Cmax} \cos[2(\omega t + \delta) + 90°)] \\ &= -VI_C \sin[2(\omega t + \delta)] \quad W \end{aligned} \tag{2.2.8}$$

The instantaneous power absorbed by a capacitor is also a double-frequency sinusoid with *zero* average value.

GENERAL RLC LOAD

For a general load composed of RLC elements under sinusoidal-steady-state excitation, the load current is of the form

$$i(t) = I_{max} \cos(\omega t + \beta) \quad A \tag{2.2.9}$$

*Use the identity: $\cos A \cos B = (1/2)[\cos(A - B) + \cos(A + B)]$.

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The instantaneous power absorbed by the load is then[†]

$$\begin{aligned} p(t) &= v(t)i(t) = \mathrm{V}_{\max}\mathrm{I}_{\max}\cos(\omega t + \delta)\cos(\omega t + \beta) \\ &= \tfrac{1}{2}\mathrm{V}_{\max}\mathrm{I}_{\max}\{\cos(\delta - \beta) + \cos[2(\omega t + \delta) - (\delta - \beta)]\} \\ &= \mathrm{VI}\cos(\delta - \beta) + \mathrm{VI}\cos(\delta - \beta)\cos[2(\omega t + \delta)] \\ &\quad + \mathrm{VI}\sin(\delta - \beta)\sin[2(\omega t - \delta)] \\ p(t) &= \mathrm{VI}\cos(\delta - \beta)\{1 + \cos[2(\omega t + \delta)]\} + \mathrm{VI}\sin(\delta - \beta) \end{aligned}$$

Letting $\mathrm{I}\cos(\delta - \beta) = \mathrm{I_R}$ and $\mathrm{I}\sin(\delta - \beta) = \mathrm{I_X}$ gives

$$p(t) = \underbrace{\mathrm{VI_R}\{1 + \cos[2(\omega t + \delta)]\}}_{p_{\mathrm{R}}(t)} + \underbrace{\mathrm{VI_X}\sin[2(\omega t + \delta)]}_{p_{\mathrm{X}}(t)} \tag{2.2.10}$$

As indicated by (2.2.10), the instantaneous power absorbed by the load has two components: One can be associated with the power $p_{\mathrm{R}}(t)$ absorbed by the resistive component of the load, and the other can be associated with the power $p_{\mathrm{X}}(t)$ absorbed by the reactive (inductive or capacitive) component of the load. The first component $p_{\mathrm{R}}(t)$ in (2.2.10) is identical to (2.2.3), where $\mathrm{I_R} = \mathrm{I}\cos(\delta - \beta)$ is the component of the load current in phase with the load voltage. The phase angle $(\delta - \beta)$ represents the angle between the voltage and current. The second component $p_{\mathrm{X}}(t)$ in (2.2.10) is identical to (2.2.6) or (2.2.8), where $\mathrm{I_x} = \mathrm{I}\sin(\delta - \beta)$ is the component of load current 90° out of phase with the voltage.

REAL POWER

Equation (2.2.10) shows that the instantaneous power $p_{\mathrm{R}}(t)$ absorbed by the resistive component of the load is a double-frequency sinusoid with average value P given by

$$\mathrm{P} = \mathrm{VI_R} = \mathrm{VI}\cos(\delta - \beta)\ \mathrm{W} \tag{2.2.11}$$

The *average power* P is also called *real power* or *active power.* All three terms indicate the same quantity P given by (2.2.11).

POWER FACTOR

The term $\cos(\delta - \beta)$ in (2.2.11) is called the *power factor.* The phase angle $(\delta - \beta)$, which is the angle between the voltage and current, is called the *power factor angle.* For dc circuits, the power absorbed by a load is the product of the dc load voltage and the dc load current; for ac circuits, the average power absorbed by a load is the product of the rms load voltage V, rms load current I, and the power factor $\cos(\delta - \beta)$, as shown by (2.2.11). For inductive loads, the current lags the voltage, which means β is less than δ, and the power factor is said to be *lagging.* For capacitive loads, the current leads the voltage, which means β is greater than δ, and the power factor is said to be *leading.* By convention, the power factor $\cos(\delta - \beta)$ is positive. If $|\delta - \beta|$

[†]Use the identity: $\cos \mathrm{A}\cos \mathrm{B} = (1/2)[\cos(\mathrm{A} - \mathrm{B}) + \cos(\mathrm{A} + \mathrm{B})]$.

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is greater than 90°, then the reference direction for current may be reversed, resulting in a positive value of $\cos(\delta - \beta)$.

REACTIVE POWER

The instantaneous power absorbed by the reactive part of the load, given by the component $p_X(t)$ in (2.2.10), is a double-frequency sinusoid with zero average value and with amplitude Q given by

$$Q = VI_X = VI \sin(\delta - \beta) \text{ var} \tag{2.2.12}$$

The term Q is given the name *reactive power.* Although it has the same units as real power, the usual practice is to define units of reactive power as volt-amperes reactive, or var.

EXAMPLE 2.1

Instantaneous, real, and reactive power and the power factor

The voltage $v(t) = 141.4 \cos(\omega t)$ is applied to a load consisting of a 10-Ω resistor in parallel with an inductive reactance $X_L = \omega L = 3.77\ \Omega$. Calculate the instantaneous power absorbed by the resistor and by the inductor. Also, calculate the real and reactive power absorbed by the load, and the power factor.

SOLUTION

The circuit and phasor diagram are shown in Figure 2.3(a). The load voltage is

$$V = \frac{141.4}{\sqrt{2}} \underline{/0°} = 100 \underline{/0°} \text{ volts}$$

The resistor current is

$$I_R = \frac{V}{R} = \frac{100}{10} \underline{/0°} = 10 \underline{/0°} \text{ A}$$

The inductor current is

$$I_L = \frac{V}{jX_L} = \frac{100}{(j3.77)} \underline{/0°} = 26.53 \underline{/-90°} \text{ A}$$

The total load current is

$$I = I_R + I_L = 10 - j26.53 = 28.35 \underline{/-69.34°} \text{ A}$$

The instantaneous power absorbed by the resistor is, from (2.2.3),

$$p_R(t) = (100)\,(10)[1 + \cos(2\omega t)]$$
$$= 1000\,[1 + \cos(2\omega t)] \quad \text{W}$$

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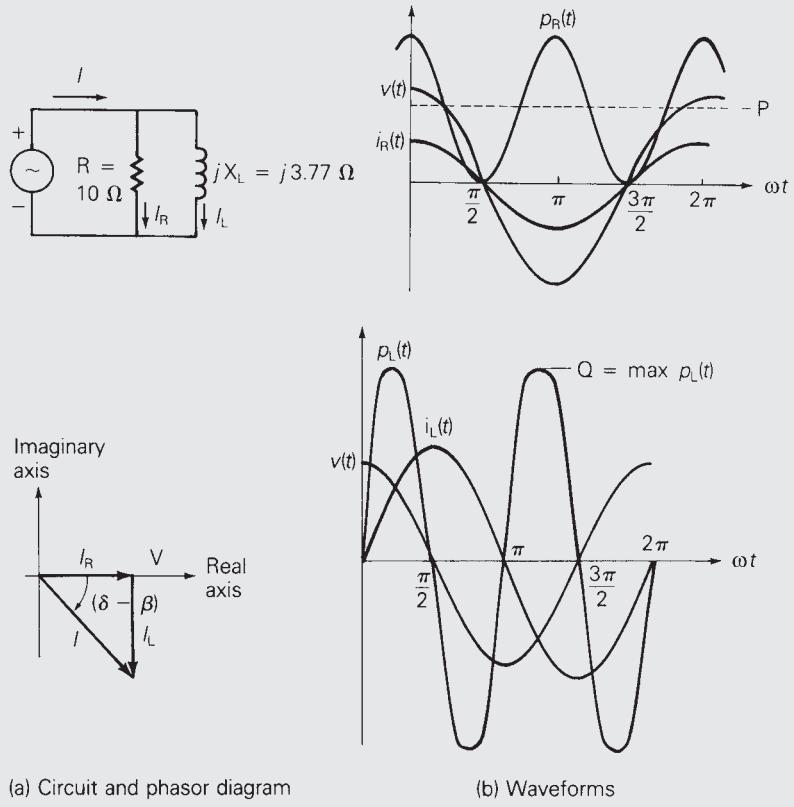
FIGURE 2.3

Circuit and phasor diagram for Example 2.1



The instantaneous power absorbed by the inductor is, from (2.2.6),

$$p_L(t) = (100)(26.53)\sin(2\omega t)$$
$$= 2653\sin(\omega t) \quad \text{W}$$

The real power absorbed by the load is, from (2.2.11),

$$P = VI\cos(\delta - \beta) = (100)(28.35)\cos(0° + 69.34°)$$
$$= 1000 \text{ W}$$

(*Note*: P is also equal to $VI_R = V^2/R$.)

The reactive power absorbed by the load is, from (2.2.12),

$$Q = VI\sin(\delta - \beta) = (100)(28.35)\sin(0° + 69.34°)$$
$$= 2653 \text{ var}$$

(*Note*: Q is also equal to $VI_L = V^2/X_L$.)

(*Continued*)

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The power factor is

p.f. $= \cos(\delta - \beta) = \cos(69.34°) = 0.3528$ lagging

Voltage, current, and power waveforms are shown in Figure 2.3(b).

As shown for this parallel RL load, the resistor absorbs real power (1000 W) and the inductor absorbs reactive power (2653 vars). The resistor current $i_R(t)$ is in phase with the load voltage, and the inductor current $i_L(t)$ lags the load voltage by 90°. The power factor is lagging for an RL load.

Note $p_R(t)$ and $p_x(t)$, given by (2.2.10), are strictly valid only for a parallel R-X load. For a general RLC load, the voltages across the resistive and reactive components may not be in phase with the source voltage $v(t)$, resulting in additional phase shifts in $p_R(t)$ and $p_x(t)$ (see Problem 2.13). However, (2.2.11) and (2.2.12) for P and Q are valid for a general RLC load.

PHYSICAL SIGNIFICANCE OF REAL AND REACTIVE POWER

The physical significance of real power P is easily understood. The total energy absorbed by a load during a time interval T, consisting of one cycle of the sinusoidal voltage, is PT watt-seconds (Ws). During a time interval of n cycles, the energy absorbed is P(nT) watt-seconds, all of which is absorbed by the resistive component of the load. A kilowatt-hour meter is designed to measure the energy absorbed by a load during a time interval $(t_2 - t_1)$, consisting of an integral number of cycles, by integrating the real power P over the time interval $(t_2 - t_1)$.

The physical significance of reactive power Q is not as easily understood. Q refers to the maximum value of the instantaneous power absorbed by the reactive component of the load. The instantaneous reactive power, given by the second term $p_x(t)$ in (2.2.10), is alternately positive and negative, and it expresses the reversible flow of energy to and from the reactive component of the load. Q may be positive or negative, depending on the sign of $(\delta - \beta)$ in (2.2.12). Reactive power Q is a useful quantity when describing the operation of power systems (this will become evident in later chapters). As one example, shunt capacitors can be used in transmission systems to deliver reactive power and thereby increase voltage magnitudes during heavy load periods (see Chapter 5).

2.3 COMPLEX POWER

For circuits operating in sinusoidal-steady-state, real and reactive power are conveniently calculated from complex power, defined below. Let the voltage across a circuit element be $V = \mathrm{V}\angle\delta$, and the current into the element be $I = \mathrm{I}\angle\beta$. Then the complex power S is the product of the voltage and the conjugate of the current:

$$\begin{aligned} S = VI^* &= [\mathrm{V}\angle\delta]\,[\mathrm{I}\angle\beta]^* = \mathrm{VI}\angle\delta - \beta \\ &= \mathrm{VI}\cos(\delta - \beta) + j\mathrm{VI}\sin(\delta - \beta) \end{aligned} \tag{2.3.1}$$

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where $(\delta - \beta)$ is the angle between the voltage and current. Comparing (2.3.1) with (2.2.11) and (2.2.12), S is recognized as

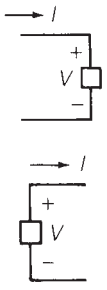
$$S = \mathrm{P} + j\mathrm{Q} \tag{2.3.2}$$

The magnitude S = VI of the complex power S is called the *apparent power.* Although it has the same units as P and Q, it is common practice to define the units of apparent power S as volt-amperes or VA. The real power P is obtained by multiplying the apparent power S = VI by the power factor p.f. = $\cos(\delta - \beta)$.

The procedure for determining whether a circuit element absorbs or delivers power is summarized in Figure 2.4. Figure 2.4(a) shows the *load convention*, where the current *enters* the positive terminal of the circuit element, and the complex power *absorbed* by the circuit element is calculated from (2.3.1). This equation shows that, depending on the value of $(\delta - \beta)$, P may have either a positive or negative value. If P is positive, then the circuit element absorbs positive real power.

However, if P is negative, the circuit element absorbs negative real power, or alternatively, it delivers positive real power. Similarly, if Q is positive, the circuit element in Figure 2.4(a) absorbs positive reactive power. However, if Q is negative, the circuit element absorbs negative reactive power, or it delivers positive reactive power.

Figure 2.4(b) shows the *generator convention*, where the current *leaves* the positive terminal of the circuit element, and the complex power *delivered* is calculated from (2.3.1). When P is positive (negative) the circuit element *delivers* positive (negative) real power. Similarly, when Q is positive (negative), the circuit element *delivers* positive (negative) reactive power.



(a) *Load convention.* Current *enters* positive terminal of circuit element. If P is positive, then positive real power is *absorbed.* If Q is positive, then positive reactive power is *absorbed.* If P (Q) is negative, then positive real (reactive) power is *delivered.*

(b) *Generator convention.* Current *leaves* positive terminal of the circuit element. If P is positive, then positive real power is *delivered.* If Q is positive, then positive reactive power is *delivered.* If P (Q) is negative, then positive real (reactive) power is *absorbed.*

FIGURE 2.4

Load and generator conventions

EXAMPLE 2.2

Real and reactive power, delivered or absorbed

A single-phase voltage source with $V = 100\angle 130°$ volts delivers a current $I = 10\angle 10°$ A, which leaves the positive terminal of the source. Calculate the source real and reactive power, and state whether the source delivers or absorbs each of these.

(*Continued*)

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SOLUTION

Since I leaves the positive terminal of the source, the generator convention is assumed, and the complex power delivered is, from (2.3.1),

$$S = VI^* = [100\angle 130^\circ]\,[10\angle 10^\circ]^*$$

$$S = 1000\angle 120^\circ = -500 + j866$$

$$\mathrm{P} = \mathrm{Re}[S] = -500 \quad \mathrm{W}$$

$$\mathrm{Q} = \mathrm{Im}[S] = +866 \quad \mathrm{var}$$

where Im denotes "imaginary part of." The source absorbs 500 W and delivers 866 var. Readers familiar with electric machines will recognize that one example of this source is a synchronous motor. When a synchronous motor operates at a leading power factor, it absorbs real power and delivers reactive power.

The *load convention* is used for the RLC elements shown in Figure 2.2. Therefore, the complex power *absorbed* by any of these three elements can be calculated as follows. Assume a load voltage $V = V\angle\delta$. Then, from (2.3.1),

$$\text{resistor:} \quad S_\mathrm{R} = VI_\mathrm{R}^* = [\mathrm{V}\angle\delta]\left[\frac{\mathrm{V}}{\mathrm{R}}\angle -\delta\right] = \frac{\mathrm{V}^2}{\mathrm{R}} \tag{2.3.3}$$

$$\text{inductor:} \quad S_\mathrm{L} = VI_\mathrm{L}^* = [\mathrm{V}\angle\delta]\left[\frac{\mathrm{V}}{-j\mathrm{X}_\mathrm{L}}\angle -\delta\right] = +j\frac{\mathrm{V}^2}{\mathrm{X}_\mathrm{L}} \tag{2.3.4}$$

$$\text{capacitor:} \quad S_\mathrm{C} = VI_\mathrm{C}^* = [\mathrm{V}\angle\delta]\left[\frac{\mathrm{V}}{j\mathrm{X}_\mathrm{C}}\angle -\delta\right] = -j\frac{\mathrm{V}^2}{\mathrm{X}_\mathrm{C}} \tag{2.3.5}$$

From these complex power expressions, the following can be stated:

A (positive-valued) resistor absorbs (positive) real power, $\mathrm{P_R} = \mathrm{V}^2/\mathrm{R}$ W, and zero reactive power, $\mathrm{Q_R} = 0$ var.

An inductor absorbs zero real power, $\mathrm{P_L} = 0$ W, and positive reactive power, $\mathrm{Q_L} = \mathrm{V}^2/\mathrm{X_L}$ var.

A capacitor absorbs zero real power, $\mathrm{P_C} = 0$ W, and *negative* reactive power, $\mathrm{Q_C} = -\mathrm{V}^2/\mathrm{X_C}$ var. Alternatively, a capacitor *delivers positive* reactive power, $+\,\mathrm{V}^2/\mathrm{X_C}$.

For a general load composed of RLC elements, complex power S is also calculated from (2.3.1). The real power $\mathrm{P} = \mathrm{Re}(S)$ absorbed by a passive load is always positive. The reactive power $\mathrm{Q} = \mathrm{Im}(S)$ absorbed by a load may be either positive or negative. When the load is inductive, the current lags the voltage, which means β is less than δ in (2.3.1), and the reactive power absorbed is positive. When

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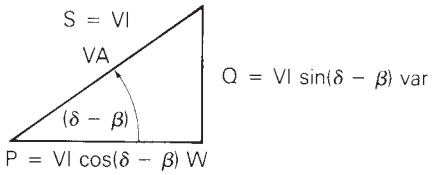

FIGURE 2.5
Power triangle

the load is capacitive, the current leads the voltage, which means β is greater than δ, and the reactive power absorbed is negative; or, alternatively, the capacitive load delivers positive reactive power.

Complex power can be summarized graphically by use of the power triangle shown in Figure 2.5. As shown, the apparent power S, real power P, and reactive power Q form the three sides of the power triangle. The power factor angle $(\delta - \beta)$ is also shown, and the following expressions can be obtained:

$$S = \sqrt{P^2 + Q^2} \tag{2.3.6}$$

$$(\delta - \beta) = \tan^{-1}(Q/P) \tag{2.3.7}$$

$$Q = P \tan(\delta - \beta) \tag{2.3.8}$$

$$\text{p.f.} = \cos(\delta - \beta) = \frac{P}{S} = \frac{P}{\sqrt{P^2 + Q^2}} \tag{2.3.9}$$

EXAMPLE 2.3

Power triangle and power factor correction

A single-phase source delivers 100 kW to a load operating at a power factor of 0.8 lagging. Calculate the reactive power to be delivered by a capacitor connected in parallel with the load in order to raise the source power factor to 0.95 lagging. Also draw the power triangle for the source and load. Assume that the source voltage is constant, and neglect the line impedance between the source and load.

SOLUTION

The circuit and power triangle are shown in Figure 2.6. The real power $P = P_S = P_R$ delivered by the source and absorbed by the load is not changed when the capacitor is connected in parallel with the load, since the capacitor delivers only reactive power Q_C. For the load, the power factor angle, reactive power absorbed, and apparent power are

$$\theta_L = (\delta - \beta_L) = \cos^{-1}(0.8) = 36.87°$$

$$Q_L = P \tan \theta_L = 100 \tan(36.87°) = 75 \quad \text{kvar}$$

$$S_L = \frac{P}{\cos \theta_L} = 125 \quad \text{kVA}$$

(*Continued*)

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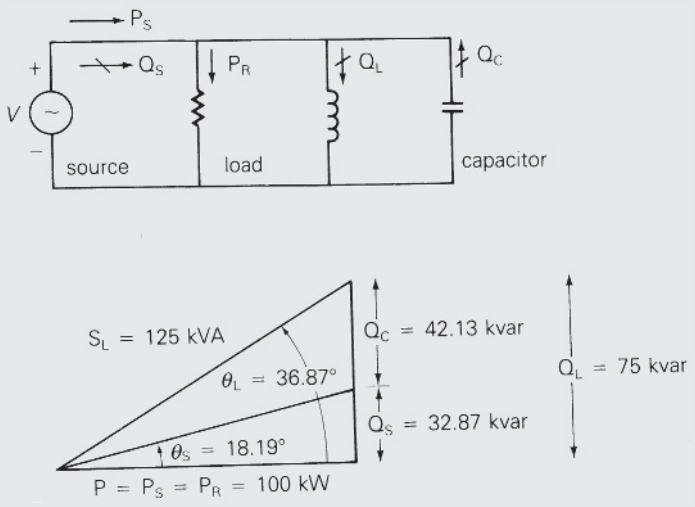
FIGURE 2.6

Circuit and power triangle for Example 2.3



After the capacitor is connected, the power factor angle, reactive power delivered, and apparent power of the source are

$$\theta_S = (\delta - \beta_S) = \cos^{-1}(0.95) = 18.19°$$

$$Q_S = P \tan \theta_S = 100 \tan(18.19°) = 32.87 \quad \text{kvar}$$

$$S_S = \frac{P}{\cos \theta_S} = \frac{100}{0.95} = 105.6 \quad \text{kVA}$$

The capacitor delivers

$$Q_C = Q_L - Q_S = 75 - 32.87 = 42.13 \text{ kVA}$$

The method of connecting a capacitor in parallel with an inductive load is known as *power factor correction.* The effect of the capacitor is to increase the power factor of the source that delivers power to the load. Also, the source apparent power S_S decreases. As shown in Figure 2.6, the source apparent power for this example decreases from 125 kVA without the capacitor to 105.3 kVA with the capacitor. The source current $I_S = S_S/V$ also decreases. When line impedance between the source and load is included, the decrease in source current results in lower line losses and lower line-voltage drops. The end result of power factor correction is improved efficiency and improved voltage regulation.

To see an animated view of this example, open PowerWorld Simulator case Example 2_3 (see Figure 2.7). From the ribbon, select the green-and-black Play button to begin the simulation. The speed and size of the green arrows are

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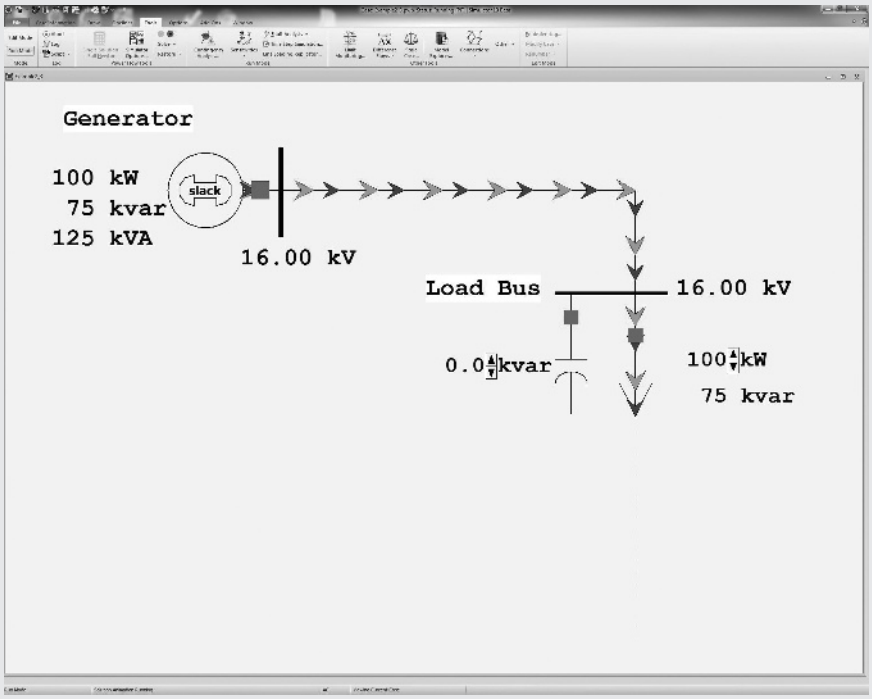
FIGURE 2.7

Screen for Example 2.3



proportional to the real power supplied to the load bus, and the blue arrows are proportional to the reactive power. Here, reactive compensation can be supplied in discrete nominal 20-kvar steps by clicking on the arrows in the capacitor's kvar field, and the load can be varied by clicking on the arrows in the load field. Notice that increasing the reactive compensation decreases both the reactive power flow on the assumed lossless supply line and the kVA power supplied by the generator; the real power flow is unchanged.

To see the impact of power factor correction when the line impedance is considered, first close the bottom line by clicking on the green circuit break symbols, and then open the top lossless line. Notice that now the load voltage is decreased and can be changed by varying the amount of compensation provided by the capacitor. Also, since the 20-kvar steps of capacitance are specified at the nominal 16-kV voltage, as indicated by (2.3.5), the amount of reactive power supplied by the capacitor is decreased since it varies with the square of its terminal voltage.

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2.4 NETWORK EQUATIONS

For circuits operating in sinusoidal-steady-state, Kirchhoff's current law (KCL) and voltage law (KVL) apply to phasor currents and voltages. Thus the sum of all phasor currents entering any node is zero, and the sum of the phasor voltage drops around any closed path is zero. Network analysis techniques based on Kirchhoff's laws, including nodal analysis, mesh or loop analysis, superposition, source transformations, and Thevenin's theorem or Norton's theorem, are useful for analyzing such circuits.

Various computer solutions of power system problems are formulated from nodal equations, which can be systematically applied to circuits. The circuit shown in Figure 2.8, which is used here to review nodal analysis, is assumed to be operating in sinusoidal-steady-state; source voltages are represented by phasors E_{S1}, E_{S2}, and E_{S3}; circuit impedances are specified in ohms. Nodal equations are written in the following three steps:

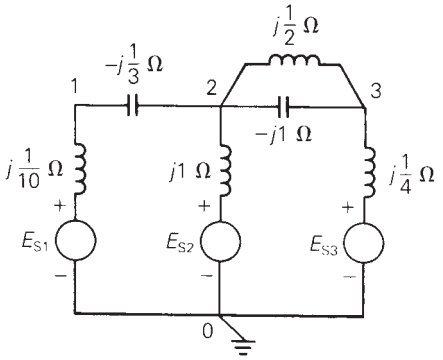
STEP 1 For a circuit with $(N + 1)$ nodes (also called buses), select one bus as the reference bus and define the voltages at the remaining buses with respect to the reference bus.

The circuit in Figure 2.8 has four buses—that is, $N + 1 = 4$ or $N = 3$. Bus 0 is selected as the reference bus, and bus voltages V_{10}, V_{20}, and V_{30} are then defined with respect to bus 0.

STEP 2 Transform each voltage source in series with an impedance to an equivalent current source in parallel with that impedance. Also, show admittance values instead of impedance values on the circuit diagram. Each current source is equal to the voltage source divided by the source impedance.

In Figure 2.9, equivalent current sources I_1, I_2, and I_3 are shown, and all impedances are converted to corresponding admittances.

FIGURE 2.8
Circuit diagram for reviewing nodal analysis



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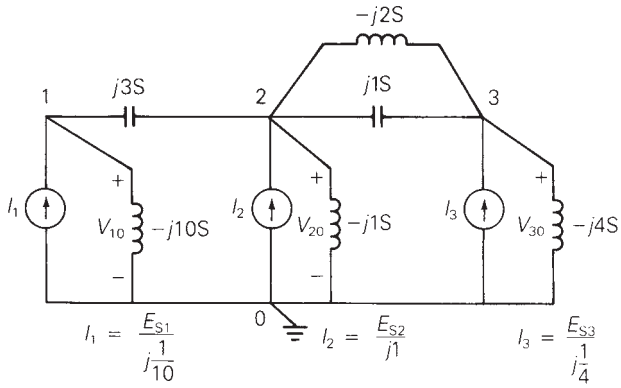

FIGURE 2.9

Circuit of Figure 2.8 with equivalent current sources replacing voltage sources. Admittance values are also shown.

STEP 3 Write nodal equations in matrix format as follows:

$$\begin{bmatrix} Y_{11} & Y_{12} & Y_{13} & \dots & Y_{1N} \\ Y_{21} & Y_{22} & Y_{23} & \dots & Y_{2N} \\ Y_{31} & Y_{32} & Y_{33} & \dots & Y_{3N} \\ \vdots & \vdots & \vdots & & \vdots \\ Y_{N1} & Y_{N2} & Y_{N3} & \dots & Y_{NN} \end{bmatrix} \begin{bmatrix} V_{10} \\ V_{20} \\ V_{30} \\ \vdots \\ V_{N0} \end{bmatrix} = \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ \vdots \\ I_N \end{bmatrix} \tag{2.4.1}$$

Using matrix notation, (2.4.1) becomes

$$\boldsymbol{YV = I} \tag{2.4.2}$$

where $\boldsymbol{Y}$ is the $N \times N$ bus admittance matrix, $\boldsymbol{V}$ is the column vector of N bus voltages, and $\boldsymbol{I}$ is the column vector of N current sources. The elements Y_{kn}, of the bus admittance matrix $\boldsymbol{Y}$, are formed as follows:

$$\text{diagonal elements:} \quad Y_{kk} = \text{sum of admittances connected to bus } k \quad (k = 1, 2, \dots, N) \tag{2.4.3}$$

$$\text{off-diagonal elements:} \quad Y_{kn} = -(\text{sum of admittances connected between buses } k \text{ and } n) \quad (k \neq n) \tag{2.4.4}$$

The diagonal element Y_{kk} is called the *self-admittance* or the *driving-point admittance* of bus k, and the off-diagonal element Y_{kn} for $k \neq n$ is called the *mutual admittance* or the *transfer admittance* between buses k and n. Since $Y_{kn} = Y_{nk}$, the matrix $\boldsymbol{Y}$ will normally be symmetric. Tap-changing transformers and phase angle regulating transformers (discussed in Chapter 3) can be represented in the matrix $\boldsymbol{Y}$ with non-symmetric elements.

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For the circuit of Figure 2.9, (2.4.1) becomes

$$\left[\begin{array}{c|c|c} (j3 - j10) & -(j3) & 0 \\ \hline -(j3) & (j3 - j1 + j1 - j2) & -(j1 - j2) \\ \hline 0 & -(j1 - j2) & (j1 - j2 - j4) \end{array}\right]\begin{bmatrix} V_{10} \\ V_{20} \\ V_{30} \end{bmatrix} = \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix}$$

$$j\begin{bmatrix} -7 & -3 & 0 \\ -3 & 1 & 1 \\ 0 & 1 & -5 \end{bmatrix}\begin{bmatrix} V_{10} \\ V_{20} \\ V_{30} \end{bmatrix} = \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} \tag{2.4.5}$$

The advantage of this method of writing nodal equations is that a digital computer can be used both to generate the admittance matrix $\boldsymbol{Y}$ and to solve (2.4.2) for the unknown bus voltage vector $\boldsymbol{V}$. Once a circuit is specified with the reference bus and other buses identified, the circuit admittances and their bus connections become computer input data for calculating the elements Y_{kn} via (2.4.3) and (2.4.4). After $\boldsymbol{Y}$ is calculated and the current source vector $\boldsymbol{I}$ is given as input, standard computer programs for solving simultaneous linear equations can then be used to determine the bus voltage vector $\boldsymbol{V}$.

When double subscripts are used to denote a voltage in this text, the voltage shall be that at the node identified by the first subscript with respect to the node identified by the second subscript. For example, the voltage V_{10} in Figure 2.9 is the voltage at node 1 with respect to node 0. Also, a current I_{ab} shall indicate the current from node a to node b. Voltage polarity marks (+/−) and current reference arrows (→) are not required when double subscript notation is employed. The polarity marks in Figure 2.9 for V_{10}, V_{20}, and V_{30}, although not required, are shown for clarity. The reference arrows for sources I_1, I_2, and I_3 in Figure 2.9 are required, however, since single subscripts are used for these currents. Matrices and vectors shall be indicated in this text by boldface type (for example, $\boldsymbol{Y}$ or $\boldsymbol{V}$).

2.5 BALANCED THREE-PHASE CIRCUITS

In this section, we introduce the following topics for balanced three-phase circuits: Y connections, line-to-neutral voltages, line-to-line voltages, line currents, Δ loads, Δ−Y conversions, and equivalent line-to-neutral diagrams.

BALANCED Y CONNECTIONS

Figure 2.10 shows a three-phase, Y-connected (or "wye-connected") voltage source feeding a balanced-Y-connected load. For a Y connection, the neutrals of each phase are connected. In Figure 2.10, the source neutral connection is labeled

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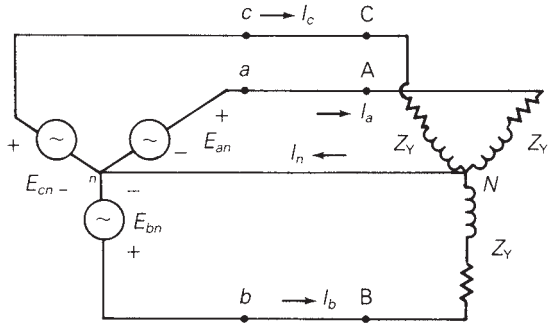


FIGURE 2.10

Circuit diagram of a three-phase Y-connected source feeding a balanced-Y load

bus *n,* and the load neutral connection is labeled bus *N.* The three-phase source is assumed to be ideal since source impedances are neglected. Also neglected are the line impedances between the source and load terminals, and the neutral impedance between buses *n* and *N.* The three-phase load is *balanced*, which means the load impedances in all three phases are identical.

BALANCED LINE-TO-NEUTRAL VOLTAGES

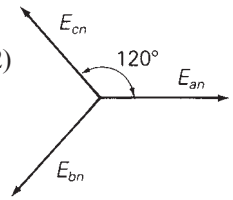
In Figure 2.10, the terminal buses of the three-phase source are labeled *a*, *b*, and c, and the source line-to-neutral voltages are labeled E_{an}, E_{bn}, and E_{cn}. The source is *balanced* when these voltages have equal magnitudes and an equal 120°-phase difference between any two phases. An example of balanced three-phase line-to-neutral voltages is

$$\begin{aligned} E_{an} &= 10\angle 0° \\ E_{bn} &= 10\angle -120° = 10\angle +240° \\ E_{cn} &= 10\angle +120° = 10\angle -240° \text{ volts} \end{aligned} \tag{2.5.1}$$

where the line-to-neutral voltage magnitude is 10 volts and E_{an} is the reference phasor. The phase sequence is called *positive sequence* or *abc sequence* when E_{an} leads E_{bn} by 120° and E_{bn} leads E_{cn} by 120°. The phase sequence is called *negative sequence* or *acb sequence* when E_{an} leads E_{cn} by 120° and E_{cn} leads E_{bn} by 120°. The voltages in (2.5.1) are positive-sequence voltages, since E_{an} leads E_{bn} by 120°. The corresponding phasor diagram is shown in Figure 2.11.

FIGURE 2.11

Phasor diagram of balanced positive-sequence line-to-neutral voltages with E_{an} as the reference



BALANCED LINE-TO-LINE VOLTAGES

The voltages E_{ab}, E_{bc}, and E_{ca} between phases are called line-to-line voltages. Writing a KVL equation for a closed path around buses *a*, *b*, and *n* in Figure 2.10.

$$E_{ab} = E_{an} - E_{bn} \tag{2.5.2}$$

For the line-to-neutral voltages of (2.5.1),

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$$E_{ab} = 10\angle 0° - 10\angle -120° = 10 - 10\left[\frac{-1 - j\sqrt{3}}{2}\right]$$
$$E_{ab} = \sqrt{3}(10)\left(\frac{\sqrt{3} + j1}{2}\right) = \sqrt{3}(10\angle 30°) \quad \text{volts} \tag{2.5.3}$$

Similarly, the line-to-line voltages E_{bc} and E_{ca} are

$$E_{bc} = E_{bn} - E_{cn} = 10\angle -120° - 10\angle +120°$$
$$= \sqrt{3}(10\angle -90°) \quad \text{volts} \tag{2.5.4}$$
$$E_{ca} = E_{cn} - E_{an} = 10\angle +120° - 10\angle 0°$$
$$= \sqrt{3}(10\angle 150°) \text{ volts} \tag{2.5.5}$$

The line-to-line voltages of (2.5.3) through (2.5.5) are also balanced, since they have equal magnitudes of $\sqrt{3}(10)$ volts and 120° displacement between any two phases. Comparison of these line-to-line voltages with the line-to-neutral voltages of (2.5.1) leads to the following conclusion:

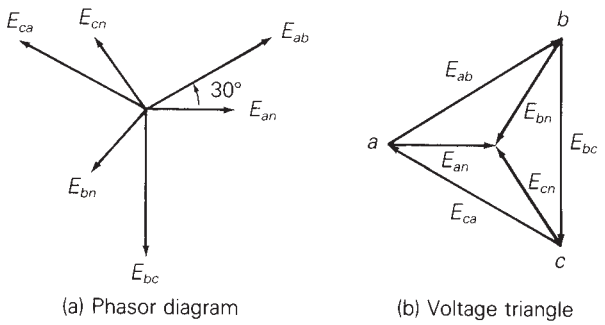
> In a balanced three-phase Y-connected system with positive-sequence sources, the line-to-line voltages are $\sqrt{3}$ times the line-to-neutral voltages and lead by 30°. That is,

$$E_{ab} = \sqrt{3}E_{an}\angle +30°$$
$$E_{bc} = \sqrt{3}E_{bn}\angle +30° \tag{2.5.6}$$
$$E_{ca} = \sqrt{3}E_{cn}\angle +30°$$

This very important result is summarized in Figure 2.12. In Figure 2.12(a), each phasor begins at the origin of the phasor diagram. In Figure 2.12(b), the line-to-line voltages form an equilateral triangle with vertices labeled *a*, *b*, *c* corresponding to buses *a*, *b*, and *c* of the system; the line-to-neutral voltages begin at the vertices and end at the center of the triangle, which is labeled *n* for neutral bus *n.* Also, the clockwise sequence of the vertices *abc* in Figure 2.12(b) indicates positive-sequence voltages. In both diagrams, E_{an} is the reference. However, the diagrams could be rotated to align with any other reference.

FIGURE 2.12

Positive-sequence line-to-neutral and line-to-line voltages in a balanced three-phase, Y-connected system



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Since the balanced line-to-line voltages form a closed triangle in Figure 2.12, their sum is zero. In fact, the sum of line-to-line voltages ($E_{ab} + E_{bc} + E_{ca}$) is *always* zero, even if the system is unbalanced, since these voltages form a closed path around buses *a*, *b*, and *c*. Also, in a balanced system, the sum of the line-to-neutral voltages ($E_{an} + E_{bn} + E_{cn}$) equals zero.

BALANCED LINE CURRENTS

Since the impedance between the source and load neutrals in Figure 2.10 is neglected, buses *n* and *N* are at the same potential, $E_{nN} = 0$. Accordingly, a separate KVL equation can be written for each phase, and the line currents can be written by inspection:

$$\begin{aligned} I_a &= E_{an}/Z_Y \\ I_b &= E_{bn}/Z_Y \\ I_c &= E_{cn}/Z_Y \end{aligned} \tag{2.5.7}$$

For example, if each phase of the Y-connected load has an impedance $Z_Y = 2\angle 30^\circ\ \Omega$, then

$$\begin{aligned} I_a &= \frac{10\angle 0^\circ}{2\angle 30^\circ} = 5\angle -30^\circ\ \text{A} \\ I_b &= \frac{10\angle -120^\circ}{2\angle 30^\circ} = 5\angle -150^\circ\ \text{A} \\ I_c &= \frac{10\angle +120^\circ}{2\angle 30^\circ} = 5\angle 90^\circ\ \text{A} \end{aligned} \tag{2.5.8}$$

The line currents are also balanced, since they have equal magnitudes of 5 A and 120° displacement between any two phases. The neutral current I_n is determined by writing a KCL equation at bus *N* in Figure 2.10.

$$I_n = I_a + I_b + I_c \tag{2.5.9}$$

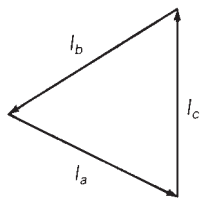
Using the line currents of (2.5.8),

$$\begin{aligned} I_n &= 5\angle -30^\circ + 5\angle -150^\circ + 5\angle 90^\circ \\ I_n &= 5\left(\frac{\sqrt{3} - j1}{2}\right) + 5\left(\frac{-\sqrt{3} - j1}{2}\right) + j5 = 0 \end{aligned} \tag{2.5.10}$$

The phasor diagram of the line currents is shown in Figure 2.13. Since these line currents form a closed triangle, their sum, which is the neutral current I_n, is zero. In general, the sum of any balanced three-phase set of phasors is zero, since balanced phasors form a closed triangle. Thus, although the impedance between neutrals *n* and *N* in Figure 2.10 is assumed to be zero, the neutral current will be zero for *any* neutral impedance ranging from short circuit (0 Ω) to open circuit (∞ Ω) as long as the system is balanced. If the system is not balanced—which could occur if the source voltages, load impedances, or line impedances were

FIGURE 2.13

Phasor diagram of line currents in a balanced three-phase system



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unbalanced—then the line currents will not be balanced, and a neutral current I_n may flow between buses n and N.

BALANCED-Δ LOADS

Figure 2.14 shows a three-phase Y-connected source feeding a balanced-Δ-connected (or "delta-connected") load. For a balanced-Δ connection, equal load impedances Z_Δ are connected in a triangle whose vertices form the buses, labeled A, B, and C in Figure 2.14. The Δ connection does not have a neutral bus.

Since the line impedances are neglected in Figure 2.14, the source line-to-line voltages are equal to the load line-to-line voltages, and the Δ-load currents I_{AB}, I_{BC} and I_{CA} are

$$\begin{aligned} I_{AB} &= E_{ab}/Z_\Delta \\ I_{BC} &= E_{bc}/Z_\Delta \\ I_{CA} &= E_{ca}/Z_\Delta \end{aligned} \tag{2.5.11}$$

For example, if the line-to-line voltages are given by (2.5.3) through (2.5.5) and if $Z_\Delta = 5\angle 30°\ \Omega$, then the Δ-load currents are

$$\begin{aligned} I_{AB} &= \sqrt{3}\left(\frac{10\angle 30°}{5\angle 30°}\right) = 3.464\angle 0°\ \text{A} \\ I_{BC} &= \sqrt{3}\left(\frac{10\angle -90°}{5\angle 30°}\right) = 3.464\angle -120°\ \text{A} \\ I_{CA} &= \sqrt{3}\left(\frac{10\angle 150°}{5\angle 30°}\right) = 3.464\angle +120°\ \text{A} \end{aligned} \tag{2.5.12}$$

Also, the line currents can be determined by writing a KCL equation at each bus of the Δ load, as follows:

$$\begin{aligned} I_a &= I_{AB} - I_{CA} = 3.464\angle 0° - 3.464\angle 120° = \sqrt{3}(3.464\angle -30°) \\ I_b &= I_{BC} - I_{AB} = 3.464\angle -120° - 3.464\angle 0° = \sqrt{3}(3.464\angle -150°) \\ I_c &= I_{CA} - I_{BC} = 3.464\angle 120° - 3.464\angle -120° = \sqrt{3}(3.464\angle +90°) \end{aligned} \tag{2.5.13}$$

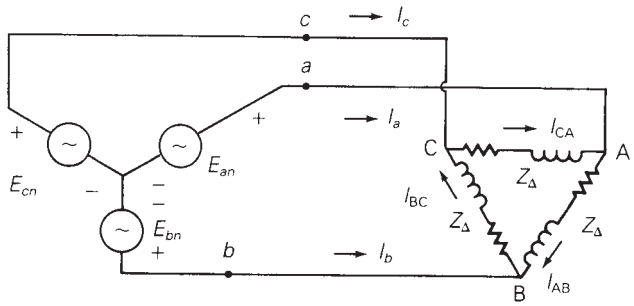

FIGURE 2.14
Circuit diagram of a Y-connected source feeding a balanced-Δ load

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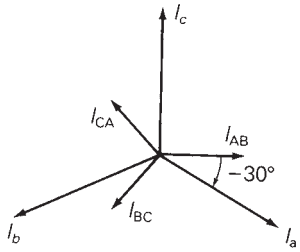

FIGURE 2.15

Phasor diagram of line currents and load currents for a balanced-Δ load

Both the Δ-load currents given by (2.5.12) and the line currents given by (2.5.13) are balanced. Thus, the sum of balanced Δ-load currents ($I_{AB} + I_{BC} + I_{CA}$) equals zero. The sum of line currents ($I_a + I_b + I_c$) is always zero for a Δ-connected load, even if the system is unbalanced, since there is no neutral wire. Comparison of (2.5.12) and (2.5.13) leads to the following conclusion:

For a balanced-Δ load supplied by a balanced positive-sequence source, the line currents into the load are $\sqrt{3}$ times the Δ-load currents and lag by 30°. That is,

$$\begin{aligned} I_a &= \sqrt{3} I_{AB} \underline{/-30^\circ} \\ I_b &= \sqrt{3} I_{BC} \underline{/-30^\circ} \\ I_c &= \sqrt{3} I_{CA} \underline{/-30^\circ} \end{aligned} \tag{2.5.14}$$

This result is summarized in Figure 2.15.

Δ−Y CONVERSION FOR BALANCED LOADS

Figure 2.16 shows the conversion of a balanced-Δ load to a balanced-Y load. If balanced voltages are applied, then these loads will be equivalent as viewed from their terminal buses A, B, and C when the line currents into the Δ load are the same as the line currents into the Y load. For the Δ load,

$$I_A = \sqrt{3} I_{AB} \underline{/-30^\circ} = \frac{\sqrt{3} E_{AB} \underline{/-30^\circ}}{Z_\Delta} \tag{2.5.15}$$

and for the Y load,

$$I_A = \frac{E_{AN}}{Z_Y} = \frac{E_{AB} \underline{/-30^\circ}}{\sqrt{3} Z_Y} \tag{2.5.16}$$

Comparison of (2.5.15) and (2.5.16) indicates that I_A will be the same for both the Δ and Y loads when

$$Z_Y = \frac{Z_\Delta}{3} \tag{2.5.17}$$

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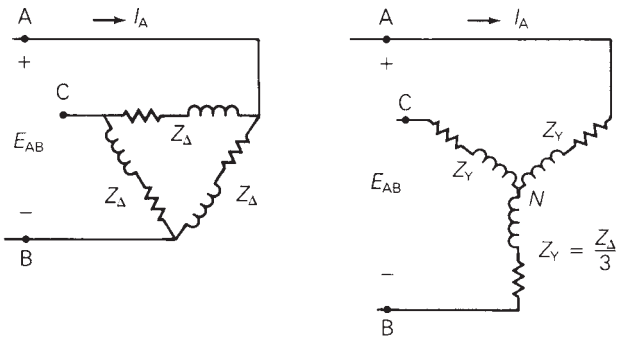
FIGURE 2.16

Δ-Y conversion for balanced loads



(a) Balanced-Δ load (b) Equivalent balanced-Y load

Also, the other line currents I_B and I_C into the Y load will equal those into the Δ load when $Z_Y = Z_\Delta/3$, since these loads are balanced. Thus a balanced-Δ load can be converted to an equivalent balanced-Y load by dividing the Δ-load impedance by 3. The angles of these Δ- and equivalent Y-load impedances are the same. Similarly, a balanced-Y load can be converted to an equivalent balanced-Δ load using $Z_\Delta = 3Z_Y$.

EXAMPLE 2.4

Balanced-Δ and -Y loads

A balanced, positive-sequence, Y-connected voltage source with $E_{ab} = 480\angle 0°$ volts is applied to a balanced-Δ load with $Z_\Delta = 30\angle 40°\ \Omega$. The line impedance between the source and load is $Z_L = 1\angle 85°\ \Omega$ for each phase. Calculate the line currents, the Δ-load currents, and the voltages at the load terminals.

SOLUTION

The solution is most easily obtained as follows. First, convert the Δ load to an equivalent Y. Then connect the source and Y-load neutrals with a zero-ohm neutral wire. The connection of the neutral wire has no effect on the circuit, since the neutral current $I_n = 0$ in a balanced system. The resulting circuit is shown in Figure 2.17. The line currents are

$$I_A = \frac{E_{an}}{Z_L + Z_Y} = \frac{\frac{480}{\sqrt{3}}\angle -30°}{1\angle 85° + \frac{30}{3}\angle 40°}$$

$$= \frac{277.1\angle -30°}{(0.0872 + j0.9962) + (7.660 + j6.428)} \qquad (2.5.18)$$

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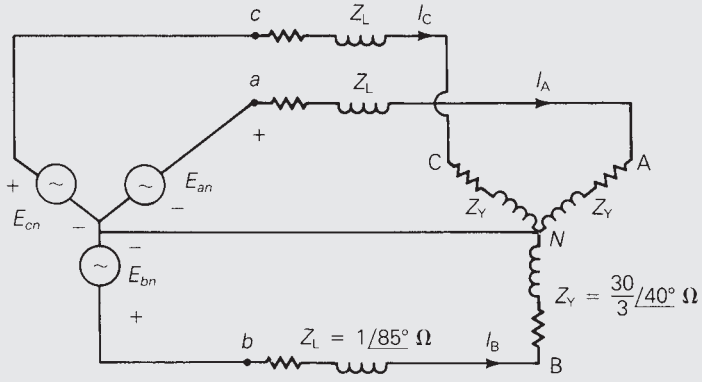
$$= \frac{277.1\angle -30^\circ}{(7.748 + j7.424)} = \frac{277.1\angle -30^\circ}{10.73\angle 43.78^\circ} = 25.83\angle -73.78^\circ \text{ A}$$

$$I_B = 25.83\angle 166.22^\circ \text{ A}$$

$$I_C = 25.83\angle 46.22^\circ \text{ A}$$

FIGURE 2.17

Circuit diagram for Example 2.4



The Δ-load currents are, from (2.5.14),

$$I_{AB} = \frac{I_A}{\sqrt{3}}\angle +30^\circ = \frac{25.83}{\sqrt{3}}\angle -73.78^\circ + 30^\circ = 14.91\angle -43.78^\circ \text{ A}$$

$$I_{BC} = 14.91\angle -163.78^\circ \text{ A} \qquad (2.5.19)$$

$$I_{CA} = 14.91\angle +76.22^\circ \text{ A}$$

The voltages at the load terminals are

$$E_{AB} = Z_\Delta I_{AB} = (30\angle 40^\circ)(14.91\angle -43.78^\circ) = 447.3\angle -3.78^\circ$$

$$E_{BC} = 447.3\angle -123.78^\circ \qquad (2.5.20)$$

$$E_{CA} = 447.3\angle 116.22^\circ \text{ volts}$$

EQUIVALENT LINE-TO-NEUTRAL DIAGRAMS

When working with balanced three-phase circuits, only one phase need be analyzed. Δ loads can be converted to Y loads, and all source and load neutrals can be connected with a zero-ohm neutral wire without changing the solution. Then one phase of the circuit can be solved. The voltages and currents in the other two phases are equal in magnitude to and ±120° out of phase with those of the solved phase. Figure 2.18 shows an equivalent line-to-neutral diagram for one phase of the circuit in Example 2.4.

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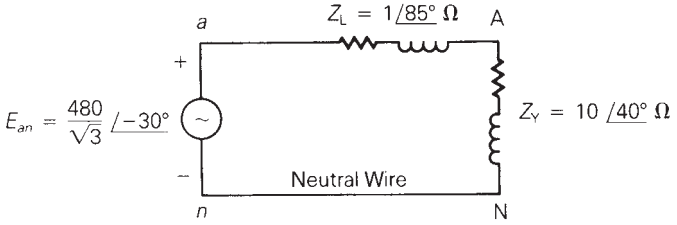
FIGURE 2.18

Equivalent line-to-neutral diagram for the circuit of Example 2.4



When discussing three-phase systems in this text, voltages shall be rms line-to-line voltages unless otherwise indicated. This is standard industry practice.

2.6 POWER IN BALANCED THREE-PHASE CIRCUITS

In this section, we discuss instantaneous power and complex power for balanced three-phase generators and motors and for balanced-Y and Δ-impedance loads.

INSTANTANEOUS POWER: BALANCED THREE-PHASE GENERATORS

Figure 2.19 shows a Y-connected generator represented by three voltage sources with their neutrals connected at bus n and by three identical generator impedances Z_g. Assume that the generator is operating under balanced steady-state conditions with the instantaneous generator terminal voltage given by

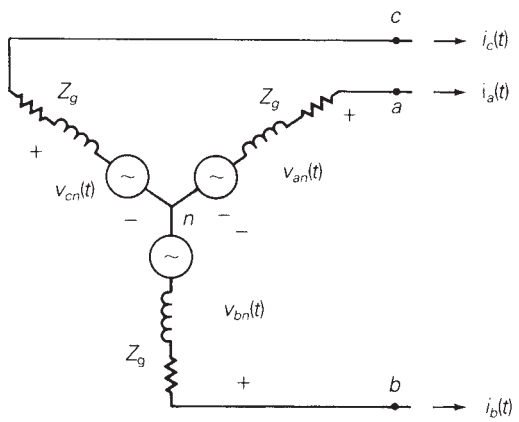
$$v_{an}(t) = \sqrt{2}\mathrm{V_{LN}}\cos(\omega t + \delta) \quad \text{volts} \tag{2.6.1}$$

and with the instantaneous current leaving the positive terminal of phase a given by

$$i_a(t) = \sqrt{2}\mathrm{I_L}\cos(\omega t + \beta)\ \mathrm{A} \tag{2.6.2}$$

FIGURE 2.19

Y-connected generator



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where V_{LN} is the rms line-to-neutral voltage and I_L is the rms line current. The instantaneous power $p_a(t)$ delivered by phase a of the generator is

$$\begin{aligned} p_a(t) &= v_{an}(t)i_a(t) \\ &= 2V_{LN}I_L \cos(\omega t + \delta)\cos(\omega t + \beta) \\ &= V_{LN}I_L \cos(\delta - \beta) + V_{LN}I_L \cos(2\omega t + \delta + \beta) \quad \text{W} \end{aligned} \tag{2.6.3}$$

Assuming balanced operating conditions, the voltages and currents of phases b and c have the same magnitudes as those of phase a and are $\pm 120°$ out of phase with phase a. Therefore, the instantaneous power delivered by phase b is

$$\begin{aligned} p_b(t) &= 2V_{LN}I_L\cos(\omega t + \delta - 120°)\cos(\omega t + \beta - 120°) \\ &= V_{LN}I_L \cos(\delta - \beta) + V_{LN}I_L \cos(2\omega t + \delta + \beta - 240°) \quad \text{W} \end{aligned} \tag{2.6.4}$$

and by phase c,

$$\begin{aligned} p_c(t) &= 2V_{LN}I_L\cos(\omega t + \delta + 120°)\cos(\omega t + \beta + 120°) \\ &= V_{LN}I_L \cos(\delta - \beta) + V_{LN}I_L \cos(2\omega t + \delta + \beta + 240°) \quad \text{W} \end{aligned} \tag{2.6.5}$$

The total instantaneous power $p_{3\phi}(t)$ delivered by the three-phase generator is the sum of the instantaneous powers delivered by each phase. Using (2.6.3) through (2.6.5):

$$\begin{aligned} p_{3\phi}(t) &= p_a(t) + p_b(t) + p_c(t) \\ &= 3V_{LN}I_L \cos(\delta - \beta) + V_{LN}I_L[\cos(2\omega t + \delta + \beta) \\ &\quad + \cos(2\omega t + \delta + \beta - 240°) \\ &\quad + \cos(2\omega t + \delta + \beta + 240°)] \ \text{W} \end{aligned} \tag{2.6.6}$$

The three cosine terms within the brackets of (2.6.6) can be represented by a *balanced* set of three phasors. Therefore, the sum of these three terms is zero for any value of δ, for any value of β, and for all values of t. Equation (2.6.6) then reduces to

$$p_{3\phi}(t) = P_{3\phi} = 3V_{LN}I_L\cos(\delta - \beta) \quad \text{W} \tag{2.6.7}$$

Equation (2.6.7) can be written in terms of the line-to-line voltage V_{LL} instead of the line-to-neutral voltage V_{LN}. Under balanced operating conditions,

$$V_{LN} = V_{LL}/\sqrt{3} \quad \text{and} \quad P_{3\phi} = \sqrt{3}V_{LL}I_L\cos(\delta - \beta) \quad \text{W} \tag{2.6.8}$$

Inspection of (2.6.8) leads to the following conclusion:

> The total instantaneous power delivered by a three-phase generator under balanced operating conditions is not a function of time, but a *constant*, $p_{3\phi}(t) = P_{3\phi}$.

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INSTANTANEOUS POWER: BALANCED THREE-PHASE MOTORS AND IMPEDANCE LOADS

The total instantaneous power absorbed by a three-phase motor under balanced steady-state conditions is also a constant. Figure 2.19 can be used to represent a three-phase motor by reversing the line currents to enter rather than leave the positive terminals. Then (2.6.1) through (2.6.8), valid for power *delivered* by a generator, are also valid for power *absorbed* by a motor. These equations are also valid for the instantaneous power absorbed by a balanced three-phase impedance load.

COMPLEX POWER: BALANCED THREE-PHASE GENERATORS

The phasor representations of the voltage and current in (2.6.1) and (2.6.2) are

$$V_{an} = \mathrm{V_{LN}}\underline{/\delta} \quad \text{volts} \tag{2.6.9}$$

$$I_a = \mathrm{I_L}\underline{/\beta} \quad \mathrm{A} \tag{2.6.10}$$

where I_a leaves positive terminal "a" of the generator. The complex power S_a delivered by phase a of the generator is

$$\begin{aligned} S_a &= V_{an}I_a^* = \mathrm{V_{LN}I_L}\underline{/(\delta - \beta)} \\ &= \mathrm{V_{LN}I_L}\cos(\delta - \beta) + j\mathrm{V_{LN}I_L}\sin(\delta - \beta) \end{aligned} \tag{2.6.11}$$

Under balanced operating conditions, the complex powers delivered by phases b and c are identical to S_a, and the total complex power $S_{3\phi}$ delivered by the generator is

$$\begin{aligned} S_{3\phi} &= S_a + S_b + S_c = 3S_a \\ &= 3\mathrm{V_{LN}I_L}\underline{/(\delta - \beta)} \\ &= 3\mathrm{V_{LN}I_L}\cos(\delta - \beta) + j3\mathrm{V_{LN}I_L}\sin(\delta - \beta) \end{aligned} \tag{2.6.12}$$

In terms of the total real and reactive powers,

$$S_{3\phi} = \mathrm{P}_{3\phi} + j\mathrm{Q}_{3\phi} \tag{2.6.13}$$

Where

$$\begin{aligned} \mathrm{P}_{3\phi} &= \mathrm{Re}(S_{3\phi}) = 3\mathrm{V_{LN}I_L}\cos(\delta - \beta) \\ &= \sqrt{3}\mathrm{V_{LL}I_L}\cos(\delta - \beta) \quad \mathrm{W} \end{aligned} \tag{2.6.14}$$

And

$$\begin{aligned} \mathrm{Q}_{3\phi} &= \mathrm{Im}(S_{3\phi}) = 3\mathrm{V_{LN}I_L}\sin(\delta - \beta) \\ &= \sqrt{3}\mathrm{V_{LL}I_L}\sin(\delta - \beta) \quad \mathrm{var} \end{aligned} \tag{2.6.15}$$

Also, the total apparent power is

$$\mathrm{S}_{3\phi} = |S_{3\phi}| = 3\mathrm{V_{LN}I_L} = \sqrt{3}\mathrm{V_{LL}I_L} \quad \mathrm{VA} \tag{2.6.16}$$

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COMPLEX POWER: BALANCED THREE-PHASE MOTORS

The preceding expressions for complex, real, reactive, and apparent power *delivered* by a three-phase generator are also valid for the complex, real, reactive, and apparent power *absorbed* by a three-phase motor.

COMPLEX POWER: BALANCED-Y AND BALANCED-Δ IMPEDANCE LOADS

Equations (2.6.13) through (2.6.16) are also valid for balanced-Y and -Δ impedance loads. For a balanced-Y load, the line-to-neutral voltage across the phase *a* load impedance and the current entering the positive terminal of that load impedance can be represented by (2.6.9) and (2.6.10). Then (2.6.11) through (2.6.16) are valid for the power absorbed by the balanced-Y load.

For a balanced-Δ load, the line-to-line voltage across the phase *a*-*b* load impedance and the current into the positive terminal of that load impedance can be represented by

$$V_{ab} = \mathrm{V_{LL}}\underline{/\delta} \quad \text{volts} \tag{2.6.17}$$

$$I_{ab} = \mathrm{I_{\Delta}}\underline{/\beta} \ \text{A} \tag{2.6.18}$$

where $\mathrm{V_{LL}}$ is the rms line-to-line voltage and $\mathrm{I_{\Delta}}$ is the rms Δ-load current. The complex power S_{ab} absorbed by the phase $a-b$ load impedance is then

$$S_{ab} = V_{ab}I_{ab}^{*} = \mathrm{V_{LL}I_{\Delta}}\underline{/(\delta - \beta)} \tag{2.6.19}$$

The total complex power absorbed by the Δ load is

$$\begin{aligned} S_{3\phi} &= S_{ab} + S_{bc} + S_{ca} = 3S_{ab} \\ &= 3\mathrm{V_{LL}I_{\Delta}}\underline{/(\delta - \beta)} \\ &= 3\mathrm{V_{LL}I_{\Delta}}\cos(\delta - \beta) + j3\mathrm{V_{LL}I_{\Delta}}\sin(\delta - \beta) \end{aligned} \tag{2.6.20}$$

Rewriting (2.6.19) in terms of the total real and reactive power,

$$S_{3\phi} = \mathrm{P_{3\phi}} + j\mathrm{Q_{3\phi}} \tag{2.6.21}$$

$$\begin{aligned} \mathrm{P_{3\phi}} &= \mathrm{Re}(S_{3\phi}) = 3\mathrm{V_{LL}I_{\Delta}}\cos(\delta - \beta) \\ &= \sqrt{3}\mathrm{V_{LL}I_{L}}\cos(\delta - \beta) \ \text{W} \end{aligned} \tag{2.6.22}$$

$$\begin{aligned} \mathrm{Q_{3\phi}} &= \mathrm{Im}(S_{3\phi}) = 3\mathrm{V_{LL}I_{\Delta}}\sin(\delta - \beta) \\ &= \sqrt{3}\mathrm{V_{LL}I_{L}}\sin(\delta - \beta) \quad \text{var} \end{aligned} \tag{2.6.23}$$

where the Δ-load current $\mathrm{I_{\Delta}}$ is expressed in terms of the line current $\mathrm{I_L} = \sqrt{3}\mathrm{I_{\Delta}}$ in (2.6.22) and (2.6.23). Also, the total apparent power is

$$\mathrm{S_{3\phi}} = |S_{3\phi}| = 3\mathrm{V_{LL}I_{\Delta}} = \sqrt{3}\mathrm{V_{LL}I_{L}} \ \text{VA} \tag{2.6.24}$$

Equations (2.6.21) through (2.6.24) developed for the balanced-Δ load are identical to (2.6.13) through (2.6.16).

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EXAMPLE 2.5

Power in a balanced three-phase system

Two balanced three-phase motors in parallel, an induction motor drawing 400 kW at 0.8 power factor lagging and a synchronous motor, drawing 150 kVA at 0.9 power factor leading, are supplied by a balanced, three-phase 4160-volt source. Cable impedances between the source and load are neglected. (a) Draw the power triangle for each motor and for the combined-motor load. (b) Determine the power factor of the combined-motor load. (c) Determine the magnitude of the line current delivered by the source. (d) A delta-connected capacitor bank is now installed in parallel with the combined-motor load. What value of capacitive reactance is required in each leg of the capacitor bank to make the source power factor unity? (e) Determine the magnitude of the line current delivered by the source with the capacitor bank installed.

SOLUTION

For the induction motor, P = 400 kW and

$S = P/\text{p.f.} = 400/0.8 = 500 \text{ kVA}$

$Q = \sqrt{S^2 - P^2} = \sqrt{(500)^2 - (400)^2} = 300 \text{ kvar absorbed}$

For the synchronous motor, S = 150 kVA and

$P = S(\text{p.f.}) = 150(0.9) = 135 \text{ kW}$

$Q = \sqrt{S^2 - P^2} = \sqrt{(150)^2 - (135)^2} = 65.4 \text{ kvar delivered}$

For the combined-motor load,

a. The power triangles for each motor and the combined-motor load are shown in Figure 2.20.

b. The power factor of the combined-motor load is p.f. = P/S = 535/584.2 = 0.916 lagging.

c. The line current delivered by the source is $I = S/(\sqrt{3}\,V)$, where S is the three-phase apparent power of the combined-motor load and V is the magnitude of the line-to-line load voltage, which is the same as the source voltage for this example. $I = 584.2/(\sqrt{3}\,4160\,V) = 0.0811 \text{ kA} = 81.1 \text{ A per phase.}$

d. For unity power factor, the three-phase reactive power supplied by the capacitor bank should equal the three-phase reactive power absorbed by the combined-motor load. That is, $Q_c = 234.6$ kvar. For a Δ-connected capacitor bank, $Q_c = 3V^2/X_\Delta$ where V is the line-to-line

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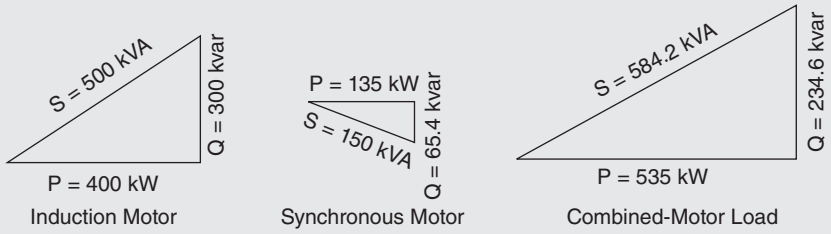
FIGURE 2.20

Power triangles for Example 2.5



voltage across the bank and X_Δ the capacitive reactance of each leg of the bank. The capacitive reactance of each leg is

$$X_\Delta = 3V^2/Q_c = 3(4160^2)/234.6 \times 10^3 = 221.3\ \Omega.$$

e. With the capacitor bank installed, the source power factor is unity, and the apparent power S delivered by the source is the same as the real power P delivered by the source. The line current magnitude is

$$I = S/(\sqrt{3}\ V) = P/(\sqrt{3}V) = 535/(\sqrt{3}\ 4160) = 0.0743\ kA = 74.3\ A \text{ per phase.}$$

In this example, the source voltage of 4160 V is not specified as a line-to-line voltage or line-to-neutral voltage, RMS or peak. Therefore, it is assumed to be an RMS line-to-line voltage, which is the convention throughout this text and a standard practice in the electric power industry. The combined-motor load absorbs 535 kW of real power. The induction motor, which operates at lagging power factor, absorbs reactive power (300 kvar) and the synchronous motor, which operates at leading power factor, delivers reactive power (65.4 kvar). The capacitor bank also delivers reactive power (234.6 kvar). Note that the line current delivered by the source is reduced from 81.1 A without the capacitor bank to 74.3 A with the capacitor bank. Any I^2R losses due to cable resistances and voltage drops due to cable reactances between the source and loads (not included in this example) would also be reduced.

2.7 ADVANTAGES OF BALANCED THREE-PHASE VERSUS SINGLE-PHASE SYSTEMS

Figure 2.21 shows three separate single-phase systems. Each single-phase system consists of the following identical components: (1) a generator represented by a voltage source and a generator impedance Z_g; (2) a forward and return conductor represented by two series line impedances Z_L; (3) a load represented by an impedance Z_Y. The three single-phase systems, although completely separated, are drawn in a Y configuration in the figure to illustrate two advantages of three-phase systems.

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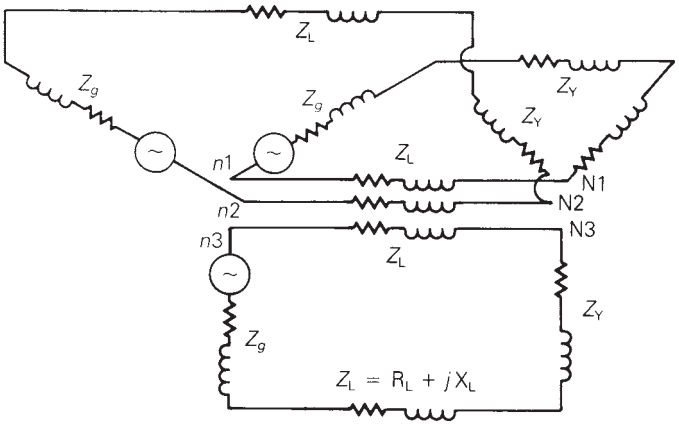
FIGURE 2.21

Three single-phase systems



Each separate single-phase system requires that *both* the forward and return conductors have a current capacity (or *ampacity*) equal to or greater than the load current. However, if the source and load neutrals in Figure 2.21 are connected to form a three-phase system, and if the source voltages are balanced with equal magnitudes and with 120° displacement between phases, then the neutral current will be zero [see (2.5.10)] and the three neutral conductors can be removed. Thus, the balanced three-phase system, while delivering the same power to the three load impedances Z_Y, requires only half the number of conductors needed for the three separate single-phase systems. Also, the total I^2R line losses in the three-phase system are only half those of the three separate single-phase systems, and the line-voltage drop between the source and load in the three-phase system is half that of each single-phase system. Therefore, one advantage of balanced three-phase systems over separate single-phase systems is reduced capital and operating costs of transmission and distribution, as well as better voltage regulation.

Some three-phase systems such as Δ-connected systems and three-wire Y-connected systems do not have any neutral conductor. However, the majority of three-phase systems are four-wire Y-connected systems, where a grounded neutral conductor is used. Neutral conductors are used to reduce transient overvoltages, which can be caused by lightning strikes and by line-switching operations, and to carry unbalanced currents, which can occur during unsymmetrical short-circuit conditions. Neutral conductors for transmission lines are typically smaller in size and ampacity than the phase conductors because the neutral current is nearly zero under normal operating conditions. Thus, the cost of a neutral conductor is substantially less than that of a phase conductor. The capital and operating costs of three-phase transmission and distribution systems with or without neutral conductors are substantially less than those of separate single-phase systems.

A second advantage of three-phase systems is that the total instantaneous electric power delivered by a three-phase generator under balanced steady-state conditions is (nearly) constant, as shown in Section 2.6. A three-phase generator (constructed with its field winding on one shaft and with its three-phase windings

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equally displaced by 120° on the stator core) will also have a nearly constant mechanical input power under balanced steady-state conditions, since the mechanical input power equals the electrical output power plus the small generator losses. Furthermore, the mechanical shaft torque, which equals mechanical input power divided by mechanical radian frequency ($T_{mech} = P_{mech}/\omega_m$) is nearly constant.

On the other hand, the equation for the instantaneous electric power delivered by a single-phase generator under balanced steady-state conditions is the same as the instantaneous power delivered by one phase of a three-phase generator, given by $p_a(t)$ in (2.6.3). As shown in that equation, $p_a(t)$ has two components: a constant and a double-frequency sinusoid. Both the mechanical input power and the mechanical shaft torque of the single-phase generator will have corresponding double-frequency components that create shaft vibration and noise, which could cause shaft failure in large machines. Accordingly, most electric generators and motors rated 5 kVA and higher are constructed as three-phase machines in order to produce nearly constant torque and thereby minimize shaft vibration and noise.

2.8 ENERGY CONVERSION

The core purpose of an electric power system is to transport energy, which involves not only transmitting electric power over a long distance, but also converting it between forms at different points along the way. These conversion functions include non-electric to electric form, electric to non-electric form, and electric to electric form.

NON-ELECTRIC TO ELECTRIC ENERGY CONVERSION

Non-electric to electric conversion is power generation. Energy is not naturally found in any usable electric form, so the first step of a power system is to convert energy from the form in which it is naturally found, such as fuel (chemical or nuclear), wind (kinetic energy of air in motion), hydro (kinetic energy of flowing water), or solar (light and heat), to electric form. This is done primarily through rotating electric machines of various types, with a notable exception being some solar generation that takes advantage of the photoelectric effect. In this text, the details of these conversions are only considered to the extent that they impact the electric power system itself, such as the modeling of synchronous machine dynamics covered in Chapters 12 and 13.

ELECTRIC TO NON-ELECTRIC ENERGY CONVERSION

Electric to non-electric conversion is found at power system loads. Most loads served by a power system need energy in a form other than electric. Various forms of conversion exist to produce useful work, such as mechanical motion (such as from a motor), light, or heat. The processes and models for such devices are numerous and are not covered in detail in this text. From a power engineering point of view, often it is sufficient to know a load's active and reactive power demand or equivalent impedance.

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ELECTRIC TO ELECTRIC ENERGY CONVERSION

Electric to electric conversion involves changing the voltage or frequency of electric power between one part of the system or another. In the 50- or 60-Hz three-phase systems that transmit the bulk of the electric energy used worldwide, the most common transformation occurs in power transformers in which voltage and current magnitudes are changed, but their frequency is unchanged. Chapter 3 covers these in more detail. Other electric-to-electric conversions involve: (1) ac to dc; (2) dc to ac; (3) dc to dc; and (4) ac to ac variable frequency transformations. Due in part to the widespread application of power electronics, electric power systems increasingly encompass the use of these conversions in the generation, storage, transmission, and consumption of electricity.

(1) ac to dc conversion

The process of converting ac into dc is known as *rectification*, with the associated devices known as *rectifiers*. Rectifiers are not new, with the first rectifier circuits (a diode) dating back to the 1880s, and they can be used with either a single-phase or three-phase ac input. The simplest single-phase circuit is a half-wave rectifier in which a single diode is used to pass half the ac waveform. A full-wave rectifier uses more diodes to pass the entire waveform, with the common bridge rectifier using four diodes for a single-phase input shown in Figure 2.22. For three-phase, the process is similar except three diodes are used for a half-wave rectifier and six for full-wave. Capacitors are often used to smooth (filter) the output voltage, and an ac transformer can be used to decrease the input voltage. When used to supply a load, this design is known as a *linear power supply*. Much more sophisticated rectifier systems are used for the conversion associated with HVDC transmission.

(2) dc to ac conversion

The conversion of dc to ac is known as *inversion* with the devices called *inverters*. Initially inversion was accomplished with motor-generator sets, but with the invention in the 1950s of thyristors, also called silicon-controlled rectifiers (SCRs), the process is now almost exclusively based on power electronics, including newer devices, such as insulated-gate bipolar transistors (IGBTs) and metal-oxide-semiconductor field-effect transistors (MOSFETs). To approximate a sinusoidal from a dc input, inverters require controlled switching, with the simplest designs just producing a square-wave output, whereas techniques such as pulse width modulation (PWM) can be used to more closely approximate a sinusoidal. There are two common types of inverters. The first are current source

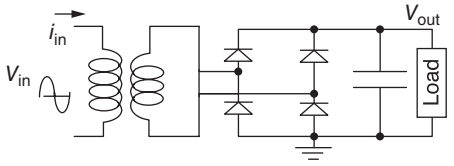

Figure 2.22
Single-phase, full-wave rectifier with a capacitor output filter

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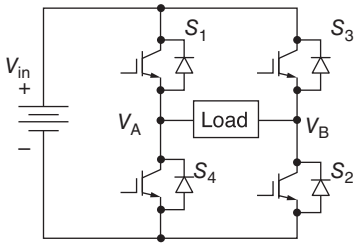

Figure 2.23
Simple single-phase inverter

inverters (CSIs), in which the inverter is fed by a constant current source, while the second are voltage source (or sourced) inverters (VSCs), in which the inverter is fed by a constant voltage source. CSIs, sometimes called line commutated converters, have long been used with HVDC since thyristors can only be turned off when their current goes through zero. VSCs, which were made possible through the introduction of newer switches, such as IGBTs and MOSFETs, are now widely used in power systems with applications, including solar photovoltaic (PV) generation, battery-based energy storage, and some newer HVDC lines. An advantage of VSC inverters is that the active and reactive power output can be controlled to provide both leading and lagging power factors. Figure 2.23 shows a simple single-phase inverter in which switch pairs S1 and S2, or S3 and S4, are alternatively switched on and off to create a square wave output.

(3) dc to dc conversion

Power electronics circuits are also now widely used for dc-dc conversion, with topologies including *buck converters* to decrease the voltage, *boost converters* to increase the voltage, and *buck-booster converters* to allow for both. With solar PVs, in which the solar panel array provides a dc output, dc-dc converters are often used for maximum power tracking to optimize the output from the solar array, with the output then inverted to connect to an ac grid.

A common application of these techniques occurs in what is known as a *switched mode power supply* in which an ac input is rectified to dc, inverted to a usually much higher frequency ac (many kHz), passed through an ac transformer that can be lighter and smaller because of the high frequency, and then rectified back to dc. Some variations also include dc to dc converters. An advantage of switched mode power supplies is they can be designed to operate with a wide variety of input voltages, such as 50-60 Hz, 100-250 V common with many power supplies for electronics.

(4) ac to ac variable frequency transformation

For ac-to-ac variable frequency transformation, frequency is first converted to dc then back to ac with a potentially different frequency. One widespread application is with electronic motor drives (sometimes called variable frequency drives [VFD]) in which the device takes a constant input frequency (e.g., 60 Hz) and produces a variable output frequency to provide speed control to an ac motor. This makes the motor load immune from most changes in the ac supply voltage magnitude and frequency and also allows for some power factor correction. Figure 2.24 shows a simple three-phase VFD in which the switching rate in the inverter circuit

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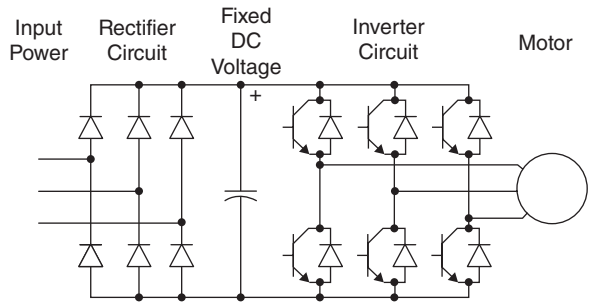

Figure 2.24
Simple three-phase variable frequency drive

is used to produce the desired output frequency. Another common application is with interfacing wind turbines with the grid, in which sometimes the full output of the wind turbine is supplied by an ac-ac (full) converter, meaning the wind generator operates asynchronous with the rest of the grid, allowing for a large amount of flexibility in its design and control.

While the integration of these newer electric energy conversion technologies provides many advantages, a key potential disadvantage is their injection of harmonics into the electric grid. In a power systems context, harmonics are defined as voltages and currents at frequencies that are integer multiples of the grid's fundamental frequency. The cause of harmonics can be understood by recalling that any periodic function can be represented by a Fourier series that consists of a sum of sinusoids that are integer multiples of the fundamental. Hence the switching and other nonlinearities inherent in these conversions introduces harmonics that can be reduced but not totally eliminated by filtering. Because these nonlinearities usually have half-wave symmetry, they only tend to introduce odd harmonics. Harmonics can cause a variety of different problems, including increased heating losses, faulty operation of equipment, resonance, and increased neutral current.

PROBLEMS

SECTION 2.1

2.1 Given the complex numbers $A_1 = 9\angle 30°$ and $A_2 = 4 + j5$: (a) convert A_1 to rectangular form; (b) convert A_2 to polar and exponential form; (c) calculate $A_3 = (A_1 + A_2)$, giving your answer in polar form; (d) calculate $A_4 = A_1 A_2$, giving your answer in rectangular form; (e) calculate $A_5 = A_1/(A_2^*)$, giving your answer in exponential form.

2.2 Convert the following instantaneous currents to phasors, using $\cos(\omega t)$ as the reference. Give your answers in both rectangular and polar form.

(a) $i(t) = 500\sqrt{2}\cos(\omega t - 30)$

(b) $i(t) = 4\sin(\omega t + 15)$

(c) $i(t) = 5\cos(\omega t - 15) + 4\sqrt{2}\sin(\omega t + 30)$

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2.3 Assume the voltage measured at one point of a circuit can be specified with the phasor $V = 12.47\angle 0°$ kV. The frequency is 60 Hz. The voltage measured at a second point has a peak magnitude of 13 kV, and that peak comes 125 microseconds after the peak of the first voltage measurement. What is the phasor expression for the second voltage measurement?

2.4 For the single-phase circuit shown in Figure 2.25, $I = 10\angle 0°$A (a) Compute the phasors I_1, I_2, and V. (b) Draw a phasor diagram showing I, I_1, I_2, and V.

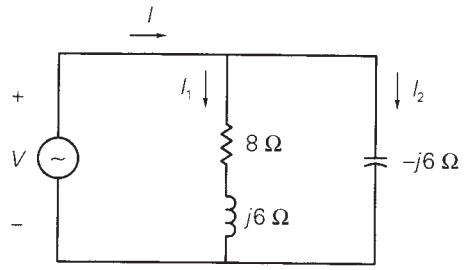

FIGURE 2.25
Circuit for Problem 2.4

2.5 A 60-Hz, single-phase source with $V = 277\angle 30°$ volts is applied to a circuit element. (a) Determine the instantaneous source voltage. Also, determine the phasor and instantaneous currents entering the positive terminal if the circuit element is (b) a 45-Ω resistor, (c) a 10-mH inductor, and (d) a capacitor with 25-Ω reactance.

2.6 (a) Transform $v(t) = 75\cos(377t - 15°)$ to phasor form. Comment on whether $\omega = 377$ appears in your answer. (b) Transform $V = 50\angle 10°$ to instantaneous form. Assume that $\omega = 377$. (c) Add the two sinusoidal functions $a(t)$ and $b(t)$ of the same frequency given as follows: $a(t) = A\sqrt{2}\cos(\omega t + \alpha)$ and $b(t) = B\sqrt{2}\cos(\omega t + \beta)$. Use phasor methods and obtain the resultant $c(t)$. Does the resultant have the same frequency?

2.7 Let a 100-V sinusoidal source be connected to a series combination of a 3-Ω resistor, an 8-Ω inductor, and a 4-Ω capacitor. (a) Draw the circuit diagram. (b) Compute the series impedance. (c) Determine the current I delivered by the source. Is the current lagging or leading the source voltage? What is the power factor of this circuit?

2.8 Consider the circuit shown in Figure 2.26 in time domain. Convert the entire circuit into phasor domain.

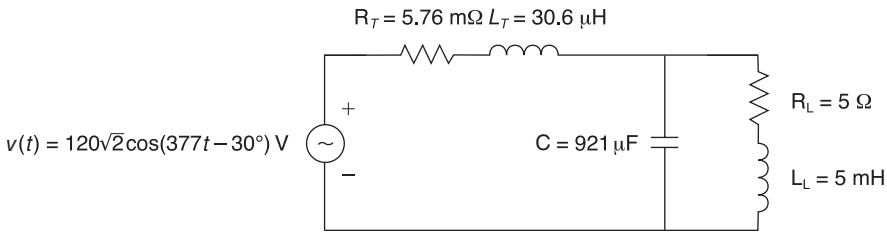

FIGURE 2.26
Circuit for Problem 2.8

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2.9 For the circuit shown in Figure 2.27, compute the voltage across the load terminals.

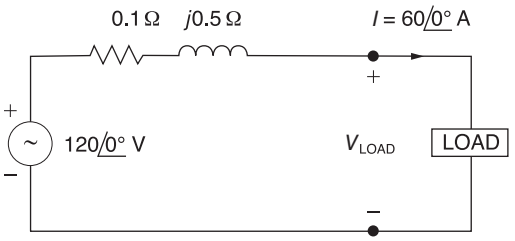

FIGURE 2.27
Circuit for Problem 2.9

SECTION 2.2

2.10 The instantaneous voltage across a circuit element is $v(t) = 400 \sin(\omega t + 30°)$ volts, and the instantaneous current entering the positive terminal of the circuit element is $i(t) = 100 \cos(\omega t + 10°)$ A. Calculate (a) the instantaneous power absorbed, (b) the real power (state whether it is delivered or absorbed), (c) the reactive power (state whether delivered or absorbed), (d) the power factor (state whether lagging or leading).
[*Note:* By convention, the power factor $\cos(\delta - \beta)$ is positive. If $|\delta - \beta|$ is greater than 90°, then the reference direction for current may be reversed, resulting in a positive value of $\cos(\delta - \beta)$].

2.11 Referring to Problem 2.5, determine the instantaneous power, real power, and reactive power absorbed by (a) the 20-Ω resistor, (b) the 10-mH inductor, and (c) the capacitor with 25-Ω reactance. Also, determine the source power factor and state whether it is lagging or leading.

2.12 The voltage $v(t) = 359.3 \cos(\omega t)$ volts is applied to a load consisting of a 10-resistor in parallel with a capacitive reactance $X_C = 25\ \Omega$. Calculate (a) the instantaneous power absorbed by the resistor, (b) the instantaneous power absorbed by the capacitor, (c) the real power absorbed by the resistor, (d) the reactive power delivered by the capacitor, and (e) the load power factor.

2.13 Repeat Problem 2.12 if the resistor and capacitor are connected in series.

2.14 A single-phase source is applied to a two-terminal, passive circuit with equivalent impedance $Z = 3.0\angle -45°\ \Omega$, measured from the terminals. The source current is $i(t) = 2\sqrt{2}\cos(\omega t)$ kA. Determine the (a) instantaneous power, (b) real power, (c) reactive power delivered by the source, and (d) source power factor.

2.15 Let a voltage source $v(t) = 4 \cos(\omega t + 60°)$ be connected to an impedance $Z = 2\angle 30°\ \Omega$. (a) Given the operating frequency to be 60 Hz, determine the expressions for the current and instantaneous power delivered by the source as functions of time. (b) Plot these functions along with $v(t)$ on a single graph for comparison. (c) Find the frequency and average value of the instantaneous power.

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2.16 A single-phase, 120-V (rms), 60-Hz source supplies power to a series R-L circuit consisting of R = 10 Ω and L = 40 mH. (a) Determine the power factor of the circuit and state whether it is lagging or leading. (b) Determine the real and reactive power absorbed by the load. (c) Calculate the peak magnetic energy W_{int} stored in the inductor by using the expression $W_{int} = L(I_{rms})^2$ and check whether the reactive power $Q = \omega W_{int}$ is satisfied. (*Note:* The instantaneous magnetic energy storage fluctuates between zero and the peak energy. This energy must be sent twice each cycle to the load from the source by means of reactive power flows.)

SECTION 2.3

2.17 Consider a load impedance of $Z = j\omega L$ connected to a voltage V and let the current drawn be I.

(a) Develop an expression for the reactive power Q in terms of ω, L, and I, from complex power considerations.
(b) Let the instantaneous current be $i(t) = \sqrt{2}I\cos(\omega t + \theta)$. Obtain an expression for the instantaneous power $p(t)$ into L, and then express it in terms of Q.
(c) Comment on the average real power P supplied to the inductor and the instantaneous power supplied.

2.18 Let a series RLC network be connected to a source voltage V, drawing a current I.

(a) In terms of the load impedance $Z = Z\angle Z$, find expressions for P and Q, from complex power considerations.
(b) Express $p(t)$ in terms of P and Q, by choosing $i(t) = \sqrt{2}I\cos\omega t$.
(c) For the case of $Z = R + j\omega L + 1/j\omega C$, interpret the result of part (b) in terms of P, Q_L, and Q_C. In particular, if $\omega^2 LC = 1$, when the inductive and capacitive reactances cancel, comment on what happens.

2.19 Consider a single-phase load with an applied voltage $v(t) = 150\cos(\omega t + 10°)$ volts and load current $i(t) = 5\cos(\omega t - 50°)$ A. (a) Determine the power triangle. (b) Find the power factor and specify whether it is lagging or leading. (c) Calculate the reactive power supplied by capacitors in parallel with the load that correct the power factor to 0.9 lagging.

2.20 A circuit consists of two impedances, $Z_1 = 20\angle 30°\,\Omega$ and $Z_2 = 25\angle 60°\,\Omega$, in parallel, supplied by a source voltage $V = 100\angle 60°$ volts. Determine the power triangle for each of the impedances and for the source.

2.21 An industrial plant consisting primarily of induction motor loads absorbs 500 kW at 0.6 power factor lagging. (a) Compute the required kVA rating of a shunt capacitor to improve the power factor to 0.9 lagging. (b) Calculate the resulting power factor if a synchronous motor rated at 500 hp with 90% efficiency

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operating at rated load and at unity power factor is added to the plant instead of the capacitor. Assume constant voltage (1 hp = 0.746 kW).

2.22 The real power delivered by a source to two impedances, $Z_1 = 4 + j5\ \Omega$ and $Z_2 = 10\ \Omega$, connected in parallel, is 1000 W. Determine (a) the real power absorbed by each of the impedances, and (b) the source current.

2.23 A single-phase source has a terminal voltage $V = 120\angle 0°$ volts and a current $I = 12\angle 30°$ A, which leaves the positive terminal of the source. Determine the real and reactive power, and state whether the source is delivering or absorbing each.

2.24 For a capacitor and resistor in parallel that, when operated at 120 V, consume 8 A of current at a power factor of 0.86 leading, (a) what are the apparent, real, and reactive power consumed by the combination, and (b) what are the values of the resistance in ohms and capacitance in farads? (Assume the system is operated at 60 Hz.)

2.25 An inductor has an inductance of 15 mH. If the ac voltages on each side of the inductor both have a magnitude of 100 V and frequency of 60 Hz, what is the maximum power that can be transferred through the inductor?

2.26 A small manufacturing plant is located 2 km down a transmission line, which has a series reactance of 0.5 Ω/km. The line resistance is negligible. The line voltage at the plant is $480\angle 0°$ V (rms), and the plant consumes 120 kW at 0.85 power factor lagging. Determine the voltage and power factor at the sending end of the transmission line by using: (a) a complex power approach; and (b) a circuit analysis approach.

2.27 An industrial load consisting of a bank of induction motors consumes 50 kW at a power factor of 0.8 lagging from a 220-V, 60-Hz, single-phase source. By placing a bank of capacitors in parallel with the load, the resultant power factor is to be raised to 0.95 lagging. Find the net capacitance of the capacitor bank in μF that is required.

2.28 Three loads are connected in parallel across a single-phase source voltage of 240 V (RMS).

Load 1 absorbs 15 kW and 6.667 kvar;
Load 2 absorbs 3 kVA at 0.96 p.f. leading;
Load 3 absorbs 15 kW at unity power factor.

Calculate the equivalent impedance, Z, for the three parallel loads, for two cases:
(a) Series combination of R and X, and (b) parallel combination of R and X.

2.29 Modeling the transmission lines as inductors, with $S_{ij} = -S_{ji}^*$, compute S_{13}, S_{31}, S_{23}, S_{32}, and S_{G3} in Figure 2.28. (*Hint:* complex power balance holds good at each bus, satisfying KCL.)

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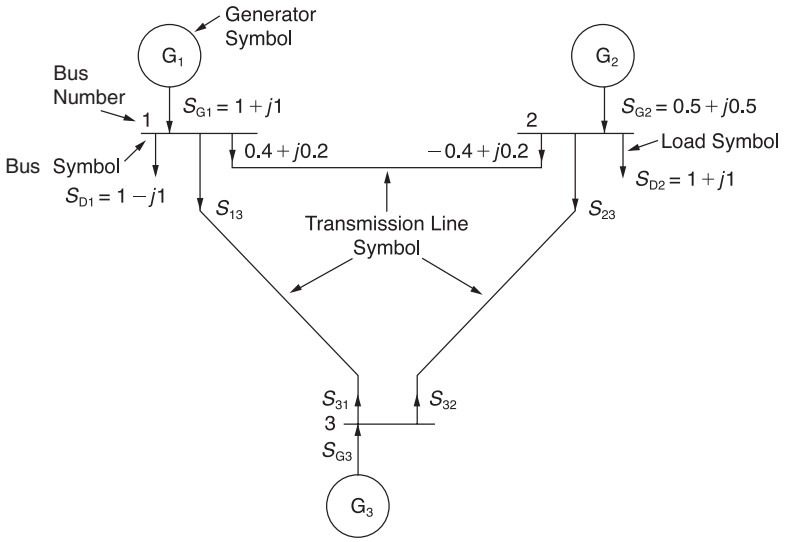

FIGURE 2.28

System diagram for Problem 2.29

2.30 Figure 2.29 shows three loads connected in parallel across a 1000-V (RMS), 60-Hz single-phase source.

Load 1: Inductive load, 125 kVA, 0.28 p.f. lagging.
Load 2: Capacitive load, 10 kW, 40 kvar.
Load 3: Resistive load, 15 kW.

(a) Determine the total kW, kvar, kva, and supply power factor.
(b) In order to improve the power factor to 0.8 lagging, a capacitor of negligible resistance is connected in parallel with the above loads. Find the kvar rating of that capacitor and the capacitance in μF.
Comment on the magnitude of the supply current after adding the capacitor.

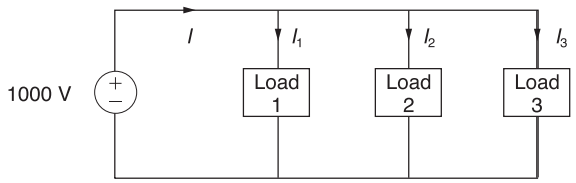

FIGURE 2.29

Circuit for Problem 2.30

2.31 Consider two interconnected voltage sources connected by a line of impedance $Z = jX\ \Omega$, as shown in Figure 2.30.

(a) Obtain expressions for P_{12} and Q_{12}.
(b) Determine the maximum power transfer and the condition for it to occur.

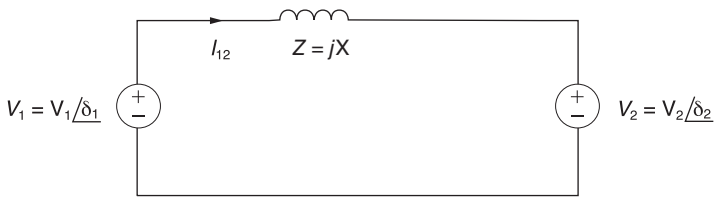

FIGURE 2.30

Circuit for Problem 2.31

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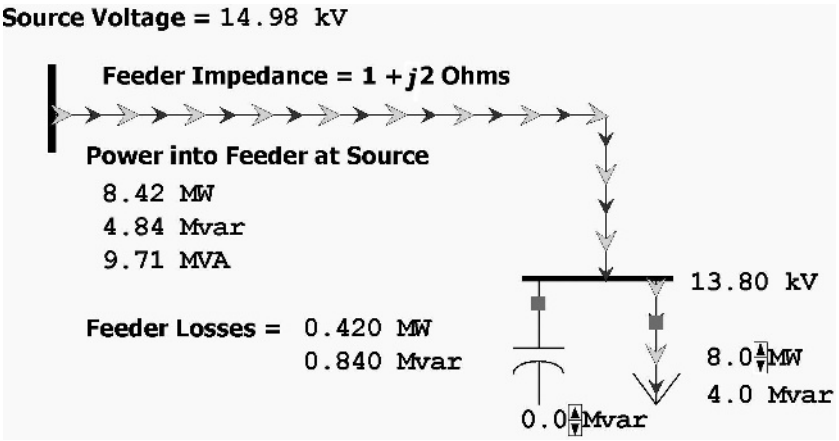
PW 2.32 In PowerWorld Simulator case Problem 2_32 (see Figure 2.31), an 8-MW and 4-Mvar load is supplied at 13.8 kV through a feeder with an impedance of $1 + j2\ \Omega$. The load is compensated with a capacitor whose output, Ω_{cap}, can be varied in 0.5 Mvar steps between 0 and 10.0 Mvars. What value of Ω_{cap} minimizes the real power line losses? What value of Ω_{cap} minimizes the MVA power flow into the feeder?

FIGURE 2.31
Screen for Problem 2.32



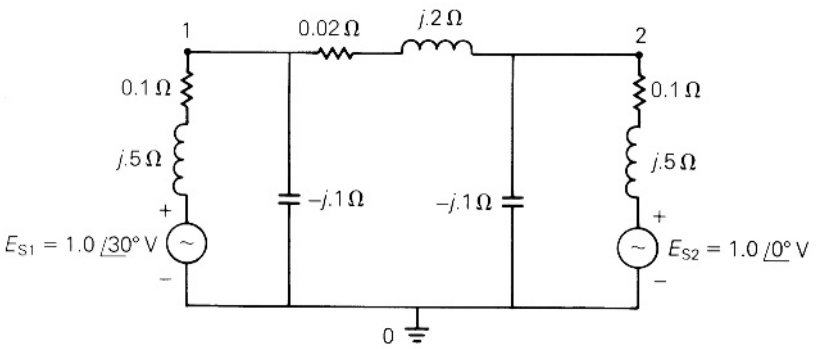
PW 2.33 For the system from Problem 2.32, plot the real and reactive line losses as Ω_{cap} is varied between 0 and 10.0 Mvars.

PW 2.34 For the system from Problem 2.32, assume that half the time the load is 10 MW and 5 Mvar, and for the other half it is 20 MW and 10 Mvar. What single value of Q_{cap} would minimize the average losses? Assume that Q_{cap} can only be varied in 0.5 Mvar steps.

SECTION 2.4

2.35 For the circuit shown in Figure 2.32, convert the voltage sources to equivalent current sources and write nodal equations in matrix format using bus 0 as the reference bus. Do not solve the equations.

FIGURE 2.32
Circuit diagram for Problems 2.35 and 2.36



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2.36 For the circuit shown in Figure 2.29, (a) determine the 2×2 bus admittance matrix $\boldsymbol{Y}_{\text{bus}}$, and (b) convert the voltage sources to current sources and determine the vector of source currents into buses 1 and 2.

2.37 Determine the 4×4 bus admittance matrix $\boldsymbol{Y}_{\text{bus}}$ and write nodal equations in matrix format for the circuit shown in Figure 2.33. Do not solve the equations.

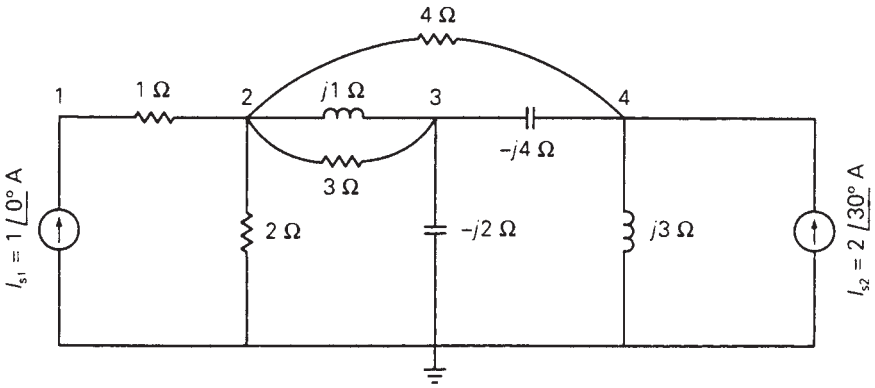

FIGURE 2.33
Circuit for Problem 2.37

2.38 Given the impedance diagram of a simple system as shown in Figure 2.34, draw the admittance diagram for the system and develop the 4×4 bus admittance matrix $\boldsymbol{Y}_{\text{bus}}$ by inspection.

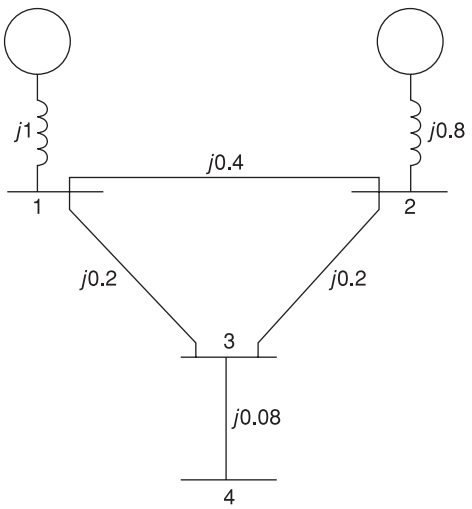

FIGURE 2.34
System diagram for Problem 2.38

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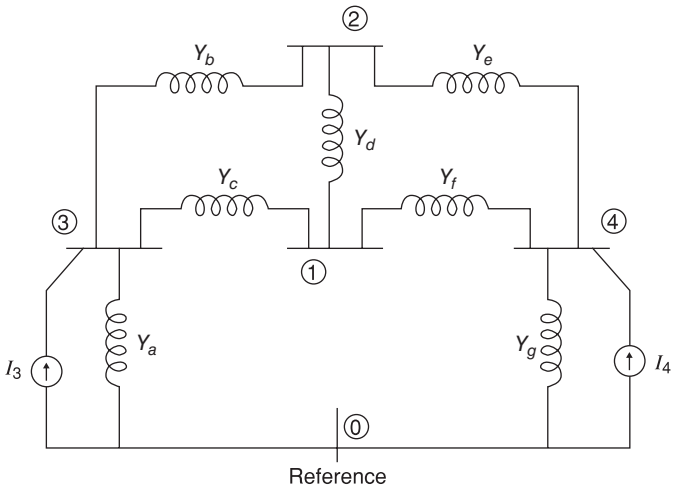
2.39 (a) Given the circuit diagram in Figure 2.35 showing admittances and current sources at nodes 3 and 4, set up the nodal equations in matrix format. (b) If the parameters are given by: $Y_a = -j0.8$ S, $Y_b = -j4.0$ S, $Y_c = -j4.0$ S, $Y_d = -j8.0$ S, $Y_e = -j5.0$ S, $Y_f = -j2.5$ S, $Y_g = -j0.8$ S, $I_3 = 1.0\angle -90°$ A, and $I_4 = 0.62\angle -135°$ A, set up the nodal equations and suggest how you would go about solving for the voltages at the nodes.

FIGURE 2.35
Circuit diagram for Problem 2.39



SECTIONS 2.5 AND 2.6

2.40 A balanced three-phase 240-V source supplies a balanced three-phase load. If the line current I_A is measured to be 12 A and is in phase with the line-to-line voltage, V_{BC}, find the per-phase load impedance if the load is (a) Y-connected, (b) Δ-connected.

2.41 A three-phase 25-kVA, 480-V, 60-Hz alternator, operating under balanced steady-state conditions, supplies a line current of 20 A per phase at a 0.8 lagging power factor and at rated voltage. Determine the power triangle for this operating condition.

2.42 A balanced Δ-connected impedance load with $(12 + j9)$ Ω per phase is supplied by a balanced three-phase 60-Hz, 208-V source. (a) Calculate the line current, the total real and reactive power absorbed by the load, the load power factor, and the apparent load power. (b) Sketch a phasor diagram showing the line currents, the line-to-line source voltages, and the Δ-load currents. Use V_{ab} as the reference.

2.43 A three-phase line, which has an impedance of $(2 + j4)$ Ω per phase, feeds two balanced three-phase loads that are connected in parallel. One of the loads is Y-connected with an impedance of $(30 + j40)$ Ω per phase, and the other is Δ-connected with an impedance of

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$(60 - j45)\ \Omega$ per phase. The line is energized at the sending end from a 60-Hz, three-phase, balanced voltage source of $120\sqrt{3}$ V (rms, line-to-line). Determine: (a) the current, real power, and reactive power delivered by the sending-end source; (b) the line-to-line voltage at the load; (c) the current per phase in each load; and (d) the total three-phase real and reactive powers absorbed by each load and by the line. Check that the total three-phase complex power delivered by the source equals the total three-phase power absorbed by the line and loads.

2.44 A Y-connected three-phase resistive load has an a-phase current of $50\angle 0°$A. What should the angle of the line-to-line voltage V_{bc} be if the system is balanced?

2.45 There are four three-phase devices connected in parallel: (a) a delta-connected capacitor bank with C = 163 μF, (b) a Y-connected bank of inductors with L = 10 mH, (c) a delta-connected inductor bank with L = 3 mH, and (4) a delta-connected resistor bank. At what frequency will the power factor of the entire combination be unity?

2.46 Three identical impedances $Z_\Delta = 30\angle 30°\Omega$ are connected in Δ to a balanced three-phase 208-V source by three identical line conductors with impedance $Z_L = (0.8 + j0.6)\ \Omega$ per line. (a) Calculate the line-to-line voltage at the load terminals. (b) Repeat part (a) when a Δ-connected capacitor bank with reactance $(-j60)\ \Omega$ per phase is connected in parallel with the load.

2.47 Two three-phase generators supply a three-phase load through separate three-phase lines. The load absorbs 30 kW at 0.8 power factor lagging. The line impedance is $(1.4 + j1.6)\ \Omega$ per phase between generator G_1 and the load, and $(0.8 + j1)\ \Omega$ per phase between generator G_2 and the load. If generator G_1 supplies 15 kW at 0.8 power factor lagging, with a terminal voltage of 460 V line-to-line, determine: (a) the voltage at the load terminals, (b) the voltage at the terminals of generator G_2, and (c) the real and reactive power supplied by generator G_2. Assume balanced operation.

2.48 Two balanced Y-connected loads in parallel, one drawing 15 kW at 0.6 power factor lagging and the other drawing 10 kVA at 0.8 power factor leading, are supplied by a balanced, three-phase, 480-volt source. (a) Draw the power triangle for each load and for the combined load. (b) Determine the power factor of the combined load and state whether lagging or leading. (c) Determine the magnitude of the line current from the source. (d) Δ-connected capacitors are now installed in parallel with the combined load. What value of capacitive reactance is needed in each leg of the Δ to make the source power factor unity? Give your answer in Ω. (e) Compute the magnitude of the current in each capacitor and the line current from the source.

2.49 Figure 2.36 gives the general Δ-Y transformation. (a) Show that the general transformation reduces to that given in Figure 2.16 for a balanced three-phase load. (b) Determine the impedances of the

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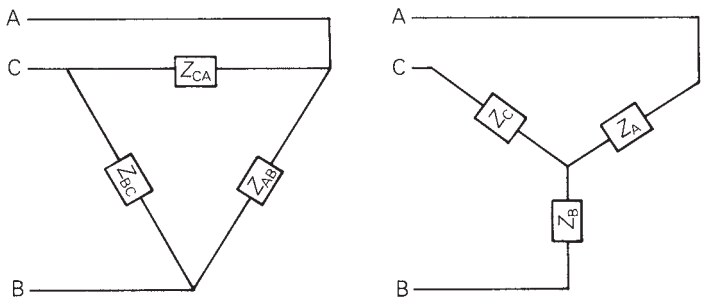
equivalent Y for the following Δ impedances: $Z_{AB} = j10$, $Z_{BC} = j20$, and $Z_{CA} = -j25\ \Omega$.

FIGURE 2.36
General Δ-Y transformation



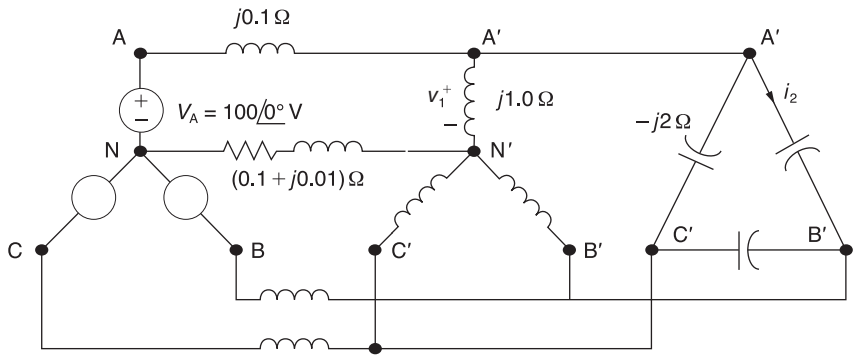
$$Z_{AB} = \frac{Z_A Z_B + Z_B Z_C + Z_C Z_A}{Z_C} \qquad Z_A = \frac{Z_{AB} Z_{CA}}{Z_{AB} + Z_{BC} + Z_{CA}}$$

$$Z_{BC} = \frac{Z_A Z_B + Z_B Z_C + Z_C Z_A}{Z_A} \qquad Z_B = \frac{Z_{AB} Z_{BC}}{Z_{AB} + Z_{BC} + Z_{CA}}$$

$$Z_{CA} = \frac{Z_A Z_B + Z_B Z_C + Z_C Z_A}{Z_B} \qquad Z_C = \frac{Z_{CA} Z_{BC}}{Z_{AB} + Z_{BC} + Z_{CA}}$$

2.50 Consider the balanced three-phase system shown in Figure 2.37. Determine $v_1(t)$ and $i_2(t)$. Assume positive phase sequence.

FIGURE 2.37
Circuit for Problem 2.50



2.51 A three-phase line with an impedance of $(0.2 + j1.0)\ \Omega$/phase feeds three balanced three-phase loads connected in parallel.

Load 1: Absorbs a total of 150 kW and 120 kvar.

Load 2: Delta connected with an impedance of $(150 - j48)\ \Omega$/phase.

Load 3: 120 kVA at 0.6 p.f. leading.

If the line-to-neutral voltage at the load end of the line is 2000 V (rms), determine the magnitude of the line-to-line voltage at the source end of the line.

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2.52 A balanced three-phase load is connected to a 4.16-kV, three-phase, four-wire, grounded-Y dedicated distribution feeder. The load can be modeled by an impedance of $Z_L = (4.7 + j9)\ \Omega$/phase, wye-connected. The impedance of the phase conductors is $(0.3 + j1)\ \Omega$. Determine the following by using the phase A to neutral voltage as a reference and assume positive phase sequence:

(a) Line currents for phases A, B, and C.
(b) Line-to-neutral voltages for all three phases at the load.
(c) Apparent, active, and reactive power dissipated per phase, and for all three phases in the load.
(d) Active power losses per phase and for all three phases in the phase conductors.

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5. Young Ngo et al., "Investing for the Future," *IEEE Power & Energy Magazine*, 18, 1 (January/February 2020), pp. 34–42.

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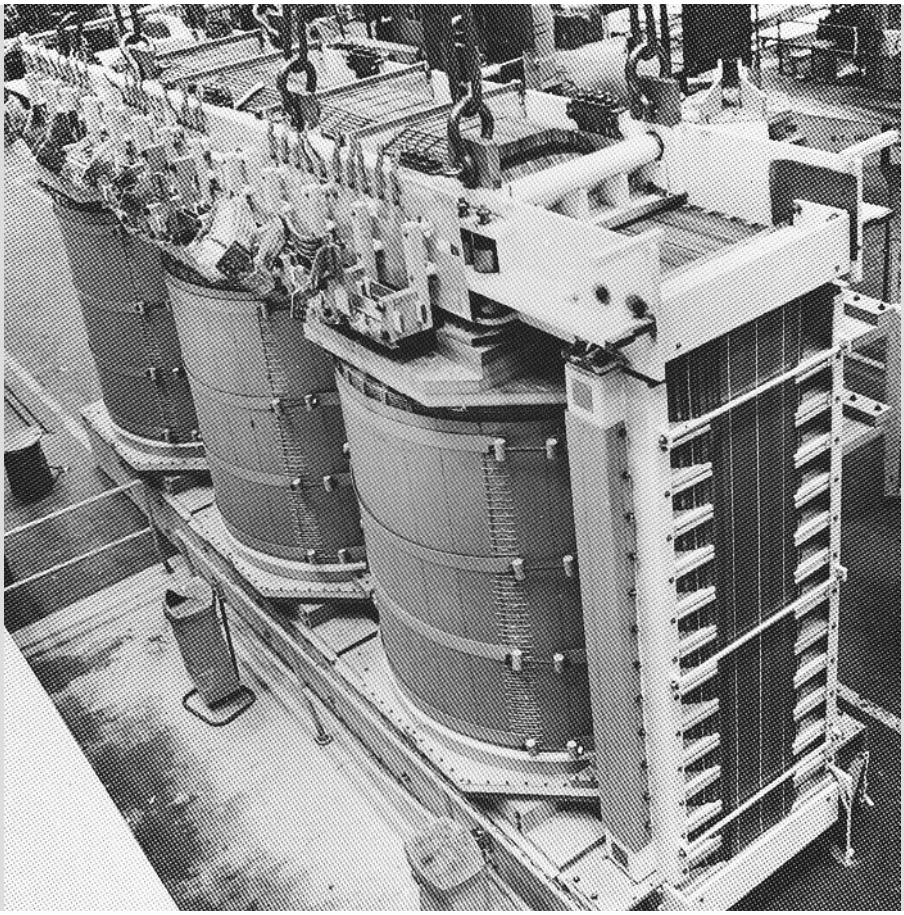
3 Power Transformers



Core and coil assemblies of a three-phase 20.3-kV Δ/345-kV Y step-up transformer. This oil-immersed transformer is rated 325 MVA self-cooled (OA)/542 MVA forced oil, forced air-cooled (FOA)/607 MVA forced oil, forced air-cooled (FOA). (Courtesy of General Electric.)

LEARNING OBJECTIVES

At the conclusion of this chapter, you should be able to:

1. Know the assumptions made for an ideal transformer and state the ideal transformer voltage and current relationships.
2. Know the convention for transformer polarity marks.
3. Be familiar with the per-unit system and state its advantages.
4. Understand per-unit equivalent circuits for practical three-phase, two-winding transformers.
5. Explain phase shift in three-phase Δ-Y and Y-Δ transformers.
6. Discuss three-winding transformers, autotransformers, tap-changing transformers, and phase-shifting transformers.

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The power transformer is a major power system component that permits economical power transmission with high efficiency and low series voltage drops. Since electric power is proportional to the product of voltage and current, low current levels (and therefore low I^2R losses and low IZ voltage drops) can be maintained for given power levels via high voltages. Power transformers transform ac voltage and current to optimum levels for generation, transmission, distribution, and utilization of electric power.

The development in 1885 by William Stanley of a commercially practical transformer was what made ac power systems more attractive than dc power systems. The ac system with a transformer overcame voltage problems encountered in dc systems as load levels and transmission distances increased. Today's modern power transformers have nearly 100% efficiency, with ratings up to and beyond 1300 MVA.

In this chapter, basic transformer theory is reviewed, and equivalent circuits for practical transformers operating under sinusoidal steady-state conditions are developed. Models of single-phase two-winding, three-phase two-winding, and three-phase three-winding transformers, as well as auto-transformers and regulating transformers, are introduced. Also, the per-unit system, which simplifies power system analysis by eliminating the ideal transformer winding in transformer equivalent circuits, is introduced in this chapter and used throughout the remainder of the text.

CASE STUDY

Transformer Innovation in a Changing Energy Landscape—Part I*

Miguel Oliva, Giuseppe Petrelli, Miguel Cuesto, and Stephane Schaal

The following paragraphs summarize the above article, which presents different cases of transformer innovation [8]. Sign in at www.cengage.com to download this product and to access its online resources.

The energy landscape is transitioning on waves of innovation that respond to today's economic, demographic, and environmental challenges. Transformers are part of this transition contributing to current megatrends like renewable energy, digitalization, sustainability, energy efficiency, circular economy, and so forth.

Ultra High Voltage Transformers

Ultra high voltage (UHV) power transformers today are reaching 1100-kV DC and 1200-kV AC, and

*Miguel Oliva et al., "Transformer Innovation in a Changing Energy Landscape–Part I," © 2020, available online, with permission, from *Transformer Magazine* (January 2020), pp. 66–76.

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are pushing power output to ever higher ratings, now exceeding 10,000 MW in the most powerful converter stations.

UHV transformer insulation structures must be prepared to cope with the increased voltage stresses by using advanced solutions for winding insulation, leads design, and turrets insulation. The nonlinear characteristics of the insulation strength result in other complexities required to meet the mechanical and thermal constraints that are exacerbated further by the larger size and dimensions that come with the higher power ratings.

UHV transformers are massive. The rating of one 1100-kV HVDC transformer developed by ABB is 607 MVA single-phase with an installed weight of more than 800 metric tons and a total length of 32 meters.

Transformer bushings are also huge at UHV levels with their increased current ratings. For example, 1100-kV dc bushings are approximately 40% longer, three times heavier, and 30% larger in diameter than 800-kV dc bushings. The increase of overall diameter comes from using larger conductors to handle the current rating, with major design steps taken to meet the higher operating requirements and test voltage levels for dc voltages. Airside corona shields are designed to match the bushing and reduce the stress in the high voltage side; and seismic requirements are evaluated together with the transformer.

Higher Power Generator Step Up Transformers

Another example of pushing boundaries comes from the power generation sector, where more powerful new generation nuclear reactors are coming into play with the EPR (European, or Evolutionary, Pressurized Power Reactor) technology. Current installations using EPR technology, under construction at different stages, are Olkiluoto in Finland with 1600 MW and two other new generation nuclear power plants, one located at Flamanville in France (1650 MW) and another at Hinkley Point in England (1630 MW), with three single-phase transformer banks.

Multiple challenges are addressed in these types of large Generator Step Up (GSU) transformers, challenges that the higher power ratings impose on the thermal, magnetic and mechanical aspects of the design. The (GSU) transformer at Hinckley Point is also prepared to operate at 15 degrees lower temperature than required by the standards to secure a longer life span and reliable operation, which means additional size and weight, but which in turn entails detailed control of hot spot temperatures and cooling requirements.

However, the development of large generator power stations is reduced worldwide with the advent of renewables. Very large GSU transformers are also used less than in the past, posing challenges to

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the transformer industry, specifically associated with knowledge management.

Extending Dry-Type Transformers to Sub-Transmission

Boundaries are also being pushed up in smaller size transformers with dry-type transformers today reaching voltage levels up to 145 kV. For years, dry-type transformers have been associated with low- and medium-voltage applications, but that has changed to expand their environmental and risk mitigation benefits. The traditional application is now extended to higher voltage and power ratings, incorporating oil-free tap changers under load, to cover (from the distribution side) buildings, hospitals, and rail traction infrastructures like metro stations, up to sub-transmission levels where they are increasingly used in metropolitan areas to step down three-phase medium-voltage to low-voltage for power distribution.

Renewables Applications

Renewable energy generation is one of the main drivers of the energy transition. There are multiple innovations associated with diverse types of renewable energy with different challenges and requirements.

- Solar transformers are exposed to the harmonic components of inverters, typically in the 2–150 kHz frequency range.
- Wind turbine transformers are following the trend of increased power ratings of turbines, with higher voltages, while maintaining efficiency and compact dimensions.
- Offshore wind transformers are subject to salty corrosive environments and vibrations with demanding size and weight requirements.

Solar

Recent developments in solar transformer applications address the need for shorter design, development and construction lead cycles, to deliver transformers quickly and ready them for faster installation and operation. One innovation is the development of standardized concepts that can be quickly implemented and customized for a particular project, providing economies of scale and process efficiency to support shorter cycles and cost reductions.

An example is containerized solutions that are adapted to any required power output, with liquid-filled or dry-type transformers. Solar transformers inside conventional containers can accommodate central inverters with an output voltage of 1000 V DC for high efficiency. There are also containerized multi-winding transformer solutions for 1500 V DC up to several MVA.

Another example is the use of ester fluids in liquid-filled transformers to enable higher-temperature or energy-efficient designs, based on different inverter voltage and power ratings. Total costs are

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optimized and environmental impacts are reduced.

Wind turbine transformers connect the turbines to the distribution system, stepping up the output voltage from the generator, while power collection transformers are installed downstream for connection to a high voltage grid. Manufacturers of wind turbines are competing to increase their power ratings (on the way to 12-14 MW each), which helps reduce leveled energy costs. Transformers are following the same trend, increasing their power and voltage ratings, with voltages up to 66 kV at the transformer exit, and with compact, lightweight design and overall efficiency gains.

Wind turbine transformer challenges include variable output and thermal cycles due to wind intermittency, harmonics associated with the interface with the drive train and other electricals, mechanical vibrations, and corrosion issues. These challenges are being addressed with compact and lightweight transformer designs. In the case of liquid-filled transformers, solutions include forced oil and forced water cooling methods, sealed and vacuum-proof moisture-free tanks, and ester fluids.

Wind turbine transformers can be located in the nacelle at the top of the tower, or at ground level inside or outside the tower. To optimize costs, transformer locations are defined by reducing the length of their connection to the lower-voltage side, which means placing the transformer as close as possible to the wind generator.

The article also presents innovations for medium frequency transformers used with converters and power electronics applications, for traction transformers used in the railway industry for electric multiple-unit trains and locomotives of high-speed trains, and for subsea transformers used in the oil and gas industry at offshore platforms. Transformer innovations are instrumental in addressing the challenges, not just with pure R&D or technology developments, but encompassing processes and operations across the whole value chain from design through transformer operation and maintenance.

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3.1 THE IDEAL TRANSFORMER

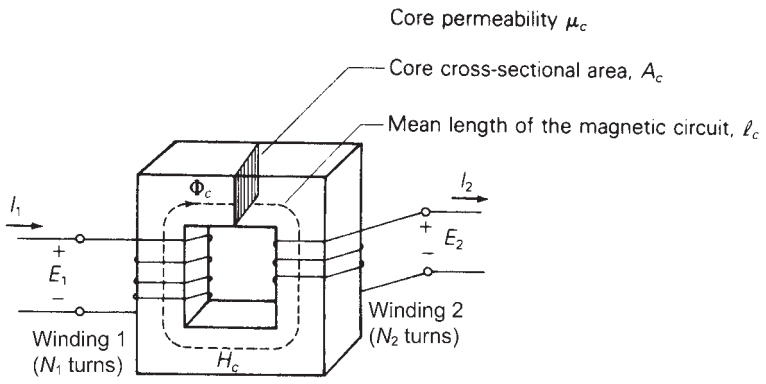
Figure 3.1 shows a basic single-phase two-winding transformer, where the two windings are wrapped around a magnetic core [1, 2, 3]. It is assumed here that the transformer is operating under sinusoidal steady-state excitation. Shown in the figure are the phasor voltages E_1 and E_2 across the windings, and the phasor currents I_1 entering winding 1, which has N_1 turns, and I_2 leaving winding 2, which has N_2 turns. A phasor flux Φ_c set up in the core and a magnetic field intensity phasor H_c are also shown. The core has a cross-sectional area denoted A_c, a mean length of the magnetic circuit l_c, and a magnetic permeability μ_c, assumed constant.

For an ideal transformer, the following are assumed:

1. The windings have zero resistance; therefore, the I^2R losses in the windings are zero.
2. The core permeability μ_c is infinite, which corresponds to zero core reluctance.
3. There is no leakage flux; that is, the entire flux Φ_c is confined to the core and links both windings.
4. There are no core losses.

A schematic representation of a two-winding transformer is shown in Figure 3.2. Ampere's and Faraday's laws can be used along with the preceding assumptions to derive the ideal transformer relationships. *Ampere's law* states that the tangential component of the magnetic field intensity vector integrated along a closed path equals the net current enclosed by that path; that is,

FIGURE 3.1
Basic single-phase two-winding transformer



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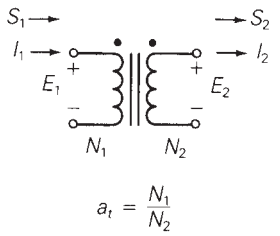

FIGURE 3.2

Schematic representation of a single-phase two-winding transformer

$$\oint H_{\tan} dl = I_{\text{enclosed}} \tag{3.1.1}$$

If the core center line shown in Figure 3.1 is selected as the closed path, and if H_c is constant along the path as well as tangent to the path, then (3.1.1) becomes

$$H_c l_c = N_1 I_1 - N_2 I_2 \tag{3.1.2}$$

Note that the current I_1 is enclosed N_1 times and I_2 is enclosed N_2 times, one time for each turn of the coils. Also, using the right-hand rule*, current I_1 contributes to clockwise flux, but current I_2 contributes to counterclockwise flux. Thus, in (3.1.2) the net current enclosed is $N_1 I_1 - N_2 I_2$. For constant core permeability μ_c, the magnetic flux density B_c within the core, also constant, is

$$B_c = \mu_c H_c \quad \text{Wb/m}^2 \tag{3.1.3}$$

and the core flux Φ_c is

$$\Phi_c = B_C \text{A}_C \quad \text{Wb} \tag{3.1.4}$$

*The right-hand rule for a coil is as follows: Wrap the fingers of your right hand around the coil in the direction of the current. Your right thumb then points in the direction of the flux.

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Using (3.1.3) and (3.1.4) in (3.1.2) yields

$$N_1 I_1 - N_2 I_2 = l_c B_c / \mu_c = \left(\frac{l_c}{\mu_c \mathrm{A}_c} \right) \Phi_c \tag{3.1.5}$$

Core reluctance R_c is defined as

$$\mathrm{R}_c = \frac{l_c}{\mu_c \mathrm{A}_c} \tag{3.1.6}$$

Then (3.1.5) becomes

$$N_1 I_1 - N_2 I_2 = \mathrm{R}_c \Phi_c \tag{3.1.7}$$

Equation (3.1.7) can be called *Ohm's law* for the magnetic circuit, wherein the net magnetomotive force mmf $= N_1 I_1 - N_2 I_2$ equals the product of the core reluctance R_c and the core flux Φ_c. Reluctance R_c, which impedes the establishment of flux in a magnetic circuit, is analogous to resistance in an electric circuit. For an ideal transformer, μ_c is assumed infinite, which from (3.1.6) means that R_c is 0, and (3.1.7) becomes

$$N_1 I_1 = N_2 I_2 \tag{3.1.8}$$

In practice, power transformer windings and cores are contained within enclosures, and the winding directions are not visible. One way of conveying winding information is to place a dot at one end of each winding such that when current enters a winding at the dot, it produces an mmf acting in the *same* direction. This dot convention is shown in the schematic of Figure 3.2. The dots are conventionally called *polarity marks.*

Equation (3.1.8) is written for current I_1 *entering* its dotted terminal and current I_2 *leaving* its dotted terminal. As such, I_1 and I_2 are *in phase,* since $I_1 = (N_2/N_1)I_2$. If the direction chosen for I_2 were reversed, such that both currents entered their dotted terminals, then I_1 would be 180° *out of phase* with I_2.

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Faraday's law states that the voltage $e(t)$ induced across an N-turn winding by a time-varying flux $\phi(t)$ linking the winding is

$$e(t) = N\frac{d\phi(t)}{dt} \tag{3.1.9}$$

Assuming a sinusoidal steady-state flux with constant frequency ω, and representing $e(t)$ and $\phi(t)$ by their phasors E and Φ, (3.1.9) becomes

$$E = N(j\omega)\Phi \tag{3.1.10}$$

For an ideal transformer, the entire flux is assumed to be confined to the core, linking both windings. From Faraday's law, the induced voltages across the windings of Figure 3.1 are

$$E_1 = N_1(j\omega)\Phi_c \tag{3.1.11}$$

$$E_2 = N_2(j\omega)\Phi_c \tag{3.1.12}$$

Dividing (3.1.11) by (3.1.12) yields

$$\frac{E_1}{E_2} = \frac{N_1}{N_2} \tag{3.1.13}$$

or

$$\frac{E_1}{N_1} = \frac{E_2}{N_2} \tag{3.1.14}$$

The dots shown in Figure 3.2 indicate that the voltages E_1 and E_2, both of which have their + polarities at the dotted terminals, are in phase. If the polarity chosen for one of the voltages in Figure 3.1 were reversed, then E_1 would be 180° out of phase with E_2.

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The turns ratio a_t is defined as follows:

$$a_t = \frac{N_1}{N_2} \tag{3.1.15}$$

Using a_t in (3.1.8) and (3.1.14), the basic relations for an ideal single-phase two-winding transformer are

$$E_1 = \left(\frac{N_1}{N_2}\right)E_2 = a_t E_2 \tag{3.1.16}$$

$$I_1 = \left(\frac{N_2}{N_1}\right)I_2 = \frac{I_2}{a_t} \tag{3.1.17}$$

Two additional relations concerning complex power and impedance can be derived from (3.1.16) and (3.1.17) as follows. The complex power entering winding 1 in Figure 3.2 is

$$S_1 = E_1 I_1^* \tag{3.1.18}$$

Using (3.1.16) and (3.1.17),

$$S_1 = E_1 I_1^* = (a_t E_2)\left(\frac{I_2}{a_t}\right)^* = E_2 I_2^* = S_2 \tag{3.1.19}$$

As shown by (3.1.19), the complex power S_1 entering winding 1 equals the complex power S_2 leaving winding 2. That is, an ideal transformer has no real or reactive power loss.

If an impedance Z_2 is connected across winding 2 of the ideal transformer in Figure 3.2, then

$$Z_2 = \frac{E_2}{I_2} \tag{3.1.20}$$

This impedance, when measured from winding 1, is

$$Z_2' = \frac{E_1}{I_1} = \frac{a_t E_2}{I_2/a_t} = a_t^2 Z_2 = \left(\frac{N_1}{N_2}\right)^2 Z_2 \tag{3.1.21}$$

Thus, the impedance Z_2 connected to winding 2 is referred to winding 1 by multiplying Z_2 by a_t^2, which is the square of the turns ratio.

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EXAMPLE 3.1

Ideal, single-phase two-winding transformer

A single-phase two-winding transformer is rated 20 kVA, 480/120 V, and 60 Hz. A source connected to the 480-V winding supplies an impedance load connected to the 120-V winding. The load absorbs 15 kVA at 0.8 p.f. lagging when the load voltage is 118 V. Assume that the transformer is ideal and calculate the following:

a. The voltage across the 480-V winding.
b. The load impedance.
c. The load impedance referred to the 480-V winding.
d. The real and reactive power supplied to the 480-V winding.

SOLUTION

a. The circuit is shown in Figure 3.3, where winding 1 denotes the 480-V winding and winding 2 denotes the 120-V winding. Selecting the load voltage E_2 as the reference,

$$E_2 = 118\ \angle 0^\circ \ \text{V}$$

The turns ratio is, from (3.1.13),

$$a_t = \frac{N_1}{N_2} = \frac{E_{1\text{rated}}}{E_{2\text{rated}}} = \frac{480}{120} = 4$$

and the voltage across winding 1 is

$$E_1 = a_t E_2 = 4(118\ \angle 0^\circ) = 472\ \angle 0^\circ \ \text{V}$$

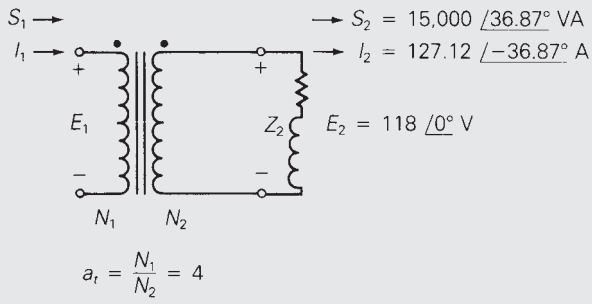

FIGURE 3.3 Circuit for Example 3.1

(*Continued*)

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b. The complex power S_2 absorbed by the load is

$$S_2 = E_2 I_2^* = 118 I_2^* = 15{,}000 \underline{/\cos^{-1}(0.8)} = 15{,}000 \underline{/36.87^\circ} \text{ VA}$$

Solving, the load current I_2 is

$$I_2 = 127.12 \underline{/-36.87^\circ} \text{ A}$$

The load impedance Z_2 is

$$Z_2 = \frac{E_2}{I_2} = \frac{118 \underline{/0^\circ}}{127.12 \underline{/-36.87^\circ}} = 0.9283 \underline{/36.87^\circ} \ \Omega$$

c. From (3.1.21), the load impedance referred to the 480-V winding is

$$Z_2' = a_t^2 Z_2 = (4)^2 (0.9283 \underline{/36.87^\circ}) = 14.85 \underline{/36.87^\circ} \ \Omega$$

d. From (3.1.19)

$$S_1 = S_2 = 15{,}000 \underline{/36.87^\circ} = 12{,}000 + j9000$$

Thus, the real and reactive powers supplied to the 480-V winding are

$$P_1 = \text{Re } S_1 = 12{,}000 \text{ W} = 12 \text{ kW}$$

$$Q_1 = \text{Im } S_1 = 9000 \text{ var} = 9 \text{ kvar}$$

Figure 3.4 shows a schematic of a conceptual single-phase, phase-shifting transformer. This transformer is not an idealization of an actual transformer since it is physically impossible to obtain a complex turns ratio. It is used later in this chapter as a mathematical model for representing phase shift of three-phase transformers. As shown in Figure 3.4, the complex turns ratio a_t is defined for the phase-shifting transformer as

$$a_t = \frac{e^{j\phi}}{1} = e^{j\phi} \tag{3.1.22}$$

where ϕ is the phase-shift angle. The transformer relations are then

$$E_1 = a_t E_2 = e^{j\phi} E_2 \tag{3.1.23}$$

$$I_1 = \frac{I_2}{a_t^*} = e^{j\phi} I_2 \tag{3.1.24}$$

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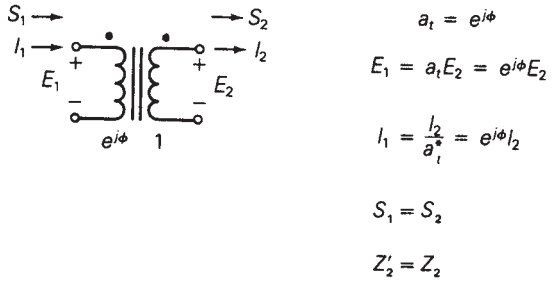

FIGURE 3.4

Schematic representation of a conceptual single-phase, phase-shifting transformer

Note that the phase angle of E_1 leads the phase angle of E_2 by ϕ. Similarly, I_1 leads I_2 by the angle ϕ. However, the magnitudes are unchanged; that is, $|E_1| = |E_2|$ and $|I_1| = |I_2|$.

From these two relations, the following two additional relations are derived:

$$S_1 = E_1 I_1^* = (a_t E_2)\left(\frac{I_2}{a_t^*}\right)^* = E_2 I_2^* = S_2 \tag{3.1.25}$$

$$Z_2' = \frac{E_1}{I_1} = \frac{a_t E_2}{\frac{1}{a_t^*} I_2} = |a_t|^2 Z_2 = Z_2 \tag{3.1.26}$$

Thus, impedance is unchanged when it is referred from one side of an ideal phase-shifting transformer to the other. Also, the ideal phase-shifting transformer has no real or reactive power losses since $S_1 = S_2$.

Note that (3.1.23) and (3.1.24) for the phase-shifting transformer are the same as (3.1.16) and (3.1.17) for the ideal physical transformer except for the complex conjugate (*) in (3.1.24). The complex conjugate for the phase-shifting transformer is required to make $S_1 = S_2$ (complex power into winding 1 equals complex power out of winding 2), as shown in (3.1.25).

3.2 EQUIVALENT CIRCUITS FOR PRACTICAL TRANSFORMERS

Figure 3.5 shows an equivalent circuit for a practical single-phase two-winding transformer, which differs from the ideal transformer as follows:

1. The windings have resistance.
2. The core permeability μ_c is finite.
3. The magnetic flux is not entirely confined to the core.
4. There are real and reactive power losses in the core.

The resistance R_1 is included in series with winding 1 of the figure to account for I^2R losses in this winding. A reactance X_1, called the *leakage reactance*

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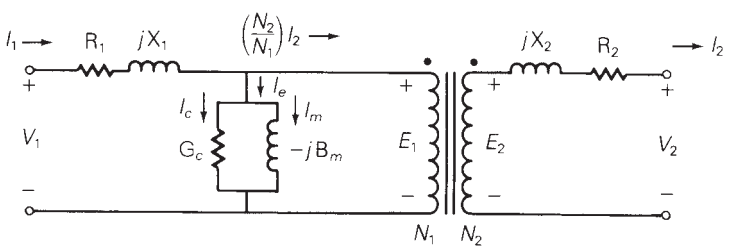

FIGURE 3.5
Equivalent circuit of a practical single-phase, two-winding transformer

of winding 1, is also included in series with winding 1 to account for the leakage flux of winding 1. This leakage flux is the component of the flux that links winding 1 but does not link winding 2; it causes a voltage drop $I_1(j\mathrm{X}_1)$, which is proportional to I_1 and leads I_1 by 90°. There is also a reactive power loss $\mathrm{I}_1^2\mathrm{X}_1$ associated with this leakage reactance. Similarly, there is a resistance R_2 and a leakage reactance X_2 in series with winding 2.

Equation (3.1.7) shows that for finite core permeability μ_c, the total mmf is not zero. Dividing (3.1.7) by N_1 and using (3.1.11), the result is

$$I_1 - \left(\frac{N_2}{N_1}\right)I_2 = \frac{\mathrm{R}_c}{N_1}\Phi_c = \frac{\mathrm{R}_c}{N_1}\left(\frac{E_1}{j\omega N_1}\right) = -j\left(\frac{\mathrm{R}_c}{\omega N_1^2}\right)E_1 \tag{3.2.1}$$

Defining the term on the right-hand side of (3.2.1) to be I_m, called *magnetizing* current, it is evident that I_m lags E_1 by 90°, and can be represented by a shunt inductor with susceptance $\mathrm{B}_m = \left(\frac{R_c}{\omega N_1^2}\right)$ mhos.* However, in reality, there is an additional shunt branch, represented by a resistor with conductance G_c mhos, which carries a current I_c, called the *core loss* current. I_c is in phase with E_1. When the core loss current I_c is included, (3.2.1) becomes

$$I_1 - \left(\frac{N_2}{N_1}\right)I_2 = I_c + I_m = (\mathrm{G}_c - j\mathrm{B}_m)E_1 \tag{3.2.2}$$

The equivalent circuit of Figure 3.5, which includes the shunt branch with admittance $(\mathrm{G}_c - j\,\mathrm{B}_m)$ mhos, satisfies the KCL equation (3.2.2). Note that when winding 2 is open ($I_2 = 0$) and when a sinusoidal voltage V_1 is applied to winding 1, then (3.2.2) indicates that the current I_1 will have two components: the core loss current I_c and the magnetizing current I_m. Associated with I_c is a real power loss $\mathrm{I}_c^2/\mathrm{G}_c = \mathrm{E}_1^2\mathrm{G}_c$ W. This real power loss accounts for both hysteresis and eddy current losses within the core. Hysteresis loss occurs because a cyclic variation of flux within the core requires energy dissipated as heat. As such, hysteresis loss can be reduced by the use of special high grades of alloy steel as core material. Eddy current loss occurs because induced currents called eddy currents flow within the

*The units of admittance, conductance, and susceptance, which in the SI system are siemens (with symbol S), are also called mhos (with symbol ℧) or ohms^{-1} (with symbol Ω^{-1}).

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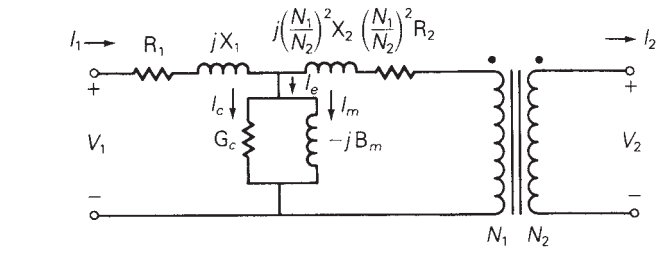
magnetic core perpendicular to the flux. As such, eddy current loss can be reduced by constructing the core with laminated sheets of alloy steel. Associated with I_m is a reactive power loss $I_m^2/B_m = E_1^2 B_m$ var. This reactive power is required to magnetize the core. The phasor sum $(I_c + I_m)$ is called the *exciting* current I_e.

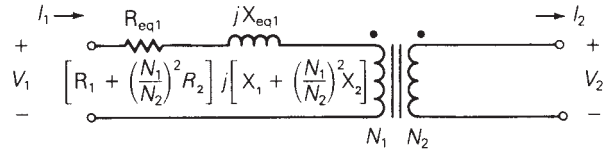
Figure 3.6 shows three alternative equivalent circuits for a practical single-phase two-winding transformer. In Figure 3.6(a), the resistance R_2 and leakage reactance X_2 of winding 2 are referred to winding 1 via (3.1.21).

In Figure 3.6(b), the shunt branch is omitted, which corresponds to neglecting the exciting current. Since the exciting current is usually less than 5% of rated current, neglecting it in power system studies is often valid unless transformer efficiency or exciting current phenomena are of particular concern. For large power transformers rated more than 500 kVA, the winding resistances, which are small compared to the leakage reactances, often can be neglected, as shown in Figure 3.6(c).

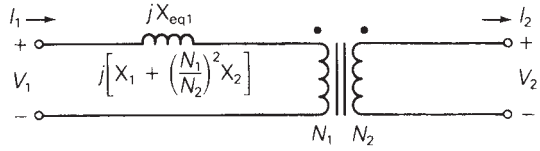
Thus, a practical transformer operating in sinusoidal steady state is equivalent to an ideal transformer with external impedance and admittance branches, as shown in Figure 3.6. The external branches can be evaluated from short-circuit and open-circuit tests, as illustrated by the following example.



(a) R_2 and X_2 are referred to winding 1



(b) Neglecting exciting current



(c) Neglecting exciting current and I^2R winding loss

FIGURE 3.6

Equivalent circuits for a practical single-phase, two-winding transformer

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EXAMPLE 3.2

Transformer short-circuit and open-circuit tests

A single-phase two-winding transformer is rated 20 kVA, 480/120 volts, and 60 Hz. During a short-circuit test, where rated current at rated frequency is applied to the 480-volt winding (denoted winding 1), with the 120-volt winding (winding 2) shorted, the following readings are obtained: $V_1 = 35$ volts, $P_1 = 300$ W. During an open-circuit test, where rated voltage is applied to winding 2, with winding 1 open, the following readings are obtained: $I_2 = 12$ A, $P_2 = 200$ W.

a. From the short-circuit test, determine the equivalent series impedance $Z_{eq1} = R_{eq1} + jX_{eq1}$ referred to winding 1. Neglect the shunt admittance.

b. From the open-circuit test, determine the shunt admittance $Y_m = G_c - jB_m$ referred to winding 1. Neglect the series impedance.

SOLUTION

a. The equivalent circuit for the short-circuit test is shown in Figure 3.7(a), where the shunt admittance branch is neglected. Rated current for winding 1 is

$$I_{1rated} = \frac{S_{rated}}{V_{1rated}} = \frac{20 \times 10^3}{480} = 41.667 \quad A$$

R_{eq1}, Z_{eq1}, and X_{eq1} are then determined as follows:

$$R_{eq1} = \frac{P_1}{I_{1rated}^2} = \frac{300}{(41.667)^2} = 0.1728 \quad \Omega$$

$$|Z_{eq1}| = \frac{V_1}{I_{1rated}} = \frac{35}{41.667} = 0.8400 \quad \Omega$$

$$X_{eq1} = \sqrt{Z_{eq1}^2 - R_{eq1}^2} = 0.8220 \quad \Omega$$

$$Z_{eq1} = R_{eq1} + jX_{eq1} = 0.1728 + j0.8220 = 0.8400 \underline{/78.13°} \quad \Omega$$

b. The equivalent circuit for the open-circuit test is shown in Figure 3.7(b), where the series impedance is neglected. From (3.1.16),

$$V_1 = E_1 = a_t E_2 = \frac{N_1}{N_2} V_{2rated} = \frac{480}{120}(120) = 480 \text{ volts}$$

G_c, Y_m, and B_m are then determined as follows:

$$G_c = \frac{P_2}{V_1^2} = \frac{200}{(480)^2} = 0.000868 \quad S$$

$$|Y_m| = \frac{\left(\frac{N_2}{N_1}\right) I_2}{V_1} = \frac{\left(\frac{120}{480}\right)(12)}{480} = 0.00625 \quad S$$

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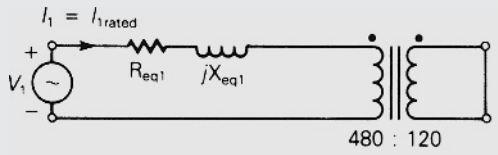
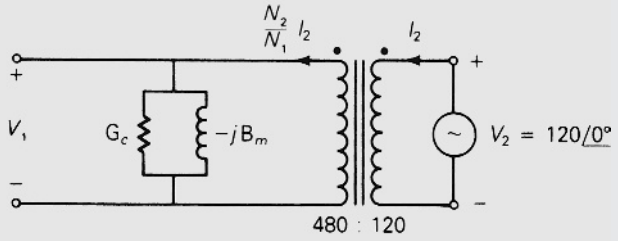
FIGURE 3.7
Circuits for Example 3.2



(a) Short-circuit test (neglecting shut admittance)



(b) Open-circuit test (neglecting series impedance)

$$B_m = \sqrt{Y_m^2 - G_c^2} = \sqrt{(0.00625)^2 - (0.000868)^2} = 0.00619 \text{ S}$$

$$Y_m = G_c - jB_m = 0.000868 - j0.00619 = 0.00625\underline{/-82.02°} \text{ S}$$

Note that the equivalent series impedance is usually evaluated at rated current from a short-circuit test, and the shunt admittance is evaluated at rated voltage from an open-circuit test. For small variations in transformer operation near rated conditions, the impedance and admittance values are often assumed constant.

The following are not represented by the equivalent circuit of Figure 3.5:

1. Saturation
2. Inrush current
3. Nonsinusoidal exciting current
4. Surge phenomena

They are briefly discussed in the following sections.

SATURATION

In deriving the equivalent circuit of the ideal and practical transformers, constant core permeability μ_c and the linear relationship $B_c = \mu_c H_c$ of (3.1.3) were assumed. However, the relationship between B and H for ferromagnetic materials used for transformer cores is nonlinear and multivalued. Figure 3.8 shows a set of B–H curves for a grain-oriented electrical steel typically used in transformers. As shown, each curve is multivalued, which is caused by hysteresis. For many

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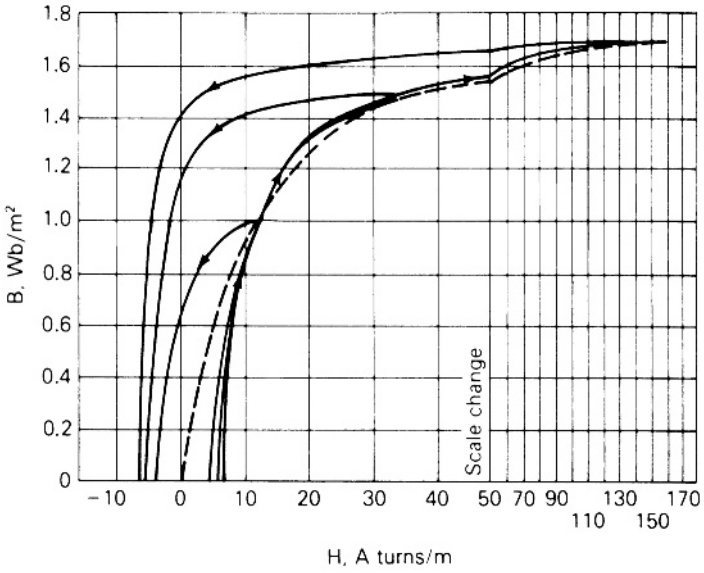

FIGURE 3.8

B–H curves for M-5 grain-oriented electrical steel 0.012 in. thick (Reprinted with permission of AK Steel Corporation.)

engineering applications, the B–H curves can be adequately described by the dashed line drawn through the curves in Figure 3.8. Note that as H increases, the core becomes saturated; that is, the curves flatten out as B increases above 1 Wb/m^2. If the magnitude of the voltage applied to a transformer is too large, the core will saturate, and a high magnetizing current will flow. In a well-designed transformer, the applied peak voltage causes the peak flux density in steady state to occur at the knee of the B–H curve, with a corresponding low value of magnetizing current.

INRUSH CURRENT

When a transformer is first energized, a transient current much larger than rated transformer current can flow for several cycles. This current, called *inrush current,* is nonsinusoidal and has a large dc component. To understand the cause of inrush, assume that before energization, the transformer core is magnetized with a residual flux density B(0) = 1.5 Wb/m^2 (near the knee of the dotted curve in Figure 3.8). If the transformer is then energized when the source voltage is positive and increasing, Faraday's law, (3.1.9) will cause the flux density B(t) to increase further, since

$$\mathrm{B}(t) = \frac{\phi(t)}{\mathrm{A}} = \frac{1}{\mathrm{NA}}\int_0^t e(t)dt + \mathrm{B}(0)$$

As B(t) moves into the saturation region of the B–H curve, large values of H(t) will occur, and, from Ampere's law, (3.1.1), corresponding large values of current $i(t)$ will flow for several cycles until it has dissipated. Since normal inrush currents can be as large as abnormal short-circuit currents in transformers, transformer protection schemes must be able to distinguish between these two types of currents.

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NONSINUSOIDAL EXCITING CURRENT

When a sinusoidal voltage is applied to one winding of a transformer with the other winding open, the flux $\phi(t)$ and flux density B(t) will, from Faraday's law, (3.1.9), be very nearly sinusoidal in steady state. However, the magnetic field intensity H(t) and the resulting exciting current is not sinusoidal in steady state, due to the nonlinear B–H curve. If the exciting current is measured and analyzed by Fourier analysis techniques, one finds that it has a fundamental component and a set of odd harmonics. The principal harmonic is the third, whose rms value is typically about 40% of the total rms exciting current. However, the nonsinusoidal nature of exciting current is usually neglected unless harmonic effects are of direct concern because the exciting current itself is usually less than 5% of rated current for power transformers.

SURGE PHENOMENA

When power transformers are subjected to transient overvoltages caused by lightning or switching surges, the capacitances of the transformer windings have important effects on transient response. Transformer winding capacitances and response to surges are discussed in Chapter 14.

3.3 THE PER-UNIT SYSTEM

Power-system quantities, such as voltage, current, power, and impedance, are often expressed in per-unit or percent of specified base values. For example, if a base voltage of 20 kV is specified, then the voltage 18 kV is (18/20) = 0.9 per unit or 90%. Calculations then can be made with per-unit quantities rather than with the actual quantities.

One advantage of the per-unit system is that by properly specifying base quantities, the transformer equivalent circuit can be simplified. The ideal transformer winding can be eliminated, such that voltages, currents, and external impedances and admittances expressed in per-unit do not change when they are referred from one side of a transformer to the other. This can be a significant advantage even in a power system of moderate size, where hundreds of transformers may be encountered. Using the per-unit system avoids the possibility of making serious calculation errors when referring quantities from one side of a transformer to the other. Another advantage of the per-unit system is that the per-unit impedances of electrical equipment of similar type usually lie within a narrow numerical range when the equipment ratings are used as base values. Because of this, per-unit impedance data can be checked rapidly for gross errors by someone familiar with per-unit quantities. In addition, manufacturers usually specify the impedances of machines and transformers in per-unit or percent of nameplate rating.

Per-unit quantities are calculated as follows:

$$\text{per-unit quantity} = \frac{\text{actual quantity}}{\text{base value of quantity}} \tag{3.3.1}$$

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where *actual quantity* is the value of the quantity in the actual units. The base value has the same units as the actual quantity, thus making the per-unit quantity dimensionless. Also, the base value is always a real number. Therefore, the angle of the per-unit quantity is the same as the angle of the actual quantity.

Two independent base values can be arbitrarily selected at one point in a power system. Usually, the base voltage V_{baseLN} and base complex power $S_{base1\phi}$ are selected for either a single-phase circuit or for one phase of a three-phase circuit. Then, in order for electrical laws to be valid in the per-unit system, the following relations must be used for other base values:

$$P_{base1\phi} = Q_{base1\phi} = S_{base1\phi} \tag{3.3.2}$$

$$I_{base} = \frac{S_{base1\phi}}{V_{baseLN}} \tag{3.3.3}$$

$$Z_{base} = R_{base} = X_{base} = \frac{V_{baseLN}}{I_{base}} = \frac{V_{baseLN}^2}{S_{base1\phi}} \tag{3.3.4}$$

$$Y_{base} = G_{base} = B_{base} = \frac{1}{Z_{base}} \tag{3.3.5}$$

In (3.3.2) through (3.3.5), the subscripts LN and 1ϕ denote "line-to-neutral" and "per-phase," respectively, for three-phase circuits. These equations are also valid for single-phase circuits, where subscripts can be omitted.

Convention requires adoption of the following two rules for base quantities:

1. The value of $S_{base1\phi}$ is the same for the entire power system of concern.
2. The ratio of the voltage bases on either side of a transformer is selected to be the same as the ratio of the transformer voltage ratings.

With these two rules, a per-unit impedance remains unchanged when referred from one side of a transformer to the other.

EXAMPLE 3.3

Per-unit impedance: single-phase transformer

A single-phase two-winding transformer is rated 20 kVA, 480/120 volts, and 60 Hz. The equivalent leakage impedance of the transformer referred to the 120-volt winding, denoted winding 2, is $Z_{eq2} = 0.0525\angle 78.13°\ \Omega$. Using the transformer ratings as base values, determine the per-unit leakage impedance referred to winding 2 and referred to winding 1.

SOLUTION

The values of S_{base}, V_{base1}, and V_{base2} are, from the transformer ratings,

$S_{base} = 20$ kVA, $V_{base1} = 480$ volts, $V_{base2} = 120$ volts

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Using (3.3.4), the base impedance on the 120-volt side of the transformer is

$$Z_{\text{base2}} = \frac{\text{V}_{\text{base2}}^2}{\text{S}_{\text{base}}} = \frac{(120)^2}{20{,}000} = 0.72 \quad \Omega$$

Then, using (3.3.1), the per-unit leakage impedance referred to winding 2 is

$$Z_{\text{eq2p.u.}} = \frac{Z_{\text{eq2}}}{Z_{\text{base2}}} = \frac{0.0525 \angle 78.13^\circ}{0.72} = 0.0729 \angle 78.13^\circ \text{ per unit}$$

If Z_{eq2} is referred to winding 1,

$$Z_{\text{eq1}} = a_{\text{t}}^2 Z_{\text{eq2}} = \left(\frac{N_1}{N_2}\right)^2 Z_{\text{eq2}} = \left(\frac{480}{120}\right)^2 (0.0525 \angle 78.13^\circ)$$
$$= 0.84 \angle 78.13^\circ \quad \Omega$$

The base impedance on the 480-volt side of the transformer is

$$Z_{\text{base1}} = \frac{\text{V}_{\text{base1}}^2}{\text{S}_{\text{base}}} = \frac{(480)^2}{20{,}000} = 11.52 \quad \Omega$$

and the per-unit leakage reactance referred to winding 1 is

$$Z_{\text{eq1p.u.}} = \frac{Z_{\text{eq1}}}{Z_{\text{base1}}} = \frac{0.84 \angle 78.13^\circ}{11.52} = 0.0729 \angle 78.13^\circ \text{ per unit} = Z_{\text{eq2p.u.}}$$

Thus, the *per-unit* leakage impedance remains unchanged when referred from winding 2 to winding 1. This has been achieved by specifying

$$\frac{\text{V}_{\text{base1}}}{\text{V}_{\text{base2}}} = \frac{\text{V}_{\text{rated1}}}{\text{V}_{\text{rated2}}} = \left(\frac{480}{120}\right)$$

Figure 3.9 shows three per-unit circuits of a single-phase two-winding transformer. The ideal transformer, shown in Figure 3.9(a), satisfies the per-unit relations $E_{\text{1p.u.}} = E_{\text{2p.u.}}$ and $I_{\text{1p.u.}} = I_{\text{2p.u.}}$, which can be derived as follows. First divide (3.1.16) by V_{base1}:

$$E_{\text{1p.u.}} = \frac{E_1}{\text{V}_{\text{base1}}} = \frac{N_1}{N_2} \times \frac{E_2}{\text{V}_{\text{base1}}} \tag{3.3.6}$$

Then, using $\text{V}_{\text{base1}}/\text{V}_{\text{base2}} = \text{V}_{\text{rated1}}/\text{V}_{\text{rated2}} = N_1/N_2$,

$$E_{\text{1p.u.}} = \frac{N_1}{N_2} \frac{E_2}{\left(\frac{N_1}{N_2}\right)\text{V}_{\text{base2}}} = \frac{E_2}{\text{V}_{\text{base2}}} = E_{\text{2p.u.}} \tag{3.3.7}$$

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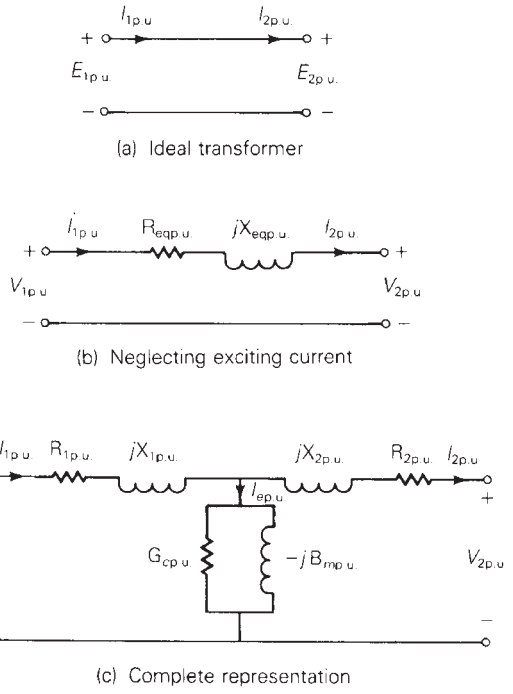
FIGURE 3.9

Per-unit equivalent circuits of a single-phase two-winding transformer



Similarly, divide (3.1.17) by $\mathrm{I}_{\mathrm{base1}}$,

$$I_{1\text{p.u.}} = \frac{I_1}{\mathrm{I}_{\mathrm{base1}}} = \frac{N_2}{N_1}\frac{I_2}{\mathrm{I}_{\mathrm{base1}}} \tag{3.3.8}$$

Then, using $\mathrm{I}_{\mathrm{base1}} = \mathrm{S}_{\mathrm{base}}/\mathrm{V}_{\mathrm{base1}} = \mathrm{S}_{\mathrm{base}}/[(N_1/N_2)\mathrm{V}_{\mathrm{base2}}] = (N_2/N_1)\mathrm{I}_{\mathrm{base2}}$,

$$I_{1\text{p.u.}} = \frac{N_2}{N_1}\frac{I_2}{\left(\frac{N_2}{N_1}\right)\mathrm{I}_{\mathrm{base2}}} = \frac{I_2}{\mathrm{I}_{\mathrm{base2}}} = I_{2\text{p.u.}} \tag{3.3.9}$$

Thus, the ideal transformer winding in Figure 3.2 is eliminated from the per-unit circuit in Figure 3.9(a). The per-unit leakage impedance is included in Figure 3.9(b), and the per-unit shunt admittance branch is added in Figure 3.9(c) to obtain the complete representation.

When only one component, such as a transformer, is considered, the nameplate ratings of that component are usually selected as base values. When several components are involved, however, the system base values may be different from the nameplate ratings of any particular device. It is then necessary to convert the per-unit impedance of a device from its nameplate ratings to the system base values. To convert a per-unit impedance from "old" to "new" base values, use

$$Z_{\text{p.u.new}} = \frac{Z_{\text{actual}}}{Z_{\text{basenew}}} = \frac{Z_{\text{p.u.old}}\,\mathrm{Z}_{\text{baseold}}}{Z_{\text{basenew}}} \tag{3.3.10}$$

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or, from (3.3.4),

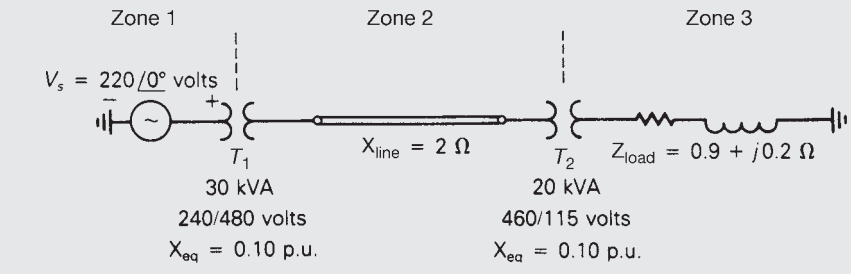
$$Z_{\text{p.u.new}} = Z_{\text{p.u.old}}\left(\frac{V_{\text{baseold}}}{V_{\text{basenew}}}\right)^2\left(\frac{S_{\text{basenew}}}{S_{\text{baseold}}}\right) \tag{3.3.11}$$

EXAMPLE 3.4

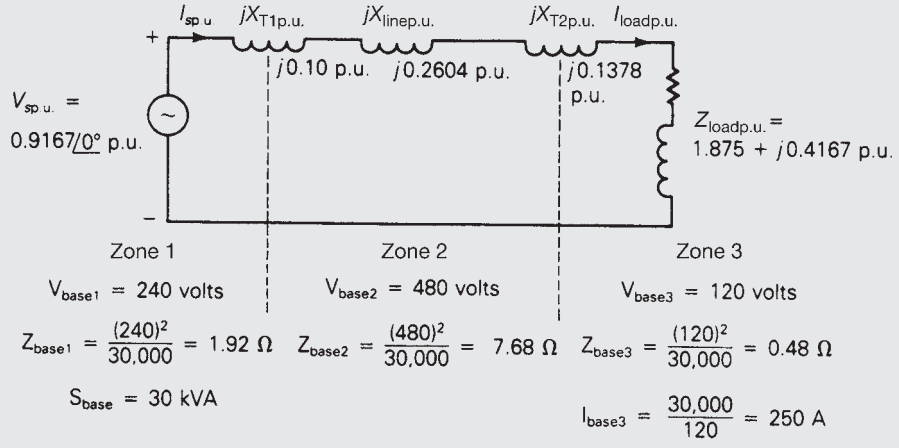
Per-unit circuit: three-zone single-phase network

Three zones of a single-phase circuit are identified in Figure 3.10(a). The zones are connected by transformers T_1 and T_2, whose ratings are also shown. Using base values of 30 kVA and 240 volts in zone 1, draw the per-unit circuit and determine the per-unit impedances and the per-unit source voltage. Then calculate the load current both in per-unit and in amperes. Transformer winding resistances and shunt admittance branches are neglected.

FIGURE 3.10
Circuit for Example 3.4



(a) Single-phase circuit



(b) Per-unit circuit

(*Continued*)

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SOLUTION

First, the base values in each zone are determined. $S_{base} = 30$ kVA is the same for the entire network. Also, $V_{base1} = 240$ volts, as specified for zone 1. When moving across a transformer, the voltage base is changed in proportion to the transformer voltage ratings. Thus,

$$V_{base2} = \left(\frac{480}{240}\right)(240) = 480 \quad \text{volts}$$

and

$$V_{base3} = \left(\frac{115}{460}\right)(480) = 120 \quad \text{volts}$$

The base impedances in zones 2 and 3 are

$$Z_{base2} = \frac{V_{base2}^2}{S_{base}} = \frac{480^2}{30{,}000} = 7.86 \quad \Omega$$

and

$$Z_{base3} = \frac{V_{base3}^2}{S_{base}} = \frac{120^2}{30{,}000} = 0.48 \quad \Omega$$

and the base current in zone 3 is

$$I_{base3} = \frac{S_{base}}{V_{base3}} = \frac{30{,}000}{120} = 250 \quad \text{A}$$

Next, the per-unit circuit impedances are calculated using the system base values. Since $S_{base} = 30$ kVA is the same as the kVA rating of transformer T_1, and $V_{base1} = 240$ volts is the same as the voltage rating of the zone 1 side of transformer T_1, the per-unit leakage reactance of T_1 is the same as its nameplate value, $X_{T1p.u.} = 0.1$ per unit. However, the per-unit leakage reactance of transformer T_2 must be converted from its nameplate rating to the system base. Using (3.3.11) and $V_{base2} = 480$ volts,

$$X_{T2p.u.} = (0.10)\left(\frac{460}{480}\right)^2\left(\frac{30{,}000}{20{,}000}\right) = 0.1378 \quad \text{per unit}$$

Alternatively, using $V_{base3} = 120$ volts,

$$X_{T2p.u.} = (0.10)\left(\frac{115}{120}\right)^2\left(\frac{30{,}000}{20{,}000}\right) = 0.1378 \quad \text{per unit}$$

which gives the same result. The line, which is located in zone 2, has a per-unit reactance

$$X_{linep.u.} = \frac{X_{line}}{Z_{base2}} = \frac{2}{7.68} = 0.2604 \quad \text{per unit}$$

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and the load, which is located in zone 3, has a per-unit impedance

$$Z_{\text{loadp.u.}} = \frac{Z_{\text{load}}}{Z_{\text{base3}}} = \frac{0.9 + j0.2}{0.48} = 1.875 + j0.4167 \quad \text{per unit}$$

The per-unit circuit is shown in Figure 3.10(b), where the base values for each zone, per-unit impedances, and the per-unit source voltage are shown. The per-unit load current is then easily calculated from Figure 3.10(b) as follows:

$$\begin{aligned} I_{\text{loadp.u.}} = I_{\text{sp.u.}} &= \frac{V_{\text{sp.u.}}}{j(X_{\text{T1p.u.}} + X_{\text{linep.u.}} + X_{\text{T2p.u.}}) + Z_{\text{loadp.u.}}} \\ &= \frac{0.9167\underline{/0^\circ}}{j(0.10 + 0.2604 + 0.1378) + (1.875 + j0.4167)} \\ &= \frac{0.9167\underline{/0^\circ}}{1.875 + j0.9149} = \frac{0.9167\underline{/0^\circ}}{2.086\underline{/26.01^\circ}} \\ &= 0.4395\underline{/-26.01^\circ} \quad \text{per unit} \end{aligned}$$

The actual load current is

$$I_{\text{load}} = (I_{\text{loadp.u.}})\text{I}_{\text{base3}} = (0.4395\underline{/-26.01^\circ})(250) = 109.9\underline{/-26.01^\circ} \quad \text{A}$$

Note that the per-unit equivalent circuit of Figure 3.10(b) is relatively easy to analyze, since ideal transformer windings have been eliminated by proper selection of base values.

Balanced three-phase circuits can be solved in per-unit on a per-phase basis after converting Δ-load impedances to equivalent Y impedances. Base values can be selected either on a per-phase basis or on a three-phase basis. Equations (3.3.1) through (3.3.5) remain valid for three-phase circuits on a per-phase basis. Usually $\text{S}_{\text{base3}\phi}$ and V_{baseLL} are selected, where the subscripts 3ϕ and LL denote "three-phase" and "line-to-line," respectively. Then the following relations must be used for other base values:

$$\text{S}_{\text{base1}\phi} = \frac{\text{S}_{\text{base3}\phi}}{3} \tag{3.3.12}$$

$$\text{V}_{\text{baseLN}} = \frac{\text{V}_{\text{baseLL}}}{\sqrt{3}} \tag{3.3.13}$$

$$\text{S}_{\text{base3}\phi} = \text{P}_{\text{base3}\phi} = \text{Q}_{\text{base3}\phi} \tag{3.3.14}$$

$$\text{I}_{\text{base}} = \frac{\text{S}_{\text{base1}\phi}}{\text{V}_{\text{baseLN}}} = \frac{\text{S}_{\text{base3}\phi}}{\sqrt{3}\text{V}_{\text{baseLL}}} \tag{3.3.15}$$

$$Z_{\text{base}} = \frac{\text{V}_{\text{baseLN}}}{\text{I}_{\text{base}}} = \frac{\text{V}^2_{\text{baseLN}}}{\text{S}_{\text{base1}\phi}} = \frac{\text{V}^2_{\text{baseLL}}}{\text{S}_{\text{base3}\phi}} \tag{3.3.16}$$

$$\text{R}_{\text{base}} = X_{\text{base}} = Z_{\text{base}} = \frac{1}{Y_{\text{base}}} \tag{3.3.17}$$

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EXAMPLE 3.5

Per-unit and actual currents in balanced three-phase networks

As in Example 2.5, a balanced-Y-connected voltage source with $E_{ab} = 480\angle 0°$ volts is applied to a balanced-Δ load with $Z_\Delta = 30\angle 40°\ \Omega$. The line impedance between the source and load is $Z_L = 1\angle 85°\ \Omega$ for each phase. Calculate the per-unit and actual current in phase a of the line using $S_{base3\phi} = 10$ kVA and $V_{baseLL} = 480$ volts.

SOLUTION

First, convert Z_Δ to an equivalent Z_Y; the equivalent line-to-neutral diagram is shown in Figure 2.17. The base impedance is, from (3.3.16),

$$Z_{base} = \frac{V_{baseLL}^2}{S_{base3\phi}} = \frac{(480)^2}{10{,}000} = 23.04 \quad \Omega$$

The per-unit line and load impedances are

$$Z_{Lp.u.} = \frac{Z_L}{Z_{base}} = \frac{10\angle 85°}{23.04} = 0.04340\angle 85° \quad \text{per unit}$$

and

$$Z_{Yp.u.} = \frac{Z_Y}{Z_{base}} = \frac{10\angle 40°}{23.04} = 0.4340\angle 40° \quad \text{per unit}$$

Also,

$$V_{baseLN} = \frac{V_{baseLL}}{\sqrt{3}} = \frac{480}{\sqrt{3}} = 277 \quad \text{volts}$$

and

$$E_{anp.u.} = \frac{E_{an}}{V_{baseLN}} = \frac{277\angle -30°}{277} = 1.0\angle -30° \quad \text{per unit}$$

The per-unit equivalent circuit is shown in Figure 3.11. The per-unit line current in phase a is then

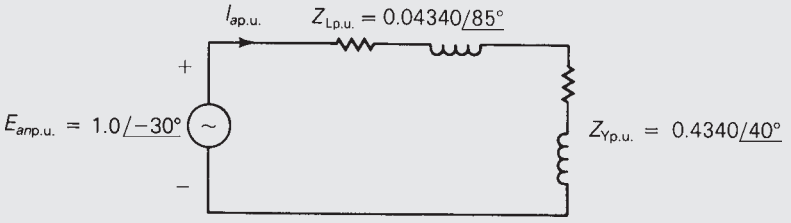

FIGURE 3.11
Circuit for Example 3.5

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$$I_{a\text{p.u.}} = \frac{E_{an\text{p.u.}}}{Z_{\text{Lp.u.}} + Z_{\text{Yp.u.}}} = \frac{1.0\angle -30^\circ}{0.04340\angle 85^\circ + 0.4340\angle 40^\circ}$$
$$= \frac{1.0\angle -30^\circ}{(0.00378 + j0.04323) + (0.3325 + j0.2790)}$$
$$= \frac{1.0\angle -30^\circ}{0.3362 + j0.3222} = \frac{1.0\angle -30^\circ}{0.4657\angle 43.78^\circ}$$
$$= 2.147\angle -73.78^\circ \text{ per unit}$$

The base current is

$$\mathrm{I_{base}} = \frac{\mathrm{S_{base3\phi}}}{\sqrt{3}\,\mathrm{V_{baseLL}}} = \frac{10{,}000}{\sqrt{3}(480)} = 12.03 \text{ A}$$

and the actual phase *a* line current is

$$I_a = (2.147\angle -73.78^\circ)(12.03) = 25.83\angle -73.78^\circ \text{ A}$$

3.4 THREE-PHASE TRANSFORMER CONNECTIONS AND PHASE SHIFT

Three identical single-phase two-winding transformers may be connected to form a three-phase bank. Four ways to connect the windings are Y–Y, Y–Δ, Δ–Y, and Δ–Δ. For example, Figure 3.12 shows a three-phase Y–Y bank. Figure 3.12(a) shows the core and coil arrangements. The American standard for marking three-phase transformers substitutes H1, H2, and H3 on the high-voltage terminals and X1, X2, and X3 on the low-voltage terminals in place of the polarity dots. Also, this text uses uppercase letters *ABC* to identify phases on the high-voltage side of the transformer and lowercase letters *abc* to identify phases on the low-voltage side of the transformer. In Figure 3.12(a), the transformer high-voltage terminals H1, H2, and H3 are connected to phases *A, B,* and *C*, and the low-voltage terminals X1, X2, and X3 are connected to phases *a, b,* and c, respectively.

Figure 3.12(b) shows a schematic representation of the three-phase Y–Y transformer. Windings on the same core are drawn in parallel, and the phasor relationship for balanced positive-sequence operation is shown. For example, high-voltage winding Hl–*N* is on the same magnetic core as low-voltage winding Xl–*n* in Figure 3.12(b). Also, V_{AN} is in phase with V_{an}. Figure 3.12(c) shows a single-line diagram of a Y–Y transformer. A single-line diagram shows one phase of a three-phase network with the neutral wire omitted and with components represented by symbols rather than equivalent circuits.

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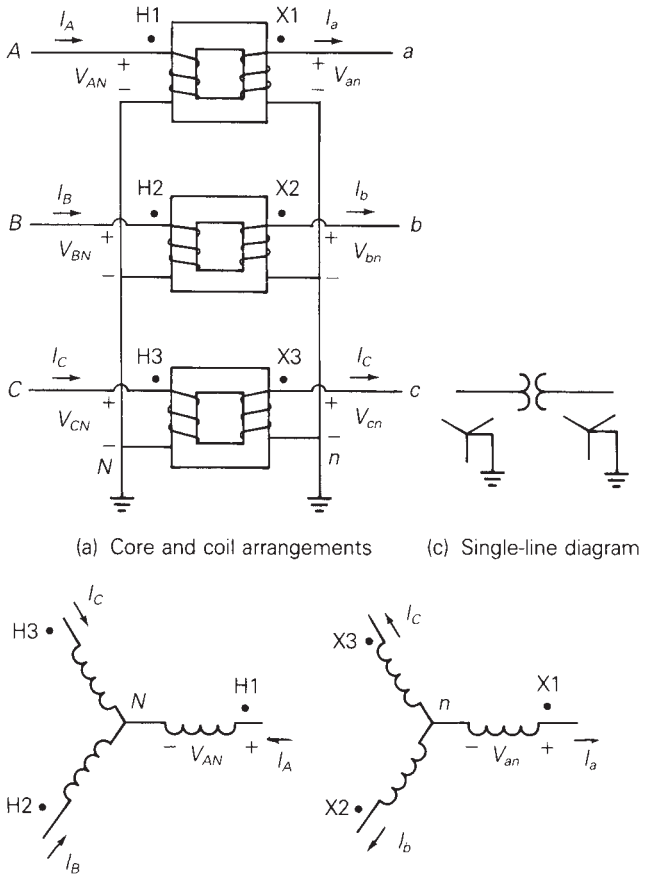
FIGURE 3.12

Three-phase, two-winding Y–Y transformer bank



The phases of a Y–Y or a Δ–Δ transformer can be labeled so there is no phase shift between corresponding quantities on the low- and high-voltage windings. However, for Y–Δ and Δ–Y transformers, there is always a phase shift. Figure 3.13 shows a Y–Δ transformer. The labeling of the windings and the schematic representation are in accordance with the American standard, which is as follows:

> In either a Y–Δ or Δ–Y transformer, positive-sequence quantities on the high-voltage side shall lead their corresponding quantities on the low-voltage side by 30°.

As shown in Figure 3.13(b), V_{AN} leads V_{an} by 30°.

The positive-sequence phasor diagram shown in Figure 3.13(b) can be constructed via the following five steps, which are also indicated in Figure 3.13.

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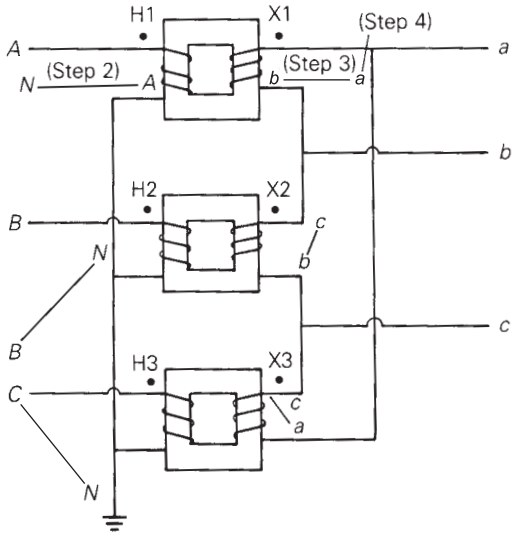
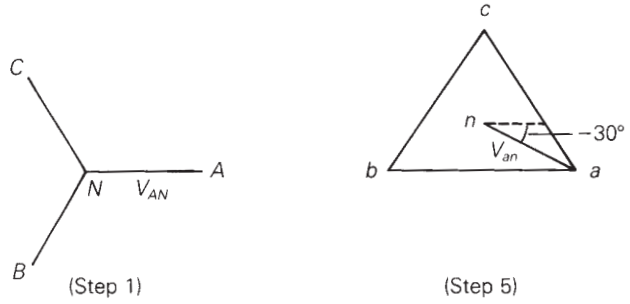


(a) Core and coil arrangement



(b) Positive-sequence phasor diagram

FIGURE 3.13

Three-phase, two-winding Y–Δ transformer bank

STEP 1 Assume that balanced positive-sequence voltages are applied to the Y winding. Draw the positive-sequence phasor diagram for these voltages.

STEP 2 Move phasor *A–N* next to terminals *A–N* in Figure 3.13(a). Identify the ends of this line in the same manner as in the phasor diagram. Similarly, move phasors *B–N* and *C–N* next to terminals *B–N* and *C–N* in Figure 3.13(a).

STEP 3 For each single-phase transformer, the voltage across the low-voltage winding must be in phase with the voltage across the high-voltage winding, assuming an ideal transformer. Therefore, draw a

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line next to each low-voltage winding parallel to the corresponding line already drawn next to the high-voltage winding.

STEP 4 Label the ends of the lines drawn in Step 3 by inspecting the polarity marks. For example, phase *A* is connected to dotted terminal H1, and *A* appears on the *right* side of line *A–N.* Therefore, phase *a,* which is connected to dotted terminal X1, must be on the *right* side, and *b* on the left side of line *a–b.* Similarly, phase *B* is connected to dotted terminal H2, and *B* is *down* on line *B–N.* Therefore, phase *b,* connected to dotted terminal X2, must be *down* on line *b–c.* Similarly, *c* is *up* on line *c–a.*

STEP 5 Bring the three lines labeled in Step 4 together to complete the phasor diagram for the low-voltage *A* winding. Note that V_{AN} leads V_{an} by 30° in accordance with the American standard.

EXAMPLE 3.6

Phase shift in Y–Δ transformers

Assume that balanced negative-sequence voltages are applied to the high-voltage windings of the Y–Δ transformer shown in Figure 3.13. Determine the negative-sequence phase shift of this transformer.

SOLUTION

The negative-sequence diagram, shown in Figure 3.14, is constructed from the following five steps, as outlined previously:

STEP 1 Draw the phasor diagram of balanced negative-sequence voltages, which are applied to the Y winding.

STEP 2 Move the phasors *A–N, B–N,* and *C–N* next to the high-voltage Y windings.

STEP 3 For each single-phase transformer, draw a line next to the low-voltage winding that is parallel to the line drawn in Step 2 next to the high-voltage winding.

STEP 4 Label the lines drawn in Step 3. For example, phase *B,* which is connected to dotted terminal H2, is shown *up* on line *B–N;* therefore phase *b,* which is connected to dotted terminal X2, must be *up* on line *b–c.*

STEP 5 Bring the lines drawn in Step 4 together to form the negative-sequence phasor diagram for the low-voltage Δ winding.

As shown in Figure 3.14, the high-voltage phasors *lag* the low-voltage phasors by 30°. Thus, the negative-sequence phase shift is the reverse of the positive-sequence phase shift.

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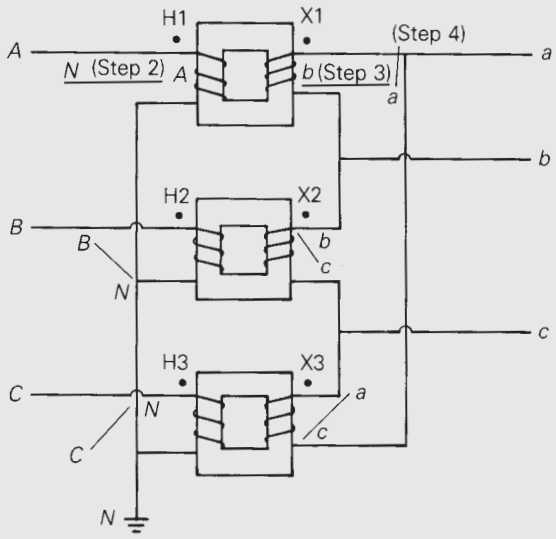
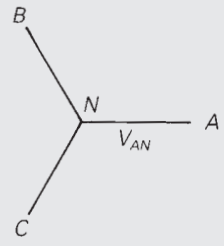
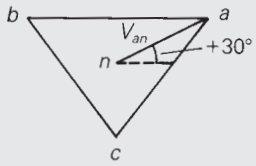
FIGURE 3.14

Example 3.6—Construction of negative-sequence phasor diagram for Y–Δ transformer bank





(Step 1)



(Step 5)

The Δ–Y transformer is commonly used as a generator step-up transformer, where the Δ winding is connected to the generator terminals and the Y winding is connected to a transmission line. One advantage of a high-voltage Y winding is that a neutral point *N* is provided for grounding on the high-voltage side. With a permanently grounded neutral, the insulation requirements for the high-voltage transformer windings are reduced. The high-voltage insulation can be graded or tapered from maximum insulation at terminals *ABC* to minimum insulation at grounded terminal *N*. One advantage of the Δ winding is that the undesirable third harmonic magnetizing current, caused by the nonlinear core *B*–*H* characteristic, remains trapped inside the Δ winding. Third harmonic currents are (triple-frequency) zero-sequence currents, which cannot enter or leave a Δ connection but can flow within the Δ. The Y–Y transformer is seldom used because of difficulties with third harmonic exciting current.

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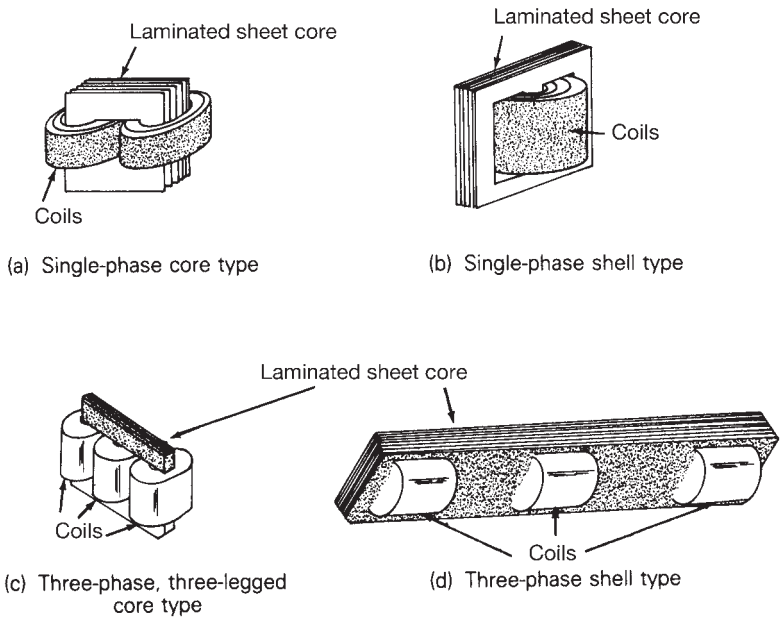
FIGURE 3.15
Transformer core configurations



The Δ–Δ transformer has the advantage that one phase can be removed for repair or maintenance while the remaining phases continue to operate as a three-phase bank. This *open*-Δ connection permits balanced three-phase operation with the kVA rating reduced to 58% of the original bank (see Problem 3.36).

Instead of a bank of three single-phase transformers, all six windings may be placed on a common three-phase core to form a three-phase transformer, as shown in Figure 3.15. The three-phase core contains less iron than the three single-phase units; therefore it costs less, weighs less, requires less floor space, and has a slightly higher efficiency. However, a winding failure would require replacement of an entire three-phase transformer, compared to replacement of only one phase of a three-phase bank.

3.5 PER-UNIT EQUIVALENT CIRCUITS OF BALANCED THREE-PHASE TWO-WINDING TRANSFORMERS

Figure 3.16(a) is a schematic representation of an ideal Y–Y transformer grounded through neutral impedances Z_N and Z_n. Figure 3.16(b) shows the per-unit equivalent circuit of this ideal transformer for balanced three-phase operation. Throughout the remainder of this text, per-unit quantities are used unless otherwise indicated. Also, the subscript "p.u.," used to indicate a per-unit quantity, is omitted in most cases.

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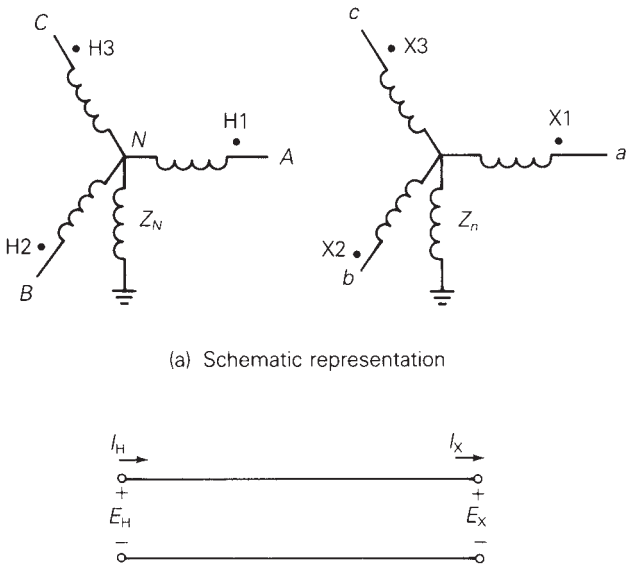

FIGURE 3.16
Ideal Y–Y transformer

The following are two conventional rules for selecting base quantities:

1. A common S_{base} is selected for both the H and X terminals.
2. The ratio of the voltage bases V_{baseH}/V_{baseX} is selected to be equal to the ratio of the rated line-to-line voltages $V_{ratedHLL}/V_{ratedXLL}$.

When balanced three-phase currents are applied to the transformer, the neutral currents are zero, and there are no voltage drops across the neutral impedances. Therefore, the per-unit equivalent circuit of the ideal Y–Y transformer, Figure 3.16(b), is the same as the per-unit single-phase ideal transformer, Figure 3.9(a).

The per-unit equivalent circuit of a practical Y–Y transformer is shown in Figure 3.17(a). This network is obtained by adding external impedances to the equivalent circuit of the ideal transformer, as in Figure 3.9(c).

The per-unit equivalent circuit of the Y–Δ transformer, shown in Figure 3.17(b), includes a phase shift. For the American standard, the positive-sequence voltages and currents on the high-voltage side of the Y–Δ transformer lead the corresponding quantities on the low-voltage side by 30°. The phase shift in the equivalent circuit of Figure 3.17(b) is represented by the phase-shifting transformer of Figure 3.4.

The per-unit equivalent circuit of the Δ–Δ transformer, shown in Figure 3.17(c), is the same as that of the Y–Y transformer. It is assumed that the windings are labeled so there is no phase shift. Also, the per-unit impedances do not depend on the winding connections, but the base voltages do.

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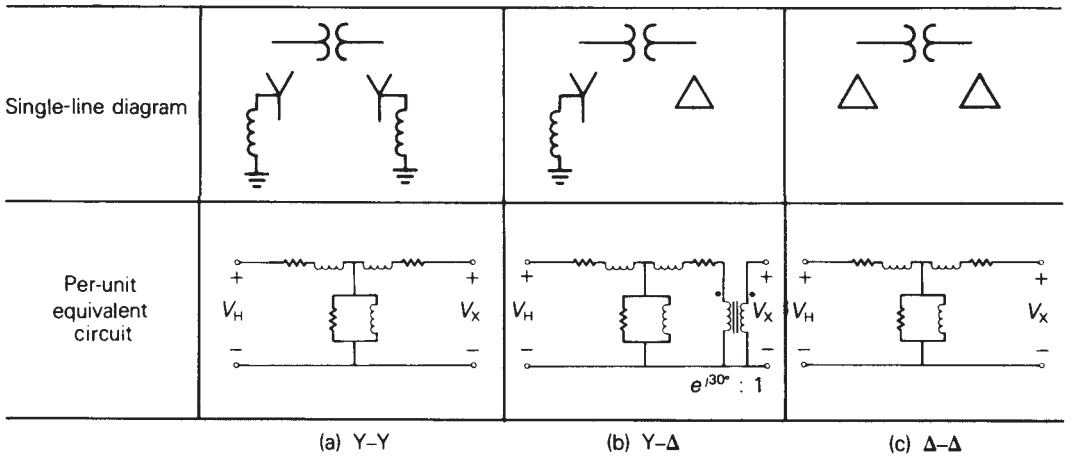

FIGURE 3.17

Per-unit equivalent circuits of practical Y–Y, Y–Δ, and Δ–Δ transformers for balanced three-phase operation

EXAMPLE 3.7

Voltage calculations: balanced Y–Y and Δ–Y transformers

Three single-phase two-winding transformers, each rated 400 MVA, 13.8/199.2 kV, with leakage reactance $X_{eq} = 0.10$ per unit, are connected to form a three-phase bank. Winding resistances and exciting current are neglected. The high-voltage windings are connected in Y. A three-phase load operating under balanced positive-sequence conditions on the high-voltage side absorbs 1000 MVA at 0.90 p.f. lagging, with $V_{AN} = 199.2 \angle 0°$ kV. Determine the voltage V_{an} at the low-voltage bus if the low-voltage windings are connected (a) in Y, and (b) in Δ.

SOLUTION

The per-unit network is shown in Figure 3.18. Using the transformer bank ratings as base quantities, $S_{base3\phi} = 1200$ MVA, $V_{baseHLL} = 345$ kV, and $I_{baseH} = 1200/(345\sqrt{3}) = 2.008$ kA. The per-unit load voltage and load current are then

$$V_{AN} = 1.0\angle 0° \text{ per unit}$$

$$I_A = \frac{1000/(345\sqrt{3})}{2.008} \angle -\cos^{-1}0.9 = 0.8333\angle -25.84° \text{ per unit}$$

a. For the Y–Y transformer, Figure 3.18(a),

$$I_a = I_A = 0.8333\angle -25.84° \text{ per unit}$$

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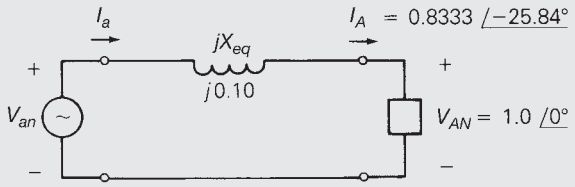
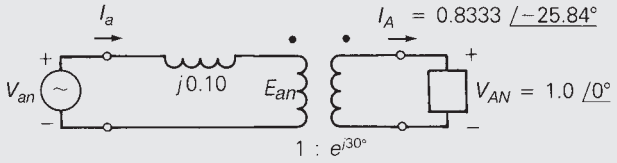
FIGURE 3.18

Per-unit network for Example 3.7



(a) Y-connected low-voltage windings



(b) Δ-connected low-voltage windings

$$V_{an} = V_{AN} + (jX_{eq})I_A$$
$$= 1.0 \angle 0° + (j0.10)(0.8333 \angle -25.84°)$$
$$= 1.0 + 0.08333 \angle 64.16° = 1.0363 + j0.0750 = 1.039 \angle 4.139°$$
$$= 1.039 \angle 4.139° \text{ per unit}$$

Further, since $V_{baseXLN} = 13.8$ kV for the low-voltage Y windings, $V_{an} = 1.039(13.8) = 14.34$ kV, and

$$V_{an} = 14.34 \angle 4.139° \text{ kV}$$

b. For the Δ–Y transformer, Figure 3.18(b),

$$E_{an} = e^{-j30°} V_{AN} = 1.0 \angle -30° \text{ per unit}$$
$$I_a = e^{-j30°} I_A = 0.8333 \angle -25.84° - 30° = 0.8333 \angle -55.84° \text{ per unit}$$
$$V_{an} = E_{an} + (jX_{eq})I_a = 1.0 \angle -30° + (j0.10)(0.8333 \angle -55.84°)$$
$$V_{an} = 1.039 \angle -25.861° \text{ per unit}$$

Further, since $V_{baseXLN} = 13.8/\sqrt{3} = 7.967$ kV for the low-voltage Δ windings,

$V_{an} = (1.039)(7.967) = 8.278$ kV, and

$$V_{an} = 8.278 \angle -25.861° \text{ kV}$$

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EXAMPLE 3.8

Per-unit voltage drop and per-unit fault current: balanced three-phase transformer

A 200-MVA, 345-kV Δ/34.5-kV Y substation transformer has an 8% leakage reactance. The transformer acts as a connecting link between 345-kV transmission and 34.5-kV distribution. Transformer winding resistances and exciting current are neglected. The high-voltage bus connected to the transformer is assumed to be an ideal 345-kV positive-sequence source with negligible source impedance. Using the transformer ratings as base values, determine:

a. The per-unit magnitudes of transformer voltage drop and voltage at the low-voltage terminals when rated transformer current at 0.8 p.f. lagging enters the high-voltage terminals.

b. The per-unit magnitude of the fault current when a three-phase-to-ground bolted short circuit occurs at the low-voltage terminals.

SOLUTION

In both parts (a) and (b), only balanced positive-sequence current will flow, since there are no imbalances. Because the only interest is in voltage and current magnitudes, the Δ–Y transformer phase shift can be omitted.

a. As shown in Figure 3.19(a),

$$V_{drop} = I_{rated}X_{ed} = (1.0)(0.08) = 0.08 \text{ per unit and}$$

$$\begin{aligned} V_{an} &= V_{AN} - (jX_{ed})I_{rated} \\ &= 1.0\angle 0° - (j0.08)(1.0\angle -36.87°) \\ &= 1.0 - (j0.08)(0.8 - j0.6) = 0.952 - j0.064 \\ &= 0.954\angle -3.85° \text{ per unit} \end{aligned}$$

b. As shown in Figure 3.19(b),

$$I_{SC} = \frac{V_{AN}}{X_{eq}} = \frac{1.0}{0.08} = 12.5 \text{ per unit}$$

Under rated current conditions [part (a)], the 0.08 per-unit voltage drop across the transformer leakage reactance causes the voltage at the low-voltage terminals to be 0.954 per unit. Also, under three-phase short-circuit conditions [part (b)], the fault current is 12.5 times the rated transformer current. This example illustrates a compromise in the design or specification of transformer leakage reactance. A low value is desired to minimize voltage drops, but a high value is desired to limit fault currents. Typical transformer leakage reactances are given in Table A.2 in the Appendix.

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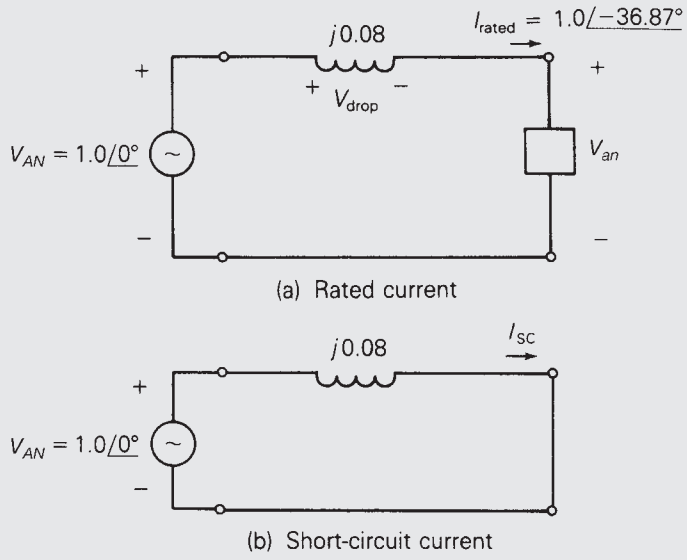
FIGURE 3.19
Circuits for Example 3.8



3.6 THREE-WINDING TRANSFORMERS

Figure 3.20(a) shows a basic single-phase three-winding transformer. The ideal transformer relations for a two-winding transformer, (3.1.8) and (3.1.14), easily can be extended to obtain corresponding relations for an ideal three-winding transformer. In actual units, these relations are

$$N_1 I_1 = N_2 I_2 + N_3 I_3 \tag{3.6.1}$$

$$\frac{E_1}{N_1} = \frac{E_2}{N_2} = \frac{E_3}{N_3} \tag{3.6.2}$$

where I_1 enters the dotted terminal, I_2 and I_3 leave dotted terminals, and E_1, E_2, and E_3 have their + polarities at dotted terminals. In per-unit, (3.6.1) and (3.6.2) are

$$I_{1\text{p.u.}} = I_{2\text{p.u.}} + I_{3\text{p.u.}} \tag{3.6.3}$$

$$E_{1\text{p.u.}} = E_{2\text{p.u.}} = E_{3\text{p.u.}} \tag{3.6.4}$$

where a common S_{base} is selected for all three windings, and voltage bases are selected in proportion to the rated voltages of the windings. These two per-unit relations are satisfied by the per-unit equivalent circuit shown in Figure 3.20(b). Also, external series impedance and shunt admittance branches are included in the practical three-winding transformer circuit shown in Figure 3.20(c). The shunt

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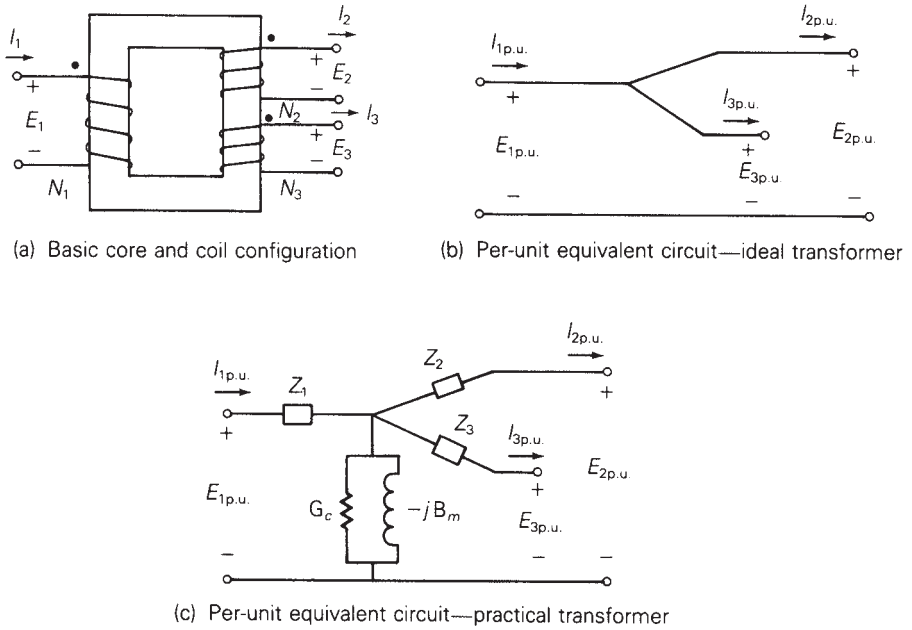
FIGURE 3.20
Single-phase, three-winding transformer



(a) Basic core and coil configuration

(b) Per-unit equivalent circuit—ideal transformer

(c) Per-unit equivalent circuit—practical transformer

admittance branch, a core loss resistor in parallel with a magnetizing inductor, can be evaluated from an open-circuit test. Also, when one winding is left open, the three-winding transformer behaves as a two-winding transformer, and standard short-circuit tests can be used to evaluate per-unit leakage impedances, which are defined as follows:

Z_{12} = per-unit leakage impedance measured from winding 1, with winding 2 shorted and winding 3 open

Z_{13} = per-unit leakage impedance measured from winding 1, with winding 3 shorted and winding 2 open

Z_{23} = per-unit leakage impedance measured from winding 2, with winding 3 shorted and winding 1 open

From Figure 3.20(c), with winding 2 shorted and winding 3 open, the leakage impedance measured from winding 1 is, neglecting the shunt admittance branch,

$$Z_{12} = Z_1 + Z_2 \tag{3.6.5}$$

Similarly,

$$Z_{13} = Z_1 + Z_3 \tag{3.6.6}$$

and

$$Z_{23} = Z_2 + Z_3 \tag{3.6.7}$$

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Solving (3.6.5) through (3.6.7),

$$Z_1 = \tfrac{1}{2}(Z_{12} + Z_{13} - Z_{23}) \tag{3.6.8}$$

$$Z_2 = \tfrac{1}{2}(Z_{12} + Z_{23} - Z_{13}) \tag{3.6.9}$$

$$Z_3 = \tfrac{1}{2}(Z_{13} + Z_{23} - Z_{12}) \tag{3.6.10}$$

Equations (3.6.8) through (3.6.10) can be used to evaluate the per-unit series impedances Z_1, Z_2, and Z_3 of the three-winding transformer equivalent circuit from the per-unit leakage impedances Z_{12}, Z_{13}, and Z_{23}, which, in turn, are determined from short-circuit tests.

Note that each of the windings on a three-winding transformer may have a *different* kVA rating. If the leakage impedances from short-circuit tests are expressed in per-unit based on winding ratings, they must first be converted to per-unit on a common S_{base} *before* they are used in (3.6.8) through (3.6.10).

EXAMPLE 3.9

Three-winding single-phase transformer: per-unit impedances

The ratings of a single-phase three-winding transformer are

winding 1: 300 MVA, 13.8 kV

winding 2: 300 MVA, 199.2 kV

winding 3: 50 MVA, 19.92 kV

The leakage reactances, from short-circuit tests, are

X_{12} = 0.10 per unit on a 300-MVA, 13.8-kV base

X_{13} = 0.16 per unit on a 50-MVA, 13.8-kV base

X_{23} = 0.14 per unit on a 50-MVA, 199.2-kV base

Winding resistances and exciting current are neglected. Calculate the impedances of the per-unit equivalent circuit using a base of 300 MVA and 13.8 kV for terminal 1.

SOLUTION

S_{base} = 300 MVA is the same for all three terminals. Also, the specified voltage base for terminal 1 is V_{base1} = 13.8 kV. The base voltages for terminals 2 and 3 are then V_{base2} = 199.2 kV and V_{base3} = 19.92 kV, which are the rated voltages of these windings. From the data given, X_{12} = 0.10 per unit was measured from terminal 1 using the same base values as those specified for

(*Continued*)

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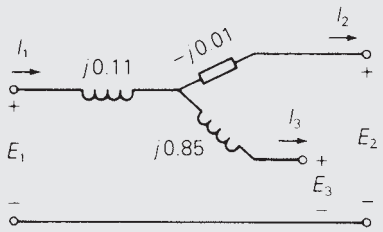
FIGURE 3.21
Circuit for Example 3.9



the circuit. However, $X_{13} = 0.16$ and $X_{23} = 0.14$ per unit on a 50-MVA base are first converted to the 300-MVA circuit base.

$$X_{13} = (0.16)\left(\frac{300}{50}\right) = 0.96 \quad \text{per unit}$$

$$X_{23} = (0.14)\left(\frac{300}{50}\right) = 0.84 \quad \text{per unit}$$

Then, from (3.6.8) through (3.6.10),

$$X_1 = \tfrac{1}{2}(0.10 + 0.96 - 0.84) = \quad 0.11 \quad \text{per unit}$$

$$X_2 = \tfrac{1}{2}(0.10 + 0.84 - 0.96) = -0.01 \quad \text{per unit}$$

$$X_3 = \tfrac{1}{2}(0.84 + 0.96 - 0.10) = \quad 0.85 \quad \text{per unit}$$

The per-unit equivalent circuit of this three-winding transformer is shown in Figure 3.21. Note that X_2 is negative. This illustrates the fact that X_1, X_2, and X_3 are *not* leakage reactances, but instead are equivalent reactances derived from the leakage reactances. Leakage reactances are always positive.

Note also that the node where the three equivalent circuit reactances are connected does not correspond to any physical location within the transformer. Rather, it is simply part of the equivalent circuit representation.

EXAMPLE 3.10

Three-winding three-phase transformer: balanced operation

Three transformers, each identical to that described in Example 3.9, are connected as a three-phase bank in order to feed power from a 900-MVA, 13.8-kV generator to a 345-kV transmission line and to a 34.5-kV distribution line. The transformer windings are connected as follows:

13.8-kV windings (X): Δ, to generator

199.2-kV windings (H): solidly grounded Y, to 345-kV line

19.92-kV windings (M): grounded Y through $Z_n = j0.10\ \Omega$, to 34.5-kV line

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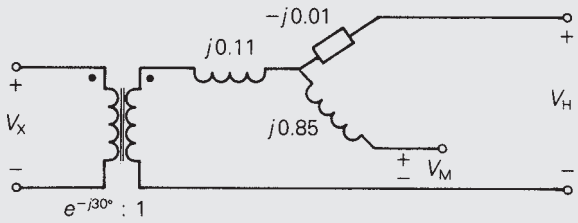
FIGURE 3.22

Per-unit network for Example 3.10



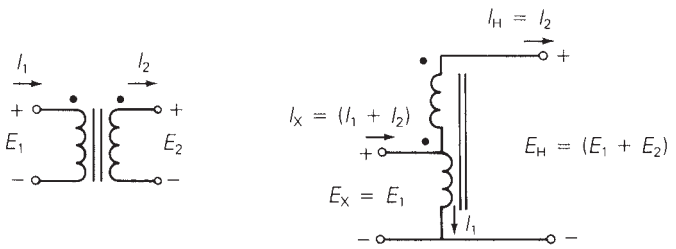
The positive-sequence voltages and currents of the high- and medium-voltage Y windings lead the corresponding quantities of the low-voltage Δ winding by 30°. Draw the per-unit network, using a three-phase base of 900 MVA and 13.8 kV for terminal X. Assume balanced positive-sequence operation.

SOLUTION

The per-unit network is shown in Figure 3.22. $V_{baseX} = 13.8$ kV, which is the rated line-to-line voltage of terminal X. Since the M and H windings are Y-connected, $V_{baseM} = \sqrt{3}(19.92) = 34.5$ kV, and $V_{baseH} = \sqrt{3}(199.2) = 345$ kV, which are the rated line-to-line voltages of the M and H windings. Also, a phase-shifting transformer is included in the network. The neutral impedance is not included in the network, since there is no neutral current under balanced operation.

3.7 AUTOTRANSFORMERS

A single-phase two-winding transformer is shown in Figure 3.23(a) with two separate windings, which is the usual two-winding transformer; the same transformer is shown in Figure 3.23(b) with the two windings connected in series, which is called an *autotransformer*. For the usual transformer [Figure 3.23(a)], the two windings are coupled magnetically via the mutual core flux. For the autotransformer [Figure 3.23(b)], the windings are both electrically and magnetically coupled. The autotransformer has smaller per-unit leakage impedances than the usual transformer; this results in both smaller series-voltage drops (an advantage) and higher short-circuit currents



(a) Two-winding transformer (b) Autotransformer

FIGURE 3.23

Ideal single-phase transformers

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(a disadvantage). The autotransformer also has lower per-unit losses (higher efficiency), lower exciting current, and lower cost if the turns ratio is not too large. The electrical connection of the windings, however, allows transient overvoltages to pass through the autotransformer more easily.

EXAMPLE 3.11

Autotransformer: single-phase

The single-phase two-winding 20-kVA, 480/120-volt transformer of Example 3.3 is connected as an autotransformer, as in Figure 3.23(b), where winding 1 is the 120-volt winding. For this autotransformer, determine (a) the voltage ratings E_X and E_H of the low- and high-voltage terminals, (b) the kVA rating, and (c) the per-unit leakage impedance.

SOLUTION

a. Since the 120-volt winding is connected to the low-voltage terminal, $E_X = 120$ volts. When $E_X = E_1 = 120$ volts is applied to the low-voltage terminal, $E_2 = 480$ volts is induced across the 480-volt winding, neglecting the voltage drop across the leakage impedance. Therefore, $E_H = E_1 + E_2 = 120 + 480 = 600$ volts.

b. As a normal two-winding transformer rated 20 kVA, the rated current of the 480-volt winding is $I_2 = I_H = 20{,}000/480 = 41.667$ A. As an autotransformer, the 480-volt winding can carry the same current. Therefore, the kVA rating $S_H = E_H I_H = (600)(41.667) = 25$ kVA. Note also that when $I_H = I_2 = 41.667$ A, a current $I_1 = (480/120)(41.667) = 166.7$ A is induced in the 120-volt winding. Therefore, $I_X = I_1 + I_2 = 208.3$ A (neglecting exciting current) and $S_X = E_X I_X = (120)(208.3) = 25$ kVA, which is the same rating as calculated for the high-voltage terminal.

c. From Example 3.3, the leakage impedance is $0.0729/\underline{78.13°}$ per unit as a normal, two-winding transformer. As an autotransformer, the leakage impedance *in ohms* is the same as for the normal transformer, since the core and windings are the same for both (only the external winding connections are different). However, the base impedances are different. For the high-voltage terminal, using (3.3.4),

$$Z_{\text{baseHold}} = \frac{(480)^2}{20{,}000} = 11.52\ \Omega \quad \text{as a normal transformer}$$

$$Z_{\text{baseHnew}} = \frac{(600)^2}{25{,}000} = 14.4\ \Omega \quad \text{as an autotransformer}$$

Therefore, using (3.3.10),

$$Z_{\text{p.u.new}} = (0.0729\underline{/78.13°})\left(\frac{11.52}{14.4}\right) = 0.05832\underline{/78.13°} \quad \text{per unit}$$

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For this example, the rating is 25 kVA, 120/600 volts as an autotransformer versus 20 kVA, 120/480 volts as a normal transformer. The autotransformer has both a larger kVA rating and a larger voltage ratio for the same cost. Also, the per-unit leakage impedance of the autotransformer is smaller. However, the increased high-voltage rating as well as the electrical connection of the windings may require more insulation for both windings.

3.8 TRANSFORMERS WITH OFF-NOMINAL TURNS RATIOS

It has been shown that models of transformers that use per-unit quantities are simpler than those that use actual quantities. The ideal transformer winding is eliminated when the ratio of the selected voltage bases equals the ratio of the voltage ratings of the windings. In some cases, however, it is impossible to select voltage bases in this manner. For example, consider the two transformers connected in parallel in Figure 3.24. Transformer T_1 is rated 13.8/345 kV, and T_2 is rated 13.2/345 kV. If $V_{baseH} = 345$ kV is selected, then transformer T_1 requires $V_{baseX} = 13.8$ kV, and T_2 requires $V_{baseX} = 13.2$ kV. It is clearly impossible to select the appropriate voltage bases for both transformers.

To accommodate this situation, develop a per-unit model of a transformer whose voltage ratings are not in proportion to the selected base voltages. Such a transformer is said to have an off-nominal turns ratio. Figure 3.25(a) shows a transformer with rated voltages V_{1rated} and V_{2rated}, which satisfy

$$V_{1rated} = a_t V_{2rated} \tag{3.8.1}$$

where a_t is assumed, in general, to be either real or complex. Suppose the selected voltage bases satisfy

$$V_{base1} = b V_{base2} \tag{3.8.2}$$

Defining $c = \dfrac{a_t}{b}$, (3.8.1) can be rewritten as

$$V_{1rated} = b\left(\frac{a_t}{b}\right) V_{2rated} = bc\ V_{2rated} \tag{3.8.3}$$

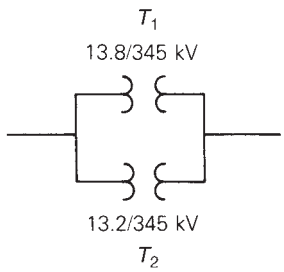

FIGURE 3.24
Two transformers connected in parallel

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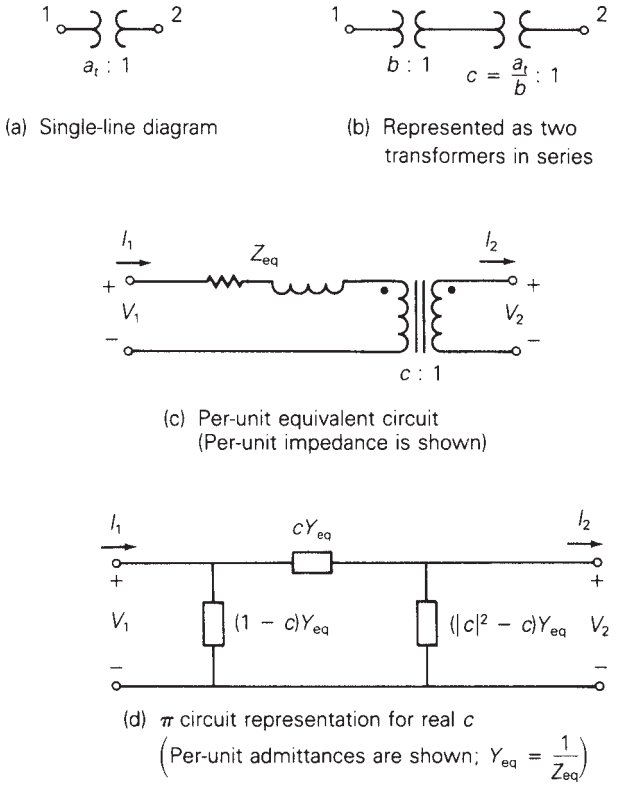
FIGURE 3.25

Transformer with off-nominal turns ratio



(a) Single-line diagram

(b) Represented as two transformers in series

(c) Per-unit equivalent circuit (Per-unit impedance is shown)

(d) π circuit representation for real c (Per-unit admittances are shown; $Y_{eq} = \frac{1}{Z_{eq}}$)

Equation (3.8.3) can be represented by two transformers in series, as shown in Figure 3.25(b). The first transformer has the same ratio of rated winding voltages as the ratio of the selected base voltages, *b*. Therefore, this transformer has a standard per-unit model, as shown in Figure 3.9 or 3.17. Assume that the second transformer is ideal, and all real and reactive losses are associated with the first transformer. The resulting per-unit model is shown in Figure 3.25(c), where, for simplicity, the shunt-exciting branch is neglected. Note that if $a_t = b$, then the ideal transformer winding shown in this figure can be eliminated, since its turns ratio $c = (a_t/b) = 1$.

The per-unit model shown in Figure 3.25(c) is perfectly valid, but it is not suitable for some of the computer programs presented in later chapters because these programs do not accommodate ideal transformer windings. An alternative representation can be developed, however, by writing nodal equations for this figure as follows:

$$\begin{bmatrix} I_1 \\ -I_2 \end{bmatrix} = \begin{bmatrix} Y_{11}\, Y_{12} \\ Y_{21}\, Y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} \tag{3.8.4}$$

where both I_1 and $-I_2$ are referenced *into* their nodes in accordance with the nodal equation method (Section 2.4). Recalling two-port network theory, the admittance parameters of (3.8.4) are, from Figure 3.23(c),

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$$Y_{11} = \left.\frac{I_1}{V_1}\right|_{V_2 = 0} = \frac{1}{Z_{eq}} = Y_{eq} \tag{3.8.5}$$

$$Y_{22} = \left.\frac{-I_2}{V_2}\right|_{V_1 = 0} = \frac{1}{Z_{eq}/|c|^2} = |c|^2 Y_{eq} \tag{3.8.6}$$

$$Y_{12} = \left.\frac{I_1}{V_2}\right|_{V_1 = 0} = \frac{-cV_2/Z_{eq}}{V_2} = -cY_{eq} \tag{3.8.7}$$

$$Y_{21} = \left.\frac{-I_2}{V_1}\right|_{V_2 = 0} = \frac{-c^*I_1}{V_1} = -c^*Y_{eq} \tag{3.8.8}$$

Equations (3.8.4) through (3.8.8) with real or complex c are convenient for representing transformers with off-nominal turns ratios in the computer programs presented later. Note that when c is complex, Y_{12} is not equal to Y_{21}, and the preceding admittance parameters cannot be synthesized with a passive RLC circuit. However, the π network shown in Figure 3.25(d), which has the same admittance parameters as (3.8.4) through (3.8.8), can be synthesized for real c. Note also that when $c = 1$, the shunt branches in this figure become open circuits (zero per unit mhos), and the series branch becomes Y_{eq} per unit mhos (or Z_{eq} per unit ohms).

EXAMPLE 3.12

Tap-changing three-phase transformer: per-unit positive-sequence network

A three-phase generator step-up transformer is rated 1000 MVA, 13.8 kV Δ/345 kV Y with $Z_{eq} = j0.10$ per unit. The transformer high-voltage winding has ±10% taps. The system base quantities are

$S_{base3\phi} = 500$ MVA

$V_{baseXLL} = 13.8$ kV

$V_{baseHLL} = 345$ kV

Determine the per-unit equivalent circuit for the following tap settings:

a. Rated tap
b. −10% tap (providing a 10% voltage decrease for the high-voltage winding)

Assume balanced positive-sequence operation. Neglect transformer winding resistance, exciting current, and phase shift.

(*Continued*)

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SOLUTION

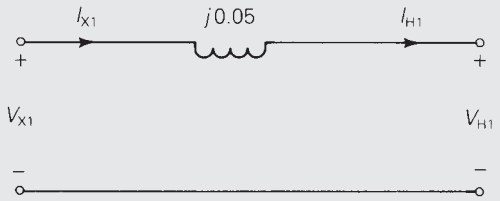
a. Using (3.8.1) and (3.8.2) with the low-voltage winding denoted winding 1,

$$a_t = \frac{13.8}{345} = 0.04 \qquad b = \frac{\mathrm{V_{baseXLL}}}{\mathrm{V_{baseHLL}}} = \frac{13.8}{345} = a_t \qquad c = 1$$

From (3.3.11),

$$Z_{\text{p.u.new}} = (j0.10)\left(\frac{500}{1000}\right) = j0.05 \quad \text{per unit}$$

The per-unit equivalent circuit, not including winding resistance, exciting current, and phase shift is



(Per-unit impedance is shown)

b. Using (3.8.1) and (3.8.2),

$$a_t = \frac{13.8}{345(0.9)} = 0.04444 \qquad b = \frac{13.8}{345} = 0.04$$

$$c = \frac{a_t}{b} = \frac{0.04444}{0.04} = 1.1111$$

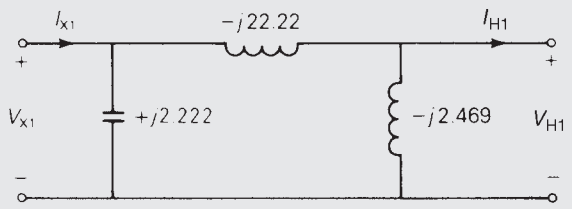
From Figure 3.25(d),

$$c\,Y_{\text{eq}} = 1.1111\left(\frac{1}{j0.05}\right) = -j22.22 \quad \text{per unit}$$

$$(1 - c)\,Y_{\text{eq}} = (-0.11111)(-j20) = +j2.222 \quad \text{per unit}$$

$$(|c|^2 - c)\,Y_{\text{eq}} = (1.2346 - 1.1)(-j20) = -j2.469 \quad \text{per unit}$$

The per-unit positive-sequence network is



(Per-unit admittances are shown)

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Open PowerWorld Simulator case Example 3_12 (see Figure 3.26) and select **Tools, Play** to see an animated view of this LTC transformer example. Initially, the generator/step-up transformer feeds a 500-MW/100-Mvar load. As is typical in practice, the transformer's taps are adjusted in discrete steps, with each step changing the tap ratio by 0.625% (hence a 10% change requires 16 steps). Click on arrows next to the transformer's tap to manually adjust the tap by one step. Note that changing the tap directly changes the load voltage.

Because of the varying voltage drops caused by changing loads, LTCs are often operated to automatically regulate a bus voltage. This is particularly true when they are used as step-down transformers. To place the example transformer on automatic control, click on the LTC Control Status "No" field. This toggles the transformer control mode to automatic. Now the transformer manually changes its tap ratio to maintain the load voltage within a specified voltage range, between 0.995 and 1.005 per unit (343.3 to 346.7 kV) in this case. To see the LTC in automatic operation, use the load arrows to vary the load, particularly the Mvar field, noting that the LTC changes to keep the load's voltage within the specified deadband.

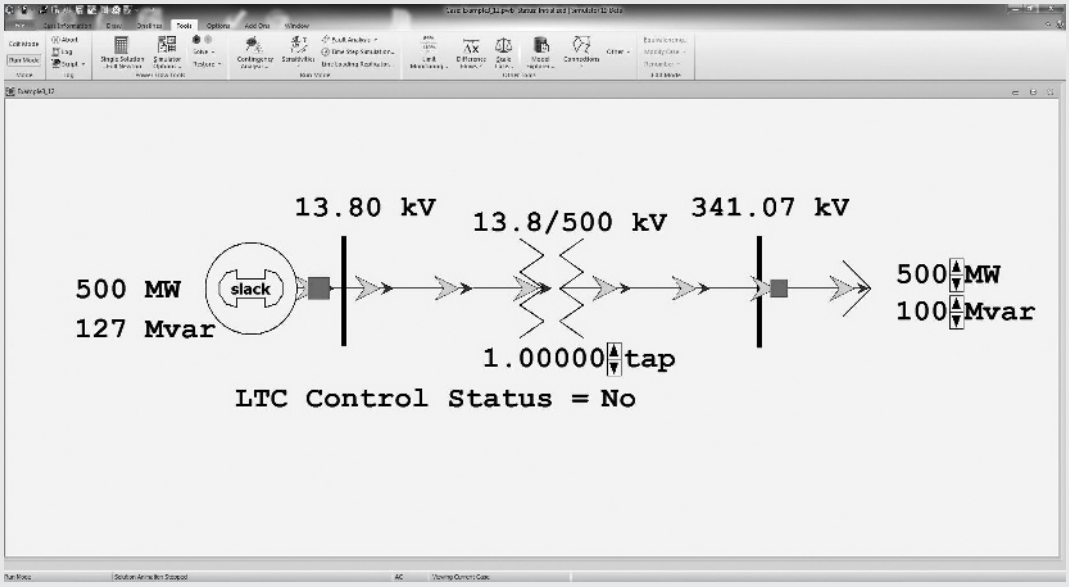

FIGURE 3.26

Screen for Example 3.12

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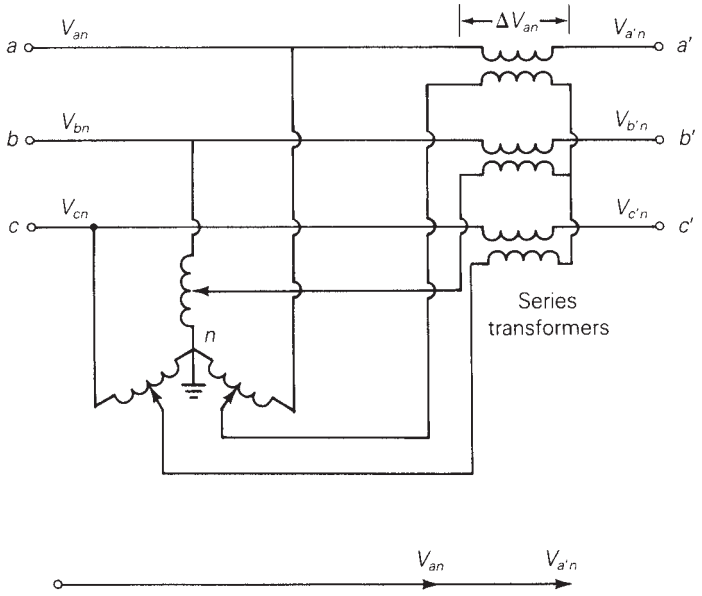
FIGURE 3.27

An example of a voltage-magnitude-regulating transformer



The three-phase regulating transformers shown in Figures 3.27 and 3.28 can be modeled as transformers with off-nominal turns ratios. For the voltage-magnitude-regulating transformer shown in Figure 3.27, adjustable voltages ΔV_{an}, ΔV_{bn}, and ΔV_{cn}, which have equal magnitudes ΔV and which are in phase with the phase voltages V_{an}, V_{bn}, and V_{cn}, are placed in the series link between buses a–a', b–b', and c–c'. Modeled as a transformer with an off-nominal turns ratio (see Figure 3.25), $c = (1 + \Delta V)$ for a voltage-magnitude increase toward bus abc, or $c = (1 + \Delta V)^{-1}$ for an increase toward bus $a'b'c'$.

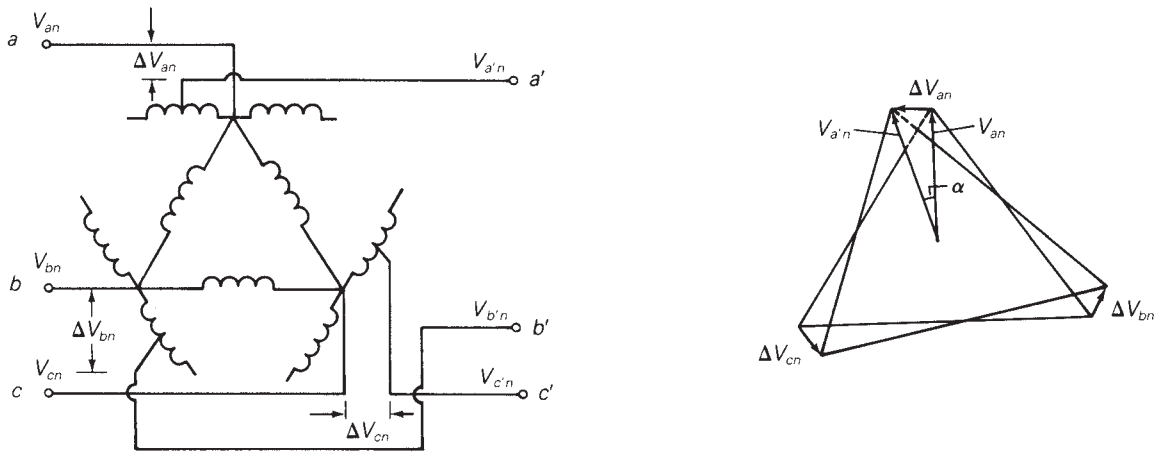


FIGURE 3.28

An example of a phase-angle-regulating transformer. Windings drawn in parallel are on the same core.

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For the phase-angle-regulating transformer in Figure 3.28, the series voltages ΔV_{an}, ΔV_{bn}, and ΔV_{cn}, are $\pm 90°$ out of phase with the phase voltages ΔV_{an}, V_{bn}, and V_{cn}. The phasor diagram in Figure 3.28 indicates that each of the bus voltages, $V_{a'n}$, $V_{b'n}$, and $V_{c'n}$, has a phase shift that is approximately proportional to the magnitude of the added series voltage. Modeled as a transformer with an off-nominal turns ratio (see Figure 3.25), $c \approx 1\underline{/\alpha}$ for a phase increase toward bus abc or $c \approx 1\underline{/-\alpha}$ for a phase increase toward bus $a'b'c'$.

EXAMPLE 3.13

Voltage-regulating and phase-shifting three-phase transformers

Two buses abc and $a'b'c'$ are connected by two parallel lines L1 and L2 with positive-sequence series reactances $X_{L1} = 0.25$ and $X_{L2} = 0.20$ per unit. A regulating transformer is placed in series with line L1 at bus $a'b'c'$. Determine the 2×2 bus admittance matrix when the regulating transformer (a) provides a 0.05 per-unit increase in voltage magnitude toward bus $a'b'c'$ and (b) advances the phase 3° toward bus $a'b'c'$. Assume that the regulating transformer is ideal. Also, the series resistance and shunt admittance of the lines are neglected.

SOLUTION

The circuit is shown in Figure 3.29.

a. For the voltage-magnitude-regulating transformer, $c = (1 + \Delta V)^{-1} = (1.05)^{-1} = 0.9524$ per unit. From (3.8.5) through (3.8.8), the admittance parameters of the regulating transformer in series with line L1 are

$$Y_{11L1} = \frac{1}{j0.25} = -j4.0$$

$$Y_{22L1} = (0.9524)^2(-j4.0) = -j3.628$$

$$Y_{12L1} = Y_{21L1} = (-0.9524)(-j4.0) = j3.810$$

For line L2 alone,

$$Y_{11L2} = Y_{22L2} = \frac{1}{j0.20} = -j5.0$$

$$Y_{12L2} = Y_{21L2} = -(-j5.0) = j5.0$$

Combining the above admittances in parallel,

$$Y_{11} = Y_{11L1} + Y_{11L2} = -j4.0 - j5.0 = -j9.0$$

$$Y_{22} = Y_{22L1} + Y_{22L2} = -j3.628 - j5.0 = -j8.628$$

$$Y_{12} = Y_{21} = Y_{12L1} + Y_{12L2} = j3.810 + j5.0 = j8.810 \quad \text{per unit}$$

(*Continued*)

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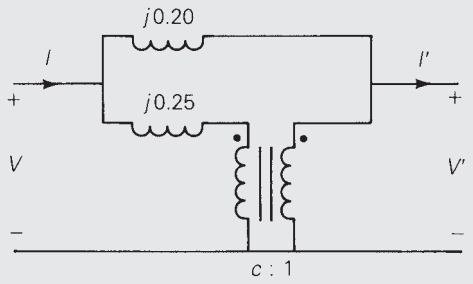
FIGURE 3.29
Positive-sequence circuit for Example 3.13



b. For the phase-angle-regulating transformer, $c = 1\angle -\alpha = 1\angle -3°$. Then, for this regulating transformer in series with line L1,

$$Y_{11L1} = \frac{1}{j0.25} = -j4.0$$

$$Y_{22L1} = |1.0\angle -3°|^2(-j4.0) = -j4.0$$

$$Y_{12L1} = -(1.0\angle -3°)(-j\,4.0) = 4.0\angle 87° = 0.2093 + j3.9945$$

$$Y_{21L1} = -(1.0\angle -3°)^*(-j4.0) = 4.0\angle 93° = -0.2093 + j3.9945$$

The admittance parameters for line L2 alone are given in part (a). Combining the admittances in parallel,

$$Y_{11} = Y_{22} = -j4.0 - j5.0 = -j9.0$$

$$Y_{12} = 0.2093 + j3.9945 + j5.0 = 0.2093 + j8.9945$$

$$Y_{21} = -0.2093 + j3.9945 + j5.0 = -0.2093 + j8.9945 \quad \text{per unit}$$

To see this example in the PowerWorld Simulator, open case Example 3_13 (see Figure 3.30). In this case, the transformer and a parallel transmission line are assumed to be supplying power from a 345-kV generator to a 345-kV load. Initially, the off-nominal turns ratio is set to the value in part (a) of the example. (PowerWorld has the off-nominal turns ratio on the load side [right-hand] so its tap value of $1.05 = c^{-1}$.) To view the PowerWorld Simulator bus admittance matrix, select the **Case Information** ribbon, then **Solution Details**, **Ybus**. To see how the system flows vary with changes to the tap, select **Tools**, **Play**, and then click on the arrows next to the tap field to change the LTC tap in 0.625% steps. Next, to verify the results from part (b), change the tap field to 1.0 and the degree field to 3.0 degrees, and then again look at the bus admittance matrix. Click on the degree field arrow to vary the phase shift angle in one-degree steps. Notice that changing the phase angle primarily changes the real power flow, whereas changing the LTC tap changes the reactive power flow. In this example, the line flow fields show the absolute value of the real or reactive power flow; the direction of the flow is indicated with arrows. Traditional power flow programs usually indicate power flow direction using a convention that flow into a transmission line or transformer is assumed to be positive. You can display

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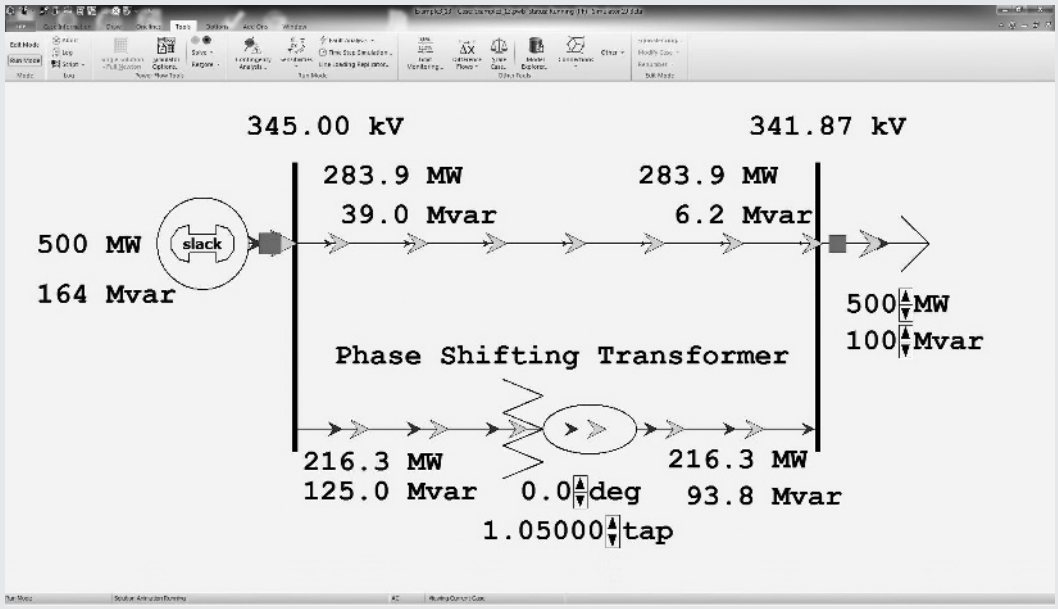

FIGURE 3.30

Screen for Example 3.13

results in PowerWorld Simulator using this convention by first clicking on the **Onelines** ribbon and then selecting **Oneline Options**. Then on the **Display Options** tab uncheck the Use Absolute Values for MW/Mvar Line Flows fields.

Note that a voltage-magnitude-regulating transformer controls the *reactive* power flow in the series link in which it is installed, whereas a phase-angle-regulating transformer controls the *real* power flow (see Problem 3.59).

The process of changing a transformer's off-nominal turns ratio also changes its series impedance. Often, particularly when the change is only to the off-nominal turns ratio magnitude, this change in impedance is rather modest, and it is ignored in most transformer models. However, sometimes the variation in the impedance with the off-nominal turns ratio is very significant, a situation that is often the case with phase-angle-regulating transformers. For example, it is common for the impedance to be 50% or more at the maximum or minimum phase angle shifts compared to at zero phase angle shift, and it's not unusual for it to be many times higher. In modeling software, this impedance variation is often represented by scaling the transformer's series impedance by a factor that is modeled as a piecewise linear function of the off-nominal turns ratio magnitude or angle. These scaling functions are sometimes called *impedance correction tables* (see Problem 3.62 for an example), and they are widely used in commercial power system analysis software.

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PROBLEMS

SECTION 3.1

3.1 For an ideal, two-winding transformer, the primary voltage is 221 kV and the primary current is 18 A. The secondary winding is connected to a load with $Z = 400\angle 15°\ \Omega$.

(a) What is the turns ratio of the transformer?
(b) Verify conservation of power in the transformer.

3.2 An ideal transformer with $N_1 = 1000$ and $N_2 = 250$ is connected with an impedance Z_{22} across winding 2. If $V_1 = 1000\ \underline{/0°}$V and $I_1 = 5\ \underline{/-30°}$ A, determine V_2, I_2, Z_2, and the impedance Z'_2, which is the value of Z_2 referred to the primary side of the transformer.

3.3 Consider an ideal transformer with $N_1 = 3000$ and $N_2 = 1000$ turns. Let winding 1 be connected to a source whose voltage is $e_1(t) = 100\ (1 - |t|)$ volts for $-1 \le t \le 1$ and $e_1(t) = 0$ for $|t| > 1$ second. A 2-farad capacitor is connected across winding 2. Sketch $e_1(t)$, $e_2(t)$, $i_1(t)$, and $i_2(t)$ versus time t.

3.4 A single-phase 100-kVA, 2400/240-volt, 60-Hz distribution transformer is used as a step-down transformer. The load, which is connected to the 240-volt secondary winding, absorbs 75 kVA at 0.8 power factor lagging and is at 230 volts. Assuming an ideal transformer, calculate the following: (a) primary voltage, (b) load impedance, (c) load impedance referred to the primary, and (d) the real and reactive power supplied to the primary winding.

3.5 Rework Problem 3.4 if the load connected to the 240-V secondary winding absorbs 125 kVA under short-term overload conditions at a 0.8 power factor leading and at 230 volts.

3.6 For a conceptual single-phase phase-shifting transformer, the primary voltage leads the secondary voltage by 30°. A load connected to the secondary winding absorbs 110 kVA at a 0.8 power factor leading and at a voltage $E_2 = 277\ \underline{/0°}$ volts. Determine (a) the primary voltage, (b) primary and secondary currents, (c) load impedance referred to the primary winding, and (d) complex power supplied to the primary winding.

3.7 Consider a source of voltage $v(t) = 10\sqrt{2}\sin(2t)$ V, with an internal resistance of 1800 Ω. A transformer that can be considered as ideal is used to couple a 50-Ω resistive load to the source. (a) Determine the transformer primary-to-secondary turns ratio required to ensure maximum power transfer by matching the load and source resistances. (b) Find the average power delivered to the load, assuming maximum power transfer.

3.8 For the circuit shown in Figure 3.31, determine $v_{out}(t)$.

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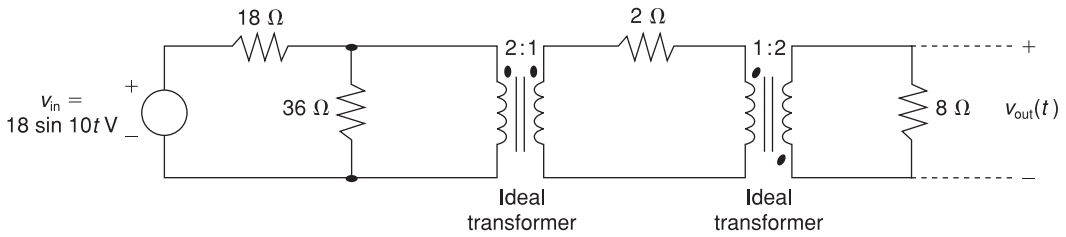

FIGURE 3.31

Problem 3.8

SECTION 3.2

3.9 For a practical, two-winding transformer with a turns ratio of 3:1, applying the connection of a 10-V dc voltage to the primary side results in a 20-A dc current in the primary terminal. Similarly, the connection of a 10-V dc voltage applied to the secondary side results in a 50-A dc current in the secondary terminal.

(a) What phenomenon is this measuring?
(b) How can this be represented as a single element on the primary side in the non-ideal transformer circuit model?

3.10 A single-phase step-down transformer is rated 13 MVA, 66 kV/11.5 kV. With the 11.5-kV winding short-circuited, rated current flows when the voltage applied to the primary is 5.5 kV. The power input is read as 100 kW. Determine R_{eq1} and X_{eq1} in ohms referred to the high-voltage winding.

3.11 For the transformer in Problem 3.10, the open-circuit test with 11.5 kV applied results in a power input of 65 kW and a current of 30 A. Compute the values for G_c and B_m in siemens referred to the high-voltage winding. Compute the efficiency of the transformer for a load of 10 MW at 0.8 p.f. lagging at rated voltage.

3.12 The following data are obtained when open-circuit and short-circuit tests are performed on a single-phase, 50-kVA, 2400/240-volt, 60-Hz distribution transformer.

	VOLTAGE (volts)	CURRENT (amperes)	POWER (watts)
Measurements on low-voltage side with high-voltage winding open	240	4.85	173
Measurements on high-voltage side with low-voltage winding shorted	52.0	20.8	650

(a) Neglecting the series impedance, determine the exciting admittance referred to the high-voltage side. (b) Neglecting the exciting admittance, determine the equivalent series impedance referred to the

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high-voltage side. (c) Assuming equal series impedances for the primary and referred secondary, obtain an equivalent T-circuit referred to the high-voltage side.

3.13 A single-phase 50-kVA, 2400/240-volt, 60-Hz distribution transformer has a 1-ohm equivalent leakage reactance and an 800-ohm magnetizing reactance referred to the high-voltage side. If rated voltage is applied to the high-voltage winding, calculate the open-circuit secondary voltage. Neglect I^2R and G_c^2V losses. Assume equal series leakage reactances for the primary and the referred secondary.

3.14 A single-phase 50-kVA, 2400/240-volt, 60-Hz distribution transformer is used as a step-down transformer at the load end of a 2400-volt feeder whose series impedance is $(1.0 + j2.0)$ ohms. The equivalent series impedance of the transformer is $(1.0 + j2.5)$ ohms referred to the high-voltage (primary) side. The transformer is delivering rated load at a 0.8 power factor lagging and at a rated secondary voltage. Neglecting the transformer exciting current, determine (a) the voltage at the transformer primary terminals, (b) the voltage at the sending end of the feeder, and (c) the real and reactive power delivered to the sending end of the feeder.

3.15 Rework Problem 3.14 if the transformer is delivering rated load at rated secondary voltage and at (a) unity power factor, (b) 0.8 power factor leading. Compare the results with those of Problem 3.14.

3.16 A single-phase, 50-kVA, 2400/240-V, 60-Hz distribution transformer has the following parameters:

Resistance of the 2400-V winding: $R_1 = 0.75\ \Omega$

Resistance of the 240-V winding: $R_2 = 0.0075\ \Omega$

Leakage reactance of the 2400-V winding: $X_1 = 1.0\ \Omega$

Leakage reactance of the 240-V winding: $X_2 = 0.01\ \Omega$

Exciting admittance on the 240-V side $= 0.003 - j0.02$ S

(a) Draw the equivalent circuit referred to the high-voltage side of the transformer.

(b) Draw the equivalent circuit referred to the low-voltage side of the transformer. Show the numerical values of impedances on the equivalent circuits.

3.17 The transformer of Problem 3.16 is supplying a rated load of 50 kVA at a rated secondary voltage of 240 V and at 0.8 power factor lagging. Neglect the transformer exciting current. (a) Determine the input terminal voltage of the transformer on the high-voltage side. (b) Sketch the corresponding phasor diagram. (c) If the transformer is used as a step-down transformer at the load end of a feeder whose impedance is $0.5 + j2.0\ \Omega$, find the voltage V_S and the power factor at the sending end of the feeder.

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SECTION 3.3

3.18 Using the transformer ratings as base quantities, work Problem 3.13 in per-unit.

3.19 Using the transformer ratings as base quantities, work Problem 3.14 in per-unit.

3.20 Using base values of 20 kVA and 115 volts in zone 3, rework Example 3.4.

3.21 Rework Example 3.5; using $S_{base3\phi} = 100$ kVA and $V_{baseLL} = 600$ volts.

3.22 A balanced Y-connected voltage source with $E_{ag} = 277\angle 0°$ volts is applied to a balanced-Y load in parallel with a balanced-Δ load where $Z_Y = 20 + j10$ and $Z_\Delta = 30 - j15$ ohms. The Y load is solidly grounded. Using base values of $S_{base1\phi} = 10$ kVA and $V_{baseLN} = 277$ volts, calculate the source current I_a in per-unit and in amperes.

3.23 Figure 3.32 shows the oneline diagram of a three-phase power system. By selecting a common base of 100 MVA and 22 kV on the generator side, draw an impedance diagram showing all impedances, including the load impedance in per-unit. The data are given as follows:

G:	90 MVA	22 kV	x = 0.18 per unit
T_1:	50 MVA	22/220 kV	x = 0.10 per unit
T_2:	40 MVA	220/11 kV	x = 0.06 per unit
T_3:	40 MVA	22/110 kV	x = 0.064 per unit
T_4:	40 MVA	110/11 kV	x = 0.08 per unit
M:	66.5 MVA	10.45 kV	x = 0.185 per unit

Lines 1 and 2 have series reactances of 48.4 and 65.43 Ω, respectively. At bus 4, the three-phase load absorbs 57 MVA at 10.45 kV and 0.6 power factor lagging.

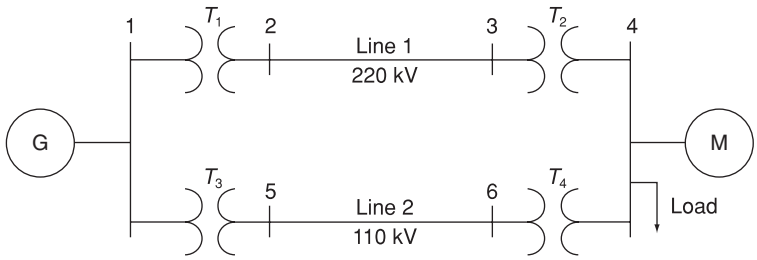

FIGURE 3.32
Problem 3.23

3.24 An ideal 1:5 single-phase transformer is connected to a 15-kV voltage source on the primary side and a 30-Ω resistor on the secondary side. (a) Calculate the load power and source current without the per-unit system, using the ideal transformer equations. (b) Create a table of the per-unit base values for this circuit, assuming a 100-MVA power base. (c) Solve for the load power and source current using the per-unit system.

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3.25 Consider a single-phase electric system shown in Figure 3.33. Transformers are rated as follows:

X–Y 15 MVA, 13.8/138 kV, leakage reactance 10%

Y–Z 15 MVA, 138/69 kV, leakage reactance 8%

With the base in circuit Y chosen as 15 MVA, 138 kV, determine the per-unit impedance of the 500-Ω resistive load in circuit Z, referred to circuits Z, Y, and X. Neglecting magnetizing currents, transformer resistances, and line impedances, draw the impedance diagram in per unit.

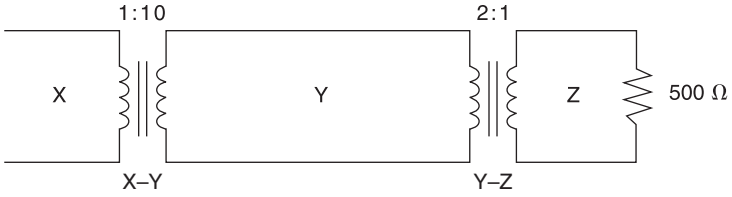

FIGURE 3.33 Single-phase electric system for Problem 3.25

3.26 A bank of three single-phase transformers, each rated 30 MVA, 38.1/3.81 kV, are connected in Y–Δ with a balanced load of three 1-Ω, Y-connected resistors. Choosing a base of 90 MVA, 66 kV for the high-voltage side of the three-phase transformer, specify the base for the low-voltage side. Compute the per-unit resistance of the load on the base for the low-voltage side. Also, determine the load resistance in ohms referred to the high-voltage side and the per-unit value on the chosen base.

3.27 A three-phase transformer is rated 1000 MVA, 220 Y/22 Δ kV. The Y-equivalent short-circuit impedance, considered equal to the leakage reactance, measured on the low-voltage side is 0.4 Ω. Compute the per-unit reactance of the transformer. In a system in which the base on the high-voltage side of the transformer is 100 MVA, 230 kV, what value of the per-unit reactance should be used to represent this transformer?

3.28 For the system shown in Figure 3.34, draw an impedance diagram in per unit by choosing 100 kVA to be the base kVA and 2400 V as the base voltage for the generators.

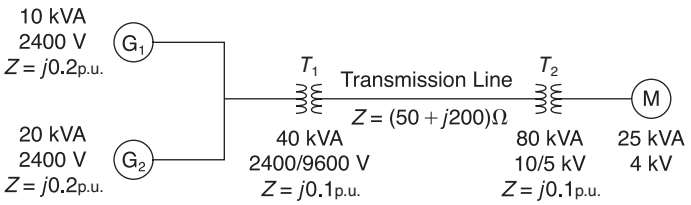

FIGURE 3.34 System for Problem 3.28

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3.29 Consider three ideal single-phase transformers (with a voltage gain of η) put together as a Δ–Y three-phase bank as shown in Figure 3.35. Assuming positive-sequence voltages for V_{an}, V_{bn}, and V_{cn}, find $V_{a'n'}$, $V_{b'n'}$, and $V_{c'n'}$ in terms of V_{an}, V_{bn}, and V_{cn}, respectively.

(a) Would such relationships hold for the line voltages as well?

(b) Looking into the current relationships, express I'_a, I'_b, and I'_c in terms of I_a, I_b, and I_c, respectively.

(c) Let S' and S be the per-phase complex power output and input, respectively. Find S′ in terms of S.

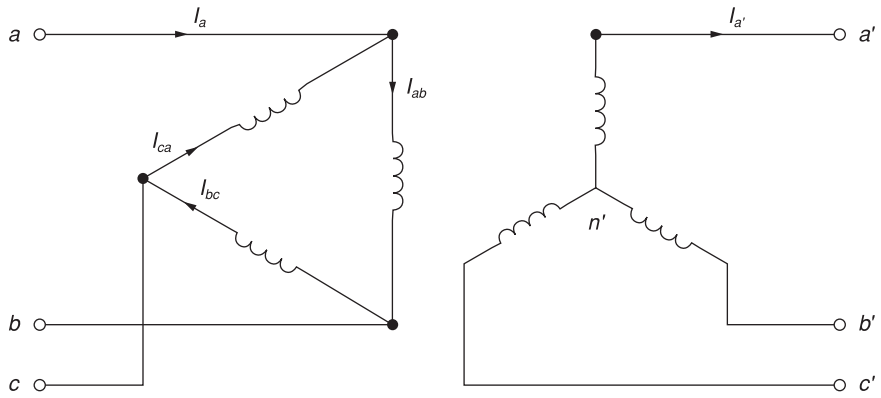

FIGURE 3.35
Δ−Y connection for Problem 3.29

3.30 Reconsider Problem 3.29. If V_{an}, V_{bn}, and V_{cn} are a negative-sequence set, how would the voltage and current relationships change?
(a) If C_1 is the complex positive-sequence voltage gain in Problem 3.29, and (b) if C_2 is the negative sequence complex voltage gain, express the relationship between C_1 and C_2.

3.31 If positive-sequence voltages are assumed and the Y-Δ connection is considered, again with ideal transformers as in Problem 3.29, find the complex voltage gain C_3.

(a) What would the gain be for a negative-sequence set?

(b) Comment on the complex power gain.

(c) When terminated in a symmetric Y-connected load, find the referred impedance Z'_L, the secondary impedance Z_L referred to primary (i.e., the per-phase driving-point impedance on the primary side) in terms of Z_L and the complex voltage gain C.

SECTION 3.4

3.32 Determine the positive- and negative-sequence phase shifts for the three-phase transformers shown in Figure 3.36.

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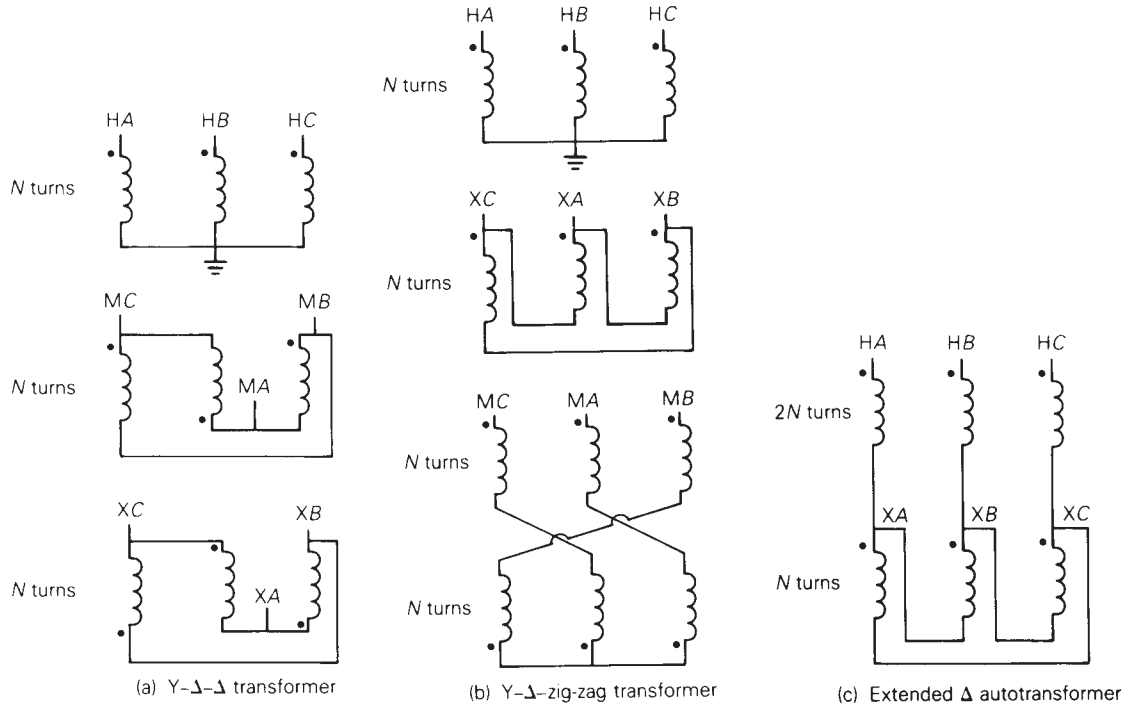


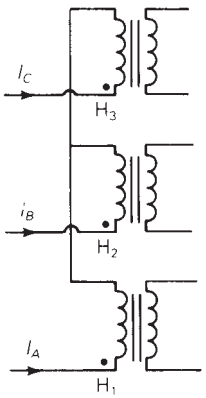
FIGURE 3.36

Problems 3.32 and 3.52 (Coils drawn on the same vertical line are on the same core.)

3.33 Consider the three single-phase two-winding transformers shown in Figure 3.37. The high-voltage windings are connected in Y. (a) For the low-voltage side, connect the windings in Δ, place the polarity marks, and label the terminals *a, b,* and *c* in accordance with the American standard. (b) Relabel the terminals a', b', and c' such that V_{AN} is 90° out of phase with $V_{a'n}$ for positive sequence.

FIGURE 3.37

Problem 3.33



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3.34 Three single-phase, two-winding transformers, each rated 450 MVA, 20 kV/288.7 kV, with leakage reactance $X_{eq} = 0.10$ per unit, are connected to form a three-phase bank. The high-voltage windings are connected in Y with a solidly grounded neutral. Draw the per-unit equivalent circuit if the low-voltage windings are connected (a) in Δ with American standard phase shift, or (b) in Y with an open neutral. Use the transformer ratings as base quantities. Winding resistances and exciting current are neglected.

3.35 An ideal Δ-Y, three-phase, 230:46-kV transformer has a voltage phasor of $V_{bc} = 50\angle 18°$ kV on the secondary side, line-to-line between phases *b* and *c*. Determine the six line-to-neutral voltages for the transformer terminals for the balanced three-phase conditions.

3.36 Three single-phase two-winding transformers, each rated 25 MVA, 34.5/13.8 kV, are connected to form a three-phase Δ–Δ bank. Balanced positive-sequence voltages are applied to the high-voltage terminals, and a balanced, resistive Y load connected to the low-voltage terminals absorbs 75 MW at 13.8 kV. If one of the single-phase transformers is removed (resulting in an open-Δ connection) and the balanced load is simultaneously reduced to 43.3 MW (57.7% of the original value), determine (a) the load voltages V_{an}, V_{bn}, and V_{cn}; (b) load currents I_a, I_b, and I_c; and (c) the MVA supplied by each of the remaining two transformers. Are balanced voltages still applied to the load? Is the open-Δ transformer overloaded?

3.37 Three single-phase two-winding transformers, each rated 25 MVA, 54.2/5.42 kV, are connected to form a three-phase Y–Δ bank with a balanced Y-connected resistive load of 0.6 Ω per phase on the low-voltage side. By choosing a base of 75 MVA (three-phase) and 94 kV (line-to-line) for the high-voltage side of the transformer bank, specify the base quantities for the low-voltage side. Determine the per-unit resistance of the load on the base for the low-voltage side. Then determine the load resistance R_L in ohms referred to the high-voltage side and the per-unit value of this load resistance on the chosen base.

3.38 Consider a three-phase generator rated 300 MVA, 23 kV, supplying a system load of 240 MVA and 0.9 power factor lagging at 230 kV through a 330 MVA, 23 Δ/ 230 Y-kV step-up transformer with a leakage reactance of 0.11 per unit. (a) Neglecting the exciting current and choosing base values at the load of 100 MVA and 230 kV, find the phasor currents I_A, I_B, and I_C supplied to the load in per unit. (b) By choosing the load terminal voltage V_A as reference, specify the proper base for the generator circuit and determine the generator voltage V, as well as the phasor currents I_A, I_B, and I_C, from the generator. (*Note:* Take into account the phase shift of the transformer.) (c) Find the generator terminal voltage in kV and the real power supplied by the generator in MW. (d) By omitting the transformer phase shift altogether, check to see whether you get the same magnitude of generator terminal voltage and real power delivered by the generator.

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SECTION 3.5

3.39 The leakage reactance of a three-phase, 300-MVA, 230 Y/23 Δ-kV transformer is 0.06 per unit based on its own ratings. The Y winding has a solidly grounded neutral. Draw the per-unit equivalent circuit. Neglect the exciting admittance and assume the American Standard phase shift.

3.40 Choosing system bases to be 240/24 kV and 100 MVA, redraw the per-unit equivalent circuit for Problem 3.39.

3.41 Consider the single-line diagram of the power system shown in Figure 3.38. Equipment ratings are

Generator 1:	1000 MVA, 18 kV, X" = 0.2 per unit
Generator 2:	1000 MVA, 18 kV, X" = 0.2 p.u.
Synchronous motor 3:	1500 MVA, 20 kV, X" = 0.2 p.u.
Three-phase Δ–Y transformers T_1,T_2, T_3, T_4:	1000 MVA, 500 kV Y/20 kV Δ, X = 0.1 p.u.
Three-phase Y–Y transformer T_5:	1500 MVA, 500 kV Y/20 kV Y, X = 0.1 p.u.

Neglecting resistance, transformer phase shift, and magnetizing reactance, draw the equivalent reactance diagram. Use a base of 1000 MVA and 500 kV for the 50-ohm line. Determine the per-unit reactances.

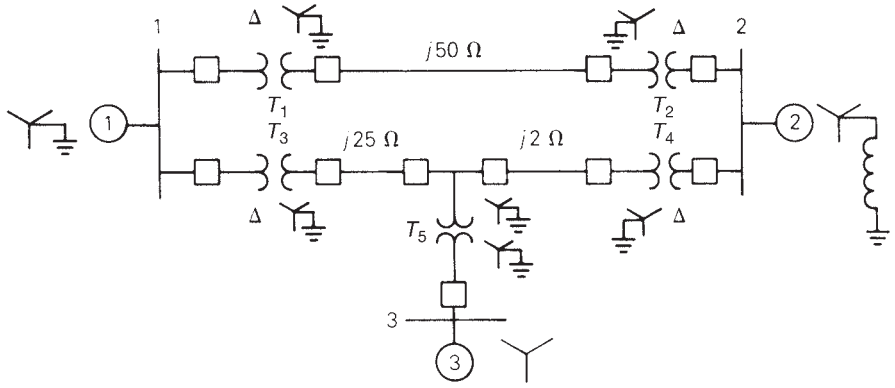

FIGURE 3.38
Problems 3.41 and 3.42

3.42 For the power system in Problem 3.41, the synchronous motor absorbs 1500 MW at 0.8 power factor leading with the bus 3 voltage at 18 kV. Determine the bus 1 and bus 2 voltages in kV. Assume that generators 1 and 2 deliver equal real powers and equal reactive powers. Also assume a balanced three-phase system with positive-sequence sources.

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3.43 Three single-phase transformers, each rated 10 MVA, 66.4/12.5 kV, 60 Hz, with an equivalent series reactance of 0.1 per unit divided equally between primary and secondary, are connected in a three-phase bank. The high-voltage windings are Y-connected, and their terminals are directly connected to a 115-kV three-phase bus. The secondary terminals are all shorted together. Find the currents entering the high-voltage terminals and leaving the low-voltage terminals if the low-voltage windings are (a) Y-connected, and (b) Δ-connected.

3.44 A 130-MVA, 13.2-kV three-phase generator, which has a positive-sequence reactance of 1.5 per unit on the generator base, is connected to a 135-MVA, 13.2 Δ/115 Y-kV step-up transformer with a series impedance of $(0.005 + j0.1)$ per unit on its own base. (a) Calculate the per-unit generator reactance on the transformer base. (b) The load at the transformer terminals is 15 MW at unity power factor and at 115 kV. Choosing the transformer high-side voltage as the reference phasor, draw a phasor diagram for this condition. (c) For the condition of part (b), find the transformer low-side voltage and the generator internal voltage behind its reactance. Also compute the generator output power and power factor.

3.45 Figure 3.39 shows a oneline diagram of a system in which the three-phase generator is rated 300 MVA, 20 kV with a subtransient reactance of 0.2 per unit and with its neutral grounded through a 0.4-Ω reactor. The transmission line is 64 km long with a series reactance of 0.5 Ω/km. The three-phase transformer T_1 is rated 350 MVA, 230/20 kV with a leakage reactance of 0.1 per unit. Transformer T_2 is composed of three single-phase transformers, each rated 100 MVA, 127/13.2 kV with a leakage reactance of 0.1 per unit. Two 13.2-kV motors M_1 and M_2 with a subtransient reactance of 0.2 per unit for each motor represent the load. M_1 has a rated input of 200 MVA with its neutral grounded through a 0.4-Ω current-limiting reactor. M_2 has a rated input of 100 MVA with its neutral not connected to ground. Neglect phase shifts associated with the transformers. Choose the generator rating as base in the generator circuit and draw the positive-sequence reactance diagram showing all reactances in per unit.

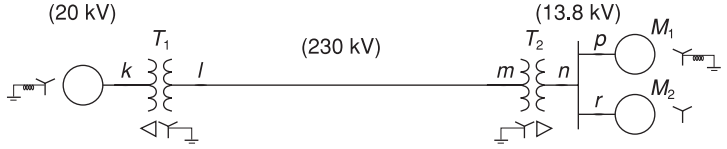

FIGURE 3.39
Problems 3.45 and 3.46

3.46 The motors M_1 and M_2 of Problem 3.45 have inputs of 120 and 60 MW, respectively, at 13.2 kV, and both operate at unity power factor. Determine the generator terminal voltage and voltage regulation of the line. Neglect transformer phase shifts.

3.47 Consider the oneline diagram shown in Figure 3.40. The three-phase transformer bank is made up of three identical single-phase transformers, each specified by $X_1 = 0.24\ \Omega$ (on the low-voltage side),

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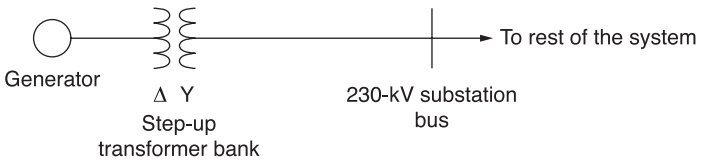
negligible resistance and magnetizing current, and turns ratio $\eta = N_2/N_1 = 10$. The transformer bank is delivering 100 MW at 0.8 p.f. lagging to a substation bus whose voltage is 230 kV.

(a) Determine the primary current magnitude, primary voltage (line-to-line) magnitude, and the three-phase complex power supplied by the generator. Choose the line-to-neutral voltage at the bus, $V_{a'n'}$ as the reference. Account for the phase shift, and assume positive-sequence operation.

(b) Find the phase shift between the primary and secondary voltages.

FIGURE 3.40
Oneline diagram for Problem 3.47

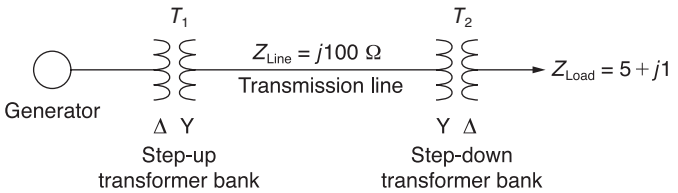


3.48 With the same transformer banks as in Problem 3.47, Figure 3.41 shows the oneline diagram of a generator, a step-up transformer bank, a transmission line, a step-down transformer bank, and an impedance load. The generator terminal voltage is 15 kV (line-to-line).

(a) Draw the per-phase equivalent circuit, accounting for phase shifts for positive-sequence operation.

(b) By choosing the line-to-neutral generator terminal voltage as the reference, determine the magnitudes of the generator current, transmission-line current, load current, and line-to-line load voltage. Also, find the three-phase complex power delivered to the load.

FIGURE 3.41
Oneline diagram for Problem 3.48



3.49 Consider the single-line diagram of a power system shown in Figure 3.42 with equipment ratings given:

Generator G_1:	50 MVA, 13.2 kV, $x = 0.15$ p.u.
Generator G_2:	20 MVA, 13.8 kV, $x = 0.15$ p.u.
Three-phase Δ–Y transformer T_1:	80 MVA, 13.2 Δ/165 Y kV, $X = 0.1$ p.u.
Three-phase Y–Δ transformer T_2:	40 MVA, 165 Y/13.8 Δ kV, $X = 0.1$ p.u.
Load:	40 MVA, 0.8 p.f. lagging, operating at 150 kV

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Choose a base of 100 MVA for the system and 132-kV base in the transmission-line circuit. Let the load be modeled as a parallel combination of resistance and inductance. Neglect transformer phase shifts. Draw a per-phase equivalent circuit of the system showing all impedances in per unit.

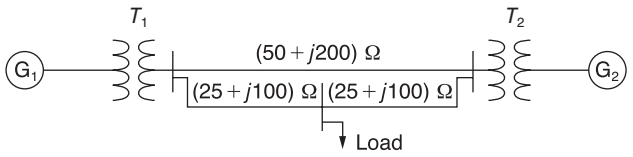

FIGURE 3.42
Oneline diagram for Problem 3.49

SECTION 3.6

3.50 A single-phase three-winding transformer has the following parameters: $Z_1 = Z_2 = Z_3 = 0 + j0.05$, $G_C = 0$, and $B_M = 0.2$ per unit. Three identical transformers, as described, are connected with their primaries in Y (solidly grounded neutral) and with their secondaries and tertiaries in Δ. Draw the per-unit sequence networks of this transformer bank.

3.51 The ratings of a three-phase three-winding transformer are

Primary (1): Y-connected, 66 kV, 15 MVA

Secondary (2): Y-connected, 13.2 kV, 10 MVA

Tertiary (3): Δ-connected, 2.3 kV, 5 MVA

Neglecting winding resistances and exciting current, the per-unit leakage reactances are

$X_{12} = 0.08$ on a 15-MVA, 66-kV base

$X_{13} = 0.10$ on a 15-MVA, 66-kV base

$X_{23} = 0.09$ on a 10-MVA, 13.2-kV base

(a) Determine the per-unit reactances X_1, X_2, X_3 of the equivalent circuit on a 15-MVA, 66-kV base at the primary terminals.

(b) Purely resistive loads of 7.5 MW at 13.2 kV and 5 MW at 2.3 kV are connected to the secondary and tertiary sides of the transformer, respectively. Draw the per-unit impedance diagram, showing the per-unit impedances on a 15-MVA, 66-kV base at the primary terminals.

3.52 Assume a 138:69:20-kV transformer is ideal. The actual voltages on the primary and secondary are at rated values of 138 kV and 69 kV, and the actual primary and secondary currents are at their rated values of 10 A and 18 A. The tertiary is rated for 20 kV and 8 A. What is the tertiary actual voltage and current?

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3.53 The ratings of a three-phase, three-winding transformer are

Primary: Y-connected, 66 kV, 15 MVA

Secondary: Y-connected, 13.2 kV, 10 MVA

Tertiary: Δ-connected, 2.3 kV, 5 MVA

Neglecting resistances and exciting current, the leakage reactances are:

X_{PS} = 0.09 per unit on a 15-MVA, 66-kV base

X_{PT} = 0.08 per unit on a 15-MVA, 66-kV base

X_{ST} = 0.05 per unit on a 10-MVA, 13.2-kV base

Determine the per-unit reactances of the per-phase equivalent circuit using a base of 15 MVA and 66 kV for the primary.

3.54 An infinite bus, which is a constant voltage source, is connected to the primary of the three-winding transformer of Problem 3.53. A 7.5-MVA, 13.2-kV synchronous motor with a subtransient reactance of 0.2 per unit is connected to the transformer secondary. A 5-MW, 2.3-kV three-phase resistive load is connected to the tertiary. Choosing a base of 66 kV and 15 MVA in the primary, draw the impedance diagram of the system showing per-unit impedances. Neglect transformer exciting current, phase shifts, and all resistances except the resistive load.

SECTION 3.7

3.55 A single-phase 10-kVA, 2300/230-volt, 60-Hz two-winding distribution transformer is connected as an autotransformer to step up the voltage from 2300 to 2530 volts. (a) Draw a schematic diagram of this arrangement, showing all voltages and currents when delivering full load at rated voltage. (b) Find the permissible kVA rating of the autotransformer if the winding currents and voltages are not to exceed the rated values as a two-winding transformer. How much of this kVA rating is transformed by magnetic induction? (c) The following data are obtained from tests carried out on the transformer when it is connected as a two-winding transformer:

Open-circuit test with the low-voltage terminals excited:
Applied voltage = 230 V, input current = 0.45 A, input power = 70 W.

Short-circuit test with the high-voltage terminals excited:
Applied voltage = 120 V, input current = 4.5 A, input power = 240 W.

Based on the data, compute the efficiency of the autotransformer corresponding to full load, rated voltage, and 0.8 power factor lagging. Comment on why the efficiency is higher as an autotransformer than as a two-winding transformer.

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3.56 Three single-phase two-winding transformers, each rated 3 kVA, 220/110 volts, 60 Hz, with a 0.10 per-unit leakage reactance, are connected as a three-phase extended Δ autotransformer bank, as shown in Figure 3.36(c). The low-voltage Δ winding has a 110-volt rating. (a) Draw the positive-sequence phasor diagram and show that the high-voltage winding has a 479.5-volt rating. (b) A three-phase load connected to the low-voltage terminals absorbs 6 kW at 110 volts and at 0.8 power factor lagging. Draw the per-unit impedance diagram and calculate the voltage and current at the high-voltage terminals. Assume positive-sequence operation.

3.57 A two-winding single-phase transformer rated 60 kVA, 240/1200 V, 60 Hz, has an efficiency of 0.96 when operated at rated load, 0.8 power factor lagging. This transformer is to be utilized as a 1440/1200-V step-down autotransformer in a power distribution system. (a) Find the permissible kVA rating of the autotransformer if the winding currents and voltages are not to exceed the ratings as a two-winding transformer. Assume an ideal transformer. (b) Determine the efficiency of the autotransformer with the kVA loading of part (a) and 0.8 power factor leading.

3.58 A single-phase two-winding transformer rated 90 MVA, 80/120 kV is to be connected as an autotransformer rated 80/200 kV. Assume that the transformer is ideal. (a) Draw a schematic diagram of the ideal transformer connected as an autotransformer, showing the voltages, currents, and dot notation for polarity. (b) Determine the permissible kVA rating of the autotransformer if the winding currents and voltages are not to exceed the rated values as a two-winding transformer. How much of the kVA rating is transferred by magnetic induction?

SECTION 3.8

3.59 The two parallel lines in Example 3.13 supply a balanced load with a load current of $1.0\angle -30^\circ$ per unit. Determine the real and reactive power supplied to the load bus from each parallel line with (a) no regulating transformer, (b) the voltage-magnitude-regulating transformer in Example 3.13(a), and (c) the phase-angle-regulating transformer in Example 3.13(b). Assume that the voltage at bus *abc* is adjusted so that the voltage at bus $a'b'c'$ remains constant at $1.0\angle 0^\circ$ per unit. Also assume positive sequence. Comment on the effects of the regulating transformers.

PW **3.60** PowerWorld Simulator case Problem 3_60 duplicates Example 3.13 except that a resistance term of 0.01 per unit has been added to the transformer and 0.03 per unit to the transmission line. Since the system is no longer lossless, a field showing the real power losses has also been added to the oneline. With the LTC tap fixed at 1.05, plot the real power losses as the phase shift angle is varied from −10 to +10 degrees. What value of phase shift minimizes the system losses?

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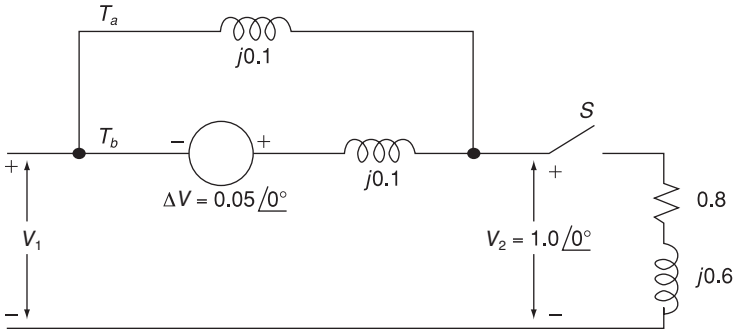
PW **3.61** Repeat Problem 3.60, except keep the phase-shift angle fixed at 3.0 degrees while varying the LTC tap between 0.9 and 1.1. What tap value minimizes the real power losses?

PW **3.62** PowerWorld Simulator cases Problem 3_62a and Problem 3_62b duplicate Example 3.13, except 1. both cases have a fixed off-nominal turns ratio magnitude of unity, 2. both cases include the resistance terms from Problem 3_60, and 3. Problem 3_62b models variation in the transformer's series impedance as a function of its phase shift using a piece-wise linear impedance correction table in which the impedance scalar changes from 2.0 at –30 degrees to 1.0 at zero degrees to 2.0 at 30 degrees (in PowerWorld the impedance correction table values can be seen by selecting **Case Information, Network, Transformer Impedance Correction Tables**). Compare the variation in the system losses between cases Problem 3_62a and Problem 3_62b by varying the phase shift over its range in 3.0-degree increments.

3.63 A 23/230-kV step-up transformer feeds a three-phase transmission line, which in turn supplies a 150-MVA, 0.8 lagging power factor load through a step-down 230/23-kV transformer. The impedance of the line and transformers at 230 kV is $18 + j60\ \Omega$. Determine the tap setting for each transformer to maintain the voltage at the load at 23 kV.

3.64 The per-unit equivalent circuit of two transformers T_a and T_b connected in parallel, with the same nominal voltage ratio and the same reactance of 0.1 per unit on the same base, is shown in Figure 3.43. Transformer T_b has a voltage-magnitude step-up toward the load of 1.05 times that of T_a (that is, the tap on the secondary winding of T_b is set to 1.05). The load is represented by $0.8 + j0.6$ per unit at a voltage $V_2 = 1.00\ 0°$ per unit. Determine the complex power in per unit transmitted to the load through each transformer. Comment on how the transformers share the real and reactive powers.

FIGURE 3.43
Problem 3.64



3.65 Reconsider Problem 3.64 with the change that now T_b includes both a transformer of the same turns ratio as T_a and a regulating transformer with a 4° phase shift. On the base of T_a, the impedance of the two components of T_b is $j0.1$ per unit. Determine the complex power in per unit transmitted to the load through each transformer. Comment on how the transformers share the real and reactive powers.

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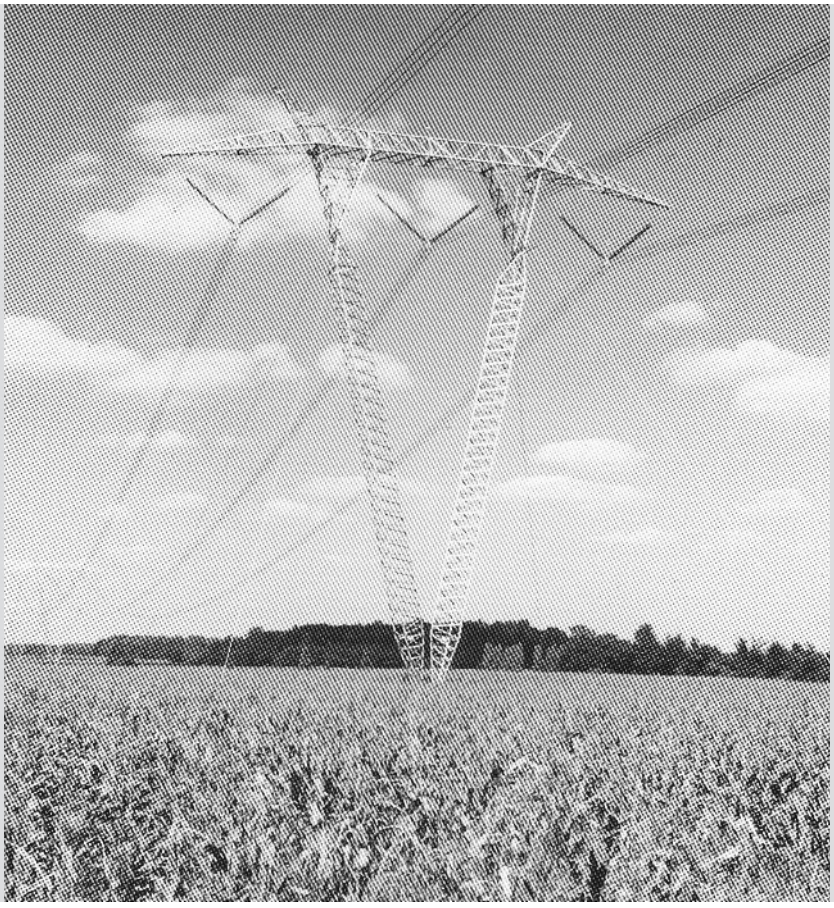
4 Transmission Line Parameters



765-kV transmission line with aluminum guyed-V towers. (Courtesy of American Electric Power Company.)

LEARNING OBJECTIVES

At the conclusion of this chapter, you should be able to:

1. Know the basic considerations of transmission line design.
2. Discuss four transmission line parameters: resistance, conductance, inductance and capacitance.
3. Explain the effects of composite conductors, unequal phase spacing, and bundled conductors on the inductive reactance and shunt admittance of overhead three-phase lines.
4. Discuss the design considerations associated with the electric fields at conductor surfaces and at ground level for an overhead transmission line.

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This chapter discusses the four basic transmission-line parameters: series resistance, series inductance, shunt capacitance, and shunt conductance and investigates transmission-line electric and magnetic fields.

Series resistance accounts for ohmic (I^2R) line losses. Series impedance, including resistance and inductive reactance, gives rise to series-voltage drops along the line. Shunt capacitance gives rise to line-charging currents. Shunt conductance accounts for V^2G line losses due to leakage currents between conductors or between conductors and ground. Shunt conductance of overhead lines is usually neglected.

Although the ideas developed in this chapter can be applied to underground transmission and distribution, the primary focus here is on overhead lines. Underground transmission in the United States presently accounts for less than 1% of total transmission and is found mostly in large cities or under waterways. There is, however, a large application for underground cable in distribution systems.

CASE STUDY 1

Renewables, Resiliency Drive Transmission Upgrades*

Kevin Ludwig, Black & Veatch

The following paragraphs summarize the above article, which describes recent trends that are driving transmission upgrades in the United States [10]. Sign in at www.cengage.com to download this product and to access its online resources.

Renewable generation is the key to enabling a zero-carbon future, but it will require a deep investment in transmission lines to transport this cleaner energy to the end user. Grid-scale renewable generation sites, for example, large wind farms and solar fields, tend to be situated where conditions are right and land is cheap and plentiful, in locations that are remote from urban and suburban hubs. As a result, renewable energy is closely tied to the success of transmission investment, and the demand is only increasing.

A 2020 Black & Veatch survey reflects the above finding. The survey polled more than 600 stakeholders across the U.S. electric industry to discover that 39% of respondents see renewables as the number one reason driving utilities to invest in beefing up their transmission capabilities over the next five years, coming in ahead of resiliency and reliability (28%), congestion (12%), interregional coordination/integration (11%), and fossil retirements (10%) (see Figure 1).

*"Renewables, Resiliency Drive Transmission Upgrades," by Kevin Ludwig, available online, with permission, from *T&D World Magazine* (January 04, 2021).

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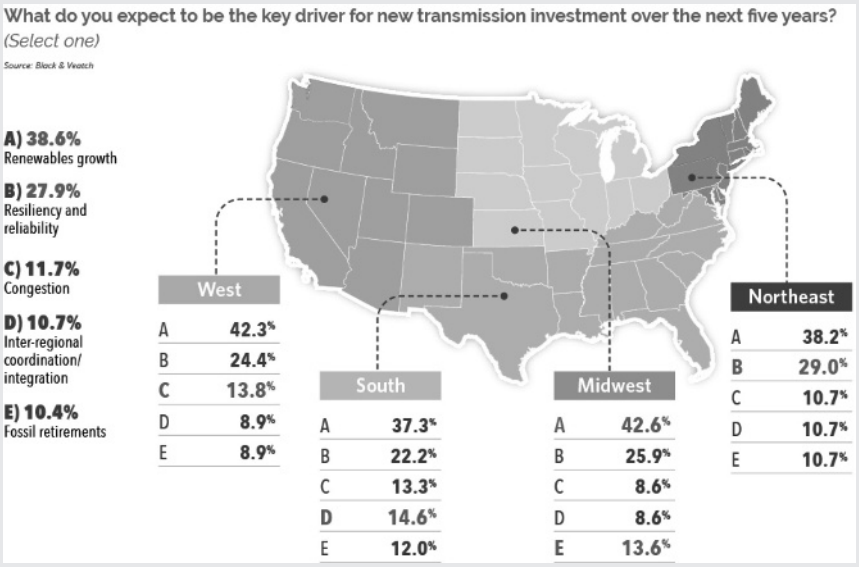

Figure 1

Utilities are actively responding to the expansion of renewable energy by changing how they allocate their investment dollars. The survey found that of the 38% of respondents who said they are increasing investment in existing assets over the previous year, almost half plan to upgrade their transmission and substations.

CASE STUDY 2

Greenlink Nevada to Drive Job Creation, Economic Recovery from Covid-19[†]

T&D World

The following paragraphs summarize the above article, which describes NV Energy's transmission infrastructure initiative to create well-paying

[†]"Greenlink Nevada to Drive Job Creation, Economic Recovery from Covid-19," available online, with permission, from *T&D World Magazine* (February 5, 2021).

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jobs for thousands of Nevadans [11]. Sign in at www.cengage.com to download this product and to access its online resources.

NV Energy proposed Greenlink Nevada to transform Nevada's clean energy landscape by tapping into resource-rich renewable energy zones throughout western and northern Nevada. It is expected to increase electric reliability for all Nevadans, and drive the creation of jobs and economic recovery from the COVID-19 pandemic. The project was approved by the Public Utilities Commission of Nevada in March 2021.

The Greenlink Nevada project includes the following (see Figure 2) [12]:

- Greenlink North, the first major segment, will be a 525-kV line that spans approximately 235 miles from Robinson Substation in Ely to Yerington, Nevada.
- Greenlink West, the second major segment, will be a 351-mile, 525-kV line from Las Vegas to Yerington, Nevada.
- Greenlink Nevada will also include three 345-kV lines from Yerington, Nevada to the Reno and Innovation Park areas to support new economic development.

NV Energy's investment of more than US$2.5 billion in Greenlink Nevada is expected to generate US$690 million in economic activity

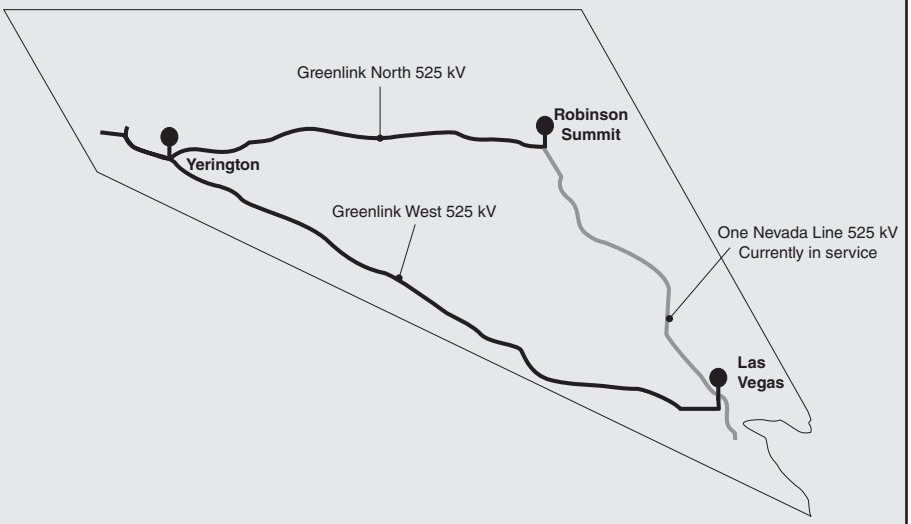

Figure 2

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and support nearly 4000 jobs. The initiative will also allow for the development of renewable energy projects throughout Nevada that will contribute significant economic growth in the form of nearly 2000 additional construction jobs and related tax revenue.

NV Energy has also announced three new solar plus storage projects. The projects, which will be built in southern Nevada, total 478 megawatts of new solar photovoltaic generation and 338 megawatts of battery storage, and have the capacity to power more than 107,000 homes [12].

4.1 TRANSMISSION LINE DESIGN CONSIDERATIONS

An overhead transmission line consists of conductors, insulators, support structures, and, in most cases, shield wires.

CONDUCTORS

Aluminum has replaced copper as the most common conductor metal for overhead transmission. Although a larger aluminum cross-sectional area is required to obtain the same loss as in a copper conductor, aluminum has a lower cost and lighter weight. Also, the supply of aluminum is abundant, whereas that of copper is limited.

One of the most common conductor types is aluminum conductor, steel-reinforced (ACSR), which consists of layers of aluminum strands surrounding a central core of steel strands (Figure 4.1). Stranded conductors are easier to manufacture, since larger conductor sizes can be obtained by simply adding successive layers of strands. Stranded conductors are also easier to handle and more flexible than solid conductors, especially in larger sizes. The use of steel strands gives ACSR conductors a high strength-to-weight ratio. For purposes of heat dissipation, overhead transmission-line conductors are bare (no insulating cover).

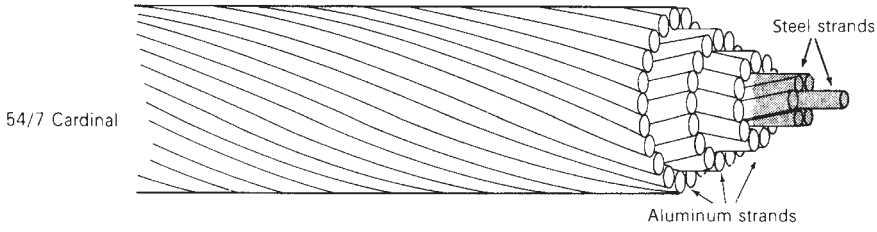

FIGURE 4.1
Typical ACSR conductor

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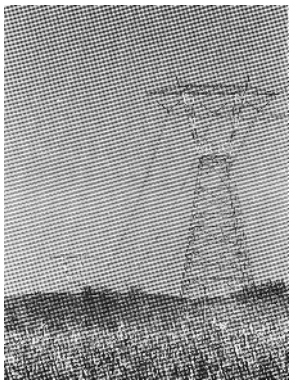

FIGURE 4.2

A 765-kV transmission line with self-supporting lattice steel towers (Courtesy of the American Electric Power Company)

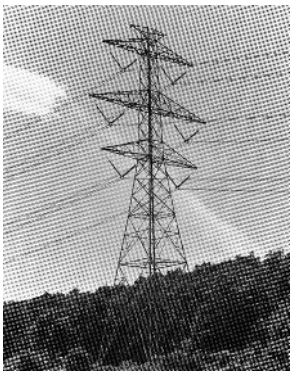

FIGURE 4.3

A 345-kV double-circuit transmission line with self-supporting lattice steel towers (Courtesy of Eversource, formerly NSTAR and Boston Edison Company)

Other conductor types include the all-aluminum conductor (AAC), all-aluminum-alloy conductor (AAAC), aluminum conductor alloy-reinforced (ACAR), and aluminum-clad steel conductor (Alumoweld). Higher-temperature conductors capable of operation in excess of 150°C include the aluminum conductor steel supported (ACSS) that uses fully annealed aluminum around a steel core, and the gap-type ZT-aluminum conductor (GZTACSR) that uses heat-resistant aluminum over a steel core with a small annular gap between the steel and first layer of aluminum strands. Advantages of higher-temperature conductors include higher capacity (ampacity) and reduced thermal sag. Newer technologies use composite materials, including the aluminum conductor fiber reinforced (ACFR), whose core is a resin matrix composite containing carbon fiber, and the aluminum conductor composite reinforced (ACCR), whose core is an aluminum matrix containing aluminum fibers [14–17].

EHV lines often have more than one conductor per phase; these conductors are called a *bundle.* The 765-kV line in Figure 4.2 has four conductors per phase, and the 345-kV double-circuit line in Figure 4.3 has two conductors per phase. Bundle conductors have a lower electric field strength at the conductor surfaces, thereby controlling corona. They also have a smaller series reactance.

INSULATORS

Insulators for transmission lines above 69 kV are typically suspension-type insulators that consist of a string of discs constructed of porcelain, toughened glass, or polymer. The standard disc (Figure 4.4) has a 0.254 m diameter, 0.146 m spacing between centers of adjacent discs, and a mechanical strength of 7500 kg. The 765-kV line in Figure 4.2 has two strings per phase in a V-shaped

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FIGURE 4.4

Cut-away view of a standard porcelain insulator disc for suspension insulator strings (Courtesy of Ohio Brass)

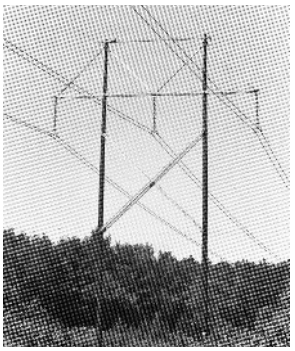

FIGURE 4.5

Wood frame structure for a 345-kV line (Courtesy of Eversource, formerly NSTAR and Boston Edison Company)

arrangement, which helps to restrain conductor swings. The 345-kV line in Figure 4.5 has one vertical string per phase. The number of insulator discs in a string increases with line voltage (Table 4.1). Other types of discs include larger units with higher mechanical strength and fog insulators for use in contaminated areas.

SUPPORT STRUCTURES

Transmission lines employ a variety of support structures. Figure 4.2 shows a self-supporting, lattice steel tower typically used for 500- and 765-kV lines. Double-circuit 345-kV lines usually have self-supporting steel towers with the phases arranged either in a triangular configuration to reduce tower height or in a vertical configuration to reduce tower width (Figure 4.3). Wood frame configurations are commonly used for voltages of 345 kV and below (Figure 4.5).

SHIELD WIRES

Shield wires located above the phase conductors protect the phase conductors against lightning. They are usually high- or extra-high-strength steel, Alumoweld, or ACSR with much smaller cross section than the phase conductors. The number and location of the shield wires are selected so that almost all lightning strikes terminate on the shield wires rather than on the phase conductors. Figures 4.2, 4.3, and 4.5 have two shield wires. Shield wires are grounded to the tower. As such, when lightning strikes a shield wire, it flows harmlessly to ground, provided the tower impedance and tower footing resistance are small.

The decision to build new transmission is based on power-system planning studies to meet future system requirements of load growth and new generation. The points of interconnection of each new line to the system, as well as the power and voltage ratings of each, are selected based on these studies. Thereafter, transmission-line design is based on optimization of electrical, mechanical, environmental, and economic factors.

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Nominal Voltage	Phase Conductors				
				Minimum Clearances	
(kV)	Number of Conductors per Bundle	Aluminum Cross-Section Area per Conductor (ACSR) (kcmil)	Bundle Spacing (cm)	Phase-to-Phase (m)	Phase-to-Ground (m)
69	1	—	—	—	—
138	1	300—700	—	4 to 5	—
230	1	400—1000	—	6 to 9	—
345	1	2000—2500	—	6 to 9	7.6 to 11
345	2	800—2200	45.7	6 to 9	7.6 to 11
500	2	2000—2500	45.7	9 to 11	9 to 14
500	3	900—1500	45.7	9 to 11	9 to 14
765	4	900—1300	45.7	13.7	12.2

Nominal Voltage	Suspension Insulator String		Shield Wires		
(kV)	Number of Strings per Phase	Number of Standard Insulator Discs per Suspension String	Type	Number	Diameter (cm)
69	1	4 to 6	Steel	0, 1 or 2	—
138	1	8 to 11	Steel	0, 1 or 2	—
230	1	12 to 21	Steel or ACSR	1 or 2	1.1 to 1.5
345	1	18 to 21	Alumoweld	2	0.87 to 1.5
345	1 and 2	18 to 21	Alumoweld	2	0.87 to 1.5
500	2 and 4	24 to 27	Alumoweld	2	0.98 to 1.5
500	2 and 4	24 to 27	Alumoweld	2	0.98 to 1.5
765	2 and 4	30 to 35	Alumoweld	2	0.98

TABLE 4.1

Typical transmission-line characteristics [1, 2] (Electric Power Research Institute (EPRI), EPRI AC Transmission Line Reference Book—200 kV and Above (Palo Alto, CA: EPRI, www.epri.com, December 2005); Westinghouse Electric Corporation, *Electrical Transmission and Distribution Reference Book*, 4th ed. (East Pittsburgh, PA, 1964).

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ELECTRICAL FACTORS

Electrical design dictates the type, size, and number of bundle conductors per phase. Phase conductors are selected to have sufficient thermal capacity to meet continuous, emergency overload, and short-circuit current ratings. For EHV lines, the number of bundle conductors per phase is selected to control the voltage gradient at conductor surfaces, thereby reducing or eliminating corona.

Electrical design also dictates the number of insulator discs, vertical or V-shaped string arrangement, phase-to-phase clearance, and phase-to-tower clearance, all selected to provide adequate line insulation. Line insulation must withstand transient overvoltages due to lightning and switching surges, even when insulators are contaminated by fog, salt, or industrial pollution. Reduced clearances due to conductor swings during wind must also be accounted for.

The number, type, and location of shield wires are selected to intercept lightning strikes that would otherwise hit the phase conductors. Also, tower footing resistance can be reduced by using driven ground rods or a buried conductor (called *counterpoise*) running parallel to the line. Line height is selected to satisfy prescribed conductor-to-ground clearances and to control ground-level electric field and its potential shock hazard.

Conductor spacings, types, and sizes also determine the series impedance and shunt admittance. Series impedance affects line-voltage drops, I^2R losses, and stability limits (Chapters 5, 14). Shunt admittance, primarily capacitive, affects line-charging currents, which inject reactive power into the power system. Shunt reactors (inductors) are often installed on lightly loaded EHV lines to absorb part of this reactive power, thereby reducing overvoltages.

MECHANICAL FACTORS

Mechanical design focuses on the strength of the conductors, insulator strings, and support structures. Conductors must be strong enough to support a specified thickness of ice and a specified wind in addition to their own weight. Suspension insulator strings must be strong enough to support the phase conductors with ice and wind loadings from tower to tower (span length). Towers that satisfy minimum strength requirements, called suspension towers, are designed to support the phase conductors and shield wires with ice and wind loadings, and, in some cases, the unbalanced pull due to breakage of one or two conductors. Dead-end towers, located every mile or so, satisfy the maximum strength requirement of breakage of all conductors on one side of the tower. Angles in the line employ angle towers with intermediate strength. Conductor vibrations, which can cause conductor fatigue failure and damage to towers, are also of concern. Vibrations are controlled by adjustment of conductor tensions, use of vibration dampers, and—for bundle conductors—large bundle spacing and frequent use of bundle spacers.

ENVIRONMENTAL FACTORS

Environmental factors include land usage and visual impact. When a line route is selected, the effect on local communities and population centers, land values,

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access to property, wildlife, and use of public parks and facilities must all be considered. Reduction in visual impact is obtained by aesthetic tower design and by blending the line with the countryside. Also, the biological effects of prolonged exposure to electric and magnetic fields near transmission lines is of concern. Extensive research has been and continues to be done in this area.

ECONOMIC FACTORS

The optimum line design meets all the technical design criteria at lowest overall cost, which includes the total installed cost of the line as well as the cost of line losses over the operating life of the line. Many design factors affect cost. Utilities and consulting organizations use digital computer programs combined with specialized knowledge and physical experience to achieve optimum line design.

4.2 RESISTANCE

The dc resistance of a conductor at a specified temperature T is

$$R_{dc,T} = \frac{\rho_T l}{A} \quad \Omega \tag{4.2.1}$$

where ρ_T = conductor resistivity at temperature T
l = conductor length
A = conductor cross-sectional area

Two sets of units commonly used for calculating resistance, SI and English units, are summarized in Table 4.2. In English units, conductor cross-sectional area is expressed in circular mils (cmil). One inch equals 1000 mils, and 1 cmil equals $\pi/4$ sq mil. A circle with diameter D in., or (D in.) (1000 mil/in.) = 1000 D mil = d mil, has an area

$$A = \left(\frac{\pi}{4} D^2 \text{ in.}^2\right)\left(1000 \frac{\text{mil}}{\text{in.}}\right)^2 = \frac{\pi}{4}(1000\ D)^2 = \frac{\pi}{4} d^2 \quad \text{sq mill}$$

or

$$A = \left(\frac{\pi}{4} d^2 \text{ sq mil}\right)\left(\frac{1 \text{ cmil}}{\pi/4 \text{ sq mil}}\right) = d^2 \quad \text{cmil} \tag{4.2.2}$$

Resistivity depends on the conductor metal. Annealed copper is the international standard for measuring resistivity ρ (or conductivity σ, where $\sigma = 1/\rho$). Resistivity of conductor metals is listed in Table 4.3. As shown, hard-drawn aluminum, which has 61% of the conductivity of the international standard, has a resistivity at 20°C of $2.83 \times 10^{-8}\ \Omega$m.

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Quantity	Symbol	SI Units	English Units
Resistivity	ρ	Ωm	Ω-cmil/ft
Length	ℓ	m	ft
Cross-sectional area	A	m^2	cmil
dc resistance	$R_{dc} = \frac{\rho\ell}{A}$	Ω	Ω

TABLE 4.2

Comparison of SI and English units for calculating conductor resistance

		$\rho_{20°C}$		T
		Resistivity at 20°C		Temperature Constant
Material	% Conductivity	$\Omega m \times 10^{-8}$	Ω-cmil/ft	°C
Copper:				
Annealed	100%	1.72	10.37	234.5
Hard-drawn	97.3%	1.77	10.66	241.5
Aluminum				
Hard-drawn	61%	2.83	17.00	228.1
Brass	20–27%	6.4–8.4	38–51	480
Iron	17.2%	10	60	180
Silver	108%	1.59	9.6	243
Sodium	40%	4.3	26	207
Steel	2–14%	12–88	72–530	180–980

TABLE 4.3

% Conductivity, resistivity, and temperature constant of conductor metals

Conductor resistance depends on the following factors:

1. Spiraling
2. Temperature
3. Frequency ("skin effect")
4. Current magnitude—magnetic conductors

These are described in the following paragraphs.

For stranded conductors, alternate layers of strands are spiraled in opposite directions to hold the strands together. Spiraling makes the strands 1 or 2% longer than the actual conductor length. As a result, the dc resistance of a stranded conductor is 1 or 2% larger than that calculated from (4.2.1) for a specified conductor length.

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Resistivity of conductor metals varies linearly over normal operating temperatures according to

$$\rho_{T2} = \rho_{T1}\left(\frac{T_2 + T}{T_1 + T}\right) \tag{4.2.3}$$

where ρ_{T2} and ρ_{T1} are resistivities at temperatures T_2 and T_1°C, respectively. T is a temperature constant that depends on the conductor material and is listed in Table 4.3.

The ac resistance or *effective* resistance of a conductor is

$$R_{ac} = \frac{P_{loss}}{|I|^2} \quad \Omega \tag{4.2.4}$$

where P_{loss} is the conductor real power loss in watts and I is the rms conductor current. For dc, the current distribution is uniform throughout the conductor cross section, and (4.2.1) is valid. However, for ac, the current distribution is nonuniform. As frequency increases, the current in a solid cylindrical conductor tends to crowd toward the conductor surface with smaller current density at the conductor center. This phenomenon is called *skin effect.* A conductor with a large radius can even have an oscillatory current density versus the radial distance from the conductor center.

With increasing frequency, conductor loss increases, which, from (4.2.4), causes the ac resistance to increase. At power frequencies (60 Hz), the ac resistance is at most a few percent higher than the dc resistance. Conductor manufacturers normally provide dc, 50-Hz, and 60-Hz conductor resistance based on test data (see Appendix Tables A.3 and A.4).

For magnetic conductors, such as steel conductors used for shield wires, resistance depends on current magnitude. The internal flux linkages, and therefore the iron or magnetic losses, depend on the current magnitude. For ACSR conductors, the steel core has a relatively high resistivity compared to the aluminum strands, and therefore the effect of current magnitude on ACSR conductor resistance is small. Tables on magnetic conductors list resistance at two current levels (see Table A.4).

EXAMPLE 4.1

Stranded conductor: dc and ac resistance

Table A.3 lists a 4/0 copper conductor with 12 strands. Strand diameter is 0.1328 in. For this conductor:

a. Verify the total copper cross-sectional area of 107.2 mm^2.

b. Verify the dc resistance at 50°C of 0.188 Ω/km. Assume a 2% increase in resistance due to spiraling.

c. From Table A.3, determine the percent increase in resistance at 60 Hz versus dc.

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SOLUTION

a. The strand diameter is $d = (0.1328 \text{ in.})(25.4 \text{ mm/in.}) = 3.373$ mm. Using four significant figures, the cross-sectional area of the 12-strand conductor is

$$A = 12\pi d^2/4 = 12\pi(3.373)^2/4 = 107.2 \text{ mm}^2$$

which agrees with the value given in Table A.3.

b. Using (4.2.3) and hard-drawn copper data from Table 4.3,

$$\rho_{50°C} = 1.77 \times 10^{-8}\left(\frac{50 + 241.5}{20 + 241.5}\right) = 1.973 \times 10^{-8} \quad \Omega\text{m}$$

From (4.2.1), the dc resistance at 50°C for a conductor length of 1 kilometer is

$$R_{dc,\,50°C} = \frac{(1.973 \times 10^{-8})(1000 \times 1.02)}{1.072 \times 10^{-4}} = 0.188 \quad \Omega/\text{km}$$

which agrees with the value listed in Table A.3.

c. From Table A.3,

$$\frac{R_{60\text{ Hz},\,50°C}}{R_{dc,\,50°C}} = \frac{0.303}{0.302} = 1.003 \qquad \frac{R_{60\text{ Hz},\,25°C}}{R_{dc,\,25°C}} = \frac{0.278}{0.276} = 1.007$$

Thus, the 60-Hz resistance of this conductor is about 0.3 to 0.7% higher than the dc resistance. The variation of these two ratios is due to the fact that resistance in Table A.3 is given to only three significant figures.

4.3 CONDUCTANCE

Conductance accounts for real power loss between conductors or between conductors and ground. For overhead lines, this power loss is due to leakage currents at insulators and to corona. Insulator leakage current depends on the amount of dirt, salt, and other contaminants that have accumulated on insulators, as well as on meteorological factors, particularly the presence of moisture. Corona occurs when a high value of electric field strength at a conductor surface causes the air to become electrically ionized and to conduct. The real power loss due to corona, called *corona loss*, depends on meteorological conditions, particularly rain, and on conductor surface irregularities. Losses due to insulator leakage and corona are usually small compared to conductor I^2R loss. Conductance is usually neglected in power system studies because it is a very small component of the shunt admittance.

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4.4 INDUCTANCE: SOLID CYLINDRICAL CONDUCTOR

The inductance of a magnetic circuit that has a constant permeability μ can be obtained by determining the following:

1. Magnetic field intensity H, from Ampere's law
2. Magnetic flux density B ($B = \mu H$)
3. Flux linkages λ
4. Inductance from flux linkages per ampere ($L = \lambda/I$)

As a step toward computing the inductances of more general conductors and conductor configurations, first compute the internal, external, and total inductance of a solid cylindrical conductor. Also compute the flux linking one conductor in an array of current-carrying conductors.

Figure 4.6 shows a 1-meter section of a solid cylindrical conductor with radius r, carrying current I. For simplicity, assume that the conductor (1) is sufficiently long that end effects are neglected, (2) is nonmagnetic ($\mu = \mu_0 = 4\pi \times 10^{-7}$ H/m), and (3) has a uniform current density (skin effect is neglected). From (3.1.1), Ampere's law states that

$$\oint H_{\tan}\, dl = I_{\text{enclosed}} \tag{4.4.1}$$

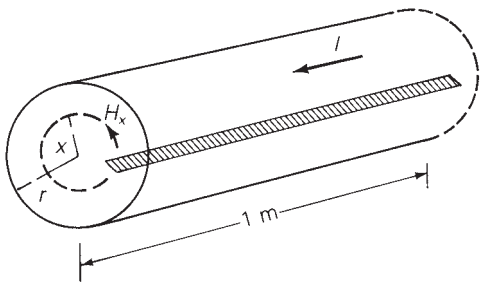
To determine the magnetic field inside the conductor, select the dashed circle of radius $x < r$ shown in Figure 4.6 as the closed contour for Ampere's law. Due to symmetry, H_x is constant along the contour. Also, there is no radial component of H_x, so H_x is tangent to the contour. That is, the conductor has a concentric magnetic field. From (4.4.1), the integral of H_x around the selected contour is

$$H_x\,(2\pi x) = I_x \qquad \text{for } x < r \tag{4.4.2}$$

where I_x is the portion of the total current enclosed by the contour. Solving (4.4.2),

$$H_x = \frac{I_x}{2\pi x} \quad \text{A/m} \tag{4.4.3}$$

FIGURE 4.6
Internal magnetic field of a solid cylindrical conductor



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Now assume a uniform current distribution within the conductor, that is,

$$I_x = \left(\frac{x}{r}\right)^2 I \quad \text{for } x < r \tag{4.4.4}$$

Using (4.4.4) in (4.4.3),

$$H_x = \frac{xI}{2\pi r^2} \quad \text{A/m} \tag{4.4.5}$$

For a nonmagnetic conductor, the magnetic flux density B_x is

$$B_x = \mu_0 H_x = \frac{\mu_0 xI}{2\pi r^2} \quad \text{Wb/m}^2 \tag{4.4.6}$$

The differential flux $d\Phi$ per-unit length of conductor in the cross-hatched rectangle of width dx shown in Figure 4.6 is

$$d\Phi = B_x\, dx \quad \text{Wb/m} \tag{4.4.7}$$

Computation of the differential flux linkage $d\lambda$ in the rectangle is tricky since only the fraction $(x/r)^2$ of the total current I is linked by the flux. That is,

$$d\lambda = \left(\frac{x}{r}\right)^2 d\Phi = \frac{\mu_0 I}{2\pi r^4}\, x^3\, dx \quad \text{Wb-t/m} \tag{4.4.8}$$

Integrating (4.4.8) from $x = 0$ to $x = r$ determines the total flux linkages λ_{int} inside the conductor

$$\lambda_{\text{int}} = \int_0^r d\lambda = \frac{\mu_0 I}{2\pi r^4} \int_0^r x^3\, dx = \frac{\mu_0 I}{8\pi} = \frac{1}{2} \times 10^{-7}\, I \quad \text{Wb-t/m} \tag{4.4.9}$$

The internal inductance L_{int} per-unit length of conductor due to this flux linkage is then

$$L_{\text{int}} = \frac{\lambda_{\text{int}}}{I} = \frac{\mu_0}{8\pi} = \frac{1}{2} \times 10^{-7} \quad \text{H/m} \tag{4.4.10}$$

Next, in order to determine the magnetic field outside the conductor, select the dashed circle of radius $x > r$ shown in Figure 4.7 as the closed contour for Ampere's law. Noting that this contour encloses the entire current I, integration of (4.4.1) yields

$$H_x(2\pi x) = I \tag{4.4.11}$$

which gives

$$H_x = \frac{I}{2\pi x} \quad \text{A/m} \quad x > r \tag{4.4.12}$$

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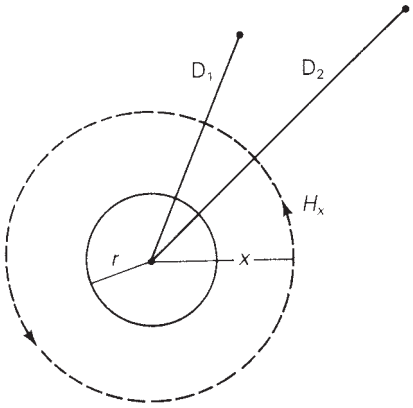
FIGURE 4.7

External magnetic field of a solid cylindrical conductor



Outside the conductor, $\mu = \mu_0$ and

$$B_x = \mu_0 H_x = (4\pi \times 10^{-7}) \frac{I}{2\pi x} = 2 \times 10^{-7} \frac{I}{x} \quad \text{Wb/m}^2 \tag{4.4.13}$$

$$d\Phi = B_x \, dx = 2 \times 10^{-7} \frac{I}{x} dx \quad \text{Wb/m} \tag{4.4.14}$$

Since the entire current I is linked by the flux outside the conductor,

$$d\lambda = d\Phi = 2 \times 10^{-7} \frac{I}{x} dx \quad \text{Wb-t/m} \tag{4.4.15}$$

Integrating (4.4.15) between two external points at distances D_1 and D_2 from the conductor center gives the external flux linkage λ_{12} between D_1 and D_2:

$$\begin{aligned} \lambda_{12} &= \int_{D_1}^{D_2} d\lambda = 2 \times 10^{-7} I \int_{D_1}^{D_2} \frac{dx}{x} \\ &= 2 \times 10^{-7} I \ln\left(\frac{D_2}{D_1}\right) \quad \text{Wb-t/m} \end{aligned} \tag{4.4.16}$$

The external inductance L_{12} per-unit length due to the flux linkages between D_1 and D_2 is then

$$L_{12} = \frac{\lambda_{12}}{I} = 2 \times 10^{-7} \ln\left(\frac{D_2}{D_1}\right) \quad \text{H/m} \tag{4.4.17}$$

The total flux λ_P linking the conductor out to external point P at distance D is the sum of the internal flux linkage, (4.4.9), and the external flux linkage, (4.4.16) from $D_1 = r$ to $D_2 = D$. That is,

$$\lambda_p = \frac{1}{2} \times 10^{-7} I + 2 \times 10^{-7} I \ln \frac{D}{r} \tag{4.4.18}$$

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Using the identity $\frac{1}{2} = 2 \ln e^{1/4}$ in (4.4.18), a more convenient expression for λ_P is obtained:

$$\begin{aligned}\lambda_p &= 2 \times 10^{-7} I \left(\ln e^{1/4} + \ln \frac{D}{r} \right) \\ &= 2 \times 10^{-7} I \ln \frac{D}{e^{-1/4} r} \\ &= 2 \times 10^{-7} I \ln \frac{D}{r'} \quad \text{Wb-t/m}\end{aligned} \tag{4.4.19}$$

Where

$$r' = e^{-1/4} r = 0.7788r \tag{4.4.20}$$

Also, the total inductance L_P due to both internal and external flux linkages out to distance D is

$$L_p = \frac{\lambda_p}{I} = 2 \times 10^{-7} \ln \left(\frac{D}{r'} \right) \quad \text{H/m} \tag{4.4.21}$$

Finally, consider the array of M solid cylindrical conductors shown in Figure 4.8. Assume that each conductor m carries current I_m referenced out of the page. Also, assume that the sum of the conductor currents is zero—that is,

$$I_1 + I_2 + \cdots + I_M = \sum_{m=1}^{M} I_m = 0 \tag{4.4.22}$$

The flux linkage λ_{kPk}, which links conductor k out to point P due to current I_k, is, from (4.4.19),

$$\lambda_{kPk} = 2 \times 10^{-7} I_k \ln \frac{D_{pk}}{r'_k} \tag{4.4.23}$$

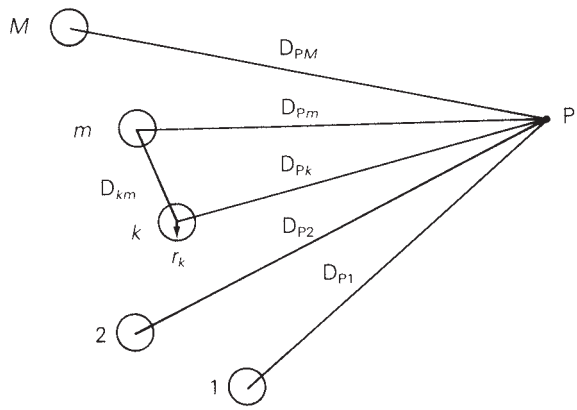

FIGURE 4.8
Array of M solid cylindrical conductors

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Note that $\lambda_{k\mathrm{P}k}$ includes both internal and external flux linkages due to I_k. The flux linkage $\lambda_{k\mathrm{P}m}$, which links conductor k out to P due to I_m, is, from (4.4.16),

$$\lambda_{k\mathrm{P}m} = 2 \times 10^{-7} I_m \ln \frac{\mathrm{D}_{\mathrm{P}m}}{\mathrm{D}_{km}} \tag{4.4.24}$$

Equation (4.4.24) uses D_{km} instead of $(\mathrm{D}_{km} - rk)$ or $(\mathrm{D}_{km} + rk)$, which is a valid approximation when D_{km} is much greater than r_k. It can also be shown that this is a good approximation even when D_{km} is small. Using superposition, the total flux linkage $\lambda_{k\mathrm{p}}$, which links conductor k out to P due to all the currents, is

$$\begin{aligned}\lambda_{k\mathrm{P}} &= \lambda_{k\mathrm{P}1} + \lambda_{k\mathrm{P}2} + \cdots + \lambda_{k\mathrm{P}M} \\ &= 2 \times 10^{-7} \sum_{m=1}^{M} I_m \ln \frac{\mathrm{D}_{\mathrm{P}m}}{\mathrm{D}_{km}}\end{aligned} \tag{4.4.25}$$

where $\mathrm{D}_{kk} = r_k' = e^{-1/4} r_k$ is defined when $m = k$ in the above summation.

Equation (4.4.25) is separated into two summations:

$$\lambda_{k\mathrm{P}} = 2 \times 10^{-7} \sum_{m=1}^{M} I_m \ln \frac{1}{\mathrm{D}_{km}} + 2 \times 10^{-7} \sum_{m=1}^{M} I_m \ln \mathrm{D}_{\mathrm{P}m} \tag{4.4.26}$$

Removing the last term from the second summation results in

$$\lambda_{k\mathrm{P}} = 2 \times 10^{-7} \left[\sum_{m=1}^{M} I_m \ln \frac{1}{\mathrm{D}_{km}} + \sum_{m=1}^{M-1} I_m \ln \mathrm{D}_{\mathrm{P}m} + I_M \ln \mathrm{D}_{\mathrm{P}M} \right] \tag{4.4.27}$$

From (4.4.22),

$$I_M = -(I_1 + I_2 + \cdots + I_{M-1}) = -\sum_{m=1}^{M-1} I_m \tag{4.4.28}$$

Using (4.4.28) in (4.4.27),

$$\begin{aligned}\lambda_{k\mathrm{P}} &= 2 \times 10^{-7} \left[\sum_{m=1}^{M} I_m \ln \frac{1}{\mathrm{D}_{km}} + \sum_{m=1}^{M-1} I_m \ln \mathrm{D}_{\mathrm{P}m} - \sum_{m=1}^{M-1} I_m \ln \mathrm{D}_{\mathrm{P}M} \right] \\ &= 2 \times 10^{-7} \left[\sum_{m=1}^{M} I_m \ln \frac{1}{\mathrm{D}_{km}} + \sum_{m=1}^{M-1} I_m \ln \frac{\mathrm{D}_{\mathrm{P}m}}{\mathrm{D}_{\mathrm{P}M}} \right]\end{aligned} \tag{4.4.29}$$

Now, let λ_k equal the total flux linking conductor k out to infinity. That is, $\lambda_k = \lim_{\mathrm{p} \to \infty} \lambda_{k\mathrm{P}}$. As $\mathrm{P} \to \infty$, all the distances $\mathrm{D}_{\mathrm{P}m}$ become equal, the ratios $\mathrm{D}_{\mathrm{P}m}/\mathrm{D}_{\mathrm{P}M}$ become unity, and $\ln(\mathrm{D}_{\mathrm{P}m}/\mathrm{D}_{\mathrm{P}M}) \to 0$. Therefore, the second summation in (4.4.29) becomes zero as $\mathrm{P} \to \infty$, and

$$\lambda_k = 2 \times 10^{-7} \sum_{m=1}^{M} I_m \ln \frac{1}{\mathrm{D}_{km}} \quad \text{Wb-t/m} \tag{4.4.30}$$

Equation (4.4.30) gives the total flux linking conductor k in an array of M conductors carrying currents $I_1, I_2, \ldots, I_M$, whose sum is zero. This equation is valid for

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either dc or ac currents. λ_k is a dc flux linkage when the currents are dc, and λ_k is a phasor flux linkage when the currents are phasor representations of sinusoids.

4.5 INDUCTANCE: SINGLE-PHASE TWO-WIRE LINE AND THREE-PHASE THREE-WIRE LINE WITH EQUAL PHASE SPACING

The results of the previous section are used here to determine the inductances of two relatively simple transmission lines: a single-phase, two-wire line and a three-phase, three-wire line with equal phase spacing.

Figure 4.9(a) shows a single-phase, two-wire line consisting of two solid cylindrical conductors x and y. Conductor x with radius r_x carries phasor current $I_x = I$ referenced out of the page. Conductor y with radius r_y carries return current $I_y = -I$. Since the sum of the two currents is zero, (4.4.30) is valid, from which the total flux linking conductor x is

$$\begin{aligned}\lambda_x &= 2 \times 10^{-7}\left(I_x \ln \frac{1}{\mathrm{D}_{xx}} + I_y \ln \frac{1}{\mathrm{D}_{xy}}\right) \\ &= 2 \times 10^{-7}\left(I \ln \frac{1}{r_x'} - I \ln \frac{1}{\mathrm{D}}\right) \\ &= 2 \times 10^{-7} I \ln \frac{\mathrm{D}}{r_x'} \quad \text{Wb-t/m}\end{aligned} \tag{4.5.1}$$

Where $r_x' = e^{-1/4} r_x = 0.7788 r_x$.

The inductance of conductor x is then

$$\mathrm{L}_x = \frac{\lambda_x}{I_x} = \frac{\lambda_x}{I} = 2 \times 10^{-7} \ln \frac{\mathrm{D}}{r_x'} \ \text{H/m per conductor} \tag{4.5.2}$$

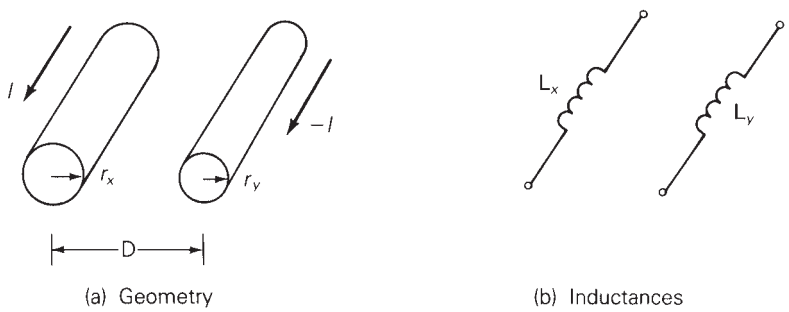

Figure 4.9
Single-phase, two-wire line

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Similarly, the total flux linking conductor y is

$$\begin{aligned}\lambda_y &= 2\times 10^{-7}\left(I_x \ln\frac{1}{\mathrm{D}_{yx}} + I_y \ln\frac{1}{\mathrm{D}_{yy}}\right)\\ &= 2\times 10^{-7}\left(I\ln\frac{1}{\mathrm{D}} - I\ln\frac{1}{r'_y}\right)\\ &= -2\times 10^{-7} I\ln\frac{\mathrm{D}}{r'_y}\end{aligned} \tag{4.5.3}$$

and

$$\mathrm{L}_y = \frac{\lambda_y}{I_y} = \frac{\lambda_y}{-I} = 2\times 10^{-7}\ln\frac{\mathrm{D}}{r'_y}\ \ \mathrm{H/m\ per\ conductor} \tag{4.5.4}$$

The total inductance of the single-phase circuit, also called *loop inductance,* is

$$\begin{aligned}\mathrm{L} = \mathrm{L}_x + \mathrm{L}_y &= 2\times 10^{-7}\left(\ln\frac{\mathrm{D}}{r'_x} + \ln\frac{\mathrm{D}}{r'_y}\right)\\ &= 2\times 10^{-7}\ln\frac{\mathrm{D}^2}{r'_x\, r'_y}\\ &= 4\times 10^{-7}\ln\frac{\mathrm{D}}{\sqrt{r'_x\, r'_y}}\ \ \mathrm{H/m\ per\ circuit}\end{aligned} \tag{4.5.5}$$

Also, if $r'_x = r'_y = r'$, the total circuit inductance is

$$\mathrm{L} = 4\times 10^{-7}\ln\frac{\mathrm{D}}{r'}\ \ \mathrm{H/m\ per\ circuit} \tag{4.5.6}$$

The inductances of the single-phase two-wire line are shown in Figure 4.9(b).

Figure 4.10(a) shows a three-phase three-wire line consisting of three solid cylindrical conductors a, b, c, each with radius r, and with equal phase spacing D between any two conductors. To determine inductance, assume balanced positive-sequence currents I_a, I_b, I_c that satisfy $I_a + I_b + I_c = 0$. Then (4.4.30) is valid, and the total flux linking the phase a conductor is

$$\begin{aligned}\lambda_a &= 2\times 10^{-7}\left(I_a\ln\frac{1}{r'} + I_b\ln\frac{1}{\mathrm{D}} + I_c\ln\frac{1}{\mathrm{D}}\right)\\ &= 2\times 10^{-7}\left[I_a\ln\frac{1}{r'} + (I_b + I_c)\ln\frac{1}{\mathrm{D}}\right]\end{aligned} \tag{4.5.7}$$

Using $(I_b + I_c) = -I_a$,

$$\begin{aligned}\lambda_a &= 2\times 10^{-7}\left(I_a\ln\frac{1}{r'} - I_a\ln\frac{1}{\mathrm{D}}\right)\\ &= 2\times 10^{-7} I_a\ln\frac{\mathrm{D}}{r'}\ \ \mathrm{Wb\text{-}t/m}\end{aligned} \tag{4.5.8}$$

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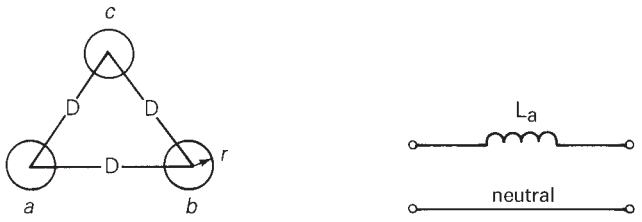

(a) Geometry (b) Phase inductance

Figure 4.10
Three-phase, three-wire line with equal phase spacing

The inductance of phase a is then

$$L_a = \frac{\lambda_a}{I_a} = 2 \times 10^{-7} \ln \frac{D}{r'} \quad \text{H/m per phase} \tag{4.5.9}$$

Due to symmetry, the same result is obtained for $L_b = \lambda_b/I_b$ and for $L_c = \lambda_c/I_c$. However, only one phase need be considered for balanced three-phase operation of this line, since the flux linkages of each phase have equal magnitudes and 120° displacement. The phase inductance is shown in Figure 4.10(b).

4.6 INDUCTANCE: COMPOSITE CONDUCTORS, UNEQUAL PHASE SPACING, BUNDLED CONDUCTORS

The results of Section 4.5 are extended here to include composite conductors, which consist of two or more solid cylindrical subconductors in parallel. A stranded conductor is one example of a composite conductor. For simplicity assume that for each conductor, the subconductors are identical and share the conductor current equally.

Figure 4.11 shows a single-phase two-conductor line consisting of two composite conductors x and y. Conductor x has N identical subconductors, each with radius r_x and with current (I/N) referenced out of the page. Similarly, conductor y consists of M identical subconductors, each with radius r_y and with return current $(-I/M)$. Since the sum of all the currents is zero, (4.4.30) is valid and the total flux Φ_k linking subconductor k of conductor x is

$$\Phi_k = 2 \times 10^{-7} \left[\frac{I}{N} \sum_{m=1}^{N} \ln \frac{1}{D_{km}} - \frac{1}{M} \sum_{m=1'}^{M} \ln \frac{1}{D_{km}} \right] \tag{4.6.1}$$

Since only the fraction $(1/N)$ of the total conductor current I is linked by this flux, the flux linkage λ_k of (the current in) subconductor k is

$$\lambda_k = \frac{\Phi_k}{N} = 2 \times 10^{-7} I \left[\frac{1}{N^2} \sum_{m=1}^{N} \ln \frac{1}{D_{km}} - \frac{1}{NM} \sum_{m=1'}^{M} \ln \frac{1}{D_{km}} \right] \tag{4.6.2}$$

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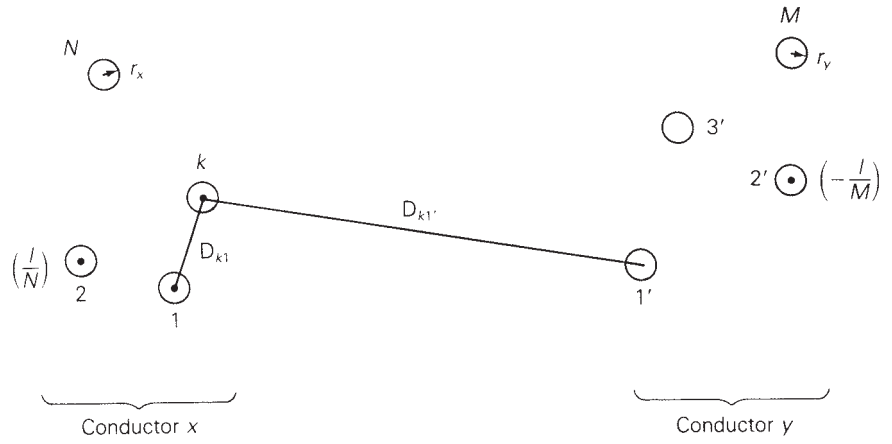
FIGURE 4.11

Single-phase, two-conductor line with composite conductors



The total flux linkage of conductor x is

$$\lambda_x = \sum_{k=1}^{N} \lambda_k = 2 \times 10^{-7} I \sum_{k=1}^{N} \left[\frac{1}{N^2} \sum_{m=1}^{N} \ln \frac{1}{\mathrm{D}_{km}} - \frac{1}{NM} \sum_{m=1'}^{M} \ln \frac{1}{\mathrm{D}_{km}} \right] \tag{4.6.3}$$

Using $\ln \mathrm{A}^{\alpha} = \alpha \ln \mathrm{A}$ and $\Sigma \ln \mathrm{A}_k = \ln \prod \mathrm{A}_k$ (sum of ln s = ln of products), (4.6.3) can be rewritten in the following form:

$$\lambda_x = 2 \times 10^{-7} I \ln \prod_{k=1}^{N} \frac{\left(\prod_{m=1'}^{M} \mathrm{D}_{km} \right)^{1/NM}}{\left(\prod_{m=1}^{N} \mathrm{D}_{km} \right)^{1/N^2}} \tag{4.6.4}$$

and the inductance of conductor x, $\mathrm{L}_x = \dfrac{\lambda_x}{I}$, can be written as

$$\mathrm{L}_x = 2 \times 10^{-7} \ln \frac{\mathrm{D}_{xy}}{\mathrm{D}_{xx}} \ \mathrm{H/m \ per \ conductor} \tag{4.6.5}$$

where

$$\mathrm{D}_{xy} = \sqrt[MN]{\prod_{k=1}^{N} \prod_{m=1'}^{M} \mathrm{D}_{km}} \tag{4.6.6}$$

$$\mathrm{D}_{xx} = \sqrt[N^2]{\prod_{k=1}^{N} \prod_{m=1}^{M} \mathrm{D}_{km}} \tag{4.6.7}$$

D_{xy}, given by (4.6.6), is the MNth root of the product of the MN distances from the subconductors of conductor x to the subconductors of conductor y. Associated with each subconductor k of conductor x are the M distances D_{k1}', D_{k2}', …, D_{kM}

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to the subconductors of conductor y. For N subconductors in conductor x, there are therefore MN of these distances. D_{xy} is called the *geometric mean distance* or GMD between conductors x and y.

Also, D_{xx}, given by (4.6.7), is the N^2 root of the product of the N^2 distances between the subconductors of conductor x. Associated with each subconductor k are the N distances $D_{k1}, D_{k2}, \ldots, D_{kk} = r', \ldots, D_{kN}$. For N subconductors in conductor x, there are therefore N^2 of these distances. D_{xx} is called the *geometric mean radius* or GMR of conductor x.

Similarly, for conductor y,

$$L_y = 2 \times 10^{-7} \ln \frac{D_{xy}}{D_{yy}} \quad \text{H/m per conductor} \tag{4.6.8}$$

where

$$D_{yy} = \sqrt[M^2]{\prod_{k=1'}^{M} \prod_{m=1'}^{M} D_{km}} \tag{4.6.9}$$

D_{yy}, the GMR of conductor y, is the M^2 root of the product of the M^2 distances between the subconductors of conductor y. The total inductance L of the single-phase circuit is

$$L = L_x + L_y \quad \text{H/m per circuit} \tag{4.6.10}$$

EXAMPLE 4.2

GMR, GMD, and inductance: single-phase two-conductor line

Expand (4.6.6), (4.6.7), and (4.6.9) for $N = 3$ and $M = 2'$. Then evaluate L_x, L_y, and L in H/m for the single-phase two-conductor line shown in Figure 4.12.

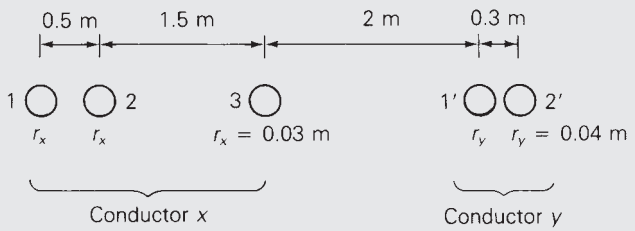

FIGURE 4.12
Single-phase, two-conductor line for Example 4.2

(*Continued*)

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SOLUTION

For $N = 3$ and $M = 2'$, (4.6.6) becomes

$$\begin{aligned} \mathrm{D}_{xy} &= \sqrt[6]{\prod_{k=1}^{3}\prod_{m=1'}^{2'} \mathrm{D}_{km}} \\ &= \sqrt[6]{\prod_{k=1}^{3} \mathrm{D}_{k1'}\,\mathrm{D}_{k2'}} \\ &= \sqrt[6]{(\mathrm{D}_{11'}\,\mathrm{D}_{12'})(\mathrm{D}_{21'}\,\mathrm{D}_{22'})(\mathrm{D}_{31'}\,\mathrm{D}_{32'})} \end{aligned}$$

Similarly, (4.6.7) becomes

$$\begin{aligned} \mathrm{D}_{xx} &= \sqrt[9]{\prod_{k=1}^{3}\prod_{m=1}^{3} \mathrm{D}_{km}} \\ &= \sqrt[9]{\prod_{k=1}^{3} \mathrm{D}_{k1}\mathrm{D}_{k2}\mathrm{D}_{k3}} \\ &= \sqrt[9]{(\mathrm{D}_{11}\mathrm{D}_{12}\mathrm{D}_{13})(\mathrm{D}_{21}\mathrm{D}_{22}\mathrm{D}_{23})(\mathrm{D}_{31}\mathrm{D}_{32}\mathrm{D}_{33})} \end{aligned}$$

and (4.6.9) becomes

$$\begin{aligned} \mathrm{D}_{yy} &= \sqrt[4]{\prod_{k=1'}^{2'}\prod_{m=1'}^{2'} \mathrm{D}_{km}} \\ &= \sqrt[4]{\prod_{k=1'}^{2'} \mathrm{D}_{k1'}\,\mathrm{D}_{k2'}} \\ &= \sqrt[4]{(\mathrm{D}_{1'1'}\,\mathrm{D}_{1'2'})(\mathrm{D}_{2'1'}\,\mathrm{D}_{2'2'})} \end{aligned}$$

Evaluating D_{xy}, D_{xx}, and D_{yy} for the single-phase two-conductor line shown in Figure 4.12,

$\mathrm{D}_{11'} = 4$ m $\quad \mathrm{D}_{12'} = 4.3$ m $\quad \mathrm{D}_{21'} = 3.5$ m

$\mathrm{D}_{22'} = 3.8$ m $\quad \mathrm{D}_{31'} = 2$ m $\quad \mathrm{D}_{32'} = 2.3$ m

$\mathrm{D}_{xy} = \sqrt[6]{(4)(4.3)(3.5)(3.8)(2)(2.3)} = 3.189$ m

$\mathrm{D}_{11} = \mathrm{D}_{22} = \mathrm{D}_{33} = r'_x = e^{-1/4}\, r_x = (0.7788)(0.03) = 0.02336$ m

$\mathrm{D}_{21} = \mathrm{D}_{12} = 0.05$ m

$\mathrm{D}_{23} = \mathrm{D}_{32} = 1.5$ m

$\mathrm{D}_{31} = \mathrm{D}_{13} = 2.0$ m

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$$D_{xx} = \sqrt[9]{(0.02336)^3(0.5)^2(1.5)^2(2.0)^2} = 0.3128 \text{ m}$$

$$D_{1'1'} = D_{2'2'} = r'_y = e^{-1/4} r_y = (0.7788)(0.04) = 0.03115 \text{ m}$$

$$D_{1'2'} = D_{2'1'} = 0.3 \text{ m}$$

$$D_{yy} = \sqrt[4]{(0.03115)^2(0.3)^2} = 0.09667 \text{ m}$$

Then, from (4.6.5), (4.6.8), and (4.6.10):

$$L_x = 2 \times 10^{-7} \ln\left(\frac{3.189}{0.3128}\right) = 4.644 \times 10^{-7} \quad \text{H/m per conductor}$$

$$L_y = 2 \times 10^{-7} \ln\left(\frac{3.189}{0.09667}\right) = 6.992 \times 10^{-7} \quad \text{H/m per conductor}$$

$$L = L_x + L_y = 1.164 \times 10^{-6} \quad \text{H/m per circuit}$$

It is seldom necessary to calculate GMR or GMD for standard lines. The GMR of standard conductors is provided by conductor manufacturers and can be found in various handbooks (see Appendix Tables A.3 and A.4). Also, if the distances between conductors are large compared to the distances between subconductors of each conductor, then the GMD between conductors is approximately equal to the distance between conductor centers.

EXAMPLE 4.3

Inductance and inductive reactance: single-phase line

A single-phase line operating at 60 Hz consists of two 4/0 12-strand copper conductors with 1.5 m spacing between conductor centers. The line length is 32 km. Determine the total inductance in H and the total inductive reactance in Ω.

SOLUTION

The GMD between conductor centers is $D_{xy} = 1.5$ m. Also, from Table A.3, the GMR of a 4/0 12-strand copper conductor is $D_{xx} = D_{yy} = 0.005334$ m. From (4.6.5) and (4.6.8),

$$L_x = L_y = 2 \times 10^{-7} \ln\left(\frac{1.5}{0.005334}\right)\frac{\text{H}}{\text{m}} \times 32{,}000 \text{ m}$$

$$= 0.03609 \quad \text{H per conductor}$$

The total inductance is

$$L = L_x + L_y = 2 \times 0.03639 = 0.07279 \quad \text{H per circuit}$$

and the total inductive reactance is

$$X_L = 2\pi f L = (2\pi)(60)(0.07279) = 27.44 \quad \Omega \text{ per circuit}$$

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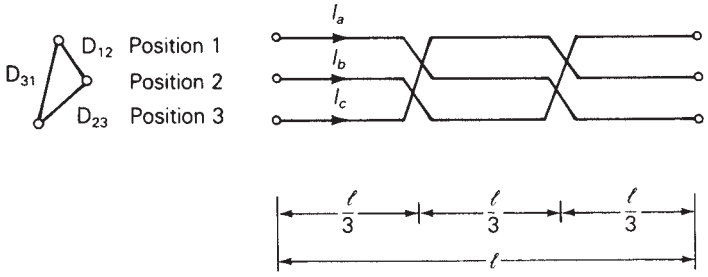
FIGURE 4.13

Completely transposed three-phase line



To calculate inductance for three-phase lines with stranded conductors and equal phase spacing, r' is replaced by the conductor GMR in (4.5.9). If the spacings between phases are unequal, then balanced positive-sequence flux linkages are not obtained from balanced positive-sequence currents. Instead, unbalanced flux linkages occur, and the phase inductances are unequal. However, balance can be restored by exchanging the conductor positions along the line, which is a technique called *transposition.*

Figure 4.13 shows a completely transposed three-phase line. The line is transposed at two locations such that each phase occupies each position for one-third of the line length. Conductor positions are denoted 1, 2, 3, with distances D_{12}, D_{23}, D_{31} between positions. The conductors are identical, each with GMR denoted D_s. To calculate inductance of this line, assume balanced positive-sequence currents I_a, I_b, I_c, for which $I_a + I_b + I_c + 0$. Again, (4.4.30) is valid, and the total flux linking the phase a conductor while it is in position 1 is

$$\lambda_{a1} = 2 \times 10^{-7}\left[I_a \ln \frac{1}{D_S} + I_b \ln \frac{1}{D_{12}} + I_c \ln \frac{1}{D_{31}}\right] \text{ Wb-t/m} \tag{4.6.11}$$

Similarly, the total flux linkage of this conductor while it is in positions 2 and 3 is

$$\lambda_{a2} = 2 \times 10^{-7}\left[I_a \ln \frac{1}{D_S} + I_b \ln \frac{1}{D_{23}} + I_c \ln \frac{1}{D_{12}}\right] \text{ Wb-t/m} \tag{4.6.12}$$

$$\lambda_{a3} = 2 \times 10^{-7}\left[I_a \ln \frac{1}{D_S} + I_b \ln \frac{1}{D_{31}} + I_c \ln \frac{1}{D_{23}}\right] \text{ Wb-t/m} \tag{4.6.13}$$

The average of the above flux linkages is

$$\lambda_a = \frac{\lambda_{a1}\left(\frac{l}{3}\right) + \lambda_{a2}\left(\frac{l}{3}\right) + \lambda_{a3}\left(\frac{l}{3}\right)}{l} = \frac{\lambda_{a1} + \lambda_{a2} + \lambda_{a3}}{3}$$

$$= \frac{2 \times 10^{-7}}{3}\left[3I_a \ln \frac{1}{D_S} + I_b \ln \frac{1}{D_{12}D_{23}D_{31}} + I_c \ln \frac{1}{D_{12}D_{23}D_{31}}\right] \tag{4.6.14}$$

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Using $(I_b + I_c) = -I_a$ in (4.6.14),

$$\lambda_a = \frac{2 \times 10^{-7}}{3}\left[3I_a \ln \frac{1}{D_S} - I_a \ln \frac{1}{D_{12}D_{23}D_{31}}\right]$$

$$= 2 \times 10^{-7} I_a \ln \frac{\sqrt[3]{D_{12}D_{23}D_{31}}}{D_S} \quad \text{Wb-t/m} \tag{4.6.15}$$

and the average inductance of phase a is

$$L_a = \frac{\lambda_a}{I_a} = 2 \times 10^{-7} \ln \frac{\sqrt[3]{D_{12}D_{23}D_{31}}}{D_S} \quad \text{H/m per phase} \tag{4.6.16}$$

The same result is obtained for $L_b = \lambda_b/I_b$ and for $L_c = \lambda_c/I_c$. However, only one phase need be considered for balanced three-phase operation of a completely transposed three-phase line. Defining

$$D_{eq} = \sqrt[3]{D_{12}D_{23}D_{31}} \tag{4.6.17}$$

results in

$$L_a = 2 \times 10^{-7} \ln \frac{D_{eq}}{D_S} \quad \text{H/m} \tag{4.6.18}$$

D_{eq}, the cube root of the product of the three-phase spacings, is the geometric mean distance between phases. Also, D_S is the conductor GMR for stranded conductors, or r' for solid cylindrical conductors.

EXAMPLE 4.4

Inductance and inductive reactance: three-phase line

A completely transposed 60-Hz three-phase line has flat horizontal phase spacing with 10 m between adjacent conductors. The conductors are 1,590,000 cmil ACSR with 54/3 stranding. Line length is 200 km. Determine the inductance in H and the inductive reactance in Ω.

SOLUTION

From Table A.4, the GMR of a 1,590,000 cmil 54/3 ACSR conductor is

$$D_S = 0.0520 \text{ ft} \frac{1 \text{ m}}{3.28 \text{ ft}} = 0.0159 \text{ m}$$

Also, from (4.6.17) and (4.6.18),

$$D_{eq} = \sqrt[3]{(10)(10)(20)} = 12.6 \text{ m}$$

$$L_a = 2 \times 10^{-7} \ln\left(\frac{12.6}{0.0159}\right) \frac{\text{H}}{\text{m}} \times \frac{1000 \text{ m}}{\text{km}} \times 200 \text{ km}$$

$$= 0.267 \text{ H}$$

The inductive reactance of phase a is

$$X_a = 2\pi f L_a = 2\pi(60)(0.267) = 101 \quad \Omega$$

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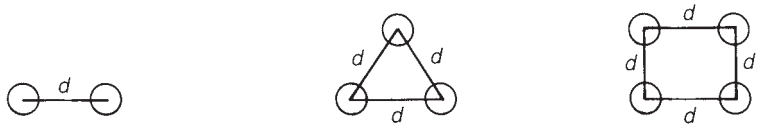
FIGURE 4.14
Bundle conductor configurations



It is common practice for EHV lines to use more than one conductor per phase, a practice called *bundling*. Bundling reduces the electric field strength at the conductor surfaces, which in turn reduces or eliminates corona and its results: undesirable power loss, communications interference, and audible noise. Bundling also reduces the series reactance of the line by increasing the GMR of the bundle.

Figure 4.14 shows common EHV bundles consisting of two, three, or four conductors. The three-conductor bundle has its conductors on the vertices of an equilateral triangle, and the four-conductor bundle has its conductors on the corners of a square. To calculate inductance, D_S in (4.6.18) is replaced by the GMR of the bundle. Since the bundle constitutes a composite conductor, calculation of bundle GMR is, in general, given by (4.6.7). If the conductors are stranded and the bundle spacing d is large compared to the conductor outside radius, each stranded conductor is first replaced by an equivalent solid cylindrical conductor with GMR $= D_S$. Then the bundle is replaced by one equivalent conductor with GMR $= D_{SL}$, given by (4.6.7) with $n = 2$, 3, or 4 as follows:

Two-conductor bundle:

$$D_{SL} = \sqrt[4]{(D_S \times d)^2} = \sqrt{D_S d} \tag{4.6.19}$$

Three-conductor bundle:

$$D_{SL} = \sqrt[9]{(D_S \times d \times d)^3} = \sqrt[3]{D_S d^2} \tag{4.6.20}$$

Four-conductor bundle:

$$D_{SL} = \sqrt[16]{(D_S \times d \times d \times d\sqrt{2})^4} = 1.091\sqrt[4]{D_S d^3} \tag{4.6.21}$$

The inductance is then

$$L_a = 2 \times 10^{-7} \ln \frac{D_{eq}}{D_{SL}} \quad \text{H/m} \tag{4.6.22}$$

If the phase spacings are large compared to the bundle spacing, then sufficient accuracy for D_{eq} is obtained by using the distances between bundle centers.

EXAMPLE 4.5

Inductive reactance: three-phase line with bundled conductors

Each of the 1,590,000 cmil conductors in Example 4.4 is replaced by two 795,000 cmil ACSR 26/2 conductors, as shown in Figure 4.15. Bundle spacing

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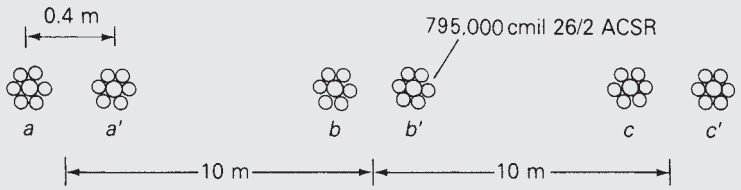
FIGURE 4.15
Three-phase bundled conductor line for Example 4.5



is 0.40 m. Flat horizontal spacing is retained, with 10 m between adjacent bundle centers. Calculate the inductive reactance of the line and compare it with that of Example 4.4.

SOLUTION

From Table A.4, the GMR of a 795,000 cmil 26/2 ACSR conductor is

$$D_S = 0.0375 \text{ ft} \times \frac{1 \text{ m}}{3.28 \text{ ft}} = 0.0114 \text{ m}$$

From (4.6.19), the two-conductor bundle GMR is

$$D_{SL} = \sqrt{(0.0114)(0.40)} = 0.0676 \text{ m}$$

Since $D_{eq} = 12.6$ m is the same as in Example 4.4,

$$L_a = 2 \times 10^{-7} \ln\left(\frac{12.6}{0.0676}\right)(1000)(200) = 0.209 \text{ H}$$

$$X_a = 2\pi f L_1 = (2\pi)(60)(0.209) = 78.8 \ \Omega$$

The reactance of the bundled line, 78.8 Ω, is 22% less than that of Example 4.4, even though the two-conductor bundle has the same amount of conductor material (that is, the same cmil per phase). One advantage of reduced series line reactance is smaller line-voltage drops. Also, the loadability of medium and long EHV lines is increased (see Chapter 5).

4.7 SERIES IMPEDANCES: THREE-PHASE LINE WITH NEUTRAL CONDUCTORS AND EARTH RETURN

This section develops equations suitable for computer calculation of the series impedances, including resistances and inductive reactances, for the three-phase overhead line shown in Figure 4.16. This line has three phase conductors *a*, *b*, and *c*, where bundled conductors, if any, have already been replaced by equivalent conductors, as described in Section 4.6. The line also has *N* neutral conductors denoted *n*1, *n*2, ... , *nN*.* All the neutral conductors are connected in parallel and

*Instead of *shield wire*, the term *neutral conductor* is used. It applies to distribution as well as transmission lines.

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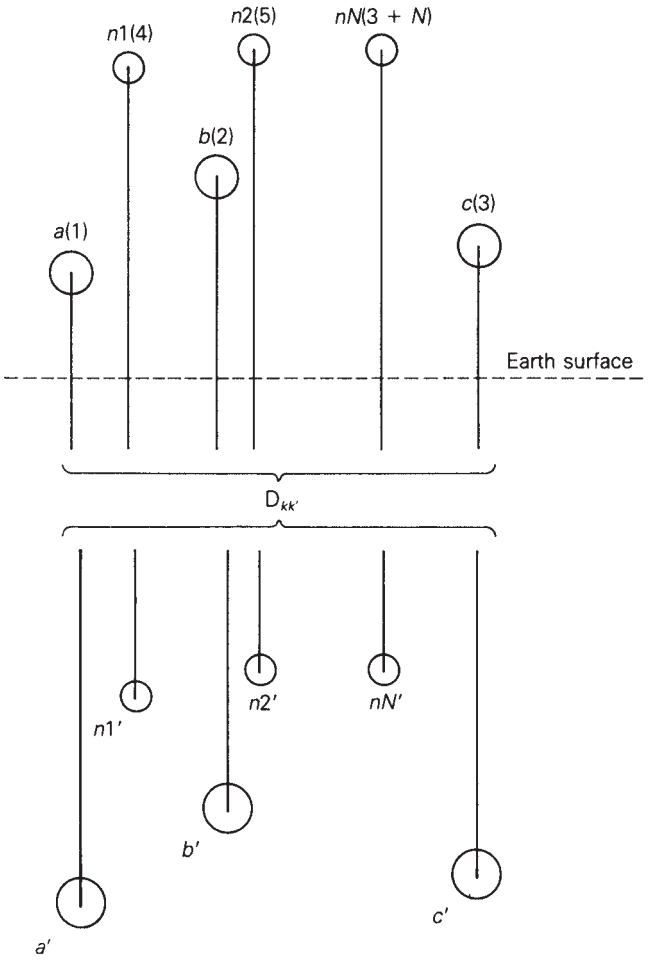
FIGURE 4.16
Three-phase transmission line with earth replaced by earth return conductors



are grounded to the earth at regular intervals along the line. Any isolated neutral conductors that carry no current are omitted. The phase conductors are insulated from each other and from the earth.

If the phase currents are not balanced, there may be a return current in the grounded neutral wires and in the earth. The earth return current will spread out under the line, seeking the lowest impedance return path. A classic paper by Carson [4], later modified by others [5, 6], shows that the earth can be replaced by a set of "earth return" conductors located directly under the overhead conductors, as shown in Figure 4.16. Each earth return conductor carries the negative of its overhead conductor current, has a GMR denoted $D_{k'k'}$, distance $D_{kk'}$ from its overhead conductor, and resistance $R_{k'}$ given by:

$$D_{k'k'} = D_{kk} \quad \text{m} \tag{4.7.1}$$

$$D_{kk'} = 658.5\sqrt{\rho/f} \quad \text{m} \tag{4.7.2}$$

$$R_{k'} = 9.869 \times 10^{-7} f \quad \Omega/\text{m} \tag{4.7.3}$$

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TABLE 4.4
Earth resistivities and 60-Hz equivalent conductor distances

Type of Earth	Resistivity (Ωm)	$D_{kk'}$ (m)
Sea water	0.01−1.0	8.50−85.0
Swampy ground	10−100	269−850
Average damp earth	100	850
Dry earth	1000	2690
Pure slate	10^7	269,000
Sandstone	10^9	2,690,000

where ρ is the earth resistivity in ohm-meters and f is frequency in hertz. Table 4.4 lists earth resistivities and 60-Hz equivalent conductor distances for various types of earth. It is common practice to select $\rho = 100$ Ωm when actual data are unavailable.

Note that the GMR of each earth return conductor, $D_{k'k'}$, is the same as the GMR of its corresponding overhead conductor, D_{kk}. Also, all the earth return conductors have the same distance $D_{kk'}$ from their overhead conductors and the same resistance $R_{k'}$.

For simplicity, renumber the overhead conductors from 1 to $(3 + N)$, beginning with the phase conductors, then overhead neutral conductors, as shown in Figure 4.16. Operating as a transmission line, the sum of the currents in all the conductors is zero.

That is,

$$\sum_{k=1}^{(6+2N)} I_k = 0 \tag{4.7.4}$$

Equation (4.4.30) is therefore valid, and the flux linking overhead conductor k is

$$\lambda_k = 2 \times 10^{-7} \sum_{m=1}^{(3+N)} I_m \ln \frac{D_{km'}}{D_{km}} \quad \text{Wb-t/m} \tag{4.7.5}$$

In matrix format, (4.7.5) becomes

$$\lambda = LI \tag{4.7.6}$$

where

λ is a $(3 + N)$ Vector

I is a $(3 + N)$ vector

L is a $(3 + N) \times (3 + N)$ matrix whose elements are

$$L_{km} = 2 \times 10^{-7} \ln \frac{D_{km'}}{D_{km}} \tag{4.7.7}$$

When $k = m$, D_{kk} in (4.7.7) is the GMR of (bundled) conductor k. When $k \neq m$, D_{km} is the distance between conductors k and m.

A circuit representation of a 1-meter section of the line is shown in Figure 4.17(a).

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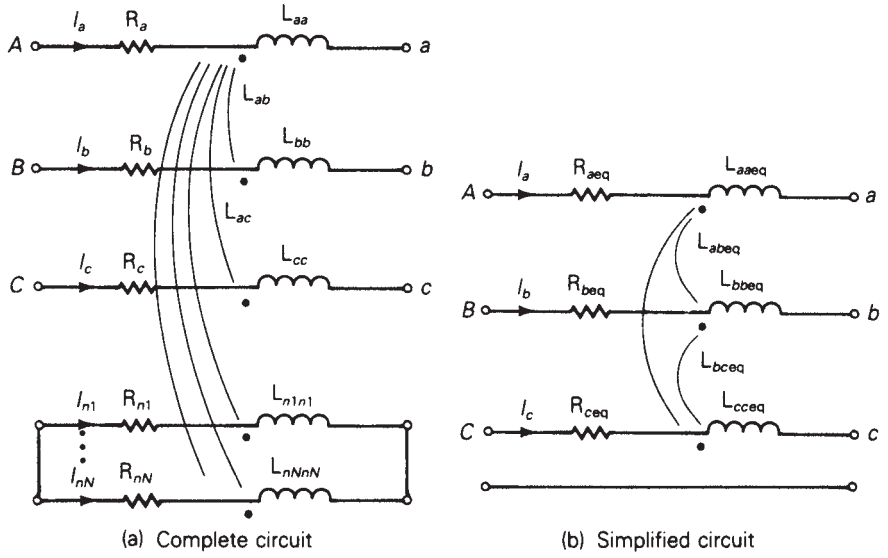
FIGURE 4.17

Circuit representation of series-phase impedances



Using this circuit, the vector of voltage drops across the conductors is

$$\begin{bmatrix} E_{Aa} \\ E_{Bb} \\ E_{Cc} \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = (\mathbf{R} + j\omega\mathbf{L}) \begin{bmatrix} I_a \\ I_b \\ I_c \\ I_{n1} \\ \vdots \\ I_{nN} \end{bmatrix} \tag{4.7.8}$$

where $\mathbf{L}$ is given by (4.7.7) and $\mathbf{R}$ is a $(3 + N) \times (3 + N)$ matrix of conductor resistances.

$$\mathbf{R} = \begin{bmatrix} (\mathrm{R}_a+\mathrm{R}_{k'})\mathrm{R}_{k'} \ldots & & \mathrm{R}_{k'} \\ \mathrm{R}_{k'}(\mathrm{R}_b+\mathrm{R}_{k'})\mathrm{R}_{k'} \ldots & & \vdots \\ (\mathrm{R}_c+\mathrm{R}_{k'})\mathrm{R}_{k'} \ldots & & \\ (\mathrm{R}_{n1}+\mathrm{R}_{k'})\mathrm{R}_{k'} \ldots & & \\ \ddots & & \\ \mathrm{R}_{k'} & (R_{nN}+R_{K'}) & \end{bmatrix} \Omega/\mathrm{m} \tag{4.7.9}$$

The resistance matrix of (4.7.9) includes the resistance R_k of each overhead conductor and a mutual resistance $\mathrm{R}_{k'}$ due to the image conductors. R_k of each overhead conductor is obtained from conductor tables, such as Appendix Table A.3 or A.4, for a specified frequency, temperature, and current. $\mathrm{R}_{k'}$ of all the image conductors is the same, as given by (4.7.3).

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Our objective now is to reduce the $(3 + N)$ equations in (4.7.8) to three equations, thereby obtaining the simplified circuit representations shown in Figure 4.17(b). We partition (4.7.8) as follows:

$$\begin{bmatrix} E_{Aa} \\ E_{Bb} \\ E_{Cc} \\ \hline 0 \\ \cdots \\ 0 \end{bmatrix} = \begin{bmatrix} \overbrace{\begin{matrix} Z_{11} & Z_{12} & Z_{13} \\ Z_{21} & Z_{22} & Z_{23} \\ Z_{31} & Z_{32} & Z_{33} \end{matrix}}^{Z_A} & \overbrace{\begin{matrix} Z_{14} & \cdots & Z_{1(3+N)} \\ Z_{24} & \cdots & Z_{2(3+N)} \\ Z_{34} & \cdots & Z_{3(3+N)} \end{matrix}}^{Z_B} \\ \hline \underbrace{\begin{matrix} Z_{41} & Z_{42} & Z_{43} \\ \\ Z_{(3+N)1} & Z_{(3+N)2} & Z_{(3+N)3} \end{matrix}}_{Z_C} & \underbrace{\begin{matrix} Z_{44} & \cdots & Z_{4(3+N)} \\ \\ Z_{(3+N)4} & \cdots & Z_{(3+N)(3+N)} \end{matrix}}_{Z_D} \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \\ \hline I_{n1} \\ I_{nN} \\ \vdots \end{bmatrix} \tag{4.7.10}$$

The diagonal elements of this matrix are

$$\mathrm{Z}_{kk} = \mathrm{R}_k + \mathrm{R}_{k'} + j\omega 2 \times 10^{-7} \ln \frac{\mathrm{D}_{kk'}}{\mathrm{D}_{kk}} \quad \Omega/\mathrm{m} \tag{4.7.11}$$

And the off-diagonal elements, for $k \neq m$, are

$$\mathrm{Z}_{km} = \mathrm{R}_{k'} + j\omega 2 \times 10^{-7} \ln \frac{\mathrm{D}_{km'}}{\mathrm{D}_{km}} \quad \Omega/\mathrm{m} \tag{4.7.12}$$

Next, (4.7.10) is partitioned as shown above to obtain

$$\left[\begin{array}{c} \boldsymbol{E}_{\mathrm{P}} \\ \hline 0 \end{array}\right] = \left[\begin{array}{c|c} \boldsymbol{Z}_A & \boldsymbol{Z}_B \\ \hline \boldsymbol{Z}_C & \boldsymbol{Z}_D \end{array}\right] \left[\begin{array}{c} \boldsymbol{I}_{\mathrm{P}} \\ \hline \boldsymbol{I}_n \end{array}\right] \tag{4.7.13}$$

where

$$\boldsymbol{E}_{\mathrm{P}} = \begin{bmatrix} E_{Aa} \\ E_{Bb} \\ E_{Cc} \end{bmatrix}; \quad \boldsymbol{I}_{\mathrm{P}} = \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix}; \quad \boldsymbol{I}_n = \begin{bmatrix} I_{n1} \\ \vdots \\ I_{nN} \end{bmatrix}$$

$\boldsymbol{E}_{\mathrm{P}}$ is the three-dimensional vector of voltage drops across the phase conductors (including the neutral voltage drop), $\boldsymbol{I}_{\mathrm{P}}$ is the three-dimensional vector of phase currents and $\boldsymbol{I}_n$ is the N vector of neutral currents. Also, the $(3 + N) \times (3 + N)$ matrix in (4.7.10) is partitioned to obtain the following matrices:

$\boldsymbol{Z_A}$ with dimension 3×3

$\boldsymbol{Z_B}$ with dimension $3 \times N$

$\boldsymbol{Z_C}$ with dimension $N \times 3$

$\boldsymbol{Z_D}$ with dimension $N \times N$

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Equation (4.7.13) is rewritten as two separate matrix equations:

$$\boldsymbol{E}_{\mathrm{P}} = \boldsymbol{Z}_A \boldsymbol{I}_{\mathrm{P}} + \boldsymbol{Z}_B \boldsymbol{I}_n \tag{4.7.14}$$

$$\boldsymbol{0} = \boldsymbol{Z}_C \boldsymbol{I}_{\mathrm{P}} + \boldsymbol{Z}_D \boldsymbol{I}_n \tag{4.7.15}$$

Solving (4.7.15) for $\boldsymbol{I}_n$,

$$\boldsymbol{I}_n = -\boldsymbol{Z}_D^{-1} \boldsymbol{Z}_C \boldsymbol{I}_{\mathrm{P}} \tag{4.7.16}$$

Using (4.7.16) in (4.7.14),

$$\boldsymbol{E}_{\mathrm{P}} = [\boldsymbol{Z}_A - \boldsymbol{Z}_B \boldsymbol{Z}_D^{-1} \boldsymbol{Z}_C] \boldsymbol{I}_{\mathrm{P}} \tag{4.7.17}$$

or

$$\boldsymbol{E}_{\mathrm{P}} = \boldsymbol{Z}_{\mathrm{P}} \boldsymbol{I}_{\mathrm{P}} \tag{4.7.18}$$

where

$$\boldsymbol{Z}_{\mathrm{P}} = [\boldsymbol{Z}_A - \boldsymbol{Z}_B \boldsymbol{Z}_D^{-1} \boldsymbol{Z}_C] \tag{4.7.19}$$

Equation (4.7.17), the desired result, relates the phase-conductor voltage drops (including neutral voltage drop) to the phase currents. $\boldsymbol{Z}_{\mathrm{P}}$ given by (4.7.19) is the 3×3 series-phase impedance matrix, whose elements are denoted

$$\boldsymbol{Z}_{\mathrm{P}} = \begin{bmatrix} Z_{aa\mathrm{eq}} & Z_{ab\mathrm{eq}} & Z_{ac\mathrm{eq}} \\ Z_{ab\mathrm{eq}} & Z_{bb\mathrm{eq}} & Z_{bc\mathrm{eq}} \\ Z_{ac\mathrm{eq}} & Z_{bc\mathrm{eq}} & Z_{cc\mathrm{eq}} \end{bmatrix} \ \Omega/\mathrm{m} \tag{4.7.20}$$

If the line is completely transposed, the diagonal and off-diagonal elements are averaged to obtain

$$\hat{\boldsymbol{Z}}_{\mathrm{P}} = \begin{bmatrix} \hat{Z}_{aa\mathrm{eq}} & \hat{Z}_{ab\mathrm{eq}} & \hat{Z}_{ab\mathrm{eq}} \\ \hat{Z}_{ab\mathrm{eq}} & \hat{Z}_{aa\mathrm{eq}} & \hat{Z}_{ab\mathrm{eq}} \\ \hat{Z}_{ab\mathrm{eq}} & \hat{Z}_{ab\mathrm{eq}} & \hat{Z}_{aa\mathrm{eq}} \end{bmatrix} \ \Omega/\mathrm{m} \tag{4.7.21}$$

where

$$\hat{Z}_{aa\mathrm{eq}} = \frac{1}{3}(Z_{aa\mathrm{eq}} + Z_{bb\mathrm{eq}} + Z_{cc\mathrm{eq}}) \tag{4.7.22}$$

$$\hat{Z}_{ab\mathrm{eq}} = \frac{1}{3}(Z_{ab\mathrm{eq}} + Z_{ac\mathrm{eq}} + Z_{bc\mathrm{eq}}) \tag{4.7.23}$$

4.8 ELECTRIC FIELD AND VOLTAGE: SOLID CYLINDRICAL CONDUCTOR

The capacitance between conductors in a medium with constant permittivity ε can be obtained by determining the following properties.

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1. Electric field strength E, from Gauss's law
2. Voltage between conductors
3. Capacitance from charge per unit volt ($C = q/V$)

As a step toward computing capacitances of general conductor configurations, first compute the electric field of a uniformly charged, solid cylindrical conductor and the voltage between two points outside the conductor. Also, compute the voltage between two conductors in an array of charged conductors.

Gauss's law states that the total electric flux leaving a closed surface equals the total charge within the volume enclosed by the surface. That is, the normal component of electric flux density integrated over a closed surface equals the charge enclosed:

$$\oiint D_{\perp}\, ds = \oiint \varepsilon E_{\perp}\, ds = Q_{\text{enclosed}} \tag{4.8.1}$$

where $D_{\perp}$ denotes the normal component of electric flux density, $E_{\perp}$ denotes the normal component of electric field strength, and ds denotes the differential surface area. From Gauss's law, electric charge is a source of electric fields. Electric field lines originate from positive charges and terminate at negative charges.

Figure 4.18 shows a solid cylindrical conductor with radius r and with charge q coulombs per meter (assumed positive in the figure), uniformly distributed on the conductor surface. For simplicity, assume that the conductor is (1) sufficiently long that end effects are negligible, and (2) a perfect conductor (that is, zero resistivity, $\rho = 0$).

Inside the perfect conductor, Ohm's law gives $E_{\text{int}} = \rho J = 0$. That is, the internal electric field E_{int} is zero. To determine the electric field outside the conductor, select the cylinder with radius $x > r$ and with 1-meter length, shown in Figure 4.18, as the closed surface for Gauss's law. Due to the uniform charge distribution, the

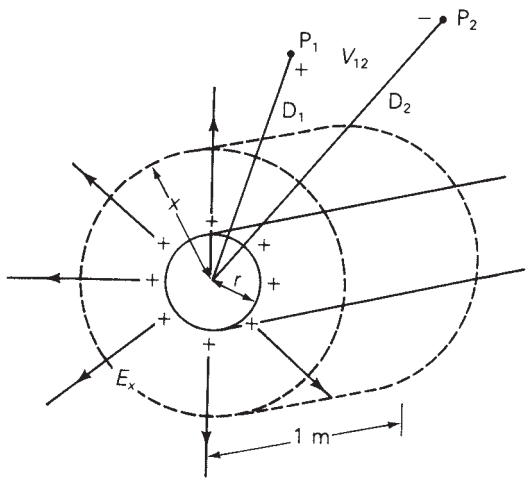

FIGURE 4.18

Perfectly conducting solid cylindrical conductor with uniform charge distribution

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electric field strength E_x is constant on the cylinder. Also, there is no tangential component of E_x, so the electric field is radial to the conductor. Then, integration of (4.8.1) yields

$$\varepsilon E_x(2\pi x)(1) = q(1)$$

$$E_x = \frac{q}{2\pi\varepsilon x} \text{ V/m} \tag{4.8.2}$$

where, for a conductor in free space, $\varepsilon = \varepsilon_0 = 8.854 \times 10^{-12}$ F/m.

A plot of the electric field lines is also shown in Figure 4.18. The direction of the field lines, denoted by the arrows, is from the positive charges where the field originates, to the negative charges, which in this case are at infinity. If the charge on the conductor surface were negative, then the direction of the field lines would be reversed.

Concentric cylinders surrounding the conductor are constant potential surfaces. The potential difference between two concentric cylinders at distances D_1 and D_2 from the conductor center is

$$V_{12} = \int_{D_1}^{D_2} E_x dx \tag{4.8.3}$$

Using (4.8.2) in (4.8.1),

$$V_{12} = \int_{D_1}^{D_2} \frac{q}{2\pi\varepsilon x} dx = \frac{q}{2\pi\varepsilon} \ln \frac{D_2}{D_1} \quad \text{volts} \tag{4.8.4}$$

Equation (4.8.4) gives the voltage V_{12} between two points, P_1 and P_2, at distances D_1 and D_2 from the conductor center, as shown in Figure 4.18. Also, in accordance with the notation, V_{12} is the voltage at P_1 with respect to P_2. If q is positive and D_2 is greater than D_1, as shown in the figure, then V_{12} is positive; that is, P_1 is at a higher potential than P_2. Equation (4.8.4) is also valid for either dc or ac. For ac, V_{12} is a phasor voltage and q is a phasor representation of a sinusoidal charge.

Now apply (4.8.4) to the array of M solid cylindrical conductors shown in Figure 4.19. Assume that each conductor m has an ac charge q_m C/m uniformly

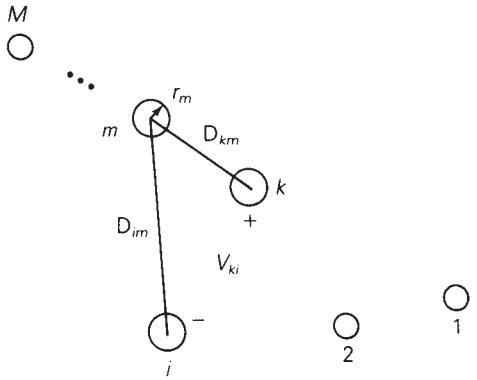

FIGURE 4.19
Array of M solid cylindrical conductors

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distributed along the conductor. The voltage V_{ki} between conductors k and i due to the charge q_m acting alone is

$$V_{ki} = \frac{q_m}{2\pi\varepsilon} \ln \frac{\mathrm{D}_{im}}{\mathrm{D}_{km}} \quad \text{volts} \tag{4.8.5}$$

where $\mathrm{D}_{mm} = r_m$ when $k = m$ or $i = m$. In (4.8.5), neglected is the distortion of the electric field in the vicinity of the other conductors, caused by the fact that the other conductors themselves are constant potential surfaces. V_{kim} can be thought of as the voltage between cylinders with radii D_{km} and D_{im} concentric to conductor m at points on the cylinders remote from conductors, where there is no distortion.

Using superposition, the voltage V_{ki} between conductors k and i due to all the charges is

$$V_{ki} = \frac{1}{2\pi\varepsilon} \sum_{m=1}^{M} q_m \ln \frac{\mathrm{D}_{im}}{\mathrm{D}_{km}} \quad \text{volts} \tag{4.8.6}$$

4.9 CAPACITANCE: SINGLE-PHASE, TWO-WIRE LINE AND THREE-PHASE, THREE-WIRE LINE WITH EQUAL PHASE SPACING

The results of the previous section are used here to determine the capacitances of the two relatively simple transmission lines considered in Section 4.5, a single-phase two-wire line and a three-phase three-wire line with equal phase spacing.

First consider the single-phase two-wire line shown in Figure 4.9. Assume that the conductors are energized by a voltage source such that conductor x has a uniform charge q C/m and, assuming conservation of charge, conductor y has an equal quantity of negative charge $-q$. Using (4.8.6) with $k = x$, $i = y$, and $m = x, y$,

$$\begin{aligned} V_{xy} &= \frac{1}{2\pi\varepsilon} \left[q \ln \frac{\mathrm{D}_{yx}}{\mathrm{D}_{xx}} - q \ln \frac{\mathrm{D}_{yy}}{\mathrm{D}_{xy}} \right] \\ &= \frac{q}{2\pi\varepsilon} \ln \frac{\mathrm{D}_{yx}\mathrm{D}_{xy}}{\mathrm{D}_{xx}\mathrm{D}_{yy}} \end{aligned} \tag{4.9.1}$$

Using $\mathrm{D}_{xy} = \mathrm{D}_{yx} = \mathrm{D}$, $\mathrm{D}_{xx} = r_x$, and $\mathrm{D}_{yy} = r_y$, (4.9.1) becomes

$$V_{xy} = \frac{q}{\pi\varepsilon} \ln \frac{\mathrm{D}}{\sqrt{r_x r_y}} \quad \text{volts} \tag{4.9.2}$$

For a 1-meter line length, the capacitance between conductors is

$$\mathrm{C}_{xy} = \frac{q}{V_{xy}} = \frac{\pi\varepsilon}{\ln\left(\dfrac{\mathrm{D}}{\sqrt{r_x r_y}}\right)} \quad \text{F/m line-to-line} \tag{4.9.3}$$

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and if $r_x = r_y = r$,

$$C_{xy} = \frac{\pi\varepsilon}{\ln (D/r)} \quad \text{F/m line-to-line} \tag{4.9.4}$$

If the two-wire line is supplied by a transformer with a grounded center tap, then the voltage between each conductor and ground is one-half that given by (4.9.2). That is,

$$V_{xn} = V_{yn} = \frac{V_{xy}}{2} \tag{4.9.5}$$

and the capacitance from either line to the grounded neutral is

$$C_n = C_{xn} = C_{yn} = \frac{q}{V_{xn}} = 2C_{xy}$$

$$= \frac{2\pi\varepsilon}{\ln (D/r)} \quad \text{F/m line-to-neutral} \tag{4.9.6}$$

Circuit representations of the line-to-line and line-to-neutral capacitances are shown in Figure 4.20. Note that if the neutral is open in Figure 4.20(b), the two line-to-neutral capacitances combine in series to give the line-to-line capacitance.

Next, consider the three-phase line with equal phase spacing shown in Figure 4.10. Neglect the effect of earth and neutral conductors here. To determine the positive-sequence capacitance, assume positive-sequence charges q_a, q_b, q_c such that $q_a + q_b + q_c = 0$. Using (4.8.6) with $k = a$, $i = b$, and $m = a, b, c$, the voltage V_{ab} between conductors a and b is

$$V_{ab} = \frac{1}{2\pi\varepsilon}\left[q_a \ln \frac{D_{ba}}{D_{aa}} + q_b \ln \frac{D_{bb}}{D_{ab}} + q_c \ln \frac{D_{bc}}{D_{ac}}\right] \tag{4.9.7}$$

Using $D_{aa} = D_{bb} = r$, and $D_{ab} = D_{ba} = D_{ca} = D_{cb} = D$, (4.9.7) becomes

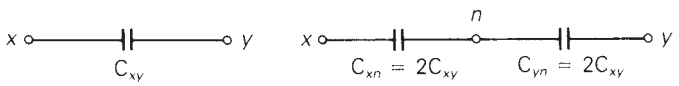
$$V_{ab} = \frac{1}{2\pi\varepsilon}\left[q_a \ln \frac{D}{r} + q_b \ln \frac{r}{D} + q_c \ln \frac{D}{D}\right]$$

$$= \frac{1}{2\pi\varepsilon}\left[q_c \ln \frac{D}{r} + q_b \ln \frac{r}{D}\right] \quad \text{volts} \tag{4.9.8}$$

Note that the third term in (4.9.8) is zero because conductors a and b are equidistant from conductor c. Thus, conductors a and b lie on a constant potential cylinder for the electric field due to q_c.

FIGURE 4.20

Circuit representation of capacitances for a single-phase, two-wire line



(a) Line-to-line capacitance (b) Line-to-neutral capacitances

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Similarly, using (4.8.6) with $k = a$, $i = c$, and $m = a,b,c$, the voltage V_{ac} is

$$V_{ac} = \frac{1}{2\pi\varepsilon}\left[q_a \ln\frac{D_{ca}}{D_{aa}} + q_b \ln\frac{D_{cb}}{D_{ab}} + q_c \ln\frac{D_{cc}}{D_{ac}}\right]$$
$$= \frac{1}{2\pi\varepsilon}\left[q_a \ln\frac{D}{r} + q_b \ln\frac{D}{D} + q_c \ln\frac{r}{D}\right]$$
$$= \frac{1}{2\pi\varepsilon}\left[q_a \ln\frac{D}{r} + q_c \ln\frac{r}{D}\right] \text{ volts} \tag{4.9.9}$$

Recall that for balanced positive-sequence voltages,

$$V_{ab} = \sqrt{3}V_{an}\angle{+30°} = \sqrt{3}V_{an}\left[\frac{\sqrt{3}}{2} + j\frac{1}{2}\right] \tag{4.9.10}$$

$$V_{ac} = -V_{ca} = \sqrt{3}V_{an}\angle{-30°} = \sqrt{3}V_{an}\left[\frac{\sqrt{3}}{2} - j\frac{1}{2}\right] \tag{4.9.11}$$

Adding (4.9.10) and (4.9.11) yields

$$V_{ab} + V_{ac} = 3V_{an} \tag{4.9.12}$$

Using (4.9.8) and (4.9.9) in (4.9.12),

$$V_{an} = \frac{1}{3}\left(\frac{1}{2\pi\varepsilon}\right)\left[2q_a \ln\frac{D}{r} + (q_b + q_c)\ln\frac{r}{D}\right] \tag{4.9.13}$$

and with $q_b + q_c = -q_a$,

$$V_{an} = \frac{1}{2\pi\varepsilon} q_a \ln\frac{D}{r} \text{ volts} \tag{4.9.14}$$

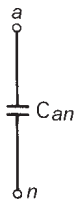
The capacitance-to-neutral per line length is

$$C_{an} = \frac{q_a}{V_{an}} = \frac{2\pi\varepsilon}{\ln\left(\frac{D}{r}\right)} \text{ F/m line-to-neutral} \tag{4.9.15}$$

Due to symmetry, the same result is obtained for $C_{bn} = q_b/V_{bn}$ and $C_{cn} = q_c/V_{cn}$. For balanced three-phase operation, however, only one phase need be considered. A circuit representation of the capacitance-to-neutral is shown in Figure 4.21.

FIGURE 4.21

Circuit representation of the capacitance-to-neutral of a three-phase line with equal phase spacing



4.10 CAPACITANCE: STRANDED CONDUCTORS, UNEQUAL PHASE SPACING, BUNDLED CONDUCTORS

Equations (4.9.6) and (4.9.15) are based on the assumption that the conductors are solid cylindrical conductors with zero resistivity. The electric field inside these conductors is zero, and the external electric field is perpendicular to the conductor

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surfaces. Practical conductors with resistivities similar to those listed in Table 4.3 have a small internal electric field. As a result, the external electric field is slightly altered near the conductor surfaces. Also, the electric field near the surface of a stranded conductor is not the same as that of a solid cylindrical conductor. However, it is normal practice when calculating line capacitance to replace a stranded conductor by a perfectly conducting solid cylindrical conductor whose radius equals the outside radius of the stranded conductor. The resulting error in capacitance is small, since only the electric field near the conductor surfaces is affected.

Also, (4.8.2) is based on the assumption that there is uniform charge distribution. But conductor charge distribution is nonuniform in the presence of other charged conductors. Therefore (4.9.6) and (4.9.15), which are derived from (4.8.2), are not exact. However, the nonuniformity of conductor charge distribution can be shown to have a negligible effect on line capacitance.

For three-phase lines with unequal phase spacing, balanced positive-sequence voltages are not obtained with balanced positive-sequence charges. Instead, unbalanced line-to-neutral voltages occur, and the phase-to-neutral capacitances are unequal. Balance can be restored by transposing the line such that each phase occupies each position for one-third of the line length. If equations similar to (4.9.7) for V_{ab} as well as for V_{ac} are written for each position in the transposition cycle, and are then averaged and used in (4.9.12) through (4.9.14), the resulting capacitance becomes

$$C_{an} = \frac{2\pi\varepsilon}{\ln (D_{eq}/r)} \quad \text{F/m} \tag{4.10.1}$$

where

$$D_{eq} = \sqrt[3]{D_{ab}D_{bc}D_{ac}} \tag{4.10.2}$$

Figure 4.22 shows a bundled conductor line with two conductors per bundle. To determine the capacitance of this line, assume balanced positive-sequence charges q_a, q_b, q_c for each phase such that $q_a + q_b + q_c = 0$. Assume that the conductors in each bundle, which are in parallel, share the charges equally. Thus, conductors a and a' each have the charge $q_a/2$. Also, assume that the phase spacings are much larger than the bundle spacings so that D_{ab} may be used instead of $(D_{ab} - d)$ or $(D_{ab} + d)$. Then, using (4.8.6) with $k = a$, $i = b$, $m = a, a', b, b', c, c'$,

$$\begin{aligned} V_{ab} &= \frac{1}{2\pi\varepsilon}\left[\frac{q_a}{2}\ln\frac{D_{ba}}{D_{aa}} + \frac{q_a}{2}\ln\frac{D_{ba'}}{D_{aa'}} + \frac{q_b}{2}\ln\frac{D_{bb}}{D_{ab}}\right. \\ &\qquad \left. + \frac{q_b}{2}\ln\frac{D_{bb'}}{D_{ab'}} + \frac{q_c}{2}\ln\frac{D_{bc}}{D_{ac}} + \frac{q_c}{2}\ln\frac{D_{bc'}}{D_{ac'}}\right] \\ &= \frac{1}{2\pi\varepsilon}\left[\frac{q_a}{2}\left(\ln\frac{D_{ab}}{r} + \ln\frac{D_{ab}}{d}\right) + \frac{q_b}{2}\left(\ln\frac{r}{D_{ab}} + \ln\frac{d}{D_{ab}}\right)\right. \\ &\qquad \left. + \frac{q_c}{2}\left(\ln\frac{D_{bc}}{D_{ac}} + \ln\frac{D_{bc}}{D_{ac}}\right)\right] \\ &= \frac{1}{2\pi\varepsilon}\left[q_a\ln\frac{D_{ab}}{\sqrt{rd}} + q_b\ln\frac{\sqrt{rd}}{D_{ab}} + q_c\ln\frac{D_{bc}}{D_{ac}}\right] \end{aligned} \tag{4.10.3}$$

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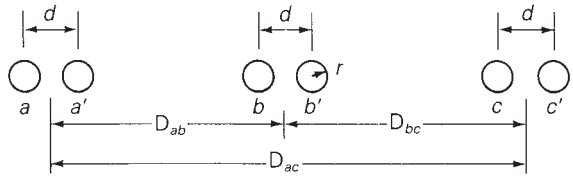

FIGURE 4.22
Three-phase line with two conductors per bundle

Equation (4.10.3) is the same as (4.9.7), except that D_{aa} and D_{bb}, in (4.9.7) are replaced by $\sqrt{rd}$ in this equation. Therefore, for a transposed line, derivation of the capacitance would yield

$$C_{an} = \frac{2\pi\varepsilon}{\ln(D_{eq}/D_{SC})} \quad \text{F/m} \tag{4.10.4}$$

where

$$D_{SC} = \sqrt{rd} \quad \text{for a two-conductor bundle} \tag{4.10.5}$$

Similarly,

$$D_{SC} = \sqrt[3]{rd^2} \quad \text{for a three-conductor bundle} \tag{4.10.6}$$

$$D_{SC} = 1.091\sqrt[4]{rd^3} \quad \text{for a four-conductor bundle} \tag{4.10.7}$$

Equation (4.10.4) for capacitance is analogous to (4.6.22) for inductance. In both cases D_{eq}, given by (4.6.17) or (4.10.2), is the geometric mean of the distances between phases. Also, (4.10.5)–(4.10.7) for D_{SC} are analogous to (4.6.19)−(4.6.21) for D_{SL}, except that the conductor outside radius r replaces the conductor GMR D_S.

The current supplied to the transmission-line capacitance is called *charging current*. For a single-phase circuit operating at line-to-line voltage $V_{xy} = V_{xy}\angle 0°$, the charging current is

$$I_{chg} = Y_{xy}V_{xy} = j\omega C_{xy}V_{xy} \quad \text{A} \tag{4.10.8}$$

As shown in Chapter 2, a capacitor delivers reactive power. From (2.3.5), the reactive power delivered by the line-to-line capacitance is

$$Q_C = \frac{V_{xy}^2}{X_c} = Y_{xy}V_{xy}^2 = \omega C_{xy}V_{xy}^2 \quad \text{var} \tag{4.10.9}$$

For a completely transposed three-phase line that has balanced positive-sequence voltages with $V_{an} = V_{LN}\angle 0°$, the phase a charging current is

$$I_{chg} = YV_{an} = j\omega C_{an}V_{LN} \quad \text{A} \tag{4.10.10}$$

and the reactive power delivered by phase a is

$$Q_{C1\phi} = YV_{an}^2 = \omega C_{an}V_{LN}^2 \quad \text{var} \tag{4.10.11}$$

The total reactive power supplied by the three-phase line is

$$Q_{C3\phi} = 3Q_{C1\phi} = 3\omega C_{an}V_{LN}^2 = \omega C_{an}V_{LL}^2 \quad \text{var} \tag{4.10.12}$$

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EXAMPLE 4.6

Capacitance, admittance, and reactive power supplied: single-phase line

For the single-phase line in Example 4.3, determine the line-to-line capacitance in F and the line-to-line admittance in S. If the line voltage is 20 kV, determine the reactive power in kvar supplied by this capacitance.

SOLUTION

From Table A.3, the outside radius of a 4/0 12-strand copper conductor is

$$r = \frac{0.552}{2}\text{ in.} \times \frac{0.0254\text{ m}}{1\text{ in.}} = 0.007010\text{ m}$$

and from (4.9.4),

$$C_{xy} = \frac{\pi(8.854 \times 10^{-12})}{\ln\left(\dfrac{1.5}{0.007010}\right)} = 5.184 \times 10^{-12} \quad \text{F/m}$$

or

$$C_{xy} = 5.184 \times 10^{-12}\frac{\text{F}}{\text{m}} \times 32{,}000\text{ m} = 1.66 \times 10^{-7} \quad \text{F}$$

and the shunt admittance is

$$Y_{xy} = j\omega C_{xy} = j(2\pi 60)(1.66 \times 10^{-7})$$
$$= j6.27 \times 10^{-5} \quad \text{S line-to-line}$$

From (4.10.9),

$$Q_C = (6.27 \times 10^{-5})(20 \times 10^3)^2 = 25.1 \quad \text{kvar}$$

EXAMPLE 4.7

Capacitance and shunt admittance; charging current and reactive power supplied: three-phase line

For the three-phase line in Example 4.5, determine the capacitance-to-neutral in F and the shunt admittance-to-neutral in S. If the line voltage is 345 kV, determine the charging current in kA per phase and the total reactive power in Mvar supplied by the line capacitance. Assume balanced positive-sequence voltages.

SOLUTION

From Table A.4, the outside radius of a 795,000 cmil 26/2 ACSR conductor is

$$r = \frac{1.108}{2}\text{ in.} \times 0.0254\frac{\text{m}}{\text{in.}} = 0.0141\text{ m}$$

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From (4.10.5), the equivalent radius of the two-conductor bundle is

$$D_{SC} = \sqrt{(0.0141)(0.40)} = 0.0750 \text{ m}$$

$D_{eq} = 12.6$ m is the same as in Example 4.5. Therefore, from (4.10.4),

$$C_{an} = \frac{(2\pi)(8.854 \times 10^{-12})}{\ln\left(\frac{12.6}{0.0750}\right)} \frac{\text{F}}{\text{m}} \times 1000 \frac{\text{m}}{\text{km}} \times 200 \text{ km}$$

$$= 2.17 \times 10^{-6} \quad \text{F}$$

The shunt admittance-to-neutral is

$$Y_{an} = j\omega C_{an} = j\,(2\pi 60)(2.17 \times 10^{-6})$$

$$= j8.19 \times 10^{-4} \quad \text{S}$$

From (4.10.10),

$$I_{chg} = |I_{chg}| = (8.19 \times 10^{-4})\left(\frac{345}{\sqrt{3}}\right) = 0.163 \text{ kA/phase}$$

and from (4.10.12),

$$Q_{C3\phi} = (8.19 \times 10^{-4})(345)^2 = 97.5 \quad \text{Mvar}$$

4.11 SHUNT ADMITTANCES: LINES WITH NEUTRAL CONDUCTORS AND EARTH RETURN

This section covers equations suitable for computer calculation of the shunt admittances for the three-phase overhead line shown in Figure 4.16. Approximate the earth's surface as a perfectly conducting horizontal plane, even though the earth under the line may have irregular terrain and resistivities as shown in Table 4.4.

The *method of images* accounts for the effect of the earth plane, described as follows. Consider a single conductor with uniform charge distribution and with height H above a perfectly conducting earth plane, as shown in Figure 4.23(a). When the conductor has a positive charge, an equal quantity of negative charge is induced on the earth. The electric field lines originate from the positive charges on the conductor and terminate at the negative charges on the earth. Also, the electric field lines are perpendicular to the surfaces of the conductor and earth.

Now replace the earth by the image conductor shown in Figure 4.23(b), which has the same radius as the original conductor, lies directly below the original conductor with conductor separation $H_{11} = 2H$, and has an equal quantity of negative charge. The electric field above the dashed line representing the location of the removed earth plane in Figure 4.23(b) is identical to the electric field above

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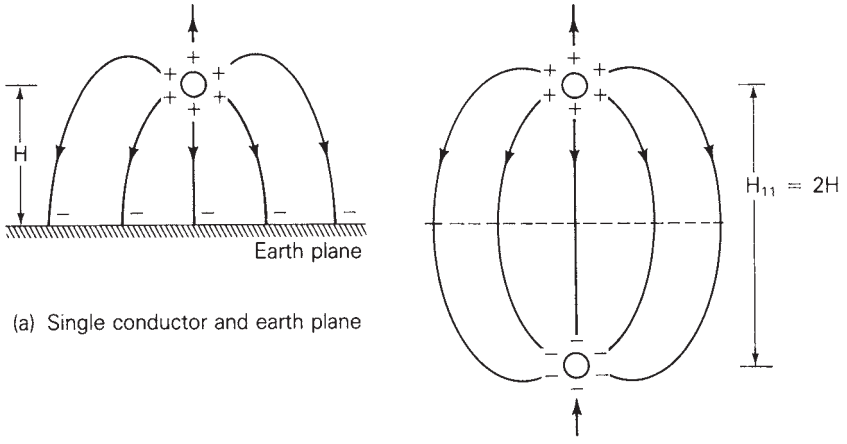
FIGURE 4.23
Method of images



(a) Single conductor and earth plane

(b) Earth plane replaced by image conductor

the earth plane in Figure 4.23(a). Therefore, the voltage between any two points above the earth is the same in both figures.

EXAMPLE 4.8

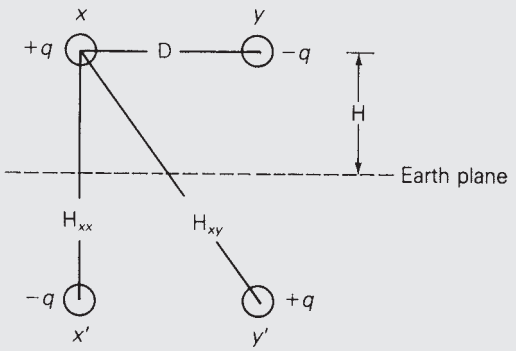
Effect of earth on capacitance: single-phase line

If the single-phase line in Example 4.6 has flat horizontal spacing with 5.5-cm average line height, determine the effect of the earth on capacitance. Assume a perfectly conducting earth plane.

SOLUTION

The earth plane is replaced by a separate image conductor for each overhead conductor, and the conductors are charged as shown in Figure 4.24. From (4.8.6), the voltage between conductors x and y is

FIGURE 4.24
Single-phase line for Example 4.8



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$$V_{xy} = \frac{q}{2\pi\varepsilon}\left[\ln\frac{D_{yx}}{D_{xx}} - \ln\frac{D_{yy}}{D_{xy}} - \ln\frac{H_{yx}}{H_{xx}} + \ln\frac{H_{yy}}{H_{xy}}\right]$$
$$= \frac{q}{2\pi\varepsilon}\left[\ln\frac{D_{yx}D_{xy}}{D_{xx}D_{yy}} - \ln\frac{H_{yx}H_{xy}}{H_{xx}H_{yy}}\right]$$
$$= \frac{q}{\pi\varepsilon}\left[\ln\frac{D}{r} - \ln\frac{H_{xy}}{H_{xx}}\right]$$

The line-to-line capacitance is

$$C_{xy} = \frac{q}{V_{xy}} = \frac{\pi\varepsilon}{\ln\dfrac{D}{r} - \ln\dfrac{H_{xy}}{H_{xx}}} \quad \text{F/m}$$

Using D = 1.5 m, r = 0.007010 m, $H_{xx} = 2H = 11.0$ m, and $H_{xy} = \sqrt{(11)^2 + (1.5)^2}$ = 11.102 m,

$$C_{xy} = \frac{\pi(8.854 \times 10^{-12})}{\ln\dfrac{1.5}{0.007010\ \text{m}} - \ln\dfrac{11.102}{11}} = 5.178 \times 10^{-12} \quad \text{F/m}$$

compared with 5.169×10^{-12} F/m in Example 4.6. The effect of the earth plane is to slightly increase the capacitance. Note that as the line height H increases, the ratio H_{xy}/H_{xx} approaches 1, $\ln(H_{xy}/H_{xx}) \to 0$, and the effect of the earth becomes negligible.

For the three-phase line with N neutral conductors shown in Figure 4.25, the perfectly conducting earth plane is replaced by a separate image conductor for each overhead conductor. The overhead conductors a, b, c, $n1$, $n2$, … , nN carry charges q_a, q_b, q_c, q_{n1}, … , q_{nN}, and the image conductors a', b', c', $n1'$ … , nN' carry charges $-q_a$, $-q_b$, $-q_c$, $-q_{n1}$, … , $-q_{nN}$. Applying (4.8.6) to determine the voltage $V_{kk'}$ between any conductor k and its image conductor k',

$$V_{kk'} = \frac{1}{2\pi\varepsilon}\left[\sum_{m=a}^{nN} q_m \ln\frac{H_{km}}{D_{km}} - \sum_{m=a}^{nN} q_m \ln\frac{D_{km}}{H_{km}}\right]$$
$$= \frac{2}{2\pi\varepsilon}\sum_{m=a}^{nN} q_m \ln\frac{H_{km}}{D_{km}} \tag{4.11.1}$$

where $D_{kk} = r_k$ and D_{km} is the distance between overhead conductors k and m. H_{km} is the distance between overhead conductor k and image conductor m. By symmetry, the voltage V_{kn} between conductor k and the earth is one-half of $V_{kk'}$.

$$V_{kn} = \frac{1}{2}V_{kk'} = \frac{1}{2\pi\varepsilon}\sum_{m=a}^{nN} q_m \ln\frac{H_{km}}{D_{km}} \tag{4.11.2}$$

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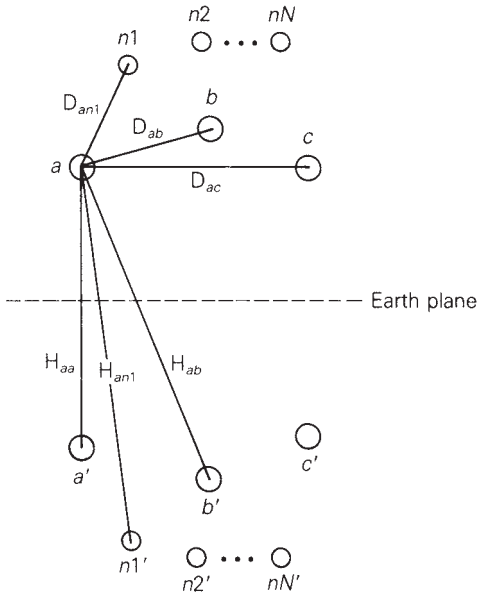
FIGURE 4.25

Three-phase line with neutral conductors and with earth plane replaced by image conductors



where

$$k = a, b, c, n1, n2, \ldots, nN$$
$$m = a, b, c, n1, n2, \ldots, nN$$

Since all the neutral conductors are grounded to the earth,

$$V_{kn} = 0 \qquad \text{for } k = n1, n2, \ldots, nN \tag{4.11.3}$$

In matrix format, (4.11.2) and (4.11.3) are

$$\begin{bmatrix} V_{an} \\ V_{bn} \\ V_{cn} \\ \hline 0 \\ \vdots \\ 0 \end{bmatrix} = \left[\begin{array}{ccc|ccc} \overbrace{\mathrm{P}_{aa} & \mathrm{P}_{ab} & \mathrm{P}_{ac}}^{\mathbf{P}_A} & \overbrace{\mathrm{P}_{an1} & \cdots & \mathrm{P}_{anN}}^{\mathbf{P}_B} \\ \mathrm{P}_{ba} & \mathrm{P}_{bb} & \mathrm{P}_{bc} & \mathrm{P}_{bn1} & \cdots & \mathrm{P}_{bnN} \\ \mathrm{P}_{ca} & \mathrm{P}_{cb} & \mathrm{P}_{cc} & \mathrm{P}_{cn1} & \cdots & \mathrm{P}_{cnN} \\ \hline \mathrm{P}_{n1a} & \mathrm{P}_{n1b} & \mathrm{P}_{n1c} & \mathrm{P}_{n1n1} & \cdots & \mathrm{P}_{n1nN} \\ \vdots & & & & & \\ \underbrace{\mathrm{P}_{nNa} & \mathrm{P}_{nNb} & \mathrm{P}_{nNc}}_{\mathbf{P}_C} & \underbrace{\mathrm{P}_{nNn1} & \cdots & \mathrm{P}_{nNnN}}_{\mathbf{P}_D} \end{array}\right] \begin{bmatrix} q_a \\ q_b \\ q_c \\ \hline q_{n1} \\ \vdots \\ q_{nN} \end{bmatrix} \tag{4.11.4}$$

The elements of the $(3 + N) \times (3 + N)$ matrix $\mathbf{P}$ are

$$\mathrm{P}_{km} = \frac{1}{2\pi\varepsilon} \ln \frac{\mathrm{H}_{km}}{\mathrm{D}_{km}} \quad \mathrm{m/F} \tag{4.11.5}$$

where

$$k = a, b, c, n1, \ldots, nN$$
$$m = a, b, c, n1, \ldots, nN$$

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Equation (4.11.4) is now partitioned as shown to obtain

$$\left[\begin{array}{c} \boldsymbol{V}_{\mathrm{P}} \\ \hline \mathbf{0} \end{array}\right] = \left[\begin{array}{c|c} \mathbf{P}_A & \mathbf{P}_B \\ \hline \mathbf{P}_C & \mathbf{P}_D \end{array}\right] \left[\begin{array}{c} \boldsymbol{q}_{\mathrm{P}} \\ \hline \boldsymbol{q}_n \end{array}\right] \tag{4.11.6}$$

$\boldsymbol{V}_{\mathrm{P}}$ is the three-dimensional vector of phase-to-neutral voltages. $\boldsymbol{q}_{\mathrm{P}}$ is the three-dimensional vector of phase-conductor charges and $\boldsymbol{q}_n$ is the N vector of neutral conductor charges. The $(3 + N) \times (3 + N)$ $\mathbf{P}$ matrix is partitioned as shown in (4.11.4) to obtain:

$\mathbf{P}_A$ with dimension 3×3

$\mathbf{P}_B$ with dimension $3 \times N$

$\mathbf{P}_C$ with dimension $N \times 3$

$\mathbf{P}_D$ with dimension $N \times N$

Equation (4.11.6) is rewritten as two separate equations:

$$\boldsymbol{V}_{\mathrm{P}} = \mathbf{P}_A \boldsymbol{q}_{\mathrm{P}} + \mathbf{P}_B \boldsymbol{q}_n \tag{4.11.7}$$

$$\mathbf{0} = \mathbf{P}_C \boldsymbol{q}_P + \mathbf{P}_D \boldsymbol{q}_n \tag{4.11.8}$$

Then (4.11.8) is solved for $\boldsymbol{q}_n$, which is used in (4.11.7) to obtain

$$\boldsymbol{V}_{\mathrm{P}} = (\mathbf{P}_A - \mathbf{P}_B \mathbf{P}_D^{-1} \mathbf{P}_C) \boldsymbol{q}_{\mathrm{P}} \tag{4.11.9}$$

or

$$\boldsymbol{q}_{\mathrm{P}} = \mathbf{C}_{\mathrm{P}} \mathbf{V}_{\mathrm{P}} \tag{4.11.10}$$

where

$$\mathbf{C}_{\mathrm{P}} = (\mathbf{P}_A - \mathbf{P}_B \mathbf{P}_D^{-1} \mathbf{P}_C)^{-1} \quad \mathrm{F/m} \tag{4.11.11}$$

Equation (4.11.10), the desired result, relates the phase-conductor charges to the phase-to-neutral voltages. $\mathbf{C}_{\mathrm{P}}$ is the 3×3 matrix of phase capacitances whose elements are denoted

$$\mathbf{C}_{\mathrm{P}} = \begin{bmatrix} \mathrm{C}_{aa} & \mathrm{C}_{ab} & \mathrm{C}_{ac} \\ \mathrm{C}_{ab} & \mathrm{C}_{bb} & \mathrm{C}_{bc} \\ \mathrm{C}_{ac} & \mathrm{C}_{bc} & \mathrm{C}_{cc} \end{bmatrix} \quad \mathrm{F/m} \tag{4.11.12}$$

It can be shown that $\mathbf{C}_{\mathrm{P}}$ is a symmetric matrix whose diagonal terms C_{aa}, C_{bb}, C_{cc} are positive, and whose off-diagonal terms C_{ab}, C_{bc}, C_{ac} are negative. This indicates that when a positive line-to-neutral voltage is applied to one phase, a positive charge is induced on that phase and negative charges are induced on the other phases, which is physically correct.

If the line is completely transposed, the diagonal and off-diagonal elements of $\mathbf{C}_{\mathrm{P}}$ are averaged to obtain

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$$\hat{\mathbf{C}}_{\mathbf{P}} = \begin{bmatrix} \hat{C}_{aa} & \hat{C}_{ab} & \hat{C}_{ab} \\ \hat{C}_{ab} & \hat{C}_{aa} & \hat{C}_{ab} \\ \hat{C}_{ab} & \hat{C}_{ab} & \hat{C}_{aa} \end{bmatrix} \quad \text{F/m} \tag{4.11.13}$$

where

$$\hat{C}_{aa} = \tfrac{1}{3}\,(C_{aa} + C_{bb} + C_{cc}) \quad \text{F/m} \tag{4.11.14}$$

$$\hat{C}_{ab} = \tfrac{1}{3}\,(C_{ab} + C_{bc} + C_{ac}) \quad \text{F/m} \tag{4.11.15}$$

$\hat{\mathbf{C}}_{\mathrm{P}}$ is a symmetrical capacitance matrix.

The shunt phase admittance matrix is given by

$$\boldsymbol{Y}_{\mathrm{P}} = j\omega\mathbf{C}_{\mathrm{P}} = j(2\pi f)\mathbf{C}_{\mathrm{P}} \quad \text{S/m} \tag{4.11.16}$$

or, for a completely transposed line,

$$\hat{\boldsymbol{Y}}_{\mathrm{P}} = j\omega\hat{\mathbf{C}}_{\mathrm{P}} = j(2\pi f)\hat{\mathbf{C}}_{\mathrm{P}} \quad \text{S/m} \tag{4.11.17}$$

4.12 ELECTRIC FIELD STRENGTH AT CONDUCTOR SURFACES AND AT GROUND LEVEL

When the electric field strength at a conductor surface exceeds the breakdown strength of air, current discharges occur. This phenomenon, called *corona*, causes additional line losses (corona loss), communications interference, and audible noise. Although breakdown strength depends on many factors, a rough value is 30 kV/cm in a uniform electric field for dry air at atmospheric pressure. The presence of water droplets or rain can lower this value significantly. To control corona, transmission lines are usually designed to maintain calculated values of conductor surface electric field strength below 20 $\mathrm{kV_{rms}}$/cm.

When line capacitances are determined and conductor voltages are known, the conductor charges can be calculated from (4.9.3) for a single-phase line or from (4.11.10) for a three-phase line. Then the electric field strength at the surface of one phase conductor, neglecting the electric fields due to charges on other phase conductors and neutral wires, is, from (4.8.2),

$$E_r = \frac{q}{2\pi\varepsilon r} \quad \text{V/m} \tag{4.12.1}$$

where r is the conductor outside radius.

For bundled conductors with N_b conductors per bundle and with charge q C/m per phase, the charge per conductor is q/N_b and

$$E_{\mathrm{rave}} = \frac{q/N_b}{2\pi\varepsilon r} \quad \text{V/m} \tag{4.12.2}$$

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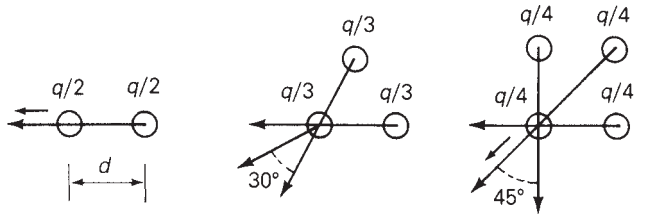

FIGURE 4.26
Vector addition of electric fields at the surface of one conductor in a bundle

Equation (4.12.2) represents an average value for an individual conductor in a bundle. The maximum electric field strength at the surface of one conductor due to all charges in a bundle, obtained by the vector addition of electric fields (as shown in Figure 4.26), is as follows:

Two-conductor bundle ($N_b = 2$):

$$E_{r\max} = \frac{q/2}{2\pi\varepsilon r} + \frac{q/2}{2\pi\varepsilon d} = \frac{q/2}{2\pi\varepsilon r}\left(1 + \frac{r}{d}\right)$$
$$= E_{\text{rave}}\left(1 + \frac{r}{d}\right) \tag{4.12.3}$$

Three-conductor bundle ($N_b = 3$):

$$E_{r\max} = \frac{q/3}{2\pi\varepsilon}\left(\frac{1}{r} + \frac{2\cos 30^\circ}{d}\right) = E_{\text{rave}}\left(1 + \frac{r\sqrt{3}}{d}\right) \tag{4.12.4}$$

Four-conductor bundle ($N_b = 4$):

$$E_{r\max} = \frac{q/4}{2\pi\epsilon}\left(\frac{1}{r} + \frac{1}{d\sqrt{2}} + \frac{2\cos 45^\circ}{d}\right) = E_{\text{rave}}\left[1 + \frac{r}{d}(2.1213)\right] \tag{4.12.5}$$

Although the electric field strength at ground level is much less than at conductor surfaces where corona occurs, there are still capacitive coupling effects. Charges are induced on ungrounded equipment, such as vehicles with rubber tires located near a line. If a person contacts the vehicle and ground, a discharge current will flow to ground. Transmission-line heights are designed to maintain discharge currents below prescribed levels for any equipment that may be on the right-of-way. Table 4.5 shows examples of maximum ground-level electric field strength.

As shown in Figure 4.27, the ground-level electric field strength due to charged conductor k and its image conductor is perpendicular to the earth plane, with a value of

$$E_k(w) = \left(\frac{q_k}{2\pi\varepsilon}\right)\frac{2\cos\theta}{\sqrt{y_k^2 + (w - x_k)^2}}$$
$$= \left(\frac{q_k}{2\pi\varepsilon}\right)\frac{2y_k}{y_k^2 + (w - x_k)^2} \quad \text{V/m} \tag{4.12.6}$$

where (x_k, y_k) are the horizontal and vertical coordinates of conductor k with respect to reference point R, w is the horizontal coordinate of the ground-level point where the electric field strength is to be determined, and q_k is the charge on

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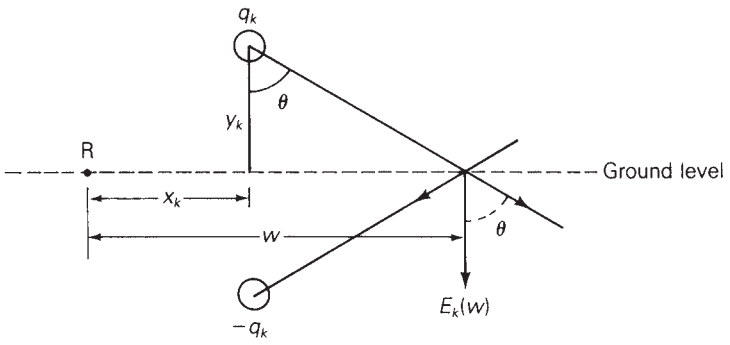
Line Voltage (kV_{rms})	Maximum Ground-Level Electric Field Strength (kV_{rms}/m)
23 (1ϕ)	0.01–0.025
23 (3ϕ)	0.01–0.05
115	0.1–0.2
345	2.3–5.0
345 (double circuit)	5.6
500	8.0
765	10.0

TABLE 4.5

Examples of maximum ground-level electric field strength versus transmission-line voltage [1] (© Copyright 1987. Electric Power Research Institute (EPRI), Publication Number EL-2500. *Transmission Line Reference Book, 345-kV and Above, Second Edition, Revised*. Reprinted with permission.)

FIGURE 4.27

Ground-level electric field strength due to an overhead conductor and its image



conductor k. The total ground-level electric field is the phasor sum of terms $E_k(w)$ for all overhead conductors. A lateral profile of ground-level electric field strength is obtained by varying w from the center of the line to the edge of the right-of-way.

EXAMPLE 4.9

Conductor surface and ground-level electric field strengths: single-phase line

For the single-phase line of Example 4.8, calculate the conductor surface electric field strength in kV_{rms}/cm. Also calculate the ground-level electric field in kV_{rms}/m directly under conductor x. The line voltage is 20 kV.

SOLUTION

From Example 4.8, $C_{xy} = 5.178 \times 10^{-12}$ F/m. Using (4.9.3) with $V_{xy} = 20\underline{/0°}$ kV,

$$q_x = -q_y = (5.178 \times 10^{-12})(20 \times 10^3\underline{/0°}) = 1.036 \times 10^{-7}\underline{/0°} \quad \text{C/m}$$

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From (4.12.1), the conductor surface electric field strength is, with $r = 0.00701$ m,

$$E_r = \frac{1.036 \times 10^{-7}}{(2\pi)(8.854 \times 10^{-12})(0.00701)} \frac{\text{V}}{\text{m}} \times \frac{\text{kV}}{1000\ \text{V}} \times \frac{\text{m}}{100\ \text{cm}}$$

$$= 2.66\ \text{kV}_{\text{rms}}/\text{cm}$$

Selecting the center of the line as the reference point R, the coordinates (x_x, y_x) for conductor x are $(-0.762$ m, 5.49 m) and $(+0.762$ m, 5.49 m) for conductor y. The ground-level electric field directly under conductor x, where $w = -0.762$ m, is, from (4.12.6),

$$E(-0.762) = E_x(-0.762) + E_y(-0.762)$$

$$= \frac{1.036 \times 10^{-7}}{(2\pi)(8.85 \times 10^{-12})} \left[\frac{(2)(5.49)}{(5.49)^2} - \frac{(2)(5.49)}{(5.49)^2 + (0.762 + 0.762)^2} \right]$$

$$= 1.862 \times 10^3 (0.364 - 0.338) = 48.5 \underline{/0^\circ}\ \text{V/m} = 0.0485\ \text{kV/m}$$

For this 20-kV line, the electric field strengths at the conductor surface and at ground level are low enough to be of relatively small concern. For EHV lines, electric field strengths and the possibility of corona and shock hazard are of more concern.

4.13 PARALLEL CIRCUIT THREE-PHASE LINES

If two parallel three-phase circuits are close together, either on the same tower as in Figure 4.3, or on the same right-of-way, there are mutual inductive and capacitive couplings between the two circuits. When calculating the equivalent series impedance and shunt admittance matrices, these couplings should not be neglected unless the spacing between the circuits is large.

Consider the double-circuit line shown in Figure 4.28. For simplicity, assume that the lines are not transposed. Since both are connected in parallel. they have the same series-voltage drop for each phase. Following the same procedure as in Section 4.7, write $2(6 + N)$ equations similar to (4.7.6) through (4.7.9): six

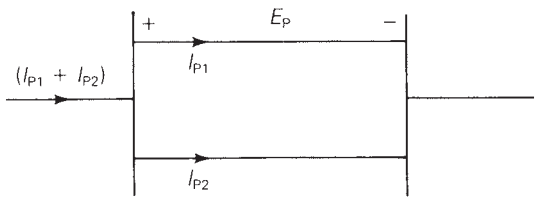

FIGURE 4.28

Single-line diagram of a double-circuit line

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equations for the overhead phase conductors, N equations for the overhead neutral conductors, and $(6 + N)$ equations for the earth return conductors. After lumping the neutral voltage drop into the voltage drops across the phase conductors, and eliminating the neutral and earth return currents, resulting in

$$\begin{bmatrix} \boldsymbol{E}_{\mathrm{P}} \\ \boldsymbol{E}_{\mathrm{P}} \end{bmatrix} = \boldsymbol{Z}_{\mathrm{P}} \begin{bmatrix} \boldsymbol{I}_{\mathrm{P1}} \\ \boldsymbol{I}_{\mathrm{P2}} \end{bmatrix} \tag{4.13.1}$$

where $\boldsymbol{E}_{\mathrm{P}}$ is the vector of phase-conductor voltage drops (including the neutral voltage drop), and $\boldsymbol{I}_{\mathrm{P1}}$ and $\boldsymbol{I}_{\mathrm{P2}}$ are the vectors of phase currents for lines 1 and 2. $\boldsymbol{Z}_{\mathrm{p}}$ is a 6×6 impedance matrix. Solving (4.13.1),

$$\left[\begin{array}{c} \boldsymbol{I}_{\mathrm{P1}} \\ \hline \boldsymbol{I}_{\mathrm{P2}} \end{array}\right] = \boldsymbol{Z}_{\mathrm{P}}^{-1} \left[\begin{array}{c} \boldsymbol{E}_{\mathrm{P}} \\ \hline \boldsymbol{E}_{\mathrm{P}} \end{array}\right] = \left[\begin{array}{c|c} \boldsymbol{Y}_A & \boldsymbol{Y}_B \\ \hline \boldsymbol{Y}_C & \boldsymbol{Y}_D \end{array}\right] \left[\begin{array}{c} \boldsymbol{E}_{\mathrm{P}} \\ \hline \boldsymbol{E}_{\mathrm{P}} \end{array}\right] = \begin{bmatrix} (\boldsymbol{Y}_A + \boldsymbol{Y}_B) \\ (\boldsymbol{Y}_C + \boldsymbol{Y}_D) \end{bmatrix} \boldsymbol{E}_{\mathrm{P}} \tag{4.13.2}$$

where $\boldsymbol{Y}_A$, $\boldsymbol{Y}_B$, $\boldsymbol{Y}_C$, and $\boldsymbol{Y}_D$ are obtained by partitioning $\boldsymbol{Z}_{\mathrm{P}}^{-1}$ into four 3×3 matrices. Adding $\boldsymbol{I}_{\mathrm{P1}}$ and $\boldsymbol{I}_{\mathrm{P2}}$,

$$(\boldsymbol{I}_{\mathrm{P1}} + \boldsymbol{I}_{\mathrm{P2}}) = (\boldsymbol{Y}_A + \boldsymbol{Y}_B + \boldsymbol{Y}_C + \boldsymbol{Y}_D)\boldsymbol{E}_{\mathrm{P}} \tag{4.13.3}$$

and solving for $\boldsymbol{E}_{\mathrm{P}}$,

$$\boldsymbol{E}_{\mathrm{P}} = \boldsymbol{Z}_{\mathrm{Peq}}(\boldsymbol{I}_{\mathrm{P1}} + \boldsymbol{I}_{\mathrm{P2}}) \tag{4.13.4}$$

where

$$\boldsymbol{Z}_{\mathrm{Peq}} = (\boldsymbol{Y}_A + \boldsymbol{Y}_B + \boldsymbol{Y}_c + \boldsymbol{Y}_D)^{-1} \tag{4.13.5}$$

$\boldsymbol{Z}_{\mathrm{Peq}}$ is the equivalent 3×3 series phase impedance matrix of the double-circuit line. Note that in (4.13.5) the matrices $\boldsymbol{Y}_B$ and $\boldsymbol{Y}_C$ account for the inductive coupling between the two circuits.

An analogous procedure can be used to obtain the shunt admittance matrix. Following the ideas of Section 4.11, write $(6 + N)$ equations similar to (4.11.4). After eliminating the neutral wire charges, the result is

$$\left[\begin{array}{c} \boldsymbol{q}_{\mathrm{P1}} \\ \hline \boldsymbol{q}_{\mathrm{P2}} \end{array}\right] = \mathbf{C}_{\mathrm{P}} \left[\begin{array}{c} \boldsymbol{V}_{\mathrm{P}} \\ \hline \boldsymbol{V}_{\mathrm{P}} \end{array}\right] = \left[\begin{array}{c|c} \mathbf{C}_A & \mathbf{C}_B \\ \hline \mathbf{C}_C & \mathbf{C}_D \end{array}\right] \left[\begin{array}{c} \boldsymbol{V}_{\mathrm{P}} \\ \hline \boldsymbol{V}_{\mathrm{P}} \end{array}\right] = \begin{bmatrix} (\mathbf{C}_A + \mathbf{C}_B) \\ (\mathbf{C}_C + \mathbf{C}_D) \end{bmatrix} \boldsymbol{V}_{\mathrm{P}} \tag{4.13.6}$$

where $\boldsymbol{V}_{\mathrm{p}}$ is the vector of phase-to-neutral voltages, and $\boldsymbol{q}_{\mathrm{Pl}}$ and $\boldsymbol{q}_{\mathrm{P2}}$ are the vectors of phase-conductor charges for lines 1 and 2. $\mathbf{C}_{\mathrm{p}}$ is a 6×6 capacitance matrix that is partitioned into four 3×3 matrices $\mathbf{C}_A$, $\mathbf{C}_B$, $\mathbf{C}_C$, and $\mathbf{C}_D$. Adding $\boldsymbol{q}_{\mathrm{Pl}}$ and $\boldsymbol{q}_{\mathrm{P2}}$,

$$(\boldsymbol{q}_{\mathrm{P1}} + \boldsymbol{q}_{\mathrm{P2}}) = \mathbf{C}_{\mathrm{Peq}} \boldsymbol{V}_{\mathrm{P}} \tag{4.13.7}$$

where

$$\mathbf{C}_{\mathrm{Peq}} = (\mathbf{C}_A + \mathbf{C}_B + \mathbf{C}_C + \mathbf{C}_D) \tag{4.13.8}$$

Also,

$$\boldsymbol{Y}_{\mathrm{Peq}} = j\omega \mathbf{C}_{\mathrm{Peq}} \tag{4.13.9}$$

$\boldsymbol{Y}_{\mathrm{Peq}}$ is the equivalent 3×3 shunt admittance matrix of the double-circuit line. The matrices $\mathbf{C}_B$ and $\mathbf{C}_C$ in (4.13.8) account for the capacitive coupling between the two circuits.

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These ideas can be extended in a straightforward fashion to more than two parallel circuits.

PROBLEMS

SECTION 4.2

4.1 A copper conductor is round with a diameter of ½ cm. What would be the dc resistance of a 3-meter long section of this conductor at 20°C? What about at 50°C? Use Table 4.3 to find the associated data for copper.

4.2 For a solid, round, hard-drawn aluminum conductor with a length of 4000 meters and diameter of 1 cm, what is the maximum temperature at which the resistance can be kept under 2Ω?

4.3 A transmission-line cable with a length of 2 km consists of 19 strands of identical copper conductors, each 1.5 mm in diameter. Because of the twist of the strands, the actual length of each conductor is increased by 5%. Determine the resistance of the cable if the resistivity of copper is 1.72 $\mu\Omega$-cm at 20°C.

4.4 One thousand circular mils or 1 kcmil is sometimes designated by the abbreviation MCM. Data for commercial bare-aluminum electrical conductors lists a 60-Hz resistance of 0.0880 ohm per kilometer at 75°C for a 795-MCM AAC conductor.

(a) Determine the cross-sectional conducting area of this conductor in square meters.

(b) Find the 60-Hz resistance of this conductor in ohms per kilometer at 50°C.

4.5 A 60-Hz, 500-kV, three-phase overhead transmission line has three ACSR 900 kcmil 54/3 conductors per phase. Determine the 60-Hz resistance of this line in ohms per kilometer per phase at 50°C.

4.6 A three-phase overhead transmission line is designed to deliver 190.5 MVA at 220 kV over a distance of 63 km, such that the total transmission line loss is not to exceed 2.5% of the rated line MVA. Given the resistivity of the conductor material to be 2.84×10^{-8} Ω-m, determine the required conductor diameter and the conductor size in circular mils. Neglect power losses due to insulator leakage currents and corona.

4.7 A single-phase transmission feeder segment has two solid, copper conductors that span a 610-meter distance. The segment needs to carry up to 18 A of current. What diameter wire would be needed to ensure the line real power losses do not exceed 12 W at 50°C?

SECTIONS 4.4 AND 4.5

4.8 A 60-Hz, single-phase two-wire overhead line has solid cylindrical copper conductors with a 1.5-cm diameter. The conductors are arranged in a horizontal configuration with 0.5-m spacing. Calculate in mH/km

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(a) the inductance of each conductor due to internal flux linkages only, (b) the inductance of each conductor due to both internal and external flux linkages, and (c) the total inductance of the line.

4.9 Rework Problem 4.8 if the diameter of each conductor is (a) increased by 20% to 1.8 cm, or (b) decreased by 20% to 1.2 cm without changing the phase spacing. Compare the results with those of Problem 4.8.

4.10 A 60-Hz, three-phase three-wire overhead line has solid cylindrical conductors arranged in the form of an equilateral triangle with 1 m conductor spacing. The conductor diameter is 1 cm. Calculate the positive-sequence inductance in H/m and the positive-sequence inductive reactance in Ω/km.

4.11 Rework Problem 4.10 if the phase spacing is (a) increased by 20% to 1.2 m or (b) decreased by 20% to 0.8 m. Compare the results with those of Problem 4.10.

4.12 Find the inductive reactance per kilometer of a single-phase overhead transmission line operating at 60 Hz given the conductors to be *Partridge* and the spacing between centers to be 10 m.

4.13 A single-phase overhead transmission line consists of two solid aluminum conductors having a radius of 3 cm with a spacing 3.5 m between centers. (a) Determine the total line inductance in mH/m. (b) Given the operating frequency to be 60 Hz, find the total inductive reactance of the line in Ω/km. (c) If the spacing is doubled to 7 m, how does the reactance change?

4.14 (a) In practice, one deals with the inductive reactance of the line per phase per kilometer and use the logarithm to the base 10. Show that Eq. (4.5.9) of the text can be rewritten as

$$x = k \log \frac{\mathrm{D}}{r'} \text{ ohms per kilometer per phase}$$

$$= x_d + x_a$$

where $x_d = k \log \mathrm{D}$ is the inductive reactance spacing factor in ohms per kilometer

$x_a = k \log \frac{1}{r'}$ is the inductive reactance at 1-m spacing in ohms per kilometer

$k = 2.893 \times 10^{-3} f = 1.736$ at 60 Hz

(b) Determine the inductive reactance per kilometer per phase at 60 Hz for a single-phase line with phase separation of 3 m and conductor radius of 2 cm. If the spacing is doubled, how does the reactance change?

SECTION 4.6

4.15 Find the GMR of a stranded conductor consisting of six outer strands surrounding and touching one central strand, all strands having the same radius r.

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4.16 A bundle configuration for UHV lines (above 1000 kV) has identical conductors equally spaced around a circle, as shown in Figure 4.29. N_b is the number of conductors in the bundle, A is the circle radius, and D_S is the conductor GMR. Using the distance D_{1n} between conductors 1 and n given by $D_{1n} = 2A \sin[(n-1)\pi/N_b]$ for $n = 1, 2, \ldots, N_b$, and the following trigonometric identity:

$$[2\sin(\pi/N_b)][2\sin(2\pi/N_b)][2\sin(3\pi/N_b)]\cdots[2\sin\{(N_b-1)\pi/N_b\}] = N_b$$

show that the bundle GMR, denoted D_{SL}, is

$$D_{SL} = [N_b D_S A^{(N_b-1)}]^{(1/N_b)}$$

Also show that the above formula agrees with (4.6.19) through (4.6.21) for EHV lines with $N_b = 2$, 3, and 4.

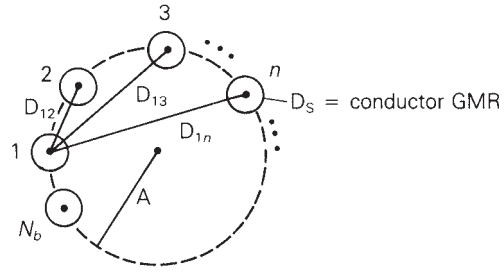

FIGURE 4.29
Bundle configuration for Problem 4.16

4.17 Determine the GMR of each of the unconventional stranded conductors shown in Figure 4.30. All strands have the same radius r.

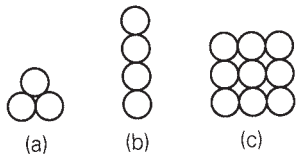

FIGURE 4.30
Unconventional stranded conductors for Problem 4.17

4.18 Calculate the per-distance series impedance (resistance and reactance) for a three-phase, completely transposed transmission line consisting of one ACSR 954 kcmil conductor per phase, arranged vertically with a phase spacing of 7.5 m. Assume this line is operated on a 60-Hz system.

4.19 How are the results of Problem 4.18 affected if (a) the phase spacing of 7.5 m is increased by 10%, (b) the phase spacing is decreased by 10%, (c) an ACSR 900 kcmil conductor is used instead, or (d) ACSR 1033.5 kcmil conductor is used?

4.20 Calculate the inductive reactance in Ω/km of a bundled 500-kV, 60-Hz, three-phase completely transposed overhead line having three ACSR 1113-kcmil conductors per bundle, with 0.5 m between conductors in the bundle. The horizontal phase spacings between bundle centers are 12, 12, and 24 m.

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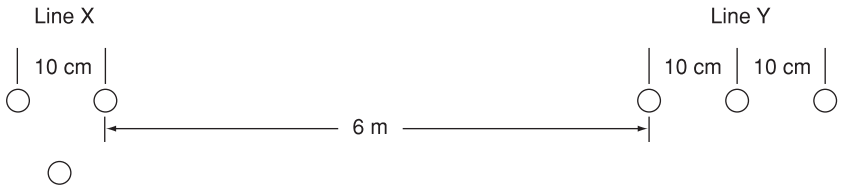
4.21 Rework Problem 4.20 if the bundled line has (a) three ACSR, 1351-kcmil conductors per phase, or (b) three ACSR, 900-kcmil conductors per phase, without changing the bundle spacing or the phase spacings between bundle centers. Compare the results with those of Problem 4.20.

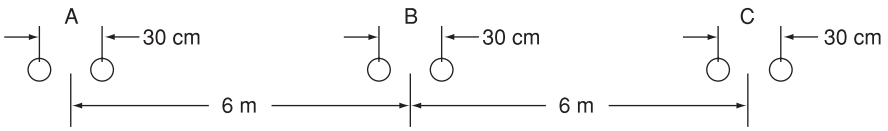
4.22 The conductor configuration of a bundled single-phase overhead transmission line is shown in Figure 4.31. Line X has its three conductors situated at the corners of an equilateral triangle with 10-cm spacing. Line Y has its three conductors arranged in a horizontal configuration with 10-cm spacing. All conductors are identical, solid-cylindrical conductors, each with a radius of 2 cm. Find the equivalent representation in terms of the geometric mean radius of each bundle and a separation that is the geometric mean distance.

FIGURE 4.31
Problem 4.22



4.23 Figure 4.32 shows the conductor configuration of a completely transposed three-phase overhead transmission line with bundled phase conductors. All conductors have a radius of 0.74 cm with a 30-cm bundle spacing. (a) Determine the inductance per phase in mH/km. (b) Find the inductive line reactance per phase in Ω/km at 60 Hz.

FIGURE 4.32
Problem 4.23



4.24 Consider a three-phase overhead line made up of three phase conductors: Linnet, 336.4 kcmil, and ACSR 26/7. The line configuration is such that the horizontal separation between center of C and that of A is 1 m, and between that of A and B is also 1 m in the same line; the vertical separation of A from the line of C–B is 0.3m. If the line is operated at 60 Hz at a conductor temperature of 75°C, determine the inductive reactance per phase in Ω/km,

(a) by using the formula given in Problem 4.14 (a), and
(b) by using (4.6.18) from the text.

4.25 For the overhead line of configuration shown in Figure 4.33 operating at 60 Hz and a conductor temperature of 70°C, determine the resistance per phase, inductive reactance in ohms/mile/phase, and the current-carrying capacity of the overhead line. Each conductor is ACSR Cardinal of Table A.4.

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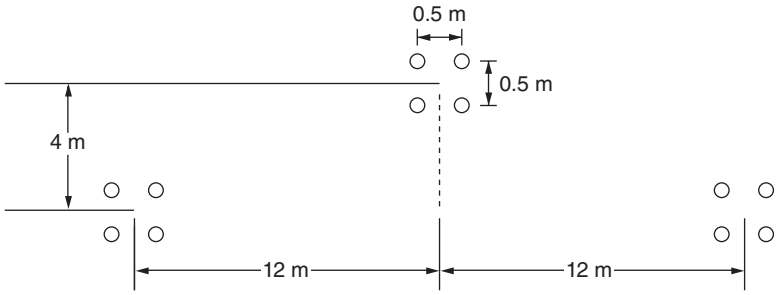

FIGURE 4.33
Line configuration for Problem 4.25

4.26 Consider a symmetrical bundle with N subconductors arranged in a circle of radius A. The inductance of a single-phase symmetrical bundle-conductor line is given by

$$L = 2 \times 10^{-7} \ln \frac{GMD}{GMR} \text{ H/m}$$

where GMR is given by $[Nr'(A)^{N-1}]^{1/N}$ and $r' = (e^{-1/4}r)$, r being the subconductor radius, and GMD is approximately the distance D between the bundle centers. Note that A is related to the subconductor spacing S in the bundle circle by $S = 2A \sin(\pi/N)$.

Now consider a 965-kV, single-phase, bundle-conductor line with eight subconductors per phase, phase spacing D = 20 m, and the subconductor spacing S = 45.72 cm. Each subconductor has a diameter of 4.572 cm. Determine the line inductance in H/m.

4.27 Figure 4.34 shows double-circuit conductors' relative positions in segment 1 of transposition of a completely transposed three-phase overhead transmission line. The inductance is given by

$$L = 2 \times 10^{-7} \ln \frac{GMD}{GMR} \text{ H/m/phase}$$

where $GMD = (D_{AB_{eq}} D_{BC_{eq}} D_{AC_{eq}})^{1/3}$

with mean distances defined by equivalent spacings

$$D_{AB_{eq}} = (D_{12}D_{1'2'}D_{12'}D_{1'2})^{1/4}$$

$$D_{BC_{eq}} = (D_{23}D_{2'3'}D_{2'3}D_{23'})^{1/4}$$

$$D_{AC_{eq}} = (D_{13}D_{1'3'}D_{13'}D_{1'3})^{1/4}$$

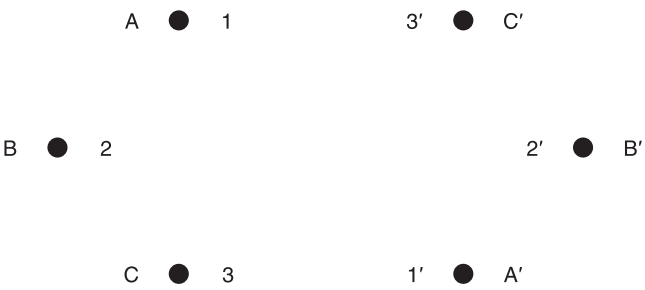

FIGURE 4.34
For Problem 4.27 (Double-circuit conductor configuration)

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And $GMR = [(GMR)_A(GMR)_B(GMR)_C]^{1/3}$ with phase GMRs defined by

$$(GMR)_A = [r'D_{11'}]^{1/2};\ (GMR)_B = [r'D_{22'}]^{1/2};\ \ (GMR)_C = [r'D_{33'}]^{1/2}$$

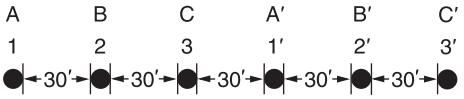
and r' is the GMR of phase conductors.

Now consider a 345-kV, three-phase, double-circuit line with phase-conductor's GMR of 0.0588 ft and the horizontal conductor configuration shown in Figure 4.35.

(a) Determine the inductance per meter per phase in Henries (H).
(b) Calculate the inductance of just one circuit and then divide by 2 to obtain the inductance of the double circuit.

FIGURE 4.35 For Problem 4.27

Find the relative error involved



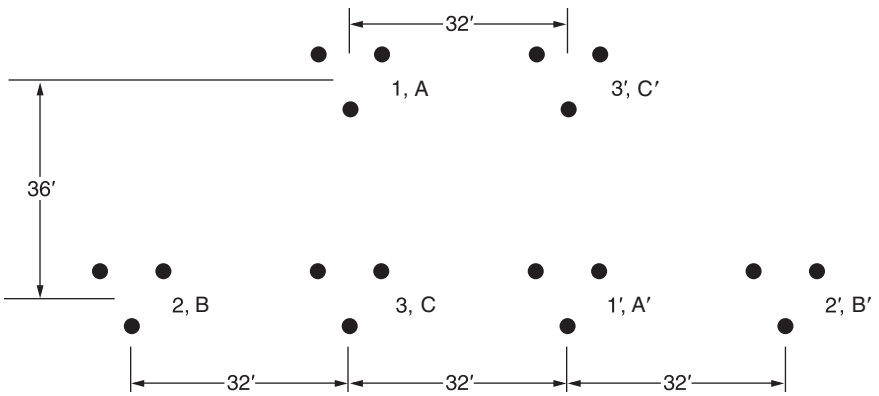
4.28 For the case of double-circuit, bundle-conductor lines, the same method indicated in Problem 4.27 applies with r' replaced by the bundle's GMR in the calculation of the overall GMR.

Now consider a double-circuit configuration shown in Figure 4.36 that belongs to a 500-kV, three-phase line with bundle conductors of three subconductors at 21-in. spacing. The GMR of each subconductor is given to be 0.0485 ft.

Determine the inductive reactance of the line in ohms per mile per phase. You may use

$$X_L = 0.2794 \log \frac{GMD}{GMR}\ \Omega/\text{mi/phase}$$

FIGURE 4.36 Configuration for Problem 4.28



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4.29 Reconsider Problem 4.28 with an alternate phase placement given below:

	Physical Position					
	1	2	3	1′	2′	3′
Phase Placement	A	B	B′	C	C′	A′

Calculate the inductive reactance of the line in Ω/km/phase.

4.30 Reconsider Problem 4.28 with still another alternate phase placement shown below.

	Physical Position					
	1	2	3	1′	2′	3′
Phase Placement	C	A	B	B′	A′	C′

Find the inductive reactance of the line in Ω/km/phase.

4.31 Figure 4.37 shows the conductor configuration of a three-phase transmission line and a telephone line supported on the same towers. The power line carries a balanced current of 250 A/phase at 60 Hz, while the telephone line is directly located below phase b. Assume balanced three-phase currents in the power line. Calculate the voltage per kilometer induced in the telephone line.

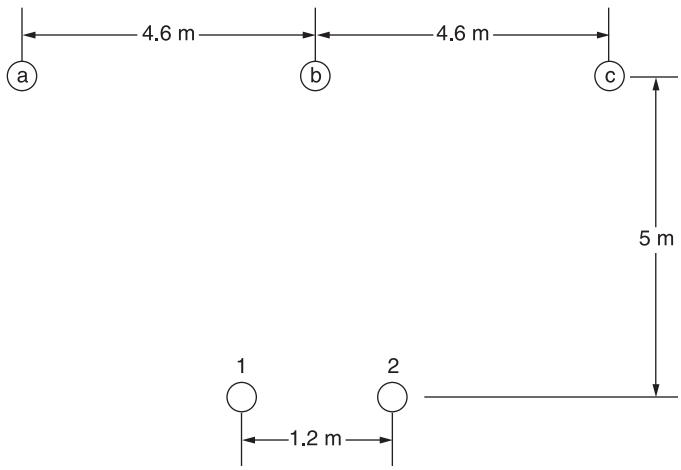

FIGURE 4.37
Conductor layout for Problem 4.31

SECTION 4.9

4.32 Calculate the capacitance-to-neutral in F/m and the admittance-to-neutral in S/km for the single-phase line in Problem 4.8. Neglect the effect of the earth plane.

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4.33 Rework Problem 4.32 if the diameter of each conductor is (a) increased by 20% to 1.8 cm, or (b) decreased by 20% to 1.2 cm. Compare the results with those of Problem 4.32.

4.34 Calculate the capacitance-to-neutral in F/m and the admittance-to-neutral in S/km for the three-phase line in Problem 4.10. Neglect the effect of the earth plane.

4.35 Rework Problem 4.34 if the phase spacing is (a) increased by 20% to 146.4 cm, or (b) decreased by 20% to 96.6 cm. Compare the results with those of Problem 4.34.

4.36 The line of Problem 4.23 as shown in Figure 4.32 is operating at 60 Hz. Determine (a) the line-to-neutral capacitance in nF/km per phase; (b) the capacitive reactance in Ω-km per phase; and (c) the capacitive reactance in Ω per phase for a line length of 100 km.

4.37 (a) In practice, one deals with the capacitive reactance of the line in ohms · km to neutral. Show that Eq. (4.9.15) of the text can be rewritten as

$$\begin{aligned} X_C &= k' \log \frac{D}{r} \quad \text{ohms} \cdot \text{km to neutral} \\ &= x'_d + x'_a \end{aligned}$$

where $x'_d = k' \log D$ is the capacitive reactance spacing factor

$x'_a = k' \log \frac{1}{r}$ is the capacitive reactance at 1-m spacing

$k' = (4.1 \times 10^6)/f = 0.06833 \times 10^6$ at $f = 60$ Hz

(b) Determine the capacitive reactance in Ω · km for a single-phase line of Problem 4.14. If the spacing is doubled, how does the reactance change?

4.38 The capacitance per phase of a balanced three-phase overhead line is given by

$$C = \frac{0.02417}{\log(GMD/r)} \mu f/\text{km/phase}$$

For the line of Problem 4.24, determine the capacitive reactance per phase in Ω · km.

SECTION 4.10

4.39 Calculate the capacitance-to-neutral in F/m and the admittance-to-neutral in S/km for the three-phase line in Problem 4.18.

4.40 Rework Problem 4.39 if the phase spacing between adjacent conductors is (a) increased by 10% to 8.25 m, or (b) decreased by 10% to 6.75 m. Compare the results with those of Problem 4.39.

4.41 Calculate the capacitance-to-neutral in F/m and the admittance-to-neutral in S/km for the line in Problem 4.20. Also, calculate the total reactive power in Mvar/km supplied by the line capacitance when it is operated at 500 kV. Neglect the effect of the earth plane.

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4.42 Rework Problem 4.41 if the bundled line has (a) three ACSR, 1351-kcmil conductors per phase, or (b) three ACSR, 900 kcmil conductors per phase without changing the bundle spacing or the phase spacings between bundle centers.

4.43 Three ACSR *Drake* conductors are used for a three-phase overhead transmission line operating at 60 Hz. The conductor configuration is in the form of an isosceles triangle with sides of 6, 6, and 12 m (a) Find the capacitance-to-neutral and capacitive reactance-to-neutral for each 1-kilometer length of line. (b) For a line length of 280 km and a normal operating voltage of 220 kV, determine the capacitive reactance-to-neutral for the entire line length as well as the charging current per kilometer and total three-phase reactive power supplied by the line capacitance.

4.44 Consider the line of Problem 4.25. Calculate the capacitive reactance per phase in $\Omega \cdot$ km.

SECTION 4.11

4.45 For an average line height of 10 m, determine the effect of the earth on capacitance for the single-phase line in Problem 4.32. Assume a perfectly conducting earth plane.

4.46 A three-phase 60-Hz, 125-km overhead transmission line has flat horizontal spacing with three identical conductors. The conductors have an outside diameter of 3.28 cm with 12 m between adjacent conductors. (a) Determine the capacitive reactance-to-neutral in Ω-m per phase and the capacitive reactance of the line in Ω per phase. Neglect the effect of the earth plane. (b) Assuming that the conductors are horizontally placed 20 m above ground, repeat part (a) while taking into account the effect of ground. Consider the earth plane to be a perfect conductor.

4.47 For the single-phase line of Problem 4.14(b), if the height of the conductor above ground is 24 m, determine the line-to-line capacitance in F/m. Neglecting earth effect, evaluate the relative error involved. If the phase separation is doubled, repeat the calculations.

4.48 The capacitance of a single-circuit, three-phase transposed line with the configuration shown in Figure 4.38, including ground effect, and with conductors not equilaterally spaced is given by

$$C_{a\eta} \frac{2\pi\varepsilon_0}{\ln \dfrac{D_{eq}}{r} - \ln \dfrac{H_m}{H_s}} \text{ F/m line-to-neutral}$$

where $D_{eq} = \sqrt[3]{D_{12}D_{23}D_{13}} = \text{GMD}$

$r =$ conductor's outside radius

$H_m = (H_{12}H_{23}H_{13})^{1/3}$

$H_s = (H_1H_2H_3)^{1/3}$

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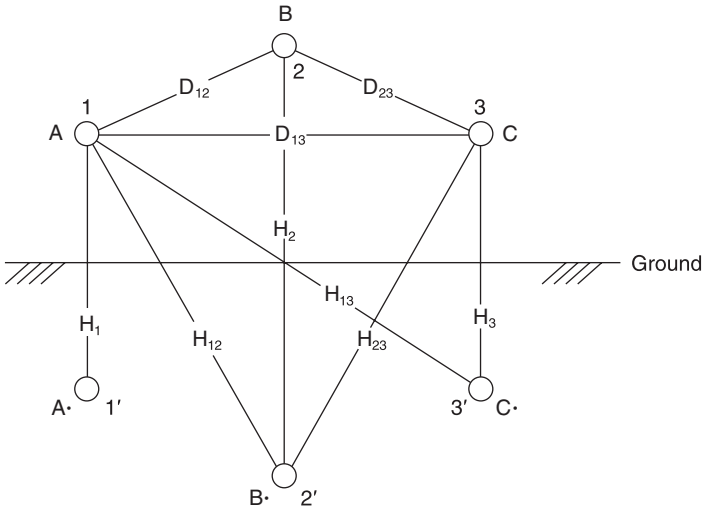
FIGURE 4.38

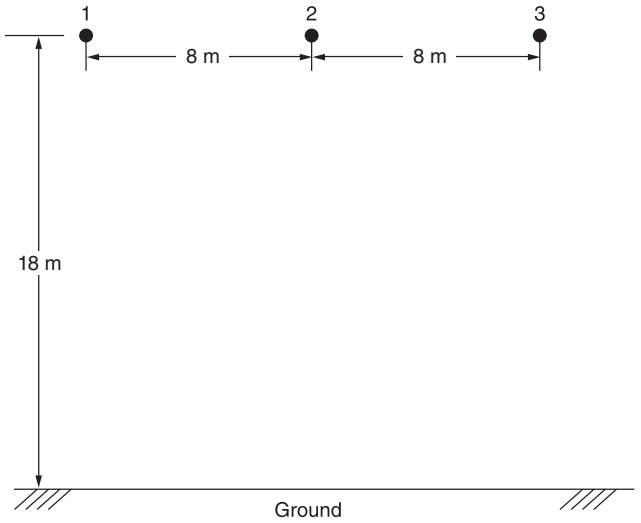
Three-phase, single-circuit line configuration including ground effect for Problem 4.48



(a) Now consider Figure 4.39 in which the configuration of a three-phase, single circuit, 345-kV line with conductors having an outside diameter of 2.5 cm is shown. Determine the capacitance to neutral in F/m, including the ground effect.

FIGURE 4.39

Configuration for Problem 4.48(a)



(b) Next, neglecting the effect of ground, see how the value changes.

4.49 The capacitance-to-neutral, neglecting the ground effect, for the three-phase, single-circuit, bundle-conductor line is given by

$$C_{a\eta} = \frac{2\pi\varepsilon_0}{\ell\eta\left(\frac{\text{GMD}}{\text{GMR}}\right)} \text{ F/m line-to-neutral}$$

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where $\text{GMD} = (\text{D}_{AB}\text{D}_{BC}\text{D}_{AC})^{1/3}$

$\text{GMR} = [rN(\text{A})^{N-1}]^{1/N}$

in which N is the number of subconductors of the bundle conductor on a circle of radius A, and each subconductor has an outside radius of r. The capacitive reactance in mega-ohms for 1 km of line at 60 Hz can be shown to be

$$\text{X}_C = 0.11 \log\left(\frac{\text{GMD}}{\text{GMR}}\right) = X'_a + \text{X}'_d$$

where $\text{X}'_a = 0.11 \log\left(\frac{1}{\text{GMR}}\right)$ and $\text{X}'_d = 0.0683 \log(\text{GMD})$.

Note that A is related to the bundle spacing S given by

$$\text{A} = \frac{S}{2 \sin\left(\frac{\pi}{N}\right)} \quad \text{for } N > 1$$

Using the above information for the configuration shown in Figure 4.40, compute the capacitance-to-neutral in F/m and the capacitive reactance in $\Omega \cdot$ km to neutral for the three-phase, 765-kV, 60-Hz, single-circuit, bundle-conductor line ($N = 4$) with subconductor's outside diameter of 3 cm and subconductor spacing (S) of 46 cm.

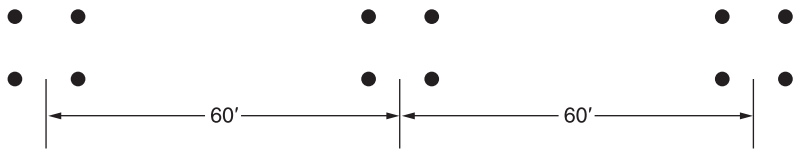

FIGURE 4.40
Configuration for Problem 4.49

SECTION 4.12

4.50 Calculate the conductor surface electric field strength in kVrms/cm for the single-phase line in Problem 4.32 when the line is operating at 20 kV. Also, calculate the ground-level electric field strength in kVrms/m directly under one conductor. Assume a line height of 10 m.

4.51 Rework Problem 4.50 if the diameter of each conductor is (a) increased by 25% to 1.875 cm, or (b) decreased by 25% to 1.125 cm without changing the phase spacings. Compare the results with those of Problem 4.50.

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5 Transmission Lines: Steady-State Operation

Series capacitor installation at Goshen Substation, Goshen, Idaho, USA rated at 395 kV, 965 Mvar. (Courtesy of PacifiCorp.)

LEARNING OBJECTIVES

At the conclusion of this chapter, you should be able to:

1. Represent a single-phase or balanced three-phase transmission line under normal steady-state operating conditions as a two-port network and as an equivalent π *circuit*.
2. Understand the concepts of voltage regulation, surge impedance loading, and wavelength.
3. Understand line loadability, including the thermal limit, voltage-drop limit, and steady-state stability limit.
4. Discuss reactive compensation techniques, including shunt reactive compensation to improve voltage regulation and series capacitive compensation to increase line loadability.

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This chapter analyzes the performance of single-phase and balanced three-phase transmission lines under normal steady-state operating conditions. Expressions for voltage and current at any point along a line are developed where the distributed nature of the series impedance and shunt admittance is taken into account. A line is treated here as a two-port network for which the *ABCD* parameters and an equivalent π circuit are derived. Also, approximations are given for a medium-length line lumping the shunt admittance, for a short line neglecting the shunt admittance, and for a lossless line, assuming zero series resistance and shunt conductance. The concepts of *surge impedance loading* and transmission-line *wavelength* are also presented.

An important issue discussed in this chapter is *voltage regulation*. Transmission-line voltages are generally high during light load periods and low during heavy load periods. Voltage regulation, defined in Section 5.1, refers to the change in line voltage as line loading varies from no-load to full load.

Another important issue discussed here is line loadability. Three major line-loading limits are (1) the thermal limit, (2) the voltage-drop limit, and (3) the steady-state stability limit. Thermal and voltage-drop limits are discussed in Section 5.1. The theoretical steady-state stability limit, discussed in Section 5.4 for lossless lines and in Section 5.5 for lossy lines, refers to the ability of synchronous machines at the ends of a line to remain in synchronism. Practical line loadability is discussed in Section 5.6.

Section 5.7 covers line compensation techniques for improving voltage regulation and for raising line loadings closer to the thermal limit.

CASE STUDY

Power system engineers and planners often consider high-voltage direct current (HVDC) systems for bulk power transmission over long distances. There are two technologies within HVDC: (1) line-commutated converters (LCC), a mature technology that uses thyristors for switching; and (2) voltage-source converters (VSC), a newer technology that uses insulated-gate bipolar transistors (IGBTs) for switching. Each technology has its own suitable range of applications. LCC, which has been available since the 1970s, is commonly applied for high-capacity power transmission between two strong power systems (high short-circuit ratio) over long distances with limited dynamic requirements. The introduction of VSC in the late 1990s provided a significant improvement in dynamic performance. The following article compares the two technologies and describes the benefits of HVDC systems embedded within a synchronous ac network [6]. Regardless of which technology is deployed, HVDC offers a technically and economically preferred solution to link asynchronous grids, cross marine environments, and enable controllable power exchange. To see Figures 1 - 5 of the article in color, download the product. Sign in at www.cengage.com to download this product and to access its online resources.

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Opportunities For Embedded High-Voltage Direct Current*

Kevin Schlönleber, et al.
Hitachi ABB Power Grids Research.

Power system engineers and planners often consider high-voltage dc (HVdc) systems for bulk power transmission over long distances. There are two technologies within HVdc: line-commutated converters (LCC-HVdc) and voltage-source converters (VSC-HVdc). Each type has its own suitable range of applications. LCC-HVdc is a mature technology that has been available since the 1970s and uses thyristor valves. LCC-HVdc is commonly applied for high-capacity power transmission between two strong power systems (high short circuit ratio) over long distances with limited dynamic requirements (e.g., features such as fast power reversal, dynamic reactive power support, or black start are not available or are available only with limited performance). Currently installed systems reach ratings of 12 GW at a transmission voltage of ±1100 kV and may cover distances beyond 3000 km. The introduction of VSC-HVdc in the late 1990s initiated the use of self-commutated valves, employing insulated-gate bipolar transistors with a significant improvement in dynamic performance. Presently, the ratings of VSC-HVdc are lower than those of LCC-HVdc. VSC-HVdc systems are now installed with a rating of 3000 MW at a transmission voltage of ±525 kV, and higher ratings are under development. For special applications such as the connection of offshore wind power plants or connection to/between weak grids, VSC-HVdc often offers the only technoeconomically viable solution.

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Using VSC-HVdc to Transform Power Systems

Today's most prominent application for bulk power transmission with HVdc is still the interconnection of asynchronous grids (see Figure 1). HVdc interconnectors might deploy either LCC or VSC technology depending on the project's needs. Regardless of which technology is used, HVdc is the technically and economically preferred solution to link asynchronous grids, cross marine environments, and enable controllable power exchange. Due to the flexibility, compactness, and dynamic performance of VSC-HVdc, transmission system operators worldwide tend to favor this technology and use it in an extended range of applications, such as

- the integration of remote offshore wind resources to the power grid, which requires long submarine cables, compact offshore substations, and a voltage source at the remote end
- power from shore for offshore installations, such as oil and gas platforms
- city center power infeed (for megacities)
- the connection of remote islands
- flexible, multiterminal dc overlay grids.

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In recent years, embedded HVdc systems within synchronous ac systems have also received significant interest and are being planned and installed all over the world. An *embedded HVdc system* is defined as

> a dc link with at least two ends being physically connected within a single synchronous ac network. With such a connection, it can perform not only the basic function of bulk power transmission, but also, importantly, some additional control functions within the ac network such as power flow control, voltage control, system stability improvement and the mitigation of system cascading failure.

(See "CIGRE," Tech. Rep. 536, in the "For Further Reading" section for the definition.)

The applications of embedded HVdc systems include

- back-to-back HVdc schemes in highly meshed systems with the goal of power flow controllability, for example, ultrafast power flow control responding to contingencies, or to avoid loop flows
- parallel HVdc links to ac lines to increase the transfer capability of a combined corridor

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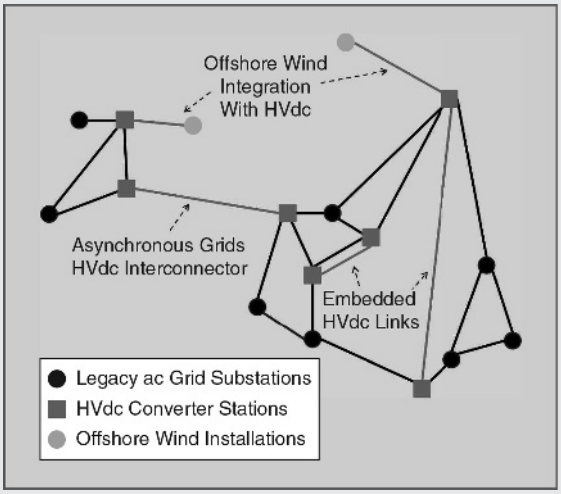

Figure 1 An example of HVdc links in a power system. The circles are substations of the legacy ac grid, the squares are HVdc converter stations, and the lines are transmission overhead lines or cables. (Go online to this article to see Figure 1 in color.)

by operating ac lines closer to their thermal limit

- long-distance HVdc corridors for the integration of remote renewable generation or to serve remote load centers within one synchronous grid.

The interesting point about embedded VSC-HVdc is not only its main functionality of bulk power transfer but also its additional features, which might increase the overall system capacity well above the actual capacity added by the HVdc.

Increasing Grid Capacity and Flexibility with Embedded VSC-HVdc

As previously discussed, an installed embedded VSC-HVdc system forms an integral part of the legacy power grid and is deployed for reasons other than its bulk power transmission capabilities. Power flow controllability, the measures used to increase power transfer capability, and ancillary services are of high interest for the secure operation of a synchronous grid. Some of the key benefits of embedded VSC-HVdc include the following:

- Dynamic/fast power flow controllability could support preventive and corrective security measures and avoid widely spreading contingencies in the ac grid.
- Reactive power support can boost the capability of the ac grid.
- Power oscillation damping controllers can free up some capacity reserve margins in the ac grid by suppressing poorly damped modes.

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These capabilities augment the power transfer capacity increase well above the actual installed rating of the HVdc system. Some ingredients of the (embedded) VSC-HVdc capabilities do not directly increase the power transfer capability but are nevertheless important features:

- Black-start capability can help energize parts of the grid with the assistance of remote generators, and HVdc could support the resynchronization of a split synchronous grid during grid restoration.
- Fast power reversal can help with responding to system contingencies, thereby avoiding cascading failures in the power system.
- Fault ride through and controlled fast fault current injection offer the converter the opportunity to stay connected during grid disturbances while providing continuous system support.
- Frequency response could be used when combined with energy storage (embedded) or when it is from an asynchronous grid (interconnector).

This article takes a closer look at the additional value proposition of steady-state reactive power support from embedded VSC-HVdc. Grid codes do have steady-state reactive power requirements for integrating HVdc links into the power system, depending on the voltage and power level (for an example, refer to the European Commission regulation in the "For Further Reading" section). On the one hand, extensive reactive power requirements increase the size and cost of converter stations. On the other hand, too little reactive power capability might lead to a local deficit of reactive power (especially in the case when local traditional generation is replaced by remote renewable sources) and unfavorable conditions for the nearby legacy ac network. It is therefore important for the system operator to understand the reactive power capabilities of HVdc converters and assess their value for power system operation.

In this article, we focus on the VSC-HVdc steady-state performance for reactive power. The inherent dynamic voltage control capabilities of VSC-HVdc are important to ensure grid stability during disturbances but are not within the scope of the article.

Assessing the Benefits of Reactive Power Support From Embedded HVdc

To compensate for the charging current of ac lines (either overhead or cables) and enable stable power transfer, reactive power compensation may be needed depending on the length of the line. Embedded VSC-HVdc does not need any reactive power to transmit active power over the HVdc line but can inject or absorb reactive power independently at each converter terminal connection to the ac grid. Reactive power can therefore be used to boost the transfer capacity over the adjacent ac grid. The main factors that influence boosting efficiency are

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- a local lack of reactive power near the HVdc terminals, for example, due to the early retirement of fossil-based power generation
- the degree of meshing along the HVdc link and near the terminals
- parallelism to existing ac lines
- the length of the HVdc link.

The greatest benefit is achieved by installing one HVdc converter in an area facing a deficit of reactive power supply. The reactive power support from the HVdc will then boost the transfer capacity of the legacy ac grid. The influence on reactive power support with respect to meshing, parallelism, and distance is depicted in Figure 2, which provides two examples of embedded HVdc (noted as ① and ②).

Embedded HVdc ① is a relatively short link, strictly in parallel to an ac line in a fairly meshed surrounding ac system. This link could obviously boost the ac system transfer capability by means of reactive power supply due to its short length, the high degree of parallelism, and meshing. The short length of the link allows each terminal to provide reactive power to the highly parallel and meshed ac grid from each side and have influence over power transfer capacity. Embedded HVdc ② is a long link between north and south, as shown in Figure 2. On the one

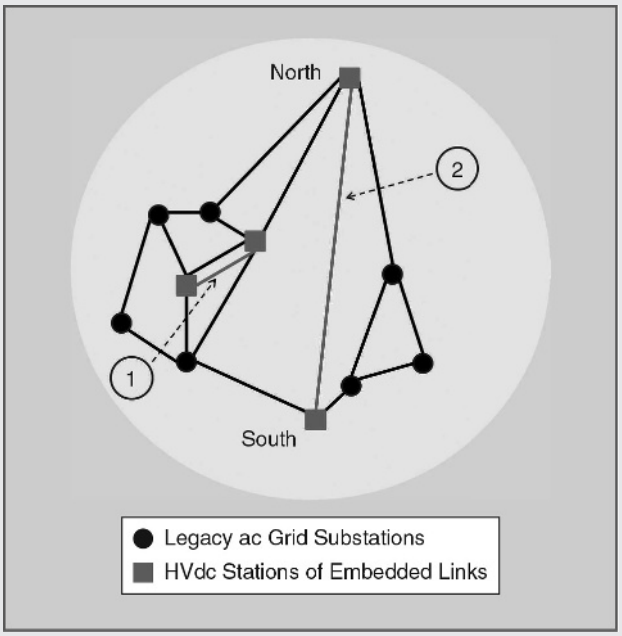

Figure 2 Examples of embedded HVdc links. The circles are substations of the legacy ac grid, the squares are HVdc converter stations, and the lines are transmission overhead lines or cables. (Go online to the article to see Figure 2 in color.)

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hand, this link can very effectively boost the power transfer over the ac system because it can reduce the loading of any of the parallel ac lines between its two terminals, which may be the bottleneck for north-to-south power transfer. In this respect, link ② is more effective in active power control. On the other hand, due to its positioning, link ② cannot effectively control the voltage in the ac nodes far away from its terminals; the expected benefit from reactive power is low, and the rating of the converters might be better invested in increasing active power transfer capabilities.

The diagram presented in Figure 3 provides an initial assessment of the system benefit of reactive power. The horizontal axis represents the configuration of the embedded HVdc system and the surrounding ac grid (e.g., including the degree of meshing of the ac grid and parallelism of the HVdc link with respect to the ac grid and the length of the corridor). The vertical axis illustrates the system benefit of reactive power. Looking at the main factors introduced to evaluate the benefit of reactive power support, it can be concluded that each grid topology must be assessed individually. Reactive power is beneficial when the HVdc link is integrated into a highly meshed ac grid with parallel lines and a rather short distance, enabling converter stations to support the legacy ac grid mutually from both sides of the link (noted as ①). In contrast, a long-distance,

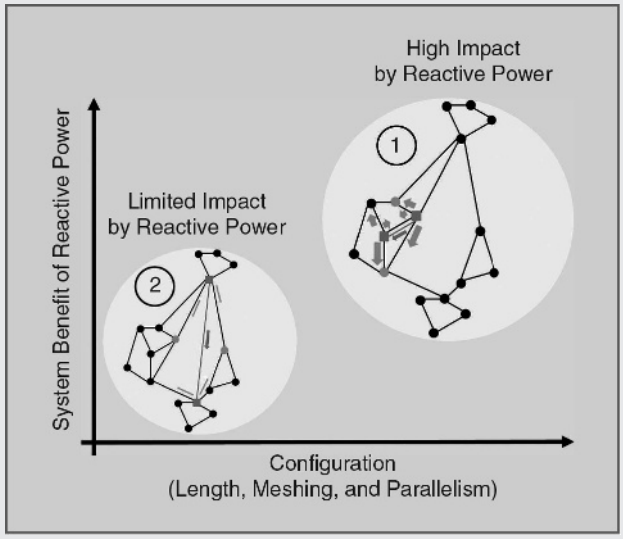

Figure 3 The benefit of reactive power support from embedded VSC-HVdc depends on the configuration (the HVdc link length, meshing of the surrounding ac grid, and parallelism of the embedded HVdc link with regard to the ac grid). (Go online to this article to see Figure 3 in color.)

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nonparallel link might not be capable of supporting the legacy ac grid, even if the converter stations have extensive reactive power ratings (noted as ②).

The Impact of Reactive Power Support From the Future Embedded HVdc Systems in Germany

Germany has ambitious goals to increase the share of renewable power sources to 80% as part of the country's total demand by 2050. The potential for large-scale renewable power is mainly located in the north with abundant on- and offshore wind resources. However, because the large demand centers are mainly located in the southern and western parts of the country, a significant upgrade of the existing transmission grid infrastructure is required. The grid infrastructure at former conventional central power plants, such as nuclear and coal-fired plants, is planned to be the entry point(s) for at least five north-to-south corridors using embedded VSC-HVdc technology with a total transmission capability of 10 GW. The increased north-to-south power transfer capacity aims to reduce the redispatch cost caused by the relatively fast-changing power generation landscape and the slower pace of transmission grid expansion. Redispatch costs occur when the dispatched active power of power plants needs to be adjusted because the transmission grid risks congestion and cannot transport the scheduled power from the generation to the demand centers. Reducing the redispatch energy volume and cost is one of the top measures used to lower the cost of electricity and avoid splitting Germany into different price zones.

Studies looking at the future need for ancillary services in changing power systems consider HVdc systems as the potential provider of such services. Shutting down conventional sources of reactive power (which, in the German case, are primarily nuclear and coal-fired power plants) leads to a deficit of reactive power in specific geographic areas. The capabilities of the HVdc converters could make up for this deficiency.

The planned embedded HVdc links in Germany cover a long distance (300–700 km) in a highly meshed legacy ac grid and are plugged into the former strong grid connection points of nuclear power plants (see Figure 4). Consequently, converters will be integrated at locations likely to have a deficiency in reactive power due to the shutdown of the nuclear plants. These locations would therefore be vital for voltage control. According to the initial assessment presented in Figure 3, this configuration would result in a medium impact, considering the favorable integration in parallel to a highly meshed system as well as the long distance of the HVdc links.

Once the desired HVdc corridor capacity is determined, the final consideration is how much additional converter capacity should

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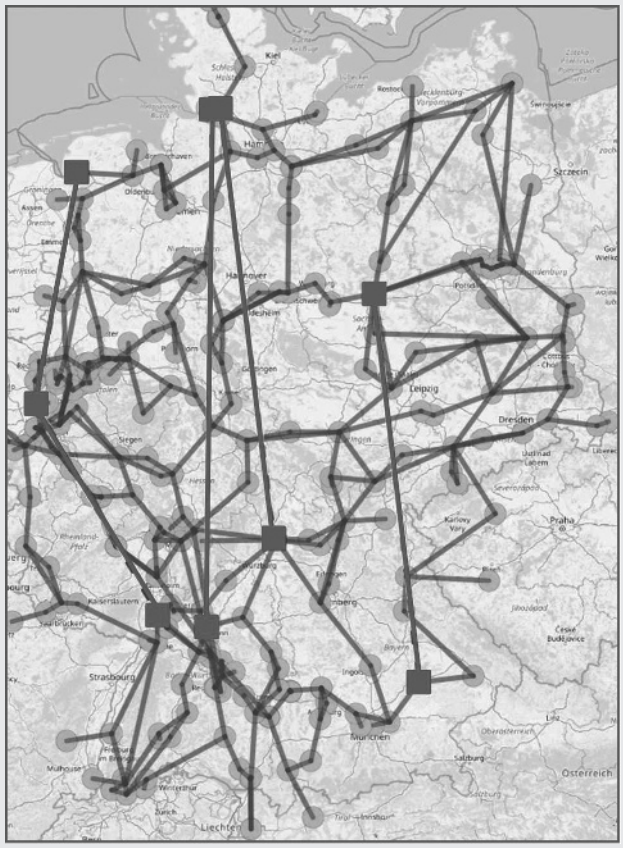

Figure 4 The German 380-kV ac transmission grid (the circles are ac substations and the squares are planned HVdc converter stations. Planned dc overhead transmission lines or dc cables are connected between the converter stations.) Image courtesy of OpenStreetMap (https://www.openstreetmap.org/). Power grid data courtesy of Network Development Plan 2017-B (https://www.netzentwicklungsplan.de/de). (Go online to this article to see Figure 4 in color.)

be reserved for reactive and active power or a combination of the two. A case study on a full-year redispatch scenario for the projected German 380-kV grid in 2030 was performed to answer this question for the three variants: maximizing reactive power, maximizing combined active/reactive power, and maximizing active power.

The *relative redispatch energy volume* is defined as the division of the annual redispatch energy calculated for each variant and the annual redispatch energy of the base case. The base case consists of the five

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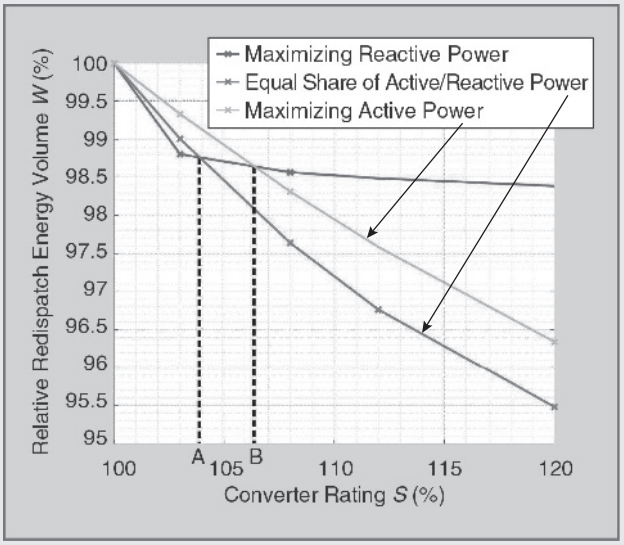

Figure 5 Case study results from the full-year simulation of the German 380-kV grid in 2030. The horizontal axis depicts the HVdc converter rating compared to the base case. The vertical axis shows the full-year redispatch energy compared to the base case. Intersections A and B highlight the converter ratings where one scenario becomes advantageous over another. (Go online to this article to see Figure 5 in color.)

HVdc links having a total of 2-GW transmission capacity without any reactive power capabilities and corresponding to a 100% converter rating. The case study is the result of a full-year simulation of the planned German 380-kV grid in 2030. Figure 5 depicts the evolution of relative redispatch energy volume (vertical axis) with respect to the converter rating (horizontal axis). On the one hand, adding 3% extra converter capacity with reactive power (Q corresponding to roughly 25% of pmax) would improve the total redispatch volume by approximately 1.2%. Beyond that point, the improvement rate lowers significantly, meaning that additional reactive power does not lead to a significant reduction of redispatch volume.

On the other hand, for 8% extra converter capacity and onward, a 50/50 share of additional reactive and active power (labeled in the Figure 5 legend as "Equal Share of Active/Reactive Power") leads to the largest redispatch volume reduction. Intersection B of the "Maximizing Active Power" and "Maximizing Reactive Power" curves is located at approximately 106% of the converter rating. The graph shows that for converter ratings higher than 106%, the active power extra capacity benefits the

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reduction of redispatch volume more than it does for the reactive power extra capacity. Intersection A displays the same effect for the comparison of the variants "Equal Share of Active/Reactive Power" and Maximizing Reactive Power," where the first becomes advantageous already at converter ratings greater than 104%.

To get to the final numbers, relevant capital expenditures for the additional converter capacity must be compared to the relevant operational expenditures, hence, a reduction in total redispatch cost over the operational lifetime. The important parameters for an initial assessment are

- the converter cost in euros per additional megavolt amp
- the redispatch volume in megawatt hours and cost in euros per megawatt hour and operational lifetime in years.

Redispatch energy volume and cost vary from year to year and strongly depend on how the actual power grid infrastructure can keep pace with changing power generation assets. For instance, in Germany, during 2014–2017, the total redispatch volume ranged from 11.5 to 18.5 TWh, and the average cost ranged from 19.40 to €26.70/MWh. In this study, a constant redispatch cost of €23/MWh is taken considering the average value of the time period from 2014 to 2017. If the grid development is fully deployed, the redispatch volume should decrease significantly (1.9–2.6 TWh is projected for 2030 in the current network development plan, when all of the HVdc links are scheduled to be in service). The HVdc links have a conservative operational lifetime of 25 years.

Additional converter ratings could be assumed to be roughly €70,000 per additional mega volt-ampere. For higher active power transfer by HVdc, the cables or overhead lines must be designed accordingly to carry the higher currents, which was not considered in our cost estimate.

From the case study the following recommendations can be made.

- In highly congested grids, as currently experienced in Germany, reactive power of up to roughly 25–30% of the converter rating would help reduce congestion-related costs. Beyond that, it is more interesting to invest in either additional active power capacity or a combination of active and reactive power.
- For less-congested grids, as projected for Germany in 2030, the benefit is lower than the additional converter costs associated with the extra reactive power capacity. However, a minimum amount of reactive power capability is required.
- In general, the case study confirmed the initial assessment that the steady-state reactive

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power capability benefits are medium to low for redispatch reduction from long, embedded HVdc systems. Depending on the generation/load/reactive power and grid scenario, extra active power capacity or a combination of both active and reactive power capability is advantageous.

Summary

This article highlighted the various features an embedded HVdc system brings to the legacy ac grid. Among others, reactive power support from embedded HVdc can boost active power transfer through the existing ac system. The possible benefits from reactive power supply by HVdc terminal stations were discussed for cases of lack of local reactive power and for cases of varying degrees of meshing, parallelism, and distance. Furthermore, the article described a full-year simulation for 2030 for the German transmission grid with the planned embedded HVdc links. The outcome was that a minimum reactive power capability is vital; however, extensive requirements should be avoided.

Acknowledgments

The authors gratefully acknowledge funding by the German Federal Ministry of Education and Research within the Kopernikus Project ENSURE. The corresponding author is Kevin Schönleber.

For Further Reading

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5.1 MEDIUM AND SHORT LINE APPROXIMATIONS

In this section, short and medium-length transmission-line approximations as a means of introducing *ABCD* parameters are presented. Some readers may prefer to start in Section 5.2, which presents the exact transmission-line equations.

It is convenient to represent a transmission line by the two-port network shown in Figure 5.1, where V_S and I_S are the sending-end voltage and current, and V_R and I_R are the receiving-end voltage and current.

The relation between the sending-end and receiving-end quantities can be written as

$$V_S = AV_R + BI_R \quad \text{volts} \tag{5.1.1}$$

$$I_S = CV_R + DI_R \quad \text{A} \tag{5.1.2}$$

or, in matrix format,

$$\begin{bmatrix} V_S \\ I_S \end{bmatrix} = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right] \begin{bmatrix} V_R \\ I_R \end{bmatrix} \tag{5.1.3}$$

where A, B, C, and D are parameters that depend on the transmission-line constants R, L, C, and G. The *ABCD* parameters are, in general, complex numbers. A and D are dimensionless. B has units of ohms, and C has units of siemens. Network theory texts [5] show that *ABCD* parameters apply to linear, passive, bilateral two-port networks, with the following general relation:

$$AD - BC = 1 \tag{5.1.4}$$

The circuit in Figure 5.2 represents a short transmission line, usually applied to overhead 60-Hz lines less than 25 km in length. Only the series resistance and reactance are included. The shunt admittance is neglected. The circuit applies to either single-phase or completely transposed three-phase lines operating under balanced conditions. For a completely transposed three-phase line, Z is the series impedance, V_S and I_R are positive-sequence line-to-neutral voltages, and I_S and I_R are positive-sequence line currents.

To avoid confusion between total series impedance and series impedance per unit length, use the following notation:

$z = \text{R} + j\omega\text{L}$ Ω/m, series impedance per unit length
$y = \text{G} + j\omega\text{C}$ S/m, shunt admittance per unit length
$Z = zl$ Ω, total series impedance
$Y = y\text{l}$ S, total shunt admittance
$l =$ line length m

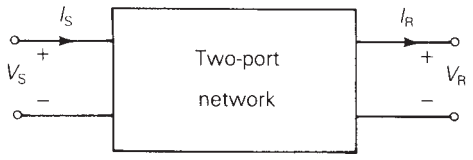

FIGURE 5.1
Representation of two-port network

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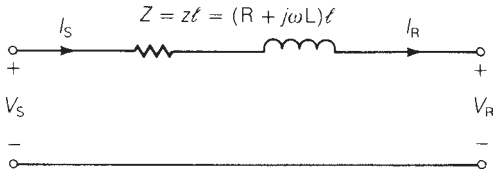

FIGURE 5.2
Short transmission line

Recall that shunt conductance G is usually neglected for overhead transmission.

The *ABCD* parameters for the short line in Figure 5.2 are easily obtained by writing a KVL and KCL equation as

$$V_S = V_R + ZI_R \tag{5.1.5}$$
$$I_S = I_R \tag{5.1.6}$$

or, in matrix format,

$$\begin{bmatrix} V_S \\ I_S \end{bmatrix} = \left[\begin{array}{c|c} 1 & Z \\ \hline 0 & 1 \end{array}\right] \begin{bmatrix} V_R \\ I_R \end{bmatrix} \tag{5.1.7}$$

Comparing (5.1.7) and (5.1.3), the *ABCD* parameters for a short line are

$$A = D = 1 \text{ per unit} \tag{5.1.8}$$
$$B = Z\ \Omega \tag{5.1.9}$$
$$C = 0\ \text{S} \tag{5.1.10}$$

For medium-length lines, typically ranging from 25 to 250 km at 60 Hz, it is common to lump the total shunt capacitance and locate half at each end of the line. Such a circuit, called a *nominal π circuit*, is shown in Figure 5.3.

To obtain the *ABCD* parameters of the nominal π circuit, note first that the current in the series branch in Figure 5.3 equals $I_R + \frac{V_R Y}{2}$. Then, writing a KVL equation,

$$\begin{aligned} V_S &= V_R + Z\left(I_R + \frac{V_R Y}{2}\right) \\ &= \left(1 + \frac{YZ}{2}\right) V_R + ZI_R \end{aligned} \tag{5.1.11}$$

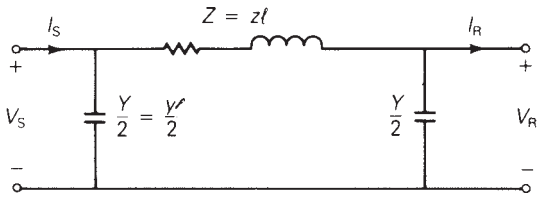

FIGURE 5.3
Medium-length transmission line—nominal π circuit

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Also, writing a KCL equation at the sending end,

$$I_S = I_R + \frac{V_R Y}{2} + \frac{V_S Y}{2} \tag{5.1.12}$$

Using (5.1.11) in (5.1.12),

$$\begin{aligned} I_S &= I_R + \frac{V_R Y}{2} + \left[\left(1 + \frac{YZ}{2}\right) V_R + Z I_R\right] \frac{Y}{2} \\ &= Y\left(1 + \frac{YZ}{4}\right) V_R + \left(1 + \frac{YZ}{2}\right) I_R \end{aligned} \tag{5.1.13}$$

Writing (5.1.11) and (5.1.13) in matrix format,

$$\begin{bmatrix} V_S \\ I_S \end{bmatrix} = \left[\begin{array}{c|c} \left(1 + \frac{YZ}{2}\right) & Z \\ \hline Y\left(1 + \frac{YZ}{4}\right) & \left(1 + \frac{YZ}{2}\right) \end{array}\right] \begin{bmatrix} V_R \\ I_R \end{bmatrix} \tag{5.1.14}$$

Thus, comparing (5.1.14) and (5.1.3),

$$A = D = 1 + \frac{YZ}{2} \quad \text{per unit} \tag{5.1.15}$$

$$B = Z \quad \Omega \tag{5.1.16}$$

$$C = Y\left(1 + \frac{YZ}{4}\right) \quad \text{S} \tag{5.1.17}$$

Note that for both the short and medium-length lines, the relation $AD - BC = 1$ is verified. Note also that since the line is the same when viewed from either end, $A = D$.

Figure 5.4 gives the *ABCD* parameters for some common networks, including a series impedance network that approximates a short line and a π circuit that approximates a medium-length line. A medium-length line could also be approximated by the T circuit shown in Figure 5.4, lumping half of the series impedance at each end of the line. Also given are the *ABCD* parameters for networks in series, which are conveniently obtained by multiplying the *ABCD* matrices of the individual networks.

ABCD parameters can be used to describe the variation of line voltage with line loading. *Voltage regulation* is the change in voltage at the receiving end of the line when the load varies from no-load to a specified full load at a specified power

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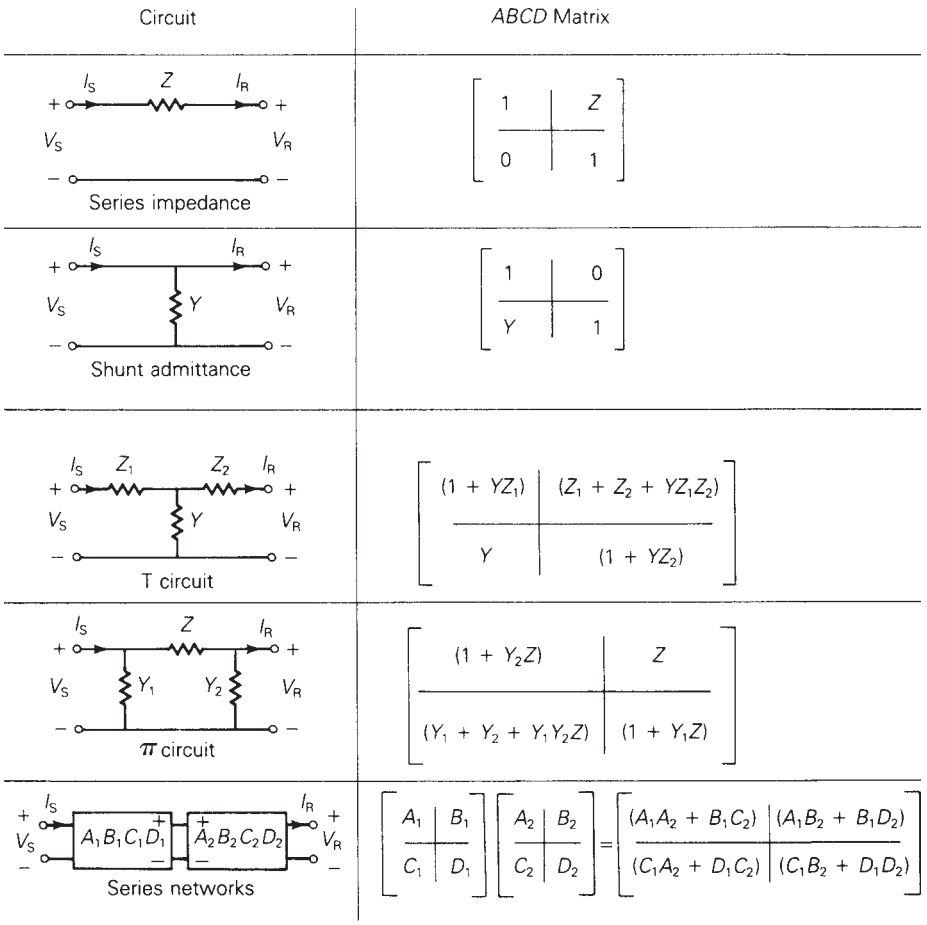

FIGURE 5.4
ABCD parameters of common networks

factor, while the sending-end voltage is held constant. Expressed in percent of full-load voltage,

$$\text{percent VR} = \frac{|V_{\text{RNL}}| - |V_{\text{RFL}}|}{|V_{\text{RFL}}|} \times 100 \tag{5.1.18}$$

where percent VR is the percent voltage regulation, $|V_{\text{RNL}}|$ is the magnitude of the no-load receiving-end voltage, and $|V_{\text{RFL}}|$ is the magnitude of the full-load receiving-end voltage.

The effect of load power factor on voltage regulation is illustrated by the phasor diagrams in Figure 5.5 for short lines. The phasor diagrams are graphical representations of (5.1.5) for lagging and leading power factor loads. Note that, from (5.1.5) at no-load, $I_{\text{RNL}} = 0$ and $V_S = V_{\text{RNL}}$ for a short line. As shown, the higher (worse) voltage regulation occurs for the lagging p.f. load, where V_{RNL}

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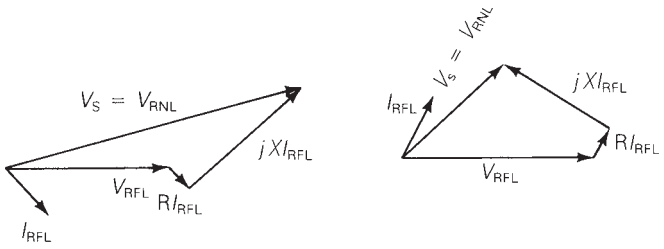
FIGURE 5.5

Phasor diagrams for a short transmission line



(a) Lagging p.f. load (b) Leading p.f. load

exceeds V_{RFL} by the larger amount. A smaller or even negative voltage regulation occurs for the leading p.f. load. In general, the no-load voltage is, from (5.1.1) with $I_{RNL} = 0$,

$$V_{RNL} = \frac{V_S}{A} \tag{5.1.19}$$

which can be used in (5.1.18) to determine voltage regulation.

In practice, transmission-line voltages decrease when heavily loaded and increase when lightly loaded. When voltages on EHV lines are maintained within $\pm 5\%$ of rated voltage, corresponding to about 10% voltage regulation, unusual operating problems are not encountered. Ten percent voltage regulation for lower voltage lines including transformer-voltage drops is also considered good operating practice.

In addition to voltage regulation, line loadability is an important issue. Three major line-loading limits are (1) the thermal limit, (2) the voltage-drop limit, and (3) the steady-state stability limit.

The maximum temperature of a conductor determines its thermal limit. Conductor temperature affects the conductor sag between towers and the loss of conductor tensile strength due to annealing. If the temperature is too high, prescribed conductor-to-ground clearances may not be met, or the elastic limit of the conductor may be exceeded such that it cannot shrink to its original length when cooled. Conductor temperature depends on the current magnitude and its time duration, as well as on ambient temperature, wind velocity, and conductor surface conditions. Appendix Tables A.3 and A.4 give approximate current-carrying capacities of copper and ACSR conductors. The loadability of short transmission lines (less than 25 km in length for 60-Hz overhead lines) is usually determined by the conductor thermal limit or by ratings of line terminal equipment, such as circuit breakers.

For longer line lengths (up to 300 km), line loadability is often determined by the voltage-drop limit. Although more severe voltage drops may be tolerated in some cases, a heavily loaded line with $V_R/V_S \geq 0.95$ is usually considered safe operating practice. For line lengths over 300 km, steady-state stability becomes a limiting factor. Stability, discussed in Section 5.4, refers to the ability of synchronous machines on either end of a line to remain in synchronism.

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EXAMPLE 5.1

ABCD parameters and the nominal π circuit: medium-length line

A three-phase, 60-Hz, completely transposed 345-kV, 200-km line has two 795,000-cmil 26/2 ACSR conductors per bundle and the following positive-sequence line constants:

$z = 0.032 + j0.35 \quad \Omega/\text{km}$
$y = j4.2 \times 10^{-6} \quad \text{S/km}$

Full load at the receiving end of the line is 700 MW at 0.99 p.f. leading and at 95% of rated voltage. Assuming a medium-length line, determine the following:

a. *ABCD* parameters of the nominal π circuit
b. Sending-end voltage V_S, current I_S, and real power P_S
c. Percent voltage regulation
d. Thermal limit, based on the approximate current-carrying capacity listed in Table A.4
e. Transmission-line efficiency at full load

SOLUTION

a. The total series impedance and shunt admittance values are

$$Z = zl = (0.032 + j0.35)(200) = 6.4 + j70 = 70.29\underline{/84.78^\circ} \quad \Omega$$
$$Y = yl = (j4.2 \times 10^{-6})(200) = 8.4 \times 10^{-4}\underline{/90^\circ} \quad \text{S}$$

From (5.1.15) through (5.1.17),

$$\begin{aligned} A = D &= 1 + (8.4 \times 10^{-4}\underline{/90^\circ})(70.29\underline{/84.78^\circ})\left(\tfrac{1}{2}\right) \\ &= 1 + 0.02952\underline{/174.78^\circ} \\ &= 0.9706 + j0.00269 = 0.9706\underline{/0.159^\circ} \quad \text{per unit} \end{aligned}$$
$$B = Z = 70.29\underline{/84.78^\circ} \quad \Omega$$
$$\begin{aligned} C &= (8.4 \times 10^{-4}\underline{/90^\circ})(1 + 0.01476\underline{/174.78^\circ}) \\ &= (8.4 \times 10^{-4}\underline{/90^\circ})(0.9853 + j0.00134) \\ &= 8.277 \times 10^{-4}\underline{/90.08^\circ} \quad \text{S} \end{aligned}$$

b. The receiving-end voltage and current quantities are

$$V_R = (0.95)(345) = 327.8 \text{ kV}_{LL}$$
$$V_R = \frac{327.8}{\sqrt{3}}\underline{/0^\circ} = 189.2\underline{/0^\circ} \quad \text{kV}_{LN}$$

(*Continued*)

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$$I_R = \frac{700\angle\cos^{-1}0.99}{(\sqrt{3})(0.95 \times 345)(0.99)} = 1.246\angle 8.11° \quad \text{kA}$$

From (5.1.1) and (5.1.2), the sending-end quantities are

$$\begin{aligned} V_S &= (0.9706\angle 0.159°)(189.2\angle 0°) + (70.29\angle 84.78)(1.246\angle 8.11°) \\ &= 183.6\angle 0.159° + 87.55\angle 92.89° \\ &= 179.2 + j87.95 = 199.6\angle 26.14° \quad \text{kV}_{LN} \end{aligned}$$

$$V_S = 199.6\sqrt{3} = 345.8 \text{ kV}_{LL} \approx 1.00 \quad \text{per unit}$$

$$\begin{aligned} I_S &= (8.277 \times 10^{-4}\angle 90.08°)(189.2\angle 0°) + (0.9706\angle 0.159°)(1.246\angle 8.11°) \\ &= 0.1566\angle 90.08° + 1.209\angle 8.27° \\ &= 1.196 + j0.331 = 1.241\angle 15.5° \quad \text{kA} \end{aligned}$$

and the real power delivered to the sending end is

$$\begin{aligned} P_S &= (\sqrt{3})(345.8)(1.241)\cos(26.14° - 15.5°) \\ &= 730.5 \quad \text{MW} \end{aligned}$$

c. From (5.1.19), the no-load receiving-end voltage is

$$V_{RNL} = \frac{V_S}{A} = \frac{345.8}{0.9706} = 356.3 \quad \text{kV}_{LL}$$

and, from (5.1.18),

$$\text{percent VR} = \frac{356.3 - 327.8}{327.8} \times 100 = 8.7\%$$

d. From Table A.4, the approximate current-carrying capacity of two 795,000-cmil 26/2 ACSR conductors is $2 \times 0.9 = 1.8$ kA.
e. The full-load line losses are $P_S - P_R = 730.5 - 700 = 30.5$ MW and the full-load transmission efficiency is

$$\text{percent EFF} = \frac{P_R}{P_S} \times 100 = \frac{700}{730.5} \times 100 = 95.8\%$$

Since $V_S = 1.00$ per unit, the full-load receiving-end voltage of 0.95 per unit corresponds to $V_R/V_S = 0.95$, considered in practice to be about the lowest operating voltage possible without encountering operating problems. Thus, for this 345-kV 200-km uncompensated line, voltage drop limits the full-load current to 1.246 kA at 0.99 p.f. leading, which is well below the thermal limit of 1.8 kA.

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5.2 TRANSMISSION-LINE DIFFERENTIAL EQUATIONS

The line constants R, L, and C are derived in Chapter 4 as per-length values having units of Ω/m, H/m, and F/m. They are not lumped, but rather are uniformly distributed along the length of the line. In order to account for the distributed nature of transmission-line constants, consider the circuit shown in Figure 5.6, which represents a line section of length Δx. $V(x)$ and $I(x)$ denote the voltage and current at position x, which is measured in meters from the right, or receiving end of the line. Similarly, $V(x + \Delta x)$ and $I(x + \Delta x)$ denote the voltage and current at position $(x + \Delta x)$. The circuit constants are

$$z = \mathrm{R} + j\omega\mathrm{L} \quad \Omega/\mathrm{m} \tag{5.2.1}$$

$$y = \mathrm{G} + j\omega\mathrm{C} \quad \mathrm{S/m} \tag{5.2.2}$$

where G is usually neglected for overhead 60-Hz lines. Writing a KVL equation for the circuit,

$$V(x + \Delta x) = V(x) + (z\Delta x)I(x) \quad \text{volts} \tag{5.2.3}$$

Rearranging (5.2.3),

$$\frac{V(x + \Delta x) - V(x)}{\Delta x} = zI(x) \tag{5.2.4}$$

and taking the limit as Δx approaches zero,

$$\frac{dV(x)}{dx} = zI(x) \tag{5.2.5}$$

Similarly, writing a KCL equation for the circuit,

$$I(x + \Delta x) = I(x) + (y\Delta x)V(x + \Delta x) \quad \mathrm{A} \tag{5.2.6}$$

Rearranging,

$$\frac{I(x + \Delta x) - I(x)}{\Delta x} = yV(x) \tag{5.2.7}$$

and taking the limit as Δx approaches zero,

$$\frac{dI(x)}{dx} = yV(x) \tag{5.2.8}$$

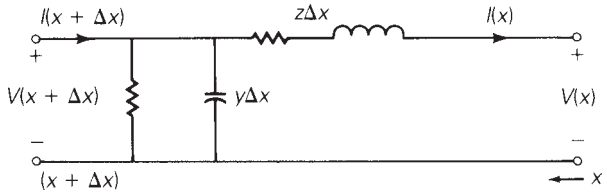

FIGURE 5.6
Transmission-line section of length Δx

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Equations (5.2.5) and (5.2.8) are two linear, first-order, homogeneous differential equations with two unknowns, $V(x)$ and $I(x)$. Eliminate $I(x)$ by differentiating (5.2.5) and using (5.2.8) as follows:

$$\frac{d^2V(x)}{dx^2} = z\frac{dI(x)}{dx} = zyV(x) \tag{5.2.9}$$

or

$$\frac{d^2V(x)}{dx^2} - zyV(x) = 0 \tag{5.2.10}$$

Equation (5.2.10) is a linear, second-order, homogeneous differential equation with one unknown, $V(x)$. By inspection, its solution is

$$V(x) = A_1e^{\gamma x} + A_2e^{-\gamma x} \quad \text{volts} \tag{5.2.11}$$

where A_1 and A_2 are integration constants and

$$\gamma = \sqrt{zy} \quad \text{m}^{-1} \tag{5.2.12}$$

γ, whose units are m^{-1}, is called the *propagation constant*. By inserting (5.2.11) and (5.2.12) into (5.2.10), the solution to the differential equation can be verified.

Next, using (5.2.11) in (5.2.5),

$$\frac{dV(x)}{dx} = \gamma A_1e^{\gamma x} - \gamma A_2e^{-\gamma x} = zI(x) \tag{5.2.13}$$

Solving for $I(x)$,

$$I(x) = \frac{A_1e^{\gamma x} - A_2e^{-\gamma x}}{z/\gamma} \tag{5.2.14}$$

Using (5.2.12), $z/\gamma = z/\sqrt{zy} = \sqrt{z/y}$, (5.2.14) becomes

$$I(x) = \frac{A_1e^{\gamma x} - A_2e^{-\gamma x}}{Z_c} \tag{5.2.15}$$

where

$$Z_c = \sqrt{\frac{z}{y}} \quad \Omega \tag{5.2.16}$$

Z_c, whose units are ohms, is called the *characteristic impedance*.

Next, the integration constants A_1 and A_2 are evaluated from the boundary conditions. At $x = 0$, which is the receiving end of the line, the receiving-end voltage and current are

$$V_R = V(0) \tag{5.2.17}$$

$$I_R = I(0) \tag{5.2.18}$$

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Also, at $x = 0$, (5.2.11) and (5.2.15) become

$$V_R = A_1 + A_2 \tag{5.2.19}$$

$$I_R = \frac{A_1 - A_2}{Z_c} \tag{5.2.20}$$

Solving for A_1 and A_2,

$$A_1 = \frac{V_R + Z_c I_R}{2} \tag{5.2.21}$$

$$A_2 = \frac{V_R - Z_c I_R}{2} \tag{5.2.22}$$

Substituting A_1 and A_2 into (5.2.11) and (5.2.15),

$$V(x) = \left(\frac{V_R + Z_c I_R}{2}\right)e^{\gamma x} + \left(\frac{V_R - Z_c I_R}{2}\right)e^{-\gamma x} \tag{5.2.23}$$

$$I(x) = \left(\frac{V_R + Z_c I_R}{2Z_c}\right)e^{\gamma x} - \left(\frac{V_R - Z_c I_R}{2Z_c}\right)e^{-\gamma x} \tag{5.2.24}$$

Rearranging (5.2.23) and (5.2.24),

$$V(x) = \left(\frac{e^{\gamma x} + e^{-\gamma x}}{2}\right)V_R + Z_c\left(\frac{e^{\gamma x} - e^{-\gamma x}}{2}\right)I_R \tag{5.2.25}$$

$$I(x) = \frac{1}{Z_c}\left(\frac{e^{\gamma x} - e^{-\gamma x}}{2}\right)V_R + \left(\frac{e^{\gamma x} + e^{-\gamma x}}{2}\right)I_R \tag{5.2.26}$$

Recognizing the hyperbolic functions cosh and sinh,

$$V(x) = \cosh(\gamma x)V_R + Z_c \sinh(\gamma x)I_R \tag{5.2.27}$$

$$I(x) = \frac{1}{Z_c}\sinh(\gamma x)\ V_R + \cosh(\gamma x)I_R \tag{5.2.28}$$

Equations (5.2.27) and (5.2.28) give the *ABCD* parameters of the distributed line. In matrix format,

$$\begin{bmatrix} V(x) \\ I(x) \end{bmatrix} = \left[\begin{array}{c|c} A(x) & B(x) \\ \hline C(x) & D(x) \end{array}\right]\begin{bmatrix} V_R \\ I_R \end{bmatrix} \tag{5.2.29}$$

where

$$A(x) = D(x) = \cosh(\gamma x) \text{ per unit} \tag{5.2.30}$$

$$B(x) = Z_c \sinh(\gamma x) \quad \Omega \tag{5.2.31}$$

$$C(x) = \frac{1}{Z_c}\sinh(\gamma x) \quad \text{S} \tag{5.2.32}$$

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Equation (5.2.29) gives the current and voltage at any point x along the line in terms of the receiving-end voltage and current. At the sending end, where $x = l$, $V(l) = V_S$ and $I(l) = I_S$. That is,

$$\begin{bmatrix} V_S \\ I_S \end{bmatrix} = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right] \begin{bmatrix} V_R \\ I_R \end{bmatrix} \tag{5.2.33}$$

where

$$A = D = \cosh(\gamma l) \text{ per unit} \tag{5.2.34}$$

$$B = Z_c \sinh(\gamma l) \quad \Omega \tag{5.2.35}$$

$$C = \frac{1}{Z_c} \sinh(\gamma l) \quad \text{S} \tag{5.2.36}$$

Equations (5.2.34) through (5.2.36) give the $ABCD$ parameters of the distributed line. In these equations, the propagation constant γ is a complex quantity with real and imaginary parts denoted α and β. That is,

$$\gamma = \alpha + j\beta \quad \text{m}^{-1} \tag{5.2.37}$$

The quantity γl is dimensionless. Also,

$$e^{\gamma l} = e^{(\alpha l + j\beta l)} = e^{\alpha l} e^{j\beta l} = e^{\alpha l} \underline{/\beta l} \tag{5.2.38}$$

Using (5.2.38), the hyperbolic functions cosh and sinh can be evaluated as follows:

$$\cosh(\gamma l) = \frac{e^{\gamma l} + e^{-\gamma l}}{2} = \frac{1}{2}(e^{\alpha l} \underline{/\beta l} + e^{-\alpha l} \underline{/-\beta l}) \tag{5.2.39}$$

and

$$\sinh(\gamma l) = \frac{e^{\gamma l} - e^{-\gamma l}}{2} = \frac{1}{2}(e^{\alpha l} \underline{/\beta l} - e^{-\alpha l} \underline{/-\beta l}) \tag{5.2.40}$$

Alternatively, the following identities can be used:

$$\cosh(\alpha l + j\beta l) = \cosh(\alpha l)\cos(\beta l) + j \sinh(\alpha l)\sin(\beta l) \tag{5.2.41}$$

$$\sinh(\alpha l + j\beta l) = \sinh(\alpha l)\cos(\beta l) + j \cosh(\alpha l)\sin(\beta l) \tag{5.2.42}$$

Note that in (5.2.39) through (5.2.42), the dimensionless quantity βl is in radians, not degrees.

The $ABCD$ parameters given by (5.2.34) through (5.2.36) are exact parameters valid for any line length. For accurate calculations, these equations must be used for overhead 60-Hz lines longer than 250 km. The $ABCD$ parameters derived in Section 5.1 are approximate parameters that are more conveniently used for hand calculations involving short- and medium-length lines. Table 5.1 summarizes the $ABCD$ parameters for short, medium, long, and lossless (see Section 5.4) lines.

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TABLE 5.1
Summary: Transmission-line *ABCD* parameters

Parameter	$A = D$	B	C
Units	per Unit	Ω	S
Short line (less than 25 km)	1	Z	0
Medium line—nominal π circuit (25 to 250 km)	$1 + \frac{YZ}{2}$	Z	$Y\left(1 + \frac{YZ}{4}\right)$
Long line—equivalent π circuit (more than 250 km)	$\cosh(\gamma\ell) = 1 + \frac{Y'Z'}{2}$	$Z_c \sinh(\gamma\ell) = Z'$	$(1/Z_c)\sinh(\gamma\ell) = Y'\left(1 + \frac{Y'Z'}{4}\right)$
Lossless line (R = G = 0)	$\cos(\beta\ell)$	$jZ_c \sin(\beta\ell)$	$\frac{j\sin(\beta\ell)}{Z_c}$

EXAMPLE 5.2

Exact *ABCD* parameters: long line

A three-phase 765-kV, 60-Hz, 300-km, completely transposed line has the following positive-sequence impedance and admittance:

$$z = 0.0165 + j0.3306 = 0.3310\angle 87.14^\circ \quad \Omega/\text{km}$$
$$y = j4.674 \times 10^{-6}\ \text{S/km}$$

Assuming positive-sequence operation, calculate the exact $ABCD$ parameters of the line. Compare the exact B parameter with that of the nominal π circuit.

SOLUTION

From (5.2.12) and (5.2.16),

$$Z_c = \sqrt{\frac{0.3310\angle 87.14^\circ}{4.674 \times 10^{-6}\angle 90^\circ}} = \sqrt{7.082 \times 10^4 \angle -2.86^\circ}$$
$$= 266.1\angle -1.43^\circ \quad \Omega$$

and

$$\gamma l = \sqrt{(0.3310\angle 87.14^\circ)(4.674 \times 10^{-6}\angle 90^\circ)} \times (300)$$
$$= \sqrt{1.547 \times 10^{-6}\angle 177.14^\circ} \times (300)$$
$$= 0.3731\angle 88.57^\circ = 0.00931 + j0.3730 \quad \text{per unit}$$

From (5.2.38),

$$e^{\gamma l} = e^{0.00931}e^{+j0.3730} = 1.0094\angle 0.3730 \quad \text{radians}$$
$$= 0.9400 + j0.3678$$

(*Continued*)

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and

$$e^{-\gamma l} = e^{-0.00931}e^{-j0.3730} = 0.9907\angle -0.3730 \quad \text{radians}$$
$$= 0.9226 - j0.3610$$

Then, from (5.2.39) and (5.2.40),

$$\cosh(\gamma l) = \frac{(0.9400 + j0.3678) + (0.9226 - j0.3610)}{2}$$
$$= 0.9313 + j0.0034 = 0.9313\angle 0.209^\circ$$
$$\sinh(\gamma l) = \frac{(0.9400 + j0.3678) - (0.9226 - j0.3610)}{2}$$
$$= 0.0087 + j0.3644 = 0.3645\angle 88.63^\circ$$

Finally, from (5.2.34) through (5.2.36),

$$A = D = \cosh(\gamma l) = 0.9313\angle 0.209^\circ \quad \text{per unit}$$

$$B = (266.1\angle -1.43^\circ)(0.3645\angle 88.63^\circ) = 97.0\angle 87.2^\circ \quad \Omega$$

$$C = \frac{0.3645\angle 88.63^\circ}{266.1\angle -1.43^\circ} = 1.37 \times 10^{-3}\angle 90.06^\circ \quad \text{S}$$

Using (5.1.16), the B parameter for the nominal π circuit is

$$B_{\text{nominal }\pi} = Z = (0.3310\angle 87.14^\circ)(300) = 99.3\angle 87.14^\circ \quad \Omega$$

which is 2% larger than the exact value.

5.3 EQUIVALENT π CIRCUIT

Many computer programs used in power system analysis and design assume circuit representations of components such as transmission lines and transformers. It is therefore convenient to represent the terminal characteristics of a transmission line by an equivalent circuit instead of its $ABCD$ parameters.

The circuit shown in Figure 5.7 is called an *equivalent π circuit*. It is identical in structure to the nominal π circuit of Figure 5.3, except that Z' and Y' are used instead of Z and Y. The objective is to determine Z' and Y' such that the equivalent π circuit has the same $ABCD$ parameters as those of the distributed line, (5.2.34) through (5.2.36). The $ABCD$ parameters of the equivalent π circuit, which has the same structure as the nominal π, are

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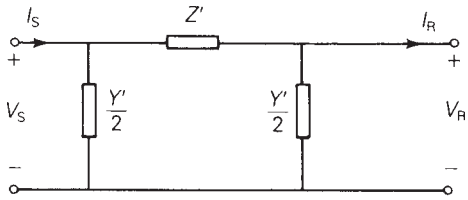

FIGURE 5.7
Transmission-line equivalent π circuit

$$Z' = Z_c \sinh(\gamma\ell) = ZF_1 = Z\frac{\sinh(\gamma\ell)}{\gamma\ell}$$

$$\frac{Y'}{2} = \frac{\tanh(\gamma\ell/2)}{Z_c} = \frac{Y}{2}F_2 = \frac{Y}{2}\frac{\tanh(\gamma\ell/2)}{(\gamma\ell/2)}$$

$$A = D = 1 + \frac{Y'Z'}{2} \quad \text{per unit} \tag{5.3.1}$$

$$B = Z' \quad \Omega \tag{5.3.2}$$

$$C = Y'\left(1 + \frac{Y'Z'}{4}\right) \quad \text{S} \tag{5.3.3}$$

where Z and Y in (5.1.15) through (5.1.17) have been replaced with Z' and Y' in (5.3.1) through (5.3.3). Equating (5.3.2) to (5.2.35),

$$Z' = Z_c \sinh(\gamma l) = \sqrt{\frac{z}{y}} \sinh(\gamma l) \tag{5.3.4}$$

Rewriting (5.3.4) in terms of the nominal π circuit impedance $Z = zl$,

$$\begin{aligned} Z' &= zl\left[\sqrt{\frac{z}{y}}\frac{\sinh(\gamma l)}{zl}\right] = zl\left[\frac{\sinh(\gamma l)}{\sqrt{zy}l}\right] \\ &= ZF_1 \quad \Omega \end{aligned} \tag{5.3.5}$$

where

$$F_1 = \frac{\sinh(\gamma l)}{\gamma l} \quad \text{per unit} \tag{5.3.6}$$

Similarly, equating (5.3.1) to (5.2.34),

$$\begin{aligned} 1 + \frac{Y'Z'}{2} &= \cosh(\gamma l) \\ \frac{Y'}{2} &= \frac{\cosh(\gamma l) - 1}{Z'} \end{aligned} \tag{5.3.7}$$

Using (5.3.4) and the identity $\tanh\left(\frac{\gamma l}{2}\right) = \frac{\cosh(\gamma l) - 1}{\sinh(\gamma l)}$, (5.3.7) becomes

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$$\frac{Y'}{2} = \frac{\cosh(\gamma l) - 1}{Z_c \sinh(\gamma l)} = \frac{\tanh(\gamma l/2)}{Z_c} = \frac{\tanh(\gamma l/2)}{\sqrt{\frac{z}{y}}} \tag{5.3.8}$$

Rewriting (5.3.8) in terms of the nominal π circuit admittance $Y = yl$,

$$\frac{Y'}{2} = \frac{yl}{2}\left[\frac{\tanh(\gamma l/2)}{\sqrt{\frac{z}{y}}\frac{yl}{2}}\right] = \frac{yl}{2}\left[\frac{\tanh(\gamma l/2)}{\sqrt{zy}\, l/2}\right]$$
$$= \frac{Y}{2}F_2 \quad \text{S} \tag{5.3.9}$$

where

$$F_2 = \frac{\tanh(\gamma l/2)}{\gamma l/2} \quad \text{per unit} \tag{5.3.10}$$

Equations (5.3.6) and (5.3.10) give the correction factors F_1 and F_2 to convert Z and Y for the nominal π circuit to Z' and Y' for the equivalent π circuit.

EXAMPLE 5.3

Equivalent π circuit: long line

Compare the equivalent and nominal π circuits for the line in Example 5.2.

SOLUTION

For the nominal π circuit,

$$Z = zl = (0.3310\underline{/87.14^\circ})(300) = 99.3\underline{/87.14^\circ} \quad \Omega$$

$$\frac{Y}{2} = \frac{yl}{2} = \left(\frac{j4.674 \times 10^{-6}}{2}\right)(300) = 7.011 \times 10^{-4}\underline{/90^\circ} \quad \text{S}$$

From (5.3.6) and (5.3.10), the correction factors are

$$F_1 = \frac{0.3645\underline{/88.63^\circ}}{0.3731\underline{/88.57^\circ}} = 0.9769\underline{/0.06^\circ} \quad \text{per unit}$$

$$F_2 = \frac{\tanh(\gamma l/2)}{\gamma l/2} = \frac{\cosh(\gamma l) - 1}{(\gamma l/2)\sinh(\gamma l)}$$

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$$= \frac{0.9313 + j0.0034 - 1}{\left(\frac{0.3731}{2}\angle 88.57°\right)(0.3645\angle 88.63°)}$$

$$= \frac{-0.0687 + j0.0034}{0.06800\angle 177.20°}$$

$$= \frac{0.06878\angle 177.17°}{0.06800\angle 177.20°} = 1.012\angle -0.03° \quad \text{per unit}$$

Then, from (5.3.5) and (5.3.9), for the equivalent π circuit,

$$Z' = (99.3\angle 87.14°)(0.9769\angle 0.06°) = 97.0\angle 87.2° \quad \Omega$$

$$\frac{Y'}{2} = (7.011 \times 10^{-4}\angle 90°)(1.012\angle -0.03°) = 7.095 \times 10^{-4}\angle 89.97° \quad \text{S}$$

$$= 3.7 \times 10^{-7} + j7.095 \times 10^{-4} \quad \text{S}$$

Comparing these nominal and equivalent π circuit values, Z' is about 2% smaller than Z, and $Y'/2$ is about 1% larger than $Y/2$. Although the circuit values are approximately the same for this line, the equivalent π circuit should be used for accurate calculations involving long lines. Note the small shunt conductance, $G' = 3.7 \times 10^{-7}$ S, introduced in the equivalent π circuit. G' is often neglected.

5.4 LOSSLESS LINES

This section discusses the following concepts for lossless lines: surge impedance, *ABCD* parameters, equivalent π circuit, wavelength, surge impedance loading, voltage profiles, and steady-state stability limit.

When line losses are neglected, simpler expressions for the line parameters are obtained and the above concepts are more easily understood. Since transmission and distribution lines for power transfer generally are designed to have low losses, the equations and concepts developed here can be used for quick and reasonably accurate hand calculations leading to seat-of-the-pants analyses and to initial designs. More accurate calculations then can be made with computer programs for follow-up analysis and design.

SURGE IMPEDANCE

For a lossless line, R = G = 0, and

$$z = j\omega \text{L} \quad \Omega/\text{m} \tag{5.4.1}$$

$$y = j\omega \text{C} \quad \text{S/m} \tag{5.4.2}$$

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From (5.2.12) and (5.2.16),

$$Z_c = \sqrt{\frac{z}{y}} = \sqrt{\frac{j\omega \mathrm{L}}{j\omega \mathrm{C}}} = \sqrt{\frac{\mathrm{L}}{\mathrm{C}}} \quad \Omega \tag{5.4.3}$$

and

$$\gamma = \sqrt{zy} = \sqrt{(j\omega \mathrm{L})(j\omega \mathrm{C})} = j\omega\sqrt{\mathrm{LC}} = j\beta \quad \mathrm{m}^{-1} \tag{5.4.4}$$

where

$$\beta = \omega\sqrt{\mathrm{LC}} \quad \mathrm{m}^{-1} \tag{5.4.5}$$

The characteristic impedance $Z_c = \sqrt{\mathrm{L}/\mathrm{C}}$, commonly called the *surge impedance* for a lossless line, is pure real—that is, resistive. The propagation constant $\gamma = j\beta$ is pure imaginary.

ABCD PARAMETERS

The *ABCD* parameters are, from (5.2.30) through (5.2.32),

$$\mathrm{A}(x) = \mathrm{D}(x) = \cosh(\gamma x) = \cosh(j\beta x)$$

$$= \frac{e^{j\beta x} + e^{-j\beta x}}{2} = \cos(\beta x) \quad \text{per unit} \tag{5.4.6}$$

$$\sinh(\gamma x) = \sinh(j\beta x) = \frac{e^{j\beta x} - e^{-j\beta x}}{2} = j\sin(\beta x) \quad \text{per unit} \tag{5.4.7}$$

$$B(x) = Z_c \sinh(\gamma x) = jZ_c \sin(\beta x) = j\sqrt{\frac{\mathrm{L}}{\mathrm{C}}}\sin(\beta x) \quad \Omega \tag{5.4.8}$$

$$C(x) = \frac{\sinh(\gamma x)}{Z_c} = \frac{j\sin(\beta x)}{\sqrt{\dfrac{\mathrm{L}}{\mathrm{C}}}} \quad \mathrm{S} \tag{5.4.9}$$

EQUIVALENT π CIRCUIT

For the equivalent π circuit, using (5.3.4),

$$Z' = j\mathrm{Z}_c \sin(\beta l) = j\mathrm{X}' \quad \Omega \tag{5.4.10}$$

or, from (5.3.5) and (5.3.6),

$$Z' = (j\omega \mathrm{L}l)\left(\frac{\sin(\beta l)}{\beta l}\right) = j\mathrm{X}' \quad \Omega \tag{5.4.11}$$

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Also, from (5.3.9) and (5.3.10),

$$\frac{Y'}{2} = \frac{Y}{2}\frac{\tanh(j\beta l/2)}{j\beta l/2} = \frac{Y}{2}\frac{\sinh(j\beta l/2)}{(j\beta l/2)\cosh(j\beta l/2)}$$
$$= \left(\frac{j\omega Cl}{2}\right)\frac{j\sin(\beta l/2)}{(j\beta l/2)\cos(\beta l/2)} = \left(\frac{j\omega Cl}{2}\right)\frac{\tan(\beta l/2)}{\beta l/2}$$
$$= \left(\frac{j\omega C'l}{2}\right) \quad \text{S} \tag{5.4.12}$$

Z' and Y' are both pure imaginary. Also, for βl less than π radians, Z' is pure inductive and Y' is pure capacitive. Thus the equivalent π circuit for a lossless line, shown in Figure 5.8, is also lossless.

WAVELENGTH

A *wavelength* is the distance required to change the phase of the voltage or current by 2π radians or 360°. For a lossless line, using (5.2.29),

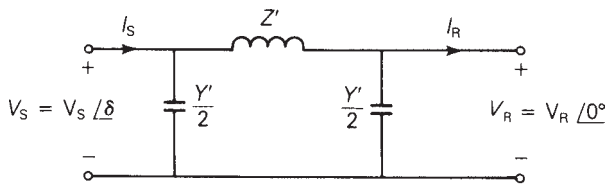
$$V(x) = \mathrm{A}(x)V_\mathrm{R} + B(x)I_\mathrm{R}$$
$$= \cos(\beta x)\,V_\mathrm{R} + jZ_c\sin(\beta x)I_\mathrm{R} \tag{5.4.13}$$

and

$$I(x) = C(x)V_\mathrm{R} + \mathrm{D}(x)I_\mathrm{R}$$
$$= \frac{j\sin(\beta x)}{Z_c}V_\mathrm{R} + \cos(\beta x)I_\mathrm{R} \tag{5.4.14}$$

From (5.4.13) and (5.4.14), $V(x)$ and $I(x)$ change phase by 2π radians when $x = 2\pi/\beta$. Denoting wavelength by λ, and using (5.4.5),

$$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{\omega\sqrt{\mathrm{LC}}} = \frac{1}{f\sqrt{\mathrm{LC}}} \quad \text{m} \tag{5.4.15}$$



$$Z' = (j\omega L\ell)\left(\frac{\sin\beta\ell}{\beta\ell}\right) = jX' \ \Omega$$

$$\frac{Y'}{2} = \left(\frac{j\omega C\ell}{2}\right)\frac{\tan(\beta\ell/2)}{(\beta\ell/2)} = \frac{j\omega C'\ell}{2} \ \text{S}$$

FIGURE 5.8

Equivalent π circuit for a lossless line ($\beta\ell$ less than π)

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or

$$f\lambda = \frac{1}{\sqrt{LC}} \tag{5.4.16}$$

The term $(1/\sqrt{LC})$ in (5.4.16) is the velocity of propagation of voltage and current waves along a lossless line, as shown in Chapter 14. For overhead lines, $(1/\sqrt{LC}) \approx 3 \times 10^8$ m/s, and for $f = 60$ Hz, (5.4.14) gives

$$\lambda \approx \frac{3 \times 10^8}{60} = 5 \times 10^6 \text{ m} = 5000 \text{ km}$$

Typical power-line lengths are only a small fraction of the above 60-Hz wavelength.

SURGE IMPEDANCE LOADING

Surge impedance loading (SIL) is the power delivered by a lossless line to a load resistance equal to the surge impedance $Z_c = \sqrt{L/C}$. Figure 5.9 shows a lossless line terminated by a resistance equal to its surge impedance. This line represents either a single-phase line or one phase-to-neutral of a balanced three-phase line. At SIL, from (5.4.13),

$$\begin{aligned} V(x) &= \cos(\beta x)\, V_R + jZ_c \sin(\beta x) I_R \\ &= \cos(\beta x) V_R + jZ_c \sin(\beta x) \left(\frac{V_R}{Z_c}\right) \\ &= (\cos \beta x + j \sin \beta x)\, V_R \\ &= e^{j\beta x} V_R \quad \text{volts} \end{aligned} \tag{5.4.17}$$

$$|V(x)| = |V_R| \quad \text{volts} \tag{5.4.18}$$

Thus, at SIL, the voltage profile is flat. That is, the voltage magnitude at any point x along a lossless line at SIL is constant. Also, from (5.4.14) at SIL,

$$\begin{aligned} I(x) &= \frac{j \sin(\beta x)}{Z_c} V_R + (\cos \beta x) \frac{V_R}{Z_c} \\ &= (\cos \beta x + j \sin \beta x) \frac{V_R}{Z_c} \\ &= (e^{j\beta x}) \frac{V_R}{Z_c} \quad \text{A} \end{aligned} \tag{5.4.19}$$

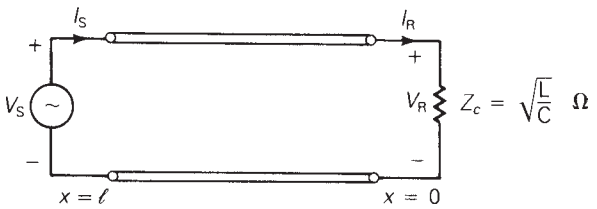

FIGURE 5.9
Lossless line terminated by its surge impedance

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Using (5.4.17) and (5.4.19), the complex power flowing at any point x along the line is

$$\begin{aligned} S(x) &= \mathrm{P}(x) + j\mathrm{Q}(x) = V(x)I^*(x) \\ &= (e^{j\beta x}V_{\mathrm{R}})\left(\frac{e^{j\beta x}V_{\mathrm{R}}}{Z_c}\right)^* \\ &= \frac{|V_{\mathrm{R}}|^2}{Z_c} \end{aligned} \tag{5.4.20}$$

Thus the real power flow along a lossless line at SIL remains constant from the sending end to the receiving end. The reactive power flow is zero.

At rated line voltage, the real power delivered, or SIL, from (5.4.20), is

$$\mathrm{SIL} = \frac{\mathrm{V}_{\mathrm{rated}}^2}{Z_c} \tag{5.4.21}$$

where rated voltage is used for a single-phase line and rated line-to-line voltage is used for the total real power delivered by a three-phase line. Table 5.2 lists surge impedance and SIL values for typical overhead 60-Hz three-phase lines.

VOLTAGE PROFILES

In practice, power lines are not terminated by their surge impedance. Instead, loadings can vary from a small fraction of SIL during light load conditions up to multiples of SIL, depending on line length and line compensation, during heavy load conditions. If a line is not terminated by its surge impedance, then the voltage profile is not flat. Figure 5.10 shows voltage profiles of lines with a fixed sending-end voltage magnitude $\mathrm{V_S}$ for line lengths l up to a quarter wavelength. This figure shows four

$\mathrm{V_{rated}}$ (kV)	$Z_c = \sqrt{\mathrm{L/C}}$ (Ω)	$\mathrm{SIL} = \mathrm{V_{rated}^2}/Z_c$ (MW)
69	366–400	12–13
138	366–405	47–52
230	365–395	134–145
345	280–366	325–425
500	233–294	850–1075
765	254–266	2200–2300

TABLE 5.2

Surge impedance and SIL values for typical 60-Hz overhead lines [1, 2]
(Source: Electric Power Research Institute (EPRI), EPRI AC Transmission Line Reference Book—200 kV and Above (Palo Alto, CA: EPRI, *www.epri.com*. December 2005); Westinghouse Electric Corporation, *Electrical Transmission and Distribution Reference Book*, 4th ed. (East Pittsburgh, PA, 1964).)

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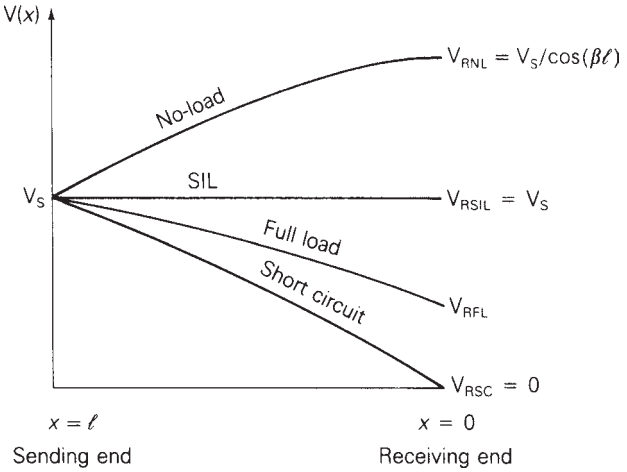

FIGURE 5.10

Voltage profiles of an uncompensated lossless line with fixed sending-end voltage for line lengths up to a quarter wavelength

loading conditions: (1) no-load; (2) SIL; (3) short circuit; and (4) full load, which are described as follows:

1. At no-load, $I_{RNL} = 0$ and (5.4.13) yields

$$V_{NL}(x) = (\cos \beta x)V_{RNL} \tag{5.4.22}$$

   The no-load voltage increases from $V_S = (\cos \beta l)V_{RNL}$ at the sending end to V_{RNL} at the receiving end (where $x = 0$).

2. From (5.4.18), the voltage profile at SIL is flat.

3. For a short circuit at the load, $V_{RSC} = 0$ and (5.4.13) yields

$$V_{SC}(x) = (Z_c \sin \beta x)I_{RSC} \tag{5.4.23}$$

   The voltage decreases from $V_S = (\sin \beta l)(Z_c I_{RSC})$ at the sending end to $V_{RSC} = 0$ at the receiving end.

4. The full-load voltage profile, which depends on the specification of full-load current, lies above the short-circuit voltage profile.

Figure 5.10 summarizes these results, showing a high receiving-end voltage at no-load and a low receiving-end voltage at full load. This voltage regulation problem becomes more severe as the line length increases. In Section 5.6, shunt compensation methods to reduce voltage fluctuations are presented.

STEADY-STATE STABILITY LIMIT

The equivalent π circuit of Figure 5.8 can be used to obtain an equation for the real power delivered by a lossless line. Assume that the voltage magnitudes V_S and V_R at the ends of the line are held constant. Also, let δ denote the voltage-phase angle at the sending end with respect to the receiving end. From KVL, the receiving-end current I_R is

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$$I_R = \frac{V_S - V_R}{Z'} - \frac{Y'}{2} V_R$$
$$= \frac{V_S e^{j\delta} - V_R}{jX'} - \frac{j\omega C' l}{2} V_R \tag{5.4.24}$$

and the complex power S_R delivered to the receiving end is

$$S_R = V_R I_R^* = V_R \left(\frac{V_S e^{j\delta} - V_R}{jX'} \right)^* + \frac{j\omega C' l}{2} V_R^2$$
$$= V_R \left(\frac{V_S e^{-j\delta} - V_R}{-jX'} \right) + \frac{j\omega C' l}{2} V_R^2$$
$$= \frac{jV_R V_S \cos\delta + V_R V_S \sin\delta - jV_R^2}{X'} + \frac{j\omega C' l}{2} V_R^2 \tag{5.4.25}$$

The real power delivered is

$$P = P_S = P_R = \mathrm{Re}(S_R) = \frac{V_R V_S}{X'} \sin\delta \quad \text{W} \tag{5.4.26}$$

Note that since the line is lossless, $P_S = P_R$.

Equation (5.4.26) is plotted in Figure 5.11. For fixed voltage magnitudes V_S and V_R, the phase angle δ increases from 0 to 90° as the real power delivered increases. The maximum power that the line can deliver, which occurs when $\delta = 90°$, is given by

$$P_{max} = \frac{V_S V_R}{X'} \quad \text{W} \tag{5.4.27}$$

P_{max} represents the theoretical *steady-state stability* limit of a lossless line. If an attempt were made to exceed this steady-state stability limit, then synchronous machines at the sending end would lose synchronism with those at the receiving end. Stability is further discussed in Chapter 12.

It is convenient to express the steady-state stability limit in terms of SIL. Using (5.4.10) in (5.4.26),

$$P = \frac{V_S V_R \sin\delta}{Z_c \sin\beta l} = \left(\frac{V_S V_R}{Z_c} \right) \frac{\sin\delta}{\sin\left(\frac{2\pi l}{\lambda} \right)} \tag{5.4.28}$$

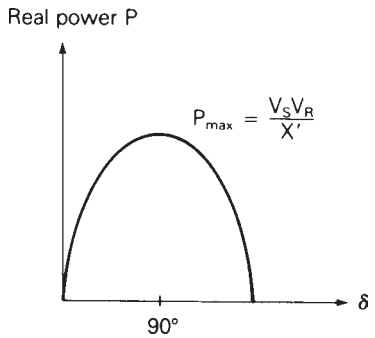

FIGURE 5.11

Real power delivered by a lossless line versus voltage angle across the line

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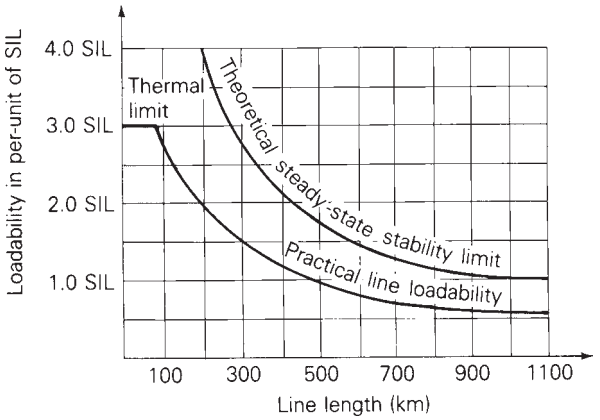
FIGURE 5.12

Transmission-line loadability curve for 60-Hz overhead lines—no series or shunt compensation



Expressing V_S and V_R in per-unit of rated line voltage,

$$P = \left(\frac{V_S}{V_{rated}}\right)\left(\frac{V_R}{V_{rated}}\right)\left(\frac{V_{rated}^2}{Z_c}\right)\frac{\sin\delta}{\sin\left(\frac{2\pi l}{\lambda}\right)}$$

$$= V_{Sp.u.}V_{Rp.u.}(SIL)\frac{\sin\delta}{\sin\left(\frac{2\pi l}{\lambda}\right)} \quad W \tag{5.4.29}$$

And for $\delta = 90°$, the theoretical steady-state stability limit is

$$P_{max} = \frac{V_{Sp.u.}V_{Rp.u.}(SIL)}{\sin\left(\frac{2\pi l}{\lambda}\right)} \quad W \tag{5.4.30}$$

Equations (5.4.27) through (5.4.30) reveal two important factors affecting the steady-state stability limit. First, from (5.4.27), it increases with the square of the line voltage. For example, a doubling of line voltage enables a fourfold increase in maximum power flow. Second, it decreases with line length. Equation (5.4.30) is plotted in Figure 5.12 for $V_{Sp.u.} = V_{Rp.u.} = 1$, $\lambda = 5000$ km, and line lengths up to 1100 km. As shown, the theoretical steady-state stability limit decreases from 4(SIL) for a 200-km line to about 2(SIL) for a 400-km line.

EXAMPLE 5.4

Theoretical steady-state stability limit: long line

Neglecting line losses, find the theoretical steady-state stability limit for the 300-km line in Example 5.2. Assume a 266.1-Ω surge impedance, a 5000-km wavelength, and $V_S = V_R = 765$ kV.

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SOLUTION

From (5.4.21),

$$\text{SIL} = \frac{(765)^2}{266.1} = 2199 \quad \text{MW}$$

From (5.4.30) with $l = 300$ km and $\lambda = 5000$ km,

$$P_{\max} = \frac{(1)(1)(2199)}{\sin\left(\frac{2\pi \times 300}{5000}\right)} = (2.716)(2199) = 5974 \quad \text{MW}$$

Alternatively, from Figure 5.12, for a 300-km line, the theoretical steady-state stability limit is (2.72)SIL = (2.72)(2199) = 5980 MW, which is about the same as the previous result (see Figure 5.13).

Open PowerWorld Simulator case Example 5_4 and select **Tools**, **Play** to see this example. When the load on a line is equal to the SIL, the voltage profile across the line is flat, and the line's net reactive power losses are zero. For loads above the SIL, the line consumes reactive power, and the load's voltage magnitude is below the sending-end value. Conversely, for loads below the SIL, the line actually generates reactive power, and the load's voltage magnitude is above the sending-end value. Use the load arrow button to vary the load to see the changes in the receiving-end voltage and the line's reactive power consumption (indicated by the reactive power supplied by the generator).

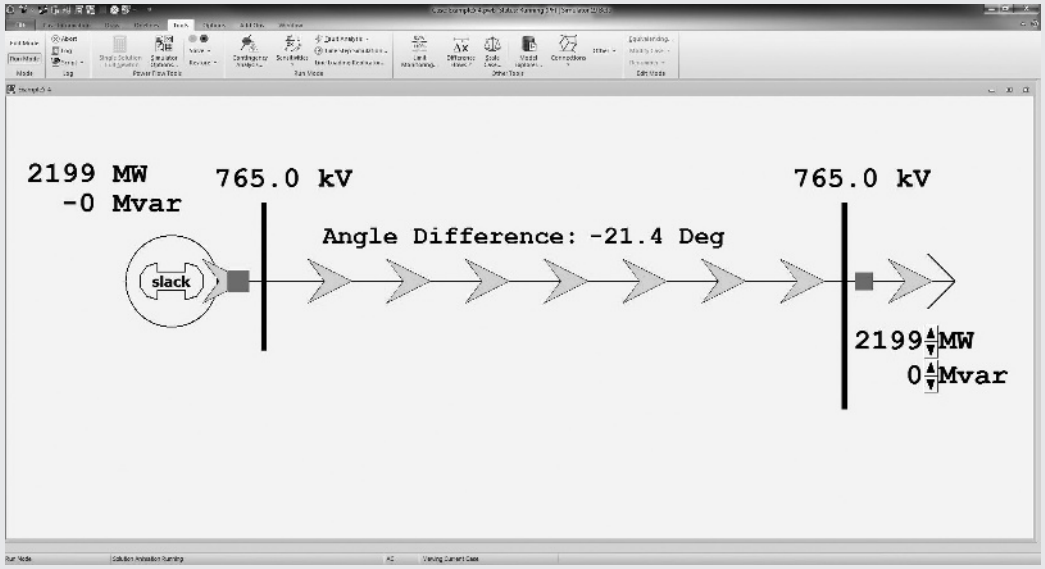

FIGURE 5.13

Screen for Example 5.4

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5.5 MAXIMUM POWER FLOW

Maximum power flow, discussed in Section 5.4 for lossless lines, is derived here in terms of the *ABCD* parameters for lossy lines. The following notation is used:

$$A = \cosh(\gamma\ell) = \mathrm{A}\underline{/\theta_\mathrm{A}}$$
$$B = Z' = Z'\underline{/\theta_Z}$$
$$V_\mathrm{S} = \mathrm{V_S}\underline{/\delta} \quad V_\mathrm{R} = \mathrm{V_R}\underline{/0^\circ}$$

Solving (5.2.33) for the receiving-end current,

$$I_\mathrm{R} = \frac{V_\mathrm{S} - AV_\mathrm{R}}{B} = \frac{\mathrm{V_S}e^{j\delta} - \mathrm{AV_R}e^{j\theta_\mathrm{A}}}{\mathrm{Z'}e^{j\theta_Z}} \tag{5.5.1}$$

The complex power delivered to the receiving end is

$$S_\mathrm{R} = \mathrm{P_R} + j\mathrm{Q_R} = V_\mathrm{R}I_\mathrm{R}^* = \mathrm{V_R}\left[\frac{\mathrm{V_S}e^{j(\delta-\theta_Z)} - \mathrm{AV_R}e^{j(\theta_\mathrm{A}-\theta_Z)}}{\mathrm{Z'}}\right]^*$$
$$= \frac{\mathrm{V_R V_S}}{\mathrm{Z'}}e^{j(\theta_Z-\delta)} - \frac{\mathrm{AV_R^2}}{\mathrm{Z'}}e^{j(\theta_Z-\theta_\mathrm{A})} \tag{5.5.2}$$

The real and reactive power delivered to the receiving end are thus

$$\mathrm{P_R} = \mathrm{Re}(S_\mathrm{R}) = \frac{\mathrm{V_R V_S}}{\mathrm{Z'}}\cos(\theta_Z - \delta) - \frac{\mathrm{AV_R^2}}{\mathrm{Z'}}\cos(\theta_Z - \theta_\mathrm{A}) \tag{5.5.3}$$

$$\mathrm{Q_R} = \mathrm{Im}(S_\mathrm{R}) = \frac{\mathrm{V_R V_S}}{\mathrm{Z'}}\sin(\theta_Z - \delta) - \frac{\mathrm{AV_R^2}}{\mathrm{Z'}}\sin(\theta_Z - \theta_\mathrm{A}) \tag{5.5.4}$$

Note that for a lossless line, $\theta_\mathrm{A} = 0^\circ$, $B = Z' = j\mathrm{X'}$, $\mathrm{Z'} = \mathrm{X'}$, $\theta_Z = 90^\circ$, and (5.5.3) reduces to

$$\mathrm{P_R} = \frac{\mathrm{V_R V_S}}{\mathrm{X'}}\cos(90 - \delta) - \frac{\mathrm{AV_R^2}}{\mathrm{X'}}\cos(90^\circ)$$
$$= \frac{\mathrm{V_R V_S}}{\mathrm{X'}}\sin\delta \tag{5.5.5}$$

which is the same as (5.4.26).

The theoretical maximum real power delivered (or steady-state stability limit) occurs when $\delta = \theta z$ in (5.5.3):

$$\mathrm{P_{Rmax}} = \frac{\mathrm{V_R V_S}}{\mathrm{Z'}} - \frac{\mathrm{AV_R^2}}{\mathrm{Z'}}\cos(\theta_Z - \theta_\mathrm{A}) \tag{5.5.6}$$

The second term in (5.5.6), and the fact that Z′ is larger than X′, reduce $\mathrm{P_{Rmax}}$ to a value somewhat less than that given by (5.4.27) for a lossless line.

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EXAMPLE 5.5

Theoretical maximum power delivered: long line

Determine the theoretical maximum power, in MW and in per-unit of SIL, that the line in Example 5.2 can deliver. Assume $V_S = V_R = 765$ kV.

SOLUTION

From Example 5.2,

$A = 0.9313$ per unit; $\theta_A = 0.209°$

$B = Z' = 97.0\ \Omega$ $\theta_Z = 87.2°$

$Z_c = 266.1\ \Omega$

From (5.5.6) with $V_S = V_R = 765$ kV,

$$P_{Rmax} = \frac{(765)^2}{97} - \frac{(0.9313)(765)^2}{97}\cos(87.2° - 0.209°)$$

$$= 6033 - 295 = 5738 \text{ MW}$$

From (5.4.20),

$$\text{SIL} = \frac{(765)^2}{266.1} = 2199 \quad \text{MW}$$

Thus,

$$P_{Rmax} = \frac{5738}{2199} = 2.61 \quad \text{per unit}$$

This value is about 4% less than that found in Example 5.4, where losses were neglected.

5.6 LINE LOADABILITY

In practice, power lines are not operated to deliver their theoretical maximum power, which is based on rated terminal voltages and an angular displacement $\delta = 90°$ across the line. Figure 5.12 shows a practical line loadability curve plotted below the theoretical steady-state stability limit. This curve is based on the voltage-drop limit $V_R/V_S \geq 0.95$ and on a maximum angular displacement of 30 to 35° across the line (or about 45° across the line and equivalent system reactances) in order to maintain stability during transient disturbances [1, 3]. The curve is valid for typical overhead 60-Hz lines with no compensation. Note that for short lines less than 25 km long, loadability is limited by the thermal rating of the conductors or by terminal equipment ratings, not by voltage drop or stability considerations. Section 5.7 investigates series and shunt compensation techniques to increase the loadability of longer lines toward their thermal limit.

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EXAMPLE 5.6

Practical line loadability and percent voltage regulation: long line

The 300-km uncompensated line in Example 5.2 has four 1,272,000 cmil 54/3 ACSR conductors per bundle. The sending-end voltage is held constant at 1.0 per unit of rated line voltage. Determine the following:

a. The practical line loadability (Assume an approximate receiving-end voltage $V_R = 0.95$ per unit and $\delta = 35°$ maximum angle across the line.)
b. The full-load current at 0.986 p.f. leading based on the above practical line loadability
c. The exact receiving-end voltage for the full-load current found in part (b)
d. Percent voltage regulation for the above full-load current
e. Thermal limit of the line, based on the approximate current-carrying capacity given in Table A.4

SOLUTION

a. From (5.5.3), with $V_S = 765$, $V_R = 0.95 \times 765$ kV, and $\delta = 35°$, using the values of Z', θ_z, A, and θ_A from Example 5.5,

$$P_R = \frac{(765)(0.95 \times 765)}{97.0}\cos(87.2° - 35°) - \frac{(0.9313)(0.95 \times 765)^2}{97.0}\cos(87.2° - 0.209°)$$
$$= 3513 - 266 = 3247 \quad \text{MW}$$

$P_R = 3247$ MW is the practical line loadability, provided the thermal and voltage-drop limits are not exceeded. Alternatively, from Figure 5.12 for a 300-km line, the practical line loadability is (1.49)SIL = (1.49)(2199) = 3277 MW, which is about the same as the previous result.

b. For the loading at 0.986 p.f. leading and at 0.95×765 kV, the full-load receiving-end current is

$$I_{RFL} = \frac{P_R}{\sqrt{3}V_R(\text{p.f.})} = \frac{3247}{(\sqrt{3})(0.95 \times 765)(0.986)} = 2.616 \quad \text{kA}$$

c. From (5.1.1) with $I_{RFL} = 2.616\underline{/\cos^{-1} 0.986} = 2.616\underline{/9.599°}$ kA, using the A and B parameters from Example 5.2,

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$$V_S = AV_{RFL} + BI_{RFL}$$

$$\frac{765}{\sqrt{3}}\underline{/\delta} = (0.9313\underline{/0.209°})\,(V_{RFL}\underline{/0°}) + (97.0\underline{/87.2°})\,(2.616\underline{/9.599°})$$

$$441.7\underline{/\delta} = (0.9313\,V_{RFL} - 30.04) + j(0.0034V_{RFL} + 251.97)$$

Taking the squared magnitude of the above equation,

$$(441.7)^2 = 0.8673V_{RFL}^2 - 54.24V_{RFL} + 64{,}391$$

Solving,

$$V_{RFL} = 420.7\ kV_{LN}$$

$$= 420.7\sqrt{3} = 728.7\ kV_{LL} = 0.953\ per\ unit$$

d. From (5.1.19), the receiving-end no-load voltage is

$$V_{RNL} = \frac{V_S}{A} = \frac{765}{0.9313} = 821.4 \quad kV_{LL}$$

And from (5.1.18),

$$\text{percent VR} = \frac{821.4 - 728.7}{728.7} \times 100 = 12.72\%$$

e. From Table A.4, the approximate current-carrying capacity of four 1,272,000-cmil 54/3 ACSR conductors is $4 \times 1.2 = 4.8$ kA.

Since the voltages $V_S = 1.0$ and $V_{RFL} = 0.953$ per unit satisfy the voltage-drop limit $V_R/V_S \geq 0.95$, the factor that limits line loadability is steady-state stability for this 300-km uncompensated line. The full-load current of 2.616 kA corresponding to loadability is also well below the thermal limit of 4.8 kA. The 12.7% voltage regulation is too high because the no-load voltage is too high. Compensation techniques to reduce no-load voltages are discussed in Section 5.7.

EXAMPLE 5.7

Selection of transmission line voltage and number of lines for power transfer

From a hydroelectric power plant, 9000 MW are to be transmitted to a load center located 500 km from the plant. Based on practical line loadability criteria, determine the number of three-phase, 60-Hz lines required to transmit this power, with one line out of service, for the following cases: (a) 345-kV lines with $Z_c = 297\ \Omega$; (b) 500-kV lines with $Z_c = 277\ \Omega$; and (c) 765-kV lines with $Z_c = 266\ \Omega$. Assume $V_S = 1.0$ per unit, $V_R = 0.95$ per unit, and $\delta = 35°$. Also,

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assume that the lines are uncompensated and widely separated such that there is negligible mutual coupling between them.

SOLUTION

a. For 345-kV lines, (5.4.21) yields

$$\text{SIL} = \frac{(345)^2}{297} = 401 \quad \text{MW}$$

Neglecting losses, from (5.4.29), with $l = 500$ km and $\delta = 35°$,

$$\text{P} = \frac{(1.0)(0.95)(401)\sin(35°)}{\sin\left(\dfrac{2\pi \times 500}{5000}\right)} = (401)(0.927) = 372 \quad \text{MW/line}$$

Alternatively, the practical line loadability curve in Figure 5.12 can be used to obtain P = (0.93)SIL for typical 500-km overhead 60-Hz uncompensated lines.

In order to transmit 9000 MW with one line out of service,

$$\#345\text{-kV lines} = \frac{9000 \text{ MW}}{372 \text{ MW/line}} + 1 = 24.2 + 1 \approx 26$$

b. For 500-kV lines,

$$\text{SIL} = \frac{(500)^2}{277} = 903 \quad \text{MW}$$

$$\text{P} = (903)(0.927) = 837 \quad \text{MW/line}$$

$$\#500\text{-kV lines} = \frac{9000}{837} + 1 = 10.8 + 1 \approx 12$$

c. For 765-kV lines,

$$\text{SIL} = \frac{(765)^2}{266} = 2200 \quad \text{MW}$$

$$\text{P} = (2200)(0.927) = 2039 \quad \text{MW/line}$$

$$\#765\text{-kV lines} = \frac{9000}{2039} + 1 = 4.4 + 1 \approx 6$$

Increasing the line voltage from 345 to 765 kV, which is a factor of 2.2, reduces the required number of lines from 26 to 6, which is a factor of 4.3.

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EXAMPLE 5.8

Effect of intermediate substations on number of lines required for power transfer

Can five instead of six 765-kV lines transmit the required power in Example 5.7 if there are two intermediate substations that divide each line into three 167-km line sections, and if only one line section is out of service?

SOLUTION

The lines are shown in Figure 5.14. For simplicity, neglect line losses. The equivalent π circuit of one 500-km, 765-kV line has a series reactance from (5.4.10) and (5.4.15),

$$X' = (266)\sin\left(\frac{2\pi \times 500}{5000}\right) = 156.35 \quad \Omega$$

Combining series/parallel reactances in Figure 5.14, the equivalent reactance of five lines with one line section out of service is

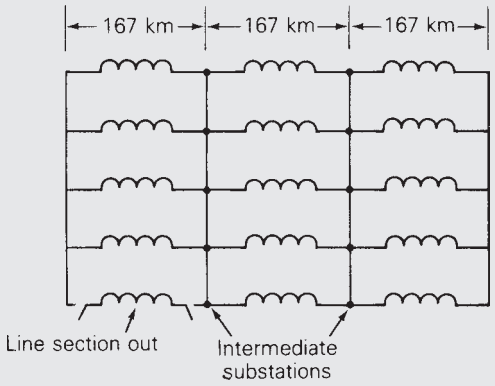
$$X_{eq} = \frac{1}{5}\left(\frac{2}{3}X'\right) + \frac{1}{4}\left(\frac{X'}{3}\right) = 0.2167X' = 33.88 \quad \Omega$$

Then, from (5.4.26) with $\delta = 35°$,

$$P = \frac{(765)(765 \times 0.95)\sin(35°)}{33.88} = 9412 \quad \text{MW}$$

Inclusion of line losses would reduce the above value by 3 or 4% to about 9100 MW. Therefore, the answer is yes. Five 765-kV, 500-km uncompensated lines with two intermediate substations and with one line section out of service

FIGURE 5.14
Transmission-line configuration for Example 5.8



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transmits 9000 MW. Intermediate substations are often economical if their costs do not outweigh the reduction in line costs.

This example is modeled in PowerWorld Simulator case Example 5_8 (see Figure 5.15). Each line segment is represented with the lossless line model from Example 5.4 with the π circuit parameters modified to exactly match those for a 167-km distributed line. The pie charts on each line segment show the percentage loading of the line, assuming a rating of 3500 MVA. The solid red squares on the lines represent closed circuit breakers, and the green squares correspond to open circuit breakers. Clicking on a circuit breaker toggles its status. The simulation results differ slightly from the simplified analysis done earlier in the example because the PowerWorld simulation includes the charging capacitance of the transmission lines. With all line segments in service, use the load's arrow to verify that the SIL for this system is 11,000 MW, which is five times that of the single circuit line in Example 5.4.

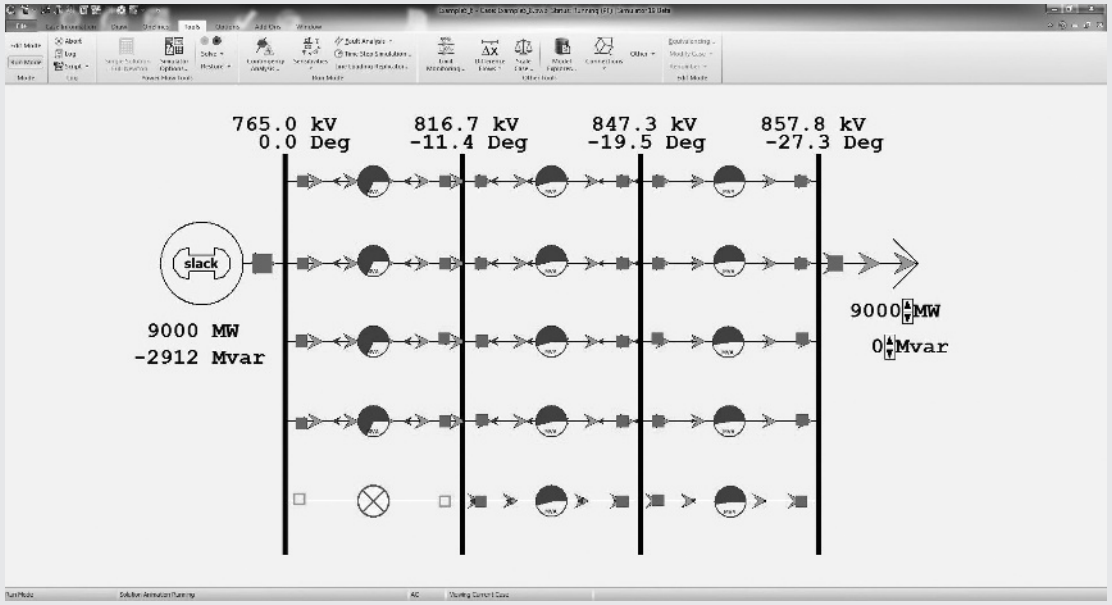

FIGURE 5.15

Screen for Example 5.8

5.7 REACTIVE COMPENSATION TECHNIQUES

Inductors and capacitors are used on medium-length and long transmission lines to increase line loadability and to maintain voltages near rated values.

Shunt reactors (inductors) are commonly installed at selected points along EHV lines from each phase to neutral. The inductors absorb reactive power and

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reduce overvoltages during light load conditions. They also reduce transient overvoltages due to switching and lightning surges. However, shunt reactors can reduce line loadability if they are not removed under full-load conditions.

In addition to shunt reactors, shunt capacitors are sometimes used to deliver reactive power and increase transmission voltages during heavy load conditions. Another type of shunt compensation includes thyristor-switched reactors in parallel with capacitors. These devices, called *static var compensators*, can absorb reactive power during light loads and deliver reactive power during heavy loads. Through automatic control of the thyristor switches, voltage fluctuations are minimized and line loadability is increased. Synchronous condensors (synchronous motors with no mechanical load) can also control their reactive power output, although more slowly than static var compensators.

Series capacitors are sometimes used on long lines to increase line loadability. Capacitor banks are installed in series with each phase conductor at selected points along a line. Their effect is to reduce the net series impedance of the line in series with the capacitor banks, thereby reducing line-voltage drops and increasing the steady-state stability limit. A disadvantage of series capacitor banks is that automatic protection devices must be installed to bypass high currents during faults and to reinsert the capacitor banks after fault clearing. Also, the addition of series capacitors can excite low-frequency oscillations, a phenomenon called *subsynchronous resonance*, which may damage turbine-generator shafts. Studies have shown, however, that series capacitive compensation can increase the loadability of long lines at only a fraction of the cost of new transmission [1].

Figure 5.16 shows a schematic and an equivalent circuit for a compensated line section, where N_C is the amount of series capacitive compensation expressed in percent of the positive-sequence line impedance and N_L is the amount of shunt reactive compensation in percent of the positive-sequence line admittance. It is assumed in Figure 5.16 that half of the compensation is installed at each end of the line section. The following two examples illustrate the effect of compensation.

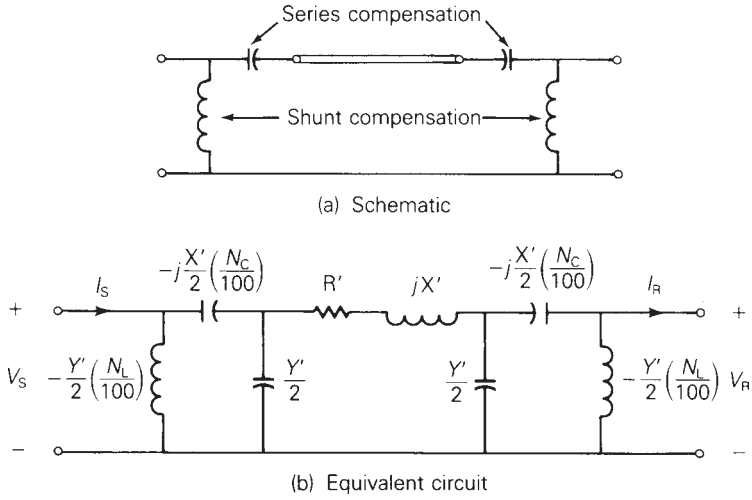

FIGURE 5.16
Compensated transmission-line section

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EXAMPLE 5.9

Shunt reactive compensation to improve transmission-line voltage regulation

Identical shunt reactors (inductors) are connected from each phase conductor to neutral at both ends of the 300-km line in Example 5.2 during light load conditions, providing 75% compensation. The reactors are removed during heavy load conditions. Full load is 1.90 kA at unity p.f. and at 730 kV. Assuming that the sending-end voltage is constant, determine the following properties.

a. Percent voltage regulation of the uncompensated line
b. The equivalent shunt admittance and series impedance of the compensated line
c. Percent voltage regulation of the compensated line

SOLUTION

a. From (5.1.1) with $I_{RFL} = 1.9\angle 0°$ kA, using the A and B parameters from Example 5.2,

$$\begin{aligned} V_S &= AV_{RFL} + BI_{RFL} \\ &= (0.9313\angle 0.209°)\left(\frac{730}{\sqrt{3}}\angle 0°\right) + (97.0\angle 87.2°)(1.9\angle 0°) \\ &= 392.5\angle 0.209° + 184.3\angle 87.2° \\ &= 401.5 + j185.5 \\ &= 442.3\angle 24.8° \quad \text{kV}_{LN} \end{aligned}$$

$$V_S = 442.3\sqrt{3} = 766.0 \quad \text{kV}_{LL}$$

The no-load receiving-end voltage is, from (5.1.19),

$$V_{RNL} = \frac{766.0}{0.9313} = 822.6 \quad \text{kV}_{LL}$$

and the percent voltage regulation for the uncompensated line is, from (5.1.18),

$$\text{percent VR} = \frac{822.6 - 730}{730} \times 100 = 12.68\%$$

b. From Example 5.3, the shunt admittance of the equivalent π circuit without compensation is

$$\begin{aligned} Y' &= 2(3.7 \times 10^{-7} + j7.094 \times 10^{-4}) \\ &= 7.4 \times 10^{-7} + j14.188 \times 10^{-4} \quad \text{S} \end{aligned}$$

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With 75% shunt compensation, the equivalent shunt admittance is

$$Y_{eq} = 7.4 \times 10^{-7} + j14.188 \times 10^{-4}\left(1 - \tfrac{75}{100}\right)$$
$$= 3.547 \times 10^{-4} \underline{/89.88°} \quad \text{S}$$

Since there is no series compensation, the equivalent series impedance is the same as without compensation:

$$Z_{eq} = Z' = 97.0 \underline{/87.2°} \quad \Omega$$

c. The equivalent A parameter for the compensated line is

$$A_{eq} = 1 + \frac{Y_{eq} Z_{eq}}{2}$$
$$= 1 + \frac{(3.547 \times 10^{-4} \underline{/89.88°})(97.0 \underline{/87.2°})}{2}$$
$$= 1 + 0.0172 \underline{/177.1°}$$
$$= 0.9828 \underline{/0.05°} \quad \text{per unit}$$

Then, from (5.1.19),

$$V_{RNL} = \frac{766}{0.9828} = 779.4 \quad kV_{LL}$$

Since the shunt reactors are removed during heavy load conditions, $V_{RFL} = 730$ kV is the same as without compensation. Therefore,

$$\text{percent VR} = \frac{779.4 - 730}{730} \times 100 = 6.77\%$$

The use of shunt reactors at light loads improves the voltage regulation from 12.68% to 6.77% for this line.

EXAMPLE 5.10

Series capacitive compensation to increase transmission-line loadability

Identical series capacitors are installed in each phase at both ends of the line in Example 5.2, providing 30% compensation. Determine the theoretical maximum power that this compensated line can deliver and compare with that of the uncompensated line. Assume $V_S = V_R = 765$ kV.

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SOLUTION

From Example 5.3, the equivalent series reactance without compensation is

$$X' = 97.0 \sin 87.2° = 96.88 \quad \Omega$$

Based on 30% series compensation, half at each end of the line, the impedance of each series capacitor is

$$Z_{cap} = -jX_{cap} = -j\left(\tfrac{1}{2}\right)(0.30)(96.88) = -j14.53 \quad \Omega$$

From Figure 5.4, the *ABCD* matrix of this series impedance is

$$\left[\begin{array}{c|c} 1 & -j14.53 \\ \hline 0 & 1 \end{array}\right]$$

As also shown in Figure 5.4, the equivalent *ABCD* matrix of networks in series is obtained by multiplying the *ABCD* matrices of the individual networks. For this example, there are three networks: the series capacitors at the sending end, the line, and the series capacitors at the receiving end. Therefore, the equivalent *ABCD* matrix of the compensated line is, using the *ABCD* parameters, from Example 5.2,

$$\left[\begin{array}{c|c} 1 & -j14.53 \\ \hline 0 & 1 \end{array}\right]\left[\begin{array}{c|c} 0.9313\angle 0.209° & 97.0\angle 87.2° \\ \hline 1.37 \times 10^{-3}\angle 90.06° & 0.9313\angle 0.209° \end{array}\right]\left[\begin{array}{c|c} 1 & -j14.53 \\ \hline 0 & 1 \end{array}\right]$$

After performing these matrix multiplications, this results in

$$\left[\begin{array}{c|c} A_{eq} & B_{eq} \\ \hline C_{eq} & D_{eq} \end{array}\right] = \left[\begin{array}{c|c} 0.9512\angle 0.205° & 69.70\angle 86.02° \\ \hline 1.37 \times 10^{-3}\angle 90.06° & 0.9512\angle 0.205° \end{array}\right]$$

Therefore,

$$A_{eq} = 0.9512 \quad \text{per unit} \qquad \theta_{Aeq} = 0.205°$$

$$B_{eq} = Z'_{eq} = 69.70 \quad \Omega \qquad \theta_{Zeq} = 86.02°$$

From (5.5.6) with $V_S = V_R = 765$ kV,

$$P_{Rmax} = \frac{(765)^2}{69.70} - \frac{(0.9512)(765)^2}{69.70}\cos(86.02° - 0.205°)$$
$$= 8396 - 583 = 7813 \quad \text{MW}$$

which is 36.2% larger than the value of 5738 MW found in Example 5.5 without compensation. Note that the practical line loadability of this series compensated line is also about 35% larger than the value of 3247 MW found in Example 5.6 without compensation.

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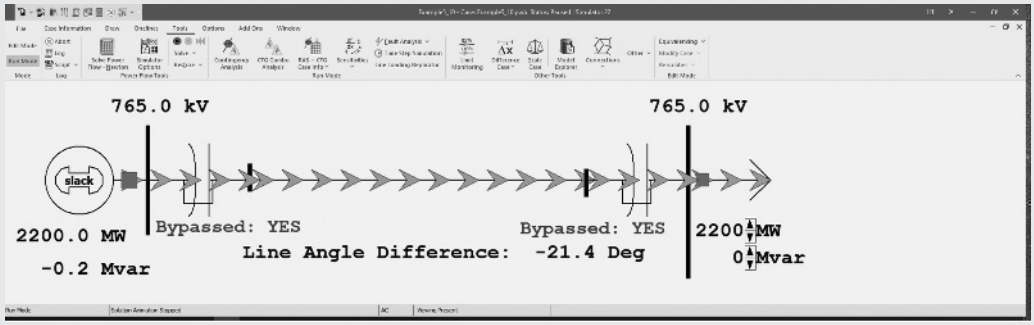

FIGURE 5.17

Screen for Example 5.10

This example is modeled in PowerWorld Simulator case Example 5_10 (see Figure 5.17). When opened, both of the series capacitors are bypassed (i.e., they are modeled as short circuits), meaning this case is initially identical to the Example 5.4 case. Click on the blue Bypassed field to place each of the series capacitors into the circuit. This decreases the angle across the line, resulting in more net power transfer.

PROBLEMS

SECTION 5.1

5.1 A 30-km, 34.5-kV, 60-Hz, three-phase line has a positive-sequence series impedance $z = 0.19 + j0.34\ \Omega$/km. The load at the receiving end absorbs 10 MVA at 33 kV. Assuming a short line, calculate: (a) the *ABCD* parameters, (b) the sending-end voltage for a load power factor of 0.9 lagging, and (c) the sending-end voltage for a load power factor of 0.9 leading.

5.2 A 200-km, 230-kV, 60-Hz, three-phase line has a positive-sequence series impedance $z = 0.08 + j0.48\ \Omega$/km and a positive-sequence shunt admittance $y = j3.33 \times 10^{-6}$ S/km. At full load, the line delivers 250 MW at 0.99 p.f. lagging and at 220 kV. Using the nominal π circuit, calculate: (a) the *ABCD* parameters, (b) the sending-end voltage and current, and (c) the percent voltage regulation.

5.3 Rework Problem 5.2 in per unit using 1000-MVA (three-phase) and 230-kV (line-to-line) base values. Calculate: (a) the per-unit *ABCD* parameters, (b) the per-unit sending-end voltage and current, and (c) the percent voltage regulation.

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5.4 Derive the *ABCD* parameters for the two networks in series, as shown in Figure 5.4.

5.5 Derive the *ABCD* parameters for the T circuit shown in Figure 5.4.

5.6 (a) Consider a medium-length transmission line represented by a nominal π circuit shown in Figure 5.3 of the text. Draw a phasor diagram for lagging power-factor condition at the load (receiving end).

(b) Now consider a nominal T circuit of the medium-length transmission line shown in Figure 5.18.

First draw the corresponding phasor diagram for lagging power-factor load condition.

Then determine the *ABCD* parameters in terms of Y and Z for the nominal T circuit and for the nominal π circuit of part (a).

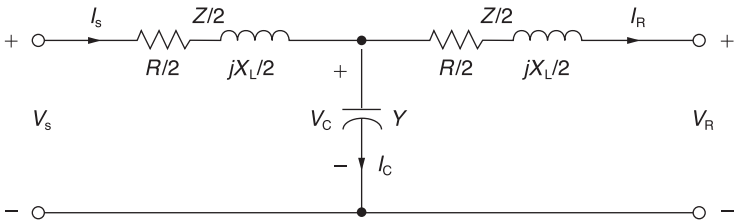

FIGURE 5.18
Nominal T-circuit for Problem 5.6

5.7 The per-phase impedance of a short three-phase transmission line is $0.5\angle 53.15°\ \Omega$. The three-phase load at the receiving end is 1200 kW at 0.8 p.f. lagging. If the line-to-line sending-end voltage is 3.3 kV, determine (a) the receiving-end line-to-line voltage in kV, and (b) the line current.

Draw the phasor diagram with the line current I, as reference.

5.8 Reconsider Problem 5.7 and find the following: (a) sending-end power factor, (b) sending-end three-phase power, and (c) the three-phase line loss.

5.9 A 100-km, 230-kV, 60 Hz, completely transposed three-phase line has one ACSR 954 kcmil conductor per phase and flat horizontal phase spacing with 7 m between adjacent conductors. The line delivers 300 MVA at 218 kV to the receiving end at full load. Using the nominal π circuit, calculate the *ABCD* parameters, sending-end voltage, and percent voltage regulation when the receiving-end power factor is (a) 0.9 lagging, (b) unity, and (c) 0.9 leading. Assume a 50°C conductor temperature to determine the resistance of this line.

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5.10 The 500-kV, 60-Hz, three-phase line in Problems 4.20 and 4.41 has a 180-km length and delivers 1600 MW at 475 kV and at 0.95 power factor leading to the receiving end at full load. Using the nominal π circuit, calculate the (a) $ABCD$ parameters, (b) sending-end voltage and current, (c) sending-end power and power factor, (d) full-load line losses and efficiency, and (e) percent voltage regulation. Assume a 50°C conductor temperature to determine the resistance of this line.

5.11 A 40-km, 220-kV, 60-Hz, three-phase overhead transmission line has a per-phase resistance of 0.15 Ω/km, a per-phase inductance of 1.3263 mH/km, and negligible shunt capacitance. Using the short line model, find the sending-end voltage, voltage regulation, sending-end power, and transmission line efficiency when the line is supplying a three-phase load of (a) 381 MVA at 0.8 power factor lagging and at 220 kV, and (b) 381 MVA at 0.8 power factor leading and at 220 kV.

5.12 A 60-Hz, 100-km, three-phase overhead transmission line, constructed of ACSR conductors, has a series impedance of $(0.1826 + j0.784)$ Ω/km per phase and a shunt capacitive reactance-to-neutral of $185.5 \times 10^3 \angle -90°$ Ω-km per phase. Using the nominal π circuit for a medium-length transmission line, (a) determine the total series impedance and shunt admittance of the line; (b) compute the voltage, the current, and the real and reactive power at the sending end if the load at the receiving end draws 200 MVA at unity power factor and at a line-to-line voltage of 230 kV; and (c) find the percent voltage regulation of the line.

SECTION 5.2

5.13 Evaluate $\cosh(\gamma l)$ and $\tanh(\gamma l/2)$ for $\gamma l = 0.40 \angle 85°$ per unit.

5.14 A 500-km, 500-kV, 60-Hz, uncompensated three-phase line has a positive-sequence series impedance $z = 0.03 + j0.35$ Ω/km and a positive-sequence shunt admittance $y = j4.4 \times 10^{-6}$ S/km. Calculate: (a) Z_c, (b) (γl), and (c) the exact $ABCD$ parameters for this line.

5.15 At full load, the line in Problem 5.14 delivers 900 MW at unity power factor and at 475 kV. Calculate: (a) the sending-end voltage, (b) the sending-end current, (c) the sending-end power factor, (d) the full-load line losses, and (e) the percent voltage regulation.

5.16 A 500-kV, 60-Hz, three-phase line has per-distance parameters of $z = 0.02 + j0.335$ Ω/km and $y = j4.807$ μS/km. It is 300-km long. Calculate: (a) Z_c, (b) (γl), and (c) the exact $ABCD$ parameters for this line. Assume a 50°C conductor temperature.

5.17 At full load, the line in Problem 5.16 delivers 1500 MVA at 480 kV to the receiving-end load. Calculate the sending-end voltage and percent voltage regulation when the receiving-end power factor is (a) 0.9 lagging, (b) unity, and (c) 0.9 leading.

5.18 A 60-Hz, 230-km, three-phase overhead transmission line has a series impedance $z = 0.8431 \angle 79.04°$ Ω/km and a shunt admittance

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$y = 5.105 \times 10^{-6} \angle 90°$ S/km. The load at the receiving end is 125 MW at unity power factor and at 215 kV. Determine the voltage, current, and both real and reactive power at the sending end and the percent voltage regulation of the line. Also, find the wavelength and velocity of propagation of the line.

5.19 Using per-unit calculations, rework Problem 5.18 to determine the sending-end voltage and current.

5.20 (a) The series expansions of the hyperbolic functions are given by

$$\cosh\theta = 1 + \frac{\theta^2}{2} + \frac{\theta^4}{24} + \frac{\theta^6}{720} + \cdots$$

$$\sinh\theta = 1 + \frac{\theta^3}{6} + \frac{\theta^5}{120} + \frac{\theta^7}{5040} + \cdots$$

For the $ABCD$ parameters of a long transmission line represented by an equivalent π circuit, apply the above expansion, considering only the first two terms, and express the result in terms of Y and Z.

(b) For the nominal π and equivalent π circuits shown in Figures 5.3 and 5.7 of the text, show that

$$\frac{A-1}{B} = \frac{Y}{2} \quad \text{and} \quad \frac{A-1}{B} = \frac{Y'}{2}$$

hold good, respectively.

5.21 Starting with (5.1.1) of the text, show that

$$A = \frac{V_S I_S + V_R I_R}{V_R I_S + V_S I_R} \quad \text{and} \quad B = \frac{V_S^2 - V_R^2}{V_R I_S + V_S I_R}$$

5.22 Consider the A parameter of the long line given by $\cosh\theta$, where $\theta = \sqrt{ZY}$. With $x = e^{-\theta} = x_1 + jx_2$ and $A = A_1 + jA_2$, show that x_1 and x_2 satisfy the following:

$$x_1^2 - x_2^2 - 2(A_1 x_1 - A_2 x_2) + 1 = 0$$

and $\quad x_1 x_2 - (A_2 x_1 + A_1 x_2) = 0.$

SECTION 5.3

5.23 Determine the equivalent π circuit for the line in Problem 5.14 and compare it with the nominal π circuit.

5.24 Determine the equivalent π circuit for the line in Problem 5.16. Compare the equivalent π circuit with the nominal π circuit.

5.25 Let the transmission line of Problem 5.12 be extended to cover a distance of 200 km. Assume conditions at the load to be the same as in Problem 5.12. Determine the (a) sending-end voltage, (b) sending-end current, (c) sending-end real and reactive powers, and (d) percent voltage regulation.

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SECTION 5.4

5.26 A 350-km, 500-kV, 60-Hz, three-phase uncompensated line has a positive-sequence series reactance $x = 0.34\ \Omega$/km and a positive-sequence shunt admittance $y = j4.5 \times 10^{-6}$ S/km. Neglecting losses, calculate: (a) Z_c, (b) γl, (c) the $ABCD$ parameters, (d) the wavelength λ of the line in kilometers, and (e) the surge impedance loading in MW.

5.27 Calculate the surge impedance for a transmission line with 0.4 μH/m of series inductance and 28.3 pF/m of shunt capacitance. Assume the line is lossless and operated at 60 Hz.

5.28 Calculate the wavelength for a transmission line with 0.4 μH/m of series inductance and 28.3 pF/m of shunt capacitance. Also calculate the propagation velocity, which is defined to be the wavelength times the frequency. Assume the line is lossless and operated at 60 Hz. Does this propagation speed seem reasonable?

5.29 Rework Problems 5.27 and 5.28 above, assuming the frequency of the line has increased to 400 Hz. How have the surge impedance, propagation velocity, and wavelength been affected?

5.30 From (4.6.22) and (4.10.4), the series inductance and shunt capacitance of a three-phase overhead line are

$$L_a = 2 \times 10^{-7} \ln(D_{eq}/D_{SL}) = \frac{\mu_0}{2\pi} \ln(D_{eq}/D_{SL}) \quad \text{H/m}$$

$$C_{an} = \frac{2\pi\varepsilon_0}{\ln(D_{eq}/D_{SC})} \quad \text{F/m}$$

where $\mu_0 = 4\pi \times 10^{-7}$ H/m and $\varepsilon_0 = \left(\frac{1}{36\pi}\right) \times 10^{-9}$ F/m.

Using these equations, determine formulas for surge impedance and velocity of propagation of an overhead lossless line. Then determine the surge impedance and velocity of propagation for the three-phase line given in Example 4.5. Assume positive-sequence operation. Neglect line losses as well as the effects of the overhead neutral wires and the earth plane.

5.31 A 500-kV, 300-km, 60-Hz, three-phase overhead transmission line, assumed to be lossless, has a series inductance of 0.97 mH/km per phase and a shunt capacitance of 0.0115 μF/km per phase. (a) Determine the phase constant β, the surge impedance Z_c, velocity of propagation v, and the wavelength λ of the line. (b) Determine the voltage, current, real and reactive power at the sending end, and the percent voltage regulation of the line if the receiving-end load is 800 MW at 0.8 power factor lagging and at 500 kV.

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5.32 The following parameters are based on a preliminary line design: $V_S = 1.0$ per unit, $V_R = 0.9$ per unit, $\lambda = 5000$ km, $Z_c = 320\,\Omega$, $\delta = 36.8°$. A three-phase power of 700 MW is to be transmitted to a substation located 315 km from the source of power. (a) Determine a nominal voltage level for the three-phase transmission line, based on the practical line-loadability equation. (b) For the voltage level obtained in part (a), determine the theoretical maximum power that can be transferred by the line.

5.33 Consider a long radial line terminated in its characteristic impedance Z_c. Determine the following:

(a) V_1/I_1, known as the driving point impedance.

(b) $|V_2|/|V_1|$, known as the voltage gain, in terms of $\alpha\ell$.

(c) $|I_2|/|I_1|$, known as the current gain, in terms of $\alpha\ell$.

(d) The complex power gain, $-S_{21}/S_{12}$, in terms of $\alpha\ell$.

(e) The real power efficiency, $(-P_{21}/P_{12}) = \eta$, in terms of $\alpha\ell$.

Note: 1 refers to sending end and 2 refers to receiving end. (S_{21}) is the complex power received at 2; S_{12} is sent from 1.

5.34 For the case of a lossless line, how would the results of Problem 5.33 change?

In terms of Z_c, which is a real quantity for this case, express P_{12} in terms $|I_1|$ and $|V_1|$.

5.35 For a lossless open-circuited line, express the sending-end voltage, V_1, in terms of the receiving-end voltage, V_2, for the three cases of short-line model, medium-length line model, and long-line model. Is it true that the voltage at the open receiving end of a long line is higher than that at the sending end, for small βl?

SECTION 5.5

5.36 For a short transmission line of impedance $(R + jX)$ ohms per phase, show that the maximum power that can be transmitted over the line is

$$P_{max} = \frac{V_R^2}{Z^2}\left(\frac{ZV_S}{V_R} - R\right) \quad \text{where } Z = \sqrt{R^2 + X^2}$$

when the sending-end and receiving-end voltages are fixed, and for the condition

$$Q = \frac{-V_R^2 X}{R^2 + X^2} \quad \text{when } dP/dQ = 0$$

5.37 (a) Consider complex power transmission via the three-phase short line for which the per-phase circuit is shown in Figure 5.19. Express S_{12}, the complex power sent by bus 1 (or V_1), and $(-S_{21})$, the complex

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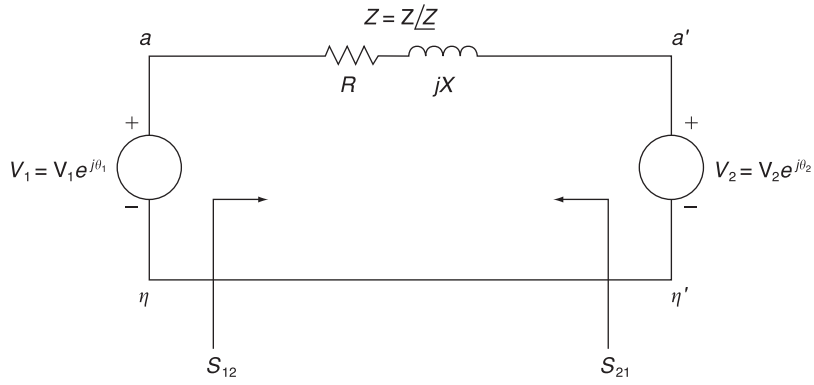

FIGURE 5.19
Per-phase circuit for Problem 5.37

power received by bus 2 (or V_2), in terms of V_1, V_2, Z, $\underline{/Z}$, and $\theta_{12} = \theta_1 - \theta_2$, which is the power angle.

(b) For a balanced three-phase transmission line in per-unit notation with $Z = 1\underline{/85°}$, $\theta_{12} = 10°$, determine S_{12} and $(-S_{21})$ for

(i) $V_1 = V_2 = 1.0$

(ii) $V_1 = 1.1$ and $V_2 = 0.9$

Comment on the changes of real and reactive powers from parts (i) to (ii).

5.38 The line in Problem 5.14 has three ACSR 1113 kcmil conductors per phase. Calculate the theoretical maximum real power that this line can deliver and compare with the thermal limit of the line. Assume $V_S = V_R = 1.0$ per unit and unity power factor at the receiving end.

5.39 Repeat Problems 5.14 and 5.38 if the line length is (a) 200 km, or (b) 600 km.

5.40 If a transmission line is modeled only as a series inductance of 12 mH, with voltage limited to 100 kV on each end and a maximum allowable angle difference of 30°, what is the theoretical maximum real power transfer along the line? Assume the line is operated at 60 Hz.

5.41 A 230-kV, 100-km, 60-Hz, three-phase overhead transmission line with a rated current of 900 A/phase has a series impedance $z = 0.088 + j0.465\ \Omega$/km and a shunt admittance $y = j3.524\ \mu$S/km. (a) Obtain the nominal π equivalent circuit in normal units and in per unit on a base of 100 MVA (three phase) and 230 kV (line-to-line). (b) Determine the three-phase rated MVA of the line. (c) Compute the *ABCD* parameters. (d) Calculate the SIL.

5.42 A three-phase power of 460 MW is transmitted to a substation located 500 km from the source of power. With $V_S = 1$ per unit, $V_R = 0.9$ per unit, $\lambda = 5000$ km, $Z_c = 500\ \Omega$, and $\delta = 36.87°$, determine a nominal voltage level for the lossless transmission line based on Eq. (5.4.29) of the text.

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Using this result, find the theoretical three-phase maximum power that can be transferred by the lossless transmission line.

PW **5.43** Open PowerWorld Simulator case Example 5_4 and graph the load bus voltage as a function of load real power (assuming unity power factor at the load). What is the maximum amount of real power that can be transferred to the load at unity power factor if the load voltage must always be greater than 0.9 per unit?

PW **5.44** Repeat Problem 5.43, but now vary the load reactive power, assuming the load real power is fixed at 1499 MW.

SECTION 5.6

5.45 For the line in Problems 5.14 and 5.38, determine (a) the practical line loadability in MW, assuming $V_S = 1.0$ per unit, $V_R \approx 0.95$ per unit, and $\delta_{max} = 35°$; (b) the full-load current at 0.99 p.f. leading, based on the above practical line loadability; (c) the exact receiving-end voltage for the full-load current in part (b); and (d) the percent voltage regulation. For this line, is loadability determined by the thermal limit, the voltage-drop limit, or steady-state stability?

5.46 Repeat Problem 5.45 for the 500-kV line given in Problem 5.10.

5.47 Determine the practical line loadability in MW and in per-unit of SIL for the line in Problem 5.14 if the line length is (a) 200 km, or (b) 600 km. Assume $V_S = 1.0$ per unit, $V_R = 0.95$ per unit, $\delta_{max} = 35°$, and 0.99 leading power factor at the receiving end.

5.48 It is desired to transmit 2000 MW from a power plant to a load center located 300 km from the plant. Determine the number of 60-Hz, three-phase, uncompensated transmission lines required to transmit this power with one line out of service for the following cases: (a) 345-kV lines, $Z_c = 300\ \Omega$, (b) 500-kV lines, $Z_c = 275\ \Omega$, (c) 765-kV lines, $Z_c = 260\ \Omega$. Assume that $V_S = 1.0$ per unit, $V_R = 0.95$ per unit, and $\delta_{max} = 35°$.

5.49 Repeat Problem 5.48 if it is desired to transmit: (a) 3200 MW to a load center located 300 km from the plant, or (b) 2000 MW to a load center located 400 km from the plant.

5.50 A three-phase power of 4000 MW is to be transmitted through four identical 60-Hz overhead transmission lines over a distance of 300 km. Based on a preliminary design, the phase constant and surge impedance of the line are $\beta = 9.46 \times 10^{-4}$ rad/km and $Z_c = 343\ \Omega$, respectively. Assuming $V_S = 1.0$ per unit, $V_R = 0.9$ per unit, and a power angle $\delta = 36.87°$, determine a suitable nominal voltage level in kV based on the practical line-loadability criteria.

5.51 The power flow at any point on a transmission line can be calculated in terms of the *ABCD* parameters. By letting $A = |A|\angle\alpha$, $B = |B|\angle\beta$, $V_R = |V_R|\angle 0°$, and $V_S = |V_S|\angle\delta$, the complex power at the receiving end can be shown to be

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$$P_R + jQ_R = \frac{|V_R||V_s|\angle\beta - \alpha}{|B|} - \frac{|\delta||V_R^2|\angle\beta - \alpha}{|B|}$$

(a) Draw a phasor diagram corresponding to the above equation. Let it be represented by a triangle O′OA with O′ as the origin and OA representing $P_R + jQ_R$.

(b) By shifting the origin from O′ to O, turn the result of part (a) into a power diagram, redrawing the phasor diagram. For a given fixed value of $|V_R|$ and a set of values for $|V_S|$, draw the loci of point A, thereby showing the so-called receiving-end circles.

(c) From the result of part (b) for a given load with a lagging power factor angle θ_R, determine the amount of reactive power that must be supplied to the receiving end to maintain a constant receiving-end voltage if the sending-end voltage magnitude decreases from $|V_{S1}|$ to $|V_{S2}|$.

5.52 (a) Consider complex power transmission via the three-phase long line for which the per-phase circuit is shown in Figure 5.20. See Problem 5.37 in which the short-line case was considered. Show that

$$\text{sending-end power} = S_{12} = \frac{Y'^*}{2}V_1^2 + \frac{V_1^2}{Z'^*} - \frac{V_1V_2}{Z'^*}e^{j\theta_{12}}$$

$$\text{received power} = -S_{21} = -\frac{Y'^*}{2}V_2^2 - \frac{V_2^2}{Z'^*} + \frac{V_1V_2}{Z'^*}e^{-j\theta_{12}}$$

where $\theta_{12} = \theta_1 - \theta_2$.

(b) For a lossless line with equal voltage magnitudes at each end, show that

$$P_{12} = -P_{21} = \frac{V_1^2 \sin\theta_{12}}{Z_c \sin\beta\ell} = P_{SIL}\frac{\sin\theta_{12}}{\sin\beta\ell}$$

(c) For $\theta_{12} = 45°$ and $\beta = 0.002$ rad/km, find (P_{12}/P_{SIL}) as a function of line length in km, and sketch it.

(d) If a thermal limit of $(P_{12}/P_{SIL}) = 2$ is set, which limit governs for short lines and long lines?

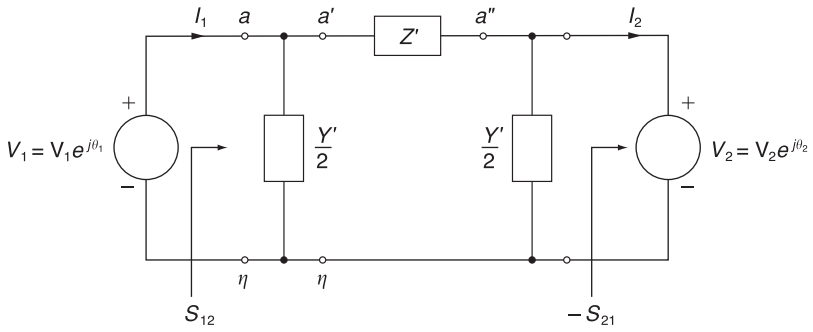

FIGURE 5.20
Per-phase circuit for Problem 5.52

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PW **5.53** Open PowerWorld Simulator case Example 5_8. If the load bus voltage is greater than or equal to 730 kV even with any line segment out of service, what is the maximum amount of real power that can be delivered to the load?

PW **5.54** Repeat Problem 5.53, but now assume any two line segments may be out of service.

SECTION 5.7

5.55 Recalculate the percent voltage regulation in Problem 5.15 when identical shunt reactors are installed at both ends of the line during light loads, providing 65% total shunt compensation. The reactors are removed at full load. Also, calculate the impedance of each shunt reactor.

5.56 Rework Problem 5.17 when identical shunt reactors are installed at both ends of the line, providing 50% total shunt compensation. The reactors are removed at full load.

5.57 Identical series capacitors are installed at both ends of the line in Problem 5.14, providing 40% total series compensation. Determine the equivalent *ABCD* parameters of this compensated line. Also, calculate the impedance of each series capacitor.

5.58 How does the solution to Problem 5.40 change if the transmission line can be operated up to 115 kV instead? What about if it can be operated at 50 Hz?

5.59 Determine the theoretical maximum real power that the series-compensated line in Problem 5.57 can deliver when $V_S = V_R = 1.0$ per unit. Compare your result with that of Problem 5.38.

5.60 What is the minimum amount of series capacitive compensation N_C in percent of the positive-sequence line reactance needed to reduce the number of 765-kV lines in Example 5.8 from five to four? Assume two intermediate substations with one line section out of service. Also, neglect line losses and assume that the series compensation is sufficiently distributed along the line so as to effectively reduce the series reactance of the equivalent π circuit to $X'(1-N_C/100)$.

5.61 Determine the equivalent *ABCD* parameters for the line in Problem 5.14 if it has 70% shunt reactive (inductors) compensation and 40% series capacitive compensation. Half of this compensation is installed at each end of the line, as in Figure 5.16.

5.62 Consider the transmission line of Problem 5.18. (a) Find the *ABCD* parameters of the line when uncompensated. (b) For a series capacitive compensation of 70% (35% at the sending end and 35% at the receiving end), determine the *ABCD* parameters. Comment on the relative change in the magnitude of the *B* parameter with respect to the relative changes in the magnitudes of the *A*, *C*, and *D* parameters. Also, comment on the maximum power that can be transmitted when series compensated.

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5.63 Given the uncompensated line of Problem 5.18, let a three-phase shunt reactor (inductor) that compensates for 70% of the total shunt admittance of the line be connected at the receiving end of the line during no-load conditions. Determine the effect of voltage regulation with the reactor connected at no load. Assume that the reactor is removed under full-load conditions.

5.64 Let the three-phase lossless transmission line of Problem 5.31 supply a load of 1000 MVA at 0.8 power factor lagging and at 500 kV. (a) Determine the capacitance/phase and total three-phase Mvars supplied by a three-phase, Δ-connected shunt-capacitor bank at the receiving end to maintain the receiving-end voltage at 500 kV when the sending end of the line is energized at 500 kV. (b) If series capacitive compensation of 40% is installed at the midpoint of the line, without the shunt capacitor bank at the receiving end, compute the sending-end voltage and percent voltage regulation.

PW **5.65** Open PowerWorld Simulator case Example 5_10 with the series capacitive compensation at both ends of the line in service. Graph the load bus voltage as a function of load real power (assuming unity power factor at the load). What is the maximum amount of real power that can be transferred to the load at unity power factor if the load voltage is always greater than 0.85 per unit?

PW **5.66** Open PowerWorld Simulator case Example 5_10 with the series capacitive compensation at both ends of the line in service. With the reactive power load fixed at 400 Mvar, graph the load bus voltage as the MW load is varied between 0 and 2600 MW in 200-MW increments. Then repeat with both of the series compensation elements out of service.

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6 Power Flows

Tennessee Valley Authority (TVA) Regional Operations Center. (Courtesy of TVA.)

LEARNING OBJECTIVES

At the conclusion of this chapter, you should be able to:

1. Know some of the basic methods, including direct and iterative techniques, for solving algebraic equations.
2. Understand the formulation of the power flow problem, which determines the voltage magnitude and angle at each bus in a power system under balanced three-phase, steady-state conditions.
3. Know two solution methods, Gauss-Seidel and Newton-Raphson, for solving system power flows.
4. Understand different techniques to control system power flows.
5. Understand some of the approximations made for fast power flow algorithms, including fast decoupled power flow and the dc power flow.
6. Use PowerWorld Simulator to calculate and visualize power flows.

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Successful power system operation under normal balanced three-phase steady-state conditions requires the following:

1. Generation supplies the demand (load) plus losses.
2. Bus voltage magnitudes remain close to rated values.
3. Generators operate within specified real and reactive power limits.
4. Transmission lines and transformers are not overloaded.

The power flow (sometimes also called the *load flow*) is the basic tool for investigating these requirements. The power flow determines the voltage magnitude and angle at each bus in a power system under balanced three-phase steady-state conditions. It also computes real and reactive power flows for all equipment interconnecting the buses, as well as equipment losses.

Both existing power systems and proposed changes, including new generation and transmission, are of interest.

Conventional nodal or loop analysis is not suitable for power flow studies because the input data for loads are normally given in terms of power, not impedance. Also, generators are considered to be power sources, not voltage or current sources. The power flow problem is therefore formulated as a set of nonlinear algebraic equations suitable for computer solution.

Sections 6.1 through 6.3 review some basic methods, including direct and iterative techniques for solving algebraic equations. Then Sections 6.4 through 6.6 formulate the power flow problem, specify input data, and present two solution methods: Gauss-Seidel and Newton-Raphson. Means for controlling power flows are discussed in Section 6.7. Sections 6.8 and 6.9 introduce sparsity techniques and a fast decoupled power flow method, while Section 6.10 discusses the dc power flow. Finally, Section 6.11 considers the power flow representation of wind turbine and solar PV generators and Section 6.12 discusses power flow solutions for large systems.

Since balanced three-phase steady-state conditions are assumed, this chapter uses only positive-sequence networks. Also, all power flow equations and input/output data are given in per unit (p.u.).

CASE STUDY

Xcel Energy Strengthens the Grid with Advanced SVCs*

Martin Cameron, Anders Böstrom, and Joe Warner

The following paragraphs summarize the above article, which describes how one utility company, Xcel Energy Inc., uses static var compensators (SVCs) to improve stability and voltage control of its

*Martin Cameron, Anders Böstrom, and Joe Warner, "Xcel Energy Strengthens Grid with Advanced SVCs," available online, with permission, from *T&D World Magazine* (January 25, 2021).

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transmission system in southeast New Mexico, U.S. [16]. Sign in at www.cengage.com to download this product and to access its online resources.

In 2010, Xcel Energy was asked by the Southwest Power Pool (SPP) to build out the transmission system in southeast New Mexico, including a transmission expansion of new 115-kV and 345-kV projects. In 2016, two static var compensators (SVCs), one at the Roadrunner Substation and the other at the China Draw Substation, were built to support SPPs request (see Figure 1).

The SVC at each of the two substations includes the following: (1) thyristor-switched capacitors (TCRs) of 70 Mvar installed on the 14.5-kV SVC bus; (2) thyristor-controlled reactors (TCRs) of 65 Mvar installed on the 14.5-kV SVC bus; (3) a 110-MVA transformer interconnecting the 14.5-kV SVC bus and 115-kV transmission bus; and (4) a 30-Mvar harmonic filter installed on the 115-kV bus (Figures 2 and 3).

The transmission system sensitivity to an SVC controller largely depends on the system fault level. When the system fault level is large compared to the SVC rating, interaction between the SVC controller and the system is usually not a concern. As a rule of thumb, the short-circuit strength of the system should be more than three times the SVC rating (i.e., the short-circuit

Figure 1 Map of southeast New Mexico

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Figure 2 Thyristor-switched capacitors (TSCs) and thyristor-controlled reactors (TCRs)

Figure 3 30-Mvar, 115-kV C-type filter at Roadrunner Substation

ratio (SCR) between the system and SVC should be greater than three). The challenge with a weak, heavily shunt-compensated transmission system, where the size of the SVC capacitive rating approaches the system fault level, is that oscillatory frequencies occur in the interaction with SVC controls. Advanced SVC control strategies in combination with a flexible SVC topology greatly improve the operability of large SVCs in weak transmission systems.

Xcel Energy's SVCs were designed to operate under two significantly different scenarios: an extremely weak transmission system in 2016 and a strong built-out system in 2020. Xcel Energy's system was fairly stable during the build-out. The SVC advanced control functionality provided more stability by decreasing interactions between the electrically coupled SVCs.

In addition to the TSCs on the 14.5-kV bus, each of the Roadrunner and China Draw substations has five mechanically switched capacitor (MSC) banks—three rated at 14.4 Mvar and two at 28.8 Mvar—on the 115-kV bus. Initially, in 2016, usage of the MSCs was limited, but by 2020, the transmission build-out enabled the usage of all MSCs. The SVC-based local control of the MSCs on the 115-kV bus preserves the SVC dynamic range by using the MSCs to absorb daily voltage changes related to normal steady-state load variation. The MSC banks can also be selected for remote operation, transferring control to the dispatch center by the supervisory control and data acquisition system.

Fault-induced delayed voltage recovery is a condition that occurs when a system with a high penetration of motor loads (e.g., pumps, heating-ventilation-air-conditioner loads, etc.) experiences a fault that causes the motors to begin stalling. Once the fault is cleared and the system reenergizes, the motors try to speed back up, but an insufficient amount of reactive power available from remote generators causes a voltage depression that delays voltage recovery. SVCs effectively reduce or eliminate the delay in voltage recovery.

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In the event of an unsymmetrical fault (e.g., single-line-to-ground fault or line-to-line fault), overvoltages on the unfaulted phases of the SVC bus trigger the SVC control to block TSC phases independently (denoted Advanced TSC blocking)—to reduce overvoltages during the unsymmetrical fault. Also, Advanced TSC blocking recognizes the moment of fault clearing and subsequently blocks all TSC phases for approximately 10 ms. These actions enable the SVC to support voltages during the fault while reducing the subsequent overvoltage immediately following fault clearing.

In 2019, a dedicated fiber optic communication link was made available between the Roadrunner and China Draw substations. Parallel mode operation of the two SVCs was then commissioned successfully and tested with parallel mode both enabled and disabled.

An alternative to SVCs for improving stability and voltage control of a transmission system is the static synchronous compensator (STATCOM). A voltage-source converter (VSC) is the basic electronic part of a STATCOM, which converts a dc voltage into a three-phase set of ac output voltages with desired amplitude, frequency, and phase.

National Grid, one of the world's largest investor-owned energy companies, deployed a STATCOM scheme totaling 975 Mvar at three substations in the southeast United Kingdom (U.K.). In November 2019, the last of the three sites was energized. With STATCOM technology in place, the Bolney, Ninfield and Richborough substations now support National Grid's ac network and enhance regional voltage stability for the U.K.–Belgium HVDC interconnection, which will deliver more than 1 GW of bidirectional power between the two countries. With less than 90-ms response time, the scheme, which consists of three separate hybrid STATCOM units, delivers dynamic reactive power in a range from −300 Mvar inductive (absorbing vars) to +675 Mvar capacitive (delivering vars) with 95% availability per customer requirement [17].

6.1 DIRECT SOLUTIONS TO LINEAR ALGEBRAIC EQUATIONS: GAUSS ELIMINATION

The power flow requires the solution of a set of nonlinear equations. Historically, two iterative approaches have been used for solving these equations, either the Gauss-Seidel method presented in Sections 6.2 and 6.5, or the Newton-Raphson method presented in Sections 6.3 and 6.6. The Newton-Raphson method is now much more common and at its core, it requires the solution of a set of linear algebraic equations. Therefore, consider the following set of linear algebraic equations in matrix format:

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$$\begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1N} \\ A_{21} & A_{22} & \cdots & A_{2N} \\ \vdots & & \vdots & \\ A_{N1} & A_{N2} & \cdots & A_{NN} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix} \tag{6.1.1}$$

or

$$\mathbf{Ax} = \mathbf{y} \tag{6.1.2}$$

where $\mathbf{x}$ and $\mathbf{y}$ are N vectors and $\mathbf{A}$ is an $N \times N$ square matrix. The components of $\mathbf{x}$, $\mathbf{y}$, and $\mathbf{A}$ may be real or complex. Given $\mathbf{A}$ and $\mathbf{y}$, the objective is to solve for $\mathbf{x}$. Assume the det($\mathbf{A}$) is nonzero, so a unique solution to (6.1.1) exists.

The solution $\mathbf{x}$ can be obtained easily when $\mathbf{A}$ is an upper triangular matrix with nonzero diagonal elements. Then (6.1.1) has the form

$$\begin{bmatrix} A_{11} & A_{12} \ldots & & A_{1N} \\ 0 & A_{22} \ldots & & A_{2N} \\ \vdots & & & \\ 0 & 0 \ldots & A_{N-1,\,N-1} & A_{N-1,N} \\ 0 & 0 \ldots 0 & & A_{NN} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{N-1} \\ x_N \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_{N-1} \\ y_N \end{bmatrix} \tag{6.1.3}$$

Since the last equation in (6.1.3) involves only x_N,

$$x_N = \frac{y_N}{A_{NN}} \tag{6.1.4}$$

After x_N is computed, the next-to-last equation can be solved:

$$x_{N-1} = \frac{y_{N-1} - A_{N-1,\,N} x_N}{A_{N-1,\,N-1}} \tag{6.1.5}$$

In general, with x_N, x_{N-1}, … , x_{k+1} already computed, the kth equation can be solved as

$$x_k = \frac{y_k - \sum_{n=k+1}^{N} A_{kn} x_n}{A_{kk}} \qquad k = N, N-1, \ldots, 1 \tag{6.1.6}$$

This procedure for solving (6.1.3) is called *back substitution.*

If $\mathbf{A}$ is not an upper triangular, (6.1.1), it can be transformed to an equivalent equation with an upper triangular matrix. The transformation, called *Gauss elimination*, is described by the following $(N - 1)$ steps. During Step 1, use the first

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equation in (6.1.1) to eliminate x_1 from the remaining equations. That is, Equation 1 is multiplied by A_{n1}/A_{11} and then subtracted from equation n, for $n = 2, 3, \ldots, N$. After completing Step 1, we have

$$\begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1N} \\ 0 & \left(A_{22} - \frac{A_{21}}{A_{11}}A_{12}\right) & \cdots & \left(A_{2N} - \frac{A_{21}}{A_{11}}A_{1N}\right) \\ 0 & \left(A_{32} - \frac{A_{31}}{A_{11}}A_{12}\right) & \cdots & \left(A_{3N} - \frac{A_{31}}{A_{11}}A_{1N}\right) \\ \vdots & \vdots & & \vdots \\ 0 & \left(A_{N2} - \frac{A_{N1}}{A_{11}}A_{12}\right) & \cdots & \left(A_{NN} - \frac{A_{N1}}{A_{11}}A_{1N}\right) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_N \end{bmatrix}$$

$$= \begin{bmatrix} y_1 \\ y_2 - \frac{A_{21}}{A_{11}}y_1 \\ y_3 - \frac{A_{31}}{A_{11}}y_1 \\ \vdots \\ y_N - \frac{A_{N1}}{A_{11}}y_1 \end{bmatrix} \tag{6.1.7}$$

Equation (6.1.7) has the following form:

$$\begin{bmatrix} A_{11}^{(1)} & A_{12}^{(1)} & \cdots & A_{1N}^{(1)} \\ 0 & A_{22}^{(1)} & \cdots & A_{2N}^{(1)} \\ 0 & A_{32}^{(1)} & \cdots & A_{3N}^{(1)} \\ \vdots & \vdots & & \vdots \\ 0 & A_{N2}^{(1)} & \cdots & A_{NN}^{(1)} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_N \end{bmatrix} = \begin{bmatrix} y_1^{(1)} \\ y_2^{(1)} \\ y_3^{(1)} \\ \vdots \\ y_N^{(1)} \end{bmatrix} \tag{6.1.8}$$

where the superscript (1) denotes Step 1 of the Gauss elimination.

During Step 2, use the second equation in (6.1.8) to eliminate x_2 from the remaining (third, fourth, fifth, and so on) equations. That is, Equation 2 is multiplied by $A_{n2}^{(1)}/A_{22}^{(1)}$ and subtracted from equation n, for $n = 3, 4, \ldots, N$.

After Step 2, there is

$$\begin{bmatrix} A_{11}^{(2)} & A_{12}^{(2)} & A_{13}^{(2)} & \cdots & A_{1N}^{(2)} \\ 0 & A_{22}^{(2)} & A_{23}^{(2)} & \cdots & A_{2N}^{(2)} \\ 0 & 0 & A_{33}^{(2)} & \cdots & A_{3N}^{(2)} \\ 0 & 0 & A_{43}^{(2)} & \cdots & A_{4N}^{(2)} \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & A_{N3}^{(2)} & \cdots & A_{NN}^{(2)} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ \vdots \\ x_N \end{bmatrix} = \begin{bmatrix} y_1^{(2)} \\ y_2^{(2)} \\ y_3^{(2)} \\ y_4^{(2)} \\ \vdots \\ y_N^{(2)} \end{bmatrix} \tag{6.1.9}$$

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During step k, start with $\mathbf{A}^{(k-1)}\mathbf{x} = \mathbf{y}^{(k-1)}$. The first k of these equations, already triangularized, are left unchanged. Also, equation k is multiplied by $A_{nk}^{(k-1)}/A_{kk}^{(k-1)}$ and then subtracted from equation n, for $n = k+1, k+2, \ldots, N$.

After $(N-1)$ steps, the equivalent equation is $\mathbf{A}^{(N-1)}\mathbf{x} = \mathbf{y}^{(N-1)}$, where $\mathbf{A}^{(N-1)}$ is upper triangular.

EXAMPLE 6.1

Gauss elimination and back substitution: direct solution to linear algebraic equations

Solve

$$\left[\begin{array}{c|c} 10 & 5 \\ \hline 2 & 9 \end{array}\right]\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$$

using Gauss elimination and back substitution.

SOLUTION

Since $N = 2$ for this example, there is $(N-1) = 1$ Gauss elimination step. Multiplying the first equation by $A_{21}/A_{11} = 2/10$ and then subtracting from the second,

$$\left[\begin{array}{c|c} 10 & 5 \\ \hline 0 & 9 - \dfrac{2}{10}(5) \end{array}\right]\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 - \dfrac{2}{10}(6) \end{bmatrix}$$

or

$$\left[\begin{array}{c|c} 10 & 5 \\ \hline 0 & 8 \end{array}\right]\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 6 \\ 1.8 \end{bmatrix}$$

which has the form $\mathbf{A}^{(1)}\mathbf{x} = \mathbf{y}^{(1)}$ where $\mathbf{A}^{(1)}$ is upper triangular. Now, using back substitution, (6.1.6) gives, for $k = 2$:

$$x_2 = \frac{y_2^{(1)}}{A_{22}^{(1)}} = \frac{1.8}{8} = 0.225$$

and, for $k = 1$,

$$x_1 = \frac{y_1^{(1)} - A_{12}^{(1)}x_2}{A_{11}^{(1)}} = \frac{6 - (5)(0.225)}{10} = 0.4875$$

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EXAMPLE 6.2

Gauss elimination: triangularizing a matrix

Use Gauss elimination to triangularize

$$\begin{bmatrix} 2 & 3 & -1 \\ -4 & 6 & 8 \\ 10 & 12 & 14 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ 9 \end{bmatrix}$$

SOLUTION

There are $(N - 1) = 2$ Gauss elimination steps. During Step 1, subtract $A_{21}/A_{11} = -4/2 = -2$ times Equation 1 from Equation 2, and subtract $A_{31}/A_{11} = 10/2 = 5$ times Equation 1 from Equation 3, to give

$$\left[\begin{array}{c|c|c} 2 & 3 & -1 \\ \hline 0 & 6-(-2)(3) & 8-(-2)(-1) \\ \hline 0 & 12-(5)(3) & 14-(5)(-1) \end{array}\right] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 7-(-2)(5) \\ 9-(5)(5) \end{bmatrix}$$

or

$$\begin{bmatrix} 2 & 3 & -1 \\ 0 & 12 & 6 \\ 0 & -3 & 19 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 17 \\ -16 \end{bmatrix}$$

which is $\mathbf{A}^{(1)}\mathbf{x} = \mathbf{y}^{(1)}$. During Step 2, subtract $A_{32}^{(1)}/A_{22}^{(1)} = -3/12 = -0.25$ times Equation 2 from Equation 3, to give

$$\left[\begin{array}{c|c|c} 2 & 3 & -1 \\ \hline 0 & 12 & 6 \\ \hline 0 & 0 & 19-(-0.25)(6) \end{array}\right] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 17 \\ -16-(-0.25)(17) \end{bmatrix}$$

or

$$\begin{bmatrix} 2 & 3 & -1 \\ 0 & 12 & 6 \\ 0 & 0 & 20.5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 17 \\ -11.75 \end{bmatrix}$$

which is triangularized. The solution $\mathbf{x}$ now can be easily obtained via back substitution.

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In general, the computer storage requirements for Gauss elimination and back substitution include N^2 memory locations for $\mathbf{A}$ and N locations for $\mathbf{y}$. If there is no further need to retain $\mathbf{A}$ and $\mathbf{y}$, then $\mathbf{A}^{(k)}$ can be stored in the location of $\mathbf{A}$, and $\mathbf{y}^{(k)}$, as well as the solution $\mathbf{x}$, can be stored in the location of $\mathbf{y}$. Additional memory is also required for iterative loops, arithmetic statements, and working space.

Computer time requirements can be evaluated by determining the number of arithmetic operations required for Gauss elimination and back substitution. One can show that Gauss elimination requires $(N^3 - N)/3$ multiplications, $(N)(N - 1)/2$ divisions, and $(N^3 - N)/3$ subtractions. Also, back substitution requires $(N)(N - 1)/2$ multiplications, N divisions, and $(N)(N - 1)/2$ subtractions. Therefore, for very large N, the approximate computer time for solving (6.1.1) by Gauss elimination and back substitution is the time required to perform $N^3/3$ multiplications and $N^3/3$ subtractions.

For example, consider a digital computer with a 2×10^{-9} s multiplication time and 1×10^{-9} s addition or subtraction time. Solving $N = 10{,}000$ equations would require

$$\frac{1}{3}N^3(2 \times 10^{-9}) + \frac{1}{3}N^3(1 \times 10^{-9}) = \frac{1}{3}(10{,}000)^3(3 \times 10^{-9}) = 1000 \quad \text{s}$$

approximately, plus some additional bookkeeping time for indexing and managing loops.

Since the power flow problem often involves solving power systems with tens of thousands of equations, by itself Gauss elimination would not be a good solution. However, for matrixes that have relatively few nonzero elements, known as sparse matrices, special techniques can be employed to significantly reduce computer storage and time requirements. Since all large power systems can be modeled using sparse matrices, these techniques are briefly introduced in Section 6.8.

6.2 ITERATIVE SOLUTIONS TO LINEAR ALGEBRAIC EQUATIONS: JACOBI AND GAUSS-SEIDEL

An alternative to a direct solution of a set of linear equations is to use an iterative approach. When the digital power flow was first introduced in the 1950s on computers, this approach was used. While the direct approach is now much more common due to the introduction of sparse matrix methods in the 1960s, there is still benefit to understanding a general iterative solution to (6.1.1), which proceeds as follows. First, select an initial guess $\mathbf{x}(0)$. Then use

$$\mathbf{x}(i+1) = \mathbf{g}[\mathbf{x}(i)] \qquad i = 0, 1, 2, \ldots \tag{6.2.1}$$

where $\mathbf{x}(i)$ is the ith guess and $\mathbf{g}$ is an N vector of functions that specify the iteration method. Continue this procedure until the following stopping condition is satisfied, as

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$$\left|\frac{x_k(i+1) - x_k(i)}{x_k(i)}\right| < \varepsilon \qquad \text{for all } k = 1, 2, \ldots, N \tag{6.2.2}$$

where $x_k(i)$ is the kth component of $\mathbf{x}(i)$ and ε is a specified *tolerance level.*

The following questions are pertinent:

1. Will the iteration procedure converge to the unique solution?
2. What is the convergence rate (how many iterations are required)?
3. When using a digital computer, what are the computer storage and time requirements?

These questions are addressed for two specific iteration methods: *Jacobi* and *Gauss-Seidel.** The Jacobi method is obtained by considering the kth equation of (6.1.1), as follows:

$$y_k = \mathrm{A}_{k1}x_1 + \mathrm{A}_{k2}x_2 + \cdots + \mathrm{A}_{kk}x_k + \cdots + \mathrm{A}_{kN}x_N \tag{6.2.3}$$

Solving for x_k,

$$\begin{aligned} x_k &= \frac{1}{\mathrm{A}_{kk}}[y_k - (\mathrm{A}_{k1}x_1 + \cdots + \mathrm{A}_{k,k-1}x_{k-1} + \mathrm{A}_{k,k+1}x_{k+1} + \cdots + \mathrm{A}_{kN}x_N)] \\ &= \frac{1}{\mathrm{A}_{kk}}\left[y_k - \sum_{n=1}^{k-1}\mathrm{A}_{kn}x_n - \sum_{n=k+1}^{N}\mathrm{A}_{kn}x_n\right] \end{aligned} \tag{6.2.4}$$

The Jacobi method uses the "old" values of $\mathbf{x}(i)$ at iteration i on the right side of (6.2.4) to generate the "new" value $x_k(i+1)$ on the left side of (6.2.4). That is,

$$x_k(i+1) = \frac{1}{\mathrm{A}_{kk}}\left[y_k - \sum_{n=1}^{k-1}\mathrm{A}_{kn}x_n(i) - \sum_{n=k+1}^{N}\mathrm{A}_{kn}x_n(i)\right] k = 1, 2, \ldots, N \tag{6.2.5}$$

The Jacobi method given by (6.2.5) can also be written in the following matrix format:

$$\mathbf{x}(i+1) = \mathbf{M}\mathbf{x}(i) + \mathbf{D}^{-1}\mathbf{y} \tag{6.2.6}$$

where

$$\mathbf{M} = \mathbf{D}^{-1}(\mathbf{D} - \mathbf{A}) \tag{6.2.7}$$

and

$$\mathbf{D} = \begin{bmatrix} \mathrm{A}_{11} & 0 & 0 & \cdots & 0 \\ 0 & \mathrm{A}_{22} & 0 & \cdots & 0 \\ 0 & \vdots & \vdots & & \vdots \\ \vdots & & & & 0 \\ 0 & 0 & 0 & \cdots & \mathrm{A}_{NN} \end{bmatrix} \tag{6.2.8}$$

*The Jacobi method is also called the Gauss method.

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For Jacobi, **D** consists of the diagonal elements of the **A** matrix.

The Gauss-Seidel method is given by

$$x_k(i+1) = \frac{1}{A_{kk}}\left[y_k - \sum_{n=1}^{k-1} \mathrm{A}_{kn}x_n(i+1) - \sum_{n=k+1}^{N} \mathrm{A}_{kn}x_n(i)\right] \tag{6.2.9}$$

Comparing (6.2.9) with (6.2.5), note that Gauss-Seidel is similar to Jacobi except that during each iteration, the "new" values, $x_n(i+1)$, for $n < k$ are used on the right side of (6.2.9) to generate the "new" value $x_k(i+1)$ on the left side.

EXAMPLE 6.3

Jacobi method: iterative solution to linear algebraic equations

Solve Example 6.1 using the Jacobi method. Start with $x_1(0) = x_2(0) = 0$ and continue until (6.2.2) is satisfied for $\varepsilon = 10^{-4}$.

SOLUTION

From (6.2.5) with $N = 2$,

$$k = 1 \quad x_1(i+1) = \frac{1}{\mathrm{A}_{11}}[y_1 - \mathrm{A}_{12}x_2(i)] = \frac{1}{10}[6 - 5x_2(i)]$$

$$k = 2 \quad x_2(i+1) = \frac{1}{\mathrm{A}_{22}}[y_2 - \mathrm{A}_{21}x_1(i)] = \frac{1}{9}[3 - 2x_1(i)]$$

Alternatively, in matrix format using (6.2.6) through (6.2.8),

$$\mathbf{D}^{-1} = \left[\begin{array}{c|c} 10 & 0 \\ \hline 0 & 9 \end{array}\right]^{-1} = \left[\begin{array}{c|c} \frac{1}{10} & 0 \\ \hline 0 & \frac{1}{9} \end{array}\right]$$

$$\mathbf{M} = \left[\begin{array}{c|c} \frac{1}{10} & 0 \\ \hline 0 & \frac{1}{9} \end{array}\right]\left[\begin{array}{c|c} 0 & -5 \\ \hline -2 & 0 \end{array}\right] = \left[\begin{array}{c|c} 0 & -\frac{5}{10} \\ \hline -\frac{2}{9} & 0 \end{array}\right]$$

$$\begin{bmatrix} x_1(i+1) \\ x_2(i+1) \end{bmatrix} = \left[\begin{array}{c|c} 0 & -\frac{5}{10} \\ \hline -\frac{2}{9} & 0 \end{array}\right]\begin{bmatrix} x_1(i) \\ x_2(i) \end{bmatrix} + \left[\begin{array}{c|c} \frac{1}{10} & 0 \\ \hline 0 & \frac{1}{9} \end{array}\right]\begin{bmatrix} 6 \\ 3 \end{bmatrix}$$

The above two formulations are identical. Starting with $x_1(0) = x_2(0) = 0$, the iterative solution is given in the following table:

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Jacobi

i	0	1	2	3	4	5	6	7	8	9	10
$x_1(i)$	0	0.60000	0.43334	0.50000	0.48148	0.48889	0.48683	0.48766	0.48743	0.48752	0.48749
$x_2(i)$	0	0.33333	0.20000	0.23704	0.22222	0.22634	0.22469	0.22515	0.22496	0.22502	0.22500

As shown, the Jacobi method converges to the unique solution obtained in Example 6.1.

The convergence criterion is satisfied at the 10th iteration, since

$$\left|\frac{x_1(10) - x_1(9)}{x_1(9)}\right| = \left|\frac{0.48749 - 0.48752}{0.48749}\right| = 6.2 \times 10^{-5} < \varepsilon$$

and

$$\left|\frac{x_2(10) - x_2(9)}{x_2(9)}\right| = \left|\frac{0.22500 - 0.22502}{0.22502}\right| = 8.9 \times 10^{-5} < \varepsilon$$

The Gauss-Seidel method of (6.2.9) can also be written in the matrix format of (6.2.6) and (6.2.7), where

$$\mathbf{D} = \begin{bmatrix} A_{11} & 0 & 0 & \cdots & 0 \\ A_{21} & A_{22} & 0 & \cdots & 0 \\ \vdots & \vdots & & & \vdots \\ A_{N1} & A_{N2} & \cdots & & A_{NN} \end{bmatrix} \tag{6.2.10}$$

For Gauss-Seidel, $\mathbf{D}$ in (6.2.10) is the lower triangular portion of $\mathbf{A}$, whereas for Jacobi, $\mathbf{D}$ in (6.2.8) is the diagonal portion of $\mathbf{A}$.

EXAMPLE 6.4

Gauss-Seidel method: iterative solution to linear algebraic equations

SOLUTION

Rework Example 6.3 using the Gauss-Seidel method.

From (6.2.9),

$$k = 1 \quad x_1(i+1) = \frac{1}{A_{11}}[y_1 - A_{12}x_2(i)] = \frac{1}{10}[6 - 5x_2(i)]$$

$$k = 2 \quad x_2(i+1) = \frac{1}{A_{22}}[y_2 - A_{21}x_1(i+1)] = \frac{1}{9}[3 - 2x_1(i+1)]$$

(*Continued*)

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Using this equation for $x_1(i+1)$, $x_2(i+1)$ also can be written as

$$x_2(i+1) = \frac{1}{9}\left\{3 - \frac{2}{10}[6 - 5x_2(i)]\right\}$$

Alternatively, in matrix format, using (6.2.10), (6.2.6), and (6.2.7),

$$\mathbf{D}^{-1} = \left[\begin{array}{c|c} 10 & 0 \\ \hline 2 & 9 \end{array}\right]^{-1} = \left[\begin{array}{c|c} \frac{1}{10} & 0 \\ \hline -\frac{2}{90} & \frac{1}{9} \end{array}\right]$$

$$\mathbf{M} = \left[\begin{array}{c|c} \frac{1}{10} & 0 \\ \hline -\frac{2}{90} & \frac{1}{9} \end{array}\right]\left[\begin{array}{c|c} 0 & -5 \\ \hline 0 & 0 \end{array}\right] = \left[\begin{array}{c|c} 0 & -\frac{1}{2} \\ \hline 0 & \frac{1}{9} \end{array}\right]$$

$$\begin{bmatrix} x_1(i+1) \\ x_2(i+1) \end{bmatrix} = \left[\begin{array}{c|c} 0 & -\frac{1}{2} \\ \hline 0 & \frac{1}{9} \end{array}\right]\begin{bmatrix} x_1(i) \\ x_2(i) \end{bmatrix} + \left[\begin{array}{c|c} \frac{1}{10} & 0 \\ \hline -\frac{2}{90} & \frac{1}{9} \end{array}\right]\begin{bmatrix} 6 \\ 3 \end{bmatrix}$$

These two formulations are identical. Starting with $x_1(0) = x_2(0) = 0$, the solution is given in the following table:

Gauss-Seidel

i	0	1	2	3	4	5	6
$x_1(i)$	0	0.60000	0.50000	0.48889	0.48765	0.48752	0.48750
$x_2(i)$	0	0.20000	0.22222	0.22469	0.22497	0.22500	0.22500

For this example, Gauss-Seidel converges in 6 iterations, compared to 10 iterations with Jacobi.

The convergence rate is faster with Gauss-Seidel for some **A** matrices, but faster with Jacobi for other **A** matrices. In some cases, one method diverges while the other converges. In other cases both methods diverge, as illustrated by the next example.

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EXAMPLE 6.5

Divergence of Gauss-Seidel method

Using the Gauss-Seidel method with $x_1(0) = x_2(0) = 0$, solve

$$\left[\begin{array}{c|c} 5 & 10 \\ \hline 9 & 2 \end{array}\right]\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$$

SOLUTION

Note that these equations are the same as those in Example 6.1, except that x_1 and x_2 are interchanged. Using (6.2.9),

$$k = 1 \qquad x_1(i+1) = \frac{1}{\mathrm{A}_{11}}[y_1 - \mathrm{A}_{12}x_2(i)] = \frac{1}{5}[6 - 10x_2(i)]$$

$$k = 2 \qquad x_2(i+1) = \frac{1}{\mathrm{A}_{22}}[y_2 - \mathrm{A}_{21}x_1(i+1)] = \frac{1}{2}[3 - 9x_1(i+1)]$$

Successive calculations of x_1 and x_2 are shown in the following table:

Gauss-Seidel

i	0	1	2	3	4	5
$x_1(i)$	0	1.2	9	79.2	711	6397
$x_2(i)$	0	−3.9	−39	−354.9	−3198	−28,786

The unique solution by matrix inversion is

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \left[\begin{array}{c|c} 5 & 10 \\ \hline 9 & 2 \end{array}\right]^{-1}\begin{bmatrix} 6 \\ 3 \end{bmatrix} = \frac{-1}{80}\left[\begin{array}{c|c} 2 & -10 \\ \hline -9 & 5 \end{array}\right]\begin{bmatrix} 6 \\ 3 \end{bmatrix} = \begin{bmatrix} 0.225 \\ 0.4875 \end{bmatrix}$$

As shown, Gauss-Seidel does not converge to the unique solution; instead it diverges. Jacobi also diverges for this example.

If any diagonal element A_{kk} equals zero, then Jacobi and Gauss-Seidel are undefined because the right-hand sides of (6.2.5) and (6.2.9) are divided by A_{kk}. Also, if any one diagonal element has too small a magnitude, these methods will diverge. In Examples 6.3 and 6.4, Jacobi and Gauss-Seidel converge, since the diagonals (10 and 9) are both large; in Example 6.5, however, the diagonals (5 and 2) are small compared to the off-diagonals, and the methods diverge.

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In general, convergence of Jacobi or Gauss-Seidel can be evaluated by recognizing that (6.2.6) represents a digital filter with input y and output $x(i)$. The z-transform of (6.2.6) may be employed to determine the filter transfer function and its poles. The output $x(i)$ converges if and only if all the filter poles have magnitudes less than 1 (see Problems 6.16 and 6.17).

Rate of convergence is also established by the filter poles. Fast convergence is obtained when the magnitudes of all the poles are small. In addition, experience with specific **A** matrices has shown that more iterations are required for Jacobi and Gauss-Seidel as the dimension N increases.

Computer storage requirements for Jacobi include N^2 memory locations for the **A** matrix and $3N$ locations for the vectors **y**, $\mathbf{x}(i)$, and $\mathbf{x}(i+1)$. Storage space is also required for loops, arithmetic statements, and working space to compute (6.2.5). Gauss-Seidel requires N fewer memory locations, since for (6.2.9) the new value $x_k(i+1)$ can be stored in the location of the old value $x_k(i)$.

Computer time per iteration is relatively small for Jacobi and Gauss-Seidel. Inspection of (6.2.5) or (6.2.9) shows that N^2 multiplications/divisions and $N(N-1)$ subtractions per iteration are required [one division, $(N-1)$ multiplications, and $(N-1)$ subtractions for each $k = 1, 2, \ldots, N$]. But as was the case with Gauss elimination, if the matrix is sparse (i.e., most of the elements are zero), special sparse matrix algorithms can be used to substantially decrease both the storage requirements and the computation time.

6.3 ITERATIVE SOLUTIONS TO NONLINEAR ALGEBRAIC EQUATIONS: NEWTON-RAPHSON

The most common method for solving the power flow problem starts with a set of nonlinear algebraic equations represented in matrix format as

$$\mathbf{f}(\mathbf{x}) = \begin{bmatrix} f_1(\mathbf{x}) \\ f_2(\mathbf{x}) \\ \vdots \\ f_N(\mathbf{x}) \end{bmatrix} = \mathbf{y} \tag{6.3.1}$$

where **y** and **x** are N vectors and $\mathbf{f}(\mathbf{x})$ is an N vector of functions. Given **y** and $\mathbf{f}(\mathbf{x})$, the objective is to solve for **x**. The iterative methods described in Section 6.2 can be extended to nonlinear equations as follows. Rewriting (6.3.1),

$$0 = \mathbf{y} - \mathbf{f}(\mathbf{x}) \tag{6.3.2}$$

Adding $\mathbf{Dx}$ to both sides of (6.3.2), where **D** is a square $N \times N$ invertible matrix.

$$\mathbf{Dx} = \mathbf{Dx} + \mathbf{y} - \mathbf{f}(\mathbf{x}) \tag{6.3.3}$$

Premultiplying by $\mathbf{D}^{-1}$,

$$\mathbf{x} = \mathbf{x} + \mathbf{D}^{-1}[\mathbf{y} - \mathbf{f}(\mathbf{x})] \tag{6.3.4}$$

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The old values $\mathbf{x}(i)$ are used on the right side of (6.3.4) to generate the new values $\mathbf{x}(i+1)$ on the left side. That is,

$$\mathbf{x}(i+1) = \mathbf{x}(i) + \mathbf{D}^{-1}\{\mathbf{y} - \mathbf{f}[\mathbf{x}(i)]\} \tag{6.3.5}$$

For linear equations, $\mathbf{f}(\mathbf{x}) = \mathbf{A}\mathbf{x}$ and (6.3.5) reduce to

$$\mathbf{x}(i+1) = \mathbf{x}(i) + \mathbf{D}^{-1}[\mathbf{y} - \mathbf{A}\mathbf{x}(i)] = \mathbf{D}^{-1}(\mathbf{D} - \mathbf{A})\mathbf{x}(i) + \mathbf{D}^{-1}\mathbf{y} \tag{6.3.6}$$

which is identical to the Jacobi and Gauss-Seidel methods of (6.2.6). For nonlinear equations, the matrix $\mathbf{D}$ in (6.3.5) must be specified.

One method for specifying $\mathbf{D}$, called *Newton-Raphson*, is based on the following Taylor series expansion of $\mathbf{f}(\mathbf{x})$ about an operating point $\mathbf{x}_0$.

$$\mathbf{y} = \mathbf{f}(\mathbf{x}_0) + \left.\frac{d\mathbf{f}}{d\mathbf{x}}\right|_{\mathbf{x}=\mathbf{x}_0} (\mathbf{x} - \mathbf{x}_0) \cdots \tag{6.3.7}$$

Neglecting the higher order terms in (6.3.7) and solving for $\mathbf{x}$,

$$\mathbf{x} = \mathbf{x}_0 + \left[\left.\frac{d\mathbf{f}}{d\mathbf{x}}\right|_{\mathbf{x}=\mathbf{x}_0}\right]^{-1} [\mathbf{y} - \mathbf{f}(\mathbf{x}_0)] \tag{6.3.8}$$

The Newton-Raphson method replaces $\mathbf{x}_0$ by the old value $\mathbf{x}(i)$ and $\mathbf{x}$ by the new value $\mathbf{x}(i+1)$ in (6.3.8). Thus,

$$\mathbf{x}(i+1) = \mathbf{x}(i) + \mathbf{J}^{-1}(i)\{\mathbf{y} - \mathbf{f}[\mathbf{x}(i)]\} \tag{6.3.9}$$

where

$$\mathbf{J}(i) = \left.\frac{d\mathbf{f}}{d\mathbf{x}}\right|_{\mathbf{x}=\mathbf{x}(i)} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_N} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_N} \\ \vdots & \vdots & & \vdots \\ \frac{\partial f_N}{\partial x_1} & \frac{\partial f_N}{\partial x_2} & \cdots & \frac{\partial f_N}{\partial x_N} \end{bmatrix}_{\mathbf{x}=\mathbf{x}(i)} \tag{6.3.10}$$

The $N \times N$ matrix $\mathbf{J}(i)$, whose elements are the partial derivatives shown in (6.3.10), is called the Jacobian matrix. The Newton-Raphson method is similar to extended Gauss-Seidel, except that $\mathbf{D}$ in (6.3.5) is replaced by $\mathbf{J}(i)$ in (6.3.9).

EXAMPLE 6.6

Newton-Raphson method: solution to polynomial equations

Solve the scalar equation $f(x) = y$, where $y = 9$ and $f(x) = x^2$. Starting with $x(0) = 1$, use (a) Newton-Raphson, and (b) extended Gauss-Seidel with $\mathbf{D} = 3$ until (6.2.2) is satisfied for $\varepsilon = 10^{-4}$. Compare the two methods.

(*Continued*)

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SOLUTION

a. Using (6.3.10) with $f(x) = x^2$,

$$\mathbf{J}(i) = \frac{d}{dx}(x^2)\bigg|_{x=x(i)} = 2x\bigg|_{x=x(i)} = 2x(i)$$

Using $\mathbf{J}(i)$ in (6.3.9),

$$x(i+1) = x(i) + \frac{1}{2x(i)}[9 - x^2(i)]$$

Starting with $x(0) = 1$, successive calculations of the Newton-Raphson equation are shown in the following table.

Newton–Raphson

i	0	1	2	3	4	5
$x(i)$	1	5.00000	3.40000	3.02353	3.00009	3.00000

b. Using (6.3.5) with $\mathbf{D} = 3$, the Gauss-Seidel method is

$$x(i+1) = x(i) + \frac{1}{3}[9 - x^2(i)]$$

The corresponding Gauss-Seidel calculations are as follows:

Gauss-Seidel (D = 3)

i	0	1	2	3	4	5	6
$x(i)$	1	3.66667	2.18519	3.59351	2.28908	3.54245	2.35945

As shown, Gauss-Seidel oscillates about the solution, slowly converging, whereas Newton-Raphson converges in five iterations to the solution $x = 3$. Note that if $x(0)$ is negative, Newton-Raphson converges to the negative solution $x = -3$. Also, it is assumed that the matrix inverse $\mathbf{J}^{-1}$ exists. Thus, the initial value $x(0) = 0$ should be avoided for this example.

EXAMPLE 6.7

Newton-Raphson method: solution to nonlinear algebraic equations

Solve

$$\begin{bmatrix} x_1 + x_2 \\ x_1 x_2 \end{bmatrix} = \begin{bmatrix} 15 \\ 50 \end{bmatrix} \qquad \mathbf{x}(0) = \begin{bmatrix} 4 \\ 9 \end{bmatrix}$$

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Use the Newton-Raphson method starting with the above $\mathbf{x}(0)$ and continue until (6.2.2) is satisfied with $\varepsilon = 10^{-4}$.

SOLUTION

Using (6.3.10) with $f_1 = (x_1 + x_2)$ and $f_2 = x_1x_2$,

$$\mathbf{J}(i)^{-1} = \begin{bmatrix} \dfrac{\partial f_1}{\partial x_1} & \dfrac{\partial f_1}{\partial x_2} \\ \dfrac{\partial f_2}{\partial x_1} & \dfrac{\partial f_2}{\partial x_2} \end{bmatrix}^{-1}_{\mathbf{x}=\mathbf{x}(i)} = \begin{bmatrix} 1 & 1 \\ x_2(i) & x_1(i) \end{bmatrix}^{-1} = \frac{\begin{bmatrix} x_1(i) & -1 \\ -x_2(i) & 1 \end{bmatrix}}{x_1(i) - x_2(i)}$$

Using $\mathbf{J}(i)^{-1}$ in (6.3.9),

$$\begin{bmatrix} x_1(i+1) \\ x_2(i+1) \end{bmatrix} = \begin{bmatrix} x_1(i) \\ x_2(i) \end{bmatrix} + \frac{\begin{bmatrix} x_1(i) & -1 \\ -x_2(i) & 1 \end{bmatrix}}{x_1(i) - x_2(i)} \begin{bmatrix} 15 - x_1(i) - x_2(i) \\ 50 - x_1(i)x_2(i) \end{bmatrix}$$

Writing the preceding as two separate equations,

$$x_1(i+1) = x_1(i) + \frac{x_1(i)[15 - x_1(i) - x_2(i)] - [50 - x_1(i)x_2(i)]}{x_1(i) - x_2(i)}$$

$$x_2(i+1) = x_2(i) + \frac{-x_2(i)[15 - x_1(i) - x_2(i)] + [50 - x_1(i)x_2(i)]}{x_1(i) - x_2(i)}$$

Successive calculations of these equations are shown in the following table.

Newton-Raphson

i	0	1	2	3	4
$x_1(i)$	4	5.20000	4.99130	4.99998	5.00000
$x_2(i)$	9	9.80000	10.00870	10.00002	10.00000

Newton-Raphson converges in four iterations for this example.

Equation (6.3.9) contains the matrix inverse $\mathbf{J}^{-1}$. Instead of computing $\mathbf{J}^{-1}$, (6.3.9) can be rewritten as follows:

$$\mathbf{J}(i)\Delta\mathbf{x}(i) = \Delta\mathbf{y}(i) \tag{6.3.11}$$

where

$$\Delta\mathbf{x}(i) = \mathbf{x}(i+1) - \mathbf{x}(i) \tag{6.3.12}$$

and

$$\Delta\mathbf{y}(i) = \mathbf{y} - \mathbf{f}[\mathbf{x}(i)] \tag{6.3.13}$$

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Then, during each iteration, the following four steps are completed:

STEP 1 Compute $\Delta\mathbf{y}(i)$ from (6.3.13).

STEP 2 Compute $\mathbf{J}(i)$ from (6.3.10).

STEP 3 Using Gauss elimination and back substitution, solve (6.3.11) for $\Delta\mathbf{x}(i)$.

STEP 4 Compute $\mathbf{x}(i+1)$ from (6.3.12).

Experience from power flow studies has shown that Newton-Raphson converges in many cases where Jacobi and Gauss-Seidel diverge. Furthermore, the number of iterations required for convergence is independent of the dimension N for Newton-Raphson, but increases with N for Jacobi and Gauss-Seidel. Most Newton-Raphson power flow problems converge in fewer than 10 iterations [1].

EXAMPLE 6.8

Newton-Raphson method in four steps

Complete the above four steps for the first iteration of Example 6.7.

SOLUTION

STEP 1 $$\Delta\mathbf{y}(0) = \mathbf{y} - \mathbf{f}[\mathbf{x}(0)] = \begin{bmatrix} 15 \\ 50 \end{bmatrix} - \begin{bmatrix} 4+9 \\ (4)(9) \end{bmatrix} = \begin{bmatrix} 2 \\ 14 \end{bmatrix}$$

STEP 2 $$\mathbf{J}(0) = \left[\begin{array}{c|c} 1 & 1 \\ \hline x_2(0) & x_1(0) \end{array}\right] = \left[\begin{array}{c|c} 1 & 1 \\ \hline 9 & 4 \end{array}\right]$$

STEP 3 Using $\Delta\mathbf{y}(0)$ and $\mathbf{J}(0)$, (6.3.11) becomes

$$\left[\begin{array}{c|c} 1 & 1 \\ \hline 9 & 4 \end{array}\right]\begin{bmatrix} \Delta x_1(0) \\ \Delta x_2(0) \end{bmatrix} = \begin{bmatrix} 2 \\ 14 \end{bmatrix}$$

Using Gauss elimination, subtract $J_{21}/J_{11} = 9/1 = 9$ times the first equation from the second equation, giving

$$\left[\begin{array}{c|c} 1 & 1 \\ \hline 0 & -5 \end{array}\right]\begin{bmatrix} \Delta x_1(0) \\ \Delta x_2(0) \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \end{bmatrix}$$

Solving by back substitution,

$$\Delta x_2(0) = \frac{-4}{-5} = 0.8$$

$$\Delta x_1(0) = 2 - 0.8 = 1.2$$

STEP 4 $$\mathbf{x}(1) = \mathbf{x}(0) + \Delta\mathbf{x}(0) = \begin{bmatrix} 4 \\ 9 \end{bmatrix} + \begin{bmatrix} 1.2 \\ 0.8 \end{bmatrix} = \begin{bmatrix} 5.2 \\ 9.8 \end{bmatrix}$$

This is the same as computed in Example 6.7.

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6.4 THE POWER FLOW PROBLEM

The power flow problem is the computation of voltage magnitude and phase angle at each bus in a power system under balanced three-phase steady-state conditions. As a byproduct of this calculation, real and reactive power flows in equipment such as transmission lines and transformers, as well as equipment losses, can be computed.

The starting point for a power flow problem is a single-line (oneline) diagram of the power system, from which the input data can be obtained. Input data consist of bus data, transmission line data, and transformer data.

As shown in Figure 6.1, the following four variables are associated with each bus k: voltage magnitude V_k, phase angle δ_k, net real power P_k, and reactive power Q_k supplied to the bus. At each bus, two of these variables are specified as input data, and the other two are unknowns to be computed by the power flow program. For convenience, the power delivered to bus k in Figure 6.1 is separated into generator and load terms. That is,

$$\begin{aligned} P_k &= P_{Gk} - P_{Lk} \\ Q_k &= Q_{Gk} - Q_{Lk} \end{aligned} \tag{6.4.1}$$

Each bus k is categorized into one of the following three bus types:

1. Swing bus (or slack bus)—There is only one swing bus, which for convenience is numbered bus 1 in this text. The swing bus is a reference bus for which $V_1 \underline{/\delta_1}$, is input data with the angle typically zero degrees and the voltage magnitude close to 1.0 per unit. The power flow computes P_1 and Q_1.

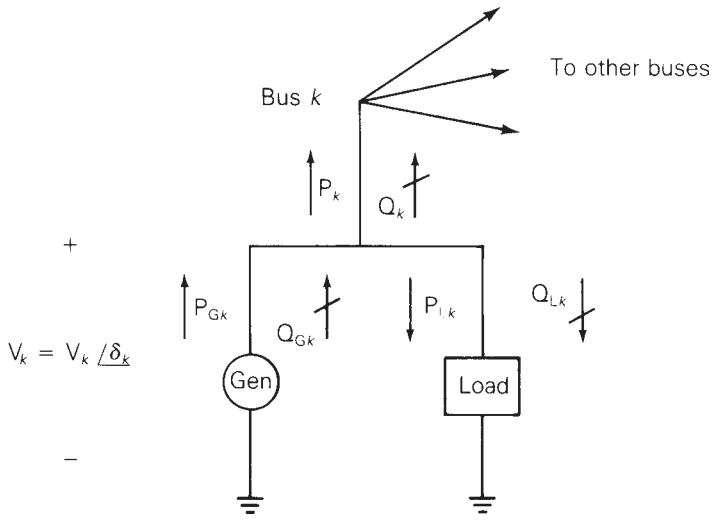

FIGURE 6.1
Bus variables V_k, δ_k, P_k, and Q_k

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2. Load (PQ) bus—P_k and Q_k are input data. The power flow computes V_k and δ_k. Most buses in typical power flows are load buses.
3. Voltage controlled (PV) bus—P_k and V_k are input data. The power flow program computes Q_k and δ_k. Examples are buses to which generators, switched shunt capacitors, or static var systems are connected. Maximum and minimum reactive power (var) limits $Q_{Gk\max}$ and $Q_{Gk\min}$ that this equipment can supply are also input data. If an upper or lower reactive power limit is reached, then the reactive power output of the generator is held at the limit, and the bus is modeled as a PQ bus. Another example is a bus to which a tap-changing transformer is connected; the power flow then computes the tap setting.

Note that when bus k is a load bus with no generation, $P_k = -P_{Lk}$ is negative; that is, the real power supplied to bus k in Figure 6.1 is negative. If the load is inductive, $Q_k = -Q_{Lk}$ is negative.

Transmission lines are represented by the equivalent π circuit, shown in Figure 5.7. Transformers are also represented by equivalent circuits, as shown in Figure 3.9 for a two-winding transformer, Figure 3.20 for a three-winding transformer, or Figure 3.25 for a tap-changing transformer.

Input data for each transmission line include the per-unit equivalent π circuit series impedance Z' and shunt admittance Y', the two buses to which the line is connected, and maximum MVA rating. Similarly, input data for each transformer include per-unit winding impedances Z, the per-unit exciting branch admittance Y, the buses to which the windings are connected, and maximum MVA ratings. Input data for tap-changing transformers also include maximum tap settings.

The bus admittance matrix $\boldsymbol{Y}_{\text{bus}}$ can be constructed from the line and transformer input data. From (2.4.3) and (2.4.4), the elements of $\boldsymbol{Y}_{\text{bus}}$ are

$$\text{Diagonal elements: } Y_{kk} = \text{sum of admittances connected to bus } k$$

$$\text{Off-diagonal elements: } Y_{kn} = -(\text{sum of admittances connected between buses } k \text{ and } n) \quad k \neq n \tag{6.4.2}$$

EXAMPLE 6.9

Power flow input data and $\boldsymbol{Y}_{\text{bus}}$

Figure 6.2 shows a single-line diagram of a five-bus power system. Input data are given in Tables 6.1, 6.2, and 6.3. As shown in Table 6.1, bus 1, to which a generator is connected, is the swing bus. Bus 3, to which a generator and a load are connected, is a voltage-controlled bus. Buses 2, 4, and 5 are load buses. Note that the loads at buses 2 and 3 are inductive since $Q_2 = -Q_{L2} = -2.8$ and $-Q_{L3} = -0.4$ are negative.

For each bus k, determine which of the variables V_k, δ_k, P_k, and Q_k are input data and which are unknowns. Also, compute the elements of the second row of $\boldsymbol{Y}_{\text{bus}}$.

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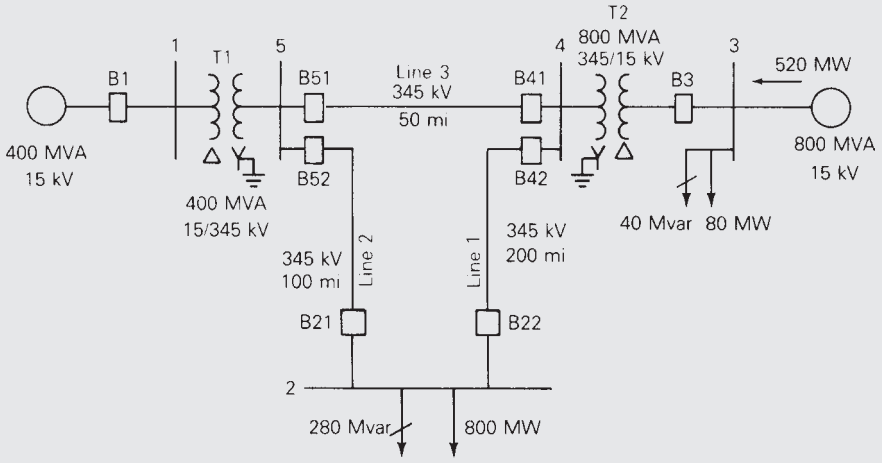
FIGURE 6.2

Single-line diagram for Example 6.9



Bus	Type	V per unit	δ degrees	P_G per unit	Q_G per unit	P_L per unit	Q_L per unit	Q_{Gmax} per unit	Q_{Gmin} per unit
1	Swing	1.0	0	—	—	0	0	—	—
2	Load	—	—	0	0	8.0	2.8	—	—
3	Constant voltage	1.05	—	5.2	—	0.8	0.4	4.0	−2.8
4	Load	—	—	0	0	0	0	—	—
5	Load	—	—	0	0	0	0	—	—

TABLE 6.1

Bus input data for Example 6.9*

*S_{base} = 100 MVA, V_{base} = 15 kV at buses 1, 3, and 345 kV at buses 2, 4, 5

Bus-to-Bus	R′ per unit	X′ per unit	G′ per unit	B′ per unit	Maximum MVA per unit
2–4	0.0090	0.100	0	1.72	12.0
2–5	0.0045	0.050	0	0.88	12.0
4–5	0.00225	0.025	0	0.44	12.0

TABLE 6.2

Line input data for Example 6.9

(*Continued*)

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Bus-to-Bus	R per unit	X per unit	G_c per unit	B_m per unit	Maximum MVA per unit	Maximum TAP Setting per unit
1–5	0.00150	0.02	0	0	6.0	—
3–4	0.00075	0.01	0	0	10.0	—

TABLE 6.3

Transformer input data for Example 6.9

Bus	Input Data	Unknowns
	$V_1 = 1.0, \delta_1 = 0$	P_1, Q_1
2	$P_2 = P_{G2} - P_{L2} = -8$ $Q_2 = Q_{G2} - Q_{L2} = -2.8$	V_2, δ_2
3	$V_3 = 1.05$ $P_3 = P_{G3} - P_{L3} = 4.4$	Q_3, δ_3
4	$P_4 = 0, Q_4 = 0$	V_4, δ_4
5	$P_5 = 0, Q_5 = 0$	V_5, δ_5

TABLE 6.4

Input data and unknowns for Example 6.9

SOLUTION

The input data and unknowns are listed in Table 6.4. For bus 1, the swing bus, P_1 and Q_1 are unknowns. For bus 3, a voltage-controlled bus, Q_3 and δ_3 are unknowns. For buses 2, 4, and 5, load buses, V_2, V_4, V_5 and δ_2, δ_4, δ_5 are unknowns.

The elements of $\boldsymbol{Y}_{\text{bus}}$ are computed from (6.4.2). Since buses 1 and 3 are not directly connected to bus 2, $Y_{21} = Y_{23} = 0$.

Using (6.4.2),

$$Y_{24} = \frac{-1}{R'_{24} + jX'_{24}} = \frac{-1}{0.009 + j0.1} = -0.89276 + j9.91964 \quad \text{per unit}$$

$$= 9.95972\underline{/95.143^\circ} \quad \text{per unit}$$

$$Y_{25} = \frac{-1}{R'_{25} + jX'_{25}} = \frac{-1}{0.0045 + j0.05} = -1.78552 + j19.83932 \quad \text{per unit}$$

$$= 19.9195\underline{/95.143^\circ} \quad \text{per unit}$$

$$Y_{22} = \frac{1}{R'_{24} + jX'_{24}} + \frac{1}{R'_{25} + jX'_{25}} + j\frac{B'_{24}}{2} + j\frac{B'_{25}}{2}$$

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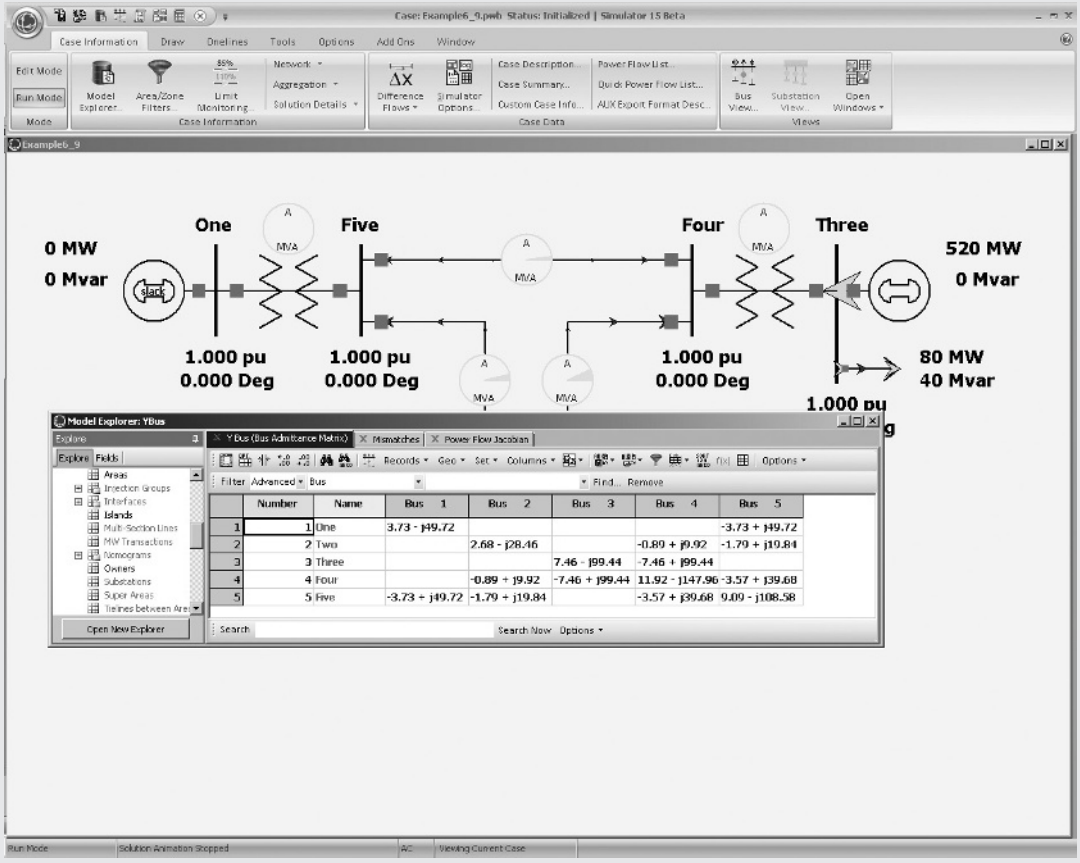

FIGURE 6.3

Screen for Example 6.9

$$= (0.89276 - j9.91964) + (1.78552 - j19.83932) + j\frac{1.72}{2} + j\frac{0.88}{2}$$

$$= 2.67828 - j28.4590 = 28.5847\underline{/-84.624^\circ} \text{ per unit}$$

where half of the shunt admittance of each line connected to bus 2 is included in Y_{22} (the other half is located at the other ends of these lines).

This five-bus power system is modeled in PowerWorld Simulator case Example 6_9 (see Figure 6.3). To view the input data, first click on the **Edit Mode** button (on the far left-hand side of the ribbon) to switch into the Edit mode (the Edit mode is used for modifying system parameters). Then by selecting the **Case Information** tab, you can view tabular displays showing the various

(*Continued*)

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parameters for the system. For example, use **Network**, **Buses** to view the parameters for each bus, and **Network**, **Lines and Transformers** to view the parameters for the transmission lines and transformers. Fields shown in blue on the screen can be directly changed simply by typing over them, and those shown in green can be toggled by clicking on them. Note that the values shown on these displays match the per unit values from Tables 6.1 to 6.3, except the power values are shown in actual MW/Mvar units.

The elements of $\boldsymbol{Y}_{\text{bus}}$ also can be displayed by selecting **Solution Details**, $\boldsymbol{Y}_{\text{bus}}$. Since the $\boldsymbol{Y}_{\text{bus}}$ entries are derived from other system parameters, they cannot be changed directly. Notice that several of the entries are blank, indicating that there is no transmission line or transformer directly connecting these two buses (a blank entry is equivalent to zero). For larger networks, most of the elements of the $\boldsymbol{Y}_{\text{bus}}$ are zero since any single bus usually only has a few incident lines (such sparse matrices are considered in Section 6.8). The elements of the $\boldsymbol{Y}_{\text{bus}}$ can be saved in a Matlab compatible format by first right-clicking within the $\boldsymbol{Y}_{\text{bus}}$ matrix to display the local menu, and then selecting **Save $\boldsymbol{Y}_{\text{bus}}$ in Matlab Format** from the local menu.

Finally, notice that no flows are shown on the oneline because the nonlinear power flow equations have not yet been solved, and the solution of these equations are covered next.

Using $\boldsymbol{Y}_{\text{bus}}$, the nodal equations for a power system network are written as

$$\boldsymbol{I} = \boldsymbol{Y}_{\text{bus}}\boldsymbol{V} \tag{6.4.3}$$

where $\boldsymbol{I}$ is the N vector of source currents injected into each bus and $\boldsymbol{V}$ is the N vector of bus voltages. For bus k, the kth equation in (6.4.3) is

$$I_k = \sum_{n=1}^{N} Y_{kn}V_n \tag{6.4.4}$$

The complex power delivered to bus k is

$$S_k = \mathrm{P}_k + j\mathrm{Q}_k = V_k I_k^* \tag{6.4.5}$$

Power flow solutions by Gauss-Seidel are based on nodal equations, (6.4.4), where each current source I_k is calculated from (6.4.5). Using (6.4.4) in (6.4.5),

$$\mathrm{P}_k + j\mathrm{Q}_k = V_k\left[\sum_{n=1}^{N} Y_{kn}V_n\right]^* \qquad k = 1, 2, \ldots, N \tag{6.4.6}$$

With the following notation,

$$V_n = \mathrm{V}_n e^{j\delta_n} \tag{6.4.7}$$

$$Y_{kn} = \mathrm{Y}_{kn}e^{j\theta_{kn}} = \mathrm{G}_{kn} + j\mathrm{B}_{kn} \qquad k,n = 1, 2, \ldots, N \tag{6.4.8}$$

(6.4.6) becomes

$$\mathrm{P}_k + j\mathrm{Q}_k = \mathrm{V}_k\sum_{n=1}^{N}\mathrm{Y}_{kn}\mathrm{V}_n e^{j(\delta_k - \delta_n - \theta_{kn})} \tag{6.4.9}$$

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Taking the real and imaginary parts of (6.4.9), the power balance equations are written as either

$$\mathrm{P}_k = \mathrm{V}_k \sum_{n=1}^{N} \mathrm{Y}_{kn} \mathrm{V}_n \cos(\delta_k - \delta_n - \theta_{kn}) \tag{6.4.10}$$

$$\mathrm{Q}_k = \mathrm{V}_k \sum_{n=1}^{N} \mathrm{Y}_{kn} \mathrm{V}_n \sin(\delta_k - \delta_n - \theta_{kn}) \qquad k = 1, 2, \ldots, N \tag{6.4.11}$$

or when the Y_{kn} is expressed in rectangular coordinates as

$$\mathrm{P}_k = \mathrm{V}_k \sum_{n=1}^{N} \mathrm{V}_n [\mathrm{G}_{kn} \cos(\delta_k - \delta_n) + \mathrm{B}_{kn} \sin(\delta_k - \delta_n)] \tag{6.4.12}$$

$$\mathrm{Q}_k = \mathrm{V}_k \sum_{n=1}^{N} \mathrm{V}_n [\mathrm{G}_{kn} \sin(\delta_k - \delta_n) - \mathrm{B}_{kn} \cos(\delta_k - \delta_n)] \quad k = 1, 2, \ldots, N \tag{6.4.13}$$

Power flow solutions by Newton-Raphson are based on the nonlinear power flow equations given by either (6.4.10) and (6.4.11), or by (6.4.12) and (6.4.13)].

6.5 POWER FLOW SOLUTION BY GAUSS-SEIDEL

Nodal equations $\boldsymbol{I} = \boldsymbol{Y}_{\text{bus}} \boldsymbol{V}$ are a set of linear equations analogous to $\mathbf{y} = \mathbf{Ax}$, solved in Section 6.2 using Gauss-Seidel. Since power flow bus data consists of P_k and Q_k for load buses or P_k and V_k for voltage-controlled buses, nodal equations do not directly fit the linear equation format; the current source vector $\boldsymbol{I}$ is unknown, and the equations are actually nonlinear. For each load bus, I_k can be calculated from (6.4.5), giving

$$I_k = \frac{\mathrm{P}_k - j\mathrm{Q}_k}{V_k^*} \tag{6.5.1}$$

Applying the Gauss-Seidel method (6.2.9) to the nodal equations with I_k given above, obtain

$$V_k(i+1) = \frac{1}{Y_{kk}} \left[\frac{\mathrm{P}_k - j\mathrm{Q}_k}{V_k^*(i)} - \sum_{n=1}^{k-1} Y_{kn} V_n(i+1) - \sum_{n=k+1}^{N} Y_{kn} V_n(i) \right] \tag{6.5.2}$$

Equation (6.5.2) can be applied twice during each iteration for load buses, first using $V_k^*(i)$, then replacing $V_k^*(i)$ by $V_k^*(i+1)$ on the right side of (6.5.2).

For a voltage-controlled bus, Q_k is unknown but can be calculated from (6.4.13), giving

$$\mathrm{Q}_k(i) = \mathrm{V}_k(i) \sum_{n=1}^{N} V_n(i) [\mathrm{G}_{kn} \sin(\delta_k(i) - \delta_n(i)) - \mathrm{B}_{kn} \cos(\delta_k(i) - \delta_n(i))] \tag{6.5.3}$$

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Also,

$$Q_{Gk} = Q_k + Q_{Lk}$$

If the calculated value of Q_{Gk} does not exceed its limits, then Q_k is used in (6.5.2) to calculate $V_k(i+1) = V_k(i+1)\underline{/\delta_k(i+1)}$. Then the magnitude $V_k(i+1)$ is changed to V_k, which is input data for the voltage-controlled bus. Thus, use (6.5.2) to compute only the angle $\delta_k(i+1)$ for voltage-controlled buses.

If the calculated value exceeds its limit $Q_{Gk\max}$ or $Q_{Gk\min}$ during any iteration, then the bus type is changed from a voltage-controlled bus to a load bus, with Q_{Gk} set to its limit value. Under this condition, the voltage-controlling device (e.g., generator, capacitor bank, static var compensator) is not capable of maintaining V_k as specified by the input data. The power flow then calculates a new value of V_k.

For the swing bus, denoted bus 1, V_1 and δ_1 *are* input data. As such, no iterations are required for the swing bus. After the iteration process has converged, one pass through (6.4.10) and (6.4.11) can be made to compute P_1 and Q_1.

EXAMPLE 6.10

Power flow solution by Gauss-Seidel

For the power system of Example 6.9, use Gauss-Seidel to calculate $V_2(1)$, the phasor voltage at bus 2 after the first iteration. Use zero initial phase angles and 1.0 per-unit initial voltage magnitudes (except at bus 3, where $V_3 = 1.05$) to start the iteration procedure.

SOLUTION

Bus 2 is a load bus. Using the input data and bus admittance values from Example 6.9 in (6.5.2),

$$V_2(1) = \frac{1}{Y_{22}}\left\{\frac{P_2 - jQ_2}{V_2^*(0)} - \left[Y_{21}V_1(1) + Y_{23}V_3(0) + Y_{24}V_4(0) + Y_{25}V_5(0)\right]\right\}$$

$$= \frac{1}{28.5847\underline{/-84.624°}}\left\{\frac{-8 - j(-2.8)}{1.0\underline{/0°}}\right.$$

$$\left. - \left[(-1.78552 + j19.83932)(1.0) + (-0.89276 + j9.91964)(1.0)\right]\right\}$$

$$= \frac{(-8 + j2.8) - (-2.67828 + j29.7589)}{28.5847\underline{/-84.624°}}$$

$$= 0.96132\underline{/-16.543°} \quad \text{per unit}$$

Next, the above value is used in (6.5.2) to recalculate $V_2(1)$;

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$$V_2(1) = \frac{1}{28.5847\angle -84.624^\circ}\left\{\frac{-8 + j2.8}{0.96132\angle 16.543^\circ} - \left[-2.67828 + j29.75829\right]\right\}$$

$$= \frac{-4.4698 - j24.5973}{28.5847\angle -84.624^\circ} = 0.87460\angle -15.675^\circ \quad \text{per unit}$$

Computations are next performed at buses 3, 4, and 5 to complete the first Gauss-Seidel iteration.

To see the complete convergence of this case, open PowerWorld Simulator case Example 6_10. By default, PowerWorld Simulator uses the Newton-Raphson method described in the next section since Gauss-Seidel, while being a useful technique for introducing the power flow to students, is now seldom used commercially. However, the free educational version of PowerWorld still allows cases to be solved with the Gauss-Seidel approach by selecting **Tools, Solve, Gauss-Seidel Power Flow**. To avoid getting stuck in an infinite loop if a case does not converge, PowerWorld Simulator places a limit on the maximum number of iterations. Usually for a Gauss-Seidel procedure, this number is rather high, perhaps equal to 100 iterations. However, in this example, to demonstrate the convergence characteristics of the Gauss-Seidel method, it has been set to a single iteration, allowing the voltages to be viewed after each iteration. To step through the solution one iteration at a time, just repeatedly select **Tools, Solve, Gauss-Seidel Power Flow**.

A common stopping criterion for the Gauss-Seidel is to use the scaled differences in the voltages from one iteration to the next (6.2.2). When the voltage differences for each bus are below a specified convergence tolerance ε, the problem is considered solved. An alternative approach, implemented in PowerWorld Simulator, is to examine the real and reactive mismatch equations, defined as the difference between the right- and left-hand sides of (6.4.10) and (6.4.11). PowerWorld Simulator continues iterating until all the bus mismatches are below an MVA (or kVA) tolerance. When single-stepping through the solution, the bus mismatches can be viewed after each iteration on the **Case Information, Mismatches** display. The solution mismatch tolerance can be changed on the Power Flow Solution page of the PowerWorld Simulator Options dialog (select **Tools, Simulator Options**, then select the **Power Flow Solution** category to view this dialog); the maximum number of iterations can also be changed from this page. A typical convergence tolerance is about 0.1 MVA. In PowerWorld Simulator, the details of the solution are available in the log, which can be seen by selecting **Tools, Log** (which is toward the left side of the ribbon).

6.6 POWER FLOW SOLUTION BY NEWTON-RAPHSON

Equations (6.4.10) and (6.4.11) are analogous to the nonlinear equation $\mathbf{y} = \mathbf{f}(\mathbf{x})$, solved in Section 6.3 by Newton-Raphson. The $\mathbf{x}$, $\mathbf{y}$, and $\mathbf{f}$ vectors for the power flow problem are defined as

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$$\mathbf{x} = \begin{bmatrix} \delta \\ \mathbf{V} \end{bmatrix} = \begin{bmatrix} \delta_2 \\ \vdots \\ \delta_N \\ \mathrm{V}_2 \\ \vdots \\ \mathrm{V}_N \end{bmatrix}; \quad \mathbf{y} = \begin{bmatrix} \mathbf{P} \\ \mathbf{Q} \end{bmatrix} = \begin{bmatrix} \mathrm{P}_2 \\ \vdots \\ \mathrm{P}_N \\ \mathrm{Q}_2 \\ \vdots \\ \mathrm{Q}_N \end{bmatrix}$$

$$\mathbf{f}(\mathbf{x}) = \begin{bmatrix} \mathbf{P}(\mathbf{x}) \\ \mathbf{Q}(\mathbf{x}) \end{bmatrix} = \begin{bmatrix} \mathrm{P}_2(\mathbf{x}) \\ \vdots \\ \mathrm{P}_N(\mathbf{x}) \\ \mathrm{Q}_2(\mathbf{x}) \\ \vdots \\ \mathrm{Q}_N(\mathbf{x}) \end{bmatrix} \tag{6.6.1}$$

where all V, P, and Q terms are in per-unit and δ terms are in radians. The swing bus variables δ_1 and V_1 are omitted from (6.6.1), since they are already known. Equations (6.4.12) and (6.4.13) then have the following form:

$$y_k = \mathrm{P}_k = \mathrm{P}_k(x) = \mathrm{V}_k \sum_{n=1}^{N} \mathrm{V}_n [\mathrm{G}_{kn} \cos(\delta_k - \delta_n) + \mathrm{B}_{kn} \sin(\delta_k - \delta_n)] \tag{6.6.2}$$

$$y_{k+N} = \mathrm{Q}_k = \mathrm{Q}_k(x) = \mathrm{V}_k \sum_{n=1}^{N} \mathrm{V}_n [\mathrm{G}_{kn} \sin(\delta_k - \delta_n) - \mathrm{B}_{kn} \cos(\delta_k - \delta_n)] \quad k = 1, 2, \ldots, N \tag{6.6.3}$$

The Jacobian matrix of (6.3.10) has the form

$$\mathbf{J} = \begin{array}{c} \begin{matrix} \quad\quad\quad \mathbf{J}1 \quad\quad\quad & & & \quad\quad\quad \mathbf{J}2 \quad\quad\quad & & \end{matrix} \\ \left[\begin{array}{ccc|ccc} \dfrac{\partial \mathrm{P}_2}{\partial \delta_2} & \cdots & \dfrac{\partial \mathrm{P}_2}{\partial \delta_N} & \dfrac{\partial \mathrm{P}_2}{\partial \mathrm{V}_2} & \cdots & \dfrac{\partial \mathrm{P}_2}{\partial \mathrm{V}_N} \\ \vdots & & & \vdots & & \\ \dfrac{\partial \mathrm{P}_N}{\partial \delta_2} & \cdots & \dfrac{\partial \mathrm{P}_N}{\partial \delta_N} & \dfrac{\partial \mathrm{P}_N}{\partial \mathrm{V}_2} & \cdots & \dfrac{\partial \mathrm{P}_N}{\partial \mathrm{V}_N} \\ \hline \dfrac{\partial \mathrm{Q}_2}{\partial \delta_2} & \cdots & \dfrac{\partial \mathrm{Q}_2}{\partial \delta_N} & \dfrac{\partial \mathrm{Q}_2}{\partial \mathrm{V}_2} & \cdots & \dfrac{\partial \mathrm{Q}_2}{\partial \mathrm{V}_N} \\ \vdots & & & \vdots & & \\ \dfrac{\partial \mathrm{Q}_N}{\partial \delta_2} & \cdots & \dfrac{\partial \mathrm{Q}_N}{\partial \delta_N} & \dfrac{\partial \mathrm{Q}_N}{\partial \mathrm{V}_2} & \cdots & \dfrac{\partial \mathrm{Q}_N}{\partial \mathrm{V}_N} \end{array}\right] \\ \begin{matrix} & & \quad\quad\quad \mathbf{J}3 \quad\quad\quad & & & \quad\quad\quad \mathbf{J}4 \quad\quad\quad \end{matrix} \end{array} \tag{6.6.4}$$

Equation (6.6.4) is partitioned into four blocks. The partial derivatives in each block, derived from (6.6.2) and (6.6.3), are given in Table 6.5.

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$n \neq k$

$$J1_{kn} = \frac{\partial P_k}{\partial \delta_n} = V_k V_n [G_{kn} \sin(\delta_k - \delta_n) - B_{kn} \cos(\delta_k - \delta_n)]$$

$$J2_{kn} = \frac{\partial P_k}{\partial V_n} = V_k [G_{kn} \cos(\delta_k - \delta_n) + B_{kn} \sin(\delta_k - \delta_n)]$$

$$J3_{kn} = \frac{\partial Q_k}{\partial \delta_n} = -V_k V_n [G_{kn} \cos(\delta_k - \delta_n) + B_{kn} \sin(\delta_k - \delta_n)]$$

$$J4_{kn} = \frac{\partial Q_k}{\partial V_n} = V_k [G_{kn} \sin(\delta_k - \delta_n) - B_{kn} \cos(\delta_k - \delta_n)]$$

$n = k$

$$J1_{kk} = \frac{\partial P_k}{\partial \delta_k} = V_k \sum_{\substack{n=1 \\ n \neq k}}^{N} V_n [-G_{kn} \sin(\delta_k - \delta_n) + B_{kn} \cos(\delta_k - \delta_n)]$$

$$J2_{kk} = \frac{\partial P_k}{\partial V_k} = 2V_k G_{kk} + \sum_{\substack{n=1 \\ n \neq k}}^{N} V_n [G_{kn} \cos(\delta_k - \delta_n) + B_{kn} \sin(\delta_k - \delta_n)]$$

$$J3_{kk} = \frac{\partial Q_k}{\partial \delta_k} = V_k \sum_{\substack{n=1 \\ n \neq k}}^{N} V_n [G_{kn} \cos(\delta_k - \delta_n) + B_{kn} \sin(\delta_k - \delta_n)]$$

$$J4_{kk} = \frac{\partial Q_k}{\partial V_k} = 2V_k B_{kk} + \sum_{\substack{n=1 \\ n \neq k}}^{N} V_n [G_{kn} \sin(\delta_k - \delta_n) - B_{kn} \cos(\delta_k - \delta_n)]$$

TABLE 6.5
Elements of the Jacobian matrix

Now apply to the power flow problem the four Newton-Raphson steps outlined in Section 6.3, starting with $\mathbf{x}(i) = \begin{bmatrix} \boldsymbol{\delta}(i) \\ \mathbf{V}(i) \end{bmatrix}$ at the ith iteration.

STEP 1 Use (6.6.2) and (6.6.3) to compute

$$\Delta \mathbf{y}(i) = \begin{bmatrix} \Delta \mathbf{P}(i) \\ \Delta \mathbf{Q}(i) \end{bmatrix} = \begin{bmatrix} \mathbf{P} - \mathbf{P}[\mathbf{x}(i)] \\ \mathbf{Q} - \mathbf{Q}[\mathbf{x}(i)] \end{bmatrix} \tag{6.6.5}$$

STEP 2 Use the equations in Table 6.5 to calculate the Jacobian matrix.

STEP 3 Use Gauss elimination and back substitution to solve

$$\left[\begin{array}{c|c} \mathbf{J1}(i) & \mathbf{J2}(i) \\ \hline \mathbf{J3}(i) & \mathbf{J4}(i) \end{array}\right] \begin{bmatrix} \Delta \boldsymbol{\delta}(i) \\ \Delta \mathbf{V}(i) \end{bmatrix} = \begin{bmatrix} \Delta \mathbf{P}(i) \\ \Delta \mathbf{Q}(i) \end{bmatrix} \tag{6.6.6}$$

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STEP 4 Compute

$$\mathbf{x}(i+1) = \begin{bmatrix} \boldsymbol{\delta}(i+1) \\ \mathbf{V}(i+1) \end{bmatrix} = \begin{bmatrix} \boldsymbol{\delta}(i) \\ \mathbf{V}(i) \end{bmatrix} + \begin{bmatrix} \Delta\boldsymbol{\delta}(i) \\ \Delta\mathbf{V}(i) \end{bmatrix} \tag{6.6.7}$$

Starting with initial value $\mathbf{x}(0)$, the procedure continues until convergence is obtained or until the number of iterations exceeds a specified maximum. Convergence criteria are often based on $\Delta\mathbf{y}(i)$ (called *power mismatches*) rather than on $\Delta\mathbf{x}(i)$ (phase angle and voltage magnitude mismatches).

For each voltage-controlled bus, the magnitude V_k is already known, and the function $Q_k(\mathbf{x})$ is not needed. Therefore, V_k from the $\mathbf{x}$ vector can be omitted, as well as Q_k from the $\mathbf{y}$ vector. The column corresponding to partial derivatives with respect to V_k and the row corresponding to partial derivatives of $Q_k(\mathbf{x})$ can also be omitted from the Jacobian matrix. Alternatively, rows and corresponding columns for voltage-controlled buses can be retained in the Jacobian matrix. Then during each iteration, the voltage magnitude $V_k(i+1)$ of each voltage-controlled bus is reset to V_k, which is input data for that bus.

At the end of each iteration, $Q_k(\mathbf{x})$ is computed using (6.6.3) and $Q_{Gk} = Q_k(\mathbf{x}) + Q_{Lk}$ for each voltage-controlled bus. If the computed value of Q_{Gk} exceeds its limits, then the bus type is changed to a load bus with Q_{Gk} set to its limit value. The power flow program also computes a new value for V_k.

EXAMPLE 6.11

Jacobian matrix and power flow solution by Newton-Raphson

Determine the dimension of the Jacobian matrix for the power system in Example 6.9. Also, calculate $\Delta P_2(0)$ in Step 1 and $Jl_{24}(0)$ in Step 2 of the first Newton-Raphson iteration. Assume zero initial phase angles and 1.0 per-unit initial voltage magnitudes (except $V_3 = 1.05$).

SOLUTION

Since there are $N = 5$ buses for Example 6.9, (6.6.2) and (6.6.3) constitute $2(N-1) = 8$ equations, for which $\mathbf{J}(i)$ has dimension 8×8. However, there is one voltage-controlled bus, bus 3. Therefore, V_3 and the equation for $Q_3(\mathbf{x})$ could be eliminated, with $\mathbf{J}(i)$ reduced to a 7×7 matrix.

From Step 1 and (6.6.2),

$$\begin{aligned} \Delta P_2(0) = P_2 - P_2(x) = P_2 - V_2(0)\{V_1(0)[G_{21}\cos(\delta_2(0) - \delta_1(0)) + B_{21}\sin(\delta_2(0) \\ - \delta_1(0))] + V_2(0)\,G_{22} \\ + V_3(0)[G_{23}\cos(\delta_2(0) - \delta_3(0)) + B_{23}\sin(\delta_2(0) - \delta_3(0))] \end{aligned}$$

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$$+ V_4(0)[G_{24}\cos(\delta_2(0) - \delta_4(0)) + B_{24}\sin(\delta_2(0) - \delta_4(0))]$$
$$+ V_5(0)[G_{25}\cos(\delta_2(0) - \delta_1(0)) + B_{25}\sin(\delta_2(0) - \delta_5(0))]\}$$

From Example 6.9 with $Y_{21} = Y_{23} = 0$, then $G_{21} = 0$, $B_{21} = 0$, $G_{23} = 0$, $B_{23} = 0$; also, $G_{22} = 2.67828$, $G_{24} = -0.89276$, $B_{24} = 9.91964$, $G_{25} = -1.78552$, $B_{25} = 19.83932$, all the initial voltage magnitudes of 1.0 and all the initial angles of zero, then

$$\Delta P_2(0) = -8.0 - 1.0\{(1.0)(2.67828) + (1.0)(-0.89276) + (1.0)(-1.78552)\}$$
$$= -8.0 - 1.0\{0\} = -8.0 \text{ per unit}$$

From Step 2 and Jl given in Table 6.5,

$$J1_{24}(0) = V_2(0)V_4(0)[G_{24}\cos(\delta_2(0) - \delta_4(0)) + B_{24}\sin(\delta_2(0) - \delta_4(0))]$$
$$= (1.0)(1.0)[-(9.91964)]$$
$$= -9.91964 \text{ per unit}$$

To see the complete convergence of this case, open PowerWorld Simulator case Example 6_11 (see Figure 6.4). Select **Case Information, Network, Mismatches** to see the initial mismatches, and **Case Information, Solution Details, Power Flow Jacobian** to view the initial Jacobian matrix. As is common in commercial power flows, PowerWorld Simulator actually includes rows in the Jacobian for voltage-controlled buses. When a generator is regulating its terminal voltage, this row corresponds to the equation setting the bus voltage magnitude equal to the generator voltage setpoint. However, if the generator hits a reactive power limit, the bus type is switched to a load bus.

To step through the Newton-Raphson solution, from the ribbon select **Solve, Polar NR Power Flow**. Ordinarily, this selection would perform a complete Newton-Raphson iteration, stopping only when all the mismatches are less than the desired tolerance. However, for this case, in order to allow you to see the solution process, the maximum number of iterations has been set to 1, allowing the voltages, mismatches, and the Jacobian to be viewed after each iteration. To complete the solution, continue to select **Solve, Polar NR Power Flow** until the solution converges to the values shown in Tables 6.6, 6.7, and 6.8 (in about three iterations). While not covered here, the power flow equations can also be formulated by expressing the complex bus voltages in rectangular coordinates; this is the default approach used with PowerWorld's **Solve, Solve Power Flow–Newton** command. While there are differences in how they converge, both the Polar and Rectangular approaches should converge to the same solution.

(*Continued*)

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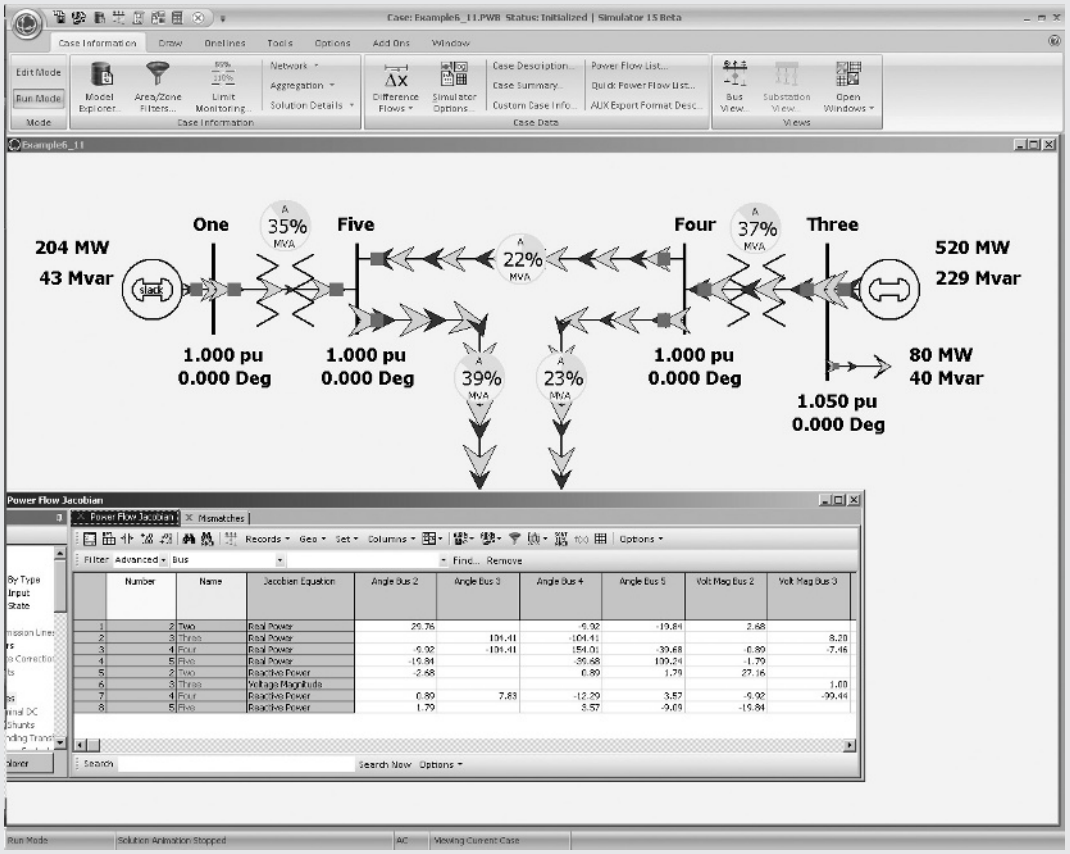

FIGURE 6.4

Screen for Example 6.11 showing Jacobian matrix at first iteration

Bus #	Voltage Magnitude (per unit)	Phase Angle (degrees)	Generation		Load	
			PG (per unit)	QG (per unit)	PL (per unit)	QL (per unit)
1	1.000	0.000	3.948	1.144	0.000	0.000
2	0.834	–22.407	0.000	0.000	8.000	2.800
3	1.050	–0.597	5.200	3.376	0.800	0.400
4	1.019	–2.834	0.000	0.000	0.000	0.000
5	0.974	–4.548	0.000	0.000	0.000	0.000
		TOTAL	9.148	4.516	8.800	3.200

TABLE 6.6

Bus output data for the power system given in Example 6.9

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Line #	Bus to Bus		P	Q	S
1	2	4	−2.920	−1.392	3.232
	4	2	3.036	1.216	3.272
2	2	5	−5.080	−1.408	5.272
	5	2	5.256	2.632	5.876
3	4	5	1.344	1.504	2.016
	5	4	−1.332	−1.824	2.260

TABLE 6.7
Line output data for the power system given in Example 6.9

Tran. #	Bus to Bus		P	Q	S
1	1	5	3.948	1.144	4.112
	5	1	−3.924	−0.804	4.004
2	3	4	4.400	2.976	5.312
	4	3	−4.380	−2.720	5.156

TABLE 6.8
Transformer output data for the power system given in Example 6.9

EXAMPLE 6.12

Power flow program: change in generation

Using the power-flow system given in Example 6.9, determine the acceptable generation range at bus 3, keeping each line and transformer loaded at or below 100% of its MVA limit.

SOLUTION

Load PowerWorld Simulator case Example 6_9. Select **Single Solution-Full Newton** to perform a single power flow solution using the Newton-Raphson approach. Then view the **Case Information** displays to verify that the PowerWorld Simulator solution matches the solution shown in Tables 6.6, 6.7, and 6.8. Additionally, the pie charts on the onelines show the percentage line and transformer loadings. Initially transformer Tl, between buses 1 and 5, is loaded at about 69% of its maximum MVA limit, while transformer T2, between buses 3 and 4, is loaded at about 53%.

Next, the bus 3 generation needs to be varied. This can be done a number of different ways in PowerWorld Simulator. The easiest (for this example) is to use the bus 3 generator MW oneline field to manually change the generation

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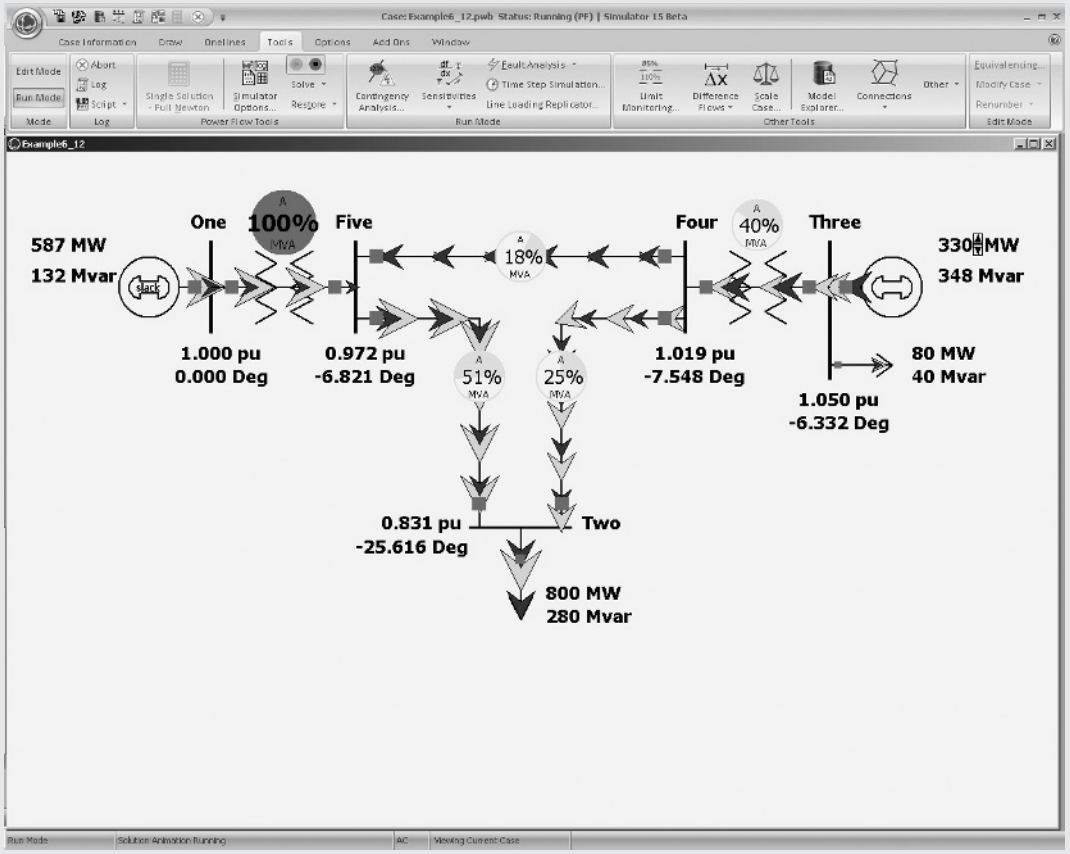

FIGURE 6.5

Screen for Example 6.12, Minimum Bus 3 Generator Loading

(see Figure 6.5). Right-click on the 520-MW field to the right of the bus 3 generator and select **Generator Field Information** dialog to view the **Generator Field Options** dialog. Set the **Delta Per Mouse Click** field to 10 and select OK. Small arrows are now visible next to this field on the oneline; clicking on the up arrow increases the generator's MW output by 10 MW, while clicking on the down arrow decreases the generation by 10 MW. Select **Tools, Play** to begin the simulation. Increase the generation until the pie chart for the transformer from bus 3 to 4 is loaded to 100%. This occurs at about 1000 MW. Notice that as the bus 3 generation is increased, the bus 1 slack generation decreases by a similar amount. Repeat the process, except now decreasing the generation. This unloads the transformer from bus 3 to 4, but increases the loading on the transformer from bus 1 to bus 5. The bus 1 to 5 transformer should reach 100% loading with the bus 3 generation equal to about 330 MW.

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Voltage-controlled buses to which tap-changing or voltage-regulating transformers are connected can be handled by various methods. One method is to treat each of these buses as a load bus. The equivalent π circuit parameters (Figure 3.25) are first calculated with tap setting $c = 1.0$ for starting. During each iteration, the computed bus voltage magnitude is compared with the desired value specified by the input data. If the computed voltage is low (or high), c is increased (or decreased) to its next setting, and the parameters of the equivalent π circuit as well as $\boldsymbol{Y}_{\text{bus}}$ are recalculated. The procedure continues until the computed bus voltage magnitude equals the desired value within a specified tolerance or until the high or low tap-setting limit is reached. Phase-shifting transformers can be handled in a similar way by using a complex turns ratio $c = 1.0\angle\alpha$ and by varying the phase-shift angle α.

A method with faster convergence makes c a variable and includes it in the $\mathbf{x}$ vector of (6.6.1). An equation is then derived to enter into the Jacobian matrix [4].

In comparing the Gauss-Seidel and Newton-Raphson algorithms, experience from power flow studies has shown that Newton-Raphson converges in many cases where Jacobi and Gauss-Seidel diverge. Furthermore, the number of iterations required for convergence is independent of the number of buses N for Newton-Raphson, but increases with N for Jacobi and Gauss-Seidel. The principal advantage of the Jacobi and Gauss-Seidel methods had been their more modest memory storage requirements and their lower computational requirements per iteration. However, with the vast increases in low-cost computer memory over the last several decades, coupled with the need to solve power flow problems with tens of thousands of buses, these advantages have been essentially eliminated. Therefore, the Newton-Raphson, or one of the derivative methods discussed in Sections 6.9 and 6.10, are the preferred power flow solution approaches.

EXAMPLE 6.13

Power flow program: 37-bus system

To see a power flow example of a larger system, open PowerWorld Simulator case Example 6_13 (see Figure 6.6). This case models a 37-bus, 9-generator power system containing three different voltage levels (345 kV, 138 kV, and 69 kV) with 57 transmission lines or transformers. The oneline can be panned by pressing the arrow keys, and it can be zoomed by pressing the <ctrl> with the up arrow key to zoom in or with the down arrow key to zoom out. Use **Tools, Play** to animate the oneline and **Tools, Pause** to stop the animation.

Determine the lowest per-unit voltage and the maximum line/transformer loading both for the initial case and for the case with the line from bus OAK69 to WALNUT69 out of service.

SOLUTION

Use single solution to initially solve the power flow, and then **Case Information, Network, Buses…** to view a listing of all the buses in the case. To quickly determine the lowest per-unit voltage magnitude, left-click on the PU Volt column header to sort the column (clicking a second time reverses the sort). The lowest

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initial voltage magnitude is 0.990 at bus LOCUST69. Next, select **Case Information, Network, Lines and Transformers…** to view the Line and Transformer Records display. Left-click on % of Max Limit to sort the lines by percentage loading. Initially, the highest percentage loading is 64.9% on the line between PEAR69 and PECAN69, circuit 1.

In the software, there are several ways to open the line between OAK69 and WALNUT69. One approach is to locate the line on the Line and Transformer Records display and then double-click on the Status field to change its value. An alternative approach is to find the line on the oneline (it is in the upper-left portion) and then click on one of its circuit breakers. Once the line is removed, use single solution to resolve the power flow. The lowest per-unit voltage is now 0.911 at LIME69, and the highest percentage line loading is 135% on the line between PEACH69 to POPLAR69. Since there are now several bus and line violations, the power system is no longer at a reliable operating point. Control actions and/or design improvements are needed to correct these problems. The design projects at the end of the chapter discuss these options.

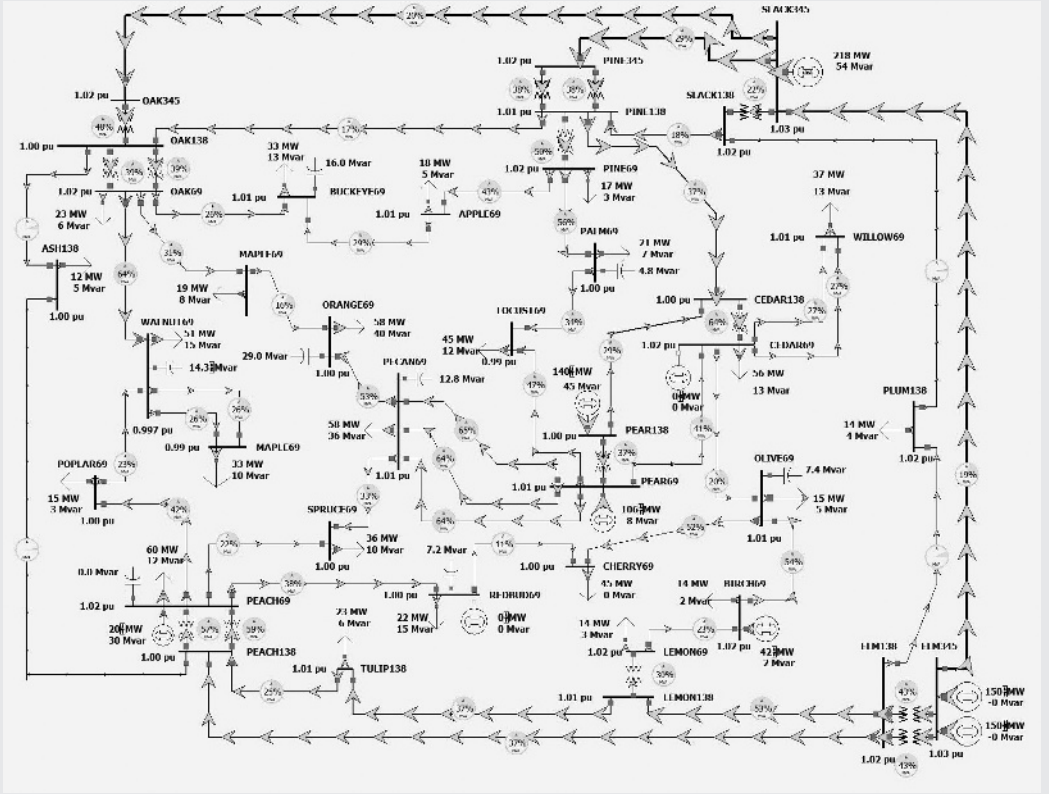

FIGURE 6.6

Screen for Example 6.13 showing the initial flows

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6.7 CONTROL OF POWER FLOW

The following means are used to control system power flows:

1. Prime mover and excitation control of generators
2. Switching of shunt capacitor banks, shunt reactors, and static var systems
3. Control of tap-changing and regulating transformers

A simple model of a generator operating under balanced steady-state conditions is the Thévenin equivalent shown in Figure 6.7. V_t is the generator terminal voltage, E_g is the excitation voltage, δ is the power angle, and X_g is the positive-sequence synchronous reactance. From the figure, the generator current is

$$I = \frac{E_g e^{j\delta} - V_t}{jX_g} \tag{6.7.1}$$

and the complex power delivered by the generator is

$$\begin{aligned} S = P + jQ = V_t I^* &= V_t\left(\frac{E_g e^{-j\delta} - V_t}{-jX_g}\right) \\ &= \frac{V_t E_g(j\cos\delta + \sin\delta) - jV_t^2}{X_g} \end{aligned} \tag{6.7.2}$$

The real and reactive powers delivered are then

$$P = \text{Re } S = \frac{V_t E_g}{X_g}\sin\delta \tag{6.7.3}$$

$$Q = \text{Im } S = \frac{V_t}{X_g}(E_g\cos\delta - V_t) \tag{6.7.4}$$

Equation (6.7.3) shows that the real power P increases when the power angle δ increases. From an operational standpoint, when the prime mover increases the power input to the generator while the excitation voltage is held constant, the rotor speed increases. As the rotor speed increases, the power angle δ also increases, causing an increase in generator real power output P. There is also a decrease in reactive power output Q, given by (6.7.4). However, when δ is less than 15°, the increase in P is much larger than the decrease in Q. From the power flow standpoint,

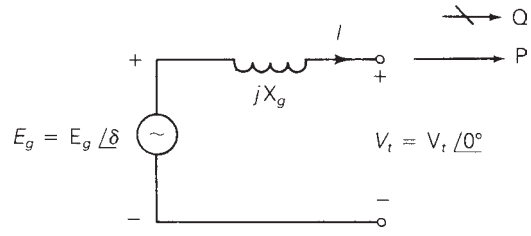

FIGURE 6.7
Generator Thévenin equivalent

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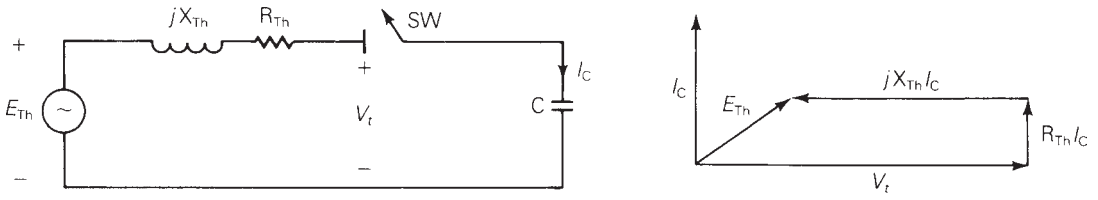
an increase in prime-mover power corresponds to an increase in P at the constant-voltage bus to which the generator is connected. The power flow program computes the increase in δ along with the small change in Q.

Equation (6.7.4) shows that reactive power output Q increases when the excitation voltage E_g increases. From the operational standpoint, when the generator exciter output increases while holding the prime-mover power constant, the rotor current increases. As the rotor current increases, the excitation voltage E_g also increases, causing an increase in generator reactive power output Q. There is also a small decrease in δ required to hold P constant in (6.7.3). From the power flow standpoint, an increase in generator excitation corresponds to an increase in voltage magnitude at the constant-voltage bus to which the generator is connected. The power flow program computes the increase in reactive power Q supplied by the generator along with the small change in δ.

Figure 6.8 shows the effect of adding a shunt capacitor bank to a power system bus. The system is modeled by its Thévenin equivalent. Before the capacitor bank is connected, the switch SW is open and the bus voltage equals E_{Th}. After the bank is connected, SW is closed, and the capacitor current I_C leads the bus voltage V_t by 90°. The phasor diagram shows that V_t is larger than E_{Th} when SW is closed. From the power flow standpoint, the addition of a shunt capacitor bank to a load bus corresponds to the addition of a negative reactive load, since a capacitor absorbs negative reactive power. The power flow program computes the increase in bus voltage magnitude along with the small change in δ. Similarly, the addition of a shunt reactor corresponds to the addition of a positive reactive load wherein the power flow program computes the decrease in voltage magnitude.

Tap-changing and voltage-magnitude-regulating transformers are used to control bus voltages as well as reactive power flows on lines to which they are connected. Similarly, phase-angle regulating transformers are used to control bus angles as well as real power flows on lines to which they are connected. Both tap-changing and regulating transformers are modeled by a transformer with an off-nominal turns ratio c (Figure 3.25). From the power flow standpoint, a change in tap setting or voltage regulation corresponds to a change in c. The power flow program computes the changes in $\boldsymbol{Y}_{\text{bus}}$, bus voltage magnitudes and angles, and branch flows.

Besides the above controls, the power flow program can be used to investigate the effect of switching in or out lines, transformers, loads, and generators.



(a) Equivalent circuit

(b) Phasor diagram with switch SW closed

FIGURE 6.8

Effect of adding a shunt capacitor bank to a power system bus

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Proposed system changes to meet future load growth, including new transmission, new transformers, and new generation can also be investigated. Power flow design studies are normally conducted by trial and error. Using engineering judgment, adjustments in generation levels and controls are made until the desired equipment loadings and voltage profile are obtained.

EXAMPLE 6.14

Power flow program: effect of shunt capacitor banks

Determine the effect of adding a 200-Mvar shunt capacitor bank at bus 2 on the power system in Example 6.9.

SOLUTION

Open PowerWorld Simulator case Example 6_14 (see Figure 6.9). This case is identical to Example 6.9 except that a 200-Mvar shunt capacitor bank has been added at bus 2. Initially, this capacitor is open. Click on the capacitor's circuit to close the capacitor and then solve the case. The capacitor increases the bus 2 voltage from 0.834 per unit to a more acceptable 0.959 per unit. The insertion of the capacitor has also substantially decreased the losses, from 34.84 to 25.37 MW.

Notice that the amount of reactive power actually supplied by the capacitor is only 184 Mvar. This discrepancy arises because a capacitor's reactive output varies with the square of the terminal voltage, $Q_{cap} = V_{cap}^2/X_c$ (see 2.3.5). A capacitor's Mvar rating is based on an assumed voltage of 1.0 per unit.

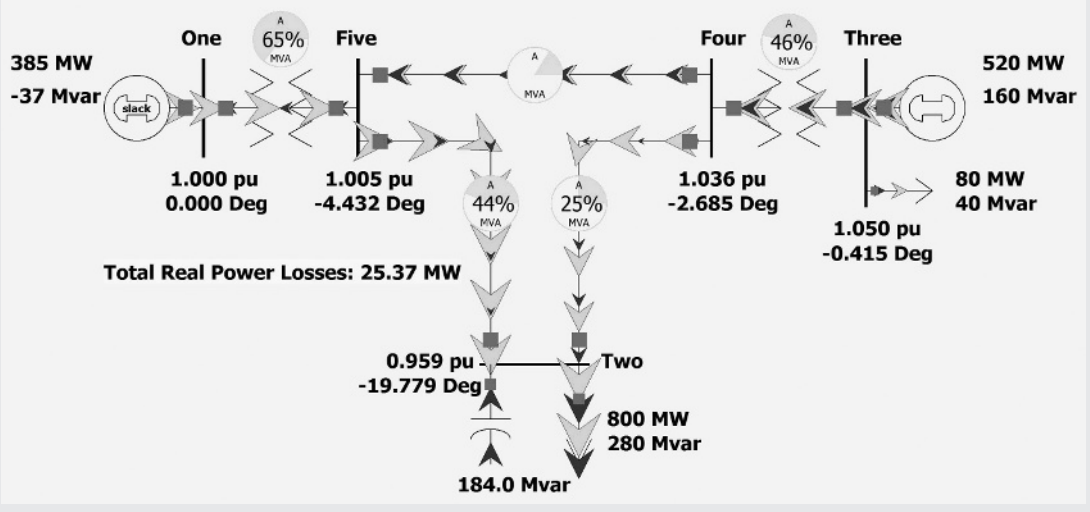

FIGURE 6.9

Screen for Example 6.14

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EXAMPLE 6.15

PowerWorld Simulator Case Example 6_15 (see Figure 6.10) modifies the Example 6.13 case by (1) opening one of the 138/69-kV transformers at the PEACH substation, and (2) opening the 69-kV transmission line between CHERRY69 and OLIVE69. This causes a flow of 116 MVA on the remaining 138/69 kV-transformer at PEACH. Since this transformer has a limit of 101 MVA, it results in an overload at 115%. Redispatch the generators in order to remove this overload.

SOLUTION

There are a number of solutions to this problem and several solution techniques. One solution technique would be to use engineering intuition, along with a trial- and-error approach (see Figure 6.11). Since the overload is from the 138-kV level to the 69-kV level and there is a generator directly connected

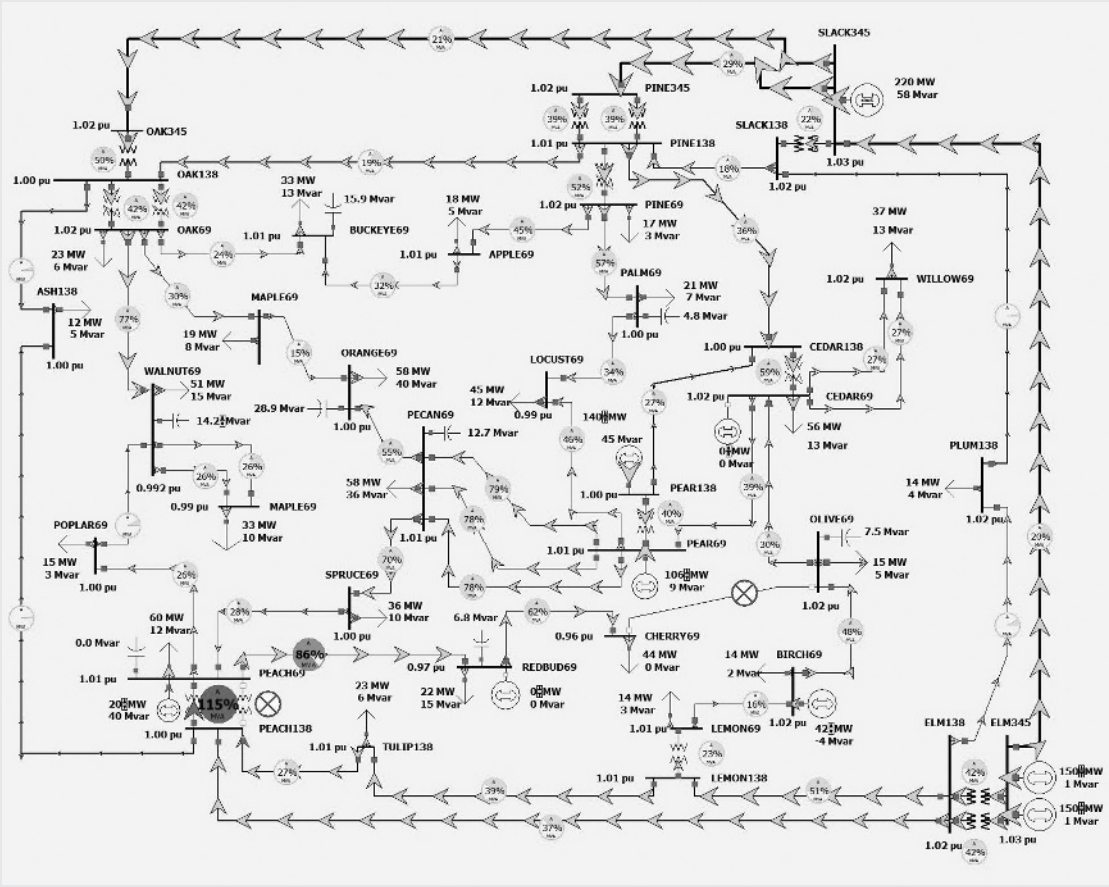

FIGURE 6.10

Screen for Example 6.15

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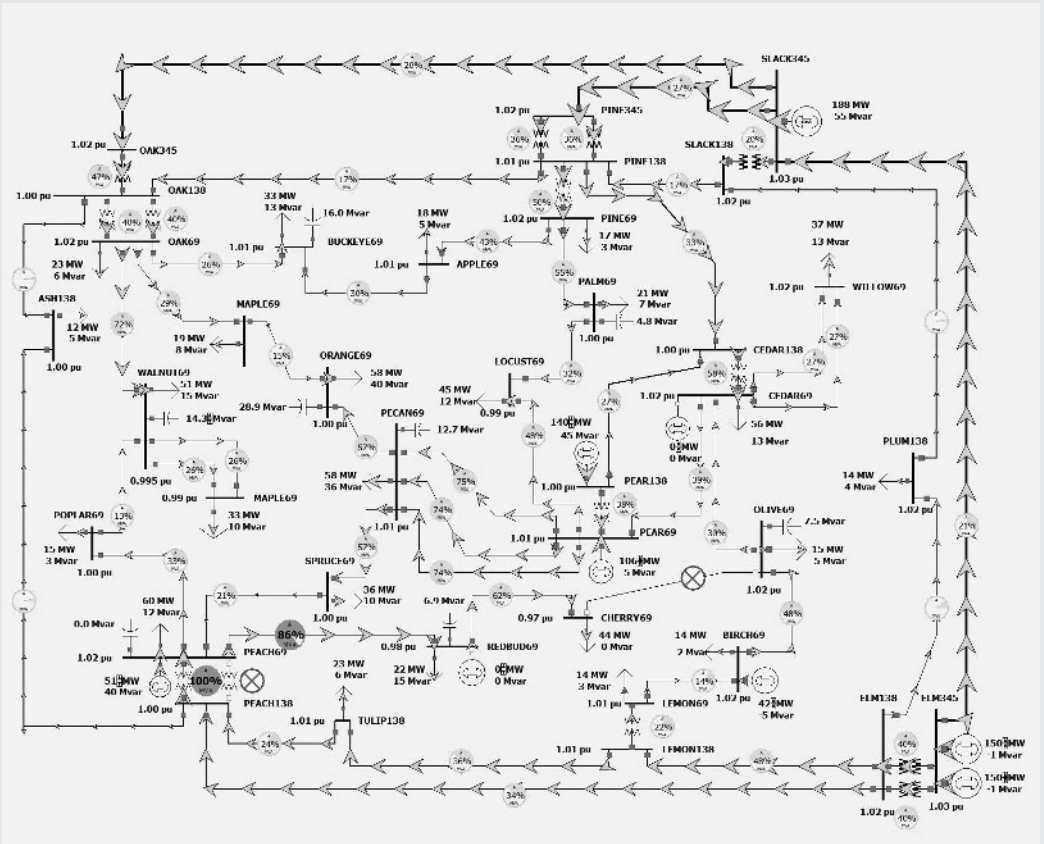

FIGURE 6.11

A solution to Example 6.15

to at the PEACH 69-kV bus, it stands to reason that increasing this generation would decrease the overload. Using this approach, the overload is removed by increasing the PEACH generation until the transformer flow is reduced to 100%. This occurs when the generation is increased from 20 MW to 51 MW. Notice that as the generation is increased, the slack bus (SLACK345) generation automatically decreases in order to satisfy the requirement that total system load plus losses must be equal to total generation.

An alternative possible solution is seen by noting that, since the overload is caused by power flowing from the 138-kV bus, decreasing the generation at ELM345 might also decrease this flow. This is indeed the case, but now the trial-and-error approach requires a substantial amount of work and ultimately doesn't solve the problem. Even when the total ELM345 generation is

(*Continued*)

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decreased from 300 MW to 0 MW, the overload is still present, albeit with its percentage decreased to 105%.

Another solution approach would be to first determine the generators with the most sensitivity to this violation and then adjust these (see Figure 6.12). This can be done in PowerWorld Simulator by selecting **Tools, Sensitivities, Flows and Voltage Sensitivities** to calculate sensitivities. On the top of the page, select the PEACH138 to PEACH69 circuit 1 transformer; then click on the **Calculate Sensitivities** button, and select the Generator Sensitivities tab toward the bottom of the dialog. The "P Sensitivity" field tells how increasing the output of each generator by one MW would affect the MVA flow on this

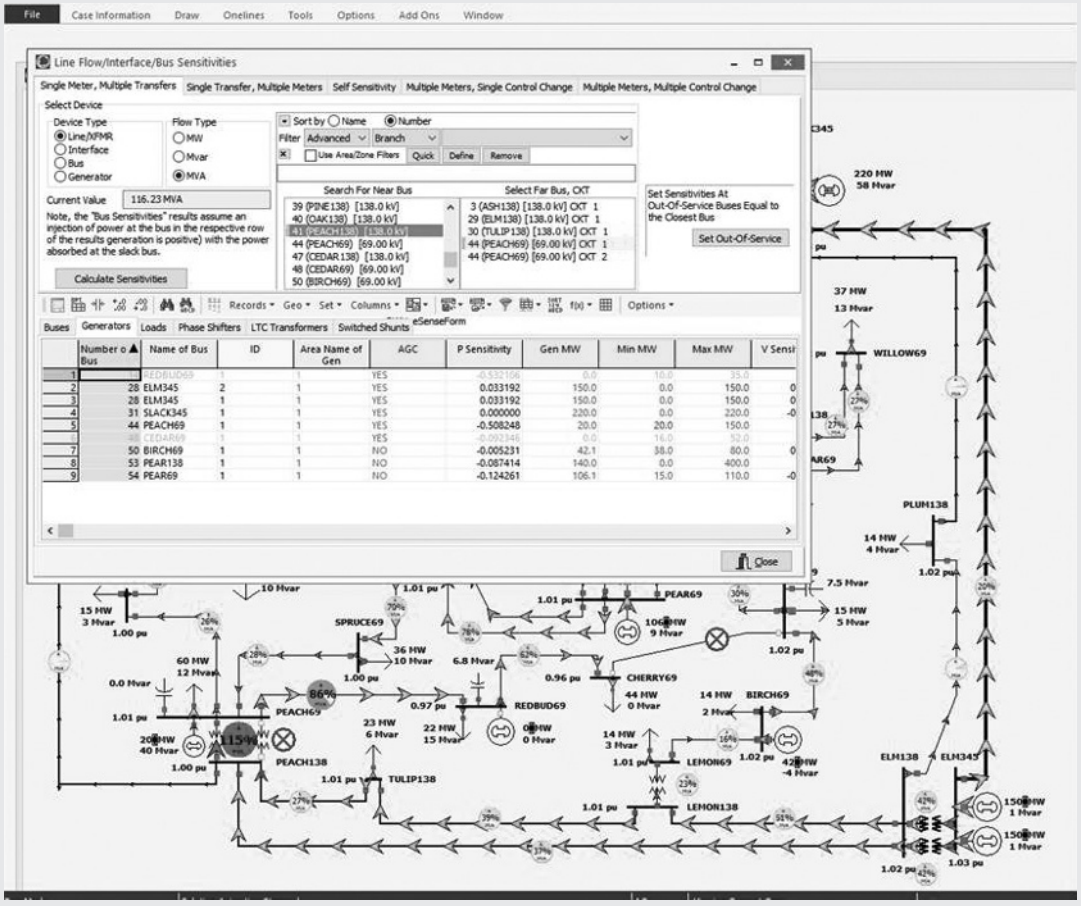

FIGURE 6.12

Example 6.15 Flow Sensitivities Dialog

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transformer. Note that the sensitivity for the PEACH69 generator is −0.508, indicating that if this generation is increased by 1 MW, the transformer MVA flow would decrease by 0.508 MVA. Hence, in order to decrease the flow by 15.2 MVA, increase the PEACH69 generator by 30 MW, which is exactly what was gotten by the trial-and-error approach. It is also clear that the ELM345 generators, with a sensitivity of just 0.033, would be relatively ineffective. In actual power system operation, these sensitivities, known as generator shift factors, are used extensively. These sensitivities are also used in the Optimal Power Flow (introduced in Section 7.2).

6.8 SPARSITY TECHNIQUES

A typical large power system has an average of fewer than three lines connected to each bus. As such, each row of $\boldsymbol{Y}_{\text{bus}}$ has an average of fewer than four nonzero elements: one off-diagonal for each line and the diagonal. Such a matrix, which has only a few nonzero elements, is said to be *sparse.*

Newton-Raphson power flow programs employ sparse matrix techniques to reduce computer storage and time requirements [2]. These techniques include compact storage of $\boldsymbol{Y}_{\text{bus}}$ and $\mathbf{J}(i)$ and reordering of buses to avoid fill-in of $\mathbf{J}(i)$ during Gauss elimination steps. Consider the following matrix:

$$\mathbf{S} = \begin{bmatrix} 1.0 & -1.1 & -2.1 & -3.1 \\ -4.1 & 2.0 & 0 & -5.1 \\ -6.1 & 0 & 3.0 & 0 \\ -7.1 & 0 & 0 & 4.0 \end{bmatrix} \tag{6.8.1}$$

One method for compact storage of S consists of the following four vectors:

$$\mathbf{DIAG} = [1.0\ 2.0\ 3.0\ 4.0] \tag{6.8.2}$$

$$\mathbf{OFFDIAG} = [-1.1\ -2.1\ -3.1\ -4.1\ -5.1\ -6.1\ -7.1] \tag{6.8.3}$$

$$\mathbf{COL} = [2\ 3\ 4\ 1\ 4\ 1\ 1] \tag{6.8.4}$$

$$\mathbf{ROW} = [3\ 2\ 1\ 1] \tag{6.8.5}$$

DIAG contains the ordered diagonal elements and **OFFDIAG** contains the nonzero off-diagonal elements of **S**. **COL** contains the column number of each off-diagonal element. For example, the *fourth* element in **COL** is 1, indicating that the *fourth* element of **OFFDIAG**, −4.1, is located in column 1. **ROW** indicates the number of off-diagonal elements in each row of **S**. For example, the *first* element

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of **ROW** is 3, indicating the *first* three elements of **OFFDIAG**, −1.1, −2.1, and −3.1, are located in the *first* row. The *second* element of **ROW** is 2, indicating the next two elements of **OFFDIAG**, −4.1 and −5.1, are located in the *second* row. The **S** matrix can be completely reconstructed from these four vectors. Note that the dimension of **DIAG** and **ROW** equals the number of diagonal elements of **S**, whereas the dimension of **OFFDIAG** and **COL** equals the number of nonzero off-diagonals.

Now assume that computer storage requirements are 4 bytes to store each magnitude and 4 bytes to store each phase entry in $\boldsymbol{Y}_{\text{bus}}$ in an N-bus power system. Also, assume $\boldsymbol{Y}_{\text{bus}}$ has an average of $3N$ nonzero off-diagonals (three lines per bus) along with its N diagonals. Using the preceding compact storage technique, $(4 + 4)3N = 24N$ bytes are required for **OFFDIAG** and $(4 + 4)N = 8N$ bytes for **DIAG**. Also, assuming 2 bytes to store each integer, $6N$ bytes are required for **COL** and $2N$ bytes for **ROW**. Total computer memory required is then $(24 + 8 + 6 + 2)N = 40N$ bytes with compact storage of $\boldsymbol{Y}_{\text{bus}}$, compared to $8N^2$ bytes without compact storage. For a 1000-bus power system, this means 40 instead of 8000 kilobytes to store $\boldsymbol{Y}_{\text{bus}}$.

The Jacobian matrix is also sparse. From Table 6.5, whenever $Y_{kn} = 0$, $\text{J1}_{kn} = \text{J2}_{kn} = \text{J3}_{kn} = \text{J4}_{kn} = 0$. Compact storage of **J** for a 30,000-bus power system requires less than 10 megabytes with the above assumptions.

The other sparsity technique is to reorder buses. Suppose Gauss elimination is used to triangularize **S** in (6.8.1). After one Gauss elimination step, as described in Section 6.1, the result is

$$\mathbf{S}^{(1)} = \begin{bmatrix} 1.0 & -1.1 & -2.1 & -3.1 \\ 0 & -2.51 & -8.61 & -7.61 \\ 0 & -6.71 & -9.81 & -18.91 \\ 0 & -7.81 & -14.91 & -18.01 \end{bmatrix} \tag{6.8.6}$$

Note that the zeros in columns 2, 3, and 4 of **S** are filled in with nonzero elements in $\mathbf{S}^{(1)}$. The original degree of sparsity is lost.

One simple reordering method is to start with those buses having the fewest connected branches and to end with those having the most connected branches. For example, **S** in (6.8.1) has three branches connected to bus 1 (three off-diagonals in row 1), two branches connected to bus 2, and one branch connected to buses 3 and 4. Reordering the buses 4, 3, 2, 1 instead of 1, 2, 3, 4 results in

$$\mathbf{S}_{\text{reordered}} = \begin{bmatrix} 4.0 & 0 & 0 & -7.1 \\ 0 & 3.0 & 0 & -6.1 \\ -5.1 & 0 & 2.0 & -4.1 \\ -3.1 & -2.1 & -1.1 & 1.0 \end{bmatrix} \tag{6.8.7}$$

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Now, after one Gauss elimination step,

$$\mathbf{S}^{(1)}_{\text{reordered}} = \begin{bmatrix} 4.0 & 0 & 0 & -7.1 \\ 0 & 3.0 & 0 & -6.1 \\ 0 & 0 & 2.0 & -13.15 \\ 0 & -2.1 & -1.1 & -4.5025 \end{bmatrix} \tag{6.8.8}$$

Note that the original degree of sparsity is not lost in (6.8.8).

Reordering buses according to the fewest connected branches can be performed once, before the Gauss elimination process begins. Alternatively, buses can be renumbered during each Gauss elimination step in order to account for changes during the elimination process.

Sparsity techniques similar to those described in this section are a standard feature of today's Newton-Raphson power flow programs. As a result of these techniques, typical 30,000-bus power flow solutions require less than 10 megabytes of storage, less than one second per iteration of computer time, and less than 10 iterations to converge.

EXAMPLE 6.16

Sparsity in a 37-bus system

To see a visualization of the sparsity of the power flow $\boldsymbol{Y}_{\text{bus}}$ and Jacobian matrices in a 37-bus system, open PowerWorld Simulator case Example 6_13. Select **Case Information, Solution Details, $\boldsymbol{Y}_{\text{bus}}$** to view the bus admittance matrix. Then press <ctrl> Page Down to zoom the display out. Blank entries in the matrix correspond to zero entries. The 37×37 $\boldsymbol{Y}_{\text{bus}}$ has a total of 1369 entries, with only about 10% nonzero (see Figure 6.13). Select **Case Information, Solution Details, Power Flow Jacobian** to view the Jacobian matrix.

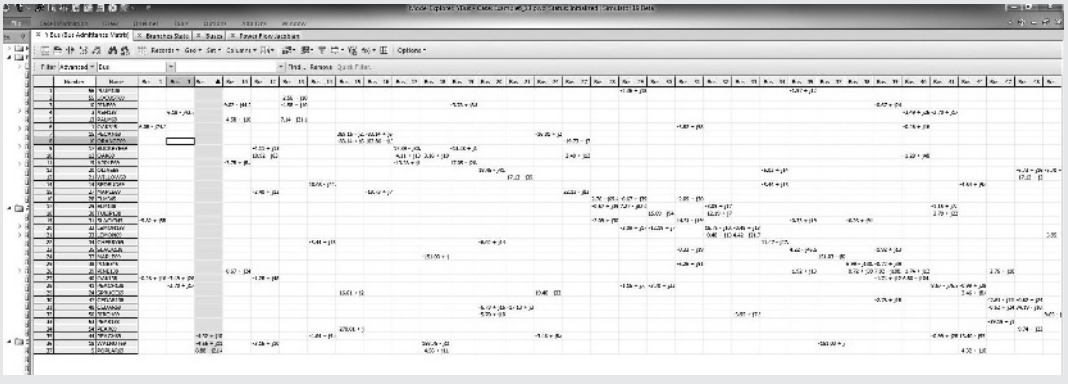

FIGURE 6.13

Screen for Example 6.16

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6.9 FAST DECOUPLED POWER FLOW

Contingencies are a major concern in power system operations. For example, operating personnel need to know what power flow changes will occur due to a particular generator outage or transmission-line outage. Contingency information, when obtained in real time, can be used to anticipate problems caused by such outages and can be used to develop operating strategies to overcome the problems.

To help with contingency solutions, a potentially faster Newton-Raphson algorithm is possible by recognizing that (1) the power flow Jacobian matrix does not need to be exact for the algorithm to converge, and (2) much of the time in a power flow solution is associated with building the Jacobian matrix and solving the equations using its inverse. One such algorithm, known as the decoupled power flow, uses the approximation that by assuming $\mathbf{J}_2(i)$ and $\mathbf{J}_3(i)$ are zero, (6.6.6) reduces to two sets of decoupled equations:

$$\mathbf{J}_1(i)\,\boldsymbol{\Delta}\delta(i) = \Delta \mathrm{P}(i) \tag{6.9.1}$$

$$\mathbf{J}_4(i)\,\boldsymbol{\Delta}\mathbf{V}(0) = \Delta\mathbf{Q}(i) \tag{6.9.2}$$

The justification for this approximation can be seen by looking at the elements in the Jacobian blocks and recognizing that (1) the conductance terms of the $\boldsymbol{Y}_{\text{bus}}$, G_{kn}, are usually much less than the susceptance terms, B_{kn}, and (2) the angle differences are usually small so the cosine terms are much larger than the sine terms. Hence, the Jacobian elements are dominated by the product of the susceptance terms and the cosine of the angle differences, elements that are only present in $\mathbf{J}_1(i)$ and $\mathbf{J}_4(i)$. The total computational time required to solve (6.9.1) and (6.9.2) is significantly less than that required to solve (6.6.6). Further reduction in computational time can be obtained from additional simplification of the Jacobian matrix. For example, assume $\mathrm{V}_k \approx \mathrm{V}_n \approx 1.0$ per unit and that the angle differences are small so the sine terms can be ignored and the cosine terms assumed to be 1.0. Then $\mathbf{J}_1$ and $\mathbf{J}_4$ are constant matrices whose elements in Table 6.5 are the negative of the imaginary components of $\boldsymbol{Y}_{\text{bus}}$. As such, $\mathbf{J}_1$ and $\mathbf{J}_4$ do not have to be recalculated during successive iterations. This approximation is used in the fast decoupled power flow [8].

The above simplifications can result in rapid power flow solutions for some systems, depending upon how well the approximations are actually met for a particular system. While the fast decoupled power flow usually takes more iterations to converge, it can be significantly faster than the Newton-Raphson algorithm since the Jacobian does not need to be recomputed each iteration. And since the mismatched equations themselves have not been modified, the solution obtained by the fast decoupled algorithm is the same as that found with the Newton-Raphson algorithm. However, in some situations in which only an approximate power flow solution is needed, the fast decoupled approach can be used with a fixed number of iterations (typically one) to give an extremely fast, albeit approximate solution.

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6.10 THE "DC" POWER FLOW

The power flow problem can be further simplified by extending the fast decoupled power flow to completely neglect the Q-V equations by assuming that the voltage magnitudes are constant at 1.0 per unit. With these simplifications, the power flow on the line from bus j to bus k with reactance X_{jk} becomes

$$P_{jk} = \frac{\delta_j - \delta_k}{X_{jk}} \tag{6.10.1}$$

and the real power balance equations reduce to a completely linear problem

$$-\mathbf{B}\delta = \mathbf{P} \tag{6.10.2}$$

where $\mathbf{B}$ is the imaginary component of the $\boldsymbol{Y}_{\text{bus}}$ calculated neglecting line resistance and excepting the slack bus row and column, and $\mathbf{P}$ is the vector of real power injections (with generation assumed positive).

Because (6.10.2) is a linear equation with a form similar to that found in solving dc resistive circuits, this technique is referred to as the dc power flow. However, in contrast to the previous power flow algorithms, the dc power flow only gives an approximate solution with the degree of approximation system dependent. Nevertheless, with the advent of power system restructuring, the dc power flow has become a commonly used analysis technique in some situations.

EXAMPLE 6.17

SOLUTION

Determine the dc power flow solution for the five bus system from Example 6.9.

With bus 1 as the system slack, the $\mathbf{B}$ matrix and $\mathbf{P}$ vector for this system are

$$\mathbf{B} = \begin{bmatrix} -30 & 0 & 10 & 20 \\ 0 & -100 & 100 & 0 \\ 10 & 100 & -150 & 40 \\ 20 & 0 & 40 & -110 \end{bmatrix} \quad \mathbf{P} = \begin{bmatrix} -8.0 \\ 4.4 \\ 0 \\ 0 \end{bmatrix}$$

$$\boldsymbol{\delta} = -\mathbf{B}^{-1}\mathbf{P} = \begin{bmatrix} -0.3263 \\ 0.0091 \\ -0.0349 \\ -0.0720 \end{bmatrix} \text{ radians} = \begin{bmatrix} -18.70 \\ 0.5214 \\ -2.000 \\ -4.125 \end{bmatrix} \text{ degrees}$$

To view this example in PowerWorld Simulator, open case Example 6_17, which has this example solved using the dc power flow (see Figure 6.14). To view the dc power flow options, select **Options, Simulator Options** to show the PowerWorld Simulator Options dialog. Then select the Power Flow Solution category and the DC Options page.

(*Continued*)

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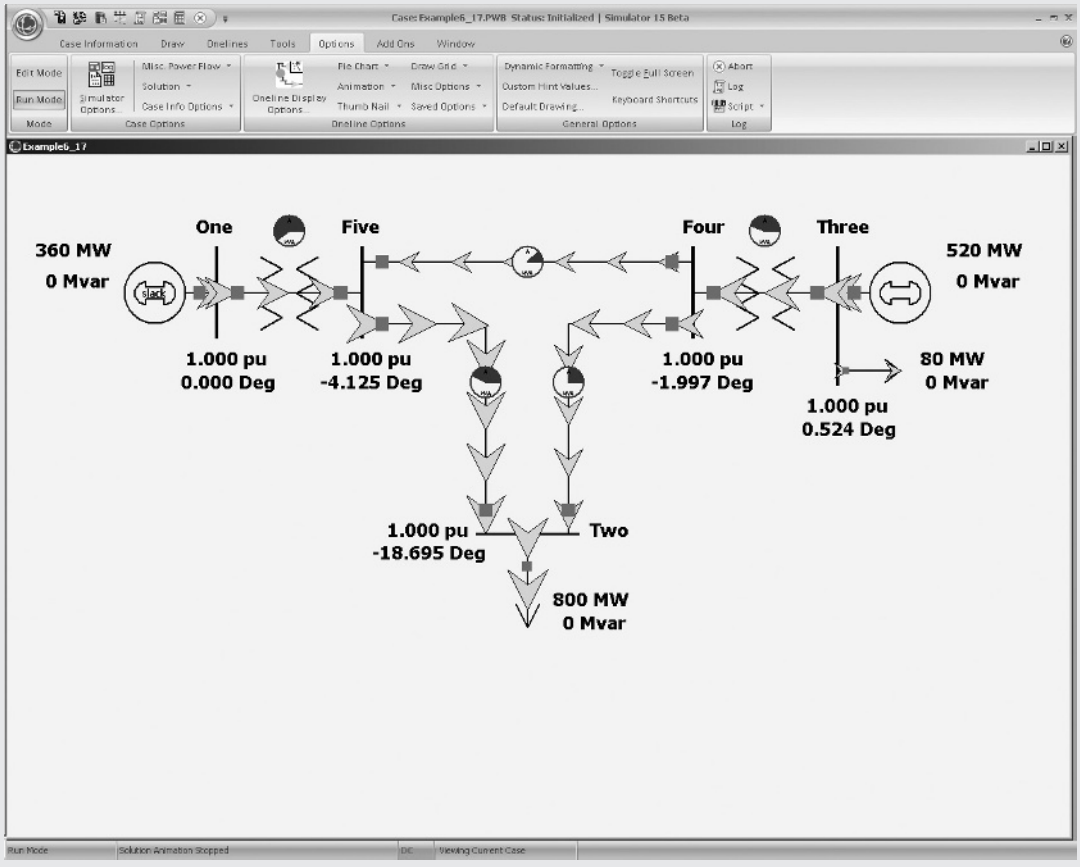

FIGURE 6.14

Screen for Example 6.17

6.11 POWER FLOW MODELING OF WIND AND SOLAR GENERATION

As was mentioned in Chapter 1, the amount of renewable generation, particularly wind and solar, being integrated into electric grids around the world is rapidly growing. For example, in 2018, Denmark obtained 41% of their total electric energy from wind while Ireland got 28%. Wind power's share of worldwide electricity usage at the end of 2018 was 4.8%. In the United States, the amount of wind capacity has been rapidly escalating from less than 35 GW in 2009 to more than 117 GW in 2020 (out of a total generation capacity of about 1000 GW), while worldwide, it was about 733 GW in 2020 [18]. Solar photovoltaic (PV) generation's

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growth is even more rapid, more than doubling in just the last four years to a total of 713 GW worldwide in 2020 [19].

Whereas most energy from traditional synchronous generators comes from large units with ratings of hundreds of MWs, comparatively speaking, individual wind turbine generator (WTG) power ratings are quite low, with common values for new land-based WTGs between one to six MWs, and offshore WTGs up to almost 10 MWs. This power is generated at low voltage (e.g., 600 V) and then usually stepped-up with a pad-mounted transformer at the base of the turbine to a distribution-level voltage (e.g., 34.5 kV). Usually dozens or even hundreds of individual WTGs are located in wind "farms" or "parks" that cover an area of many square kilometers with most of the land still available for other uses, such as farming. An underground and/or overhead collector system is used to transmit the power to a single interconnection point at which its voltage is stepped-up to a transmission-level voltage (>100 kV). The layout of such a system is shown in Figure 6.15.

From a power system analysis perspective for large-scale studies, the entire wind farm can usually be represented as a single equivalent generator that is either directly connected at the interconnection point transmission system bus or connected to this bus through an equivalent impedance that represents the impedance of the collector system and the step-up transformers. The parameters associated with the equivalent generator are usually just scaled values of the parameters for the individual WTGs.

There are four main types of WTGs [13-15]; more details on each type are provided in Chapter 12. Here, the focus is on their power flow characteristics. As is the case with traditional synchronous generators, the real power outputs for all the WTG types are considered to be a constant value in power flow studies. Of course, how much real power a wind farm can actually produce at any moment depends upon the wind speed, with a typical wind speed versus power curve shown in Figure 6.16.

Type 1 WTGs are squirrel-cage induction machines. Since induction machines consume reactive power and their reactive power output cannot be

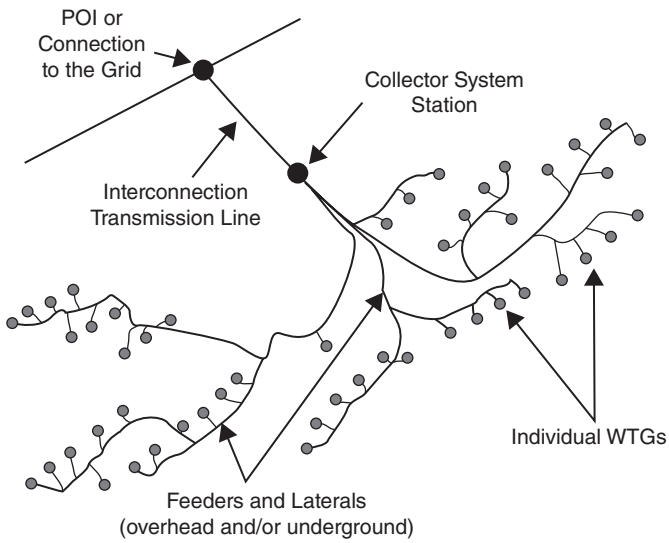

FIGURE 6.15

Wind power plant collector system topology [14] (Figure 1 from WECC Wind Generation Modeling Group, "WECC Wind Power Plant Power Flow Model Guide," WECC, May 2008, p. 2.)

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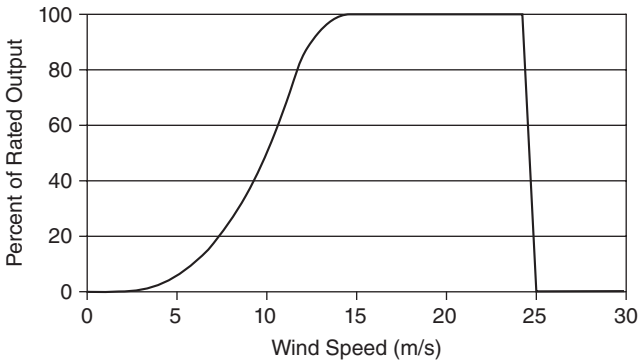
FIGURE 6.16

Typical wind speed versus power curve



independently controlled, typically these machines are modeled as a constant power factor PQ bus. By themselves, these machines have under-excited (consuming reactive power) power factors of between 0.85 and 0.9, but banks of switched capacitors are often used to correct the wind farm power factor. Type 2 WTGs are wound rotor induction machines in which the rotor resistance can be controlled. The advantages of this approach are discussed in Chapter 12; from a power flow perspective, they perform like Type 1 WTGs.

Most of the installed wind capacity and almost all new WTGs are either Type 3 or Type 4. Type 3 wind turbines are used to represent doubly-fed asynchronous generators (DFAGs), also sometimes referred to as doubly-fed induction generators (DFIGs). This type models induction machines in which the rotor circuit is also connected to the ac network through an ac-dc-ac converter, allowing for much greater control of the WTG. Type 4 wind turbines are fully asynchronous machines in which the full power output of the machine is coupled to the ac network through an ac-dc-ac converter. From a power flow perspective, both types are capable of full voltage control like a traditional bus generator with reactive power control within a power factor range. However, like traditional synchronous generators, how their reactive power is actually controlled depends on commercial considerations. One common approach is to use a model in which the reactive power output decreases as a linear function of voltage, possibly with a deadband, in an approach known as voltage droop control.

In the power flow, solar PV is often modeled differently depending upon whether it is representing a utility-scale solar (also known as solar power plants), in which hundreds or thousands of individual solar panels are connected together to act as a single central station, or whether it is represented by much smaller locations distributed in the lower voltage distribution system often located behind the customer meter (behind-the-meter or distributed generation). Solar power plants are modeled similarly to Type 4 wind farms with sizes ranging up to 100s of MWs. Behind-the-meter solar is often modeled as a distributed generation component of the load.

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6.12 REALISTIC AND LARGE-SCALE POWER FLOW MODELS

The power flow is one of, if not the most, common power system analysis algorithms, and the previous sections have provided a solid coverage of the basics of power system modeling. Commercial power flow models come in all levels of complexity and sizes, but often what is encountered in engineering practice is substantially larger and more complex than the previous examples. The purpose of this section is to ease the transition to such models by introducing some of the common complexities and data structures that are encountered in practice.

As has been previously mentioned, the power flow seeks to determine the steady-state operating point for an electric grid, or perhaps more accurately for most power systems, its quasi-steady-state operating point, since an actual system is almost always slowly changing. Modern power systems often include a variety of automatic controls, and also controls that are manually changed by a human operator within the power flow time frame (i.e., many seconds to minutes). To determine the quasi-steady-state operating point, the power flow needs to incorporate such control actions.

There are two main types: those that control the voltage magnitude and reactive power flow, and those that control the real power flow. The first group includes the voltage control systems for the generators, which has been previously introduced through the concept of a PV bus, in which the generator's reactive power output varies continuously within a range to maintain a setpoint voltage. The simplest form of a PV bus is a single generator controlling its terminal voltage magnitude, but sometimes they can be more complex, ranging up to generators at several buses controlling the voltage magnitude at a remote bus. This can be used to model the common situation of a set of generators controlling the high side voltage of their generator step-up transformer (GSU). Static var compensators, in which power electronics are used with capacitors and inductors to provide continuously varying reactive power output, are modeled using a similar approach. The other voltage regulation devices include switched shunts, in which the reactive power output changes in discrete steps modeling the switching of capacitors or reactors, or LTC transformers, in which the off-nominal turns ratio is changed in discrete increments. Because these controls are discrete, their setpoint voltage needs to be specified as a range with the magnitude of the range, dependent upon the discrete control increments. One of the challenges with designing a voltage control system, and with including it in the power flow, is these controls can interact. Currently for many utilities, at least part of this system is manually controlled.

Much of the control of real power is accomplished by generation changes. In most power flow models, this is represented by assigning devices (e.g., buses, loads, generators) to areas. While the term *area* is now used in several different contexts in the electric industry, in the power flow, it retains its historical usage as a control area in which the generation is varied to meet the load plus losses, while allowing for scheduled power transfers to other areas. The transmission lines and transformers that join areas are known as tie lines, and the area's net interchange

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is the algebraic sum of its tie line flows, with power exports assumed as positive. When solving a power flow with multiple areas, the generation in each area needs to be adjusted to insure the correct area interchange. There are several ways to accomplish this, with one common approach to assume; when there is a generation mismatch, it is proportioned to all the online generators in the area based on a participation factor assigned to each generator. The alternative, covered in the next chapter, is to dispatch the generation economically. Another type of real power control device is phase angle regulators (PARs) (sometimes known as phase-shifting transformers), in which a discrete off-nominal phase shift is varied to control the real power flow through the transformer.

EXAMPLE 6.18

Additional realism in the 37-bus system

An example of some of this additional realism is illustrated in PowerWorld Simulator case Example 6_18, shown in Figure 6.17. Most apparent, the system has now been divided into two areas with the magenta line showing the division with Area One on right and Area Two on the left. Each area is set to control its interchange with a tolerance of 0.5 MW, using generator participation factor control. Three additional buses have also been added. The first is to model a phase shifting transformer at the SLACK345 bus that is set to automatically control the flow on the line to ELM345. This transformer also uses an impedance correction table (see Section 3.8) to represent that its series impedance changes substantially as its phase shift is varied. The other two buses are to model the GSUs for the ELM345 generators now, with each of the generators regulating the ELM345 bus and the reactive power sharing set so the top generator contributes 2/3 of the reactive power. There are also several LTC regulating transformers and switched shunts.

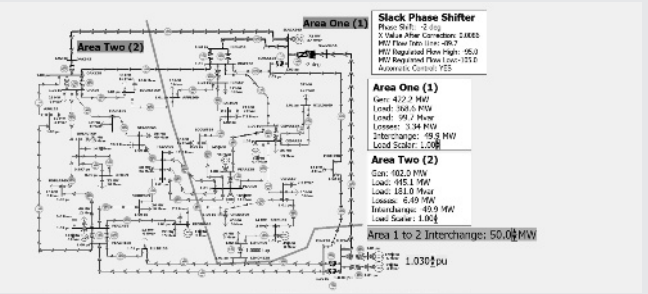

FIGURE 6.17
PowerWorld Simulator case Example 6-18

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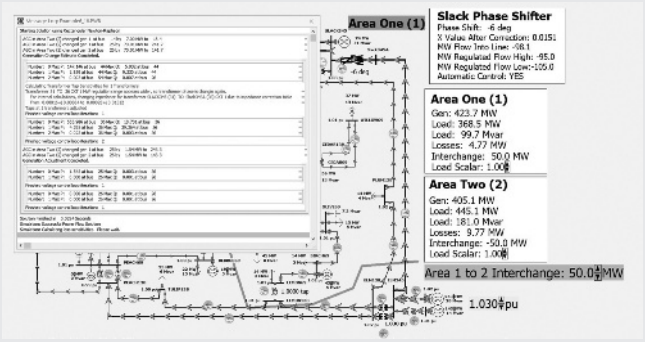
FIGURE 6.18

System after assumed loss of generator at PEACH69



This case can be used to demonstrate many of the previously mentioned power flow complexities. For example, Figure 6.18 shows the system after an assumed loss of the generator at PEACH69, which had been providing 147 MW. Because of the area interchange control all the lost generation was made up in Area Two, with only a small change in the Area One generation due to a change in losses. Also, the phase shifter has moved to keep the MW flow on the line between SLACK345 and ELM345 within its range.

In the early days of the power flow the size of the models was limited by computer memory, with extremely effective memory management allowing a system with 949 buses to be solved using the Netwon-Raphson approach with 32 KB in 1967. Current large-scale models range in size up to about 100,000 buses, albeit being solved on computers that might now have GBs of memory. The challenge in solving these larger power flows is no longer memory or even processing speed, but rather in the complexity of the interactions of the controls that have been previously mentioned. While most portions of the grid may solve fine, the power flow may not converge if some of the control tolerances are not correctly specified, such as two tight of control ranges for discrete controls. Nevertheless, commercial software has been designed to handle such issues, and hence power flows with tens of thousands of buses are routinely solved.

One challenge with the use of models of real electric grids for education and research is that in many locations around the world, some of the engineering data needed to solve a power flow might be restricted and hence not publicly available. In the U.S., some real grid power flow information is considered to be critical energy/electric infrastructure information (CEII), and so power flow models are not publicly available. To address this need, smaller scale synthetic electric grids, defined as any model not designed to represent a real grid, have been used for many decades and include all of the previous examples in this book. Over the last several years, there has been great progress in algorithms for the creation of large-scale, geographically based synthetic grids for use in education and research [21].

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FIGURE 6.19
Example of geographically based synthetic grid

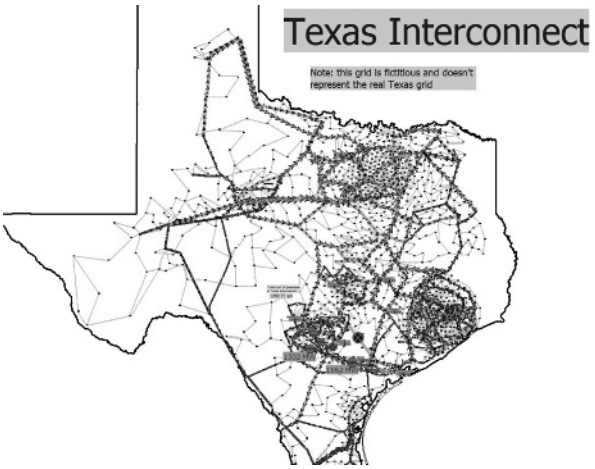

Figure 6.19 shows an example of one such grid that contains 2000 buses, eight areas, and covers a geographic footprint corresponding to the U.S. state of Texas. While the educational version of PowerWorld associated with this book is limited to 42 buses, PowerWorld provides a free Viewer software package (available at www.powerworld.com/gloveroverbye) that can be used to view, but not solve or save, these larger grids. There, a pre-solved version of this grid is included as PowerWorld case Example 6_19; additional details on this grid are available at [22].

PROBLEMS

SECTION 6.1

6.1 Using Gauss elimination, solve the following linear algebraic equations:

$$\begin{aligned} -25x_1 + 10x_2 + 10x_3 + 10x_4 &= 0 \\ 5x_1 - 10x_2 + 10x_3 &= 2 \\ 10x_1 + 5x_2 - 10x_3 + 10x_4 &= 1 \\ 10x_1 - 20x_4 &= -2 \end{aligned}$$

6.2 Using Gauss elimination and back substitution, solve

$$\begin{bmatrix} 8 & 2 & 1 \\ 4 & 6 & 2 \\ 3 & 4 & 14 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 2 \end{bmatrix}$$

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6.3 Rework Problem 6.2 with the value of 8 changed to 4.

6.4 What is the difficulty in applying Gauss elimination to the following linear algebraic equations?

$$-5x_1 + 5x_2 = 5$$
$$10x_1 - 10x_2 = -5$$

6.5 Show that, after triangularizing $\mathbf{Ax} = \mathbf{y}$, the back substitution method of solving $\mathbf{A}^{(N-1)}\mathbf{x} = \mathbf{y}^{(N-1)}$ requires N divisions, $N(N-1)/2$ multiplications, and $N(N-1)/2$ subtractions. Assume that all the elements of $\mathbf{A}^{(N-1)}$ and $\mathbf{y}^{(N-1)}$are nonzero and real.

SECTION 6.2

6.6 Solve Problem 6.2 using the Jacobi iterative method. Start with $x_1(0) = x_2(0) = x_3(0) = 0$, and continue until (6.2.2) is satisfied with $\varepsilon = 0.01$.

6.7 Repeat Problem 6.6 using the Gauss-Seidel iterative method. Which method converges more rapidly?

6.8 Express the following set of equations in the form of (6.2.6), and then solve using the Jacobi iterative method with $\varepsilon = 0.05$ and with $x_1(0) = 1$, and $x_2(0) = 1$, $x_3(0) = 0$.

$$\begin{bmatrix} 10 & -2 & -4 \\ -2 & 6 & -2 \\ -4 & -2 & 10 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2 \\ 3 \\ -1 \end{bmatrix}$$

6.9 Solve for x_1 and x_2 in the system of equations given by

$$x_2 - 3x_1 + 1.9 = 0$$
$$x_2 + x_1^2 - 3.0 = 0$$

using the Gauss method with an initial guess of $x_1 = 1$ and $x_2 = 1$.

6.10 Solve $x^2 - 4x + 1 = 0$ using the Jacobi iterative method with $x(0) = 1$. Continue until (6.2.2) is satisfied with $\varepsilon = 0.01$. Check using the quadratic formula.

6.11 Try to solve Problem 6.2 using the Jacobi and Gauss-Seidel iterative methods with the value of A_{33} changed from 14 to 0.14 and with $x_1(0) = x_2(0) = x_3(0) = 0$. Show that neither method converges to the unique solution.

6.12 Using the Jacobi method (also known as the Gauss method), solve for x_1 and x_2 in the following system of equations.

$$x_2 - 3x_1 + 1.9 = 0$$
$$x_2 + x_1^2 - 1.8 = 0$$

Use an initial guess of $x_1(0) = 1.0$ and $x_2 = (0) = 1.0$. Also, see what happens when you choose an uneducated initial guess of $x_1(0) = x_2(0) = 100$.

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6.13 Use the Gauss-Seidel method to solve the following equations that contain terms that are often found in power flow equations.

$$x_1 = (1/(-20j)) * [(-1 + 0.5j)/(x_1)^* - (j10) * x_2 - (j10)]$$
$$x_2 = (1/(-20j)) * [(-3 + j)/(x_2)^* - (j10) * x_1 - (j10)]$$

Use an initial estimate of $x_1(0) = 1$ and $x_2(0) = 1$, and a stopping of $\varepsilon = 0.05$.

6.14 Find a root of the following equation by using the Gauss-Seidel method: (use an initial estimate of $x = 2$) $f(x) = x^3 - 6x^2 + 9x - 4 = 0$.

6.15 Use the Jacobi method to find a solution to $x^2 \cos x - x + 0.5 = 0$. Use $x(0) = 1$ and $\varepsilon = 0.01$. Experimentally determine the range of initial values that results in convergence.

6.16 Take the z-transform of (6.2.6) and show that $\mathbf{X}(z) = \mathbf{G}(z)\mathbf{Y}(z)$, where $\mathbf{G}(z) = (z\mathbf{U} - \mathbf{M})^{-1}\,\mathbf{D}^{-1}$ and $\mathbf{U}$ is the unit matrix.

Note: $\mathbf{G}(z)$ is the matrix transfer function of a digital filter that represents the Jacobi or Gauss-Seidel methods. The filter poles are obtained by solving $\det(z\mathbf{U} - \mathbf{M}) = 0$. The filter is stable if and only if all the poles have magnitudes less than 1.

6.17 Determine the poles of the Jacobi and Gauss-Seidel digital filters for the general two-dimensional problem ($N = 2$):

$$\left[\begin{array}{c|c} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \hline \mathbf{A}_{21} & \mathbf{A}_{22} \end{array}\right] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

Then determine a necessary and sufficient condition for convergence of these filters when $N = 2$.

SECTION 6.3

6.18 Use Newton-Raphson to find a solution to the polynomial equation $f(x) = y$ where $y = 0$ and $f(x) = x^3 + 8x^2 + 2x - 40$. Start with $x(0) = 1$ and continue until (6.2.2) is satisfied with $\varepsilon = 0.001$.

6.19 Repeat 6.19 using $x(0) = -2$.

6.20 Use Newton-Raphson to find one solution to the polynomial equation $f(x) = y$, where $y = 7$ and $f(x) = x^4 + 3x^3 - 15x^2 - 19x + 30$. Start with $x(0) = 0$ and continue until (6.2.2) is satisfied with $\varepsilon = 0.001$.

6.21 Repeat Problem 6.20 with an initial guess of $x(0) = 4$.

6.22 For Problem 6.20, plot the function $f(x)$ between $x = 0$ and 4. Then provide a graphical interpretation why points close to $x = 2.2$ would be poorer initial guesses.

6.23 Use Newton-Raphson to find a solution to

$$\begin{bmatrix} e^{x_1 x_2} \\ \cos(x_1 + x_2) \end{bmatrix} = \begin{bmatrix} 1.2 \\ 0.5 \end{bmatrix}$$

where x_1 and x_2 are in radians. (a) Start with $x_1(0) = 1.0$ and $x_2(0) = 0.5$, and continue until (6.2.2) is satisfied with $\varepsilon = 0.005$. (b) Show that Newton-Raphson diverges for this example if $x_1(0) = 1.0$ and $x_2(0) = 2.0$.

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6.24 Solve the following equations by the Newton-Raphson method:

$$2x_1 + x_2^2 - 8 = 0$$

$$x_1^2 - x_2^2 + x_1x_2 - 3 = 0$$

Start with an initial guess of $x_1 = 1$ and $x_2 = 1$.

6.25 The following nonlinear equations contain terms that are often found in the power flow equations:

$$f_1(x) = 10x_1 \sin x_2 + 2 = 0$$

$$f_2(x) = 10\,(x_1)^2 - 10x_1 \cos x_2 + 1 = 0$$

Solve using the Newton-Raphson method starting with an initial guess of $x_1(0) = 1$ and $x_2(0) = 0$ radians and a stopping criteria of $\varepsilon = 10^{-4}$.

6.26 Repeat Problem 6.25 except using $x_1(0) = 0.25$ and $x_2(0) = 0$ radians as an initial guess.

6.27 For the Newton-Raphson method, the *region of attraction* (or *basin of attraction*) for a particular solution is the set of all initial guesses that converge to that solution. Usually initial guesses close to a particular solution will converge to that solution. However, for all but the simplest of multi-dimensional, nonlinear problems, the region of attraction boundary is often fractal. This makes it impossible to quantify the region of attraction and hence to guarantee convergence. Problem 6.25 has two solutions when x_2 is restricted to being between $-\pi$ and π. With the x_2 initial guess fixed at 0 radians, numerically determine the values of the x_1 initial guesses that converge to the Problem 6.25 solution. Restrict your search to values of x_1 between 0 and 1.

SECTION 6.4

6.28 Consider the simplified electric power system shown in Figure 6.20 for which the power flow solution can be obtained without resorting to iterative techniques. (a) Compute the elements of the bus admittance matrix $\boldsymbol{Y}_{\text{bus}}$. (b) Calculate the phase angle δ_2 by using the real power equation at bus 2 (voltage-controlled bus). (c) Determine $|V_3|$ and δ_3 by

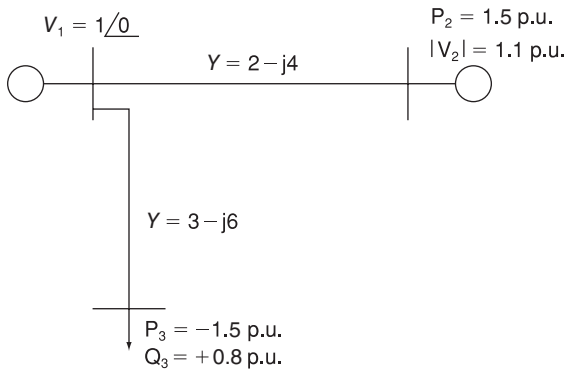

FIGURE 6.20
Problem 6.28

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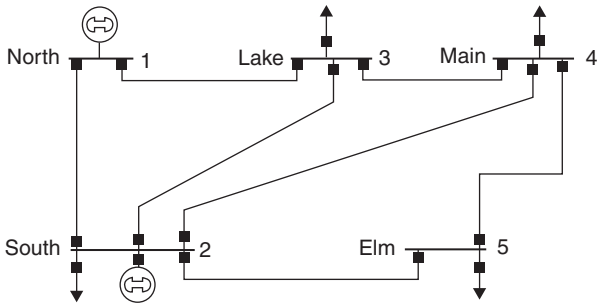
using both the real and reactive power equations at bus 3 (load bus). (d) Find the real power generated at bus 1 (swing bus). (e) Evaluate the total real power losses in the system.

6.29 In Example 6.9, double the impedance on the line from bus 2 to bus 5. Determine the new values for the second row of $\boldsymbol{Y}_{bus}$. Verify your result using PowerWorld Simulator case Example 6_9.

6.30 Determine the bus admittance matrix ($\boldsymbol{Y}_{bus}$) for the three-phase power system shown in Figure 6.21 with input data given in Table 6.9 and partial results in Table 6.10. Assume a three-phase 100-MVA per unit base.

FIGURE 6.21
Problem 6.30



Bus-to-Bus	R per unit	X per unit	B per unit
1-2	0.02	0.06	0.06
1-3	0.08	0.24	0.05
2-3	0.06	0.18	0.04
2-4	0.08	0.24	0.05
2-5	0.02	0.06	0.02
3-4	0.01	0.04	0.01
4-5	0.03	0.10	0.04

TABLE 6.9
Bus input data for Problem 6.30

$6.25 - j18.695$	$-5.00 + j15.00$	$-1.25 + j3.75$	0	0
$-5.00 + j15.00$				

TABLE 6.10
Partially Completed Bus Admittance Matrix ($\boldsymbol{Y}_{bus}$) for Problem 6.30

6.31 For the system from Problem 6.30, assume that a 75-Mvar shunt capacitance (three-phase assuming one per unit bus voltage) is added at bus 4. Calculate the new value of Y_{44}.

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SECTION 6.5

6.32 For a two-bus power system, a $0.7 + j0.4$ per unit load at bus 2 is supplied by a generator at bus 1 through a transmission line with series impedance of $0.05 + j0.1$ per unit. With bus 1 as the slack bus with a fixed per-unit voltage of $1.0\angle 0$, use the Gauss-Seidel method to calculate the voltage at bus 2 after three iterations.

6.33 Repeat Problem 6.32 with the slack bus voltage changed to $1.0\angle 30°$ per unit.

6.34 For the three-bus system whose $\boldsymbol{Y}_{\text{bus}}$ is given, calculate the second iteration value of V_3 using the Gauss-Seidel method. Assume bus 1 as the slack (with $V_1 = 1.0\angle 0°$), and buses 2 and 3 are load buses with a per-unit load of $S_2 = 1 + j0.5$ and $S_3 = 1.5 + j0.75$. Use voltage guesses of $1.0\angle 0°$ at both buses 2 and 3. The bus admittance matrix for a three-bus system is

$$\boldsymbol{Y}_{\text{bus}} = \begin{bmatrix} -j10 & j5 & j5 \\ j5 & -j10 & j5 \\ j5 & j5 & -j10 \end{bmatrix}$$

6.35 Repeat Problem 6.34, except assume the bus 1 (slack bus) voltage of $V_1 = 1.05\angle 0°$.

6.36 The bus admittance matrix for the power system shown in Figure 6.22 is given by

$$\boldsymbol{Y}_{\text{bus}} = \begin{bmatrix} 3 - j9 & -2 + j6 & -1 + j3 & 0 \\ -2 + j6 & 3.666 - j11 & -0.666 + j2 & -1 + j3 \\ -1 + j3 & -0.666 + j2 & 3.666 - j11 & -2 + j6 \\ 0 & -1 + j3 & -2 + j6 & 3 - j9 \end{bmatrix} \text{ per unit}$$

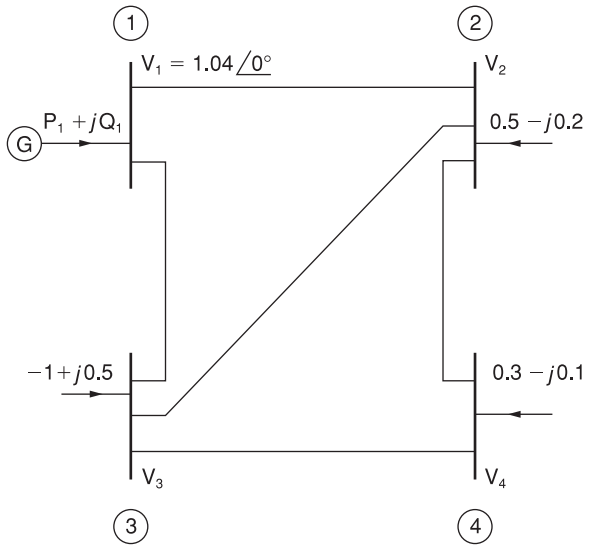

FIGURE 6.22
Problem 6.36

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With the complex powers on load buses 2, 3, and 4 as shown in Figure 6.22, determine the value for V_2 that is produced by the first and second iterations of the Gauss-Seidel procedure. Choose the initial guess $V_2(0) = V_3(0) = V_4(0) = 1.0\angle 0°$ per unit.

6.37 The bus admittance matrix of a three-bus power system is given by

$$\boldsymbol{Y}_{\text{bus}} = -j\begin{bmatrix} 7 & -2 & -5 \\ -2 & 6 & -4 \\ -5 & -4 & 9 \end{bmatrix} \text{ per unit}$$

with $V_1 = 1.0\angle 0°$ per unit; $V_2 = 1.0$ per unit; $P_2 = 60$ MW; $P_3 = -80$ MW; $Q_3 = -60$ Mvar (lagging) as a part of the power flow solution of the system. Find V_2 and V_3 within a tolerance of 0.01 per unit by using the Gauss-Seidel iteration method. Start with $\delta_2 = 0$, $V_3 = 1.0$ per unit, and $\delta_3 = 0$.

SECTION 6.6

6.38 A generator bus (with a 1.0 per unit voltage) supplies a 180-MW, 60-Mvar load through a lossless transmission line with per-unit (100-MVA base) impedance of $j0.1$ and no line charging. Starting with an initial voltage guess of $1.0\angle 0°$, iterate until converged using the Newton-Raphson power flow method. For convergence criteria, use a maximum power flow mismatch of 0.1 MVA.

6.39 Repeat Problem 6.38, except use an initial voltage guess of $1.0\angle 30°$.

6.40 Repeat Problem 6.38, except use an initial voltage guess of $0.25\angle 0°$.

6.41 Determine the initial Jacobian matrix for the power system described in Problem 6.34.

6.42 Use the Newton-Raphson power flow method to solve the power system described in Problem 6.34. For convergence criteria, use a maximum power flow mismatch of 0.1 MVA.

6.43 For a three-bus power system, assume bus 1 is the slack with a per unit voltage of $1.0\angle 0°$, bus 2 is a PQ bus with a per-unit load of $2.0 + j0.5$, and bus 3 is a PV bus with 1.0 per-unit generation and a 1.0 voltage setpoint. The per-unit line impedances are $j0.1$ between buses 1 and 2, $j0.4$ between buses 1 and 3, and $j0.2$ between buses 2 and 3. Using a flat start, use the Newton-Raphson approach to determine the first iteration phasor voltages at buses 2 and 3.

6.44 Repeat Problem 6.43, except with the bus 2 real power load changed to 1.0 per unit.

PW **6.45** Load PowerWorld Simulator case Example 6_11; this case is set to perform a single iteration of the Newton-Raphson power flow method each time **Single Solution** is selected. Verify that initially the Jacobian element J_{33} is 104.41. Then give and verify the value of this element after each of the next three iterations (until the case converges).

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PW **6.46** Load PowerWorld Simulator case Problem 6_46. Using a 100-MVA base, each of the three transmission lines have an impedance of $0.05 + j0.1$ p.u. There is a single 180-MW load at bus 3, while bus 2 is a PV bus with generation of 80 MW and a voltage setpoint of 1.0 p.u. Bus 1 is the slack bus with a voltage setpoint of 1.0 p.u. Manually solve this case using the Newton-Raphson approach with a convergence criteria of 0.1 MVA. Show all your work. Then verify your solution by solving the case with PowerWorld Simulator.

PW **6.47** As was mentioned in Section 6.4, if a generator's reactive power output reaches its limit, then it is modeled as though it were a PQ bus. Repeat Problem 6.46, except assume the generator at bus 2 is operating with its reactive power limited to a maximum of 50 Mvar. Then verify your solution by solving the case with PowerWorld Simulator. To increase the reactive power output of the bus 2 generator, select **Tools, Play** to begin the power flow simulation, then click on the up arrow on the bus 2 magenta voltage setpoint field until the reactive power output reaches its maximum.

PW **6.48** Load PowerWorld Simulator case Problem 6_46. Plot the reactive power output of the generator at bus 2 as a function of its voltage setpoint value in 0.005 p.u. voltage steps over the range between its lower limit of -50 Mvar and its upper limit of 50 Mvar. To change the generator 2 voltage set point, first select **Tools, Play** to begin the power flow simulation, and then click on the up/down arrows on the bus 2 magenta voltage setpoint field.

SECTION 6.7

PW **6.49** Open PowerWorld Simulator case Problem 6_49. This case is identical to Example 6.9, except that the transformer between buses 1 and 5 is now a tap-changing transformer with a tap range between 0.9 and 1.1 and a tap step size of 0.00625. The tap is on the high side of the transformer. As the tap is varied between 0.975 and 1.1, show the variation in the reactive power output of generator 1, V_5, V_2, and the total real power losses.

PW **6.50** Use PowerWorld Simulator to determine the Mvar rating of the shunt capacitor bank in the Example 6_14 case that increases V_2 to 1.0 per unit. Also, determine the effect of this capacitor bank on line loadings and the total real power losses (shown immediately below bus 2 on the oneline). To vary the capacitor's nominal Mvar rating, right-click on the capacitor symbol to view the Switched Shunt Dialog, and then change Nominal Mvar field.

PW **6.51** Use PowerWorld Simulator to modify the Example 6_9 case by inserting a second line between bus 2 and bus 5. Give the new line a circuit identifier of "2" to distinguish it from the existing line. The line parameters of the added line should be identical to those of the existing lines 2 to 5. Determine the new line's effect on V_2, the line loadings, and on the total real power losses.

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PW **6.52** Open PowerWorld Simulator case Problem 6_52. Open the 69-kV line between buses REDBUD69 and PEACH69 (shown toward the bottom of the oneline). With the line open, determine the amount of Mvar (to the nearest 1 Mvar) needed from the REDBUD69 capacitor bank to correct the REDBUD69 voltage to at least 1.0 p.u.

PW **6.53** Open PowerWorld Simulator case Problem 6_53. Plot the variation in the total system real power losses as the generation at bus PEAR138 is varied in 20-MW blocks between 0 MW and 400 MW. What value of PEAR138 generation minimizes the total system losses?

PW **6.54** Repeat Problem 6.53, except first remove the 69-kV line between LOCUST69 and PEAR69.

SECTION 6.8

6.55 Using the compact storage technique described in Section 6.8, determine the vectors **DIAG**, **OFFDIAG**, **COL**, and **ROW** for the following matrix:

$$\mathbf{S} = \begin{bmatrix} 17 & -9.1 & 0 & 0 & -2.1 & -7.1 \\ -9.1 & 25 & -8.1 & -1.1 & -6.1 & 0 \\ 0 & -8.1 & 9 & 0 & 0 & 0 \\ 0 & -1.1 & 0 & 2 & 0 & 0 \\ -2.1 & -6.1 & 0 & 0 & 14 & -5.1 \\ -7.1 & 0 & 0 & 0 & -5.1 & 15 \end{bmatrix}$$

6.56 For the triangular factorization of the corresponding $\boldsymbol{Y}_{\text{bus}}$, number the nodes of the graph shown in Figure 6.9 in an optimal order.

SECTION 6.10

6.57 Compare the angles and line flows between the Example 6_17 case and results shown in Tables 6.6, 6.7, and 6.8.

6.58 Redo Example 6.17 with the assumption that the per-unit reactance on the line between buses 2 and 5 is changed from 0.05 to 0.03.

PW **6.59** Open PowerWorld Simulator case Problem 6_59, which models a seven-bus system using the dc power flow approximation. Bus 7 is the system slack. The real power generation/load at each bus is as shown, while the per-unit reactance of each of the lines (on a 100-MVA base) is as shown in yellow on the oneline. (a) Determine the six-by-six **B** matrix for this system and the **P** vector. (b) Use a matrix package such as Matlab to verify the angles as shown on the oneline.

PW **6.60** Using the PowerWorld Simulator case from Problem 6.59, if the rating on the line between buses 1 and 2 is 150 MW, the current flow is 101 MW (from bus 1 to bus 3), and the bus 1 generation is 160 MW, analytically determine the amount this generation can increase until this line reaches 100% flow. Assume any change in the bus 1 generation is absorbed at the system slack.

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SECTION 6.11

PW **6.61** PowerWorld Simulator cases Problem 6_61_PQ and 6_61_PV model a seven-bus power system in which the generation at bus 4 is modeled as a Type 1 or 2 wind turbine in the first case and as a Type 3 or 4 wind turbine in the second. A shunt capacitor is used to make the net reactive power injection at the bus the same in both cases. Compare the bus 4 voltage between the two cases for a contingency in which the line between buses 2 and 4 is opened. What is an advantage of a Type 3 or 4 wind turbine with respect to voltage regulation following a contingency? What is the variation in the Mvar output of a shunt capacitor with respect to bus voltage magnitude?

DESIGN PROJECT 1: NEW SOLAR

The starting point for this design project is the power flow given in PowerWorld Simulator case Chapter6_Design1Start, which is modified version of the 37-bus system introduced in Example 6.13 (see Figure 6.23). The goal of the design project is for you to develop recommendations on the least-cost design for the construction of new transmission lines and/or transformers to integrate in 75 MW of new utility solar PV at the location given on the oneline (at the NEWSOLAR

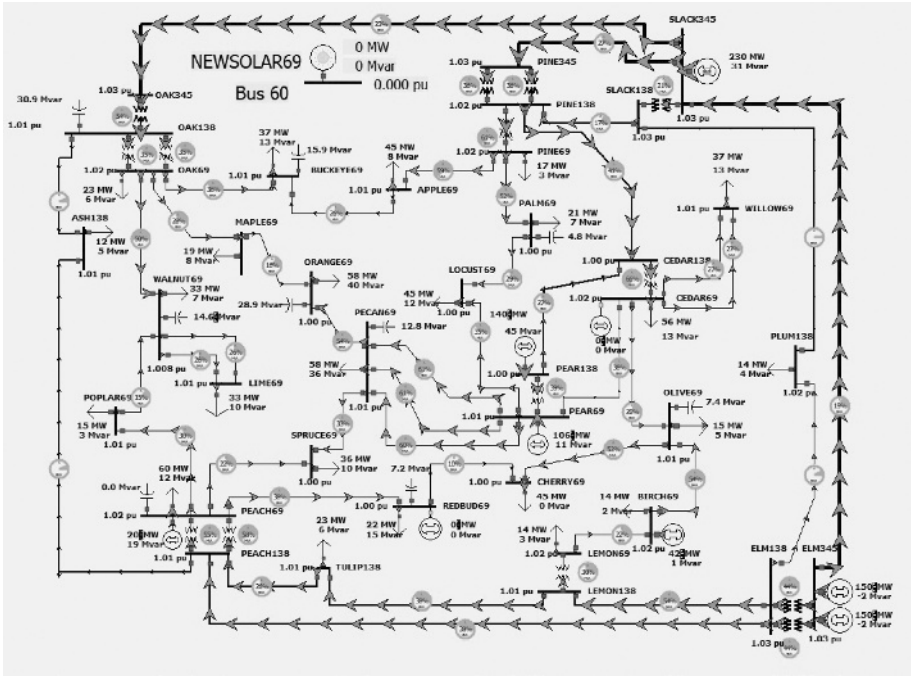

FIGURE 6.23
Design Project 1 System Oneline Diagram

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substation), which as a starting point has a 69-kV bus (NEWSOLAR69, bus number 60). This new generation also provides an opportunity to correct some existing problems. The design requirements are that the entire transmission system has no voltage or line loading violations for both the base case and for the first contingency loading situations when both the new solar is operating at its full 75-MW capacity and when it is offline.

You can model the new solar in the power flow as a single equivalent traditional PV bus generator with an output of 75 MW and reactive power limits of ±35 Mvar—the setpoint voltage is a design parameter. The solar is already connected to the provided bus NEWSOLAR69 and for reliability purposes, there must be two separate transmission lines into the NEWSOLAR substation. This substation is large enough to also accommodate a 138-kV bus and a 138/69-kV transformer.

Table 6.11 shows the right-of-way distances that are available for the construction of new 69-kV and/or new 138-kV lines. All existing 69-kV-only substations are large enough to accommodate 138 kV as well, and all the existing 69- and 138-kV buses can accommodate new transmission lines. Table 6.12 provides cost information for the different conductor types.

Right-of-Way/Substation	Right-of-Way Distance (km)
NEWSOLAR to OAK	13
NEWSOLAR to BUCKEY	6
NEWSOLAR to APPLE	6
NEWSOLAR to PINE	14
NEWSOLAR to MAPLE	15
BUCKEYE to OAK	8
BUCKEYE to ORANGE	8
APPLE to PINE	10
APPLE to ORANGE	9
APPLE to LOCUST	10

TABLE 6.11
Available New Rights-of-Way

Design Procedure

1. Load Chapter6_Design1Start into PowerWorld Simulator. Perform an initial power-flow solution to verify the base case system operation without the NEWSOLAR generation. Note that the entire line flows and bus voltage magnitudes are within their limits. Assume all line MVA flows must be at or below 100% of the limit A values, and all voltages must be between 0.95 and 1.10 per unit.

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2. Repeat the above analysis, considering the impact of any single transmission line or transformer outage. This is known as contingency analysis. To simplify this analysis, PowerWorld Simulator has the ability to automatically perform a contingency analysis study. Select **Tools, Contingency Analysis** to show the Contingency Analysis display. Note that the 56 single line/transformer contingencies are already defined. Select **Start Run** to automatically see the impact of removing any single element. This system has line violations for several different contingencies due to recent new load growth at some of the substations by NEWSOLAR. Your design must correct these existing problems as well.
3. Using the rights-of-way and the transmission line parameters/costs given in the below cost section, determine the least expensive system additions so that the base case and all the contingences result in reliable operation points (i.e., one with no violations) with the NEWSOLAR generation operating both at 75 MW and offline. The parameters of the new transmission lines(s) need to be derived using the tower configurations and conductor types. Tower configurations are often provided by the instructor with default values given with the cost data. Several different conductor types are available in the cost section. The total cost of an addition is defined as the construction costs minus the savings associated with any decrease in system losses over the next 5 years.
4. Write a detailed report discussing the initial system problems, your approach to optimally solving the system problems, and the justification for your final recommendation.

Simplifying Assumptions

To simplify the analysis, several assumptions are made:

1. You need only consider the base case loading level given in Chapter6_Design1Start. In a real design, typically a number of different operating points/loading levels would need to be considered.
2. You should consider the existing generator outputs as fixed values; as a simplification, any generation changes are by the system slack generator.
3. You should not modify the status of the capacitors or the transformer taps.
4. You should assume that the system losses remain constant over the 5-year period and need only consider the impact the new design has on the base case losses. Electricity is priced at $60/MWh.

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DESIGN PROJECT 1: TRANSMISSION SYSTEM DESIGN COSTS

Transmission Lines (69 kV, 138 kV)

New transmission lines include a fixed cost and a variable cost. The fixed cost is for the design work, the purchase/installation of the three-phase circuit breakers, associated relays, and changes to the substation bus structure. The fixed costs are $1,250,000 for a 138-kV line, and $750,000 for a 69-kV line. The variable costs depend on the type of conductor and the length of the line. The assumed cost in $/km is given in Table 6.12.

Conductor Type	Current Rating (Amps)	138-kV Lines	69-kV Lines
Rook	770		$370,000/km
Crow	830	$500,000/km	$390,000/km
Condor	900	$525,000/km	$410,000/km
Cardinal	1110	$540,000/km	

TABLE 6.12

Assumed costs

Line impedance data and MVA ratings are determined based on the conductor type and tower configuration. The conductor characteristics are given in Table A.4 of the book. For these design problems, assume a symmetric tower configuration. Often the spacings between conductors are provided by the instructor and may be student-specific. If no values are given, assume a GMD of 2 m for 69 kV, 4 m for 138 kV.

Transformers

Transformer costs include associated circuit breakers, relaying, and installation.

138/69 kV, 101 MVA	$1,500,000
138/69 kV, 168 MVA	$1,800,000

Assume any new 138/69-kV transformer has 0.0025 per unit resistance and 0.07 per unit reactance.

Bus Work

Upgrade 69-kV substation to also include 138 kV $900,000

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DESIGN PROJECT 2: ELECTRIC GRID VOLTAGE CONTROL DESIGN

The starting point for this design project is the power flow given in PowerWorld Simulator case Chapter6_Design2Start, which is a 39-bus 345/161-kV synthetic grid modeling the Island Electric Company (IEC). It is a nicely designed system, except the person who created this synthetic grid decided (for whatever reason) to not include any capacitors (switched shunts). Hence, it has unacceptably low voltages not only in first contingency situations, but also in the base case. Your job is to provide a least-cost solution for fixing these voltage problems when the system has a total load equal to both its initial value (3535 MW, 805 Mvar) and a value of 70% of the initial (2475 MW, 564 Mvar).

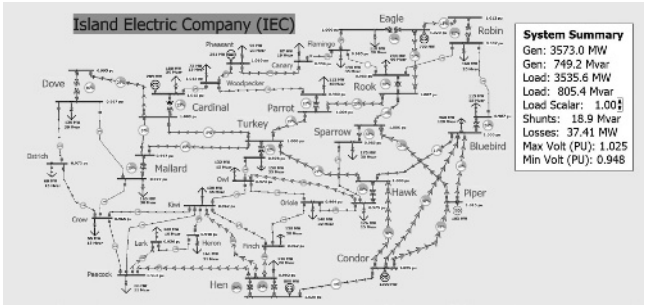

FIGURE 6.24
Design Project 2 System Oneline Diagram

Design Procedure

1. Load Chapter6_Design2Start into PowerWorld Simulator. Perform an initial power-flow solution to verify the base case system operation has some initial voltage violations. For this design project, assume that in the base case, all the voltage magnitudes must be between 0.98 and 1.06 per unit. Throughout the design project, you may ignore transmission line or transformer MVA limit violations.
2. Repeat the above analysis considering the impact of any single transmission line or transformer outage. This is known as contingency analysis. To simplify this analysis, PowerWorld Simulator has the ability to automatically perform a contingency analysis study. Select **Tools, Contingency Analysis** to show the Contingency Analysis display. Note that the 60 single line/transformer contingencies are already defined. Select **Start Run** to automatically see the impact of removing any single element. This system has many different voltage magnitude violations. For contingency analysis, all the contingent voltage magnitudes must be between 0.95 and 1.08 per unit.
3. With the simulation animating, use the Load Scalar field (shown in the white System Summary box on the right) to gradually decrease

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the load to 70% of nominal. Rerun the power flow and contingency analysis at this load level, and note that there are still some violations even at this lower load level. For convenience, this starting case is saved as Chapter6_Design2Start70Percent.

4. Using your power systems knowledge, iteratively add new capacitance (switched shunts) at the 161-kV buses to develop a low-cost solution to fix these voltage problems. For convenience, the case is already set up with switched shunts at all the 161-kV buses with zero values. The single exception is two 10-Mvar capacitor blocks have been added at bus 5 as an example. You can see the switched shunts by selected **Case Information, Network, Switched Shunts** and then right-clicking and selecting **Show Dialog** to see the associated dialog. You should add capacitors in blocks of equal Mvar size. Then for each switched shunt set its **Auto Control** field to "Yes," its Control Mode field to "Discrete," and then select an appropriate **High Value** and **Low Value** for the **Control Regulation Settings** fields. Adding capacitance has a fixed cost of $10,000 per Mvar.
5. If desired you may also adjust the generator voltage setpoint values. These values are shown on the Generator Information dialog. There is no cost for changing these fields.
6. Your job is to select capacitors that minimize overall system costs. While a real system operates at many different load levels, for this analysis assume that the system spends 50% of its time at the 100% loading level and 50% at the 70% loading level. While adding capacitors costs money, there are savings associated with a potential decrease in system losses. Therefore, total costs are given by the following equation:

$$\text{Total Cost} = Q_{new} * (\$10{,}000) - ((\Delta Loss_{100\%}) + (\Delta Loss_{70\%})) * (\$657{,}000)$$

where

Q_{new} is the total amount of new capacitance in Mvar added to the system

$\Delta Loss_{100\%}$ is the decrease in base case losses in MW with 100% loading (i.e., 37.41 new losses)

$\Delta Loss_{70\%}$ is the decrease in base case losses in MW with 70% loading (i.e., 37.41 new losses)

$10,000 is the total cost for new capacitance including installation

$657,000 accounts for the loss reduction savings over 5 years (0.5*5*8760*$30/MWh)

7. Write a detailed report discussing the initial system problems, your approach to optimally solving the system problems, and the justification for your final recommendation.

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Simplifying Assumptions

To simplify the analysis, several assumptions are made:

1. You do not need to be concerned with the transmission line and transformer loadings. None should be overloaded, but if some are, you can ignore them.
2. When the load is changed the generation is also changed using the area control discussed in the text. You can ignore the details of how this is occurring (the system is actually being dispatched using an economic dispatch algorithm discussed in Chapter 7).
3. Like many power system design problems, it could be very difficult to prove that a solution is actually optimal, and there could be many solutions that give very similar overall total costs that are near optimal. Given all the uncertainties associated with actual power system operation, such solutions are usually perfectly acceptable.

DESIGN PROJECT 3: POWER FLOW/SHORT CIRCUITS

Time given: 3 weeks

Approximate time required: 15 hours

Each student is assigned one of the single-line diagrams shown in Figures 6.25 and 6.26. Also, the length of line 2 in these figures is varied for each student.

Assignment I: Power Flow Preparation

For the single-line diagram that you have been assigned (Figure 6.25 or 6.26), convert all positive-sequence impedance, load, and voltage data to per unit using the given system base quantities. Then using the PowerWorld Simulator, create three input data files: bus input data, line input data, and transformer input data. Note that bus 1 is the swing bus. Your output for this assignment consists of three power-flow input data files.

The purpose of this assignment is to get started and to correct errors before going to the next assignment. It requires a knowledge of the per-unit system, which was covered in Chapter 3 but may need review.

Assignment 2: Power Flow

Case 1. Run the power flow program and obtain the bus, line, and transformer input/output data files that you prepared in Assignment 1.

Case 2. Suggest one method of increasing the voltage magnitude at bus 4 by 5%. Demonstrate the effectiveness of your method by making appropriate changes to the input data of case 1 and by running the power flow program.

Your output for this assignment consists of 12 data files, 3 input, and 3 output data files for each case, along with a one-paragraph explanation of your method for increasing the voltage at bus 4 by 5%.

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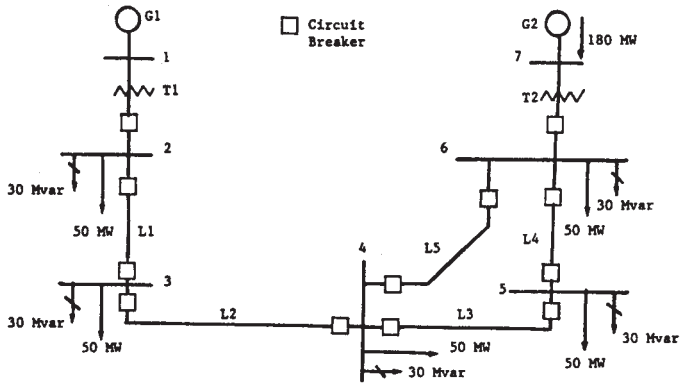
During this assignment, course material contains voltage control methods, including use of generator excitation control, tap changing and regulating transformers, static capacitors, static var systems, and parallel transmission lines.

This project continues in Chapters 7 and 9.

FIGURE 6.25

Single-line diagram for Design Project 3—transmission loop

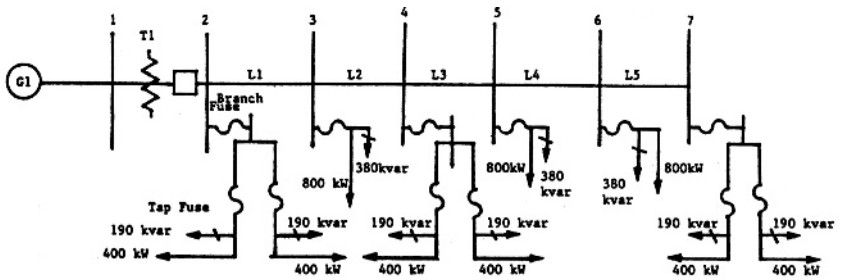


Generator Ratings
G1: 100 MVA,13.8 kV, $x'' = 0.12, x_2 = 0.14, x_0 = 0.05$ per unit
G2: 200 MVA,15.0 kV, $x'' = 0.12, x_2 = 0.14, x_0 = 0.05$ per unit
The generator neutrals are solidly grounded
Transformer Ratings
T1: 100 MVA,13.8 kVΔ/230 kVY, x=0.1 per unit
T2: 200 MVA, 15 kVΔ/230 kVY, x=0.1 per unit
The transformer neutrals are solidly grounded
Transmission Line Ratings
All Lines: 230 kV, $z_1 = 0.08 + j0.5\ \Omega/km$,
$z_0 = 0.2 + j1.5\ \Omega/km$, $y_1 = j3.3$ E-6 S/km,
Maximum MVA = 400
Line Lengths: $L_1 = 15$ km, L_2 assigned by the instructor (20 to 50 km), $L_3 = 40$ km, $L_4 = 15$ km, $L_5 = 50$ km.

Power Flow Data
Bus 1 : Swing bus, $V_1 = 13.8$ kV, $\partial_1 = 0°$
Bus 2,3,4,5,6 : Load buses
Bus 7 : Constant voltage magnitude bus, $V_7 = 15$ kV,
$P_{G7} = 180$ MW, -87 Mvar $< Q_{G7} < +87$ Mvar
System Base Quantities
$S_{base} = 100$ MVA (three-phase)
$V_{base} = 13.8$ kV (line-to-line) in the zone of G1

FIGURE 6.26

Single-line diagram for Design Project 3—radial distribution feeder



Generator Ratings
G1 (infinite bus): 50 MVA, 345 kV, $x'' = x_2 = 0.15$ per unit
The generator neutrals are solidly grounded
Transformer Ratings
T1: 5 MVA, 345 kVΔ/13.8 kVY, x = 0.1 per unit
The transformer neutrals are solidly grounded
Transmission Line Ratings
All Lines: 13.8 kV, $z_1 = 0.19 + j0.38\ \Omega/km$,
$z_0 = 0.6 + j1.0\ \Omega/km$, $y_1 = j4.0$ E-6 S/km,
Maximum MVA = 5
Line Lengths: $L_1 = 2$ km, L_2 assigned by the instructor (1 to 5 km), $L_3 = L_4 = L_5 = 2$ km.

Power Flow Data
Bus 1 : Swing bus, $V_1 = 345$ kV, $\partial_1 = 0°$
Bus 2,3,4,5,6,7 : Load buses
System Base Quantities
$S_{base} = 10$ MVA (three-phase)
$V_{base} = 13.8$ kV (line-to-line) in the zone of the lines

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7 Power System Economics and Optimization

Texas A&M University Electric Control Center Training Center. (Courtesy of Wei Trinh.)

LEARNING OBJECTIVES

At the conclusion of this chapter, you should be able to:

1. Understand the basic formulation of economic dispatch for fossil-fuel generating units, including the effects of generator output limits and transmission line losses.
2. Briefly describe the extension of the basic formulation of economic dispatch to other types of generating units, including wind and solar generation, nuclear units, pumped-storage hydro units, and hydro units along a river.
3. Understand optimal power flow as a combination of economic dispatch and power flow, considering transmission line constraints.
4. Use PowerWorld Simulator's optimal power flow algorithm to examine the effect of line constraints on LMPs.

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Economic Dispatch is the operation of generation facilities to produce energy at the lowest cost to reliably serve consumers, recognizing transmission line losses and operational limits of generation. The main idea is that, in order to satisfy the load at a minimum total cost, the set of generators with the lowest marginal costs must be used first, with the marginal cost of the final generator needed to meet load setting the system marginal cost, which is the cost of delivering one additional MWh of energy to the system.

Optimal Power Flow is an extension of conventional economic dispatch, where economic dispatch and power flow are solved simultaneously, considering transmission line constraints. In a competitive electricity market, an important function of the market operator is to determine prices of generator energy sales and associated transmission services, requiring the calculation of spot or real-time prices. Due to transmission constraints, cost levels may vary at different locations within a power grid. These different cost levels are identified as *Locational Marginal Prices* (LMPs).

CASE STUDY

The introduction to a U.S. electricity market environment resulted from Federal Energy Regulatory Commission (FERC) Order 888, issued in 1996, which initiated a restructuring of the U.S. power industry. The following article provides an overview of U.S. wholesale electric markets, their implementation via independent system operators, challenges being faced, and possible solutions. The article describes the main features of present-day U.S. wholesale electricity markets, including day-ahead and real-time markets, locational marginal pricing, co-optimization of energy and ancillary services, zonal reserve, financial transmission rights, and forward-capacity markets. The article also describes the market management system components, which comprise the internal architecture of the system operator, allowing it to function effectively. The market management system is complex but made up of basic building blocks including unit commitment, automatic generation control, economic dispatch, supervisory control and data acquisition, security analysis, and pricing. The authors identify challenges facing U.S. markets, including those pertaining to the grid, and ideas to address them [9].

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Electricity Markets in the United States*

Eugene Litvinov, Feng Zhao, and Tongxin Zheng—ISO New England

Following order 888 from the Federal Regulatory Energy Commission (FERC) in 1996, the U.S. power industry began its restructuring process. This undertaking involved a transition from a vertically integrated utility structure where generation, transmission, and distribution are combined to serve consumers. Through advancing to a market environment, generation companies (GenCos) can compete with each other to provide energy through open transmission and distribution systems managed by transmission (TransCos) and distribution companies (DisCos). The goals of the restructuring are the efficient production of electricity and efficient generation investment through competition. Under the restructured environment shown in Figure 1, wholesale electricity markets are managed by independent system operators (ISOs) or regional transmission organizations (RTOs), whose responsibility is reliable system operation, market administration, and system planning. GenCos, large consumers, load aggregators, marketers, and load serving entities (LSEs) buy and sell electricity through FERC-regulated wholesale electricity markets. Small consumers like individual households, some large consumers, LSEs, marketers, and load aggregators can transact energy either at a regulated retail rate overseen by the public utility commission (PUC) or through retail competition. The restructuring process has been complex, especially in trying to harmonize the wholesale and retail sides, and involves many stakeholders from both public and private sectors (see Figure 1).

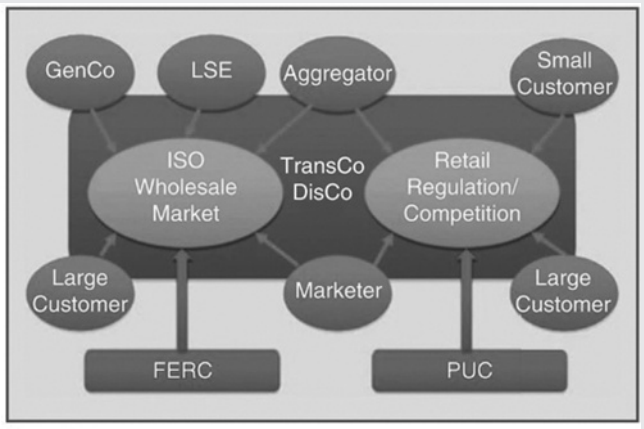

Figure1 The deregulated power industry in the United States

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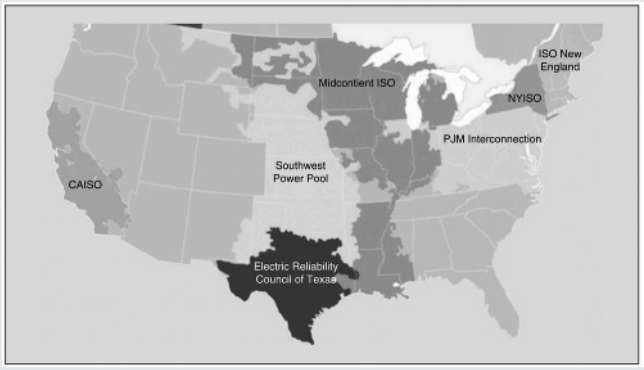

Figure 2 The U.S. ISOs/RTOs (Online: https://www.ferc.gov/industries/electric/indus-act/rto.asp)

U.S. electricity markets have been considered largely successful, and the RTO footprint continues to grow. Today there are seven RTOs operating in the United States (see Figure 2), bringing substantial benefits to end users. The benefits achieved by ISO New England's (ISO-NE's) wholesale electricity markets are summarized in Table l, as one example.

Wholesale electricity markets evolved with changes in regulatory policy, technological innovations, and economic conditions since the start of electric restructuring. During the past two decades, the U.S. power industry has experienced a transformation with the retirement of coal, oil, and nuclear resources; increased electricity production and investment in natural gas facilities due to environmental and siting factors as well as relatively low gas prices; increased penetration of renewable resources as a result of state environmental policies; and an increase in demand resources and distributed generation technologies. These changes have created challenges to power system operation and planning and electricity market design. We will review the main features of existing wholesale electricity markets and several key components. As pricing is particularly important for an effective market design, we will discuss ongoing pricing issues. Finally, we will take a brief look at the evolution of the electric grid and future challenges and then identify and discuss several key market design initiatives stemming from these challenges.

Main Features of U.S. Wholesale Electricity Markets

Wholesale electricity markets emerged in the United States in the late 1990s with different models. PJM and New York ISO (NYISO) started locational marginal pricing (LMP)-based energy markets with a two-settlement system in 1998

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Generation Availability	Increased from 75% in 1997 to 88% in 2016
Emissions reductions between 2001 and 2016	29% lower carbon dioxide emissions 73% lower nitrogen oxide emissions 98% reduction in sulfur dioxide emissions
Wholesale energy prices	2016 average annual energy market prices were lowest since 2003
Private investment	More than US$8 billion
New resources	More than 15 GW of new generation since 1999 and about 2.7 GW of new demand

Table 1

ISO-NE wholesale market benefits

and 1999, respectively. California ISO (CAISO) adopted a day-ahead power exchange with the real-time energy market in 1998. ISO-NE opened its real-time energy market with a single market clearing price and an uplift payment-based congestion management model in l999. After the California energy crisis in the early 2000s, the power exchange model was considered to be prone to market power abuse and gradually abandoned in the United States. FERC's standard market design efforts, coupled with the inefficiency of the uplift payment-based congestion management model, led the wholesale electricity markets to gradually converge to a pool-based model similar to those of PJM and NY-ISO. Today this model involves a set of common features, including the two-settlement system [day-ahead market (DAM) and real-time market (RTM)], LMP, co-optimization of energy and ancillary services, zonal reserve products, financial transmission rights, and a forward capacity market (FCM).

The goal of wholesale electricity markets is to increase the power system's economic efficiency without compromising its reliability. Therefore, market products traded in most U.S. electricity markets are designed according to the needs of bulk power system operation and planning. As a result, electricity markets always interact with the physical power system planning, scheduling, and operation. Figure 3 presents markets with different time frames and their corresponding physical systems. We will discuss the main market products and major features associated with each of the markets in more detail.

Energy

Energy is the main product traded in U.S. electricity markets, which typically include the DAM and the RTM. Although market designs vary in different ISO regions, all have structured the two markets in a two-settlement system with the forward DAM and the spot RTM. All markets also have a similar objective

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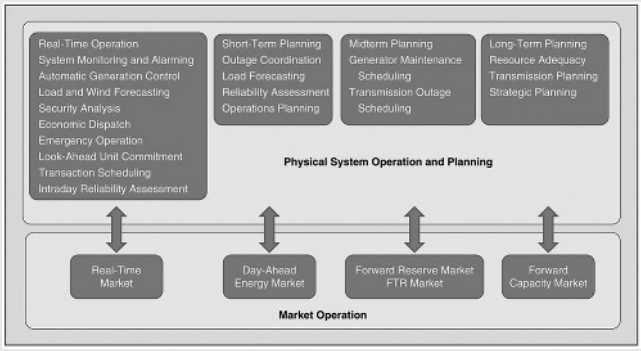

Figure 3 The markets and systems. FTR: financial transmission rights

of maximizing the total social surplus, and LMP is used to manage the transmission congestion.

Electricity producers and consumers may be located at different areas but are connected through the grid. Delivering energy from a generator to a consumer at a different location not only incurs energy losses but also may overload the transmission lines. This, in turn, prevents low-cost generation from being delivered to the customer, a phenomenon often called *congestion*. LMP recognizes the locational differences in energy production and consumption and has been the main approach to pricing electricity and managing congestion in the transmission system. LMP reflects the cost of serving an increment of load at a specific location in the transmission system. Electricity producers and consumers are paid or charged by the corresponding LMP at their locations. As a type of uniform pricing, LMP provides strong incentives for truthful bidding in energy markets and long-term investment signals for building new generators at the locations where they are needed (indicated by higher LMPs), including the effect of marginal losses. Mathematically, LMP can be decomposed into energy, congestion, and loss components, where the congestion component in the DAM is used to settle FTRs.

LMP calculations solve an economic dispatch (ED) problem. Due to the nonconvex features such as unit commitment (UC) decisions that are not modeled in the ED, out-of-market side payments to the generators are often used by ISOs in addition to the uniform LMP-based market payment. The total payments then allow generators to recover their start-up and no-load costs. The out-of-market payments distort the market and are not transparent, therefore creating problems in market participants' risk hedging. Reducing these out-of-market payments has been a difficult problem that will be discussed in later sections.

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Because ISO systems are connected by interties to facilitate the power exchange among neighboring ISOs, specific locations are used for pricing of imports or exports of energy. Currently, most ISOs assign a proxy bus for pricing energy imports and exports. The selection of these proxy buses is often arbitrary and may not conform to the actual power flow. As a result, the interchange schedule between ISOs is often inefficient.

The DAM is often considered a forward financial market where participants submit bids to buy or offers to sell electricity at each market location. The bids and offers can be physical and associated with generating units or demands or virtual and not associated with any physical assets. Bids and offers come in the form of price–quantity pairs in a descending or increasing price order at any market location. In addition, each physical asset submits its physical operating parameters, such as economic minimum and maximum, ramp rate, start-up and no-load costs, and minimum up and down times. The market clears at the point where supply meets demand at a maximal social surplus subject to net-work security constraints. The market clearing problem is called the *security-constrained UC* (*SCUC*) problem. DAM results are financially binding, which leaves only the real-time deviations from the day-ahead schedules to be exposed to the real-time price (two-settlement system). As a result, the ISO's DAM design provides a way for market participants to hedge against RTM price volatility or arbitrage between the two markets.

The DAM closes at a particular time of the day prior to the delivery date. Market participants submit their bids into the market before the closing time. Once the market closes, the market administrator runs software solving a 24-h (or longer) SCUC problem. The problem is typically a mixed integer linear program (MILP) because it involves both hourly binary on/off decisions for physical generating units and continuous unit output level decisions. DAM clearing prices, or locational market prices, are determined from the constraint shadow prices calculated by the ED process for each hour.

The RTM is a spot market where electricity is physically delivered to the consumers in a least-cost and reliable fashion. It is closely coupled with the power system operations by including all of the system and security constraints into the market clearing problem, that is security-constrained ED (SCED). Different from the DAM, the RTM usually does not require demand bids due to loads' lack of response to RTM prices.

In practice, after the DAM is cleared and prior to the operating day, ISOs often run a 24-hour or longer reliability UC (RUC) process committing additional units that have not been selected in the DAM to meet the forecasted load (which may exceed the DAM cleared as-bid load). The reliability commitment problem is similar to the DAM UC problem, except that only physical

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units are considered and no market clearing prices are produced. The RUC process may be run again throughout the operating day to look ahead a few hours and determine possible needs for additional resources.

With the committed units from the DAM and RUC, ISOs run the RTM, which determines the generation schedule and RTM clearing prices. Market participants are allowed to submit RTM bids that differ from their DAM bids. ISOs solve an SCED problem that minimizes the total dispatch cost subject to network and individual unit constraints. Different ISOs may model SCED differently. ISO-NE implements a single-interval dispatch for its RTM, whereas NYISO solves a multi-interval dispatch problem. Special pricing rules for the fast-start resources are often considered in the RTM to reflect their deployment costs in addition to the incremental costs in the LMP calculation. These rules often differ among RTOs but share the similar principle of eliminating the nonconvexity of the resources by relaxing certain characteristics, such as relaxing the minimum output level to be zero or amortizing the commitment costs into the dispatch costs.

Energy trading in the ISO-managed wholesale markets often creates two financial positions for each market participant: DAM and RTM (cleared transaction quantity). A two-settlement system is adopted to settle each financial position. The DAM settles on the DAM position with DAM LMPs, and the RTM settles on the energy deviations from the cleared DAM position with RTM LMPs. Under such design, physical players, such as generators and loads, are able to hedge their risk by locking in the price of their transactions with DAM locational market prices. Financial players who place virtual transactions in the DAM without delivering in RTM can arbitrage the price difference between DAM and RTM for their cleared DAM quantity. The two-settlement system has worked well in U.S. electricity markets, closing the gap between DAM and RTM prices.

Although LMP reflects the marginal cost of producing an increment of energy, it does not capture the entire cost of generation, such as the start-up cost of an online unit. As a result, the generation as-bid cost of a resource may not be fully recovered by the LMP payment. An illustration of cost and payment is shown in Figure 4, where the as-bid cost is represented by the area under the generation offer curve (i.e., areas A and B) and the LMP payment received by the generator is LMP multiplied by the generation output (i.e., areas A and C). When the cost exceeds the payment (i.e., $A + B > A + C$), the generator has a revenue shortage of $B - C$. Compensation for such revenue shortage, often termed a *make-whole payment* or an *uplift payment*, is needed and has been widely used in existing ISO markets to incentivize generators to follow ISO dispatch instructions.

Uplift payments are caused by the nonconvexity of the bids,

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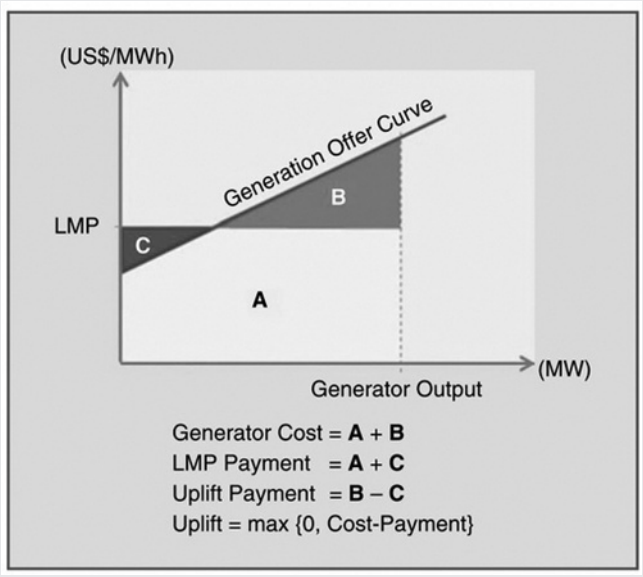

Figure 4 An illustration of uplift payment

for example, the integer commitment costs and the presence of minimum generation level. They are unit specific as opposed to the uniform market clearing price LMP and are often opaque to the market participants as being very difficult to hedge. ISOs have been striving to reduce such out-of-market payments, but, due to generators' nonconvex physical characteristics, they are unlikely to be eliminated. An alternative approach is to force one-part bidding and thereby eliminate the physical parameters that cause nonconvexity in their bids. This effectively removes the uplift payment at the cost of increased risk for participants that have to embed the nonconvexity-related costs into their convex bids, and it presents increased difficulty for market power monitoring. Currently, all U.S. markets have adopted the three-part bidding (i.e., incremental cost, start-up, and no-load cost) and uplift payments approach instead of the convex bidding seen in some other countries.

Ancillary Services

In addition to meeting the system load, power system operators have to prepare for generation contingencies and load fluctuations in real time. These additional reliability needs can be satisfied by ancillary service products besides energy, for example, regulation, reserves, and ramping. Resources with these additional capabilities are compensated through the ancillary service markets. Ancillary service products could be quite different among ISOs because the ISO systems may have different reliability needs. Consequently, the ancillary

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service markets are diverse. CAISO, for example, considers a five-minute ramping product, Midcontinent ISO projects a ten-minute ramping product, and ISO-NE does not currently have a ramping product. With the increasing penetration of renewables, new system features arise and may create a need for new ancillary services.

Because energy and ancillary services share the capacity of generation units and transmission network , they are often co-optimized to achieve the most efficient capacity allocation of resources, leading to an SCED optimization problem that simultaneously determines the energy and ancillary services designations for resources. Similar to energy LMPs, the clearing prices for ancillary services are obtained from the dual variables of the SCED problem.

Financial Transmission Rights

An FTR is a financial instrument for hedging against transmission congestion; it has a source and a sink location as well as a megawatt quantity and entitles its holder to collect congestion revenue based on the difference of LMP congestion components between the sink and source locations. FTRs have been implemented in all of the U.S. ISOs. Existing FTRs are point-to-point annual or multiyear products and can be obtained through corresponding FTR auctions or allocated to market participants based on grandfathered rights.

ISOs sell the FTRs and collect the auction revenue in the annual FTR auction. Depending on the region, the FTR auction revenue is allocated to the LSEs, transmission owners, or other entities in accordance with the market rules. FTRs are usually settled on the DAM prices. Due to the discrepancy between the transmission model used in the FTR and the model used in the DAM, the congestion revenue in the DAM and RTM may not cover 100% of the entitled payments to FTR holders, that is, FTR underfunding. To remedy the problem, ISOs often auction a reduced transmission capacity in the FTR auctions.

Forward Capacity Product

Generation investment is a key step to secure the long-term supply of electricity to meet demand growth in the future. In some regions, responsibility for resource adequacy has remained with vertically integrated utilities. In other regions, such as ISO-NE and PJM, where utilities divested their generation assets, resource adequacy is generally met through the wholesale markets. Due to the presence of price caps in the energy markets and the manual actions of system operators that may depress the energy price, revenues from the energy market alone may not be sufficient to cover a unit's capital investment cost. This phenomenon is termed the *missing money problem*. Facing the revenue shortage risk, generating units may not be available when needed, potentially causing a shortage of capacity and, therefore, reliability issues. One solution to the problem that

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has been implemented in ISO-NE is the FCM, where units' generating capacity is obtained in advance and compensated through the capacity product and the corresponding market clearing price. An important enhancement to the FCM, the pay-for-performance mechanism, reduces revenues for resources that do not perform and rewards resources that perform during critical system events. The FCM provides a steady revenue stream in advance for existing resources to recover the missing money and for new resources to reduce the investment risk.

As a forward market, FCM auctions run several years to several weeks ahead prior to the real-time operation of the capacity commitment period, depending on different market designs. Qualified resources, existing or new, are required to submit their capacity offers to the ISO, which, in turn, clears the FCM auction amounts and determines the clearing price. Local capacity zones may be modeled in the FCM to reflect local reliability needs. Capacity requirements are administratively set by the ISO through reliability simulations that are similar to other loss of load expectation resource adequacy studies.

With fixed capacity requirements, the FCM auction clearing prices may be volatile, that is, the market clearing price tends to substantially increase when there is any capacity shortage from the fixed requirement and plummet when there is a capacity surplus. Such price volatility increases the risk for capacity investment and load payment and impedes the formation of adequate price signals for long-term investment. Furthermore, such a market may be prone to market power. To remedy this issue, ISO-NE has replaced the fixed requirements with demand curves based on the trade-off between reliability and cost of capacity.

Interchange Scheduling and Market Coordination

The evolution of the power grid has led to today's large interconnected power systems, each operated by an ISO. An ISO has complete information of its own control area but may not have information from other ISOs, such as network model, generation offers, and system condition. Furthermore, ISOs may run electricity markets with different designs. As a result, the dispatch solutions of individual areas may not be optimal for the entire interconnected system, that is, economic efficiency could be impaired. Such a problem caused by the heterogeneous market designs and insufficient information sharing is termed the *seams issue*. To achieve higher economic efficiency, coordination among neighboring ISOs is needed. The purpose of such coordination is to drive each ISO's area dispatch solution toward the optimal solution of the entire region's dispatch problem.

The coordination between two markets can be performed through interchange transaction scheduling and generation redispatch. There are two market-based interchange scheduling approaches. One is scheduled by market participants

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by submitting bids to buy from one market and offers to sell to another market. Both buy bids and sell offers must be cleared in both markets to schedule the physical interchange flow. This scheduling process is generally conducted on an hourly basis. Another is the coordinated transaction scheduling (CTS) implemented by ISO-NE, PJM, and NYISO. In this approach, market participants submit CTS bids that represent their willingness to buy or sell energy based on the price difference between two markets. The CTS market commonly runs on fifteen-minute intervals to increase the scheduling efficiency. In addition to interchange scheduling, a market-to-market coordination scheme is also available for coordinated congestion management. Under this scheme, two markets exchange the cost of redispatch in their markets and relieve the congestion through dispatching more efficient generation.

Market Management System Components

The market management system (MMS) components are central to the operation of wholesale electricity markets in ISO control centers. The aforementioned market features are all implemented in this system. Different from financial market operation, electricity market operation, especially real time, requires constant interaction with the underlying physical power system, making the MMS an essential part of system control. A typical MMS is shown in Figure 5, where it interacts with the physical control system, such as the energy management system (EMS), and the financial settlement system. The physical system provides essential operating parameters, such as load forecast, system topology, and transfer limits to MMS, which determines generation schedules (a type of control signal) for the EMS and market clearing prices for the financial settlement system. MMS is complex but made up of several basic building blocks, for example, UC, ED, security analysis, and pricing.

Architecture

Typical electricity market systems produce generation schedules and clearing prices by solving an optimization problem. The common components in the optimization problem are UC, ED, security analysis, and pricing. The UC determines the integer on/off decisions for generation units. The preceding components can be combined to solve different problems. The security analysis, for example, is iterated with ED until there are no security violations. The components can be organized in different ways depending on the market requirements. Figure 6 shows an implementation of the SCUC with these components. Notice that the architecture has two loops. The inner loop is often termed *SCED*. The individual components are explained as follows.

UC

The UC component is an optimization problem that determines each unit's on/off status for each time interval by respecting the unit-level

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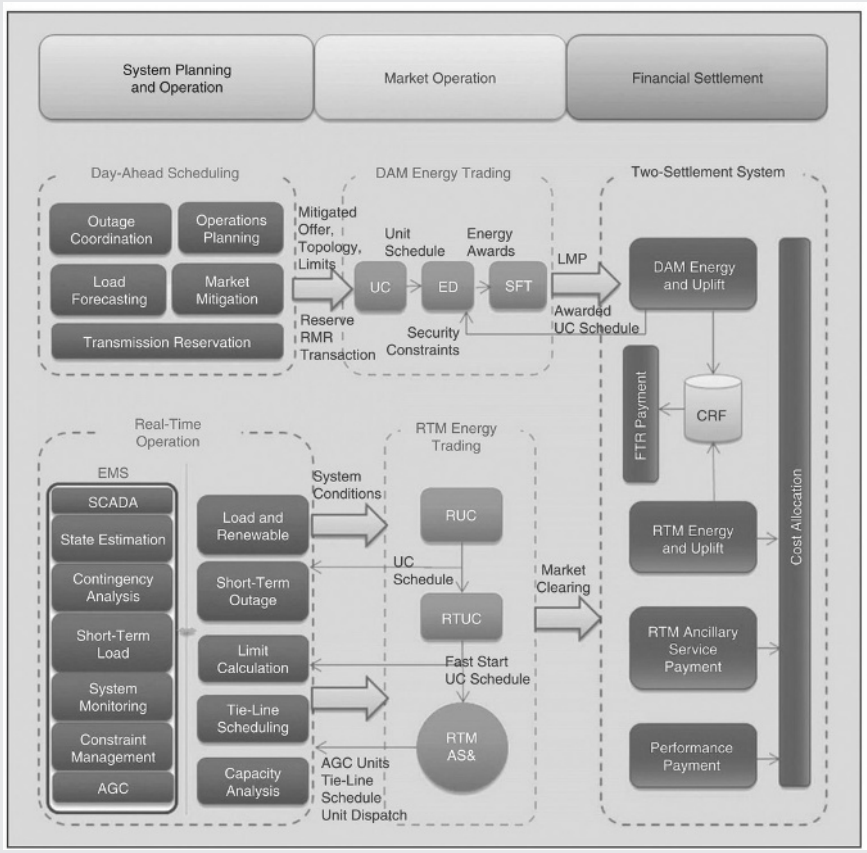

Figure 5 The MMS architecture. RTUC: real-time UC; AGC: automatic generation control; SCADA: supervisory control and data acquisition; CRF: congestion revenue fund; RMR: reliability must run units; SFT: simultaneous feasibility test

constraints, such as minimum up/down time, ramping limit, and system constraints, such as energy balance and security. Typically, the DAM involves a 24-hour or longer UC module for hourly on or off decisions. The UC problem is generally hard to solve because its integer solution space grows exponentially as the number of generation units increases. The most commonly used solution techniques for the UC are Lagrange relaxation and MILP. Enabled by the increasing computing power in recent years, the MILP solution that allows more modeling flexibility has become a dominant approach to UC. The MILP performance can be much affected by the problem formulation. Therefore, an efficient MILP formulation is very important. Studies on tightening UC formulations can be found in the literature. However, it is

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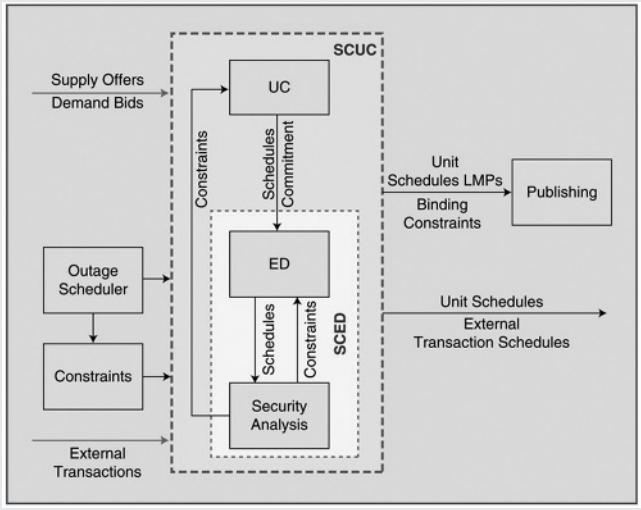

Figure 6 The SCUC architecture

generally considered that there is no universally good formulation for all UC problems due to the various factors that affect the path of a MILP solver.

ED

The ED component is an optimization problem that determines each unit's generation output, subject to unit and system constraints. The objective of the optimization problem is to minimize the total dispatch cost based on units' submitted cost curves. The UC statuses determined by the UC problem are fixed, so there are generally no integer decisions for the ED component. Depending on the offer curve characteristics, the ED problem can be formulated as a linear or quadratic program.

Security Analysis

The security analysis component examines the dispatch solution against N-1 contingencies. For each contingency, a power flow problem is solved to check and identify any network constraint violations. The violated constraints are linearized and fed back to the next iteration of the ED problem, forming the inner loop of SCED. If the SCED cannot resolve all security constraint violations, then the remaining violated constraints are fed into the next iteration of UC until all violations are resolved, forming the SCUC problem.

Security analysis could be time-consuming due to the large number of contingencies. Because the simulation of different contingencies is independent, parallel computing techniques are often adopted.

Pricing and Settlement

Whereas all of the preceding components are intended to obtain a secure commitment and dispatch

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solution for all units, the clearing price is critical for a good market design because it acts as the signal to the market participants. If properly priced, a unit will have no incentive to deviate from the ISO's schedule, which is a desired market design property that allows market participants to bid truthfully. As we will discuss next, pricing could be particularly challenging in the presence of market nonconvexity or multiple-interval dispatch.

With the market clearing prices, settlement is defined by a set of market rules of who pays or gets paid and by how much. The settlement component collects information from the MMS as well as EMS to calculate uplift payments and so-called settlement positions for each market product and time frame.

Electricity Pricing Challenges

Pricing is a critical part of a market because proper market prices provide the right signals to market participants for achieving an efficient market outcome. Pricing in electricity markets could be complex in the presence of transmission congestion and losses. Furthermore, nonconvexity in a market is known to cause problems for conventional linear prices. Unfortunately, nonconvexities are abundant in electricity markets as a result of physical unit characteristics. Pricing with nonconvexity has therefore become one of the central topics in electricity markets. A less discussed pricing problem, but attracting more interest recently due to the emerging ramping needs in several ISOs, is the appropriate pricing and settlement design with multiple scheduling intervals. This problem is further complicated by the practical implementation of multi-interval scheduling with rolling horizons. We will discuss the solution of a multisettlement system as an extension to the existing two-settlement system.

Nonconvexity

Wholesale electricity prices, or LMPs, are typically produced from the shadow prices of the SCED problem. Because the SCED problem is often convex, that is, being a linear program or a quadratic objective function with linear constraints, the constraint shadow prices exist and can be conveniently used to calculate market clearing prices. These prices are consistent with the optimal dispatch solutions in the sense that, given the prices, each unit will not have financial incentives to deviate from the dispatch results. In reality, however, a unit's decisions are also based on the commitment costs in addition to the incremental costs. The clearing prices from the convex dispatch problem reflect the marginal cost of dispatch but do not capture commitment costs and thus may not be consistent with the unit's commitment decisions. As a result of the fundamental mathematical properties of nonconvex problems, there's no such set of perfectly consistent prices for the UC problem. Consequently, for any given market clearing prices, some units may have incentives to deviate from their optimal schedule. Recognizing this nonexistence of perfect

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prices, in reality, markets are designed to pursue a second-best set of prices with certain criteria.

Most of the existing markets use the uplift payment as the additional compensation alongside the market clearing prices for units to follow the ISO's optimal commitment and dispatch instructions. The total uplift payments are then allocated to demands in the market. Different markets may have different uplift calculation and allocation rules. Most ISOs pay the make-whole payment for online units to cover the revenue shortage, although some ISOs (e.g., ISO-NE) also pay the so-called lost opportunity cost (LOC) for certain situations even when the unit has no revenue shortage. Nevertheless, the uplift payments cannot be efficiently allocated through a market mechanism and are often considered nontransparent. Many ISOs have explored ways to reduce the uplift payments by adopting alternative market clearing prices in place of existing LMPs. One approach, termed *convex-hull pricing*, uses the dual variable solutions of the nonconvex UC problem to arrive at the minimal level of total uplift and LOC (although the uplift payment may actually increase). Other approaches typically relax the nonconvex features of the problem, such as removing the minimal generation output parameter or allowing partial commitments of units for obtaining a price that at least relates to the commitment costs. A variety of different relaxations have been implemented by the ISOs; however, in theory, there is no perfect price that can eliminate the need for uplift payments due to the presence of the duality gap for nonconvex problems. The practical question remains: what are the objectives and desired properties of the pricing? Different ISOs may have different answers. The decision on the trade-off between the level of nonconvexity and appropriate set of prices is central to a good market design.

Pricing Multi-Interval Solutions

Given that UC and dispatch decisions are temporally coupled through ramp limits and minimum up/down time constraints, the market clearing prices for different time intervals are naturally related. Although the multi-interval dispatch problem can be convex, and the prices for each interval conveniently produced out of the dual variables, the challenge of pricing with multiple intervals lies in its real-time applications that run on the rolling horizon. NYISO and CAISO, for example, solve a multi-interval dispatch problem but only use the first interval prices for settlement; prices for other intervals are only advisory and will be determined when the interval becomes first in later runs. With the rolling horizon, each multi-interval dispatch run includes new time intervals that are not modeled in the previous intervals. As a result, the prices produced from different runs for the same interval could be different. The ISO's settlement typically uses the prices from the latest run where the interval being priced is the first. These prices for different intervals are from different runs, and so they may not

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be consistent with the dispatch solutions for different intervals. This may cause the incentive compatibility problem, that is, deviation from the ISO instructions, as the units chase the settlement prices.

A natural way to remedy this problem is to make all prices from different runs financially binding, that is, one interval would be settled multiple times instead of just for the last time. This effectively requires a multisettlement design where multiple levels of markets in addition to the existing DAM and RTM are involved. ISO-NE is developing a new approach to appropriately price ramping constraints.

New Market Initiatives

The existing market designs are shaped by the historical, physical, and regulatory features of the system as manifested by the difference in the designs of different ISO markets. Currently, the power industry is undergoing a major transformation toward a system with a lower carbon footprint, more decentra lized controls, and an increasingly flexible load. These changes bring fundamental challenges to power system operations, planning, and market design. In the following, we highlight the new challenges associated with the new electric grid, wrapping up with a discussion on future market design to accommodate the challenges.

A New Electric Grid

Modern power systems went through different stages of evolution driven by technical, economic, and regulatory events. They went from a decentralized, loosely coupled grid to large, highly interconnected, and centrally controlled systems. The sheer size, large number of components, and proliferation of power electronics have made the electric grid a complex cyberphysical system.

Currently, the power industry is facing yet another revolutionary change. Government directives to lower the carbon footprint and, as a consequence, high penetration of renewable energy resources and smart grid technologies are transforming power system planning and operational paradigms. Distributed energy resources (DERs) are being built deeply into the distribution networks, and the boundary between transmission and distribution is blurring. A significant part of DERs is unobservable to the transmission system operators, introducing an unprecedented level of uncertainty not only spatially but also temporally due to the intermittent nature of renewable resources. Even the concept of contingency is changing from being binary (e.g., on and off events) to continuous, e.g., the system load or generation can swing by several gigawatts in a comparatively short period of time. This behavior, previously considered as abnormal or emergency, becomes part of normal operation. As a result, system operators will be challenged to deal with completely different patterns in the new grid architecture, such as the one shown in Figure 7. Another property of large cyberphysical systems is high interdependence with different infrastructures. Not only do we have to monitor the

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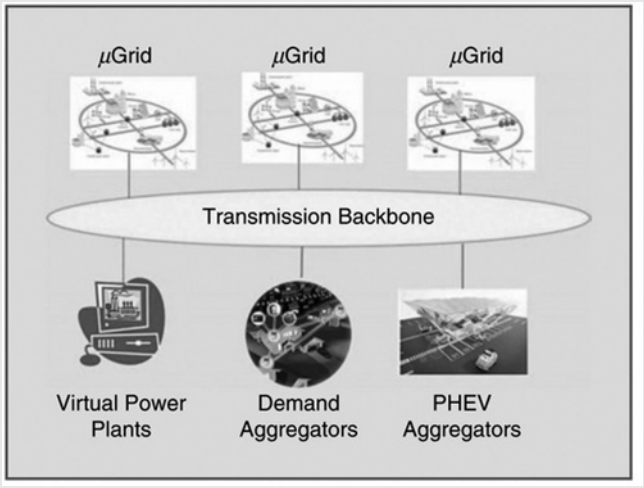

Figure 7 A new power grid. PHEV: plug-in hybrid electric vehicle

electric grid contingencies but also the failures in communication, information technology, and fuel delivery systems. Today's preventive approach to power systems planning and operation, that is, every contingency is treated with equal probability and no contingency is allowed to cause violation of the N-1 reliability criteria, becomes economically prohibitive in the new grid era. More corrective actions must be introduced to make power system planning and operation less expensive. This all requires a significantly different control paradigm, new models, and new algorithms. In response, the market design needs to be changed as well to accommodate these new changes in physical systems.

Future Market Design

Although the changes in the electric grid impose significant challenges to power system operation, planning, and control, adapting existing electricity market designs and developing new design solutions to address these challenges is an even more daunting task, given the relatively short history of electricity markets. In this section, we discuss several key market design problems that are likely to occur in the near future and suggest possible solutions.

The unprecedented level of uncertainty due to the high penetration of renewable resources and DERs requires more flexibility to deal with uncertainty. This leads to the need for a new flexibility market mechanism to create incentives for investment in flexible technologies. Some ISOs, such as CAISO, have introduced a five-minute ramping product to accommodate the steep change of the net load curve due to significant swings in photovoltaic production. The design of such a new product should consider its

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interaction with existing products, such as reserves. It is important that the new products provide unique and different services from the existing products so as to attract targeted flexible resources.

An alternative way of addressing these uncertainty challenges is to manage them through more sophisticated commitment and dispatch processes. Instead of the existing deterministic commitment and dispatch problems currently solved by the ISOs, stochastic programming or robust optimization that explicitly models the uncertainty within their formulation through scenario tree and uncertainty set seem to be promising. However, a bigger issue is how to appropriately price the market clearing decision and compensate the selected resources based on multiple possible realizations of uncertain parameters. From the market design perspective, transparent compensation schemes, proper information policy, and incentive compatibility issues should be resolved before we adopt these advanced risk management techniques.

Direct demand participation (where demand bids are submitted to the market) in the RTMs can bring significant benefits to system operation. It enhances system reliability and also improves the price formation in the electricity market. However, an increasing level of indirect demand participation (demand responds to the price on its own) can create challenges in load forecasting, system control, and, thus, real-time pricing. To overcome these issues, more and more ISOs start incorporating ISO-developed demand curves into the market. The Electric Reliability Council of Texas, for example, has applied the operating reserve demand curve to its real-time reserve market. ISO-NE has recently developed system and zonal capacity demand curves for its FCM based on tradeoffs between the costs of reliability and capacity. The development of these demand curves based on different reliability needs is a promising direction of the new market design.

The electric industry is moving into a new era with new carbon emission limits, new resilience needs, and the zero marginal cost economy where the incremental cost of producing a good approaches zero. All of these new features bring significant challenges for electricity pricing. With the carbon limits and resilience needs, prices for these features need to be defined for the market. The existing marginal cost pricing may lead to zero or negative market clearing prices, when more and more renewable resources with zero marginal costs and state/federal subsidies participate in the market. Therefore, the marginal pricing principle may need to be augmented with some new constructs, including new scarcity pricing rules.

Acknowledgment

The views expressed in this article do not represent those of ISO New England, Inc.

For Further Reading

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7.1 GENERATOR AND LOAD ECONOMICS

The technical problems of power flow from the last chapter are inevitably intertwined with economic problems. How much does it cost to produce electricity, and what price are people and industries willing to pay for it? Everyone benefits when a power system is operated with optimal economic efficiency—and even a 1% improvement in operations can add billions of dollars of surplus over time, which will lead to lower costs for consumers and incentives for investment in transmission infrastructure and new generation. This chapter is about designing and analyzing a power system for smooth economic operation.

Electric energy has near-zero shelf life. Energy storage—a collection of technologies, such as pumped hydroelectric, compressed air, batteries, and flywheels—is for now only a small portion of the grid's total energy throughput. Most studies assume electric energy must be consumed the moment it is produced. On the other hand, generators can take months or years to build, hours or days to start up, and several minutes to increase or decrease their energy production (a process known

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as *ramping up* or *ramping down*). The varying time scales that are relevant in grid economic planning are part of the challenge of this field.

Generator costs have two main components. There is a *fixed* cost to build and maintain the unit and a *variable* cost that depends on the amount of energy being produced. Technologies that have a low variable cost—even if the fixed cost is high—incentivize dispatching these units to their maximum capacity. The *variable* cost, for many generators, consists of the *fuel cost* divided by a turbine-dependent function known as the incremental heat rate that measures fuel input needed for a unit of electric output. Generators with a lower heat rate are more efficient; they take less fuel to produce a MWh of energy.

The variety of technologies used in today's generating fleet have diverse economic characteristics. Nuclear power plants have the longest construction time, and highest fixed costs. But the variable costs are low. Normally, nuclear units are dispatched to their maximum capacity for more than 90% of the year.

Power generated by burning fossil fuels—coal, natural gas, or petroleum—is characterized by a large variable cost component. A cost curve as a function of power being generated, $C(P)$, is formulated by either a polynomial or, more commonly, a piecewise-linear function. These costs are proportional to the associated fuel costs. Lower construction costs, fuel costs, and the short construction timespan have, in recent years, incentivized investment in natural gas generation. Coal and petroleum generation have higher costs and comprise a decreasing share of the generating fleet in many places.

Hydroelectric power has no variable fuel cost, meaning that the overall cost is usually low. But it does have complex constraints that are relevant over multi-hour scheduling, considering reservoir level needs, water cycles, ecological requirements, and coordination with other facilities upstream and downstream. There is only a limited amount of water that can be passed through the turbines over time. The result is that hydroelectric units are commonly used during peak load times, when there is the greatest need.

Wind and solar generation are growing rapidly due to modest fixed capital costs and quite low operating and variable costs. These units have a variable maximum capacity. Forecasting the wind and solar energy that will be available at any time period is an essential initial precursor step to determining the optimal system operation. Normally, these types of units are dispatched to their maximum available capacity.

While the majority of generation comes from large-scale transmission-connected generating units, some generation comes from small-scale generation, which is distributed among customers at medium- and low-voltage connection points. The costs and optimization of these units are often not considered in system analysis—many of them are solar units, dispatched to their full capacity, which can sometimes be treated as a negative load.

Economic load modeling involves evaluating the price customers are willing to pay to consume electric energy during a time period. The *sensitivity* or *elasticity* of demand measures how much additional power would be consumed if the price were lower. In the long term, customers will switch between electric and non-electric energy sources for some tasks if prices are too high or too low. Some

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industrial customers, and increasingly some residential and commercial customers, have the ability to pay a more variable price for their electric consumption in the short term. Certain tasks (such as data center calculations, manufacturing processes, or home clothes-drying) are not crucially time-sensitive, and so access to a variable price allows performance of these tasks while prices are low. For many customers, however, there is low short-term price elasticity because they often see only an averaged price over a period of months and years. A first-order economic model is to assume a constant, zero elasticity demand curve and dispatch the generation to match this load. More realistic models will include the price sensitivity that some loads will show.

Energy storage, mentioned before, consists of a number of technologies that allow the system to store a small amount of electric energy during off-peak hours and put it back on the system when it is needed at peak hours. Hence, the advantage of increased energy storage is *peak shaving*.

Economic analysis and planning is usually done on an area basis. Thus, another factor is the import or export to neighboring areas through *area transactions*. In some time periods, an area may be seeking to sell excess generation at a low price through ac or dc interconnections to neighboring systems, and in other time periods, the neighboring systems may be providing power to help meet demand. These potential *imports* and *exports* should be considered when planning in conjunction with local area generation.

The electric transmission and distribution systems also should be taken into consideration. First, there are transmission and distribution losses, which add to the demand when calculating the total power that must be generated or imported. Second, the losses can be dependent on which generators are operated. If local generation near the load is used, the losses may be lower compared to importing power from generators hundreds of kilometers away. Finally, there are a number of transmission-related constraints, such as transmission line and transformer limits, voltage limits, and reactive power balance, which must be satisfied. Sometimes, the most economic dispatch of generation and load does not correspond to a feasible operating point.

The rest of this chapter explores key calculations involved in designing and planning for power system economics. In Section 7.2, we examine the economic dispatch problem, which determines how to dispatch a fleet of generating units to meet demand. In Section 7.3, we consider optimal power flow and security constrained optimal power flow (OPF and SCOPF), which integrate power flow with economic considerations and seek to optimize grid planning with transmission constraints. Section 7.4 briefly discusses the unit commitment problem to schedule generators over a longer time span. In Section 7.5, we discuss electric energy markets, which is how many of these concepts are applied in actual systems.

7.2 ECONOMIC DISPATCH

This section describes how the real power output of a controlled generating unit is selected to meet a given load and to minimize the total operating costs. This is the *economic dispatch* problem [1]. In interconnected power systems,

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economic dispatch is often solved for areas, in which the total generation in each area is controlled to match the total area load; further details are provided in Chapter 13.

The economic dispatch problem is first solved for the idealized case, in which generators are assumed to have a given cost curve with no minimum or maximum limits or effect on losses. Then it is expanded to include inequality constraints on generator outputs and to consider the impact of transmission losses.

UNCONSTRAINED ECONOMIC DISPATCH

Figure 7.1 shows the cost C_i of a generating unit versus its real power output P_i. Fuel cost is the major portion of the variable cost of operation, although other variable costs, such as maintenance, could have been included in the figure. Fixed costs, such as the capital cost of installing the unit, are not included. Only those costs that are a function of unit power output—that is, those costs that can be controlled by operating strategy—enter into the economic dispatch formulation.

In practice, C_i is constructed of piecewise continuous functions valid for ranges of output P_i based on empirical data. The discontinuities in Figure 7.1 may be due to the firing of equipment, such as additional boilers or condensers, as power output is increased. It is often convenient to express C_i in terms of BTU/hr, which is relatively constant over the lifetime of the unit, rather than \$/hr, which can change monthly or daily. C_i can be converted to \$/hr by multiplying the fuel input in BTU/hr by the cost of fuel in \$/BTU.

Figure 7.2 shows the unit incremental operating cost dC_i/dP_i versus unit output P_i, which is the slope or derivative of the C_i versus P_i curve in Figure 7.1. When C_i consists of only fuel costs, dC_i/dP_i is the ratio of the incremental fuel energy input in BTU to incremental energy output in kWh, which is called the incremental *heat rate.* Note that the reciprocal of the heat rate, which is the ratio of output energy to input energy, gives a measure of fuel efficiency for the unit. For the unit shown in Figure 7.1, maximum efficiency occurs at $P_i = 600$ MW, where

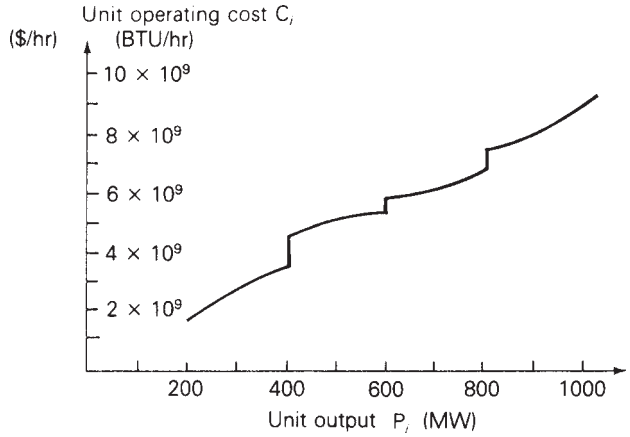

FIGURE 7.1

Unit operating cost versus real power output

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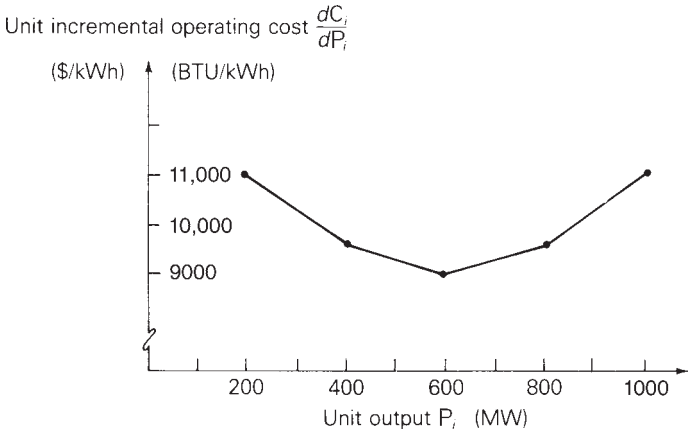
FIGURE 7.2

Unit incremental operating cost versus real power output



the heat rate is $C_i/P_i = 5.4 \times 10^9/600 \times 10^3 = 9000$ BTU/kWh. The efficiency at this output is

$$\text{percentage efficency} = \left(\frac{1}{9000}\ \frac{\text{kWh}}{\text{BTU}}\right)\left(3413\ \frac{\text{BTU}}{\text{kWh}}\right) \times 100 = 37.92\%$$

The dC_i/dP_i curve in Figure 7.2 is also represented by piecewise continuous functions valid for ranges of output P_i. For analytical work, the actual curves are often approximated by straight lines. The ratio dC_i/dP_i can also be converted to \$/kWh by multiplying the incremental heat rate in BTU/kWh by the cost of fuel in \$/BTU.

For the area of an interconnected power system consisting of N units operating on economic dispatch, the total variable cost C_T of operating these units is

$$\begin{aligned} C_T &= \sum_{i=i}^{N} C_i \\ &= C_1(P_1) + C_2(P_2) + \cdots + C_N(P_N) \quad \$/\text{hr} \end{aligned} \tag{7.1.1}$$

where C_i, expressed in \$/hr, includes fuel cost as well as any other variable costs of unit *i*. Let P_T equal the total load demand in the area. Neglecting transmission losses,

$$P_1 + P_2 + \cdots + P_N = P_T \tag{7.1.2}$$

Due to relatively slow changes in load demand, P_T may be considered constant for periods of 2 to 10 minutes. The economic dispatch problem can be stated as follows:

Find the values of unit outputs $P_1, P_2, \ldots, P_N$ that minimize C_T given by (7.1.1), subject to the equality constraint given by (7.1.2).

A criterion for the solution to this problem is: All units on economic dispatch should operate at equal incremental operating cost. That is,

$$\frac{dC_1}{dP_1} = \frac{dC_2}{dP_2} = \cdots = \frac{dC_N}{dP_N} \tag{7.1.3}$$

An intuitive explanation of this criterion is the following. Suppose one unit is operating at a higher incremental operating cost than the other units. If the output power of that unit is reduced and transferred to units with lower incremental

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operating costs, then the total operating cost C_T decreases. That is, reducing the output of the unit with the *higher* incremental cost results in a *greater cost decrease* than the cost increase of adding that same output reduction to units with lower incremental costs. Therefore, all units must operate at the same incremental operating cost (the economic dispatch criterion).

A mathematical solution to the economic dispatch problem also can be given. The minimum value of C_T occurs when the total differential dC_T is zero. That is,

$$dC_T = \frac{\partial C_T}{\partial P_1} dP_1 + \frac{\partial C_T}{\partial P_2} dP_2 + \cdots + \frac{\partial C_T}{\partial P_N} dP_N = 0 \tag{7.1.4}$$

Using (7.1.1), (7.1.4) becomes

$$dC_T = \frac{dC_1}{dP_1} dP_1 + \frac{dC_2}{dP_2} dP_2 + \cdots + \frac{dC_N}{dP_N} dP_N = 0 \tag{7.1.5}$$

Also, assuming P_T is constant, the differential of (7.1.2) is

$$dP_1 + dP_2 + \cdots + dP_N = 0 \tag{7.1.6}$$

Multiplying (7.1.6) by λ and subtracting the resulting equation from (7.1.5),

$$\left(\frac{dC_1}{dP_1} - \lambda\right) dP_1 + \left(\frac{dC_2}{dP_2} - \lambda\right) dP_2 + \cdots + \left(\frac{dC_N}{dP_N} - \lambda\right) dP_N = 0 \tag{7.1.7}$$

Equation (7.1.7) is satisfied when each term in parentheses equals zero. That is,

$$\frac{dC_1}{dP_1} = \frac{dC_2}{dP_2} = \cdots = \frac{dC_N}{dP_N} = \lambda \tag{7.1.8}$$

Therefore, all units have the same incremental operating cost, denoted here by λ, in order to minimize the total operating cost C_T.

EXAMPLE 7.1

Economic dispatch solution neglecting generator limits and line losses

An area of an interconnected power system has two fossil-fuel units operating on economic dispatch. The variable operating costs of these units are given by

$C_1 = 10P_1 + 8 \times 10^{-3}P_1^2$ \$/hr

$C_2 = 8P_2 + 9 \times 10^{-3}P_2^2$ \$/hr

where P_1 and P_2 are in megawatts. Determine the power output of each unit, the incremental operating cost, and the total operating cost C_T that minimizes C_T as the total load demand P_T varies from 500 to 1500 MW. Generating unit inequality constraints and transmission losses are neglected.

(*Continued*)

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SOLUTION

The incremental operating costs of the units are

$$\frac{dC_1}{dP_1} = 10 + 16 \times 10^{-3}P_1 \quad \$/MWh$$

$$\frac{dC_2}{dP_2} = 8 + 18 \times 10^{-3}P_2 \quad \$/MWh$$

Using (7.1.8), the minimum total operating cost occurs when

$$\frac{dC_1}{dP_1} = 10 + 16 \times 10^{-3}P_1 = \frac{dC_2}{dP_2} = 8 + 18 \times 10^{-3}P_2$$

Using $P_2 = P_T - P_1$, the preceding equation becomes

$$10 + 16 \times 10^{-3}P_1 = 8 + 18 \times 10^{-3}(P_T - P_1)$$

Solving for P_1,

$$P_1 = \frac{18 \times 10^{-3}P_T - 2}{34 \times 10^{-3}} = 0.5294P_T - 58.82 \quad MW$$

Also, the incremental operating cost when C_T is minimized is

$$\frac{dC_2}{dP_2} = \frac{dC_1}{dP_1} = 10 + 16 \times 10^{-3}P_1 = 10 + 16 \times 10^{-3}(0.5294P_T - 58.82)$$
$$= 9.0589 + 8.4704 \times 10^{-3}P_T \ \$/MWh$$

and the minimum total operating cost is

$$C_T = C_1 + C_2 = (10P_1 + 8 \times 10^{-3}P_1^2) + (8P_2 + 9 \times 10^{-3}P_2^2) \quad \$/hr$$

The economic dispatch solution is shown in Table 7.1 for values of P_T from 500 to 1500 MW.

TABLE 7.1
Economic dispatch solution for Example 7.1

P_T MW	P_1 MW	P_2 MW	dC_1/dP_1 \$/MWh	C_T \$/hr
500	206	294	13.29	5529
600	259	341	14.14	6901
700	312	388	14.99	8358
800	365	435	15.84	9899
900	418	482	16.68	11,525
1000	471	529	17.53	13,235
1100	524	576	18.38	15,030
1200	576	624	19.22	16,910
1300	629	671	20.07	18,875
1400	682	718	20.92	20,924
1500	735	765	21.76	23,058

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EFFECT OF INEQUALITY CONSTRAINTS

Each generating unit must not operate above its rating or below some minimum value. That is,

$$P_{i\min} < P_i < P_{i\max} \qquad i = 1, 2, \ldots, N \tag{7.1.9}$$

Other inequality constraints also may be included in the economic dispatch problem. For example, some unit outputs may be restricted so that certain transmission lines or other equipment are not overloaded. Also, under adverse weather conditions, generation at some units may be limited to reduce emissions.

When inequality constraints are included, modify the economic dispatch solution as follows. If one or more units reach their limit values, then these units are held at their limits, and the remaining units operate at equal incremental operating cost λ. The incremental operating cost of the area equals the common λ for the units that are not at their limits.

EXAMPLE 7.2

Economic dispatch solution including generator limits

Rework Example 7.1 if the units are subject to the following inequality constraints:

$100 \leqq P_1 \leqq 600$ MW

$400 \leqq P_2 \leqq 1000$ MW

SOLUTION

At light loads, unit 2 operates at its lower limit of 400 MW, where its incremental operating cost is $dC_2/dP_2 = 15.2$ \$/MWh. Additional load comes from unit 1 until $dC_1/dP_1 = 15.2$ \$/MWh, or

$$\frac{dC_1}{dP_1} = 10 + 16 \times 10^{-3} P_1 = 15.2$$

$$P_1 = 325 \text{ MW}$$

For P_T less than 725 MW, where P_1 is less than 325 MW, the incremental operating cost of the area is determined by unit 1 alone.

At heavy loads, unit 1 operates at its upper limit of 600 MW, where its incremental operating cost is $dC_1/dP_1 = 19.60$ \$/MWh. Additional load comes from unit 2 for all values of dC_2/dP_2 greater than 19.60 \$/MWh. At $dC_2/dP_2 = 19.60$ \$/MWh,

$$\frac{dC_2}{dP_2} = 8 + 18 \times 10^{-3} P_2 = 19.60$$

$$P_2 = 644 \text{ MW}$$

(*Continued*)

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For P_T greater than 1244 MW, where P_2 is greater than 644 MW, the incremental operating cost of the area is determined by unit 2 alone.

For $725 < P_T < 1244$ MW, neither unit has reached a limit value, and the economic dispatch solution is the same as that given in Table 7.1.

The solution to this example is summarized in Table 7.2 for values of P_T from 500 to 1500 MW.

P_T MW	P_1 MW	P_2 MW	dC/dP \$/MWh	C_T \$/hr
500	100	400	11.60	5720
600	200	400	13.20	6960
700	300	400	14.80	8360
725	325	400	15.20	8735
800	365	435	15.84	9899
900	418	482	16.68	11,525
1000	471	529	17.53	13,235
1100	524	576	18.38	15,030
1200	576	624	19.22	16,910
1244	600	644	19.60	17,765
1300	600	700	20.60	18,890
1400	600	800	22.40	21,040
1500	600	900	24.20	23,370

$\frac{dC_1}{dP_1}$ (rows 500–725) $\qquad$ $\frac{dC_2}{dP_2}$ (rows 1244–1500)

TABLE 7.2
Economic dispatch solution for Example 7.2

EXAMPLE 7.3

PowerWorld Simulator—economic dispatch, including generator limits

PowerWorld Simulator case Example 7_3 uses a five-bus, three-generator lossless case to show the interaction between economic dispatch and the transmission system (see Figure 7.3). The variable operating costs for each of the units are given by

$$C_1 = 10P_1 + 0.016P_1^2 \quad \$/hr$$
$$C_2 = 8P_2 + 0.018P_2^2 \quad \$/hr$$
$$C_4 = 12P_4 + 0.018P_4^2 \quad \$/hr$$

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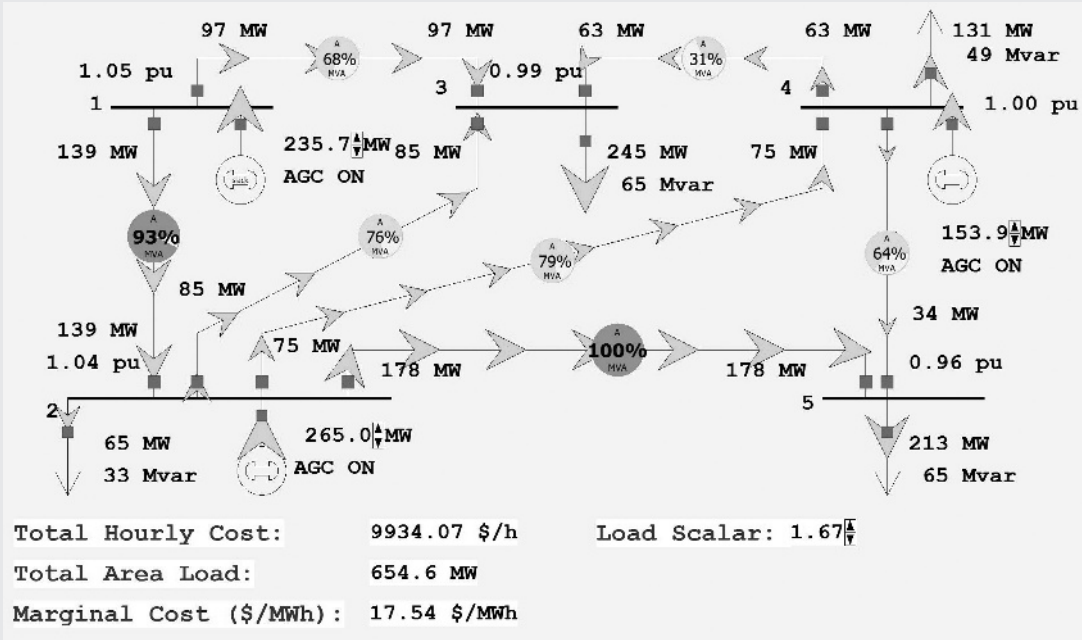

FIGURE 7.3

Example 7.3 with maximum economic loading

where P_1, P_2, and P_4 are the generator outputs in megawatts. Each generator has minimum/maximum limits of

$$100 \leq P_1 \leq 400 \text{ MW}$$

$$150 \leq P_2 \leq 500 \text{ MW}$$

$$50 \leq P_4 \leq 300 \text{ MW}$$

In addition to solving the power flow equations, PowerWorld Simulator can simultaneously solve the economic dispatch problem to optimally allocate the generation in an area. To turn on this option, select **Case Information, Aggregation, Areas…** to view a list of each of the control areas in a case (just one in this example). Then toggle the AGC Status field to ED. Now anytime the power flow equations are solved, the generator outputs are also changed using the economic dispatch.

Initially, the case has a total load of 392 MW with an economic dispatch of $P_1 = 141$ MW, $P_2 = 181$ MW, and $P_4 = 70$ MW, and an incremental operating cost, λ, of 14.52 \$/MWh. To view a graph showing the incremental cost curves for all of the area generators, right-click on any generator to display the

(*Continued*)

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generator's local menu, and then select All Area Gen IC Curves (right-click on the graph's axes to change their scaling).

To see how changing the load impacts the economic dispatch and power flow solutions, first select **Tools, Play** to begin the simulation. Then, on the oneline, click on the up/down arrows next to the Load Scalar field. This field is used to scale the load at each bus in the system. Notice that the change in the Total Hourly Cost field is well approximated by the change in the load multiplied by the incremental operating cost.

Determine the maximum amount of load this system can supply without overloading any transmission line with the generators dispatched using economic dispatch.

SOLUTION

The maximum system economic loading is determined numerically to be 655 MW (which occurs with a Load Scalar of 1.67) with the line from bus 2 to bus 5 being the critical element.

EFFECT OF TRANSMISSION LOSSES

Although one unit may be very efficient with a low incremental operating cost, it also may be located far from the load center. The transmission losses associated with this unit may be so high that the economic dispatch solution requires the unit to decrease its output, while other units with higher incremental operating costs but lower transmission losses increase their outputs.

When transmission losses are included in the economic dispatch problem, (7.1.2) becomes

$$P_1 + P_2 + \cdots + P_N - P_L = P_T \tag{7.1.10}$$

where P_T is the total load demand and P_L is the total transmission loss in the area. In general, P_L is not constant but depends on the unit outputs $P_1, P_2, \ldots, P_N$, The total differential of (7.1.10) is

$$(dP_1 + dP_2 + \cdots + dP_N) - \left(\frac{\partial P_L}{\partial P_1}dP_1 + \frac{\partial P_L}{\partial P_2}dP_2 + \cdots + \frac{\partial P_L}{\partial P_N}dP_N\right) = 0 \tag{7.1.11}$$

Multiplying (7.1.11) by λ and subtracting the resulting equation from (7.1.5),

$$\left(\frac{dC_1}{dP_1} + \lambda\frac{\partial P_L}{\partial P_1} - \lambda\right)dP_1 + \left(\frac{dC_2}{dP_2} + \lambda\frac{\partial P_L}{\partial P_2} - \lambda\right)dP_2 + \cdots + \left(\frac{dC_N}{dP_N} + \lambda\frac{\partial P_L}{\partial P_N} - \lambda\right)dP_N = 0 \tag{7.1.12}$$

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Equation (7.1.12) is satisfied when each term in parentheses equals zero. That is,

$$\frac{dC_i}{dP_i} + \lambda \frac{\partial P_L}{\partial P_i} - \lambda = 0$$

or

$$\lambda = \frac{dC_i}{dP_i}(L_i) = \frac{dC_i}{dP_i}\left(\frac{1}{1 - \dfrac{\partial P_L}{\partial P_i}}\right) \quad i = 1, 2, \ldots, N \tag{7.1.13}$$

Equation (7.1.13) gives the economic dispatch criteria, including transmission losses. Each unit that is not at a limit value operates such that its incremental operating cost dC_i/dP_i multiplied by the *penalty factor* L_i is the same. Note that when transmission losses are negligible, $\partial P_L/\partial P_i = 0$, $L_i = 1$, and (7.1.13) reduces to (7.1.8).

EXAMPLE 7.4

Economic dispatch solution including generator limits and line losses

Total transmission losses for the power system area given in Example 7.1 are given by

$$P_L = 1.5 \times 10^{-4}P_1^2 + 2 \times 10^{-5}P_1P_2 + 3 \times 10^{-5}P_2^2 \text{ MW}$$

where P_1 and P_2 are given in megawatts. Determine the output of each unit, total transmission losses, total load demand, and total operating cost C_T when the area $\lambda = 16.00$ \$/MWh.

SOLUTION

Using the incremental operating costs from Example 7.1 in (7.1.13),

$$\frac{dC_1}{dP_1}\left(\frac{1}{1 - \dfrac{\partial P_L}{\partial P_1}}\right) = \frac{10 + 16 \times 10^{-3}P_1}{1 - (3 \times 10^{-4}P_1 + 2 \times 10^{-5}P_2)} = 16.00$$

$$\frac{dC_2}{dP_2}\left(\frac{1}{1 - \dfrac{\partial P_L}{\partial P_2}}\right) = \frac{8 + 18 \times 10^{-3}P_2}{1 - (6 \times 10^{-5}P_2 + 2 \times 10^{-5}P_1)} = 16.00$$

Rearranging the two equations,

$$20.8 \times 10^{-3}P_1 + 32 \times 10^{-5}P_2 = 6.00$$
$$32 \times 10^{-5}P_1 + 18.96 \times 10^{-3}P_2 = 8.00$$

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Solving,

$P_1 = 282 \quad MW \qquad\qquad P_2 = 417 \quad MW$

Using the equation for total transmission losses,

$$P_L = 1.5 \times 10^{-4}(282)^2 + 2 \times 10^{-5}(282)(417) + 3 \times 10^{-5}(417)^2 = 19.5 \quad MW$$

From (7.1.10), the total load demand is

$$P_T = P_1 + P_2 - P_L = 282 + 417 - 19.5 = 679.5 \quad MW$$

Also, using the cost formulas given in Example 7.1, the total operating cost is

$$C_T = C_1 + C_2 = 10(282) + 8 \times 10^{-3}(282)^2 + 8(417) + 9 \times 10^{-3}(417)^2$$
$$= 8357 \ \$/h$$

Note that when transmission losses are included, λ given by (7.1.13) is no longer the incremental operating cost of the area. Instead, λ is the unit incremental operating cost dC_i/dP_i multiplied by the unit penalty factor L_i.

EXAMPLE 7.5

PowerWorld Simulator—economic dispatch, including generator limits and line losses

This example repeats the Example 7.3 power system, except that now losses are included with each transmission line modeled with an R/X ratio of 1/3 (see Figure 7.4). The current value of each generator's loss sensitivity, $\partial P_L/\partial P_i$, is shown immediately below the generator's MW output field. Calculate the penalty factors L_i, and verify that the economic dispatch shown in the figure is optimal. Assume a Load Scalar of 1.0.

SOLUTION

From (7.1.13), the condition for optimal dispatch is

$$\lambda = dC_i/dP_i(1/(1 - \partial P_L/\partial P_i) = dC_i/dP_i L_i \qquad i = 1, 2, \ldots, N$$

with

$$L_i = 1/(1 - \partial P_L/\partial P_i)$$

Therefore, $L_1 = 1.0$, $L_2 = 0.9733$, and $L_4 = 0.9238$,

with $P_1 = 130.1$ MW, $\quad dC_1/dP_1 \cdot L_1 = (10 + 0.032 \cdot 130.1) \cdot 1.0$

$= 14.16 \ \$/MWh$

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With $P_2 = 181.8$ MW, $dC_2/dP_2 \cdot L_2 = (8 + 0.036 \cdot 181.8) \cdot 0.9733$
$= 14.16$ \$/MWh

With $P_4 = 92.4$ MW, $dC_4/dP_4 \cdot L_4 = (12 + 0.036 \cdot 92.4) \cdot 0.9238$
$= 14.16$ \$/MWh

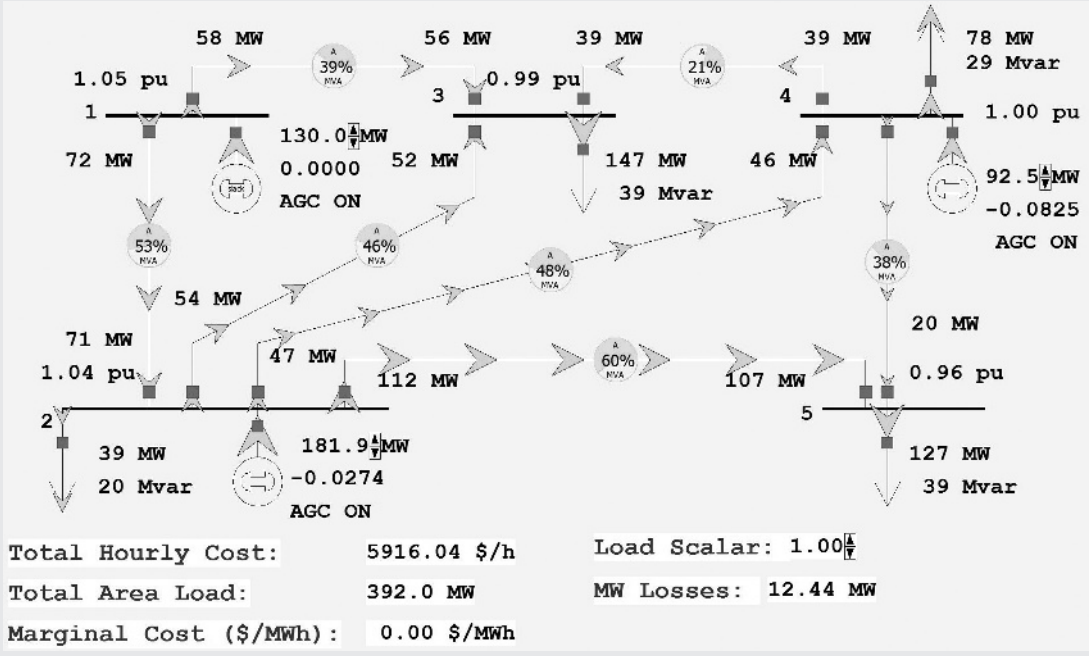

FIGURE 7.4

Example 7.5 five-bus case with transmission line losses

In Example 7.4, total transmission losses are expressed as a quadratic function of unit output powers. For an area with N units, this formula generalizes to

$$P_L = \sum_{i=1}^{N} \sum_{j=1}^{N} P_i B_{ij} P_j \tag{7.1.14}$$

where the B_{ij} terms are called *loss coefficients* or B *coefficients.* The B coefficients are not truly constant but vary with unit loadings. However, the B coefficients are often assumed constant in practice since the calculation of $\partial P_L/\partial P_i$ is thereby simplified. Using (7.1.14),

$$\frac{\partial P_L}{\partial P_i} = 2 \sum_{j=1}^{N} B_{ij} P_j \tag{7.1.15}$$

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This equation can be used to compute the penalty factor L_i, in (7.1.13).

Various methods of evaluating B coefficients from power flow studies are available [3]. In practice, more than one set of B coefficients may be used during the daily load cycle.

When the unit incremental cost curves are linear, an analytic solution to the economic dispatch problem is possible, as illustrated by Examples 7.1 through 7.3. However, in practice, the incremental cost curves are nonlinear and contain discontinuities. In this case, an iterative solution can be obtained. Given the load demand P_T, the unit incremental cost curves, generator limits, and B coefficients, such an iterative solution can be obtained by the following nine steps. Assume that the incremental cost curves are stored in tabular form, such that a unique value of P_i can be read for each dC_i/dP_i.

STEP 1 Set iteration index $m = 1$.

STEP 2 Estimate mth value of λ.

STEP 3 Skip this step for all $m > 1$. Determine initial unit outputs P_i, ($i = 1, 2, \ldots, N$). Use $dC_i/dP_i = \lambda$ and read P_i from each incremental operating cost table. Transmission losses are neglected here.

STEP 4 Compute $\partial P_L/\partial P_i$ from (7.1.15) ($i = 1, 2, \ldots, N$).

STEP 5 Compute dC_i/dP_i from (7.1.13) ($i = 1, 2, \ldots, N$).

STEP 6 Determine updated values of unit output P_i ($i = 1, 2, \ldots, N$). Read P_i from each incremental operating cost table. If P_i exceeds a limit value, set P_i to the limit value.

STEP 7 Compare P_i determined in Step 6 with the previous value ($i = 1, 2, \ldots, N$). If the change in each unit output is less than a specified tolerance ε_1, go to Step 8. Otherwise, return to Step 4.

STEP 8 Compute P_L from (7.1.14).

STEP 9 If $\left|\left(\sum_{i=1}^{N} P_i\right) - P_L - P_T\right|$ is less than a specified tolerance ε_2, stop. Otherwise, set $m = m + 1$ and return to Step 2.

Instead of having their values stored in tabular form for this procedure, the incremental cost curves instead could be represented by nonlinear functions such as polynomials. Then, in Step 3 and Step 5, each unit output P_i would be computed from the nonlinear functions instead of being read from a table. Note that this procedure assumes that the total load demand P_T is constant. In practice, this economic dispatch program is executed every few minutes with updated values of P_T.

OTHER ECONOMIC DISPATCH CONSIDERATIONS

There are several other considerations when applying economic dispatch to a large system containing a mix of different types of units.

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Wind and solar generation, which have no fuel costs, are represented with very low or negative cost (with the negative costs due in part to production tax credits in some countries). As such, they are preferred sources for economic dispatch and are used by system operators whenever possible, unless there are generator operating limits or transmission constraints.

Although the fixed costs of a nuclear unit may be high, their operating costs are low due to inexpensive nuclear fuel. As such, nuclear units are normally also represented with low cost, and as a result, are dispatched to the maximum capacity possible.

Pumped-storage hydro is a form of energy storage. During off-peak hours, these units are operated as synchronous motors to pump water to a higher elevation. Then during peak-load hours the water is released, and the units are operated as synchronous generators to supply power. As such, pumped-storage hydro units are used for light-load build-up and peak-load shaving. Economic operation of the area is improved by pumping during off-peak hours when the area λ is low, and by generating during peak-load hours when λ is high. Techniques are available for incorporating pumped-storage hydro units into economic dispatch of fossil-fuel units [6].

In an area consisting of hydro plants located along a river, the objective is to maximize the value of energy generated over the yearly water cycle rather than to minimize total operating costs. Reservoirs are used to store water during high-water or light-load periods, although some water may have to be released through spillways. Also, there are constraints on water levels due to river transportation, irrigation, or fishing requirements. Optimal strategies are available for coordinating outputs of plants along a river [7]. Economic dispatch strategies for mixed fossil-fuel/hydro systems are also available [2, 4, 8].

Techniques are also available for including reactive power flows in the economic dispatch formulation, whereby both active and reactive powers are selected to minimize total operating costs. In particular, reactive injections from generators, switched capacitor banks, and static var systems, along with transformer tap settings, can be selected to minimize transmission-line losses [4, 5]. However, electric utility companies usually control reactive power locally. That is, the reactive power output of each generator is selected to control the generator terminal voltage, and the reactive power output of each capacitor bank or static var system located at a power system bus is selected to control the voltage magnitude at that bus. In this way, the reactive power flows on transmission lines are low, and the need for central dispatch of reactive power is eliminated.

7.3 OPTIMAL POWER FLOW

Economic dispatch has one significant shortcoming—it ignores the limits imposed by the devices in the transmission system. Each transmission line and transformer has a limit on the amount of power that can be transmitted through it, with the limits arising because of thermal, voltage, or stability considerations (Section 5.6). Traditionally, the transmission system was designed so that when the generation was dispatched economically, there would be no limit violations.

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Hence, just solving economic dispatch was usually sufficient. However, with the worldwide trend toward deregulation of the electric utility industry, transmission systems have suffered from constraints associated with active transmission line and transformer limit violations (with these constraints sometimes called congestion). Recently, many organizations that oversee electric utility markets have been successfully reducing congestion costs. For example, in the PJM power market in the eastern United States, total congestion costs decreased from $1.39 billion in 2015 to $396 million during the first nine months of 2020 [10]. For systems where there is a potential for transmission congestion, a simple economic dispatch will not be sufficient.

The optimal power flow (OPF) addresses this problem by combining economic dispatch with power flow analysis. It is a more complex optimization procedure, but it allows for a much wider array of system constraints to be included. There are many variations in how the OPF can be formulated—we will look at three.

FORMULATION AND SOLUTIONS OF THE DC OPF

The "dc" OPF (DCOPF) uses the linearized dc power flow to approximate the transmission network with the assumptions introduced in Section 6.10. In particular, reactive power is ignored, shunt and resistive impedances are omitted, and bus voltage magnitudes are assumed to be equal to 1.0 per unit. All that is left is a linear relationship between the vector of bus real power injections, $\mathbf{P}$, and the vector of bus voltage angles, δ. That relationship is characterized by the network susceptance matrix $\mathbf{B}$ using the equation $-\mathbf{B}\boldsymbol{\delta} = \mathbf{P}$. Each branch power flow F_{jk} is approximated using branch susceptance $B_{jk} = 1/X_{jk}$ as $F_{jk} = B_{jk}(\delta_j - \delta_k)$. This fully linear system can be added to an economic dispatch problem to create a DCOPF formulation that requires transmission line and transformer limits to be satisfied, as follows:

$$\min C_T = \sum_{i=1}^{n} C_i(P_{gi})$$

subject to $P_{gi,\,min} \le P_{gi} \le P_{gi,\,max}$ for all generators i

$P_i = P_{gi} - P_{di}$ for all buses i where P_{di} is the load demand at bus i

$F_{jk} = B_{jk}(\delta_j - \delta_k)$ for all branches jk

$-F_{ik,\,max} \le F_{jk} \le F_{jk,\,max}$ for all branches jk

$\mathbf{P} = -\mathbf{B}\,\boldsymbol{\delta}$ matrix network equations

In this formulation, the generator costs as a function of real power $C_i(P_{gi})$ and generator limits $P_{gi,\,min}$, $P_{gi,\,max}$ are formulated the same as the economic dispatch problem. Instead of simply summing the generation to equal the load, now each bus must maintain power balance as specified by the dc matrix network equations. With the voltage angle variables δ now part of the optimization problem, line and transformer flows F_{jk} can become variables, and maximum flow constraints can be included in the formulation.

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If the cost functions $C_i(P_{gi})$ are linear or piecewise-linear functions, then the DCOPF as formulated above is a linear program (LP). LPs have well-known solution methods that tend to be robust and lead to globally optimal solutions. Two common approaches are the simplex method—which operates by traversing corners of the convex polyhedron of the feasible space—and interior point methods—which use barrier functions for the constraints and calculate the most direct gradient to the optimal corner. Even if the cost functions are polynomial, these methods can be adapted by either approximating the cost functions as piecewise linear, solving the problem iteratively, or using quadratic programming methods.

When the problem is solved, there will be Lagrange multipliers associated with each of the power balance equality constraints in the matrix calculation $\mathbf{P = -B\delta}$. These multipliers may form a vector $\boldsymbol{\lambda} = [\lambda_1, \lambda_2, \ldots \lambda_n]$. They represent the marginal cost to serve electricity at each bus in the system. Unlike in the economic dispatch problem, there is no longer only one marginal cost λ for the system, but the cost may vary by location as transmission constraints require more expensive generators to be used in some areas. These variables are sometimes known as *locational marginal prices* of electricity (LMPs).

The DCOPF is very common in actual system dispatch calculations because it is relatively easy, fast, and robust to solve. It is well-suited to situations where the key constraints of concern, beyond what can be handled by an economic dispatch, are real power flow limits on transmission lines and transformers.

EXAMPLE 7.6

DCOPF formulation

Formulate the following DCOPF problem as a linear program. There are three buses in the system, each with a generator and a load. The generators have minimum and maximum MW limits and a linear cost as specified in Table 7.3, which also specifies the load at each bus. There are three transmission lines, with reactance and flow limits given in the Table 7.4.

Bus Number	Minimum Generation (MW)	Maximum Generation (MW)	Load (MW)	Generation Cost ($/MWh)
1	100	400	140	18
2	150	500	265	22
3	50	300	415	31

TABLE 7.3
Bus input data for Example 7.6

(*Continued*)

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Branch from Bus	Branch to Bus	Branch Per-unit Reactance on 100-MVA Base	Branch Flow Limit (MW)
1	2	0.07	180
1	3	0.05	120
2	3	0.18	140

TABLE 7.4
Branch input data for Example 7.6

SOLUTION

The problem will be to minimize the total generation cost:

$$\min 18P_{g1} + 22P_{g2} + 31P_{g3}$$

Subject to generation max and min values:

$$100 \le P_{g1} \le 400$$
$$150 \le P_{g2} \le 500$$
$$-50 \le P_{g3} \le 300$$

Also subject to branch flow limits:

$$-180 \le 100 \cdot \frac{1}{0.07}(\delta_1 - \delta_2) \le 180$$

$$-120 \le 100 \cdot \frac{1}{0.05}(\delta_1 - \delta_3) \le 120$$

$$-140 \le 100 \cdot \frac{1}{0.18}(\delta_2 - \delta_3) \le 140$$

And subject to the dc power balance equations (with power values multiplied by 100 to convert from per-unit),

$$-100 \cdot \begin{bmatrix} -24.28 & 14.28 & 20.00 \\ 14.28 & -19.83 & 5.55 \\ 20.00 & 5.55 & -25.55 \end{bmatrix} \begin{bmatrix} \delta_1 \\ \delta_2 \\ \delta_3 \end{bmatrix} = \begin{bmatrix} P_{g1} - 140 \\ P_{g2} - 265 \\ P_{g3} - 415 \end{bmatrix}$$

FORMULATION AND SOLUTIONS OF THE AC OPF

The full or "ac" OPF (ACOPF) does not make the severe approximations that the DCOPF uses. Instead, it includes variables for the voltage magnitude and angle at each bus, $V_i < \delta_i$ and both real and reactive power injections at each bus, $S_i = P_i + jQ_i$, which are related by the full, non-linear ac power balance equations given in Section 6.4 [see (6.4.12) and (6.4.13)].

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$$P_k = V_k \sum_{n=1}^{N} V_n [G_{kn} \cos(\delta_k - \delta_n) + B_{kn} \sin(\delta_k - \delta_n)]$$

$$Q_k = V_k \sum_{n=1}^{n} V_n [G_{kn} \sin(\delta_k - \delta_n) - B_{kn} \cos(\delta_k - \delta_n)]$$

Then more precise line and transformer flows can be calculated and included in the problem, as well as bounds on voltage magnitude and generator reactive power output. Loads are specified as both a real and reactive power. ACOPF may also include discrete variables, such as tap-changing transformers, switchable shunt reactors and capacitors, phase-shifting transformers, or branches which can be switched in and out of service.

The full ACOPF is a complex optimization problem that is very non-linear. There are several methods for solving the OPF with [11] providing a nice summary. One common approach is sequential linear programming (LP); this is the technique used with the PowerWorld Simulator. The LP OPF solution algorithm iterates between solving the power flow to determine the flow of power in the system devices and solving an LP to economically dispatch the generation (and possibly other controls), subject to the transmission system limits. Other approaches use nonlinear programming methods, such as variations on Newton's method. For handling discrete variables, sometimes mixed integer programming is required.

EXAMPLE 7.7

PowerWorld Simulator—optimal power flow

PowerWorld Simulator case Example 7_7 duplicates the five-bus case from Example 7.3, except that the case is solved using PowerWorld Simulator's LP OPF algorithm (see Figure 7.5). To turn on the OPF option, first select **Case Information, Aggregation, Areas…,** and toggle the AGC Status field to OPF. Then, rather than solving the case with the "Single Solution" button, select **Add-ons, Primal LP** to solve using the LP OPF. Initially, the OPF solution matches the ED solution from Example 7.3 since there are no overloaded lines. The green-colored fields on the screen immediately to the right of the buses show the marginal cost of supplying electricity to each bus in the system (i.e., the bus LMPs). With the system initially unconstrained, the bus marginal prices are all identical at \$14.5/MWh, with a Load Scalar of 1.0.

Now increase the Load Scalar field from 1.00 to the maximum economic loading value, determined to be 1.67 in Example 7.3, and again select **Add-ons, Primal LP**. The bus marginal prices are still all identical, now at a value of \$17.5/MWh, and with the line from bus 2 to 5 just reaching its maximum value. For load scalar values above 1.67, the line from bus 2 to bus 5 becomes constrained, with a result that the bus marginal prices on the constrained side of the line become higher than those on the unconstrained side.

With the load scalar equal to 1.80, numerically verify that the price of power at bus 5 is approximately \$40.60/MWh.

(*Continued*)

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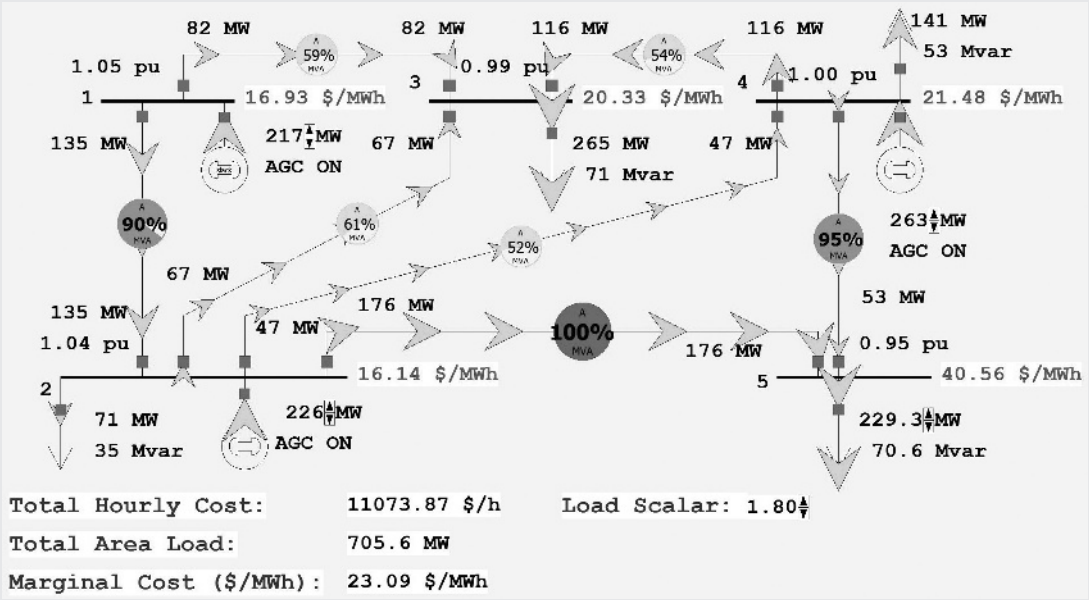

FIGURE 7.5

Example 7.7 optimal power flow solution with load multiplier = 1.80

SOLUTION

The easiest way to numerically verify the bus 5 price is to increase the load at bus 5 by a small amount and compare the change in total system operating cost. With a load scalar of 1.80, the bus 5 MW load is 229.3 MW with a case hourly cost of $11,073.90. Increasing the bus 5 load by 0.9 MW and resolving the LP OPF gives a new cost of $11,110.40, which is a change of about $40.60/MWh (note that this increase in load also increases the bus 5 price to over $41/MWh). Because of the constraint, the price of power at bus 5 is actually more than double the incremental cost of the most expensive generator!

FORMULATION AND SOLUTIONS OF THE SCOPF

The security-constrained OPF (SCOPF) can come in two forms: DC-SCOPF and AC-SCOPF, depending on whether the DCOPF or ACOPF is used as the underlying model. The SCOPF adds additional constraints to the ordinary OPF that are associated with a number of contingency conditions. That is, the problem requires that the resulting system operating point associated with the chosen generator dispatch is able to satisfy line and transformer limits, voltage magnitude, and other constraints both in the base operating point and after a pre-defined set

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of contingency conditions. An example contingency set would be the set of all single element outages (requiring N-1 security).

For a large system with many contingencies, the SCOPF adds so many constraints it becomes very computationally intensive to formulate a solution by explicitly including all of the constraints for all of the contingencies. A common heuristic takes advantage of the fact that typically most contingencies do not have any binding constraints to the problem. The technique would then be to first solve the base DCOPF or ACOPF, then check contingency conditions, iteratively adding critical constraints until all relevant constraints have been added, which is often a small fraction of the overall set of constraints.

7.4 UNIT COMMITMENT AND LONGER TERM OPTIMIZATION

A key requirement of a power system is there must always be sufficient generation available to meet the total load plus losses along with providing enough reserves in case of an unexpected loss of some generation. However, in most electric grids, the load varies substantially over the course of a year, sometimes by more than a 100% increase between the minimum and maximum load, and the load often has strong daily, weekly, and seasonal cycles. The result is most of the time there is much more generator capacity available than is needed to meet the supply needs. Hence, there are often substantial savings if some of these unused generators are turned off. The process of determining which generating units to turn on is known as unit commitment, with the results of unit commitment used as an input for the previously mentioned economic dispatch and optimal power flow algorithms.

How much lead time is needed to turn on (commit) a generator or to turn it off (decommit) depends on the generator technology. Some, such as solar, wind, and some hydro, can be committed and decommitted quickly, almost instantaneously in the case of some solar, to minutes for wind, and no more than ten minutes for most hydro. Others, such as nuclear power generators and some large coal, can take more than 12 hours. Natural gas generators tend to be in-between, ranging from less than an hour to several hours. There are also costs potentially associated with both startup and shutdown of generators, and there can be minimum time limits on how long the generator needs to be on once started, or off once shut down. How fast generators can change their outputs, known as ramping, also needs to be considered along with recognizing that some generators, once committed, can have quite high minimum output limits.

Unit commitment seeks to optimize the operation of an electric grid over a specified time period, with optimization over a day common. Inputs to the unit commitment include the initial generator statuses, a forecast of the load over the time period, weather forecasts that could impact wind and solar capacities, any generators unavailable, such as due to maintenance, operating costs for all of the available generators, and any other constraints and costs associated changing a generator's status.

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Over the years, a number of different solution algorithms have been used. The simplest is to combine a priority list of the overall least expensive generators with heuristics and experience, recognizing that electric grid operations often have the previously mentioned strong daily, weekly, and seasonal patterns. Other algorithms that have been used include dynamic programming and Lagrangian relaxation with the details for both beyond the scope of the text. Today, traditional unit commitment is often combined with transmission system limits in an approach known as security-constrained unit commitment (SCUC). Currently, the SCUC is often solved using mixed integer programming (MIP) algorithms, recognizing that over the last decades there have been many improvements in the MIP algorithms.

In trying to optimize the electric grid over even longer time periods, going from weeks to years, there are often several other constraints that need to be considered, including needs for generator maintenance and overall fuel supply limitations. A quite common fuel limitation is associated with hydro systems, in which only so much energy (water) is available over the course of a year, and there needs to be coordination between the different hydro facilities on a watershed and other watershed requirements, such as reservoir levels for flood control, navigation, irrigation, and recreation. Seasonal patterns in renewable resources also need to be considered.

7.5 MARKETS

When it comes to actual implementation of economic planning and operations, there are several different frameworks, with many variations, that are used worldwide. The oldest, most traditional structure for a power system is to have one utility organization own everything from the generating plants to the customer meters. These *vertically-integrated utilities* can be either government subsidiaries (such as a municipal utility), consumer owned electric cooperatives, or for-profit corporations with a natural monopoly regulated by the government. In these structures, the utility is incentivized to optimize the system, both in long-term system planning and in day-to-day dispatch operations, to keep costs down. When the economic dispatch or OPF is run, the actual cost of the generators to the utility is used. Since this structure has no competition, the consumer prices will be set through a government oversight process that allows the utility to cover its costs and make a reasonable profit. Even with a vertically-integrated utility framework, there is a market that consists of bilateral transactions between multiple interconnected utilities. Utilities buy or sell from their neighbors if there is a mutual agreement to help reduce costs for both.

In many places, this vertically-integrated framework has been restructured into a *wholesale electricity market*. An independent system operator (ISO) or equivalent body is responsible for maintaining the reliability of its portion of the grid, but it does not itself own most of the system assets—generation, transmission, and distribution. Generators are owned by independent investment companies and submit *bids* to sell energy to a regulated market. Transmission and distribution companies purchase this energy as the market is *cleared* by the ISO and provide it to their customers. There are often other entities that participate in the

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electricity markets, trading financial instruments to mitigate risk and incentivize investment in system reliability. In this framework, competition and scarcity drive the wholesale price of energy. There are seven such markets in the United States and many others throughout the world.

Due to the complex constraints associated with unit commitment and scheduling, wholesale energy markets operate on a day-ahead schedule. Market participants submit bids for twenty-four hours at a time, and the ISO clears the market, deciding which bids are accepted at each time period based on a co-optimization analysis that seeks to choose the lowest bids while maintaining all system operational constraints. In most cases, a DCOPF is at the core of this optimization routine. For market solutions, rather than generator costs, the generator bids are used in the objective function of the OPF. Generator owners are not required to reveal their costs and often submit bids different from their costs.

When the market clears, the ISO will produce a locational marginal price (LMP) for each system node at each time period. These LMPs are a byproduct of the DCOPF sensitivity calculations. Participants buying or selling energy at that time period from that location will pay or receive the LMP, rather than their costs or bids. Whether individual consumers will see the LMP depends on the market—in many places, most customers are charged a more consistent rate, with other entities absorbing the risk associated with variations in LMPs.

The day-ahead market relies on accurate forecasting of the load demand and the availability of renewable generation, such as wind and solar. But the actual load and renewable generation will always vary at least slightly from the forecasted load. To resolve this, a real-time market operates in parallel with the day-ahead market, allowing participants to bid for the difference in energy demand and reconciling the day-ahead projections. The real-time market also compensates for other unexpected events, such as the failure of a generator that had been scheduled to run.

One way in which wholesale markets differ significantly from each other is in what services are sold on the market. All markets buy and sell real power (MW or energy market). For some markets, this is all that is sold. But generators provide other services: energy reserves, peak capacity, ramping capability, synchronizing inertia, and reactive power support. In certain markets, one or more of these *ancillary services* are part of the generator bidding process, so that units can be compensated for their additional services.

There are several challenges when it comes to market structure and design. One is how to ensure that there is sufficient competition, which can be threatened when one entity gains too large of a market share, enough to artificially inflate prices. Usually, this is monitored through market regulation and oversight. Another challenge is how to properly incentivize investment in the transmission grid and in future generation capacity. A third challenge is what to do in times of extreme scarcity, when there is so much demand relative to available supply that essentially any generation offered for sale will be accepted. One opportunity in market design is the implementation of ACOPF-based market-clearing tools, which require more intricate optimization techniques but has the potential to improve the efficiency of market operations by coordinating more system control variables for an optimal dispatch.

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PROBLEMS

SECTION 7.2

7.1 The fuel-cost curves for two generators are given as follows:

$$C_1(P_1) = 600 + 18 \cdot P_1 + 0.04 \cdot (P_1)^2$$
$$C_2(P_2) = 700 + 20 \cdot P_2 + 0.03 \cdot (P_2)^2$$

Assuming the system is lossless, calculate the optimal dispatch values of P_1 and P_2 for a total load of 1000 MW, the incremental operating cost, and the total operating cost.

7.2 Rework Problem 7.1, except assume that the limit outputs are subject to the following inequality constraints:

$$200 \leq P_1 \leq 800 \text{ MW}$$
$$100 \leq P_2 \leq 400 \text{ MW}$$

7.3 Rework Problem 7.1, except assume the 1000-MW value also includes losses, and the penalty factor for the first unit is 1.0 and for the second unit 0.95.

7.4 The fuel-cost curves for a two-generator power system are given as follows:

$$C_1(P_1) = 600 + 15 \cdot P_1 + 0.05 \cdot (P_1)^2$$
$$C_2(P_2) = 700 + 20 \cdot P_2 + 0.04 \cdot (P_2)^2$$

while the system losses can be approximated as

$$P_L = 2 \times 10^{-4}(P_1)^2 + 3 \times 10^{-4}(P_2)^2 - 4 \times 10^{-4}P_1P_2 \text{ MW}$$

If the system is operating with a marginal cost (λ) of \$60/hr, determine the output of each unit, the total transmission losses, the total load demand, and the total operating cost.

7.5 Expand the summations in (7.1.14) for $N = 2$, and verify the formula for $\partial P_L/\partial P_i$ given by (7.1.15). Assume $B_{ij} = B_{ji}$.

7.6 Given two generating units with their respective variable operating costs as

$$C_1 = 0.01\,P_{G1}^2 + 2P_{G1} + 100 \text{ \$/hr} \qquad \text{for } 25 \leq P_{G1} \leq 150 \text{ MW}$$
$$C_2 = 0.004\,P_{G2}^2 + 2.6P_{G2} + 80 \text{ \$/hr} \qquad \text{for } 30 \leq P_{G2} \leq 200 \text{ MW}$$

determine the economically optimal division of generation for $55 \leq P_L \leq 350$ MW. In particular, for $P_L = 282$ MW, compute P_{G1} and P_{G2}. Neglect transmission losses.

PW 7.7 Resolve Example 7.3, except with the generation at bus 2 set to a fixed value (i.e., modeled as off of Automatic Generation Control). Plot the variation in the total hourly cost as the generation at bus 2 is varied between 1000 and 200 MW in 5 MW steps, resolving the economic dispatch at each step. What is the relationship between bus 2 generation

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at the minimum point on this plot and the value from economic dispatch in Example 7.3? Assume a Load Scalar of 1.0.

PW 7.8 Using PowerWorld case Example 7_5 with the Load Scalar equal to 1.0, determine the generation dispatch that minimizes system losses. (*Hint:* Manually vary the generation at buses 2 and 4 until their loss sensitivity values are zero.) Compare the operating cost between this solution and the Example 7_5 economic dispatch result. Which is better?

PW 7.9 Repeat Problem 7.8, except with the Load Scalar equal to 1.4.

SECTION 7.3

PW 7.10 Using LP OPF with PowerWorld Simulator case Example 7_6, plot the variation in the bus 5 marginal price as the Load Scalar is increased from 1.0 in steps of 0.02. What is the maximum possible load scalar without overloading any transmission line? Why is it impossible to operate without violations above this value?

PW 7.11 Load PowerWorld Simulator case Problem 7_11. This case models a slightly modified, lossless version of the 37-bus case from Example 6.13 with generator cost information, but also with the transformer between buses PEAR138 and PEAR69 open. When the case is loaded, the "Total Cost" field shows the economic dispatch solution, which results in an overload on several lines. Before solving the case, select **Add-Ons, OPF Case Info, OPF Buses** to view the bus LMPs, noting that they are all identical. Then Select **Add-Ons, Primal LP** to solve the case using the OPF, and again view the bus LMPs. Verify the LMP at the PECAN69 bus by manually changing the load at the bus by one MW, and then noting the change in the Total Cost field. Repeat for the LOCUST69 bus. *Note*: Because of solution convergence tolerances, the manually calculated results may not exactly match the OFP calculated bus LMPs.

7.12 In Example 7.6, what is the economic dispatch solution? Why is this not a feasible solution for the DCOPF?

7.13 In Example 7.6, the solution is $P_{G1} = 394$ MW, $P_{G2} = 150$ MW, and $P_{G3} = 276$ MW. The resulting LMPs are $\lambda_1 = 18$ \$/MWh, $\lambda_2 = 9.72$ \$/MWh, $\lambda_3 = 31$ \$/MWh, Why is there this variation in LMPs?

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8 Symmetrical Faults

345-kV SF6 circuit breaker installation at Goshen Substation, Idaho Falls, Idaho, USA. This circuit breaker has a continuous current rating of 2000 A and an interrupting current rating of 40 kA. (Courtesy of PacifiCorp.)

LEARNING OBJECTIVES

At the conclusion of this chapter, you should be able to:

1. Understand the components of short-circuit current in an R-L series circuit supplied by a sinusoidal voltage source, including the symmetrical short-circuit current, the dc offset current, and the total (asymmetrical) short-circuit current.
2. Understand the components of short-circuit current for a three-phase fault at the terminals of an unloaded synchronous machine, including subtransient fault current, transient fault current, steady-state fault current, dc offset current, and total (asymmetrical) fault current.
3. Know the method of calculating the subtransient fault current for a three-phase fault at any bus in an *N*-bus power system using the bus impedance matrix.
4. Use PowerWorld Simulator to calculate symmetrical fault currents.
5. Know the basic considerations for selecting circuit breakers and fuses in an ac power system.

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Short circuits occur in power systems when equipment insulation fails due to system overvoltages caused by lightning or switching surges, to insulation contamination (salt spray or pollution), or to other mechanical causes. The resulting short circuit or "fault" current is determined by the internal voltages of the synchronous machines and by the system impedances between the machine voltages and the fault. Short-circuit currents may be several orders of magnitude larger than normal operating currents and, if allowed to persist, may cause thermal damage to equipment. Windings and bus bars may also suffer mechanical damage due to high magnetic forces during faults. It is therefore necessary to remove faulted sections of a power system from service as soon as possible. Standard EHV protective equipment is designed to clear faults within 3 cycles (50 ms at 60 Hz). Lower-voltage protective equipment operate more slowly (for example, 5 to 20 cycles).

Section 8.1 begins by reviewing series R–L circuit transients, followed in Section 8.2 by a description of three-phase short-circuit currents at unloaded synchronous machines, analyzing both the ac component, including subtransient, transient, and steady-state currents, and the dc component of fault current. These results are extended in Sections 8.3 and 8.4 to power system three-phase short circuits by means of the superposition principle. We observe that the bus impedance matrix is the key to calculating fault currents. Section 8.5 discusses circuit breaker and fuse selection.

Balanced three-phase power systems are assumed throughout this chapter, working in per unit.

CASE STUDY

Pumped Storage Hydro: Then and Now*

Peter J. Donalek

The following paragraphs summarize the above article, which describes pumped storage hydro as part of modern electric grids. Sign in at www.cengage.com to download this product and to access its online resources.

The article begins with a short history of hydropower and some of the early pumped storage hydro plants. Pumped storage plants that have been built in a number of locations, including the United States, Japan, China, and Europe, are also described. The synergies between pumped storage and nuclear generation are discussed, and technologies related to efficient, adjustable-speed, pumped storage units are presented.

A list of energy storage technologies includes mechanical, electrical,

*Donalek, Peter J., "Pumped Storage Hydro: Then and Now," © 2020 IEEE. Available online, with permission, from *IEEE Power & Energy Magazine* (September/October 2020), pp. 49–57.

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electromechanical, chemical, and thermal. Among these technologies, pumped storage hydro and electrochemical battery storage are the dominant storage technologies in use today. Each of these storage technologies involves a charge-and-discharge cycle that can convert three-phase ac energy to another form of energy for storage and then reverses the process to return the stored energy as three-phase ac electric energy. With mechanical storage, energy is stored in the form of potential energy and then returned as kinetic energy.

A pumped storage plant requires two bodies of water (reservoirs) separated vertically and connected by water conductors. With this configuration, sufficient hydraulic head (pressure) can be provided to effectively convert potential energy into kinetic energy in the form of water flow that drives hydro turbines and synchronous generators. Unlike chemical-based storage technologies, which require the procurement of chemicals (involving cost and environmental consequences), pumped storage hydro uses water and gravity, which are available, plentiful, and convenient (Figure 1).

In the charge mode, a pump driven by a synchronous motor converts surplus three-phase alternating power, whenever it is available, to pump water from a lower reservoir to an upper one, where the water, with its potential energy, is stored until it is required in the discharge mode. In the discharge mode, water stored in the upper reservoir flows to a lower elevation, converting potential energy into rotating mechanical torque that drives a synchronous generator and delivers three-phase ac power to the grid.

The U.S. National Hydropower Association reported that in 2018, there were 42 pumped storage hydro projects in the United States, with a total installed capacity of approximately

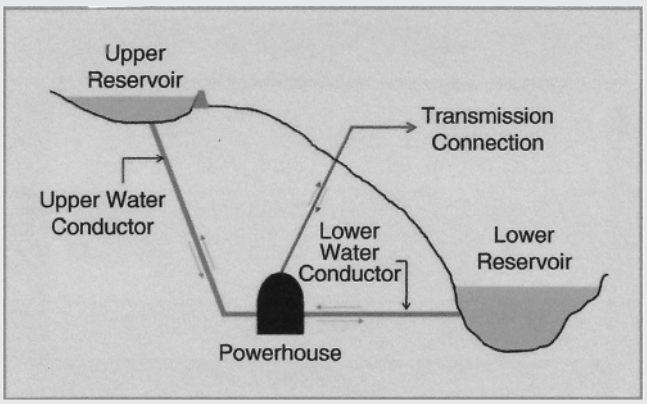

Figure 1 The pumped storage concept (Courtesy of Argonne National Laboratory, managed and operated by UChicago, Argonne, LLC, for the U.S. Department of Energy under Contract DE-AC02-O6CH11357.) (Courtesy of Argonne National Laboratory, managed and operated by UChicago, Argonne, LLC, for the U.S. Department of Energy under Contract DE-AC02-O6CH11357.)

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Figure 2 The Bath County pumped storage project, Virginia (Source: Dominion Energy; used with permission.)

22 GW. Many of these plants were constructed from the 1960s through the 1980s to complement large base-load nuclear and coal-powered plants by providing fixed pump loads during the early morning hours, with peaking power and load following in generation mode during the daytime peak. An example of a pumped storage plant that was built to accommodate the needs of nuclear plants is the largest U.S. pumped storage plant, the 3-GW (3000 MW) six-unit, Bath County project in Virginia, with an energy storage capacity of 24,000 MWh. The plant is jointly owned by Dominion Generation and First Energy (Figure 2). The U.S. Department of Energy (DOE) Energy Storage Database reports that as of June 2020, worldwide, there were 352 pumped storage plants, with an installed capacity of 185 GW.

Japanese utilities pioneered the application of adjustable-speed technology to pumped storage hydro units. The motivation for adjustable-speed units in Japan was the need for frequency regulation capability, as base-loaded nuclear units were added, and a reduction in imported fuel (oil and liquid natural gas) for diesel and gas turbine-generators that provided frequency regulation. In the early 1990s, Kansai Electric and Tokyo Electric installed adjustable-speed pumped storage units. One of the first is the Yagisawa pumped storage 85-MW unit that was converted from a single-speed, 150-r/min synchronous motor/generator to a doubly fed induction motor/generator (DFIM) with an adjustable speed range of 130-156 r/min.

A motivating factor for new adjustable-speed pumped storage units is the need to address the impact that increasing amounts of wind and solar photovoltaics have on grid system inertia. Throughout the world, there are numerous new pumped storage project sites in various stages of planning, design, and construction. Many of these projects will include adjustable-speed pumped storage units. DFIM, adjustable-speed pumped storage units with fast response capabilities, are expected to play a significant role in grid system dynamic performance [12].

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8.1 SERIES R–L CIRCUIT TRANSIENTS

Consider the series R–L circuit shown in Figure 8.1. The closing of switch SW at $t = 0$ represents to a first approximation a three-phase short circuit at the terminals of an unloaded synchronous machine. For simplicity, assume zero fault impedance; that is, the short circuit is a solid or "bolted" fault. The current is assumed to be zero before SW closes, and the source angle α determines the source voltage at $t = 0$. Writing a KVL equation for the circuit,

$$\frac{\mathrm{L}di(t)}{dt} + \mathrm{R}i(t) = \sqrt{2}\mathrm{V}\sin(\omega t + \alpha)\ t \geq 0 \tag{8.1.1}$$

The solution to (8.1.1) is

$$\begin{aligned} i(t) &= i_{\mathrm{ac}}(t) + i_{\mathrm{dc}}(t) \\ &= \frac{\sqrt{2}\mathrm{V}}{\mathrm{Z}}[\sin(\omega t + \alpha - \theta) - \sin(\alpha - \theta)e^{-t/\mathrm{T}}] \quad \mathrm{A} \end{aligned} \tag{8.1.2}$$

where

$$i_{\mathrm{ac}}(t) = \frac{\sqrt{2}\mathrm{V}}{\mathrm{Z}}\sin(\omega t + \alpha - \theta) \quad \mathrm{A} \tag{8.1.3}$$

$$i_{\mathrm{dc}}(t) = -\frac{\sqrt{2}\mathrm{V}}{\mathrm{Z}}\sin(\alpha - \theta)e^{-t/T} \quad \mathrm{A} \tag{8.1.4}$$

$$Z = \sqrt{\mathrm{R}^2 + (\omega\mathrm{L})^2} = \sqrt{\mathrm{R}^2 + \mathrm{X}^2} \quad \Omega \tag{8.1.5}$$

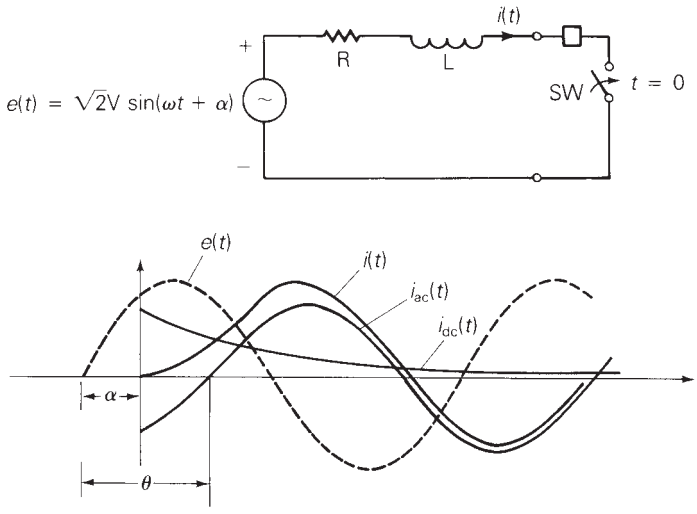

FIGURE 8.1

Current in a series R–L circuit with ac voltage source

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$$\theta = \tan^{-1}\frac{\omega \mathrm{L}}{\mathrm{R}} = \tan^{-1}\frac{\mathrm{X}}{\mathrm{R}} \tag{8.1.6}$$

$$\mathrm{T} = \frac{\mathrm{L}}{\mathrm{R}} = \frac{\mathrm{X}}{\omega \mathrm{R}} = \frac{\mathrm{X}}{2\pi f \mathrm{R}} \quad \mathrm{s} \tag{8.1.7}$$

The total fault current in (8.1.2), called the *asymmetrical fault current*, is plotted in Figure 8.1 along with its two components. The ac fault current (also called *symmetrical* or *steady-state fault current*), given by (8.1.3), is a sinusoid. The *dc offset current*, given by (8.1.4), decays exponentially with time constant T = L/R.

The rms ac fault current is $\mathrm{I_{ac}} = \mathrm{V/Z}$. The magnitude of the dc offset, which depends on α, varies from 0 when $\alpha = \theta$ to $\sqrt{2}\mathrm{I_{ac}}$ when $\alpha = (\theta \pm \pi/2)$. Note that a short circuit may occur at any instant during a cycle of the ac source; that is, α can have any value. To find the largest fault current, choose $\alpha = (\theta - \pi/2)$. Then (8.1.2) becomes

$$i(t) = \sqrt{2}\mathrm{I_{ac}}\left[\sin\left(\omega t - \pi/2\right) + e^{-t/\mathrm{T}}\right] \quad \mathrm{A} \tag{8.1.8}$$

where

$$\mathrm{I_{ac}} = \frac{\mathrm{V}}{\mathrm{Z}} \quad \mathrm{A} \tag{8.1.9}$$

The rms value of $i(t)$ is of interest. Since $i(t)$ in (8.1.8) is not strictly periodic, its rms value is not strictly defined. However, to calculate the rms asymmetrical fault current with maximum dc offset, treat the exponential term as a constant, stretching the rms concept as follows:

$$\begin{aligned}\mathrm{I_{rms}}(t) &= \sqrt{[\mathrm{I_{ac}}]^2 + [\mathrm{I_{dc}}(t)]^2} \\ &= \sqrt{[\mathrm{I_{ac}}]^2 + [\sqrt{2}\mathrm{I_{ac}}e^{-t/\mathrm{T}}]^2} \\ &= \mathrm{I_{ac}}\sqrt{1 + 2e^{-2t/\mathrm{T}}} \quad \mathrm{A}\end{aligned} \tag{8.1.10}$$

It is convenient to use $\mathrm{T} = \mathrm{X}/(2\pi f\,\mathrm{R})$ and $t = \tau/f$, where τ is time in cycles, and write (8.1.10) as

$$\mathrm{I_{rms}}(\tau) = \mathrm{K}(\tau)\mathrm{I_{ac}} \quad \mathrm{A} \tag{8.1.11}$$

where

$$\mathrm{K}(\tau) = \sqrt{1 + 2e^{-4\pi\tau/(\mathrm{X/R})}} \quad \text{per unit} \tag{8.1.12}$$

From (8.1.11) and (8.1.12), the rms asymmetrical fault current equals the rms ac fault current times an "asymmetry factor," $\mathrm{K}(\tau)$. $\mathrm{I_{rms}}(\tau)$ decreases from $\sqrt{3}\mathrm{I_{ac}}$ when $\tau = 0$ to $\mathrm{I_{ac}}$ when τ is large. Also, higher X to R ratios (X/R) give higher values of $\mathrm{I_{rms}}(\tau)$. The above series R–L short-circuit currents are summarized in Table 8.1.

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Component	Instantaneous Current (A)	rms Current (A)
Symmetrical (ac)	$i_{ac}(t) = \frac{\sqrt{2}V}{Z}\sin(\omega t + \alpha - \theta)$	$I_{ac} = \frac{V}{Z}$
dc offset	$i_{dc}(t) = \frac{-\sqrt{2}V}{Z}\sin(\alpha - \theta)e^{-t/T}$	
Asymmetrical (total)	$i(t) = i_{ac}(t) + i_{dc}(t)$	$I_{rms}(t) = \sqrt{I_{ac}^2 + i_{dc}(t)^2}$ with maximum dc offset: $I_{rms}(\tau) = K(\tau)I_{ac}$

TABLE 8.1

Short-circuit current—series R–L circuit*

*See Figure 8.1 and Equations (8.1.1) through (8.1.12).

EXAMPLE 8.1

Fault currents: R–L circuit with ac source

A bolted short circuit occurs in the series R–L circuit of Figure 8.1, with V = 20 kV, X = 8 Ω, R = 0.8 Ω, and with maximum dc offset. The circuit breaker opens 3 cycles after fault inception. Determine (a) the rms ac fault current, (b) the rms "momentary" current at $\tau = 0.5$ cycle, which passes through the breaker before it opens, and (c) the rms asymmetrical fault current that the breaker interrupts.

SOLUTION

a. From (8.1.9),

$$I_{ac} = \frac{20 \times 10^3}{\sqrt{(8)^2 + (0.8)^2}} = \frac{20 \times 10^3}{8.040} = 2.488 \quad \text{kA}$$

b. From (8.1.11) and (8.1.12) with (X/R) = 8/(0.8) = 10 and $\tau = 0.5$ cycle,

$$K(0.5 \text{ cycle}) = \sqrt{1 + 2e^{-4\pi(0.5)/10}} = 1.438$$

$$I_{momentary} = K(0.5 \text{ cycle})I_{ac} = (1.438)(2.488) = 3.576 \quad \text{kA}$$

c. From (8.1.11) and (8.1.12) with (X/R) = 10 and $\tau = 3$ cycles,

$$K(3 \text{ cycles}) = \sqrt{1 + 2e^{-4\pi(3)/10}} = 1.023$$

$$I_{rms}(3 \text{ cycles}) = (1.023)(2.488) = 2.544 \quad \text{kA}$$

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8.2 THREE-PHASE SHORT CIRCUIT—UNLOADED SYNCHRONOUS MACHINE

One way to investigate a three-phase short circuit at the terminals of a synchronous machine is to perform a test on an actual machine. Figure 8.2 shows an oscillogram of the ac fault current in one phase of an unloaded synchronous machine during such a test. The dc offset has been removed from the oscillogram. As shown, the amplitude of the sinusoidal waveform decreases from a high initial value to a lower steady-state value.

A physical explanation for this phenomenon is that the magnetic flux caused by the short-circuit armature currents (or by the resultant armature MMF) is initially forced to flow through high-reluctance paths that do not link the field winding or damper circuits of the machine. This is a result of the theorem of constant flux linkages that states that the flux linking a closed winding cannot change instantaneously. The armature inductance, which is inversely proportional to reluctance, is therefore initially low. As the flux then moves toward the lower reluctance paths, the armature inductance increases.

The ac fault current in a synchronous machine can be modeled by the series R–L circuit of Figure 8.1 if a time-varying inductance $L(t)$ or reactance $X(t) = \omega L(t)$ is employed. In standard machine theory texts [3, 4], the following reactances are defined:

X''_d = direct axis subtransient reactance
X'_d = direct axis transient reactance
X_d = direct axis synchronous reactance

where $X''_d < X'_d < X_d$. The subscript d refers to the direct axis. There are similar quadrature axis reactances $X''_q < X'_q$, and X_q [3, 4]. However, if the armature resistance is small, the quadrature axis reactances do not significantly affect the

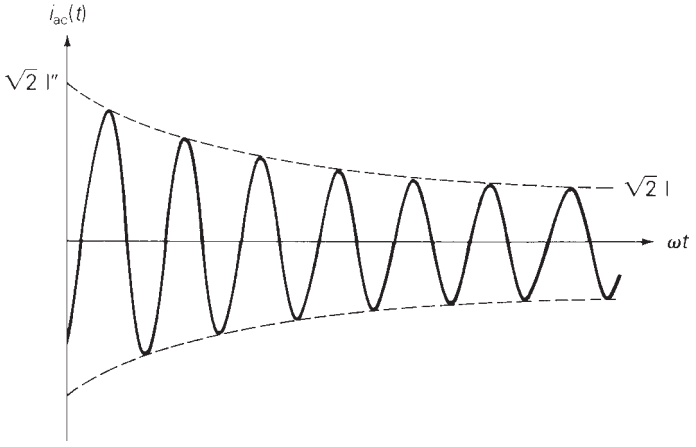

FIGURE 8.2
The ac fault current in one phase of an unloaded synchronous machine during a three-phase short circuit (the dc offset current is removed)

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short-circuit current. Using the above direct axis reactances, the instantaneous ac fault current can be written as

$$i_{\text{ac}}(t) = \sqrt{2}\text{E}_g\left[\left(\frac{1}{\text{X}_d''} - \frac{1}{\text{X}_d'}\right)e^{-t/\text{T}_d''} + \left(\frac{1}{\text{X}_d'} - \frac{1}{\text{X}_d}\right)e^{-t/\text{T}_d'} + \frac{1}{\text{X}_d}\right]\sin\left(\omega t + \alpha - \frac{\pi}{2}\right) \tag{8.2.1}$$

where E_g is the rms line-to-neutral prefault terminal voltage of the unloaded synchronous machine. Armature resistance is neglected in (8.2.1). Note that at $t = 0$, when the fault occurs, the rms value of $i_{\text{ac}}(t)$ in (8.2.1) is

$$\text{I}_{\text{ac}}(0) = \frac{\text{E}_g}{\text{X}_d''} = \text{I}'' \tag{8.2.2}$$

which is called the rms *subtransient fault current*, I″. The duration of I″ is determined by the time constant T_d'', called the *direct axis short-circuit sub-transient time constant*.

At a later time, when t is large compared to T_d'' but small compared to the *direct axis short-circuit transient time constant* T_d', the first exponential term in (8.2.1) has decayed almost to zero, but the second exponential has not decayed significantly. The rms ac fault current then equals the rms *transient fault current*, given by

$$\text{I}' = \frac{\text{E}_g}{\text{X}_d'} \tag{8.2.3}$$

When t is much larger than T_d', the rms ac fault current approaches its steady-state value, given by

$$\text{I}_{\text{ac}}(\infty) = \frac{\text{E}_g}{\text{X}_d} = \text{I} \tag{8.2.4}$$

Since the three-phase no-load voltages are displaced 120° from each other, the three-phase ac fault currents are also displaced 120° from each other. In addition to the ac fault current, each phase has a different dc offset. The maximum dc offset in any one phase, which occurs when $\alpha = 0$ in (8.2.1), is

$$i_{\text{dcmax}}(t) = \frac{\sqrt{2}\text{E}_g}{\text{X}_d''}e^{-t/\text{T}_\text{A}} = \sqrt{2}\text{I}''e^{-t/\text{T}_\text{A}} \tag{8.2.5}$$

where T_A is called the *armature time constant*. Note that the magnitude of the maximum dc offset depends only on the rms subtransient fault current I″. The above synchronous machine short-circuit currents are summarized in Table 8.2.

Machine reactances X_d'', X_d', and X_d, as well as time constants T_d'', T_d', and T_A, are usually provided by synchronous machine manufacturers. They also can be

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Component	Instantaneous Current (A)	rms Current (A)
Symmetrical (ac)	(8.2.1)	$I_{ac}(t) = E_g\left[\left(\frac{1}{X''_d} - \frac{1}{X'_d}\right)e^{-t/T''_d} + \left(\frac{1}{X'_d} - \frac{1}{X_d}\right)e^{-t/T'_d} + \frac{1}{X_d}\right]$
Subtransient		$I'' = E_g/X''_d$
Transient		$I' = E_g/X'_d$
Steady-state		$I = E_g/X_d$
Maximum dc offset	$i_{dc}(t) = \sqrt{2}I''e^{-t/T_A}$	
Asymmetrical (total)	$i(t) = i_{ac}(t) + i_{dc}(t)$	$I_{rms}(t) = \sqrt{I_{ac}(t)^2 + i_{dc}(t)^2}$ with maximum dc offset: $I_{rms}(t) = \sqrt{I_{ac}(t)^2 + [\sqrt{2}I''\,e^{-t/T_A}]^2}$

TABLE 8.2

Short-circuit current—unloaded synchronous machine*

*See Figure 8.2 and Equations (8.2.1) through (8.2.5).

obtained from a three-phase short-circuit test by analyzing an oscillogram such as that in Figure 8.2 [2]. Typical values of synchronous machine reactances and time constants are given in Appendix Table A.1.

EXAMPLE 8.2

Three-phase short-circuit currents, unloaded synchronous generator

A 500-MVA, 20-kV, 60-Hz synchronous generator with reactances $X''_d = 0.15$, $X'_d = 0.24$, and $X_d = 1.1$ per unit, and time constants $T''_d = 0.035$, $T'_d = 2.0$, $T_A = 0.20$ s is connected to a circuit breaker. The generator is operating at 5% above rated voltage and at no-load when a bolted three-phase short circuit occurs on the load side of the breaker. The breaker interrupts the fault three cycles after fault inception. Determine (a) the subtransient fault current in per-unit and kA rms; (b) maximum dc offset as a function of time; and (c) rms asymmetrical fault current that the breaker interrupts, assuming maximum dc offset.

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SOLUTION

a. The no-load voltage before the fault occurs is $E_g = 1.05$ per unit. From (8.2.2), the subtransient fault current that occurs in each of the three phases is

$$I'' = \frac{1.05}{0.15} = 7.0 \text{ per unit}$$

The generator base current is

$$I_{base} = \frac{S_{rated}}{\sqrt{3}V_{rated}} = \frac{500}{(\sqrt{3})(20)} = 14.43 \quad \text{kA}$$

The rms subtransient fault current in kA is the per-unit value multiplied by the base current:

$$I'' = (7.0)(14.43) = 101.0 \quad \text{kA}$$

b. From (8.2.5), the maximum dc offset that may occur in any one phase is

$$i_{dcmax}(t) = \sqrt{2}(101.0)e^{-t/0.20} = 142.9e^{-t/0.20} \quad \text{kA}$$

c. From (8.2.1), the rms ac fault current at $t = 3$ cycles $= 0.05$ s is

$$I_{ac}(0.05 \text{ s}) = 1.05\left[\left(\frac{1}{0.15} - \frac{1}{0.24}\right)e^{-0.05/0.035} + \left(\frac{1}{0.24} - \frac{1}{1.1}\right)e^{-0.05/2.0} + \frac{1}{1.1}\right]$$

$$= 4.920 \text{ per unit}$$

$$= (4.920)(14.43) = 71.01 \quad \text{kA}$$

Modifying (8.1.10) to account for the time-varying symmetrical component of fault current,

$$I_{rms}(0.05) = \sqrt{[I_{ac}(0.05)]^2 + [\sqrt{2}I''e^{-t/T_a}]^2}$$

$$= I_{ac}(0.05)\sqrt{1 + 2\left[\frac{I''}{I_{ac}(0.05)}\right]^2 e^{-2t/T_a}}$$

$$= (71.01)\sqrt{1 + 2\left[\frac{101}{71.01}\right]^2 e^{-2(0.05)/0.20}}$$

$$= (71.01)(1.8585)$$

$$= 132 \quad \text{kA}$$

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8.3 POWER SYSTEM THREE-PHASE SHORT CIRCUITS

In order to calculate the subtransient fault current for a three-phase short circuit in a power system, make the following assumptions:

1. Transformers are represented by their leakage reactances. Winding resistances, shunt admittances, and $\Delta - Y$ phase shifts are neglected.
2. Transmission lines are represented by their equivalent series reactances. Neglect series resistances and shunt admittances.
3. Synchronous machines are represented by constant-voltage sources behind subtransient reactances. Neglect armature resistance, saliency, and saturation.
4. Neglect all nonrotating impedance loads.
5. Especially for small motors rated less than 50 hp, either neglect induction motors or represent them in the same manner as synchronous machines.

These assumptions are made for simplicity in this text, and in practice they should not be made for all cases. For example, in distribution systems, resistances of primary and secondary distribution lines may in some cases significantly reduce fault current magnitudes.

Figure 8.3 shows a single-line diagram consisting of a synchronous generator feeding a synchronous motor through two transformers and a transmission line. Consider a three-phase short circuit at bus 1. The positive-sequence equivalent circuit is shown in Figure 8.4(a), where the voltages E_g'' and E_m'' are the prefault internal voltages behind the subtransient reactances of the machines, and the closing of switch SW represents the fault. For purposes of calculating the subtransient fault current, E_g'' and E_m'' are assumed to be constant-voltage sources.

In Figure 8.4(b), the fault is represented by two opposing voltage sources with equal phasor values V_F. Using superposition, the fault current then can be calculated from the two circuits shown in Figure 8.4(c). However, if V_F equals the prefault voltage at the fault, then the second circuit in Figure 8.4(c) represents the system before the fault occurs. As such, $I_{F2}'' = 0$ and V_F, which has no effect, can be removed from the second circuit, as shown in Figure 8.4(d). The subtransient fault current is then determined from the first circuit in Figure 8.4(d), $I_F'' = I_{F1}''$. The contribution to the fault from the generator is $I_g'' = I_{g1}'' + I_{g2}'' = I_{g1}'' + I_L$, where I_L is the prefault generator current. Similarly, $I_m'' = I_{m1}'' - I_L$.

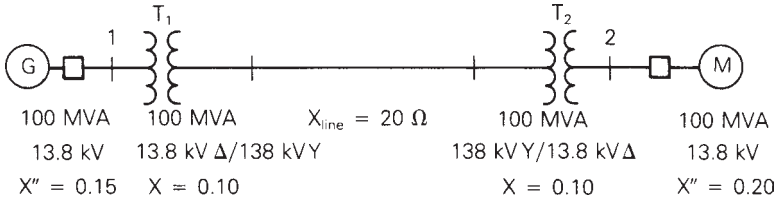

FIGURE 8.3
Single-line diagram of a synchronous generator feeding a synchronous motor

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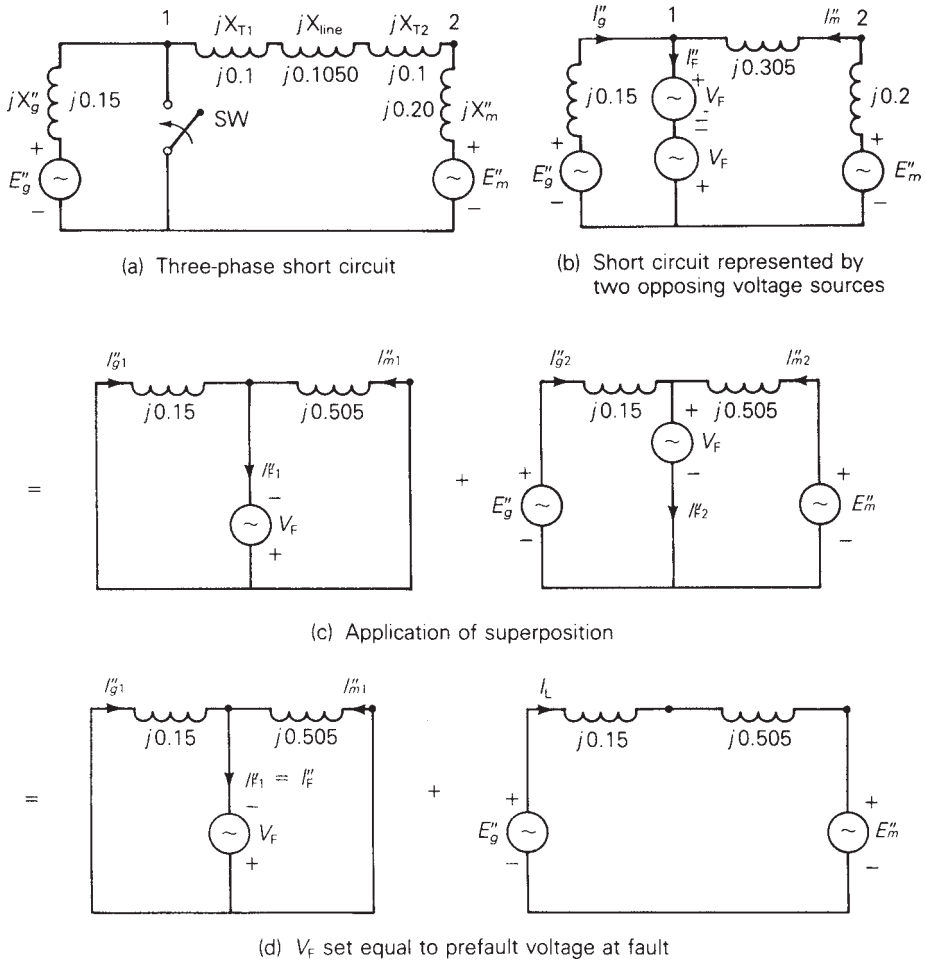

FIGURE 8.4

Application of superposition to a power system three-phase short circuit

EXAMPLE 8.3

Three-phase short-circuit currents, power system

The synchronous generator in Figure 8.3 is operating at rated MVA, 0.95 p.f. lagging and at 5% above rated voltage when a bolted three-phase short circuit occurs at bus 1. Calculate the per-unit values of (a) subtransient fault current; (b) subtransient generator and motor currents neglecting prefault current; and (c) subtransient generator and motor currents including prefault current.

(*Continued*)

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SOLUTION

a. Using a 100-MVA base, the base impedance in the zone of the transmission line is

$$Z_{\text{base, line}} = \frac{(138)^2}{100} = 190.44 \quad \Omega$$

and

$$X_{\text{line}} = \frac{20}{190.44} = 0.1050 \quad \text{per unit}$$

Figure 8.4 shows the per-unit reactances. From the first circuit in Figure 8.4(d), the Thévenin impedance as viewed from the fault is

$$Z_{\text{Th}} = jX_{\text{Th}} = j\frac{(0.15)(0.505)}{(0.15 + 0.505)} = j0.11565 \quad \text{per unit}$$

and the prefault voltage at the generator terminals is

$$V_{\text{F}} = 1.05 \angle 0^\circ \quad \text{per unit}$$

The subtransient fault current is then

$$I''_{\text{F}} = \frac{V_{\text{F}}}{Z_{\text{Th}}} = \frac{1.05\angle 0^\circ}{j0.11565} = -j9.079 \quad \text{per unit}$$

b. Using current division in the first circuit of Figure 8.4(d),

$$I''_{g1} = \left(\frac{0.505}{0.505 + 0.15}\right) I''_{\text{F}} = (0.7710)(-j9.079) = -j7.000 \quad \text{per unit}$$

$$I''_{m1} = \left(\frac{0.15}{0.505 + 0.15}\right) I''_{\text{F}} = (0.2290)(-j9.079) = -j2.079 \quad \text{per unit}$$

c. The generator base current is

$$I_{\text{base,gen}} = \frac{100}{(\sqrt{3})(13.8)} = 4.1837 \quad \text{kA}$$

and the prefault generator current is

$$I_{\text{L}} = \frac{100}{(\sqrt{3})(1.05 \times 13.8)} \angle -\cos^{-1}0.95 = 3.9845 \angle -18.19^\circ \quad \text{kA}$$

$$= \frac{3.9845 \angle -18/19^\circ}{4.1837} = 0.9524 \angle -18.19^\circ$$

$$= 0.9048 - j0.2974 \quad \text{per unit}$$

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The subtransient generator and motor currents, including prefault current, are then

$$I''_g = I''_{g1} + I_L = -j7.000 + 0.9048 - j0.2974$$
$$= 0.9048 - j7.297 = 7.353\underline{/-82.9^\circ} \quad \text{per unit}$$
$$I''_m = I''_{m1} - I_L = -j2.079 - 0.9048 + j0.2974$$
$$= -0.9048 - j1.782 = 1.999\underline{/243.1^\circ} \quad \text{per unit}$$

An alternate method of solving Example 8.3 is to first calculate the internal voltages E''_g and E''_m using the prefault load current I_L. Then, instead of using superposition, the fault currents can be resolved directly from the circuit in Figure 8.4(a) (see Problem 8.11). However, in a system with many synchronous machines, the superposition method has the advantage that all machine voltage sources are shorted, and the prefault voltage is the only source required to calculate the fault current. Also, when calculating the contributions to fault current from each branch, prefault currents are usually small, and hence can be neglected. Otherwise, prefault load currents could be obtained from a power-flow program.

8.4 BUS IMPEDANCE MATRIX

Now to extend the results of the previous section to calculate subtransient fault currents for three-phase faults in an N-bus power system, the system is modeled by its positive-sequence network, where lines and transformers are represented by series reactances and synchronous machines are represented by constant-voltage sources behind subtransient reactances. As before, all resistances, shunt admittances, and nonrotating impedance loads, and also for simplicity, prefault load currents, are neglected.

Consider a three-phase short circuit at any bus n. To analyze two separate circuits, use the superposition method described in Section 8.3. [For example, see Figure 8.4(d).] In the first circuit, all machine-voltage sources are short-circuited, and the only source is due to the prefault voltage at the fault. Writing nodal equations for the first circuit,

$$\boldsymbol{Y}_{\text{bus}}\boldsymbol{E}^{(1)} = \boldsymbol{I}^{(1)} \tag{8.4.1}$$

where $\boldsymbol{Y}_{\text{bus}}$ is the positive-sequence bus admittance matrix, $\boldsymbol{E}^{(1)}$ is the vector of bus voltages, and $\boldsymbol{I}^{(1)}$ is the vector of current sources. The superscript (1) denotes the first circuit. Solving (8.4.1),

$$\boldsymbol{Z}_{\text{bus}}\boldsymbol{I}^{(1)} = \boldsymbol{E}^{(1)} \tag{8.4.2}$$

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where

$$\boldsymbol{Z}_{\text{bus}} = \boldsymbol{Y}_{\text{bus}}^{-1} \tag{8.4.3}$$

$\boldsymbol{Z}_{\text{bus}}$, the inverse of $\boldsymbol{Y}_{\text{bus}}$, is called the positive-sequence *bus impedance matrix*. Both $\boldsymbol{Z}_{\text{bus}}$ and $\boldsymbol{Y}_{\text{bus}}$ are symmetric matrices.

Since the first circuit contains only one source, located at faulted bus n, the current source vector contains only one nonzero component, $I_n^{(1)} = -I''_{\text{F}n}$. Also, the voltage at faulted bus n in the first circuit is $E_n^{(1)} = -V_{\text{F}}$. Rewriting (8.4.2),

$$\begin{bmatrix} Z_{11} & Z_{12} & \cdots & Z_{1n} & \cdots & Z_{1N} \\ Z_{21} & Z_{22} & \cdots & Z_{2n} & \cdots & Z_{2N} \\ \vdots & & & & & \\ Z_{n1} & Z_{n2} & \cdots & Z_{nn} & \cdots & Z_{nN} \\ \vdots & & & & & \\ Z_{N1} & Z_{N2} & \cdots & Z_{Nn} & \cdots & Z_{NN} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \vdots \\ -I''_{\text{F}n} \\ \vdots \\ 0 \end{bmatrix} - \begin{bmatrix} E_1^{(1)} \\ E_2^{(1)} \\ \vdots \\ -V_{\text{F}} \\ \vdots \\ E_N^{(1)} \end{bmatrix} \tag{8.4.4}$$

The minus sign associated with the current source in (8.4.4) indicates that the current injected into bus n is the negative of I''_{Fn}, since I''_{Fn} flows away from bus n to the neutral. From (8.4.4), the subtransient fault current is

$$I''_{\text{F}n} = \frac{V_{\text{F}}}{Z_{nn}} \tag{8.4.5}$$

Also, from (8.4.4) and (8.4.5), the voltage at any bus k in the first circuit is

$$E_k^{(1)} = Z_{kn}(-I''_{\text{F}n}) = \frac{-Z_{kn}}{Z_{nn}} V_{\text{F}} \tag{8.4.6}$$

The second circuit represents the prefault conditions. Neglecting prefault load current, all voltages throughout the second circuit are equal to the prefault voltage; that is, $E^{(2)}{}_k = V_{\text{F}}$ for each bus k. Applying superposition,

$$\begin{aligned} E_k &= E_k^{(1)} + E_k^{(2)} = \frac{-Z_{kn}}{Z_{nn}} V_{\text{F}} + V_{\text{F}} \\ &= \left(1 - \frac{Z_{kn}}{Z_{nn}}\right) V_{\text{F}} \quad k = 1, 2, \ldots, N \end{aligned} \tag{8.4.7}$$

EXAMPLE 8.4

Using Z_{bus} to compute three-phase short-circuit currents in a power system

The faults at bus 1 and 2 in Figure 8.3 are of interest. The prefault voltage is 1.05 per unit, and prefault load current is neglected. (a) Determine the 2 × 2

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positive-sequence bus impedance matrix. (b) For a bolted three-phase short circuit at bus 1, use $\boldsymbol{Z}_{\text{bus}}$ to calculate the subtransient fault current and the contribution to the fault current from the transmission line. (c) Repeat part (b) for a bolted three-phase short circuit at bus 2.

SOLUTION

a. The circuit of Figure 8.4(a) is redrawn in Figure 8.5 showing per-unit admittance rather than per-unit impedance values. Neglecting prefault load current, $E_g'' = E_m'' = V_{\text{F}} = 1.05\underline{/0^\circ}$ per unit. From Figure 8.5, the positive-sequence bus admittance matrix is

$$\boldsymbol{Y}_{\text{bus}} = -j\begin{bmatrix} 9.9454 & -3.2787 \\ -3.2787 & 8.2787 \end{bmatrix} \quad \text{per unit}$$

Inverting $\boldsymbol{Y}_{\text{bus}}$,

$$\boldsymbol{Z}_{\text{bus}} = \boldsymbol{Y}_{\text{bus}}^{-1} = +j\begin{bmatrix} 0.11565 & 0.04580 \\ 0.04580 & 0.13893 \end{bmatrix} \quad \text{per unit}$$

b. Using (8.4.5), the subtransient fault current at bus 1 is

$$I_{\text{F1}}'' = \frac{V_{\text{F}}}{Z_{11}} = \frac{1.05\underline{/0^\circ}}{j0.11565} = -j9.079 \quad \text{per unit}$$

which agrees with the result in Example 8.3, part (a). The voltages at buses 1 and 2 during the fault are, from (8.4.7),

$$E_1 = \left(1 - \frac{Z_{11}}{Z_{11}}\right)V_{\text{F}} = 0$$

$$E_2 = \left(1 - \frac{Z_{21}}{Z_{11}}\right)V_{\text{F}} = \left(1 - \frac{j0.04580}{j0.11565}\right)1.05\underline{/0^\circ} = 0.6342\underline{/0^\circ}$$

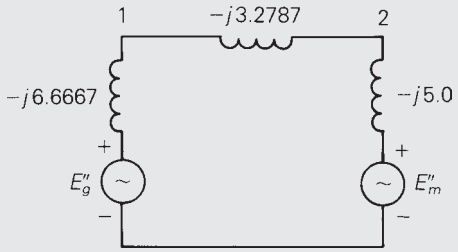

FIGURE 8.5
Circuit of Figure 8.4(a) showing per-unit admittance values

(*Continued*)

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The current to the fault from the transmission line is obtained from the voltage drop from bus 2 to 1 divided by the impedance of the line and transformers T_1 and T_2:

$$I_{21} = \frac{E_2 - E_1}{j(X_{line} + X_{T1} + X_{T2})} = \frac{0.6342 - 0}{j0.3050} = -j2.079 \quad \text{per unit}$$

which agrees with the motor current calculated in Example 8.3, part (b), where prefault load current is neglected.

c. Using (8.4.5), the subtransient fault current at bus 2 is

$$I''_{F2} = \frac{V_F}{Z_{22}} = \frac{1.05\angle 0^\circ}{j0.13893} = -j7.558 \quad \text{per unit}$$

and from (8.4.7),

$$E_1 = \left(1 - \frac{Z_{12}}{Z_{22}}\right)V_F = \left(1 - \frac{j0.04580}{j0.13893}\right)1.05\angle 0^\circ = 0.7039\angle 0^\circ$$

$$E_2 = \left(1 - \frac{Z_{22}}{Z_{22}}\right)V_F = 0$$

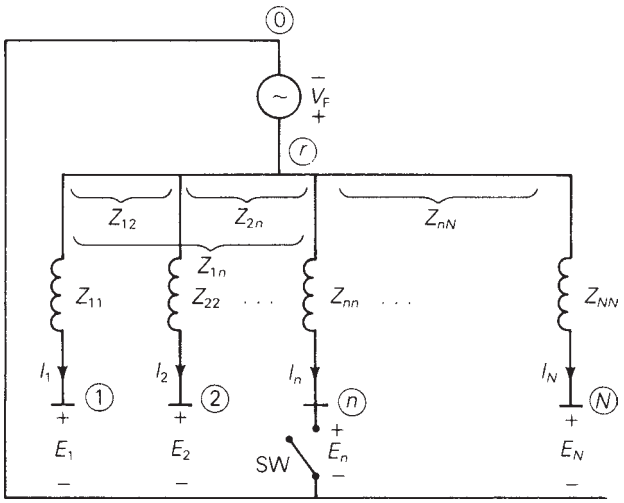
The current to the fault from the transmission line is

$$I_{12} = \frac{E_1 - E_2}{j(X_{line} + X_{T1} + X_{T2})} = \frac{0.7039 - 0}{j0.3050} = -j2.308 \quad \text{per unit}$$

Figure 8.6 shows a bus impedance equivalent circuit that illustrates the short-circuit currents in an N-bus system. This circuit is given the name *rake equivalent* in Neuenswander [5] due to its shape, which is similar to a garden rake.

FIGURE 8.6

Bus impedance equivalent circuit (*rake equivalent*)



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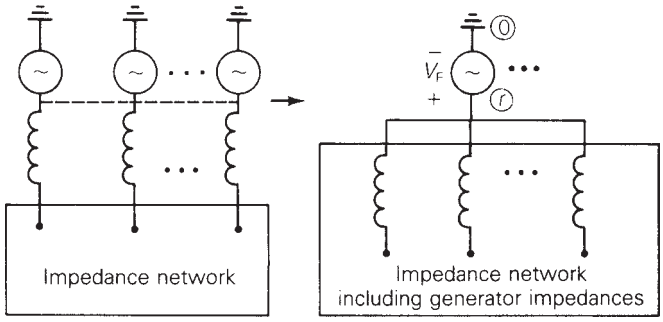

FIGURE 8.7

Parallel connection of unloaded synchronous machine internal-voltage sources

The diagonal elements Z_{11}, Z_{22}, …, Z_{NN} of the bus impedance matrix, which are the *self-impedances*, are shown in Figure 8.6. The off-diagonal elements, or the *mutual impedances*, are indicated by the brackets in the figure.

Neglecting prefault load currents, the internal voltage sources of all synchronous machines are equal both in magnitude and phase. As such, they can be connected, as shown in Figure 8.7, and replaced by one equivalent source V_F from neutral bus 0 to a references bus, denoted r. This equivalent source is also shown in the rake equivalent of Figure 8.6.

Using Z_{bus}, the fault currents in Figure 8.6 are given by

$$\begin{bmatrix} Z_{11} & Z_{12} & \cdots & Z_{1n} & \cdots & Z_{1N} \\ Z_{21} & Z_{22} & \cdots & Z_{2n} & \cdots & Z_{2N} \\ \vdots & & & & & \\ Z_{n1} & Z_{n2} & \cdots & Z_{nn} & \cdots & Z_{nN} \\ \vdots & & & & & \\ Z_{N1} & Z_{N2} & \cdots & Z_{Nn} & \cdots & Z_{NN} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_n \\ \vdots \\ I_N \end{bmatrix} = \begin{bmatrix} V_F - E_1 \\ V_F - E_2 \\ \vdots \\ V_F - E_n \\ \vdots \\ V_F - E_N \end{bmatrix} \tag{8.4.8}$$

where I_1, I_2, … are the branch currents and $(V_F - E_1)$, $(V_F - E_2)$, … are the voltages across the branches.

If switch SW in Figure 8.6 is open, all currents are zero and the voltage at each bus with respect to the neutral equals V_F. This corresponds to prefault conditions, neglecting prefault load currents. If switch SW is closed, corresponding to a short circuit at bus n, $E_n = 0$ and all currents except I_n remain zero. The fault current is $I''_{Fn} = I_n = V_F/Z_{nn}$, which agrees with (8.4.5). This fault current also induces a voltage drop $Z_{kn}I_n = (Z_{kn}/Z_{nn})V_F$ across each branch k. The voltage at bus k with respect to the neutral then equals V_F minus this voltage drop, which agrees with (8.4.7).

As shown by Figure 8.6 as well as (8.4.5), subtransient fault currents throughout an N-bus system can be determined from the bus impedance matrix and the prefault voltage. $\boldsymbol{Z}_{bus}$ can be computed by first constructing $\boldsymbol{Y}_{bus}$, via nodal equations, and then inverting $\boldsymbol{Y}_{bus}$. Once $\boldsymbol{Z}_{bus}$ has been obtained, these fault currents are easily computed.

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EXAMPLE 8.5

PowerWorld Simulator case Example 8_5 models the 5-bus power system whose oneline diagram is shown in Figure 6.2. Machine, line, and transformer data are given in Tables 8.3, 8.4, and 8.5. This system is initially unloaded. Prefault voltages at all the buses are 1.05 per unit. Use PowerWorld Simulator to determine the fault current for three-phase faults at each of the buses.

SOLUTION

To fault a bus from the oneline, first right-click on the bus symbol to display the local menu, and then select **Fault** to display the Fault Analysis dialog (see Figure 8.8). In the list on the left side of the dialog, set the **Fault Definitions** to Single Fault; the Faulted Bus should be automatically selected to the selected bus location. Verify that the Fault Location field is Bus Fault and the Fault Type is

Bus	Machine Subtransient Reactance—X''_d
1	0.045
3	0.0225

TABLE 8.3

Synchronous machine data for the SYMMETRICAL SHORT CIRCUITS program*

*S_{base} = 100 MVA
V_{base} = 15 kV at buses 1, 3
= 345 kV at buses 2, 4, 5

Bus-to-Bus	Equivalent Positive-Sequence Series Reactance (per unit)
2–4	0.1
2–5	0.05
4–5	0.025

TABLE 8.4

Line data for the SYMMETRICAL SHORT CIRCUITS program

Bus-to-Bus	Leakage Reactance—X (per unit)
1–5	0.02
3–4	0.01

TABLE 8.5

Transformer data for the SYMMETRICAL SHORT CIRCUITS program

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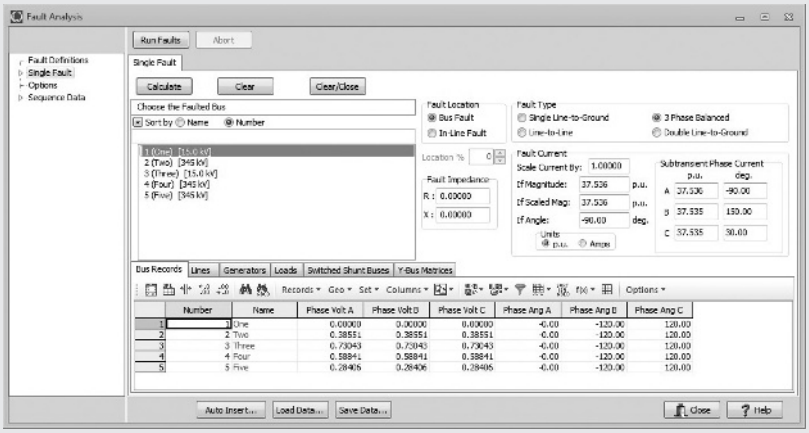

FIGURE 8.8
Fault Analysis Dialog for Example 8.5—fault at bus 1

	0.0279725	0.0177025	0.0085125	0.0122975	0.020405
	0.0177025	0.0569525	0.0136475	0.019715	0.02557
j	0.0085125	0.0136475	0.0182425	0.016353	0.012298
	0.0122975	0.019715	0.016353	0.0236	0.017763
	0.020405	0.02557	0.012298	0.017763	0.029475

TABLE 8.6
Z_{bus} for Example 8.5

3-Phase Balanced (unbalanced faults are covered in Chapter 10). Then select **Calculate**, located in the upper left portion of the dialog, to determine the fault currents and voltages. The results are shown in the tables at the bottom of the dialog. Additionally, the fault currents can be shown on the oneline by selecting **Options** in the left list, and changing the Oneline Display Field value. Since with a three-phase fault the system remains balanced, the magnitudes of the a phase, b phase, and c phase values are identical. The 5×5 $\boldsymbol{Z}_{bus}$ matrix for this system is shown in Table 8.6, and the fault currents and bus voltages for faults at each of the buses are given in Table 8.7. Note that these fault currents are subtransient fault currents, since the machine reactance input data consist of direct axis subtransient reactances.

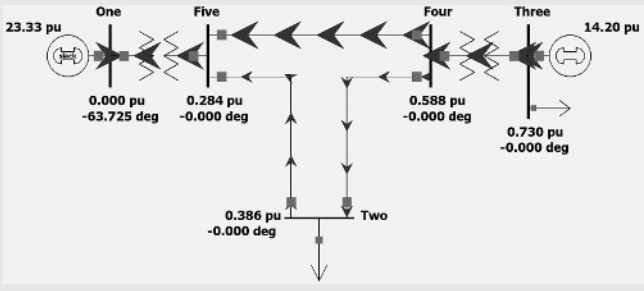

FIGURE 8.9

(*Continued*)

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Fault Bus	Fault Current (per unit)	Contributions to Fault Current		
		Gen Line or TRSF	Bus-to-Bus	Current (per unit)
1	37.536			
		G 1	GRND–1	23.332
		T 1	5–1	14.204
2	18.436			
		L 1	4–2	6.864
		L 2	5–2	11.572
3	57.556			
		G 2	GRND-3	46.668
		T 2	4–3	10.888
4	44.456			
		L 1	2–4	1.736
		L 3	5–4	10.412
		T 2	3–4	32.308
5	35.624			
		L 2	2–5	2.78
		L 3	4–5	16.688
		T 1	1–5	16.152

$V_F = 1.05$ Fault Bus	Per-Unit Bus Voltage Magnitudes During the Fault				
	Bus 1	Bus 2	Bus 3	Bus 4	Bus 5
1	0.0000	0.7236	0.5600	0.5033	0.3231
2	0.3855	0.0000	0.2644	0.1736	0.1391
3	0.7304	0.7984	0.0000	0.3231	0.6119
4	0.5884	0.6865	0.1089	0.0000	0.4172
5	0.2840	0.5786	0.3422	0.2603	0.0000

TABLE 8.7

Fault currents and bus voltages for Example 8.5

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EXAMPLE 8.6

Redo Example 8.5 with an additional line installed between buses 2 and 4. This line, whose reactance is 0.075 per unit, is not mutually coupled to any other line.

SOLUTION

The modified system is contained in PowerWorld Simulator case Example 8_6. The $\boldsymbol{Z}_{\text{bus}}$ along with the fault currents and bus voltages are shown in Tables 8.8 and 8.9. Notice the fault currents have increased, particularly at bus 2, since with the new line the total impedance between the faulted bus and the generators has decreased.

$$\boldsymbol{Z}_{\text{bus}} = j \begin{bmatrix} 0.027723 & 0.01597 & 0.00864 & 0.01248 & 0.02004 \\ 0.01597 & 0.04501 & 0.01452 & 0.02097 & 0.02307 \\ 0.00864 & 0.01452 & 0.01818 & 0.01626 & 0.01248 \\ 0.01248 & 0.02097 & 0.01626 & 0.02349 & 0.01803 \\ 0.02004 & 0.02307 & 0.01248 & 0.01803 & 0.02895 \end{bmatrix}$$

TABLE 8.8

$\boldsymbol{Z}_{\text{bus}}$ for Example 8.6

		Contributions to Fault Current		
Fault Bus	Fault Current (per unit)	Gen Line or TRSF	Bus-to-Bus	Current (per unit)
1	37.872			
		G 1	GRND–1	23.332
		T 1	5–1	14.544
2	23.328			
		L 1	4–2	5.608
		L 2	5–2	10.24
		L 4	4–2	7.48
3	57.756			
		G 2	GRND–3	46.668
		T 2	4–3	11.088
4	44.704			
		L 1	2–4	1.128
		L 3	5–4	9.768
		L 4	2–4	1.504
		T 2	3–4	32.308
5	36.268			
		L 2	2–5	4.268
		L 3	4–5	15.848
		T 1	1–5	16.152

TABLE 8.9 ***(Continued)***

(*Continued*)

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$V_F = 1.05$	Per-Unit Bus Voltage Magnitudes During the Fault				
Fault Bus	Bus 1	Bus 2	Bus 3	Bus 4	Bus 5
1	0.0000	0.6775	0.5510	0.4921	0.3231
2	0.4451	0.0000	0.2117	0.1127	0.2133
3	0.7228	0.7114	0.0000	0.3231	0.5974
4	0.5773	0.5609	0.1109	0.0000	0.3962
5	0.2909	0.5119	0.3293	0.2442	0.0000

TABLE 8.9
Fault currents and bus voltages for Example 8.6

8.5 CIRCUIT BREAKER AND FUSE SELECTION

A SHORT CIRCUITS computer program may be utilized in power system design to select, set, and coordinate protective equipment, such as circuit breakers, fuses, relays, and instrument transformers. In this section, basic principles of circuit breaker and fuse selection are discussed.

AC CIRCUIT BREAKERS

A *circuit breaker* is a mechanical switch capable of interrupting fault currents and of reclosing. When circuit-breaker contacts separate while carrying current, an arc forms. The breaker is designed to extinguish the arc by elongating and cooling it. The fact that ac arc current naturally passes through zero twice during its 60-Hz cycle aids the arc extinction process.

Circuit breakers are classified as *power* circuit breakers when they are intended for service in ac circuits above 1500 V and as *low-voltage* circuit breakers in ac circuits up to 1500 V. There are different types of circuit breakers depending on the medium—air, oil, SF_6 gas, green gas or vacuum—in which the arc is elongated. Also, the arc can be elongated either by a magnetic force or by a blast of air.

Some circuit breakers are equipped with a high-speed automatic reclosing capability. Since most faults are temporary and self-clearing, reclosing is based on the idea that, if a circuit is deenergized for a short time, it is likely that whatever caused the fault has disintegrated and the ionized arc in the fault has dissipated.

When reclosing breakers are employed in EHV systems, standard practice is to reclose only once, approximately 15 to 50 cycles (depending on operating voltage) after the breaker interrupts the fault. If the fault persists and the EHV breaker recloses into it, the breaker reinterrupts the fault current and then "locks out," requiring operator resetting. Multiple-shot reclosing in EHV systems is not standard practice because transient stability (Chapter 12) may be compromised.

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However, for distribution systems (2.4 to 46 kV) where customer outages are of concern, standard reclosers are equipped for two or more reclosures.

For low-voltage applications, molded case circuit breakers with dual trip capability are available. There is a magnetic instantaneous trip for large fault currents above a specified threshold and a thermal trip with time delay for smaller fault currents.

Modern circuit-breaker standards are based on symmetrical interrupting current (Table 8.10). It is usually necessary to calculate only symmetrical fault current at a system location, and then select a breaker with a symmetrical interrupting capability equal to or above the calculated current. The breaker has the additional capability to interrupt the asymmetrical (or total) fault current if the dc offset is not too large.

Recall from Section 8.1 that the maximum asymmetry factor K ($\tau = 0$) is $\sqrt{3}$, which occurs at fault inception ($\tau = 0$). After fault inception, the dc fault current decays exponentially with time constant $T = (L/R) = (X/\omega R)$, and the asymmetry factor decreases. Power circuit breakers with a 2-cycle rated interruption time are designed for an asymmetrical interrupting capability up to 1.4 times their symmetrical interrupting capability, whereas slower circuit breakers have a lower asymmetrical interrupting capability.

A simplified method for breaker selection is called the "E/X simplified method" [1, 7]. A revision to the E/X simplified method [10] is described as follows. The maximum symmetrical short-circuit currents for both three-phase and single-line-to-ground short circuits at the system location in question are calculated from the prefault voltage and system reactance characteristics using computer programs. Resistances, shunt admittances, nonrotating impedance loads, and prefault load currents are neglected. Then, if the X/R ratios for both the maximum three-phase and single-line-to-ground short circuits at the system location are both less than 17, a breaker with a symmetrical interrupting capability equal to or above the calculated symmetrical currents at the given operating voltage is satisfactory. In this case, the breaker has sufficient capability to interrupt both the maximum symmetrical fault current and the maximum dc offset current.

However, if an X/R ratio is greater than 17, the dc offset may not have decayed to a sufficiently low value. In this case, a method for correcting the calculated fault current to account for dc and ac time constants as well as breaker speed can be used [10]. If X/R is unknown, the simplified E/X method may also be used by selecting a circuit breaker such that the calculated maximum three-phase symmetrical fault current is not greater than 80% of the breaker's symmetrical interrupting capability, and the calculated maximum single-line-to-ground symmetrical fault current is not greater than 70% of the breaker's symmetrical interrupting capability.

When selecting circuit breakers for generators, two cycle breakers are employed in practice, and the subtransient fault current is calculated; therefore, subtransient machine reactances X_d'' are used in fault calculations. For synchronous motors, subtransient reactances X_d'' or transient reactances X_d' are used, depending on breaker speed. Also, induction motors can momentarily contribute to fault

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Line No.	Rated Maximum Voltage U_r kV, rms	Rated Continuous Current A, rms	Rated Short-Circuit and Short-Time Current kA, rms	Rated Interrupting Time ms	Maximum Permissible Tripping Time Delay Y, s	Rated Closing and Latching Current kA, peak
	Col 1	Col 2	Col 3	Col 4	Col 5	Col 6
1	123	1200,2000	31.5	50	1	82
2	123	2000, 3000, 4000	40	50	1	104
3	123	2000, 3000, 4000	50	50	1	130
4	123	2000, 3000, 4000	63	50	1	164
5	145	1200, 2000	31.5	33 or 50	1	82
6	145	1600, 2000, 3000	40	33 or 50	1	104
7	145	2000, 3000	50	33 or 50	1	130
8	145	2000, 3000	63	33 or 50	1	164
9	145	2000, 3000, 4000	80	33 or 50	1	208
10	170	1600, 2000	31.5	33 or 50	1	82
11	170	2000, 3000	40	33 or 50	1	104
12	170	2000, 3000, 4000, 5000	50	33 or 50	1	130
13	170	2000, 3000, 4000, 5000	63	33 or 50	1	164
14	170	3000, 4000, 5000	80	33 or 50	1	208
15	170	4000, 5000	100	33 or 50	1	260
16	245	1600, 2000, 3000	31.5	33 or 50	1	82
17	245	2000, 3000	40	33 or 50	1	104
18	245	2000, 3000	50	33 or 50	1	130
19	245	2000, 3000, 4000, 5000	63	33 or 50	1	164
20	245	3000, 4000, 5000	80	33 or 50	1	208
21	362	2000, 3000	40	33 or 50	1	104
22	362	2000, 3000	50	33 or 50	1	130
23	362	2000, 3000, 4000	63	33 or 50	1	164
24	550	2000, 3000	40	33	1	104
25	550	3000, 4000	50	33	1	130
26	550	3000, 4000	63	33	1	164
27	800	2000, 3000	40	33	1	104
28	800	3000, 4000	50	33	1	130
29	800	3000, 4000	63	33	1	164

TABLE 8.10

Preferred ratings for outdoor circuit breakers rated 100 kV and above, including circuit breakers applied in gas-insulated substations (symmetrical current basis of rating)

Source: © 2009 IEEE. Reprinted, with permission from IEEE Std C37.06-2009, IEEE Standard for AC High-Voltage Circuit Breakers Rated on a Symmetrical Current Basis - Preferred Ratings and Related Required Capabilities for Voltages Above 1000 V.

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current. Large induction motors are usually modeled as sources in series with X_d'' or X_d', depending on breaker speed. Smaller induction motors (below 50 hp) are often neglected entirely.

Table 8.10 shows a schedule of preferred ratings for outdoor power circuit breakers [11]. The ratings shown in Table 8.10 are described next.

Rated maximum voltage: Designates the maximum rms line-to-line operating voltage. The breaker should be used in systems with an operating voltage less than or equal to this rating.

Rated continuous current: The maximum 60-Hz rms current that the breaker can carry continuously while it is in the closed position without overheating.

Rated short-circuit current: The maximum rms symmetrical current that the breaker can safely interrupt at rated maximum voltage.

Rated short-time current: The maximum short-circuit current that the breaker can carry in the closed position (without tripping) for a specified period of time. As shown in Table 8.10, the rated short-time current is equal to the rated short-circuit current.

Rated interrupting time: The time in ms from the instant the trip coil is energized to the instant that all three phases of the fault current are cleared.

Maximum permissible tripping time delay: The maximum delay time in seconds for protective relaying to trip the circuit breaker during short-circuit conditions, based upon the rated short-time current and short-time current-carrying time period.

Rated closing and latching current: The peak 60-Hz current that the breaker is capable of closing and latching. As shown in Table 8.10, the rated closing and latching current is 2.6 times the rated short-circuit current.

EXAMPLE 8.7

Circuit breaker selection with known X/R ratios

The calculated maximum symmetrical fault currents are 35 kA for a three-phase short circuit and 32 kA for a single-line-to-ground short circuit at a three-phase bus where the maximum operating voltage is 360 kV. The X/R ratios at the bus are 16.5 for the three-phase short circuit and 14.8 for the single-line-to-ground short circuit. The maximum continuous current, which occurs during either the most recent peak load conditions or any peak forecasted for the next five years, is 2400 A. Select a circuit breaker from Table 8.10 for this bus.

SOLUTION

Since the maximum operating voltage is 360 kV, select a circuit breaker from Lines 21 to 23 in Table 8.10, where the rated maximum voltage is 362 kV

(*Continued*)

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(line-to-line rms). Also, for a maximum continuous current of 2400 A, select a breaker in Lines 21-23 with a 3000-A (rms) rated continuous current. Since the X/R ratios for the three-phase and single line-to-ground short circuits are both less than 17, select a breaker with a 40-kA rated short-circuit current, which exceeds both the maximum three-phase symmetrical fault current (35 kA) and the maximum single-line-to-ground symmetrical fault current (32 kA). Thus, select a breaker from Line 21 with a 3000-A continuous current rating, a 362-kV rated maximum voltage, and a 40-kA rated short-circuit current.

EXAMPLE 8.8

Circuit breaker selection with unknown X/R ratios

Repeat Example 8.7 if the X/R ratios are unknown.

SOLUTION

We still select a circuit breaker with 3000-A continuous current from Lines 21–23 in Table 8.10, where the rated maximum voltage is 362 kV. Since the X/R ratios are unknown, select the circuit breaker in Line 22 with a 50-kA rated short-circuit current, such that the calculated maximum three-phase symmetrical fault current (35 kA) is not greater than 80% of the breaker's symmetrical interrupting capability (i.e., 35 kA is not greater than $0.8 \times 50 = 40$ kA), and the calculated maximum single-line-to-ground symmetrical fault current (32 kA) is not greater than 70% of the breaker's symmetrical interrupting capability (i.e., 32 kA is not greater than $0.7 \times 50 = 35$ kA). Thus, select a breaker from Line 22 with a 3000-A continuous current rating, a 362-kV rated maximum voltage, and a 50-kA rated short-circuit current.

FUSES

Figure 8.10(a) shows a cutaway view of a fuse, which is one of the simplest overcurrent devices. The fuse consists of a metal "fusible" link or links encapsulated in a tube, packed in filler material, and connected to contact terminals. Silver is a typical link metal, and sand is a typical filler material.

During normal operation, when the fuse is operating below its continuous current rating, the electrical resistance of the link is so low that it simply acts as a conductor. If an overload current from one to about six times its continuous current rating occurs and persists for more than a short interval of time, the

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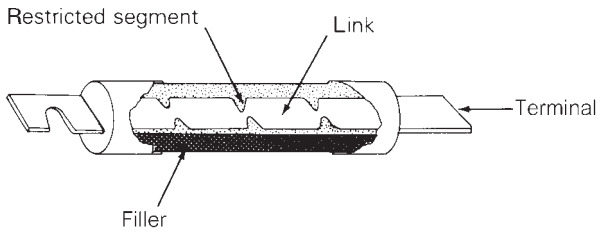

(a) Cutaway view

(b) The link melts and an arc is established under sustained overload current

(c) The "open" link after clearing the overload current

FIGURE 8.10
Typical fuse

temperature of the link eventually reaches a level that causes a restricted segment of the link to melt. As shown in Figure 8.10(b), a gap is then formed, and an electric arc is established. As the arc causes the link metal to burn back, the gap width increases. The resistance of the arc eventually reaches such a high level that the arc cannot be sustained, and it is extinguished, as in Figure 8.10(c). The current flow within the fuse is then completely cut off.

If the fuse is subjected to fault currents higher than about six times its continuous current rating, several restricted segments melt simultaneously, resulting in rapid arc suppression and fault clearing. Arc suppression is accelerated by the filler material in the fuse.

Many modern fuses are current limiting. As shown in Figure 8.11, a current-limiting fuse has such a high speed of response that it cuts off a high fault current in less than a half cycle—before it can build up to its full peak value. By limiting fault currents, these fuses permit the use of motors, transformers, conductors, and bus structures that could not otherwise withstand the destructive forces of high fault currents.

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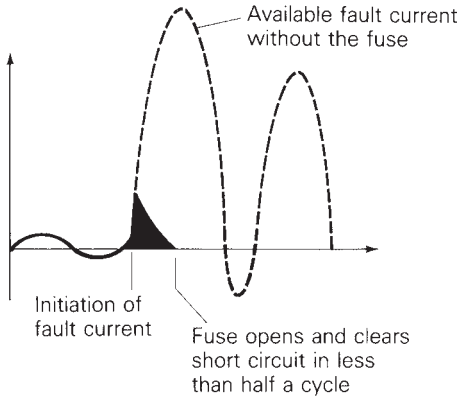
FIGURE 8.11

Operation of a current-limiting fuse



Fuse specification is normally based on the following four factors.

1. *Voltage rating*. This rms voltage determines the ability of a fuse to suppress the internal arc that occurs after the fuse link melts. A blown fuse should be able to withstand its voltage rating. Most low-voltage fuses have 250 or 600 V ratings. Ratings of medium-voltage fuses range from 2.4 to 34.5 kV.
2. *Continuous current rating*. The fuse should carry this rms current indefinitely, without melting and clearing.
3. *Interrupting current rating*. This is the largest rms asymmetrical current that the fuse can safely interrupt. Most modern, low-voltage current-limiting fuses have a 200-kA interrupting rating. Standard interrupting ratings for medium-voltage current-limiting fuses include 65, 80, and 100 kA.
4. *Time response*. The melting and clearing time of a fuse depends on the magnitude of the overcurrent or fault current and is usually specified by a "time-current" curve. Figure 8.12 shows the time-current curve of a 15.5-kV, 100-A (continuous) current-limiting fuse. As shown, the fuse link melts within 2 s and clears within 5 s for a 500-A current. For a 5-kA current, the fuse link melts in less than 0.01 s and clears within 0.015 s.

It is usually a simple matter to coordinate fuses in a power circuit such that only the fuse closest to the fault opens the circuit. In a radial circuit, fuses with larger continuous-current ratings are located closer to the source, such that the fuse closest to the fault clears before other, upstream fuses melt.

Fuses are inexpensive, fast operating, easily coordinated, and reliable, and they do not require protective relays or instrument transformers. Their chief disadvantage is that the fuse or the fuse link must be manually replaced after it melts. They are basically one-shot devices that are, for example, incapable of high-speed reclosing.

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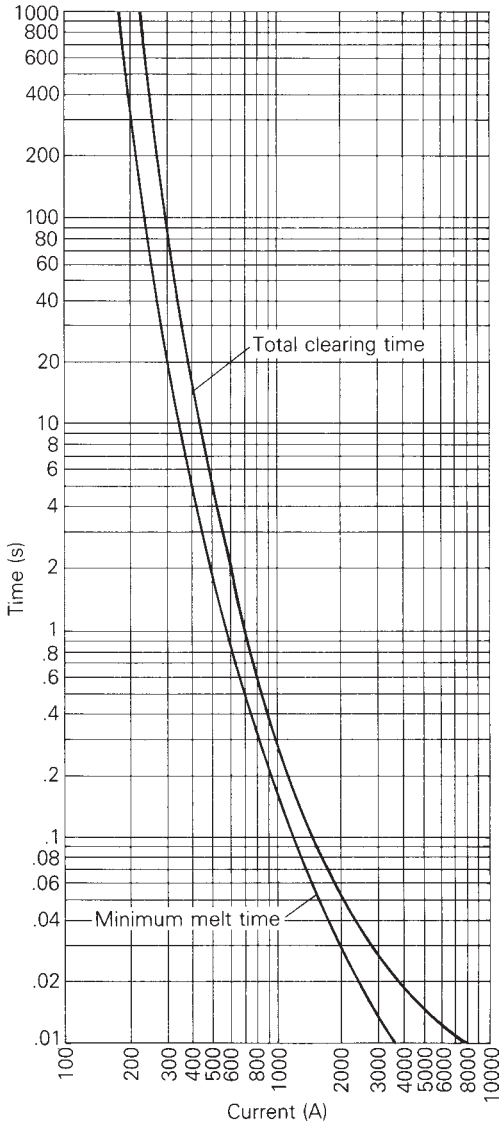

FIGURE 8.12

Time-current curves for a 15.5-kV, 100-A current-limiting fuse

PROBLEMS

SECTION 8.1

8.1 In the circuit of Figure 8.1, V = 277 volts, L = 3 mH, R = 0.4 Ω, and $\omega = 2\pi 60$ rad/s. Determine (a) the rms symmetrical fault current; (b) the rms asymmetrical fault current at the instant the switch closes, assuming maximum dc offset; (c) the rms asymmetrical fault current five cycles after the switch closes, assuming maximum dc offset; and

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(d) the dc offset as a function of time if the switch closes when the instantaneous source voltage is 300 volts.

8.2 Repeat Example 8.1 with V = 4 kV, X = 3 Ω, and R = 1 Ω.

8.3 In the circuit of Figure 8.1, let R = 0.125 Ω, L = 10 mH, and the source voltage is $e(t) = 151 \sin(377t + \alpha)$ V. Determine the current response after closing the switch for the following cases: (a) no dc offset, or (b) maximum dc offset. Sketch the current waveform up to t = 0.10 s corresponding to parts (a) and (b).

8.4 Consider the expression for $i(t)$ given by

$$i(t) = \sqrt{2}\mathrm{I}_{\mathrm{rms}}[\sin(\omega t - \theta_z) + \sin\theta_z e^{-(\omega R/X)t}]$$

where $\theta_z = \tan^{-1}(\omega L/R)$.

(a) For (X/R) equal to zero and infinity, plot $i(t)$ as a function of (ωt)

(b) Comment on the dc offset of the fault current waveforms.

(c) Find the asymmetrical current factor and the time of peak, t_p, in milliseconds, for (X/R) ratios of zero and infinity.

8.5 If the source impedance at a 13.2-kV distribution substation bus is $(0.5 + j1.5)$ Ω per phase, compute the rms and maximum peak instantaneous value of the fault current for a balanced three-phase fault. For the system (X/R) ratio of 3.0, the asymmetrical factor is 1.9495 and the time of peak is 7.1 ms (see Problem 8.4). Comment on the withstanding peak current capability to which all substation electrical equipment need to be designed.

SECTION 8.2

8.6 A 1000-MVA, 20-kV, 60-Hz, three-phase generator is connected through a 1000-MVA, 20-kV, Δ/345-kV, Y transformer to a 345-kV circuit breaker and a 345-kV transmission line. The generator reactances are $\mathrm{X}_d'' = 0.17$, $\mathrm{X}_d' = 0.30$, and $\mathrm{X}_d = 1.5$ per unit, and its time constants are $\mathrm{T}_d'' = 0.05$, $\mathrm{T}_d' = 1.0$, and $\mathrm{T}_\mathrm{A} = 0.10$s. The transformer series reactance is 0.10 per unit; transformer losses and exciting current are neglected. A three-phase short circuit occurs on the line side of the circuit breaker when the generator is operated at rated terminal voltage and at no-load. The breaker interrupts the fault three cycles after fault inception. Determine (a) the subtransient current through the breaker in per-unit and in kA rms, and (b) the rms asymmetrical fault current the breaker interrupts, assuming maximum dc offset. Neglect the effect of the transformer on the time constants.

8.7 For Problem 8.6, determine (a) the instantaneous symmetrical fault current in kA in phase *a* of the generator as a function of time, assuming maximum dc offset occurs in this generator phase, and (b) the maximum dc offset current in kA as a function of time that can occur in any one generator phase.

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8.8 A 300-MVA, 13.8-kV, three-phase, 60-Hz, Y-connected synchronous generator is adjusted to produce rated voltage on open circuit. A balanced three-phase fault is applied to the terminals at t = 0. After analyzing the raw data, the symmetrical transient current is obtained as

$$i_{ac}(t) = 10^4(1 + e^{-t/\tau_1} + 6e^{-t/\tau_2}) \cdot \sin\left(120\pi t - \frac{\pi}{4}\right) \text{ A}$$

where $\tau_1 = 200$ ms and $\tau_2 = 15$ ms.

(a) Sketch $i_{ac}(t)$ as a function of time for $0 \le t \le 500$ ms. (b) Determine X_d'' and X_d in per unit based on the machine ratings.

8.9 Two identical synchronous machines, each rated 60 MVA and 15 kV with a subtransient reactance of 0.1 p.u., are connected through a line of reactance 0.1 p.u. on the base of the machine rating. One machine is acting as a synchronous generator, while the other is working as a motor drawing 40 MW at 0.8 p.f. leading with a terminal voltage of 14.5 kV, when a symmetrical three-phase fault occurs at the motor terminals. Determine the subtransient currents in the generator, the motor, and the fault by using the internal voltages of the machines. Choose a base of 60 MVA and 15 kV in the generator circuit.

SECTION 8.3

8.10 Recalculate the subtransient current through the breaker in Problem 8.6 if the generator is initially delivering rated MVA at 0.80 p.f. lagging and at rated terminal voltage.

8.11 Solve Example 8.3 parts (a) and (c) without using the superposition principle. First calculate the internal machine voltages E_g'' and E_m'' using the prefault load current. Then determine the subtransient fault, generator, and motor currents directly from Figure 8.4(a). Compare your answers with those of Example 8.3.

8.12 Equipment ratings for the four-bus power system shown in Figure 8.13 are as follows:

Generator Gl:	500 MVA, 13.8 kV, X″ = 0.20 per unit
Generator G2:	750 MVA, 18 kV, X″ = 0.18 per unit
Generator G3:	1000 MVA, 20 kV, X″ = 0.17 per unit
Transformer Tl:	500 MVA, 13.8 Δ/500 Y kV, X = 0.12 per unit
Transformer T2:	750 MVA, 18 Δ/500 Y kV, X = 0.10 per unit
Transformer T3:	1000 MVA, 20 Δ/500 Y kV, X = 0.10 per unit
Each 500-kV line:	$X_1 = 50\ \Omega$

A three-phase short circuit occurs at bus 1, where the prefault voltage is 525 kV. Prefault load current is neglected. Draw the positive-sequence reactance diagram in per unit on a 1000-MVA, 20-kV base in the zone of generator G3. Determine (a) the Thévenin reactance in per unit at the fault, (b) the subtransient fault current in per unit and in kA rms, and (c) contributions to the fault current from generator Gl and from line 1–2.

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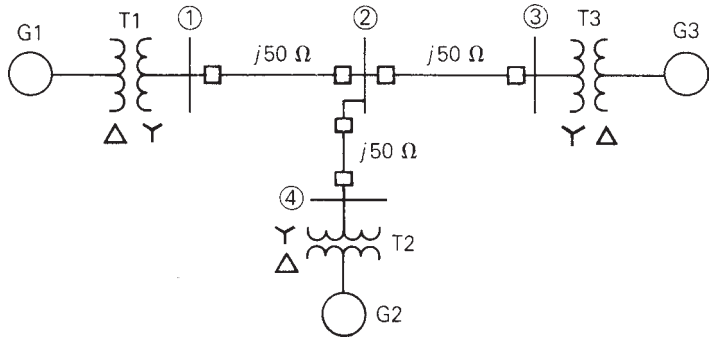
FIGURE 8.13
Problems 8.12, 8.13, 8.19, 8.24, 8.25, and 8.26



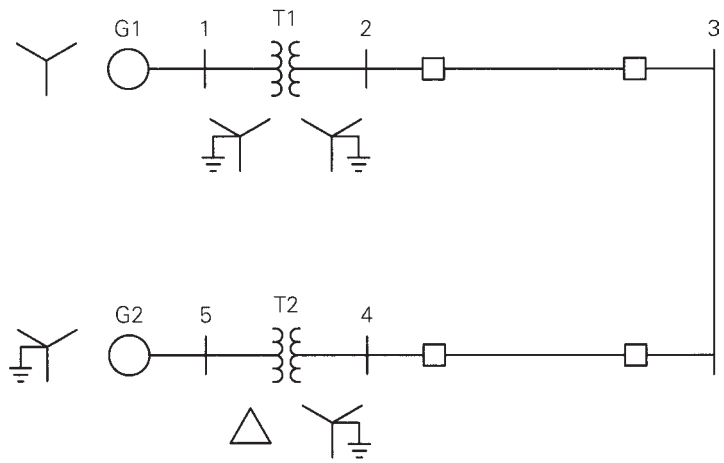
8.13 For the power system given in Problem 8.12, a three-phase short circuit occurs at bus 2, where the prefault voltage is 525 kV. Prefault load current is neglected. Determine the (a) Thévenin equivalent at the fault, (b) subtransient fault current in per unit and in kA rms, and (c) contributions to the fault from lines 1–2, 2–3, and 2–4.

8.14 Equipment ratings for the five-bus power system shown in Figure 8.14 are as follows:

Generator Gl:	50 MVA, 12 kV, X″ = 0.2 per unit
Generator G2:	100 MVA, 15 kV, X″ = 0.2 per unit
Transformer Tl:	50 MVA, 10 kV Y/138 kV Y, X = 0.10 per unit
Transformer T2:	100 MVA, 15 kV Δ/138 kV Y, X = 0.10 per unit
Each 138-kV line:	$X_1 = 40\ \Omega$.

A three-phase short circuit occurs at bus 5, where the prefault voltage is 15 kV. Prefault load current is neglected. (a) Draw the positive-sequence reactance diagram in per unit on a 100-MVA, 15-kV base in the zone of generator G2. Determine (b) the Thévenin equivalent at the fault, (c) the subtransient fault current in per unit and in kA rms, and (d) contributions to the fault from generator G2 and from transformer T2.

FIGURE 8.14
Problems 8.14, 8.15, and 8.20



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8.15 For the power system given in Problem 8.14, a three-phase short circuit occurs at bus 4, where the prefault voltage is 138 kV. Prefault load current is neglected. Determine (a) the Thévenin equivalent at the fault, (b) the subtransient fault current in per unit and in kA rms, and (c) contributions to the fault from transformer T2 and from line 3–4.

8.16 In the system shown in Figure 8.15, a three-phase short circuit occurs at point F. Assume that prefault currents are zero and that the generators are operating at rated voltage. Determine the symmetrical fault current.

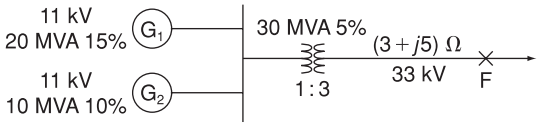

FIGURE 8.15
Problem 8.16

8.17 A three-phase short circuit occurs at the generator bus (bus 1) for the system shown in Figure 8.16. Neglecting prefault currents and assuming that the generator is operating at its rated voltage, determine the subtransient fault current using superposition. Percent direct axis subtransient reactances are shown in Figure 8.16.

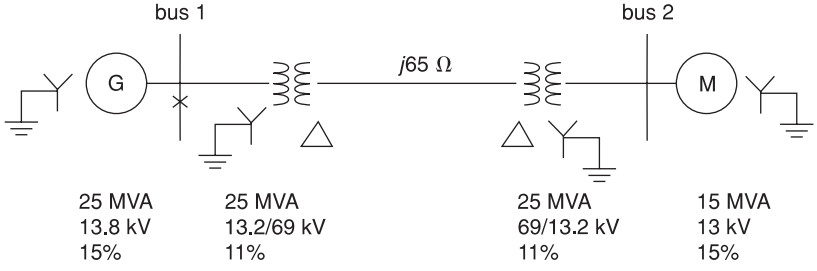

FIGURE 8.16
Problem 8.17

SECTION 8.4

8.18 First, the bus impedance matrix for a three-bus power system is

$$\boldsymbol{Z}_{\text{bus}} = j\begin{bmatrix} 0.12 & 0.08 & 0.04 \\ 0.08 & 0.12 & 0.06 \\ 0.04 & 0.06 & 0.08 \end{bmatrix} \text{ per unit}$$

where subtransient reactances were used to compute $\boldsymbol{Z}_{\text{bus}}$. Prefault voltage is 1.0 per unit and prefault current is neglected. (a) Draw the bus impedance matrix equivalent circuit (rake equivalent). Identify the per-unit self- and mutual impedances as well as the prefault voltage in the circuit. (b) A three-phase short circuit occurs at bus 2. Determine the subtransient fault current and the voltages at buses 1, 2, and 3 during the fault.

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Second, repeat for the case of

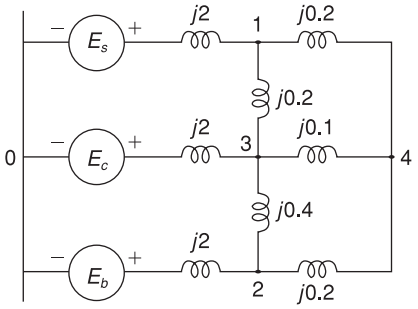
$$\boldsymbol{Z}_{\text{bus}} = j\begin{bmatrix} 0.4 & 0.1 & 0.3 \\ 0.1 & 0.8 & 0.5 \\ 0.3 & 0.5 & 1.2 \end{bmatrix} \text{ per unit}$$

8.19 Determine $\boldsymbol{Y}_{\text{bus}}$ in per unit for the circuit in Problem 8.12. Then invert $\boldsymbol{Y}_{\text{bus}}$ to obtain $\boldsymbol{Z}_{\text{bus}}$.

8.20 Determine $\boldsymbol{Y}_{\text{bus}}$ in per unit for the circuit in Problem 8.14. Then invert $\boldsymbol{Y}_{\text{bus}}$ to obtain $\boldsymbol{Z}_{\text{bus}}$.

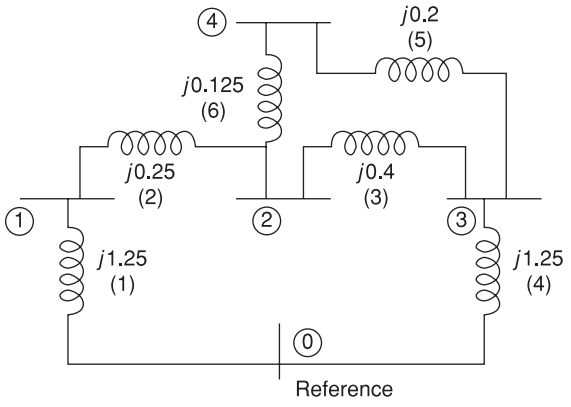
8.21 Figure 8.17 shows a system reactance diagram. (a) Draw the admittance diagram for the system by using source transformations. (b) Find the bus admittance matrix $\boldsymbol{Y}_{\text{bus}}$. (c) Find the bus impedance $\boldsymbol{Z}_{\text{bus}}$ matrix by inverting $\boldsymbol{Y}_{\text{bus}}$.

FIGURE 8.17
Problem 8.21



8.22 For the network shown in Figure 8.18, impedances labeled 1 through 6 are in per unit. (a) Determine $\boldsymbol{Y}_{\text{bus}}$, preserving all buses. (b) Using MATLAB or a similar computer program, invert $\boldsymbol{Y}_{\text{bus}}$ to obtain $\boldsymbol{Z}_{\text{bus}}$.

FIGURE 8.18
Problem 8.22



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8.23 A single-line diagram of a four-bus system is shown in Figure 8.19, for which $\boldsymbol{Z}_{\text{bus}}$ is given below:

$$\boldsymbol{Z}_{\text{bus}} = j\begin{bmatrix} 0.25 & 0.2 & 0.16 & 0.14 \\ 0.2 & 0.23 & 0.15 & 0.151 \\ 0.16 & 0.15 & 0.196 & 0.1 \\ 0.14 & 0.151 & 0.1 & 0.195 \end{bmatrix} \text{ per unit}$$

Let a three-phase fault occur at bus 2 of the network.

(a) Calculate the initial symmetrical rms current in the fault.
(b) Determine the voltages during the fault at buses 1, 3, and 4.
(c) Compute the fault currents contributed to bus 2 by the adjacent unfaulted buses 1, 3, and 4.
(d) Find the current flow in the line from bus 3 to bus 1. Assume the prefault voltage V_f at bus 2 to be $1\angle 0°$ p.u., and neglect all prefault currents.

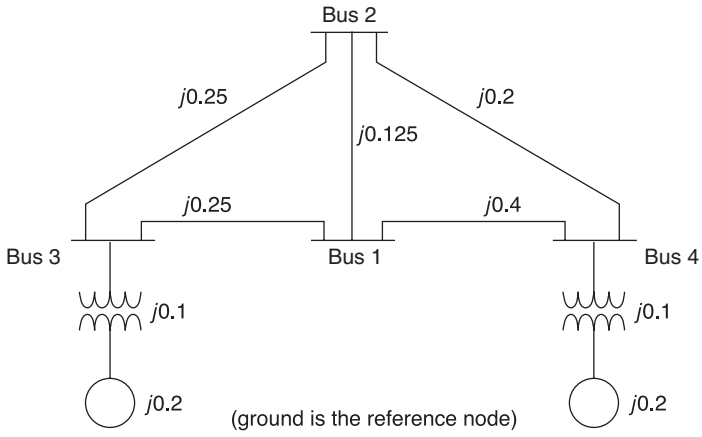

FIGURE 8.19
Single-line diagram for Problem 8.23

PW 8.24 PowerWorld Simulator case Problem 8_24 models the system shown in Figure 8.13 with all data on a 1000-MVA base. Using PowerWorld Simulator, determine the current supplied by each generator and the per-unit bus voltage magnitudes at each bus for a three-phase fault at bus 3.

PW 8.25 Repeat Problem 8.24, except place the fault at bus 4.

PW 8.26 Repeat Problem 8.24, except place the fault midway between buses 2 and 3. Determining the values for line faults requires that the line be split with a fictitious bus added at the point of the fault. The original line's impedance is then allocated to the two new lines based on the fault location, which is 50% each for this problem. Fault calculations are then the same as for a bus fault. This is done automatically in

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PowerWorld Simulator by first right-clicking on a line and then selecting Fault. The Fault dialog appears as before, except now the fault type is changed to In-Line Fault. Set the location percentage field to 50% to model a fault midway between buses 2 and 4.

PW 8.27 One technique for limiting fault current is to place reactance in series with the generators. Such reactance can be modeled in PowerWorld Simulator by increasing the value of the generator's positive sequence internal impedance. For the Problem 8.24 case, how much per-unit reactance must be added to G2 to limit its maximum fault current to 2.5 per unit for all three-phase bus faults? Where is the location of the most severe bus fault?

PW 8.28 Using PowerWorld Simulator case Example 6_13, determine the per-unit current and actual current in amps supplied by each of the generators for a fault at the POPLAR69 bus. During the fault, what percentage of the system buses have voltage magnitudes below 0.75 per unit?

PW 8.29 Repeat Problem 8.28, except place the fault at the REDBUD69 bus.

PW 8.30 Using PowerWorld Simulator case Example 8_5, open the line connecting buses 4 and 5. Then, determine the per unit current supplied by the generator at bus 3 due to a fault at bus 2.

SECTION 8.5

8.31 A three-phase circuit breaker is to be installed on a system with a maximum operating voltage of 110 kV, which should be able to carry at least 1800 A of continuous current. The X/R ratio is measured to be 8.4 for both three-phase and single-line-to-ground faults. If the short-circuit current is calculated to be 37 kA for a three-phase fault and 32 kA for a single-line-to-ground fault, which breaker from Table 8.10 should be selected?

8.32 A 345-kV, three-phase transmission line has a 2.2-kA continuous current rating and a 2.5-kA maximum short-time overload rating with a 356-kV maximum operating voltage. The maximum calculated symmetrical fault current on the line is 30 kA for a three-phase short circuit and 27 kA for a single-line-to-ground short circuit. Select a circuit breaker for this line from Table 8.10. Assume the X/R ratios are less than 17.

8.33 Rework Problem 8.31, assuming the X/R ratio is 17.2.

8.34 As shown in Figure 8.20, a 25-MVA, 13.8-kV, 60-Hz, synchronous generator with $X_d'' = 0.15$ per unit is connected through a transformer to a bus that supplies four identical motors. The rating of the three-phase transformer is 25 MVA and 13.8/6.9 kV with a leakage reactance of 0.1 per unit. Each motor has a subtransient reactance $X_d'' = 0.2$ per unit on a base of 5 MVA and 6.9 kV. A three-phase fault occurs at point P, when the bus voltage at the motors is 6.9 kV. Determine (a) the subtransient fault current, (b) the subtransient current

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through breaker A, and (c) the symmetrical short-circuit interrupting current (as defined for circuit breaker applications) in the fault and in breaker A.

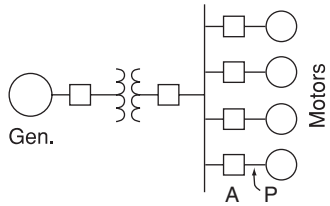

FIGURE 8.20
Problem 8.34

DESIGN PROJECT 3 (*CONTINUED*): POWER FLOW/ SHORT CIRCUITS

Additional time given: 3 weeks

Additional time required: 10 hours

This is a continuation of Design Project 3. Assignments 1 and 2 are given in Chapter 6.

Assignment 3: Symmetrical Short Circuits

For the single-line diagram that you have been assigned (Figure 6.25 or 6.26), convert the positive-sequence reactance data to per unit using the given base quantities. For synchronous machines, use subtransient reactance. Then using PowerWorld Simulator, create the machine, transmission line, and transformer input data files. Next, run the program to compute subtransient fault currents for a bolted three-phase-to-ground fault at bus 1, then at bus 2, then at bus 3, and so on. Also, compute bus voltages during the faults and the positive-sequence bus impedance matrix. Assume 1.0 per-unit prefault voltage. Neglect prefault load currents and all losses.

Your output for this assignment consists of three input data files and three output data (fault currents, bus voltages, and the bus impedance matrix) files.

This project continues in Chapter 10.

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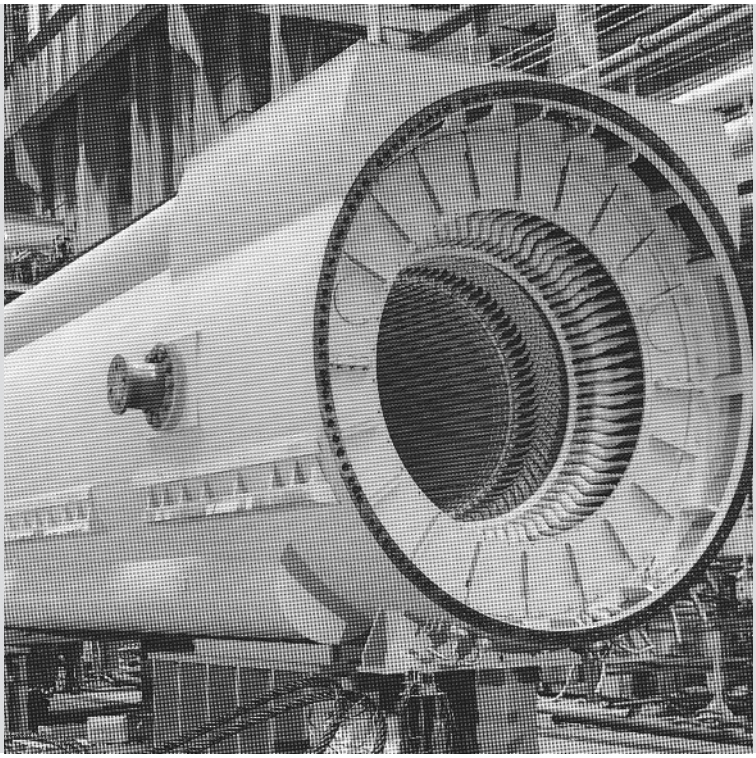
9 Symmetrical Components



Generator stator showing completed windings for a 757-MVA, 3600-rpm, 60-Hz synchronous generator. (Courtesy of General Electric.)

LEARNING OBJECTIVES

At the conclusion of this chapter, you should be able to:

1. Know the definition of symmetrical components including the zero-, positive-, and negative-sequence components, as well as the definition of the complex number a.
2. Understand the per-unit sequence networks of the following: three-phase impedance loads, three-phase series impedances, three-phase lines, rotating machines, and three-phase transformers.
3. Know the relation between power in three-phase networks and power in sequence networks.

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The method of symmetrical components, first developed by C. L. Fortescue in 1918, is a powerful technique for analyzing unbalanced three-phase systems. Fortescue defined a linear transformation from phase components to a new set of components called *symmetrical components.* The advantage of this transformation is that for balanced three-phase networks the equivalent circuits obtained for the symmetrical components, called *sequence networks*, are separated into three uncoupled networks. Furthermore, for unbalanced three-phase systems, the three sequence networks are connected only at points of unbalance. As a result, sequence networks for many cases of unbalanced three-phase systems are relatively easy to analyze.

The symmetrical component method is basically a modeling technique that permits systematic analysis and design of three-phase systems. Decoupling a detailed three-phase network into three simpler sequence networks reveals complicated phenomena in more simplistic terms. Sequence network results then can be superposed to obtain three-phase network results. As an example, the application of symmetrical components to unsymmetrical short-circuit studies (see Chapter 10) is indispensable.

The objective of this chapter is to introduce the concept of symmetrical components in order to lay a foundation and provide a framework for later chapters covering both equipment models as well as power system analysis and design methods. Section 9.1 defines symmetrical components. In Sections 9.2 through 9.7, sequence networks of loads, series impedances, transmission lines, rotating machines, and transformers are presented. Complex power in sequence networks is presented in Section 9.8. Although Fortescue's original work is valid for polyphase systems with n phases, this chapter considers only three-phase systems.

CASE STUDY

The case study in the previous chapter highlights pumped storage hydro, which has long dominated energy storage in modern electric power grids. But pumped storage hydro requires two large reservoirs of water at different elevations. The following case study describes an alternative energy storage method currently under development using the same basic principle, lifting a mass and letting it drop, with the advantage that it can fit almost anywhere [4].

The Ups and Downs of Gravity Energy Storage*

Samuel K. Moore

Cranes are a familiar fixture of practically any city skyline, but one in the Swiss City of Ticino, near the Italian border, would stand out anywhere: It has six arms. This 110-meter-high starfish of the skyline isn't intended for construction. It's meant to prove that renewable energy can

*©2021 IEEE. Reprinted with permission from Samuel K. Moore, "The Ups and Downs of Gravity Energy Storage," *IEEE Spectrum*, January 2021, pp. 38–39.

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SKYLINE STARFISH: Energy Vault's concept demonstrator has been hooked to the grid in Ticino, Switzerland, since July 2020. By raising and lowering 35-metric-ton blocks [not shown] the tower stores and releases energy.

be stored by hefting heavy loads and dispatched by releasing them.

Energy Vault, the Swiss company that built the structure, has already begun a test program that will lead to its first commercial deployments in 2021. At least one competitor, Gravitricity, in Scotland, is nearing the same point. And there are at least two companies with similar ideas, New Energy Let's Go and Gravity Power, that are searching for the funding to push forward.

To be sure, nearly all the world's currently operational energy storage facilities, which can generate a total of 174 gigawatts, rely on gravity. Pumped hydro storage, where water is pumped to a higher elevation and then run back through a turbine to generate electricity, has long dominated the energy-storage landscape. But pumped hydro requires some very specific geography—two big reservoirs of water at elevations with a vertical separation that's large, but not too large. So building new sites is difficult.

Energy Vault, Gravity Power, and their competitors seek to use the same basic principle—lifting a mass and letting it drop—while making an energy-storage facility that can fit almost anywhere. At the same time they hope to best batteries—the new darling of renewable-energy storage—by offering lower long-term costs and fewer environmental issues.

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In action, Energy Vault's towers are constantly stacking and unstacking 35-metric-ton bricks arrayed in concentric rings. Bricks in an inner ring, for example, might be stacked up to store 35 megawatt-hours of energy. Then the system's six arms would systematically disassemble it, lowering the bricks to build an outer ring and discharging energy in the process.

This joule-storing Jenga game can be complicated. To maintain a constant output, one block needs to be accelerating while another is decelerating. "That's why we use six arms," explains Robert Piconi, the company's CEO and cofounder.

What's more, the control system has to compensate for gusts of wind, the deflection of the crane as it picks up and sets down bricks, the elongation of the cable, pendulum effects, and more, he says.

Piconi sees several advantages over batteries. Advantage No.1 is environmental. Instead of chemically reactive and difficult-to- recycle lithium-ion batteries, Energy Vault's main expenditure is the bricks themselves, which can be made on-site using available dirt and waste material mixed with a new polymer from the Mexico-based cement giant Cemex.

Another advantage, according to Piconi, is the lower operating expense, which the company calculates to be about half that of a battery installation with equivalent storage capacity. Battery-storage facilities must continually replace cells as they degrade. But that's not the case for Energy Vault's infrastructure.

The startup is confident enough in its numbers to claim that 2021 will see the start of multiple commercial installations. Energy Vault raised US$110 million in 2019 to build the demonstration unit in Ticino and prepare for a "multicontinent build-out," says Piconi.

Compared with Energy Vault's effort, Gravitricity's energy-storage scheme seems simple. Instead of a six-armed crane shuttling blocks, Gravitricity plans to pull one or just a few much heavier weights up and down abandoned, kilometer-deep mine shafts.

These great masses, each one between 500 and 5000 metric tons, need only move at mere centimeters per second to produce megawatt-level outputs. Using a single weight lends itself to applications that need high power quickly and for a short duration, such as dealing with second-by-second fluctuations in the grid and maintaining grid frequency, explains Chris Yendell, Gravitricity's project development manager. Multiple-weight systems would be more suited to storing more energy and generating for longer periods, he says.

Proving the second-to-second response is a primary goal of a 250-kilowatt concept demonstrator that Gravitricity is building in Scotland. Its 50-metric-ton weight will be suspended 7 meters up on a lattice tower. Testing should start during the first quarter of 2021. "We expect to be able to achieve full generation within less than one second of receiving a signal," says Yendell.

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The company will also be developing sites for a full-scale prototype during 2021. "We are currently liaising with mine owners in Europe and in South Africa, [and we're] certainly interested in the United States as well," says Yendell. Such a full-scale system would then come on line in 2023.

Gravity Power and its competitor New Energy Let's Go, which acquired its technology from the now bankrupt Heindl Energy, are also looking underground for energy storage, but they are more closely inspired by pumped hydro. Instead of storing energy using reservoirs at different elevations, they pump water underground to lift an extremely heavy piston. Allowing the piston to fall pushes water through a turbine to generate electricity.

"Reservoirs are the Achilles' heel of pumped hydro," says Jim Fiske, the company's founder. "The whole purpose of a Gravity Power plant is to remove the need for reservoirs. [Our plants] allow us to put pumped-hydro-scale power and storage capacity in 3 to 5 acres [1 to 2 hectares] of flat land."

Fiske estimates that a 400-megawatt plant with 16 hours of storage (or 6.4 gigawatt-hours of energy) would have a piston that's more than 8 million metric tons. That might sound ludicrous, but it's well within the lifting abilities of today's pumps and the constraints of construction processes, he says.

While these companies expect such underground storage sites to be more economical than battery installations, they will still be expensive. But nations concerned about the changing climate may be willing to pay for storage options like these when they recognize the gravity of the crisis.

9.1 DEFINITION OF SYMMETRICAL COMPONENTS

Assume that a set of three-phase voltages designated V_a, V_b, and V_c is given. In accordance with Fortescue, these phase voltages are resolved into the following three sets of sequence components:

1. *Zero-sequence* components, consisting of three phasors with equal magnitudes and with zero phase displacement, as shown in Figure 9.1(a)
2. *Positive-sequence* components, consisting of three phasors with equal magnitudes, $\pm 120°$ phase displacement, and positive sequence, as in Figure 9.1(b)
3. *Negative-sequence* components, consisting of three phasors with equal magnitudes, $\pm 120°$ phase displacement, and negative sequence, as in Figure 9.1(c)

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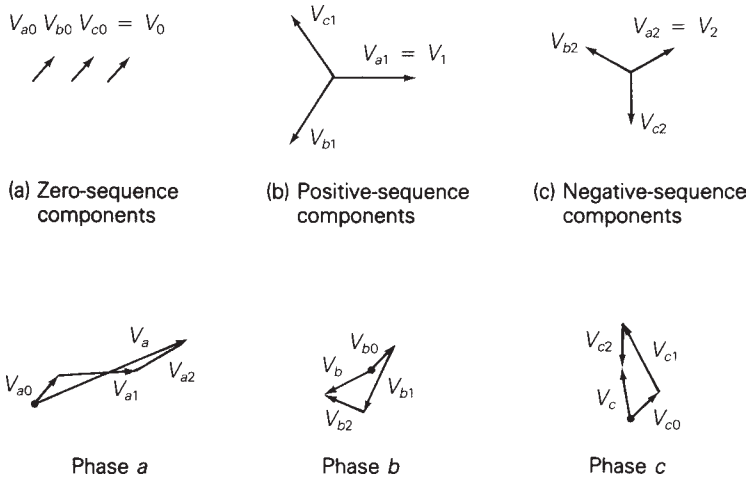
FIGURE 9.1

Resolving phase voltages into three sets of sequence components



The zero-, positive-, and negative-sequence components of phase *a*, which are V_{a0}, V_{a1}, and V_{a2}, respectively, are presented in this section. For simplicity, drop the subscript *a* and denote these sequence components as V_0, V_1, and V_2. They are defined by the following transformation:

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} \tag{9.1.1}$$

where

$$a = 1\angle 120^\circ = \frac{-1}{2} + j\frac{\sqrt{3}}{2} \tag{9.1.2}$$

Writing (9.1.1) as three separate equations:

$$V_a = V_0 + V_1 + V_2 \tag{9.1.3}$$

$$V_b = V_0 + a^2 V_1 + aV_2 \tag{9.1.4}$$

$$V_c = V_0 + aV_1 + a^2 V \tag{9.1.5}$$

In (9.1.2), *a* is a complex number with unit magnitude and a 120° phase angle. When any phasor is multiplied by *a*, that phasor rotates by 120° (counterclockwise). Similarly, when any phasor is multiplied by $a^2 = (1\angle 120^\circ)\ (1\angle 120^\circ) = 1\angle 240^\circ$, the phasor rotates by 240°. Table 9.1 lists some common identities involving *a*.

The complex number *a* is similar to the well-known complex number $j = \sqrt{-1} = 1\angle 90^\circ$. Thus, the only difference between *j* and *a* is that the angle of *j* is 90°, and that of *a* is 120°.

Equation (9.1.1) can be rewritten more compactly using matrix notation. Define the following vectors $\boldsymbol{V}_P$ and $\boldsymbol{V}_S$, and matrix $\boldsymbol{A}$:

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$a^4 = a = 1\angle 120°$
$a^2 = 1\angle 240°$
$a^3 = 1\angle 0°$
$1 + a + a^2 = 0$
$1 - a = \sqrt{3}\angle -30°$
$1 - a^2 = \sqrt{3}\angle +30°$
$a^2 - a = \sqrt{3}\angle 270°$
$ja = 1\angle 210°$
$1 + a = -a^2 = 1\angle 60°$
$1 + a^2 = -a = 1\angle -60°$
$a + a^2 = -1 = 1\angle 180°$

TABLE 9.1

Common identities involving $a = 1\angle 120°$

$$\boldsymbol{V}_{\mathrm{P}} = \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} \tag{9.1.6}$$

$$\boldsymbol{V}_{\mathrm{S}} = \begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} \tag{9.1.7}$$

$$\boldsymbol{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \tag{9.1.8}$$

where $\boldsymbol{V}_{\mathrm{P}}$ is the column vector of phase voltages, $\boldsymbol{V}_{\mathrm{S}}$ is the column vector of sequence voltages, and $\boldsymbol{A}$ is a 3×3 transformation matrix. Using these definitions, (9.1.1) becomes

$$\boldsymbol{V}_{\mathrm{P}} = \boldsymbol{A}\boldsymbol{V}_{\mathrm{S}} \tag{9.1.9}$$

The inverse of the $\boldsymbol{A}$ matrix is

$$\boldsymbol{A}^{-1} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \tag{9.1.10}$$

Equation (9.1.10) can be verified by showing that the product $\boldsymbol{A}\boldsymbol{A}^{-1}$ is the unit matrix. Also, premultiplying (9.1.9) by $\boldsymbol{A}^{-1}$ gives

$$\boldsymbol{V}_{\mathrm{S}} = \boldsymbol{A}^{-1}\boldsymbol{V}_{\mathrm{P}} \tag{9.1.11}$$

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Using (9.1.6), (9.1.7), and (9.1.10), then (9.1.11) becomes

$$\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} \tag{9.1.12}$$

Writing (9.1.12) as three separate equations,

$$V_0 = \tfrac{1}{3}(V_a + V_b + V_c) \tag{9.1.13}$$

$$V_1 = \tfrac{1}{3}(V_a + aV_b + a^2V_c) \tag{9.1.14}$$

$$V_2 = \tfrac{1}{3}(V_a + a^2V_b + aV_c) \tag{9.1.15}$$

Equation (9.1.13) shows that there is no zero-sequence voltage in a *balanced* three-phase system because the sum of three balanced phasors is zero. In an unbalanced three-phase system, line-to-neutral voltages may have a zero-sequence component. However, line-to-line voltages never have a zero-sequence component, since by KVL, their sum is always zero.

The symmetrical component transformation also can be applied to currents, as follows. Let

$$\boldsymbol{I}_{\mathrm{P}} = \boldsymbol{A}\boldsymbol{I}_{\mathrm{S}} \tag{9.1.16}$$

where $\boldsymbol{I}_{\mathrm{P}}$ is a vector of phase currents,

$$\boldsymbol{I}_{\mathrm{P}} = \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} \tag{9.1.17}$$

and $\boldsymbol{I}_{\mathrm{S}}$ is a vector of sequence currents,

$$\boldsymbol{I}_{\mathrm{S}} = \begin{bmatrix} I_0 \\ I_1 \\ I_2 \end{bmatrix} \tag{9.1.18}$$

Also,

$$\boldsymbol{I}_{\mathrm{S}} = \boldsymbol{A}^{-1}\boldsymbol{I}_{\mathrm{P}} \tag{9.1.19}$$

Equations (9.1.16) and (9.1.19) can be written as separate equations as follows. The phase currents are

$$I_a = I_0 + I_1 + I_2 \tag{9.1.20}$$

$$I_b = I_0 + a^2I_1 + aI_2 \tag{9.1.21}$$

$$I_c = I_0 + aI_1 + a^2I_2 \tag{9.1.22}$$

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and the sequence currents are

$$I_0 = \tfrac{1}{3}(I_a + I_b + I_c) \tag{9.1.23}$$

$$I_1 = \tfrac{1}{3}(I_a + aI_b + a^2 I_c) \tag{9.1.24}$$

$$I_2 = \tfrac{1}{3}(I_a + a^2 I_b + aI_c) \tag{9.1.25}$$

In a three-phase Y-connected system, the neutral current I_n is the sum of the line currents:

$$I_n = I_a + I_b + I_c \tag{9.1.26}$$

Comparing (9.1.26) and (9.1.23),

$$I_n = 3I_0 \tag{9.1.27}$$

The neutral current equals three times the zero-sequence current. In a balanced Y-connected system, line currents have no zero-sequence component, since the neutral current is zero. Also, in any three-phase system with no neutral path, such as a Δ-connected system or a three-wire Y-connected system with an ungrounded neutral, line currents have no zero-sequence component.

The following three examples further illustrate symmetrical components.

EXAMPLE 9.1

Sequence components: balanced line-to-neutral voltages

Calculate the sequence components of the following balanced line-to-neutral voltages with *abc* sequence:

$$\boldsymbol{V}_{\mathrm{P}} = \begin{bmatrix} V_{an} \\ V_{bn} \\ V_{cn} \end{bmatrix} = \begin{bmatrix} 277\angle 0^\circ \\ 277\angle -120^\circ \\ 277\angle +120^\circ \end{bmatrix} \text{ volts}$$

SOLUTION

Using (9.1.13) through (9.1.15),

$$V_0 = \tfrac{1}{3}[277\angle 0^\circ + 277\angle -120^\circ + 277\angle +120^\circ] = 0$$

$$V_1 = \tfrac{1}{3}[277\angle 0^\circ + 277\angle(-120^\circ + 120^\circ) + 277\angle(120^\circ + 240^\circ)]$$

$$= 277\angle 0^\circ \text{ volts} = V_{an}$$

$$V_2 = \tfrac{1}{3}[277\angle 0^\circ + 277\angle(-120^\circ + 240^\circ) + 277\angle(120^\circ + 120^\circ)]$$

$$= \tfrac{1}{3}[277\angle 0^\circ + 277\angle 120^\circ + 277\angle 240^\circ] = 0$$

(*Continued*)

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This example illustrates the fact that balanced three-phase systems with *abc* sequence (or positive sequence) have no zero-sequence or negative-sequence components. For this example, the positive-sequence voltage V_1 equals V_{an}, and the zero-sequence and negative-sequence voltages are both zero.

EXAMPLE 9.2

Sequence components: balanced *acb* currents

A Y-connected load has balanced currents with *acb* sequence given by

$$\boldsymbol{I}_{\mathrm{P}} = \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 10\angle 0^\circ \\ 10\angle +120^\circ \\ 10\angle -120^\circ \end{bmatrix} \text{ A}$$

Calculate the sequence currents.

SOLUTION

Using (9.1.23) through (9.1.25),

$$I_0 = \tfrac{1}{3}[10\angle 0^\circ + 10\angle 120^\circ + 10\angle -120^\circ] = 0$$

$$I_1 = \tfrac{1}{3}[10\angle 0^\circ + 10\angle (120^\circ + 120^\circ) + 10\angle (-120^\circ + 240^\circ)]$$

$$= \tfrac{1}{3}[10\angle 0^\circ + 10\angle 240^\circ + 10\angle 120^\circ] = 0$$

$$I_2 = \tfrac{1}{3}[10\angle 0^\circ + 10\angle (120^\circ + 240^\circ) + 10\angle (-120^\circ + 120^\circ)]$$

$$= 10\angle 0^\circ \text{ A} = I_a$$

This example illustrates the fact that balanced three-phase systems with *acb* sequence (or negative sequence) have no zero-sequence or positive-sequence components. For this example, the negative-sequence current I_2 equals I_a, and the zero-sequence and positive-sequence currents are both zero.

EXAMPLE 9.3

Sequence components: unbalanced currents

A three-phase line feeding a balanced-Y load has one of its phases (phase *b*) open. The load neutral is grounded, and the unbalanced line currents are

$$\boldsymbol{I}_{\mathrm{P}} = \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 10\angle 0^\circ \\ 0 \\ 10\angle 120^\circ \end{bmatrix} \text{ A}$$

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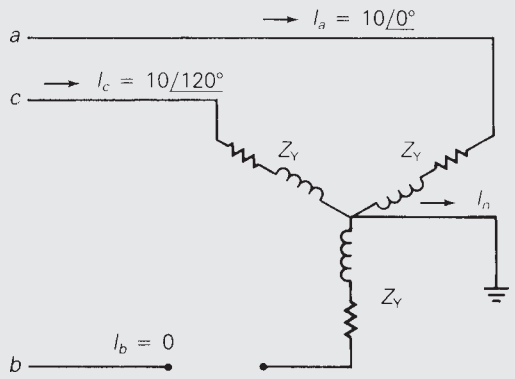
FIGURE 9.2
Circuit for Example 9.3



Calculate the sequence currents and the neutral current.

SOLUTION

The circuit is shown in Figure 9.2. Using (9.1.23) through (9.1.25),

$$I_0 = \tfrac{1}{3}[10/0° + 0 + 10/120°]$$
$$= 3.333/60° \text{ A}$$
$$I_1 = \tfrac{1}{3}[10/0° + 0 + 10/(120° + 240°)] = 6.667/0° \text{ A}$$
$$I_2 = \tfrac{1}{3}[10/0° + 0 + 10/(120° + 120°)]$$
$$= 3.333/-60° \text{ A}$$

Using (9.1.26), the neutral current is

$$I_n = (10/0° + 0 + 10/120°)$$
$$= 10/60° \text{ A} = 3I_0$$

This example illustrates the fact that *unbalanced* three-phase systems may have nonzero values for all sequence components. Also, the neutral current equals three times the zero-sequence current, as given by (9.1.27).

9.2 SEQUENCE NETWORKS OF IMPEDANCE LOADS

Figure 9.3 shows a balanced-Y impedance load. The impedance of each phase is designated Z_Y, and a neutral impedance Z_n is connected between the load neutral and ground. Note from Figure 9.3 that the line-to-ground voltage V_{ag} is

$$\begin{aligned} V_{ag} &= Z_Y I_a + Z_n I_n \\ &= Z_Y I_a + Z_n(I_a + I_b + I_c) \\ &= (Z_Y + Z_n)I_a + Z_n I_b + Z_n I_c \end{aligned} \tag{9.2.1}$$

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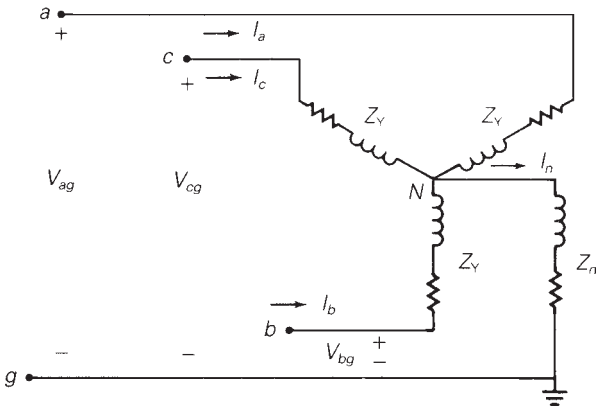
FIGURE 9.3
Balanced-Y impedance load



Similar equations can be written for V_{bg} and V_{cg}:

$$V_{bg} = Z_n I_a + (Z_Y + Z_n) I_b + Z_n I_c \tag{9.2.2}$$

$$V_{cg} = Z_n I_a + Z_n I_b + (Z_Y + Z_n) I_c \tag{9.2.3}$$

Equations (9.2.1) through (9.2.3) can be rewritten in matrix format:

$$\begin{bmatrix} V_{ag} \\ V_{bg} \\ V_{cg} \end{bmatrix} = \begin{bmatrix} (Z_Y + Z_n) & Z_n & Z_n \\ Z_n & (Z_Y + Z_n) & Z_n \\ Z_n & Z_n & (Z_Y + Z_n) \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} \tag{9.2.4}$$

Equation (9.2.4) is written more compactly as

$$\boldsymbol{V}_P = \boldsymbol{Z}_P \boldsymbol{I}_P \tag{9.2.5}$$

where $\boldsymbol{V}_P$ is the vector of line-to-ground voltages (or phase voltages), $\boldsymbol{I}_P$ is the vector of line currents (or phase currents), and $\boldsymbol{Z}_P$ is the 3×3 phase impedance matrix shown in (9.2.4). Equations (9.1.9) and (9.1.16) can now be used in (9.2.5) to determine the relationship between the sequence voltages and currents, as follows:

$$\boldsymbol{A}\boldsymbol{V}_S = \boldsymbol{Z}_P \boldsymbol{A} \boldsymbol{I}_S \tag{9.2.6}$$

Premultiplying both sides of (9.2.6) of $\boldsymbol{A}^{-1}$ gives

$$\boldsymbol{V}_S = (\boldsymbol{A}^{-1} \boldsymbol{Z}_P \boldsymbol{A}) \boldsymbol{I}_S \tag{9.2.7}$$

or

$$\boldsymbol{V}_S = \boldsymbol{Z}_S \boldsymbol{I}_S \tag{9.2.8}$$

where

$$\boldsymbol{Z}_S = \boldsymbol{A}^{-1} \boldsymbol{Z}_P \boldsymbol{A} \tag{9.2.9}$$

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The impedance matrix $\boldsymbol{Z}_S$ defined by (9.2.9) is called the *sequence impedance matrix.* Using the definition of $\boldsymbol{A}$, its inverse $\boldsymbol{A}^{-1}$, and $\boldsymbol{Z}_P$ given by (9.1.8), (9.1.10), and (9.2.4), the sequence impedance matrix $\boldsymbol{Z}_S$ for the balanced-Y load is

$$\boldsymbol{Z}_S = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} (Z_Y + Z_n) & Z_n & Z_n \\ Z_n & (Z_Y + Z_n) & Z_n \\ Z_n & Z_n & (Z_Y + Z_n) \end{bmatrix} \times \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \tag{9.2.10}$$

Performing the indicated matrix multiplications in (9.2.10), and using the identity $(1 + a + a^2) = 0$,

$$\boldsymbol{Z}_S = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} (Z_Y + 3Z_n) & Z_Y & Z_Y \\ (Z_Y + 3Z_n) & a^2 Z_Y & aZ_Y \\ (Z_Y + 3Z_n) & aZ_Y & a^2 Z_Y \end{bmatrix} = \begin{bmatrix} (Z_Y + 3Z_n) & 0 & 0 \\ 0 & Z_Y & 0 \\ 0 & 0 & Z_Y \end{bmatrix} \tag{9.2.11}$$

As shown in (9.2.11), the sequence impedance matrix $\boldsymbol{Z}_S$ for the balanced-Y load of Figure 9.3 is a diagonal matrix. Since $\boldsymbol{Z}_S$ is diagonal, (9.2.8) can be written as three *uncoupled* equations. Using (9.1.7), (9.1.18), and (9.2.11) in (9.2.8),

$$\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} (Z_Y + 3Z_n) & 0 & 0 \\ 0 & Z_Y & 0 \\ 0 & 0 & Z_Y \end{bmatrix}\begin{bmatrix} I_0 \\ I_1 \\ I_2 \end{bmatrix} \tag{9.2.12}$$

Rewriting (9.2.12) as three separate equations,

$$V_0 = (Z_Y + 3Z_n)I_0 = Z_0 I_0 \tag{9.2.13}$$

$$V_1 = Z_Y I_1 = Z_1 I_1 \tag{9.2.14}$$

$$V_2 = Z_Y I_2 = Z_2 I_2 \tag{9.2.15}$$

As shown in (9.2.13), the zero-sequence voltage V_0 depends only on the zero-sequence current I_0 and the impedance $(Z_Y + 3Z_n)$. This impedance is called the *zero-sequence impedance* and is designated Z_0. Also, the positive-sequence voltage V_1 depends only on the positive-sequence current I_1 and an impedance $Z_1 = Z_Y$ called the *positive-sequence impedance.* Similarly, V_2 depends only on I_2 and the *negative-sequence impedance* $Z_2 = Z_Y$.

Equations (9.2.13) through (9.2.15) can be represented by the three networks shown in Figure 9.4. These networks are called the *zero-sequence*,

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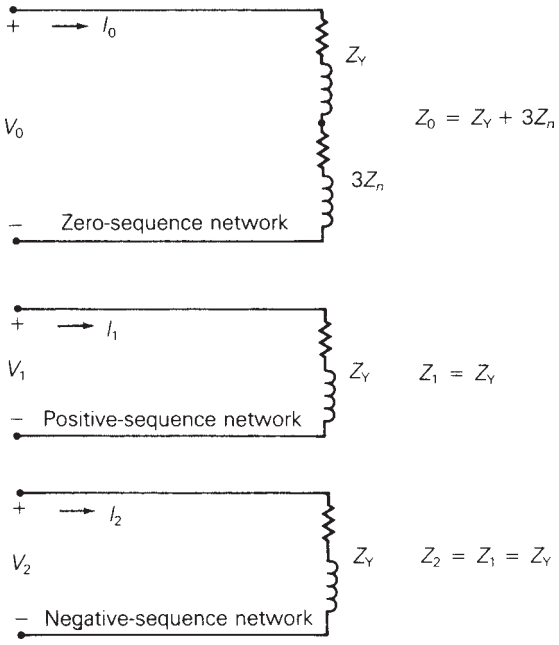
FIGURE 9.4

Sequence networks of a balanced-Y load



positive-sequence, and *negative-sequence networks.* As shown, each sequence network is separate, uncoupled from the other two. The separation of these sequence networks is a consequence of the fact that $\boldsymbol{Z}_S$ is a diagonal matrix for a balanced-Y load. This separation underlies the advantage of symmetrical components.

Note that the neutral impedance does not appear in the positive- and negative-sequence networks of Figure 9.4. This illustrates the fact that positive- and negative-sequence currents do not flow in neutral impedances. However, the neutral impedance is multiplied by 3 and placed in the zero-sequence network of the figure. The voltage $I_0(3Z_n)$ across the impedance $3Z_n$ is the voltage drop (I_nZ_n) across the neutral impedance Z_n in Figure 9.3, since $I_n = 3I_0$.

When the neutral of the Y load in Figure 9.3 has no return path, then the neutral impedance Z_n is infinite and the term $3Z_n$ in the zero-sequence network of Figure 9.4 becomes an open circuit. Under this condition of an open neutral, no zero-sequence current exists. However, when the neutral of the Y load is solidly grounded with a zero-ohm conductor, then the neutral impedance is zero and the term $3Z_n$ in the zero-sequence network becomes a short circuit. Under this condition of a solidly grounded neutral, zero-sequence current I_0 can exist when there is a zero-sequence voltage caused by unbalanced voltages applied to the load.

Figure 2.16 shows a balanced-Δ load and its equivalent balanced-Y load. Since the Δ load has no neutral connection, the equivalent Y load in Figure 2.16 has an open neutral. The sequence networks of the equivalent Y load corresponding to a balanced-Δ load are shown in Figure 9.5. As shown, the equivalent Y impedance $Z_Y = Z_\Delta/3$ appears in each of the sequence networks. Also, the zero-sequence

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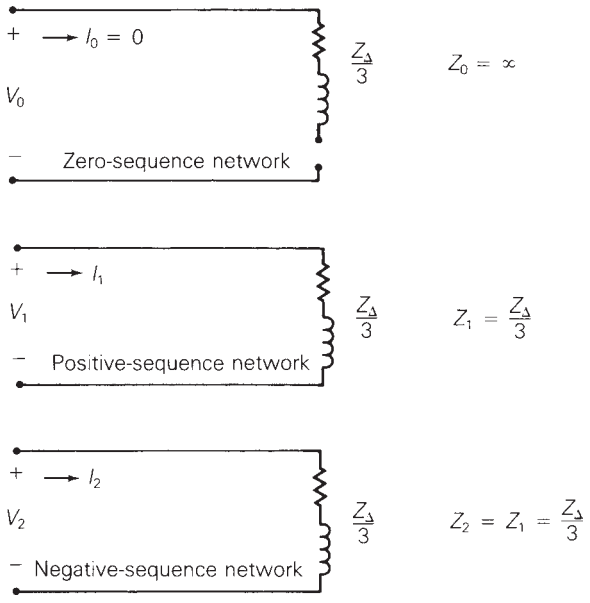

FIGURE 9.5

Sequence networks for an equivalent Y representation of a balanced-Δ load

network has an open circuit, since $Z_n = \infty$ corresponds to an open neutral. No zero-sequence current occurs in the equivalent Y load.

The sequence networks of Figure 9.5 represent the balanced-Δ load as viewed from its terminals, but they do not represent the internal load characteristics. The currents I_0, I_1, and I_2 in Figure 9.5 are the sequence components of the line currents feeding the Δ load, not the load currents within the Δ. The Δ load currents, which are related to the line currents by (2.5.14), are not shown in Figure 9.5.

EXAMPLE 9.4

Sequence networks: balanced-Y and balanced-Δ loads

A balanced-Y load is in parallel with a balanced-Δ-connected capacitor bank. The Y load has an impedance $Z_Y = (3 + j4)\ \Omega$ per phase, and its neutral is grounded through an inductive reactance $X_n = 2\ \Omega$. The capacitor bank has a reactance $X_c = 30\ \Omega$ per phase. Draw the sequence networks for this load and calculate the load-sequence impedances.

SOLUTION

The sequence networks are shown in Figure 9.6. As shown, the Y-load impedance in the zero-sequence network is in series with three times the neutral impedance. Also, the Δ-load branch in the zero-sequence network is open,

(*Continued*)

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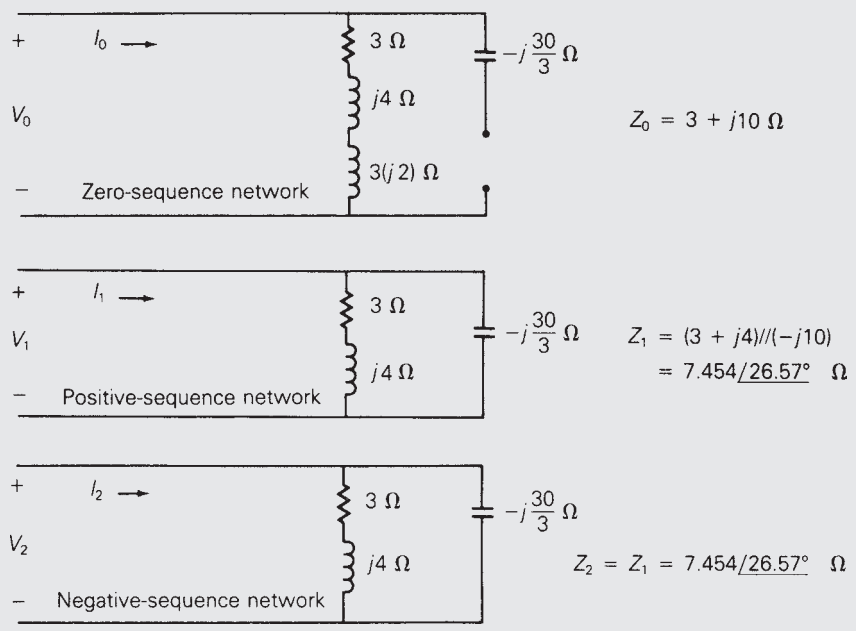
FIGURE 9.6
Sequence networks for Example 9.4



since no zero-sequence current flows into the Δ load. In the positive- and negative-sequence circuits, the Δ-load impedance is divided by 3 and placed in parallel with the Y-load impedance. The equivalent sequence impedances are

$$Z_0 = Z_Y + 3Z_n = 3 + j4 + 3(j2) = 3 + j10 \quad \Omega$$

$$Z_1 = Z_Y//(Z_\Delta/3) = \frac{(3 + j4)(-j30/3)}{3 + j4 - j(30/3)}$$

$$= \frac{(5\angle 53.13°)(10\angle -90°)}{6.708\angle -63.43°} = 7.454\angle 26.57° \quad \Omega$$

$$Z_2 = Z_1 = 7.454\angle 26.57° \quad \Omega$$

Figure 9.7 shows a general three-phase linear impedance load. The load could represent a balanced load such as the balanced-Y or balanced-Δ load, or an unbalanced impedance load. The general relationship between the line-to-ground voltages and line currents for this load can be written as

$$\begin{bmatrix} V_{ag} \\ V_{bg} \\ V_{cg} \end{bmatrix} = \begin{bmatrix} Z_{aa} & Z_{ab} & Z_{ac} \\ Z_{ab} & Z_{bb} & Z_{bc} \\ Z_{ac} & Z_{bc} & Z_{cc} \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} \tag{9.2.16}$$

or

$$\boldsymbol{V}_P = \boldsymbol{Z}_P \boldsymbol{I}_P \tag{9.2.17}$$

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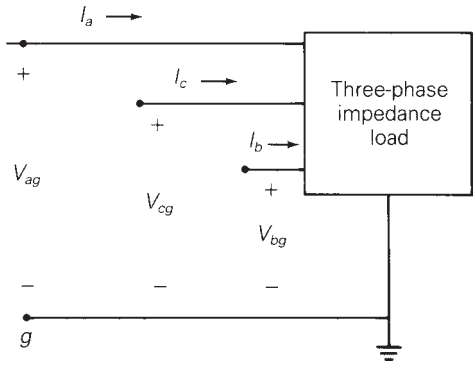

FIGURE 9.7

General three-phase impedance load (linear, bilateral network, nonrotating equipment)

where $\boldsymbol{V}_{\mathrm{P}}$ is the vector of line-to-neutral (or phase) voltages, $\boldsymbol{I}_{\mathrm{P}}$ is the vector of line (or phase) currents, and $\boldsymbol{Z}_{\mathrm{P}}$ is a 3×3 phase impedance matrix. It is assumed here that the load is nonrotating, and that $\boldsymbol{Z}_{\mathrm{P}}$ is a symmetric matrix, which corresponds to a bilateral network.

Since (9.2.17) has the same form as (9.2.5), the relationship between the sequence voltages and currents for the general three-phase load of Figure 9.6 is the same as that of (9.2.8) and (9.2.9), which are rewritten here:

$$\boldsymbol{V}_{\mathrm{S}} = \boldsymbol{Z}_{\mathrm{S}}\boldsymbol{I}_{\mathrm{S}} \tag{9.2.18}$$

$$\boldsymbol{Z}_{\mathrm{S}} = \boldsymbol{A}^{-1}\boldsymbol{Z}_{\mathrm{P}}\boldsymbol{A} \tag{9.2.19}$$

The sequence impedance matrix $\boldsymbol{Z}_{\mathrm{S}}$ given by (9.2.19) is a 3×3 matrix with nine sequence impedances, defined as follows:

$$\boldsymbol{Z}_{\mathrm{S}} = \begin{bmatrix} Z_0 & Z_{01} & Z_{02} \\ Z_{10} & Z_1 & Z_{12} \\ Z_{20} & Z_{21} & Z_2 \end{bmatrix} \tag{9.2.20}$$

The diagonal impedances Z_0, Z_1, and Z_2 in this matrix are the self-impedances of the zero-, positive-, and negative-sequence networks. The off-diagonal impedances are the mutual impedances between sequence networks. Using the definitions of $\boldsymbol{A}$, $\boldsymbol{A}^{-1}$, $\boldsymbol{Z}_{\mathrm{P}}$, and $\boldsymbol{Z}_{\mathrm{S}}$, (9.2.19) is

$$\begin{bmatrix} Z_0 & Z_{01} & Z_{02} \\ Z_{10} & Z_1 & Z_{12} \\ Z_{20} & Z_{21} & Z_2 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} Z_{aa} & Z_{ab} & Z_{ac} \\ Z_{ab} & Z_{bb} & Z_{bc} \\ Z_{ac} & Z_{cb} & Z_{cc} \end{bmatrix}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \tag{9.2.21}$$

Performing the indicated multiplications in (9.2.21), and using the identity $(1 + a + a^2) = 0$, the following separate equations can be obtained (see Problem 9.18).

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DIAGONAL SEQUENCE IMPEDANCES

$$Z_0 = \tfrac{1}{3}\,(Z_{aa} + Z_{bb} + Z_{cc} + 2Z_{ab} + 2Z_{ac} + 2Z_{bc}) \tag{9.2.22}$$

$$Z_1 = Z_2 = \tfrac{1}{3}\,(Z_{aa} + Z_{bb} + Z_{cc} - Z_{ab} - Z_{ac} - Z_{bc}) \tag{9.2.23}$$

OFF-DIAGONAL SEQUENCE IMPEDANCES

$$Z_{01} = Z_{20} = \tfrac{1}{3}\,(Z_{aa} + a^2Z_{bb} + aZ_{cc} - aZ_{ab} - a^2Z_{ac} - Z_{bc}) \tag{9.2.24}$$

$$Z_{02} = Z_{10} = \tfrac{1}{3}\,(Z_{aa} + aZ_{bb} + a^2Z_{cc} - a^2Z_{ab} - aZ_{ac} - Z_{bc}) \tag{9.2.25}$$

$$Z_{12} = \tfrac{1}{3}\,(Z_{aa} + a^2Z_{bb} + aZ_{cc} + 2aZ_{ab} + 2a^2Z_{ac} + 2Z_{bc}) \tag{9.2.26}$$

$$Z_{21} = \tfrac{1}{3}\,(Z_{aa} + aZ_{bb} + a^2Z_{cc} + 2a^2Z_{ab} + 2aZ_{ac} + 2Z_{bc}) \tag{9.2.27}$$

A *symmetrical load* is defined as a load whose sequence impedance matrix is diagonal; that is, all the mutual impedances in (9.2.24) through (9.2.27) are zero. Equating these mutual impedances to zero and solving, the following conditions for a symmetrical load are determined. When both

$$\left.\begin{aligned} Z_{aa} &= Z_{bb} = Z_{cc} \\ \text{and} \qquad & \\ Z_{ab} &= Z_{ac} = Z_{bc} \end{aligned}\right\} \text{ conditions for a symmetrical load} \tag{9.2.28, 9.2.29}$$

then

$$Z_{01} = Z_{10} = Z_{02} = Z_{20} = Z_{12} = Z_{21} = 0 \tag{9.2.30}$$

$$Z_0 = Z_{aa} + 2Z_{ab} \tag{9.2.31}$$

$$Z_1 = Z_2 = Z_{aa} - Z_{ab} \tag{9.2.32}$$

The conditions for a symmetrical load are that the diagonal phase impedances be equal and that the off-diagonal phase impedances be equal. These conditions can be verified by using (9.2.28) and (9.2.29) with the identity $(1 + a + a^2) = 0$ in (9.2.24) through (9.2.27) to show that all the mutual sequence impedances are zero. Note that the positive- and negative-sequence impedances are equal for a symmetrical load, as shown by (9.2.32), and for a nonsymmetrical load, as shown by (9.2.23). This is always true for linear, symmetric impedances that represent nonrotating equipment, such as transformers and transmission lines. However, the positive- and negative- sequence impedances of rotating equipment such as generators and motors are generally not equal. Note also that the zero-sequence impedance Z_0 is not equal to the positive- and negative-sequence impedances of a symmetrical load unless the mutual phase impedances $Z_{ab} = Z_{ac} = Z_{bc}$ are zero.

The sequence networks of a symmetrical impedance load are shown in Figure 9.8. Since the sequence impedance matrix $\boldsymbol{Z}_S$ is diagonal for a symmetrical load, the sequence networks are separate or uncoupled.

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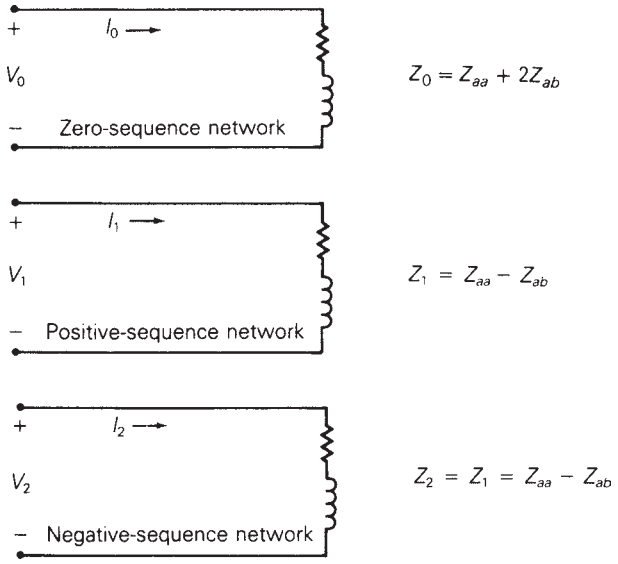

FIGURE 9.8

Sequence networks of a three-phase symmetrical impedance load (linear, bilateral network, nonrotating equipment)

9.3 SEQUENCE NETWORKS OF SERIES IMPEDANCES

Figure 9.9 shows series impedances connected between two three-phase buses denoted *abc* and $a'b'c'$. Self-impedances of each phase are denoted Z_{aa}, Z_{bb}, and Z_{cc}. In general, the series network also may have mutual impedances between phases. The voltage drops across the series-phase impedances are given by

$$\begin{bmatrix} V_{an} - V_{a'n} \\ V_{bn} - V_{b'n} \\ V_{cn} - V_{c'n} \end{bmatrix} = \begin{bmatrix} V_{aa'} \\ V_{bb'} \\ V_{cc'} \end{bmatrix} = \begin{bmatrix} Z_{aa} & Z_{ab} & Z_{ac} \\ Z_{ab} & Z_{bb} & Z_{bc} \\ Z_{ac} & Z_{cb} & Z_{cc} \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} \tag{9.3.1}$$

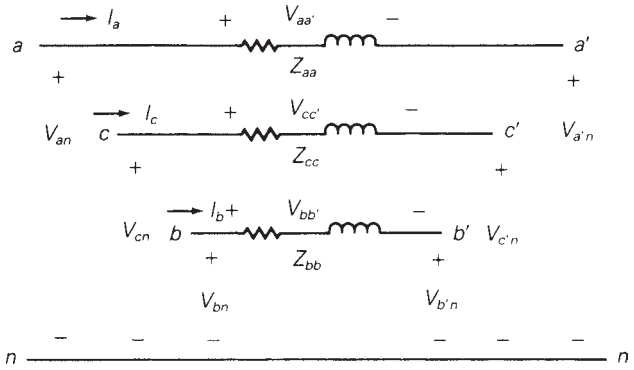

FIGURE 9.9

Three-phase series impedances (linear, bilateral network, nonrotating equipment)

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Both self-impedances and mutual impedances are included in (9.3.1). It is assumed that the impedance matrix is symmetric, which corresponds to a bilateral network. It is also assumed that these impedances represent nonrotating equipment. Typical examples are series impedances of transmission lines and of transformers. Equation (9.3.1) has the following form:

$$\boldsymbol{V}_{\mathrm{P}} - \boldsymbol{V}_{\mathrm{P}'} = \boldsymbol{Z}_{\mathrm{P}}\boldsymbol{I}_{\mathrm{P}} \tag{9.3.2}$$

where $\boldsymbol{V}_{\mathrm{P}}$ is the vector of line-to-neutral voltages at bus abc, $\boldsymbol{V}_{\mathrm{P}'}$ the vector of line-to-neutral voltages at bus $a'b'c'$, $\boldsymbol{I}_{\mathrm{P}}$ is the vector of line currents, and $\boldsymbol{Z}_{\mathrm{P}}$ is the 3×3 phase impedance matrix for the series network. Equation (9.3.2) is now transformed to the sequence domain in the same manner that the load-phase impedances were transformed in Section 9.2. Thus,

$$\boldsymbol{V}_{\mathrm{S}} - \boldsymbol{V}_{\mathrm{S}'} = \boldsymbol{Z}_{\mathrm{S}}\boldsymbol{I}_{\mathrm{S}} \tag{9.3.3}$$

where

$$\boldsymbol{Z}_{\mathrm{S}} = \boldsymbol{A}^{-1}\,\boldsymbol{Z}_{\mathrm{P}}\boldsymbol{A} \tag{9.3.4}$$

From the results of Section 9.2, this sequence impedance $\boldsymbol{Z}_{\mathrm{S}}$ matrix is diagonal under the following conditions:

$$\left.\begin{array}{rl} & Z_{aa} = Z_{bb} = Z_{cc} \\ \text{and} & \\ & Z_{ab} = Z_{ac} = Z_{bc} \end{array}\right\} \begin{array}{l} \text{conditions for} \\ \text{symmetrical} \\ \text{series impedances} \end{array} \tag{9.3.5}$$

When the phase impedance matrix $\boldsymbol{Z}_{\mathrm{P}}$ of (9.3.1) has both equal self-impedances and equal mutual impedances, then (9.3.4) becomes

$$Z_{\mathrm{S}} = \begin{bmatrix} Z_0 & 0 & 0 \\ 0 & Z_1 & 0 \\ 0 & 0 & Z_2 \end{bmatrix} \tag{9.3.6}$$

where

$$Z_0 = Z_{aa} + 2Z_{ab} \tag{9.3.7}$$

and

$$Z_1 = Z_2 = Z_{aa} - Z_{ab} \tag{9.3.8}$$

and (9.3.3) becomes three uncoupled equations, written as follows:

$$V_0 - V_{0'} = Z_0 I_0 \tag{9.3.9}$$

$$V_1 - V_{1'} = Z_1 I_1 \tag{9.3.10}$$

$$V_2 - V_{2'} = Z_2 I_2 \tag{9.3.11}$$

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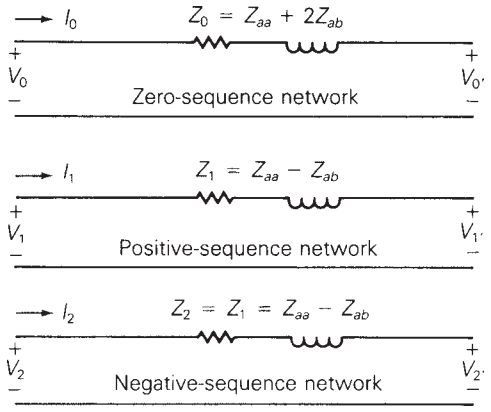

FIGURE 9.10

Sequence networks of three-phase symmetrical series impedances (linear, bilateral network, nonrotating equipment)

Equations (9.3.9) through (9.3.11) are represented by the three uncoupled sequence networks shown in Figure 9.10. From the figure, it is apparent that, for symmetrical series impedances, positive-sequence currents produce only positive-sequence voltage drops. Similarly, negative-sequence currents produce only negative-sequence voltage drops, and zero-sequence currents produce only zero-sequence voltage drops. However, if the series impedances are not symmetrical, then $\boldsymbol{Z}_S$ is not diagonal, the sequence networks are coupled, and the voltage drop across any one sequence network depends on all three sequence currents.

9.4 SEQUENCE NETWORKS OF THREE-PHASE LINES

Section 4.7 develops equations suitable for computer calculation of the series phase impedances, including resistances and inductive reactances, of three-phase overhead transmission lines. The series phase impedance matrix $\boldsymbol{Z}_P$ for an untransposed line is given by Equation (4.7.19) and $\hat{\boldsymbol{Z}}_P$ for a completely transposed line is given by (4.7.21) through (4.7.23). Equation (4.7.19) can be transformed to the sequence domain to obtain

$$\boldsymbol{Z}_S = \boldsymbol{A}^{-1}\,\boldsymbol{Z}_P\boldsymbol{A} \tag{9.4.1}$$

$\boldsymbol{Z}_S$ is the 3×3 series sequence impedance matrix whose elements are

$$\boldsymbol{Z}_S = \begin{bmatrix} Z_0 & Z_{01} & Z_{02} \\ Z_{10} & Z_1 & Z_{12} \\ Z_{20} & Z_{21} & Z_2 \end{bmatrix} \quad \Omega/\text{m} \tag{9.4.2}$$

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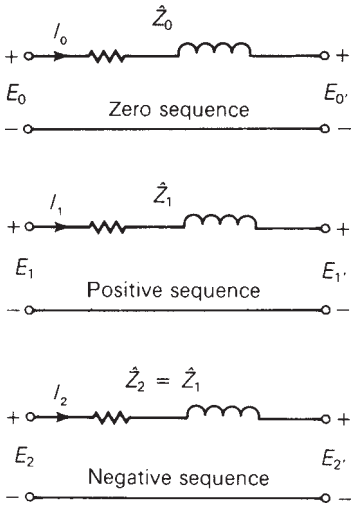
FIGURE 9.11

Circuit representation of the series sequence impedances of a completely transposed three-phase line



In general, $\boldsymbol{Z}_S$ is not diagonal. However, if the line is completely transposed,

$$\hat{\boldsymbol{Z}}_S = \boldsymbol{A}^{-1}\hat{\boldsymbol{Z}}_P\boldsymbol{A} = \begin{bmatrix} \hat{Z}_0 & 0 & 0 \\ 0 & \hat{Z}_1 & 0 \\ 0 & 0 & \hat{Z}_2 \end{bmatrix} \tag{9.4.3}$$

where, from (9.3.7) and (9.3.8),

$$\hat{Z}_0 = \hat{Z}_{aaeq} + 2\hat{Z}_{abeq} \tag{9.4.4}$$

$$\hat{Z}_1 = \hat{Z}_2 = \hat{Z}_{aaeq} - \hat{Z}_{abeq} \tag{9.4.5}$$

A circuit representation of the series sequence impedances of a completely transposed three-phase line is shown in Figure 9.11.

Section 4.11 develops equations suitable for computer calculation of the shunt phase admittances of three-phase overhead transmission lines. The shunt admittance matrix $\boldsymbol{Y}_P$ for an untransposed line is given by Equation (4.11.16), and $\hat{\boldsymbol{Y}}_P$ for a completely transposed three-phase line is given by (4.11.17).

Equation (4.11.16) can be transformed to the sequence domain to obtain

$$\boldsymbol{Y}_S = \boldsymbol{A}^{-1}\boldsymbol{Y}_P\boldsymbol{A} \tag{9.4.6}$$

where

$$\boldsymbol{Y}_S = \mathbf{G}_S + j(2\pi f)\mathbf{C}_S \tag{9.4.7}$$

$$\mathbf{C}_S = \begin{bmatrix} C_0 & C_{01} & C_{02} \\ C_{10} & C_1 & C_{12} \\ C_{20} & C_{21} & C_2 \end{bmatrix} \text{ F/m} \tag{9.4.8}$$

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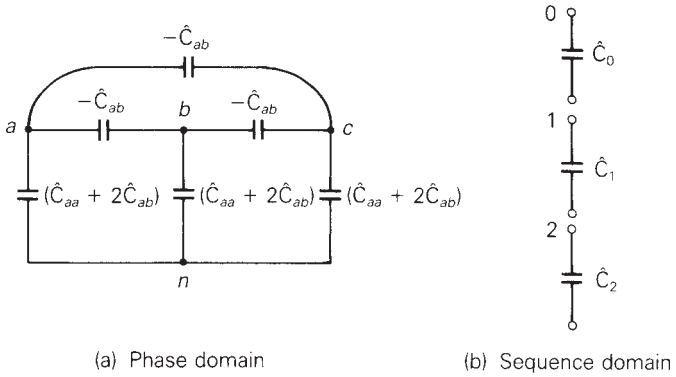

FIGURE 9.12

Circuit representations of the capacitances of a completely transposed three-phase line

In general, $\mathbf{C}_S$ is not diagonal. However, for the completely transposed line,

$$\hat{\mathbf{Y}}_S = \boldsymbol{A}^{-1}\,\hat{\boldsymbol{Y}}_P\boldsymbol{A} = \begin{bmatrix} \hat{y}_0 & 0 & 0 \\ 0 & \hat{y}_1 & 0 \\ 0 & 0 & \hat{y}_2 \end{bmatrix} = j(2\pi f)\begin{bmatrix} \hat{C}_0 & 0 & 0 \\ 0 & \hat{C}_1 & 0 \\ 0 & 0 & \hat{C}_2 \end{bmatrix} \tag{9.4.9}$$

where

$$\hat{C}_0 = \hat{C}_{aa} + 2\hat{C}_{ab} \quad \text{F/m} \tag{9.4.10}$$

$$\hat{C}_1 = \hat{C}_2 = \hat{C}_{aa} - \hat{C}_{ab} \quad \text{F/m} \tag{9.4.11}$$

Since $\hat{C}_{ab}$ is negative, the zero-sequence capacitance $\hat{C}_0$ is usually much less than the positive- or negative-sequence capacitance.

Circuit representations of the phase and sequence capacitances of a completely transposed three-phase line are shown in Figure 9.12.

9.5 SEQUENCE NETWORKS OF ROTATING MACHINES

A Y-connected synchronous generator grounded through a neutral impedance Z_n is shown in Figure 9.13. The internal generator voltages are designated E_a, E_b, and E_c, and the generator line currents are designated I_a, I_b, and I_c.

The sequence networks of the generator are shown in Figure 9.14. Since a three-phase synchronous generator is designed to produce balanced internal phase voltages E_a, E_b, and E_c with only a positive-sequence component, a source voltage E_{g1} is included only in the positive-sequence network. The sequence components of the line-to-ground voltages at the generator terminals are denoted V_0, V_1, and V_2 in Figure 9.14.

The voltage drop in the generator neutral impedance is Z_nI_n, which can be written as $(3Z_n)I_0$, since, from (9.1.27), the neutral current is three times the zero-sequence current. Since this voltage drop is due only to zero-sequence current, an impedance $(3Z_n)$ is placed in the zero-sequence network of Figure 9.14 in series with the generator zero-sequence impedance Z_{g0}.

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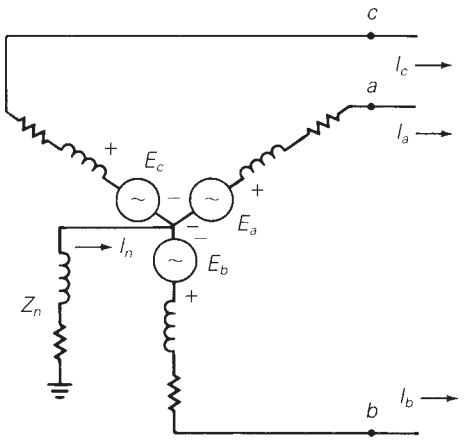
FIGURE 9.13

Y-connected synchronous generator



The sequence impedances of rotating machines are generally not equal. A detailed analysis of machine-sequence impedances is given in machine theory texts. Here is a brief explanation.

When a synchronous generator stator has balanced three-phase positive-sequence currents under steady-state conditions, the net mmf produced by these positive-sequence currents rotates at the synchronous rotor speed in the same direction as that of the rotor. Under this condition, a high value of magnetic flux penetrates the rotor, and the positive-sequence impedance Z_{g1} has a high value. Under steady-state conditions, the positive-sequence generator impedance is called the *synchronous impedance*.

When a synchronous generator stator has balanced three-phase negative-sequence currents, the net mmf produced by these currents rotates at synchronous speed in the direction opposite to that of the rotor. With respect to the rotor, the net mmf is not stationary but rotates at twice synchronous speed. Under this condition, currents are induced in the rotor windings that prevent the magnetic flux from penetrating the rotor. As such, the negative-sequence impedance Z_{g2} is less than the positive-sequence synchronous impedance.

When a synchronous generator has only zero-sequence currents, which are line (or phase) currents with equal magnitude and phase, then the net mmf produced

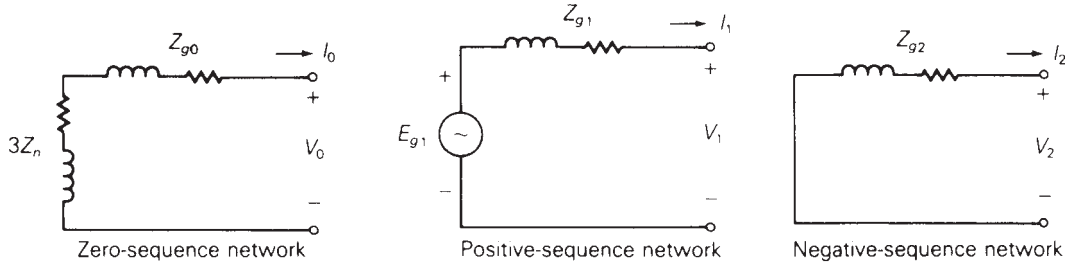


FIGURE 9.14

Sequence networks of a Y-connected synchronous generator

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by these currents is theoretically zero. The generator zero-sequence impedance Z_{g0} is the smallest sequence impedance and is due to leakage flux, end turns, and harmonic flux from windings that do not produce a perfectly sinusoidal mmf.

Typical values of machine-sequence impedances are listed in Table A.1 in the Appendix. The positive-sequence machine impedance is synchronous, transient, or subtransient. *Synchronous* impedances are used for steady-state conditions, such as in power-flow studies, which are described in Chapter 6. *Transient* impedances are used for stability studies, which are described in Chapter 12, and *subtransient* impedances are used for short-circuit studies, which are described in Chapters 8 and 10. Unlike the positive-sequence impedances, a machine has only one negative-sequence impedance and only one zero-sequence impedance.

The sequence networks for three-phase synchronous motors and for three-phase induction motors are shown in Figure 9.15. Synchronous motors have the same sequence networks as synchronous generators, except that the sequence currents for synchronous motors are referenced *into* rather than out of the sequence networks. Also, induction motors have the same sequence networks as synchronous motors, except that

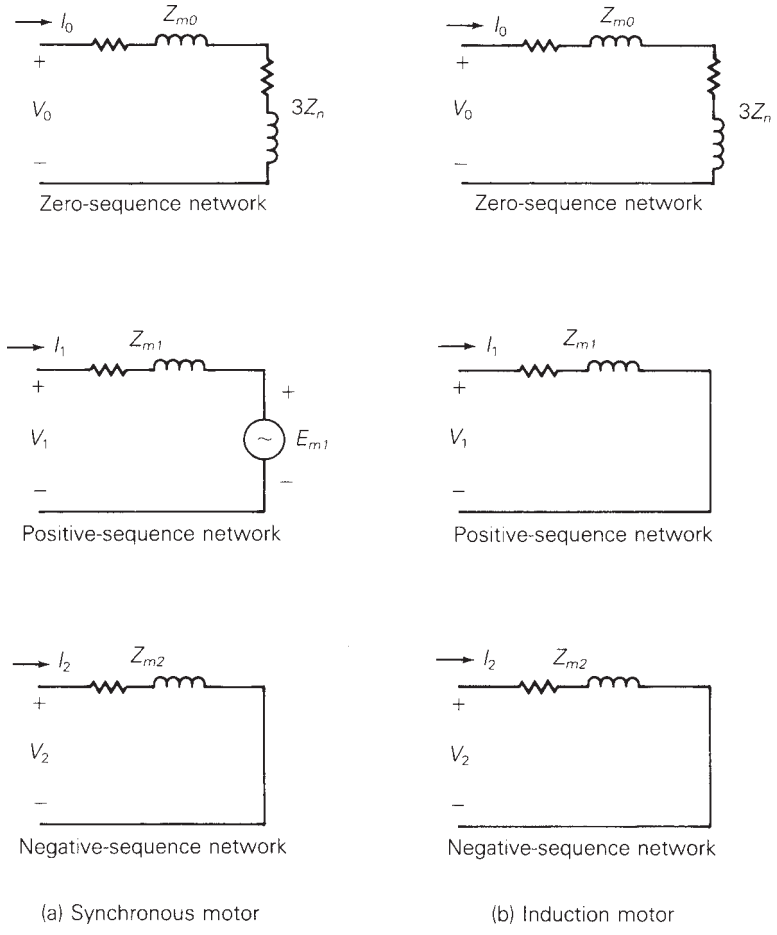

FIGURE 9.15

Sequence networks of three-phase motors

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the positive-sequence voltage source E_{m1} is removed. Induction motors do not have a dc source of magnetic flux in their rotor circuits, and therefore E_{m1} is zero (or a short circuit). The sequence networks shown in Figures 9.14 and 9.15 are simplified networks for rotating machines. The networks do not take into account such phenomena as machine saliency, saturation effects, and more complicated transient effects. These simplified networks, however, are in many cases accurate enough for power system studies.

EXAMPLE 9.5

Currents in sequence networks

Draw the sequence networks for the circuit of Example 2.4 and calculate the sequence components of the line current. Assume that the generator neutral is grounded through an impedance $Z_n = j10\ \Omega$, and that the generator sequence impedances are $Z_{g0} = j1\ \Omega$, $Z_{g1} = j15\ \Omega$, and $Z_{g2} = j3\ \Omega$.

SOLUTION

The sequence networks are shown in Figure 9.16. They are obtained by interconnecting the sequence networks for a balanced-Δ load, for series-line

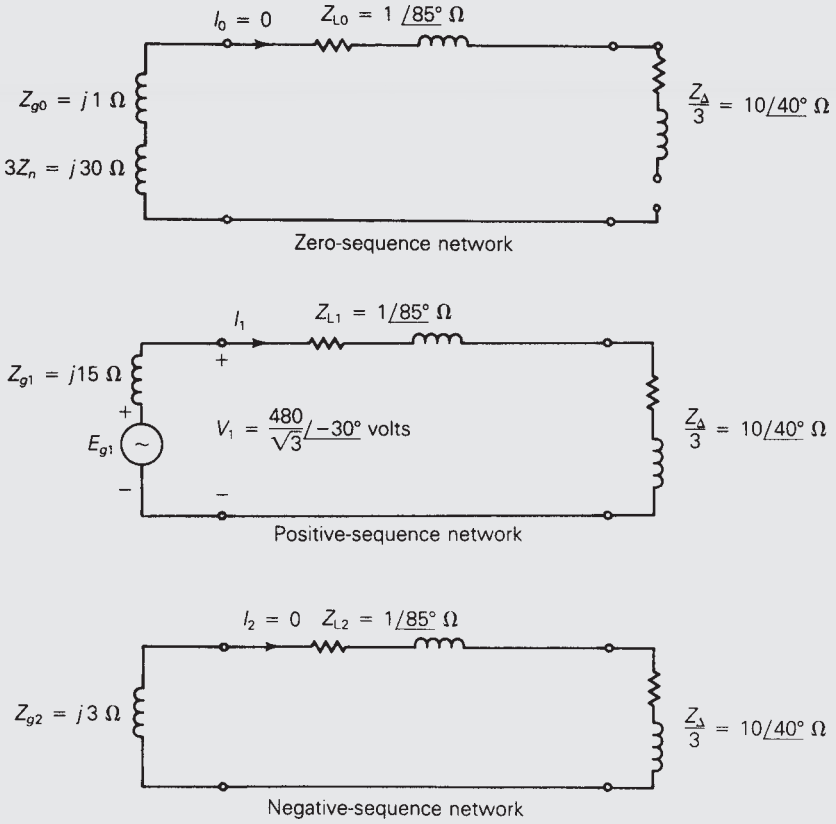

FIGURE 9.16
Sequence networks for Example 9.5

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impedances, and for a synchronous generator, which are given in Figures 9.5, 9.10, and 9.14.

It is clear from Figure 9.16 that $I_0 = I_2 = 0$ since there are no sources in the zero- and negative-sequence networks. Also, the positive-sequence generator terminal voltage V_1 equals the generator line-to-neutral terminal voltage. Therefore, from the positive-sequence network shown in the figure and from the results of Example 2.4,

$$I_1 = \frac{V_1}{\left(Z_{L1} + \frac{1}{3}Z_\Delta\right)} = 25.83\angle -73.78^\circ \text{ A} = I_a$$

Note that from (9.1.20), I_1 equals the line current I_a, since $I_0 = I_2 = 0$.

The following example illustrates the superiority of using symmetrical components for analyzing unbalanced systems.

EXAMPLE 9.6

Solving unbalanced three-phase networks using sequence components

A Y-connected voltage source with the following unbalanced voltage is applied to the balanced line and load of Example 2.4.

$$\begin{bmatrix} V_{ag} \\ V_{bg} \\ V_{cg} \end{bmatrix} = \begin{bmatrix} 277\angle 0^\circ \\ 260\angle -120^\circ \\ 295\angle +115^\circ \end{bmatrix} \text{ volts}$$

The source neutral is solidly grounded. Using the method of symmetrical components, calculate the source currents I_a, I_b, and I_c.

SOLUTION

Using (9.1.13) through (9.1.15), the sequence components of the source voltages are

$$V_0 = \tfrac{1}{3}(277\angle 0^\circ + 260\angle -120^\circ + 295\angle 115^\circ)$$
$$= 7.4425 + j14.065 = 15.912\angle 62.11^\circ \text{ volts}$$
$$V_1 = \tfrac{1}{3}(277\angle 0^\circ + 260\angle -120^\circ + 120^\circ + 295\angle 115^\circ + 240^\circ)$$
$$= \tfrac{1}{3}(277\angle 0^\circ + 260\angle 0^\circ + 295\angle -5^\circ)$$
$$= 276.96 - j8.5703 = 277.1\angle -1.772^\circ \text{ volts}$$

(*Continued*)

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$$V_2 = \tfrac{1}{3}(277\angle 0° + 260\angle -120° + 240° + 295\angle 115° + 120°)$$
$$= \tfrac{1}{3}(277\angle 0° + 260\angle 120° + 295\angle 235°)$$
$$= -7.4017 - j5.4944 = 9.218\angle 216.59° \text{ volts}$$

These sequence voltages are applied to the sequence networks of the line and load, as shown in Figure 9.17. The sequence networks of this figure are uncoupled, and the sequence components of the source currents are easily calculated as follows:

$$I_0 = 0$$
$$I_1 = \frac{V_1}{Z_{L1} + \frac{Z_\Delta}{3}} = \frac{277.1\angle -1.772°}{10.73\angle 43.78°} = 25.82\angle -45.55° \text{ A}$$
$$I_2 = \frac{V_2}{Z_{L2} + \frac{Z_\Delta}{3}} = \frac{9.218\angle 216.59°}{10.73\angle 43.78°} = 0.8591\angle 172.81° \text{ A}$$

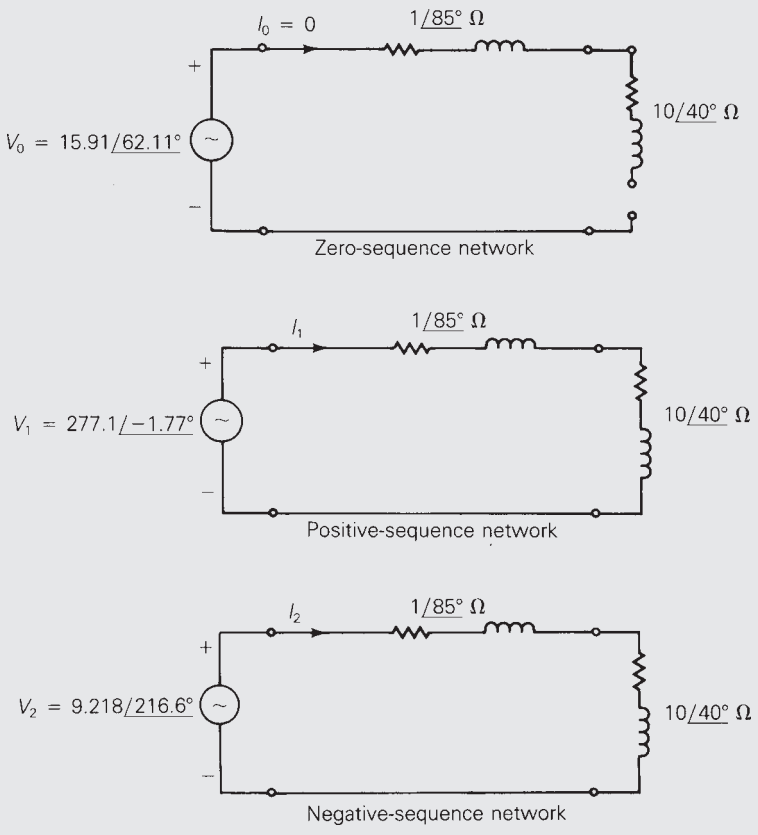

FIGURE 9.17
Sequence networks for Example 9.6

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Using (9.1.20) through (9.1.22), the source currents are

$$
\begin{aligned}
I_a &= (0 + 25.82\angle -45.55^\circ + 0.8591\angle 172.81^\circ) \\
&= 17.23 - j18.32 = 25.15\angle -46.76^\circ \text{ A} \\
I_b &= (0 + 25.82\angle -45.55^\circ + 240^\circ + 0.8591\angle 172.81^\circ + 120^\circ) \\
&= (25.82\angle 194.45^\circ + 0.8591\angle 292.81^\circ) \\
&= -24.67 - j7.235 = 25.71\angle 196.34^\circ \text{ A} \\
I_c &= (0 + 25.82\angle -45.55^\circ + 120^\circ + 0.8591\angle 172.81^\circ + 240^\circ) \\
&= (25.82\angle 74.45^\circ + 0.8591\angle 52.81^\circ) \\
&= 7.441 + j25.56 = 26.62\angle 73.77^\circ \text{ A}
\end{aligned}
$$

You should calculate the line currents for this example without using symmetrical components in order to verify this result and to compare the two solution methods (see Problem 9.33). Without symmetrical components, coupled KVL equations must be solved. With symmetrical components, the conversion from phase to sequence components decouples the networks as well as the resulting KVL equations, as shown above.

9.6 PER-UNIT SEQUENCE MODELS OF THREE-PHASE, TWO-WINDING TRANSFORMERS

Figure 9.18(a) is a schematic representation of an ideal Y–Y transformer grounded through neutral impedances Z_N and Z_n. Figures 9.18(b through d) show the per-unit sequence networks of this ideal transformer.

When balanced positive-sequence currents or balanced negative-sequence currents are applied to the transformer, the neutral currents are zero, and there are no voltage drops across the neutral impedances. Therefore, the per-unit positive- and negative-sequence networks of the ideal Y–Y transformer, Figures 9.18(b) and (c), are the same as the per-unit single-phase ideal transformer, Figure 3.9(a).

Zero-sequence currents have equal magnitudes and equal phase angles. When per-unit sequence currents $I_{A0} = I_{B0} = I_{C0} = I_0$ are applied to the high-voltage windings of an ideal Y–Y transformer, the neutral current I_0 flows through the neutral impedance Z_N, with a voltage drop $(3Z_N)I_0$. Also, per-unit zero-sequence current I_0 flows in each low-voltage winding [from Equation (3.3.9)], and therefore, $3I_0$ flows through neutral impedance Z_n, with a voltage drop $(3I_0)Z_n$.

The per-unit zero-sequence network, which includes the impedances $(3Z_N)$ and $(3Z_n)$, is shown in Figure 9.18(b).

Note that if either one of the neutrals of an ideal transformer is ungrounded, then no zero sequence can flow in either the high- or low-voltage windings. For example, if the high-voltage winding has an open neutral, then $I_N = 3I_0 = 0$, which in turn forces $I_0 = 0$ on the low-voltage side. This can be shown in the

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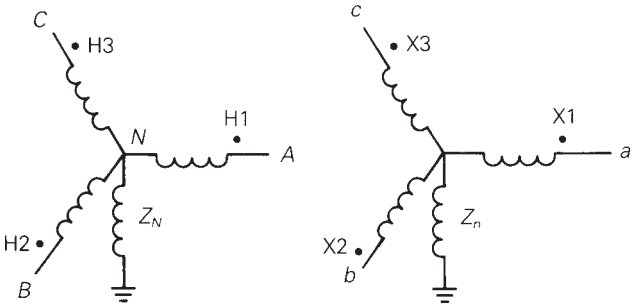
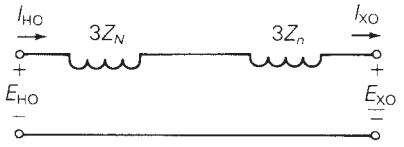
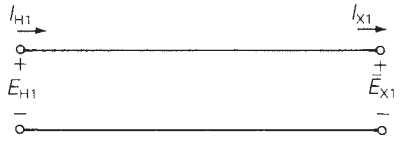
FIGURE 9.18
Ideal Y–Y transformer



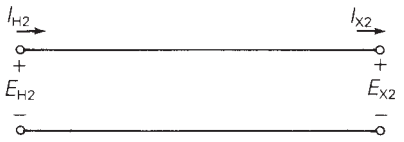
(a) Schematic representation



(b) Per-unit zero-sequence network



(c) Per-unit positive-sequence network



(d) Per-unit negative-sequence network

zero-sequence network of Figure 9.18(b) by making $Z_N = \infty$, which corresponds to an open circuit.

The per-unit sequence networks of a practical Y–Y transformer are shown in Figure 9.19(a). These networks are obtained by adding external impedances to the sequence networks of the ideal transformer, as follows. The leakage impedances of the high-voltage windings are series impedances like the series impedances shown in Figure 9.9, with no coupling between phases ($Z_{ab} = 0$). If the phase *a*, *b*, and *c* windings have equal leakage impedances $Z_H = R_H + jX_H$, then the series impedances are *symmetrical* with sequence networks, as shown in Figure 9.10, where $Z_{H0} = Z_{H1} = Z_{H2} = Z_H$. Similarly, the leakage impedances of the low-voltage windings are symmetrical series impedances with $Z_{X0} = Z_{X1} = Z_{X2} = Z_X$. These series leakage impedances are shown in per unit in the sequence networks of Figure 9.19(a).

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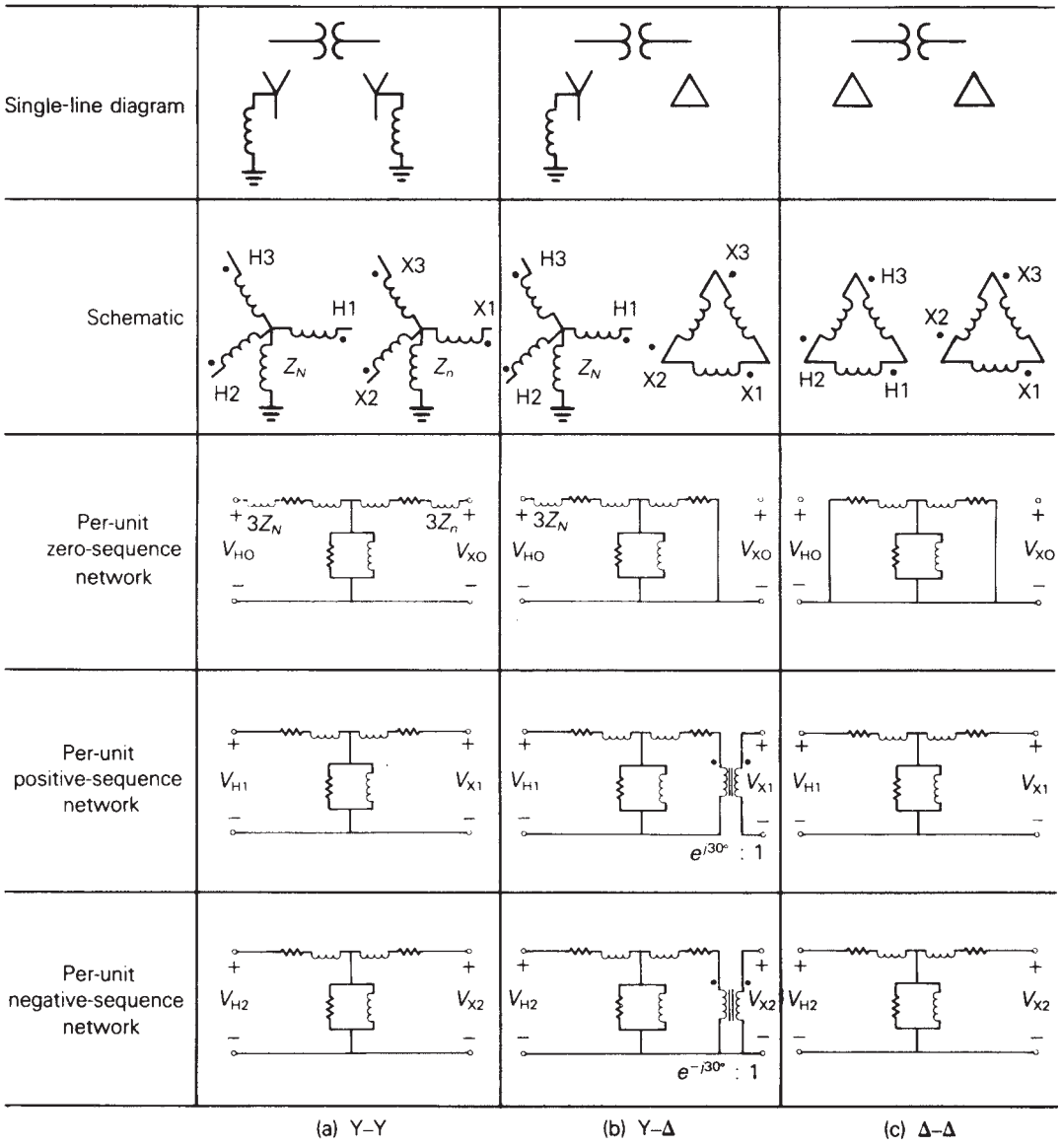

FIGURE 9.19

Per-unit sequence networks of practical Y–Y, Y–Δ, and Δ–Δ transformers

The shunt branches of the practical Y–Y transformer, which represent exciting current, are equivalent to the Y load of Figure 9.3. Each phase in Figure 9.3 represents a core loss resistor in parallel with a magnetizing inductance. Assuming these are the same for each phase, then the Y load is *symmetrical*, and the sequence networks are shown in Figure 9.4. These shunt branches are also shown in Figure 9.19(a). Note that $(3Z_N)$ and $(3Z_n)$ have already been included in the zero-sequence network.

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The per-unit positive- and negative-sequence transformer impedances of the practical Y–Y transformer in Figure 9.19(a) are identical, which is always true for nonrotating equipment. The per-unit zero-sequence network, however, depends on the neutral impedances Z_N and Z_n.

The per-unit sequence networks of the Y–Δ transformer, shown in Figure 9.19(b), have the following features:

1. The per-unit impedances do not depend on the winding connections. That is, the per-unit impedances of a transformer that is connected Y–Y, Y–Δ, Δ–Y, or Δ–Δ are the same. However, the base voltages do depend on the winding connections.
2. A phase shift is included in the per-unit positive- and negative-sequence networks. For the American standard, the positive-sequence voltages and currents on the high-voltage side of the Y–Δ transformer lead the corresponding quantities on the low-voltage side by 30°. For negative sequence, the high-voltage quantities lag by 30°.
3. Zero-sequence currents can flow in the Y winding if there is a neutral connection, and corresponding zero-sequence currents flow within the Δ winding. However, no zero-sequence current enters or leaves the Δ winding.

The phase shifts in the positive- and negative-sequence networks of Figure 9.19(b) are represented by the phase-shifting transformer of Figure 3.4. Also, the zero-sequence network of Figure 9.19(b) provides a path on the Y side for zero-sequence current to flow, but no zero-sequence current can enter or leave the Δ side.

The per-unit sequence networks of the Δ–Δ transformer, shown in Figure 9.19(c), have the following features:

1. The positive- and negative-sequence networks, which are identical, are the same as those for the Y–Y transformer. It is assumed that the windings are labeled so there is no phase shift. Also, the per-unit impedances do not depend on the winding connections, but the base voltages do.
2. Zero-sequence currents *cannot* enter or leave either Δ winding, although they can circulate within the Δ windings.

EXAMPLE 9.7

Solving unbalanced three-phase networks with transformers using per-unit sequence components

A 75-kVA, 480-volt Δ/208-volt Y transformer with a solidly grounded neutral is connected between the source and line of Example 9.6. The transformer leakage reactance is $X_{eq} = 0.10$ per unit; winding resistances and

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exciting current are neglected. Using the transformer ratings as base quantities, draw the per-unit sequence networks and calculate the phase *a* source current I_a.

SOLUTION

The base quantities are $S_{base1\phi} = 75/3 = 25$ kVA, $V_{baseHLN} = 480/\sqrt{3} = 277.1$ volts, $V_{baseXLN} = 208/\sqrt{3} = 120.1$ volts, and $Z_{baseX} = (120.1)^2/25{,}000 = 0.5770\,\Omega$. The sequence components of the actual source voltages are given in Figure 9.17. In per unit, these voltages are

$$V_0 = \frac{15.91\angle 62.11^\circ}{277.1} = 0.05742\angle 62.11^\circ \text{ per unit}$$
$$V_1 = \frac{277.1\angle -1.772^\circ}{277.1} = 1.0\angle -1.772^\circ \text{ per unit}$$
$$V_2 = \frac{9.218\angle 216.59^\circ}{277.1} = 0.03327\angle 216.59^\circ \text{ per unit}$$

The per-unit line and load impedances, which are located on the low-voltage side of the transformer, are

$$Z_{L0} = Z_{L1} = Z_{L2} = \frac{1\angle 85^\circ}{0.577} = 1.733\angle 85^\circ \text{ per unit}$$
$$Z_{load1} = Z_{load2} = \frac{Z_\Delta}{3(0.577)} = \frac{10\angle 40^\circ}{0.577} = 17.33\angle 40^\circ \text{ per unit}$$

The per-unit sequence networks are shown in Figure 9.20. Note that the per-unit line and load impedances, when referred to the high-voltage side of the phase-shifting transformer, do not change [see Equation (3.1.26)]. Therefore, from Figure 9.20, the sequence components of the source currents are

$$I_0 = 0$$
$$I_1 = \frac{V_1}{jX_{eq} + Z_{L1} + Z_{load1}} = \frac{1.0\angle -1.772^\circ}{j0.10 + 1.733\angle 85^\circ + 17.33\angle 40^\circ}$$
$$= \frac{1.0\angle -1.772^\circ}{13.43 + j12.97} = \frac{1.0\angle -1.772^\circ}{18.67\angle 44.0^\circ} = 0.05356\angle -45.77^\circ \text{ per unit}$$

$$I_2 = \frac{V_2}{jX_{eq} + Z_{L2} + Z_{load2}} = \frac{0.03327\angle 216.59^\circ}{18.67\angle 44.0^\circ}$$
$$= 0.001782\angle 172.59^\circ \text{ per unit}$$

(*Continued*)

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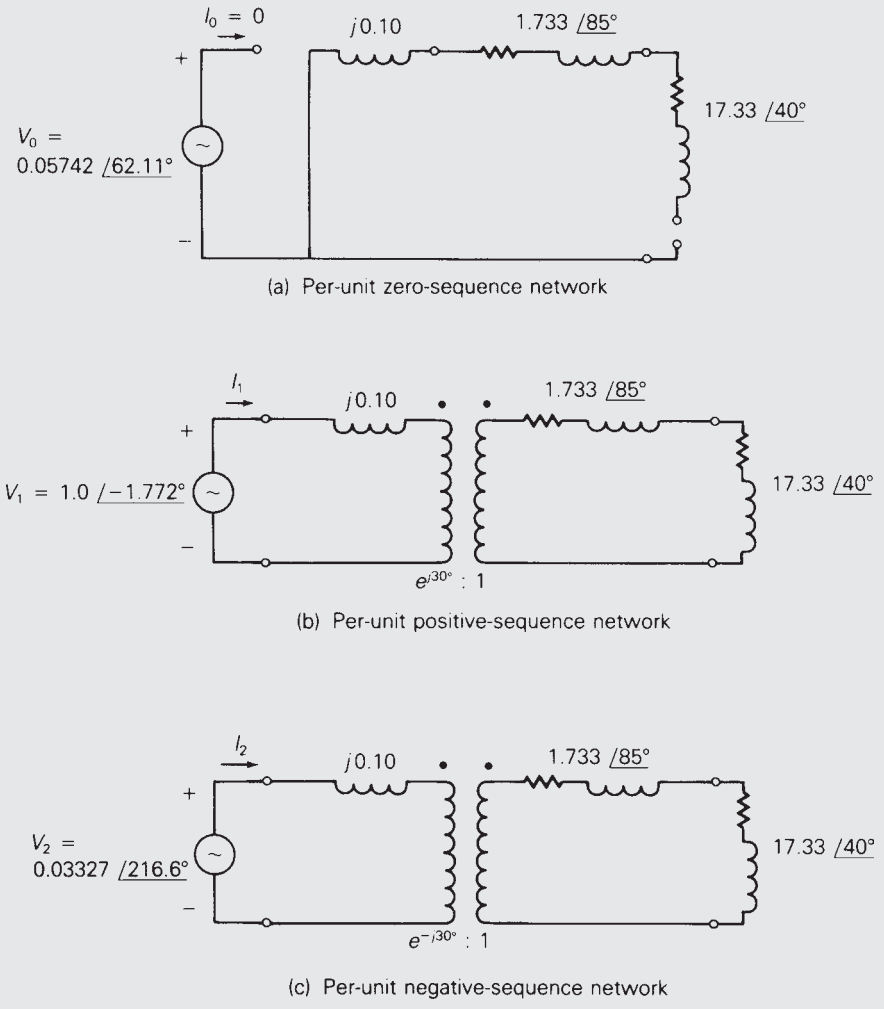
FIGURE 9.20

Per-unit sequence networks for Example 9.7



The phase *a* source current is then, using (9.1.20),

$$I_a = I_0 + I_1 + I_2$$
$$= 0 + 0.05356\angle -45.77° + 0.001782\angle 172.59°$$
$$= 0.03511 - j0.03764 = 0.05216\angle -46.19° \text{ per unit}$$

$$\text{Using } I_{\text{baseH}} = \frac{75{,}000}{480\sqrt{3}} = 90.21 \text{ A},$$

$$I_a = (0.05216)(90.21)\angle -46.19° = 4.705\angle -46.19° \text{ A}$$

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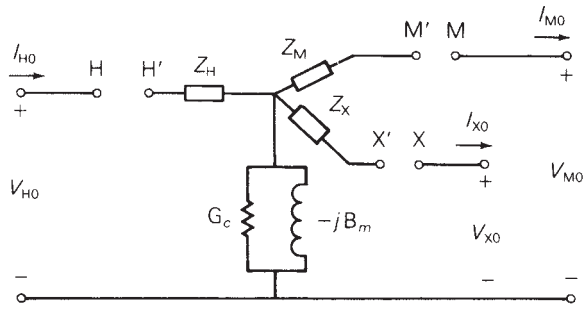
9.7 PER-UNIT SEQUENCE MODELS OF THREE-PHASE, THREE-WINDING TRANSFORMERS

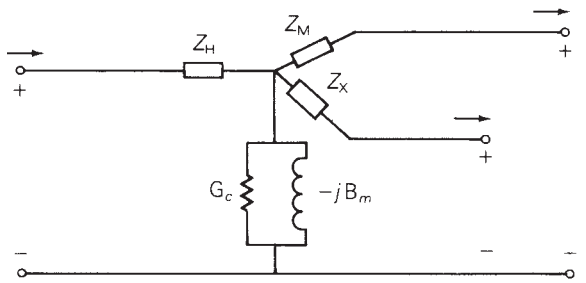
Three identical single-phase, three-winding transformers can be connected to form a three-phase bank. Figure 9.21 shows the general per-unit sequence networks of a three-phase, three-winding transformer. Instead of labeling the windings 1, 2, and 3, as was done for the single-phase transformer, the letters *H*, *M*, and *X* are used to denote the high-, medium-, and low-voltage windings, respectively. By convention, a common S_{base} is selected for the H, M, and X terminals, and voltage bases V_{baseH}, V_{baseM}, and V_{baseX} are selected in proportion to the rated line-to-line voltages of the transformer.

For the general zero-sequence network, Figure 9.21(a), the connection between terminals H and H′ depends on how the high-voltage windings are connected, as follows:

1. Solidly grounded Y—Short H to H′.
2. Grounded Y through Z_N—Connect $(3Z_N)$ from H to H′.
3. Ungrounded Y—Leave H–H′ open as shown.
4. Δ—Short H′ to the reference bus.



(a) Per-unit zero-sequence network



(b) Per-unit positive- or negative-sequence network (phase shift not shown)

FIGURE 9.21
Per-unit sequence networks of a three-phase three-winding transformer

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Terminals X–X′ and M–M′ are connected in a similar manner.

The impedances of the per-unit negative-sequence network are the same as those of the per-unit positive-sequence network, which is always true for nonrotating equipment. Phase-shifting transformers, not shown in Figure 9.21(b), can be included to model phase shift between Δ and Y windings.

EXAMPLE 9.8

Three-winding three-phase transformer: per-unit sequence networks

Three transformers, each identical to that described in Example 3.9, are connected as a three-phase bank in order to feed power from a 900-MVA, 13.8-kV generator to a 345-kV transmission line and to a 34.5-kV distribution line. The transformer windings are connected as follows:

13.8-kV windings (X):	Δ, to generator
199.2-kV windings (H):	solidly grounded Y, to 345-kV line
19.92-kV windings (M):	grounded Y through $Z_n = j01.0\ \Omega$, to 34.5-kV line

The positive-sequence voltages and currents of the high- and medium-voltage Y windings lead the corresponding quantities of the low-voltage Δ winding by 30°. Draw the per-unit sequence networks, using a three-phase base of 900 MVA and 13.8 kV for terminal X.

SOLUTION

The per-unit sequence networks are shown in Figure 9.22. Since $V_{baseX} = 13.8$ kV is the rated line-to-line voltage of terminal X, $V_{baseM} = \sqrt{3}\,(19.92) = 34.5$ kV, which is the rated line-to-line voltage of terminal M. The base impedance of the medium-voltage terminal is then

$$Z_{baseM} = \frac{(34.5)^2}{900} = 1.3225 \quad \Omega$$

Therefore, the per-unit neutral impedance is

$$Z_n = \frac{j0.10}{1.3225} = j0.07561 \quad \text{per unit}$$

and $(3Z_n) = j0.2268$ is connected from terminal M to M′ in the per-unit zero-sequence network. Since the high-voltage windings have a solidly grounded neutral, H to H′ is shorted in the zero-sequence network. Also, phase-shifting transformers are included in the positive- and negative-sequence networks.

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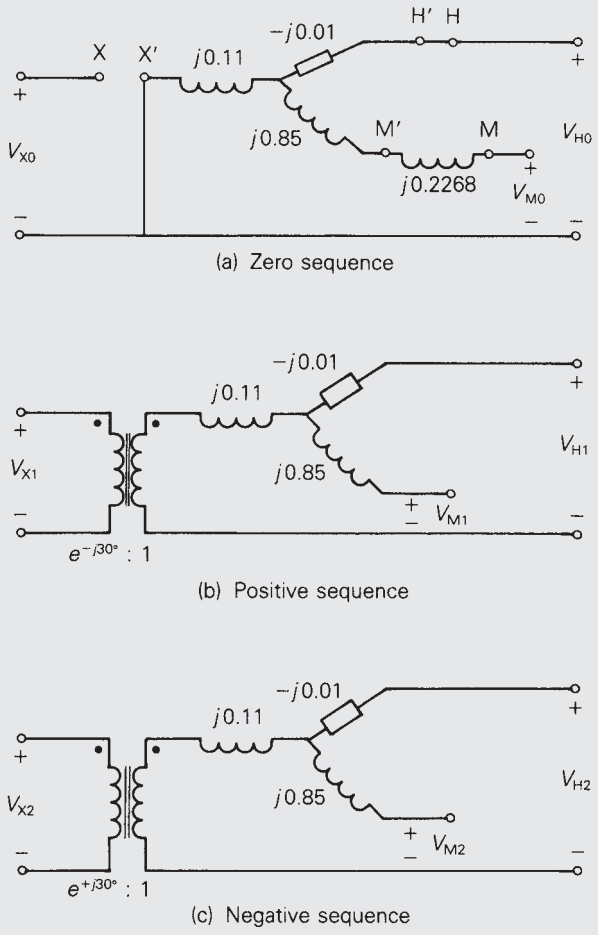
FIGURE 9.22

Per-unit sequence networks for Example 9.8



9.8 POWER IN SEQUENCE NETWORKS

The power delivered to a three-phase network can be determined from the power delivered to the sequence networks. Let S_P denote the total complex power delivered to the three-phase load of Figure 9.7, which can be calculated from

$$S_P = V_{ag}I_a^* + V_{bg}I_b^* + V_{cg}I_c^* \tag{9.8.1}$$

Equation (9.8.1) is also valid for the total complex power delivered by the three-phase generator of Figure 9.13, or for the complex power delivered to any three-phase bus. Rewriting (9.8.1) in matrix format,

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$$S_{\mathrm{P}} = [V_{ag} V_{bg} V_{cg}] \begin{bmatrix} I_a^* \\ I_b^* \\ I_c^* \end{bmatrix}$$
$$= \boldsymbol{V}_{\mathrm{P}}^{\mathrm{T}} \boldsymbol{I}_{\mathrm{P}}^* \tag{9.8.2}$$

where T denotes transpose and * denotes complex conjugate. Now, using (9.1.9) and (9.1.16),

$$\begin{aligned} S_{\mathrm{P}} &= (\boldsymbol{A}\boldsymbol{V}_s)^{\mathrm{T}} \, (\boldsymbol{A}\boldsymbol{I}_s)^* \\ &= \boldsymbol{V}_s^{\mathrm{T}} \, [\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}^*]\boldsymbol{I}_s^* \end{aligned} \tag{9.8.3}$$

Using the definition of $\boldsymbol{A}$, which is (9.1.8), to calculate the term within the brackets of (9.8.3), and noting that a and a^2 are conjugates,

$$\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}^* = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix}^{\mathrm{T}} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix}^*$$
$$= \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}$$
$$= \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} = 3\mathbf{U} \tag{9.8.4}$$

Equation (9.8.4) can now be used in (9.8.3) to obtain

$$S_{\mathrm{P}} = 3\boldsymbol{V}_{\mathrm{S}}^{\mathrm{T}} \boldsymbol{I}_{\mathrm{S}}^*$$
$$= 3[V_0 + V_1 + V_2] \begin{bmatrix} I_0^* \\ I_1^* \\ I_2^* \end{bmatrix} \tag{9.8.5}$$

$$S_{\mathrm{P}} = 3(V_0 I_0^* + V_1 I_1^* + V_2 I_2^*)$$
$$= 3S_{\mathrm{S}} \tag{9.8.6}$$

Thus, the total complex power S_{P} delivered to a three-phase network equals *three* times the total complex power S_{S} delivered to the sequence networks.

The factor of 3 occurs in (9.8.6) because $\boldsymbol{A}^T\boldsymbol{A}^* = 3\mathbf{U}$ as shown by (9.8.4). It is possible to eliminate this factor of 3 by defining a new transformation matrix $\boldsymbol{A}_1 = (1/\sqrt{3})\, \boldsymbol{A}$ such that $\boldsymbol{A}_1^{\mathrm{T}}\boldsymbol{A}_1^* = \mathbf{U}$, which means that $\boldsymbol{A}_1$ is a *unitary* matrix. Using $\boldsymbol{A}_1$ instead of $\boldsymbol{A}$, the total complex power delivered to three-phase networks would equal the total complex power delivered to the sequence networks. However, standard industry practice for symmetrical components is to use $\boldsymbol{A}$, defined by (9.1.8).

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EXAMPLE 9.9

Power in sequence networks

Calculate S_P and S_S delivered by the three-phase source in Example 9.6. Verify that $S_P = 3S_S$.

SOLUTION

Using (9.5.1),

$$\begin{aligned} S_P &= (277\angle 0°)(25.15\angle +46.76°) + (260\angle -120°)(25.71\angle -196.34°) \\ &\quad + (295\angle 115°)(26.62\angle -73.77°) \\ &= 6967\angle 46.76° + 6685\angle 43.66° + 7853\angle 41.23° \\ &= 15{,}520 + j14{,}870 = 21{,}490\angle 43.78° \text{ VA} \end{aligned}$$

In the sequence domain,

$$\begin{aligned} S_S &= V_0 I_0^* + V_1 I_1^* + V_2 I_2^* \\ &= 0 + (277.1\angle -1.77°)(25.82\angle 45.55°) \\ &\quad + (9.218\angle 216.59°)(0.8591\angle -172.81°) \\ &= 7155\angle 43.78° + 7.919\angle 43.78° \\ &= 5172 + j4958 = 7163\angle 43.78° \text{ VA} \end{aligned}$$

Also,

$$3S_S = 3(7163\angle 43.78°) = 21{,}490\angle 43.78° = S_P$$

PROBLEMS

SECTION 9.1

9.1 Using the operator $a = 1\angle 120°$, evaluate the following in polar form: (a) $(a - 1)/(1 + a - a^2)$, (b) $(a^2 + a + j)/(ja + a^2)$, (c) $(1 + a)(1 + a^2)$, (d) $(a - a^2)(a^2 - 1)$, and (e) $\dfrac{-j}{a}$.

9.2 Using $a = 1\angle 120°$, evaluate the following in rectangular form:

a. a^{10}
b. $(ja)^{10}$
c. $(1 - a)^3$
d. e^a

Hint for (d): $e^{(x+jy)} = e^x e^{jy} = e^x\angle y$, where y is in radians.

e. $-\left(\dfrac{j}{a}\right)^{10}$

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9.3 Determine the symmetrical components of the following line currents: (a) $I_a = 6\angle 90°$, $I_b = 6\angle 320°$, $I_c = 6\angle 220°$ A, and (b) $I_a = j40$, $I_b = 40$, $I_c = 0$ A.

9.4 Find the phase voltages V_{an}, V_{bn}, and V_{cn}, whose sequence components are $V_0 = 45\angle 80°$, $V_1 = 90\angle 0°$, $V_2 = 45\angle 90°$ V.

9.5 For the unbalanced three-phase system described by $I_a = 9\angle 0°$ A

$I_b = 8\angle -90°$ A, $I_c = 6\angle 150°$ A,

compute the symmetrical components I_0, I_1, and I_2.

9.6 (a) Given the symmetrical components to be

$$V_0 = 10\angle 0° V,\ V_1 = 80\angle 30° V,\ V_2 = 40\angle -30° V,$$

determine the unbalanced phase voltages V_a, V_b, and V_c. (b) Using the results of part (a), calculate the line-to-line voltages V_{ab}, V_{bc}, and V_{ca}. Then determine the symmetrical components of these line-to-line voltages, the symmetrical components of the corresponding phase voltages, and the phase voltages. Compare them with the result of part (a). Comment on why they are different, even though either set results in the same line-to-line voltages.

9.7 One line of a three-phase generator is open-circuited, while the other two are short-circuited to ground. The line currents are $I_a = 0$, $I_b = 1500\angle 150°$, and $I_c = 1500\angle 30°$. Find the symmetrical components of these currents. Also, find the current into the ground.

9.8 Let an unbalanced, three-phase, Y-connected load (with phase impedances of Z_a, Z_b, and Z_c) be connected to a balanced three-phase supply, resulting in phase voltages of V_a, V_b, and V_c across the corresponding phase impedances. Choosing V_{ab} as the reference, show that

$$V_{ab,0} = 0; \qquad V_{ab,1} = \sqrt{3}V_{a,1}e^{j30°}; \qquad V_{ab,2} = \sqrt{3}V_{a,2}e^{-j30°}$$

9.9 Reconsider Problem 9.8 and choosing V_{bc} as the reference, show that

$$V_{bc,0} = 0; \qquad V_{bc,1} = -j\sqrt{3}V_{a,1}; \qquad V_{bc,2} = j\sqrt{3}V_{a,2}$$

9.10 Given the line-to-ground voltages $V_{ag} = 280\angle 0°$, $V_{bg} = 250\angle -110°$, and $V_{cg} = 290\angle 130°$ volts, calculate (a) the sequence components of the line-to-ground voltages, denoted V_{Lg0}, V_{Lg1}, and V_{Lg2}; (b) line-to-line voltages V_{LL0}, V_{LL1}, and V_{LL2}; and (c) sequence components of the line-to-line voltages V_{LL0}, V_{LL1}, and V_{LL2}. Also, verify the following general relation: $V_{LL0} = 0$, $V_{LL1} = \sqrt{3}V_{Lg1}\angle +30°$, and $V_{LL2} = \sqrt{3}V_{Lg2}\angle -30°$ volts.

9.11 A balanced Δ-connected load is fed by a three-phase supply for which phase C is open and phase A is carrying a current of $10\angle 0°$ A. Find the symmetrical components of the line currents. (Note that zero-sequence currents are not present for any three-wire system.)

9.12 A Y-connected load bank with a three-phase rating of 500 kVA and 2300 V consists of three identical resistors of 10.58 Ω. The load bank has the following applied voltages: $V_{ab} = 1840\angle 82.8°$, $V_{bc} = 2760\angle -41.4°$,

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and $V_{ca} = 2300\underline{/180°}$ V. Determine the symmetrical components of (a) the line-to-line voltages V_{ab0}, V_{ab1}, and V_{ab2}; (b) the line-to-neutral voltages V_{an0}, V_{an1}, and V_{an2}; and (c) the line currents I_{a0}, I_{a1}, and I_{a2}. (Note that the absence of a neutral connection means that zero-sequence currents are not present.)

SECTION 9.2

9.13 The currents in a Δ load are $I_{ab} = 10\underline{/0°}$, $I_{bc} = 12\underline{/-90°}$, and $I_{ca} = 15\underline{/90°}$ A. Calculate (a) the sequence components of the Δ-load currents, denoted $I_{\Delta 0}$, $I_{\Delta 1}$, and $I_{\Delta 2}$; (b) the line currents I_a, I_b, and I_c, which feed the Δ load; and (c) sequence components of the line currents I_{L0}, I_{L1}, and I_{L2}. Also, verify the following general relation: $I_{L0} = 0$, $I_{L1} = \sqrt{3}I_{\Delta 1}\underline{/-30°}$, and $I_{L2} = \sqrt{3}I_{\Delta 2}\underline{/+30°}$.

9.14 The voltages given in Problem 9.10 are applied to a balanced-Y load consisting of $(12 + j16)$ ohms per phase. The load neutral is solidly grounded. Draw the sequence networks and calculate I_0, I_1, and I_2, the sequence components of the line currents. Then calculate the line currents I_a, I_b, and I_c.

9.15 Repeat Problem 9.14 with the load neutral open.

9.16 Repeat Problem 9.14 for a balanced-Δ load consisting of $(12 + j16)$ ohms per phase.

9.17 Repeat Problem 9.14 for the load shown in Example 9.4 (Figure 9.6).

9.18 Perform the indicated matrix multiplications in (9.2.21) and verify the sequence impedances given by (9.2.22) through (9.2.27).

9.19 The following unbalanced line-to-ground voltages are applied to the balanced-Y load shown in Figure 3.3: $V_{ag} = 100\underline{/0°}$, $V_{bg} = 75\underline{/180°}$, and $V_{cg} = 50\underline{/90°}$ volts. The Y load has $Z_Y = 3 + j4\ \Omega$ per phase with neutral impedance $Z_n = j1\ \Omega$. (a) Calculate the line currents I_a, I_b, and I_c without using symmetrical components. (b) Calculate the line currents I_a, I_b, and I_c using symmetrical components. Which method is easier?

9.20 (a) Consider three equal impedances of $(j27)\ \Omega$ connected in Δ. Obtain the sequence networks.

(b) Now, with a mutual impedance of $(j6)\ \Omega$ between each pair of adjacent branches in the Δ-connected load of part (a), how would the sequence networks change?

9.21 Consider a three-phase impedance load with the following phase impedance matrix:

$$\boldsymbol{Z}_P = \begin{bmatrix} 3 + j7 & 0 & 0 \\ 0 & 3 + j7 & 0 \\ 0 & 0 & 3 + j7 \end{bmatrix}\ \Omega$$

Determine the sequence impedance matrix Z_S for this load. Is the load symmetrical?

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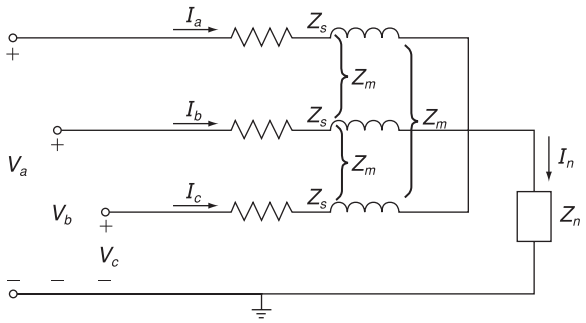
9.22 The three-phase impedance load shown in Figure 9.7 has the following sequence impedance matrix:

$$\boldsymbol{Z}_{\mathrm{P}} = \begin{bmatrix} (6 + j10) & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} \ \Omega$$

Determine the phase impedance matrix Z_{P} for this load. Is the load symmetrical?

9.23 Consider a three-phase balanced Y-connected load with self and mutual impedances as shown in Figure 9.23. Let the load neutral be grounded through an impedance Z_n. Using Kirchhoff's laws, develop the equations for line-to-neutral voltages, and then determine the elements of the phase impedance matrix. Also, find the elements of the corresponding sequence impedance matrix.

FIGURE 9.23
Problem 9.23



9.24 A three-phase balanced voltage source is applied to a balanced Y-connected load with ungrounded neutral. The Y-connected load consists of three mutually coupled reactances, where the reactance of each phase is $j12\ \Omega$, and the mutual coupling between any two phases is $j4\ \Omega$. The line-to-line source voltage is $100\sqrt{3}$ V. Determine the line currents (a) by mesh analysis without using symmetrical components, and (b) using symmetrical components.

9.25 A three-phase balanced Y-connected load with series impedances of $(6+ j24)\ \Omega$ per phase and mutual impedance between any two phases of $j3\ \Omega$ is supplied by a three-phase unbalanced source with line-to-neutral voltages of $V_{an} = 200\underline{/25°}$, $V_{bn} = 100\underline{/-155°}$, and $V_{cn} = 80\underline{/100°}$ V. The load and source neutrals are both solidly grounded. Determine: (a) the load sequence impedance matrix, (b) the symmetrical components of the line-to-neutral voltages, (c) the symmetrical components of the load currents, and (d) the load currents.

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SECTION 9.3

9.26 Repeat Problem 9.14, but include balanced three-phase line impedances of $(3 + j4)$ ohms per phase between the source and load.

9.27 Consider the flow of unbalanced currents in the symmetrical three-phase line section with neutral conductor as shown in Figure 9.24. (a) Express the voltage drops across the line conductors given by V_{aa}, V_{bb}, and V_{cc} in terms of line currents, self-impedances defined by $Z_S = Z_{aa} + Z_{nn} - 2Z_{an}$, and mutual impedances defined by $Z_m = Z_{ab} + Z_{nn} = 2Z_{an}$. (b) Show that the sequence components of the voltage drops between the ends of the line section can be written as $V_{aa'0} = Z_0 I_{a0}$, $V_{aa'1} = Z_1 I_{a1}$, and $V_{aa'2} = Z_2 I_{a2}$, where $Z_0 = Z_S + 2Z_m = Z_{aa} + 2Z_{ab} + 3Z_{nn} - 6Z_{an}$ and $Z_1 = Z_2 = Z_S = Z_m = Z_{aa} - Z_{ab}$.

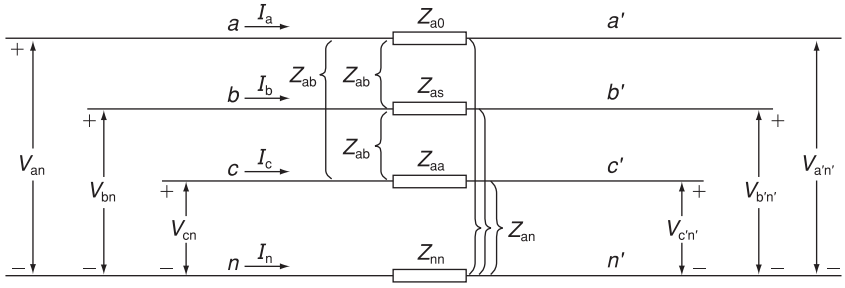

FIGURE 9.24
Problem 9.27

9.28 Let the terminal voltages at the two ends of the line section shown in Figure 9.24 be given by

$$V_{an} = (182 + j70)\ \text{kV} \qquad V_{an'} = (154 + j28)\ \text{kV}$$
$$V_{bn} = (72.24 - j32.62)\ \text{kV} \qquad V_{bn'} = (44.24 + j74.62)\ \text{kV}$$
$$V_{cn} = (-170.24 + j88.62)\ \text{kV} \qquad V_{cn'} = (-198.24 + j46.62)\ \text{kV}$$

The line impedances are given by:

$$Z_{aa} = j60\ \Omega \qquad Z_{ab} = j20\ \Omega \qquad Z_{nn} = j80\ \Omega \qquad Z_{an} = 0$$

(a) Compute the line currents using symmetrical components. *(Hint:* See Problem 9.27.) (b) Compute the line currents without using symmetrical components.

9.29 A completely transposed three-phase transmission line of 200 km in length has the following symmetrical sequence impedances and sequence admittances:

$$z_1 = z_2 = j0.5\ \Omega/\text{km}; \quad z_0 = j2\ \Omega/\text{km}$$
$$y_1 = y_2 = j3 \times 10^{-9}\ \text{s/m}; \quad y_0 = j1 \times 10^{-9}\ \text{s/m}$$

Set up the nominal π sequence circuits of this medium-length line.

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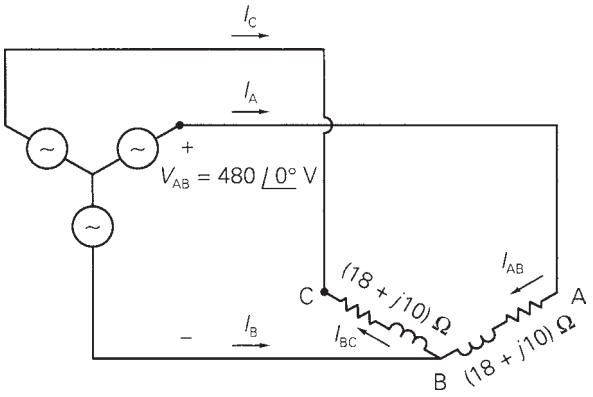
SECTION 9.5

9.30 As shown in Figure 9.25, a balanced three-phase, positive-sequence source with $V_{AB} = 480\angle 0°$ volts is applied to an unbalanced Δ load. Note that one leg of the Δ is open. Determine (a) the load currents I_{AB} and $I_{BC;}$ (b) the line currents I_A, I_B, and I_C, which feed the Δ load; and (c) the zero-, positive-, and negative-sequence components of the line currents.

FIGURE 9.25
Problem 9.30



9.31 A balanced Y-connected generator with terminal voltage $V_{bc} = 200\angle 0°$ volts is connected to a balanced-Δ load whose impedance is $10\angle 40°$ ohms per phase. The line impedance between the source and load is $0.5\angle 80°$ ohm for each phase. The generator neutral is grounded through an impedance of $j5$ ohms. The generator sequence impedances are given by $Z_{g0} = j7$, $Z_{g1} = j15$, and $Z_{g2} = j10$ ohms. Draw the sequence networks for this system and determine the sequence components of the line currents.

9.32 In a three-phase system, a synchronous generator supplies power to a 200-volt synchronous motor through a line having an impedance of $0.5\angle 80°$ ohm per phase. The motor draws 5 kW at 0.8 p.f. leading and at rated voltage. The neutrals of both the generator and motor are grounded through impedances of $j5$ ohms. The sequence impedances of both machines are $Z_0 = j5$, $Z_1 = j15$, and $Z_2 = j10$ ohms. Draw the sequence networks for this system and find the line-to-line voltage at the generator terminals. Assume balanced three-phase operation.

9.33 Calculate the source currents in Example 9.6 without using symmetrical components. Compare your solution method with that of Example 9.6. Which method is easier?

9.34 A Y-connected synchronous generator rated 20 MVA at 13.8 kV has a positive-sequence reactance of $j2.38\ \Omega$, negative-sequence reactance of $j3.33\ \Omega$, and zero-sequence reactance of $j0.95\ \Omega$. The generator neutral is solidly grounded. With the generator operating unloaded

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at rated voltage, a so-called single line-to-ground fault occurs at the machine terminals. During this fault, the line-to-ground voltages at the generator terminals are $V_{ag} = 0$, $V_{bg} = 8.071\angle -102.25°$, and $V_{cg} = 8.071\angle 102.25°$ kV. Determine the sequence components of the generator fault currents and the generator fault currents. Draw a phasor diagram of the prefault and postfault generator terminal voltages. (*Note:* For this fault, the sequence components of the generator fault currents are all equal to each other.)

9.35 Figure 9.26 shows a single-line diagram of a three-phase, interconnected generator-reactor system, in which the given per-unit reactances are based on the ratings of the individual pieces of equipment. If a three-phase short circuit occurs at fault point F, obtain the fault MVA and fault current in kA if the prefault bus bar line-to-line voltage is 13.2 kV. Choose 100 MVA as the base MVA for the system.

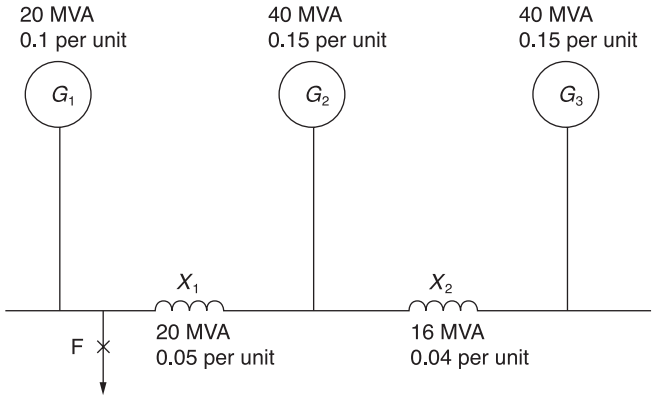

FIGURE 9.26
Oneline diagram for Problem 9.35

9.36 Consider Figures 9.13 and 9.14 with reference to a Y-connected synchronous generator (grounded through a neutral impedance Z_n) operating at no load. For a line-to-ground fault occurring on phase *a* of the generator, list the constraints on the currents and voltages in the phase domain, transform those into the sequence domain, and then obtain a sequence-network representation. Also, find the expression for the fault current in phase *a*.

9.37 Reconsider the synchronous generator of Problem 9.36. Obtain sequence-network representations for the following fault conditions.
(a) A short-circuit between phases *b* and *c*.
(b) A double line-to-ground fault with phases *b* and *c* grounded.

SECTION 9.6

9.38 Three single-phase, two-winding transformers, each rated 450 MVA, 20 kV/288.7 kV, with leakage reactance $X_{eq} = 0.12$ per unit, are

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connected to form a three-phase bank. The high-voltage windings are connected in Y with a solidly grounded neutral. Draw the per-unit zero-, positive-, and negative-sequence networks if the low-voltage windings are connected (a) in Δ with American standard phase shift, or (b) in Y with an open neutral. Use the transformer ratings as base quantities. Winding resistances and exciting current are neglected.

9.39 The leakage reactance of a three-phase, 500-MVA, 345 Y/23 Δ-kV transformer is 0.09 per unit based on its own ratings. The Y winding has a solidly grounded neutral. Draw the sequence networks. Neglect the exciting admittance and assume American standard phase shift.

9.40 Choosing system bases to be 360/24 kV and 100 MVA, redraw the sequence networks for Problem 9.39.

9.41 Draw the zero-sequence reactance diagram for the power system shown in Figure 3.38. The zero-sequence reactance of each generator and of the synchronous motor is 0.05 per unit based on equipment ratings. Generator 2 is grounded through a neutral reactor of 0.06 per unit on a 100-MVA, 18-kV base. The zero-sequence reactance of each transmission line is assumed to be three times its positive-sequence reactance. Use the same base as in Problem 3.41.

9.42 Three identical Y-connected resistors of $1.0\angle 0°$ per unit form a load bank that is supplied from the low-voltage Y-side of a Y–Δ transformer. The neutral of the load is not connected to the neutral of the system. The positive- and negative-sequence currents flowing toward the resistive load are given by

$$I_{a,1} = 1\angle 4.5° \text{ per unit;} \qquad I_{a,2} = 0.5\angle 250° \text{ per unit}$$

and the corresponding voltages on the low-voltage Y-side of the transformer are

$$V_{an,1} = 1\angle 45° \text{ per unit (Line-to-neutral voltage base)}$$

$$V_{an,2} = 0.5\angle 250° \text{ per unit (Line-to-neutral voltage base)}$$

Determine the line-to-line voltages and the line currents in per unit on the high-voltage side of the transformer. Account for the phase shift.

SECTION 9.7

9.43 Draw the positive-, negative-, and zero-sequence circuits for the transformers shown in Figure 3.36. Include ideal phase-shifting transformers showing phase shifts determined in Problem 3.32. Assume that all windings have the same kVA rating and that the equivalent leakage reactance of any two windings with the third winding open is 0.10 per unit. Neglect the exciting admittance.

9.44 A single-phase three-winding transformer has the following parameters: $Z_1 = Z_2 = Z_3 = 0 + j0.05$, $G_c = 0$, and $B_m = 0.2$ per unit. Three identical transformers, as described, are connected with their primaries in Y (solidly grounded neutral) and with their secondaries and

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tertiaries in Δ. Draw the per-unit sequence networks of this transformer bank.

SECTION 9.8

9.45 For Problem 9.14, calculate the real and reactive power delivered to the three-phase load.

9.46 A three-phase impedance load consists of a balanced-Δ load in parallel with a balanced-Y load. The impedance of each leg of the Δ load is $Z_\Delta = 6 + j6\ \Omega$, and the impedance of each leg of the Y load is $Z_Y = 2 + j2\ \Omega$. The Y load is grounded through a neutral impedance $Z_n = j1\ \Omega$. Unbalanced line-to-ground source voltages V_{ag}, V_{bg}, and V_{cg} with sequence components $V_0 = 10\underline{/60°}$, $V_1 = 100\underline{/0°}$, and $V_2 = 15\underline{/200°}$ volts are applied to the load. (a) Draw the zero-, positive-, and negative-sequence networks. (b) Determine the complex power delivered to each sequence network. (c) Determine the total complex power delivered to the three-phase load.

9.47 For Problem 9.12, compute the power absorbed by the load using symmetrical components. Then verify the answer by computing directly without using symmetrical components.

9.48 For Problem 9.25, determine the complex power delivered to the load in terms of symmetrical components. Verify the answer by adding up the complex power of each of the three phases.

9.49 Using the voltages of Problem 9.6(a) and the currents of Problem 9.5, compute the complex power dissipated based on (a) phase components, and (b) symmetrical components.

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10 Unsymmetrical Faults

The converter switchyard at Bonneville Power Administration's Celilo Converter Station in 2009. This station converts ac power to HVDC for transmission up to 3150 MW at ±500kV over a 1360-km bipolar line in the USA between The Dalles, OR and Los Angeles, CA. (© Greg Wahl-Stephens/Getty Images.)

LEARNING OBJECTIVES

At the end of this chapter, you will be able to:

1. Understand the use of per-unit zero-sequence, positive-sequence, and negative-sequence networks to calculate fault currents for balanced three-phase, single-line-to-ground, line-to-line, and double line-to-ground short circuits.
2. Understand the use of per-unit zero-sequence, positive-sequence, and negative-sequence bus impedance matrices to compute symmetrical fault currents for balanced and unbalanced short circuits.
3. Use PowerWorld Simulator to compute symmetrical fault currents for balanced and unbalanced short circuits.

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Short circuits occur in three-phase power systems as follows, in order of frequency of occurrence: single line-to-ground, line-to-line, double line-to-ground, and balanced three-phase faults. The path of the fault current may have either zero impedance, which is called a *bolted* short circuit, or nonzero impedance. Other types of faults include one-conductor-open and two-conductors-open, which can occur when conductors break or when one or two phases of a circuit breaker inadvertently open.

Although the three-phase short circuit occurs the least, it was covered first, in Chapter 8, because of its simplicity. When a balanced three-phase fault occurs in a balanced three-phase system, there is only positive-sequence fault current; the zero-, positive-, and negative-sequence networks are completely uncoupled.

When an unsymmetrical fault occurs in an otherwise balanced system, the sequence networks are interconnected only at the fault location. As such, the computation of fault currents is greatly simplified by the use of sequence networks.

As in the case of balanced three-phase faults, unsymmetrical faults have two components of fault current: an ac or symmetrical component—including subtransient, transient, and steady-state currents—and a dc component. The simplified E/X method for breaker selection described in Section 8.5 is also applicable to unsymmetrical faults. The dc offset current need not be considered unless it is too large—for example, when the X/R ratio for a single-line-to-ground fault is too large.

This chapter begins by using the per-unit zero-, positive-, and negative-sequence networks to represent a three-phase system. Also, certain assumptions are made to simplify fault-current calculations, and the balanced three-phase fault is briefly reviewed. Single line-to-ground, line-to-line, and double line-to-ground faults are presented in Sections 10.2, 10.3, and 10.4. The use of the positive-sequence bus impedance matrix for three-phase fault calculations in Section 8.4 is extended in Section 10.5 to unsymmetrical fault calculations by considering a bus impedance matrix for each sequence network. Examples using PowerWorld Simulator, which is based on the use of bus impedance matrices, are also included. The PowerWorld Simulator computes symmetrical fault currents for both three-phase and unsymmetrical faults. The Simulator may be used in power system design to select, set, and coordinate protective equipment.

Sulphur hexafluoride (SF6) has been used for decades as an insulating medium in transmission and distribution switchgear worldwide. SF6 is a man-made fluorinated gas with excellent dielectric properties. The compactness and durability of SF6-insulated switchgear are key reasons for its popularity. However, it is also considered harmful to the environment because of its greenhouse gas effect. SF6 has a global warming potential up to 23,500 times greater than carbon dioxide (CO_2) and can remain in the atmosphere for 3200 years.

Recently, switchgear manufacturers have been developing alternatives to SF6 switchgear that offer the same advantages without harmful environmental effects. The following paragraphs summarize case studies

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of SF6-free switchgear ordered by electric utility companies from three manufacturers: ABB, GE, and Siemens [10–12]. Sign in at www.cengage.com to download this product and to access its online resources.

CASE STUDY 1

ABB Commissions Switchgear Installation with New Eco-Efficient Gas[1]

T&D World Magazine

ABB's eco-efficient switchgear uses an SF6-alternative gas mixture as the insulation medium and additionally as a switching medium for high voltage switchgear. The fluoroketone-based SF6-alternative gas mixture is a chemical compound developed for switchgear applications in collaboration with 3M. The global warming potential of the new gas mixture is almost 100 percent lower than that of SF6.

170/24-kV switchgear bays have been supplied to the Swiss utility ewz for their newly energized 3 × 50-MVA substation in Zurich, Switzerland that supplies electricity to the northern and most populous part of the city—home to approximately 50,000 inhabitants as well as Zurich's largest event hall and trade fairgrounds. Low-noise and high efficiency power transformers and substation automation protection and control systems from ABB have also been deployed in the substation.

[1]© ABB Commissions Switchgear with New Eco-Efficient Gas": ABB, Zurich, Switzerland 2015-08-24. https://new.abb.com/news/detail/13211/abb-commissions-worlds-first-switchgear-installation-with-new-eco-efficient-gas

CASE STUDY 2

Transforming the Transmission Industry: The rapid adoption of Green Gas for Grid (g3) signals a global change in environmental responsibility[2]

Juan de Bedout

In about a year, nine countries decided to adopt GE's recently developed technology in the transmission and distribution sector—one that can reduce greenhouse gas emissions by up to 99 percent. *Green Gas for Grid* (denoted "g cubed" or "g3") is an insulating medium for high voltage (HV) switchgear designed by GE Power's Grid Solutions business in

[2]"Transforming the Transmission Industry", by Juan de Bedout, *T&D World Magazine* (April 27, 2018). With Permission of GE Renewable Energy.

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collaboration with 3M. G3 matches the dielectric strength of SF6 with no voltage limitation, and is compatible with the standard materials used in HV equipment, offering the same technical performance, the same dimensional footprint, and the same ability to operate in ambient conditions as SF6. The following three utilities have ordered GE's g3 substation equipment.

1. CERIUS, Denmark's largest cooperative energy company, chose g3 at the Haslev substation in the south of Sjaelland, located approximately 60 kilometers from Copenhagen, and it was expected to go into commercial operation in 2018. The project includes six full g3-gas-insulated substation (GIS) bays that will be part of the distribution network and supply power to more than 11,000 homes. It is expected to save about 1850 tons of CO_2 equivalent over a 40-year period.
2. A group of utilities from across Europe gathered in late 2017 at the Etzel substation near Zurich, Switzerland to witness the successful HV testing of g3 on a 145-kV, g3-enabled substation. The meeting was requested by GE's customers, Swiss utilities, Axpo Power AG, and the Elektrizitätswerke des Kantons Zurich (EKZ), to confirm g3 as a viable alternative to SF6. Following the tests, site works continued to connect the high voltage transformers and the substation to the network. Energization of the full site was scheduled for Spring 2018 (see Figure 1).
3. In 2017, National Grid became the world's first utility to energize a SF6-free 420-kV gas-insulated line, located in their Southeast England Sellindge substation using g3. The 230-meter-long, gas-insulated circuits, which connect the substation to the bushings and overhead lines, will increase the United Kingdom's energy security, reliability, and capacity by connecting with the French electricity transmission network (see Figure 2).

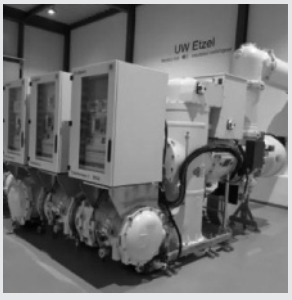

Figure 1 145-kV g3 substation in Switzerland (Axpo)

Figure 2 420-kV g3-insulated lines in England (National Grid)

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CASE STUDY 3

PG&E to Use SF6-Free Products From Siemens[3]

T&D World Magazine

In the United States, Pacific Gas and Electric (PG&E) ordered SF6-free dead-tank circuit breakers and gas-insulated switchgear from Siemens for California's 115-kV high-voltage grid. The orders comprise dead-tank circuit-breakers for 72.5 kV and 115 kV as well as gas-insulated substations for 115 kV. These circuit breakers use vacuum interrupters for switching functions and treated air known as "clean air" as the insulating medium. It's now possible to eliminate the use of sulfur hexafluoride, and this order is a major step forward for Siemens' SF6-free portfolio. The first delivery was scheduled for mid-2019.

Siemens' SF6-free circuit-breakers and gas-insulated substations are a further development of insulation using SF6, which has been proven in practice for years. A modern vacuum interrupter unit handles the switching and arc-extinguishing functions. Industrially prepared and purified air in a mixing ratio of 80 percent nitrogen to 20 percent oxygen ("clean air") provides the insulation for the current-carrying conductors inside the housing. The dead-tank circuit breakers have a rated voltage of 115 kV, a rated continuous current of 3000 amperes (A), and rated short-circuit breaking currents of 40 and 50 kiloamperes (kA).

The benefits over conventional SF6-insulated switchgear are significant: high switching power with the same footprint for outdoor and dead-tank circuit breakers. The products can be operated at lower temperatures without their own heating systems, and the switching unit requires no maintenance with a 40-year circuit-breaker lifecycle. "Clean air" can be released into the atmosphere with no harmful effects.

[3]"PG&E to use SF6-Free Products from Siemens", available online, with permission, from *T&D World Magazine* (November 15, 2018).

10.1 SYSTEM REPRESENTATION

A three-phase power system is represented by its sequence networks in this chapter. The zero-, positive-, and negative-sequence networks of system components—generators, motors, transformers, and transmission lines—as developed in Chapter 9, can be used to construct system zero-, positive-, and negative-sequence networks. The following assumptions are made:

1. The power system operates under balanced steady-state conditions before the fault occurs. Thus, the zero-, positive-, and negative-sequence

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networks are uncoupled before the fault occurs. During unsymmetrical faults, they are interconnected only at the fault location.

2. Prefault load current is neglected. Because of this, the positive-sequence internal voltages of all machines are equal to the prefault voltage V_F. Therefore, the prefault voltage at each bus in the positive-sequence network equals V_F.
3. Transformer winding resistances and shunt admittances are neglected.
4. Transmission-line series resistances and shunt admittances are neglected.
5. Synchronous machine armature resistance, saliency, and saturation are neglected.
6. All nonrotating impedance loads are neglected.
7. Induction motors are either neglected (especially for motors rated 50 hp or less) or represented in the same manner as synchronous machines.

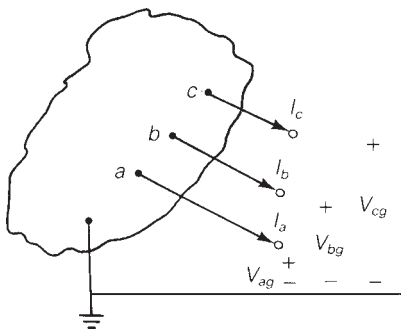
Note that these assumptions are made for simplicity in this text, and in practice should not be made for all cases. For example, in primary and secondary distribution systems, prefault currents may be in some cases comparable to short-circuit currents, and in other cases line resistances may significantly reduce fault currents.

Although fault currents as well as contributions to fault currents on the fault side of Δ–Y transformers are not affected by Δ–Y phase shifts, contributions to the fault from the other side of such transformers are affected by Δ–Y phase shifts for unsymmetrical faults. Therefore, Δ–Y phase-shift effects are included in this chapter.

Consider faults at the general three-phase bus shown in Figure 10.1. Terminals *abc,* denoted the *fault terminals*, are brought out in order to make external connections that represent faults. Before a fault occurs, the currents I_a, I_b, and I_c are zero.

Figure 10.2(a) shows general sequence networks as viewed from the fault terminals. Since the prefault system is balanced, these zero-, positive-, and negative-sequence networks are uncoupled. Also, the sequence components of the fault currents, I_0, I_1, and I_2, are zero before a fault occurs. The general sequence networks in Figure 10.2(a) are reduced to their Thévenin equivalents as viewed from the fault terminals in Figure 10.2(b). Each sequence network has a Thévenin equivalent impedance. Also, the positive-sequence network has a Thévenin equivalent voltage source, which equals the prefault voltage V_F.

FIGURE 10.1
General three-phase bus



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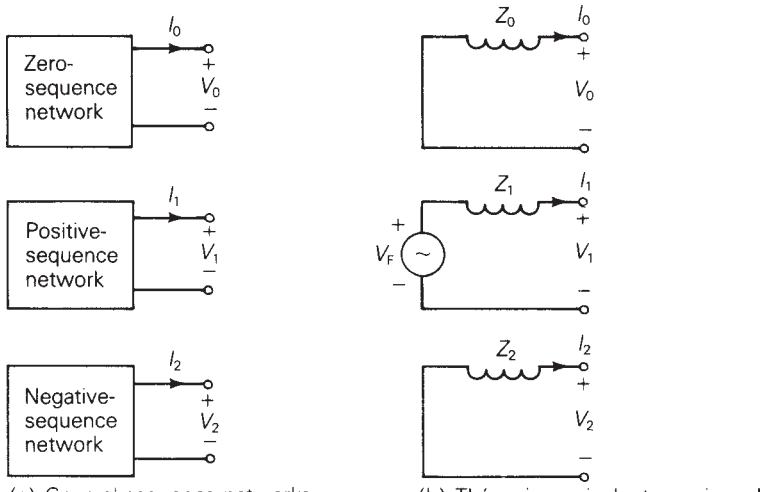


FIGURE 10.2

Sequence networks at a general three-phase bus in a balanced system

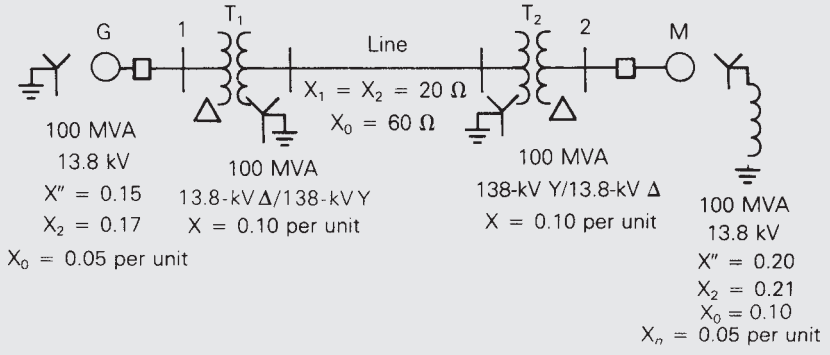
EXAMPLE 10.1

Power-system sequence networks and their Thévenin equivalents

A single-line diagram of the power system considered in Example 8.3 is shown in Figure 10.3, where negative- and zero-sequence reactances are also given. The neutrals of the generator and Δ–Y transformers are solidly grounded. The motor neutral is grounded through a reactance $X_n = 0.05$ per unit on the motor base. (a) Draw the per-unit zero-, positive-, and negative-sequence networks on a 100-MVA, 13.8-kV base in the zone of the generator. (b) Reduce the sequence networks to their Thévenin equivalents, as viewed from bus 2. Prefault voltage is $V_F = 1.05\angle 0°$ per unit. The prefault load current and Δ–Y transformer phase shift are neglected.

FIGURE 10.3

Single-line diagram for Example 10.1



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SOLUTION

a. The sequence networks are shown in Figure 10.4. The positive-sequence network is the same as that shown in Figure 8.4(a). The negative-sequence network is similar to the positive-sequence network, except that it shows no sources but does show negative-sequence machine reactances. Δ–Y phase shifts are omitted from the positive- and negative-sequence networks for this example. In the zero-sequence network, the zero-sequence generator, motor, and transmission-line reactances are shown. Since the motor neutral is grounded through a neutral reactance X_n, $3X_n$ is included in the zero-sequence motor circuit. Also, the zero-sequence Δ–Y transformer models are taken from Figure 9.19.

b. Figure 10.5 shows the sequence networks reduced to their Thévenin equivalents, as viewed from bus 2. For the positive-sequence equivalent, the Thévenin voltage source is the prefault voltage $V_F = 1.05\angle 0°$ per unit.

FIGURE 10.4
Sequence networks for Example 10.1

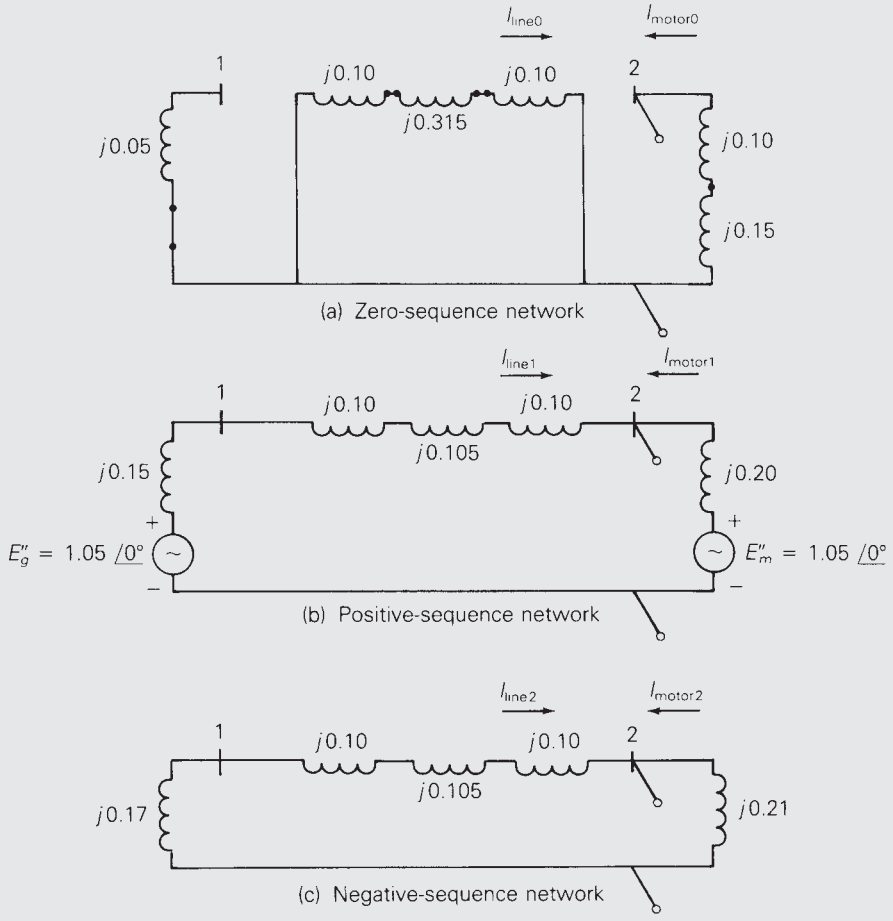

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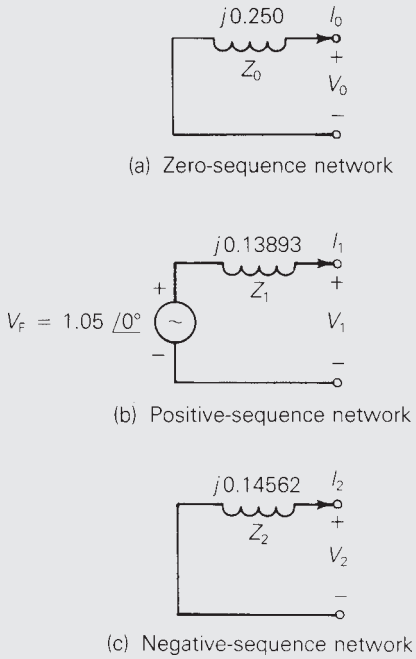
FIGURE 10.5

Thévenin equivalents of sequence networks for Example 10.1



From Figure 10.4, the positive-sequence Thévenin impedance at bus 2 is the motor impedance $j0.20$, as seen to the right of bus 2, in parallel with $j(0.15 + 0.10 + 0.105 + 0.10) = j0.455$, as seen to the left; the parallel combination is $j0.20 // j0.455 = j0.13893$ per unit. Similarly, the negative-sequence Thévenin impedance is $j0.21 // j(0.17 + 0.10 + 0.105 + 0.10) = j0.21 // j0.475 = j0.14562$ per unit. In the zero-sequence network of Figure 10.4, the Thévenin impedance at bus 2 consists only of $j(0.10 + 0.15) = j0.25$ per unit, as seen to the right of bus 2; due to the Δ connection of transformer T_2, the zero-sequence network looking to the left of bus 2 is open.

Recall that for three-phase faults, as considered in Chapter 8, the fault currents are balanced and have only a positive-sequence component. Therefore, work only with the positive-sequence network when calculating three-phase fault currents.

EXAMPLE 10.2

Three-phase short-circuit calculations using sequence networks

Calculate the per-unit subtransient fault currents in phases *a, b,* and *c* for a bolted three-phase-to-ground short circuit at bus 2 in Example 10.1.

(*Continued*)

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SOLUTION

The terminals of the positive-sequence network in Figure 10.5(b) are shorted, as shown in Figure 10.6. The positive-sequence fault current is

$$I_1 = \frac{V_F}{Z_1} = \frac{1.05\angle 0^\circ}{j0.13893} = -j7.558 \quad \text{per unit}$$

which is the same result as obtained in part (c) of Example 8.4. Note that since subtransient machine reactances are used in Figures 10.4 through 10.6, the current calculated above is the positive-sequence subtransient fault current at bus 2. Also, the zero-sequence current I_0 and negative-sequence current I_2 are both zero. Therefore, the subtransient fault currents in each phase are, from (9.1.16),

$$\begin{bmatrix} I_a'' \\ I_b'' \\ I_c'' \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} 0 \\ -j7.558 \\ 0 \end{bmatrix} = \begin{bmatrix} 7.558\angle -90^\circ \\ 7.558\angle 150^\circ \\ 7.558\angle 30^\circ \end{bmatrix} \quad \text{per unit}$$

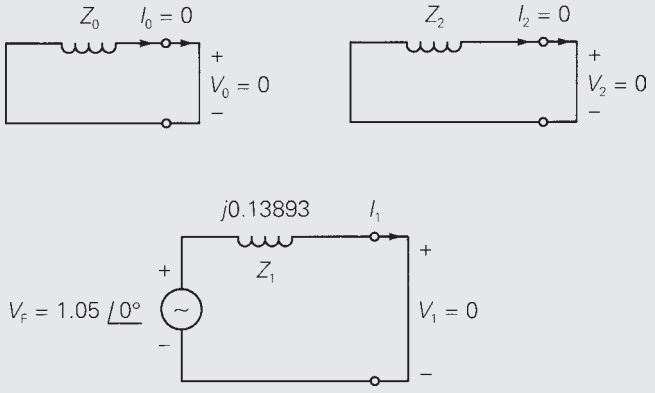

FIGURE 10.6
Example 10.2: Bolted three-phase-to-ground fault at bus 2

The sequence components of the line-to-ground voltages at the fault terminals are, from Figure 10.2(b),

$$\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 0 \\ V_F \\ 0 \end{bmatrix} - \begin{bmatrix} Z_0 & 0 & 0 \\ 0 & Z_1 & 0 \\ 0 & 0 & Z_2 \end{bmatrix} \begin{bmatrix} I_0 \\ I_1 \\ I_2 \end{bmatrix} \tag{10.1.1}$$

During a bolted three-phase fault, the sequence fault currents are $I_0 = I_2 = 0$ and $I_1 = V_F/Z_1$; therefore, from Equation (10.1.1), the sequence fault voltages are $V_0 = V_1 = V_2 = 0$, which must be true since $V_{ag} = V_{bg} = V_{cg} = 0$. However, fault voltages need not be zero during unsymmetrical faults, which is considered next.

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10.2 SINGLE LINE-TO-GROUND FAULT

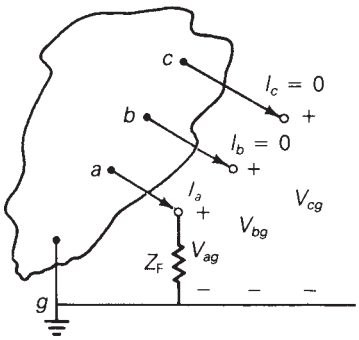
Consider a single line-to-ground fault from phase *a* to ground at the general three-phase bus shown in Figure 10.7(a). For generality, a fault impedance Z_F is included. In the case of a bolted fault, $Z_F = 0$, whereas for an arcing fault, Z_F is the arc impedance. In the case of a transmission-line insulator flashover, Z_F includes the total fault impedance between the line and ground, including the impedances of the arc and the transmission tower, as well as the tower footing if there are no neutral wires.

The relations to be derived here apply only to a single line-to-ground fault on phase *a.* However, since any of the three phases can be arbitrarily labeled phase *a,* single line-to-ground faults on other phases are not considered.

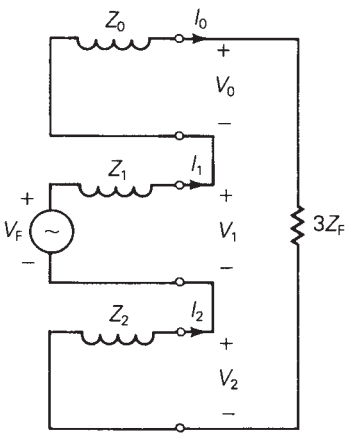
From Figure 10.7(a),

Fault conditions in phase domain: $I_b = I_c = 0$ (10.2.1)

Single line-to-ground fault: $V_{ag} = Z_F I_a$ (10.2.2)



(a) General three-phase bus



(b) Interconnected sequence networks

FIGURE 10.7

Single line-to-ground fault

Fault conditions in phase domain:

$V_{ag} = Z_F I_a$

$I_b = I_c = 0$

Fault conditions in sequence domain:

$I_0 = I_1 = I_2$

$(V_0 + V_1 + V_2) = 3Z_F I_1$

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Now transform (10.2.1) and (10.2.2) to the sequence domain. Using (10.2.1) in (9.1.19),

$$\begin{bmatrix} I_0 \\ I_1 \\ I_2 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} I_a \\ 0 \\ 0 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} I_a \\ I_a \\ I_a \end{bmatrix} \tag{10.2.3}$$

Also, using (9.1.3) and (9.1.20) in (10.2.2),

$$(V_0 + V_1 + V_2) = Z_F\,(I_0 + I_1 + I_2) \tag{10.2.4}$$

From (10.2.3) and (10.2.4),

Fault conditions in sequence domain: $I_0 = I_1 = I_2$ (10.2.5)

Single line-to-ground fault: $(V_0 + V_1 + V_2) = (3Z_F)I_1$ (10.2.6)

Equations (10.2.5) and (10.2.6) can be satisfied by interconnecting the sequence networks in series at the fault terminals through the impedance $(3Z_F)$, as shown in Figure 10.7(b). From this figure, the sequence components of the fault currents are

$$I_0 = I_1 = I_2 = \frac{V_F}{Z_0 + Z_1 + Z_2 + (3Z_F)} \tag{10.2.7}$$

Transforming (10.2.7) to the phase domain via (9.1.20),

$$I_a = I_0 + I_1 + I_2 = 3I_1 = \frac{3V_F}{Z_0 + Z_1 + Z_2 + (3Z_F)} \tag{10.2.8}$$

Note also from (9.1.21) and (9.1.22),

$$I_b = (I_0 + a^2 I_1 + aI_2) = (1 + a^2 + a)I_1 = 0 \tag{10.2.9}$$

$$I_c = (I_0 + aI_1 + a^2 I_2) = (1 + a + a^2)I_1 = 0 \tag{10.2.10}$$

These are obvious, since the single line-to-ground fault is on phase *a,* not phase *b* or *c.*

The sequence components of the line-to-ground voltages at the fault are determined from (10.1.1). The line-to-ground voltages at the fault can then be obtained by transforming the sequence voltages to the phase domain.

EXAMPLE 10.3

Single line-to-ground short-circuit calculations using sequence networks

Calculate the subtransient fault current in per-unit and in kA for a bolted single line-to-ground short circuit from phase *a* to ground at bus 2 in Example 10.1. Also, calculate the per-unit line-to-ground voltages at faulted bus 2.

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SOLUTION

The zero-, positive-, and negative-sequence networks in Figure 10.5 are connected in series at the fault terminals, as shown in Figure 10.8.

Since the short circuit is bolted, $Z_F = 0$. From (10.2.7), the sequence currents are:

$$I_0 = I_1 = I_2 = \frac{1.05\angle 0^\circ}{j(0.25 + 0.13893 + 0.14562)}$$
$$= \frac{1.05}{j0.53455} = -j1.96427 \quad \text{per unit}$$

From (10.2.8), the subtransient fault current is

$$I_a'' = 3(-j1.96427) = -j5.8928 \quad \text{per unit}$$

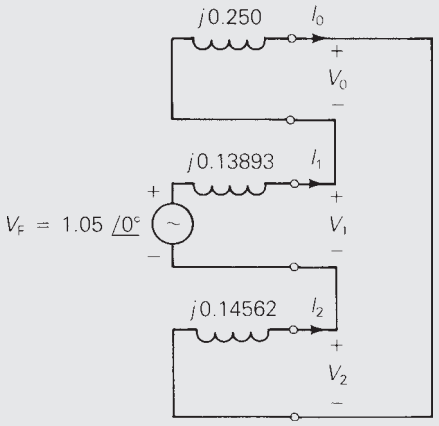
The base current at bus 2 is $100/(13.8\sqrt{3}) = 4.1837$ kA. Therefore,

$$I_a'' = (-j5.8928)(4.1837) = 24.65\angle -90^\circ \quad \text{kA}$$

From (10.1.1), the sequence components of the voltages at the fault are

$$\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1.05\angle 0^\circ \\ 0 \end{bmatrix} - \begin{bmatrix} j0.25 & 0 & 0 \\ 0 & j0.13893 & 0 \\ 0 & 0 & j0.14562 \end{bmatrix} \begin{bmatrix} -j1.96427 \\ -j1.96427 \\ -j1.96427 \end{bmatrix}$$
$$= \begin{bmatrix} -0.49107 \\ 0.77710 \\ -0.28604 \end{bmatrix} \quad \text{per unit}$$

FIGURE 10.8
Example 10.3: Single line-to-ground fault at bus 2



(*Continued*)

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Transforming to the phase domain, the line-to-ground voltages at faulted bus 2 are

$$\begin{bmatrix} V_{ag} \\ V_{bg} \\ V_{cg} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} -0.49107 \\ 0.77710 \\ -0.28604 \end{bmatrix} = \begin{bmatrix} 0 \\ 1.179\angle 231.3^\circ \\ 1.179\angle 128.7^\circ \end{bmatrix} \quad \text{per unit}$$

Note that for the single-line-to-ground fault on phase *a* at bus 2, there is an overvoltage of 1.179 per unit on unfaulted phases *b* and *c*.

Open PowerWorld Simulator case Example 10_3 to see this example. This case includes the positive-, negative-, and zero-sequence values for the system devices. To see these values, right-click on an object on the oneline (e.g., a generator), open the object's Information dialog, and select the **Fault** tab. The process for simulating an unsymmetrical fault is almost identical to that for a balanced fault. That is, from the oneline, first right-click on the bus symbol corresponding to the fault location. This displays the local menu. Select

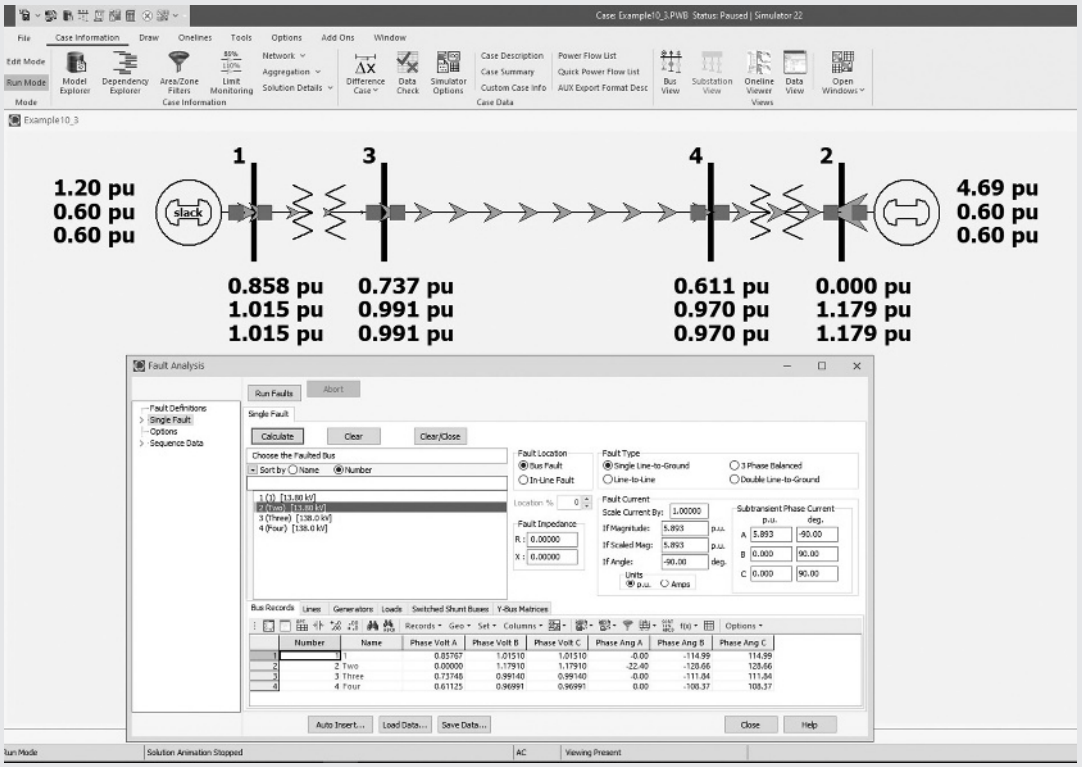

FIGURE 10.9

Screen for Example 10.3—fault at bus 2

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"Fault.." to display the Fault Analysis dialog, and in the list on the left side of the dialog, set the **Fault Definitions** to Single Fault. Verify that the correct bus is selected, and then set the Fault Type field to Single Line-to-Ground. Finally, click on **Calculate** to determine the fault currents and voltages. The results are shown in the tables at the bottom of the dialog. Notice that with an unsymmetrical fault, the phase magnitudes are no longer identical and the phase angles no longer differ by 120°. The fault values can be shown on the oneline by selecting **Options** in the left list, and changing the **Oneline Display** field value.

10.3 LINE-TO-LINE FAULT

Consider a line-to-line fault from phase *b* to *c,* shown in Figure 10.10(a). Again, include a fault impedance Z_F for generality. From Figure 10.10(a),

Fault conditions in phase domain
Line-to-line fault

$$I_a = 0 \tag{10.3.1}$$
$$I_c = -I_b \tag{10.3.2}$$
$$V_{bg} - V_{cg} = Z_F I_b \tag{10.3.3}$$

Transform (10.3.1) through (10.3.3) to the sequence domain. Using (10.3.1) and (10.3.2) in (9.1.19),

$$\begin{bmatrix} I_0 \\ I_1 \\ I_2 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} 0 \\ I_b \\ -I_b \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{1}{3}(a - a^2)I_b \\ \frac{1}{3}(a^2 - a)I_b \end{bmatrix} \tag{10.3.4}$$

Using (9.1.4), (9.1.5), and (9.1.21) in (10.3.3),

$$(V_0 + a^2V_1 + aV_2) - (V_0 + aV_1 + a^2V_2) = Z_F(I_0 + a^2I_1 + aI_2) \tag{10.3.5}$$

Noting from (10.3.4) that $I_0 = 0$ and $I_2 = -I_1$, (10.3.5) simplifies to

$$(a^2 - a)V_1 - (a^2 - a)V_2 = Z_F(a^2 - a)I_1$$

or

$$V_1 - V_2 = Z_F I_1 \tag{10.3.6}$$

Therefore, from (10.3.4) and (10.3.6),

Fault conditions in sequence domain
Line-to-line fault

$$I_0 = 0 \tag{10.3.7}$$
$$I_2 = -I_1 \tag{10.3.8}$$
$$V_1 - V_2 = Z_F I_1 \tag{10.3.9}$$

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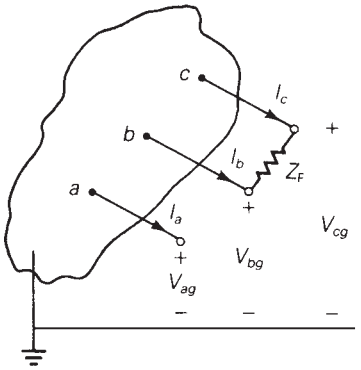
FIGURE 10.10
Line-to-line fault



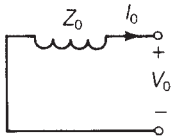
(a) General three-phase bus

Fault conditions in phase domain:

$I_a = 0$

$I_c = -I_b$

$(V_{bg} - V_{cg}) = Z_F I_b$

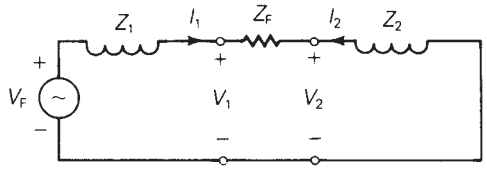


Fault conditions in sequence domain:

$I_0 = 0$

$I_2 = -I_1$

$(V_1 - V_2) = Z_F I_1$



(b) Interconnected sequence networks

Equations (10.3.7) through (10.3.9) are satisfied by connecting the positive- and negative-sequence networks in parallel at the fault terminals through the fault impedance Z_F, as shown in Figure 10.10(b). From this figure, the fault currents are

$$I_1 = -I_2 = \frac{V_F}{(Z_1 + Z_2 + Z_F)} \quad I_0 = 0 \tag{10.3.10}$$

Transforming (10.3.10) to the phase domain and using the identity $(a^2 - a) = -j\sqrt{3}$, the fault current in phase b is

$$I_b = I_0 + a^2 I_1 + a I_2 = (a^2 - a) I_1$$

$$= -j\sqrt{3} I_1 = \frac{-j\sqrt{3} V_F}{(Z_1 + Z_2 + Z_F)} \tag{10.3.11}$$

Note also from (9.1.20) and (9.1.22) that

$$I_a = I_0 + I_1 + I_2 = 0 \tag{10.3.12}$$

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and

$$I_c = I_0 + aI_1 + a^2I_2 = (a - a^2)I_1 = -I_b \tag{10.3.13}$$

which verify the fault conditions given by (10.3.1) and (10.3.2). The sequence components of the line-to-ground voltages at the fault are given by (10.1.1).

EXAMPLE 10.4

Line-to-line short-circuit calculations using sequence networks

Calculate the subtransient fault current in per-unit and in kA for a bolted line-to-line fault from phase b to c at bus 2 in Example 10.1.

SOLUTION

The positive- and negative-sequence networks in Figure 10.5 are connected in parallel at the fault terminals, as shown in Figure 10.11. From (10.3.10) with $Z_F = 0$, the sequence fault currents are

$$I_1 = -I_2 = \frac{1.05\angle 0°}{j(0.13893 + 0.14562)} = 3.690\angle -90°$$

$$I_0 = 0$$

From (10.3.11), the subtransient fault current in phase b is

$$I_b'' = (-j\sqrt{3})(3.690\angle -90°) = -6.391 = 6.391\angle 180° \quad \text{per unit}$$

Using 4.1837 kA as the base current at bus 2,

$$I_b'' = (6.391\angle 180°)(4.1837) = 26.74\angle 180° \quad \text{kA}$$

Also, from (10.3.12) and (10.3.13),

$$I_a'' = 0 \qquad I_c'' = 26.74\angle 0° \quad \text{kA}$$

The line-to-line fault results for this example can be shown in PowerWorld Simulator by repeating the Example 10.3 procedure, with the exception that the Fault Type field value should be Line-to-Line.

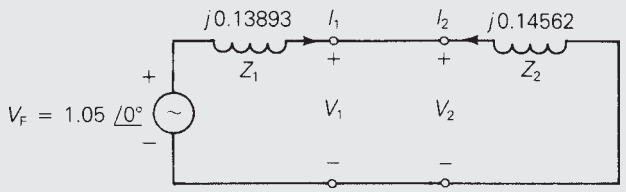

FIGURE 10.11
Example 10.4: Bolted three-phase-to-ground fault at bus 2

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10.4 DOUBLE LINE-TO-GROUND FAULT

A double line-to-ground fault from phase b to phase c through fault impedance Z_F to ground is shown in Figure 10.12(a). From this figure,

Fault conditions in the phase domain
Double line-to-ground fault

$$I_a = 0 \tag{10.4.1}$$
$$V_{cg} = V_{bg} \tag{10.4.2}$$
$$V_{bg} = Z_F(I_b + I_c) \tag{10.4.3}$$

Transforming (10.4.1) to the sequence domain via (9.1.20),

$$I_0 + I_1 + I_2 = 0 \tag{10.4.4}$$

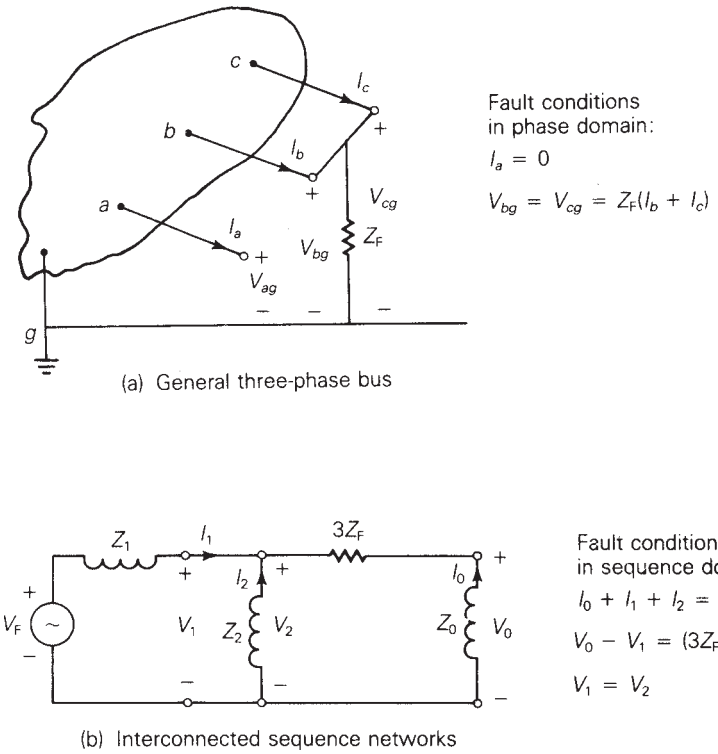
Also, using (9.1.4) and (9.1.5) in (10.4.2),

$$(V_0 + aV_1 + a^2V_2) = (V_0 + a^2V_1 + aV_2)$$

Simplifying,

$$(a^2 - a)V_2 = (a^2 - a)V_1$$

FIGURE 10.12
Double line-to-ground fault



(a) General three-phase bus

(b) Interconnected sequence networks

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or

$$V_2 = V_1 \tag{10.4.5}$$

Now, using (9.1.4), (9.1.21), and (9.1.22) in (10.4.3),

$$(V_0 + a^2V_1 + aV_2) = Z_F(I_0 + a^2I_1 + aI_2 + I_0 + aI_1 + a^2I_2) \tag{10.4.6}$$

Using (10.4.5) and the identity $a^2 + a = -1$ in (10.4.6),

$$(V_0 - V_1) = Z_F(2I_0 - I_1 - I_2) \tag{10.4.7}$$

From (10.4.4), $I_0 = -(I_1 + I_2)$; therefore, (10.4.7) becomes

$$V_0 - V_1 = (3Z_F)I_0 \tag{10.4.8}$$

In summary, from (10.4.4), (10.4.5), and (10.4.8),

Fault conditions in the sequence domain
Double line-to-ground fault

$$\left.\begin{aligned} I_0 + I_1 + I_2 &= 0 \quad (10.4.9)\\ V_2 = V_1 &\quad (10.4.10)\\ V_0 - V_1 &= (3Z_F)I_0 \quad (10.4.11)\end{aligned}\right\}$$

Equations (10.4.9) through (10.4.11) are satisfied by connecting the zero-, positive-, and negative-sequence networks in parallel at the fault terminal; additionally, $(3Z_F)$ is included in series with the zero-sequence network. This connection is shown in Figure 10.12(b). From this figure, the positive-sequence fault current is

$$I_1 = \frac{V_F}{Z_1 + [Z_2 /\!/ (Z_0 + 3Z_F)]} = \frac{V_F}{Z_1 + \left[\dfrac{Z_2(Z_0 + 3Z_F)}{Z_2 + Z_0 + 3Z_F}\right]} \tag{10.4.12}$$

Using current division in Figure 10.12(b), the negative- and zero-sequence fault currents are

$$I_2 = (-I_1)\left(\frac{Z_0 + 3Z_F}{Z_0 + 3Z_F + Z_2}\right) \tag{10.4.13}$$

$$I_0 = (-I_1)\left(\frac{Z_2}{Z_0 + 3Z_F + Z_2}\right) \tag{10.4.14}$$

These sequence fault currents can be transformed to the phase domain via (9.1.16). Also, the sequence components of the line-to-ground voltages at the fault are given by (10.1.1).

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EXAMPLE 10.5

Double line-to-ground short-circuit calculations using sequence networks

Calculate (a) the subtransient fault current in each phase, (b) neutral fault current, and (c) contributions to the fault current from the motor and from the transmission line, for a bolted double line-to-ground fault from phase b to c to ground at bus 2 in Example 10.1. Neglect the Δ–Y transformer phase shifts.

SOLUTION

a. The zero-, positive-, and negative-sequence networks in Figure 10.5 are connected in parallel at the fault terminals in Figure 10.13. From (10.4.12) with $Z_F = 0$,

$$I_1 = \frac{1.05\angle 0°}{j\left[0.13893 + \dfrac{(0.14562)(0.25)}{0.14562 + 0.25}\right]} = \frac{1.05\angle 0°}{j0.23095}$$

$$= -j4.5464 \quad \text{per unit}$$

From (10.4.13) and (10.4.14),

$$I_2 = (+j4.5464)\left(\frac{0.25}{0.25 + 0.14562}\right) = j2.8730 \quad \text{per unit}$$

$$I_0 = (+j4.5464)\left(\frac{0.14562}{0.25 + 0.14562}\right) = j1.6734 \quad \text{per unit}$$

Transforming to the phase domain, the subtransient fault currents are:

$$\begin{bmatrix} I_a'' \\ I_b'' \\ I_c'' \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} +j1.6734 \\ -j4.5464 \\ +j2.8730 \end{bmatrix} = \begin{bmatrix} 0 \\ 6.8983\angle 158.66° \\ 6.8983\angle 21.34° \end{bmatrix} \quad \text{per unit}$$

Using the base current of 4.1837 kA at bus 2,

$$\begin{bmatrix} I_a'' \\ I_b'' \\ I_c'' \end{bmatrix} = \begin{bmatrix} 0 \\ 6.8983\angle 158.66° \\ 6.8983\angle 21.34° \end{bmatrix} (4.1837) = \begin{bmatrix} 0 \\ 28.86\angle 158.66° \\ 28.86\angle 21.34° \end{bmatrix} \quad \text{kA}$$

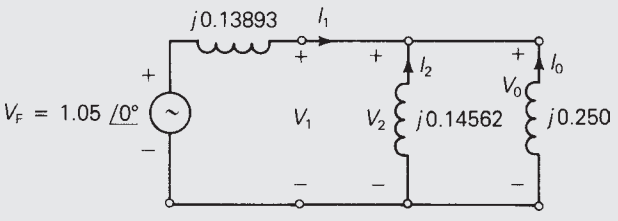

FIGURE 10.13
Example 10.5: Double line-to-ground fault at bus 2

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b. The neutral fault current is

$$I_n = (I_b'' + I_c'') = 3I_0 = j5.0202 \quad \text{per unit}$$
$$= (j5.0202)(4.1837) = 21.00\angle 90° \quad \text{kA}$$

c. Neglecting Δ–Y transformer phase shifts, the contributions to the fault current from the motor and transmission line can be obtained from Figure 10.4. From the zero-sequence network, Figure 10.4(a), the contribution to the zero-sequence fault current from the line is zero, due to the transformer connection. That is,

$$I_{\text{line }0} = 0$$
$$I_{\text{motor }0} = I_0 = j1.6734 \quad \text{per unit}$$

From the positive-sequence network, Figure 10.4(b), the positive terminals of the internal machine voltages can be connected, since $E_g'' = E_m''$. Then, by current division,

$$I_{\text{line }1} = \frac{\text{X}_m''}{\text{X}_m'' + (\text{X}_g'' + \text{X}_{\text{T1}} + \text{X}_{\text{line }1} + \text{X}_{\text{T2}})} I_1$$
$$= \frac{0.20}{0.20 + (0.455)}(-j4.5464) = -j1.3882 \quad \text{per unit}$$
$$I_{\text{motor }1} = \frac{0.455}{0.20 + 0.455}(-j4.5464) = -j3.1582 \quad \text{per unit}$$

From the negative-sequence network, Figure 10.4(c), using current division,

$$I_{\text{line }2} = \frac{0.21}{0.21 + 0.475}(j2.8730) = j0.8808 \quad \text{per unit}$$
$$I_{\text{motor }2} = \frac{0.475}{0.21 + 0.475}(j2.8730) = j1.9922 \quad \text{per unit}$$

Transforming to the phase domain with base currents of 0.41837 kA for the line and 4.1837 kA for the motor,

$$\begin{bmatrix} I''_{\text{line }a} \\ I''_{\text{line }b} \\ I''_{\text{line }c} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} 0 \\ -j1.3882 \\ j0.8808 \end{bmatrix}$$
$$= \begin{bmatrix} 0.5074\angle -90° \\ 1.9813\angle 172.643° \\ 1.9813\angle 7.357° \end{bmatrix} \quad \text{per unit}$$
$$= \begin{bmatrix} 0.2123\angle -90° \\ 0.8289\angle 172.643° \\ 0.8289\angle 7.357° \end{bmatrix} \quad \text{kA}$$

(*Continued*)

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$$\begin{bmatrix} I''_{\text{motor } a} \\ I''_{\text{motor } b} \\ I''_{\text{motor } c} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} j1.6734 \\ -j3.1582 \\ j1.9922 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5074\angle 90° \\ 4.9986\angle 153.17° \\ 4.9986\angle 26.83° \end{bmatrix} \text{ per unit}$$

$$= \begin{bmatrix} 2.123\angle 90° \\ 20.91\angle 153.17° \\ 20.91\angle 26.83° \end{bmatrix} \text{ kA}$$

The double line-to-line fault results for this example can be shown in PowerWorld Simulator by repeating the Example 10.3 procedure, with the exception that the Fault Type field value should be Double Line-to-Ground.

EXAMPLE 10.6

Effect of Δ–Y transformer phase shift on fault currents

Rework Example 10.5, with the Δ–Y transformer phase shifts included. Assume American standard phase shift.

SOLUTION

The sequence networks of Figure 10.4 are redrawn in Figure 10.14 with ideal phase-shifting transformers representing Δ–Y phase shifts. In accordance with the American standard, positive-sequence quantities on the high-voltage side of the transformers lead their corresponding quantities on the low-voltage side by 30°. Also, the negative-sequence phase shifts are the reverse of the positive-sequence phase shifts.

a. Recall from Section 3.1 and (3.1.26) that per-unit impedance is unchanged when it is referred from one side of an ideal phase-shifting transformer to the other. Accordingly, the Thévenin equivalents of the sequence networks in Figure 10.14, as viewed from fault bus 2, are the same as those given in Figure 10.5. Therefore, the sequence components as well as the phase components of the fault currents are the same as those given in Example 10.5(a).

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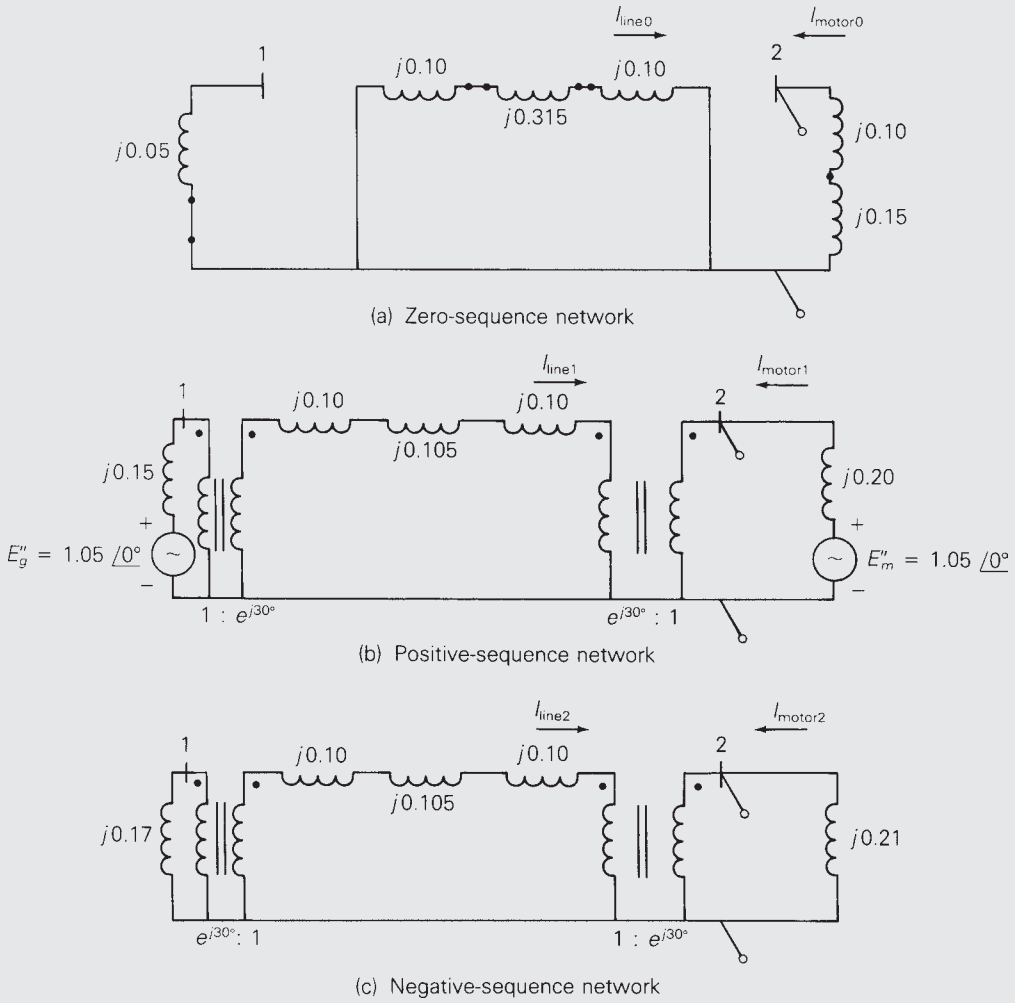

FIGURE 10.14

Sequence networks for Example 10.6

b. The neutral fault current is the same as that given in Example 10.5(b).

c. The zero-sequence network, Figure 10.14(a), is the same as that given in Figure 10.4(a). Therefore, the contributions to the zero-sequence fault current from the line and motor are the same as those given in Example 10.5(c).

$$I_{\text{line }0} = 0 \qquad I_{\text{motor }0} = I_0 = j1.6734 \quad \text{per unit}$$

(*Continued*)

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The contribution to the positive-sequence fault current from the line in Figure 10.14(b) leads that in Figure 10.4(b) by 30°. That is,

$$I_{\text{line }1} = (-j1.3882)(1\angle 30°) = 1.3882\angle -60° \quad \text{per unit}$$

$$I_{\text{motor }1} = -j3.1582 \quad \text{per unit}$$

Similarly, the contribution to the negative-sequence fault current from the line in Figure 10.14(c) lags that in Figure 10.4(c) by 30°. That is,

$$I_{\text{line }2} = (j0.8808)(1\angle -30°) = 0.8808\angle 60° \quad \text{per unit}$$

$$I_{\text{motor }2} = j1.9922 \quad \text{per unit}$$

Thus, the sequence currents as well as the phase currents from the motor are the same as those given in Example 10.5(c). Also, the sequence currents from the line have the same magnitudes as those given in Example 10.5(c), but the positive- and negative-sequence line currents are shifted by +30° and −30°, respectively. Transforming the line currents to the phase domain,

$$\begin{bmatrix} I''_{\text{line }a} \\ I''_{\text{line }b} \\ I''_{\text{line }c} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} 0 \\ 1.3882\angle -60° \\ 0.8808\angle +60° \end{bmatrix}$$

$$= \begin{bmatrix} 1.2166\angle -21.17° \\ 2.2690\angle 180° \\ 1.2166\angle 21.17° \end{bmatrix} \quad \text{per unit}$$

$$= \begin{bmatrix} 0.5090\angle -21.17° \\ 0.9492\angle 180° \\ 0.5090\angle 21.17° \end{bmatrix} \quad \text{kA}$$

In conclusion, Δ–Y transformer phase shifts have no effect on the fault currents and no effect on the contribution to the fault currents on the fault side of the Δ–Y transformers. However, on the other side of the Δ–Y transformers, the positive- and negative-sequence components of the contributions to the fault currents are shifted by ±30°, which affects both the magnitude as well as the angle of the phase components of these fault contributions for unsymmetrical faults.

Figure 10.15 summarizes the sequence network connections for both the balanced three-phase fault and the unsymmetrical faults that have been considered. Sequence network connections for two additional faults, one-conductor-open and two-conductors-open, are also shown in Figure 10.15 and are left as an exercise for you to verify (see Problems 10.28 and 10.29).

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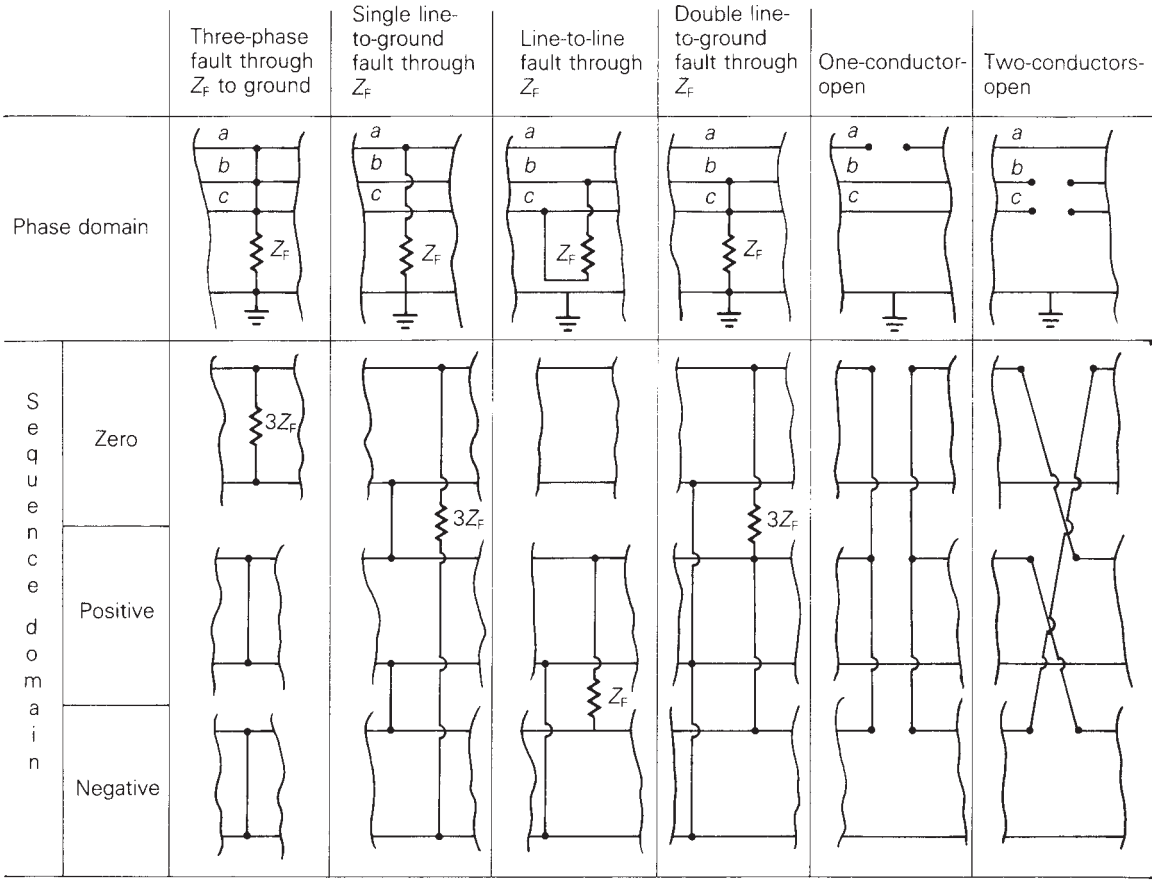

FIGURE 10.15

Summary of faults

10.5 SEQUENCE BUS IMPEDANCE MATRICES

The positive-sequence bus impedance matrix in Section 8.4 is used for calculating currents and voltages during balanced three-phase faults. This method is extended here to unsymmetrical faults by representing each sequence network as a bus impedance equivalent circuit (or as a rake equivalent). A bus impedance matrix can be computed for each sequence network by inverting the corresponding bus admittance network. For simplicity, resistances, shunt admittances, nonrotating impedance loads, and prefault load currents are neglected.

Figure 10.16 shows the connection of sequence rake equivalents for both symmetrical and unsymmetrical faults at bus n of an N-bus three-phase power system. Each bus impedance element has an additional subscript, 0, 1, or 2, that identifies the sequence rake equivalent in which it is located. Mutual impedances are not shown in the figure. The prefault voltage V_F is included in the

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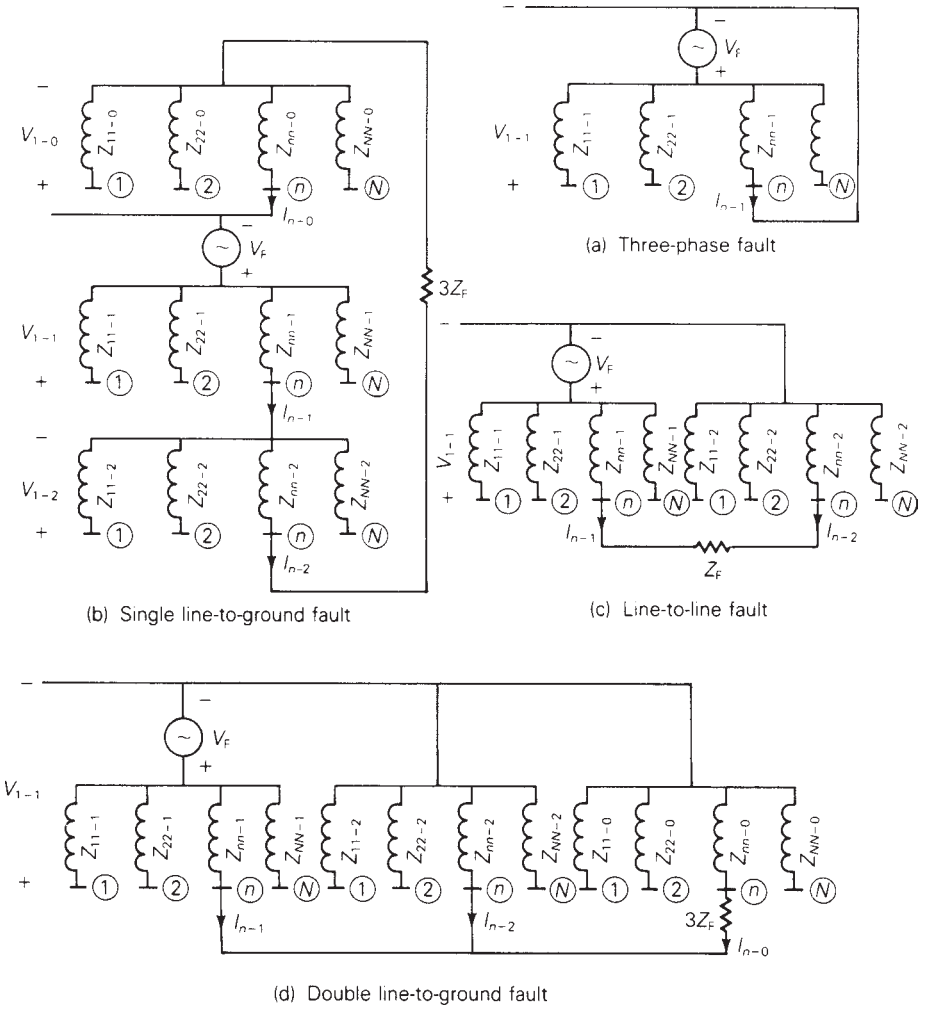
FIGURE 10.16

Connection of rake equivalent sequence networks for three-phase system faults (mutual impedances not shown)



positive-sequence rake equivalent. From the figure the sequence components of the fault current for each type of fault at bus n are as follows:

BALANCED THREE-PHASE FAULT:

$$I_{n-1} = \frac{V_{\mathrm{F}}}{Z_{nn-1}} \tag{10.5.1}$$

$$I_{n-0} = I_{n-2} = 0 \tag{10.5.2}$$

SINGLE LINE-TO-GROUND FAULT (PHASE *a* TO GROUND):

$$I_{n-0} = I_{n-1} = I_{n-2} = \frac{V_{\mathrm{F}}}{Z_{nn-0} + Z_{nn-1} + Z_{nn-2} + 3Z_{\mathrm{F}}} \tag{10.5.3}$$

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LINE-TO-LINE FAULT (PHASE *b* TO *c*):

$$I_{n-1} = -I_{n-2} = \frac{V_F}{Z_{nn-1} + Z_{nn-2} + Z_F} \tag{10.5.4}$$

$$I_{n-0} = 0 \tag{10.5.5}$$

DOUBLE LINE-TO-GROUND FAULT (PHASE *b* TO *c* TO GROUND):

$$I_{n-1} = \frac{V_F}{Z_{nn-1} + \left[\dfrac{Z_{nn-2}(Z_{nn-0} + 3Z_F)}{Z_{nn-2} + Z_{nn-0} + 3Z_F}\right]} \tag{10.5.6}$$

$$I_{n-2} = (-I_{n-1})\left(\frac{Z_{nn-0} + 3Z_F}{Z_{nn-0} + 3Z_F + Z_{nn-2}}\right) \tag{10.5.7}$$

$$I_{n-0} = (-I_{n-1})\left(\frac{Z_{nn-2}}{Z_{nn-0} + 3Z_F + Z_{nn-2}}\right) \tag{10.5.8}$$

Also from Figure 10.16, the sequence components of the line-to-ground voltages at any bus k during a fault at bus n are:

$$\begin{bmatrix} V_{k-0} \\ V_{k-1} \\ V_{k-2} \end{bmatrix} = \begin{bmatrix} 0 \\ V_F \\ 0 \end{bmatrix} - \begin{bmatrix} Z_{kn-0} & 0 & 0 \\ 0 & Z_{kn-1} & 0 \\ 0 & 0 & Z_{kn-2} \end{bmatrix} \begin{bmatrix} I_{n-0} \\ I_{n-1} \\ I_{n-2} \end{bmatrix} \tag{10.5.9}$$

If bus k is on the unfaulted side of a Δ–Y transformer, then the phase angles of V_{k-1} and V_{k-2} in (10.5.9) are modified to account for Δ–Y phase shifts. Also, the above sequence fault currents and sequence voltages can be transformed to the phase domain via (9.1.16) and (9.1.9).

EXAMPLE 10.7

Single line-to-ground short-circuit calculations using $Z_{bus\,0}$, $Z_{bus\,1}$, and $Z_{bus\,2}$

Faults at buses 1 and 2 for the three-phase power system given in Example 10.1 are of interest. The prefault voltage is 1.05 per unit. Prefault load current is neglected. (a) Determine the per-unit zero-, positive-, and negative-sequence bus impedance matrices. (b) Find the subtransient fault current in per-unit for a bolted single line-to-ground fault current from phase *a* to ground at bus 1, and (c) at bus 2. Find the per-unit line-to-ground voltages at (d) bus 1, and (e) bus 2 during the single line-to-ground fault at bus 1.

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SOLUTION

a. Referring to Figure 10.4(a), the zero-sequence bus admittance matrix is

$$\boldsymbol{Y}_{\text{bus }0} = -j\left[\begin{array}{c|c} 20 & 0 \\ \hline 0 & 4 \end{array}\right] \quad \text{per unit}$$

Inverting $\boldsymbol{Y}_{\text{bus }0}$,

$$\boldsymbol{Z}_{\text{bus }0} = j\left[\begin{array}{c|c} 0.05 & 0 \\ \hline 0 & 0.25 \end{array}\right] \quad \text{per unit}$$

Note that the transformer leakage reactances and the zero-sequence transmission-line reactance in Figure 10.4(a) have no effect on $\boldsymbol{Z}_{\text{bus }0}$. The transformer Δ connections block the flow of zero-sequence current from the transformers to bus 1 and 2.

The positive-sequence bus admittance matrix, from Figure 10.4(b), is

$$\boldsymbol{Y}_{\text{bus }1} = -j\left[\begin{array}{c|c} 9.9454 & -3.2787 \\ \hline -3.2787 & 8.2787 \end{array}\right] \quad \text{per unit}$$

Inverting $\boldsymbol{Y}_{\text{bus }1}$,

$$\boldsymbol{Z}_{\text{bus }1} = j\left[\begin{array}{c|c} 0.11565 & 0.04580 \\ \hline 0.04580 & 0.13893 \end{array}\right] \quad \text{per unit}$$

Similarly, from Figure 10.4(c),

$$\boldsymbol{Y}_{\text{bus }2} = -j\left[\begin{array}{c|c} 9.1611 & -3.2787 \\ \hline -3.2787 & 8.0406 \end{array}\right] \quad \text{per unit}$$

Inverting $\boldsymbol{Y}_{\text{bus }2}$,

$$\boldsymbol{Z}_{\text{bus }2} = j\left[\begin{array}{c|c} 0.12781 & 0.05212 \\ \hline 0.05212 & 0.14562 \end{array}\right] \quad \text{per unit}$$

b. From (10.5.3), with $n = 1$ and $Z_F = 0$, the sequence fault currents are

$$I_{1-0} = I_{1-1} = I_{1-2} = \frac{V_F}{Z_{11-0} + Z_{11-1} + Z_{11-2}}$$

$$= \frac{1.05\angle 0^\circ}{j(0.05 + 0.11565 + 0.12781)} = \frac{1.05}{j0.29346} = -j3.578 \quad \text{per unit}$$

The subtransient fault currents at bus 1 are, from (9.1.16),

$$\begin{bmatrix} I''_{1a} \\ I''_{1b} \\ I''_{1c} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} -j3.578 \\ -j3.578 \\ -j3.578 \end{bmatrix} = \begin{bmatrix} -j10.73 \\ 0 \\ 0 \end{bmatrix} \quad \text{per unit}$$

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c. Again from (10.5.3), with $n = 2$ and $Z_F = 0$,

$$I_{2-0} = I_{2-1} = I_{2-2} = \frac{V_F}{Z_{22-0} + Z_{22-1} + Z_{22-2}}$$

$$= \frac{1.05\angle 0^\circ}{j(0.25 + 0.13893 + 0.14562)} = \frac{1.05}{j0.53455}$$

$$= -j1.96427 \quad \text{per unit}$$

and

$$\begin{bmatrix} I''_{2a} \\ I''_{2b} \\ I''_{2c} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} -j1.96427 \\ -j1.96427 \\ -j1.96427 \end{bmatrix} = \begin{bmatrix} -j5.8928 \\ 0 \\ 0 \end{bmatrix} \quad \text{per unit}$$

This is the same result as obtained in Example 10.3.

d. The sequence components of the line-to-ground voltages at bus 1 during the fault at bus 1 are, from (10.5.9), with $k = 1$ and $n = 1$,

$$\begin{bmatrix} V_{1-0} \\ V_{1-1} \\ V_{1-2} \end{bmatrix} = \begin{bmatrix} 0 \\ 1.05\angle 0^\circ \\ 0 \end{bmatrix} - \begin{bmatrix} j0.05 & 0 & 0 \\ 0 & j0.11565 & 0 \\ 0 & 0 & j0.12781 \end{bmatrix} \begin{bmatrix} -j3.578 \\ -j3.578 \\ -j3.578 \end{bmatrix}$$

$$= \begin{bmatrix} -0.1789 \\ 0.6362 \\ -0.4573 \end{bmatrix} \quad \text{per unit}$$

and the line-to-ground voltages at bus 1 during the fault at bus 1 are

$$\begin{bmatrix} V_{1-ag} \\ V_{1-bg} \\ V_{1-cg} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} -0.1789 \\ +0.6362 \\ -0.4573 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0.9843\angle 254.2^\circ \\ 0.9843\angle 105.8^\circ \end{bmatrix} \quad \text{per unit}$$

(*Continued*)

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e. The sequence components of the line-to-ground voltages at bus 2 during the fault at bus 1 are, from (10.5.9), with $k = 2$ and $n = 1$,

$$\begin{bmatrix} V_{2-0} \\ V_{2-1} \\ V_{2-2} \end{bmatrix} = \begin{bmatrix} 0 \\ 1.05\underline{/0^\circ} \\ 0 \end{bmatrix} - \begin{bmatrix} 0 & 0 & 0 \\ 0 & j0.04580 & 0 \\ 0 & 0 & j0.05212 \end{bmatrix} \begin{bmatrix} -j3.578 \\ -j3.578 \\ -j3.578 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0.8861 \\ -0.18649 \end{bmatrix} \text{ per unit}$$

Note that since both bus 1 and 2 are on the low-voltage side of the Δ–Y transformers in Figure 10.3, there is no shift in the phase angles of these sequence voltages. From the above, the line-to-ground voltages at bus 2 during the fault at bus 1 are

$$\begin{bmatrix} V_{2-ag} \\ V_{2-bg} \\ V_{2-cg} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} 0 \\ 0.8861 \\ -0.18649 \end{bmatrix}$$

$$= \begin{bmatrix} 0.70 \\ 0.9926\underline{/249.4^\circ} \\ 0.9926\underline{/110.6^\circ} \end{bmatrix} \text{ per unit}$$

PowerWorld Simulator computes the symmetrical fault current for each of the following faults at any bus in an N-bus power system: balanced three-phase fault, single line-to-ground fault, line-to-line fault, or double line-to-ground fault. For each fault, the Simulator also computes bus voltages and contributions to the fault current from transmission lines and transformers connected to the fault bus.

Input data for the Simulator include machine, transmission-line, and transformer data, as illustrated in Tables 10.1, 10.2, and 10.3 as well as the prefault voltage V_F and fault impedance Z_F. When the machine positive-sequence reactance input data consist of direct axis subtransient reactances, the computed symmetrical fault currents are subtransient fault currents. Alternatively, transient or steady-state fault currents are computed when these input data consist of direct axis transient or synchronous reactances. Transmission-line positive- and zero-sequence series reactances are those of the equivalent π circuits for long lines or of the nominal π circuit for medium or short lines. Also, recall that the negative-sequence transmission-line reactance equals the positive-sequence transmission-line reactance. All machine, line, and transformer reactances are given in per-unit on a common MVA base. Prefault load currents are neglected.

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Bus	X_0 per unit	$X_1 = X_d''$ per unit	X_2 per unit	Neutral Reactance X_n per unit
1	0.0125	0.045	0.045	0
3	0.005	0.0225	0.0225	0.0025

TABLE 10.1

Synchronous machine data for Example 10.8

Bus-to-Bus	X_0 per unit	X_1 per unit
2–4	0.3	0.1
2–5	0.15	0.05
4–5	0.075	0.025

TABLE 10.2

Line data for Example 10.8

Low-Voltage (connection) Bus	High-Voltage (connection) Bus	Leakage Reactance per unit	Neutral Reactance per unit
1(Δ)	5(Y)	0.02	0
3(Δ)	4(Y)	0.01	0

TABLE 10.3

Transformer data for Example 10.8

$S_{base} = 100$ MVA

$$V_{base} = \begin{cases} 15 \text{ kV at buses } 1, 3 \\ 345 \text{ kV at buses } 2, 4, 5 \end{cases}$$

The Simulator computes (but does not show) the zero-, positive-, and negative-sequence bus impedance matrices $\boldsymbol{Z}_{\text{bus }0}$, $\boldsymbol{Z}_{\text{bus }1}$, and $\boldsymbol{Z}_{\text{bus }2}$, by inverting the corresponding bus admittance matrices.

After $\boldsymbol{Z}_{\text{bus }0}$, $\boldsymbol{Z}_{\text{bus }1}$, and $\boldsymbol{Z}_{\text{bus }2}$ are computed, (10.5.1) through (10.5.9) are used to compute the sequence fault currents and the sequence voltages at each bus during a fault at bus 1 for the fault type selected by the program user (for example, three-phase fault, or single line-to-ground fault, and so on). Contributions to the sequence fault currents from each line or transformer branch connected to the fault bus are computed by dividing the sequence voltage across the branch by the branch sequence impedance. The phase angles of positive- and negative-sequence voltages are also modified to account for Δ–Y transformer phase shifts. The sequence currents and sequence voltages are then transformed to the phase domain via (9.1.16) and (9.1.9). All these computations are then repeated for a fault at bus 2, then bus 3, and so on to bus *N*.

Output data for the fault type and fault impedance selected by the user consist of the fault current in each phase, contributions to the fault current from each branch connected to the fault bus for each phase, and the line-to-ground voltages at each bus—for a fault at bus 1, then bus 2, and so on to bus *N*.

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EXAMPLE 10.8

PowerWorld Simulator

Consider the five-bus power system whose single-line diagram is shown in Figure 6.2. Machine, line, and transformer data are given in Tables 10.1, 10.2, and 10.3. Note that the neutrals of both transformers and generator 1 are solidly grounded, as indicated by a neutral reactance of zero for this equipment. However, a neutral reactance equal to 0.0025 per unit is connected to the generator 2 neutral. The prefault voltage is 1.05 per unit. Using PowerWorld Simulator, determine the fault currents and voltages for a bolted single line-to-ground fault at bus 1, then bus 2, and so on to bus 5.

SOLUTION

Open PowerWorld Simulator case Example 10_8 to see this example. Tables 10.4 and 10.5 summarize the PowerWorld Simulator results for each of the faults. Note that these fault currents are subtransient currents, since the machine positive-sequence reactance input consists of direct axis subtransient reactances. Also, in Table 10.5, note that for the single-line-to-ground fault on phase *a* at bus 2, there is an overvoltage of 1.1915 per unit on unfaulted phases *b* and *c*.

			Contributions to Fault Current			
				Current		
Fault Bus	Single Line-to-Ground Fault Current (Phase *a*) per unit/degrees	GEN LINE OR TRSF	Bus-to-Bus	Phase *a*	Phase *b*	Phase *c*
				per unit/degrees		
1	46.02/−90.00	G1	GRND–1	34.41/−90.00	5.804/−90.00	5.804/−90.00
		T1	5−1	11.61/−90.00	5.804/90.00	5.804/90.00
2	14.14/−90.00	L1	4−2	5.151/−90.00	0.1124/90.00	0.1124/90.00
		L2	5−2	8.984/−90.00	0.1124/−90.00	0.1124/−90.00
3	64.30/−90.00	G2	GRND−3	56.19/−90.00	4.055/−90.00	4.055/−90.00
		T2	4−3	8.110/−90.00	4.055/90.00	4.055/90.00
4	56.07/−90.00	L1	2−4	1.742/−90.00	0.4464/90.00	0.4464/90.00
		L3	5−4	10.46/−90.00	2.679/90.00	2.679/90.00
		T2	3−4	43.88/−90.00	3.125/−90.00	3.125/−90.00
5	42.16/−90.00	L2	2−5	2.621/−90.00	0.6716/90.00	0.6716/90.00
		L3	4−5	15.72/−90.00	4.029/90.00	4.029/90.00
		T1	1−5	23.82/−90.00	4.700/−90.00	4.700/−90.00

TABLE 10.4
Fault currents for Example 10.8

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$V_{prefault} = 1.05 \angle 0$		Bus Voltages During Fault		
Fault Bus	Bus	Phase *a*	Phase *b*	Phase *c*
1	1	0.0000 ∠ 0.00	0.9537 ∠ −107.55	0.9537 ∠ 107.55
	2	0.5069 ∠ 0.00	0.9440 ∠ −105.57	0.9440 ∠ 105.57
	3	0.7888 ∠ 0.00	0.9912 ∠ −113.45	0.9912 ∠ 113.45
	4	0.6727 ∠ 0.00	0.9695 ∠ −110.30	0.9695 ∠ 110.30
	5	0.4239 ∠ 0.00	0.9337 ∠ −103.12	0.9337 ∠ 103.12
2	1	0.8832 ∠ 0.00	1.0109 ∠ −115.90	1.0109 ∠ 115.90
	2	0.0000 ∠ 0.00	1.1915 ∠ −130.26	1.1915 ∠ 130.26
	3	0.9214 ∠ 0.00	1.0194 ∠ −116.87	1.0194 ∠ 116.87
	4	0.8435 ∠ 0.00	1.0158 ∠ −116.47	1.0158 ∠ 116.47
	5	0.7562 ∠ 0.00	1.0179 ∠ −116.70	1.0179 ∠ 116.70
3	1	0.6851 ∠ 0.00	0.9717 ∠ −110.64	0.9717 ∠ 110.64
	2	0.4649 ∠ 0.00	0.9386 ∠ −104.34	0.9386 ∠ 104.34
	3	0.0000 ∠ 0.00	0.9942 ∠ −113.84	0.9942 ∠ 113.84
	4	0.3490 ∠ 0.00	0.9259 ∠ −100.86	0.9259 ∠ 100.86
	5	0.5228 ∠ 0.00	0.9462 ∠ −106.04	0.9462 ∠ 106.04
4	1	0.5903 ∠ 0.00	0.9560 ∠ −107.98	0.9560 ∠ 107.98
	2	0.2309 ∠ 0.00	0.9401 ∠ −104.70	0.9401 ∠ 104.70
	3	0.4387 ∠ 0.00	0.9354 ∠ −103.56	0.9354 ∠ 103.56
	4	0.0000 ∠ 0.00	0.9432 ∠ −105.41	0.9432 ∠ 105.41
	5	0.3463 ∠ 0.00	0.9386 ∠ −104.35	0.9386 ∠ 104.35
5	1	0.4764 ∠ 0.00	0.9400 ∠ −104.68	0.9400 ∠ 104.68
	2	0.1736 ∠ 0.00	0.9651 ∠ −109.57	0.9651 ∠ 109.57
	3	0.7043 ∠ 0.00	0.9751 ∠ −111.17	0.9751 ∠ 111.17
	4	0.5209 ∠ 0.00	0.9592 ∠ −108.55	0.9592 ∠ 108.55
	5	0.0000 ∠ 0.00	0.9681 ∠ −110.07	0.9681 ∠ 110.07

TABLE 10.5

Bus voltages for Example 10.8

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PROBLEMS

SECTION 10.1

10.1 The single-line diagram of a three-phase power system is shown in Figure 10.17. Equipment ratings are given as follows:

Synchronous generators:

G1	1000 MVA	15 kV	$X''_d = X_2 = 0.18$, $X_0 = 0.07$ per unit
G2	1000 MVA	15 kV	$X''_d = X_2 = 0.20$, $X_0 = 0.10$ per unit
G3	500 MVA	13.8 kV	$X''_d = X_2 = 0.15$, $X_0 = 0.05$ per unit
G4	750 MVA	13.8 kV	$X''_d = 0.30$, $X_2 = 0.40$, $X_0 = 0.10$ per unit

Transformers:

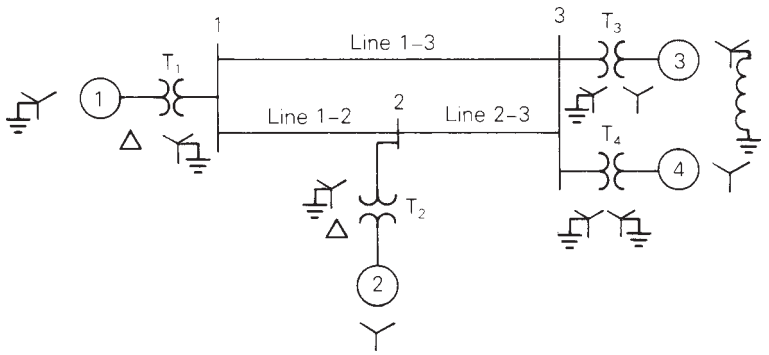
T1	1000 MVA	15 kV Δ/765 kV Y	X = 0.10 per unit
T2	1000 MVA	15 kV Δ/765 kV Y	X = 0.10 per unit
T3	500 MVA	15 kV Y/765 kV Y	X = 0.12 per unit
T4	750 MVA	15 kV Y/765 kV Y	X = 0.11 per unit

Transmission lines:

1–2	765 kV	$X_1 = 50\ \Omega$, $X_0 = 150\ \Omega$
1–3	765 kV	$X_1 = 40\ \Omega$, $X_0 = 100\ \Omega$
2–3	765 kV	$X_1 = 40\ \Omega$, $X_0 = 100\ \Omega$

The inductor connected to generator 3 neutral has a reactance of 0.05 per unit using generator 3 ratings as a base. Draw the zero-, positive-, and negative-sequence reactance diagrams using a 1000-MVA, 765-kV base in the zone of line 1–2. Neglect the Δ–Y transformer phase shifts.

FIGURE 10.17
Problem 10.1



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10.2 Faults at bus n in Problem 10.1 are of interest (the instructor selects $n = 1$, 2, or 3). Determine the Thévenin equivalent of each sequence network as viewed from the fault bus. Prefault voltage is 1.0 per unit. Prefault load currents and Δ–Y transformer phase shifts are neglected. (*Hint:* Use the Y–Δ conversion in Figure 2.33.)

10.3 Determine the subtransient fault current in per-unit and in kA during a bolted three-phase fault at the fault bus selected in Problem 10.2.

10.4 In Problem 10.1 and Figure 10.17, let 765 kV be replaced by 500 kV, keeping the rest of the data to be the same. Repeat Problems (a) 10.1, (b) 10.2, and (c) 10.3. Use a 1000-MVA, 500-kV base in the zone of the lines.

10.5 Equipment ratings for the four-bus power system shown in Figure 8.13 are given as follows:

Generator G1:	500 MVA, 13.8 kV, $X''_d = X_2 = 0.20$, $X_0 = 0.10$ per unit
Generator G2:	750 MVA, 18 kV, $X''_d = X_2 = 0.18$, $X_0 = 0.09$ per unit
Generator G3:	1000 MVA, 20 kV, $X''_d = 0.17$, $X_2 = 0.20$, $X_0 = 0.09$ per unit
Transformer T1:	500 MVA, 13.8 kV Δ/500 kV Y, X = 0.12 per unit
Transformer T2:	750 MVA, 18 kV Δ/500 kV Y, X = 0.10 per unit
Transformer T3:	1000 MVA, 20 kV Δ/500 kV Y, X = 0.10 per unit
Each line:	$X_1 = 50$ ohms, $X_0 = 150$ ohms

The inductor connected to generator G3 neutral has a reactance of 0.028 Ω. Draw the zero-, positive-, and negative-sequence reactance diagrams using a 1000-MVA, 20-kV base in the zone of generator G3. Neglect Δ–Y transformer phase shifts.

10.6 Faults at bus n in Problem 10.5 are of interest (the instructor selects $n = 1$, 2, 3, or 4). Determine the Thévenin equivalent of each sequence network as viewed from the fault bus. Prefault voltage is 1.0 per unit. Prefault load currents and Δ–Y phase shifts are neglected.

10.7 Determine the subtransient fault current in per-unit and in kA during a bolted three-phase fault at the fault bus selected in Problem 10.6.

10.8 Equipment ratings for the five-bus power system shown in Figure 8.14 are given as follows:

Generator G1:	50 MVA, 12 kV, $X''_d = X_2 = 0.20$, $X_0 = 0.10$ per unit
Generator G2:	100 MVA, 15 kV, $X''_d = 0.2$, $X_2 = 0.23$, $X_0 = 0.1$ per unit
Transformer T1:	50 MVA, 10 kV Y/138 kV Y, X = 0.10 per unit
Transformer T2:	100 MVA, 15 kV Δ/138 kV Y, X = 0.10 per unit
Each 138-kV line:	$X_1 = 40$ ohms, $X_0 = 100$ ohms

Draw the zero-, positive-, and negative-sequence reactance diagrams using a 100-MVA, 15-kV base in the zone of generator G2. Neglect Δ–Y transformer phase shifts.

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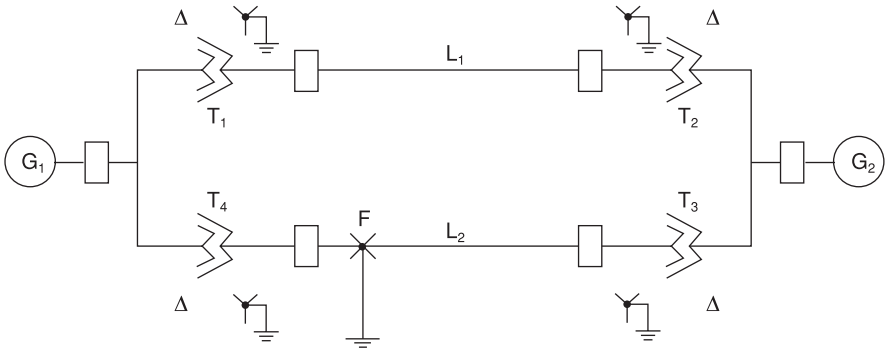
10.9 Faults at bus n in Problem 10.8 are of interest (the instructor selects $n = 1, 2, 3, 4$, or 5). Determine the Thévenin equivalent of each sequence network as viewed from the fault bus. Prefault voltage is 1.0 per unit. Prefault load currents and Δ–Y phase shifts are neglected.

10.10 Determine the subtransient fault current in per-unit and in kA during a bolted three-phase fault at the fault bus selected in Problem 10.9.

10.11 Consider the system shown in Figure 10.18. (a) As viewed from the fault at F, determine the Thévenin equivalent of each sequence network. Neglect Δ–Y phase shifts. (b) Compute the fault currents for a balanced three-phase fault at fault point F through three fault impedances $Z_{FA} = Z_{FB} = Z_{FC} = j0.5$ per unit. Equipment data in per-unit on the same base are given as follows:

FIGURE 10.18
Problem 10.11



Synchronous generators:

G1 $X_1 = 0.2$ $X_2 = 0.12$ $X_0 = 0.06$

G2 $X_1 = 0.33$ $X_2 = 0.22$ $X_0 = 0.066$

Transformers:

T1 $X_1 = X_2 = X_0 = 0.2$

T2 $X_1 = X_2 = X_0 = 0.225$

T3 $X_1 = X_2 = X_0 = 0.27$

T4 $X_1 = X_2 = X_0 = 0.16$

Transmission lines:

L1 $X_1 = X_2 = 0.14$ $X_0 = 0.3$

L1 $X_1 = X_2 = 0.35$ $X_0 = 0.6$

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10.12 Equipment ratings and per-unit reactances for the system shown in Figure 10.19 are given as follows:

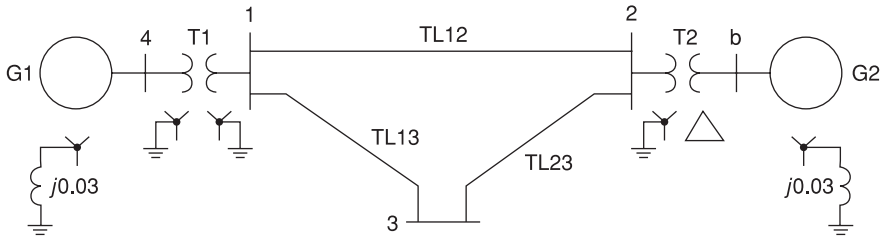

FIGURE 10.19
Problem 10.12

Synchronous generators:

G1	100 MVA	25 kV	$X_1 = X_2 = 0.2$	$X_0 = 0.05$
G2	100 MVA	13.8 kV	$X_1 = X_2 = 0.2$	$X_0 = 0.05$

Transformers:

T1	100 MVA	25/230 kV	$X_1 = X_2 = X_0 = 0.05$
T2	100 MVA	13.8/230 kV	$X_1 = X_2 = X_0 = 0.05$

Transmission lines:

TL12	100 MVA	230 kV	$X_1 = X_2 = 0.1$	$X_0 = 0.3$
TL13	100 MVA	230 kV	$X_1 = X_2 = 0.1$	$X_0 = 0.3$
TL23	100 MVA	230 kV	$X_1 = X_2 = 0.1$	$X_0 = 0.3$

Using a 100-MVA, 230-kV base for the transmission lines, draw the per-unit sequence networks and reduce them to their Thévenin equivalents, "looking in" at bus 3. Neglect Δ–Y phase shifts. Compute the fault currents for a bolted three-phase fault at bus 3.

10.13 Consider the oneline diagram of a simple power system shown in Figure 10.20. System data in per-unit on a 100-MVA base are given as follows:

Synchronous generators:

G1	100 MVA	20 kV	$X_1 = X_2 = 0.15$	$X_0 = 0.05$
G2	100 MVA	20 kV	$X_1 = X_2 = 0.15$	$X_0 = 0.05$

Transformers:

T1	100 MVA	20/220 kV	$X_1 = X_2 = X_0 = 0.1$
T2	100 MVA	20/220 kV	$X_1 = X_2 = X_0 = 0.1$

Transmission lines:

L12	100 MVA	220 kV	$X_1 = X_2 = 0.125$	$X_0 = 0.3$
L13	100 MVA	220 kV	$X_1 = X_2 = 0.15$	$X_0 = 0.35$
L23	100 MVA	220 kV	$X_1 = X_2 = 0.25$	$X_0 = 0.7125$

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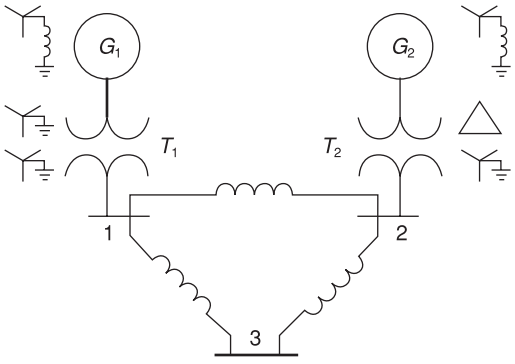
FIGURE 10.20
Problem 10.13



The neutral of each generator is grounded through a current-limiting reactor of 0.08333 per unit on a 100-MVA base. All transformer neutrals are solidly grounded. The generators are operating no-load at their rated voltages and rated frequency with their EMFs in phase. Determine the fault current for a balanced three-phase fault at bus 3 through a fault impedance $Z_F = 0.1$ per unit on a 100-MVA base. Neglect Δ–Y phase shifts.

SECTIONS 10.2–10.4

10.14 Determine the subtransient fault current in per-unit and in kA, as well as the per-unit line-to-ground voltages at the fault bus for a bolted single line-to-ground fault at the fault bus selected in Problem 10.2.

10.15 Repeat Problem 10.14 for a single line-to-ground arcing fault with arc impedance $Z_F = 15 + j0\ \Omega$.

10.16 Repeat Problem 10.14 for a bolted line-to-line fault.

10.17 Repeat Problem 10.14 for a bolted double line-to-ground fault.

10.18 Repeat Problems 10.1 and 10.14 including Δ–Y transformer phase shifts. Assume American standard phase shift. Also, calculate the sequence components and phase components of the contribution to the fault current from generator n ($n = 1$, 2, or 3, as specified by the instructor in Problem 10.2).

10.19 (a) Repeat Problem 10.14 for the case of Problem 10.4(b).
(b) Repeat Problem 10.19(a) for a single line-to-ground arcing fault with arc impedance $Z_F = (15 + j0)\ \Omega$.
(c) Repeat Problem 10.19(a) for a bolted line-to-line fault.
(d) Repeat Problem 10.19(a) for a bolted double line-to-ground fault.
(e) Repeat Problems 10.4(a) and 10.19(a) including Δ–Y transformer phase shifts. Assume American standard phase shift. Also, calculate the sequence components and phase components of the contribution to the fault current from generator n ($n = 1$, 2, or 3) as specified by the instructor in Problem 10.4(b).

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10.20 A 500-MVA, 13.8-kV synchronous generator with $X''_d = X_2 = 0.20$ and $X_0 = 0.05$ per unit is connected to a 500-MVA, 13.8-kV Δ/500-kV Y transformer with 0.10 per-unit leakage reactance. The generator and transformer neutrals are solidly grounded. The generator is operated at no-load and rated voltage, and the high-voltage side of the transformer is disconnected from the power system. Compare the subtransient fault currents for the following bolted faults at the transformer high-voltage terminals: three-phase fault, single line-to-ground fault, line-to-line fault, and double line-to-ground fault.

10.21 Determine the subtransient fault current in per-unit and in kA, as well as contributions to the fault current from each line and transformer connected to the fault bus for a bolted single line-to-ground fault at the fault bus selected in Problem 10.6.

10.22 Repeat Problem 10.21 for a bolted line-to-line fault.

10.23 Repeat Problem 10.21 for a bolted double line-to-ground fault.

10.24 Determine the subtransient fault current in per-unit and in kA, as well as contributions to the fault current from each line, transformer, and generator connected to the fault bus for a bolted single line-to-ground fault at the fault bus selected in Problem 10.9.

10.25 Repeat Problem 10.24 for a single line-to-ground arcing fault with arc impedance $Z_F = 0 + j0.1$ per unit.

10.26 Repeat Problem 10.24 for a bolted line-to-line fault.

10.27 Repeat Problem 10.24 for a bolted double line-to-ground fault.

10.28 As shown in Figure 10.21(a), two three-phase buses *abc* and $a'b'c'$ are interconnected by short circuits between phases *b* and b' and between *c* and c', with an open circuit between phases *a* and a'. The fault conditions in the phase domain are $I_a = I_{a'} = 0$ and $V_{bb'} = V_{cc'} = 0$. Determine the fault conditions in the sequence domain and verify the interconnection of the sequence networks as shown in Figure 10.15 for this one-conductor-open fault.

10.29 Repeat Problem 10.28 for the two-conductors-open fault shown in Figure 10.21(b). The fault conditions in the phase domain are

$$I_b = I_{b'} = I_c = I_{c'} = 0 \quad \text{and} \quad V_{aa'} = 0$$

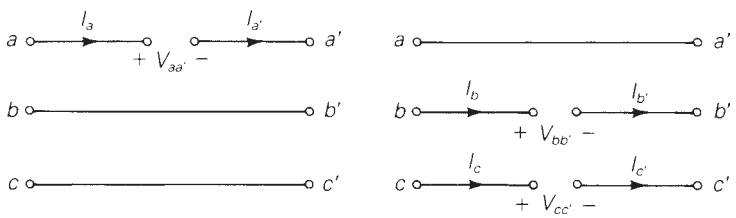

FIGURE 10.21
Problems 10.28 and 10.29: open conductor faults

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10.30 For the system of Problem 10.11, compute the fault current and voltages at the fault for the following faults at point F: (a) a bolted single line-to-ground fault; (b) a line-to-line fault through a fault impedance $Z_F = j0.05$ per unit; (c) a double line-to-ground fault from phase b to c to ground, where phase b has a fault impedance $Z_F = j0.05$ per unit, phase c also has a fault impedance $Z_F = j0.05$ per unit, and the common line-to-ground fault impedance is $Z_G = j0.033$ per unit.

10.31 For the system of Problem 10.12, compute the fault current and voltages at the fault for the following faults at bus 3: (a) a bolted single line-to-ground fault, (b) a bolted line-to-line fault, and (c) a bolted double line-to-ground fault. Also, for the single line-to-ground fault at bus 3, determine the currents and voltages at the terminals of generators Gl and G2.

10.32 For the system of Problem 10.13, compute the fault current for the following faults at bus 3: (a) a single line-to-ground fault through a fault impedance $Z_F = j0.1$ per unit, (b) a line-to-line fault through a fault impedance $Z_F = j0.1$ per unit, and (c) a double line-to-ground fault through a common fault impedance to ground $Z_F = j0.1$ per unit.

10.33 For the three-phase power system with single-line diagram shown in Figure 10.22, equipment ratings and per-unit reactances are given as follows:

Machines 1 and 2:	100 MVA	20 kV	$X_1 = X_2 = 0.2$
	$X_0 = 0.04$	$X_n = 0.04$	
Transformers 1 and 2:	100 MVA	20Δ/345Y kV	
	$X_1 = X_2 = X_0 = 0.08$		

Select a base of 100 MVA, 345 kV for the transmission line. On that base, the series reactances of the line are $X_1 = X_2 = 0.15$ and $X_0 = 0.5$ per unit. With a nominal system voltage of 345 kV at bus 3, machine 2 is operating as a motor drawing 50 MVA at 0.8 power factor lagging. Compute the change in voltage at bus 3 when the transmission line undergoes (a) a one-conductor-open fault, and (b) a two-conductor-open fault along its span between buses 2 and 3.

FIGURE 10.22
Problem 10.33

10.34 At the general three-phase bus shown in Figure 10.7(a), consider a simultaneous single line-to-ground fault on phase a and line-to-line fault between phases b and c, with no fault impedances. Obtain the

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sequence-network interconnection satisfying the current and voltage constraints.

10.35 Thévenin equivalent sequence networks looking into the faulted bus of a power system are given with $Z_1 = j0.15$, $Z_2 = j0.15$, $Z_0 = j0.2$, and $E_1 = 1\angle 0°$ per unit. Compute the fault currents and voltages for the following faults occurring at the faulted bus:

(a) Balanced three-phase fault
(b) Single line-to-ground fault
(c) Line-to-line fault
(d) Double line-to-ground fault

Which is the worst fault from the viewpoint of the fault current?

10.36 The single-line diagram of a simple power system is shown in Figure 10.23 with per-unit values. Determine the fault current at bus 2 for a three-phase fault. Ignore the effect of phase shift.

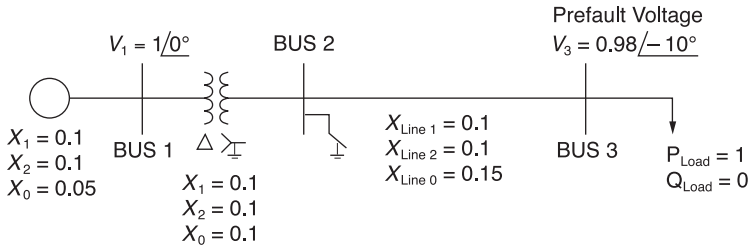

FIGURE 10.23
Problem 10.36

10.37 Consider a simple circuit configuration shown in Figure 10.24 to calculate the fault currents I_1, I_2, and I with the switch closed.

(a) Compute E_1 and E_2 prior to the fault based on the prefault voltage $V = 1\angle 0°$ and then, with the switch closed, determine I_1, I_2, and I.
(b) Start by ignoring prefault currents, with $E_1 = E_2 = 1\angle 0°$. Then superimpose the load currents, which are the prefault currents, $I_1 = -I_2 = 1\angle 0°$. Compare the results with those of part (a).

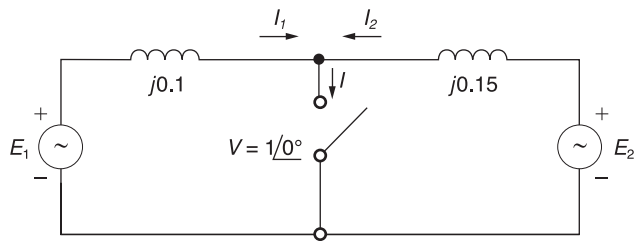

FIGURE 10.24
Problem 10.37

SECTION 10.5

10.38 The zero-, positive-, and negative-sequence bus impedance matrices for a three-bus three-phase power system are

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$$\boldsymbol{Z}_{\text{bus }0} = j\begin{bmatrix} 0.10 & 0 & 0 \\ 0 & 0.20 & 0 \\ 0 & 0 & 0.10 \end{bmatrix} \quad \text{per unit}$$

$$\boldsymbol{Z}_{\text{bus }1} = \boldsymbol{Z}_{\text{bus }2} = j\begin{bmatrix} 0.12 & 0.08 & 0.04 \\ 0.08 & 0.12 & 0.06 \\ 0.04 & 0.06 & 0.08 \end{bmatrix}$$

Determine the per-unit fault current and per-unit voltage at bus 2 for a bolted three-phase fault at bus 1. The prefault voltage is 1.0 per unit.

10.39 Repeat Problem 10.38 for a bolted single line-to-ground fault at bus 1.

10.40 Repeat Problem 10.38 for a bolted line-to-line fault at bus 1.

10.41 Repeat Problem 10.38 for a bolted double line-to-ground fault at bus 1.

10.42 (a) Compute the 3×3 per-unit zero-, positive-, and negative-sequence bus impedance matrices for the power system given in Problem 10.1. Use a base of 1000 MVA and 765 kV in the zone of line 1–2.
(b) Using the bus impedance matrices determined in Problem 10.42, verify the fault currents for the faults given in Problems 10.3, 10.14, 10.15, 10.16, and 10.17.

10.43 The zero-, positive-, and negative-sequence bus impedance matrices for a two-bus three-phase power system are

$$\boldsymbol{Z}_{\text{bus }0} = j\left[\begin{array}{c|c} 0.10 & 0 \\ \hline 0 & 0.10 \end{array}\right] \quad \text{per unit}$$

$$\boldsymbol{Z}_{\text{bus }1} = \boldsymbol{Z}_{\text{bus }2} = j\left[\begin{array}{c|c} 0.20 & 0.10 \\ \hline 0.10 & 0.30 \end{array}\right] \quad \text{per unit}$$

Determine the per-unit fault current and per-unit voltage at bus 2 for a bolted three-phase fault at bus 1. The prefault voltage is 1.03 per unit.

10.44 Repeat Problem 10.43 for a bolted single line-to-ground fault at bus 1.

10.45 Repeat Problem 10.43 for a bolted line-to-line fault at bus 1.

10.46 Repeat Problem 10.43 for a bolted double line-to-ground fault at bus 1.

10.47 Compute the 3×3 per-unit zero-, positive-, and negative-sequence bus impedance matrices for the power system given in Problem 10.4(a). Use a base of 1000 MVA and 500 kV in the zone of line 1–2.

10.48 Using the bus impedance matrices determined in Problem 10.47, verify the fault currents for the faults given in Problems 10.4(b), 10.4(c), and 10.19(a through d).

10.49 Compute the 4×4 per-unit zero-, positive-, and negative-sequence bus impedance matrices for the power system given in Problem 10.5. Use a base of 1000 MVA and 20 kV in the zone of generator G3.

10.50 Using the bus impedance matrices determined in Problem 10.42, verify the fault currents for the faults given in Problems 10.7, 10.21, 10.22, and 10.23.

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10.51 Compute the 5 × 5 per-unit zero-, positive-, and negative-sequence bus impedance matrices for the power system given in Problem 10.8. Use a base of 100 MVA and 15 kV in the zone of generator G2.

10.52 Using the bus impedance matrices determined in Problem 10.51, verify the fault currents for the faults given in Problems 10.10, 10.24, 10.25, 10.26, and 10.27.

10.53 The positive-sequence impedance diagram of a five-bus network with all values in per-unit on a 100-MVA base is shown in Figure 10.25. The generators at buses 1 and 3 are rated 270 and 225 MVA, respectively. Generator reactances include subtransient values plus reactances of the transformers connecting them to the buses. The turns ratios of the transformers are such that the voltage base in each generator circuit is equal to the voltage rating of the generator. (a) Develop the positive-sequence bus admittance matrix $\boldsymbol{Y}_{\text{bus 1}}$. (b) Using MATLAB or another computer program, invert $\boldsymbol{Y}_{\text{bus 1}}$ to obtain $\boldsymbol{Z}_{\text{bus 1}}$. (c) Determine the subtransient current for a three-phase fault at bus 4 and the contributions to the fault current from each line. Neglect prefault currents and assume a prefault voltage of 1.0 per unit.

10.54 For the five-bus network shown in Figure 10.25, a bolted single-line-to-ground fault occurs at the bus 2 end of the transmission line between buses 1 and 2. The fault causes the circuit breaker at the bus 2 end of the line to open, but all other breakers remain closed. The fault is shown in Figure 10.26. Compute the subtransient fault current with the circuit breaker at the bus-2 end of the faulted line open. Neglect prefault current and assume a prefault voltage of 1.0 per unit.

10.55 A single-line diagram of a four-bus system is shown in Figure 10.27. Equipment ratings and per-unit reactances are given as follows.

Machines 1 and 2:	100 MVA	20 kV	$X_1 = X_2 = 0.2$
	$X_0 = 0.04$	$X_n = 0.05$	
Transformers T_1 and T_2:	100 MVA	20Δ/345Y kV	
	$X_1 = X_2 = X_0 = 0.08$		

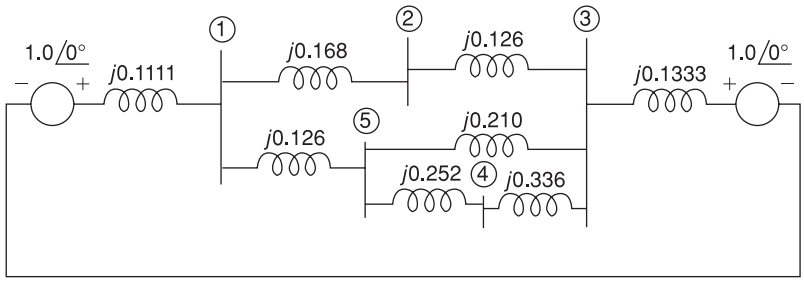

FIGURE 10.25
Problems 10.53 and 10.54

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FIGURE 10.26
Problem 10.54

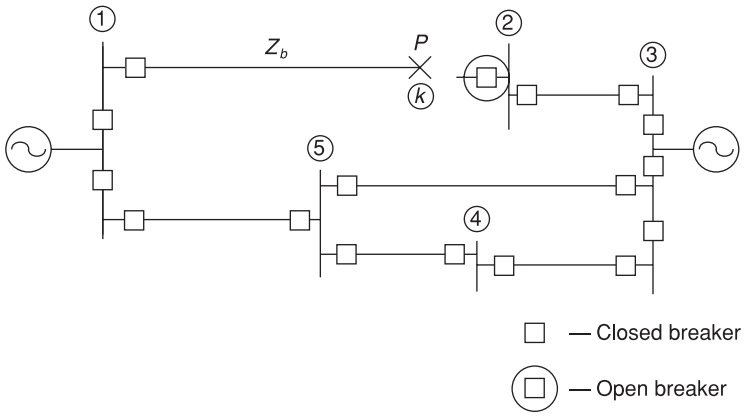
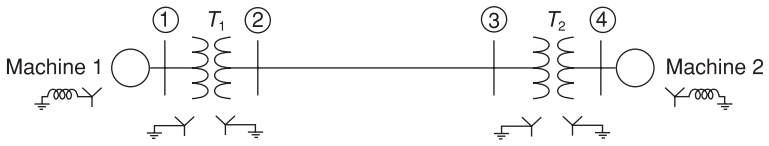


FIGURE 10.27
Problem 10.55

On a base of 100 MVA and 345 kV in the zone of the transmission line, the series reactances of the transmission line are $X_1 = X_2 = 0.15$ and $X_0 = 0.5$ per unit. (a) Draw each of the sequence networks and determine the bus impedance matrix for each of them. (b) Assume the system to be operating at nominal system voltage without prefault currents when a bolted line-to-line fault occurs at bus 3. Compute the fault current, the line-to-line voltages at the faulted bus, and the line-to-line voltages at the terminals of machine 2. (c) Assume the system to be operating at nominal system voltage without prefault currents, when a bolted double line-to-ground fault occurs at the terminals of machine 2. Compute the fault current and the line-to-line voltages at the faulted bus.

10.56 The system shown in Figure 10.28 is the same as in Problem 10.48, except that the transformers are now Y–Y connected and solidly grounded on both sides. (a) Determine the bus impedance matrix for each of the three sequence networks. (b) Assume the system to be operating at nominal system voltage without prefault currents when a bolted single-line-to-ground fault occurs on phase *a* at bus 3. Compute the fault current, the current out of phase *c* of machine 2 during the fault, and the line-to-ground voltages at the terminals of machine 2 during the fault.

FIGURE 10.28
Problem 10.56



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10.57 The results in Table 10.5 show that during a phase *a* single line-to-ground fault, the phase angle on phase *a* voltages is always zero. Explain why we would expect this result.

PW **10.58** The results in Table 10.5 show that during the single line-to-ground fault at bus 2, the *b* and *c* phase voltage magnitudes at bus 2 actually rise above the prefault voltage of 1.05 per unit. Use PowerWorld Simulator with case Example 10_8 to determine the type of in-line fault midway between bus 2 and 4 fault that gives the highest per-unit voltage magnitude.

PW **10.59** Using PowerWorld Simulator case Example 10_8, plot the variation in the bus 2 phase *a*, *b*, *c* voltage magnitudes during a single line-to-ground fault at bus 2 as the fault reactance is varied from 0 to 0.30 per unit in 0.05 per-unit steps (the fault impedance is specified on the Fault Options page of the Fault Analysis dialog).

PW **10.60** Using the Example 10_8 case, determine the fault current in amps, except with a line-to-line fault at each of the buses. Compare the fault currents with the values given in Table 10.4.

PW **10.61** Using the Example 10_8 case, determine the fault current in amps, except with a bolted double line-to-ground fault at each of the buses. Compare the fault currents with the values given in Table 10.4.

PW **10.62** Re-determine the Example 10_8 fault currents, except with a new line installed between buses 2 and 5. The parameters for this new line should be identical to those of the existing line between buses 2 and 5. The new line is not mutually coupled to any other line. Are the fault currents larger or smaller than the Example 10_8 values?

PW **10.63** Re-determine the Example 10_8 fault currents, except with a second generator added at bus 3. The parameters for the new generator should be identical to those of the existing generator at bus 3. Are the fault currents larger or smaller than the Example 10_8 values?

PW **10.64** Using PowerWorld Simulator case Problem10_64, calculate the per-unit fault current and the current supplied by each of the generators for a single line-to-ground fault at the ORANGE69 bus. During the fault, what percentage of buses have voltage magnitude below 0.75 per unit?

PW **10.65** Repeat Problem 10.64, except place the fault at the POPLAR69 bus.

DESIGN PROJECT 3 (*CONTINUED*): POWER FLOW/ SHORT CIRCUITS

Additional time given: 3 weeks

Additional time required: 10 hours

This is a continuation of Design Project 3. Assignments 1 and 2 are given in Chapter 6. Assignment 3 is given in Chapter 8.

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Assignment 4: Short Circuits—Breaker/Fuse Selection

For the single-line diagram that you have been assigned (Figure 6.25 or 6.26), convert the zero-, positive-, and negative-sequence reactance data to per-unit using the given system base quantities. Use subtransient machine reactances. Then using PowerWorld Simulator, create the generator, transmission line, and transformer input data files. Next, run the Simulator to compute sub-transient fault currents for (1) single-line-to-ground, (2) line-to-line, and (3) double line-to-ground bolted faults at each bus. Also, compute the zero-, positive-, and negative-sequence bus impedance matrices. Assume 1.0 per-unit prefault voltage. Also, neglect prefault load currents and all losses.

For students assigned to Figure 6.25: Select a suitable circuit breaker from Table 8.10 for *each* location shown on your single-line diagram. Each breaker that you select should: (1) have a rated voltage larger than the maximum system operating voltage, (2) have a rated continuous current at least 30% larger than normal load current (normal load currents are computed in Assignment 2), and (3) have a rated short-circuit current larger than the maximum fault current for any type of fault at the bus where the breaker is located (fault currents are computed in Assignments 3 and 4). This conservative practice of selecting a breaker to interrupt the entire fault current, not just the contribution to the fault through the breaker, allows for future increases in fault currents. *Note:* Assume that the (X/R) ratio at each bus is less than 17 for both a three-phase fault and a single-line-to-ground fault such that the breakers are capable of interrupting the dc offset in addition to the subtransient fault current. Circuit breaker cost should also be a factor in your selection. Do *not* select a breaker that interrupts 63 kA if a 40-kA or a 31.5-kA breaker will do the job.

For students assigned to Figure 6.26: Enclosed [8, 9] are "melting time" and "total clearing time" curves for K rated fuses with continuous current ratings from 15 to 200 A. Select suitable branch and tap fuses from these curves for each of the following three locations on your single-line diagram: bus 2, bus 4, and bus 7. Each fuse you select should have a continuous current rating that is at least 15% higher but not more than 50% higher than the normal load current at that bus (normal load currents are computed in Assignment 2). Assume that cables to the load can withstand 50% continuous overload currents. Also, branch fuses should be coordinated with tap fuses; that is, for every fault current, the tap fuse should clear before the branch fuse melts. For each of the three buses, assume a reasonable X/R ratio and determine the asymmetrical fault current for a three-phase bolted fault (subtransient current is computed in Assignment 3). Then for the fuses that you select from [8, 9], determine the clearing time CT of tap fuses and the melting time MT of branch fuses. The ratio MT/CT should be less than 0.75 for good coordination.

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DESIGN PROJECT 4: CIRCUIT BREAKER SELECTION

Time given: 3 weeks

Approximate time required: 10 hours

As a protection engineer for Metropolis Light and Power (MLP), your job is to ensure that the transmission line and transformer circuit breaker ratings are sufficient to interrupt the fault current associated with any type of fault (balanced three-phase, single line-to-ground, line-to-line, and double line-to-ground). The MLP power system is modeled in case Chapter10_DesignCase4. This case models the positive-, negative-, and zero-sequence values for each system device. Note that the 69/138-kV transformers are grounded Y on the low side and Δ on the high side; the 138/345-kV transformers are grounded Y on both sides. In this design problem, your job is to evaluate the circuit breaker ratings for the three 345-kV transmission lines and the six 345/138-kV transformers. You need not consider the 138- or 69-kV transmission lines, or the 138/69-kV transformers.

Design Procedure

1. Load Chapter10_DesignCase4 into PowerWorld Simulator. Perform an initial power flow solution to get the base case system operating point.
2. Apply each of the four fault types to each of the 345-kV buses and to the 138-kV buses attached to 345/138-kV transformers to determine the maximum fault current that each of the 345-kV lines and 345/138-kV transformers will experience.
3. For each device, select a suitable circuit breaker from Table 8.10. Each breaker that you select should (a) have a rated voltage larger than the maximum system operating voltage, (b) have a rated continuous current at least 30% larger than the normal rated current for the line, and (c) have a rated short circuit current larger than the maximum fault current for any type of fault at the bus where the breaker is located. This conservative practice of selecting a breaker to interrupt the entire fault current, not just the contribution to the fault current through the breaker allows for future increases in fault currents. Since higher-rated circuit breakers cost more, you should select the circuit breaker with the lowest rating that satisfies the design constraints.

Simplifying Assumptions

1. You need only consider the base case conditions given in the Chapter10_DesignCase4 case.
2. You may assume that the X/R ratios at each bus are sufficiently small (less than 17) so that the dc offset has decayed to a sufficiently low value (see Section 8.5 for details).
3. As is common with commercial software, including PowerWorld Simulator, the Δ–Y transformer phase shifts are neglected.

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11 System Protection

Lightning slices through rainy skies above a city, in a time-exposure view. (Jhaz Photography /Shutterstock .com.)

LEARNING OBJECTIVES

At the conclusion of this chapter, you should be able to:

1. Know the components of protection systems, including instrument transformers, relays, circuit breakers, reclosers, fuses, and sectionalizers.
2. Be familiar with different types of relays, including overcurrent, directional, distance, differential, pilot, and multifunction numeric relays.
3. Understand zones of protection and their characteristics.

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Short circuits occur in power systems when equipment insulation fails, due to system overvoltages caused by lightning or switching surges, to insulation contamination, or to other mechanical and natural causes. Careful design, operation, and maintenance can minimize the occurrence of short circuits but cannot eliminate them. Methods for calculating short-circuit currents for balanced and unbalanced faults were discussed in Chapters 8 and 10. Such currents can be several orders of magnitude larger than normal operating currents and, if allowed to persist, may cause insulation damage, conductor melting, fire, and explosion. Windings and bus bars may also suffer mechanical damage due to high magnetic forces during faults. Clearly, faults must be quickly removed from a power system. Standard EHV protective equipment is designed to clear faults within 3 cycles, whereas lower-voltage protective equipment typically operates within 5–20 cycles.

This chapter provides an introduction to power system protection. Blackburn defines protection as "the science, skill, and art of applying and setting relays and/or fuses to provide maximum sensitivity to faults and undesirable conditions, but to avoid their operation on all permissible or tolerable conditions" [1]. The basic idea is to define the undesirable conditions and look for differences between the undesirable and permissible conditions that relays or fuses can sense. It is also important to remove only the faulted equipment from the system while maintaining as much of the unfaulted system as possible in service, in order to continue to supply as much of the load as possible.

Although fuses and reclosers (circuit breakers with built-in instrument transformers and relays) are widely used to protect primary distribution systems (with voltages in the 2.4–46 kV range), this chapter is primarily focused on circuit breakers and relays, which are used to protect HV (115–230 kV) and EHV (345–765 kV) power systems. The IEEE defines a relay as "a device whose function is to detect defective lines or apparatus or other power system conditions of an abnormal or dangerous nature and to initiate appropriate control action" [1]. In practice, a relay is a device that closes or opens a contact when energized. Relays are also used in low-voltage (600-V and below) power systems and almost anywhere that electricity is used. They are used in heating, air conditioning, stoves, clothes washers and dryers, refrigerators, dishwashers, telephone networks, traffic controls, airplane and other transportation systems, and robotics, as well as many other applications.

Problems with the protection equipment itself can occur. A second line of defense, called *backup* relays, may be used to protect the first line of defense, called *primary* relays. In HV and EHV systems, separate current- or voltage-measuring devices, separate trip coils on the circuit breakers, and separate batteries for the trip coils may be used. Also, the various protective devices must be properly coordinated such that primary relays assigned to protect equipment in a particular zone operate first. If the primary relays fail, then backup relays should operate after a specified time delay.

This chapter begins with a discussion of the basic system-protection components.

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CASE STUDY

On Good Behavior:

Inverter-Grid Protections for Integrating Distributed Photovoltaics*

Tom Key, Gefei "Derek" Kou, & Michael Jensen

The following paragraphs summarize the above article, which describes system protection impacts resulting from the integration of inverter-based distributed energy resources (DERs), including photovoltaic (PV) power plants [14]. Sign in at www.cengage.com to download this product and to access its online resources.

The article addresses several practical protection issues with inverter-based PV plants, including temporary overvoltage, fault and arc flash contributions, response to feeder open-phase events, and islanding protection and reclosing practices.

Temporary Overvoltage and PV Plants

DERs can contribute to a ground-fault overvoltage (GFO) on unfaulted phases in a four-wire system. Effective grounding is intended to limit a line-to-ground overvoltage to 80% of rated voltage. For DERs, the concern for GFO is a key factor in selecting the grid-connection transformer. While transformer connections are normally required to provide supplemental grounding for synchronous-machine DERs, the need is less clear for inverter-connected DERs because of significantly different fault behavior. When supplemental grounding is required, it is via the transformer interconnection or a special grounding transformer. A reactor may also be required in the transformer neutral for protection coordination.

Figure 1 shows the results of a 2013 Electric Power Research Institute (EPRI) survey identifying the Yg-Yg transformer configuration as the most common DER connection. A 2020 EPRI survey provided similar results. Given that the Yg-Yg connection does not provide effective grounding, the question of supplemental requirements for PV plants is raised. In the most recent 2020 survey, there was a 50/50 split among twenty large utilities on requiring supplemental grounding for PV plants.

Another overvoltage concern for inverters is their response to a sudden loss of load. Load-rejection overvoltage (LRO) happens when a grid-following inverter is disconnected, for example, due to breaker operation. Figure 2 shows an example captured at a typical utility-scale, 5-MW solar power plant at 34.5 kV. The transient overvoltage at the plant reached 159% of the nominal voltage and lasted 12 cycles, a

* "On Good Behavior: Inverter-Grid Protections for Integrating Distributed Photovoltaics," by Tom Key, Gefei "Derek" Kou, & Michael Jensen, available online, with permission, from *IEEE Power & Energy Magazine* (November/December 2020), pp. 75–85.

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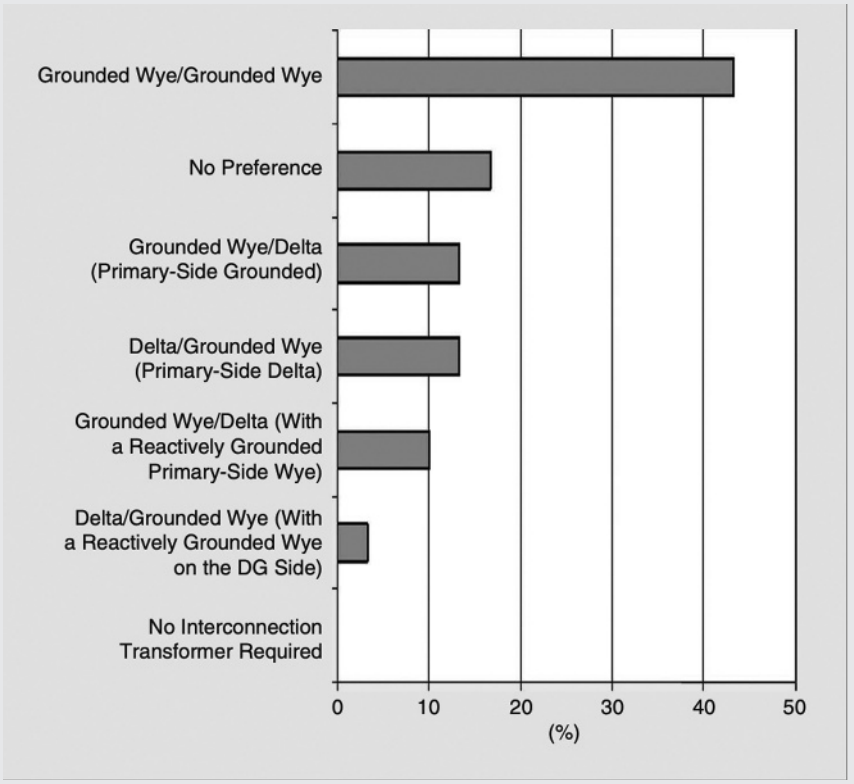

Figure 1 Preferred transformer connections for inverter-based systems
EPRI 3002001277

phenomenon that is not uncommon with rapid load rejection.

Looking ahead, inverter GFO and LRO events are expected to be largely mitigated with the application of the new IEEE Standard 1547-2018. An inverter certified under this standard "shall not cause fundamental frequency line-to-ground voltages to exceed 138% for a duration exceeding one cycle."

Fault and Arc-Flash Contributions

Accurate short-circuit representation of solar inverters is key to protection coordination and relay settings. Unlike synchronous generators, solar inverters act as a positive-sequence source and, depending on design, have insignificant negative- and zero-sequence fault currents. Figure 3 shows a single-phase-to-ground fault near a 20-MW PV power plant. The faulted phase dropped to 35% of nominal voltage. As shown, the fault current from the inverter was dominated by a positive-sequence current (I_1). Negative- and zero-sequence components (I_2 and I_0) are less than 5% of rated current.

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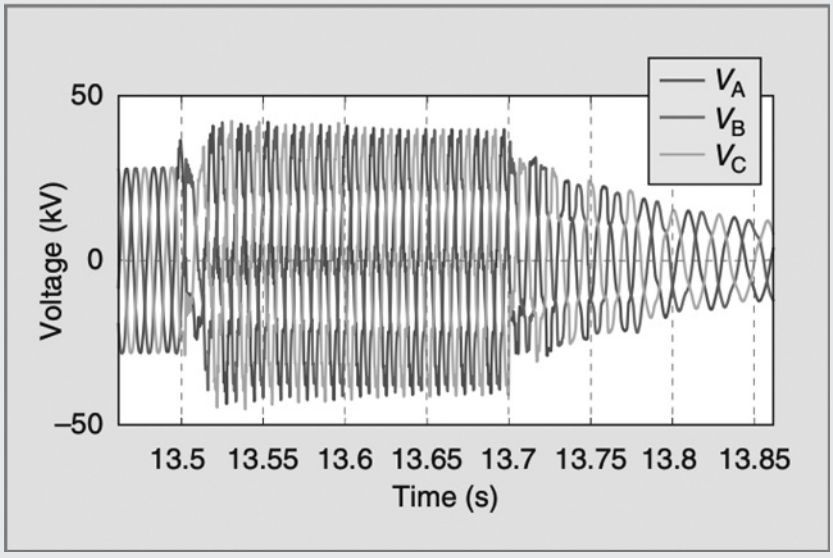

Figure 2 The LRO measured at a utility-scale solar power plant. Grid disconnection happened at 13.5 s.

The collective efforts of the industry have greatly improved the modeling techniques of inverter-based generation. Currently, most short-circuit simulation platforms, including ASPEN, CAPE, and ETAP, provide for inverter modeling, where PV inverters are modeled as voltage-controlled current sources.

When field crews work on energized distribution feeders, there is a concern about accidental faults and arc flash. Arc flash correlates with line voltage, fault current magnitude, and fault-clearing time. DERs can increase arc-flash incident energy in two major ways. First, the step-up transformer windings, such as medium-voltage wye-grounded and delta connection to the DER, provide a zero-sequence current, which can reduce ground current from the substation and delay fault clearing in the upstream protective device. Second, after loss of the grid source and prior to plant tripping, DERs can back-feed into a fault.

PV plant mitigations include neutral grounding resistors and recloser/relay settings at the point of interconnection. Many utilities also enable high-speed fault clearing in feeder relays when crews are working on a line, so as to reduce the grid's contribution to arc-flash exposure.

Feeder Open-Phase Events

Incidents of open phase (or single-phasing) on three-phase distribution circuits can happen. This brings several concerns for PV plants. Depending on the transformer configuration and DER output to load,

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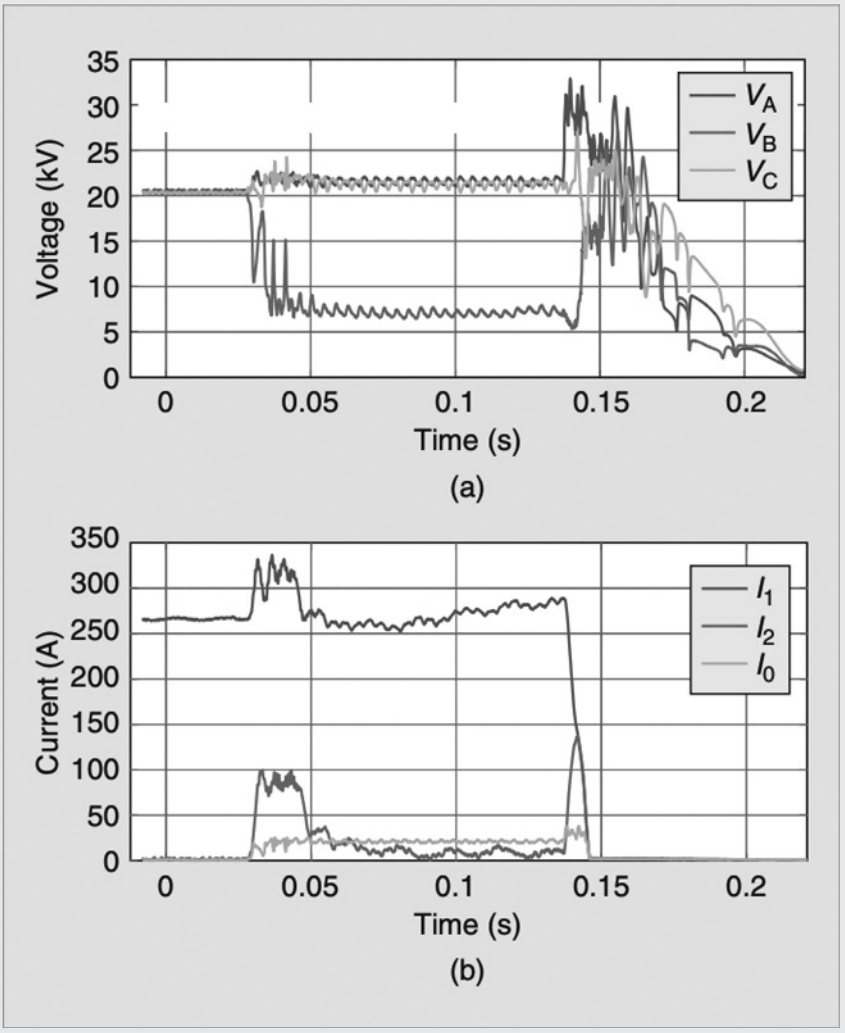

Figure 3 PV inverter fault (a) phase voltage, and (b) symmetrical component current contribution

voltage on the open phase might be regenerated. During the open-phase incident, this leads to poor voltage regulation or elevated voltages, and can potentially expose the public to a live downed conductor.

Although open-phase events are a concern without DERs, the addition of three-phase PV plants has led to an increase in reported incidents. One explanation is the increased number of fuses and plant-scale generation sources. Plant design, control of back-feed, and transformer connections all play a role. Behaviors vary among

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Reasons for Failures	Ceased but restared in <5 min	Did not cease to generate	Cease delayed >2 s	Open-phase voltage >1.1 per unit	Other reasons
Number of Failures	1	2	16	3	12
IEEE Standard 1547 Section	4.10.3	6.2.2			—

TABLE 1

A summary of open-phase test results showing 34 failures at 82 large PV plants sites (Source: Xcel Energy)

underground cables, different transformer windings, and the addition of supplemental grounding.

IEEE Standard 1547-2018 prohibits DERs from back-feeding into abnormal grid conditions, including open-phase events. However, detecting a plant high-side, open-phase event from the DER terminals can be challenging, since the DER terminal voltage might remain within the normal operating range.

Table 1 shows field experience of open-phase results from the commissioning of PV plants reported by EPRI in 2019. These plants were built prior to the IEEE 1547 standard, which requires DERs to "cease to energize and trip" immediately upon detection of an open phase. Also, a trip requires a 5-minute delay time after voltage is restored.

Inverter-connected DERs can detect primary-side, open-phase events using negative-sequence components. EPRI has shown that such inverter detection methods are effective depending on the plant transformer connection and relative generation to load levels.

Island-Prevention and Reclosing Practices

Unintended islanding has been a long-standing concern in the United States, beginning with the U.S. Public Utility Regulatory Policies Act (PURPA) in 1978. Although interconnection rules for qualifying facilities have come a long way, the idea of unsupervised DER operation is still a key consideration for grid safety and reliability.

Since PURPA and, for the most part, still today, utility-grade relaying and communications have been required for synchronous machine-based and large PV plants. Conventional relay protections passively detect island conditions and are limited to voltage and/or frequency measurements without positive feedback. The common island-prevention practice has been to augment local detection with direct-transfer trip (DTT) when an upstream protective device operates. DTT (or transfer-trip) systems are used to quickly disconnect DERs when a feeder breaker opens, usually at the substation (see Figure 4).

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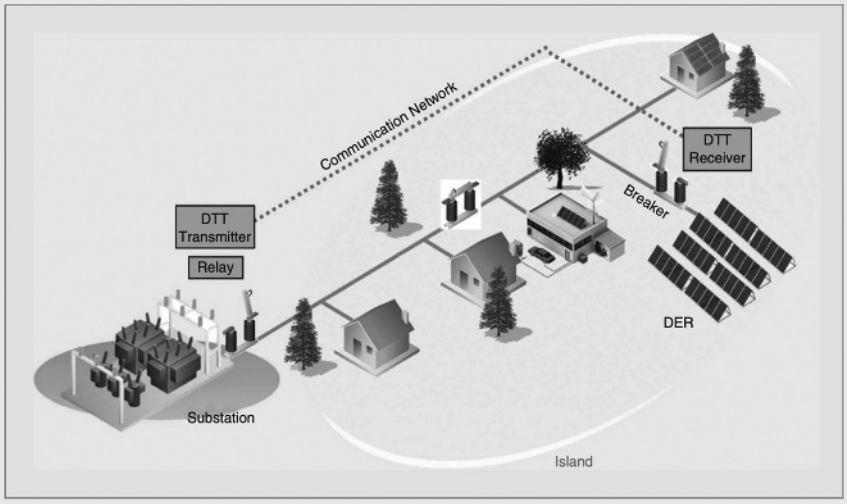

Figure 4 The basic components of a direct-transfer-trip (DTT) scheme

EPRI 3002011008

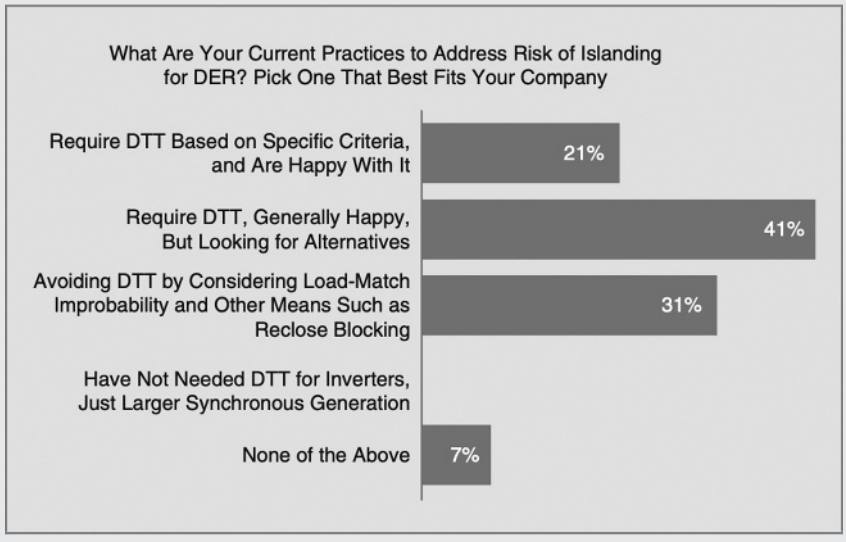

Figure 5 A recent EPRI survey of 76 utilities regarding the application of DTT

Island prevention using utility-controlled DTT is reliable and effective, but it is typically expensive for individual sites. Communication alternatives to copper and fiber are being considered, such as power-line carrier and cellular. Figure 5 shows current practices to address the risk of DER islanding among 76 utilities.

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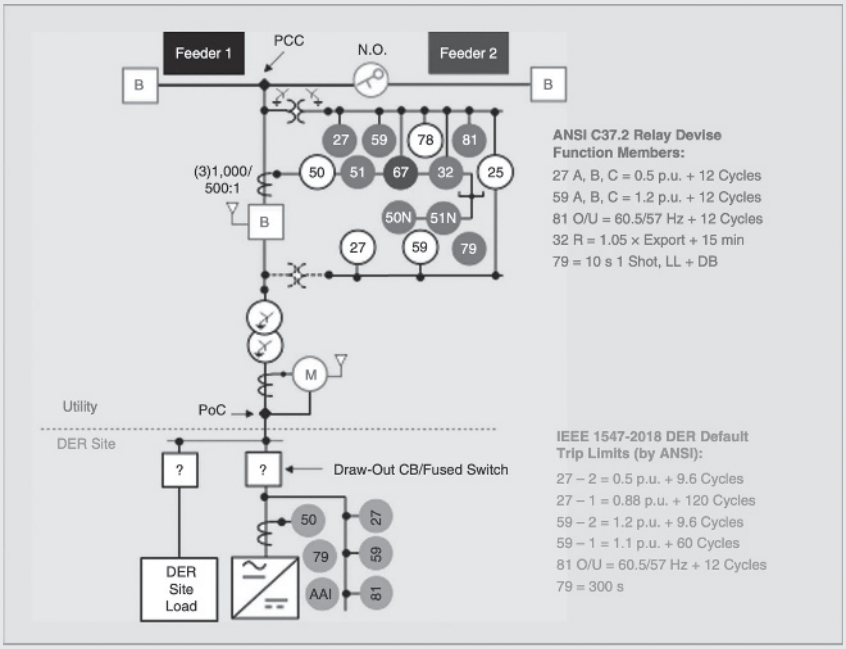

Figure 6 The relays per ANSI Standard C37.2. Note that site relays might be integrated in inverter DERs. AAI: Active anti-islanding; PCC: point of common coupling with utility feeder; PoC: point of DER connection; N.O.: normally open; M: meter; CB: circuit breaker; O/U: over/under; LL: live line; DB: dead bus; B: breaker

Southern Company Services

Onboard active islanding detection is an important alternative to DTT for inverter-based DERs. Recent studies by EPRI and Sandia National Laboratories indicate that two things need to happen for the acceptance of onboard protection as an alternative to DTT. The first is for inverter manufacturers to identify which generic detection methods they are using. The second is for utilities to update island risk-assessment methods to account for new fault ride-through requirements.

Directly related to DTT and DER ride-through, there is the need to coordinate relays that control plant-to-grid interconnection reclosers. PV plant site multifunction relays installed with reclosers have the full range of protection setting options. Typically, these reclosers are required for operation on/off control of larger DEWR facilities. In many cases, they are also used as the disconnection means for a protection-grade DTT scheme. A good example of coordinating site recloser relays with DER site intertie protection is shown in Figure 6.

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11.1 SYSTEM PROTECTION COMPONENTS

Protection systems have three basic components:

1. Instrument transformers
2. Relays
3. Circuit breakers

Figure 11.1 shows a simple overcurrent protection schematic with: (1) one type of instrument transformer—the current transformer (CT), (2) an overcurrent relay (OC), and (3) a circuit breaker (CB) for a single-phase line. The function of the CT is to reproduce in its secondary winding a current I' that is proportional to the primary current I. The CT converts primary currents in the kiloamp range to secondary currents in the 0–5 ampere range for convenience of measurement, with the following advantages:

Safety: Instrument transformers provide electrical isolation from the power system so that personnel working with relays will work in a safer environment.

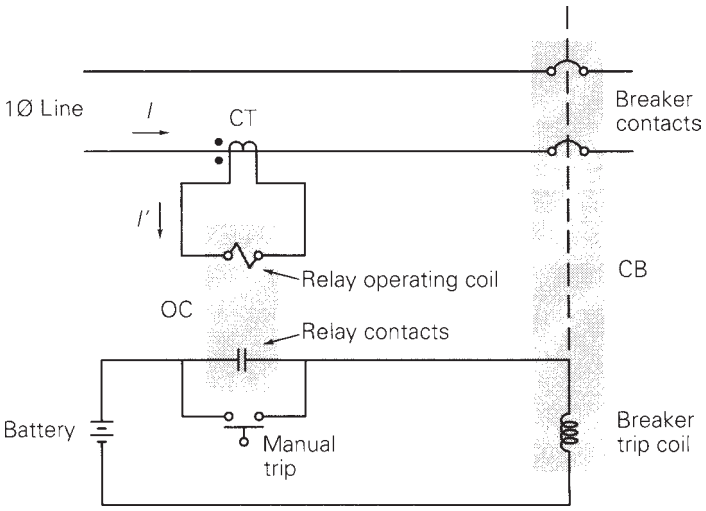
Economy: Lower-level relay inputs enable relays to be smaller, simpler, and less expensive.

Accuracy: Instrument transformers accurately reproduce power system currents and voltages over wide operating ranges.

The function of the relay is to discriminate between normal operation and fault conditions. The OC relay in Figure 11.1 has an operating coil, which is connected to the CT secondary winding, and a set of contacts.

When $|I'|$ exceeds a specified "pickup" value, the operating coil causes the normally open contacts to close. When the relay contacts close, the trip coil of the circuit breaker is energized, which then causes the circuit breaker to open.

FIGURE 11.1
Overcurrent protection schematic



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Note that the circuit breaker does not open until its operating coil is energized, either manually or by relay operation. Based on information from instrument transformers, a decision is made and "relayed" to the trip coil of the breaker, which actually opens the power circuit—hence the name *relay*.

System-protection components have the following design criteria [2]:

Reliability: Operate dependably when fault conditions occur, even after remaining idle for months or years. Failure to do so may result in costly damages.

Selectivity: Avoid unnecessary, false trips.

Speed: Operate rapidly to minimize fault duration and equipment damage. Any intentional time delays should be precise.

Economy: Provide maximum protection at minimum cost.

Simplicity: Minimize protection equipment and circuitry.

Since it is impossible to satisfy all these criteria simultaneously, compromises must be made in system protection.

11.2 INSTRUMENT TRANSFORMERS

There are two basic types of instrument transformers: voltage transformers (VTs), formerly called potential transformers (PTs), and current transformers (CTs). Figure 11.2 shows a schematic representation for the VT and CT.

The transformer primary is connected to or into the power system and is insulated for the power system voltage. The VT reduces the primary voltage, and the CT reduces the primary current to much lower, standardized levels suitable for operation of relays. Photos of VTs and CTs are shown in Figures 11.3–11.6.

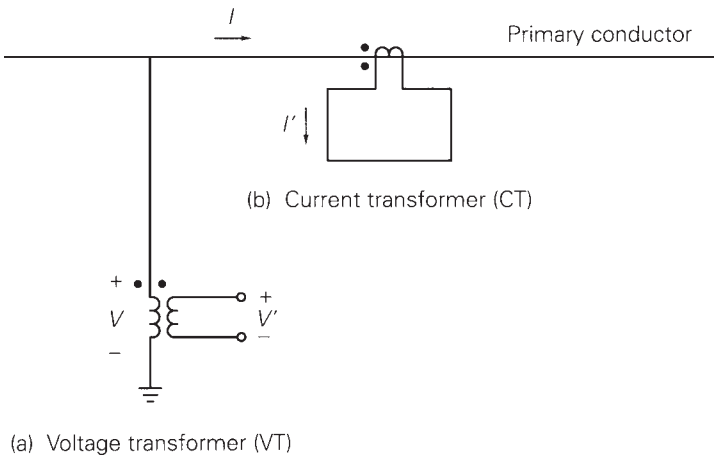

FIGURE 11.2
VT and CT schematic

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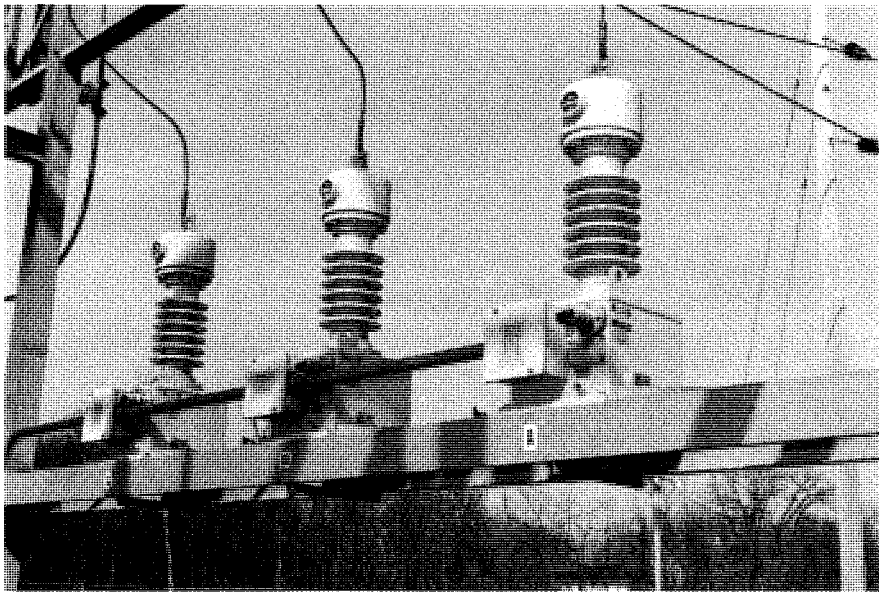

FIGURE 11.3

Three 34.5-kV voltage transformers with 34.5 kV: 115/67-volt VT ratios, at Lisle substation, Lisle, Illinois (Courtesy of Commonwealth Edison, an Exelon Company.)

For system-protection purposes, VTs are generally considered to be sufficiently accurate. Therefore, the VT is usually modeled as an ideal transformer, where

$$V' = (1/\mathrm{n})V \tag{11.2.1}$$

V' is a scaled-down representation of V and is in phase with V. A standard VT secondary voltage rating is 115 V (line-to-line). Standard VT ratios are given in Table 11.1.

Ideally, the VT secondary is connected to a voltage-sensing device with infinite impedance, such that the entire VT secondary voltage is across the sensing device. In practice, the secondary voltage divides across the high-impedance sensing device and the VT series leakage impedances. VT leakage impedances are kept low in order to minimize voltage drops and phase-angle differences from primary to secondary.

The primary winding of a current transformer usually consists of a single turn, obtained by running the power system's primary conductor through the CT core. The normal current rating of CT secondaries is standardized at 5 A in the United States, whereas 1 A is standard in Europe and some other regions. Currents of 10 to 20 times (or greater) normal rating often occur in CT windings for a few cycles during short circuits. Standard CT ratios are given in Table 11.2.

Ideally, the CT secondary is connected to a current-sensing device with zero impedance, such that the entire CT secondary current flows through the sensing device. In practice, the secondary current divides, with most flowing through the

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FIGURE 11.4

Three 500-kV coupling capacitor voltage transformers with 303.1 kV: 115/67-volt VT ratios, Westwing 500-kV Switching Substation (Courtesy of Arizona Public Service.)

low-impedance sensing device and some flowing through the CT shunt excitation impedance. CT excitation impedance is kept high in order to minimize excitation current.

An approximate equivalent circuit of a CT is shown in Figure 11.7, where

Z' = CT secondary leakage impedance

X_e = (Saturable) CT excitation reactance

Z_B = Impedance of terminating device (relay, including leads)

The total impedance Z_B of the terminating device is called the *burden* and is typically expressed in values of less than an ohm. The burden on a CT may also be expressed as volt-amperes at a specified current.

Associated with the CT equivalent circuit is an excitation curve that determines the relationship between the CT secondary voltage E′ and excitation current I_e. Excitation curves for a multiratio bushing CT with ANSI classification C100 are shown in Figure 11.8.

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FIGURE 11.5

Three 25-kV class current transformers—window design (Courtesy of Arizona Public Service.)

FIGURE 11.6

500-kV class current transformers with 2000:5 CT ratios in front of 500-kV SF6 circuit breakers. Westwing 500-kV Switching Substation (Courtesy of Arizona Public Service.)

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Voltage Ratios						
1:1	2:1	2.5:1	4:1	5:1	20:1	40:1
60:1	100:1	200:1	300:1	400:1	600:1	800:1
1000:1	2000:1	3000:1	4500:1			

TABLE 11.1
Standard VT ratios

Current Ratios						
50:5	100:5	150:5	200:5	250:5	300:5	400:5
450:5	500:5	600:5	800:5	900:5	1000:5	1200:5
1500:5	1600:5	2000:5	2400:5	2500:5	3000:5	3200:5
4000:5	5000:5	6000:5				

TABLE 11.2
Standard CT ratios

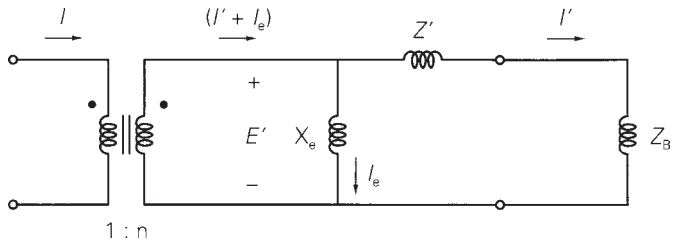

FIGURE 11.7
CT equivalent circuit

Current transformer performance is based on the ability to deliver a secondary output current I' that accurately reproduces the primary current I. Performance is determined by the highest current that can be reproduced without saturation to cause large errors. Using the CT equivalent circuit and excitation curves, the following procedure can be used to determine CT performance.

STEP 1 Assume a CT secondary output current I'.

STEP 2 Compute $E' = (Z' + Z_B)I'$.

STEP 3 Using E', find I_e from the excitation curve.

STEP 4 Compute $I = n(I' + I_e)$.

STEP 5 Repeat Steps 1–4 for different values of I', then plot I' versus I.

For simplicity, approximate computations are made with magnitudes rather than with phasors. Also, the CT error is the percentage difference between $(I' + I_e)$ and I', given by:

$$\text{CT error} = \frac{I_e}{I' + I_e} \times 100\% \tag{11.2.2}$$

The following examples illustrate the procedure.

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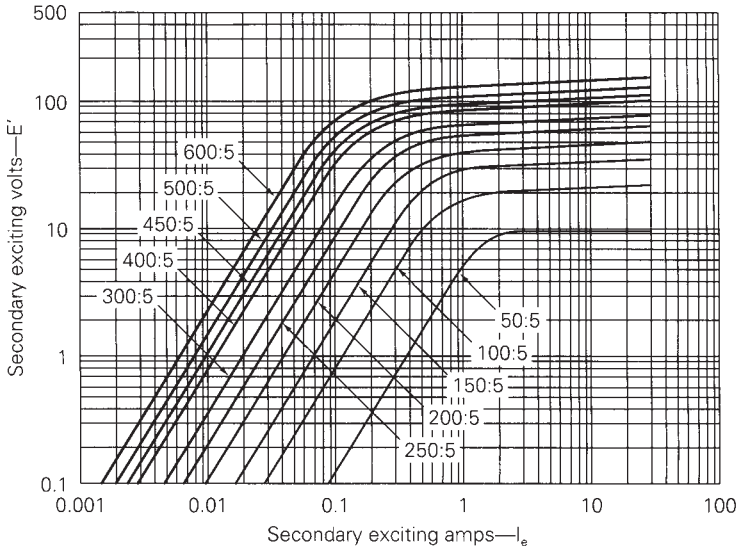

CT ratio	Secondary resistance Ω
50:5	0.061
100:5	0.082
150:5	0.104
200:5	0.125
250:5	0.146
300:5	0.168
400:5	0.211
450:5	0.230
500:5	0.242
600:5	0.296

FIGURE 11.8

Excitation curves for a bushing CT with a C100 ANSI accuracy classification [3] (*Westinghouse Relay Manual*, A New Silent Sentinels Publication (Newark, NJ: Westinghouse Electric Corporation, 1972).)

EXAMPLE 11.1

Current transformer (CT) performance

Evaluate the performance of the multiratio CT in Figure 11.8 with a 100:5 CT ratio, for the following secondary output currents and burdens: (a) $I' = 5$ A and $Z_B = 0.5\ \Omega$; (b) $I' = 8$ A and $Z_B = 0.8\ \Omega$; and (c) $I' = 15$ A and $Z_B = 1.5\ \Omega$. Also, compute the CT error for each output current.

SOLUTION

From Figure 11.8, the CT with a 100:5 CT ratio has a secondary resistance $Z' = 0.082\ \Omega$. Completing the above steps:

a. **STEP 1** $I' = 5$ A

STEP 2 From Figure 11.7,

$$E' = (Z' + Z_B)I' = (0.082 + 0.5)(5) = 2.91 \text{ V}$$

STEP 3 From Figure 11.8, $I_e = 0.25$ A

STEP 4 From Figure 11.7, $I = (100/5)(5 + 0.25) = 105$ A

$$\text{CT error} = \frac{0.25}{5.25} \times 100 = 4.8\%$$

b. **STEP 1** $I' = 8$ A

STEP 2 From Figure 11.7,

$$E' = (Z' + Z_B)I' = (0.082 + 0.8)(8) = 7.06 \text{ V}$$

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STEP 3 From Figure 11.8, $I_e = 0.4$ A

STEP 4 From Figure 11.7, $I = (100/5)(8 + 0.4) = 168$ A

$$\text{CT error} = \frac{0.4}{8.4} \times 100 = 4.8\%$$

c. **STEP 1** $I' = 15$ A

STEP 2 From Figure 11.7,

$$E' = (Z' + Z_B)I' = (0.082 + 1.5)(15) = 23.73 \text{ V}$$

STEP 3 From Figure 11.8, $I_e = 20$ A

STEP 4 From Figure 11.7, $I = (100/5)(15 + 20) = 700$ A

$$\text{CT error} = \frac{20}{35} \times 100 = 57.1\%$$

Note that for the 15-A secondary current in (c), high CT saturation causes a large CT error of 57.1%. Standard practice is to select a CT ratio to give a little less than 5-A secondary output current at maximum normal load. From (a), the 100:5 CT ratio and 0.5-Ω burden are suitable for a maximum primary load current of about 100 A. This example is extended in Problem 11.2 to obtain a plot of I' versus I.

EXAMPLE 11.2

Relay operation versus fault current and CT burden

An overcurrent relay set to operate at 8 A is connected to the multiratio CT in Figure 11.8 with a 100:5 CT ratio. Will the relay detect a 200-A primary fault current if the burden Z_B is (a) 0.8 Ω, (b) 3.0 Ω?

SOLUTION

Note that if an ideal CT is assumed, $(100/5) \times 8 = 160$-A primary current would cause the relay to operate.

a. From Example 11.1(b), a 168-A primary current with $Z_B = 0.8$ Ω produces a secondary output current of 8 A, which would cause the relay to operate. Therefore, the higher 200-A fault current also causes the relay to operate.

b. **STEP 1** $I' = 8$ A

STEP 2 From Figure 11.7,

$$E' = (Z' + Z_B)I' = (0.082 + 3.0)(8) = 24.7 \text{ V}$$

STEP 3 From Figure 11.8, $I_e = 30$ A

STEP 4 From Figure 11.7, $I = (100/5)(8 + 30) = 760$ A

With a 3.0-Ω burden, 760 A is the lowest primary current that causes the relay to operate. Therefore, the relay will not operate for the 200-A fault current.

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11.3 OVERCURRENT RELAYS

As shown in Figure 11.1, the CT secondary current I' is the input to the overcurrent relay operating coil. Instantaneous overcurrent relays respond to the magnitude of their input current, as shown by the trip and block regions in Figure 11.9. If the current magnitude $I' = |I'|$ exceeds a specified adjustable current magnitude I_p, called the *pickup* current, then the relay contacts close "instantaneously" to energize the circuit breaker trip coil. If I' is less than the pickup current I_p, then the relay contacts remain open, blocking the trip coil.

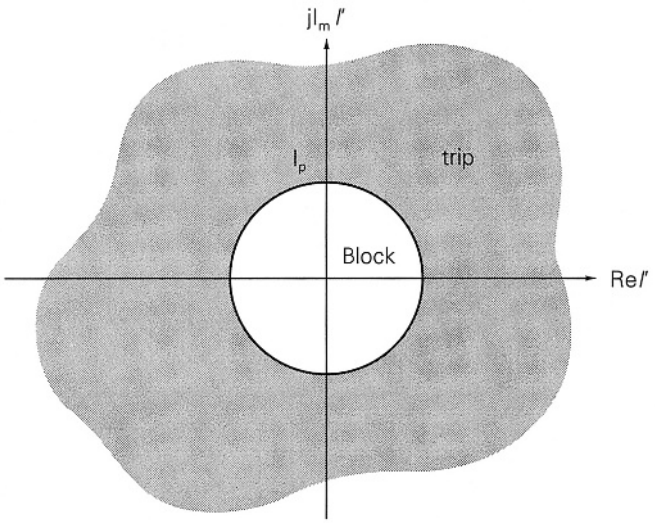
Time-delay overcurrent relays also respond to the magnitude of their input current, but with an intentional time delay. As shown in Figure 11.10, the time delay depends on the magnitude of the relay input current. If I' is a large multiple of the pickup current I_p, then the relay operates (or trips) after a small time delay. For smaller multiples of pickup, the relay trips after a longer time delay. And if $I' < I_p$, the relay remains in the blocking position.

Figure 11.11 shows two examples of a time-delay overcurrent relay: (a) Westinghouse electromechanical CO relay; and (b) Basler Electric digital relay. Characteristic curves of the Westinghouse CO-8 relay are shown in Figure 11.12. These relays have two settings:

Current tap setting: The pickup current in amperes.
Time-dial setting: The adjustable amount of time delay.

The characteristic curves are usually shown with operating time in seconds versus relay input current as a multiple of the pickup current. The curves are asymptotic to the vertical axis and decrease with some inverse power of current magnitude for values exceeding the pickup current. This inverse time characteristic can be shifted up or down by adjustment of the time-dial setting. Although discrete time-dial settings are shown in Figure 11.12, intermediate values can be obtained by interpolating between the discrete curves.

FIGURE 11.9
Instantaneous overcurrent relay block and trip regions



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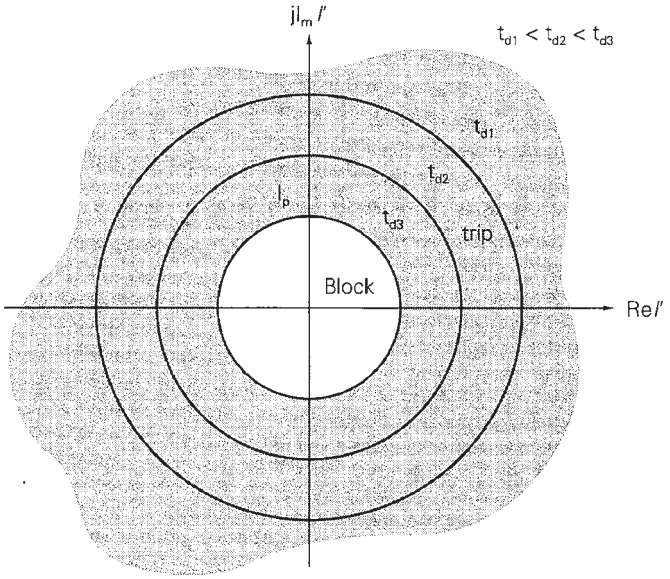

FIGURE 11.10

Time-delay overcurrent relay block and trip regions

FIGURE 11.11

Time-delay overcurrent relays: (a) Westinghouse Electromechanical (Courtesy of ABB-Westinghouse.) (b) Basler Electric Digital (Courtesy Danvers Electric.)

Figure 11.13 shows the time-current characteristics of five CO time-delay overcurrent relays used in transmission and distribution lines. The time-dial settings are selected in the figure so that all relays operate in 0.2 seconds at 20 times the pickup current. The choice of relay time-current characteristic depends on the sources, lines, and loads. The definite (CO-6) and moderately inverse (CO-7)

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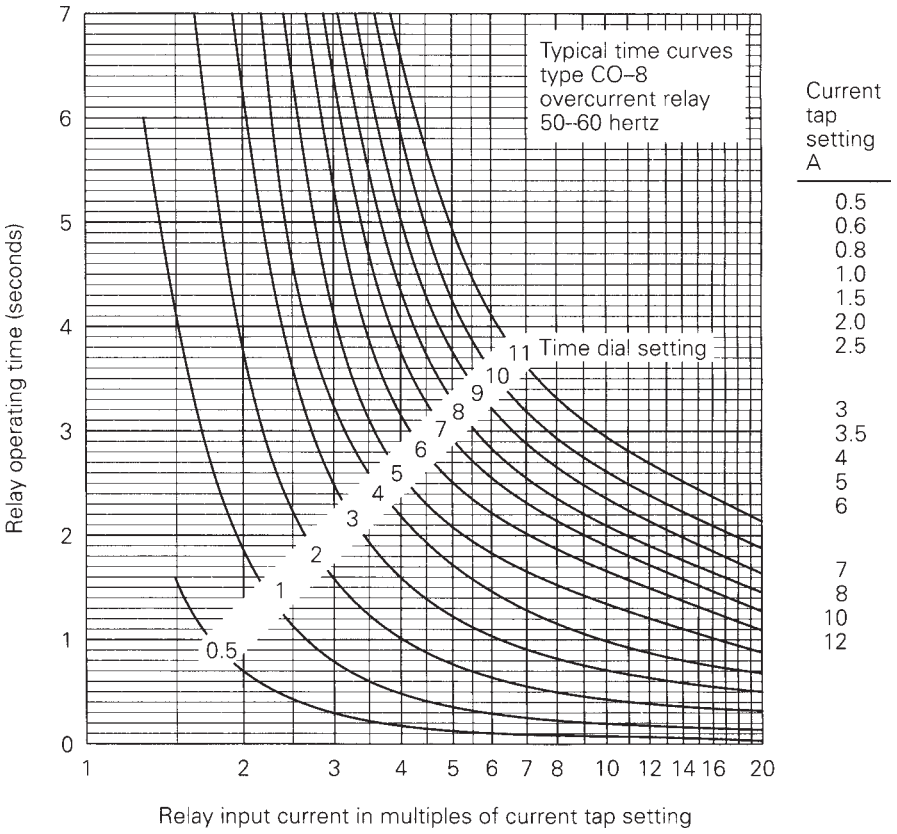
FIGURE 11.12

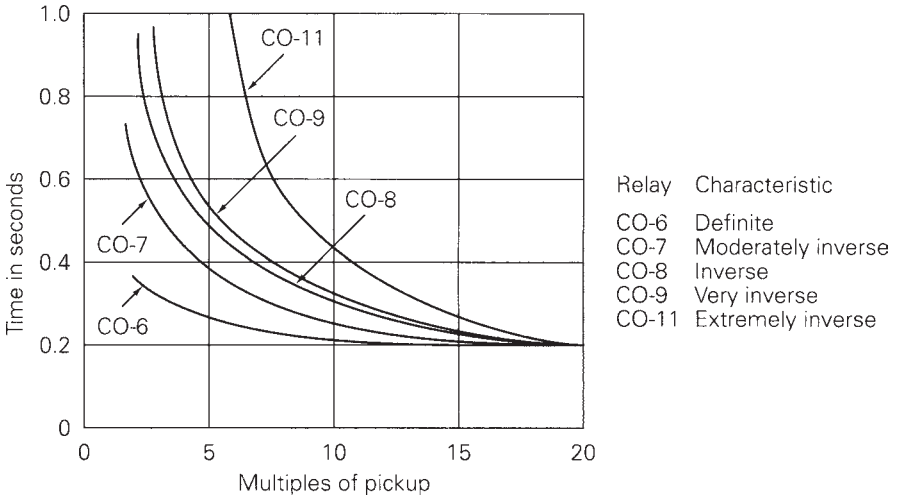
CO-8 time-delay overcurrent relay characteristics (Courtesy of Westinghouse Electric Corporation.)



relays maintain a relatively constant operating time above 10 times pickup. The inverse (CO-8), very inverse (CO-9), and extremely inverse (CO-11) relays operate respectively faster on higher fault currents.

FIGURE 11.13

Comparison of CO relay characteristics (Courtesy of Westinghouse Electric Corporation.)



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EXAMPLE 11.3

Operating time for a CO-8 time-delay overcurrent relay

The CO-8 relay with a current tap setting of 6 amperes and a time-dial setting of 1 is used with the 100:5 CT in Example 11.1. Determine the relay operating time for each case.

SOLUTION

a. From Example 11.1(a),

$I' = 5\ A \qquad \frac{I'}{I_p} = \frac{5}{6} = 0.83$

The relay does not operate. It remains in the blocking position.

b. $I' = 8\ A \qquad \frac{I'}{I_p} = \frac{8}{6} = 1.33$

Using curve 1 in Figure 11.12, $t_{operating} = 6$ seconds.

c. $I' = 15\ A \qquad \frac{I'}{I_p} = \frac{15}{6} = 2.5$

From curve 1, $t_{operating} = 1.2$ seconds.

Figure 11.14 illustrates the operating principle of an electromechanical time-delay overcurrent relay. The ac input current to the relay operating coil sets up a magnetic field that is perpendicular to a conducting aluminum disc. The disc can rotate and is restrained by a spiral spring. Current is induced in the disc, interacts with the magnetic field, and produces a torque. If the input current exceeds the pickup current, the disc rotates through an angle θ to close the relay contacts. The larger the input current, the larger is the torque and the faster the contact closing. After the current is removed or reduced below the pickup, the spring provides reset of the contacts.

A solid-state relay panel between older-style electromechanical relays is shown in Figure 11.15.

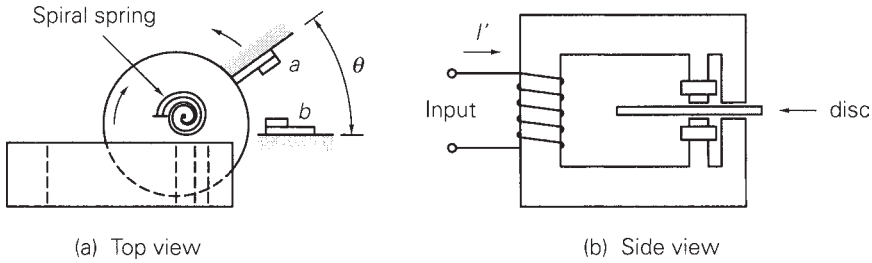

FIGURE 11.14
Electromechanical time-delay overcurrent relay—induction disc type

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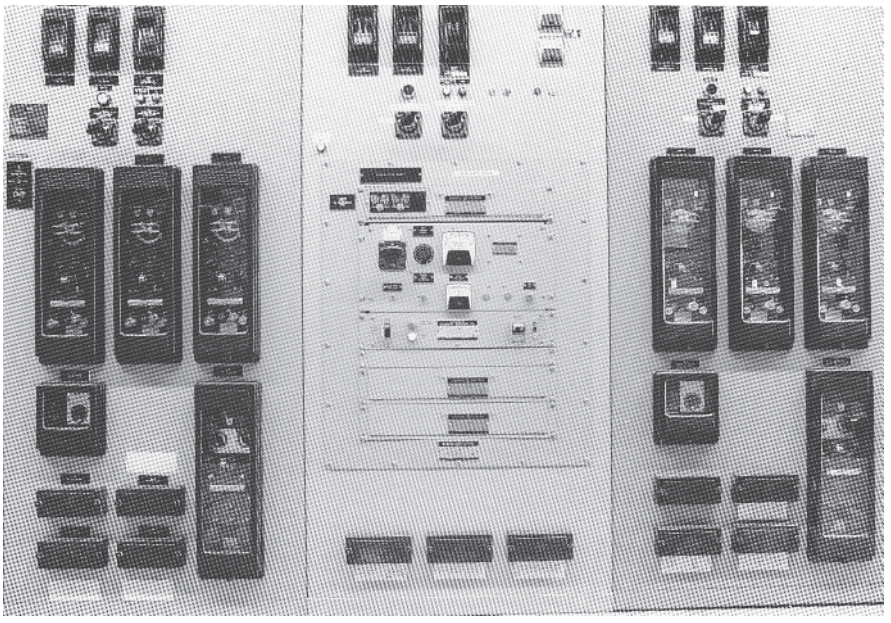

FIGURE 11.15

Solid-state relay panel (center) for a 345-kV transmission line, with electromechanical relays on each side, at Electric Junction Substation, Naperville, Illinois (Courtesy of Commonwealth Edison, an Exelon Company.)

11.4 RADIAL SYSTEM PROTECTION

Many radial systems are protected by time-delay overcurrent relays. Adjustable time delays can be selected such that the breaker closest to the fault opens, while other upstream breakers with larger time delays remain closed. That is, the relays can be coordinated to operate in sequence so as to interrupt minimum load during faults. Successful relay coordination is obtained when fault currents are much larger than normal load currents. Also, coordination of overcurrent relays usually limits the maximum number of breakers in a radial system to five or less; otherwise, the relay closest to the source may have an excessive time delay.

Consider a fault at P_1 to the right of breaker B3 for the radial system of Figure 11.16. For this fault, breaker B3 should open while B2 (and B1) remains closed. Under these conditions, only load L3 is interrupted. Select a longer time delay for the relay at B2, so that B3 operates first. Thus, for any fault to the right of B3, B3 provides primary protection. Only if B3 fails to open will B2 open, after a time delay, thus providing backup protection.

Similarly, consider a fault at P_2 between B2 and B3. B2 should open while B1 remains closed. Under these conditions, loads L2 and L3 are interrupted. Since the fault is closer to the source, the fault current is larger than for the previous fault considered. B2, set to open for the previous, smaller fault current after a time delay, will open more rapidly for this fault. Also selected is the B1 relay with a longer time delay than B2 so that B2 opens first. Thus, B2 provides primary protection for faults between B2 and B3, as well as backup protection for faults to the

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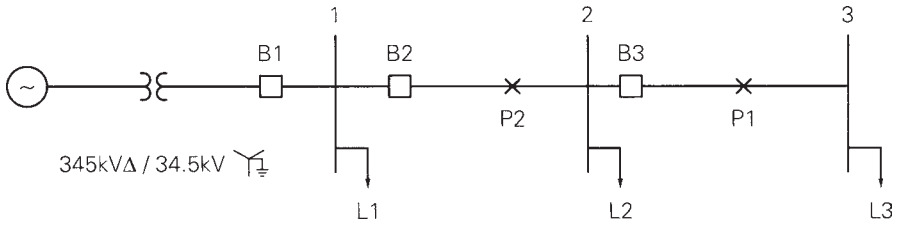

FIGURE 11.16
Single-line diagram of a 34.5-kV radial system

right of B3. Similarly, B1 provides primary protection for faults between B1 and B2, as well as backup protection for further downstream faults.

The *coordination time interval* is the time interval between the primary and remote backup protective devices. It is the difference between the time that the backup relaying operates and the time that circuit breakers clear the fault under primary relaying. Precise determination of relay operating times is complicated by several factors, including CT error, dc offset component of fault current, and relay overtravel. Therefore, typical coordination time intervals from 0.2 to 0.5 seconds are selected to account for these factors in most practical applications.

EXAMPLE 11.4

Coordinating time-delay overcurrent relays in a radial system

Data for the 60-Hz radial system of Figure 11.16 are given in Tables 11.3, 11.4, and 11.5. Select current tap settings (TSs) and time-dial settings (TDSs)

Bus	S MVA	Lagging p.f.
1	11.0	0.95
2	4.0	0.95
3	6.0	0.95

TABLE 11.3
Maximum loads—Example 11.4

Bus	Maximum Fault Current (Bolted Three-Phase) A	Minimum Fault Current (L–G or L–L) A
1	3000	2200
2	2000	1500
3	1000	700

TABLE 11.4
Symmetrical fault currents—Example 11.4

(*Continued*)

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Breaker	Breaker Operating Time	CT Ratio	Relay
B1	5 cycles	400:5	CO-8
B2	5 cycles	200:5	CO-8
B3	5 cycles	200:5	CO-8

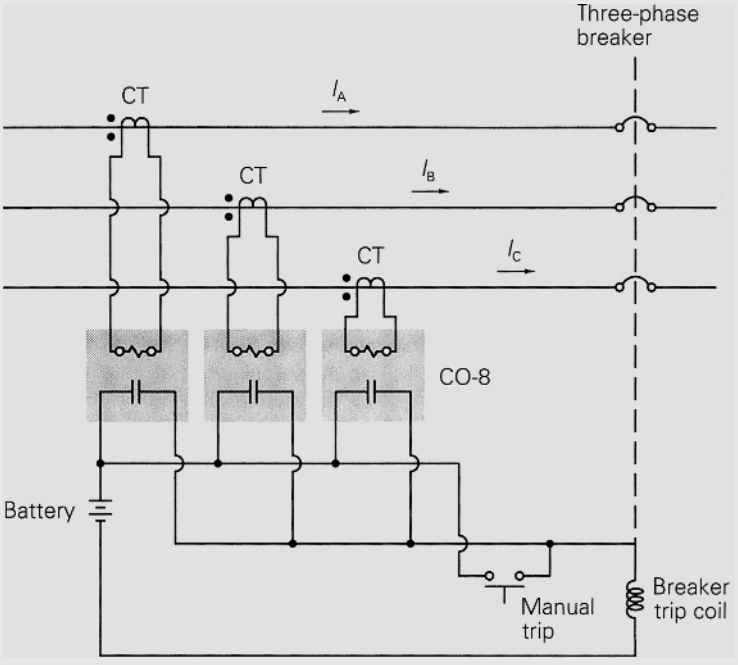
TABLE 11.5

Breaker, CT, and relay data—Example 11.4

to protect the system from faults. Assume three CO-8 relays for each breaker, one for each phase, with a 0.3-second coordination time interval. The relays for each breaker are connected as shown in Figure 11.17, so that all three phases of the breaker open when a fault is detected on any one phase. Assume a 34.5-kV (line-to-line) voltage at all buses during normal operation. Also, future load growth is included in Table 11.3 such that maximum loads over the operating life of the radial system are given in this table.

FIGURE 11.17

Relay connections to trip all three phases



SOLUTION

First, select TSs such that the relays do not operate for maximum load currents. Starting at B3, the primary and secondary CT currents for maximum load L3 are

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$$I_{L3} = \frac{S_{L3}}{V_3\sqrt{3}} = \frac{6 \times 10^6}{(34.5 \times 10^3)\sqrt{3}} = 100.4 \text{ A}$$

$$I'_{L3} = \frac{100.4}{(200/5)} = 2.51\text{A}$$

From Figure 11.12, relay a 3-A TS selected for B3, which is the lowest TS above 2.51 A.

Note that $|S_{L2} + S_{L3}| = |S_{L2}| + |S_{L3}|$ because the load power factors are identical. Thus, at B2, the primary and secondary CT currents for maximum load are

$$I_{L2} = \frac{S_{L2} + S_{L3}}{V_2\sqrt{3}} = \frac{(4 + 6) \times 10^6}{(34.5 \times 10^3)\sqrt{3}} = 167.3 \text{ A}$$

$$I'_{L2} = \frac{167.3}{(200/5)} = 4.18 \text{ A}$$

From Figure 11.12, select for the B2 relay a 5-A TS, the lowest TS above 4.18 A. At B1,

$$I_{L1} = \frac{S_{L1} + S_{L2} + S_{L3}}{V_1\sqrt{3}} = \frac{(11 + 4 + 6) \times 10^6}{(34.5 \times 10^3)\sqrt{3}} = 351.4 \text{ A}$$

$$I'_{L1} = \frac{351.4}{(400/5)} = 4.39 \text{ A}$$

Select a 5-A TS for the B1 relay.

Next, select the TDSs. First, coordinate for the maximum fault currents in Table 11.4, checking coordination for minimum fault currents later. Starting at B3, the largest fault current through B3 is 2000 A, which occurs for the three-phase fault at bus 2 (just to the right of B3). Neglecting CT saturation, the fault-to-pickup current ratio at B3 for this fault is

$$\frac{I'_{3\text{Fault}}}{\text{TS3}} = \frac{2000/(200/5)}{3} = 16.7$$

To clear faults as rapidly as possible, select a 1/2 TDS for the B3 relay. Then, from the 1/2 TDS curve in Figure 11.12, the relay operating time is T3 = 0.05 seconds. Adding the breaker operating time (5 cycles = 0.083 s), primary protection clears this fault in $\text{T3} + \text{T}_{\text{breaker}} = 0.05 + 0.083 = 0.133$ seconds.

For this same fault, the fault-to-pickup current ratio at B2 is

$$\frac{I'_{2\text{Fault}}}{\text{TS2}} = \frac{2000/(200/5)}{5} = 10.0$$

Adding the B3 relay operating time (T3 = 0.05 s), breaker operating time (0.083 s), and 0.3 s coordination time interval, a B2 relay operating time is desirable:

$$\text{T2} = \text{T3} + \text{T}_{\text{breaker}} + \text{T}_{\text{coordination}} = 0.05 + 0.083 + 0.3 \approx 0.43 \text{ s}$$

(*Continued*)

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Breaker	Relay	TS	TDS
B1	CO-8	5	3
B2	CO-8	5	2
B3	CO-8	3	1/2

TABLE 11.6

Solution—Example 11.4

From Figure 11.12, select TDS2 = 2.

Next select the TDS at Bl. The largest fault current through B2 is 3000 A, for a three-phase fault at bus 1 (just to the right of B2). The fault-to-pickup current ratio at B2 for this fault is

$$\frac{I'_{2Fault}}{TS2} = \frac{3000/(200/5)}{5} = 15.0$$

From the 2 TDS curve in Figure 11.12, T2 = 0.38 s. For this same fault,

$$\frac{I'_{1Fault}}{TS1} = \frac{3000/(400/5)}{5} = 7.5$$

$$T1 = T2 + T_{breaker} + T_{coordination} = 0.38 + 0.083 + 0.3 \approx 0.76 \text{ s}$$

From Figure 11.12, select TDS1 = 3. The relay settings are shown in Table 11.6. Note that for reliable relay operation, the fault-to-pickup current ratios with minimum fault currents should be greater than 2. Coordination for minimum fault currents listed in Table 11.4 is evaluated in Problem 11.11.

Note that separate relays are used for each phase in Example 11.4, and therefore, these relays will operate for three-phase as well as line-to-line, single line-to-ground, and double line-to-ground faults. However, in many cases, single line-to-ground fault currents are much lower than three-phase fault currents, especially for distribution feeders with high zero-sequence impedances. In these cases, a separate ground relay with a lower current tap setting than the phase relays is used. The ground relay is connected to operate on zero-sequence current from three of the phase CTs connected in parallel or from a CT in the grounded neutral.

11.5 RECLOSERS, FUSES, AND SECTIONALIZERS

Automatic circuit reclosers are commonly used for distribution circuit protection. A *recloser* is a self-controlled device for automatically interrupting and reclosing an ac circuit with a preset sequence of openings and reclosures. Unlike circuit breakers, which have separate relays to control breaker opening and reclosing, reclosers have built-in controls. More than 80% of faults on overhead distribution

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circuits are temporary, caused by tree limb contact, by animal interference, by wind bringing bare conductors in contact, or by lightning. The automatic tripping-reclosing sequence of reclosers clears these temporary faults and restores service with only momentary outages, thereby significantly improving customer service. A disadvantage of reclosers is the increased hazard when a circuit is physically contacted by people—for example, in the case of a broken conductor at ground level that remains energized. Also, reclosing should be locked out during live-line maintenance by utility personnel.

Figure 11.18 shows a common protection scheme for radial distribution circuits utilizing fuses, reclosers, and time-delay overcurrent relays. Data for the 13.8-kV feeder in this figure is given in Table 11.7. There are three load taps protected by fuses. The recloser ahead of the fuses is set to open and reclose for faults up to and beyond the fuses. For temporary faults, the recloser can be set for one or more instantaneous or time-delayed trips and reclosures in order to clear the faults and restore service. If faults persist, the fuses operate for faults to their right (downstream), or the recloser opens after a time delay and locks out for faults between the recloser and fuses. Separate time-delay overcurrent phase and ground relays open the substation breaker after multiple reclosures of the recloser.

Coordination of the fuses, recloser, and time-delay overcurrent relays is shown via the time-current curves in Figure 11.19. Type T (slow) fuses are selected because their time-current characteristics coordinate well with reclosers. The fuses are selected on the basis of maximum loads served from the taps. A 65-T fuse is selected for the bus 1 tap, which has a 60-A maximum load current, and 100 T is selected for the bus 2 and 3 taps, which have 95-A maximum load currents. The fuses should also have a rated voltage larger than the maximum bus voltage and an interrupting current rating larger than the maximum asymmetrical fault current at the fuse location. Type T fuses with voltage ratings of 15 kV and interrupting current ratings of 10 kA and higher are standard.

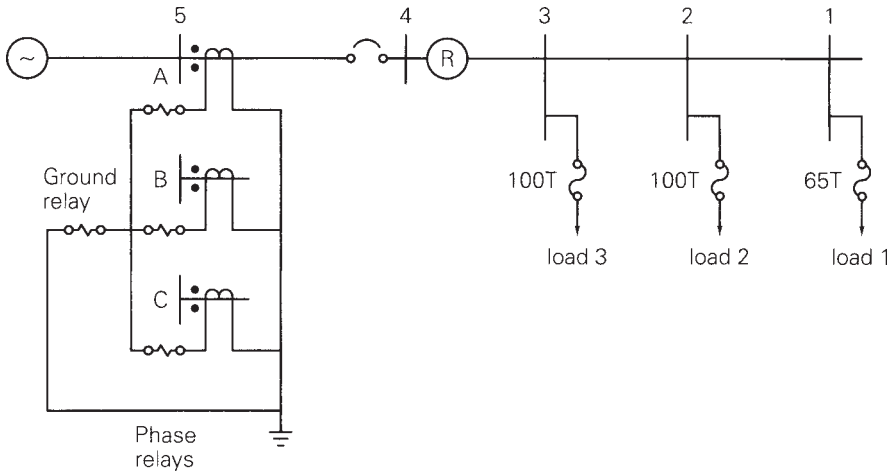

FIGURE 11.18

Single-line diagram of a 13.8-kV radial distribution feeder with fuse/recloser/relay protection

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Bus	Maximum Load Current A	3ϕ Fault Current A	IL-G Fault Current A
1	60	1000	850
2	95	1500	1300
3	95	2000	1700
4	250	3000	2600
5	250	4000	4050

TABLE 11.7

Data for Figure 11.18

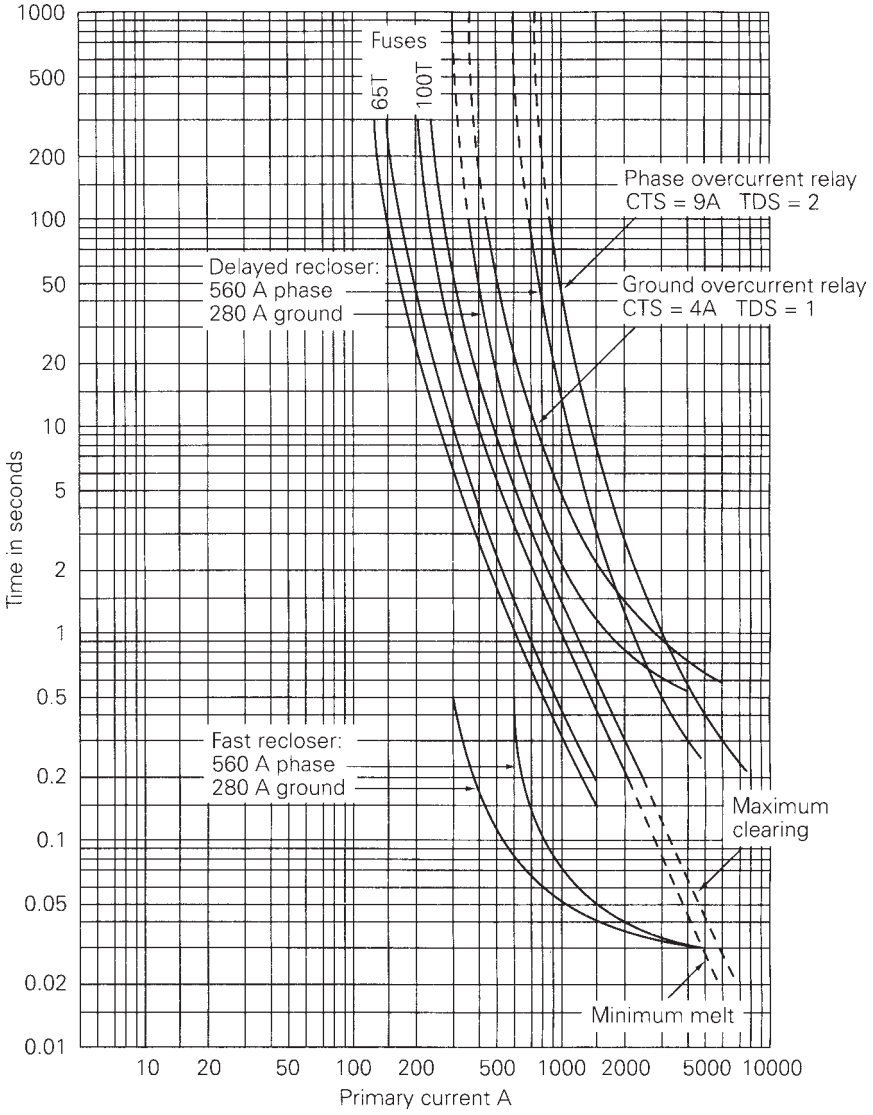

FIGURE 11.19

Time-current curves for the radial distribution circuit of Figure 11.18

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Standard reclosers have normal (continuous) current ratings up to 1250 A, with voltage ratings up to 38 kV and maximum interrupting currents up to 16 kA. A minimum trip rating of 200–250% of maximum load current is typically selected for the phases in order to override cold load pickup with a safety factor. The minimum trip rating of the ground unit is typically set at maximum load and should be higher than the maximum allowable load unbalance. For the recloser in Figure 11.18 that carries a 250-A maximum load, minimum trip ratings of 560 A for each phase and 280 A for the ground unit are selected.

A popular operation sequence for reclosers is one or two fast operations without intentional time delay, followed by two delayed operations. The fast operation(s) allows temporary faults to self-clear, whereas the delayed operations allow downstream fuses to clear permanent faults. Note that the time-current curves of the fast recloser lie below the fuse curves in Figure 11.19 such that the recloser opens before the fuses melt. The fuse curves lie below the delayed recloser curves such that the fuses clear before the recloser opens. The recloser is typically programmed to reclose $\frac{1}{2}$ s after the first fast trip, 2 s after the second fast trip, and 5–10 s after a delayed trip.

Time-delay overcurrent relays with an extremely inverse characteristic coordinate with both reclosers and type T fuses. A 300:5 CT ratio is selected to give a secondary current of $250 \times (5/300) = 4.17$ A at maximum load. Relay settings are selected to allow the recloser to operate effectively to clear faults before relay operation. A current tap setting of 9 A is selected for the CO-11 phase relays so that minimum pickup exceeds twice the maximum load. A time-dial setting of 2 is selected so that the delayed recloser trips at least 0.2 s before the relay. The ground relay is set with a current tap setting of 4 A and a time-dial setting of 1.

EXAMPLE 11.5

Fuse/recloser coordination

For the system of Figure 11.18, describe the operating sequence of the protective devices for the following faults: (a) a self-clearing, temporary, three-phase fault on the load side of tap 2; and (b) a permanent three-phase fault on the load side of tap 2.

SOLUTION

a. From Table 11.7, the three-phase fault current at bus 2 is 1500 A. From Figure 11.19, the 560-A fast recloser opens 0.05 s after the 1500-A fault current occurs, and then recloses $\frac{1}{2}$ s later. Assuming the fault has self-cleared, normal service is restored. During the 0.05-s fault duration, the 100 T fuse does not melt.

b. For a permanent fault, the fast recloser opens after 0.05 s, recloses $\frac{1}{2}$ s later into the permanent fault, opens again after $\frac{1}{2}$ s, and recloses into the fault a second time after a 2-s delay. Then the 560-A delayed recloser opens 3 seconds later. During this interval, the 100-T fuse clears the fault. The delayed recloser then recloses 5 to 10 s later, restoring service to loads 1 and 3.

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Besides reclosers, fuses, and overcurrent relays, sectionalizing switches, denoted "sectionalizers," are commonly used in distribution systems. A sectionalizer is a device that automatically isolates faulted sections of a distribution circuit after an upstream circuit breaker or recloser has interrupted the fault current. Since sectionalizers have no capacity to interrupt fault currents, they must be used with a backup device, such as a recloser, that has the capacity to interrupt fault currents. Sectionalizers count the number of recloser operations during fault conditions. After a pre-set number of recloser openings, and while the recloser is open, the sectionalizer opens and isolates the faulted section of the line. This permits the recloser to close and reenergize the unfaulted section of the line before the recloser locks out. If a temporary fault is cleared during recloser operations, before the sectionalizer reaches its pre-set count, the operating mechanism of the sectionalizer is reset [20].

Since sectionalizers do not have time-current operating characteristics, coordination between sectionalizers and reclosers does not require any analysis of time-current curves. Instead, coordination is based on the number of recloser operations. The recloser can be set for any combination of fast or delayed-trip operations. The sectionalizer can then be set for one trip operation less than the total number of recloser trip operations. If a permanent fault occurs in a radial circuit downstream of the sectionalizer, the sectionalizer opens and isolates the fault after the next-to-last recloser trip. For example, the sectionalizer opens on the third operation if the recloser is programmed for two fast and two delayed-trip operations.

If two sectionalizers are installed in series in a radial circuit, the downstream sectionalizer can be adjusted for a smaller count. A fault results in recloser operation and the start of the counters in both sectionalizers. For a permanent fault, the downstream sectionalizer with the smaller count trips first. If the fault is located between the sectionalizers, the upstream sectionalizer will trip next. If the fault is upstream of both sectionalizers, the recloser will trip on its final operation and lock out.

11.6 DIRECTIONAL RELAYS

Directional relays are designed to operate for fault currents in only one direction. Consider the directional relay D in Figure 11.20, which is required to operate only for faults to the right of the CT. Since the line impedance is mostly reactive, a fault at P_1 to the right of the CT will have a fault current I from bus 1 to bus 2 that lags the bus voltage V by an angle of almost 90°. This fault current is said to be in the forward direction. On the other hand, a fault at P_2, to the left of the CT, will have a fault current I that leads V by almost 90°. This fault current is said to be in the reverse direction.

The directional relay has two inputs: the reference voltage $V = V\angle 0°$ and current $I = I\angle\phi$. The relay trip and block regions, shown in Figure 11.21, can be described by

$$-180° < (\phi - \phi_1) < 0° \quad \text{(Trip)} \qquad (11.6.1)$$
$$\text{Otherwise (Block)}$$

where ϕ is the angle of the current with respect to the voltage and ϕ_1, typically 2° to 8°, defines the boundary between the trip and block regions.

The contacts of the overcurrent relay OC and the directional relay D are connected in series in Figure 11.20 so that the breaker trip coil is energized only

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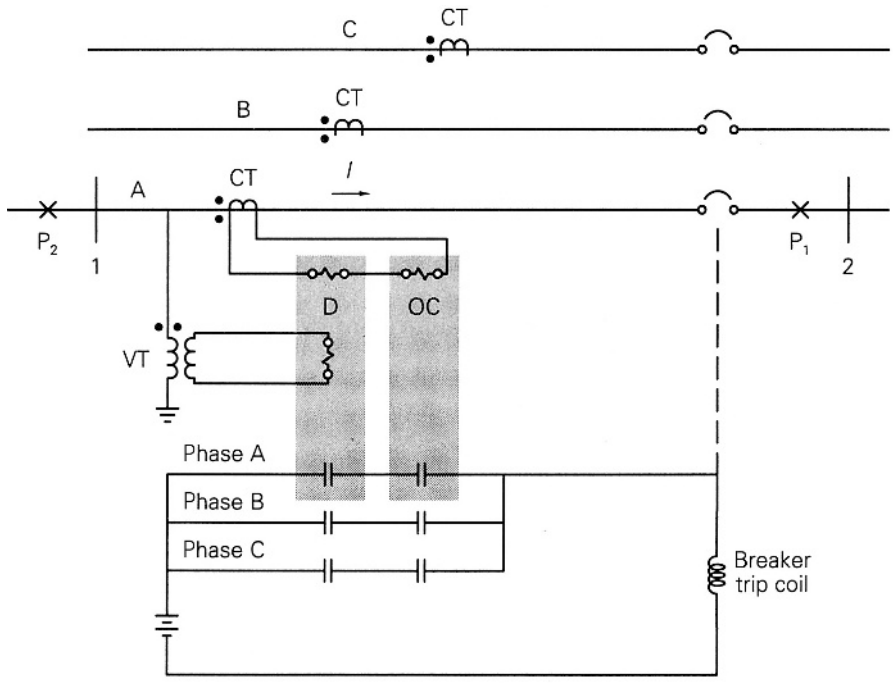

FIGURE 11.20

Directional relay in series with over-current relay (only phase A is shown)

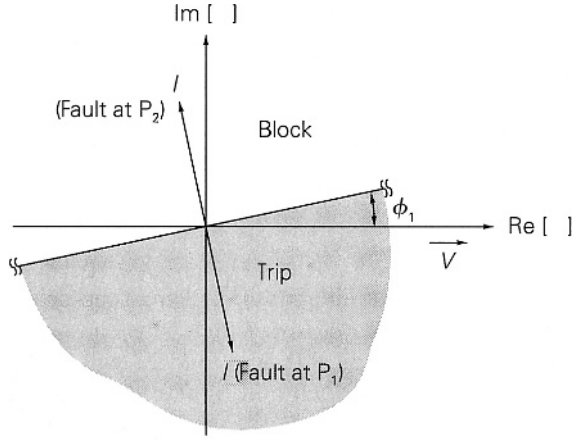

FIGURE 11.21

Directional relay block and trip regions in the complex plane

when the CT secondary current (1) exceeds the OC relay pickup value, and (2) is in the forward tripping direction.

Although construction details differ, the operating principle of an electromechanical directional relay is similar to that of an electromechanical-type watt-hour meter. There are two input coils, a voltage coil and a current coil, both located on a stator, and there is a rotating disc element. Suppose that the reference voltage is passed through a phase-shifting element to obtain $V_1 = \mathrm{V}\underline{/\phi_1 - 90°}$. If V_1 and $I = \mathrm{I}\underline{/\phi}$ are applied to a watt-hour meter, the torque on the rotating element is

$$\mathrm{T} = \mathrm{kVI}\cos(\phi_1 - \phi - 90°) = \mathrm{kVI}\sin(\phi_1 - \phi) \qquad (11.6.2)$$

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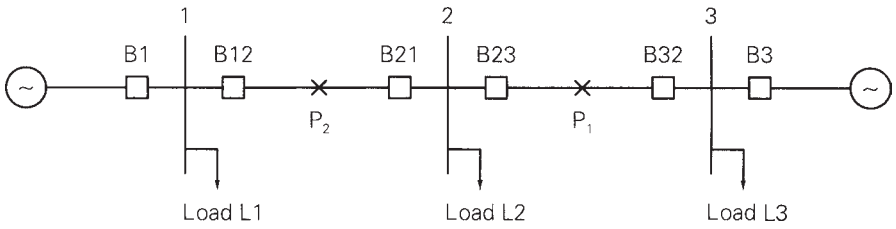
Note that for faults in the forward direction, the current lags the voltage, and the angle $(\phi_1 - \phi)$ in (11.6.2) is close to 90°. This results in maximum positive torque on the rotating disc, which would cause the relay contacts to close. On the other hand, for faults in the reverse direction, the current leads the voltage, and $(\phi_1 - \phi)$ is close to −90°. This results in maximum negative torque tending to rotate the disc element in the backward direction. Backward motion can be restrained by mechanical stops.

11.7 PROTECTION OF A TWO-SOURCE SYSTEM WITH DIRECTIONAL RELAYS

It becomes difficult and in some cases impossible to coordinate overcurrent relays when there are two or more sources at different locations. Consider the system with two sources shown in Figure 11.22. Suppose there is a fault at P_1. B23 and B32 should clear the fault so that service to the three loads continues without interruption. Using time-delay overcurrent relays, set B23 faster than B21. Now consider a fault at P_2 instead. Breaker B23 will open faster than B21, and load L2 is disconnected. When a fault can be fed from both the left and right, overcurrent relays cannot be coordinated. However, directional relays can be used to overcome this problem.

FIGURE 11.22
System with two sources



EXAMPLE 11.6

Two-source system protection with directional and time-delay overcurrent relays

Explain how directional and time-delay overcurrent relays can be used to protect the system in Figure 11.22. Which relays should be coordinated for a fault (a) at P_1, (b) at P_2? (c) Is the system also protected against bus faults?

SOLUTION

Breakers B12, B21, B23, and B32 should respond only to faults on their "forward" or "line" sides. Directional overcurrent relays connected as shown in Figure 11.20 can be used for these breakers. Overcurrent relays alone can be used for breakers B1 and B3, which do not need to be directional.

a. For a fault at P_1, the B21 relay would not operate; B12 should coordinate with B23 so that B23 trips before B12 (and B1). Also, B3 should coordinate with B32.

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b. For a fault at P_2, B23 would not operate; B32 should coordinate with B21 so that B21 trips before B32 (and B3). Also, B1 should coordinate with B12.
c. Yes, the directional overcurrent relays also protect the system against bus faults. If the fault is at bus 2, relays at B21 and B23 will not operate, but B12 and B32 will operate to clear the fault. B1 and B21 will operate to clear a fault at bus 1. B3 and B23 will clear a fault at bus 3.

11.8 ZONES OF PROTECTION

Protection of simple systems has been discussed so far. For more general power system configurations, a fundamental concept is the division of a system into protective zones [1]. If a fault occurs anywhere within a zone, action is taken to isolate that zone from the rest of the system. Zones are defined for:

generators,
transformers,
buses,
transmission and distribution lines, and
motors.

Figure 11.23 illustrates the protective zone concept. Each zone is defined by a closed, dashed line. Zone 1, for example, contains a generator and connecting leads to a transformer. In some cases a zone may contain more than one component. For example, zone 3 contains a generator-transformer unit and connecting leads to a bus, and zone 10 contains a transformer and a line. Protective zones have the following characteristics:

(1) Zones are overlapped.
(2) Circuit breakers are located in the overlap regions.

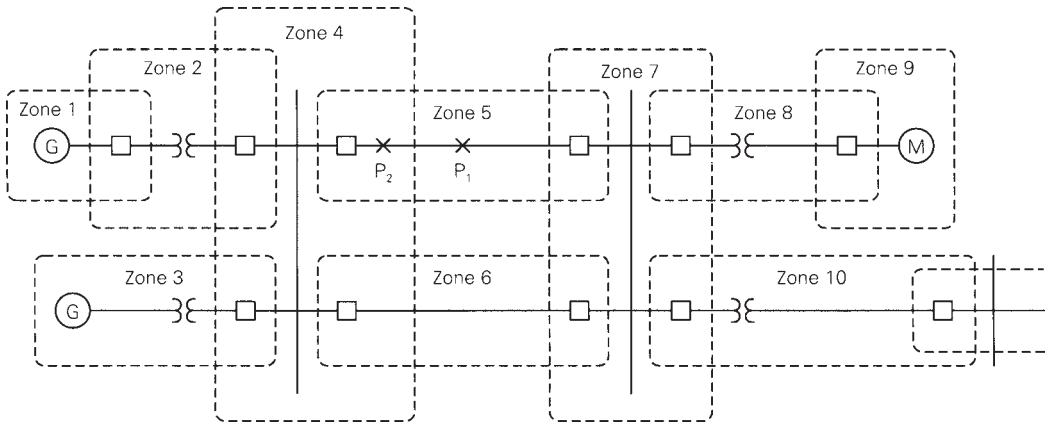

FIGURE 11.23

Power system protective zones

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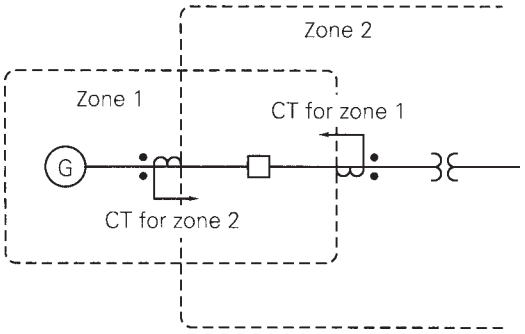
FIGURE 11.24

Overlapping protection around a circuit breaker



(3) For a fault anywhere in a zone, all circuit breakers in that zone open to isolate the fault.

Neighboring zones are overlapped to avoid the possibility of unprotected areas. Without overlap the small area between two neighboring zones would not be located in any zone and thus would not be protected.

Since isolation during faults is done by circuit breakers, they should be inserted between equipment in a zone and each connection to the system. That is, breakers should be inserted in each overlap region. As such, they identify the boundaries of protective zones. For example, zone 5 in Figure 11.23 is connected to zones 4 and 7. Therefore, a circuit breaker is located in the overlap region between zones 5 and 4, as well as between zones 5 and 7.

If a fault occurs anywhere within a zone, action is taken to open all breakers in that zone. For example, if a fault occurs at P_1 on the line in zone 5, then the two breakers in zone 5 should open. If a fault occurs at P_2 within the overlap region of zones 4 and 5, then all five breakers in zones 4 and 5 should open. Clearly, if a fault occurs within an overlap region, two zones are isolated and a larger part of the system is lost from service. To minimize this possibility, overlap regions are kept as small as possible.

Overlap is accomplished by having two sets of instrument transformers and relays for each circuit breaker. For example, the breaker in Figure 11.24 shows two CTs, one for zone 1 and one for zone 2. Overlap is achieved by the order of the arrangement: first the equipment in the zone, second the breaker, and then the CT for that zone.

EXAMPLE 11.7

Zones of protection

Draw the protective zones for the power system shown in Figure 11.25. Which circuit breakers should open for a fault at P_1? at P_2?

SOLUTION

Noting that circuit breakers identify zone boundaries, protective zones are drawn with dashed lines as shown in Figure 11.26. For a fault at P_1, located in zone 5, breakers B24 and B42 should open. For a fault at P_2, located in the overlap region of zones 4 and 5, breakers B24, B42, B21, and B23 should open.

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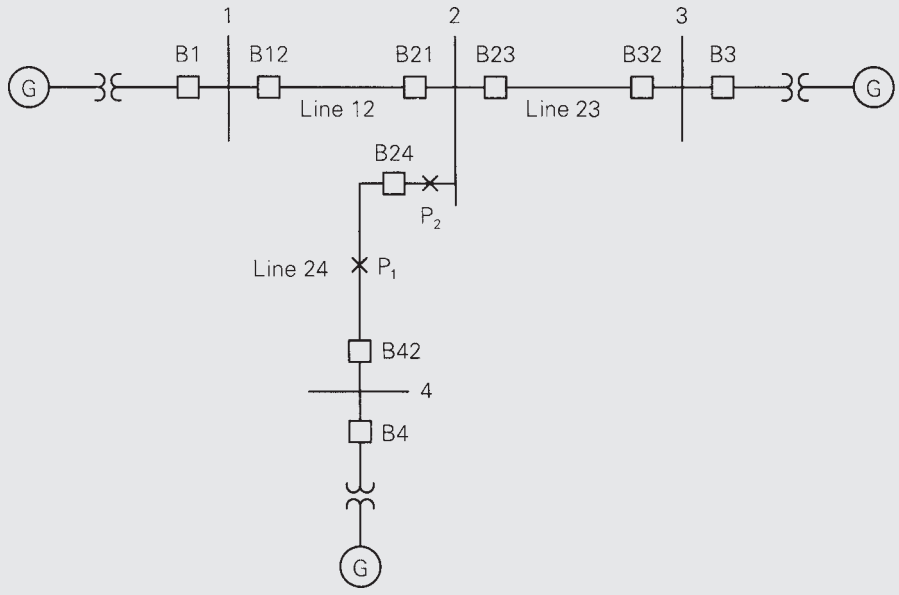
FIGURE 11.25

Power system for Example 11.7



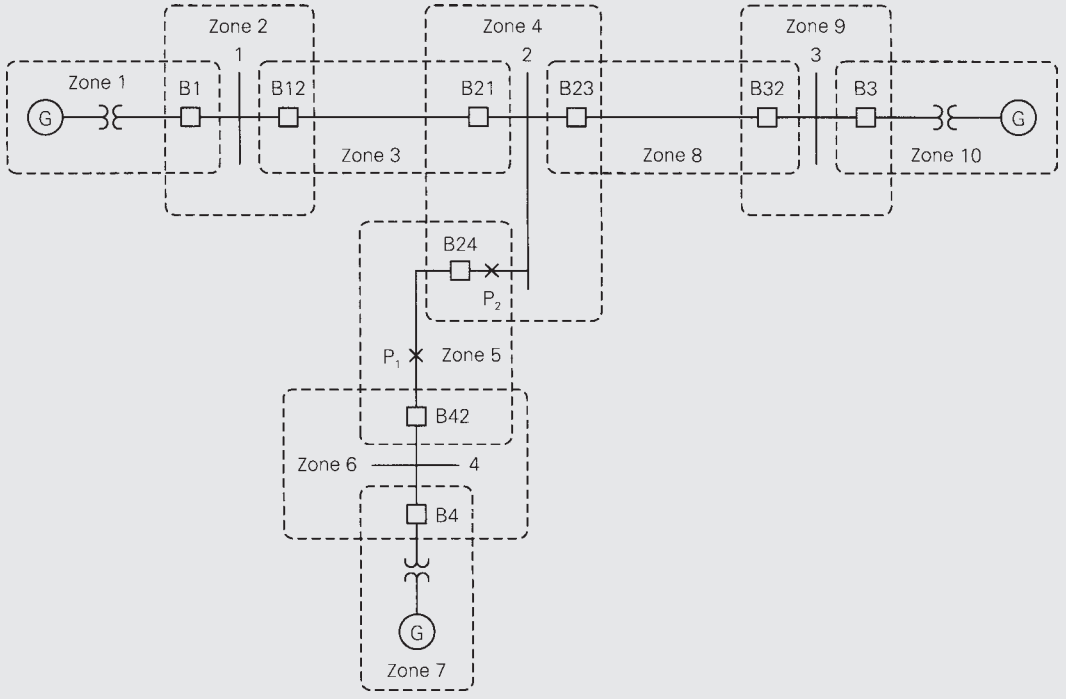


FIGURE 11.26

Protective zones for Example 11.7

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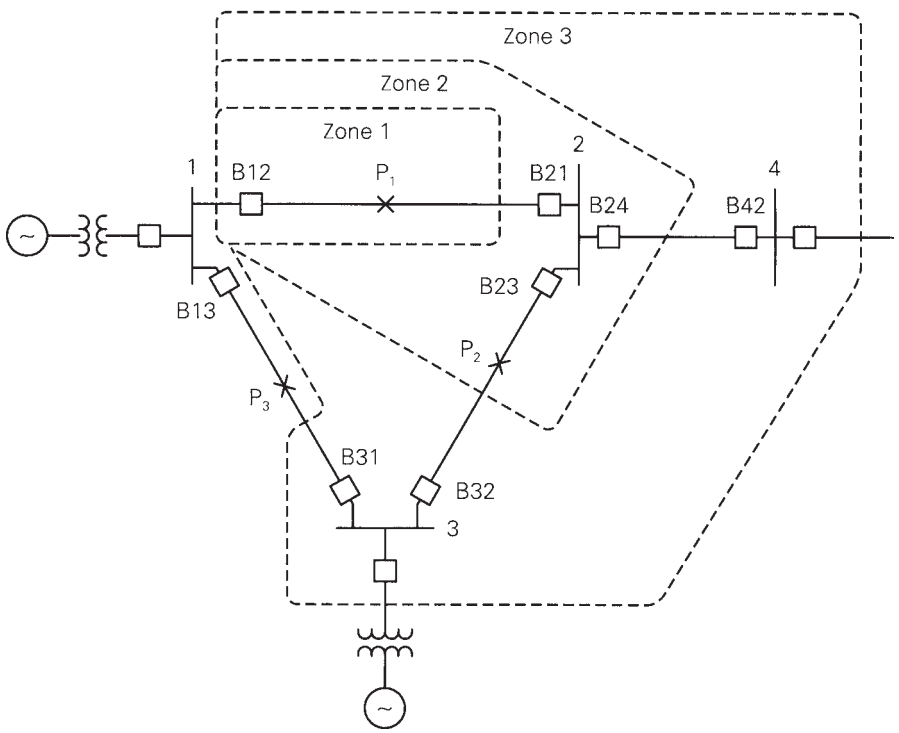
11.9 LINE PROTECTION WITH IMPEDANCE (DISTANCE) RELAYS

Coordinating time-delay overcurrent relays can also be difficult for some radial systems. If there are too many radial lines and buses, the time delay for the breaker closest to the source becomes excessive.

Also, directional overcurrent relays are difficult to coordinate in transmission loops with multiple sources. Consider the use of these relays for the transmission loop shown in Figure 11.27. For a fault at P_1, set the B21 relay to operate faster than the B32 relay. For a fault at P_2, B32 should be faster than B13. And for a fault at P_3, set B13 faster than B21. Proper coordination, which depends on the magnitudes of the fault currents, becomes a tedious process. Furthermore, when consideration is given to various lines or sources out of service, coordination becomes extremely difficult.

To overcome these problems, relays that respond to a voltage-to-current ratio can be used. Note that during a three-phase fault, current increases while bus voltages close to the fault decrease. If, for example, current increases by a factor of 5 while voltage decreases by a factor of 2, then the voltage-to-current ratio decreases by a factor of 10. That is, the voltage-to-current ratio is more sensitive to faults than current alone. A relay that operates on the basis of

FIGURE 11.27
345-kV transmission loop



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voltage-to-current ratio is called an *impedance* relay. It is also called a *distance* relay or a *ratio* relay.

Impedance relay block and trip regions are shown in Figure 11.28, where the impedance Z is defined as the voltage-to-current ratio at the relay location. The relay trips for $|Z| < |Z_r|$, where Z_r is an adjustable relay setting. The impedance circle that defines the border between the block and trip regions passes through Z_r.

A straight line called the *line impedance locus* is shown for the impedance relay in Figure 11.28. This locus is a plot of positive sequence line impedances, predominantly reactive, as viewed between the relay location and various points along the line. The relay setting Z_r is a point in the R-X plane through which the impedance circle that defines the trip-block boundary must pass.

Consider an impedance relay for breaker B12 in Figure 11.27, for which $Z = V_1/I_{12}$. During normal operation, load currents are usually much smaller than fault currents, and the ratio Z has a large magnitude (and some arbitrary phase angle). Therefore, Z will lie outside the circle of Figure 11.28, and the relay will not trip during normal operation.

During a three-phase fault at P_1, however, Z appears to relay B12 to be the line impedance from the B12 relay to the fault. If $|Z_r|$ in Figure 11.28 is set to be larger than the magnitude of this impedance, then the B12 relay will trip. Also, during a three-phase fault at P_3, Z appears to relay B12 to be the negative of the line impedance from the relay to the fault. If $|Z_r|$ is larger than the magnitude of this impedance, the B12 relay will trip. Thus, the impedance relay of Figure 11.28 is not directional; a fault to the left or right of the relay can cause a trip.

Two ways to include directional capability with an impedance relay are shown in Figure 11.29. In Figure 11.29(a), an impedance relay with directional restraint is obtained by including a directional relay in series with an impedance relay, just as was done previously with an overcurrent relay. In Figure 11.29(b),

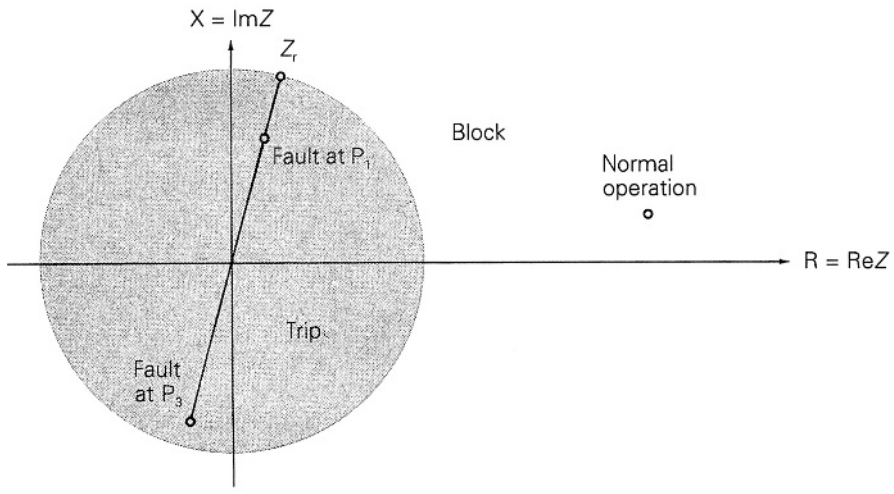

FIGURE 11.28

Impedance relay block and trip regions

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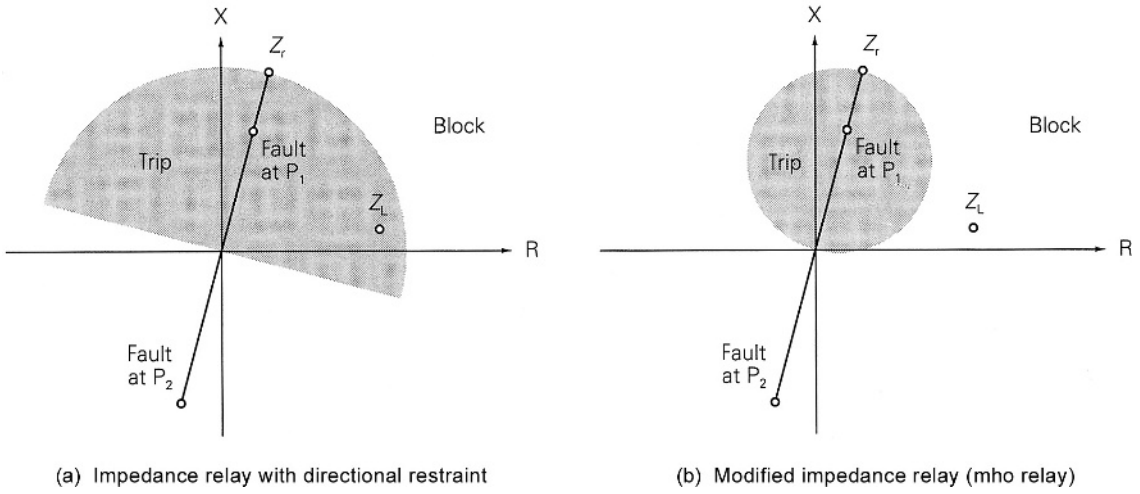

FIGURE 11.29

Impedance relays with directional capability

a modified impedance relay is obtained by offsetting the center of the impedance circle from the origin. This modified impedance relay is sometimes called an *mho* relay. If either of these relays is used at B12 in Figure 11.27, a fault at P_1 will result in a trip decision, but a fault at P_3 will result in a block decision.

Note that the radius of the impedance circle for the modified impedance relay is half of the corresponding radius for the impedance relay with directional restraint. The modified impedance relay has the advantage of better selectivity for high power factor loads. For example, the high power factor load Z_L lies outside the trip region of Figure 11.29(b) but inside the trip region of Figure 11.29(a).

The *reach* of an impedance relay denotes how far down the line the relay detects faults. For example, an 80% reach means that the relay will detect any (solid three-phase) fault between the relay and 80% of the line length. This explains the term *distance* relay.

It is common practice to use three directional impedance relays per phase, with increasing reaches and longer time delays. For example, Figure 11.27 shows three protection zones for B12. The zone 1 relay is typically set for an 80% reach and instantaneous operation in order to provide primary protection for line 1–2. The zone 2 relay is set for about 120% reach, extending beyond bus 2, with a typical time delay of 0.2 to 0.3 seconds. The zone 2 relay provides backup protection for faults on line 1–2 as well as remote backup for faults on line 2–3 or 2–4 in zone 2.

Note that in the case of a fault on line 2–3, it is desirable to set the B23 relay to trip, not the B12 relay. Since the impedance seen by B12 for faults near bus 2, either on line 1–2 or line 2–3, is essentially the same, do not set the B12 zone 1 relay for 100% reach. Instead, select an 80% reach to avoid instantaneous operation

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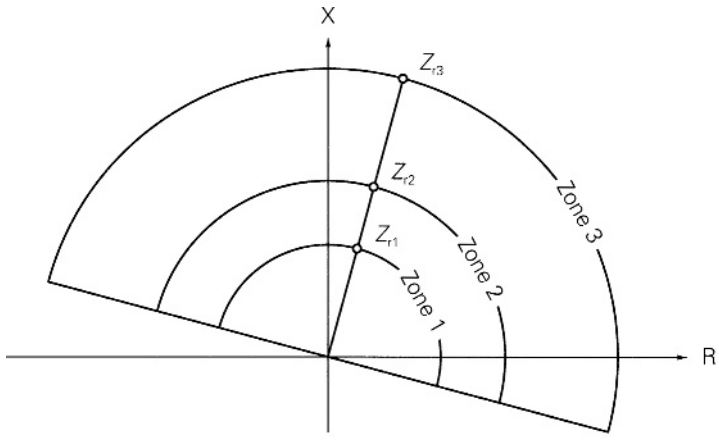
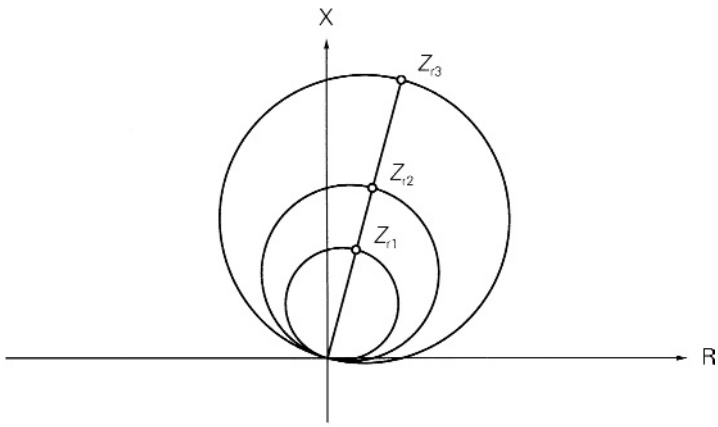
of B12 for a fault on line 2–3 near bus 2. For example, if there is a fault at P_2 on line 2–3, B23 should trip instantaneously; if it fails, B12 will trip after a time delay. Other faults at or near bus 2 also cause tripping of the B12 zone 2 relay after a time delay.

Reach for the zone 3 B12 relay is typically set to extend beyond buses 3 and 4 in Figure 11.27 in order to provide remote backup for neighboring lines. As such, the zone 3 reach is set for 100% of line 1–2 plus 120% of either line 2–3 or 2–4, whichever is longer, with an even larger time delay, typically one second.

Typical block and trip regions are shown in Figure 11.30 for both types of three-zone, directional impedance relays. Relay connections for a three-zone impedance relay with directional restraint are shown in Figure 11.31.



(a) Impedance relay with directional restraint



(b) Modified impedance relay (mho relay)

FIGURE 11.30

Three-zone, directional impedance relay

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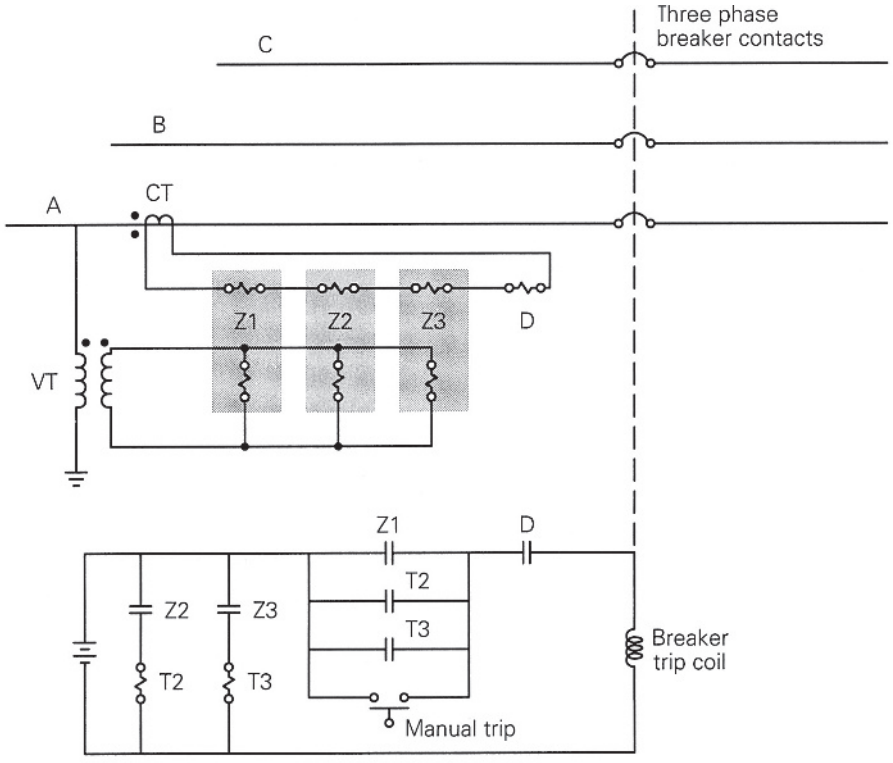
FIGURE 11.31

Relay connections for a three-zone directional impedance relay (only phase A is shown)



EXAMPLE 11.8

Three-zone impedance relay settings

Table 11.8 gives positive-sequence line impedances as well as CT and VT ratios at B12 for the 345-kV system shown in Figure 11.27.

(a) Determine the settings Z_{rl}, Z_{r2}, and Z_{r3} for the B12 three-zone, directional impedance relays connected as shown in Figure 11.31. Consider only solid, three-phase faults.

(b) Maximum current for line 1–2 during emergency loading conditions is 1500 A at a power factor of 0.95 lagging. Verify that B12 does not trip during normal and emergency loadings.

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Line	Positive-Sequence Impedance Ω	
1–2	$8 + j50$	
2–3	$4 + j25$	
2–4	$5.3 + j33$	
1–3	$4.3 + j27$	
Breaker	**CT Ratio**	**VT Ratio**
B12	1500:5	3000:1

TABLE 11.8
Data for Example 11.8

SOLUTION

a. Denoting V_{LN} as the line-to-neutral voltage at bus 1 and I_L as the line current through B12, the primary impedance Z viewed at B12 is

$$Z = \frac{V_{LN}}{I_L}\ \Omega$$

Using the CT and VT ratios given in Table 11.8, the secondary impedance viewed by the B12 impedance relays is

$$Z' = \frac{V_{LN}\Big/\left(\frac{3000}{1}\right)}{I_L\Big/\left(\frac{1500}{5}\right)} = \frac{Z}{10}$$

Setting the B12 zone 1 relay for 80% reach, that is, 80% of the line 1–2 (secondary) impedance:

$Z_{r1} = 0.80(8 + j50)/10 = 0.64 + j4 = 4.05\underline{/80.9°}\ \Omega$ secondary

Setting the B12 zone 2 relay for 120% reach:

$Z_{r2} = 1.2(8 + j50)/10 = 0.96 + j6 = 6.08\underline{/80.9°}\ \Omega$ secondary

From Table 11.8, line 2–4 has a larger impedance than line 2–3. Therefore, set the B12 zone 3 relay for 100% reach of line 1–2 plus 120% reach of line 2–4.

$Z_{r3} = 1.0(8 + j50)/10 + 1.2(5.3 + j33)/10$

$= 1.44 + j8.96 = 9.07\underline{/80.9°}\ \Omega$ secondary

b. The secondary impedance viewed by B12 during emergency loading, using $V_{LN} = 345/\sqrt{3}\underline{/0°} = 199.2\underline{/0°}$ kV and $I_L = 1500\underline{/-\cos^{-1}(0.95)} = 1500\underline{/-18.19°}$ A, is

(*Continued*)

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$$Z' = Z/10 = \left(\frac{199.2 \times 10^3}{1500\underline{/-18.19^\circ}}\right)\Big/10 = 13.28\underline{/18.19^\circ}\ \Omega \quad \text{secondary}$$

Since this impedance exceeds the zone 3 setting of $9.07\underline{/80.9^\circ}\ \Omega$, the impedance during emergency loading lies outside the trip regions of the three-zone, directional impedance relay. Also, lower line loadings during normal operation will result in even larger impedances farther away from the trip regions. B12 will trip during faults but not during normal and emergency loadings.

Remote backup protection of adjacent lines using zone 3 of an impedance relay may be ineffective. In practice, buses have multiple lines of different lengths with sources at their remote ends. Contributions to fault currents from the multiple lines may cause the zone 3 relay to underreach. This "infeed effect" is illustrated in Problem 11.21.

The impedance relays considered so far use line-to-neutral voltages and line currents and are called *ground fault relays*. They respond to three-phase, single line-to-ground, and double line-to-ground faults very effectively. The impedance seen by the relay during unbalanced faults will generally not be the same as seen during three-phase faults and will not be truly proportional to the distance to the fault location. However, the relay can be accurately set for any fault location after computing impedance to the fault using fault currents and voltages. For other fault locations farther away (or closer), the impedance to the fault will increase (or decrease).

Ground fault relays are relatively insensitive to line-to-line faults. Impedance relays that use line-to-line voltages V_{ab}, V_{bc}, V_{ca} and line-current differences $I_a - I_b$, $I_b - I_c$, $I_c - I_a$ are called *phase relays*. Phase relays respond effectively to line-to-line faults and double line-to-ground faults but are relatively insensitive to single line-to-ground faults. Therefore, both phase and ground fault relays need to be used.

11.10 DIFFERENTIAL RELAYS

Differential relays are commonly used to protect generators, buses, and transformers. Figure 11.32 illustrates the basic method of differential relaying for generator protection. The protection of only one phase is shown. The method is repeated for the other two phases. When the relay in any one phase operates, all three phases of the main circuit breaker will open, as well as the generator neutral and field breakers (not shown).

For the case of no internal fault within the generator windings, $I_1 = I_2$, and, assuming identical CTs, $I_1' = I_2'$. For this case, the current in the relay operating coil is zero, and the relay does not operate. On the other hand, for an internal fault such as a phase-to-ground or phase-to-phase short within the generator winding,

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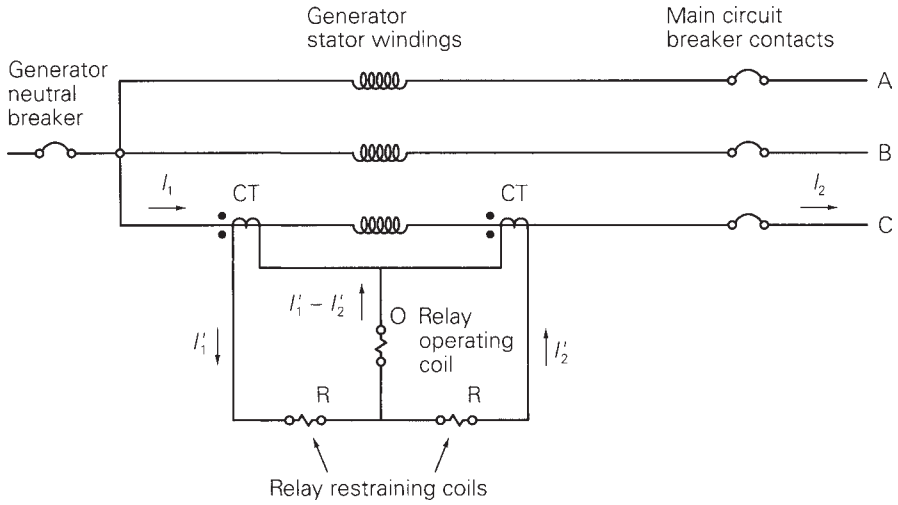

FIGURE 11.32

Differential relaying for generator protection (protection for one phase shown)

$I_1 \neq I_2$, and $I_1' \neq I_2'$. Therefore, a difference current $I_1' - I_2'$ flows in the relay operating coil, which may cause the relay to operate. Since this relay operation depends on a *difference* current, it is called a *differential* relay.

An electromechanical differential relay called a *balance beam* relay is shown in Figure 11.33. The relay contacts close if the downward force on the right side exceeds the downward force on the left side. The electromagnetic force on the right, operating coil is proportional to the square of the operating coil mmf—that is, to $[N_0(I_1' - I_2')]^2$. Similarly, the electromagnetic force on the left, restraining coil is proportional to $[N_r(I_1' + I_2')/2]^2$. The condition for relay operation is then

$$[N_0(I_1' - I_2')]^2 > [N_r(I_1' + I_2')/2]^2 \tag{11.10.1}$$

Taking the square root:

$$|I_1' - I_2'| > k|(I_1' + I_2')/2| \tag{11.10.2}$$

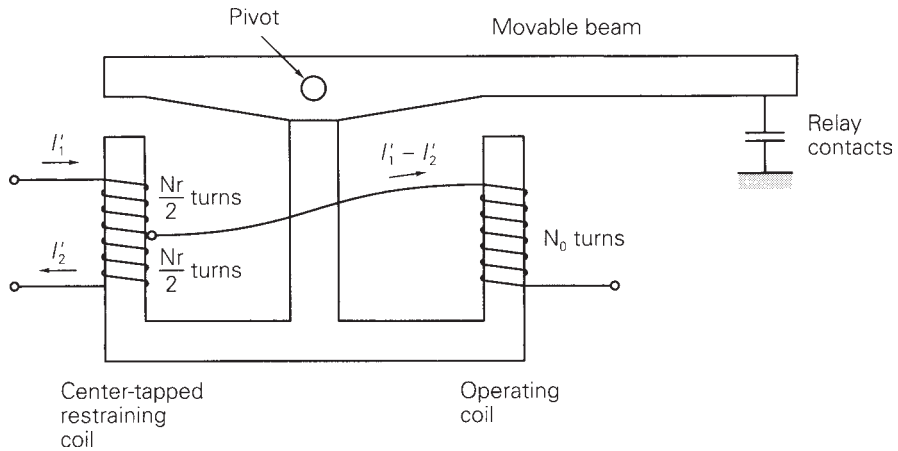

FIGURE 11.33

Balance beam differential relay

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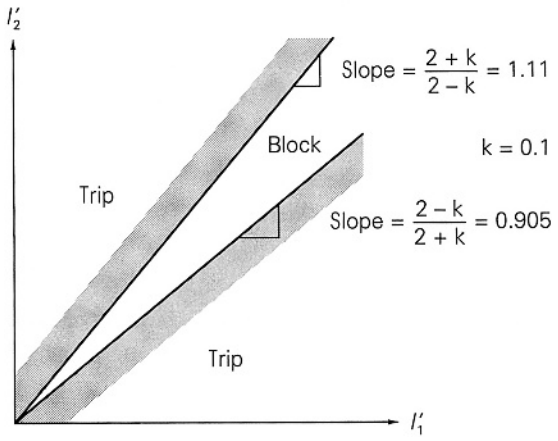
FIGURE 11.34
Differential relay block and trip regions



where

$$k = N_r/N_0 \tag{11.10.3}$$

Assuming I_1' and I_2' are in phase, (11.10.2) is solved to obtain

$$\begin{aligned} I_2' &> \frac{2+k}{2-k} I_1' \quad \text{for } I_2' > I_1' \\ I_2' &< \frac{2-k}{2+k} I_1' \quad \text{for } I_2' < I_1' \end{aligned} \tag{11.10.4}$$

Equation (11.10.4) is plotted in Figure 11.34 to obtain the block and trip regions of the differential relay for k = 0.1. Note that as k increases, the block region becomes larger; that is, the relay becomes less sensitive. In practice, no two CTs are identical, and the differential relay current $I_1' - I_2'$ can become appreciable during external faults, even though $I_1 = I_2$. The balanced beam relay solves this problem without sacrificing sensitivity during normal currents, since the block region increases as the currents increase, as shown in Figure 11.34. Also, the relay can be easily modified to enlarge the block region for very small currents near the origin, in order to avoid false trips at low currents.

Note that differential relaying provides primary zone protection without backup. Coordination with protection in adjacent zones is eliminated, which permits high-speed tripping. Precise relay settings are unnecessary. Also, the need to calculate system fault currents and voltages is avoided.

11.11 BUS PROTECTION WITH DIFFERENTIAL RELAYS

Differential bus protection is illustrated by the single-line diagram of Figure 11.35. In practice, three differential relays are required, one for each phase. Operation of any one relay would cause all of the three-phase circuit breakers connected to the bus to open, thereby isolating the three-phase bus from service.

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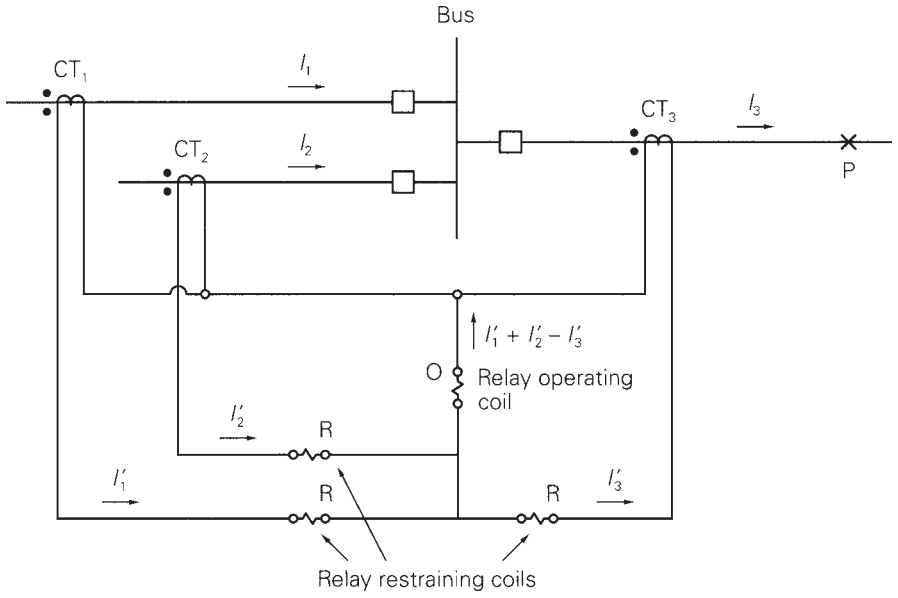

FIGURE 11.35

Single-line diagram of differential bus protection

For the case of no internal fault between the CTs—that is, no bus fault—$I_1 + I_2 = I_3$. Assuming identical CTs, the differential relay current $I'_1 + I'_2 - I'_3$ equals zero, and the relay does not operate. However, if there is a bus fault, the differential current $I'_1 + I'_2 - I'_3$, which is not zero, flows in the operating coil to operate the relay. Use of the restraining coils overcomes the problem of nonidentical CTs.

A problem with differential bus protection can result from different levels of fault currents and varying amounts of CT saturation. For example, consider an external fault at point P in Figure 11.35. Each of the CT_1 and CT_2 primaries carries part of the fault current, but the CT_3 primary carries the sum $I_3 = I_1 + I_2$. CT_3, energized at a higher level, will have more saturation, such that $I'_3 \neq I'_1 + I'_2$. If the saturation is too high, the differential current in the relay operating coil could result in a false trip. This problem becomes more difficult when there are large numbers of circuits connected to the bus. Various schemes have been developed to overcome this problem [1].

11.12 TRANSFORMER PROTECTION WITH DIFFERENTIAL RELAYS

The protection method used for power transformers depends on the transformer MVA rating. Fuses are often used to protect transformers with small MVA ratings, whereas differential relays are commonly used to protect transformers with ratings larger than 10 MVA.

The differential protection method is illustrated in Figure 11.36 for a single-phase, two-winding transformer. Denoting the turns ratio of the primary and secondary CTs by $1/n_1$ and $1/n_2$, respectively (a CT with 1 primary turn and n secondary turns has a turns ratio a = 1/n), the CT secondary currents are

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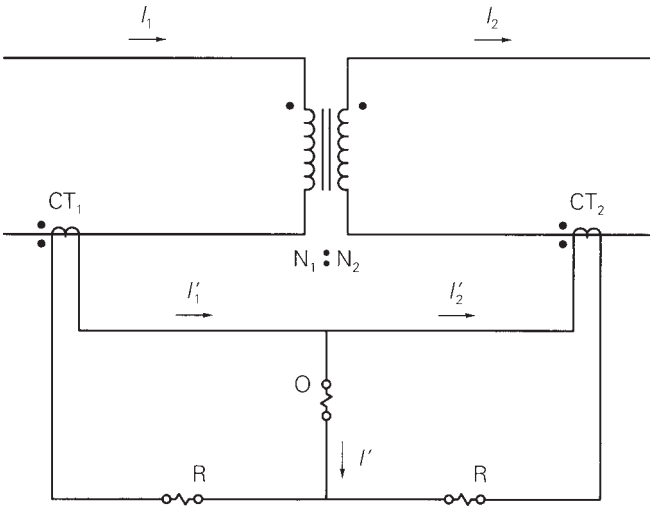
FIGURE 11.36

Differential protection of a single-phase, two-winding transformer



$$I'_1 = \frac{I_1}{n_1} \qquad I'_2 = \frac{I_2}{n_2} \tag{11.12.1}$$

and the current in the relay operating coil is

$$I' = I'_1 - I'_2 = \frac{I_1}{n_1} - \frac{I_2}{n_2} \tag{11.12.2}$$

For the case of no fault between the CTs—that is, no internal transformer fault—the primary and secondary currents for an ideal transformer are related by

$$I_2 = \frac{N_1 I_1}{N_2} \tag{11.12.3}$$

Using (11.12.3) in (11.12.2),

$$I' = \frac{I_1}{n_1}\left(1 - \frac{N_1/N_2}{n_2/n_1}\right) \tag{11.12.4}$$

To prevent the relay from tripping for the case of no internal transformer fault, where (11.12.3) and (11.12.4) are satisfied, the differential relay current I′ must be zero. Therefore, from (11.12.4), select

$$\frac{n_2}{n_1} = \frac{N_1}{N_2} \tag{11.12.5}$$

If an internal transformer fault between the CTs does occur, (11.12.3) is not satisfied and the differential relay current $I' = I'_1 - I'_2$ is not zero. The relay will trip if the operating condition given by (11.10.4) is satisfied. Also, the value of k in (11.10.4) can be selected to control the size of the block region shown in Figure 11.34, thereby controlling relay sensitivity.

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EXAMPLE 11.9

Differential relay protection for a single-phase transformer

A single-phase two-winding, 10-MVA, 80-kV/20-kV transformer has differential relay protection. Select suitable CT ratios. Also, select k such that the relay blocks for up to 25% mismatch between I_1' and I_2'.

SOLUTION

The transformer-rated primary current is

$$I_{1\text{rated}} = \frac{10 \times 10^6}{80 \times 10^3} = 125 \quad \text{A}$$

From Table 11.2, select a 150:5 primary CT ratio to give $I_1' = 125(5/150) = 4.17$ A at rated conditions. Similarly, $I_{2\text{rated}} = 500$ A. Select a 600:5 secondary CT ratio to give $I_2' = 500(5/600) = 4.17$ A and a differential current $I = I_1' - I_2' = 0$ (neglecting magnetizing current) at rated conditions. Also, for a 25% mismatch between I_1' and I_2', select a 1.25 upper slope in Figure 11.34. That is,

$$\frac{2 + k}{2 - k} = 1.25 \qquad k = 0.2222$$

A common problem in differential transformer protection is the mismatch of relay currents that occurs when standard CT ratios are used. If the primary winding in Example 11.9 has a 138-kV instead of 80-kV rating, then $I_{1\text{rated}} = 10 \times 10^6/138 \times 10^3 = 72.46$ A, and a 100:5 primary CT would give $I_1' = 72.46(5/100) = 3.62$ A at rated conditions. This current does not balance $I_2' = 4.17$ A using a 5:600 secondary CT, nor $I_2' = 3.13$ A using a 5:800 secondary CT. The mismatch is about 15%.

One solution to this problem is to use auxiliary CTs, which provide a wide range of turns ratios. A 5:5.76 auxiliary CT connected to the 5:600 secondary CT in the above example would reduce I_2' to $4.17(5/5.76) = 3.62$ A, which does balance I_1'. Unfortunately, auxiliary CTs add their own burden to the main CTs and also increase transformation errors. A better solution is to use tap settings on the relays themselves, which have the same effect as auxiliary CTs. Most transformer differential relays have taps that provide for differences in restraining windings in the order of 2 or 3 to 1.

When a transformer is initially energized, it can draw a large "inrush" current, a transient current that flows in the shunt magnetizing branch and decays after a few cycles to a small steady-state value. Inrush current appears as a differential current since it flows only in the primary winding. If a large inrush current does occur upon transformer energization, a differential relay will see a large differential current and trip out the transformer unless the protection method is modified to detect inrush current.

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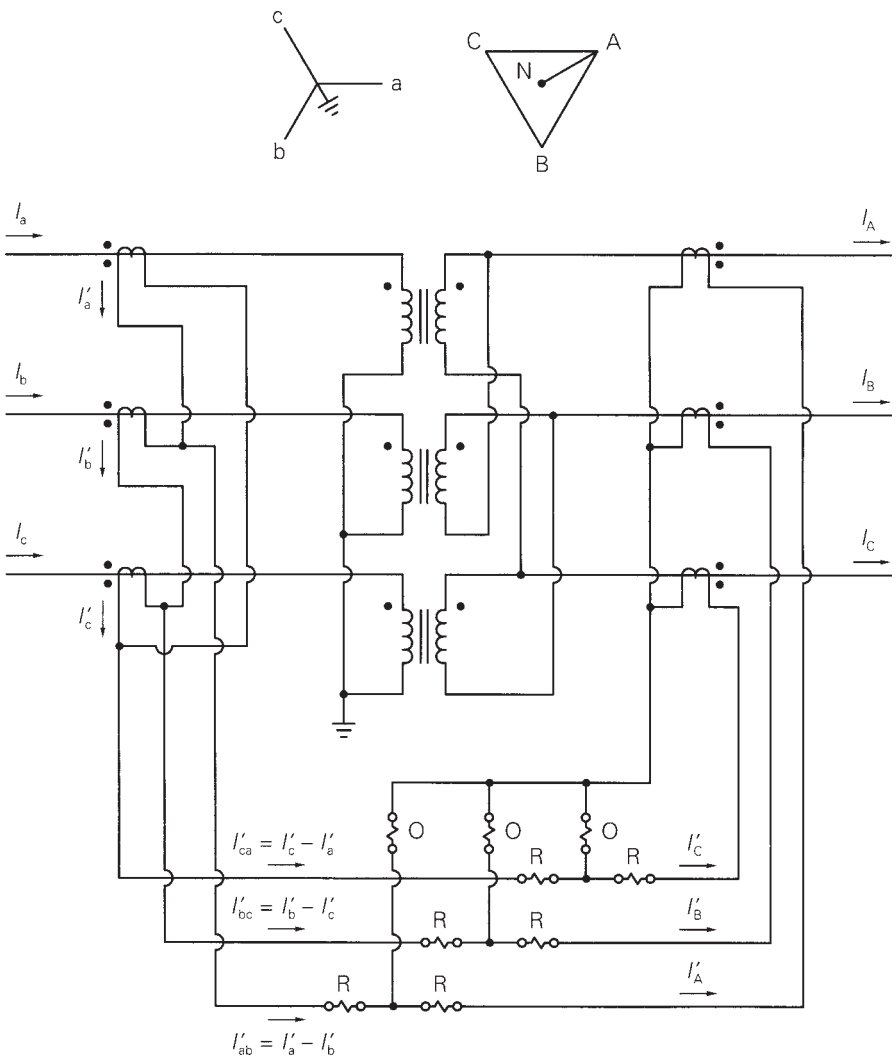
One method to prevent tripping during transformer inrush is based on the fact that inrush current is nonsinusoidal with a large second-harmonic component. A filter can be used to pass fundamental and block harmonic components of the differential current I' to the relay operating coil. Another method is based on the fact that inrush current has a large dc component, which can be used to desensitize the relay. Time-delay relays may also be used to temporarily desensitize the differential relay until the inrush current has decayed to a low value.

Figure 11.37 illustrates differential protection of a three-phase Y–Δ two-winding transformer. Note that a Y–Δ transformer produces 30° phase shifts in the line currents. The CTs must be connected to compensate for the 30° phase

FIGURE 11.37

Differential protection of a three-phase, Y–Δ, two-winding transformer



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shifts such that the CT secondary currents as seen by the relays are in phase. The correct phase-angle relationship is obtained by connecting CTs on the Y side of the transformer in Δ, and CTs on the Δ side in Y.

EXAMPLE 11.10

Differential relay protection for a three-phase transformer

A 30-MVA, 34.5-kV Y/138-kV Δ transformer is protected by differential relays with taps. Select CT ratios, CT connections, and relay tap settings. Also, determine currents in the transformer and in the CTs at rated conditions. Assume that the available relay tap settings are 5:5, 5:5.5, 5:6.6, 5:7.3, 5:8, 5:9, and 5:10, giving relay tap ratios of 1.00, 1.10, 1.32, 1.46, 1.60, 1.80, and 2.00.

SOLUTION

As shown in Figure 11.37, CTs are connected in Δ on the (34.5 kV) Y side of the transformer, and CTs are connected in Y on the (138 kV) Δ side, in order to obtain the correct phasing of the relay currents.

Rated current on the 138-kV side of the transformer is

$$I_{A\ rated} = \frac{30 \times 10^6}{\sqrt{3}(138 \times 10^3)} = 125.51 \quad A$$

Select a 150:5 CT on the 138-kV side to give $I'_A = 125.51(5/150) = 4.184$ A in the 138-kV CT secondaries and in the righthand restraining windings of Figure 11.37.

Next, rated current on the 34.5-kV side of the transformer is

$$I_{a\ rated} = \frac{30 \times 10^6}{\sqrt{3}(34.5 \times 10^3)} = 502.04 \quad A$$

Select a 500:5 CT on the 34.5-kV side to give $I'_a = 502.0(5/500) = 5.02$ A in the 34.5-kV CT secondaries and $I'_{ab} = 5.02\sqrt{3} = 8.696$ A in the lefthand restraining windings of Figure 11.37.

Finally, select relay taps to balance the currents in the restraining windings. The ratio of the currents in the left- to righthand restraining windings is

$$\frac{I'_{ab}}{I'_A} = \frac{8.696}{4.184} = 2.078$$

The closest relay tap ratio is $T'_{AB}/T'_A = 2.0$, corresponding to a relay tap setting of $T'_A{:}T'_{ab} = 5{:}10$. The percentage mismatch for this tap setting is

$$\left|\frac{(I'_A/T'_A) - (I'_{ab}/T'_{ab})}{(I'_{ab}/T'_{ab})}\right| \times 100 = \left|\frac{(4.184/5) - (8.696/10)}{(8.696/10)}\right| \times 100 = 3.77\%$$

(*Continued*)

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This is a good mismatch; since transformer differential relays typically have their block regions adjusted between 20% and 60% (by adjusting k in Figure 11.34), a 3.77% mismatch gives an ample safety margin in the event of CT and relay differences.

For three-phase transformers (Y–Y, Y–Δ, Δ–Y, Δ–Δ), the general rule is to connect CTs on the Y side in Δ and CTs on the Δ side in Y. This arrangement compensates for the 30° phase shifts in Y–Δ or Δ–Y banks. Note also that zero-sequence current cannot enter a Δ side of a transformer or the CTs on that side, and zero-sequence current on a grounded Y side cannot enter the Δ-connected CTs on that side. Therefore, this arrangement also blocks zero-sequence currents in the differential relays during external ground faults. For internal ground faults, however, the relays can operate from the positive- and negative-sequence currents involved in these faults.

Differential protection methods have been modified to handle multi-winding transformers, voltage-regulating transformers, phase-angle regulating transformers, power-rectifier transformers, transformers with special connections (such as zig-zag), and other, special-purpose transformers. Also, other types of relays, such as gas-pressure detectors for liquid-filled transformers, are used.

11.13 PILOT RELAYING

Pilot relaying refers to a type of differential protection that compares the quantities at the terminals via a communication channel rather than by a direct wire interconnection of the relays. Differential protection of generators, buses, and transformers considered in previous sections does not require pilot relaying because each of these devices is at one geographical location where CTs and relays can be directly interconnected. However, differential relaying of transmission lines requires pilot relaying because the terminals are widely separated (often by many kilometers). In actual practice, pilot relaying is typically applied to short transmission lines (up to 80 km) with 69- to 115-kV ratings.

Types of communication channels used for pilot relaying include the following:

1. *Pilot wires*: Separate electrical circuits operating at dc, 50 to 60 Hz, or audio frequencies. These could be owned by the power company or leased from the telephone company.
2. *Power-line carrier*: The transmission line itself is used as the communication circuit, with frequencies between 30 and 300 kHz being transmitted. The communication signals are applied to all three

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phases using an L–C voltage divider and are confined to the line under protection by blocking filters called *line traps* at each end.

3. *Microwave*: A 2- to 12-GHz signal transmitted by line-of-sight paths between terminals using dish antennas.
4. *Fiber optic cable*: Signals transmitted by light modulation through electrically nonconducting cable. This cable eliminates problems due to electrical insulation, inductive coupling from other circuits, and atmospheric disturbances.

Two common fault detection methods are *directional comparison*, where the power flows at the line terminals are compared, and *phase comparison*, where the relative phase angles of the currents at the terminals are compared. Also, the communication channel can either be required for trip operations, which is known as a *transfer trip system*, or not be required for trip operations, known as a *blocking system*. A particular pilot-relaying method is usually identified by specifying the fault-detection method and the channel use. The four basic combinations are directional comparison blocking, directional comparison transfer trip, phase comparison blocking, and phase comparison transfer trip.

Like differential relays, pilot relays provide primary zone protection without backup. Thus, coordination with protection in adjacent zones is eliminated, resulting in high-speed tripping. Precise relay settings are unnecessary. Also, the need to calculate system fault currents and voltages is eliminated.

11.14 NUMERIC RELAYING

Previous sections covered the operating principle of relays built with electromechanical components, including the induction disc time-delay over-current relay, as in Figure 11.14; the directional relay, similar in operation to an electromechanical-type watt-hour meter; and the balance-beam differential relay, as in Figure 11.33. These electromechanical relays, introduced in the early 1900s, have performed well over the years and continue in relatively maintenance-free operation today. Solid-state relays using analog circuits and logic gates, with block-trip regions similar to those of electromechanical relays and with newer types of block/trip regions, have been available since the late 1950s. Such relays, widely used in HV and EHV systems, offer the reliability and ruggedness of their electromechanical counterparts at a competitive price. Beyond solid-state analog relays, a new generation of numeric relays based on digital computer technology has been under development since the 1980s.

The numeric relay (or digital relay) is a protective relay that uses a microprocessor with software for purposes of fault detection. Benefits of numeric relays include the following [16, 17]:

- Compact size
- Programmable
- Low burden

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- Multi-function capability
- Flexibility and reliability
- Sensitivity and speed
- Permits storage of fault data and disturbance records
- Self-checking diagnostics
- Adaptive relaying schemes
- Digital communications capabilities
- Costs

In North America, protective relays are usually referred to by standard device numbers [19]. Table 11.9 lists commonly used relay device numbers.

Device No.	Description	Device No.	Description
1	Master Element	39	Mechanical Conduction Monitor
2	Time Delay Starting or Closing Relay	40	Field Relay
12	Overspeed Device	41	Field Circuit Breaker
13	Synchronous-speed Device	42	Running Circuit Breaker
14	Underspeed Device	43	Manual Transfer or Selector Device
15	Speed - or Frequency-Matching Device	46	Reverse-Phase or Phase-Balance Relay
21	Distance Relay	47	Phase-Sequence Voltage Relay
23	Temperature Control Device	48	Incomplete-Sequence Relay
25	Synchronizing or Synchronism-Check Device	49	Machine or Transformer Thermal Relay
26	Apparatus Thermal Device	50	Instantaneous Overcurrent
27	Undervoltage Relay	51	AC Time Overcurrent Relay
29	Isolating Contactor	52	AC Circuit Breaker
30	Annunciator Relay	53	Exciter or DC Generator Relay
32	Directional Power Relay	54	High-Speed DC Circuit Breaker
36	Polarity or Polarizing Voltage Devices	55	Power Factor Relay
37	Undercurrent or Underpower Relay	56	Field Application Relay
38	Bearing Protective Device	59	Overvoltage Relay

(*Continued*)

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Device No.	Description	Device No.	Description
60	Voltage or Current Balance Relay	79	AC-Reclosing Relay
61	Machine Split Phase Current Balance	81	Frequency Relay
62	Time-Delay Stopping or Opening Relay	83	Automatic Selective Control or Transfer Relay
63	Pressure Switch	84	Operating Mechanism
64	Ground Detector Relay	85	Carrier or Pilot-Wire Receiver Relay
65	Governor	86	Lockout Relay
66	Starts per Hour	87	Differential Protective Relay
67	AC Directional Overcurrent Relay	89	Line Switch
68	Blocking Relay	90	Regulating Device
69	Permissive Control Device	91	Voltage Directional Relay
71	Level Switch	92	Voltage and Power Directional Relay
72	DC Circuit Breaker	94	Tripping or Trip-Free Relay
74	Alarm Relay	95	Reluctance Torque Synchrocheck
75	Position Changing Mechanism	96	Autoloading Relay
76	DC Overcurrent Relay		
78	Phase-Angle Measuring or Out-of-Step Protective Relay		

TABLE 11.9

GE's Protection & Control Reference Guide. www.gegridsolutions.com

As an example of a multifunction numeric relay, Figure 11.38 shows a Basler BE1-11f relay for a feeder protection system, including Figure 11.38(a), the BE1.11-f in a rack-mount case, and Figure 11.38(b), the BE1.11-f relay schematic. As shown in the relay schematic, relay function options include undervoltage (27), overvoltage (59), directional or reverse power (32), instantaneous overcurrent (50), inverse time overcurrent (51), ac directional overcurrent (67), under/over frequency (81), as well as other options. The relay also offers multiple-shot breaker reclosing, breaker- and trip-circuit monitoring, and oscillography and sequential events recording.

Numeric relays can be physically smaller and normally require less panel wiring than electromechanical or solid-state relays. Numeric relays also have the

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advantage that modifications to tripping characteristics, either changes in conventional settings or shaping of entirely new block/trip regions, can be made by updating the software from a remote computer terminal. For example, the relay engineer can reprogram tripping characteristics of field-installed, in-service

FIGURE 11.38a

Basler BE1-11f multifunction relay for a feeder protection system: Rack-mount case. (Basler Electric Company)

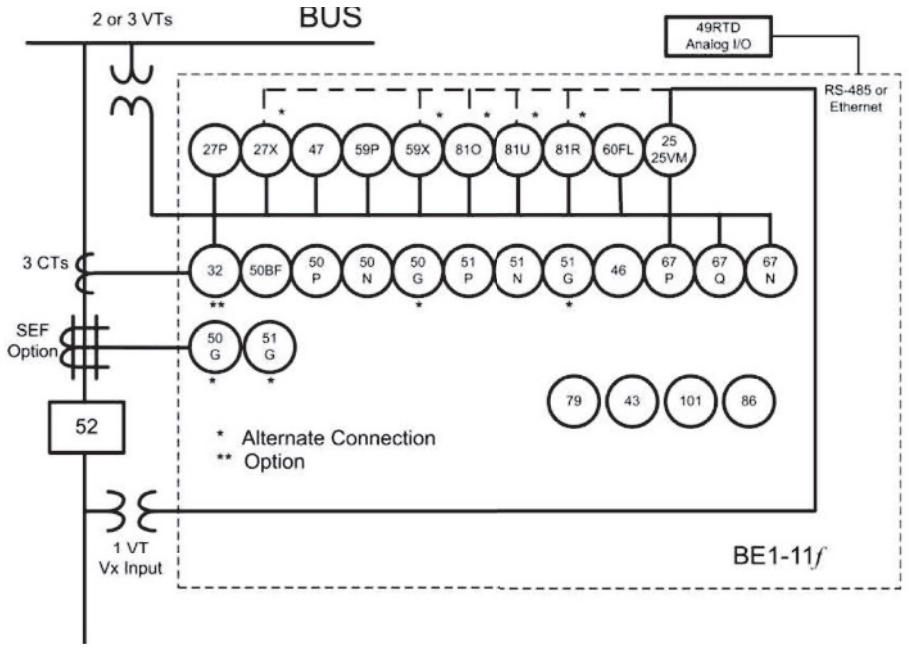

FIGURE 11.38b

Basler BE1-11f multifunction relay for a feeder protection system: Relay schematic. (Basler Electric Company)

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relays without leaving the engineering office. Numeric relays have minimum burdens on instrument transformers. Numeric relays offer multiple functions and multiple characteristics with a wide range of operation, compared to electromechanical and solid-state relays with single functions and narrower ranges of operation. For numeric relays, software can be modified with the same hardware to achieve a flexible variety of protection functions, and significant improvement in reliability is obtained through use of fewer components and fewer interconnections.

Numeric relays offer greater sensitivity with high pickup ratio, and tripping times (not including circuit breaker opening times) of ½ cycle or less. Numeric relays can assist in post-fault analysis by recording fault data, including the type of fault, duration and magnitude of fault, circuit breaker problem, and CT saturation. Numeric relays have the capability to check whether return to a normal condition has been attained and then automatically reset. Numeric relay settings can be changed instantaneously to adapt to varying power-system requirements. Numeric relays offer secure communications capabilities for interface with smart-grid controls, supervisory control, and data acquisition systems. By combining several functions that share hardware in one numeric relay, capital and maintenance costs are reduced compared to electromechanical and solid-state relays.

Numeric relays do have limitations. With digital communications, numeric relays are exposed to the potential risk of hacking, and therefore should be properly managed for cybersecurity. In North America, numeric relays should comply with NERC security requirements. Relay engineers should be aware of common failure modes that can affect multiple elements of protection. For example, the failure of a power supply or an input signal processor could disable a numeric relay that provides many different protection functions. Numeric relays may be exposed to external power-system transients that would not normally affect electromechanical or solid-state relays. And while numeric relays offer multiple functions as well as greater flexibility, sensitivity, and speed, that does not necessarily translate into better protection [16].

An important feature of power system protection is the decentralized, local nature of relays. Except for pilot relaying, each relay receives information from nearby local CTs and VTs and trips only local breakers. Interest in numeric relaying is not directed at replacing local relays by a central computer. Instead, local electromechanical or solid-state relays would be replaced by dedicated, local numeric relays with similar operating principles, such as time-delay overcurrent, impedance, or differential relaying. The central computer would interact with local numeric relays in a supervisory role.

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PROBLEMS

SECTION 11.2

11.1 The primary conductor in Figure 11.2 is one phase of a three-phase transmission line operating at 345 kV, 700 MVA, 0.95 power factor lagging. The CT ratio is 1200:5, and the VT ratio is 3000:1. Determine the CT secondary current I′ and the VT secondary voltage V′. Assume zero CT error.

11.2 A CO-8 relay with a current tap setting of 5 amperes is used with the 100:5 CT in Example 11.1. The CT secondary current I′ is the input to the relay operating coil. The CO-8 relay burden is shown in the following table for various relay input currents.

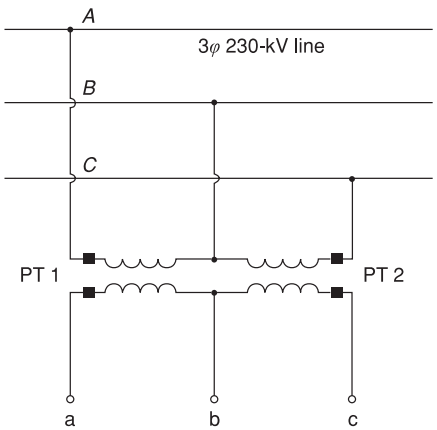
CO-8 relay input current I′, A	5	8	10	13	15
CO-8 relay burden Z_B, Ω	0.5	0.8	1.0	1.3	1.5

Primary current and CT error are computed in Example 11.1 for the 5-, 8-, and 15-A relay input currents. Compute the primary current and CT error for (a) I′ = 10 A and Z_B = 1.0 Ω, and for (b) I′ = 13 A and Z_B = 1.3 Ω. (c) Plot I′ versus I for the above five values of I′. (d) For reliable relay operation, the fault-to-pickup current ratio with minimum fault current should be greater than two. Determine the minimum fault current for application of this CT and relay with 5-A tap setting.

11.3 An overcurrent relay set to operate at 10 A is connected to the CT in Figure 11.8 with a 500:5 CT ratio. Determine the minimum primary fault current that the relay will detect if the burden Z_B is (a) 1.0 Ω, (b) 4.0 Ω, and (c) 5.0 Ω.

11.4 Given the open-delta VT connection shown in Figure 11.39, both VTs having a voltage rating of 240 kV:120 V, the voltages are specified as $V_{AB} = 230\angle 0°$, $V_{BC} = 230\angle -120°$, and $V_{CA} = 230\angle 120°$ kV. Determine V_{ab}, V_{bc}, and V_{ca} for the following cases: (a) The dots are as shown in Figure 11.39. (b) The dot near c is moved to b in Figure 11.39.

FIGURE 11.39
Problem 11.4



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11.5 A CT with an excitation curve given in Figure 11.40 has a rated current ratio of 500:5 A and a secondary leakage impedance of $0.1 + j0.5\ \Omega$. Calculate the CT secondary output current and the CT error for the following cases: (a) The impedance of the terminating device is $4.9 + j0.5\ \Omega$ and the primary CT load current is 400 A. (b) The impedance of the terminating device is $4.9 + j0.5\ \Omega$ and the primary CT fault current is 1200 A. (c) The impedance of the terminating device is $14.9 + j1.5\ \Omega$ and the primary CT load current is 400 A. (d) The impedance of the terminating device is $14.9 + j1.5\ \Omega$ and the primary CT fault current is 1200 A.

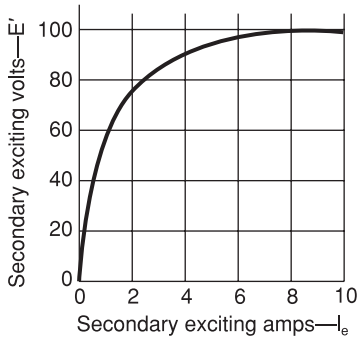

FIGURE 11.40
Problem 11.5

11.6 The CT of Problem 11.5 is utilized in conjunction with a current-sensitive device that will operate at current levels of 8 A or above. Check whether the device will detect the 1300-A fault current for cases (b) and (d) in Problem 11.5.

SECTION 11.3

11.7 The input current to a CO-8 relay is 10 A. Determine the relay operating time for the following current tap settings (TS) and time dial settings (TDS): (a) TS = 1.0, TDS = 1/2; (b) TS = 2.0, TDS = 1.5; (c) TS = 2.0, TDS = 7; (d) TS = 3.0, TDS = 7; and (e) TS = 12.0, TDS = 1.

11.8 The relay in Problem 11.2 has a time-dial setting of 4. Determine the relay operating time if the primary fault current is 400 A.

11.9 An RC circuit used to produce time delay is shown in Figure 11.41. For a step input voltage $v_i(t) = 2u_{-1}(t)$ and $C = 10\ \mu F$, determine T_{delay} for the following cases: (a) $R = 100\ k\Omega$; and (b) $R = 1\ M\Omega$. Sketch the output $v_o(t)$ versus time for cases (a) and (b).

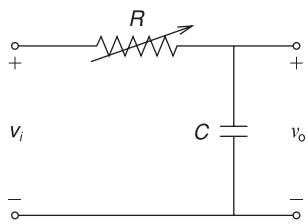

FIGURE 11.41
Problem 11.9

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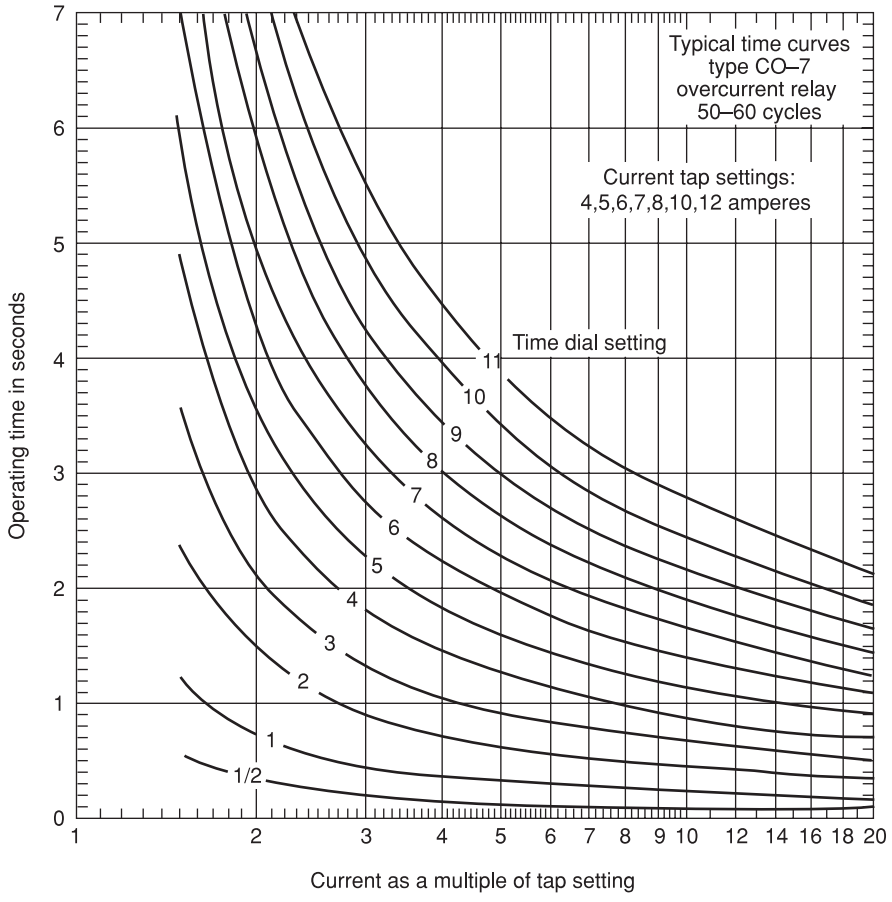
11.10 Reconsider case (b) of Problem 11.5. Let the load impedance $4.9 + j0.5\ \Omega$ be the input impedance to a CO-7 time-delay overcurrent relay. The CO-7 relay characteristic is shown in Figure 11.42. For a tap setting of 4 A and a time dial setting of 2, determine the relay operating time.

FIGURE 11.42

Problems 11.10 and 11.14



SECTION 11.4

11.11 Evaluate relay coordination for the minimum fault currents in Example 11.4. For the selected current tap settings and time dial settings, (a) determine the operating time of relays at B2 and B3 for the 700-A fault current. (b) Determine the operating time of relays at Bl and B2 for the 1500-A fault current. Are the fault-to-pickup current ratios ≥ 2.0 (a requirement for reliable relay operation) in all cases? Are the coordination time intervals ≥ 0.3 seconds in all cases?

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11.12 Repeat Example 11.4 for the following system data. Coordinate the relays for the maximum fault currents.

	Maximum Load		Symmetrical Fault Current	
Bus	MVA	Lagging p.f.	Maximum A	Minimum A
1	9.0	0.95	5000	3750
2	9.0	0.95	3000	2250
3	9.0	0.95	2000	1500

Breaker	Breaker Operating Time	CT Ratio	Relay
B1	5 cycles	600:5	CO-8
B2	5 cycles	400:5	CO-8
B3	5 cycles	200:5	CO-8

11.13 Using the current tap settings and time dial settings that you have selected in Problem 11.12, evaluate relay coordination for the minimum fault currents. Are the fault-to-pickup current ratios $\geqslant 2.0$, and are the coordination time delays $\geqslant 0.3$ seconds in all cases?

11.14 An 11-kV radial system is shown in Figure 11.43. Assuming a CO-7 relay with relay characteristic given in Figure 11.42 and the same power factor for all loads, select relay settings to protect the system.

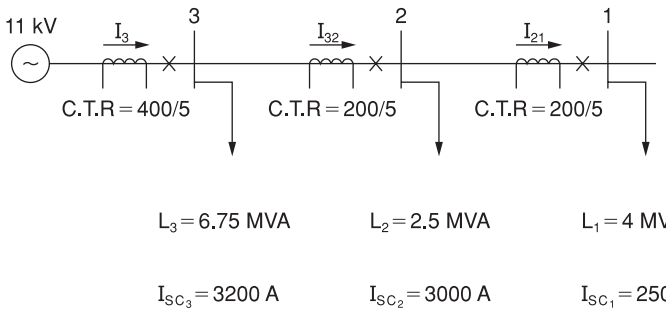

FIGURE 11.43
Problem 11.14

SECTION 11.5

11.15 Rework Example 11.5 for the following faults: (a) a three-phase, permanent fault on the load side of tap 3; (b) a single line-to-ground, permanent fault at bus 4 on the load side of the recloser; and (c) a three-phase, permanent fault at bus 4 on the source side of the recloser.

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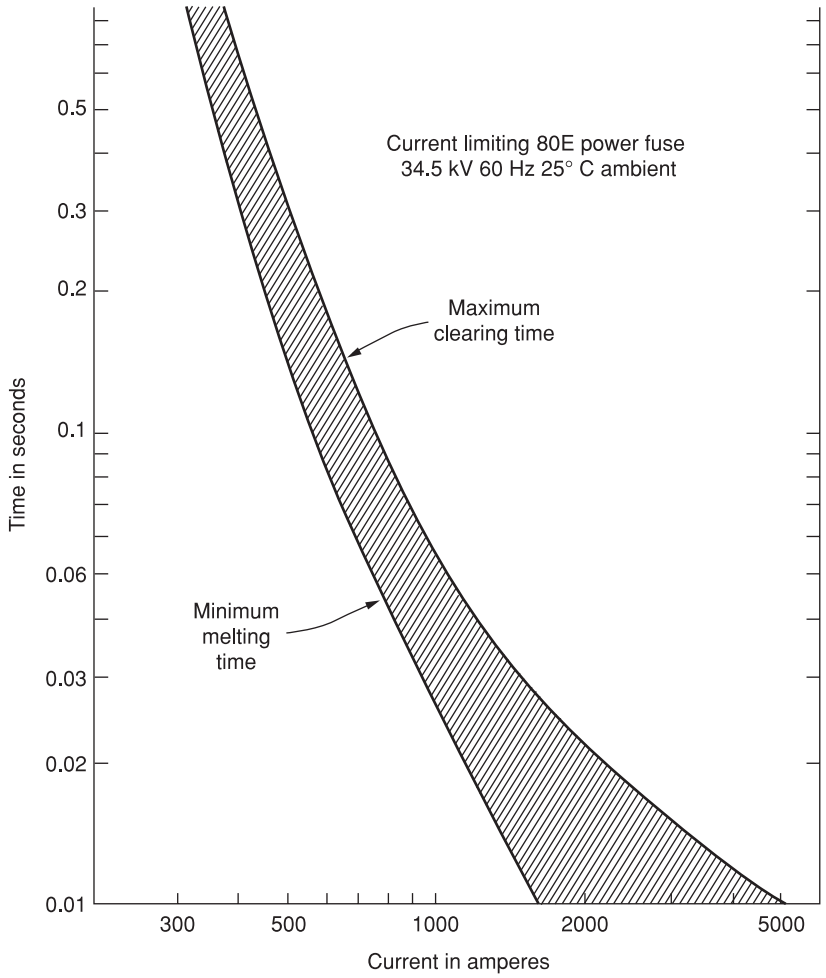
11.16 A three-phase 34.5-kV feeder supplying a 3.5-MVA load is protected by 80E power fuses in each phase, in series with a recloser. The time-current characteristic of the 80E fuse is shown in Figure 11.44. Analysis yields maximum and minimum fault currents of 1000 and 500 A, respectively. (a) To have the recloser clear the fault, find the maximum clearing time necessary for recloser operation; and (b) to have the fuses clear the fault, find the minimum recloser clearing time. Assume that the recloser operating time is independent of fault current magnitude. (c) Draw the recloser fast clearing curve and the recloser time-delay clearing curve on the fuse time-current characteristic.

FIGURE 11.44
Problem 11.16



SECTION 11.7

11.17 For the system shown in Figure 11.45, directional overcurrent relays are used at breakers B12, B21, B23, B32, B34, and B43. Overcurrent relays alone are used at B1 and B4. (a) For a fault at P_1, which breakers

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do not operate? Which breakers should be coordinated? Repeat (a) for a fault at (b) P_2, and (c) P_3. (d) Explain how the system is protected against bus faults.

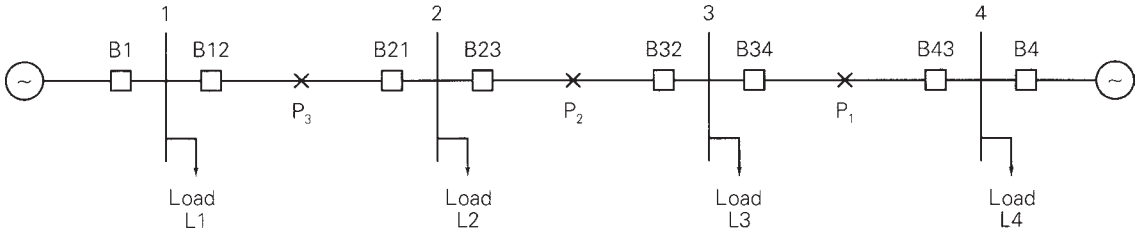

FIGURE 11.45

Problem 11.17

SECTION 11.8

11.18 (a) Draw the protective zones for the power system shown in Figure 11.46. Which circuit breakers should open for a fault at (a) P_1, (b) P_2, and (c) P_3?

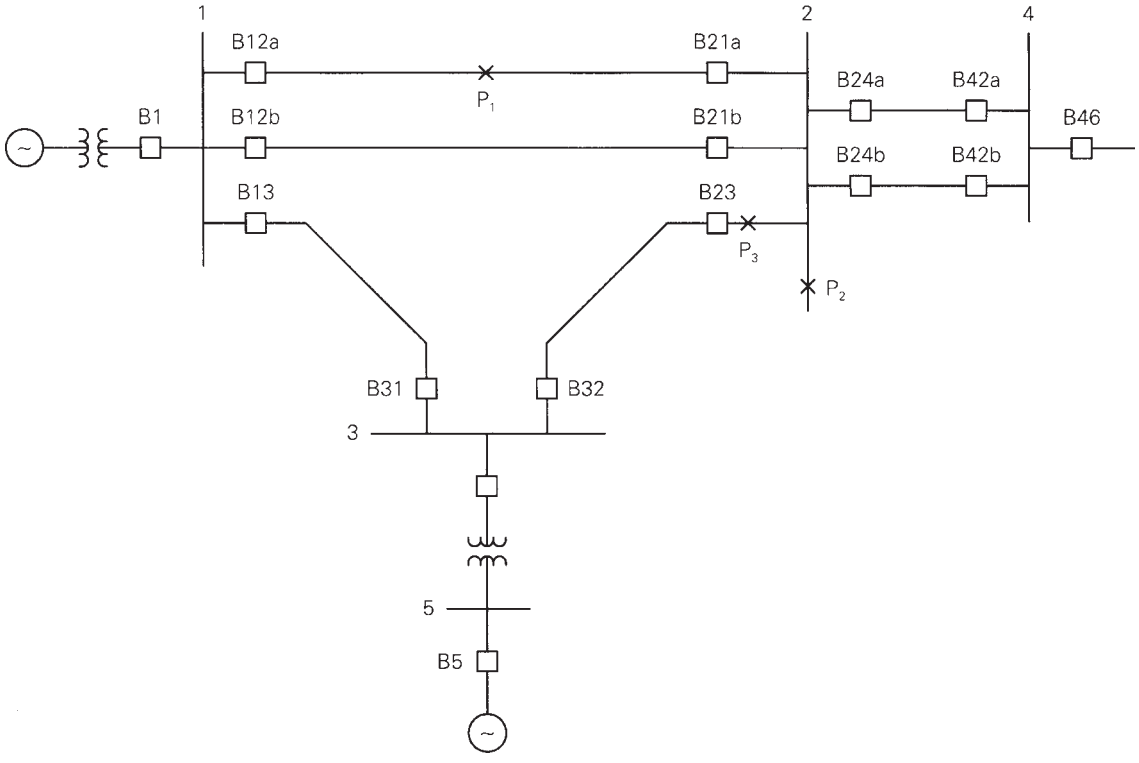

FIGURE 11.46

Problem 11.18

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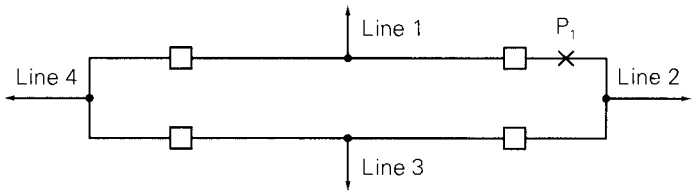
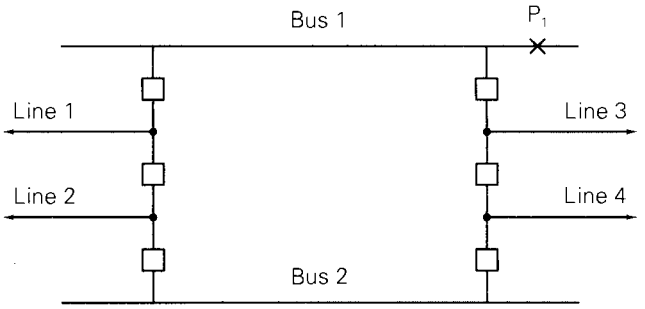
11.19 Figure 11.47 shows three typical bus arrangements. Although the number of lines connected to each arrangement varies widely in practice, four lines are shown for convenience and comparison. Note that the required number of circuit breakers per line is 1 for the ring bus, $1\frac{1}{2}$ for the breaker-and-a-half double-bus, and 2 for the double-breaker double-bus arrangement. For each arrangement: (a) Draw the protective zones. (b) Identify the breakers that open under primary protection for a fault on line 1. (c) Identify the lines that are removed from service under primary protection during a bus fault at P_1. (d) Identify the breakers that open under backup protection in the event a breaker fails to clear a fault on line 1 (that is, a stuck breaker during a fault on line 1).

FIGURE 11.47

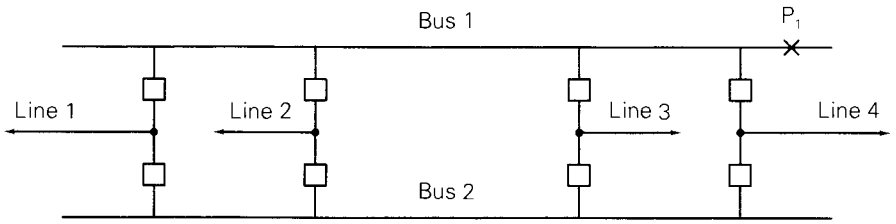
Problem 11.19—typical bus arrangements



(a) Ring bus



(b) Breaker-and-a-half double bus



(c) Double-breaker double bus

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SECTION 11.9

11.20 Three-zone mho relays are used for transmission line protection of the power system shown in Figure 11.25. Positive-sequence line impedances are given as follows.

Line	Positive-Sequence Impedance, Ω
1–2	$6 + j60$
2–3	$4 + j40$
2–4	$5 + j50$

Rated voltage for the high-voltage buses is 500 kV. Assume a 2500:5 CT ratio and a 4500:1 VT ratio at B12. (a) Determine the settings Z_{t1}, Z_{t2}, and Z_{t3} for the mho relay at B12. (b) Maximum current for line 1–2 under emergency loading conditions is 1400 A at 0.90 power factor lagging. Verify that B12 does not trip during emergency loading conditions.

11.21 Line impedances for the power system shown in Figure 11.48 are $Z_{12} = Z_{23} = 3.0 + j40.0\ \Omega$, and $Z_{24} = 6.0 + j80.0\ \Omega$. Reach for the zone 3 B12 impedance relays is set for 100% of line 1–2 plus 120% of line 2–4. (a) For a bolted three-phase fault at bus 4, show that the apparent primary impedance "seen" by the B12 relays is

$$Z_{\text{apparent}} = Z_{12} + Z_{24} + (\mathrm{I}_{32}/\mathrm{I}_{12})Z_{24}$$

where(I_{32}/I_{12}) is the line 2–3 to line 1–2 fault current ratio. (b) If $|I_{32}/I_{12}| > 0.20$, does the B12 relay see the fault at bus 4?

Note: This problem illustrates the "infeed effect." Fault currents from line 2–3 can cause the zone 3 B12 relay to underreach. As such, remote backup of line 2–4 at B12 is ineffective.

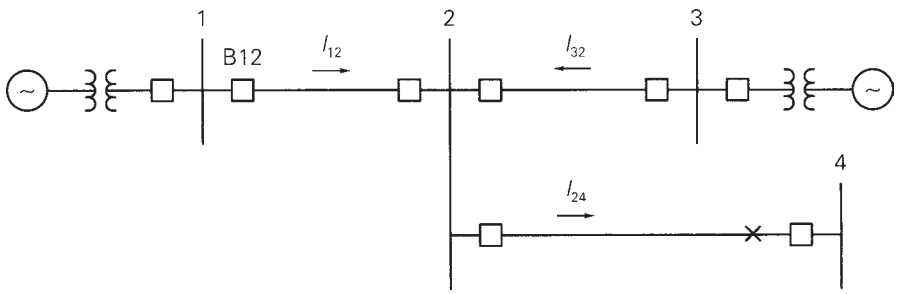

FIGURE 11.48
Problem 11.21

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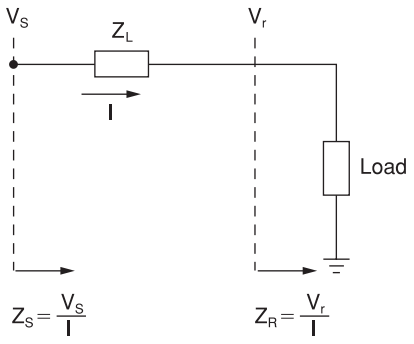
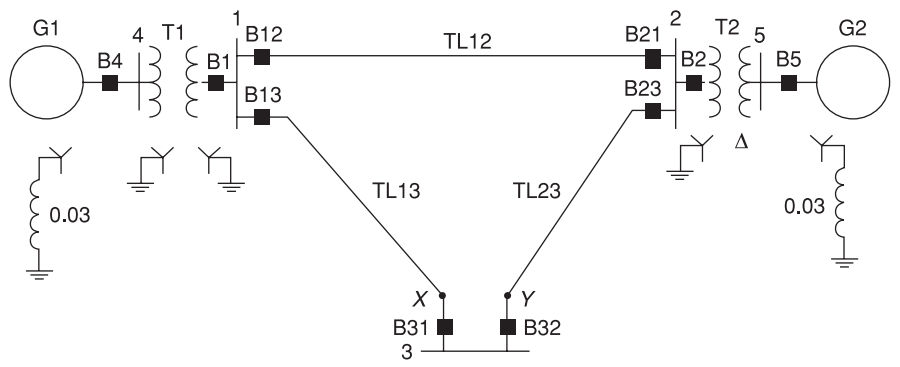
11.22 Consider the transmission line shown in Figure 11.49 with series impedance Z_L, negligible shunt admittance, and a load impedance Z_R at the receiving end. (a) Determine Z_R for the given conditions of $V_R = 1.0$ per unit and $S_R = 2 + j0.8$ per unit. (b) Construct the impedance diagram in the R-X plane for $Z_L = 0.1 + j0.3$ per unit. (c) Find Z_S for this condition and the angle δ between Z_S and Z_R.

FIGURE 11.49
Problem 11.22



11.23 A simple system with circuit breaker-relay locations is shown in Figure 11.50. The six transmission-line circuit breakers are controlled by zone distance and directional relays, as shown in Figure 11.51. The three transmission lines have the same positive-sequence impedance of $j0.1$ per unit. The reaches for zones 1, 2, and 3 are 80%, 120%, and 250%, respectively. Consider only three-phase faults. (a) Find the settings Z_r in per unit for all distance relays. (b) Convert the settings in Ω if the VTs are rated 133 kV:115 V and the CTs are rated 400:5 A. (c) For a fault at location X, which is 10% down line TL31 from bus 3, discuss relay operations.

FIGURE 11.50
Problem 11.23



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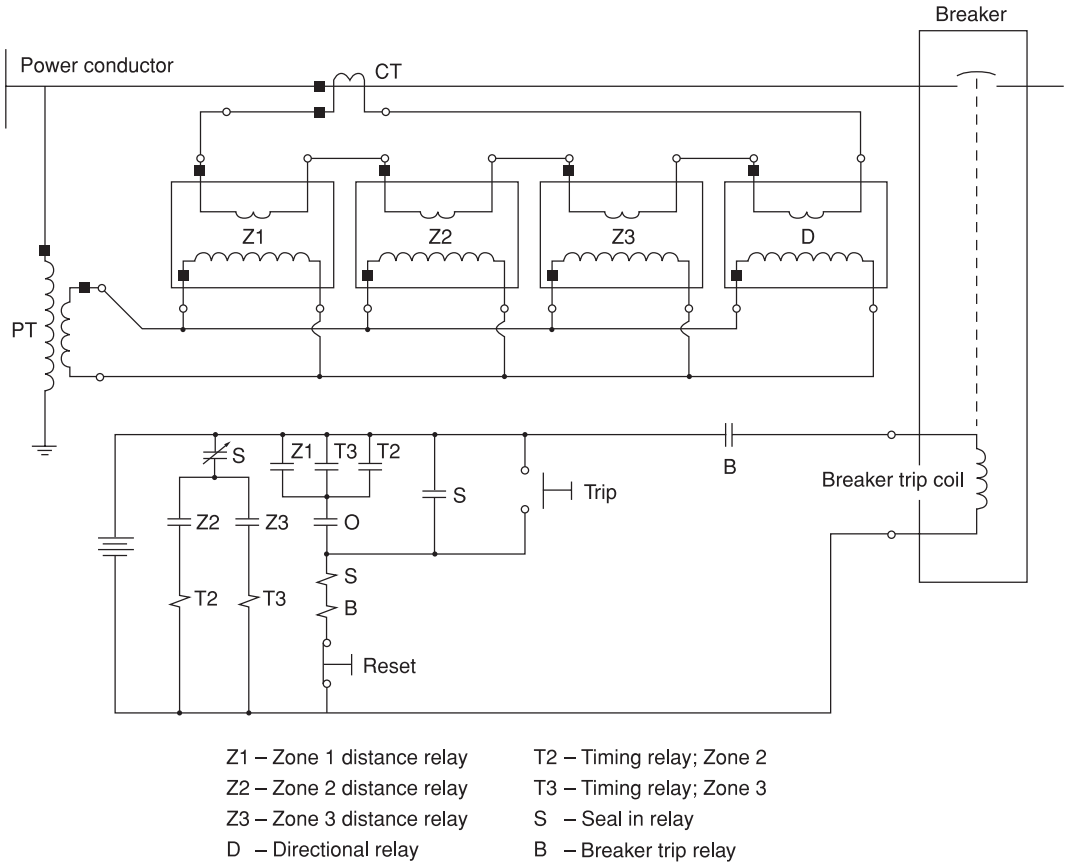

FIGURE 11.51

Three-zone, distance-relay scheme (shown for one phase only) for Problem 11.23

SECTION 11.10

11.24 Select k such that the differential relay characteristic shown in Figure 11.34 blocks for up to 20% mismatch between I'_1 and I'_2. Assume I'_1 and I'_2 are in phase.

SECTION 11.11

11.25 Consider a protected bus that terminates four lines, as shown in Figure 11.52. Assume that the linear couplers have the standard $X_m = 5\ m\Omega$ and a three-phase fault externally located on line 3 causes the fault currents shown in Figure 11.52. Note that the infeed current on line 3 to the fault is $-j10$ kA. (a) Determine V_0. (b) Let the fault be moved to an internal location on the protected bus between lines 3 and 4. Find V_0 and discuss what happens. (c) By moving the external fault from line 3 to a corresponding point on (i) line 2, and (ii) line 4, determine V_0 in each case.

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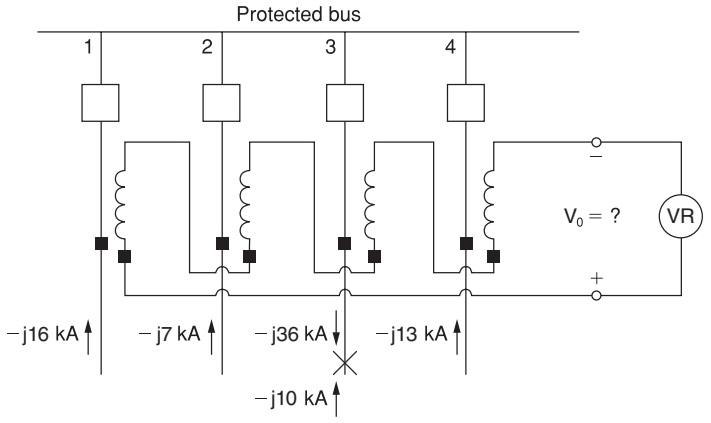
FIGURE 11.52

Problem 11.25—Bus differential protection using linear couplers



SECTION 11.12

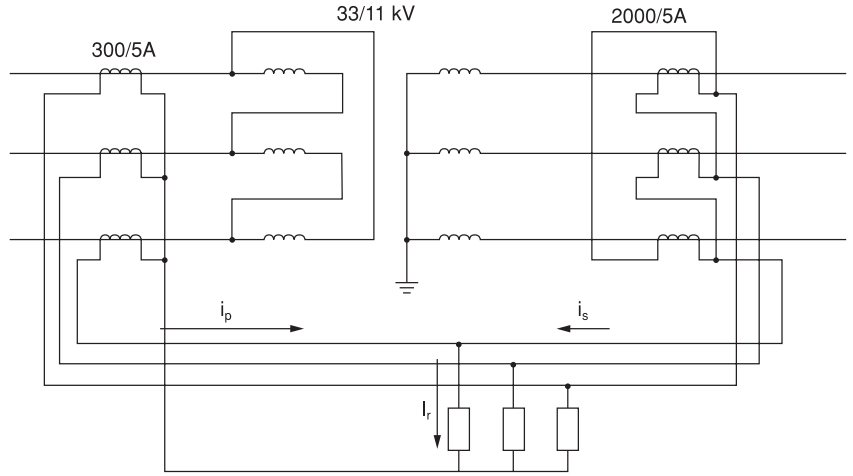
11.26 A single-phase, 5-MVA, 20/8.66-kV transformer is protected by a differential relay with taps. Available relay tap settings are 5:5, 5:5.5, 5:6.6, 5:7.3, 5:8, 5:9, and 5:10, giving tap ratios of 1.00, 1.10, 1.32, 1.46, 1.60, 1.80, and 2.00. Select CT ratios and relay tap settings. Also, determine the percentage mismatch for the selected tap setting.

11.27 A three-phase, 500-MVA, 345-kV Δ /500-kV Y transformer is protected by differential relays with taps. Select CT ratios, CT connections, and relay tap settings. Determine the currents in the transformer and in the CTs at rated conditions. Also, determine the percentage mismatch for the selected relay tap settings. Available relay tap settings are given in Problem 11.26.

11.28 For a Δ–Y connected, 15-MVA, 33:11-kV transformer with differential relay protection and CT ratios shown in Figure 11.53, determine the relay currents at full load and calculate the minimum relay current setting to allow 125% overload.

FIGURE 11.53

Problem 11.28



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11.29 Consider a three-phase Δ–Y connected, 30-MVA, 33:11-kV transformer with differential relay protection. If the CT ratios are 500:5 A on the primary side and 2000:5 A on the secondary side, compute the relay current setting for faults drawing up to 200% of rated transformer current.

11.30 Determine the CT ratios for differential protection of a three-phase, Δ–Y connected, 10-MVA, 33:11-kV transformer, such that the circulating current in the transformer Δ does not exceed 5 A.

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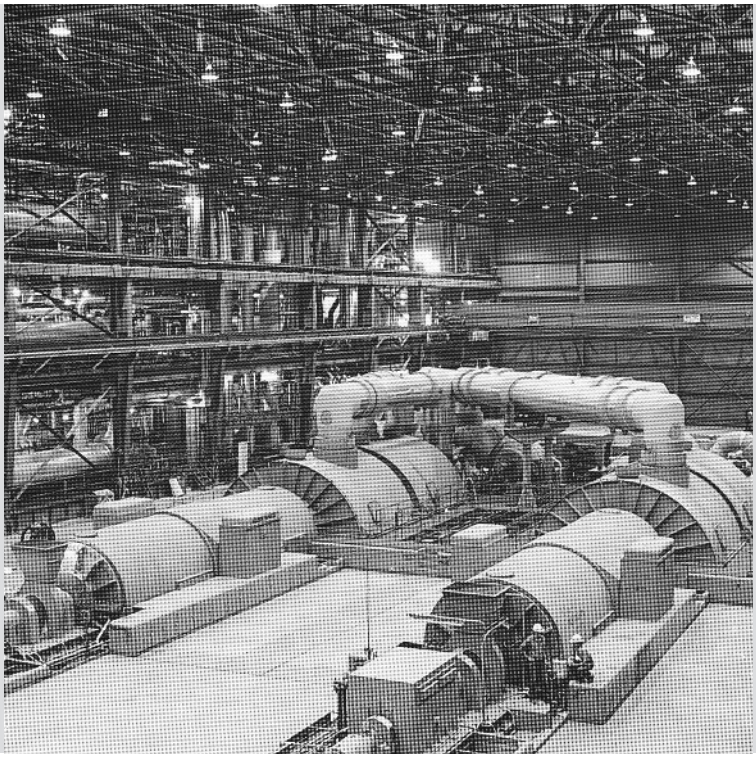
12 Power System Stability



1300-MW generating unit consisting of a cross-compound steam turbine and two 722-MVA synchronous generators. (Courtesy of American Electric Power.)

LEARNING OBJECTIVES

At the conclusion of this chapter, you should be able to:

1. Understand the different types of power system stability.
2. Understand the swing equation, which is the basic equation for the rotor dynamics of a synchronous machine.
3. Be familiar with simplified models of synchronous generators, aspects of more detailed models, and models of wind-turbine generators.
4. Understand the equal area criteria, which gives a direct method of determining transient stability of one machine connected to a system equivalent.
5. Be familiar with numerical integration techniques for solving swing equations in a multi-machine network.
6. Use PW Simulator to evaluate power system stability.
7. Be familiar with design methods for improving power system stability.

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According to [16], "Power system stability is the ability of an electric power system, for a given initial operating condition, to regain a state of operating equilibrium after being subjected to a physical disturbance, with most system variables bounded so that practically the entire system remains intact." As has been noted in earlier chapters, power systems are seldom at a true equilibrium point since the generation and load are always changing, albeit perhaps slowly to a human observer. However, usually it can be thought of as being at a quasi-steady-state operating point, with its voltages and flows well approximated by a power flow solution (Chapter 6). In real time, human operators ensure that bus voltages are close to nominal values, phase angles across transmission lines are not too large, and generators, transmission lines, transformers, and other equipment are not overloaded. The power flow can then be used to check if following some contingency (e.g., opening a transmission line) the system will return to a new acceptable quasi-steady-state equilibrium point.

Power system stability, the main focus of this chapter, involves considering the dynamics associated with major disturbances, such as loss of generation, line-switching operations, faults, and sudden load changes. That is, following a disturbance, how will the power system dynamics respond, and will the system actually return to a new equilibrium point? According to [14], there are five types of power system stability: (1) rotor angle stability; (2) frequency stability; (3) voltage stability; (4) converter-driven stability; and (5) resonance stability. The first three are the focus of this chapter, with the last two beyond the scope of this book.

In classical power system stability following a disturbance, synchronous machine frequencies undergo transient deviations from synchronous frequency (60 Hz in North America, 50 Hz in many other locations), and machine power angles change. The objective of a stability study is to determine (1) whether or not the machines will return to synchronous frequency with new steady-state power angles (rotor angle stability), (2) whether there will excessive changes in the bus frequencies and whether they will recover to their nominal value (frequency stability), and (3) whether there will be excessive changes in the bus voltage magnitudes, potentially leading to a voltage collapse (voltage stability). With the rapid growth in wind and solar generation, their dynamics also need to be considered, along with the dynamics of the load.

Elgerd [2] gives an interesting mechanical analogy to the power system transient stability. As shown in Figure 12.1, a number of masses representing synchronous machines are interconnected by a network of elastic strings representing transmission lines. Assume that this network is initially at rest in steady state,

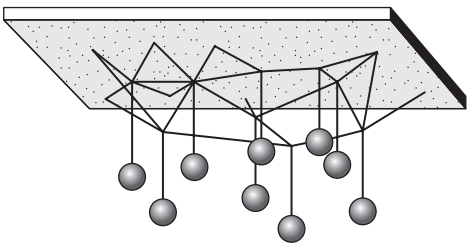

FIGURE 12.1

Mechanical analog of power system transient stability [2] (Based on Elgerd, Electric Energy Systems Theory: An Introduction, 1982, McGraw-Hill.)

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with the net force on each string below its break point when one of the strings is cut, representing the loss of a transmission line. As a result, the masses undergo transient oscillations, and the forces on the strings fluctuate. The system will then either settle down to a new steady-state operating point with a new set of string forces, or additional strings will break, resulting in an even weaker network and eventual system collapse. That is, for a given disturbance, the system is either transiently stable or unstable.

Power system stability studies are solved, using a time-domain simulation for a specific disturbance that at each time point involves a solution of algebraic equations, representing (primarily) the network power balance constraints, and differential equations, representing the generator and sometimes the load dynamics. There are two main approaches for solving the system at each time point—either simultaneous, in which the algebraic and differential equations are solved simultaneously or partitioned in the program, alternately solving the algebraic and differential equations [15]. Predisturbance, disturbance, and postdisturbance computations are performed. The program output includes power angles and frequencies of synchronous machines, bus voltages, and power flows versus time. Large-scale system studies can often involve many thousands of algebraic equations, and sometimes, hundreds of thousands of differential equations.

Historically, power system stability studies were limited to just the first swing of machine power angles following a disturbance. During the first swing, which typically lasts no more than a few seconds, the mechanical output power and the internal voltage of a generating unit are often assumed constant. However, now stability studies usually last at least a dozen seconds and sometimes up to several minutes to better consider multiple generator swings, along with frequency and voltage stability. Hence, models of turbine-governors and excitation systems (for example, see Figures 13.3 and 13.5), as well as more detailed machine models and sometimes dynamic load models, must be used to obtain accurate results over this longer time period. For studies lasting minutes, the actions of tap-changing transformers, the protection system, and controls from a power system dispatch center could be considered since they can interact to stabilize or destabilize the system.

To simplify stability studies, the following assumptions are commonly made:

1. Only balanced three-phase systems and balanced disturbances are considered. Therefore, only positive-sequence networks are employed.
2. Deviations of machine frequencies from synchronous frequency (60 Hz) are relatively small, and dc offset currents and harmonics are neglected.

Section 12.1 introduces the swing equation, which determines synchronous machine rotor dynamics. In Section 12.2, a simplified model of a synchronous machine and a Thévenin equivalent of a system consisting of lines, transformers, loads, and other machines are given. Then the equal-area criterion that gives a direct method for determining the transient stability of one machine connected to a system equivalent is presented in Section 12.3. Numerical integration techniques for solving swing equations step by step are covered in Section 12.4, and they are

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used in Section 12.5 to determine multi-machine stability. Section 12.6 introduces a more detailed synchronous generator model. Section 12.7 discusses how wind turbines and solar PV are modeled in stability studies, while Section 12.8 considers models for the electric load. Finally, Section 12.9 discusses design methods for improving power system stability.

CASE STUDY

The Impact of Renewables on Operational Security*

Ivan M. Dudurych

The following paragraphs summarize the above article, which describes the challenges to and impacts on power system operational security being experienced as higher levels of renewable energy sources operate within a system [11]. Sign in at www.cengage.com to download this product and to access its online resources.

The article describes the different operating states of a transmission system and proposes a structure for operational security.

Power Transmission System States

Figure 1 shows the following five possible transmission system states:

(1) Normal State: In the normal state, the balance between generation and load (plus losses) is met, there is no overloaded equipment, and voltages and frequencies are within normal operational limits. Also, the system's reserve margins are sufficient, and the system can withstand any of the credible contingencies so that it will remain within its operational security bounds and be brought back to normal operation following remedial actions. The goal of managing operational security is to keep a system in its normal state 99.99% of the time.

(2) Alert State: When a system's security level is weakened and the probability of a disturbance increases, a credible contingency would cause the system to reach a state where some network equipment would be overloaded or voltages at key busses or system frequencies would exceed limits. In this state, all constraints are satisfied, but there are no sufficient reserve margins to make sure the parameters will be satisfied when a severe contingency occurs. In this state, preventive actions can be taken to restore the system to normal state.

(3) Emergency State: A system can move from Alert State to Emergency State if a severe contingency occurs before any preventive action is taken. In the Emergency State, certain network

* "The Impact of Renewables on Operational Security" by Ivan M. Dudurych, available online, with permission, from *IEEE Power & Energy Magazine* (March/April 2021), pp. 37–45.

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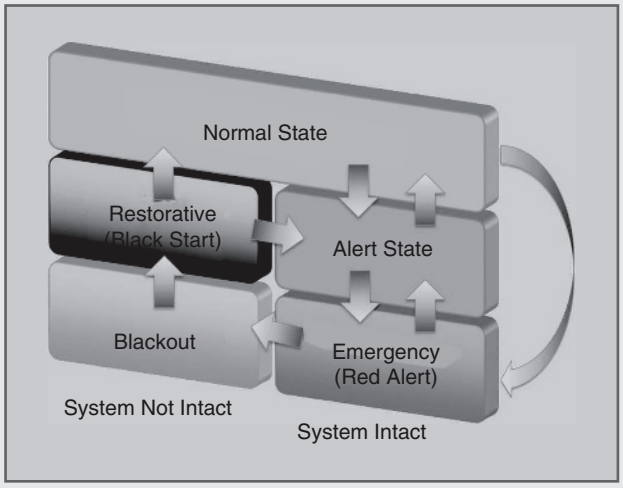

Figure 1 Power transmission system states

components are overloaded or voltages at key busses or system frequencies are outside their limits. Nevertheless, the system remains intact and can be restored to its Normal State or at least to its Alert State if appropriate corrective actions are taken.

(4) Blackout State: In the Blackout State, many generators may lose synchronism, leading to cascading transmission component outages. A system may split into islands, and significant parts of the transmission network may shut down. Control actions, such as load shedding and system separation, are used for saving as much of the network as possible from a widespread blackout. A system can move from the Emergency State to the Blackout State if no appropriate corrective action is taken, and it can move from the Alert State to the Blackout State when a severe disturbance occurs.

(5) Restorative (Black-Start) State: In the Restorative State, an operator performs control actions to reconnect facilities and restore system load. Control actions are performed according to a black-start restoration plan to gradually bring frequency, voltage, and other parameters within operational security limits.

Operational Security Definitions and Criteria

Figure 2 shows a proposed structure for operational security components. As shown, operational security is defined in terms of steady-state and dynamic security criteria. Steady-state criteria (also called *static limits*) refer to thermal, short-circuit current, voltage, and frequency bounds. Active power reserves and ramping reserves are also included. Dynamic criteria (also called *stability limits*) refer to transient stability (rotor-angle stability for synchronous generators and fault ride-through capability for

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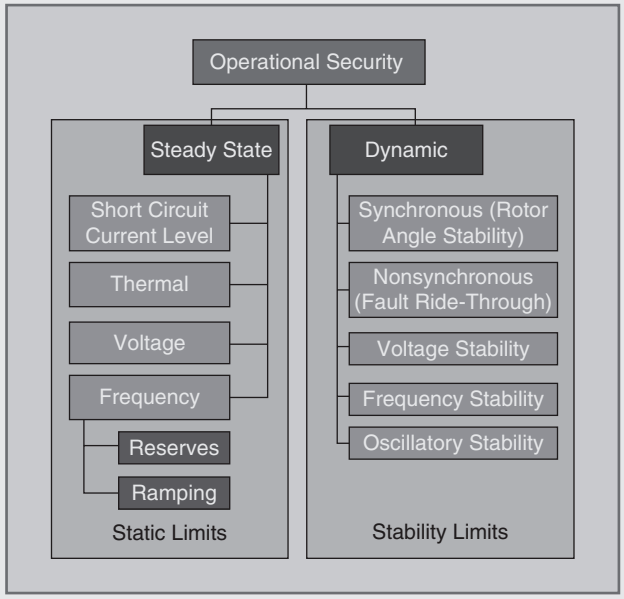

Figure 2 Operational security components

nonsynchronous plants), voltage stability, and frequency stability. Also included is oscillatory stability, which describes the ability of a power system to damp oscillations.

The following set of operational security criteria is proposed:

Thermal criterion: Transmission lines, cables, and transformers must be loaded at less than 100% of their nominal thermal ratings in base case conditions and at less than 110% in N-1 contingency conditions. For transformers, which have a greater short-term load capability, 130% may be acceptable.

Voltage stability criterion: Voltage stability must be maintained in base-case conditions and all studied contingency conditions. Steady-state voltage levels at all HV busses within the transmission system (e.g., 110 kV and higher) must be within limits: 0.95–1.1 per unit in base-case conditions and 0.9–1.12 per unit in N-1 contingency conditions.

Synchronous plant transient stability criterion: The maximum rotor angles of synchronous generators must be within limits to stay in synchronism. This corresponds to a requirement that the maximum angle separation of any two synchronous generators in one island must be less than 360°.

Frequency stability criterion: In the event of tripping one generator or a three-phase fault with a forced outage of an element (transmission line or cable), the frequency deviation must be lower than 1 Hz when the duration of the frequency deviation is longer than 0.5 s.

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Oscillatory stability criterion: The oscillation damping coefficient can be assessed through small-signal analysis and as an additional outcome of time-domain simulation analysis. The damping factor must be greater than 0.05.

Additional criteria:

System nonsynchronous penetration (SNSP) level: The maximum SNSP level can be established based on the comprehensive analysis of the operational security of the synchronous system and revised in line with improving flexibility of system components (mainly generators' ability to provide inertial and fast frequency response). The maximum SNSP is maintained by curtailment of nonsynchronous renewable generation.

Rate of change of frequency (RoCoF): This is the measure of frequency through time following the disconnection of a generator or an interconnector. At present, both maximum and minimum RoCoF are monitored. Minimum RoCoF could occur due to the trip of a large generator or an importing connector. Maximum RoCoF could result from the trip of an exporting interconnector.

System inertia: Monitoring system inertia from synchronous generation ensures that operators always maintain a minimum number of conventional generating units synchronized to the power system.

Active power operational reserve: This is a classical requirement for a rotating reserve that can be complemented by new technologies (e.g., batteries).

Available megawatt ramping: This is an assessment of the total available ramping availability of an online plant and an offline plant across various time horizons (e.g., 3 h, 8 h, etc.).

Impacts of Extremely High Penetration of Renewable Sources on Operational Security

Extremely high SNSP levels have impacts on the following: resource variability, transient stability, voltage stability, and frequency stability.

The Effects of Resource Variability

To illustrate the impact of an extremely high penetration of renewable sources on operational security, the power system in Ireland and Northern Ireland is selected as an example. As of May 2018, the system's generation fleet had some 9500 MW of conventional sources, 4500 MW of wind resources, a peak demand of 6500 MW, and a minimum demand of 2500 MW. The system connects to the United Kingdom through a 500-MW HVDC line in the north and another 500-MW HVDC line in the south. The relatively small size of the island's synchronous system versus the increasing penetration of intermittent renewables creates a "large-scale renewable energy penetration laboratory."

Figure 3 displays the percentage of nonsynchronous generation (mostly wind) on the island of Ireland, where the SNSP is shown as a percentage of total generation for April–June 2019. As shown, the renewable generation is extremely

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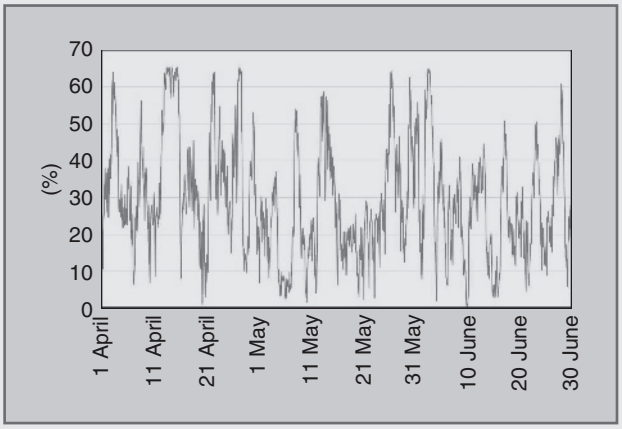

Figure 3 The percentage of System Nonsynchronous Penetration (SNSP) in Ireland's total generation during April–June 2019

(Source: Eirgrid)

volatile and changes randomly from almost zero to 65%. This means that the scheduling of a conventional plant will be significantly different from day to day and hour to hour, causing the plant to cycle and even stop and start much more than during the past. More frequent cycling and operation outside the plant's most efficient range (which is normally near the plant's full rated output minus the reserve provision) reduces efficiency and leads to increased wear, and thus it can reduce the plant's reliability.

With the advent of renewable and other distributed generation, transmission and distribution, systems now experience multidirectional power flows, depending on the amount of instant renewable generation in different parts of the system. When wind generation is low, power flows from the transmission system to the distribution system, while during periods of high wind generation, power may flow in the opposite direction, from the distribution system to the transmission system.

It is well known that conventional power stations have been built close to load centers, whereas renewable generation is typically located far from load centers. Therefore, if there is a lot of wind generation, power flows can reverse and reach significant values that were not originally anticipated for the transmission circuits. Such circuits can be highly loaded and can overload if there are outages on parallel circuits. Therefore, thermal security must be monitored in real time to avoid cascading outages due to overload conditions.

Effect of Renewables on Transient Stability

Transient stability concerns the ability of a system to keep its components connected following large disturbances, such as faults in the transmission system that are cleared by protection, sudden trips of large generators, the disconnection of

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large loads, etc. Transient stability further includes:

- *Synchronous plant stability*: the ability of a system to maintain its synchronous generators in synchronism (rotor angle stability).
- *Nonsynchronous plant stability*: The ability of a system to keep nonsynchronous sources of generation connected (also called *ride-through capability*).

Figure 4 qualitatively illustrates how synchronous machine rotor angle stability is affected by a significant SNSP. As shown, the electromagnetic power output of a synchronous machine changes sinusoidally with rotor angle. In steady state, the power output is balanced by the turbine mechanical power (operating point A in Figure 4). During a disturbance (power imbalance), the rotor angle changes as the rotor speed deviates from synchronous speed. As a measure of rotor angle stability, we can use the concept of "synchronizing torque," which shows how much the rotor angle changes for the same variation in power from the steady-state operating point of the machine. Higher synchronizing torques give more stability to the synchronous plant because it encourages the synchronous machines to spin together at the same frequency. As the machine gets more loaded (the operating point moves from A to B in Figure 4), the synchronizing torque decreases, which contributes to a reduction in transient stability due to disturbances that perturb the rotor angle. That is, as the machine moves to the higher operating point B, the forces trying to resynchronize it to that operating point will be weaker. Similarly, if the maximum electromagnetic power output of the machine decreases (see the lower sinusoidal curve in Figure 4), it also has the effect of decreasing the synchronizing torque resulting in a reduction of transient stability.

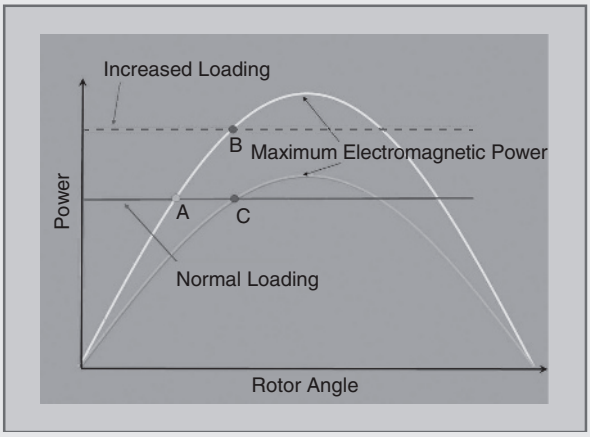

Figure 4 The power angle characteristic of synchronous machines with normal and increased operating points (A and B) and a reduced power angle characteristic (operating point C)

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Now consider what initially happens when only a small portion of the synchronous generation is replaced by converter-based renewables. If the remaining synchronous generators are less loaded than normal, their synchronizing torques increase, thereby improving transient stability of the system. However, a further increase of renewables requires a progressive decommissioning of conventional plants. The equivalent power-angle characteristic will mirror the lower sinusoidal curve in Figure 4, with operating point C. It will be more difficult to "push" power from the remaining synchronous machines to the loads, due to the increased reactances of the now longer paths and the poorer voltage support from the now prevailing nonsynchronous generation. At very high renewable penetration levels, transient stability deteriorates rapidly because the remaining synchronous units are forced to be highly loaded. Note, however, that recent advances in grid connection applications based on insulated gate bipolar transistor (IGBT) technology will enable engineers to counteract these phenomena.

Figure 5 shows the synchronizing power versus nonsynchronous plant penetration based on an estimated reduction in critical clearing time in a 2010 "All Island Transmission System Operator (TSO) Facilitation of Renewables Study" in Ireland. Critical clearing time is the maximum time allowed to clear a three-phase fault at the terminals of a synchronous generator before the generator starts slipping poles. As shown, the rotor angle stability of Ireland's system slightly improves until the wind penetration increases up to 50%. Then it starts declining, and it quickly deteriorates at wind penetration levels of 80% and higher.

To investigate the transient security of nonsynchronous plants (and

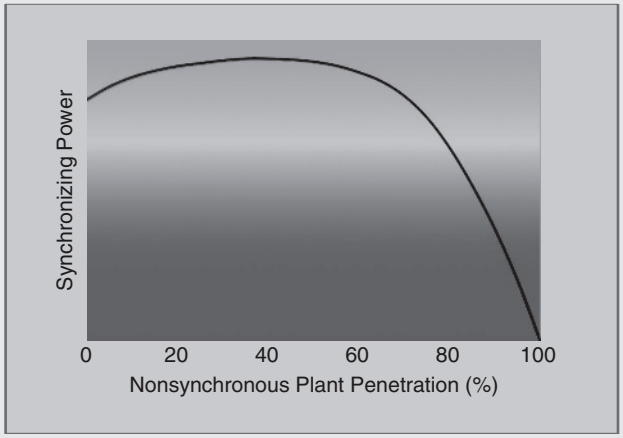

Figure 5 The synchronizing power versus nonsynchronous plant penetration level (derived from critical clearance time studies in a 2010 "All Island TSO Facilitation of Renewables" in Ireland)

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their ability to ride through three-phase faults), consider how three-phase fault characteristics evolve. The most economical way of connecting renewable sources to a synchronous system is via power electronic converters. This has two main implications: (1) the mechanical system inertia is increasingly depleted when renewable penetration increases; and (2) the short-circuit current contribution from renewable sources is limited to their maximum load current. The first implication will affect a system's frequency security. The second causes voltage dips associated with faults in the transmission system to propagate more widely in a system with reduced short-circuit levels, which has a direct impact on fault ride-through capabilities of nonsynchronous sources. As the bigger part of a system sees a voltage dip, a larger portion of nonsynchronous generation can trip on undervoltage protection.

Effect of Renewables on Voltage Stability

In the early days, renewable generators did not have any voltage and reactive power capability. Soon, as renewable nonsynchronous generators displaced conventional ones, it was recognized that conventional generators' voltage control capabilities were being depleted. Grid codes began to be updated to include at least some conventional generation-like functionality into a nonsynchronous generation portfolio. However, most distribution-connected renewable sources are still fixed-power-factor machines that do not regulate voltage, which means that voltage regulation becomes increasingly difficult in a system with high SNSP levels. In areas with significant renewable generation, voltage collapses become more probable.

Effect of Renewables on Frequency Stability

The main effect of renewable generation on frequency stability is related to the fact that with the progressive replacement of conventional generation with nonsynchronous renewable generation, the available system inertia decreases. Figure 6 shows that frequency deviation will be larger in a system that has less inertia available. To an extent, the issue can be alleviated by the introduction of fast frequency response mechanisms, but the genuine inertia provided by a synchronous plant can be fully replicated by a converter-based plant only if the power imbalance on the system is directly assessed and compensated in real time. Curve C in Figure 6 corresponds to the activation of an underfrequency load shedding scheme, which is the system's defense mechanism designed to automatically disconnect a portion of the load. This is a last defense and not accepted for routine operation.

Assessing the operational security of a power system becomes compulsory for TSOs that integrate large amounts of renewables. Such assessments can be performed by using online dynamic security assessment technology: software that takes snapshots of power system conditions, performs security evaluations in near real time and for future time frames, and provides warnings about, and remedial measures for, abnormal conditions.

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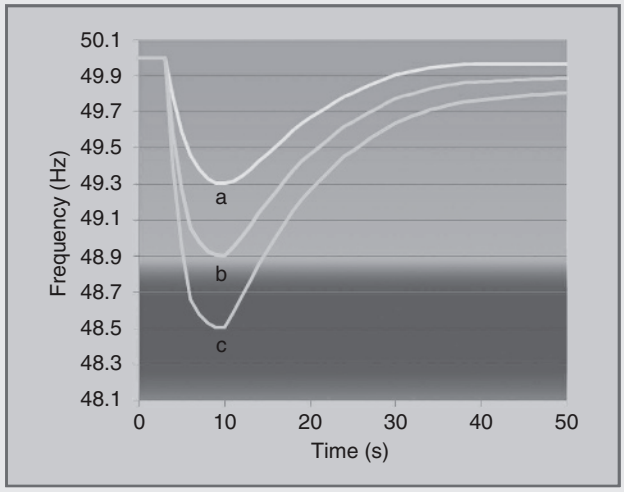

Figure 6 Frequency deviations due to the sudden trip of a large generator for (a) a traditional generation portfolio; (b) a portfolio with a large nonsynchronous generation penetration; and (c) a portfolio with a significant nonsynchronous generation penetration

12.1 THE SWING EQUATION

Consider a generating unit consisting of a three-phase synchronous generator and its prime mover. The rotor motion is determined by Newton's second law, given by

$$J\alpha_m(t) = T_m(t) - T_e(t) = T_a(t) \tag{12.1.1}$$

where J = total moment of inertia of the rotating masses, kg-m^2
α_m = rotor angular acceleration, rad/s^2
T_m = mechanical torque supplied by the prime mover minus the retarding torque due to mechanical losses, N-m
T_e = electrical torque that accounts for the total three-phase electrical power output of the generator, plus electrical losses, N-m
T_a = net accelerating torque, N-m

Also, the rotor angular acceleration is given by

$$\alpha_m(t) = \frac{d\omega_m(t)}{dt} = \frac{d^2\theta_m(t)}{dt^2} \tag{12.1.2}$$

$$\omega_m(t) = \frac{d\theta_m(t)}{dt} \tag{12.1.3}$$

where ω_m = rotor angular velocity, rad/s
θ_m = rotor angular position with respect to a stationary axis, rad

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T_m and T_e *are* positive for generator operation. In steady state, T_m equals T_e, the accelerating torque T_a is zero, and, from (12.1.1), the rotor acceleration α_m is zero, resulting in a constant rotor velocity called *synchronous speed.* When T_m is greater than T_e, T_a is positive and α_m is therefore positive, resulting in increasing rotor speed. Similarly, when T_m is less than T_e, the rotor speed is decreasing.

It is convenient to measure the rotor angular position with respect to a synchronously rotating reference axis instead of a stationary axis. Accordingly,

$$\theta_m(t) = \omega_{msyn}t + \delta_m(t) \tag{12.1.4}$$

where ω_{msyn} = synchronous angular velocity of the rotor, rad/s

δ_m = rotor angular position with respect to a synchronously rotating reference, rad

Using (12.1.2) and (12.1.4), (12.1.1) becomes

$$J\frac{d^2\theta_m(t)}{dt^2} = J\frac{d^2\delta_m(t)}{dt^2} = T_m(t) - T_e(t) = T_a(t) \tag{12.1.5}$$

It is also convenient to work with power rather than torque, and to work in per-unit rather than in actual units. Accordingly, multiply (12.1.5) by $\omega_m(t)$ and divide by S_{rated}, the three-phase volt-ampere rating of the generator:

$$\begin{aligned}\frac{J\omega_m(t)}{S_{rated}}\frac{d^2\delta_m(t)}{dt^2} &= \frac{\omega_m(t)T_m(t) - \omega_m(t)T_e(t)}{S_{rated}} \\ &= \frac{p_m(t) - p_e(t)}{S_{rated}} = p_{mp.u.}(t) - p_{ep.u.}(t) = p_{ap.u.}(t)\end{aligned} \tag{12.1.6}$$

where $p_{mp.u.}$ = mechanical power supplied by the prime mover minus mechanical losses, per unit

$p_{ep.u.}$ = electrical power output of the generator plus electrical losses, per unit

Finally, it is convenient to work with a normalized inertia constant, called the H constant, which is defined as

$$\begin{aligned}H &= \frac{\text{stored kinetic energy at synchronous speed}}{\text{generator volt-ampere rating}} \\ &= \frac{\frac{1}{2}J\omega_{msyn}^2}{S_{rated}} \text{ joules/VA or per unit-seconds}\end{aligned} \tag{12.1.7}$$

The H constant has the advantage that it falls within a fairly narrow range, normally between 1 and 10 p.u.-s, whereas J varies widely, depending on generating unit size and type. Solving (12.1.7) for J and using in (12.1.6),

$$2H\frac{\omega_m(t)}{\omega_{msyn}^2}\frac{d^2\delta_m(t)}{dt^2} = p_{mp.u.}(t) - p_{ep.u.}(t) = p_{ap.u.}(t) \tag{12.1.8}$$

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Defining per-unit rotor angular velocity,

$$\omega_{\text{p.u.}}(t) = \frac{\omega_m(t)}{\omega_{m\text{syn}}} \tag{12.1.9}$$

Equation (12.1.8) becomes

$$\frac{2\text{H}}{\omega_{m\text{syn}}}\omega_{\text{p.u.}}(t)\frac{d^2\delta_m(t)}{dt^2} = p_{m\text{p.u.}}(t) - p_{e\text{p.u.}}(t) = p_{a\text{p.u.}}(t) \tag{12.1.10}$$

For a synchronous generator with P poles, the electrical angular acceleration α, electrical radian frequency ω, and power angle δ are

$$\alpha(t) = \frac{\text{P}}{2}\alpha_m(t) \tag{12.1.11}$$

$$\omega(t) = \frac{\text{P}}{2}\omega_m(t) \tag{12.1.12}$$

$$\delta(t) = \frac{\text{P}}{2}\delta_m(t) \tag{12.1.13}$$

Similarly, the synchronous electrical radian frequency is

$$\omega_{\text{syn}} = \frac{\text{P}}{2}\omega_{m\text{syn}} \tag{12.1.14}$$

The per-unit electrical frequency is

$$\omega_{\text{p.u.}}(t) = \frac{\omega(t)}{\omega_{\text{syn}}} = \frac{\frac{2}{\text{P}}\omega(t)}{\frac{2}{\text{P}}\omega_{\text{syn}}} = \frac{\omega_m(t)}{\omega_{m\text{syn}}} \tag{12.1.15}$$

Therefore, using (12.1.13–12.1.15), (12.1.10) can be written as

$$\frac{2\text{H}}{\omega_{\text{syn}}}\omega_{\text{p.u.}}(t)\frac{d^2\delta(t)}{dt^2} = p_{m\text{p.u.}}(t) - p_{e\text{p.u.}}(t) = p_{a\text{p.u.}}(t) \tag{12.1.16}$$

Frequently (12.1.16) is modified to also include a term that represents a damping torque any time the generator deviates from its synchronous speed, with its value proportional to the speed deviation

$$\begin{aligned} &2\text{H}/\omega_{\text{syn}}\omega_{\text{p.u.}}(t)(d^2\delta(t)/(dt^2)) \\ &\quad = p_{m\text{p.u.}}(t) - p_{e\text{p.u.}}(t) - \text{D}/\omega_{\text{syn}}(d\delta(t)/(dt)) \\ &\quad = p_{a\text{p.u.}}(t) \end{aligned} \tag{12.1.17}$$

where D is either zero or a relatively small positive number with typical values between 0 and 2. The units of D are per-unit power divided by per-unit speed deviation.

Equation (12.1.17), called the per-unit *swing equation*, is the fundamental equation that determines rotor dynamics in stability studies. Note that it is nonlinear due to $p_{e\text{p.u.}}(t)$, which is shown in Section 12.2 to be a nonlinear function

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of δ. Equation (12.1.17) is also nonlinear due to the $\omega_{\text{p.u.}}(t)$ term. While the rotor speed can certainly vary during transients, a reasonable approximation for hand calculations is that $\omega_{\text{p.u.}}(t) \simeq 1.0$.

Equation (12.1.17) is a second-order differential equation that can be rewritten as two first-order differential equations. Differentiating (12.1.4), and then using (12.1.3) and (12.1.12) through (12.1.14), result in

$$\frac{d\delta(t)}{dt} = \omega(t) - \omega_{\text{syn}} \tag{12.1.18}$$

Using (12.1.18) in (12.1.17),

$$\frac{2\text{H}}{\omega_{\text{syn}}}\omega_{\text{p.u.}}(t)\frac{d\omega(t)}{dt} = p_{m\text{p.u.}}(t) - p_{e\text{p.u.}}(t) - D/\omega_{\text{syn}}\frac{d\delta(t)}{dt} = p_{a\text{p.u.}}(t) \tag{12.1.19}$$

Equations (12.1.18) and (12.1.19) are two first-order differential equations.

EXAMPLE 12.1

Generator per-unit swing equation and power angle during a short circuit

A three-phase, 60-Hz, 500-MVA, 15-kV, 32-pole hydroelectric generating unit has an H constant of 2.0 p.u.-s and D = 0. (a) Determine ω_{syn} and $\omega_{m\text{syn}}$. (b) Give the per-unit swing equation for this unit. (c) The unit is initially operating at $p_{m\text{p.u.}} = p_{e\text{p.u.}} = 1.0$, $\omega = \omega_{\text{syn}}$, and $\delta = 10°$ when a three-phase-to-ground bolted short circuit at the generator terminals causes $p_{e\text{p.u.}}$ to drop to zero for $t \geqslant 0$. Determine the power angle 3 cycles after the short circuit commences. Assume $p_{m\text{p.u.}}$ remains constant at 1.0 per unit. Also, assume $\omega_{\text{p.u.}}(t) = 1.0$ in the swing equation.

SOLUTION

a. For a 60-Hz generator,

$\omega_{\text{syn}} = 2\pi 60 = 377$ rad/s

and, from (12.1.14), with P = 32 poles,

$$\omega_{m\text{syn}} = \frac{2}{\text{P}}\omega_{\text{syn}} = \left(\frac{2}{32}\right)377 = 23.56 \quad \text{rad/s}$$

b. From (12.1.16), with H = 2.0 p.u.-s,

$$\frac{4}{2\pi 60}\omega_{\text{p.u.}}(t)\frac{d^2\delta(t)}{dt^2} = p_{m\text{p.u.}}(t) - p_{e\text{p.u.}}(t)$$

c. The initial power angle is

$\delta(0) = 10° = 0.1745$ radian

(*Continued*)

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Also, from (12.1.17), at $t = 0$,

$$\frac{d\delta(0)}{dt} = 0$$

Using $p_{m\text{p.u.}}(t) = 1.0$, $p_{e\text{p.u.}} = 0$, and $\omega_{\text{p.u.}}(t) = 1.0$, the swing equation from (b) is

$$\left(\frac{4}{2\pi 60}\right)\frac{d^2\delta(t)}{dt^2} = 1.0 \quad t \geq 0$$

Integrating twice and using the above initial conditions,

$$\frac{d\delta(t)}{dt} = \left(\frac{2\pi 60}{4}\right)t + 0$$

$$\delta(t) = \left(\frac{2\pi 60}{8}\right)t^2 + 0.1745$$

$$\text{At } t = 3 \text{ cycles} = \frac{3 \text{ cycles}}{60 \text{ cycles/second}} = 0.05 \text{ second},$$

$$\delta(0.05) = \left(\frac{2\pi 60}{8}\right)(0.05)^2 + 0.1745$$

$$= 0.2923 \text{ radian} = 16.75°$$

EXAMPLE 12.2

Equivalent swing equation: two generating units

A power plant has two three-phase, 60-Hz generating units with the following ratings:

Unit 1: 500 MVA, 15 kV, 0.85 power factor, 32 poles, $H_1 = 2.0$ p.u.-s, D = 0
Unit 2: 300 MVA, 15 kV, 0.90 power factor, 16 poles, $H_2 = 2.5$ p.u.-s, D = 0

a. Give the per-unit swing equation of each unit on a 100-MVA system base.
b. If the units are assumed to "swing together," that is, $\delta_1(t) = \delta_2(t)$, combine the two swing equations into one equivalent swing equation.

SOLUTION

a. If the per-unit powers on the right-hand side of the swing equation are converted to the system base, then the H constant on the left-hand side must also be converted. That is,

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$$H_{new} = H_{old}\frac{S_{old}}{S_{new}} \quad \text{per unit}$$

Converting H_1 from its 500-MVA rating to the 100-MVA system base,

$$H_{1new} = H_{1old}\frac{S_{old}}{S_{new}} = (2.0)\left(\frac{500}{100}\right) = 10 \quad \text{p.u.-s}$$

Similarly, converting H_2,

$$H_{2new} = (2.5)\left(\frac{300}{100}\right) = 7.5 \quad \text{p.u.-s}$$

The per-unit swing equations on the system base are then

$$\frac{2H_{1new}}{\omega_{syn}}\omega_{1p.u.}(t)\frac{d^2\delta_1(t)}{dt^2} = \frac{20.0}{2\pi 60}\omega_{1p.u.}(t)\frac{d^2\delta_1(t)}{dt^2}$$

$$= p_{m1p.u.}(t) - p_{e1p.u.}(t)$$

$$\frac{2H_{2new}}{\omega_{syn}}\omega_{2p.u.}(t)\frac{d^2\delta_2(t)}{dt^2} = \frac{15.0}{2\pi 60}\omega_{2p.u.}(t)\frac{d^2\delta_2(t)}{dt^2} = p_{m2p.u.}(t) - p_{e2p.u.}$$

b. Letting:

$$\delta(t) = \delta_1(t) = \delta_2(t)$$

$$\omega_{p.u.}(t) = \omega_{1p.u.}(t) = \omega_{2p.u.}(t)$$

$$p_{mp.u.}(t) = p_{m1p.u.}(t) + p_{m2p.u.}(t)$$

$$p_{ep.u.}(t) = p_{e1p.u.}(t) + p_{e2p.u.}(t)$$

and adding the above swing equations

$$\frac{2(H_{1new} + H_{2new})}{\omega_{syn}}\omega_{p.u.}(t)\frac{d^2\delta(t)}{dt^2}$$

$$= \frac{35.0}{2\pi 60}\omega_{p.u.}(t)\frac{d^2\delta(t)}{dt^2} = p_{mp.u.}(t) - p_{ep.u.}(t)$$

When stability studies involving large-scale power systems with many generating units are performed, computation time can be reduced by combining the swing equations of those units that swing together. Such units, which are called *coherent machines*, usually are connected to the same bus or are electrically close, and they are usually remote from the network disturbances under study.

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12.2 SIMPLIFIED SYNCHRONOUS MACHINE MODEL AND SYSTEM EQUIVALENTS

Figure 12.2 shows a simplified model of a synchronous machine, called the classical model, that can be used in shorter-term transient stability studies. As shown, the synchronous machine is represented by a constant internal voltage E' behind its direct axis transient reactance X'_d. This model is based on the following assumptions:

1. The machine is operating under balanced three-phase positive-sequence conditions.
2. Machine excitation is constant.
3. Machine losses, saturation, and saliency are neglected.

In commercial stability programs, more detailed models are be used to represent exciters, losses, saturation, and saliency. However, the simplified model reduces model complexity while maintaining reasonable accuracy for some stability calculations.

Each generator in the model is connected to a system consisting of transmission lines, transformers, loads, and other machines. To a first approximation, the system can be represented by an "infinite bus" behind a system reactance. An infinite bus is an ideal voltage source that maintains constant voltage magnitude, constant phase, and constant frequency.

Figure 12.3 shows a synchronous generator connected to a system equivalent. The voltage magnitude V_{bus} and 0° phase of the infinite bus are constant. The phase angle δ of the internal machine voltage is the machine power angle with respect to the infinite bus.

FIGURE 12.2

Simplified synchronous machine model for transient stability studies

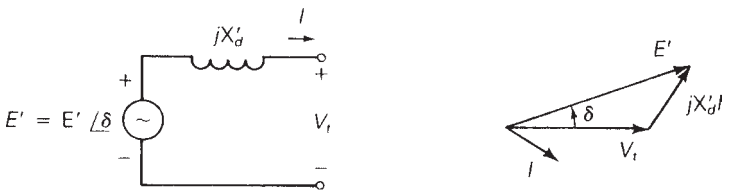
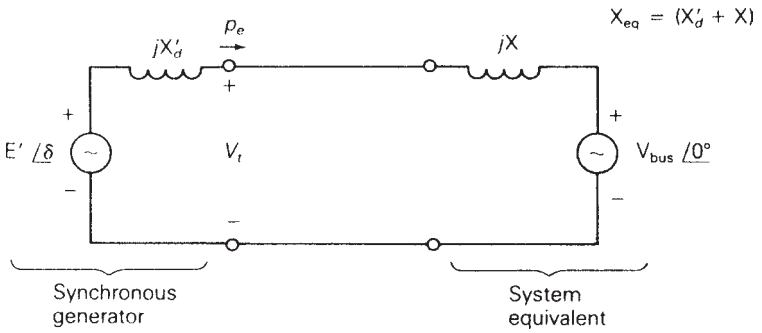


FIGURE 12.3

Synchronous generator connected to a system equivalent



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The equivalent reactance between the machine internal voltage and the infinite bus is $X_{eq} = (X'_d + X)$. From (6.7.3), the real power delivered by the synchronous generator to the infinite bus is

$$p_e = \frac{E'V_{bus}}{X_{eq}} \sin\delta \tag{12.2.1}$$

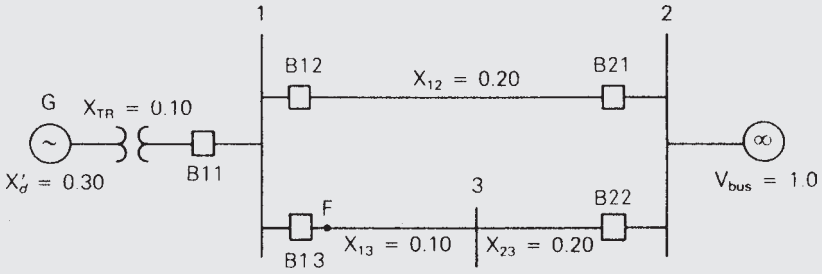
During transient disturbances, both E' and V_{bus} are considered constant in (12.2.1). Thus, p_e is a sinusoidal function of the machine power angle δ.

EXAMPLE 12.3

Generator internal voltage and real power output versus power angle

Figure 12.4 shows a single-line diagram of a three-phase, 60-Hz synchronous generator, connected through a transformer and parallel transmission lines to an infinite bus. All reactances are given in per-unit on a common system base. If the infinite bus receives 1.0 per unit real power at 0.95 p.f. lagging, determine (a) the internal voltage of the generator, and (b) the equation for the electrical power delivered by the generator versus its power angle δ.

FIGURE 12.4
Single-line diagram for Example 12.3



SOLUTION

a. The equivalent circuit is shown in Figure 12.5, from which the equivalent reactance between the machine internal voltage and infinite bus is

$$\begin{aligned} X_{eq} &= X'_d + X_{TR} + X_{12} \| (X_{13} + X_{23}) \\ &= 0.30 + 0.10 + 0.20 \| (0.10 + 0.20) \\ &= 0.520 \text{ per unit} \end{aligned}$$

The current into the infinite bus is

$$I = \frac{P}{V_{bus}(\text{p.f.})} \underline{/-\cos^{-1}(\text{p.f.})} = \frac{(1.0)}{(1.0)(0.95)} \underline{/-\cos^{-1} 0.95}$$

$$= 1.05263 \underline{/-18.195^\circ} \text{ per unit}$$

(*Continued*)

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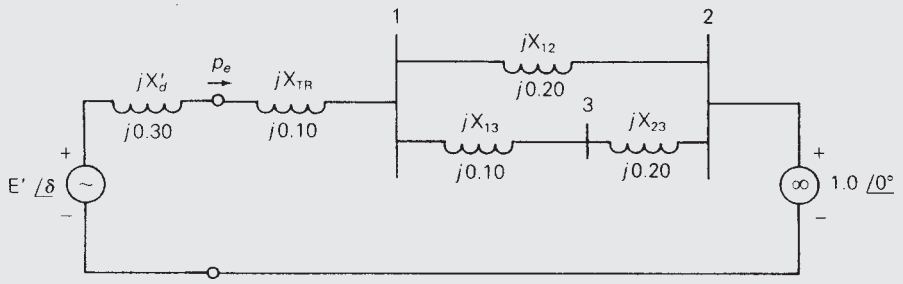
FIGURE 12.5
Equivalent circuit for Example 12.3



and the machine internal voltage is

$$E' = \mathrm{E}'\angle\delta = V_{\text{bus}} + j\mathrm{X}_{\text{eq}}I$$

$$= 1.0\angle 0° + (j0.520)(1.05263\angle -18.195°)$$

$$= 1.0\angle 0° + 0.54737\angle 71.805°$$

$$= 1.1709 + j0.5200$$

$$= 1.2812\angle 23.946° \quad \text{per unit}$$

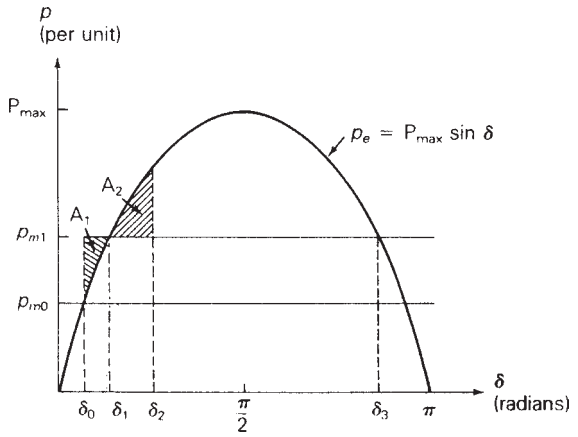
b. From (12.2.1),

$$p_e = \frac{(1.2812)(1.0)}{0.520}\sin\delta = 2.4628\sin\delta \quad \text{per unit}$$

12.3 THE EQUAL-AREA CRITERION

Consider a synchronous generating unit connected through a reactance to an infinite bus. Plots of electrical power p_e and mechanical power p_m versus power angle δ are shown in Figure 12.6. p_e is a sinusoidal function of δ, as given by (12.2.1).

FIGURE 12.6
p_e and p_m versus δ



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Suppose the unit is initially operating in steady state at $p_e = p_m = p_{m0}$ and $\delta = \delta_0$, when a step change in p_m from p_{m0} to p_{m1} occurs at $t = 0$. Due to rotor inertia, the rotor position cannot change instantaneously. That is, $\delta_m(0^+) = \delta_m(0^-)$; therefore, $\delta(0^+) = \delta(0^-) = \delta_0$ and $p_e(0^+) = p_e(0^-)$. Since $p_m(0^+) = p_{m1}$ is greater than $p_e(0^+)$, the acceleration power $p_a(0^+)$ is positive and, from (12.1.16), $(d^2\delta)/(dt^2)(0^+)$ is positive. The rotor accelerates and δ increases. When δ reaches δ_1, $p_e = p_{m1}$ and $(d^2\delta)\cdot(dt^2)$ becomes zero. However, $d\delta/dt$ is still positive and δ continues to increase, overshooting its final steady-state operating point. When δ is greater than δ_1, p_m is less than p_e, p_a is negative, and the rotor decelerates. Eventually, δ reaches a maximum value δ_2 and then swings back toward δ_1. Using (12.1.16), which has no damping, δ would continually oscillate around δ_1. However, damping due to mechanical and electrical losses causes δ to stabilize at its final steady-state operating point δ_1. Note that if the power angle exceeded δ_3, then p_m would exceed p_e and the rotor would accelerate again, causing a further increase in δ and loss of stability.

One method for determining stability and maximum power angle is to solve the nonlinear swing equation via numerical integration techniques using a digital computer. This method, which is applicable to multi-machine systems, is described in Section 12.4. However, there is also a direct method for determining stability that does not involve solving the swing equation; this method is applicable for one machine connected to an infinite bus or for two machines. The method, called the *equal-area criterion*, is described in this section.

In Figure 12.6, p_m is greater than p_e during the interval $\delta_0 < \delta < \delta_1$, and the rotor is accelerating. The shaded area A_1 between the p_m and p_e curves is called the accelerating area. During the interval $\delta_1 < \delta < \delta_2$, p_m is less than p_e, the rotor is decelerating, and the shaded area A_2 is the decelerating area. At both the initial value $\delta = \delta_0$ and the maximum value $\delta = \delta_2$, $d\delta/dt = 0$. The equal-area criterion states that $A_1 = A_2$.

To derive the equal-area criterion for one machine connected to an infinite bus, assume $\omega_{\text{p.u.}}(t) = 1$ in (12.1.16), giving

$$\frac{2H}{\omega_{syn}}\frac{d^2\delta}{dt^2} = p_{m\text{p.u.}} - p_{e\text{p.u.}} \tag{12.3.1}$$

Multiplying by $d\delta/dt$ and using

$$\frac{d}{dt}\left[\frac{d\delta}{dt}\right]^2 = 2\left(\frac{d\delta}{dt}\right)\left(\frac{d^2\delta}{dt^2}\right)$$

(12.3.1) becomes

$$\frac{2H}{\omega_{syn}}\left(\frac{d^2\delta}{dt^2}\right)\left(\frac{d\delta}{dt}\right) = \frac{H}{\omega_{syn}}\frac{d}{dt}\left[\frac{d\delta}{dt}\right]^2 = (p_{m\text{p.u.}} - p_{e\text{p.u.}})\frac{d\delta}{dt} \tag{12.3.2}$$

Multiplying (12.3.2) by dt and integrating from δ_0 to δ,

$$\frac{H}{\omega_{syn}}\int_{\delta_0}^{\delta} d\left[\frac{d\delta}{dt}\right]^2 = \int_{\delta_0}^{\delta}(p_{m\text{p.u.}} - p_{e\text{p.u.}})d\delta$$

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or

$$\frac{\mathrm{H}}{\omega_{\text{syn}}}\left[\frac{d\delta}{dt}\right]^2\Bigg|_{\delta_0}^{\delta} = \int_{\delta_0}^{\delta}(p_{m\text{p.u.}} - p_{e\text{p.u.}})\,d\delta \tag{12.3.3}$$

The above integration begins at δ_0 where $d\delta/dt = 0$, and continues to an arbitrary δ. When δ reaches its maximum value, denoted δ_2, $d\delta/dt$ again equals zero. Therefore, the left-hand side of (12.3.3) equals zero for $\delta = \delta_2$ and

$$\int_{\delta_0}^{\delta_2}(p_{m\text{p.u.}} - p_{e\text{p.u.}})d\delta = 0 \tag{12.3.4}$$

Separating this integral into positive (accelerating) and negative (decelerating) areas, results in the equal-area criterion

$$\int_{\delta_0}^{\delta_1}(p_{m\text{p.u.}} - p_{e\text{p.u.}})d\delta + \int_{\delta_1}^{\delta_2}(p_{m\text{p.u.}} - p_{e\text{p.u.}})d\delta = 0$$

or

$$\int_{\delta_0}^{\delta_1}\underbrace{(p_{m\text{p.u.}} - p_{e\text{p.u.}})d\delta}_{\mathrm{A}_1} = \int_{\delta_1}^{\delta_2}\underbrace{(p_{e\text{p.u.}} - p_{m\text{p.u.}})d\delta}_{\mathrm{A}_2} \tag{12.3.5}$$

In practice, sudden changes in mechanical power usually do not occur, since the time constants associated with prime mover dynamics are on the order of seconds. However, stability phenomena similar to that described above can also occur from sudden changes in electrical power, due to system faults and line switching. The following three examples are illustrative.

EXAMPLE 12.4

Equal-area criterion: transient stability during a three-phase fault

The synchronous generator shown in Figure 12.4 is initially operating in the steady-state condition given in Example 12.3, when a temporary three-phase-to-ground bolted short circuit occurs on line 1–3 at bus 1, shown as point F in Figure 12.4. Three cycles later, the fault extinguishes by itself. Due to a relay fault, all circuit breakers remain closed. Determine whether stability is or is not maintained and determine the maximum power angle. The inertia constant of the generating unit is 3.0 per unit-seconds on the system base. Assume p_m remains constant throughout the disturbance. Also, assume $\omega_{\text{p.u.}} = (t) = 1.0$ in the swing equation.

SOLUTION

Plots of p_e and p_m versus δ are shown in Figure 12.7. From Example 12.3, the initial operating point is $p_e(0^-) = p_m = 1.0$ per unit and $\delta(0^+) = \delta(0^-) = \delta_0 = 23.95° = 0.4179$ radian. At $t = 0$, when the short circuit occurs, p_e instantaneously

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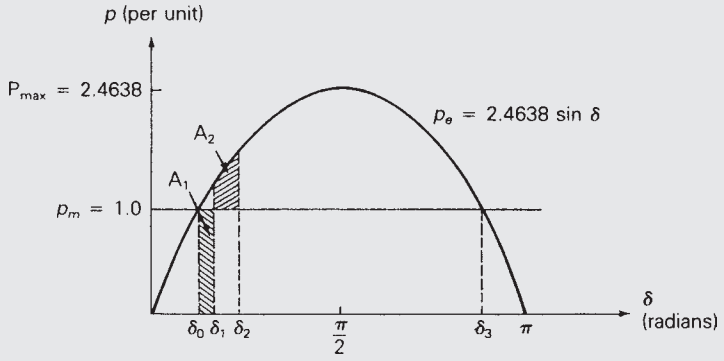
FIGURE 12.7

p–δ plot for Example 12.4



drops to zero and remains at zero during the fault since power cannot be transferred past faulted bus 1. From (12.1.16), with $\omega_{\text{p.u.}}(t) = 1.0$,

$$\frac{2\text{H}}{\omega_{\text{syn}}}\frac{d^2\delta(t)}{dt^2} = p_{m\text{p.u.}} \qquad 0 \leqslant t \leqslant 0.05 \quad \text{s}$$

Integrating twice with initial condition $\delta(0) = \delta_0$ and $\dfrac{d\delta(0)}{dt} = 0$,

$$\frac{d\delta(t)}{dt} = \frac{\omega_{\text{syn}}\, p_{m\text{p.u.}}}{2\text{H}}t + 0$$

$$\delta(t) = \frac{\omega_{\text{syn}}\, p_{m\text{p.u.}}}{4\text{H}}t^2 + \delta_0$$

At $t = 3$ cycles $= 0.05$ second,

$$\delta_1 = \delta(0.05\text{ s}) = \frac{2\pi 60}{12}(0.05)^2 + 0.4179$$

$$= 0.4964 \text{ radian} = 28.44°$$

The accelerating area A_1, shaded in Figure 12.7, is

$$\text{A}_1 = \int_{\delta_0}^{\delta_1} p_m d\delta = \int_{\delta_0}^{\delta_1} 1.0 d\delta = (\delta_1 - \delta_0) = 0.4964 - 0.4179 = 0.0785$$

At $t = 0.05$ s, the fault extinguishes and p_e instantaneously increases from zero to the sinusoidal curve in Figure 12.7. δ continues to increase until the decelerating area A_2 equals A_1. That is,

$$\text{A}_2 = \int_{\delta_1}^{\delta_2} (p_{\max}\sin\delta - p_m)d\delta$$

$$= \int_{0.4964}^{\delta_2} (2.4638\sin\delta - 1.0)d\delta = \text{A}_1 = 0.0785$$

Integrating,

$$2.4638[\cos(0.4964) - \cos\delta_2] - (\delta_2 - 0.4964) = 0.0785$$

$$2.4638\cos\delta_2 + \delta_2 = 2.5843$$

(*Continued*)

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The above nonlinear algebraic equation can be solved iteratively to obtain $\delta_2 = 0.7003$ radian $= 40.12°$.

Since the maximum angle δ_2 does not exceed $\delta_3 = (180° - \delta_0) = 156.05°$, stability is maintained. In steady state, the generator returns to its initial operating point $p_{ess} = p_m = 1.0$ per unit and $\delta_{ss} = \delta_0 = 23.95°$.

Note that as the fault duration increases, the risk of instability also increases. The *critical clearing time*, denoted t_{cr}, is the longest fault duration allowable for stability.

To see this case modeled in PowerWorld Simulator, open case Example 12_4 (see Figure 12.8). Then select **Add-Ons, Transient Stability**, which displays the Transient Stability Analysis Form. Notice that in the Transient Stability Contingency Elements list, a fault is applied to bus 1 at $t = 0$ s and cleared at $t = 0.05$ s (three cycles later). To see the time variation in the generator angle (modeled at bus 4 in PowerWorld Simulator), click the **Run Transient Stability** button. When the simulation is finished, a graph showing this angle automatically appears, as shown in the figure. More detailed results are also available by clicking on **Results** in the list on the left side of the form. To rerun the example with a different fault duration, modify the Time (second) field in the Transient Contingency Elements list, and then again click the **Run Transient Stability** button.

Notice that because this system is modeled without damping (i.e., D = 0), the angle oscillations do not damp out with time. To extend the example, right-click on the Bus 4 generator on the one-line diagram and select **Generator Information Dialog**. Then click on the **Stability**, **Machine Models** tab to see

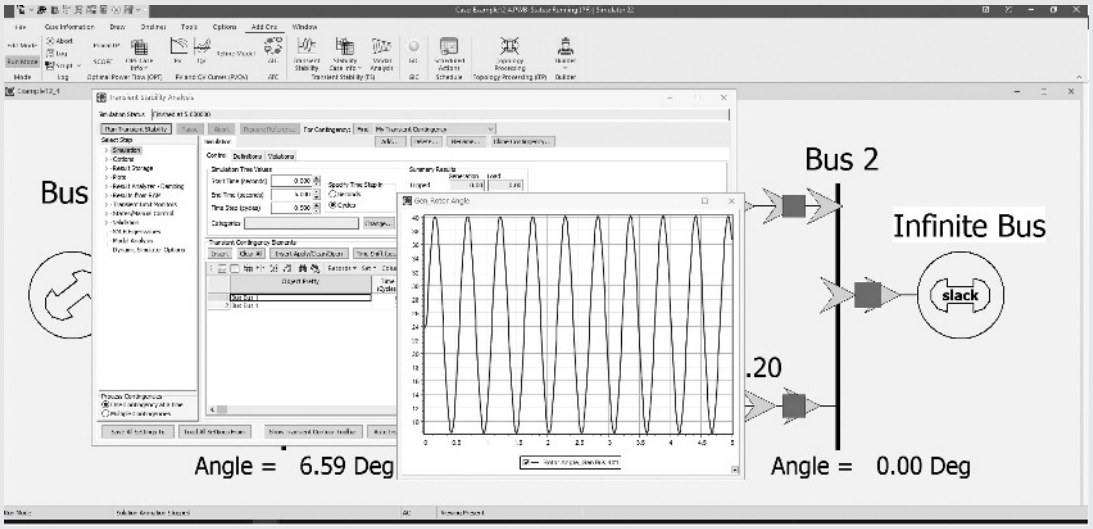

FIGURE 12.8

Variation in $\delta(t)$ without damping

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the parameters associated with the GENCLS model (i.e., a classical model—a more detailed machine model is introduced in Section 12.6). Change the D field to 1.0, select **OK** to close the dialog, and then rerun the transient stability case. The results are as shown in Figure 12.9. While the inclusion of damping did not significantly alter the maximum for δ(t), the magnitude of the angle oscillations is now decreasing with time. For convenience, this modified example is contained in PowerWorld Simulator case Example 12_4b.

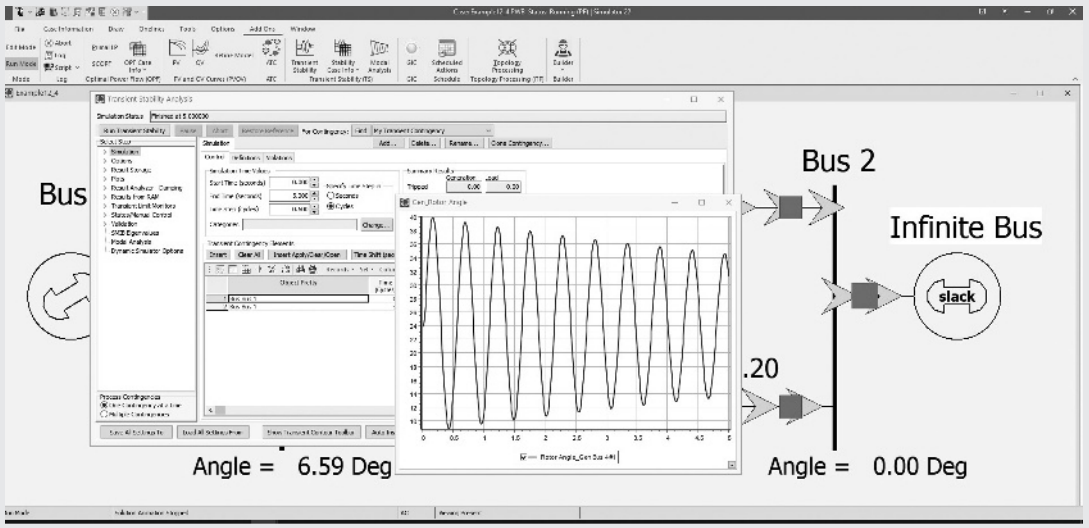

FIGURE 12.9

Variation in $\delta(t)$ with damping

EXAMPLE 12.5

Equal-area criterion: critical clearing time for a temporary three-phase fault

Assuming the temporary short circuit in Example 12.4 lasts longer than 3 cycles, calculate the critical clearing time.

(*Continued*)

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SOLUTION

The p–δ plot is shown in Figure 12.10. At the critical clearing angle, denoted δ_{cr}, the fault is extinguished. The power angle then increases to a maximum value $\delta_3 = 180° - \delta_0 = 156.05° = 2.7236$ radians, which gives the maximum decelerating area. Equating the accelerating and decelerating areas,

$$A_1 = \int_{\delta_0}^{\delta_{cr}} p_m \, d\delta = A_2 = \int_{\delta_{cr}}^{\delta_3} (P_{max} \sin\delta - p_m) d\delta$$

$$\int_{0.4179}^{\delta_{cr}} 1.0 \, d\delta = \int_{\delta_{cr}}^{2.7236} (2.4638 \sin\delta - 1.0) \, d\delta$$

Solving for δ_{cr},

$$(\delta_{cr} - 0.4179) = 2.4638[\cos\delta_{cr} - \cos(2.7236)] - (2.7236 - \delta_{cr})$$
$$2.4638 \cos\delta_{cr} = +0.05402$$
$$\delta_{cr} = 1.5489 \text{ radians} = 88.74°$$

From the solution to the swing equation given in Example 12.4,

$$\delta(t) = \frac{\omega_{syn} p_{mp.u.}}{4H} t^2 + \delta_0$$

Solving,

$$t = \sqrt{\frac{4H}{\omega_{syn} p_{mp.u.}} (\delta(t) - \delta_0)}$$

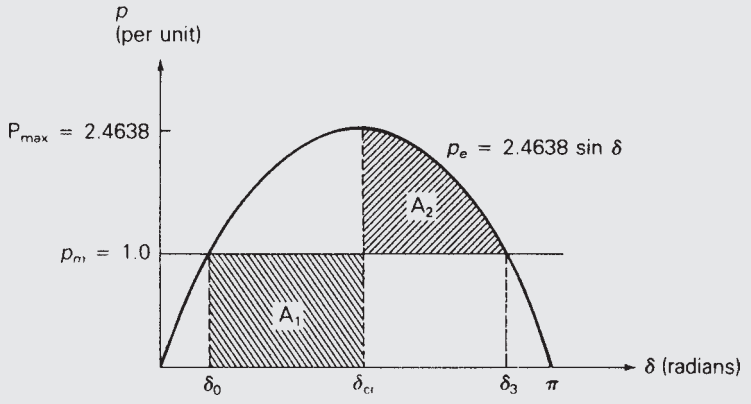

FIGURE 12.10
p–δ plot for Example 12.5

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Using $\delta(t_{cr}) = \delta_{cr} = 1.5489$ and $\delta_0 = 0.4179$ radian,

$$t_{cr} = \sqrt{\frac{12}{(2\pi 60)(1.0)}(1.5489 - 0.4179)}$$
$$= 0.1897 \text{ s} = 11.38 \text{ cycles}$$

If the fault is cleared before $t = t_{cr} = 11.38$ cycles, stability is maintained. Otherwise, the generator goes out of synchronism with the infinite bus; that is, stability is lost.

To see a time-domain simulation of this case, open Example 12_5 in PowerWorld Simulator (see Figure 12.11). Again select **Add-Ons, Transient Stability** to view the Transient Stability Analysis Form. In order to better visualize the results on a PowerWorld one-line diagram, there is an option to transfer the transient stability results to the one-line every n timesteps. To access this option, select the **Options** page from the list on the left side of the display, then the **General** tab, then check the **Transfer Results to Power Flow after Interval Check** field. With this option checked, click on the **Run Transient Stability**, which will run the case with a critical clearing time of 0.1895 seconds. The plot is set to dynamically update as well. The final results are shown in Figure 12.11. Because the oneline is reanimated every n time-steps (4 in this case), a potential downside to this option is it takes longer to run. Uncheck the option to restore full solution speed.

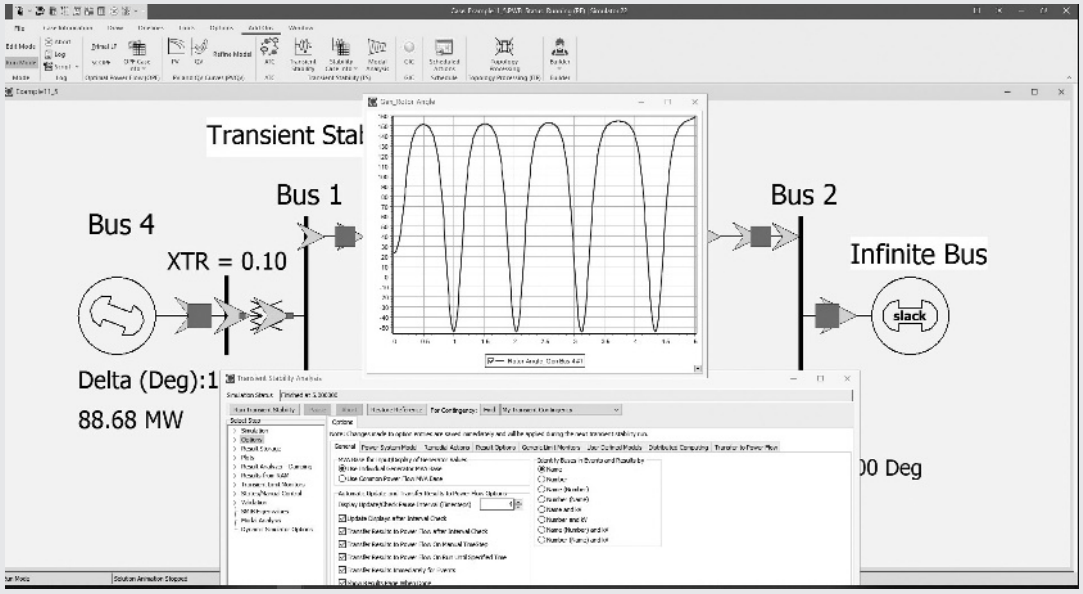

FIGURE 12.11

Variation in $\delta(t)$ for Example 12.5

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EXAMPLE 12.6

Equal-area criterion: critical clearing angle for a cleared three-phase fault

The synchronous generator in Figure 12.4 is initially operating in the steady-state condition given in Example 12.3 when a permanent three-phase-to-ground bolted short circuit occurs on line 1–3 at bus 3. The fault is cleared by opening the circuit breakers at the ends of line 1–3 and line 2–3. These circuit breakers then remain open. Calculate the critical clearing angle. As in previous examples, H = 3.0 p.u.-s, $p_m = 1.0$ per unit, and $\omega_{\text{p.u.}} = 1.0$ in the swing equation.

SOLUTION

From Example 12.3, the equation for the prefault electrical power, denoted p_{e1} here, is $p_{e1} = 2.4638 \sin\delta$ per unit. The faulted network is shown in Figure 12.12(a), and the Thévenin equivalent of the faulted network, as viewed from the generator internal voltage source, is shown in Figure 12.12(b). The Thévenin reactance is

$$X_{\text{Th}} = 0.40 + 0.20 \| 0.10 = 0.46666 \quad \text{per unit}$$

and the Thévenin voltage source is

$$V_{\text{Th}} = 1.0\angle 0^\circ \left[\frac{X_{13}}{X_{13} + X_{12}}\right] = 1.0\angle 0^\circ \frac{0.10}{0.30}$$

$$= 0.33333\angle 0^\circ \quad \text{per unit}$$

From Figure 12.12(b), the equation for the electrical power delivered by the generator to the infinite bus during the fault, denoted p_{e2}, is

$$p_{e2} = \frac{E' V_{\text{Th}}}{X_{\text{Th}}} \sin\delta = \frac{(1.2812)(0.3333)}{0.46666} \sin\delta = 0.9152 \sin\delta \quad \text{per unit}$$

The postfault network is shown in Figure 12.12(c), where circuit breakers have opened and removed lines 1–3 and 2–3. From this figure, the postfault electrical power delivered, denoted p_{e3}, is

$$p_{e3} = \frac{(1.2812)(1.0)}{0.60} \sin\delta = 2.1353 \sin\delta \quad \text{per unit}$$

The p–δ curves as well as the accelerating area A_1 and decelerating area A_2 corresponding to critical clearing are shown in Figure 12.12(d). Equating A_1 and A_2,

$$A_1 = \int_{\delta_0}^{\delta_{cr}} (p_m - P_{2\max} \sin\delta) d\delta = A_2 = \int_{\delta_{cr}}^{\delta_3} (P_{3\max} \sin\delta - p_m) d\delta$$

$$\int_{0.4179}^{\delta_{cr}} (1.0 - 0.9152 \sin\delta) d\delta = \int_{\delta_{cr}}^{2.6542} (2.1353 \sin\delta - 1.0) d\delta$$

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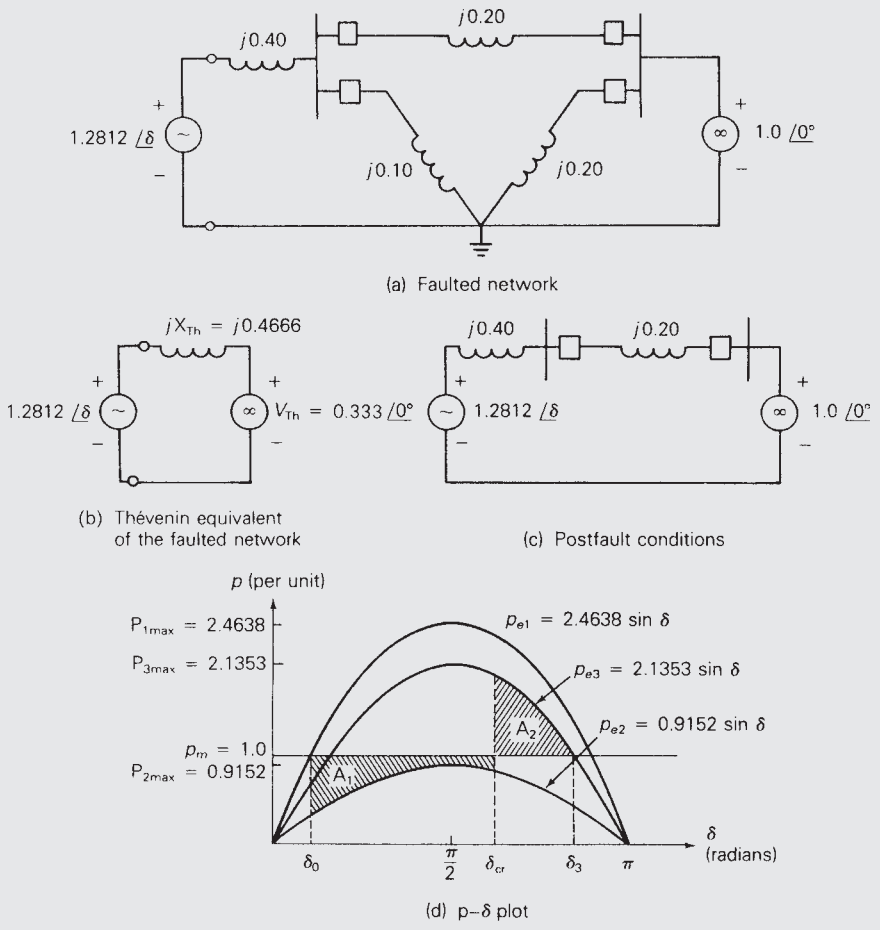
FIGURE 12.12
Example 12.6



Solving for δ_{cr},

$$(\delta_{cr} - 0.4179) + 0.9152(\cos\delta_{cr} - \cos 0.4179)$$

$$= 2.1353(\cos\delta_{cr} - \cos 2.6542) - (2.6542 - \delta_{cr})$$

$$-1.2201\cos\delta_{cr} = 0.4868$$

$$\delta_{cr} = 1.9812 \text{ radians} = 113.5°$$

If the fault is cleared before $\delta = \delta_{cr} = 113.5°$, stability is maintained. Otherwise, stability is lost. To see this case in PowerWorld Simulator, open case Example 12_6.

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12.4 NUMERICAL INTEGRATION OF THE SWING EQUATION

The equal-area criterion is applicable to one machine and an infinite bus or to two machines. For multi-machine stability problems, however, numerical integration techniques can be employed to solve the swing equation for each machine.

Given a first-order differential equation,

$$\frac{dx}{dt} = f(x) \tag{12.4.1}$$

one relatively simple integration technique is Euler's method [1], illustrated in Figure 12.13. The integration step size is denoted Δt. Calculating the slope at the beginning of the integration interval, from (12.4.1),

$$\frac{dx_t}{dt} = f(x_t) \tag{12.4.2}$$

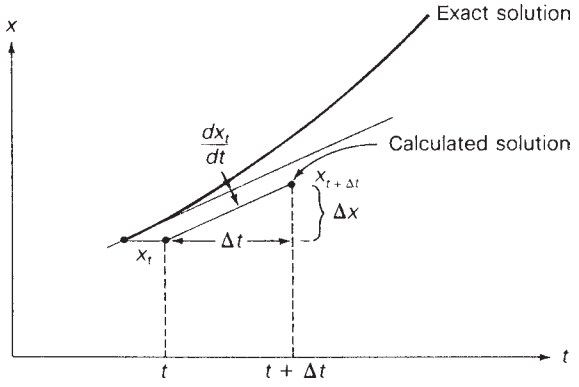
The new value $x_{t+\Delta t}$ is calculated from the old value x_t by adding the increment Δx,

$$x_{t+\Delta t} = x_t + \Delta x = x_t + \left(\frac{dx_t}{dt}\right)\Delta t \tag{12.4.3}$$

As shown in the figure, Euler's method assumes that the slope is constant over the entire interval Δt. An improvement can be obtained by calculating the slope at both the beginning and end of the interval, and then averaging these slopes. The modified Euler's method is illustrated in Figure 12.14. First, the slope at the beginning of the interval is calculated from (12.4.1) and used to calculate a preliminary value $\tilde{x}$ given by

$$\tilde{x} = x_t + \left(\frac{dx_t}{dt}\right)\Delta t \tag{12.4.4}$$

FIGURE 12.13
Euler's method



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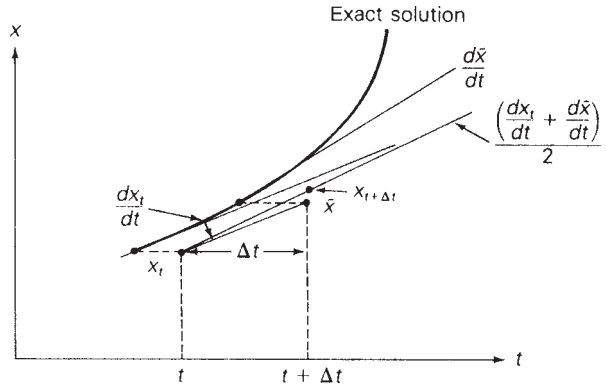

FIGURE 12.14
Modified Euler's method

Next, the slope at $\tilde{x}$ is calculated:

$$\frac{d\tilde{x}}{dt} = f(\tilde{x}) \tag{12.4.5}$$

Then, the new value is calculated using the average slope:

$$x_{t+\Delta t} = x_t + \frac{\left(\frac{dx_t}{dt} + \frac{d\tilde{x}}{dt}\right)}{2}\Delta t \tag{12.4.6}$$

Now apply the modified Euler's method to calculate machine frequency ω and power angle δ. Letting x be either δ or ω, the old values at the beginning of the interval are denoted δ_t and ω_t. From (12.1.17) and (12.1.18), the slopes at the beginning of the interval are

$$\frac{d\delta_t}{dt} = \omega_t - \omega_{\text{syn}} \tag{12.4.7}$$

$$\frac{d\omega_t}{dt} = \frac{p_{a\text{p.u.}t}\omega_{\text{syn}}}{2\text{H}\omega_{\text{p.u.}t}} \tag{12.4.8}$$

where $p_{a\text{p.u.}t}$ is the per-unit accelerating power calculated at $\delta = \delta_t$ and $\omega_{\text{p.u.}t} = \omega_t/\omega_{\text{syn}}$. Applying (12.4.4), preliminary values are

$$\tilde{\delta} = \delta_t + \left(\frac{d\delta_t}{dt}\right)\Delta t \tag{12.4.9}$$

$$\tilde{\omega} = \omega_t + \left(\frac{d\omega_t}{dt}\right)\Delta t \tag{12.4.10}$$

Next, the slopes at $\tilde{\delta}$ and $\tilde{\omega}$ *are* calculated, again using (12.1.17) and (12.1.18):

$$\frac{d\tilde{\delta}}{dt} = \tilde{\omega} - \omega_{\text{syn}} \tag{12.4.11}$$

$$\frac{d\tilde{\omega}}{dt} = \frac{\tilde{p}_{a\text{p.u.}}\omega_{\text{syn}}}{2\text{H}\tilde{\omega}_{\text{p.u.}}} \tag{12.4.12}$$

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where $\tilde{p}_{a\text{p.u.}}$ is the per-unit accelerating power calculated at $\delta = \tilde{\delta}$, and $\tilde{\omega}_{\text{p.u.}} = \tilde{\omega}/\omega_{\text{syn}}$. Applying (12.4.6), the new values at the end of the interval are

$$\delta_{t+\Delta t} = \delta_t + \frac{\left(\frac{d\delta_t}{dt} + \frac{d\tilde{\delta}}{dt}\right)}{2}\Delta t \tag{12.4.13}$$

$$\omega_{t+\Delta t} = \omega_t + \frac{\left(\frac{d\omega_t}{dt} + \frac{d\tilde{\omega}}{dt}\right)}{2}\Delta t \tag{12.4.14}$$

This procedure, given by (12.4.7) through (12.4.13), begins at $t = 0$ with specified initial values δ_0 and ω_0, and continues iteratively until $t = T$, a specified final time.

In addition to Euler's method, there are many other numerical integration techniques, such as Runge-Kutta, Picard's method, and Milne's predictor-corrector method [1]. Comparison of the methods shows a trade-off of accuracy versus computation complexity. The Euler method is a relatively simple method to compute, but requires a small step size Δt for accuracy. Some of the other methods can use a larger step size for comparable accuracy, but the computations are more complex.

To see this case in PowerWorld Simulator, open case Example 12_7, which plots both the generator angle and speed. However, rather than showing the speed in radians per second, Hertz is used. Also, in addition to plotting the angle and speed versus time, the case includes a "phase portrait" in which the speed is plotted as a function of the angle. Numeric results are available by clicking on the **Results** page. The results shown in PowerWorld differ slightly from those in the table because PowerWorld uses a more exact second order integration method.

EXAMPLE 12.7

Euler's method: computer solution to swing equation and critical clearing time

Verify the critical clearing angle determined in Example 12.6, and calculate the critical clearing time by applying the modified Euler's method to solve the swing equation for the following two cases:

Case 1 The fault is cleared at $\delta = 1.95$ radians $= 112°$ (which is less than δ_{cr})

Case 2 The fault is cleared at $\delta = 2.09$ radians $= 120°$ (which is greater than δ_{cr})

For calculations, use a step size $\Delta t = 0.01$ s, and solve the swing equation from $t = 0$ to $t = T = 0.85$ s.

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SOLUTION

Equations (12.4.7) through (12.4.14) are solved by a computer program written in BASIC. From Example 12.6, the initial conditions at $t = 0$ are

$\delta_0 = 0.4179$ rad
$\omega_0 = \omega_{syn} = 2\pi 60$ rad/s

Also, the H constant is 3.0 p.u.-s, and the faulted accelerating power is

$p_{a\text{p.u.}} = 1.0 - 0.9152 \sin \delta$

The postfault accelerating power is

$p_{a\text{p.u.}} = 1.0 - 2.1353 \sin \delta$ per unit

The computer program and results at 0.02 s printout intervals are listed in Table 12.1. As shown, these results agree with Example 12.6, since the system is stable for Case 1 and unstable for Case 2. Also from Table 12.1, the critical clearing time is between 0.34 and 0.36 s.

Case 1 Stable			Case 2 Unstable			
Time s	Delta rad	Omega rad/s	Time s	Delta rad	Omega rad/s	Program Listing
0.000	0.418	376.991	0.000	0.418	376.991	
0.020	0.426	377.778	0.020	0.426	377.778	10 REM EXAMPLE 12.7
0.040	0.449	378.547	0.040	0.449	378.547	20 REM SOLUTION TO SWING EQUATION
0.060	0.488	379.283	0.060	0.488	379.283	30 REM THE STEP SIZE IS DELTA
0.080	0.541	379.970	0.080	0.541	379.970	40 REM THE CLEARING ANGLE IS DLTCLR
0.100	0.607	380.599	0.100	0.607	380.599	50 DELTA = 0.01
0.120	0.685	381.159	0.120	0.685	381.159	60 DLTCLR = 1.95
0.140	0.773	381.646	0.140	0.773	381.646	70 J = 1
0.160	0.870	382.056	0.160	0.870	382.056	80 PMAX = 0.9152
0.180	0.975	382.392	0.180	0.975	382.392	90 PI = 3.1415927 #
0.200	1.086	382.660	0.200	1.086	382.660	100 T = 0
0.220	1.202	382.868	0.220	1.202	382.868	110 X1 = 0.4179
0.240	1.321	383.027	0.240	1.321	383.027	120 X2 = 2*PI*60
0.260	1.443	383.153	0.260	1.443	383.153	130 LPRINT "TIME DELTA OMEGA"
0.280	1.567	383.262	0.280	1.567	383.262	140 LPRINT"s rad rad/s"
0.300	1.694	383.370	0.300	1.694	383.370	150 LPRINT USING "#####.###"; T;X1;X2
0.320	1.823	383.495	0.320	1.823	383.495	160 FOR K = 1 TO 86
0.340	1.954	383.658	0.340	1.954	383.658	170 REM LINE 180 IS EQ (12.4.7)
	Fault Cleared		0.360	2.090	383.876	180 X3 = X2 – (2*PI*60)
0.360	2.076	382.516		Fault Cleared		190 IF J = 2 THEN GOTO 240
0.380	2.176	381.510	0.380	2.217	382.915	200 IF X1 > DLTCLR OR X1 = DLTCLR THEN
0.400	2.257	380.638	0.400	2.327	382.138	PMAX = 2.1353
0.420	2.322	379.886	0.420	2.424	381.546	210 IF X1 > DLTCLR OR X1 = DLTCLR THEN
0.440	2.373	379.237	0.440	2.511	381.135	LPRINT "FAULT CLEARED"
0.460	2.413	378.674	0.460	2.591	380.902	220 IF X1 > DLTCLR OR X1 = DLTCLR THEN
0.480	2.441	378.176	0.480	2.668	380.844	J = 2

(*Continued*)

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0.500	2.460	377.726	0.500	2.746	380.969	230 REM LINES 240 AND 250 ARE EQ (12.4.8)
0.520	2.471	377.307	0.520	2.828	381.288	240 X4=1- PMAX*SIN(X1)
0.540	2.473	376.900	0.540	2.919	381.824	250 X5 = X4 * (2 * PI * 60) * (2 * PI * 60)/(6 * X2)
0.560	2.467	376.488	0.560	3.022	382.609	260 REM LINE 270 IS EQ (12.4.9)
0.580	2.453	376.056	0.580	3.145	383.686	270 X6 = X1 +X3* DELTA
0.600	2.429	375.583	0.600	3.292	385.111	280 REM LINE 290 IS EQ (12.4.10)
0.620	2.396	375.053	0.620	3.472	386.949	290 X7 = X2 + X5* DELTA
0.640	2.351	374.446	0.640	3.693	389.265	300 REM LINE 310 IS EQ (12.4.11)
0.660	2.294	373.740	0.660	3.965	392.099	310 X8 = X7-2*PI*60
0.680	2.221	372.917	0.680	4.300	395.426	320 REM LINES 330 AND 340 ARE EQ (12.4.12)
0.700	2.130	371.960	0.700	4.704	399.079	330 X9 = 1-PMAX*SIN (X6)
0.720	2.019	370.855	0.720	5.183	402.689	340 X10 = X9 * (2 * PI * 60) * (2 * PI * 60)/(6 * X7)
0.740	1.884	369.604	0.740	5.729	405.683	350 REM LINE 360 IS EQ (12.4.13)
0.760	1.723	368.226	0.760	6.325	407.477	360 X1 = X1 + (X3 + X8) * (DELTA/2)
0.780	1.533	366.773	0.780	6.941	407.812	370 REM LINE 380 IS EQ (12.4.14)
0.800	1.314	365.341	0.800	7.551	406.981	380 X2 = X2 + (X5 + X10) * (DELTA/2)
0.820	1.068	364.070	0.820	8.139	405.711	390 T = K* DELTA
0.840	0.799	363.143	0.840	8.702	404.819	400 Z = K/2
0.860	0.516	362.750	0.860	9.257	404.934	410 M = INT (Z)
						420 IF M = Z THEN LPRINT USING "#####.###"; T;X1;X2
						430 NEXT K
						440 END

TABLE 12.1

Computer calculation of swing curves for Example 12.7

12.5 MULTIMACHINE STABILITY

The numerical integration methods discussed in Section 12.4 can be used to solve the swing equations for a multi-machine stability problem. However, a method is required for computing machine output powers for a general network. Figure 12.15 shows a general N-bus power system with M synchronous machines. Each machine is the same as that represented by the simplified model of Figure 12.2, and the internal machine voltages are denoted $E'_1, E'_2, \ldots, E'_M$. The M machine terminals

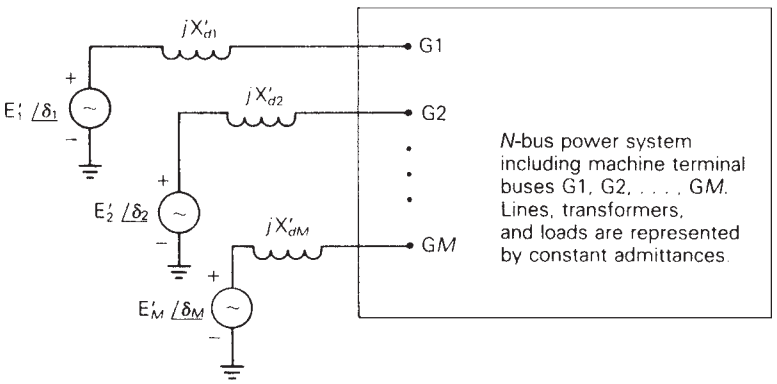

FIGURE 12.15

N-bus power-system representation for transient stability studies

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are connected to system buses denoted Gl, G2,…, GM in Figure 12.15. Initially, for simplicity, all the loads are modeled as constant admittances. Writing nodal equations for this network,

$$\begin{bmatrix} \boldsymbol{Y}_{11} & \boldsymbol{Y}_{12} \\ \boldsymbol{Y}_{12}^{\mathrm{T}} & \boldsymbol{Y}_{22} \end{bmatrix} \begin{bmatrix} \boldsymbol{V} \\ \boldsymbol{E} \end{bmatrix} = \begin{bmatrix} \boldsymbol{0} \\ \boldsymbol{I} \end{bmatrix} \tag{12.5.1}$$

where

$$\boldsymbol{V} = \begin{bmatrix} V_1 \\ V_2 \\ \vdots \\ V_N \end{bmatrix} \text{ is the } N \text{ vector of bus voltages} \tag{12.5.2}$$

$$\boldsymbol{E} = \begin{bmatrix} E'_1 \\ E'_2 \\ \vdots \\ E'_M \end{bmatrix} \text{ is the } M \text{ vector of machine voltages} \tag{12.5.3}$$

$$\boldsymbol{I} = \begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_M \end{bmatrix} \text{ is the } M \text{ vector of machine currents (these are current sources)} \tag{12.5.4}$$

$$\begin{bmatrix} \boldsymbol{Y}_{11} & \boldsymbol{Y}_{12} \\ \boldsymbol{Y}_{12}^{\mathrm{T}} & \boldsymbol{Y}_{22} \end{bmatrix} \text{ is an } (N + M) \times (N + M) \text{ admitance matrix} \tag{12.5.5}$$

The admittance matrix in (12.5.5) is partitioned in accordance with the N system buses and M internal machine buses, as follows:

$\boldsymbol{Y}_{11}$ is $N \times N$
$\boldsymbol{Y}_{12}$ is $N \times M$
$\boldsymbol{Y}_{22}$ is $M \times M$

$\boldsymbol{Y}_{11}$ is similar to the bus admittance matrix used for power flows in Chapter 6, except that load admittances and inverted generator impedances are included. That is, if a load is connected to bus n, then that load admittance is added to the diagonal element Y_{11nn}. Also, $(1/jX'_{dn})$ is added to the diagonal element Y_{11GnGn}.

$\boldsymbol{Y}_{22}$ is a diagonal matrix of inverted generator impedances; that is,

$$\boldsymbol{Y}_{22} = \begin{bmatrix} \dfrac{1}{j\mathrm{X}'_{d1}} & & & 0 \\ & \dfrac{1}{j\mathrm{X}'_{d2}} & & \\ & & \ddots & \\ 0 & & & \dfrac{1}{j\mathrm{X}'_{dM}} \end{bmatrix} \tag{12.5.6}$$

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Also, the kmth element of $\boldsymbol{Y}_{12}$ is

$$Y_{12km} = \begin{cases} \dfrac{-1}{jX'_{dn}} & \text{if } k = Gn \text{ and } m = n \\ 0 & \text{otherwise} \end{cases} \tag{12.5.7}$$

Writing (12.5.1) as two separate equations,

$$\boldsymbol{Y}_{11}\boldsymbol{V} + \boldsymbol{Y}_{12}\boldsymbol{E} = \boldsymbol{0} \tag{12.5.8}$$

$$\boldsymbol{Y}_{12}^{\mathrm{T}}\boldsymbol{V} + \boldsymbol{Y}_{22}\boldsymbol{E} = \boldsymbol{I} \tag{12.5.9}$$

Assuming $\boldsymbol{E}$ is known, (12.5.8) is a linear equation in $\boldsymbol{V}$ that can be solved either iteratively or by Gauss elimination. Using the Gauss-Seidel iterative method given by (6.2.9), the kth component of $\boldsymbol{V}$ is

$$V_k(i+1) = \frac{1}{Y_{11kk}}\left[-\sum_{n=1}^{M} Y_{12kn}E_n - \sum_{n=1}^{k-1} Y_{11kn}V_n(i+1) - \sum_{n=k+1}^{N} Y_{11kn}V_n(i)\right] \tag{12.5.10}$$

After $\boldsymbol{V}$ is computed, the machine currents can be obtained from (12.5.9). That is,

$$\boldsymbol{I} = \begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_M \end{bmatrix} = \boldsymbol{Y}_{12}^{T}\boldsymbol{V} + \boldsymbol{Y}_{22}\boldsymbol{E} \tag{12.5.11}$$

The (real) electrical power output of machine n is then

$$p_{en} = \mathrm{Re}[E_n I_n^*] \qquad n = 1,2,\ldots,M \tag{12.5.12}$$

The computation procedure for solving a power system stability simulation alternately solves the differential equations (currently just the swing equations representing the machines) and the above algebraic power-flow equations representing the network. Use the modified Euler method of Section 12.4 to solve the swing equations and the Newton-Raphson iterative method to solve the algebraic equations. The procedure is outlined in the following 11 steps.

POWER SYSTEM STABILITY COMPUTATION PROCEDURE

STEP 1 Run a prefault power-flow program to compute initial bus voltages V_k, $k = 1, 2,\ldots,N$, initial machine currents I_n, and initial machine electrical power outputs p_{en}, $n = 1, 2,\ldots,M$. Set machine mechanical

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power outputs, $p_{mn} = p_{en}$. Set initial machine frequencies, $\omega_n = \omega_{\text{syn}}$. Compute the load admittances.

STEP 2 Compute the internal machine voltages:

$$E_n = E_n \angle \underline{\delta_n} = V_{Gn} + (jX'_{dn})I_n \quad n = 1, 2, \ldots, M$$

where V_{Gn} and I_n are computed in Step 1. The magnitudes E_n will remain constant throughout the study. The angles δ_n *are* the initial power angles.

STEP 3 Compute $\boldsymbol{Y}_{11}$. Modify the $(N \times N)$ power-flow bus admittance matrix by including the load admittances and inverted generator impedances.

STEP 4 Compute $\boldsymbol{Y}_{22}$ from (12.5.6) and Y_{12} from (12.5.7).

STEP 5 Set time $t = 0$.

STEP 6 Is there a switching operation, change in load, short circuit, or change in data? For a switching operation or change in load, modify the bus admittance matrix. For a short circuit, set the faulted bus voltage [in (12.5.10)] to zero.

STEP 7 Using the internal machine voltages $E_n = E_n \angle \underline{\delta_n}$, $n = 1, 2, \ldots, M$, with the values of δ_n at time t, compute the machine electrical powers p_{en} at time t from (12.5.10) to (12.5.12).

STEP 8 Using p_{en} computed in Step 7 and the values of δ_n and ω_n at time t, compute the preliminary estimates of power angles δ_n and machine speeds $\bar{\omega}_n$ at time $(t + \Delta t)$ from (12.4.7) to (12.4.10).

STEP 9 Using $E_n = E_n \angle \underline{\bar{\delta}_n}$, $n = 1, 2, \ldots, M$, compute the preliminary estimates of the machine electrical powers p_{en} at time $(t + \Delta t)$ from (12.5.10) to (12.5.12).

STEP 10 Using $\bar{p}_{en}$ computed in Step 9, as well as $\bar{\delta}_n$ and $\bar{\omega}_n$ computed in Step 8, compute the final estimates of power angles δ_n and machine speeds ω_n at time $(t + \Delta t)$ from (12.4.11) to (12.4.14).

STEP 11 Set time $t = t + \Delta t$. Stop if $t > T$. Otherwise, return to Step 6.

An important simulation parameter is the step size (time step), Δt, used in the numerical integration. Because the time required to solve a simulation varies inversely with the time step, a larger value would be preferred. However, if too large a value is chosen, then the solution accuracy may suffer, and for some integration methods, such as Euler's, the solution can experience numeric instability. To see an example of numeric instability, re-do the PowerWorld Simulator Example 12_7, except change the time step to 0.02 seconds. A typical time step for commercial stability simulations is 1/2 cycle (0.00833 seconds for a 60-Hz system).

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EXAMPLE 12.8

Modifying power-flow Y_{bus} for application to multi-machine stability

Consider a stability study for the power system given in Example 6.9, with the 184-Mvar shunt capacitor of Example 6.14 installed at bus 2. Machine transient reactances are $X'_{d1} = 0.05$ and $X'_{d2} = 0.025$ per unit on the system base. Determine the admittance matrices $\boldsymbol{Y}_{11}$, $\boldsymbol{Y}_{22}$, and $\boldsymbol{Y}_{12}$.

SOLUTION

From Example 6.9, the power system has $N = 5$ buses and $M = 2$ machines. The second row of the 5×5 bus admittance matrix used for power flows is calculated in Example 6.9. Calculating the other rows in the same manner, results in

$$Y_{\text{bus}} = \begin{bmatrix} (3.728 - j49.72) & 0 & 0 & 0 & (-3.728 + j49.72) \\ 0 & (2.68 - j26.46) & 0 & (-0.892 + j9.92) & (-1.784 + j19.84) \\ 0 & 0 & (7.46 - j99.44) & (-7.46 + j99.44) & 0 \\ 0 & (-0.892 + j9.92) & (-7.46 + j99.44) & (11.92 - j148.) & (-3.572 + j39.68) \\ (-3.728 + j49.72) & (-1.784 + j19.84) & 0 & (-3.572 + j39.68) & (9.084 - j108.6) \end{bmatrix} \text{ per unit}$$

To obtain $\boldsymbol{Y}_{11}$, $\boldsymbol{Y}_{\text{bus}}$ is modified by including load admittances and inverted generator impedances. From Table 6.1, the load at bus 3 is $P_{L3} + jQ_{L3} = 0.8 + j0.4$ per unit and the voltage at bus 3 is $V_3 = 1.05$ per unit. Representing this load as a constant admittance,

$$Y_{\text{load 3}} = \frac{P_{L3} - jQ_{L3}}{V_3^2} = \frac{0.8 - j0.4}{(1.05)^2} = 0.7256 - j0.3628 \quad \text{per unit}$$

Similarly, the load admittance at bus 2 is

$$Y_{\text{load 2}} = \frac{P_{L2} - jQ_{L2}}{V_2^2} = \frac{8 - j2.8 + j1.84}{(0.959)^2} = 8.699 - j1.044$$

where V_2 is obtained from Example 6.14 and the 184-Mvar (1.84 per unit) shunt capacitor bank is included in the bus 2 load.

The inverted generator impedances are, for machine 1 connected to bus 1,

$$\frac{1}{jX'_{d1}} = \frac{1}{j0.05} = -j20.0 \quad \text{per unit}$$

and for machine 2 connected to bus 3,

$$\frac{1}{jX'_{d2}} = \frac{1}{j0.025} = -j40.0 \quad \text{per unit}$$

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To obtain $\boldsymbol{Y}_{11}$, add $(1/jX'_{d1})$ to the first diagonal element of $\boldsymbol{Y}_{\text{bus}}$, add $\boldsymbol{Y}_{\text{load2}}$ to the second diagonal element, and add $Y_{\text{load 3}} + (1/jX'_{d2})$ to the third diagonal element. The 5×5 matrix $\boldsymbol{Y}_{11}$ is then

$$\boldsymbol{Y}_{11} = \begin{bmatrix} (3.728 - j69.72) & 0 & 0 & 0 & (-3.728 - j49.72) \\ 0 & (11.38 - j29.50) & 0 & (-0.892 + j9.92) & (-1.784 + j19.84) \\ 0 & 0 & (8.186 - j139.80) & (-7.46 + j99.44) & 0 \\ 0 & (-0.892 + j9.92) & (-7.46 + j99.44) & (11.92 - j148) & (-3.572 - j39.68) \\ (-3.728 + j49.72) & (-1.784 + j19.84) & 0 & (-3.572 + j39.68) & (9.084 - j108.6) \end{bmatrix} \text{ per unit}$$

From (12.5.6), the 2×2 matrix $\boldsymbol{Y}_{22}$ is

$$\boldsymbol{Y}_{22} = \begin{bmatrix} \dfrac{1}{jX'_{d1}} & 0 \\ 0 & \dfrac{1}{jX'_{d2}} \end{bmatrix} = \begin{bmatrix} -j20.0 & 0 \\ 0 & -j40.0 \end{bmatrix} \text{ per unit}$$

From Figure 6.2, generator 1 is connected to bus 1 (therefore, bus G1 = 1 and generator 2 is connected to bus 3 (therefore G2 = 3). From (12.5.7), the 5×2 matrix $\boldsymbol{Y}_{12}$ is

$$\boldsymbol{Y}_{12} = \begin{bmatrix} j20.0 & 0 \\ 0 & 0 \\ 0 & j40.0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \text{ per unit}$$

To see this case in PowerWorld Simulator, open case Example 12_8. To see the $\boldsymbol{Y}_{11}$ matrix entries, first display the Transient Stability Analysis Form, and then select **States/Manual Control, Transient Stability $\boldsymbol{Y}_{\text{bus}}$**. By default, this case is set to solve a self-clearing fault at bus 4 that extinguishes itself after three cycles (0.05 s). Both generators are modeled with H = 5.0 p.u.-s and D = 1.0 p.u.

For the bus 4 fault, Figure 12.16 shows the variation in the rotor angles for the two generators with respect to a 60-Hz synchronous reference frame. The angles are increasing with time because neither of the generators is modeled with a governor, and there is no infinite bus. While it is clear that the generator angles remain together, it is very difficult to tell from Figure 12.16 the exact variation in the angle differences. Therefore, power system stability programs usually report angle differences, either with respect to the angle at a specified bus or with respect to the average of all the generator angles. The latter is shown in Figure 12.17, which displays the results from the PowerWorld Simulator Example 12_8A case.

(*Continued*)

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FIGURE 12.16

Variation in the rotor angles with respect to a synchronous speed reference frame

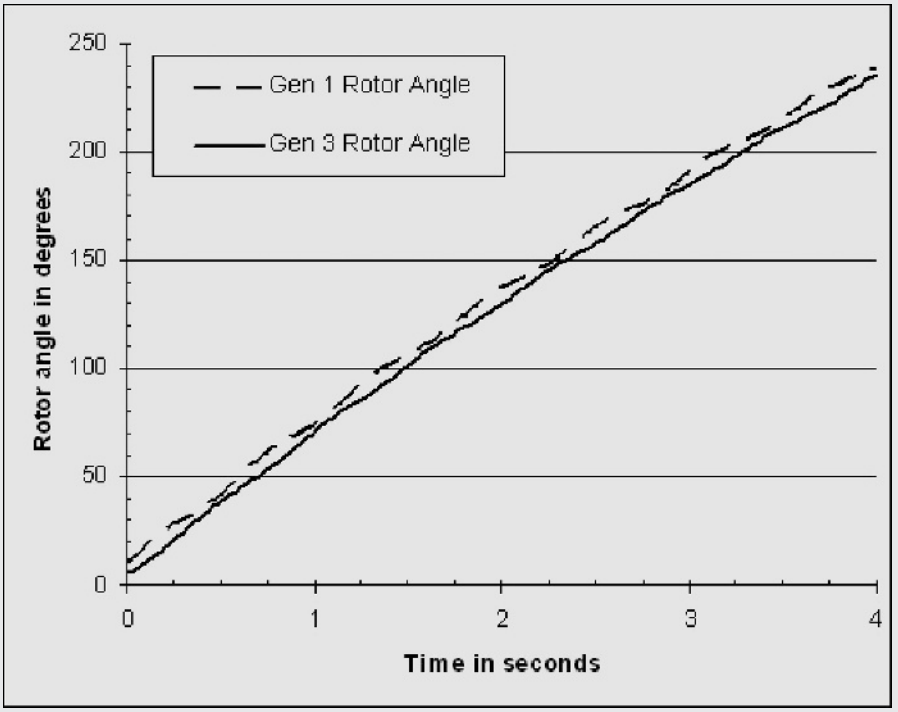
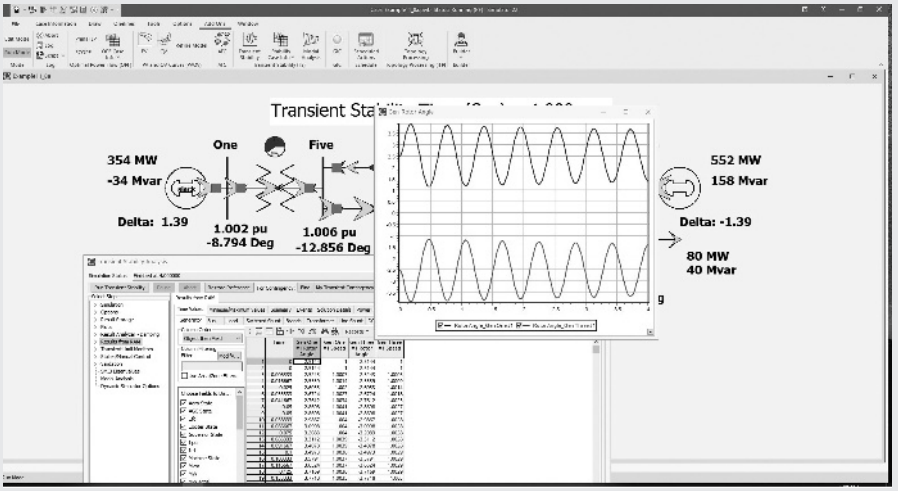


FIGURE 12.17

Relative variation of the rotor angles



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EXAMPLE 12.9

Stability results for 37-bus, 9-generator system

PowerWorld Simulator case Example 12_9 demonstrates a stability solution using the 37-bus system introduced in Chapter 6 with the system augmented to include classical models for each of the generators. By default, the case models a transmission line fault on the 69-kV line between bus 44 (PEACH69) and bus 14 (REDBUD69), with the fault at the bus 44 end of the line. The fault is cleared after 0.1 seconds by opening this transmission line. The results from this simulation are shown in Figure 12.18, with the largest generator angle variation occurring (not surprisingly) at the bus 44 generator. Notice that during and initially after the fault, the bus 44 generator's angle increases relative to all the other angles in the system. The critical clearing time for this fault is about 0.262 seconds.

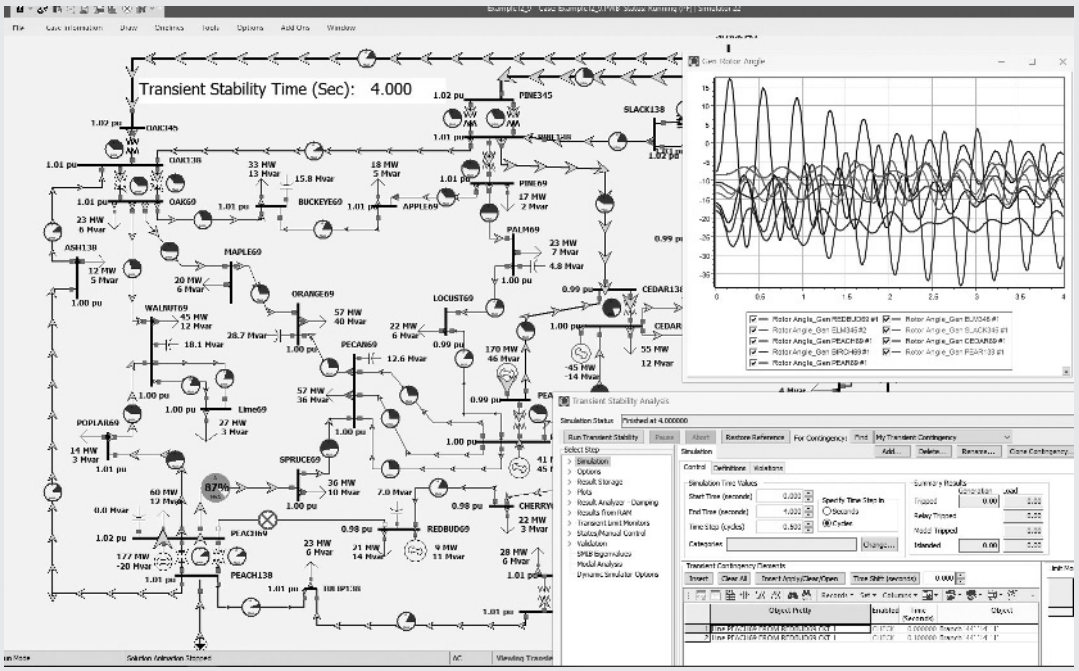

FIGURE 12.18

Rotor angles for Example 12.9 case

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12.6 A TWO-AXIS SYNCHRONOUS MACHINE MODEL

While the classical model for a synchronous machine provides a useful mechanism for introducing power system stability concepts, it is only appropriate for the most basic of system studies. Also, it is usually not coupled with the exciter and governor models that are introduced in the next chapter. In this section, a more realistic synchronous machine model is introduced.

The analysis of more detailed synchronous machine models requires that each machine model be expressed in a frame of reference that rotates at the same speed as its rotor. The standard approach is to use a *d-q* reference frame in which the major "direct" (*d*) axis is aligned with the rotor poles and the quadrature (*q*) axis leads the direct axis by 90°. The rotor angle δ is then defined as the angle by which the *q*-axis leads the network reference frame (see Figure 12.19). The equation for transforming the network quantities to the *d-q* reference frame is given by (12.6.1) and from the *d-q* reference frame by (12.6.2),

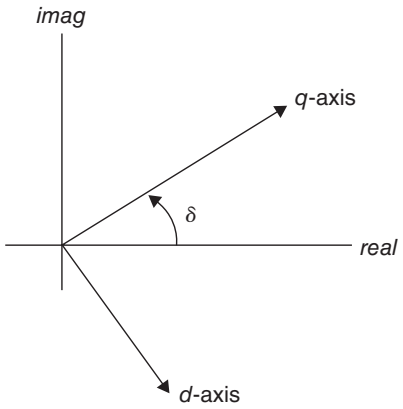
$$\begin{bmatrix} V_r \\ V_i \end{bmatrix} = \begin{bmatrix} \sin\delta & \cos\delta \\ -\cos\delta & \sin\delta \end{bmatrix} \begin{bmatrix} V_d \\ V_q \end{bmatrix} \tag{12.6.1}$$

$$\begin{bmatrix} V_d \\ V_q \end{bmatrix} = \begin{bmatrix} \sin\delta & -\cos\delta \\ \cos\delta & \sin\delta \end{bmatrix} \begin{bmatrix} V_r \\ V_i \end{bmatrix} \tag{12.6.2}$$

where the terminal voltage in the network reference frame is $V_T = V_r + jV_i$. A similar conversion is done for the currents.

Numerous different stability models exist for synchronous machines, most of which are beyond the scope of this text. The two-axis model, which models

FIGURE 12.19
Reference frame transformations



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the dynamics associated with the synchronous generator field winding and one damper winding, while neglecting the faster subtransient damper dynamics and stator transients, provides a nice compromise. For accessibility, machine saturation is not considered. With the two-axis model, the electrical behavior of the generator is represented by two algebraic equations and two differential equations

$$\mathrm{E}'_q = \mathrm{V}_q + \mathrm{R}_a\mathrm{I}_q + \mathrm{X}'_d\,\mathrm{I}_d \tag{12.6.3}$$

$$\mathrm{E}'_d = \mathrm{V}_d + \mathrm{R}_d\mathrm{I}_d - \mathrm{X}'_q\,\mathrm{I}_q \tag{12.6.4}$$

$$\frac{d\mathrm{E}'_q}{dt} = \frac{1}{\mathrm{T}'_{do}}\left(-\mathrm{E}'_q - (\mathrm{X}_d - \mathrm{X}'_d)\mathrm{I}_d + \mathrm{E}_{fd}\right) \tag{12.6.5}$$

$$\frac{d\mathrm{E}'_d}{dt} = \frac{1}{\mathrm{T}'_{qo}}\left(-\mathrm{E}'_d + (\mathrm{X}_q - \mathrm{X}'_q)\mathrm{I}_q\right) \tag{12.6.6}$$

where $\mathrm{V}_d + j\mathrm{V}_q$ and $\mathrm{I}_d + j\mathrm{I}_q$ are the generator's terminal voltage and current shifted into the generator's reference frame, and E_{fd} is proportional to the field voltage. The per-unit electrical torque, T_{elec} is then

$$\mathrm{T}_e = \mathrm{V}_d\mathrm{I}_d + \mathrm{V}_q\mathrm{I}_q + \mathrm{R}_a(\mathrm{I}_d^2 + \mathrm{I}_q^2) \tag{12.6.7}$$

While $p_e = \mathrm{T}_e\omega_{\mathrm{p.u.}}$, it is often assumed that $\omega_{\mathrm{p.u.}} = 1.0$ [10] with the result being an assumption that $p_e = \mathrm{T}_e$. When (12.6.5) and (12.6.6) are combined with generator mechanical equations presented in (12.1.18) and (12.1.19), substituting (12.6.7) for $p_{e\mathrm{p.u.}}$, the result is a synchronous generator model containing four first-order differential equations.

The initial value for δ can be determined by noting that in steady state, δ is the same as the angle of the internal voltage [10],

$$E = V_T + j\mathrm{X}_q I \tag{12.6.8}$$

Hence the initial value of δ is the angle on *E.* Once δ has been determined, (12.6.2) is used to transfer the generator terminal voltage and current into the generator's reference frame, and then (12.6.3), (12.6.4), and (12.6.5) (assuming the left-hand side is zero) are used to determine the initial values of E'_q, E'_d, and E_{fd}. In this chapter, the field voltage, E_{fd}, will be assumed constant. (The use of the generator exciters to control the field voltage is a topic covered in the next chapter.)

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EXAMPLE 12.10

Two-axis model example

For the system from Example 12.3, with the synchronous generator modeled using a two-axis model, determine (a) the initial conditions, and then (b) use PowerWorld Simulator to determine the critical clearing time for the Example 12.6 fault (three-phase fault at bus 3, cleared by opening lines 1–3 and 2–3). Assume H = 3.0 per unit-seconds, $R_a = 0$, $X_d = 2.1$, $X_q = 2.0$, $X'_d = 0.3$, $X'_q = 0.5$, all per unit using the 100-MVA system base.

SOLUTION

a. From Example 12.3, the current out of the generator is

$$I = 1.0526\angle -18.20^\circ = 1 - j0.3288$$

which gives a generator terminal voltage of

$$V_T = 1.0\angle 0^\circ + (j0.22)(1.0526\angle -18.20^\circ) = 1.0946\angle 11.59^\circ = 1.0723 + j0.220$$

From (12.6.8),

$$E = 1.0946\angle 11.59^\circ + (j2.0)(1.052\angle -18.2^\circ) = 2.814\angle 52.1^\circ$$

$$\rightarrow \delta = 52.1^\circ$$

Using (12.6.2) gives

$$\begin{bmatrix} V_d \\ V_q \end{bmatrix} = \begin{bmatrix} 0.7889 & -0.6146 \\ 0.6146 & 0.7889 \end{bmatrix} \begin{bmatrix} 1.0723 \\ 0.220 \end{bmatrix} = \begin{bmatrix} 0.7107 \\ 0.8326 \end{bmatrix}$$

and

$$\begin{bmatrix} I_d \\ I_q \end{bmatrix} = \begin{bmatrix} 0.7889 & -0.6146 \\ 0.6146 & 0.7889 \end{bmatrix} \begin{bmatrix} 1.000 \\ -0.3287 \end{bmatrix} = \begin{bmatrix} 0.9909 \\ 0.3553 \end{bmatrix}$$

Then, solving (12.6.3), (12.6.4), and (12.6.5) gives

$$E'_q = 0.8326 + (0.3)(0.9909) = 1.1299$$
$$E'_d = 0.7107 - (0.5)(0.3553) = 0.5330$$
$$E_{fd} = 1.1299 + (2.1 - 0.3)(0.9909) = 2.9135$$

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b. Open PowerWorld Simulator case Example 12_10 (see Figure 12.20). Select **Add-Ons**, **Transient Stability** to view the Transient Stability Analysis Form. Initially, the bus 3 fault is set to clear at 0.05 seconds. Select **Run Transient Stability** to create the results shown in Figure 12.20. In comparing these results with those from Example 12.4, notice that while the initial value of δ is different, the initial angle increase of about 13° is similar to the increase of 16° in Example 12.4. A key difference between the two is the substantial amount of damping in the two-axis model case. This damping arises because of the explicit modeling of the field and damper windings with the two-axis model. The critical clearing time can be determined by gradually increasing the clearing time until the generator loses synchronism. This occurs at about 0.30 seconds, with the almost critically cleared angle shown in Figure 12.21. Since there are now two additional state variables for generator 4, E'_q, and E'_d, their values can also be shown. This is done in Figure 12.22, again for the 0.30-second clearing time.

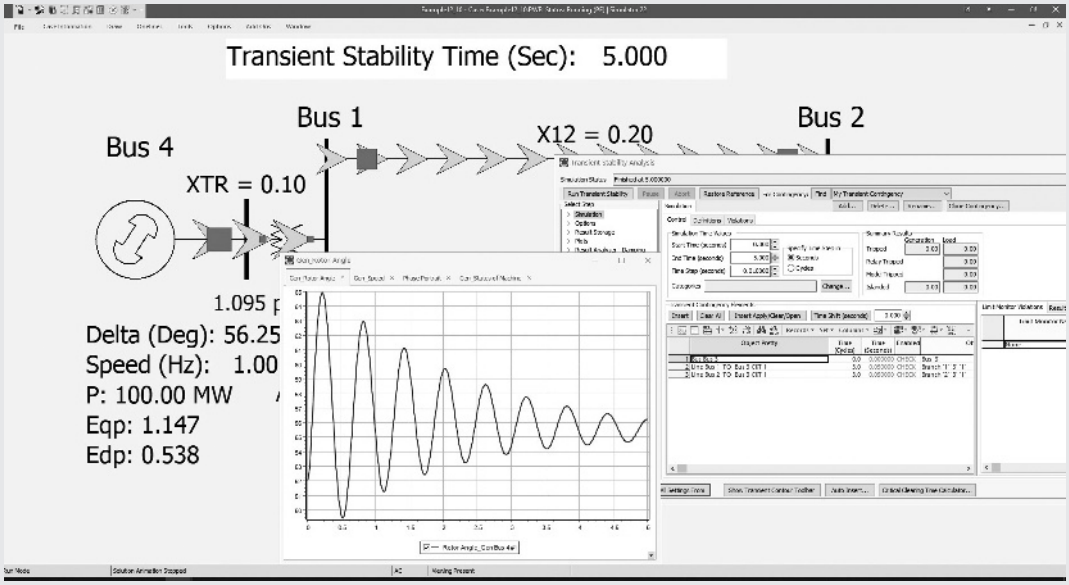

FIGURE 12.20

Variation in generator 4 rotor angle with a fault clearing time of 0.05 seconds

(*Continued*)

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FIGURE 12.21

Variation in generator 4 rotor angle with a fault clearing time of 0.30 seconds

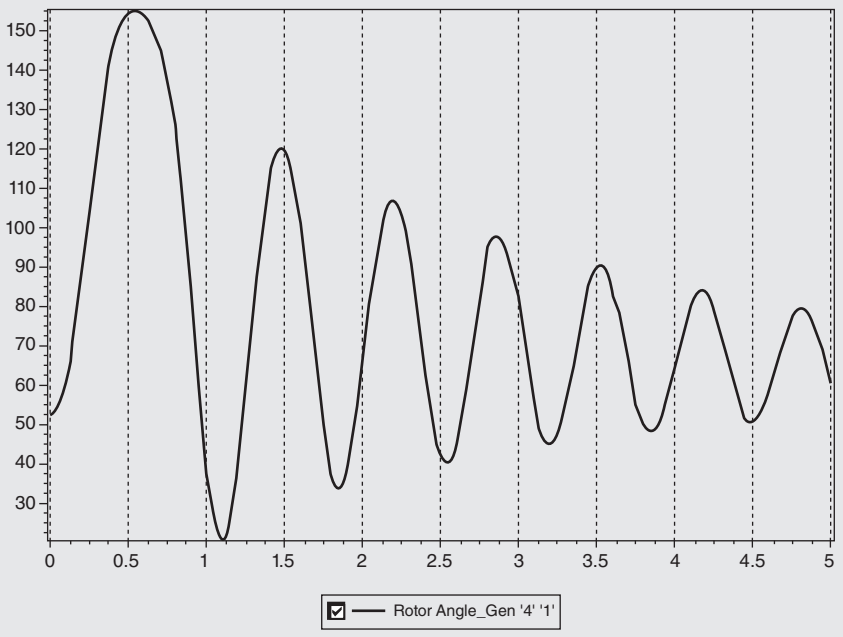
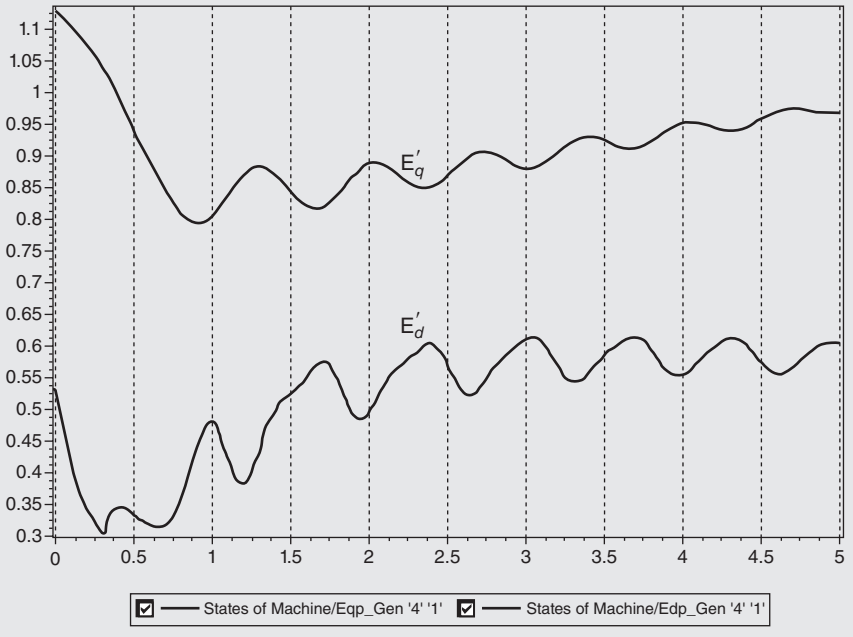


FIGURE 12.22

Variation in generator 4, E'_q and E'_d, with a fault clearing time of 0.30 seconds



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12.7 WIND TURBINE AND SOLAR PV MACHINE MODELS

As wind and solar PV energy continue their rapid growth, models for these devices need to be included in dynamic analysis studies. These models take two general forms. The first are the quite detailed models provided by the manufacturers for specific generator interconnection studies. These vendor-specific models are used to study dynamics that are often much faster than what are considered in this chapter, and they can also be used to study unbalanced system operation; such models are beyond the scope of this book. The second are the more generic models that have been developed for use in the positive sequence, large-scale interconnected grid studies that are for the focus of this chapter. This section begins to present these models by covering their generator/converter models (similar to the synchronous machine models presented in the previous sections). More information on their electrical controls and (for wind) mechanical controls is given in the next chapter.

As was introduced in Chapter 6, for wind there are four generic types of turbines that should be considered [14], [17]. Model types 1 and 2 are based on an induction machine model. As is the case with a synchronous machine, the stator windings of the induction machine are connected to the rest of the electric network. However, rather than having a dc field winding on the rotor, the ac rotor currents are induced by the relative motion between the rotating magnetic field setup by the stator currents, and the rotor. Usually the difference between the per-unit synchronous speed, n_s, and the per-unit rotor speed, n_r, is quantified by the slip (S), defined (using the standard motor convention) as

$$\mathrm{S} = \frac{n_s - n_r}{n_s} \tag{12.7.1}$$

From (12.7.1), it is clear that if the machine were operating at synchronous speed, its slip would be 0, with positive values when it is operating as a motor, and negative values when it is operating as a generator. Expressing all values in per unit, the mechanical equation for an induction machine is

$$\frac{d\mathrm{S}}{dt} = \frac{1}{2\mathrm{H}}(\mathrm{T}_m - \mathrm{T}_e) \tag{12.7.2}$$

where H is the inertia constant, T_m is the mechanical torque, and T_e the electrical torque, defined in (12.7.10).

The simplified electric circuit for a single-cage induction machine is shown in Figure 12.23, using the generator convention in which current out of the machine is assumed to be positive. Similar to what is done for synchronous machines, an induction machine can be modeled as an equivalent voltage behind the stator resistance and a transient reactance X'. Referring to Figure 12.23, the values used in this representation are

$$\mathrm{X}' = \mathrm{X}_a + \frac{\mathrm{X}_1\mathrm{X}_m}{\mathrm{X}_1 + \mathrm{X}_m} \tag{12.7.3}$$

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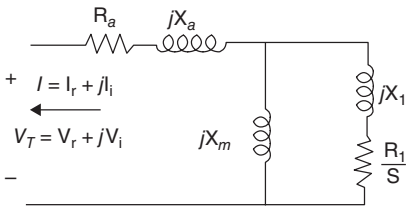
FIGURE 12.23

Equivalent circuit for a single cage induction machine



where X′ is the apparent reactance seen when the rotor is locked (i.e., slip is 1),

$$X = X_a + X_m \tag{12.7.4}$$

X is the synchronous reactance, and

$$T_0' = \frac{(X_1 + X_m)}{\omega_0 R_1} \tag{12.7.5}$$

is the open-circuit time constant for the rotor. Also, X_a is commonly called the leakage reactance.

Electrically, the induction machine is modeled using two algebraic and two differential equations. However, in contrast to synchronous machines, because the reactances of induction machines do not depend upon the rotor position values, they are specified in the network reference frame. The equations are

$$V_r = E_r' - R_a I_r + X' I_i \tag{12.7.6}$$

$$V_i = E_i' - R_a I_i - X' I_r \tag{12.7.7}$$

$$\frac{dE_r'}{dt} = \omega_o S E_i' - \frac{1}{T_o'}\left((E_r' - (X - X')I_i\right) \tag{12.7.8}$$

$$\frac{dE_i'}{dt} = -\omega_o S E_r' - \frac{1}{T_o'}\left((E_i' + (X - X')I_r\right) \tag{12.7.9}$$

The induction machine electric torque is then given by

$$T_e = (E_r' I_r + E_i' I_i)/\omega_o \tag{12.7.10}$$

and the terminal real power injection by

$$P_e = (V_r I_r + V_i I_i) \tag{12.7.11}$$

The initial conditions are determined by setting (12.7.8) and (12.7.9) to zero, and then using the power flow real power injection and terminal voltage as inputs to solve (12.7.6), (12.7.7), (12.7.8), (12.7.9), and (12.7.11) for the other variables. The Newton-Raphson approach (Section 6.3) is commonly used. Since the induction machine reactive power injection will not normally match the power flow value, the difference is modeled by including a shunt capacitor whose susceptance is determined to match the initial power flow conditions. The reactive power produced by the machine is given by

$$Q_e = (-V_r I_i + V_i I_r) \tag{12.7.12}$$

with the value negative since induction machines consume reactive power.

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EXAMPLE 12.11

Induction generator example

For the system from Example 12.3, assume the synchronous generator is replaced with an induction generator and shunt capacitor in order to represent a wind farm with the same initial real and reactive power output as in Example 12.3. The induction generator parameters are H = 0.9 per unit-seconds, $R_a = 0.013$, $X_a = 0.067$, $X_m = 3.8$, $R_1 = 0.0124$, $X_1 = 0.17$ (all per unit using the 100-MVA system base). This system is modeled in PowerWorld Simulator case Example 12_11. (a) Use the previous equations to verify the initial conditions of S = −0.0111, $E'_r = 0.9314$, $E'_i = 0.4117$, $I_r = 0.7974$, $I_i = 0.6586$. (b) Plot the terminal voltage for the fault sequence from Example 12.6.

SOLUTION

a. Using (12.7.3) to (12.7.5), the values of X', X, and T'_0 are determined to be 0.2297, 3.867 per unit, and 0.85 seconds, respectively. With $V_T = 1.0723 + j0.220$ and $P_e = 1.0$ per unit, (12.7.6) and (12.7.7) are verified as

$$V_r = 0.9314 - (0.013)(0.7974) + (0.2297)(0.6586) = 1.0723$$
$$V_i = 0.4117 - (0.013)(0.6586) - (0.2297)(0.7974) = 0.2200$$

and (12.7.8), (12.7.9) as

$$\frac{dE'_r}{dt} = 2\pi 60(-0.0111)(0.4117) - \frac{1}{0.85}(0.9314 - (3.637)(0.6586)) = 0.0$$
$$\frac{dE'_i}{dt} = -2\pi 60(-0.0111)(0.9314) - \frac{1}{0.85}(0.4117 + (3.637)(0.7974)) = 0.0$$
$$P_e = (1.0723)(0.7974) + (0.220)(0.6586) = 1.000$$
$$Q_e = -(1.0723)(0.6586) + (0.220)(0.7974) = -0.531$$

You can see the initial values in PowerWorld Simulator by first displaying the **States/Manual Control** page of the Transient Stability Analysis form, which initializes the stability simulation. Then from the one-line diagram, view the Generator Information Dialog for the generator at bus 4, and select **Stability**, **Terminal and State**, **Terminal Values**. Because the generator in the power flow is producing 57.2 Mvar, and the induction machine is consuming 53.1 Mvar, a shunt capacitor that produces 110.3 Mvar with a 1.0946 terminal voltage must be modeled.

b. Figure 12.24 plots the terminal voltage for the three-cycle fault.

(*Continued*)

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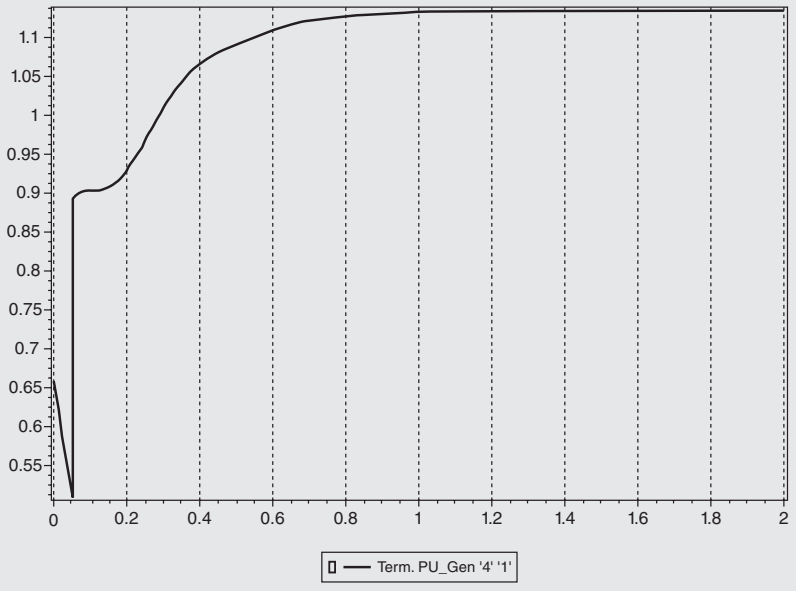
FIGURE 12.24

Example 12.11 Generator 4 voltage magnitude for a fault clearing time of 0.05 seconds



Both the Type 1 and 2 wind turbine models utilize induction generators, but whereas the Type 1 models have a conventional squirrel cage rotor with fixed rotor resistance, the Type 2 models are wound rotor induction machines that utilize a control system to vary the rotor resistance. The reason for this is to provide a more steady power output from the wind turbine during wind variation. From (12.7.5), it is clear that increasing this external resistance has the effect of decreasing the open-circuit time constant. The inputs to the rotor resistance control system are turbine speed and electrical power output, while the output is the external resistance that is in series with R_1 from Figure 12.22. Figure 12.25 plots the variation in the real power output for the Example 12.11 generator as a function of speed for the original rotor resistance of 0.0124 and for a total rotor resistance of 0.05 per unit. With a total resistance of 0.05, the operating point slip changes to about -0.045, which corresponds to a per-unit speed of 1.045.

Most new wind turbines are either Type 3 or Type 4. Type 3 wind turbines are used to represent doubly fed asynchronous generators (DFAGs), also sometimes referred to as doubly fed induction generators (DFIGs) [13,14]. A DFAG consists of a traditional wound rotor induction machine, but with the rotor windings connected to the ac network through an ac-dc-ac converter—the machine is "doubly fed" through both the stator and rotor windings (see Figure 12.26). The advantages of this arrangement are that it allows for separate control of both the real and reactive power (like a synchronous machine), and the ability to transfer power both ways through the rotor converter allows for a much wider speed range. Because the stator is directly connected to the ac grid, the rotor circuit converter need only be sized to about 30% of the machine's rated capacity. Another consequence

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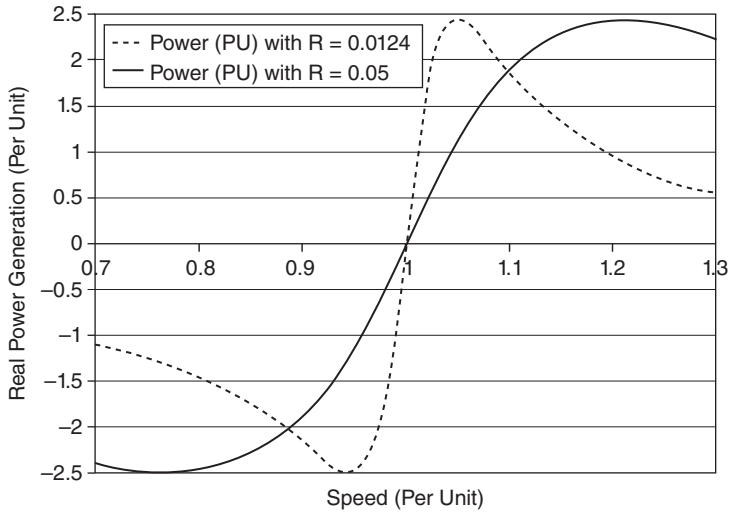

FIGURE 12.25

Effect of varying external resistance on an induction machine torque–speed curve

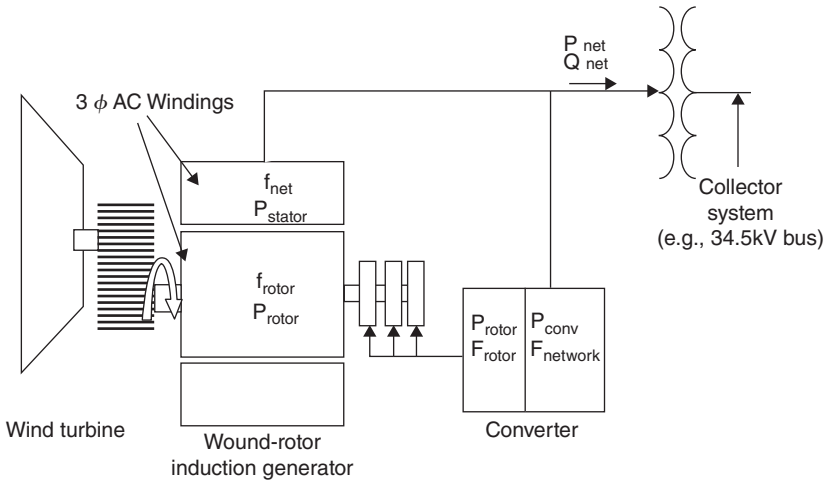

FIGURE 12.26

Doubly fed asynchronous generator components

of this design is the absence of an electrical coupling with the mechanical equation such as was seen with (12.1.10) for the synchronous machines and in (12.7.2) for the Type 1 and 2 induction machine models.

From a stability perspective, the DFAG dynamics are driven by the converter, with the result that the machine can be well approximated as a voltage-source converter (VSC). A VSC can be approximated as a synthesized current injection in parallel with an effective reactance, X_{eq}, (Figure 12.27) in which the current in phase with the terminal voltage, I_p, and the reactive power current, I_q, can be controlled independently. Low and high voltage current management is used to limit these values during system disturbances. With a terminal voltage angle of 0, the current injection on the network reference is

$$I_{sorc} = (I_p + jI_q)\ (1\angle\theta) \tag{12.7.13}$$

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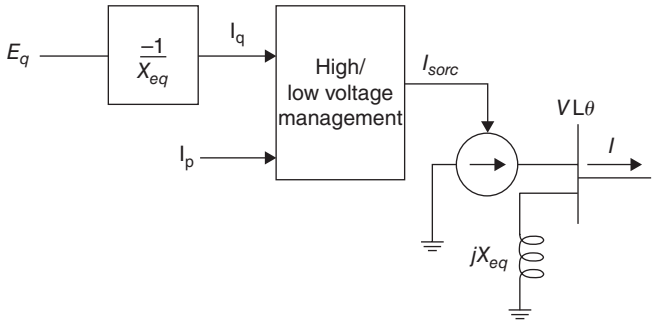
FIGURE 12.27

Type 3 DFAG model circuit diagram



And the reactive voltage is

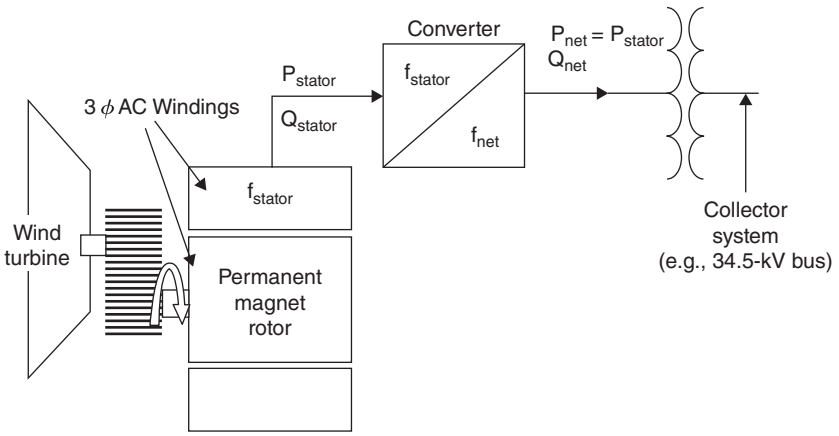
$$E_q = -\mathrm{I}_q X_{eq} \tag{12.7.14}$$

Type 4 wind turbines utilize a completely asynchronous design in which the full output of the machine is connected to the ac network through an ac-dc-ac converter (see Figure 12.28) [13, 14]. Because the converter completely decouples the electric generator from the rest of the network, there is considerable freedom in selecting the electric machine type: for example, a conventional synchronous generator, a permanent magnet synchronous generator, or even a squirrel cage induction machine.

From a stability perspective, electrically, the Type 4 model is similar to the Type 3 in that it can also be represented as a VSC. The key difference is lack of the effective reactance, with I_p and I_q being the direct control variables for the Type 4 model. As is the case with a DFAG, there is no electrical coupling with the turbine dynamics.

FIGURE 12.28

Type 4 full converter components



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EXAMPLE 12.12

Doubly fed asynchronous generator example

For the system from Example 12.3, assume the synchronous generator is replaced with a Type 3 DFAG generator in order to represent a wind farm with the initial current into the infinite bus set to 1.0 (unity power factor). The DFAG reactance $X_{eq} = 0.8$ per unit using a 100-MVA system base. Determine the initial values for I_p, I_q, and E_q.

SOLUTION

With $I = 1.0$ and an impedance of $j0.22$ between the machine's terminal and the infinite bus, the terminal voltage is

$$V_T = 1.0 + (1.0)(j0.22) = 1.0 + j0.22 = 1.0239\angle 12.41°$$

The amount supplied by I_{sorc} is I plus the amount modeled as going into X_{eq}:

$$I_{sorc} = I - \frac{V_T}{jX_{eq}} = 1.00 + \frac{1.0 + j0.220}{j0.8}$$

$$I_{sorc} = 1.275 - j1.25$$

The values of I_p and I_q are then calculated by shifting these values backward by the angle of the terminal voltage

$$I_p + jI_q = (1.275 - j1.25) * (1\angle -12.41°) = 0.977 - j1.495$$

and then

$$E_q = -(-1.495)(0.8) = 1.196$$

You can see the initial values in PowerWorld Simulator by opening case Example 12_12 and then displaying the **States/Manual Control** page of the Transient Stability Analysis form and selecting **Reset to Start Time**, which initializes the stability simulation. Then from the one-line diagram, view the Generator Information Dialog for the generator at bus 4, and select **Stability**, **Terminal and State**, **Terminal Values**.

Grid-connected photovoltaic systems come in a wide variety of different sizes, ranging from residential systems of perhaps several kW up to central station installations that often have 100s of MW, some of which are now over 1000 MW. In dynamic studies, the residential systems are usually considered as part of the load models (covered in the next section). For central stations, because PV systems are dc and therefore require an inverter, their generator/converter models are the same as those used for Type 4 wind turbines.

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12.8 LOAD MODELS

During power system stability studies the bus frequencies and voltage magnitudes can change substantially, which in turn impacts the loads. How the loads respond during a simulation depends upon their models. As a good starting point, the previous sections have assumed a constant admittance (impedance) load model, allowing the loads to be directly included in the bus admittance matrix. This approach was actually quite common for many years and was somewhat accurate when simulations only covered a few seconds, and much of the load consisted of resistor-based incandescent light and heating devices. This is no longer the case and now quite involved load models are common in power system stability studies.

However, there are significant challenges associated with accurately modeling the load. In contrast to generator models that represent specific equipment, load models almost always are used to represent an aggregation of many individual loads. The load is also both continually changing in terms of what portion of the installed load is turned on at any particular point in time (e.g., streetlights are only on at night, air conditioning is mostly used when the ambient temperature is high, commercial offices are more empty on the weekends, etc.), and changing more slowly as technology evolves and new loads are installed (e.g., incandescent lighting being replaced by LEDs, newer appliances).

There are three general classes of load models used in stability studies: (1) static models; (2) dynamic models; and (3) composite models. Static models are the simplest since they treat the load as a static function of its bus voltage magnitude sometimes adding a frequency-dependent model. The simplest is the previously mentioned constant impedance model, but since both its real and reactive power vary with the square of the supply voltage, it tends to underestimate the severity of a contingency. The advantages are that it is simple and it requires no model parameters, just the initial bus load available from the power flow. The slightly more complex ZIP model treats part of the load as constant impedance (Z), part as constant current so the load varies linearly with voltage magnitude (I), and part as constant power (P) with different coefficients commonly used for the real and reactive parts of the load. A constant power model is reasonable for loads with switched mode power supplies provided the voltage magnitude is not too low; if the voltage is below a cutoff value, the load is dropped (i.e., modeled as zero). One common ZIP model used in dynamic studies is to treat all the real power as constant current and all the reactive power as constant impedance. Exponential models are also used in which the power is treated as an exponential function of the bus voltage magnitude.

The second class is dynamic models in which the load's behavior includes at least some differential equations so its real and reactive power becomes a function of both the voltage magnitude and time. The most common types of dynamic models are used to represent the behavior of induction motors (IM), with the equivalent circuit for a single cage induction machine given in Figure 12.23. IM models are particularly important when studying voltage stability since if their supply voltage is decreased, such as during a fault, the speed of the IMs causing their slip to increase which in turns increases their reactive power consumption

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potentially delaying voltage recovery. How quickly this occurs is dependent on the parameters of each IM. Hence the modeling challenge is how to best represent a potentially quite diverse load.

The challenge of load diversity is partially addressed by the third class, known as composite models. The idea of a composite models is to use several separate models to try to approximate the behavior of the aggregate load at a bus. Figure 12.29 gives an example of one common composite model in which the load is represented as a combination of two types of static load and up to four different types of motors. This model also includes an approximation of the step-down transformer and the distribution system that is supplying the load. To fully represent the load using this model requires more than 120 parameters and involves the solution of about 15 differential equations! A further complication with load modeling is with the advent of much more distributed generation in the distribution system (e.g., solar PV). Often this generation is represented as negative load, and hence its behavior needs to be included in these composite load models. While in practice, similar sets of parameters are used to represent the load at different buses, there is clearly a tradeoff between potentially enhanced load modeling accuracy and modeling complexity. Load modeling has been an active area of research for many years, and will surely continue to be for years into the future.

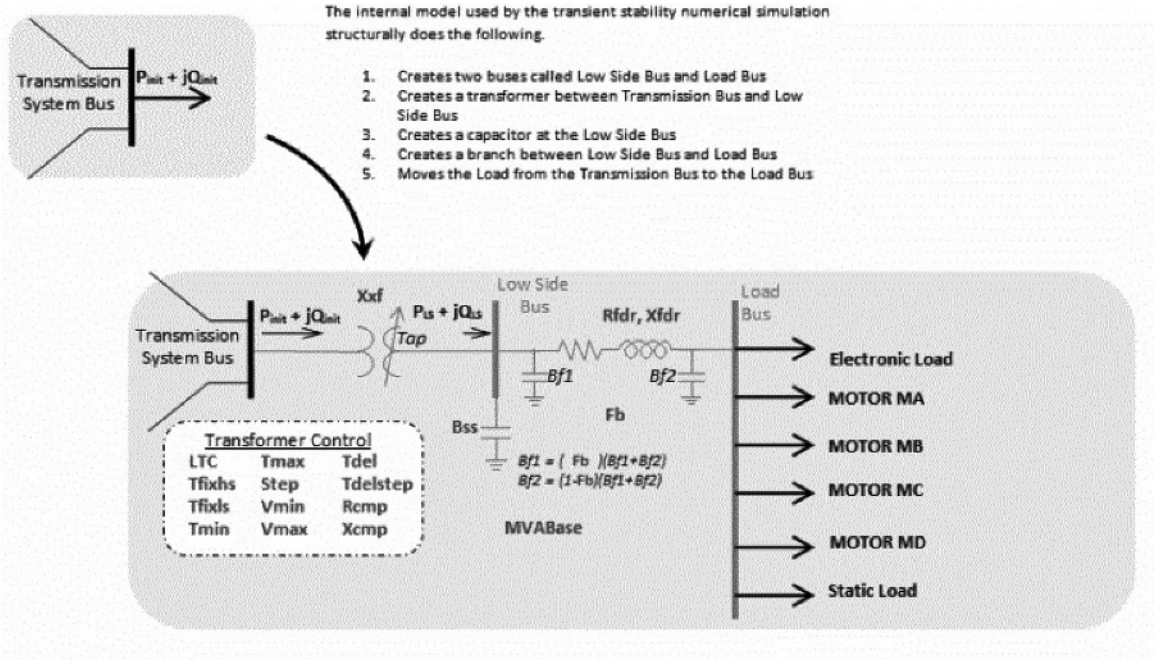

FIGURE 12.29

An example composite load model that requires more than 100 parameters

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12.9 DESIGN METHODS FOR IMPROVING POWER SYSTEM STABILITY

Design methods for improving power system stability include the following:

1. Improved steady-state stability
   - **a.** Higher system voltage levels
   - **b.** Additional transmission lines
   - **c.** Smaller transmission-line series reactances
   - **d.** Smaller transformer leakage reactances
   - **e.** Series capacitive transmission-line compensation
   - **f.** Static var compensators and flexible ac transmission systems (FACTS)
2. High-speed fault clearing
3. High-speed reclosure of circuit breakers
4. Single-pole switching
5. Larger machine inertia, lower transient reactance, use of synthetic inertia
6. Fast responding, high-gain exciters
7. Fast valving
8. Braking resistors
9. Remedial Action Schemes (RASs)
10. Power system stabilizers (covered in Chapter 13)

These design methods are discussed in the following paragraphs.

1. Increasing the maximum power transfer in steady state can also improve transient stability, allowing for increased power transfer through the unfaulted portion of a network during disturbances. Upgrading voltage on existing transmission or opting for higher voltages on new transmission increases line loadability (5.5.6). Additional parallel lines increase power-transfer capability. Reducing system reactances also increases power-transfer capability. Lines with bundled phase conductors have lower series reactances than lines that are not bundled. Oversized transformers with lower leakage reactances also help. Series capacitors reduce the total series reactances of a line by compensating for the series line inductance.
2. High-speed fault clearing is fundamental to power system stability. Standard practice for EHV systems is 1-cycle relaying and 2-cycle circuit breakers, allowing for fault clearing within 3 cycles (0.05 s). Ongoing research is presently aimed at reducing these to one-half cycle relaying and 1-cycle circuit breakers.

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3. The majority of transmission-line short circuits are temporary, with the fault arc self-extinguishing within 5–40 cycles (depending on system voltage) after the line is deenergized. High-speed reclosure of circuit breakers can increase postfault transfer power, thereby improving transient stability. Conservative practice for EHV systems is to employ high-speed reclosure only if stability is maintained when reclosing into a permanent fault with subsequent reopening and lockout of breakers.
4. Since the majority of short circuits are single line-to-ground, relaying schemes and independent-pole circuit breakers can be used to clear a faulted phase while keeping the unfaulted phases of a line operating, thereby maintaining some power transfer across the faulted line. Studies have shown that single line-to-ground faults are self-clearing even when only the faulted phase is deenergized. Capacitive coupling between the energized unfaulted phases and the deenergized faulted phase is, in most cases, not strong enough to maintain an arcing short circuit [5].
5. Inspection of the swing equation, (12.1.16), shows that increasing the per-unit inertia constant H of a synchronous machine reduces angular acceleration, thereby slowing down angular swings and increasing critical clearing times. Stability is also improved by reducing machine transient reactances, which increases power-transfer capability during fault or postfault periods [see (12.2.1)]. Current trends show increasing levels of renewable energy sources along with retirements of conventional synchronous generators that are uneconomical. Renewable energy sources (e.g., solar photovoltaics, as well as Type 4 wind turbine generators) that use converter-based technologies to connect to the grid do not inherently provide an inertial contribution, which is crucial for stability of the synchronous machines. As a result, systems with high penetrations of renewables are facing stability challenges (see the Case Study at the beginning of this chapter). However, if there is fast-responding, inverter-based storage on the system, such as from batteries, it can be controlled to mimic the inertia provided by the rotating synchronous machines, providing synthetic inertia.
6. Modern machine excitation systems with fast thyristor controls and high amplifier gains (to overcome generator saturation) can rapidly increase generator field excitation after sensing low terminal voltage during faults. The effect is to rapidly increase internal machine voltages during faults, thereby increasing generator output power during fault and postfault periods. Critical clearing times are also increased [6].
7. Some steam turbines are equipped with fast valving to divert steam flows and rapidly reduce turbine mechanical power outputs. During faults near the generator, when electrical power output is reduced, fast valving action acts to balance mechanical and electrical power, providing reduced acceleration and longer critical clearing times.

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The turbines are designed to withstand thermal stresses due to fast valving [7].

8. In power systems with generation areas that can be temporarily separated from load areas, braking resistors can improve stability. When separation occurs, the braking resistor is inserted into the generation area for a second or two, preventing or slowing acceleration in the generation area. Shelton et al. [8] describe a 1.4-GW, 3-GW-s braking resistor that has now been used for almost 50 years.
9. Remedial Action Schemes (RASs), also previously known as Special Protection Schemes (SPSs), are "a scheme designed to detect predetermined system conditions and automatically take corrective actions that may include, but are not limited to, adjusting or tripping generation (MW and Mvar), tripping load, or reconfiguring a system(s)" [19]. RASs, which are more involved than the protection systems installed to detect faults and isolated faulted equipment, can be configured to operate within seconds thereby helping to improve power system stability.

PROBLEMS

SECTION 12.1

12.1 A three-phase, 60-Hz, 500-MVA, 11.8-kV, 4-pole steam turbine-generating unit has an H constant of 4 p.u.-s. Determine: (a) ω_{syn} and ω_{msyn}; (b) the kinetic energy in joules stored in the rotating masses at synchronous speed; (c) the mechanical angular acceleration α_m and electrical angular acceleration α if the unit is operating at synchronous speed with an accelerating power of 500 MW.

12.2 Calculate J in kg-m^2 for the generating unit given in Problem 12.1.

12.3 A synchronous generator has a rated power of 500 MW and stores 1.2 J in its rotating mass, what is the per-unit inertia constant H?

12.4 The generating unit in Problem 12.1 is initially operating at $p_{mp.u.} = p_{ep.u.} = 0.7$ per unit, $\omega = \omega_{syn}$, and $\delta = 12°$ when a fault reduces the generator electrical power output by 60%. Determine the power angle five cycles after the fault commences. Assume that the accelerating power remains constant during the fault. Also, assume that $\omega_{p.u.}(t) = 1.0$ in the swing equation.

12.5 How would the value of H change if a generator's assumed operating frequency is changed from 60 Hz to 56 Hz?

12.6 Repeat Example 12.1 except assume the number of poles is changed from 32 to 16, H is changed from 2.0 p.u.-s to 1.5 p.u.-s, and the unit is initially operating with an electrical and mechanical power of 0.5 p.u.

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SECTION 12.2

12.7 Given that for a moving mass $W_{\text{kinetic}} = 1/2\ \text{Mv}^2$, how fast would a 120,000-kg diesel locomotive need to go to equal the energy stored in a 60-Hz, 100-MVA, 2-pole generator spinning at synchronous speed with an H of 4.0 p.u.-s?

12.8 The synchronous generator in Figure 12.4 delivers 0.8 per-unit real power at 1.05 per-unit terminal voltage. Determine: (a) the reactive power output of the generator; (b) the generator internal voltage; and (c) an equation for the electrical power delivered by the generator versus power angle δ.

12.9 The generator in Figure 12.4 is initially operating in the steady-state condition given in Problem 12.8 when a three-phase-to-ground bolted short circuit occurs at bus 3. Determine an equation for the electrical power delivered by the generator versus power angle δ during the fault.

12.10 For the five-bus system from Example 6.9, assume the transmission lines and transformers are modeled with just their per-unit reactance (e.g., neglect their resistance and B shunt values). If bus one is assumed to be an infinite bus, what is the equivalent (Thévenin) reactance looking into the system from the bus three terminal? Neglect any impedances associated with the loads.

12.11 Repeat Problem 12.10, except assume there is a three-phase-to-ground bolted short circuit at bus five.

SECTION 12.3

12.12 The generator in Figure 12.4 is initially operating in the steady-state condition given in Example 12.3 when circuit breaker B12 inadvertently opens. Use the equal-area criterion to calculate the maximum value of the generator power angle δ. Assume $\omega_{\text{p.u.}}(t) = 1.0$ in the swing equation.

12.13 The generator in Figure 12.4 is initially operating in the steady-state condition given in Example 12.3 when a temporary three-phase-to-ground short circuit occurs at point F. Three cycles later, circuit breakers B13 and B22 permanently open to clear the fault. Use the equal-area criterion to determine the maximum value of the power angle δ.

12.14 If breakers B13 and B22 in Problem 12.13 open later than 3 cycles after the fault commences, determine the critical clearing time.

12.15 Building upon Problem 12.11, assume a 60-Hz nominal system frequency, that the bus fault actually occurs on the line between buses five and two but at the bus two end, and that the fault is cleared by opening breakers B21 and B52. Again, neglecting the loads, assume that the generator at bus three is modeled with the classical generator model having a per-unit value (on its 800-MVA base) of $X'_d = 0.24$, and H = 3 p.u.-s. Before the fault occurs the generator is delivering 300 MW into the infinite bus at unity power factor (hence its terminal voltage is not 1.05 as was assumed in Example 6.9). Further, assume the fault is cleared after 3 cycles. Determine: (a) the initial generator

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one power angle, (b) the power angle when the fault is cleared, and (c) the maximum value of the power angle using the equal area criteria.

12.16 Analytically determine whether there is a critical clearing time for Problem 12.15.

SECTION 12.4

12.17 Consider the first-order differential equation, $\frac{dx}{dt} = -x$, with an initial value $x(0) = 10$. With an integration step size of 0.1 seconds, determine the value of $x(0.5)$ using (a) Euler's method, (b) the modified Euler's method.

12.18 The following set of differential equations can be used to represent that behavior of a simple spring-mass system, with $x_1(t)$ the mass's position and $x_2(t)$ its velocity:

$$\frac{dx_1}{dt} = x_2$$

$$\frac{dx_2}{dt} = -x_1$$

For the initial condition of $x_1(0) = 1.0$, $x_2(0) = 0$, and a step size 0.1 seconds, determine the values $x_1(0.3)$ and $x_2(0.3)$ using (a) Euler's method, (b) the modified Euler's method.

12.19 A 60-Hz generator is supplying 500 MW (and 0 Mvar) to an infinite bus (with 1.0 per unit voltage) through two parallel transmission lines. Each transmission line has a per-unit impedance (100-MVA base) of $0.08j$. The per-unit transient reactance for the generator is $0.0375j$, the per-unit inertia constant for the generator (H) is 20 seconds, and damping is 0.1 per unit (all with a 100-MVA base). At time = 0, one of the transmission lines experiences a balanced three-phase short to ground one third (1/3) of the way down the line from the generator to the infinite bus. (a) Using the classical generator model, determine the prefault internal voltage magnitude and angle of the generator. (b) Express the system dynamics during the fault as a set of first-order differential equations. (c) Using Euler's method, determine the generator internal angle at the end of the second timestep. Use an integration step size of one cycle.

PW 12.20 Open PowerWorld Simulator case Problem 12_20. This case models a slightly modified Example 12.4 system with damping at the bus 1 generator, and with a line fault midway between buses 1 and 2. The fault is cleared by opening the line. Determine the critical clearing time for this fault (to the nearest 0.01 second).

PW 12.21 Open PowerWorld Simulator case Problem 12_21. This case models a slightly modified Example 12.4 system with damping at the bus 1 generator, and with a line fault midway between buses 2 and 3. The fault is cleared by opening the line. Determine the critical clearing time (to the nearest 0.01 second) for this fault.

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SECTION 12.5

12.22 Consider the six-bus power system shown in Figure 12.30, where all data are given in per unit on a common 100-MVA system base. All resistances as well as transmission-line capacitances are neglected. (a) Determine the 6×6 per-unit bus admittance matrix $\boldsymbol{Y}_{\text{bus}}$ suitable for a power-flow computer program. (b) Determine the per-unit admittance matrices $\boldsymbol{Y}_{11}$, $\boldsymbol{Y}_{12}$, and $\boldsymbol{Y}_{22}$ given in (12.5.5), which are suitable for a transient stability study.

12.23 Modify the matrices $\boldsymbol{Y}_{11}$, $\boldsymbol{Y}_{12}$, and $\boldsymbol{Y}_{22}$ determined in Problem 12.22 for (a) the case when circuit breakers B52 and B41 open to remove line 4 (between buses 4 and 5); and (b) the case when the load $P_{L4} + jQ_{L4}$ is removed.

PW 12.24 Open PowerWorld Simulator case Problem 12_24, which models the Figure 12.30 six-bus system with a fault occurring on line 1 close to bus 2. The fault is cleared by opening the line. Determine the critical clearing time (to the nearest 0.01 second) for this fault.

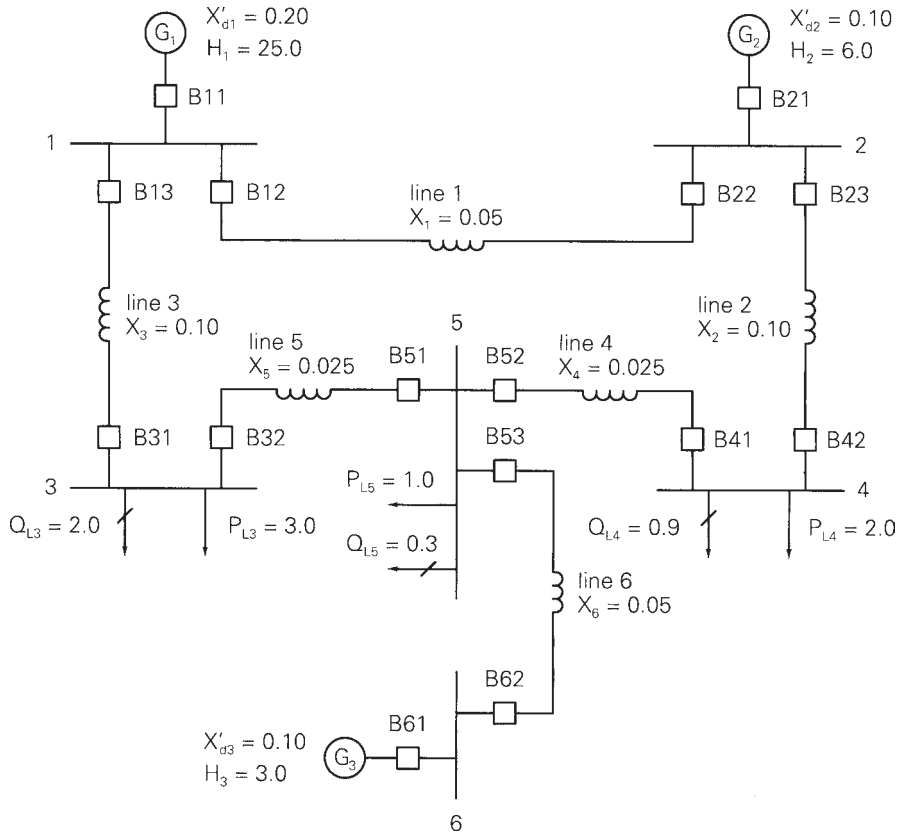

FIGURE 12.30
Problems 12.22, 12.23, 12.24, and 12.28 Single-line diagram of a six-bus power system (per-unit values are shown)

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PW 12.25 Open PowerWorld Simulator case Problem 12_25, which models the Example 6.9 with transient stability data added for the generators. Determine the critical clearing time (to the nearest 0.01 second) for a fault on the line between buses 4 and 5 at the bus 4 end, which is cleared by opening the line.

PW 12.26 With PowerWorld Simulator, using the Example 12_9 case, determine the critical clearing time (to the closest 0.01 second) for a transmission line fault on the transmission line between bus 39 (PINE138) and bus 47 (CEDAR138), with the fault occurring near bus 39.

SECTION 12.6

PW 12.27 PowerWorld Simulator case Problem 12_27 duplicates Example 12.10, except with the synchronous generator initially supplying 80 MW at unity power factor to the infinite bus. (a) Derive the initial values for δ, E'_q, E'_d, and E_{fd}. (b) Determine the critical clearing time for the Example 12.10 fault.

PW 12.28 PowerWorld Simulator case Problem 12_28 duplicates the system from Problem 12.24, except the generators are modeled using a two-axis model, with the same X'_d and H parameters as are in Problem 12.24. Compare the critical clearing time between this case and the Problem 12.24 case.

PW 12.29 PowerWorld Simulator case Problem 12_29 duplicates the system from Problem 12.25, except the generators are modeled using a two-axis model, with the same X'_d and H parameters as are in Problem 12.25. Compare the critical clearing time between this case and the Problem 12.25 case.

SECTION 12.7

PW 12.30 PowerWorld Simulator case Problem 12_30 duplicates Example 12.11 except the wind turbine generator is set so it is initially supplying 100 MW to the infinite bus at unity power factor. (a) Use the induction machine equations to verify the initial conditions of $S = -0.0129$, $E'_r = 0.8475$, $E'_i = 0.4230$, $I_r = 0.8433$, and $I_i = 0.7119$. (b) Plot the terminal voltage for the fault sequence from Example 12.6.

12.31 Redo Example 12.12 with the assumption the generator is supplying $100 + j10$ MVA to the infinite bus.

SECTION 12.8

12.32 If a load's power at unity voltage is P = 100 MW and Q = 50 Mvar, and it is represented with normalized ZIP model parameters of $P_Z = 0.3$, $P_I = 0.4$, $P_P = 0.3$, $Q_Z = 0.5$, $Q_I = 0.4$, $Q_P = 0.1$, what will be its power consumption if the terminal voltage is reduced to 0.85 per unit?

12.33 Using the induction model from Figure 12.23 and induction machine parameters of $R_a = 0$, $X_a = 0.067$, $X_m = 3.8$, $R_1 = 0.014$, and $X_1 = 0.17$ (all per unit on a 100-MVA base), if the induction machine is operated with a terminal voltage of $1.0\angle 0°$ per unit and a slip of 0.01, what will

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be its total real and reactive power consumption? How does the load change if the slip is changed to 0.02?

12.34 Building on Problem 12.33, again assume the Figure 12.23 induction machine model, the Problem 12.33 parameters, and the $1.0\angle 0°$ per unit terminal voltage. If the real power load at the bus is 80 MW and it is completely represented using the induction machine model, what is the associated slip? Hint: You may need to solve for this numerically; your answer should within 0.001 per unit of the actual value; note that at a slip of zero (synchronous speed), the real power consumption will be zero. When induction motor models are included in stability studies in order to exactly match the power flow reactive power results, a reactive shunt needs to be included in parallel with the motor.

PW 12.35 Open PowerWorld Simulator case Problem 12_35. This two-bus case has an infinite bus supplying a load bus in which the load bus's power is modeled using the induction motor load model from Problem 12.31. As is common with commercial models, a parallel capacitor is included with the induction motor so that the real and reactive power consumption match the power flow value of 100 MW and zero Mvar. Initially, there is a 0.01-s duration self-clearing, three-phase fault at the motor's terminal. Increase the duration of the fault to determine the fault time duration at which the motor stalls. Stalling will be indicated by the voltage failing to recover, the motor speed continually decreasing, and the motor looking like a large reactive load. Induction motor load models usually have a built-in protection system that will trip the motor if its voltage stays low for too long. That is disabled here.

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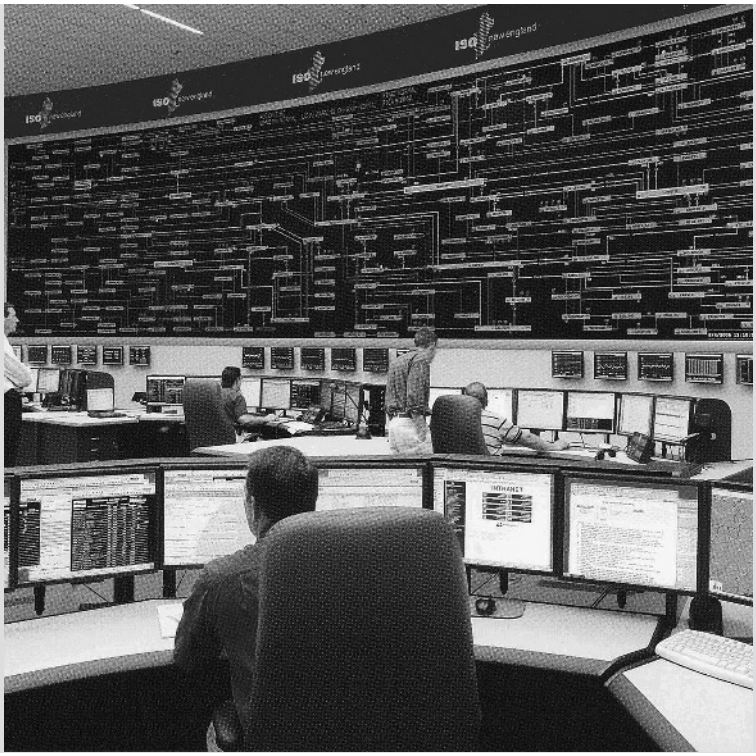
13 Power System Controls



ISO New England's state-of-the-art control center helps to ensure the reliable operation of New England's bulk power generation and transmission system. (Photograph © Adam Laipson.)

LEARNING OBJECTIVES

At the conclusion of this chapter, you should be able to:

1. Understand basic generator-voltage and turbine-governor controls for conventional synchronous machines as well as for wind-turbine generators.
2. Understand area control error (ACE) and the objectives of load-frequency control (LFC) for an interconnected power system.
3. Understand how economic dispatch is coordinated with LFC.
4. Understand the use of power system stabilizers.

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Automatic control systems are used extensively in power systems. Local controls are employed at turbine-generator units of conventional synchronous machines, at nonsynchronous machines (e.g., Type 4 wind-turbine generators and solar photovoltaic plants), and at selected voltage-controlled buses. Central controls are employed at area control centers.

Figure 13.1 shows two basic controls of a steam turbine-generator for a conventional synchronous machine: the voltage regulator and turbine-governor. The voltage regulator adjusts the power output of the generator exciter in order to control the magnitude of generator terminal voltage V_t. When a reference voltage V_{ref} is raised (or lowered), the output voltage V_r of the regulator increases (or decreases) the exciter voltage E_{fd} applied to the generator field winding, which in turn acts to increase (or decrease) V_t. Also, a voltage transformer and rectifier monitor V_t to use as a feedback signal in the voltage regulator. If V_t decreases, the voltage regulator increases V_r to increase E_{fd}, which in turn acts to increase V_t.

The turbine-governor shown in Figure 13.1 adjusts the steam valve position to control the mechanical power output p_m of the turbine. When a reference power level p_{ref} is raised (or lowered), the governor moves the steam valve in the open (or close) direction to increase (or decrease) p_m. The governor also monitors rotor speed ω_m, which is used as a feedback signal to control the balance between p_m and the electrical power output p_e of the generator. Neglecting losses, if p_m is greater than p_e, ω_m increases, the governor moves the steam valve in the close direction to reduce p_m. Similarly, if p_m is less than p_e, ω_m decreases, and the governor moves the valve in the open direction.

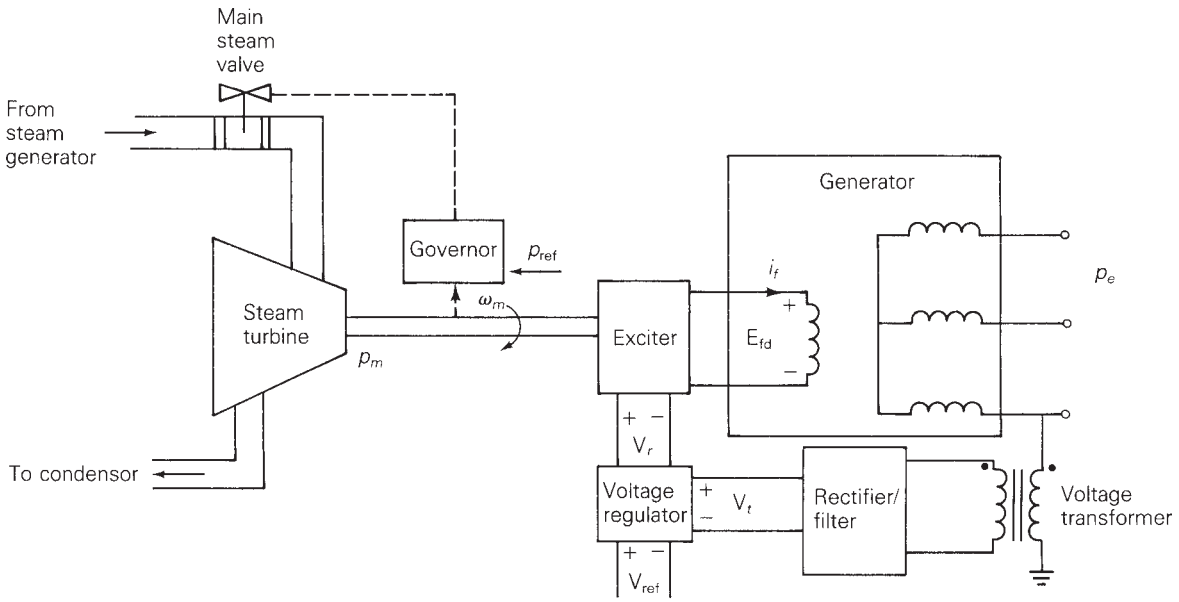

FIGURE 13.1

Voltage regulator and turbine-governor controls for a steam-turbine generator

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In addition to voltage regulators at generator buses, equipment is used to control voltage magnitudes at other selected buses. Tap-changing transformers, switched capacitor banks, and static var systems can be automatically regulated for rapid voltage control.

Central controls also play an important role in modern power systems. Today's systems are composed of interconnected areas, where each area has its own control center. There are many advantages to interconnections. For example, interconnected areas can share their reserve power to handle anticipated load peaks and unanticipated generator outages. Interconnected areas can also tolerate larger load changes with smaller frequency deviations than an isolated area.

Figure 13.2 shows how a typical area meets its daily load cycle. The base load is carried by base-loaded generators running at 100% of their rating for 24 hours. Nuclear units and large fossil-fuel units are typically base-loaded. The variable part of the load is carried by units that are controlled from the central control center. Medium-sized fossil-fuel units and hydro units are used for control. During peak load hours, smaller, less efficient units, such as gas-turbine or diesel-generating units, are employed. Renewable generators, such as wind and solar, are usually operated to maximize their outputs for the given wind or solar conditions since their "fuel" is essentially free. In addition, generators operating at partial output (with *spinning reserve*) and standby generators provide a reserve margin.

The central control center monitors information including area frequency, generating unit outputs, and tie-line power flows to interconnected areas. This information is used by automatic *load-frequency control* (LFC) in order to maintain area frequency at its scheduled value (60 Hz) and net tie-line power flow out of the area at its scheduled value. Raise and lower reference power signals are dispatched to the turbine-governors of controlled units.

This chapter covers automatic controls employed in power systems under normal operation. Sections 13.1 and 13.2 describe the operation of the two generator controls: generator voltage control and turbine-governor control. Load-frequency control is discussed in Section 13.3, and power system stabilizers are discussed in Section 13.4.

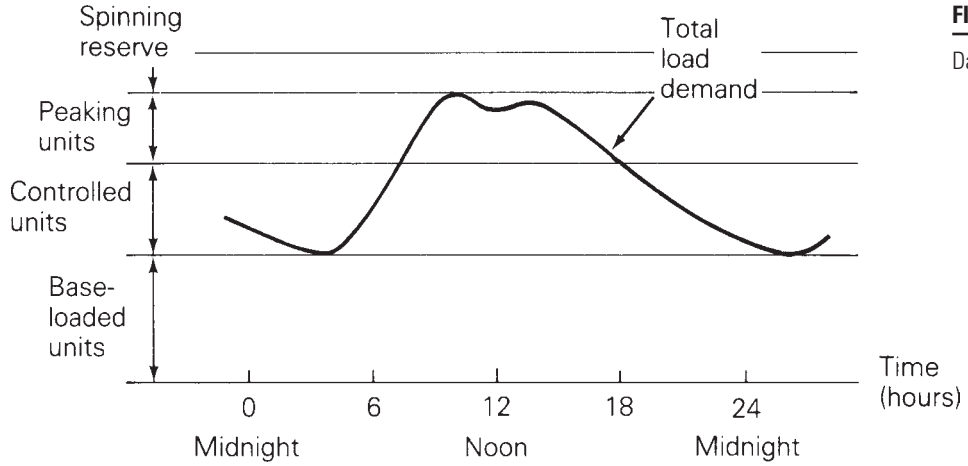

FIGURE 13.2
Daily load cycle

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CASE STUDY

The Software-Defined Power Grid: How software and sensors are bringing century-old grid technology into the modern age*

Patrick T. Lee

The following paragraphs summarize the above article, which describes how sensors and algorithms enable real-time balancing of solar photovoltaic plants and wind-turbine generators [12]. Sign in at www.cengage.com to download this product and to access its online resources.

The article describes how one company, PXiSE Energy Solutions, uses phasor measurement units (PMUs) to better integrate renewable energy sources in power grids. The PMU measures voltage and current signals at various points on the grid and then computes the magnitude and phase of the signals, with each digitized measurement receiving a time stamp accurate to within one microsecond of true time. PMUs, first developed in the 1980s, were commercialized in 1992, at which point utilities began deploying the sensors to help identify outages and other grid events. Today there are tens of thousands of PMUs installed at major substations throughout the United States. Today's high-speed PMUs provide data 60 times per second, which is more than 200 times as fast as the sampling rate of conventional supervisory control and data acquisition (SCADA) systems used on most electric grids. Furthermore, PMU data are time-stamped with great precision using the Global Positioning System (GPS), which synchronizes all measurements worldwide. For that reason, PMU data are sometimes called synchrophasors.

Power electronics used in inverter-based systems like solar photovoltaic plants, wind-turbine generators, and battery storage units operate on the millisecond level. As such, they require high-speed control to maximize their benefits. PXiSE developed machine-learning algorithms so that its high-speed controller can act quickly and autonomously to changes in generation and consumption, and also predict likely future conditions on the network. This intelligent system mitigates any grid disturbances while continuously balancing solar generation, battery power, and other available energy resources, making the grid more efficient and reliable. The system can be integrated into virtually any type of power grid, regardless of size, age, or mix of generation and loads.

Figure 1 shows the power grid operating system envisioned

*"The Software-Defined Power Grid," by Patrick T. Lee, available online, with permission, from *IEEE Spectrum Magazine* (July 2020), pp. 40–46.

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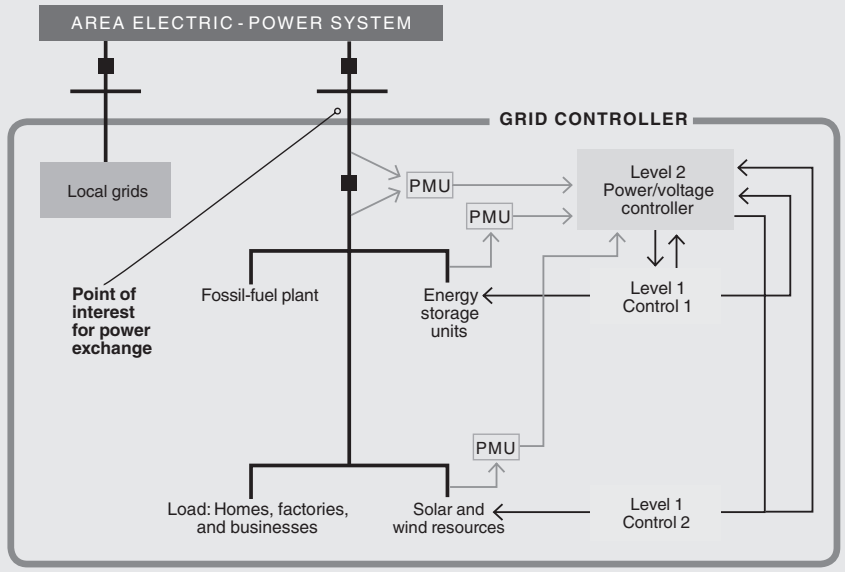

Figure 1 A power grid operating system

by PXiSE. In the microgrid shown in the figure, solar and wind power, energy storage, and fossil-fuel power plant supply electricity to homes, factories and businesses. The software-based controller manages the network's resources. Using data from PMUs, the controller continuously balances generation and load within the microgrid and keeps the microgrid's connection to the main grid stable. The technology has been deployed at various locations including the following: (1) a controller for a 21-MW wind farm and battery storage facility on an island in Hawaii; (2) a microgrid in Sempra Energy's high-rise office building in downtown San Diego; (3) a grid controller for Jeju Island in South Korea that includes local battery storage, a 500-kW solar photovoltaic plant and 600 kW of wind power; and (4) a microgrid controller for a 1-MW solar photovoltaic plant and 2.5-MW battery storage facility near Onslow in Western Australia.

(1) Wind Farm and Battery Storage in Hawaii

In recent years, Hawaii has dramatically increased solar and wind generation, but this particular island's grid had reached a limit to integrating more renewable energy, due to the variability of wind resources. It was less complicated for the grid operator to rely on fossil-fuel generators than to mitigate the intermittent power from the wind farm, which was periodically shut off, or curtailed, at night to ensure grid reliability.

With the PXiSE control system in place, inherent power fluctuations of the wind farm were

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smoothed out by real-time dispatch from the battery storage facility.

(2) Sempra Energy High-Rise Office Building in San Diego

PXiSE installed a controller running the PMU-based algorithm to create a microgrid in Sempra-Energy's office building. The controller was designed to optimize the use of the building's electric-vehicle chargers, solar panels, and storage batteries. It would also reduce the reliance on grid power in the late afternoon and early evening, when demand and prices typically spike. At such times, the optimization algorithm automatically determines the proper resource mix and schedule, enabling one floor to be served solely by local renewable resources and stored energy rather than the main grid, which reduced the building's energy bill by 20% (typical for similar microgrids).

(3) Jeju Island in South Korea

Prior to the installation of the grid controller, Jeju Island's electricity came from the mainland via high-voltage underwater cables. The system was designed to interconnect two local battery-storage systems—with capacities of 224 kWh and 776 kWh—with a 500-kW solar photovoltaic farm and 600 kW of wind power (see Figure 2). To meet the island's renewable energy goals, wind and solar power are stored in the batteries during the day, when power demand and grid prices are low. The stored electricity is then used at night, when power demand and grid prices are high.

PXiSE's controller uses the batteries to provide what is known as "ramp control" for the wind turbines. Most grids can tolerate only a gradual change in power flow, typically no more than 1 MW per minute, so ramp control acts like a shock absorber. It smooths out the

Figure 2 Wind turbine-generator on Jeju Island in South Korea.
(© Bloomberg/Getty Images.)

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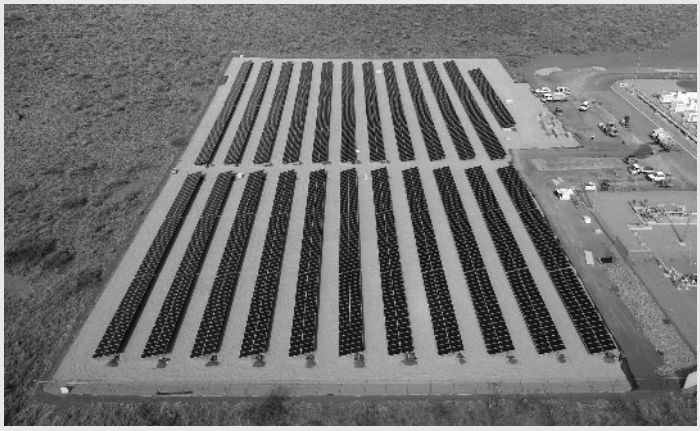

Figure 3 1-MW solar photovoltaic farm near Onslow, Western Australia

turbine-generator's power output when the wind suddenly changes, which would otherwise cause reliability problems for the grid.

Every 16.7 milliseconds (one cycle of the 60-Hz waveform), the controller looks for any change in wind power output. If the wind stops blowing, the system instantly compensates by discharging the batteries to the grid, gradually decreasing the amount of discharged battery power to zero, as the power from the main grid gradually increases to match demand. When the wind starts blowing again, the batteries immediately respond by absorbing a portion of the wind power, gradually reducing the amount being stored to zero as the controller ramps up the amount of wind power feeding the grid. The controller also creates its own forecasts of load and generation based on weather forecasts and artificial intelligence (AI) processing of historical load data. The forecasting models are updated in real time to compute optimal settings for power and energy storage in the system.

(4) Solar Photovoltaic Plant and Battery Storage Facility in Western Australia

Western Australia's utility Horizon Power faces an enormous challenge in providing reliable electricity to hundreds of small communities scattered across a wide area. Actually, calling this a "wide area" is an understatement: Horizon's territory covers some 2.3 million square kilometers, about one and a half times the size of Alaska. As the country tries to shrink its carbon footprint, Horizon is working with its customers to deploy renewables such as photovoltaics and wind turbines.

PXiSE's controller continuously balances electricity demand and generation near Onslow, in Western Australia. It monitors grid conditions at the millisecond scale, allowing efficient use of a 1-MW solar photovoltaic farm (Figure 3) and a 2.5-MW battery storage facility (Figure 4). The time-stamped PMU data are particularly helpful when resources are scattered across an area as wide as Horizon's power grid.

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Figure 4 2.5-MW battery storage facility near Onslow, Western Australia

13.1 GENERATOR-VOLTAGE CONTROL

SYNCHRONOUS GENERATORS

The *exciter* delivers dc power to the field winding on the rotor of a synchronous generator. For older generators, the exciter consists of a dc generator driven by the rotor. The dc power is transferred to the rotor via slip rings and brushes. For newer generators, *static* or *brushless* exciters are often employed.

For static exciters, ac power is obtained directly from the generator terminals or a nearby station service bus. The ac power is then rectified via thyristors and transferred to the rotor of the synchronous generator via slip rings and brushes.

For brushless exciters, ac power is obtained from an "inverted" synchronous generator whose three-phase armature windings are located on the main generator rotor and whose field winding is located on the stator. The ac power from the armature windings is rectified via diodes mounted on the rotor and is transferred directly to the field winding. For this design, slip rings and brushes are eliminated.

Block diagrams of several standard types of generator-voltage control systems have been developed by the IEEE Power and Energy Society, beginning in 1968 with [1] and most recently updated in 2016 with IEEE Std 421.5-2016. A block diagram for what is commonly known as the IEEE Type 1 exciter, which uses a shaft-driven dc generator to create the field current, is shown in Figure 13.3 (neglecting saturation) [1].

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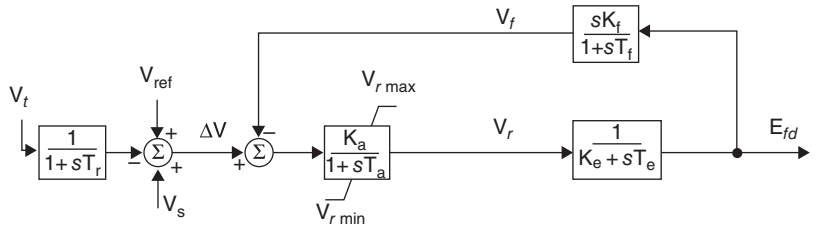

FIGURE 13.3

Block diagram for the IEEE Type 1 exciter (neglecting saturation)

In Figure 13.3, the leftmost block, $1/(1 + sT_r)$, represents the delay associated with measuring the terminal voltage V_t where s is the Laplace operator and T_r is the measurement time constant. Note that if a unit step is applied to a $1/(1 + sT_r)$ block, the output rises exponentially to unity with time constant T_r. The measured generator terminal voltage V_t is compared with a voltage reference V_{ref} to obtain a voltage error, ΔV, which in turn is applied to the voltage regulator. The voltage regulator is modeled as an amplifier with gain K_a and a time constant T_a, while the last forward block represents the dynamics of the exciter's dc generator. The output is the field voltage E_{fd}, which is applied to the generator field winding and acts to adjust the generator terminal voltage, as in (12.6.5). The feedback block in Figure 13.3 is used to improve the dynamic response of the exciter by reducing excessive overshoot. This feedback is represented by $(sK_f)/(1 + sT_f)$, which provides a filtered first derivative negative feedback.

For any transient stability study, the initial values for the state variables need to be determined. This is done by assuming that the system is initially operating in steady state, and recognizing that in steady state, all the derivatives will be zero. Then, by knowing the initial field voltage (found as in Example 12.10) and terminal voltage, all the other variables can be determined.

EXAMPLE 13.1

Synchronous generator exciter response

Using the system from Example 12.10, assume the two-axis generator is augmented with an IEEE Type 1 exciter with $T_r = 0$, $K_a = 100$, $T_a = 0.05$, $V_{rmax} = 5$, $V_{rmin} = -5$, $K_e = 1$, $T_e = 0.26$, $K_f = 0.01$, and $T_f = 1.0$. (a) Determine the initial values of V_r, V_f, and V_{ref}. (b) Using the fault sequence from Example 12.10, determine the bus 4 terminal voltage after 1 second and then after 5 seconds.

SOLUTION

a. The initial field voltage and terminal voltage, E_{fd} and V_t, do not depend on the exciter, so their values are equal to those found in Example 12.10, that is, 2.9135 and 1.0946, respectively. Since the system is initially in steady state,

$$V_r = (K_e)(E_{fd}) = (1.0)(2.9135) = 2.9135$$

(*Continued*)

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Because V_f is the output of the filtered derivative feedback, its initial value is zero. Finally, writing the equation for the second summation block in Figure 13.3,

$$(V_{ref} - V_t - V_f)(K_a) = V_r$$

$$V_{ref} = \frac{V_r}{K_a} + V_t + V_f = \frac{2.9135}{100} + 1.0946 = 1.1237$$

b. Open PowerWorld Simulator case Example 13_1 and display the Transient Stability Analysis Form (see Figure 13.4). To see the initial conditions, select the **States/Manual Control** page, and then select the **Transfer Present State to Power Flow** button to update the oneline display. From this page, it is also possible to just do a specified number of timesteps by selecting the **Do Specified Number of Timesteps** button or to run to a specified simulation time using the **Run Until Specified Time** button. To determine the terminal voltage after one second, select the **Run Until Specified Time** button. The value is 1.10 p.u. To finish the simulation, select the **Continue** button. The terminal voltage at five seconds is 1.0946 p.u., which is close to the prefault voltage, indicating the exciter is restoring the voltage to its set-point value. In contrast, the bus 4 terminal voltage after five seconds in the Example 12.10 case, which does not include an exciter, is 1.116 p.u.

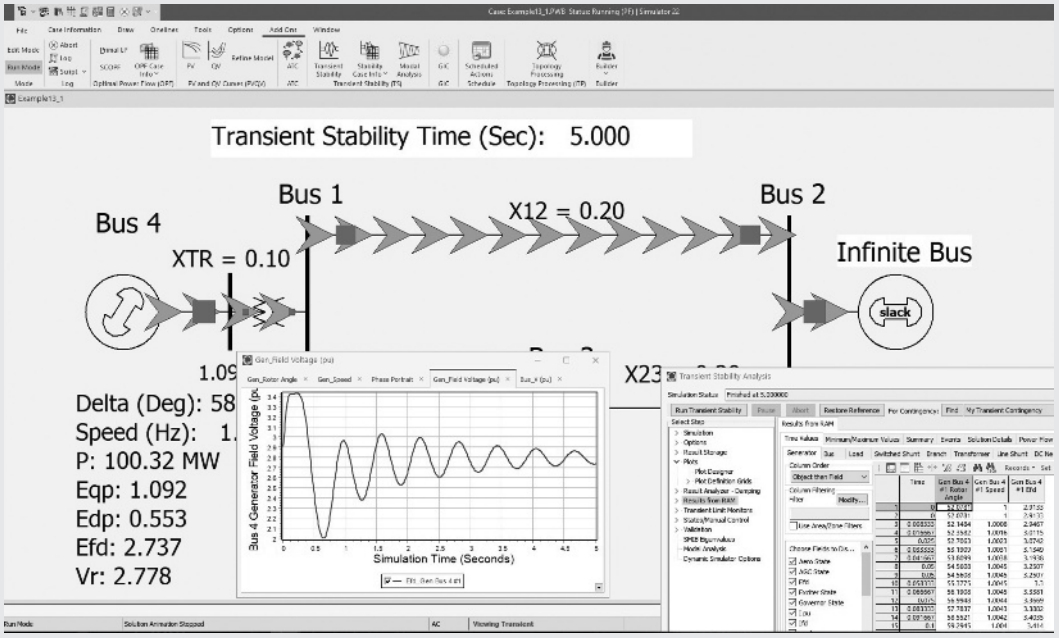

FIGURE 13.4

Example 13.1 results

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Block diagrams, such as those shown in Figure 13.3, are used for computer representation of generator-voltage control in transient stability computer programs (see Chapter 12). In practice, high-gain, fast-responding exciters provide large, rapid increases in field voltage E_{fd} during short circuits at the generator terminals in order to improve transient stability after fault clearing. Equations represented in the block diagram can be used to compute the transient response of generator-voltage control.

WIND TURBINE-GENERATORS

For wind turbine-generators, how their voltage is controlled depends upon the type of the wind turbines. Type 1 wind turbine-generators, squirrel cage induction machines, have no direct voltage control. Type 2 wind turbine-generators are wound rotor induction machines with variable external resistance. While they do not have direct voltage control, the external resistance control system is usually modeled as a type of exciter. The block diagram for such a model is shown in Figure 13.5. The purpose for this control is to allow for a more constant power output from the wind turbine. For example, if a wind gust were to cause the turbine blades to accelerate, this controller would quickly increase the external resistance, flattening the torque-speed curve, as shown in Figure 12.25.

Similar to synchronous machines, the Type 3 and 4 wind turbine-generators have the ability to perform voltage or reactive power control. Common control modes include constant power factor control, coordinated control across a wind farm to maintain a constant voltage at the interconnection point, and constant reactive power control. Figure 13.6 shows a simplified version of a Type 3 wind turbine exciter, in which Q_{cmd} is the commanded reactive power, V_t is the terminal voltage, and the output, E_q, is the input to the DFAG model shown in Figure 12.27. For fixed reactive power Q_{cmd} is a constant, while for power factor control, Q_{cmd} varies linearly with the real power output.

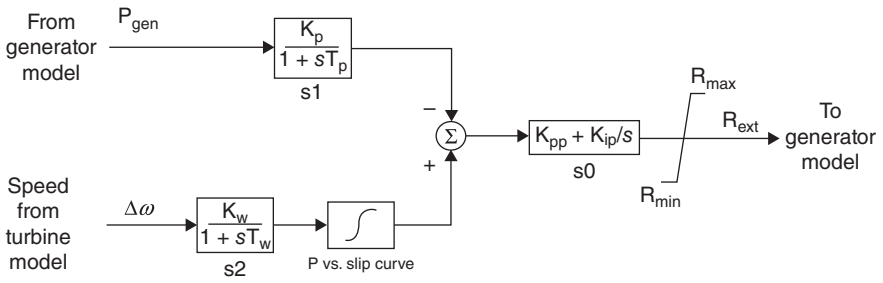

FIGURE 13.5

Simplified block diagram for a Type 2 wind turbine R_{ext} control system

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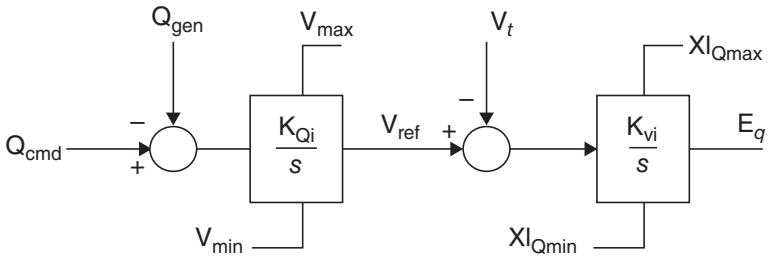
FIGURE 13.6

Simplified block diagram for a Type 3 wind turbine reactive power control system



EXAMPLE 13.2

Type 3 wind turbine-generator reactive power control

Assume the Type 3 wind turbine-generator from Example 12.12 has a Figure 13.6 reactive power control system with $K_{Qi} = 0.4$, $K_{Vi} = 40$, $XI_{Qmax} = 1.45$, $XI_{Qmin} = 0.5$, $V_{max} = 1.1$, $V_{min} = 0.9$ (per unit using a 100-MVA base). For the Example 12.12 system conditions, determine the initial values for V_{ref}, Q_{cmd}, and estimate the maximum amount of reactive power this system could supply during a fault that depresses the terminal voltage to 0.48 p.u.

SOLUTION

Since in steady state the inputs to each of the two integrator blocks in Figure 13.6 must be zero, V_{ref} is just equal to the initial terminal voltage magnitude from Example 12.12, that is, 1.024 p.u., and Q_{cmd} is the initial reactive power output, which is 0.22 per unit (22 Mvar), found from the imaginary part of the product of the terminal voltage and the conjugate of the terminal current. During the fault with its low terminal voltage, the positive input into the K_{Vi} integration block will cause E_q to rapidly rise to its limit $XI_{Qmax} = 1.45$. The reactive component of I_{sorc} will then be $-1.45/0.8 = -1.8125$ p.u. The total per-unit reactive power injection with $V_t = 0.48$ during the fault is then

$$Q_{net} = (0.48)(1.8125) - \frac{0.48^2}{0.8} = 0.582 \text{ p.u.} = 58.2 \quad \text{Mvar}$$

This result can be confirmed by opening PowerWorld Simulator case Example 13_2, which models such a fault condition (see Figure 13.7). After the fault is cleared, the reactive power controller restores the machine's reactive power output to its prefault value.

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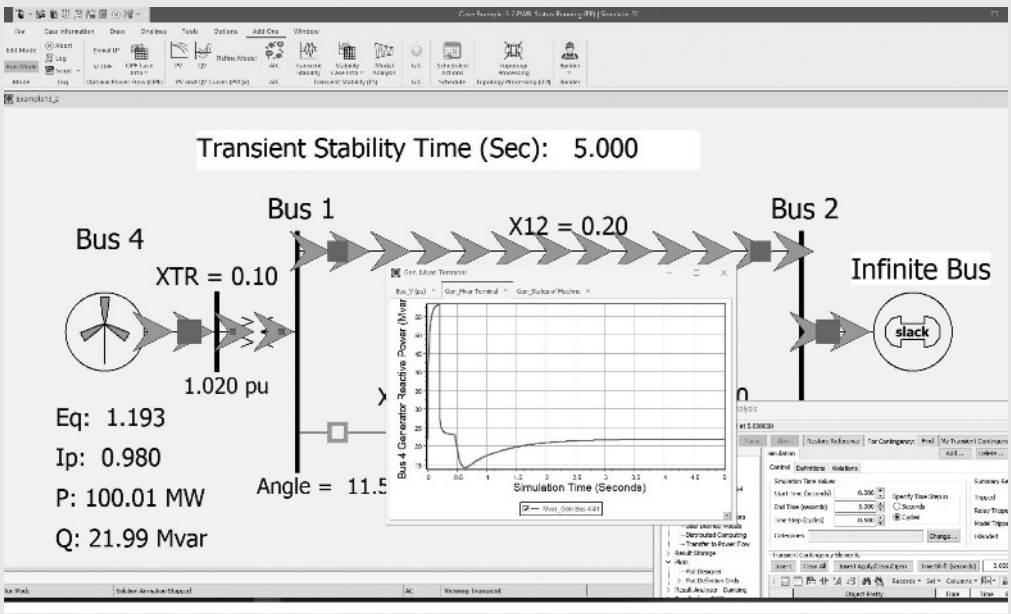

FIGURE 13.7

Example 13.2 variation in generator reactive power output

13.2 TURBINE-GOVERNOR CONTROL

SYNCHRONOUS GENERATORS

Turbine-generator units for synchronous machines operating in a power system contain stored kinetic energy due to their rotating masses. If the system load suddenly increases, stored kinetic energy is released to initially supply the load increase. Also, the electrical torque T_e of each turbine-generating unit increases to supply the load increase, while the mechanical torque T_m of the turbine initially remains constant. From Newton's second law, $J\alpha = T_m - T_e$, the acceleration α is therefore negative. That is, each turbine-generator decelerates and the rotor speed drops as kinetic energy is released to supply the load increase. The electrical frequency of each generator, which is proportional to rotor speed for synchronous machines, also drops.

From this, we conclude that either rotor speed or generator frequency indicates a balance or imbalance of generator electrical torque T_e and turbine mechanical torque T_m. If speed or frequency is decreasing, then T_e is greater than T_m (neglecting generator losses). Similarly, if speed or frequency is increasing,

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T_e is less than T_m. Accordingly, generator frequency is an appropriate control signal for governing the mechanical output power of the turbine.

The steady-state frequency-power relation for turbine-governor control is

$$\Delta p_m = \Delta p_{\text{ref}} - \frac{1}{\text{R}}\Delta f \tag{13.2.1}$$

where Δf is the change in frequency, Δp_m is the change in turbine mechanical power output, and Δp_{ref} is the change in a reference power setting. R is called the *regulation constant.* The equation is plotted in Figure 13.8 as a family of curves, with Δp_{ref} as a parameter. Note that when Δp_{ref} is fixed, Δp_m is directly proportional to the drop in frequency.

Figure 13.8 illustrates a steady-state frequency-power relation. When an electrical load change occurs, the turbine-generator rotor accelerates or decelerates, and frequency undergoes a transient disturbance. Under normal operating conditions, the rotor acceleration eventually becomes zero, and the frequency reaches a new steady state, shown in the figure.

The regulation constant R in (13.2.1) is the negative of the slope of the Δf versus Δp_m curves shown in Figure 13.8. The units of R are Hz/MW when Δf is in Hz and Δp_m is in MW. When Δf and Δp_m are given in per unit, however, R is also in per unit.

This type of control is often referred to as a *droop speed control* or just *droop control*. The word droop is used because as the terminal bus frequency increases, the steady-state MW output of the generator will decrease or droop. The per-unit input parameter R tells the amount of frequency change that would result in a rated power change in the generator's output. For example, with a 60-Hz, 200-MVA generator, an R value of 0.05 per unit means that for a 3-Hz (5%) increase in frequency, the generator's steady-state MW output would decrease by 200 MW. Usually, the bus frequency deviation from nominal is less than 0.1%.

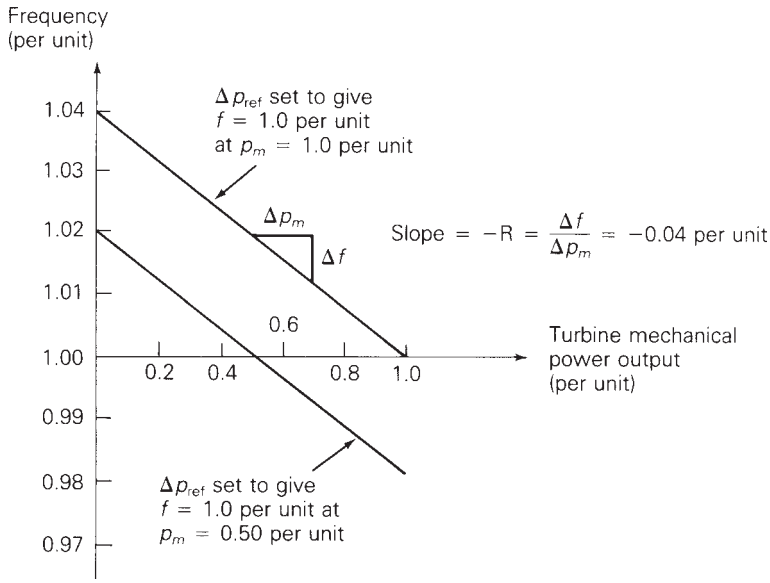

FIGURE 13.8
Steady-state frequency–power relation for a turbine-governor

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EXAMPLE 13.3

Turbine-governor response to a frequency change at a synchronous generating unit

A 500-MVA, 60-Hz turbine-generator for a synchronous machine has a regulation constant R = 0.05 per unit based on its own rating. If the generator frequency increases by 0.01 Hz in steady state, what is the decrease in turbine mechanical power output? Assume a fixed reference power setting.

SOLUTION

The per-unit change in frequency is

$$\Delta f_{\text{p.u.}} = \frac{\Delta f}{f_{\text{base}}} = \frac{0.01}{60} = 1.6667 \times 10^{-4} \quad \text{per unit}$$

Then, from (13.2.1), with $\Delta p_{\text{ref}} = 0$,

$$\Delta p_{m\text{p.u.}} = \left(\frac{-1}{0.05}\right)(1.6667 \times 10^{-4}) = -3.3333 \times 10^{-3} \quad \text{per unit}$$

$$\Delta p_m = (\Delta p_{m\text{p.u.}})\text{S}_{\text{base}} = (-3.3333 \times 10^{-3})(500) = -1.6667 \quad \text{MW}$$

The turbine mechanical power output decreases by 1.67 MW.

The steady-state frequency-power relation for one area of an interconnected power system can be determined by summing (13.2.1) for each turbine-generating unit in the area. Noting that Δf is the same for each unit,

$$\begin{aligned}\Delta p_m &= \Delta p_{m1} + \Delta p_{m2} + \Delta p_{m3} + \cdots \\ &= (\Delta p_{\text{ref1}} + \Delta p_{\text{ref2}} + \cdots) - \left(\frac{1}{\text{R}_1} + \frac{1}{\text{R}_2} + \cdots\right)\Delta f \\ &= \Delta p_{\text{ref}} - \left(\frac{1}{\text{R}_1} + \frac{1}{\text{R}_2} + \cdots\right)\Delta f \end{aligned} \tag{13.2.2}$$

where Δp_m is the total change in turbine mechanical powers and Δp_{ref} is the total change in reference power settings within the area. We define the *area frequency response characteristic* β as

$$\beta = \left(\frac{1}{\text{R}_1} + \frac{1}{\text{R}_2} \cdots\right) \tag{13.2.3}$$

Using (13.2.3) in (13.2.2),

$$\Delta p_m = \Delta p_{\text{ref}} - \beta \Delta f \tag{13.2.4}$$

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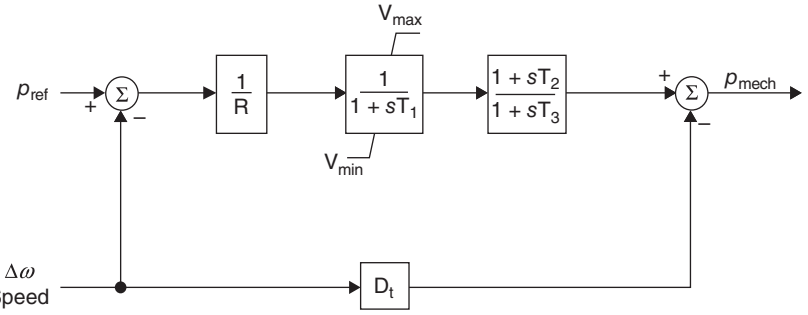
FIGURE 13.9

Turbine-governor block diagram



Equation (13.2.4) is the area steady-state frequency-power relation. The units of β are MW/Hz when Δf is in Hz and Δp_m is in MW. β can also be given in per-unit. In practice, β is somewhat higher than that given by (13.2.3) due to system losses and the frequency dependence of loads.

A standard value for the regulation constant is R = 0.05 per unit. When all turbine-generating units have the same per-unit value of R based on their own ratings, then each unit shares total power changes in proportion to its own rating. Figure 13.9 shows a block diagram for a simple steam turbine governor commonly known as the TGOV1 model. The $1/(1 + sT_1)$ models the time delays associated with the governor, where s is again the Laplace operator and T_1 is the time constant. The limits on the output of this block account for the fact that turbines have minimum and maximum outputs. The second block diagram models the delays associated with the turbine; for non-reheat turbines, T_2 should be zero. Typical values are R = 0.05 p.u., T_1 = 0.5 seconds, T_2 = 0, and T_3 = 0.5 seconds for a non-reheat turbine, or T_2 = 2.5 seconds and T_3 = 7.5 seconds otherwise. D_t is a turbine damping coefficient that is usually 0.02 or less (often zero). Additional turbine block diagrams are available in [3].

EXAMPLE 13.4

Response of turbine-governors for synchronous machines to a load change in an interconnected power system

An interconnected 60-Hz power system consists of one area with three turbine-generator units (synchronous machines) rated 1000, 750, and 500 MVA, respectively. The regulation constant of each unit is R = 0.05 per unit based on its own rating. Each unit is initially operating at one-half of its own rating when the system load suddenly increases by 200 MW. Determine (a) the per-unit area frequency response characteristic β on a 1000-MVA system base, (b) the steady-state drop in area frequency, and (c) the increase in turbine mechanical power output of each unit. Assume that the reference power setting of each turbine-generator remains constant. Neglect losses and the dependence of load on frequency.

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SOLUTION

a. The regulation constants are converted to per-unit on the system base using

$$R_{p.u.new} = R_{p.u.old}\frac{S_{base(new)}}{S_{base(old)}}$$

We obtain

$$R_{1p.u.new} = R_{1p.u.old} = 0.05$$

$$R_{2p.u.new} = (0.05)\left(\frac{1000}{750}\right) = 0.06667$$

$$R_{3p.u.new} = (0.05)\left(\frac{1000}{550}\right) = 0.10 \quad \text{per unit}$$

Using (13.2.3),

$$\beta = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{0.05} + \frac{1}{0.06667} + \frac{1}{0.10} = 45.0 \quad \text{per unit}$$

b. Neglecting losses and dependence of load on frequency, the steady-state increase in total turbine mechanical power equals the load increase, 200 MW or 0.20 per unit. Using (13.2.4) with $\Delta p_{ref} = 0$,

$$\Delta f = \left(\frac{-1}{\beta}\right)\Delta p_m = \left(\frac{-1}{45}\right)(0.20) = -4.444 \times 10^{-3} \quad \text{per unit}$$

$$= (-4.444 \times 10^{-3})\,(60) = -0.2667 \quad \text{Hz}$$

The steady-state frequency drop is 0.2667 Hz.

c. From (13.2.1), using $\Delta f = -4.444 \times 10^{-3}$ per unit,

$$\Delta p_{m1} = \left(\frac{-1}{0.05}\right)(-4.444 \times 10^{-3}) = 0.08888 \quad \text{per unit}$$

$$= 88.88 \quad \text{MW}$$

$$\Delta p_{m2} = \left(\frac{-1}{0.06667}\right)(-4.444 \times 10^{-3}) = 0.06666 \quad \text{per unit}$$

$$= 66.66 \quad \text{MW}$$

$$\Delta p_{m3} = \left(\frac{-1}{0.10}\right)(-4.444 \times 10^{-3}) = 0.04444 \quad \text{per unit}$$

$$= 44.44 \quad \text{MW}$$

Note that unit 1, whose MVA rating is 33% larger than that of unit 2 and 100% larger than that of unit 3, picks up 33% more load than unit 2 and 100% more

(*Continued*)

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load than unit 3. That is, each unit shares the total load change in proportion to its own rating.

PowerWorld Simulator case Example 13_4 contains a lossless nine-bus, three-generator system that duplicates the conditions from this example (see Figure 13.10). The synchronous generators at buses 1, 2 and 3 have ratings of 500, 1000, and 750 MVA, respectively, with initial outputs of 300, 600, 500 MWs. Each is modeled with a two-axis synchronous machine model (see Section 12.6), an IEEET1 exciter and a TGOV1 governor model with the parameters equal to the defaults given earlier. At time t = 0.5 seconds, the load at bus 8 is increased from 200 to 400 MW. Figure 13.10 shows the results of a 10-second transient stability simulation. The final generator outputs are 344.5, 689.0, and 566.7 MWs, while the final frequency decline is 0.272 Hz, closely matching the results predicted in the example (the frequency decline exactly matches the 0.266-Hz prediction if the simulation is extended to 20 seconds).

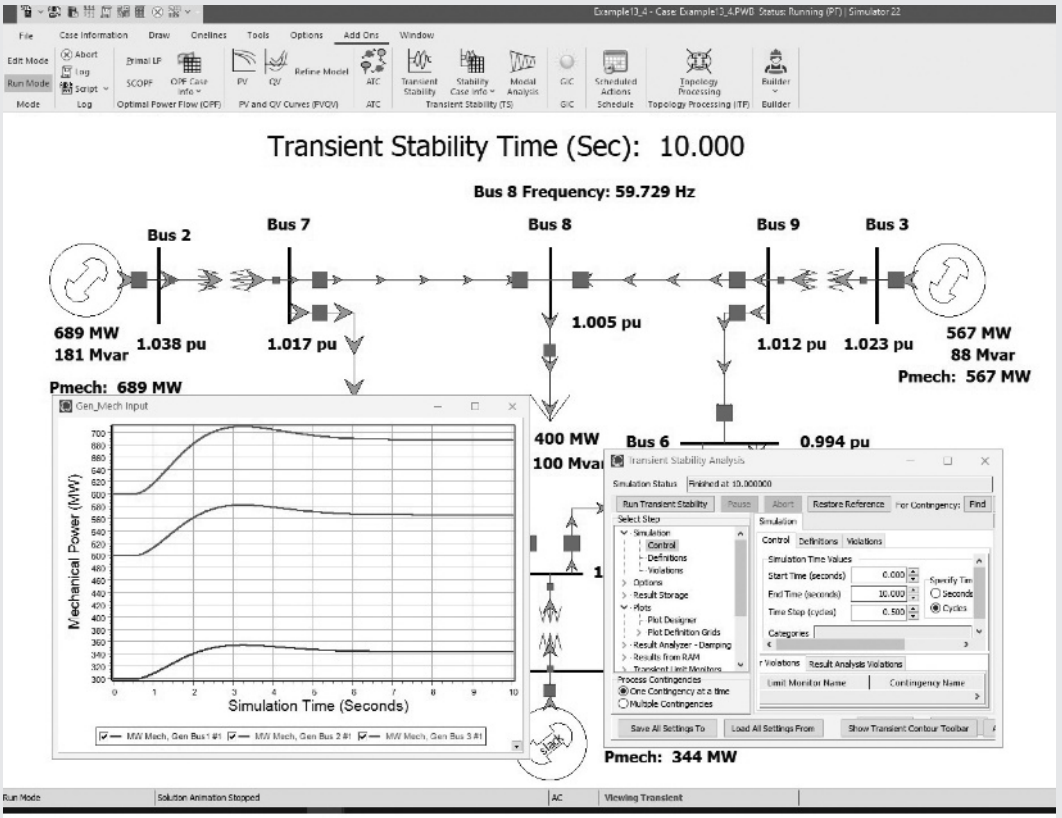

FIGURE 13.10

System oneline with generator mechanical power variation for Example 13.4

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In general, the larger the size of the interconnected system, the better the frequency response since there are more generators to share the task. However, the amount of online generation capability that is actually available to respond to a frequency disturbance, defined in [14] as *frequency responsive reserve* (FRR), depends on the type of generators that are available and how they are operated. Some generators, such as hydro, can respond very quickly and often have available capability. For others, there can be strong economic incentives for them to operate at their maximums, leaving little capacity for FRR. In general, "drawing frequency response from a large pool of geographically diverse resources makes frequency response faster, more reliable, and more effective than drawing from select isolated resources" [13]. How FRR is ensured varies from system to system, with some now allowing controllable load and select energy storage devices to also count.

WIND TURBINE-GENERATORS

The power output from wind turbine-generators can be controlled by changing the pitch angle of the blades [15]. When the available power in the wind is above the rating for the turbine, its blades are pitched to limit the mechanical power delivered to the electric machine. When the available power is less than the machine's rating, the pitch angle is set to its minimum. Figure 13.11 shows the generic pitch control model for Type 3 and 4 wind turbines, with the inputs being the per-unit speed of the turbine, ω_t, the desired speed, ω_{ref}, the ordered per-unit electrical output, P_{ord}, and a setpoint power, P_{ref0}. These signals are combined as shown on the figure to produce a commanded angle for the blades, θ, expressed in degrees. The right side of the block diagram models the dynamics and limits associated with changing the pitch angle of the blades; R_θ is the rate at which the blades change their angle in degrees per second. Typical values are T_{pi} = 0.3 seconds, minimum and maximum pitch angles (pi_{min}, pi_{max}) between 0° and 27°, rate limits (piratmin, piratmax) of ±10°/s, K_{iw} = 25, K_{pw} = 150 (both for pitch control), K_{ic} = 30, K_{pc} = 3 (both for pitch compensation) [15].

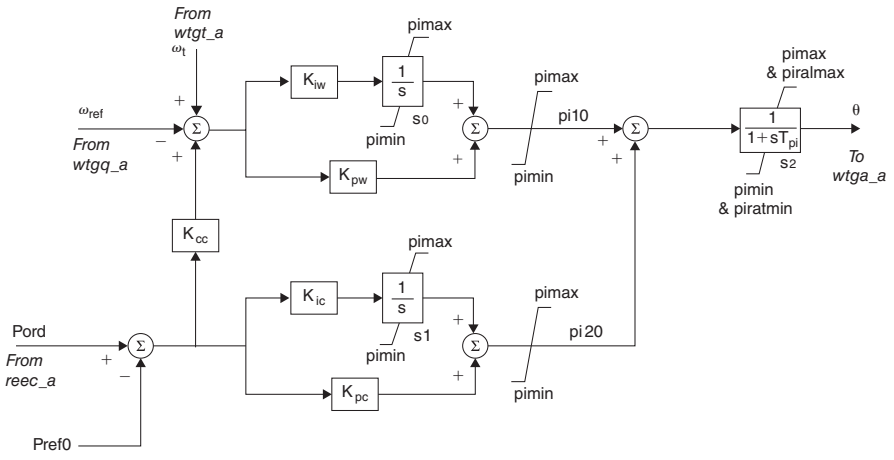

FIGURE 13.11

Pitch control for a Type 3 or 4 wind turbine model

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Note that Type 4 wind turbine-generators (see Figure 12.28) contribute no inertia to a power system grid because the ac-dc-ac converter completely decouples the turbine's mechanical power dynamics from the electrical power delivered to the grid. Also, Type 3 wind-turbine generators (Figures 12.26 and 12.27) contribute little or no inertia to the grid. With the increasing replacement of conventional (synchronous) generators with nonsynchronous renewable generators (including wind-turbine generators and solar photovoltaic plants), the available system inertia decreases. When a system power imbalance occurs, the frequency deviation will be larger in a system with high penetrations of renewables and less available inertia.

13.3 LOAD-FREQUENCY CONTROL

As shown in Section 13.2, turbine-governor control eliminates rotor accelerations and decelerations following load changes during normal operation. However, there is a steady-state frequency error Δf when the change in turbine-governor reference setting Δp_{ref} is zero. One of the objectives of load-frequency control (LFC), therefore, is to return Δf to zero.

In a power system consisting of interconnected areas, each area agrees to export or import a scheduled amount of power through transmission-line interconnections, or tie-lines, to its neighboring areas. Thus, a second LFC objective is to have each area absorb its own load changes during normal operation. This objective is achieved by maintaining the net tie-line power flow out of each area at its scheduled value.

The following summarizes the two basic LFC objectives for an interconnected power system:

1. Following a load change, each area should assist in returning the steady-state frequency error Δf to zero.
2. Each area should maintain the net tie-line power flow out of the area at its scheduled value, in order for the area to absorb its own load changes.

The following control strategy developed by N. Cohn [4] meets these LFC objectives. We first define the *area control error* (ACE) as follows:

$$\begin{aligned} \text{ACE} &= (p_{\text{tie}} - p_{\text{tie,sched}}) + \text{B}_f(f - 60) \\ &= \Delta p_{\text{tie}} + \text{B}_f \Delta_f \end{aligned} \tag{13.3.1}$$

where Δp_{tie} is the deviation in net tie-line power flow out of the area from its scheduled value $p_{\text{tie, sched}}$, and Δf is the deviation of area frequency from its scheduled value (60 Hz). Thus, the ACE for each area consists of a linear combination of tie-line error Δp_{tie} and frequency error Δf. The constant B_f is called a *frequency bias constant.*

The change in reference power setting $\Delta p_{\text{ref}i}$ of each turbine-governor operating under LFC is proportional to the integral of the area control error. That is,

$$\Delta p_{\text{ref}i} = -\text{K}_i \int \text{ACE}\, dt \tag{13.3.2}$$

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Each area monitors its own tie-line power flows and frequency at the area control center. The ACE given by (13.3.1) is computed and a percentage of the ACE is allocated to each controlled turbine-generator unit. Raise or lower commands are dispatched to the turbine-governors at discrete time intervals of two or more seconds in order to adjust the reference power settings. As the commands accumulate, the integral action in (13.3.2) is achieved.

The constant K_i in (13.3.2) is an integrator gain. The minus sign in (13.3.2) indicates that if either the net tie-line power flow out of the area or the area frequency is low—that is, if the ACE is negative—then the area should increase its generation.

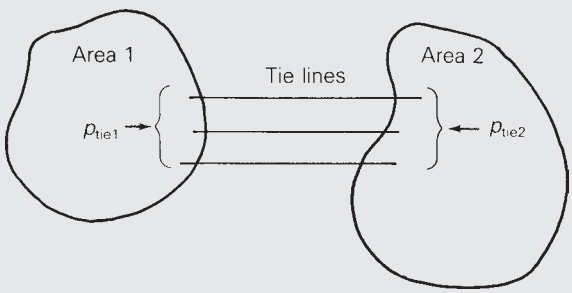
When a load change occurs in any area, a new steady-state operation can be obtained only after the power output of every turbine-generating unit in the interconnected system reaches a constant value. This occurs only when all reference power settings are zero, which in turn occurs only when the ACE of every area is zero. Furthermore, the ACE is zero in every area only when both Δp_{tie} and Δf are zero. Therefore, in steady state, both LFC objectives are satisfied.

EXAMPLE 13.5

Response of LFC to a load change in an interconnected power system

As shown in Figure 13.12, a 60-Hz power system consists of two interconnected areas. Area 1 has 2000 MW of total generation and an area frequency response characteristic $\beta_1 = 700$ MW/Hz. Area 2 has 4000 MW of total generation and $\beta_2 = 1400$ MW/Hz. Each area is initially generating one-half of its total generation at $\Delta p_{\text{tie1}} = \Delta p_{\text{tie2}} = 0$ and at 60 Hz when the load in area 1 suddenly increases by 100 MW. Determine the steady-state frequency error Δf and the steady-state tie-line error Δp_{tie} of each area for the following two cases: (a) without LFC, and (b) with LFC given by (13.3.1) and (13.3.2). Neglect losses and the dependence of load on frequency.

FIGURE 13.12
Example 13.5



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SOLUTION

a. Since the two areas are interconnected, the steady-state frequency error Δf is the same for both areas. Adding (13.2.4) for each area,

$$(\Delta p_{m1} + \Delta p_{m2}) = (\Delta p_{\text{ref1}} + \Delta p_{\text{ref2}}) - (\beta_1 + \beta_2)\Delta f$$

Neglecting losses and the dependence of load on frequency, the steady-state increase in total mechanical power of both areas equals the load increase, 100 MW. Also, without LFC, Δp_{ref1} and Δp_{ref2} *are* both zero. The above equation then becomes

$$100 = -(\beta_1 + \beta_2)\Delta f = -(700 + 1400)\Delta f$$
$$\Delta f = -100/2100 = -0.0476 \quad \text{Hz}$$

Next, using (13.2.4) for each area, with $\Delta p_{\text{ref}} = 0$,

$$\Delta p_{m1} = -\beta_1 \Delta f = -(700)(-0.0476) = 33.33 \quad \text{MW}$$
$$\Delta p_{m2} = -\beta_2 \Delta f = -(1400)(-0.0476) = 66.67 \quad \text{MW}$$

In response to the 100-MW load increase in area 1, area 1 picks up 33.33 MW and area 2 picks up 66.67 MW of generation. The 66.67-MW increase in area 2 generation is transferred to area 1 through the tie-lines. Therefore, the change in net tie-line power flow out of each area is

$$\Delta p_{\text{tie2}} = +66.67 \quad \text{MW}$$
$$\Delta p_{\text{tie1}} = -66.67 \quad \text{MW}$$

b. From (13.3.1), the area control error for each area is

$$\text{ACE}_1 = \Delta p_{\text{tie1}} + \text{B}_1 \Delta f_1$$
$$\text{ACE}_2 = \Delta p_{\text{tie2}} + \text{B}_2 \Delta f_2$$

Neglecting losses, the sum of the net tie-line flows must be zero; that is, $\Delta p_{\text{tie1}} + \Delta p_{\text{tie2}} = 0$ or $\Delta p_{\text{tie2}} = -\Delta p_{\text{tie1}}$. Also, in steady state, $\Delta f_1 = \Delta f_2 = \Delta f$.

Using these relations in the above equations,

$$\text{ACE}_1 = \Delta p_{\text{tie1}} + \text{B}_1 \Delta f$$
$$\text{ACE}_2 = -\Delta p_{\text{tie1}} + \text{B}_2 \Delta f$$

In steady state, $\text{ACE}_1 = \text{ACE}_2 = 0$; otherwise, the LFC given by (13.3.2) would be changing the reference power settings of turbine-governors on LFC. Adding the above two equations,

$$\text{ACE}_1 + \text{ACE}_2 = 0 = (\text{B}_1 + \text{B}_2)\Delta f$$

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Therefore, $\Delta f = 0$ and $\Delta p_{\text{tie1}} = \Delta p_{\text{tie2}} = 0$. That is, in steady state, the frequency error is returned to zero, area 1 picks up its own 100-MW load increase, and area 2 returns to its original operating condition—that is, the condition before the load increase occurred.

Note that the turbine-governor controls act almost instantaneously, subject only to the time delays shown in Figure 13.9. However, LFC acts more slowly. LFC raise and lower signals are dispatched from the area control center to turbine-governors at discrete time intervals of 2 or more seconds. Also, it takes time for the raise or lower signals to accumulate. Thus, case (a) represents the first action. Turbine-governors in both areas rapidly respond to the load increase in area 1 in order to stabilize the frequency drop. Case (b) represents the second action. As LFC signals are dispatched to turbine-governors, Δf and Δp_{tie} are slowly returned to zero.

The choice of the B_f and K_i constants in (13.3.1) and (13.3.2) affects the transient response to load changes—for example, the speed and stability of the response. The frequency bias B_f should be high enough such that each area adequately contributes to frequency control. Cohn [4] has shown that choosing B_f equal to the area frequency response characteristic, $B_f = \beta$, gives satisfactory performance of the interconnected system. The integrator gain K_i should not be too high; otherwise, instability may result. Also, the time interval at which LFC signals are dispatched, 2 or more seconds, should be long enough so that LFC does not attempt to follow random or spurious load changes. A detailed investigation of the effect of B_f, K_i, and LFC time interval on the transient response of LFC and turbine-governor controls is beyond the scope of this text.

Two additional LFC objectives are to return the integral of frequency error and the integral of net tie-line error to zero in steady state. By meeting these objectives, LFC controls both the time of clocks that are driven by 60-Hz motors, and energy transfers out of each area. These two objectives are achieved by making temporary changes in the frequency schedule and tie-line schedule in (13.3.1).

Finally, note that LFC maintains control during normal changes in load and frequency—that is, changes that are not too large. During emergencies, when large imbalances between generation and load occur, LFC is bypassed and other, emergency controls are applied.

COORDINATION OF ECONOMIC DISPATCH WITH LFC

Both the load-frequency control (LFC) and economic dispatch objectives are achieved by adjusting the reference power settings of turbine-governors on control. Figure 13.13 shows an *automatic generation control* strategy for achieving both objectives in a coordinated manner. In this figure, P_{iD} is the desired output of each generator as computed from an economic dispatch program, which is discussed in Section 7.1. As shown in Figure 13.13, the area control error (ACE) is

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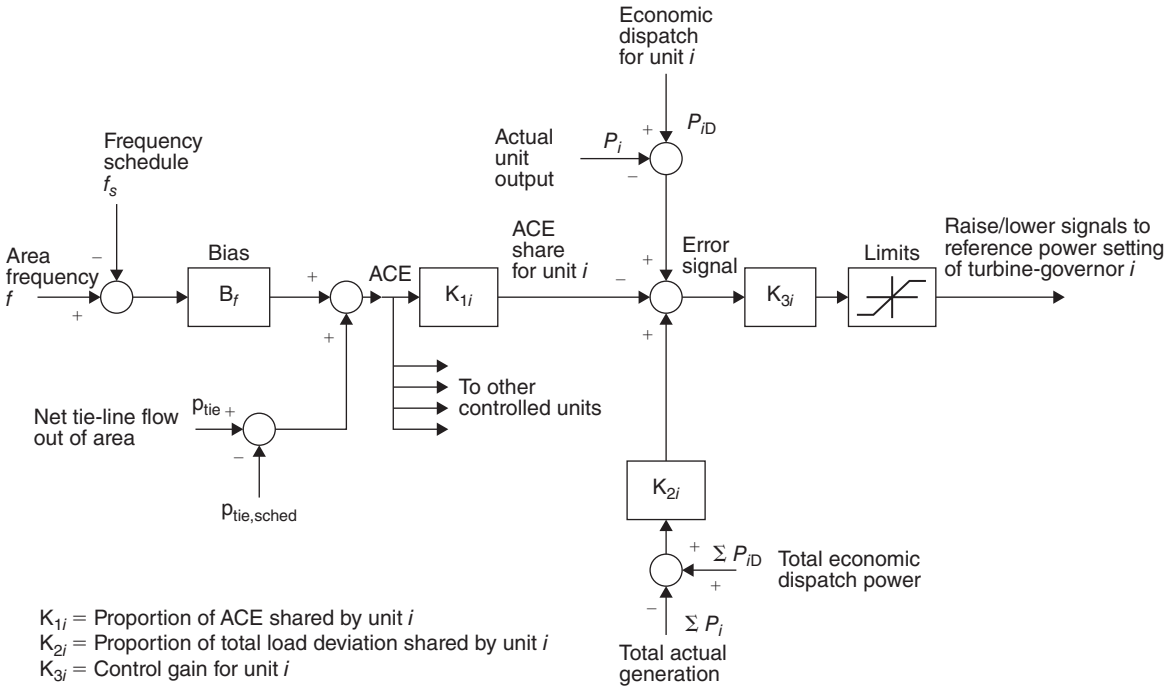

FIGURE 13.13

Automatic generation control [11] (Based on A.J. Wood and B.F. Wollenberg, *Power Generation, Operation, and Control*, 1989, John Wiley & Sons)

first computed, and a share K_{1i} of the ACE is allocated to each unit. Second, the deviation of total actual generation from total desired generation is computed, and a share $K_{2i}\Sigma(P_{iD} - P_i)$ is allocated to unit *i*. Third, the deviation of actual generation from desired generation of unit *i* is computed, and $(P_{iD} - P_i)$ is allocated to unit *i.* An error signal formed from these three components and multiplied by a control gain K_{3i} determines the raise or lower signals that are sent to the turbine-governor of each unit *i* on control.

In practice, raise or lower signals are dispatched to the units at discrete time intervals of 2 to 10 seconds. The desired outputs P_{iD} of units on control, determined from the economic dispatch program, are updated at slower intervals, typically every 2 to 10 minutes.

13.4 POWER SYSTEM STABILIZER CONTROL

One of the challenges with power system design is that beneficial changes in one type of control can also result in detrimental impacts elsewhere. As noted in [16], an example of this began to occur in the late 1950s as faster, continuously acting exciters started to be used, resulting in the development of power system

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oscillations at frequencies in the range of between 0.1 and 5 Hz. An oscillation is just a repetitive motion of a set of system values (signals) that can be either positively damped (decaying with time), negatively damped (growing with time), or undamped. Such oscillations are usually subdivided into inter-area oscillations with frequencies between 0.1 and 1.0 Hz that affect all or large portions of an interconnect, and local or plant oscillations with frequencies between 1.0 and 5.0 Hz that affect just a small portion of the grid. These oscillations may be due to either the natural response of the grid or they may be forced due to external inputs or malfunctioning equipment, and they may be apparent only after a large contingency occurs (ring-down oscillations) or they may always be apparent as an ambient response due to the small random changes continually affecting grids [18]. Also, multiple oscillations may occur simultaneously. Regardless of the cause, the design goal is to ensure that they are sufficiently well damped.

When considering just the oscillatory part of a power system signal, $y(t)$, such as a generator rotor angle or bus voltage frequency, it can be written as a summation of q modes,

$$y(t) = \sum_{i=1}^{q} A_i e^{\sigma_i t} \cos(\omega_i t + \phi_i) \tag{13.4.1}$$

where σ_i is the mode's damping factor, ω_i its angular frequency, A_i its mode shape magnitude, and ϕ_i its phase angle. The mode's percent damping is then defined as

$$\zeta_i = \frac{-\sigma_i}{\sqrt{\sigma_i^2 + \omega_i^2}} \times 100 \tag{13.4.2}$$

with $\zeta_i > 0$, indicating a positively damped mode that decays with time.

A now common way to increase the modal damping is to modulate the synchronous generator exciter voltage input through the use of a power system stabilizer (PSS). The block diagram for a simple PSS is shown in Figure 13.14, with the output of the PSS, V_s, shown as one of the exciter voltage inputs in Figure 13.3. In order to provide increased damping, the PSS output must provide torque that is in phase with the speed variation. To do this since the PSS input is applied as an exciter input, the PSS must account for the gain and phase lag characteristics of the generator, exciter, and power system, which is denoted in [16] as the GEP(s). PSSs can be designed for different types of inputs, and now often have two different input signals. These include generator speed, generator power, generator accelerating power, bus voltage frequency, and bus voltage magnitude.

Referring to the single input PSS in Figure 13.14, the first block models the input signal sensing delay with T_r almost always <0.05 seconds and which is often set to zero. The next two blocks represent the washout (high frequency) filter that is used to quickly remove low frequency (below 0.1 Hz) input components; usually T_w is between 2 and 10 seconds. The output limits are used to avoid large values during transients and usually range between ± 0.05 and ± 0.1 per unit. The remaining parameters, K_s and T_1 to T_4, are set via a process known as tuning to provide the designed phase lead compensation at a particular frequency and associated gain. Usually $T_1 = T_3$, and $T_2 = T_4$, so each block provides half of the

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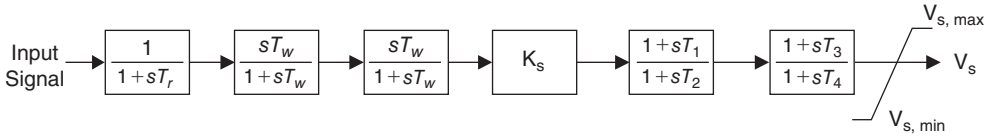

FIGURE 13.14

Block diagram of a simple Power System Stabilizer (PSS)

phase lead compensation. If the goal for the first block is to provide phase lead compensation of ϕ degrees at an angular frequency ω_d (with $-90° < \phi < 90°$), then

$$\alpha = \frac{1 - \sin\phi}{1 + \sin\phi} \tag{13.4.3}$$

$$T_1 = \frac{1}{\omega_d\sqrt{\alpha}},\ T_2 = \alpha T_1 \tag{13.4.4}$$

PSS turning is usually done through testing with the goal to determine the phase lag at the desired angular frequency ω_d and set a value for K_S. Typically, ω_d is associated with a plant mode of between 1 and 3 Hz. During testing, the generator is operated close to its maximum output level. Then K_s is increased until there is an undamped oscillation. For good stability, the gain should be set to 1/3 to ½ this value [21].

EXAMPLE 13.6

PSS application example

PowerWorld Simulator case Example13_6 demonstrates the tuning of a stabilizer using a modified version of the nine-bus system used in Example 13.4, with the contingency being a self-clearing fault at bus 8. Initially, the case does not have an active PSS; to see the initial buses frequencies without the PSS, open the case and select **Transient Stability**, **Run Transient Stability**. This response is shown in Figure 13.15. To determine the benefits of adding a PSS, a first step is to quantify the existing modal frequencies and damping. Based on research over the last several decades, there are now several methods for doing this from measurements, with PowerWorld Simulator including a version of the Matrix Pencil Method [22]. To calculate the modes for this example, select **Add Ons**, **Modal Analysis** to display the Modal Analysis Form; then click on **Do Modal Analysis**. The modes are shown in the upper right, with the largest mode having a frequency of 1.36 Hz and damping of 5.8%.

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To improve the damping, a PSS1A stabilizer has been added to the generator at bus 2; this stabilizer is similar to the one given in Figure 13.14, except it only has one washout block. To tune the PSS, assume $\omega_d = 2\pi \times 1.36$ Hz and total phase lead of 100° corresponding to 50° per lead-lag (with this phase value determined in actual tuning through testing and here by injecting a sample test signal and measuring the phase lag). Using (13.4.3) and (13.4.4),

$$\alpha = \frac{1 - \sin 50^\circ}{1 + \sin 50^\circ} = 0.1325$$

$$T_1 = \frac{1}{1.36 \times 2\pi\sqrt{0.1325}} = 0.321 = T_3$$

$$T_2 = \alpha T_1 = 0.042 = T_4$$

and then K_s is set by first iteratively determining the value at which sustained oscillations start ($K_s = 57$), and then setting its value to about 1/3 of this ($K_s = 19$). To see this in PowerWorld Simulator, right-click on the bus 2 generator symbol and display the Generator Information Dialog. Then select **Stability**, **Stabilizer** and set the PSS1A stabilizer as Active; its initial values

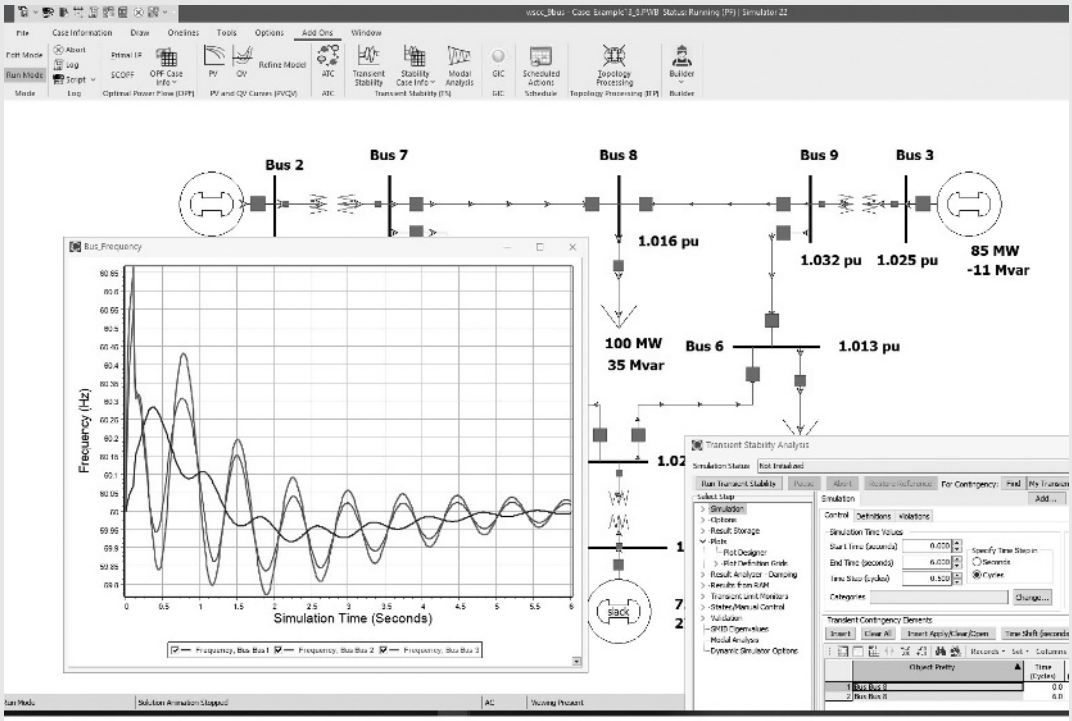

FIGURE 13.15

Bus frequency variations in Example 13.6 without power system stabilizer control

(*Continued*)

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match these with $K_s = 57$. Run the transient stability simulation and note the small, sustained oscillations (which is easiest to see by showing the Generator_V_s plot). Then change K_s to 19, rerun the stability simulation, and redo the modal analysis. Note that the damping has increased to 17.3%, with the result shown in Figure 13.16, and is also stored in the PowerWorld Simulator case Example13_6_WithPSSActive.

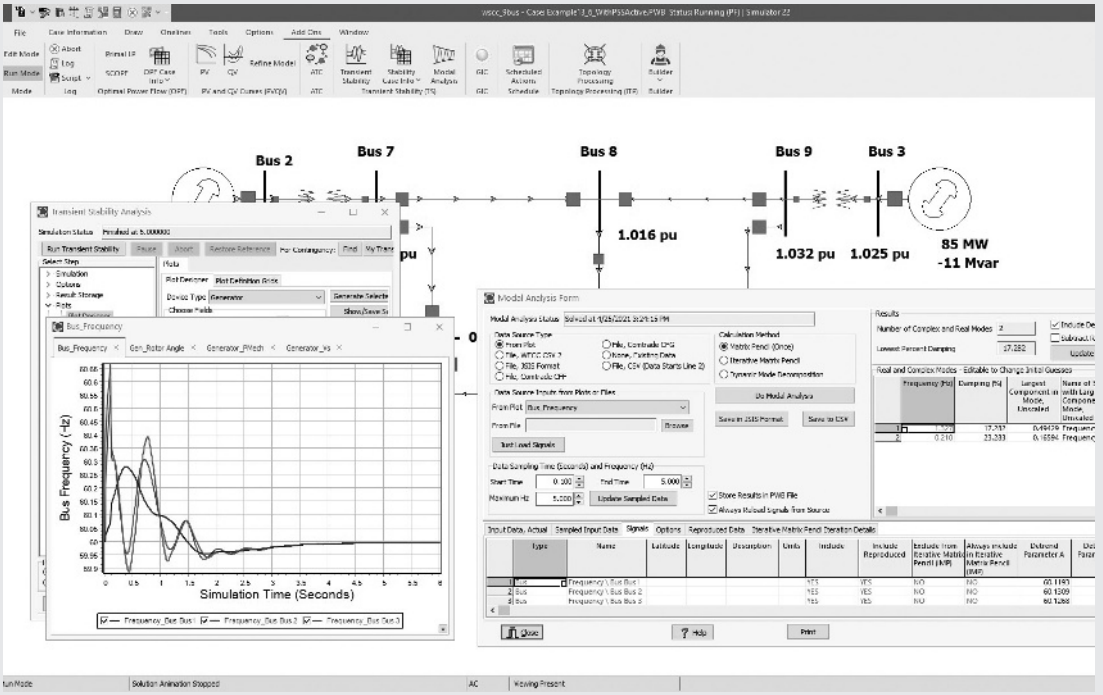

FIGURE 13.16

Bus frequency variations in Example 13.6 with power system stabilizer control

PROBLEMS

SECTION 13.1

13.1 The block-diagram representation of a closed-loop automatic regulating system, in which voltage control of a synchronous generator is accomplished by controlling the exciter voltage, is shown in Figure 13.17. T_a, T_e, and T_f are the time constants associated with the amplifier, exciter, and generator field circuit, respectively. (a) Find the open-loop transfer function G(s). (b) Evaluate the minimum open-loop gain such that the steady-state error Δe_{ss} does not exceed 1%. (c) Discuss the nature of the dynamic response of the system to a step change in the reference input voltage.

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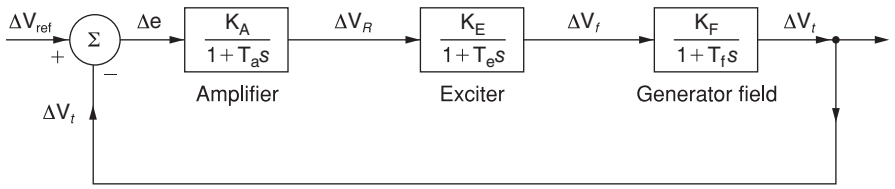

FIGURE 13.17
Problem 13.1

13.2 The Automatic Voltage Regulator (AVR) system of a synchronous generator is represented by the simplified block diagram shown in Figure 13.18, in which the sensor is modeled by a simple first-order transfer function. The voltage is sensed through a voltage transformer and then rectified through a bridge rectifier. Parameters of the AVR system are given as follows.

	Gain	Time Constant (seconds)
Amplifier	K_A	$T_A = 0.1$
Exciter	$K_E = 1$	$T_E = 0.4$
Generator	$K_G = 1$	$T_G = 1.0$
Sensor	$K_R = 1$	$T_R = 0.05$

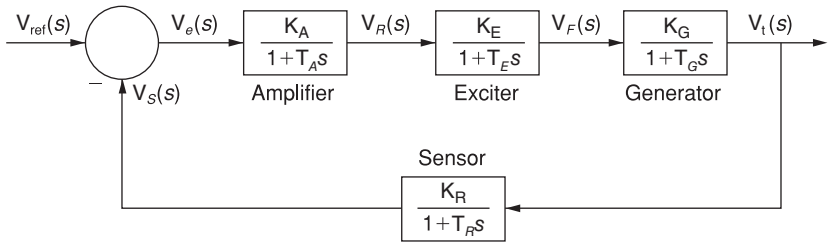

FIGURE 13.18
Problem 13.2

(a) Determine the open-loop transfer function of the block diagram and the closed-loop transfer function relating the generator terminal voltage $V_t(s)$ to the reference voltage $V_{ref}(s)$. (b) For the range of K_A from 0 to 12.16, comment on the stability of the system. (c) For $K_A = 10$, evaluate the steady-state step response and steady-state error.

PW 13.3 Open PowerWorld Simulator case Problem 13_3. This case models the system from Example 13.1, except that the rate feedback gain constant, K_f, has been set to zero and the simulation end time was increased to 30 seconds. Without rate feedback, the system voltage response will become unstable if the amplifier gain, K_a, becomes too large. In the simulation, this instability will be indicated by undamped oscillations in the terminal voltage (because of the limits on V_r, the response does not grow to infinity but rather bounces between the limits). Using transient stability simulations, iteratively determine

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the approximate value of K_a at which the system becomes unstable. The value of K_a can be on the **Generator Information Dialog**, **Stability**, **Exciters** page.

PW **13.4** One of the disadvantages of the IEEET1 exciter is that following a fault, the terminal voltage does not necessarily return to its prefault value. Using PowerWorld Simulator case Problem 13_3, determine the prefault bus 4 terminal voltage and field voltage. Then use the simulation to determine the final, postfault values for these fields for K_a = 100, 300, 30, and 10. Referring to Figure 13.3, what is the relationship between the reference voltage, the steady-state terminal voltage, and the field voltage?

SECTION 13.2

13.5 An area of an interconnected 60-Hz power system has three turbine-generator units (synchronous machines) rated 200, 300, and 500 MVA. The regulation constants of the units are 0.04, 0.05, and 0.06 per unit, respectively, based on their ratings. Each unit is initially operating at one-half its own rating when the load suddenly decreases by 150 MW. Determine (a) the unit area frequency response characteristic β on a 100-MVA base, (b) the steady-state increase in area frequency, and (c) the MW decrease in mechanical power output of each turbine. Assume that the reference power setting of each turbine-governor remains constant. Neglect losses and the dependence of load on frequency.

13.6 Each unit in Problem 13.5 is initially operating at one-half its own rating when the load suddenly increases by 100 MW. Determine (a) the steady-state decrease in area frequency, and (b) the MW increase in mechanical power output of each turbine. Assume that the reference power setting of each turbine-generator remains constant. Neglect losses and the dependence of load on frequency.

13.7 Each unit in Problem 13.5 is initially operating at one-half its own rating when the frequency increases by 0.005 per unit. Determine the MW decrease of each unit. The reference power setting of each turbine-governor is fixed. Neglect losses and the dependence of load on frequency.

13.8 Repeat Problem 13.7 if the frequency decreases by 0.005 per unit. Determine the MW increase of each unit.

13.9 An interconnected 60-Hz power system consisting of one area has two turbine-generator units (synchronous machines), rated 500 and 750 MVA, with regulation constants of 0.04 and 0.06 per unit, respectively, based on their respective ratings. When each unit carries a 300-MVA steady-state load, let the area load suddenly increase by 250 MVA. (a) Compute the area frequency response characteristic β on a 1000-MVA base. (b) Calculate Δf in per unit on a 60-Hz base and in Hz.

PW **13.10** Open PowerWorld Simulator case Problem 13_10. The case models the system from Example 13.4, except (1) the load increase is a 40% rise at bus 6 for a total increase of 200 MW (from 500 MW to 700 MW), and

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(2) the value of R for generator 1 is changed from 0.05 to 0.04 per unit. Repeat Example 13.4 using these modified values.

PW **13.11** Open PowerWorld Simulator case Problem 13_11, which includes a transient stability representation of the system from Example 6.13. Each generator is modeled using a two-axis machine model, an IEEE Type 1 exciter, and a TGOV1 governor with R = 0.05 per unit. (A summary of the generator models is available by selecting either **Stability Case Info**, **Transient Stability Generator Model Use**, which includes the generator MVA base, or **Stability Case Info**, **Models In Use**.) The contingency is the loss of the generator at bus PEAR69, which initially has 65 MW of generation. Analytically determine the steady-state frequency error in Hz following this contingency. Use PowerWorld Simulator to confirm this result; also, determine the magnitude and time of the largest bus frequency deviation.

PW **13.12** Repeat Problem 13.11, except first double the H value for each of the machines. This can be most easily accomplished by selecting **Stability Case Info**, **Models in Use** to view a form showing all the stability models. Right-click on the line corresponding to the Machine Model class, and then select Show Dialog to view an editable form of the model parameters. Compare the magnitude and time of the largest bus frequency deviations between Problem 13.12 and 13.11.

13.13 For a large, 60-Hz, interconnected electrical system, assume that following the loss of two 1400-MW generators (for a total generation loss of 2800 MW), the change in frequency is −0.12 Hz. If all the on-line generators that are available to participate in frequency regulation have an R of 0.04 per unit (on their own MVA base), estimate the total MVA rating of these units.

SECTION 13.3

13.14 A 60-Hz power system consists of two interconnected areas. Area 1 has 1200 MW of generation and an area frequency response characteristic β_1 = 400 MW/Hz. Area 2 has 1800 MW of generation and β_2 = 600 MW/Hz. Each area is initially operating at one-half its total generation, at $\Delta p_{\text{tie1}} = \Delta p_{\text{tie2}} = 0$ and at 60 Hz, when the load in area 1 suddenly increases by 400 MW. Determine the steady-state frequency error and the steady-state tie-line error Δp_{tie} of each area. Assume that the reference power settings of all turbine-governors are fixed. That is, LFC is not employed in any area. Neglect losses and the dependence of load on frequency.

13.15 Repeat Problem 13.14 if LFC is employed in area 2 alone. The area 2 frequency bias coefficient is set at $B_{f2} = \beta_2 = 600$ MW/Hz. Assume that LFC in area 1 is inoperative due to a computer failure.

13.16 Repeat Problem 13.14 if LFC is employed in both areas. The frequency bias coefficients are $B_{f1} = \beta_1 = 400$ MW/Hz and $B_{f2} = \beta_2 = 600$ MW/Hz.

13.17 Rework Problems 13.15 and 13.16 when the load in area 2 suddenly decreases by 300 MW. The load in area 1 does not change.

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13.18 On a 1000-MVA common base, a two-area system interconnected by a tie line has the following parameters:

Area	1	2
Area Frequency Response Characteristic	$\beta_1 = 0.05$ per unit	$\beta_2 = 0.0625$ per unit
Frequency-Dependent Load Coefficient	$D_1 = 0.6$ per unit	$D_2 = 0.9$ per unit
Base Power	1000 MVA	1000 MVA
Governor Time Constant	$\tau_{g1} = 0.25$ s	$\tau_{g2} = 0.3$ s
Turbine Time Constant	$\tau_{t1} = 0.5$ s	$\tau_{t2} = 0.6$ s

The two areas are operating in parallel at the nominal frequency of 60 Hz. The areas are initially operating in steady state with each area supplying 1000 MW when a sudden load increase of 187.5 MW occurs in area 1. Compute the new steady-state frequency and change in tie-line power flow (a) without LFC, and (b) with LFC.

SECTION 13.4

13.19 Using the Figure 13.14 model, if the goal is to provide a total of 60° of phase lead compensation at a frequency of 2 Hz, select the values T_1 and T_2. (Assume $T_1 = T_3$ and $T_2 = T_4$.) You can ignore the impact of the washout filters and can assume $T_r = 0$.

13.20 In a model using Figure 13.14, if the value of T1 = T3 is 0.4 seconds and T2 = T4 is 0.1 seconds, what is the corresponding angular frequency ω_d and total phase lead? Ignore any phase shift associated with the washout filters.

PW **13.21** Repeat Example 13.6, except assume that the desired total phase lead is zero degrees at 1.36 Hz. Give the values of T_1 and T_2 (with $T_3 = T_1$ and $T_4 = T_2$), the maximum percent damping, and the associated value of K_s.

PW **13.22** Repeat Example 13.6, except assume that the desired total phase lead is 30 degrees (15 degrees per block) at 1.36 Hz. Give the values of T_1 and T_2 (with $T_3 = T_1$ and $T_4 = T_2$), the maximum percent damping, and the associated value of K_s.

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14 Transmission Lines: Transient Operation

Power engineers with electric gird protection equipment. (Courtesy of Wei Trinh.)

LEARNING OBJECTIVES

At the conclusion of this chapter, you should be able to:

1. Understand the concept of traveling waves on single-phase, lossless lines.
2. Determine traveling wave solutions on single-phase lossless lines energized by voltage or current sources and terminated by RLC elements.
3. Organize reflections and refractions on single-phase, lossless lines using the Bewley lattice diagram.
4. Be familiar with discrete-time models of transmission lines and RLC elements suitable for digital computer solution of traveling waves on transmission lines.
5. Be familiar with different types of overvoltages encountered by power system equipment.
6. Understand the concepts of Basic Insulation Level (BIL) and insulation coordination, and be familiar with protective devices used to guard against equipment insulation failure.

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Transient overvoltages caused by lightning strikes to transmission lines and by switching operations are of fundamental importance in selecting equipment insulation levels and surge-protection devices. It is important, therefore, to understand the nature of transmission-line transients.

During our study of the steady-state performance of transmission lines in Chapter 5, the line constants R, L, G, and C were recognized as distributed rather than lumped constants. When a line with distributed constants is subjected to a disturbance, such as a lightning strike or a switching operation, voltage and current waves arise and travel along the line at a velocity near the speed of light. When these waves arrive at the line terminals, reflected voltage and current waves arise and travel back down the line, superimposed on the initial waves.

Because of line losses, traveling waves are attenuated and essentially die out after a few reflections. Also, the series inductances of transformer windings effectively block the disturbances, thereby preventing them from entering generator windings. However, due to the reinforcing action of several reflected waves, it is possible for voltage to build up to a level that could cause transformer insulation or line insulation to arc over and suffer damage.

Circuit breakers, which can operate within 50 ms, are too slow to protect against lightning or switching surges. Lightning surges can rise to peak levels within a few microseconds and switching surges within a few hundred microseconds—fast enough to destroy insulation before a circuit breaker could open. However, protective devices are available. Called surge arresters, these can be used to protect equipment insulation against transient overvoltages. These devices limit voltage to a ceiling level and absorb the energy from lightning and switching surges.

This chapter begins with a discussion of traveling waves on single-phase lossless lines (Section 14.1). Section 14.2 presents boundary conditions, and the Bewley lattice diagram for organizing reflections is covered in Section 14.3. Discrete-time models of single-phase lines and of lumped RLC elements are derived in Section 14.4, and the effects of line losses and multi-conductor lines are discussed in Sections 14.5 and 14.6. In Section 14.7, power system overvoltages, including lightning surges, switching surges, and power-frequency overvoltages, followed by an introduction to insulation coordination, are discussed in Section 14.8.

CASE STUDY

The following case study describes metal oxide varistor (MOV) arresters used by electric utilities to protect transmission and distribution equipment against transient overvoltages in power systems with rated voltages through 345 kV [22].

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Surge Arresters

VariSTAR Station-Class Type AZE Surge Arresters for Systems through 345 kV IEEE Certified*

General

Eaton's Cooper Power™ series VariSTAR® AZE Surge Arrester offers the latest in metal oxide varistor (MOV) technology for the economical protection of power and substation equipment. This arrester is gapless and constructed of a single series column of MOV disks. The arrester is designed and tested to the requirements of IEEE Std C62.11™-1993 standard and is available in ratings suitable for the transient overvoltage protection of electrical equipment on systems through 345 kV.

Eaton assures the design integrity of the AZE arrester through a rigorous testing program conducted at our manufacturing and testing facility in Olean, New York. The availability of complete in-house testing facilities assures that as continuous process improvements are made, they are professionally validated to high technical standards.

Table 1, shown below, contains information on some of the specific ratings and characteristics of AZE Series S (AZES) surge arresters.

Arrester Characteristic	Rating
System Application Voltages	3-345 kV
Arrester Voltage Ratings	3-360 kV
Rated Discharge Energy, (kJ/kV of MCOV)	
Arrester Ratings: 3-108 kV	3.4
120-240 kV	5.6
259-360 kV	8.9
System Frequency	50/60 Hz
Impulse Classifying Current	10 kA
High Current Withstand	100 kA
Pressure Relief Rating, kA rms sym	
Metal-Top Designs	65 kA
Cubicle-Mount Designs	40 kA
Cantilever Strength (in-lbs)*	
Metal-Top Designs:	
3-240 kV	90,000
258-360 kV	120,000

TABLE 1

AZE Series S (AZES) Ratings and Characteristics

*Maximum working load should not exceed 40% of this value.

Construction

External

The Type AZE station class arrester is available in two design configurations—a metal-top design in ratings 3-360 kV and a cubicle-mount design in ratings 3-48 kV. Cubicle-mount designs are ideally suited for confined spaces where clearances between live parts are limited.

*VariSTAR Type AZE Station-Class Surge Arresters for Systems through 345 kV IEEE Certified. https://www.eaton.com/content/dam/eaton/products/medium-voltage-power-distribution-control-systems/line-installation-and-protective-equipment/varistar-type-aze-stat

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The wet-process porcelain housing features an alternating shed design (ratings >48 kV) that provides excellent resistance to the effects of atmospheric housing contamination. AZE arresters are available with optional extra creepage porcelains for use in areas with extreme natural atmospheric and man-made pollution.

The dielectric properties of the porcelain are coordinated with the electrical protective characteristics of the arrester. The unit end castings are of a corrosion-resistant aluminum alloy configured for interchangeable mounting with other manufacturers' arresters for ease in upgrading to the VariSTAR arrester technology. This three-footed mounting is provided on a 21.6- to 25.4 cm diameter pattern for customer supplied 1.3-cm diameter hardware.

High Cantilever strength assures mechanical integrity (Table 1 lists the cantilever strength of metal-top AZES arresters). Eaton recommends that a load limit of 100 pounds not be exceeded on the line terminal of cubicle mount designs. Loads exceeding this limit could cause a shortening of arrester life. Housings are available in standard grey or optional brown glaze color.

Standard line and ground terminal connectors accommodate up to a 1.9 cm diameter conductor. Insulating bases and discharge counters are optionally available for in-service monitoring of arrester discharge activity.

Figure 1 120-kV rated VariSTAR Type AZE surge arrester

The end fittings and porcelain housing of each arrester unit are sealed and tested by means of a sensitive helium mass spectrometer;

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this assures that the quality and insulation protection provided by the arrester is never compromised over its lifetime by the entrance of moisture. A corrosion-resistant nameplate is provided and contains all information required by standards. In addition, stacking arrangement information is provided for multi-unit arresters. Voltage grading rings are included for arresters rated 172 kV and above.

Internal

VariSTAR AZE arresters are a totally gapless MOV design. Gapless construction makes a significant contribution to the performance of arresters through the elimination of gap reseal as a consideration associated with the discharge of switching surge currents. The specially formulated metal-oxide varistors, manufactured under Eaton's exclusive quality

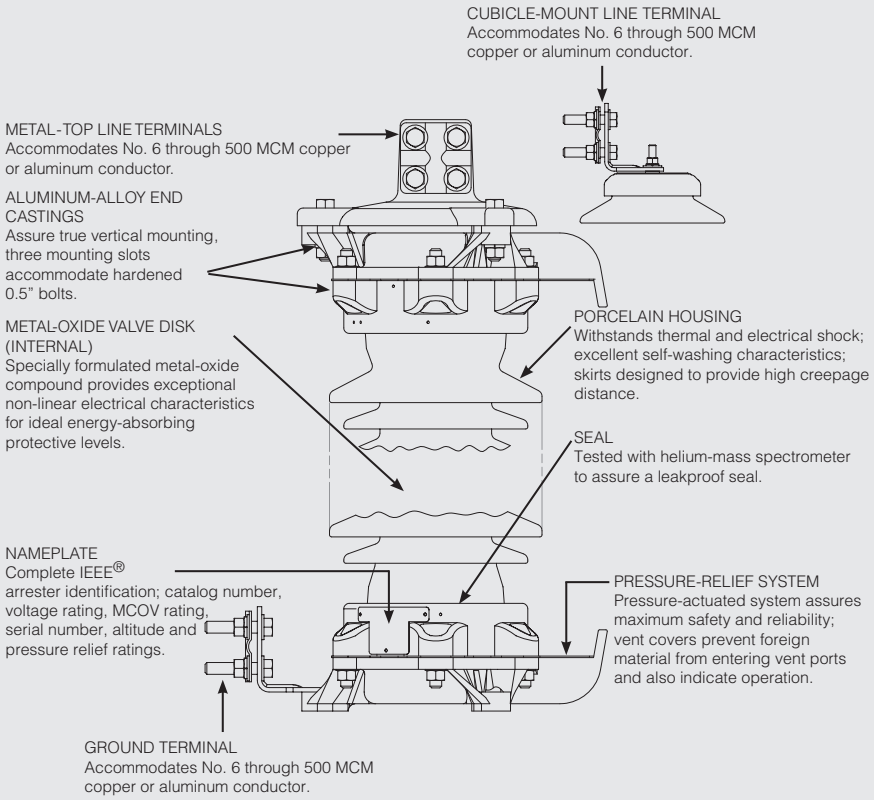

Figure 2 VariSTAR Type AZE arrester construction details. ***Note:*** *Multi-section arresters include an additional unit nameplate showing the order of assembly. For example, a 2-section arrester will have a bottom section with a unit nameplate labeled unit 1 of 2 and the top section with a unit nameplate labeled unit 2 of 2.*

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control, provide exceptional nonlinear protective characteristics, durability, and dependable energy dissipation capabilities.

Operation

The VariSTAR AZE arrester conducts only a few milliamperes of leakage current when energized at normal system voltage. When a surge event occurs, the arrester conducts only the current, and consequently the energy necessary to limit the overvoltage. It provides precise and predictable protection, minimizes the absorbed energy, and does not discharge power frequency systems currents.

A controlled and directed pressure relief system is incorporated in the VariSTAR AZE arrester design. In the unlikely event of an arrester failure, this pressure venting system rapidly relieves internal pressure and transfers the internal arc to the outside of the arrester porcelain through vent ports in the end castings.

When called upon to operate, this mechanism vents internal pressures in fractions of a cycle—preventing violent arrester failure. This mechanism is effective to system fault currents up to 65 kA in metal-top designs and 40 kA in cubicle-mount designs.

General Application Recommendations

The rating of an arrester is the power-frequency line-to-ground voltage at which the arrester is designed to pass the IEEE Std C62.11™-1993 standard duty-cycle test. Table 2 provides a general guide for the selection of the proper arrester for a given system voltage. Eaton application engineers are available to make specific system application recommendations.

Selection of Arrester Rating

In arrester rating selection, it is preferable to determine the lowest arrester rating that will ensure satisfactory operation. This is the optimum solution because the arrester selected will not only provide the greatest margin of insulation protection but also be the most economical choice.

Increasing the arrester rating above the minimum increases the likelihood of arrester survival during potential system contingencies, but compromises the protection of equipment insulation.

Rating selection should begin with consideration of the maximum system operating voltage. The maximum power frequency voltage expected under normal system conditions (line-to-ground) should not exceed the selected arrester's maximum continuous operating voltage (MCOV).

The temporary overvoltage (TOV) capability of the VariSTAR AZE arrester is shown in Figure 3. The curves indicate the arrester's ability to withstand abnormal system power frequency (sinusoidal) overvoltages for various durations. The values shown assume that the arrester has been energized at MCOV prior to

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System Voltage (kV rms)		Suggested Arrester Rating (kV rms)	
Nominal	Maximum	Solidly Grounded Neutral Circuits	High Impedance Grounded, Ungrounded, or Temporarily Ungrounded Circuits
2.4	2.52	—	3
4.16	4.37	3	6
4.8	5.04	—	6
6.9	7.24	—	9
12.47	13.2	9–10	—
13.2	14.0	10	15–18
13.8	14.5	10–12	15–18
20.7	21.8	15	—
23.0	24.2	—	24–27
24.9	26.4	18–21	—
27.6	29.0	21–24	27–30
34.5	36.5	27–30	36–39
46.0	48.3	—	48
69.0	72.5	54–60	66–72
115	121	90–96	108–120
138	145	108–120	132–144
161	169	120–144	144–168
230	242	172–192	228–240
345	362	258–312	288–360

TABLE 2
Commonly Applied Voltage Ratings of the VariSTAR Type AZE Arrester

an overvoltage event and that the arrester is in an ambient temperature of 60°C. After the overvoltage durations shown, the arrester will thermally recover when once again energized at MCOV.

Figure 3 also illustrates the arrester's TOV capabilities with and without prior switching surge duties of up to the maximum capability of the arrester as listed in Table 1 (Rated Discharge Energy).

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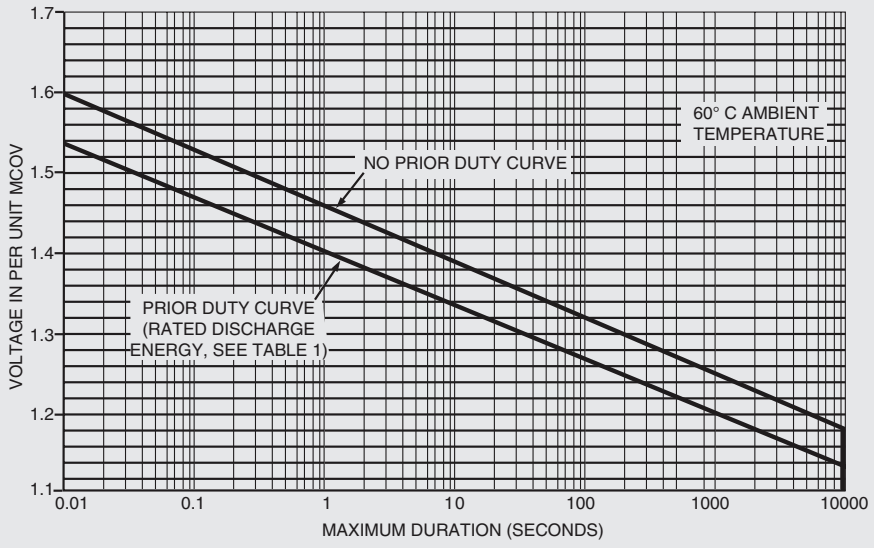

Figure 3 Temporary overvoltage capability of VariSTAR AZE surge arresters

It is not recommended that the TOV curve be extended for periods in excess of 10,000 seconds (2.8 hrs).

For ungrounded systems, systems utilizing high impedance or resonant grounding and other systems where the line-to-ground voltage may be elevated to line-to-line voltages for extended periods, arresters having an MCOV equal to line-to-line voltage may be required.

For non-sinusoidal transient voltages caused by system switching operations, a transient network analyzer (TNA) study is recommended; Eaton engineers are available to make these studies.

To assure proper application, the following information is normally required:

1. Maximum system operating voltage.
2. System grounding conditions.
   - **A.** For four-wire circuits, grounding conditions depend upon whether the system is multi-grounded, whether it has neutral impedance, and whether common primary and secondary neutrals are used.
   - **B.** For three-wire circuits, grounding conditions depend upon whether the system is solidly grounded at the source, grounded through neutral impedance at the source, grounded through transformers, or ungrounded.

Where unusual conditions exist (high ground resistance, high capacitive load, unusual switching surge duty, etc.), the following supplementary information is required:

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1. Type of unusual condition.
2. BIL of equipment and separation distance to protected equipment.
3. Type of construction (phase spacing, length of line, conductor size, etc.).
4. Grounding and phase-sequence components of source impedances.
5. Phase-sequence components of load impedances.
6. Available fault current.
7. Potential for loss of neutral grounding during system events.

Routine Tests

A complete production test program assures the quality of every VariSTAR AZE surge arrester. Each completed arrester is required to satisfactorily pass the following test regimen conducted in accordance with the procedures established in IEEE Std C62.11™-1993 standard.

- Partial Discharge Test at 1.05 times MCOV;
- Power Frequency Test at 1.20 times MCOV;
- Discharge Voltage Test; and
- Sealing Effectiveness Test of Housing by helium mass spectrometer.

Standards

The VariSTAR AZE surge arrester has been tested and certified to IEEE Std C62.11™-1993 standard. Guaranteed performance characteristics are specified in this catalog section and in the relevant 95028 "Design Certification Test Report" 95028.

Dimensions and Mounting

Figure 4 illustrates an in-line mounting arrangement; the applicable minimum values of B and C may be found in Tables 4 and 5. Line and ground terminal details are shown in Figure 5; the supplied terminals accommodate aluminum and copper conductors to a maximum size of 1.9 cm.

For other conductors, the terminal drilling pattern shown will accommodate industry standard two (2) and four (4) hole flat

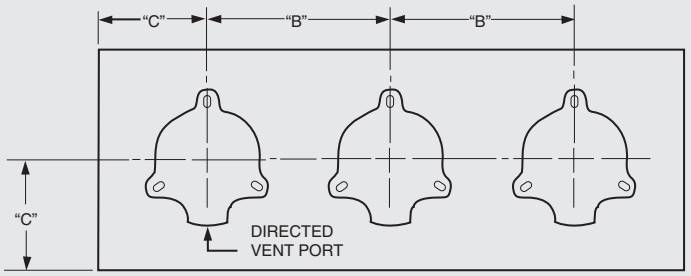

Figure 4 Three-phase in-line mounting, top view. ***Note:*** *Refer to Tables 4 and 5 for Dimensions B and C.*

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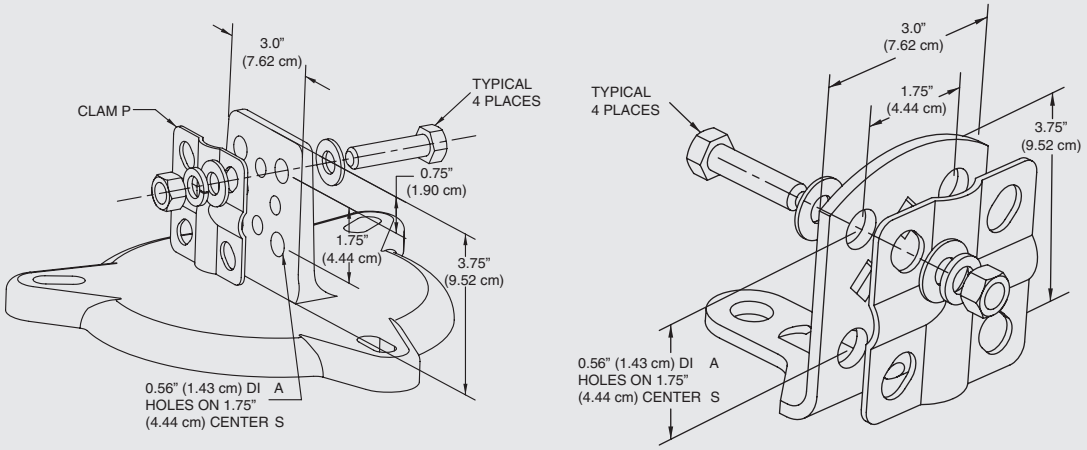

Figure 5 Line and ground terminals (suitable for copper or aluminum conductors up to 1.9-cm diameter)

Arrester Rating (kV rms)	Arrester MCOV (kV rms)	TOV* (kV rms)		Front-of-wave Protective Level** (kV Crest)	Maximum Discharge Voltage (kV crest) 8/20 μs Current Wave						Switching Surge Protective Level*** (kV crest)				
		1 sec	10 sec		1.5 kA	3 kA	5 kA	10 kA	20 kA	40 kA	125 A	250 A	500 A	1000 A	2000A
3	2.55	3.73	3.56	9.3	7.0	7.4	7.7	8.4	9.4	11.0	6.1	6.3	6.5	6.7	—
6	5.10	7.47	7.11	18.2	13.9	14.7	15.4	16.7	18.6	21.4	12.2	12.6	13.0	13.5	—
9	7.65	11.2	10.7	27.2	20.9	22.0	23.1	25.0	27.7	31.7	18.3	18.9	19.5	20.2	—
10	8.40	12.3	11.7	29.7	23.0	24.2	25.4	27.4	30.4	34.8	20.1	20.7	21.4	22.2	—
12	10.2	14.9	14.2	36.0	27.9	29.4	30.8	33.3	36.9	42.1	24.4	25.2	26.0	25.9	—
15	12.7	18.6	17.7	44.7	34.7	36.6	38.3	41.4	45.9	52.2	30.4	31.3	32.4	33.5	—
18	15.3	22.4	21.3	53.7	41.8	44.0	46.2	49.8	55.2	62.8	36.6	37.7	39.0	40.4	—
21	17.0	24.9	23.7	59.7	46.4	48.9	51.3	55.4	61.3	69.7	40.7	41.9	43.4	44.9	—
24	19.5	28.6	27.2	68.4	53.3	56.1	58.8	63.5	70.3	79.9	46.7	48.1	49.8	51.5	—
27	22.0	32.2	30.7	77.0	60.1	63.3	66.3	71.6	79.3	90.0	52.7	54.3	56.1	58.1	—
30	24.4	35.7	34.0	85.4	66.6	70.2	73.6	79.4	87.9	99.8	58.4	60.2	62.3	64.4	—
33	27.5	40.3	38.4	96.2	75.1	79.1	82.9	89.5	99.1	112	65.9	67.8	70.2	72.6	—
36	29.0	42.5	40.5	101	79.2	83.4	87.4	94.4	105	119	69.5	71.5	74.0	76.6	—

TABLE 3

Discharge Voltages—Maximum Guaranteed Protective Characteristics for AZES Surge Arresters

*Temporary overvoltage with prior duty energy surge.
**Based on a 10-kA current impulse that results in a discharge voltage cresting in 0.5 μs.
***45–60 μs rise time current surge.

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39	31.5	46.1	43.9	111	86.0	90.6	95.0	103	113	129	75.4	77.7	80.4	83.1	—
42	34.0	49.8	47.4	119	92.8	97.8	103	111	122	139	81.4	83.9	86.8	89.7	—
45	36.5	53.4	50.9	128	99.7	105	110	119	131	149	87.4	90.0	93.1	95.3	—
48	39.0	57.1	54.4	136	107	112	118	127	140	159	93.4	96.2	99.5	103	—
54	42.0	61.5	58.6	147	115	121	127	137	151	171	101	104	107	111	—
60	48.0	70.3	67.0	167	131	138	145	156	173	196	115	118	123	127	—
66	53.0	77.6	73.9	186	145	153	160	173	191	217	127	131	135	140	—
72	57.0	83.5	79.5	200	156	164	172	186	205	233	137	141	145	151	—
78	62.0	90.8	86.5	217	169	178	187	202	223	253	149	153	158	164	—
84	68.0	99.6	94.9	237	186	196	205	221	245	278	163	168	174	179	—
90	70.0	102.5	97.7	245	191	201	211	228	252	286	168	173	179	185	—
96	76.0	111.3	106.0	265	208	219	229	247	274	310	182	188	194	201	—
108	84.0	123.0	117.2	293	229	242	253	273	302	343	201	207	214	222	—
120	98.0	143.5	136.7	321	255	269	279	298	328	366	226	232	238	247	—
132	106	155.2	147.9	349	276	290	302	323	355	396	244	250	258	267	—
138	111	162.5	154.9	365	289	304	316	338	372	415	256	262	270	280	—
144	115	168.4	160.4	378	300	315	327	350	385	430	265	272	279	290	—
162	130	190.3	181.4	427	339	356	370	396	435	486	300	307	316	327	—
168	131	191.8	182.8	430	341	359	373	399	438	489	302	309	318	330	—
172	140	205.0	195.3	459	365	384	398	426	468	523	323	331	340	352	—
180	144	210.8	200.9	472	375	395	410	438	482	538	332	340	350	362	—
192	152	222.5	212.0	499	396	417	432	463	509	568	350	359	369	383	—
198	160	234.2	223.2	525	417	439	455	487	536	598	369	378	389	403	—
204	165	241.6	230.2	541	430	452	469	502	552	617	380	390	401	415	—
216	174	254.7	242.7	570	453	477	495	529	582	650	401	411	423	438	—
228	180	263.5	251.1	591	469	493	512	548	602	672	415	425	437	453	—
240	190	278.2	265.1	623	495	521	540	578	636	709	438	449	462	478	—
258	209	—	—	684	547	568	580	605	666	771	—	—	502	526	535
264	212	—	—	693	555	576	588	613	675	782	—	—	509	533	543
276	220	—	—	720	575	598	611	637	701	811	—	—	528	553	563
288	230	—	—	751	602	625	639	665	732	848	—	—	552	578	589
294	235	—	—	767	615	639	652	679	748	866	—	—	564	591	602
300	239	—	—	781	625	650	663	691	761	881	—	—	574	601	612
312	245	—	—	801	630	655	669	709	780	903	—	—	578	606	617
330	267	—	—	872	698	726	741	772	850	985	—	—	641	671	683
336	269	—	—	879	704	731	747	778	856	991	—	—	645	676	689
360	289	—	—	945	756	785	802	836	920	1064	—	—	693	727	740

TABLE 3
(*Continued*)

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Arrester Rating (kV, rms)	Arrester MCOV (kV, rms)	Catalog Number	Figure 8 Dim. A (in.)	Figure 4 Dim. B Minimum Phase-to-Phase Clearance (in.)	Figure 4 Dim. C Minimum Phase-to-Ground Clearance (in.)	Housing Leakage Distance (in.)	Housing Strike Distance (in.)	Insulation Withstand Voltages			Weight (lbs.)
								1.2/50 μs Impulse (kV Crest)	60 Hz Dry, 60 sec (kV, rms)	60 Hz Wet, 10 sec (kV, rms)	
3	2.55	AZES001G002003	18.6	12	6	9.2	5.2	130	75	40	42
6	5.10	AZES001G005006	18.6	12	7	9.2	5.2	130	75	40	42
9	7.65	AZES001G007009	18.6	13	7	9.2	5.2	130	75	40	43
10	8.40	AZES002G008010	21.1	13	7	16.0	7.7	170	95	65	48
12	10.2	AZES002G010012	21.1	14	8	16.0	7.7	170	95	65	49
15	12.7	AZES002G012015	21.1	14	9	16.0	7.7	170	95	65	49
18	15.3	AZES003G015018	24.8	15	10	26.2	11.4	230	125	95	56
21	17.0	AZES003G017021	24.8	16	11	26.2	11.4	230	125	95	57
24	19.5	AZES003G019024	24.8	16	11	26.2	11.4	230	125	95	58
27	22.0	AZES004G022027	28.6	17	12	36.3	15.2	265	150	125	65
30	24.4	AZES004G024030	28.6	18	13	36.3	15.2	265	150	125	65
33	27.5	AZES004G027033	28.6	20	14	36.3	15.2	265	150	125	66
36	29.0	AZES004G029036	28.6	20	14	36.3	15.2	265	150	125	66
39	31.5	AZES005G031039	33.6	21	15	49.9	20.2	320	190	165	77
42	34.0	AZES005G034042	33.6	22	16	49.9	20.2	320	190	165	78
45	36.5	AZES005G036045	33.6	23	17	49.9	20.2	320	190	165	78
48	39.0	AZES005G039048	33.6	24	18	49.9	20.2	320	190	165	78
54	42.0	AZES006G042054	36.6	25	20	64.8	23.6	365	200	170	86
60	48.0	AZES006G048060	36.6	28	22	64.8	23.6	365	200	170	87
66	53.0	AZES007G053066	39.4	30	24	73.7	26.5	385	235	200	96
72	57.0	AZES007G057072	39.4	31	26	73.7	26.5	385	235	200	97
78	62.0	AZES008G062078	48.0	33	28	100.0	35.0	505	305	260	115
84	68.0	AZES008G068084	48.0	36	30	100.0	35.0	505	305	260	118
90	70.0	AZES008G070090	48.0	36	31	100.0	35.0	505	305	260	118
96	76.0	AZES008G076096	48.0	39	33	100.0	35.0	505	305	260	119
108	84.0	AZES009G084108	56.5	42	36	127.0	43.6	650	370	300	161
120	98.0	AZES009G098120	56.5	43	37	127.0	43.6	650	370	300	175
132	106	AZES012G106132	71.5	46	40	138.5	50.1	735	425	360	191
138	111	AZES012G111138	71.5	47	42	138.5	50.1	735	425	360	193
144	115	AZES013G115144	74.4	49	43	147.4	52.9	770	475	400	202
162	130	AZES014G130162	80.1	54	48	164.8	58.6	865	490	415	217
168	131	AZES015G131169	82.9	54	48	173.7	61.5	880	545	455	225
172	140	AZES021G140172	83.2	72	59	173.7	59.4	920	535	450	237
180	144	AZES022G144180	88.9	73	60	191.8	60.9	920	515	440	270
192	152	AZES022G152192	88.9	76	63	191.8	60.9	920	515	440	273
198	160	AZES023G160198	91.8	79	66	200.7	63.8	930	535	480	283
204	165	AZES024G165204	100.3	81	68	227.0	72.3	1065	595	545	303

TABLE 4

Catalog Numbers, Dimensional Information, Insulation Withstand Voltages, and Weights for AZES Metal-Top Surge Arresters

Note: All arresters are available in grey (standard) or brown porcelain glaze. For brown glaze, substitute B for G in the eighth position of the catalog number.

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216	174	AZES024G174216	100.3	84	71	227.0	72.3	1065	595	545	306
228	182	AZES025G182228	108.9	86	73	254.0	80.9	1185	655	580	348
240	190	AZES025G190240	108.9	89	76	254.0	80.9	1185	655	580	351
258	209	AZES067G209258	123.9	105	84	265.5	83.9	1265	690	625	472
264	212	AZES067G212264	123.9	106	85	265.5	83.9	1265	690	625	473
276	220	AZES069G220276	132.5	108	87	291.8	92.4	1300	750	675	499
288	230	AZES069G230288	132.5	111	90	291.8	92.4	1300	750	675	503
294	235	AZES070G235294	135.3	113	92	300.7	95.3	1405	765	725	524
300	239	AZES070G239300	135.3	114	93	300.7	95.3	1405	765	725	526
312	245	AZES071G245312	141.0	116	95	318.8	100.9	1475	805	730	558
330	267	AZES074G267330	152.4	136	108	354.0	109.5	1440	810	790	605
336	269	AZES074G269336	152.4	136	109	354.0	109.5	1440	810	790	606
360	289	AZES075G289360	161.0	143	115	381.0	118.0	1535	860	840	655

TABLE 4

(*Continued*)

Arrester MCOV (kV, rms)	Arrester MCOV (kV, rms)	Catalog Number	Figure 7 Dim. "A" (in.)	Figure 4 Dim. "B" Minimum Phase-to-Phase Clearance (in.)	Figure 4 Dim. "C" Minimum Phase-to-Ground Clearance (in.)	Housing Leakage Distance (in.)	Housing Strike Distance (in.)	Insulation Withstand Voltages			Weight (lbs.)
								1.2/50 μs Impulse (kV Crest)	60 Hz Dry, 60 sec (kV, rms)	60 Hz Wet, 10 sec (kV, rms)	
3	2.55	AZES091G002003	13.4	8	4	9.0	4.7	95	65	32	23
6	5.10	AZES091G005006	13.4	8	5	9.0	4.7	95	65	32	24
9	7.65	AZES091G007009	13.4	8	5	9.0	4.7	95	65	32	24
10	8.40	AZES092G008010	15.9	8	5	16.0	7.2	128	93	46	29
12	10.2	AZES092G010012	15.9	9	6	16.0	7.2	128	93	46	30
15	12.7	AZES092G012015	15.9	10	7	16.0	7.2	128	93	46	30
18	15.3	AZES093G015018	20.2	11	8	26.4	11.0	182	119	75	38
21	17.0	AZES093G017021	20.2	12	8	26.4	11.0	182	119	75	38
24	19.5	AZES093G019024	20.2	12	9	26.4	11.0	182	119	75	39
27	22.0	AZES094G022027	23.4	13	10	36.8	14.7	230	148	115	47
30	24.4	AZES094G024030	23.4	14	11	36.8	14.7	230	148	115	47
33	27.5	AZES094G027033	23.4	15	12	36.8	14.7	230	148	115	47
36	29.0	AZES094G029036	23.4	16	12	36.8	14.7	230	148	115	48
39	31.5	AZES095G031039	28.4	17	13	50.7	19.7	294	174	143	59
42	34.0	AZES095G034042	28.4	18	14	50.7	19.7	294	174	143	60
45	36.5	AZES095G036045	28.4	19	15	50.7	19.7	294	174	143	60
48	39.0	AZES095G039048	28.4	20	16	50.7	19.7	294	174	143	60

TABLE 5

Catalog Numbers, Dimensional Information, Insulation Withstand Voltages, and Weights for AZES Cubicle-Mount Surge Arresters

Note: All arresters are available in grey (standard) or brown porcelain glaze. For brown glaze, substitute B for G in the eighth position of the catalog number.

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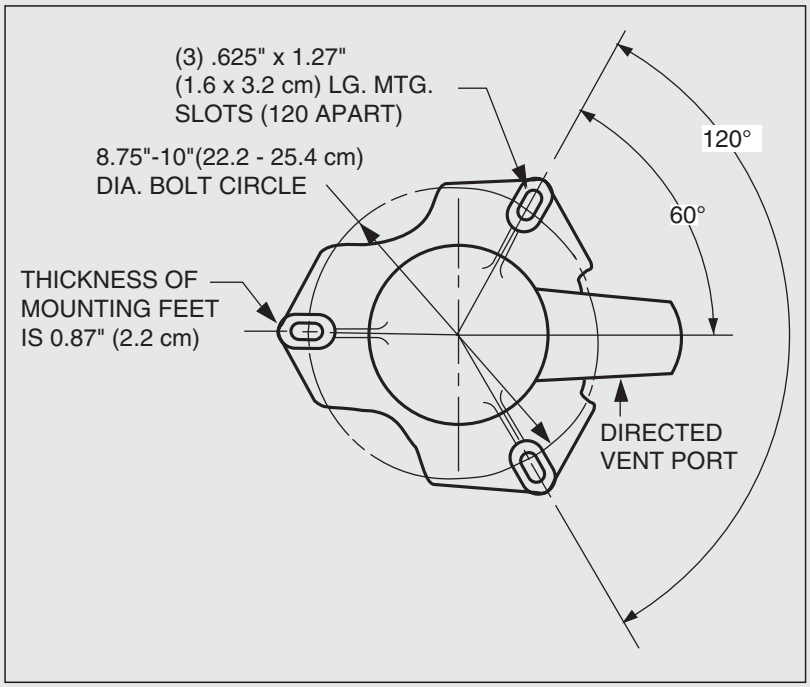

Figure 6 Base mounting. ***Note:*** *To develop rated cantilever strength use 25-cm bolt circle mounting diameter and 1.3-cm hardened bolts and flat washers.*

pad connectors having a 4.4 cm spacing. Figure 6 provides the dimensional details for universal base mounting.

The vent port in the base must be directed away from any adjacent equipment to control and prevent ionized gases from damaging other equipment in the unlikely event of arrester failure.

Performance and Protective Characteristics

Table 3 displays the Arrester Rating, Maximum Continuous Operating Voltage (MCOV) and the guaranteed protective characteristics of the AZES surge arrester.

The Front-of-Wave protective level is the maximum discharge voltage for a 10-kA current impulse that results in a discharge voltage cresting in 0.5 microseconds. Lightning Impulse Discharge Voltages represent the maximum voltage levels generated by the arrester when discharging lightning currents of the standard 8/20 microsecond wave-shape. The maximum Switching Impulse Discharge Voltages are based on a switching surge current having a time to crest of 45 to 60 microseconds.

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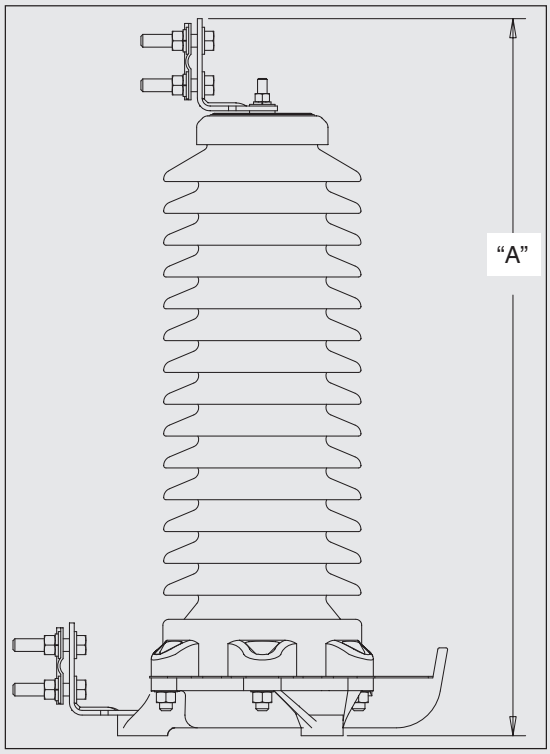

Figure 7 Dimensions of VariSTAR Type AZE cubicle-mount surge arrester. ***Note:*** *Refer to Table 5 for dimension A.*

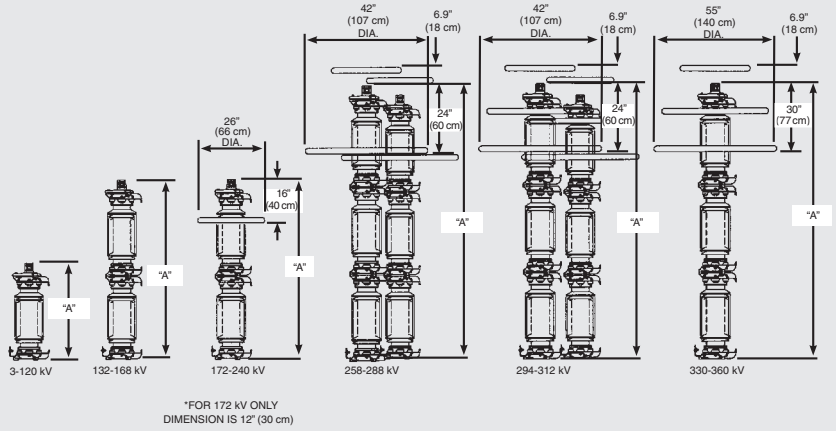

Figure 8 Dimensions of VariSTAR Type AZE metal-top surge arresters. ***Note:*** *Refer to Table 4 for dimension A.*

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14.1 TRAVELING WAVES ON SINGLE-PHASE LOSSLESS LINES

First, consider a single-phase, two-wire lossless transmission line. Figure 14.1 shows a line section of length Δx meters. If the line has a loop inductance L H/m and a line-to-line capacitance C F/m, then the line section has a series inductance L Δx H and shunt capacitance C Δx F, as shown. In Chapter 5, the direction of line position x was selected to be from the receiving end ($x = 0$) to the sending end ($x = l$); this selection was unimportant, since the variable x was subsequently eliminated when relating the steady-state sending-end quantities V_s and I_s to the receiving-end quantities V_R and I_R. Here, however, the interest is in voltages and current waveforms traveling along the line. Therefore, the direction of increasing x as being from the sending end ($x = 0$) toward the receiving end ($x = l$) is selected.

Writing a KVL and KCL equation for the circuit in Figure 14.1,

$$v(x + \Delta x, t) - v(x, t) = -\mathrm{L}\Delta x \frac{\partial i(x, t)}{\partial t} \tag{14.1.1}$$

$$i(x + \Delta x, t) - i(x, t) = -\mathrm{C}\Delta x \frac{\partial v(x, t)}{\partial t} \tag{14.1.2}$$

Dividing (14.1.1) and (14.1.2) by Δx and taking the limit as $\Delta x \to 0$, results in

$$\frac{\partial v(x, t)}{\partial x} = -\mathrm{L}\frac{\partial i(x, t)}{\partial t} \tag{14.1.3}$$

$$\frac{\partial i(x, t)}{\partial x} = -\mathrm{C}\frac{\partial v(x, t)}{\partial t} \tag{14.1.4}$$

Use partial derivatives here because $v(x, t)$ and $i(x, t)$ are differentiated with respect to both position x and time t. Also, the negative signs in (14.1.3) and (14.1.4) are due to the reference direction for x. For example, with a positive value of $\partial i/\partial t$ in Figure 14.1, $v(x, t)$ decreases as x increases. Taking the Laplace transform of (14.1.3) and (14.1.4),

$$\frac{d\mathrm{V}(x,s)}{dx} = -s\mathrm{LI}(x, s) \tag{14.1.5}$$

$$\frac{d\mathrm{I}(x,s)}{dx} = -s\mathrm{CV}(x, s) \tag{14.1.6}$$

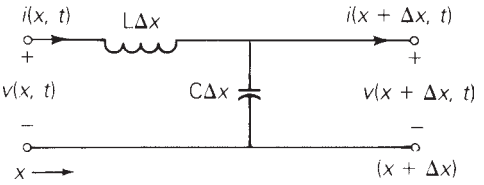

FIGURE 14.1
Single-phase two-wire lossless line section of length Δx

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where zero initial conditions are assumed. $\mathrm{V}(x, s)$ and $\mathrm{I}(x, s)$ are the Laplace transforms of $v(x, t)$ and $i(x, t)$. Also, ordinary rather than partial derivatives are used since the derivatives are now with respect to only one variable, x.

Next, differentiate (14.1.5) with respect to x and use (14.1.6) in order to eliminate $i(x, s)$:

$$\frac{d^2\mathrm{V}(x,s)}{dx^2} = -s\mathrm{L}\frac{d\mathrm{I}(x,s)}{dx} = s^2\mathrm{LCV}(x,s)$$

or

$$\frac{d^2\mathrm{V}(x,s)}{dx^2} - s^2\mathrm{LCV}(x,s) = 0 \tag{14.1.7}$$

Similarly, (14.1.6) can be differentiated in order to obtain

$$\frac{d^2\mathrm{I}(x,s)}{dx^2} - s^2\mathrm{LCI}(x,s) = 0 \tag{14.1.8}$$

Equation (14.1.7) is a linear, second-order homogeneous differential equation. By inspection, its solution is

$$\mathrm{V}(x,s) = \mathrm{V}^+(s)e^{-sx/v} + \mathrm{V}^-(s)e^{+sx/v} \tag{14.1.9}$$

where

$$v = \frac{1}{\sqrt{\mathrm{LC}}} \ \mathrm{m/s} \tag{14.1.10}$$

Similarly, the solution to (14.1.8) is

$$\mathrm{I}(x,s) = \mathrm{I}^+(s)e^{-sx/v} + \mathrm{I}^-(s)e^{+sx/v} \tag{14.1.11}$$

You can quickly verify that these solutions satisfy (14.1.7) and (14.1.8). The "constants" $\mathrm{V}^+(s)$, $\mathrm{V}^-(s)$, $\mathrm{I}^+(s)$, and $\mathrm{I}^-(s)$, which in general are functions of s but are independent of x, can be evaluated from the boundary conditions at the sending and receiving ends of the line. The superscripts + and − refer to waves traveling in the positive x and negative x directions, soon to be explained.

Taking the inverse Laplace transform of (14.1.9) and (14.1.11), and recalling the time shift properly, $\mathscr{L}[f(t-\tau)] = \mathrm{F}(s)e^{-s\tau}$, results in

$$v(x, t) = v^+\left(t - \frac{x}{v}\right) + v^-\left(t + \frac{x}{v}\right) \tag{14.1.12}$$

$$i(x, t) = i^+\left(t - \frac{x}{v}\right) + i^-\left(t + \frac{x}{v}\right) \tag{14.1.13}$$

where the functions $v^+(\,)$, $v^-(\,)$, $i^+(\,)$, and $i^-(\,)$, can be evaluated from the boundary conditions.

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$v^+(t - x/v)^+$ represents a voltage wave traveling in the positive x direction with velocity $v = 1/\sqrt{LC}$ m/s. Consider any wave $f^+(u)$, where $u = t - x/v$. Suppose that this wave begins at $u = u_0$, as shown in Figure 14.2(a). At time $t = t_1$, the wavefront is at $u_0 = (t_1 - x_1/v)$, or at $x_1 = v(t_1 - u_0)$. At a later time, t_2, the wavefront is at $u_0 = (t_2 - x_2/v)$ or at $x_2 = v(t_2 - u_0)$. As shown in Figure 14.2(b), the wavefront has moved in the positive x direction a distance $(x_2 - x_1) = v(t_2 - t_1)$ during time $(t_2 - t_1)$. The velocity is therefore $(x_2 - x_1)/(t_2 - t_1) = v$.

Similarly, $i^+(t - x/v)$ represents a current wave traveling in the positive x direction with velocity v. The expressions $v^+(t - x/v)$ and $i^+(t - x/v)$ are called the *forward* traveling voltage and current waves. It can be shown analogously that $v^-(t + x/v)$ and $i^-(t + x/v)$ travel in the negative x direction with velocity v. Also, $v^-(t + x/v)$ and $i^-(t + x/v)$ are called the *backward* traveling voltage and current waves.

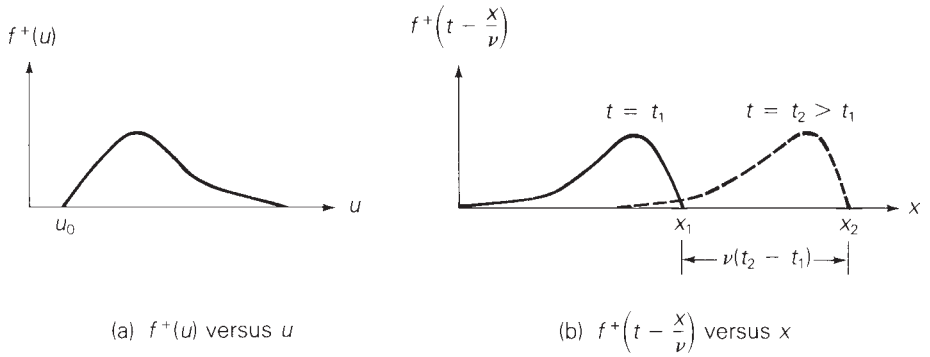
Recall from (5.4.16) that for a lossless line $f\lambda = 1/\sqrt{LC}$. It is now evident that the term $1/\sqrt{LC}$ in this equation is v, the velocity of propagation of voltage and current waves along the lossless line. Also, recall from Chapter 4 that L is proportional to μ and C is proportional to ε. For overhead lines, $v = 1/\sqrt{LC}$ is approximately equal to $1/\sqrt{\mu\varepsilon} = 1/\sqrt{\mu_0\varepsilon_0} = 3 \times 10^8$ m/s, the speed of light in free space. For cables, the relative permittivity $\varepsilon/\varepsilon_0$ may be 3 to 5 or even higher, resulting in a value of v lower than that for overhead lines.

Next, evaluate the terms $I^+(s)$ and $I^-(s)$. Using (14.1.9) and (14.1.11) in (14.1.6),

$$\frac{s}{v}[-I^+(s)e^{-sx/v} + I^-(s)e^{+sx/v}] = -sC[V^+(s)e^{-sx/v} + V^-(s)e^{+sx/v}]$$

Equating the coefficients of $e^{-sx/v}$ on both sides of this equation,

$$I^+(s) = (vC)V^+(s) = \frac{V^+(s)}{\sqrt{\frac{L}{C}}} = \frac{V^+(s)}{Z_c} \tag{14.1.14}$$



(a) $f^+(u)$ versus u (b) $f^+\left(t - \frac{x}{v}\right)$ versus x

FIGURE 14.2
The function $f^+(u)$, where $u = \left(t - \frac{x}{v}\right)$

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where

$$Z_c = \sqrt{\frac{\mathrm{L}}{\mathrm{C}}} \ \Omega \tag{14.1.15}$$

Similarly, equating the coefficients of $e^{+sx/v}$,

$$\mathrm{I}^-(s) = \frac{-\mathrm{V}^-(s)}{Z_c} \tag{14.1.16}$$

Thus, (14.1.11) and (14.1.13) can be rewritten as

$$\mathrm{I}(x,s) = \frac{1}{Z_c}[\mathrm{V}^+(s)e^{-sx/v} - \mathrm{V}^-(s)e^{+sx/v}] \tag{14.1.17}$$

$$i(x,\,t) = \frac{1}{Z_c}\left[v^+\left(t - \frac{x}{v}\right) - v^-\left(t + \frac{x}{v}\right)\right] \tag{14.1.18}$$

Recall from (5.4.3) that $Z_c = \sqrt{\mathrm{L}/\mathrm{C}}$ is the characteristic impedance (also called surge impedance) of a lossless line.

14.2 BOUNDARY CONDITIONS FOR SINGLE-PHASE LOSSLESS LINES

Figure 14.3 shows a single-phase two-wire lossless line terminated by an impedance $\mathrm{Z_R}(s)$ at the receiving end and by a source with Thévenin voltage $\mathrm{E_G}(s)$ and Thévenin impedance $\mathrm{Z_G}(s)$ at the sending end. $\mathrm{V}(x, s)$ and $\mathrm{I}(x, s)$ are the Laplace transforms of the voltage and current at position x. The line has length l, surge impedance $Z_c = \sqrt{\mathrm{L}/\mathrm{C}}$, and velocity $v = 1/\sqrt{\mathrm{LC}}$. Assume that the line is initially unenergized.

From Figure 14.3, the boundary condition at the receiving end is

$$V(l, s) = Z_{\mathrm{R}}(s)\mathrm{I}(l, s) \tag{14.2.1}$$

Using (14.1.9) and (14.1.17) in (14.2.1),

$$\mathrm{V}^+(s)e^{-sl/v} + \mathrm{V}^-(s)e^{+sl/v} = \frac{Z_{\mathrm{R}}(s)}{Z_c}[\mathrm{V}^+(s)e^{-sl/v} - \mathrm{V}^-(s)e^{+sl/v}]$$

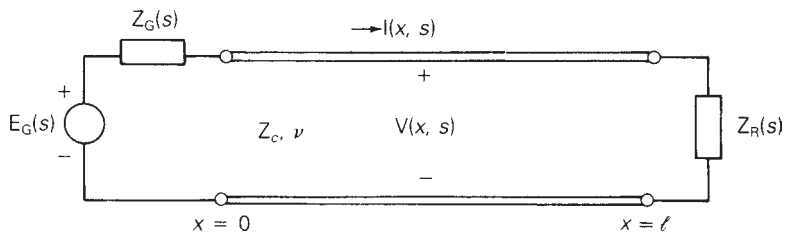

FIGURE 14.3

Single-phase two-wire lossless line with source and load terminations

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Solving for $V^-(l, s)$,

$$V^-(l, s) = \Gamma_R(s)V^+(s)e^{-2s\tau} \tag{14.2.2}$$

where

$$\Gamma_R(s) = \frac{\dfrac{Z_R(s)}{Z_c} - 1}{\dfrac{Z_R(s)}{Z_c} + 1} \quad \text{per unit} \tag{14.2.3}$$

$$\tau = \frac{l}{v} \quad \text{seconds} \tag{14.2.4}$$

$\Gamma_R(s)$ is called the *receiving-end voltage reflection coefficient.* Also, τ, called the *transit time* of the line, is the time it takes a wave to travel the length of the line.

Using (14.2.2) in (14.1.9) and (14.1.17),

$$V(x, s) = V^+(s)[e^{-sx/v} + \Gamma_R(s)e^{s[(x/v)-2\tau]}] \tag{14.2.5}$$

$$I(x, s) = \frac{V^+(s)}{Z_c}[e^{-sx/v} - \Gamma_R(s)e^{s[(x/v)-2\tau]}] \tag{14.2.6}$$

From Figure 14.3, the boundary condition at the sending end is

$$V(0, s) = E_G(s) - Z_G(s)I(0, s) \tag{14.2.7}$$

Using (14.2.5) and (14.2.6) in (14.2.7),

$$V^+(s)[1 + \Gamma_R(s)e^{-2s\tau}] = E_G(s) - \left[\frac{Z_G(s)}{Z_c}\right]V^+(s)[1 - \Gamma_R(s)e^{-2s\tau}]$$

Solving for $V^+(s)$,

$$V^+(s)\left\{\left[\frac{Z_G(s)}{Z_c} + 1\right] - \Gamma_R(s)e^{-2s\tau}\left[\frac{Z_G(s)}{Z_c} - 1\right]\right\} = E_G(s)$$

$$V^+(s)\left[\frac{Z_G(s)}{Z_c} + 1\right]\{1 - \Gamma_R(s)\Gamma_S(s)e^{-2s\tau}\} = E_G(s)$$

or

$$V^+(s) = E_G(s)\left[\frac{Z_c}{Z_G(s) + Z_c}\right]\left[\frac{1}{1 - \Gamma_R(s)\Gamma_S(s)e^{-2s\tau}}\right] \tag{14.2.8}$$

where

$$\Gamma_S(s) = \frac{\dfrac{Z_G(s)}{Z_c} - 1}{\dfrac{Z_G(s)}{Z_c} + 1} \tag{14.2.9}$$

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$\Gamma_S(s)$ is called the *sending-end voltage reflection coefficient.* Using (14.2.9) in (14.2.5) and (14.2.6), the complete solution is

$$V(x,s) = E_G(s)\left[\frac{Z_c}{Z_G(s)+Z_c}\right]\left[\frac{e^{-sx/v}+\Gamma_R(s)e^{s[(x/v)-2\tau]}}{1-\Gamma_R(s)\Gamma_S(s)e^{-2s\tau}}\right] \tag{14.2.10}$$

$$I(x,s) = \left[\frac{E_G(s)}{Z_G(s)+Z_c}\right]\left[\frac{e^{-sx/v}-\Gamma_R(s)e^{s[(x/v)-2\tau]}}{1-\Gamma_R(s)\Gamma_S(s)e^{-2s\tau}}\right] \tag{14.2.11}$$

where

$$\Gamma_R(s) = \frac{\dfrac{Z_R(s)}{Z_c}-1}{\dfrac{Z_R(s)}{Z_c}+1} \quad \text{per unit}$$

$$\Gamma_S(s) = \frac{\dfrac{Z_G(s)}{Z_c}-1}{\dfrac{Z_G(s)}{Z_c}+1} \quad \text{per unit} \tag{14.2.12}$$

$$Z_c = \sqrt{\frac{L}{C}}\ \Omega \qquad v = \frac{1}{\sqrt{LC}}\ \text{m/s} \qquad \tau = \frac{l}{v}\ \text{s} \tag{14.2.13}$$

The following four examples illustrate this general solution. All four examples refer to the line shown in Figure 14.3, which has length l, velocity v, characteristic impedance Z_c, and is initially unenergized.

EXAMPLE 14.1

Single-phase lossless-line transients: step-voltage source at sending end, matched load at receiving end

Let $Z_R = Z_c$ and $Z_G = 0$. The source voltage is a step, $e_G(t) = Eu_{-1}(t)$. (a) Determine $v(x, t)$ and $i(x, t)$. Plot the voltage and current versus time t at the center of the line and at the receiving end.

SOLUTION

a. From (14.2.12) with $Z_R = Z_c$ and $Z_G = 0$,

$$\Gamma_R(s) = \frac{1-1}{1+1} = 0 \qquad \Gamma_S(s) = \frac{0-1}{0+1} = -1$$

The Laplace transform of the source voltage is $E_G(s) = E/s$. Then, from (14.2.10) and (14.2.11),

(*Continued*)

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$$\mathrm{V}(x,s) = \left(\frac{\mathrm{E}}{s}\right)(1)(e^{-sx/v}) = \frac{\mathrm{E}e^{-sx/v}}{s}$$

$$\mathrm{I}(x,s) = \frac{(\mathrm{E}/Z_c)}{s}\, e^{-sx/v}$$

Taking the inverse Laplace transform,

$$v(x,\, t) = \mathrm{E}u_{-1}\left(t - \frac{x}{v}\right)$$

$$i(x,\, t) = \frac{\mathrm{E}}{Z_c}u_{-1}\left(t - \frac{x}{v}\right)$$

b. At the center of the line, where $x = l/2$,

$$v\left(\frac{l}{2},\, t\right) = \mathrm{E}u_{-1}\left(t - \frac{\tau}{2}\right) \qquad i\left(\frac{l}{2},\, t\right) = \frac{\mathrm{E}}{Z_c}u_{-1}\left(t - \frac{\tau}{2}\right)$$

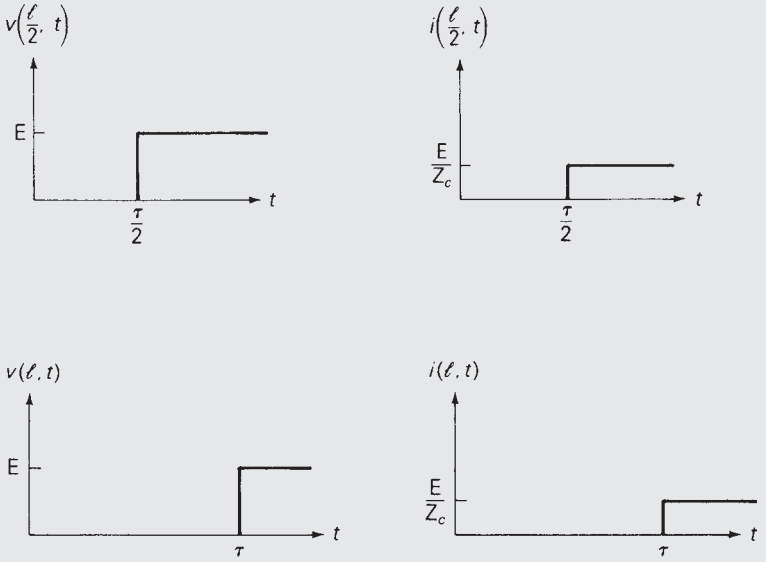
At the receiving end, where $x = l$,

$$v(l,\, t) = \mathrm{E}u_{-1}(t - \tau) \qquad i(l,\, t) = \frac{\mathrm{E}}{Z_c}u_{-1}(t - \tau)$$

These waves, plotted in Figure 14.4, can be explained as follows. At $t = 0$, the ideal step voltage of E volts, applied to the sending end, encounters Z_c, the characteristic impedance of the line. Therefore, a forward traveling step voltage

FIGURE 14.4

Voltage and current waveforms for Example 14.1



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wave of E volts is initiated at the sending end. Also, since the ratio of the forward traveling voltage to current is Z_c, a forward traveling step current wave of (E/Z_c) amperes is initiated. These waves travel in the positive x direction, arriving at the center of the line at $t = \tau/2$, and at the end of the line at $t = \tau$. The receiving-end load is *matched* to the line; that is, $Z_R = Z_c$, for a matched load, $\Gamma_R = 0$, and therefore no backward traveling waves are initiated. In steady state, the line with matched load is energized at E volts with current E/Z_c amperes.

EXAMPLE 14.2

Single-phase lossless-line transients: step-voltage source matched at sending end, open receiving end

The receiving end is open. The source voltage at the sending end is a step $e_G(t) = Eu_{-1}(t)$, with $Z_G(s) = Z_c$. (a) Determine $v(x, t)$ and $i(x, t)$. (b) Plot the voltage and current versus time t at the center of the line.

SOLUTION

a. From (14.2.12),

$$\Gamma_R(s) = \lim_{Z_R \to \infty} \frac{\dfrac{Z_R}{Z_c} - 1}{\dfrac{Z_R}{Z_c} + 1} = 1 \qquad \Gamma_S(s) = \frac{1 - 1}{1 + 1} = 0$$

The Laplace transform of the source voltage is $E_G(s) = E/s$. Then, from (14.2.10) and (14.2.11),

$$V(x, s) = \frac{E}{s}\left(\frac{1}{2}\right)[e^{-sx/v} + e^{s[(x/v)-2\tau]}]$$

$$I(x, s) = \frac{E}{s}\left(\frac{1}{2Z_c}\right)[e^{-sx/v} - e^{s[(x/v)-2\tau]}]$$

Taking the inverse Laplace transform,

$$v(x, t) = \frac{E}{2}u_{-1}\left(t - \frac{x}{v}\right) + \frac{E}{2}u_{-1}\left(t + \frac{x}{v} - 2\tau\right)$$

$$i(x, t) = \frac{E}{2Z_c}u_{-1}\left(t - \frac{x}{v}\right) - \frac{E}{2Z_c}u_{-1}\left(t + \frac{x}{v} - 2\tau\right)$$

(*Continued*)

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b. At the center of the line, where $x = l/2$,

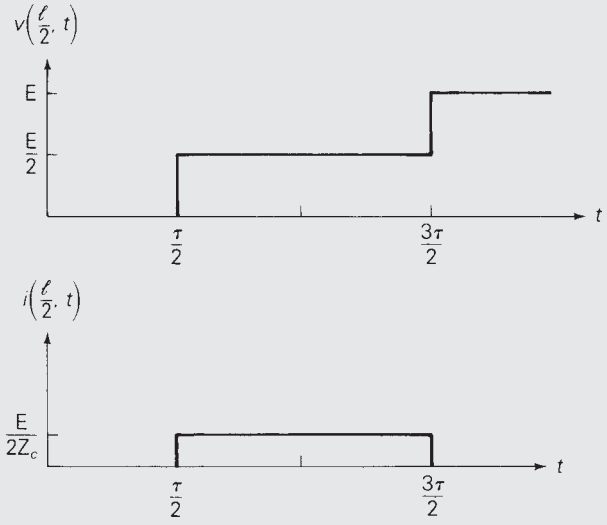
$$v\left(\frac{l}{2}, t\right) = \frac{\mathrm{E}}{2} u_{-1}\left(t - \frac{\tau}{2}\right) + \frac{\mathrm{E}}{2} u_{-1}\left(t - \frac{3\tau}{2}\right)$$

$$i\left(\frac{l}{2}, t\right) = \frac{\mathrm{E}}{2Z_c} u_{-1}\left(t - \frac{\tau}{2}\right) - \frac{\mathrm{E}}{2Z_c} u_{-1}\left(t - \frac{3\tau}{2}\right)$$

These waves are plotted in Figure 14.5. At $t = 0$, the step voltage source of E volts encounters the source impedance $\mathrm{Z_G} = \mathrm{Z}_c$ in series with the characteristic impedance of the line, Z_c. Using voltage division, the sending-end voltage at $t = 0$ is E/2. Therefore, a forward traveling step voltage wave of E/2 volts and a forward traveling step current wave of $\mathrm{E}/(2\mathrm{Z}_c)$ amperes are initiated at the sending end. These waves arrive at the center of the line at $t = \tau/2$. Also, with $\Gamma_\mathrm{R} = 1$, the backward traveling voltage wave equals the forward traveling voltage wave, and the backward traveling current wave is the negative of the forward traveling current wave. These backward traveling waves, which are initiated at the receiving end at $t = \tau$ when the forward traveling waves arrive there, arrive at the center of the line at $t = 3\tau/2$ and are superimposed on the forward traveling waves. No additional forward or backward traveling waves are initiated because the source impedance is matched to the line; that is, $\Gamma_\mathrm{S}(s) = 0$. In steady state, the line, which is open at the receiving end, is energized at E volts with zero current.

FIGURE 14.5

Voltage and current waveforms for Example 14.2



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EXAMPLE 14.3

Single-phase lossless-line transients: step-voltage source matched at sending end, capacitive load at receiving end

The receiving end is terminated by a capacitor with C_R farads, which is initially unenergized. The source voltage at the sending end is a unit step $e_G(t) = Eu_{-1}(t)$, with $Z_G = Z_c$. Determine and plot $v(x, t)$ versus time t at the sending end of the line.

SOLUTION

From (14.2.12) with $Z_R = \dfrac{1}{sC_R}$ and $Z_G = Z_c$,

$$\Gamma_R(s) = \frac{\dfrac{1}{sC_RZ_c} - 1}{\dfrac{1}{sC_RZ_c} + 1} = \frac{-s + \dfrac{1}{Z_cC_R}}{s + \dfrac{1}{Z_cC_R}}$$

$$\Gamma_S(s) = \frac{1 - 1}{1 + 1} = 0$$

Then, from (14.2.10), with $E_G(s) = E/s$,

$$V(x, s) = \frac{E}{s}\left(\frac{1}{2}\right)\left[e^{-sx/v} + \left(\frac{-s + \frac{1}{Z_cC_R}}{s + \frac{1}{Z_cC_R}}\right)e^{s[(x/v)-2\tau]}\right]$$

$$= \frac{E}{2}\left[\frac{e^{-sx/v}}{s} + \frac{1}{s}\left(\frac{-s + \frac{1}{Z_cC_R}}{s + \frac{1}{Z_cC_R}}\right)e^{s[(x/v)-2\tau]}\right]$$

Using partial fraction expansion of the second term above,

$$V(x, s) = \frac{E}{2}\left[\frac{e^{-sx/v}}{s} + \left(\frac{1}{s} - \frac{2}{s + \frac{1}{Z_cC_R}}\right)e^{s[(x/v)-2\tau]}\right]$$

The inverse Laplace transform is

$$v(x, t) = \frac{E}{2}u_{-1}\left(t - \frac{x}{v}\right) + \frac{E}{2}[1 - 2e^{(-1/Z_cC_R)(t+x/v-2\tau)}]u_{-1}\left(t + \frac{x}{v} - 2\tau\right)$$

At the sending end, where $x = 0$,

$$v(0, t) = \frac{E}{2}u_{-1}(t) + \frac{E}{2}[1 - 2e^{(-1/Z_cC_R)(t-2\tau)}]u_{-1}(t - 2\tau)$$

(*Continued*)

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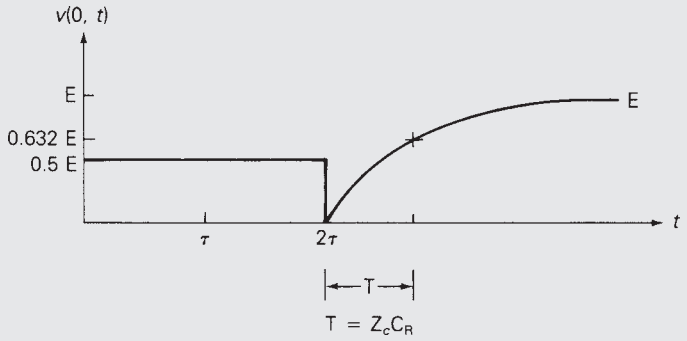
FIGURE 14.6
Voltage waveform for Example 14.3



$v(0, t)$ is plotted in Figure 14.6. As in Example 14.2, a forward traveling step voltage wave of E/2 volts is initiated at the sending end at $t = 0$. At $t = \tau$, when the forward traveling wave arrives at the receiving end, a backward traveling wave is initiated. The backward traveling voltage wave, an exponential with initial value $-E/2$, steady-state value $+E/2$, and time constant Z_cC_R, arrives at the sending end at $t = 2\tau$, where it is superimposed on the forward traveling wave. No additional waves are initiated, since the source impedance is matched to the line. In steady state, the line and the capacitor at the receiving end are energized at E volts with zero current.

The capacitor at the receiving end can also be viewed as a short circuit at the instant $t = \tau$, when the forward traveling wave arrives at the receiving end. For a short circuit at the receiving end, $\Gamma_R = -1$, and therefore the backward traveling voltage wavefront is $-E/2$, the negative of the forward traveling wave. However, in steady state, the capacitor is an open circuit, for which $\Gamma_R = +1$, and the steady-state backward traveling voltage wave equals the forward traveling voltage wave.

EXAMPLE 14.4

Single-phase lossless-line transients: step-voltage source with unmatched source resistance at sending end, unmatched resistive load at receiving end

SOLUTION

At the receiving end, $Z_R = Z_c/3$. At the sending end, $e_G(t) = Eu_{-1}(t)$ and $Z_G = 2Z_c$. Determine and plot the voltage versus time at the center of the line.

From (14.2.12),

$$\Gamma_R = \frac{\frac{1}{3} - 1}{\frac{1}{3} + 1} = -\frac{1}{2} \qquad \Gamma_S = \frac{2 - 1}{2 + 1} = \frac{1}{3}$$

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From (14.2.10), with $E_G(s) = E/s$,

$$V(x, s) = \frac{E}{s}\left(\frac{1}{3}\right)\frac{\left[e^{-sx/v} - \frac{1}{2}e^{s[(x/v)-2\tau]}\right]}{1 + \left(\frac{1}{6}e^{-2s\tau}\right)}$$

The preceding equation can be rewritten using the following geometric series:

$$\frac{1}{1+y} = 1 - y + y^2 - y^3 + y^4 - \cdots$$

with $y = \frac{1}{6}e^{-2s\tau}$,

$$V(x, s) = \frac{E}{3s}\left[e^{-sx/v} - \frac{1}{2}e^{s[(x/v)-2\tau]}\right] \times \left[1 - \frac{1}{6}e^{-2s\tau} + \frac{1}{36}e^{-4s\tau} - \frac{1}{216}e^{-6s\tau} + \cdots\right]$$

Multiplying the terms within the brackets,

$$V(x, s) = \frac{E}{3s}\left[e^{-sx/v} - \frac{1}{2}e^{s[(x/v)-2\tau]} - \frac{1}{6}e^{-s[(x/v)+2\tau]} + \frac{1}{12}e^{s[(x/v)-4\tau]} + \frac{1}{36}e^{-s[(x/v)+4\tau]} - \frac{1}{72}e^{s[(x/v)-6\tau]} + \cdots\right]$$

Taking the inverse Laplace transform,

$$v(x, t) = \frac{E}{3}\left[u_{-1}\left(t - \frac{x}{v}\right) - \frac{1}{2}u_{-1}\left(t + \frac{x}{v} - 2\tau\right) - \frac{1}{6}u_{-1}\left(t - \frac{x}{v} - 2\tau\right) + \frac{1}{12}u_{-1}\left(t + \frac{x}{v} - 4\tau\right) + \frac{1}{36}u_{-1}\left(t - \frac{x}{v} - 4\tau\right) - \frac{1}{72}u_{-1}\left(t + \frac{x}{v} - 6\tau\right)\cdots\right]$$

At the center of the line, where $x = l/2$,

$$v\left(\frac{l}{2}, t\right) = \frac{E}{3}\left[u_{-1}\left(t - \frac{\tau}{2}\right) - \frac{1}{2}u_{-1}\left(t - \frac{3\tau}{2}\right) - \frac{1}{6}u_{-1}\left(t - \frac{5\tau}{2}\right) + \frac{1}{12}u_{-1}\left(t - \frac{7\tau}{2}\right) + \frac{1}{36}u_{-1}\left(t - \frac{9\tau}{2}\right) - \frac{1}{72}u_{-1}\left(t - \frac{11\tau}{2}\right)\cdots\right]$$

$v(l/2, t)$ is plotted in Figure 14.7(a). Since neither the source nor the load is matched to the line, the voltage at any point along the line consists of an

(*Continued*)

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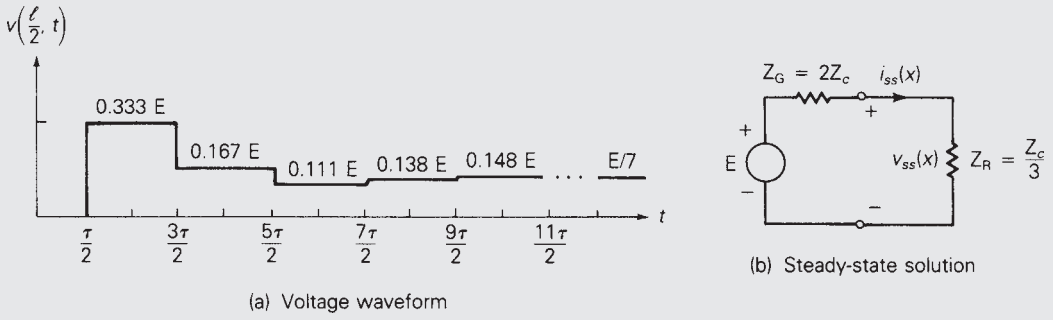

FIGURE 14.7

Example 14.4

infinite series of forward and backward traveling waves. At the center of the line, the first forward traveling wave arrives at $t = \tau/2$; then a backward traveling wave arrives at $3\tau/2$, another forward traveling wave arrives at $5\tau/2$, another backward traveling wave at $7\tau/2$, and so on.

The steady-state voltage can be evaluated from the final value theorem. That is,

$$v_{ss}(x) = \lim_{t\to\infty} v(x,t) = \lim_{s\to 0} sV(x,s)$$

$$= \lim_{s\to 0}\left\{s\left(\frac{E}{s}\right)\left(\frac{1}{3}\right)\frac{[e^{-sx/v} - \frac{1}{2}e^{s[(x/v)-2\tau]}]}{1+\frac{1}{6}e^{-2s\tau}}\right\}$$

$$= E\left(\frac{1}{3}\right)\left(\frac{1-\frac{1}{2}}{1+\frac{1}{6}}\right) = \frac{E}{7}$$

The steady-state solution can also be evaluated from the circuit in Figure 14.7(b). Since there is no steady-state voltage drop across the lossless line when a dc source is applied, the line can be eliminated, leaving only the source and load. The steady-state voltage is then, by voltage division,

$$v_{ss}(x) = E\left(\frac{Z_R}{Z_R + Z_G}\right) = E\left(\frac{\frac{1}{3}}{\frac{1}{3}+2}\right) = \frac{E}{7}$$

14.3 BEWLEY LATTICE DIAGRAM

A lattice diagram developed by L. V. Bewley [2] conveniently organizes the reflections that occur during transmission-line transients. For the Bewley lattice diagram, shown in Figure 14.8, the vertical scale represents time and is scaled in units of τ, the transient time of the line. The horizontal scale represents line position x, and the diagonal lines represent traveling waves. Each reflection is determined by

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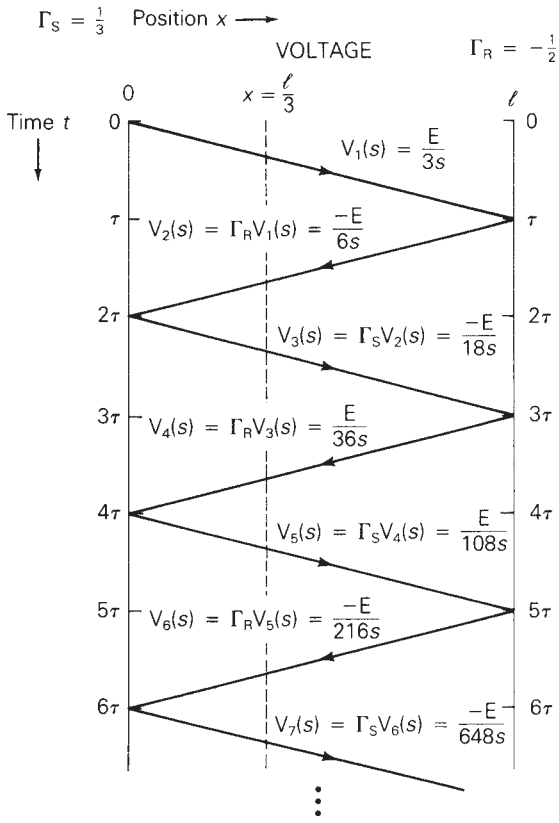

FIGURE 14.8
Bewley lattice diagram for Example 14.5

multiplying the incident wave arriving at an end by the reflection coefficient at that end. The voltage $v(x, t)$ at any point x and t on the diagram is determined by adding all the terms directly above that point.

EXAMPLE 14.5

Lattice diagram: single-phase lossless line

For the line and terminations given in Example 14.4, draw the lattice diagram and plot $v(l/3, t)$ versus time t.

SOLUTION

The lattice diagram is shown in Figure 14.8. At $t = 0$, the source voltage encounters the source impedance and the line characteristic impedance, and the first forward traveling wave is determined by voltage division:

$$V_1(s) = E_G(s)\left[\frac{Z_c}{Z_c + Z_G}\right] = \frac{E}{s}\left[\frac{1}{1+2}\right] = \frac{E}{3s}$$

(*Continued*)

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which is a step with magnitude (E/3) volts. The next traveling wave, a backward one, is $V_2(s) = \Gamma_R(s)V_1(s) = \left(-\frac{1}{2}\right)V_1(s) = -E/(6s)$, and the next wave, a forward one, is $V_3(s) = \Gamma_s(s)V_2(s) = \left(\frac{1}{3}\right)V_2(s) = -E/(18s)$. Subsequent waves are calculated in a similar manner.

The voltage at $x = l/3$ is determined by drawing a vertical line at $x = l/3$ on the lattice diagram, shown dashed in Figure 14.8. Starting at the top of the dashed line, where $t = 0$, and moving down, each voltage wave is added at the time it intersects the dashed line. The first wave v_1 arrives at $t = \tau/3$, the second v_2 arrives at $5\tau/3$, v_3 at $7\tau/3$, and so on. $v(l/3, t)$ is plotted in Figure 14.9.

FIGURE 14.9
Voltage waveform for Example 14.5

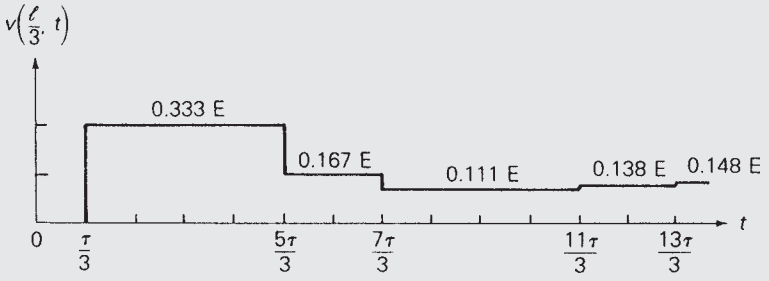


Figure 14.10 shows a forward traveling voltage wave $\mathbf{V}_A^+$ arriving at the junction of two lossless lines A and B with characteristic impedances Z_A and Z_B, respectively. This could be, for example, the junction of an overhead line and a cable. When $\mathbf{V}_A^+$ arrives at the junction, both a reflection $\mathbf{V}_A^-$ on line A and a refraction $\mathbf{V}_B^+$ on line B will occur. Writing a KVL and KCL equation at the junction,

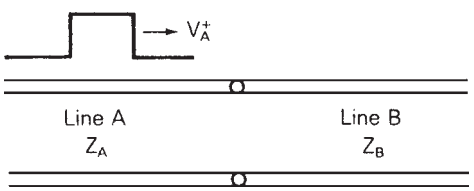
$$\mathbf{V}_A^+ + \mathbf{V}_A^- = \mathbf{V}_B^+ \tag{14.3.1}$$

$$\mathbf{I}_A^+ + \mathbf{I}_A^- = \mathbf{I}_B^+ \tag{14.3.2}$$

Recall that $\mathbf{I}_A^+ = \mathbf{V}_A^+/Z_A$, $\mathbf{I}_A^- = -\mathbf{V}_A^-/Z_A$, and $\mathbf{I}_B^+ = \mathbf{V}_B^+/Z_B$. Using these relations in (14.3.2),

$$\frac{\mathbf{V}_A^+}{Z_A} - \frac{\mathbf{V}_A^-}{Z_A} = \frac{\mathbf{V}_B^+}{Z_B} \tag{14.3.3}$$

FIGURE 14.10
Junction of two single-phase lossless lines



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Solving (14.3.1) and (14.3.3) for $\mathbf{V}_A^-$ and $\mathbf{V}_B^+$ in terms of $\mathbf{V}_A^+$ yields

$$V_A^- = \Gamma_{AA} V_A^+ \tag{14.3.4}$$

where

$$\Gamma_{AA} = \frac{\dfrac{Z_B}{Z_A} - 1}{\dfrac{Z_B}{Z_A} + 1} \tag{14.3.5}$$

and

$$V_B^+ = \Gamma_{BA} V_A \tag{14.3.6}$$

where

$$\Gamma_{BA} = \frac{2\left(\dfrac{Z_B}{Z_A}\right)}{\dfrac{Z_B}{Z_A} + 1} \tag{14.3.7}$$

Note that Γ_{AA}, given by (14.3.5), is similar to Γ_R, given by (14.2.12), except that Z_B replaces Z_R. Thus, for waves arriving at the junction from line A, the "load" at the receiving end of line A is the characteristic impedance of line B.

EXAMPLE 14.6

Lattice diagram: overhead line connected to a cable, single-phase lossless lines

As shown in Figure 14.10, a single-phase lossless overhead line with $Z_A = 400\ \Omega$, $v_A = 3 \times 10^8$ m/s, and $l_A = 30$ km is connected to a single-phase lossless cable with $Z_B = 100\ \Omega$, $v_B = 2 \times 10^8$ m/s, and $l_B = 20$ km. At the sending end of line A, $e_g(t) = Eu_{-1}(t)$ and $Z_G = Z_A$. At the receiving end of line B, $Z_R = 2Z_B = 200\ \Omega$. Draw the lattice diagram for $0 \le t \le 0.6$ ms and plot the voltage at the junction versus time. The line and cable are initially unenergized.

SOLUTION

From (14.2.13),

$$\tau_A = \frac{30 \times 10^3}{3 \times 10^8} = 0.1 \times 10^{-3} \text{ s} \qquad \tau_B = \frac{20 \times 10^3}{2 \times 10^8} = 0.1 \times 10^{-3} \text{ s}$$

From (14.2.12), with $Z_G = Z_A$ and $Z_R = 2Z_B$,

$$\Gamma_S = \frac{1 - 1}{1 + 1} = 0 \qquad \Gamma_R = \frac{2 - 1}{2 + 1} = \frac{1}{3}$$

(*Continued*)

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From (14.3.5) and (14.3.6), the reflection and refraction coefficients for waves arriving at the junction from line A are

$$\left.\begin{aligned}\Gamma_{AA} &= \frac{\dfrac{100}{400} - 1}{\dfrac{100}{400} + 1} = \frac{-3}{5} \qquad \Gamma_{BA} = \frac{2\dfrac{100}{400}}{\dfrac{100}{400} + 1} = \frac{2}{5}\end{aligned}\right\} \text{from line A}$$

Reversing A and B, the reflection and refraction coefficients for waves returning to the junction from line B are

$$\left.\begin{aligned}\Gamma_{BB} &= \frac{\dfrac{400}{100} - 1}{\dfrac{400}{100} + 1} = \frac{3}{5} \qquad \Gamma_{AB} = \frac{2\dfrac{400}{100}}{\dfrac{400}{100} + 1} = \frac{8}{5}\end{aligned}\right\} \text{from line B}$$

The lattice diagram is shown in Figure 14.11. Using voltage division, the first forward traveling voltage wave is

$$V_1(s) = E_G(s)\left(\frac{Z_A}{Z_A + Z_G}\right) = \frac{E}{s}\left(\frac{1}{2}\right) = \frac{E}{2s}$$

When v_1 arrives at the junction, a reflected wave v_2 and refracted wave v_3 are initiated. Using the reflection and refraction coefficients for line A,

$$V_2(s) = \Gamma_{AA}V_1(s) = \left(\frac{-3}{5}\right)\left(\frac{E}{2s}\right) = \frac{-3E}{10s}$$

$$V_3(s) = \Gamma_{BA}V_1(s) = \left(\frac{2}{5}\right)\left(\frac{E}{2s}\right) = \frac{E}{5s}$$

When v_2 arrives at the receiving end of line B, a reflected wave $V_4(s) = \Gamma_R V_3(s) = \frac{1}{3}(E/5s) = (E/15s)$ is initiated. When v_4 arrives at the junction, reflected wave v_5 and refracted wave v_6 are initiated. Using the reflection and refraction coefficients for line B,

$$V_5(s) = \Gamma_{BB}V_4(s) = \left(\frac{3}{5}\right)\left(\frac{E}{15s}\right) = \frac{E}{25s}$$

$$V_6(s) = \Gamma_{AB}V_4(s) = \left(\frac{8}{5}\right)\left(\frac{E}{15s}\right) = \frac{E}{75s}$$

Subsequent reflections and refractions are calculated in a similar manner.

The voltage at the junction is determined by starting at $x = l_A$ at the top of the lattice diagram, where $t = 0$. Then, moving down the lattice diagram, voltage waves either just to the left or just to the right of the junction are added when they occur. For example, looking just to the right of the junction at $x = l_A^+$, the voltage wave v_3, a step of magnitude E/5 volts occurs at $t = \tau_A$.

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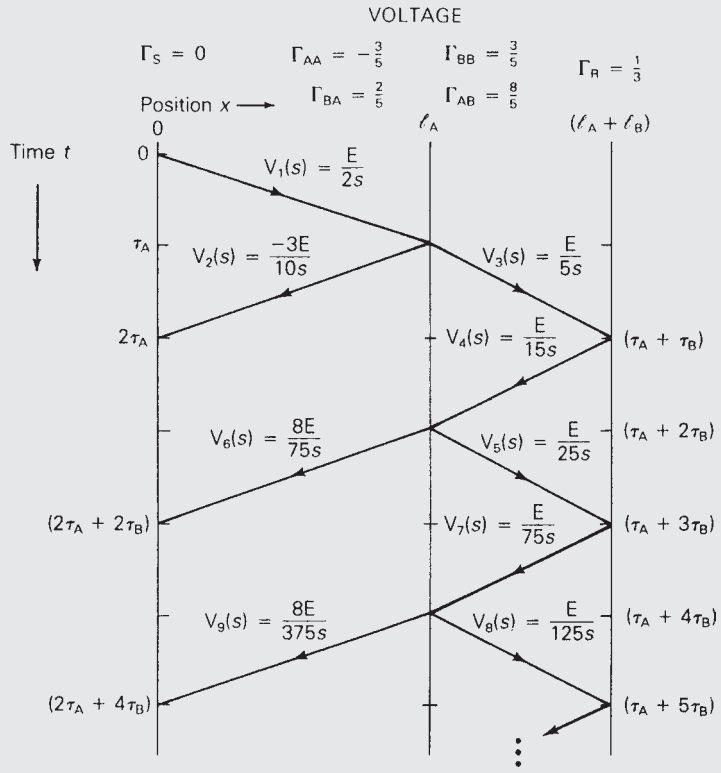
FIGURE 14.11

Lattice diagram for Example 14.6



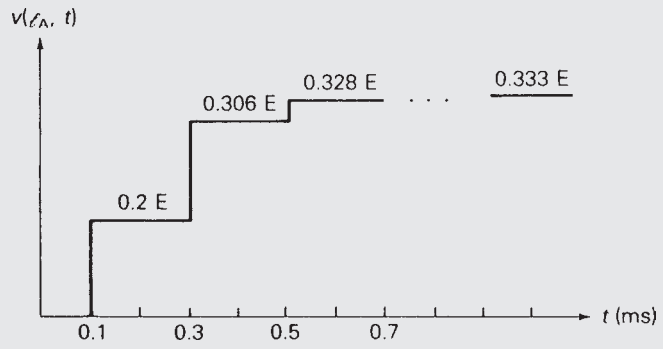
Then at $t = (\tau_A + 2\tau_B)$, two waves v_4 and v_5, which are steps of magnitude E/15 and E/25, are added to v_3. $v(l_A, t)$ is plotted in Figure 14.12.

The steady-state voltage is determined by removing the lossless lines and calculating the steady-state voltage across the receiving-end load:

$$v_{ss}(x) = \mathrm{E}\left(\frac{\mathrm{Z_R}}{\mathrm{Z_R} + \mathrm{Z_G}}\right) = \mathrm{E}\left(\frac{200}{200 + 400}\right) = \frac{\mathrm{E}}{3}$$

FIGURE 14.12

Junction voltage for Example 14.6



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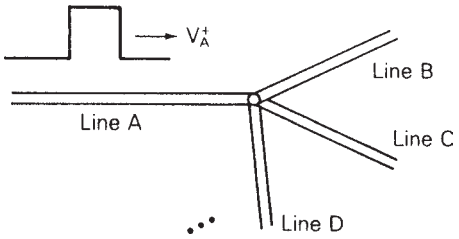

FIGURE 14.13
Junction of lossless lines A, B, C, D, and so on

The preceding analysis can be extended to the junction of more than two lossless lines, as shown in Figure 14.13. Writing a KVL and KCL equation at the junction for a voltage $\mathbf{V}_A^+$ arriving at the junction from line A,

$$V_A^+ + V_A^- = V_B^+ = V_C^+ = V_D^+ = \cdots \tag{14.3.8}$$

$$I_A^+ + I_A^- = I_B^+ + I_C^+ + I_D^+ + \cdots \tag{14.3.9}$$

Using $I_A^+ = V_A^+/Z_A$, $I_A^- = -V_A^-/Z_A$, $I_B^+ = V_B^+/Z_B$, and so on in (14.3.9),

$$\frac{V_A^+}{Z_A} - \frac{V_A^-}{Z_A} = \frac{V_B^+}{Z_B} + \frac{V_C^+}{Z_C} + \frac{V_D^+}{Z_D} + \cdots \tag{14.3.10}$$

Equations (14.3.8) and (14.3.10) can be solved for V_A^+, V_B^+, V_C^+, V_D^+, and so on in terms of $\mathbf{V}_A^+$. (See Problem 14.14.)

14.4 DISCRETE-TIME MODELS OF SINGLE-PHASE LOSSLESS LINES AND LUMPED RLC ELEMENTS

Our objective in this section is to develop discrete-time models of single-phase lossless lines and of lumped RLC elements suitable for computer calculation of transmission-line transients at discrete-time intervals $t = \Delta t$, $2\Delta t$, $3\Delta t$, and so on. The discrete-time models are presented as equivalent circuits consisting of lumped resistors and current sources. The current sources in the models represent the past history of the circuit—that is, the history at times $t - \Delta t$, $t - 2\Delta t$, and so on. After interconnecting the equivalent circuits of all the components in any given circuit, nodal equations can then be written for each discrete time. Discrete-time models, first developed by L. Bergeron [3], are presented first.

SINGLE-PHASE LOSSLESS LINE

From the general solution of a single-phase lossless line, given by (14.1.12) and (14.1.18):

$$v(x, t) + Z_c i(x, t) = 2v^+\left(t - \frac{x}{v}\right) \tag{14.4.1}$$

$$v(x, t) - Z_c i(x, t) = 2v^-\left(t + \frac{x}{v}\right) \tag{14.4.2}$$

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In (14.4.1), the left side $(v + Z_c i)$ remains constant when the argument $(t - x/v)$ is constant. Therefore, to a fictitious observer traveling at velocity v in the positive x direction along the line, $(v + Z_c i)$ remains constant. If τ is the transit time from terminal k to terminal m of the line, the value of $(v + Z_c i)$ observed at time $(t - \tau)$ at terminal k must equal the value at time t at terminal m. That is,

$$v_k(t - \tau) + Z_c i_k(t - \tau) = v_m(t) + Z_c i_m(t) \tag{14.4.3}$$

where k and m denote terminals k and m, as shown in Figure 14.14(a).

Similarly, $(v - Z_c i)$ in (14.4.2) remains constant when $(t + x/v)$ is constant. To a fictitious observer traveling at velocity v in the negative x direction, $(v - Z_c i)$ remains constant. Therefore, the value of $(v - Z_c i)$ at time $(t - \tau)$ at terminal m must equal the value at time t at terminal k. That is,

$$v_m(t - \tau) - Z_c i_m(t - \tau) = v_k(t) - Z_c i_k(t) \tag{14.4.4}$$

Equation (14.4.3) is rewritten as

$$i_m(t) = I_m(t - \tau) - \frac{1}{Z_c} v_m(t) \tag{14.4.5}$$

where

$$I_m(t - \tau) = i_k(t - \tau) + \frac{1}{Z_c} v_k(t - \tau) \tag{14.4.6}$$

Similarly, (14.4.4) is rewritten as

$$i_k(t) = I_k(t - \tau) + \frac{1}{Z_c} v_k(t) \tag{14.4.7}$$

where

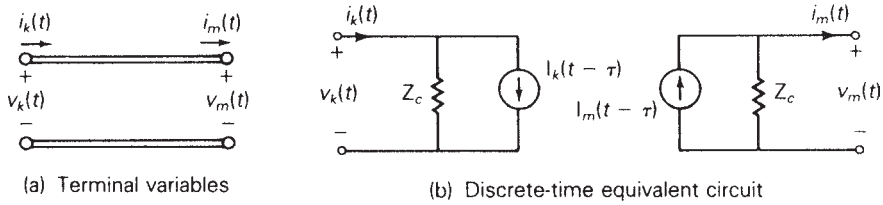
$$I_k(t - \tau) = i_m(t - \tau) - \frac{1}{Z_c} v_m(t - \tau) \tag{14.4.8}$$

Also, using (14.4.7) in (14.4.6),

$$I_m(t - \tau) = I_k(t - 2\tau) + \frac{2}{Z_c} v_k(t - \tau) \tag{14.4.9}$$

and using (14.4.5) in (14.4.8),

$$I_k(t - \tau) = I_m(t - 2\tau) - \frac{2}{Z_c} v_m(t - \tau) \tag{14.4.10}$$



(a) Terminal variables

(b) Discrete-time equivalent circuit

FIGURE 14.14
Single-phase two-wire lossless line

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Equations (14.4.5) and (14.4.7) are represented by the circuit shown in Figure 14.14(b). The current sources $I_m(t - \tau)$ and $I_k(t - \tau)$ shown in this figure, which are given by (14.4.9) and (14.4.10), represent the past history of the transmission line.

Note that in Figure 14.14(b), terminals *k* and *m* are not directly connected. The conditions at one terminal are "felt" indirectly at the other terminal after a delay of τ seconds.

LUMPED INDUCTANCE

As shown in Figure 14.15(a), for a constant lumped inductance L,

$$v(t) = \mathrm{L}\frac{di(t)}{dt} \tag{14.4.11}$$

Integrating this equation from time $(t - \Delta t)$ to t,

$$\int_{t-\Delta t}^{t} di(t) = \frac{1}{\mathrm{L}}\int_{t-\Delta t}^{t} v(t)dt \tag{14.4.12}$$

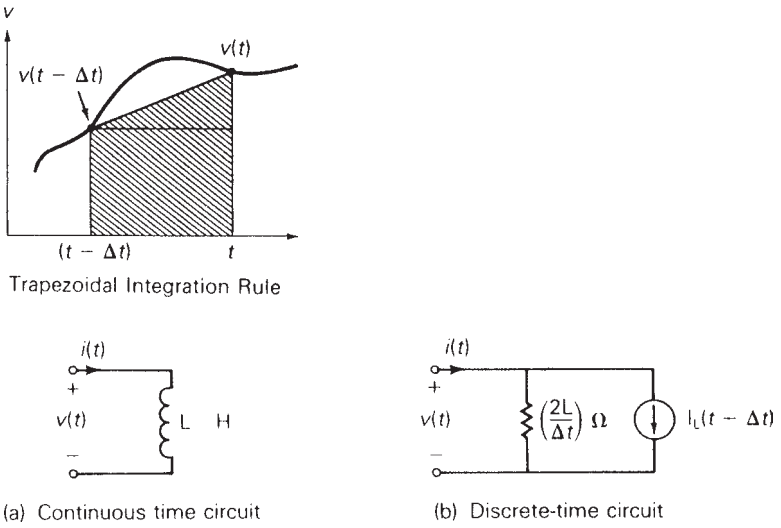
Using the trapezoidal rule of integration,

$$i(t) - i(t - \Delta t) = \left(\frac{1}{\mathrm{L}}\right)\left(\frac{\Delta t}{2}\right)[v(t) + v(t - \Delta t)]$$

Rearranging gives

$$i(t) = \frac{v(t)}{(2\mathrm{L}/\Delta t)} + \left[i(t - \Delta t) + \frac{v(t - \Delta t)}{(2\mathrm{L}/\Delta t)}\right]$$

FIGURE 14.15
Lumped inductance



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or

$$i(t) = \frac{v(t)}{(2L/\Delta t)} + I_L(t - \Delta t) \tag{14.4.13}$$

where

$$I_L(t - \Delta t) = i(t - \Delta t) + \frac{v(t - \Delta t)}{(2L/\Delta t)} = I_L(t - 2\Delta t) + \frac{v(t - \Delta t)}{(L/\Delta t)} \tag{14.4.14}$$

Equations (14.4.13) and (14.4.14) are represented by the circuit shown in Figure 14.15(b). As shown, the inductor is replaced by a resistor with resistance $(2L/\Delta t)\,\Omega$. A current source $I_L(t - \Delta t)$ given by (14.4.14) is also included. $I_L(t - \Delta t)$ represents the past history of the inductor. Note that the trapezoidal rule introduces an error of the order $(\Delta t)^3$.

LUMPED CAPACITANCE

As shown in Figure 14.16(a), for a constant lumped capacitance C,

$$i(t) = C\frac{dv(t)}{dt} \tag{14.4.15}$$

Integrating from time $(t - \Delta t)$ to t,

$$\int_{t-\Delta t}^{t} dv(t) = \frac{1}{C}\int_{t-\Delta t}^{t} i(t)\,dt \tag{14.4.16}$$

Using the trapezoidal rule of integration,

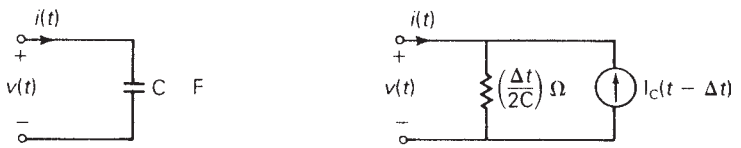
$$v(t) - v(t - \Delta t) = \frac{1}{C}\left(\frac{\Delta t}{2}\right)[i(t) + i(t - \Delta t)]$$

Rearranging gives

$$i(t) = \frac{v(t)}{(\Delta t/2C)} - I_C(t - \Delta t) \tag{14.4.17}$$

where

$$I_C(t - \Delta t) = i(t - \Delta t) + \frac{v(t - \Delta t)}{(\Delta t/2C)} = -I_C(t - 2\Delta t) + \frac{v(t - \Delta t)}{(\Delta t/4C)} \tag{14.4.18}$$



(a) Continuous time circuit (b) Discrete-time circuit

FIGURE 14.16
Lumped capacitance

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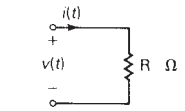
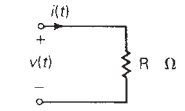


(a) Continuous time circuit



(b) Discrete-time circuit

FIGURE 14.17
Lumped resistance

Equations (14.4.17) and (14.4.18) are represented by the circuit in Figure 14.16(b). The capacitor is replaced by a resistor with resistance $(\Delta t/2C)\ \Omega$. A current source $I_C(t - \Delta t)$, which represents the capacitor's past history, is also included.

LUMPED RESISTANCE

The discrete model of a constant lumped resistance R, as shown in Figure 14.17, is the same as the continuous model. That is,

$$v(t) = Ri(t) \tag{14.4.19}$$

NODAL EQUATIONS

A circuit consisting of single-phase lossless transmission lines and constant lumped RLC elements can be replaced by the equivalent circuits given in Figures 14.14(b), 14.15(b), 14.16(b), and 14.17(b). Then, writing nodal equations, the result is a set of linear algebraic equations that determine the bus voltages at each instant t.

EXAMPLE 14.7

Discrete-time equivalent circuit, single-phase lossless line transients, computer solution

For the circuit given in Example 14.3, replace the circuit elements by their discrete-time equivalent circuits and write the nodal equations that determine the sending-end and receiving-end voltages. Then, using a digital computer, compute the sending-end and receiving-end voltages for $0 \le t \le 9$ ms. For numerical calculations, assume $E = 100$ V, $Z_c = 400\ \Omega$, $C_R = 5\ \mu$F, $\tau = 1.0$ ms, and $\Delta t = 0.1$ ms.

SOLUTION

The discrete model is shown in Figure 14.18, where $v_k(t)$ represents the sending-end voltage $v(0, t)$ and $v_m(t)$ represents the receiving-end voltage $v(l, t)$. Also, the sending-end voltage source $e_G(t)$ in series with Z_G is converted to an equivalent current source in parallel with Z_G. Writing nodal equations for this circuit,

$$\left[\begin{array}{c|c} \left(\dfrac{1}{400}+\dfrac{1}{400}\right) & 0 \\ \hline 0 & \left(\dfrac{1}{400}+\dfrac{1}{10}\right) \end{array}\right] \begin{bmatrix} v_k(t) \\ v_m(t) \end{bmatrix} = \begin{bmatrix} \frac{1}{4} - I_k(t-1.0) \\ I_m(t-1.0) + I_C(t-0.1) \end{bmatrix}$$

Solving,

$$v_k(t) = 200[\tfrac{1}{4} - I_k(t-1.0)] \tag{a}$$

$$v_m(t) = 9.75610[I_m(t-1.0) + I_C(t-0.1)] \tag{b}$$

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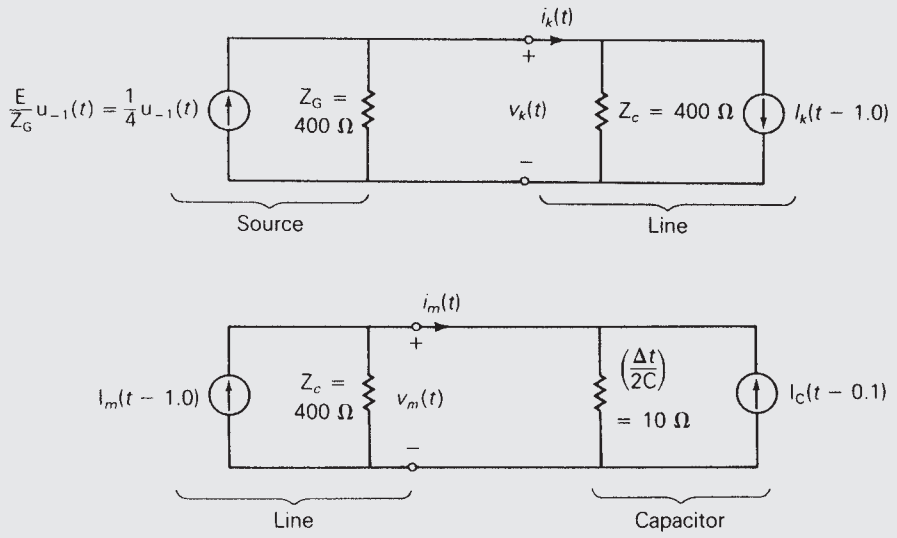
FIGURE 14.18

Discrete-time equivalent circuit for Example 14.7



The current sources in these equations are, from (14.4.9), (14.4.10), and (14.4.18), with the argument $(t - \tau)$ replaced by t,

$$\mathrm{I}_m(t) = \mathrm{I}_k(t - 1.0) + \frac{2}{400}\upsilon_k(t) \tag{c}$$

$$\mathrm{I}_k(t) = \mathrm{I}_m(t - 1.0) - \frac{2}{400}\upsilon_m(t) \tag{d}$$

$$\mathrm{I}_C(t) = -\mathrm{I}_C(t - 0.1) + \frac{1}{5}\upsilon_m(t) \tag{e}$$

Equations (a) through (e) above are in a form suitable for digital computer solution. A scheme for iteratively computing υ_k and υ_m is as follows, starting at $t = 0$:

1. Compute $\upsilon_k(t)$ and $\upsilon_m(t)$ from equations (a) and (b).
2. Compute $\mathrm{I}_m(t)$, $\mathrm{I}_k(t)$, and $\mathrm{I}_C(t)$ from equations (c), (d), and (e). Store $\mathrm{I}_m(t)$ and $\mathrm{I}_k(t)$.
3. Change t to $(t + \Delta t) = (t + 0.1)$ and return to (1) above.

Note that since the transmission line and capacitor are unenergized for time t less than zero, the current sources $\mathrm{I}_m(\,)$, $\mathrm{I}_k(\,)$, and $\mathrm{I}_C(\,)$ are zero whenever their arguments $(\,)$ are negative. Note also from equations (a) through (e) that it is necessary to store the past ten values of $\mathrm{I}_m(\,)$ and $\mathrm{I}_k(\,)$.

(*Continued*)

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FIGURE 14.19

Example 14.7

Output			
Time ms	VK Volts	VM Volts	Computer Program Listing
0.00	50.00	0.00	10 REM EXAMPLE 14.7
0.20	50.00	0.00	20 LPRINT "TIME VK VM"
0.40	50.00	0.00	30 LPRINT "ms Volts Volts"
0.60	50.00	0.00	40 IC = 0
0.80	50.00	0.00	50 T = 0
1.00	50.00	2.44	60 KPRINT = 2
1.20	50.00	11.73	65 REM T IS TIME. KPRINT DETERMINES THE PRINTOUT INTERVAL.
1.40	50.00	20.13	70 REM LINES 110 to 210 COMPUTE EQS(a)−(e) FOR THE FIRST
1.60	50.00	27.73	80 REM TEN TIME STEPS (A TOTAL OF ONE ms) DURING WHICH
1.80	50.00	34.61	90 REM THE CURRENT SOURCES ON THE RIGHT HAND SIDE
2.00	2.44	40.83	100 REM OF THE EQUATIONS ARE ZERO. TEN VALUES OF
2.20	11.73	46.46	105 REM CURRENT SOURCES IK(J) AND IM(J) ARE STORED.
2.40	20.13	51.56	110 FOR J = 1 TO 10
2.60	27.73	56.17	120 VK = 200/4
2.80	34.61	60.34	130 VM = 9.7561*IC
3.00	40.83	64.12	140 IM(J) = (2/400)*VK
3.20	46.46	67.53	150 IK(J) = (−2/400)*VM
3.40	51.56	70.62	160 IC=−IC + (1/5)*VM
3.60	56.17	73.42	170 Z = (J−1)/KPRINT
3.80	60.34	75.95	180 M = INT (Z)
4.00	64.12	78.24	190 IF M = Z THEN LPRINT USING "*****"; T,VK,VM
4.20	67.53	80.31	200 T = T + .1
4.40	70.62	82.18	210 NEXT J
4.60	73.42	83.88	220 REM LINES 250 to 420 COMPUTE EQS(a)−(e) FOR TIME T
4.80	75.95	85.41	230 REM EQUAL TO AND GREATER THAN 1.0 ms. THE PAST TEN
5.00	78.24	86.80	240 REM VALUES OF IK(J) AND IM(J) ARE STORED
5.20	80.31	88.06	250 FOR J = 1 TO 10
5.40	82.18	89.20	260 REM LINE 270 IS EQ(a).
5.60	83.88	90.22	270 VK = 200*((1/4)−IK (J))
5.80	85.41	91.15	280 REM LINE 290 IS EQ (b).
6.00	86.80	92.00	290 VM = 9.7561*(IM (J) + IC)
6.20	88.06	92.76	300 REM LINE 310 IS EQ (e).
6.40	89.20	93.45	310 IC=−IC + (1/5)*VM
6.60	90.22	94.07	320 REM LINES 330−360 ARE EQS (c) and (d).
6.80	91.15	94.64	330 C1 = IK(J) + (2/400)*VK
7.00	92.00	95.15	340 C2 = IM(J)−(2/400)*VM
7.20	92.76	95.61	350 IM(J) = C1
7.40	93.45	96.03	360 IK(J) = C2
7.60	94.07	96.40	370 Z = (J−1)/KPRINT
7.80	94.64	96.75	380 M = INT(Z)
8.00	95.15	97.06	390 IF M = Z THEN LPRINT USING "*****"; T,VK,VM
8.20	95.61	97.34	400 T = T +.1
8.40	96.03	97.59	410 NEXT J
8.60	96.40	97.82	420 IF T < 9.0 THEN GOTO 250
8.80	96.75	98.03	430 STOP
9.00	97.06	98.22	

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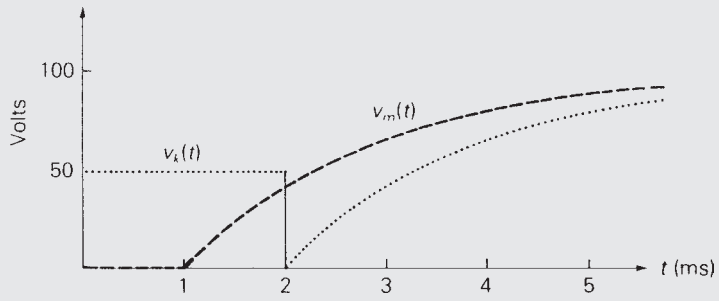
FIGURE 14.19
(*Continued*)



A personal computer program written in BASIC that executes the above scheme and the computational results are shown in Figure 14.19. The plotted sending-end voltage $v_k(t)$ can be compared with the results of Example 14.3.

Example 14.7 can be generalized to compute bus voltages at discrete-time intervals for an arbitrary number of buses, single-phase lossless lines, and lumped RLC elements. When current sources instead of voltage sources are employed, the unknowns are all bus voltages, for which nodal equations $\mathbf{YV} = \mathbf{I}$ can be written at each discrete-time instant. Also, the dependent current sources in $\mathbf{I}$ are written in terms of bus voltages and current sources at prior times. For computational convenience, the time interval Δt can be chosen constant so that the bus admittance matrix $\mathbf{Y}$ is a constant real symmetric matrix as long as the RLC elements are constant.

14.5 LOSSY LINES

Transmission-line series resistance or shunt conductance causes the following:

1. Attenuation
2. Distortion
3. Power losses

A brief discussion of these effects follows.

ATTENUATION

When constant series resistance R Ω/m and shunt conductance G S/m are included in the circuit of Figure 14.1 for a single-phase two-wire line, (14.1.3) and (14.1.4) become

$$\frac{\partial v(x, t)}{\partial x} = -\mathrm{R}\, i(x, t) - \mathrm{L}\frac{\partial i(x, t)}{\partial t} \tag{14.5.1}$$

$$\frac{\partial i(x, t)}{\partial x} = -\mathrm{G} v(x, t) - \mathrm{C}\frac{\partial v(x, t)}{\partial t} \tag{14.5.2}$$

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Taking the Laplace transform of these, equations analogous to (14.1.7) and (14.1.8) are

$$\frac{d^2\mathrm{V}(x, s)}{dx^2} - \gamma^2(s)\,\mathrm{V}(x, s) = 0 \tag{14.5.3}$$

$$\frac{d^2\mathrm{I}(x, s)}{dx^2} - \gamma^2(s)\mathrm{I}(x, s) = 0 \tag{14.5.4}$$

where

$$\gamma(s) = \sqrt{(\mathrm{R} + s\mathrm{L})(\mathrm{G} + s\mathrm{C})} \tag{14.5.5}$$

The solution to these equations is

$$\mathrm{V}(x, s) = \mathrm{V}^+(s)e^{-\gamma(s)x} + \mathrm{V}^-(s)e^{+\gamma(s)x} \tag{14.5.6}$$

$$\mathrm{I}(x, s) = \mathrm{I}^+(s)e^{-\gamma(s)x} + \mathrm{I}^-(s)e^{+\gamma(s)x} \tag{14.5.7}$$

In general, it is impossible to obtain a closed form expression for $v(x, t)$ and $i(x, t)$, which are the inverse Laplace transforms of these equations. However, for the special case of a *distortionless* line, which has the property $\mathrm{R/L} = \mathrm{G/C}$, the inverse Laplace transform can be obtained as follows. Rewrite (14.5.5) as

$$\gamma(s) = \sqrt{\mathrm{LC}[(s + \delta)^2 - \sigma^2]} \tag{14.5.8}$$

where

$$\delta = \frac{1}{2}\left(\frac{\mathrm{R}}{\mathrm{L}} + \frac{\mathrm{G}}{\mathrm{C}}\right) \tag{14.5.9}$$

$$\sigma = \frac{1}{2}\left(\frac{\mathrm{R}}{\mathrm{L}} - \frac{\mathrm{G}}{\mathrm{C}}\right) \tag{14.5.10}$$

For a distortionless line, $\sigma = 0$, $\delta = \mathrm{R/L}$, and (14.5.6) and (14.5.7) become

$$\mathrm{V}(x, s) = \mathrm{V}^+(s)e^{-\sqrt{\mathrm{LC}}[s+(\mathrm{R/L})]x} + \mathrm{V}^-(s)e^{+\sqrt{\mathrm{LC}}[s+(\mathrm{R/L})]x} \tag{14.5.11}$$

$$\mathrm{I}(x, s) = \mathrm{I}^+(s)e^{-\sqrt{\mathrm{LC}}[s+(\mathrm{R/L})]x} + \mathrm{I}^-(s)e^{+\sqrt{\mathrm{LC}}[s+(\mathrm{R/L})]x} \tag{14.5.12}$$

Using $v = 1/\sqrt{\mathrm{LC}}$ and $\sqrt{\mathrm{LC}}(\mathrm{R/L}) = \sqrt{\mathrm{RG}} = \alpha$ for the distortionless line, the inverse transform of these equations is

$$v(x, t) = e^{-\alpha x}v^+\left(t - \frac{x}{v}\right) + e^{+\alpha x}v^-\left(t + \frac{x}{v}\right) \tag{14.5.13}$$

$$i(x, t) = e^{-\alpha x}i^+\left(t - \frac{x}{v}\right) + e^{\alpha x}i^-\left(t + \frac{x}{v}\right) \tag{14.5.14}$$

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These voltage and current waves consist of forward and backward traveling waves similar to (14.1.12) and (14.1.13) for a lossless line. However, for the lossy distortionless line, the waves are attenuated versus x due to the $e^{\pm \alpha x}$ terms. Note that the attenuation term $\alpha = \sqrt{RG}$ is constant. Also, the attenuated waves travel at constant velocity $v = 1/\sqrt{LC}$. Therefore, waves traveling along the distortionless line do not change their shape; only their magnitudes are attenuated.

DISTORTION

For sinusoidal steady-state waves, the propagation constant $\gamma(j\omega)$ is, from (14.5.5), with $s = j\omega$,

$$\gamma(j\omega) = \sqrt{(R + j\omega L)(G + j\omega C)} = \alpha + j\beta \tag{14.5.15}$$

For a lossless line, $R = G = 0$; therefore, $\alpha = 0$, $\beta = \omega\sqrt{LC}$, and the phase velocity $v = \omega/\beta = 1/\sqrt{LC}$ is constant. Thus, sinusoidal waves of all frequencies travel at constant velocity v without attenuation along a lossless line.

For a distortionless line $(R/L) = (G/C)$, and $\gamma(j\omega)$ can be rewritten, using (14.5.8) through (14.5.10), as

$$\begin{aligned}\gamma(j\omega) &= \sqrt{LC\left(j\omega + \frac{R}{L}\right)^2} = \sqrt{LC}\left(j\omega + \frac{R}{L}\right) \\ &= \sqrt{RG} + j\frac{\omega}{v} = \alpha + j\beta\end{aligned} \tag{14.5.16}$$

Since $\alpha = \sqrt{RG}$ and $v = 1/\sqrt{LC}$ are constant, sinusoidal waves of all frequencies travel along the distortionless line at constant velocity with constant attenuation—that is, without distortion.

It can also be shown that for frequencies above 1 MHz, practical transmission lines with typical constants R, L, G, and C tend to be distortionless. Above 1 MHz, α and β can be approximated by

$$\alpha \simeq \frac{R}{2}\sqrt{\frac{C}{L}} + \frac{G}{2}\sqrt{\frac{L}{C}} \tag{14.5.17}$$

$$\beta \simeq \omega\sqrt{LC} = \frac{\omega}{v} \tag{14.5.18}$$

Therefore, sinusoidal waves with frequencies above 1 MHz travel along a practical line undistorted at constant velocity $v = 1/\sqrt{LC}$ with attenuation α given by (14.5.17).

At frequencies below 1 MHz, these approximations do not hold, and lines are generally not distortionless. For typical transmission and distribution lines, (R/L) is much greater than (G/C) by a factor of 1000 or so. Therefore, the condition $(R/L) = (G/C)$ for a distortionless line does not hold.

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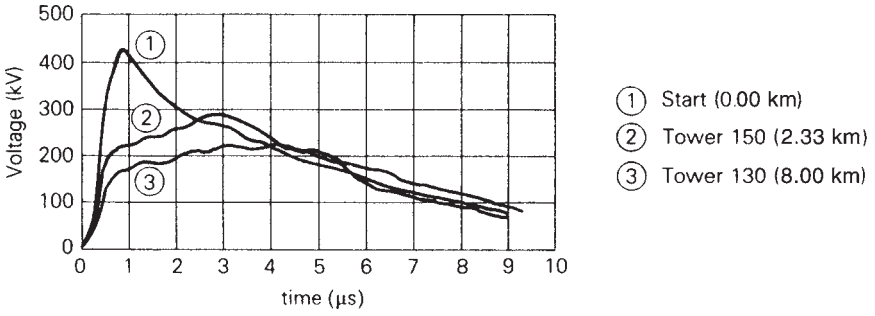

FIGURE 14.20

Distortion and attenuation of surges on a 132-kV overhead line [4] (Lacey, H.M., "The lightning protection of high-voltage overhead transmission and distribution systems," *Proceedings of the IEE - Part II: Power Engineering*, vol. 96, no. 50, pp. 287–299, April 1949. Reprinted by permission of Institution of Engineering and Technology.)

Figure 14.20 shows the effect of distortion and attenuation of voltage surges based on experiments on a 132-kV overhead transmission line [4]. The shapes of the surges at three points along the line are shown. Note how distortion reduces the front of the wave and builds up the tail as it travels along the line.

POWER LOSSES

Power losses are associated with series resistance R and shunt conductance G. When a current I flows along a line, I^2R losses occur, and when a voltage V appears across the conductors, V^2G losses occur. V^2G losses are primarily due to insulator leakage and corona for overhead lines, and to dielectric losses for cables. For practical lines operating at rated voltage and rated current, I^2R losses are much greater than V^2G losses.

As discussed above, the analysis of transients on single-phase two-wire lossy lines with constant parameters R, L, G, and C is complicated. The analysis becomes more complicated when skin effect is included, which means that R is not constant but frequency-dependent. Additional complications arise for a single-phase line consisting of one conductor with earth return, where Carson [5] has shown that both series resistance and inductance are frequency-dependent.

In view of these complications, the solution of transients on lossy lines is best handled via digital computation techniques. A single-phase line of length l can be approximated by a lossless line with half the total resistance $(Rl/2)$ Ω lumped in series with the line at both ends. For improved accuracy, the line can be divided into various line sections, and each section can be approximated by a lossless line section, with a series resistance lumped at both ends. Simulations have shown that accuracy does not significantly improve with more than two line sections.

Discrete-time equivalent circuits of a single-phase lossless line, shown in Figure 14.14, together with a constant lumped resistance, shown in Figure 14.17, can be used to approximate a lossy line section, as shown in Figure 14.21. Also, digital techniques for modeling frequency-dependent line parameters [6, 7] are available but are beyond the scope of this book.

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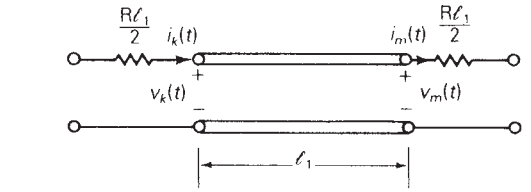
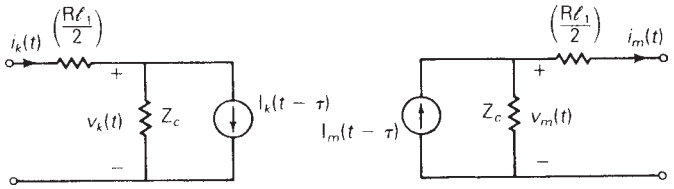


(a) Lossless line segment of length ℓ_1 with lumped line resistance



(b) Discrete-time model

FIGURE 14.21

Approximate model of a lossy line segment

14.6 MULTICONDUCTOR LINES

Transients on single-phase two-wire lines have been considered until now. For a transmission line with n conductors above a ground plane, waves travel in n "modes," where each mode has its own wave velocity and its own surge impedance. In this section "model analysis" for a relatively simple three-phase line [8] is illustrated.

Given a three-phase, lossless, completely transposed line consisting of three conductors above a perfectly conducting ground plane, the transmission-line equations are

$$\frac{d\mathbf{V}(x, s)}{dx} = -s\mathbf{L}\mathbf{I}(x, s) \tag{14.6.1}$$

$$\frac{d\mathbf{I}(x, s)}{dx} = -s\mathbf{C}\mathbf{V}(x, s) \tag{14.6.2}$$

where

$$\mathbf{V}(x, s) = \begin{bmatrix} \mathrm{V}_{ag}(x, s) \\ \mathrm{V}_{bg}(x, s) \\ \mathrm{V}_{cg}(x, s) \end{bmatrix} \qquad \mathbf{I}(x, s) = \begin{bmatrix} \mathrm{I}_a(x, s) \\ \mathrm{I}_b(x, s) \\ \mathrm{I}_c(x, s) \end{bmatrix} \tag{14.6.3}$$

Equations (14.6.1) and (14.6.2) are identical to (14.1.5) and (14.1.6) except that scalar quantities are replaced by vector quantities. $\mathbf{V}(x, s)$ is the vector of line-to-ground voltages and $\mathbf{I}(x, s)$ is the vector of line currents. For a completely transposed line, the 3×3 inductance matrix $\mathbf{L}$ and capacitance matrix $\mathbf{C}$ are given by

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$$\mathbf{L} = \begin{bmatrix} L_s & L_m & L_m \\ L_m & L_s & L_m \\ L_m & L_m & L_s \end{bmatrix} \quad \text{H/m} \tag{14.6.4}$$

$$\mathbf{C} = \begin{bmatrix} C_s & C_m & C_m \\ C_m & C_s & C_m \\ C_m & C_m & C_s \end{bmatrix} \quad \text{F/m} \tag{14.6.5}$$

For any given line configuration, **L** and **C** can be computed from the equations given in Sections 4.7 and 4.11. Note that L_s, L_m, and C_s are positive, whereas C_m is negative.

Now transform the phase quantities to modal quantities. First, define

$$\begin{bmatrix} V_{ag}(x, s) \\ V_{bg}(x, s) \\ V_{cg}(x, s) \end{bmatrix} = \mathbf{T}_V \begin{bmatrix} V^0(x, s) \\ V^+(x, s) \\ V^-(x, s) \end{bmatrix} \tag{14.6.6}$$

$$\begin{bmatrix} I_a(x, s) \\ I_b(x, s) \\ I_c(x, s) \end{bmatrix} = \mathbf{T}_I \begin{bmatrix} I^0(x, s) \\ I^+(x, s) \\ I^-(x, s) \end{bmatrix} \tag{14.6.7}$$

$V^0(x, s)$, $V^+(x, s)$, and $V^-(x, s)$ are denoted *zero-mode*, *positive-mode*, and *negative-mode* voltages, respectively. Similarly, $I^0(x, s)$, $I^+(x, s)$, and $I^-(x, s)$ are *zero-*, *positive-*, and *negative-mode* currents. $\mathbf{T}_V$ and $\mathbf{T}_I$ are 3×3 constant transformation matrices, soon to be specified. Denoting $\mathbf{V}_m(x, s)$ and $\mathbf{I}_m(x, s)$ as the modal voltage and modal current vectors,

$$\mathbf{V}(x, s) = \mathbf{T}_V \mathbf{V}_m(x, s) \tag{14.6.8}$$

$$\mathbf{I}(x, s) = \mathbf{T}_I \mathbf{I}_m(x, s) \tag{14.6.9}$$

Using (14.6.8) and (14.6.9) in (14.6.1),

$$\mathbf{T}_V \frac{d\mathbf{V}_m(x, s)}{dx} = -s\mathbf{L}\mathbf{T}_I \mathbf{I}_m(x, s)$$

or

$$\frac{d\mathbf{V}_m(x, s)}{dx} = -s(\mathbf{T}_V^{-1}\mathbf{L}\mathbf{T}_I)\mathbf{I}_m(x, s) \tag{14.6.10}$$

Similarly, using (14.6.8) and (14.6.9) in (14.6.2),

$$\frac{d\mathbf{I}_m(x, s)}{dx} = -s(\mathbf{T}_I^{-1}\mathbf{C}\mathbf{T}_V)\mathbf{V}_m(x, s) \tag{14.6.11}$$

The objective of the modal transformation is to diagonalize the matrix products within the parentheses of (14.6.10) and (14.6.11), thereby decoupling these vector equations. For a three-phase completely transposed line, $\mathbf{T}_V$ and $\mathbf{T}_I$ are given by

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$$\mathbf{T}_{\mathrm{V}} = \mathbf{T}_{\mathrm{I}} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix} \tag{14.6.12}$$

Also, the inverse transformation matrices are

$$\mathbf{T}_{\mathrm{V}}^{-1} = \mathbf{T}_{\mathrm{I}}^{-1} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \tag{14.6.13}$$

Substituting (14.6.12), (14.6.13), (14.6.4), and (14.6.5) into (14.6.10) and (14.6.11) yields

$$\frac{d}{dx}\begin{bmatrix} \mathrm{V}^0(x,s) \\ \mathrm{V}^+(x,s) \\ \mathrm{V}^-(x,s) \end{bmatrix} = \begin{bmatrix} -s(\mathrm{L}_s + 2\mathrm{L}_m) & 0 & 0 \\ 0 & -s(\mathrm{L}_s - \mathrm{L}_m) & 0 \\ 0 & 0 & -s(\mathrm{L}_s - \mathrm{L}_m) \end{bmatrix} \times \begin{bmatrix} \mathrm{I}^0(x,s) \\ \mathrm{I}^+(x,s) \\ \mathrm{I}^-(x,s) \end{bmatrix} \tag{14.6.14}$$

$$\frac{d}{dx}\begin{bmatrix} \mathrm{I}^0(x,s) \\ \mathrm{I}^+(x,s) \\ \mathrm{I}^-(x,s) \end{bmatrix} = \begin{bmatrix} -s(\mathrm{C}_s + 2\mathrm{C}_m) & 0 & 0 \\ 0 & -s(\mathrm{C}_s - \mathrm{C}_m) & 0 \\ 0 & 0 & -s(\mathrm{C}_s - \mathrm{C}_m) \end{bmatrix} \times \begin{bmatrix} \mathrm{V}^0(x,s) \\ \mathrm{V}^+(x,s) \\ \mathrm{V}^-(x,s) \end{bmatrix} \tag{14.6.15}$$

From (14.6.14) and (14.6.15), the zero-mode equations are

$$\frac{d\mathrm{V}^0(x,s)}{dx} = -s(\mathrm{L}_s + 2\mathrm{L}_m)\mathrm{I}^0(x,s) \tag{14.6.16}$$

$$\frac{d\mathrm{I}^0(x,s)}{dx} = -s(\mathrm{C}_s + 2\mathrm{C}_m)\mathrm{V}^0(x,s) \tag{14.6.17}$$

These equations are identical in form to those of a two-wire lossless line, (14.1.5) and (14.1.6). By analogy, the zero-mode waves travel at velocity

$$v^0 = \frac{1}{\sqrt{(\mathrm{L}_s + 2\mathrm{L}_m)\,(\mathrm{C}_s + 2\mathrm{C}_m)}} \quad \text{m/s} \tag{14.6.18}$$

and the zero-mode surge impedance is

$$\mathrm{Z}_c^0 = \sqrt{\frac{\mathrm{L}_s + 2\mathrm{L}_m}{\mathrm{C}_s + 2\mathrm{C}_m}} \quad \Omega \tag{14.6.19}$$

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Similarly, the positive- and negative-mode velocities and surge impedances are

$$v^+ = v^- = \frac{1}{\sqrt{(L_s - L_m)(C_s - C_m)}} \quad \text{m/s} \tag{14.6.20}$$

$$Z_c^+ = Z_c^- = \sqrt{\frac{L_s - L_m}{C_s - C_m}} \quad \Omega \tag{14.6.21}$$

These equations can be extended to more than three conductors—for example, to a three-phase line with shield wires or to a double-circuit three-phase line. Although the details are more complicated, the modal transformation is straightforward. There is one mode for each conductor, and each mode has its own wave velocity and its own surge impedance.

The solution of transients on multi-conductor lines is best handled via digital computer methods, and such programs are available [9, 10]. Digital techniques are also available to model the following effects:

1. Nonlinear and time-varying RLC elements [8]
2. Lossy lines with frequency-dependent line parameters [6, 7, 12]

14.7 POWER SYSTEM OVERVOLTAGES

Overvoltages encountered by power system equipment are of three types:

1. Lightning surges
2. Switching surges
3. Power frequency (50 or 60 Hz) overvoltages

LIGHTNING

Cloud-to-ground (CG) lightning is the greatest single cause of overhead transmission and distribution line outages. Data obtained over a 14-year period from electric utility companies in the United States and Canada and covering 40,234 kilometers of transmission show that CG lightning accounted for about 26% of outages on 230-kV circuits and about 65% of outages on 345-kV circuits [13]. A similar study in Britain, also over a 14-year period, covering 50,000 faults on distribution lines, shows that CG lightning accounted for 47% of outages on circuits up to and including 33 kV [14].

The electrical phenomena that occur within clouds leading to a lightning strike are complex and not totally understood. Several theories [15, 16, 17] generally agree, however, that charge separation occurs within clouds. Wilson [15] postulates that falling raindrops attract negative charges and therefore leave behind masses of positively charged air. The falling raindrops bring the negative charge to the bottom of the cloud, and upward air drafts carry the positively charged air and ice crystals to the top of the cloud, as shown in Figure 14.22. Negative charges at the bottom of the cloud induce a positively charged region, or "shadow," on the earth directly below the cloud. The electric field lines shown in Figure 14.22 originate from the positive charges and terminate at the negative charges.

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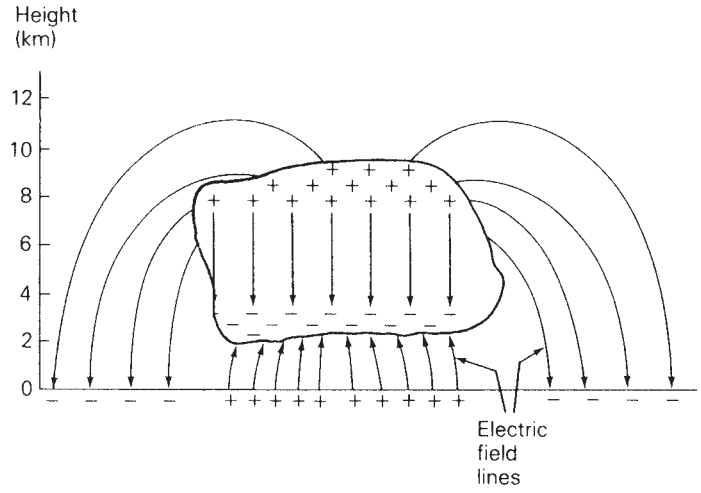

FIGURE 14.22

Postulation of charge separation within clouds [16] (Based on G. B. Simpson and F. J. Scrase, "The Distribution of Electricity in Thunderclouds", *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 161 (1937), p. 309.)

When voltage gradients reach the breakdown strength of the humid air within the cloud, typically 5 to 15 kV/cm, an ionized path or downward *leader* moves from the cloud toward the earth. The leader progresses somewhat randomly along an irregular path, in steps. These leader steps, about 50 m long, move at a velocity of about 10^5 m/s. As a result of the opposite charge distribution under the cloud, another upward leader may rise to meet the downward leader. When the two leaders meet, a lightning discharge occurs, which neutralizes the charges.

The current involved in a CG lightning stroke typically rises to a peak value within 1 to 10 μs, and then diminishes to one-half the peak within 20 to 100 μs. The distribution of peak currents is shown in Figure 14.23 [20]. This curve represents the percentage of strokes that exceed a given peak current. For example, 50% of all strokes

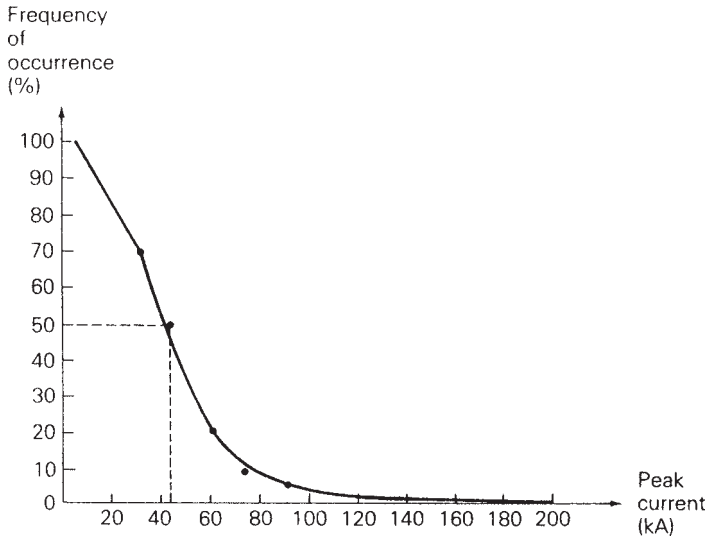

FIGURE 14.23

Frequency of occurrence of lightning currents that exceed a given peak value [20] (© 1997 IEEE. Reprinted, with permission from IEEE Std C62.22-1997: IEEE Guide for the Application of Metal-Oxide Surge Arresters for Alternating-Current Systems.)

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have a peak current greater than 45 kA. In extreme cases, the peak current can exceed 200 kA. Also, test results indicate that approximately 90% of all strokes are negative.

It has also been shown that what appears to the eye as a single flash of lightning is often the cumulative effect of many strokes. A typical flash consists of typically 3 to 5, and occasionally as many as 40, strokes, at intervals of 50 ms.

The U.S. National Lightning Detection Network® (NLDN), owned and operated by Vaisala, Inc., is a system that senses the electromagnetic fields radiated by individual return strokes in CG flashes. As of 2001, the NLDN employed more than 100 remote, ground-based sensors geographically distributed throughout the 48 contiguous United States. The sensors transmit lightning data to a network control center in Tucson, Arizona, via a satellite communication system. Data from the remote sensors are recorded and processed in real time at the network control center to provide the time, location, polarity, and an estimate of the peak current in each return stroke. The real-time data are then sent back through the communications network for satellite broadcast dissemination to real-time users, all within 30−40 seconds of each CG lightning flash. Recorded data are also reprocessed offline within a few days of acquisition and stored in a permanent database for access by users who do not require real-time information. Since 1989, the NLDN has reported more than 20 million cloud-to-ground lightning flashes that occur every year, and currently reports about 85 million cloud-to-ground strokes annually. [24, 25].

Figure 14.24 shows a lightning flash density contour map providing a representation of measured annual CG flash density detected by the NLDN from

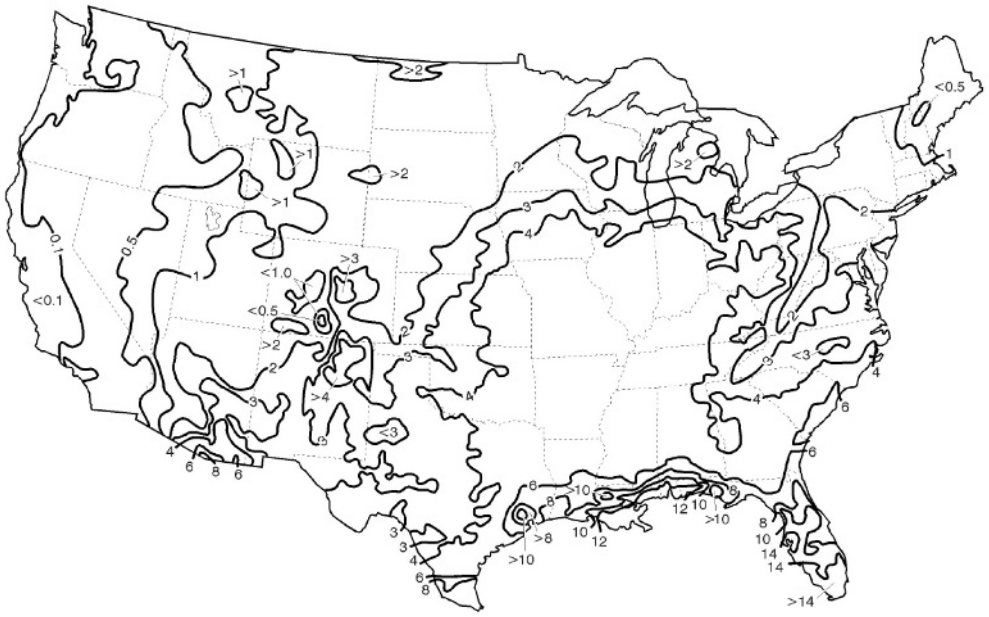

FIGURE 14.24

Lightning flash density contour map showing annual CG flash densities (#flashes/km^2/year) in the contiguous United States, as detected by the NLDN from 1989 to 1998 (Courtesy of Vaisala, Inc.)

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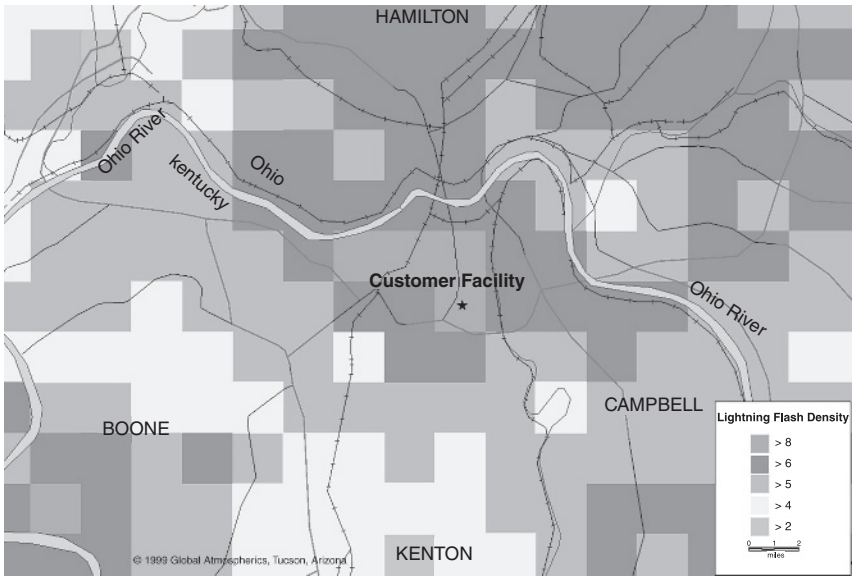

FIGURE 14.25

High-resolution CG flash density map in grid format (Courtesy of Vaisala, Inc.)

1989 to 1998. As shown, average annual CG lightning flash densities range from about 0.1 flashes/km^2/year near the West Coast to more than 14 flashes/km^2/year in portions of the Florida peninsula. Figure 14.25 shows a high-resolution, 3.2 km $\times$ 3.2 km, CG flash density map in grid format.

Figure 14.26 shows an "asset exposure map" of all CG strikes in 1995 in a region that contains a 24 kilometer 69-kV transmission line. This map provides an indication of the level of exposure to lightning within an exposure area that surrounds the transmission line. By combining this data with estimates of peak

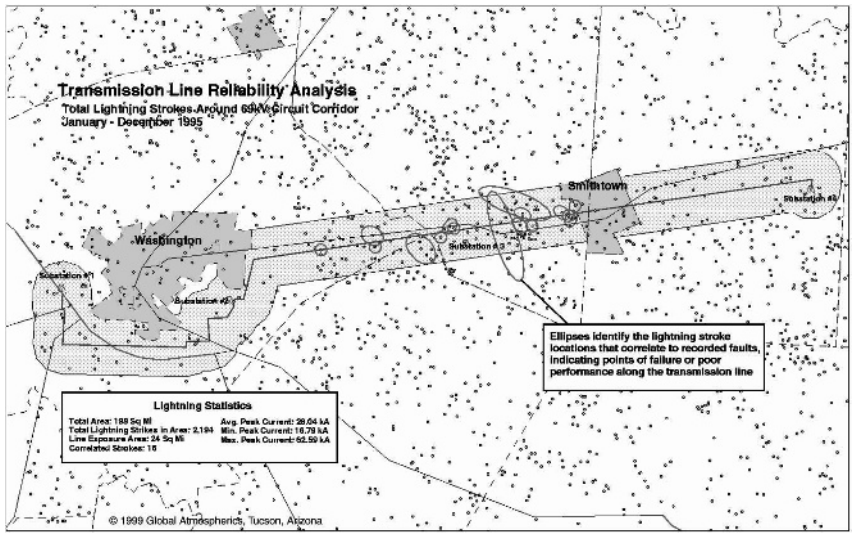

FIGURE 14.26

Asset exposure map showing CG strikes in 1995 in a region that contains a 24 kilometer 69-kV transmission line (Courtesy of Vaisala, Inc.)

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stroke currents and transmission line fault records, the lightning performance of the line and individual line segments can be quantified. Improvements in line design and line protection can also be evaluated.

Electric utilities use real-time lightning maps to monitor the approach of lightning storms, estimate their severity, and then either position repair crews in advance of storms, call them out, or hold them over as required. Utilities also use real-time lightning data together with on-line monitoring of circuit breakers, relays, and/or substation alarms to improve operations, minimize or avert damage, and speed up the restoration of their systems.

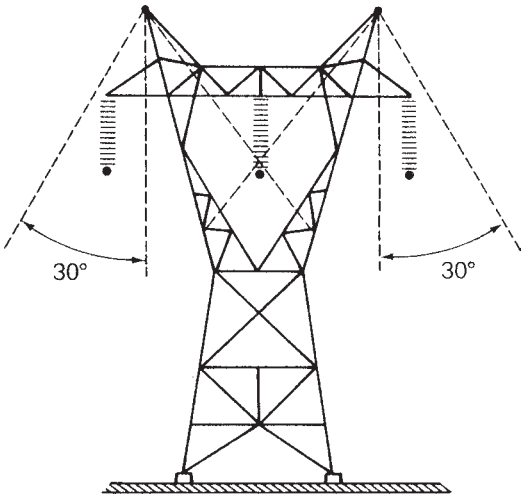
A typical transmission-line design goal is to have an average of less than 0.50 lightning outages per year per 160 kilometer of transmission. For a given overhead line with a specified voltage rating, the following factors affect this design goal:

1. Tower height
2. Number and location of shield wires
3. Number of standard insulator discs per phase wire
4. Tower impedance and tower-to-ground impedance

It is well known that lightning strikes tall objects. Thus, shorter, H-frame structures are less susceptible to lightning strokes than taller, lattice towers. Also, shorter span lengths with more towers per kilometer can reduce the number of strikes.

Shield wires installed above phase conductors can effectively shield the phase conductors from direct lightning strokes. Figure 14.27 illustrates the effect of shield wires. Experience has shown that the chance of a direct hit to phase conductors located within $\pm 30°$ arcs beneath the shield wires is reduced by a factor of 1000 [18]. Some lightning strokes are, therefore, expected to hit these overhead shield wires. When this occurs, traveling voltage and current waves propagate in both directions along the shield wire that is hit. When a wave arrives at a tower, a

FIGURE 14.27
Effect of shield wires



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reflected wave returns toward the point where the lightning hit, and two refracted waves occur. One refracted wave moves along the shield wire into the next span. Since the shield wire is electrically connected to the tower, the other refracted wave moves down the tower, its energy being harmlessly diverted to ground.

However, if the tower impedance or tower-to-ground impedance is too high, IZ voltages that are produced could exceed the breakdown strength of the insulator discs that hold the phase wires. The number of insulator discs per string (see Table 4.1) is selected to avoid insulator flashover. Also, tower impedances and tower footing resistances are designed to be as low as possible. If the inherent tower construction does not give a naturally low resistance to ground, driven ground rods can be employed. Sometimes buried conductors running under the line (called *counterpoise*) are employed.

SWITCHING SURGES

The magnitudes of overvoltages due to lightning surges are not significantly affected by the power system voltage. On the other hand, overvoltages due to switching surges are directly proportional to system voltage. Consequently, lightning surges are less important for EHV transmission above 345 kV and for UHV transmission, which has improved insulation. Switching surges become the limiting factor in insulation coordination for system voltages above 345 kV.

One of the simplest and largest overvoltages can occur when an open-circuited line is energized, as shown in Figure 14.28. Assume that the circuit breaker closes at the instant the sinusoidal source voltage has a peak value $\sqrt{2}$ V. Assuming zero source impedance, a forward traveling voltage wave of magnitude $\sqrt{2}$ V occurs. When this wave arrives at the open-circuited receiving end, where $\Gamma_R = +1$, the reflected voltage wave superimposed on the forward wave results in a maximum voltage of $2\sqrt{2}$ V = 2.83 V. Even higher voltages can occur when a line is reclosed after momentary interruption.

In order to reduce overvoltages due to line energizing or reclosing, resistors are almost always pre-inserted in circuit breakers at 345 kV and above. Resistors ranging from 200 to 800 Ω are pre-inserted when EHV circuit breakers are closed and subsequently bypassed. When a circuit breaker closes, the source voltage divides across the pre-inserted resistors and the line, thereby reducing the initial line voltage. When the resistors are shorted out, a new transient is initiated, but the maximum line voltage can be substantially reduced by careful design.

Dangerous overvoltages can also occur during a single line-to-ground fault on one phase of a transmission line. When such a fault occurs, a voltage equal

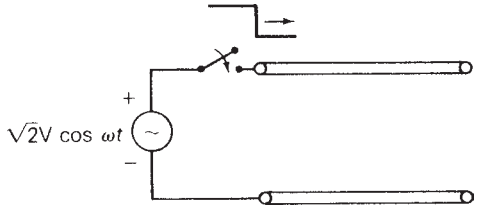

FIGURE 14.28
Energizing an open-circuited line

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and opposite to that on the faulted phase occurs at the instant of fault inception. Traveling waves are initiated on both the faulted phase and, due to capacitive coupling, the unfaulted phases. At the line ends, reflections are produced and are superimposed on the normal operating voltages of the unfaulted phases. Kimbark and Legate [19] show that a line-to-ground fault can create an overvoltage on an unfaulted phase as high as 2.1 times the peak line-to-neutral voltage of the three-phase line.

POWER FREQUENCY OVERVOLTAGES

Sustained overvoltages at the fundamental power frequency (60 Hz in the United States) or at higher harmonic frequencies (such as 120 Hz, 180 Hz, and so on) occur due to load rejection, to ferroresonance, or to permanent faults. These overvoltages are normally of long duration, seconds to minutes, and are weakly damped.

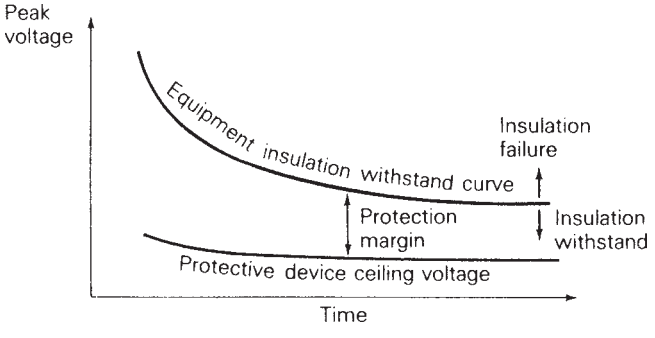
14.8 INSULATION COORDINATION

Insulation coordination is the process of correlating electric equipment insulation strength with protective device characteristics so that the equipment is protected against expected overvoltages. The selection of equipment insulation strength and the protected voltage level provided by protective devices depends on engineering judgment and cost.

As shown by the top curve in Figure 14.29, equipment insulation strength is a function of time. Equipment insulation can generally withstand high transient overvoltages only if they are of sufficiently short duration. However, determination of insulation strength is somewhat complicated. During repeated tests with identical voltage waveforms under identical conditions, equipment insulation may fail one test and withstand another.

For purposes of insulation testing, a standard impulse voltage wave, as shown in Figure 14.30, is defined. The impulse wave shape is specified by giving

FIGURE 14.29
Equipment insulation strength



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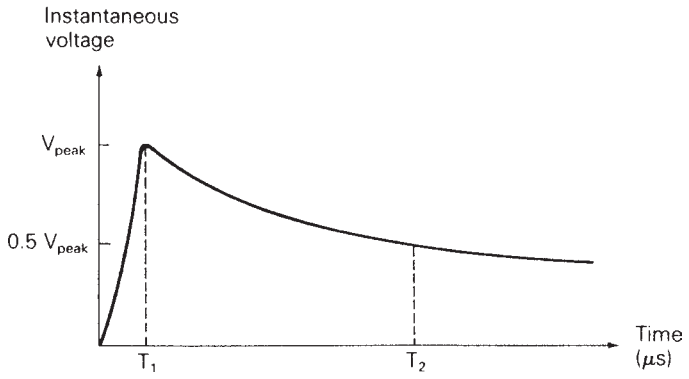

FIGURE 14.30
Standard impulse voltage waveform

the time T_1 in microseconds for the voltage to reach its peak value and the time T_2 for the voltage to decay to one-half its peak. One standard wave is a 1.2×50 wave, which rises to a peak value at $T_1 = 1.2$ μs and decays to one-half its peak at $T_2 = 50$ μs.

Basic insulation level, or BIL, is defined as the peak value of the standard impulse voltage wave in Figure 14.30. Standard BILs adopted by the IEEE are shown in Table 14.1. Equipment conforming to these BILs must be capable of withstanding repeated applications of the standard waveform of positive or negative polarity without insulation failure. Also, these standard BILs apply to equipment regardless of how it is grounded. For nominal system voltages 115 kV and above, solidly grounded equipment with the reduced BILs shown in the table have been used.

BILs are often expressed in per unit, where the base voltage is the maximum value of nominal line-to-ground system voltage. Consider, for example, a 345-kV system, for which the maximum value of nominal line-to-ground voltage is $\sqrt{2}(345/\sqrt{3}) = 281.7$ kV. The 1550-kV standard BIL shown in Table 14.1 is then $(1550/281.7) = 5.5$ per unit.

Note that overhead-transmission-line insulation, which is external insulation, is usually self-restoring. When a transmission-line insulator string flashes over, a short circuit occurs. After circuit breakers open to deenergize the line, the insulation of the string usually recovers, and the line can be rapidly reenergized. However, transformer insulation, which is internal, is not self-restoring. When transformer insulation fails, the transformer must be removed for repair or replaced.

To protect equipment, such as a transformer, against overvoltages higher than its BIL, a protective device, such as that shown in Figure 14.31, is employed. Such protective devices are generally connected in parallel with the equipment from each phase to ground. As shown in Figure 14.29, the function of the protective device is to maintain its voltage at a ceiling voltage below the BIL of the equipment it protects. The difference between the equipment breakdown voltage and the protective device ceiling voltage is the *protection margin.*

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Nominal System		
Voltage kV rms	Standard BIL kV	Reduced BIL* kV
1.2	45	
2.5	60	
5.0	75	
8.7	95	
15	110	
23	150	
34.5	200	
46	250	
69	350	
92	450	
115	550	450
138	650	550
161	750	650
196	900	750
230	1050	825—900
287	1300	1000—1100
345	1550	1175—1300
500		1300—1800
765		1675—2300

TABLE 14.1

Standard and reduced basic insulation levels [18]

*For solidly grounded systems
These BILs are based on $1.2 \times 50\ \mu$s voltage waveforms. They apply to internal (or non-self-restoring) insulation such as transformer insulation, as well as to external (or self-restoring) insulation, such as transmission-line insulation, on a statistical basis.
(Based on Westinghouse Electric Corporation, *Electrical Transmission and Distribution Reference Book*, 4th ed. (East Pittsburgh, PA: 1964).)

Protective devices should satisfy the following four criteria:

1. Provide a high or infinite impedance during normal system voltages, to minimize steady-state losses;
2. Provide a low impedance during surges, to limit voltage;
3. Dissipate or store the energy in the surge without damage to itself; and
4. Return to open-circuit conditions after the passage of a surge.

One of the simplest protective devices is the rod gap, two metal rods with a fixed air gap, which is designed to spark over at specified overvoltages. Although it satisfies the first two protective device criteria, it dissipates very little energy, and it cannot clear itself after arcing over.

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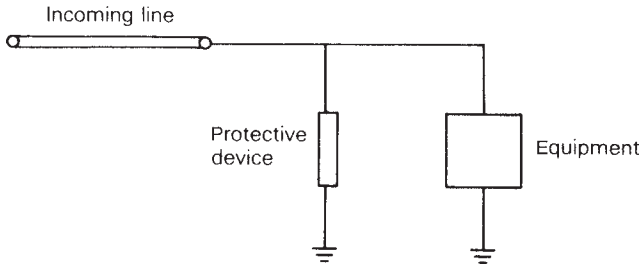

FIGURE 14.31
Single-line diagram of equipment and protective device

A surge arrester, consisting of an air gap in series with a nonlinear silicon carbide resistor, satisfies all four criteria. The gap eliminates losses at normal voltages and arcs during overvoltages. The resistor has the property that its resistance decreases sharply as the current through it increases, thereby limiting the voltage across the resistor to a specified ceiling. The resistor also dissipates the energy in the surge. Finally, following the passage of a surge, various forms of arc control quench the arc within the gap and restore the surge arrester to normal open-circuit conditions.

The "gapless" surge arrester, consisting of a nonlinear metal oxide resistor with no air gap, also satisfies all four criteria. At normal voltages, the resistance is extremely high, limiting steady-state currents to microamperes and steady-state losses to a few watts. During surges, the resistance sharply decreases, thereby limiting overvoltage while dissipating surge energy. After the surge passes, the resistance naturally returns to its original high value. One advantage of the gapless arrester is that its ceiling voltage is closer to its normal operating voltage than is the arrester with a gap, thus permitting reduced BILs and potential savings in the capital cost of equipment insulation.

There are four classes of surge arresters: station, intermediate, distribution, and secondary. Station arresters, which have the heaviest construction, are designed for the greatest range of ratings and have the best protective characteristics. Intermediate arresters, which have moderate construction, are designed for systems with nominal voltages of 138 kV and below. Distribution arresters are employed with lower-voltage transformers and lines where there is a need for economy. Secondary arresters are used for nominal system voltages below 1000 V. A summary of the protective characteristics of station- and intermediate-class metal-oxide surge arresters is given in Table 14.2 [20]. This summary is based on manufacturers' catalog information.

Note that arrester currents due to lightning surges are generally less than the lightning currents shown in Figure 14.23. In the case of direct strokes to transmission-line phase conductors, traveling waves are set up in two directions from the point of the stroke. Flashover of line insulation diverts part of the lightning current from the arrester. Only in the case of a direct stroke to a phase conductor very near an arrester, where no line flash-over occurs, does the arrester discharge the full lightning current. The probability of this occurrence can be significantly reduced by using overhead shield wires to shield transmission lines and

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Station Class									
Steady-state operation system voltage and arrester ratings effectively grounded systems (NOTE 1)				Protective levels range of industry maxima per unit (crest of 60 Hz) of MCOV			Durability characteristics IEEE Std. C62.11-2005		
Max system rms L-L voltage kV*	Max system rms L-G voltage kV*	Min rms MCOV rating kV	Duty-cycle rms voltage rating kV	0.5 μs FOW protective level (NOTE 2)	8/20 μs protective level (NOTE 3)	Switching surge protective level (NOTE 4)	High crest current withstand A	Trans. line discharge Miles	Pressure relief rms (symmetrical) current kA (NOTE 5)
4.37	2.52	2.55	3	2.32–2.48	2.10–2.20	1.70–1.85	65 000	150	40–80
8.73	5.04	5.1	6–9	2.33–2.48	1.97–2.23	1.70–1.85	65 000	150	40–80
13.1	7.56	7.65	9–12	2.33–2.48	1.97–2.23	1.70–1.85	65 000	150	40–80
13.9	8.00	8.4	10–15	2.33–2.48	1.97–2.23	1.70–1.85	65 000	150	40–80
14.5	8.37	8.4	10–15	2.33–2.48	1.97–2.23	1.70–1.85	65 000	150	40–80
26.2	15.1	15.3	18–27	2.33–2.48	1.97–2.23	1.70–1.85	65 000	150	40–80
36.2	20.9	22	27–36	2.43–2.48	1.97–2.23	1.70–1.85	65 000	150	40–80
48.3	27.8	29	36–48	2.43–2.48	1.97–2.23	1.70–1.85	65 000	150	40–80
72	41.8	42	54–72	2.19–2.40	1.97–2.18	1.64–1.84	65 000	150	40–80
121	69.8	70	90–120	2.19–2.40	1.97–2.18	1.64–1.84	65 000	150	40–80
145	83.7	84	108–144	2.19–2.39	1.97–2.17	1.64–1.84	65 000	150	40–80
169	97.5	98	120–172	2.19–2.39	1.97–2.17	1.64–1.84	65 000	175	40–80
242	139	140	172–240	2.19–2.36	1.97–2.15	1.64–1.84	65 000	175	40–80
362	209	209	258–312	2.19–2.36	1.97–2.15	1.71–1.85	65 000	200	40–80
550	317	318	396–564	2.01–2.47	2.01–2.25	1.71–1.85	65 000	200	40–80
800	461	462	576–612	2.01–2.47	2.01–2.25	1.71–1.85	65 000	200	40–80
Intermediate class									
4.37–145	2.52–83.7	2.8–84	3–144	2.38–2.85	2.28–2.55	1.71–1.85	65 000	100	16.1

TABLE 14.2

Typical characteristics of station- and intermediate-class metal-oxide surge arresters [20]

*Voltage Range A, ANSI C84.1-2006

NOTE 1—See Table 8 and 6.4 for typical arrester ratings for systems with a noneffectively grounded neutral.

NOTE 2—Equivalent FOW producing a voltage wave cresting in 0.5 μs.

NOTE 3—The protective level is the maximum discharge voltage for a 10-kA impulse current wave on arrester duty-cycle rating through 312 kV, 15 kA for duty-cycle ratings 396 kV to 564 kV, and 20 kA for duty-cycle ratings 576 kV to 612 kV, per IEEE Std. C62.11-2005.

NOTE 4—Switching surge characteristics based on maximum switching surge classifying current (based on an impulse current wave with a time to actual crest of 45 μs to 60 μs) of 500 A on arrester duty-cycle ratings 3 kV to 108 kV, 1000 A on duty-cycle ratings 120 kV to 240 kV, and 2000 A on duty-cycle ratings above 240 kV, per IEEE Std. C62.11-2005.

NOTE 5—Test values for arresters with porcelain tops have not been standardized, Pressure relief classification is in 5-kA. steps.

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EXAMPLE 14.8

Metal-oxide surge arrester selection

Consider the selection of a station-class metal-oxide surge arrester for a 345-kV system in which the maximum 60-Hz voltage under normal system conditions is 1.04 per unit. (a) Select a station-class arrester from Table 14.2 with a maximum continuous operating voltage (MCOV) rating that exceeds the 1.04 per-unit maximum 60-Hz voltage of the system under normal system conditions. (b) For the selected arrester, determine the protective margin for equipment in the system with a 1300-kV BIL, based on a 10-kA impulse current wave cresting in 0.5 μs.

SOLUTION

a. The maximum 60-Hz line-to-neutral voltage under normal system conditions is $1.04(345/\sqrt{3}) = 207$ kV. From Table 14.2, select a station-class surge arrester with a 209-kV MCOV. This is the lowest MCOV rating that exceeds the 207 kV providing the greatest protective margin as well as economy.

b. From Table 14.2 for the selected surge arrester, the maximum discharge voltage for a 10-kA impulse current wave cresting in 0.5 μs ranges from 2.19 to 2.36 in per unit of MCOV, or 457–493 kV, depending on the arrester manufacturer. Therefore, the protective margin ranges from $(1300 - 493) = 807$ kV to $(1300 - 457) = 843$ kV.

substations. Recommended practice for substations with unshielded lines is to select an arrester discharge current of at least 20 kA (even higher if the isokeraunic level is above 40 thunderstorm days per year). For substations with shielded lines, lower arrester discharge currents, from 5 to 20 kA, have been found satisfactory in most situations [20].

When selecting a metal-oxide surge arrester, it is important that the arrester MCOV exceeds the maximum 60-Hz system voltage (line-to-neutral) under normal conditions. In addition to considerations affecting the selection of arrester MCOV, metal-oxide surge arresters should also be selected to withstand temporary overvoltages in the system at the arrester location—for example, the voltage rise on unfaulted phases during line-to-ground faults. That is, the temporary overvoltage (TOV) capability of metal-oxide surge arresters should not be exceeded. Additional considerations in the selection of metal-oxide surge arresters are discussed in reference [22] (see www.eaton.com).

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PROBLEMS

SECTION 14.2

14.1 From the results of Example 14.2, plot the voltage and current profiles along the line at times $\tau/2$, τ, and 2τ. That is, plot $v(x, \tau/2)$ and $i(x, \tau/2)$ versus x for $0 \leq x \leq l$; then plot $v(x, \tau)$, $i(x, \tau)$, $v(x, 2\tau)$, and $i(x, 2\tau)$ versus x.

14.2 Rework Example 14.2 if the source voltage at the sending end is a ramp, $e_G(t) = Eu_{-2}(t) = Etu_{-1}(t)$, with $Z_G = 2Z_c$.

14.3 Referring to the single-phase two-wire lossless line shown in Figure 14.3, the receiving end is terminated by an inductor with $2L_R$ henries. The source voltage at the sending end is a step, $e_G(t) = Eu_{-1}(t)$ with $Z_G = Z_c$. Both the line and inductor are initially unenergized. Determine and plot the voltage at the center of the line $v(l/2, t)$ versus time t.

14.4 Rework Problem 14.3 if $Z_R = Z_c$ at the receiving end and the source voltage at the sending end is $e_G(t) = Eu_{-1}(t)$, with an inductive source impedance $Z_G(s) = s2L_G$. Both the line and source inductor are initially unenergized.

14.5 Rework Example 14.4 with $Z_R = 5Z_c$ and $Z_G = Z_c/3$.

14.6 The single-phase, two-wire lossless line in Figure 14.3 has a series inductance $L = (1/3) \times 10^{-6}$ H/m, a shunt capacitance $C = (1/3) \times 10^{-10}$ F/m, and a 50-km line length. The source voltage at the sending end is a step $e_G(t) = 100u_{-1}(t)$ kV with $Z_G(s) = 100\ \Omega$. The receiving-end load consists of a 100-Ω resistor in parallel with a 2-mH inductor. The line and load are initially unenergized. Determine (a) the characteristic impedance in ohms, the wave velocity in m/s, and the transit time in ms for this line; (b) the sending- and receiving-end voltage reflection coefficients in per unit; (c) the Laplace transform of the receiving-end current, $I_R(s)$; and (d) the receiving-end current $i_R(t)$ as a function of time.

14.7 The single-phase, two-wire lossless line in Figure 14.3 has a series inductance $L = 2 \times 10^{-6}$ H/m, a shunt capacitance $C = 1.25 \times 10^{-11}$ F/m, and a 100-km line length. The source voltage at the sending end is a step $e_G(t) = 100u_{-1}(t)$ kV with a source impedance equal to the characteristic impedance of the line. The receiving-end load consists of a 100-mH inductor in series with a 1-μF capacitor. The line and load are initially unenergized. Determine (a) the characteristic impedance in Ω, the wave velocity in m/s, and the transit time in ms for this line; (b) the sending- and receiving-end voltage reflection coefficients in per unit; (c) the receiving-end voltage $v_R(t)$ as a function of time; and (d) the steady-state receiving-end voltage.

14.8 The single-phase, two-wire lossless line in Figure 14.3 has a series inductance $L = 0.999 \times 10^{-6}$ H/m, a shunt capacitance $C = 1.112 \times 10^{-11}$ F/m, and a 60-km line length. The source voltage at the sending

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end is a ramp $e_G(t) = Etu_{-1}(t) = Eu_{-2}(t)$ kV with a source impedance equal to the characteristic impedance of the line. The receiving-end load consists of a 150-Ω resistor in parallel with a 1-μF capacitor. The line and load are initially unenergized. Determine (a) the characteristic impedance in Ω, the wave velocity in m/s, and the transit time in ms for this line; (b) the sending- and receiving-end voltage reflection coefficients in per unit; (c) the Laplace transform of the sending-end voltage, $V_S(s)$; and (d) the sending-end voltage $v_S(t)$ as a function of time.

SECTION 14.3

14.9 Draw the Bewley lattice diagram for Problem 14.5.

14.10 Rework Problem 14.9 if the source voltage is a pulse of magnitude E and duration $\tau/10$; that is, $e_G(t) = E[u_{-1}(t) - u_{-1}(t - \tau/10)]$. $Z_R = 5Z_c$ and $Z_G = Z_c/3$ are the same as in Problem 14.9. Also plot $v(l/3, t)$ versus time t for $0 \le t < 6\tau$.

14.11 Rework Example 14.6 if the source impedance at the sending end of line A is $Z_G = Z_A/4 = 100\ \Omega$, and the receiving end of line B is short-circuited, $Z_R = 0$.

14.12 Rework Example 14.6 if the overhead line and cable are interchanged. That is, $Z_A = 100\ \Omega$, $v_A = 2 \times 10^8$ m/s, $l_A = 20$ km, $Z_B = 400\ \Omega$, $v_B = 3 \times 10^8$ m/s, and $l_B = 30$ km. The step voltage source $e_G(t) = Eu_{-1}(t)$ is applied to the sending end of line A with $Z_G = Z_A = 100\ \Omega$, and $Z_R = 2Z_B = 800\ \Omega$, at the receiving end. Draw the lattice diagram for $0 \le t \le 0.6$ ms and plot the junction voltage versus time t.

14.13 As shown in Figure 14.32, a single-phase two-wire lossless line with $Z_c = 400\ \Omega$, $v = 3 \times 10^8$ m/s, and $l = 100$ km has a 400-Ω resistor, denoted R_J, installed across the center of the line, thereby dividing the line into two sections, A and B. The source voltage at the sending end is a pulse of magnitude 100 V and duration 0.1 ms. The source impedance is $Z_G = Z_c = 400\ \Omega$, and the receiving end of the line is short-circuited. (a) Show that for an incident voltage wave arriving at the center of the line from either line section, the voltage reflection and refraction coefficients are given by

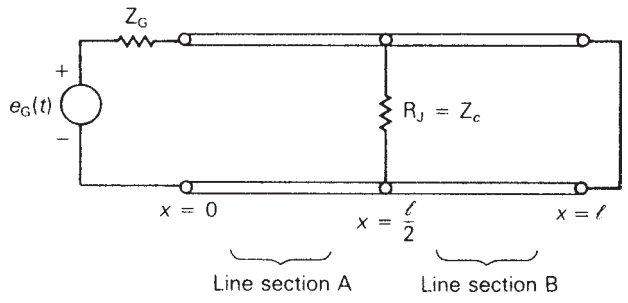

FIGURE 14.32
Circuit for Problem 14.13

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$$\Gamma_{BB} = \Gamma_{AA} = \frac{\left(\frac{Z_{eq}}{Z_c}\right) - 1}{\left(\frac{Z_{eq}}{Z_c}\right) + 1} \qquad \Gamma_{AB} = \Gamma_{BA} = \frac{2\left(\frac{Z_{eq}}{Z_c}\right)}{\left(\frac{Z_{eq}}{Z_c}\right) + 1}$$

where

$$Z_{eq} = \frac{R_J Z_c}{R_J + Z_c}$$

(b) Draw the Bewley lattice diagram for $0 \le t \le 6\tau$. (c) Plot $v(1/2, t)$ versus time t for $0 \le t \le 6\,\tau$ and plot $v(x, 6\tau)$ versus x for $0 \le x \le l$.

14.14 The junction of four single-phase two-wire lossless lines, denoted A, B, C, and D, is shown in Figure 14.13. Consider a voltage wave v_A^+ arriving at the junction from line A. Using (14.3.8) and (14.3.9), determine the voltage reflection coefficient Γ_{AA} and the voltage refraction coefficients Γ_{BA}, Γ_{CA}, and Γ_{DA}.

14.15 Referring to Figure 14.3, the source voltage at the sending end is a step $e_G(t) = Eu_{-1}(t)$ with an inductive source impedance $Z_G(s) = sL_G$, where $L_G/Z_c = \tau/3$. At the receiving end, $Z_R = Z_c/4$. The line and source inductance are initially unenergized. (a) Draw the Bewley lattice diagram for $0 \le t \le 5\tau$. (b) Plot $v(l, t)$ versus time t for $0 \le t \le 5\tau$.

14.16 As shown in Figure 14.33, two identical, single-phase, two-wire, lossless lines are connected in parallel at both the sending and receiving ends. Each line has a 400-Ω characteristic impedance, 3×10^8 m/s velocity of propagation, and 100-km line length. The source voltage at the sending end is a 100-kV step with source impedance $Z_G = 100\ \Omega$. The receiving end is shorted ($Z_R = 0$). Both lines are initially unenergized. (a) Determine the first forward traveling voltage waves that start at time $t = 0$ and travel on each line toward the receiving end. (b) Determine the sending- and receiving-end voltage reflection coefficients in per unit. (c) Draw the Bewley lattice diagram for $0 \le t \le 2.0$ ms. (d) Plot the voltage at the center of one line versus time t for $0 \le t \le 2.0$ ms.

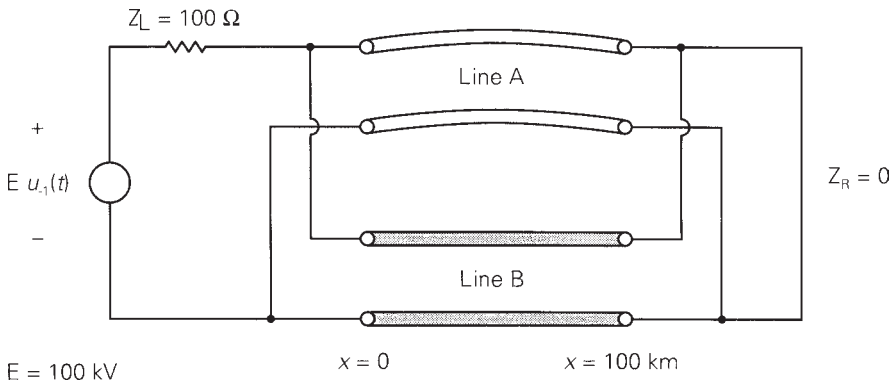

FIGURE 14.33
Circuit for Problem 14.16

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14.17 As shown in Figure 14.34, an ideal current source consisting of a 10-kA pulse with 50-μs duration is applied to the junction of a single-phase, lossless cable and a single-phase, lossless overhead line. The cable has a 200-Ω characteristic impedance, 2×10^8 m/s velocity of propagation, and 20-km length. The overhead line has a 300-Ω characteristic impedance, 3×10^8 m/s velocity of propagation, and 60-km length. The sending end of the cable is terminated by a 400-Ω resistor, and the receiving end of the overhead line is terminated by a 100-Ω resistor. Both the line and cable are initially unenergized. (a) Determine the voltage reflection coefficients Γ_S, Γ_R, Γ_{AA}, Γ_{AB}, Γ_{BA}, and Γ_{BB}. (b) Draw the Bewley lattice diagram for $0 \leq t \leq 0.8$ ms. (c) Determine and plot the voltage $v(0, t)$ at $x = 0$ versus time t for $0 \leq t \leq 0.8$ ms.

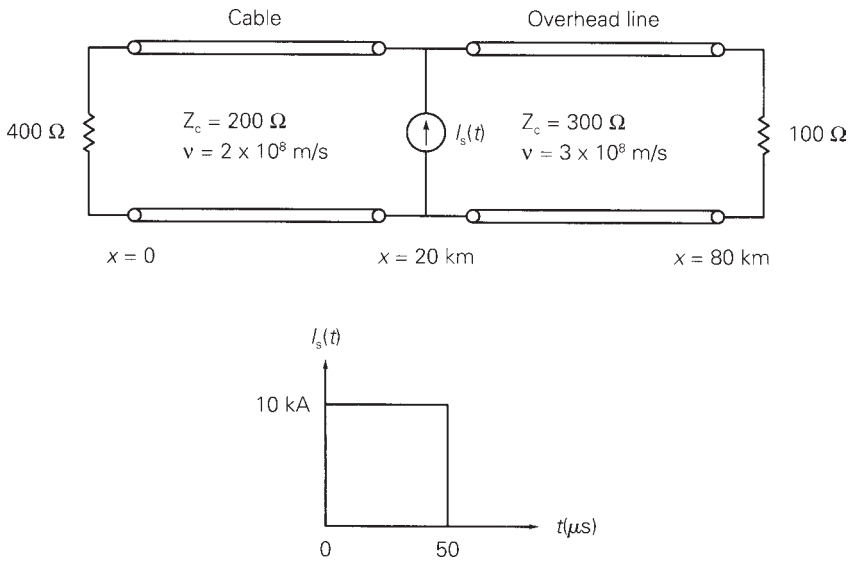

FIGURE 14.34
Circuit for Problem 14.17

SECTION 14.4

14.18 For the circuit given in Problem 14.3, replace the circuit elements by their discrete-time equivalent circuits and write nodal equations in a form suitable for computer solution of the sending-end and receiving-end voltages. Give equations for all dependent sources. Assume E = 1000 V, L_R = 20 mH, Z_c = 100 Ω, $v = 2 \times 10^8$ m/s, l = 40 km, and Δt = 0.02 ms.

14.19 Repeat Problem 14.18 for the circuit given in Problem 14.13. Assume Δt = 0.03333 ms.

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14.20 For the circuit given in Problem 14.7, replace the circuit elements by their discrete-time equivalent circuits. Use $\Delta t = 100\ \mu s = 1 \times 10^{-4}$ s. Determine and show all resistance values on the discrete-time circuit. Write nodal equations for the discrete-time circuit, giving equations for all dependent sources. Then solve the nodal equations and determine the sending- and receiving-end voltages at the following times: $t = 100$, 200, 300, 400, 500, and 600 μs.

14.21 For the circuit given in Problem 14.8, replace the circuit elements by their discrete-time equivalent circuits. Use $\Delta t = 50\ \mu s = 5 \times 10^{-5}$ s and E = 100 kV. Determine and show all resistance values on the discrete-time circuit. Write nodal equations for the discrete-time circuit, giving equations for all dependent sources. Then solve the nodal equations and determine the sending- and receiving-end voltages at the following times: $t = 50$, 100, 150, 200, 250, and 300 μs.

SECTION 14.5

14.22 Rework Problem 14.18 for a lossy line with a constant series resistance R = 0.3 Ω/km. Lump half of the total resistance at each end of the line.

SECTION 14.8

14.23 Repeat Example 14.8 for a 500-kV system with a 1.08 per-unit maximum 60-Hz voltage under normal operating conditions and with a 2000-kV BIL.

14.24 Select a station-class metal-oxide surge arrester from Table 14.2 for the high-voltage side of a three-phase 400-MVA, 345-kV Y/13.8-kV Δ transformer. The maximum 60-Hz operating voltage of the transformer under normal operating conditions is 1.03 per unit. The high-voltage windings of the transformer have a BIL of 1300 kV and a solidly grounded neutral. A minimum protective margin of 1.4 per unit based on a 10-kA impulse current wave cresting in 0.5 μs is required. (*Note:* Additional considerations for the selection of metal-oxide surge arresters are given in reference [22] (www.eaton.com).)

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15 Power Distribution

Distribution substation fed by two 22.9-kV (one overhead and one underground) lines through two 15-MVA, 22.9Δ/4.16Y-kV distribution substation transformers. The transformers are located on either side of the switchgear building shown in the center. This substation feeds six 4.16-kV radial primary feeders through 5-kV, 1200-A vacuum circuit breakers. (Courtesy of Danvers Electric.)

LEARNING OBJECTIVES

At the conclusion of this chapter, you should be able to:

1. Understand the basic components of electric power systems.
2. Be familiar with the basic types of primary distribution systems, including primary radial, primary loop, and primary network systems.
3. Be familiar with the basic types of secondary distribution systems, including individual distribution transformer per customer, common secondary main, secondary network, and spot network.
4. Be familiar with the basic types of transformers in distribution systems, including distribution substation transformers and distribution transformers.
5. Understand the use of capacitors in distribution systems.
6. Be familiar with the basics of distribution reliability, distribution automation, and smart grids.

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Major components of an electric power system are generation, transmission, and distribution. Distribution, including primary and secondary distribution, is that portion of a power system that runs from distribution substations to a customer's service entrance equipment. In 2019, in the United States, distribution systems served approximately 155 million customers that consumed 3.95 trillion kWh [31].

This chapter provides an overview of distribution. Sections 15.1–15.3 introduce the basic configurations and characteristics of distribution, including primary and secondary distribution. Sections 15.4 and 15.5 discuss the application of transformers and capacitors in distribution systems. Then in Sections 15.6–15.9, distribution software, distribution reliability, distribution automation, and smart grids are introduced.

CASE STUDY

The case study discusses opportunities, technical barriers, and a road map of high-frequency power electronics for smart grid applications. Increasing numbers of power-electronics applications are currently being located at the distribution level near the edge of the electric grid. These applications include renewable energy, transportation electrification, data communication, and industrial electrification. Increasing the switching frequency of power electronic devices reduces passive component size, weight and cost. Optimally designed high-performance power electronics leveraging state-of-the-art techniques promise significantly improved efficiency, power density, and new system functionality [20].

High-Frequency Power Electronics at the Grid Edge: A Bottom-Up Approach Toward the Smart Grid*

Minjie Chen and H. Vincent Poor

Currently, solar energy systems are supplying 2% of U.S. electricity generation and their use is rapidly growing. Data centers make up 2% of the U.S. total energy consumption. By 2030, 40% of the overall worldwide energy consumption is expected to be in the form of electricity. The future development of transportation electrification and smart grids requires grid-scale energy storage. These emerging and high-impact applications locate in the distribution grid. They represent exciting opportunities in developing fundamentally new principles for

*"High-Frequency Power Electronics at the Grid Edge: A Bottom-Up Approach Toward the Smart Grid," *IEEE Electrification Magazine*, September 2020, pp. 6–17. ©2020 IEEE. Reprinted with permission.

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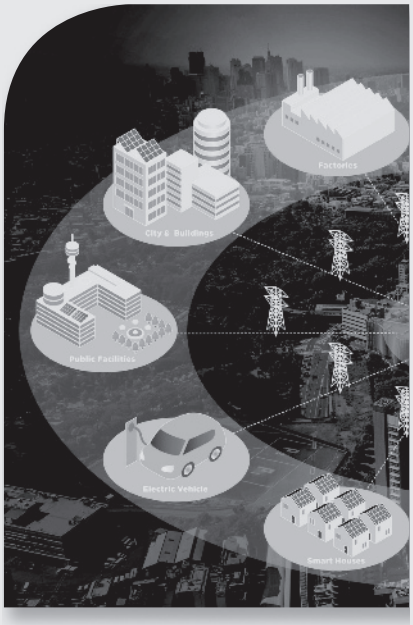

high-performance power electronics that are smaller, smarter, more efficient, and capable of performing new functions (Figure 1).

High Frequency Power Electronics at the Grid Edge: Motivations

Power electronics systems are pervasive in many aspects of our daily lives, including renewable energy, transportation electrification, data communication, and industrial electrification. These applications all locate at the distribution level, near the edge of the electric grid. "Grid edge" is an emerging and important opportunity for power electronics. Power converters usually do not perform active gridforming functions at the distribution level. This, however, is rapidly changing following the emergence of renewable energy, transportation electrification, and smart grids, where power electronics systems serve as the main energy gateway. One common attribute of many of these grid-edge energy systems is that they comprise numerous classic power converters designed simply for voltage conversion. They also do not typically contribute to grid stability or resiliency. Moreover, conventional designs are simply considered as "grid followers," which passively extract energy from the grid. Grid-interface and internal energy management are separately considered. More holistically integrated solutions are needed to connect distributed sources to loads and "form" the grid (Figure 2).

Increasing the switching frequency and thus reducing the passive component size is a fundamental way to improve the performance of power electronics. Power semiconductor devices, control techniques, magnetics, packaging, and thermal management all pose

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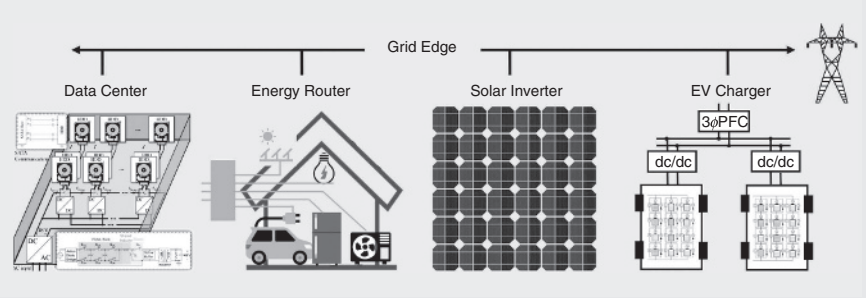

Figure 1 The new applications open new opportunities. Data centers, smart homes, solar farms, and electric vehicle (EV) charging stations all need advanced power electronics at the grid edge. PFC: power factor correction.

difficulties in high-frequency designs. Advancements are enabled by rapid developments in wideband-gap (WBG) materials and passive components as well as supported by new architectures and fabrication techniques. Optimally designed high-performance power electronics leveraging these state-of-the-art techniques promises significantly improved efficiency and power density. Both existing and emerging applications at the grid edge—ranging from power supplies for low-power Internet-of-Things (IoT) devices to kilowatt or higher power-level systems in electric vehicles (EVs), grid-scale energy storage, data centers, and renewable energy integration—require specialized power electronics to manage the sophisticated power flow.

High-frequency power electronics are distributed "sensors" and "actuators" functioning at the grid edge. They replace the traditional centralized power plants and support the grid. By developing high-performance, high-frequency power electronics, the distribution

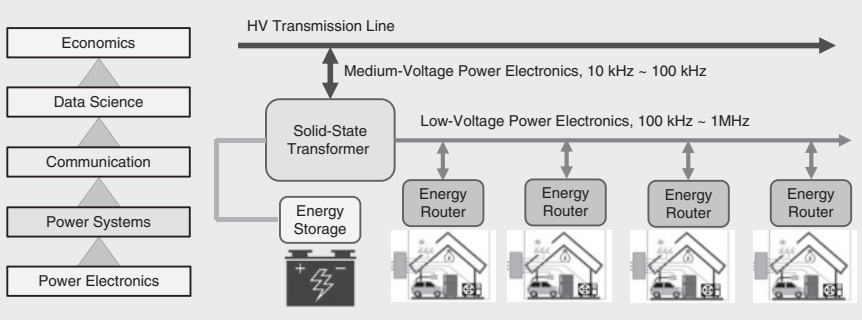

Figure 2 The architecture of the future grid with pervasive power electronics at the edge. Innovations across disciplines, from power electronics and power systems to data science and economics, are needed to enable the future smart grid (see Divan et al.). HV: high voltage.

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grid's capability of hosting renewable energy resources and intermittent loads can be significantly enhanced, yielding a robust and resilient future grid with transformative societal impact and energy security benefits. High-frequency power electronics will also improve the energy efficiency and power density of future energy systems. Research toward high-frequency power electronics will enable the development of a new generation of energy systems that are intelligent and responsive while performing sophisticated functions, such as demand response, or peer-to-peer packetized energy trading. These new operation modes and cost allocation mechanisms will offset the steep high initial cost of high-frequency, high-performance power electronics before the price is further driven down by large-scale commercial adoption.

High-Frequency Power Electronics at the Grid Edge: Technical Barriers

The switching frequency of power electronics has continuously increased over the past few decades. The performance of a power electronics system is usually measured by its efficiency, power density, and functionality. Designing smaller, smarter, and more efficient power electronics is the goal. Higher switching frequency reduces the energy storage requirements and passive component sizes, which will benefit almost all power electronics–enabled applications. The bulky nature of traditional power electronics means that miniaturization can often greatly reduce overall system size, weight, and cost as well as providing new system functionality. Miniaturization requires lower energy storage, smaller component size, tighter system integration, higher efficiency, and better thermal management. This is especially true in relatively high-voltage, low-power applications (e.g., voltages of up to a few hundred volts and power levels of up to tens of watts), such as offline power supplies, LED drivers, converters, and inverters for photovoltaic (PV) panels and battery interface converters.

Energy efficiency, on the other hand, determines the lifetime average cost of electricity. In applications like EVs, improved energy efficiency in motor drives translates to extended drive ranges for EVs. For solar inverters, improved energy efficiency leads to reduced average cost of electrical energy. Energy efficiency also sets a fundamental limit for the power density to be achieved due to thermal constraints.

In a future smart grid, grid-forming power electronics are no longer simply designed for voltage conversion, but are used for routing the power among numerous distributed sources and loads. Multiway energy routers are needed to realize packetized energy management and peer-to-peer energy trading. Bottom-up innovations from materials, devices, packaging, modeling, and control techniques drive improvements in switching frequency. Higher switching frequencies enable faster sensing and actuation at

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the grid edge. Wide-bandgap semiconductor devices facilitate high-frequency operation with smaller footprints, faster response times, and lower device losses. It is reasonable to expect that if appropriate system architectures, passive components, and circuit designs are optimally combined, high-frequency power electronics will achieve significant performance improvements over contemporary designs while offering unprecedented functions at the grid edge. The advanced functions and improved efficiency will justify the additional cost.

In addition to the needs of the electric grid, other technologies and markets for high-frequency power electronics are beginning to emerge. The vision toward a future smart grid offers top-down motivations for high-frequency power electronics. Power electronics are playing increasingly important roles in future energy systems, from supporting the future grid with more than 50% renewable energy, to powering data centers with millions of modular servers and microprocessors and managing thousands of battery cells in an EV (Figure 3). Novel architectures open new challenges and enable application-driven innovation that cannot simply be achieved by increasing the switching frequency.

From the grid perspective, power converters become more "ideal" at higher frequencies. The gap between the switching frequency of the power converter and the 60-Hz grid frequency makes the inverters and power factor correction (PFC) circuits look more like ideal voltage or current sources. As noted previously, high-frequency power converters are distributed sensors and actuators located at the grid edge; they probe the dynamic operating conditions of the grid and provide the needed response to contribute to grid stability and power quality. These converters will gradually replace bulky central power plants and ultimately support the future grid as distributed generators.

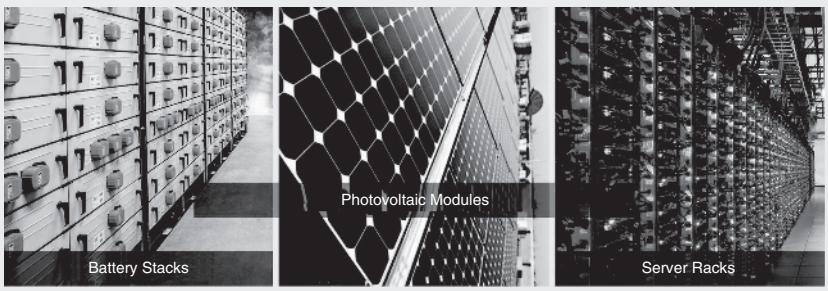

Figure 3 The battery stacks, PV modules, and server racks all comprise a large number of modular units functioning together at the grid edge.

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Devices and Components

High-frequency wide-bandgap semiconductor devices are ready to be adopted by grid-interface power electronics. The voltage ratings of gallium nitride (GaN) devices have reached 650 V or even higher, enabling them to be widely used in ac-dc adapters for consumer electronics, EV chargers, and server power supplies, where miniaturization is the key. The switching frequency of commercial GaN-based ac-dc adapter products have exceeded 1 MHz and are approaching 5 MHz. With their cost rapidly dropping, silicon carbide (SiC) devices are becoming commercially competitive in the medium voltage market. The switching frequencies of commercial SiC-based inverter products are approaching megahertz. The switching frequency of low-power grid-interface power electronics (e.g., adapters, onboard chargers, rooftop inverters) is likely to approach 10 MHz in 10 years, and the switching frequency of medium-voltage power grid-interface systems (e.g., PV inverters, fast chargers, and data center uninterrupted power supply) is likely to exceed 1 MHz in 10 years (Figure 4).

Magnetics components remain bulky and lossy. They present an even greater bottleneck for high-frequency power electronics than do semiconductor devices, especially at high power. It is not uncommon for sophisticated, modern power converter designs to have approximately half of their volume and power losses arising from inductors and transformers. Although GaN and SiC devices offer superior performance at higher frequencies, the material performance and design techniques for magnetics are still limiting the path toward high-frequency grid-interface power electronics.

Despite the challenges, recent research has made significant advances in miniaturized power electronics operating in the high-frequency and very-high-frequency ranges (3–300 MHz), well above typical modern designs operating from hundreds of kilohertz to a few megahertz (Figure 5). Research shows that low-permeability

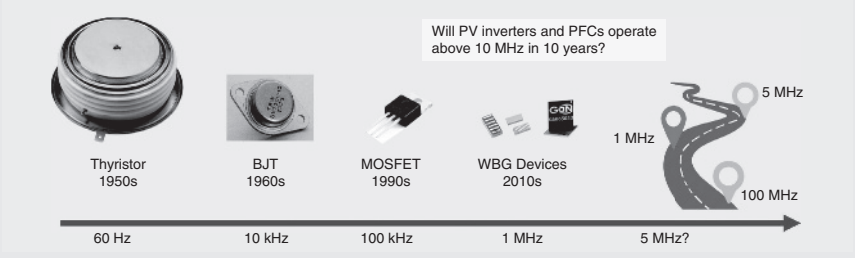

Figure 4 The road map toward high-frequency power electronics. BJT: bipolar junction transistor; WBG: wide-bandgap.

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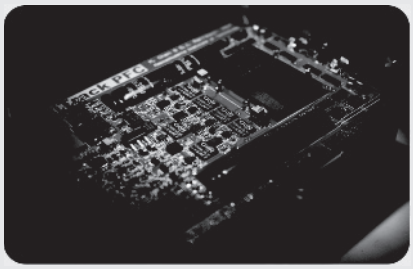

Figure 5 A 1- to 5-MHz PFC circuit
(Source: Chen et al., 2019)

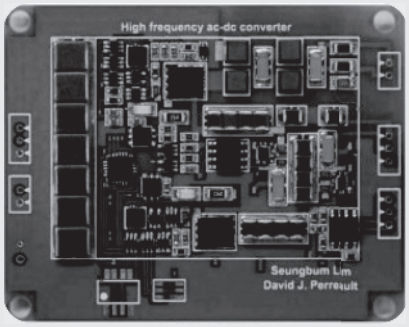

Figure 6 A 5- to 10-MHz LED driver
(Source: Lim et al.)

radio-frequency magnetic materials can offer higher performance factors than traditional ferrite materials in the megahertz range (Figure 6). Although such advances have been substantial, the optimal design methods for using low-permeability magnetic materials are still unclear. Multidisciplinary efforts are still needed to model magnetic core losses and winding losses with holistic system integration.

Circuits and Topology

Single-phase PFC converters in the hundreds of watts range typically need to meet harmonic current specifications, operate over a universal input voltage, provide holdup energy for line transients, and create large voltage step-down and isolation to the output. Meeting the harmonic specification requires precise control of the converter input current to provide a waveform with low harmonic content. Converters for many applications are often expected to deliver a full load during a transient event in which the ac line is disconnected—a typical holdup time is one full line cycle or 20 ms for a 50-Hz ac input. The converter needs to have an energy storage element to provide said energy and this usually requires significant volume. Simultaneously meeting all these requirements while achieving high efficiency and power density is a challenge in miniaturized PFC converters. To function together with intermittent renewable energy sources, future PFC converters will likely include energy storage elements that can buffer energy for a longer duration.

The size of single-phase grid-interface converters cannot simply be reduced by increasing the switching frequency, as many of the challenges are determined by the line frequency (i.e., 50 Hz or 60 Hz) instead of the switching frequency. Innovations in advanced architectures are needed to 1) enable the converter to operate efficiently across a wide range and 2) enable the converter to handle the line frequency pulsating power with enough energy storage. Active energy buffer technologies are critical in future grid-edge power electronics. One emerging trend to enable the miniaturization of single-phase PFC converters is to leverage capacitor-based topologies. Recent progress in flying capacitor multilevel (FCML) converters, switched capacitor circuits,

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high-voltage monothetic integrated circuits, and multicell distributed architectures create opportunities for improving the system performance through circuit and topology innovations (Figure 7).

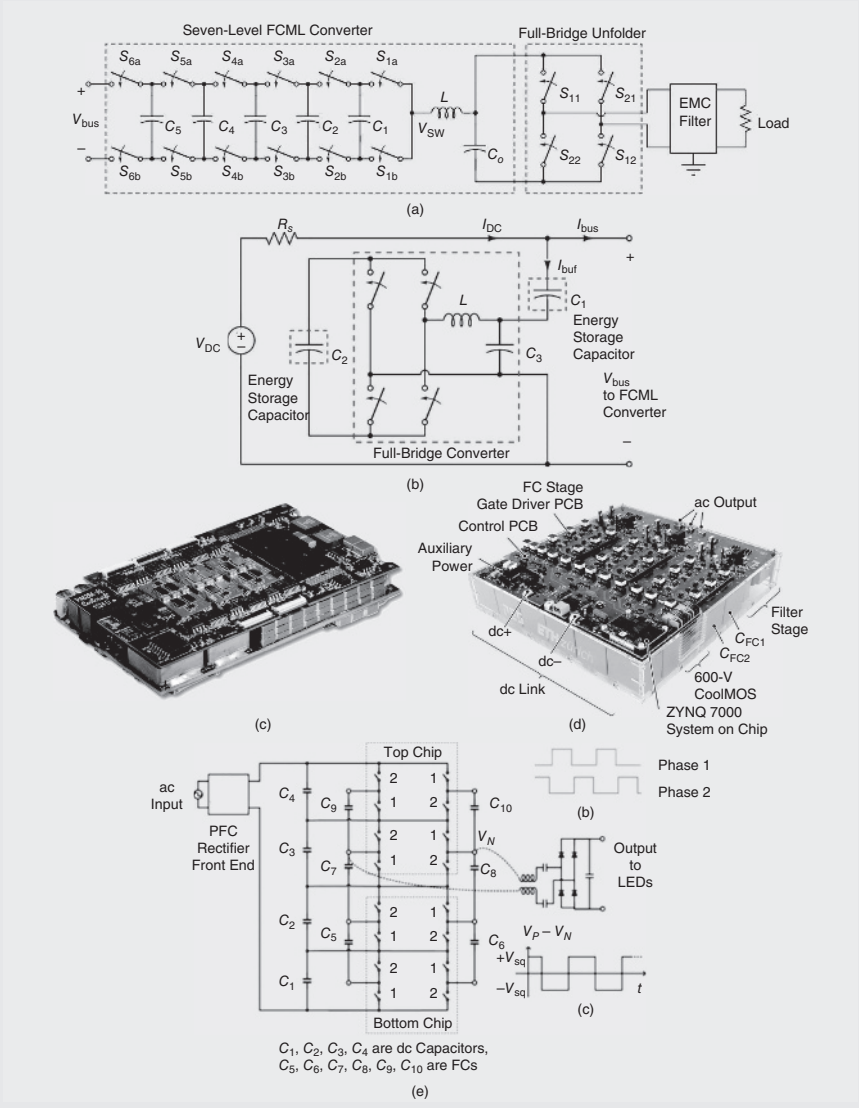

Figure 7 Hybrid switched-capacitor converters leveraging the higher power density of capacitors over magnetics. EMC: electromagnetic compatibility; FC: flying capacitor; FCML: FC multilevel; PCB: printed circuit board.

(Sources: Fig 7a and b–© 2017 IEEE. Reprinted, with permission from Y. Lei et al., "A 2-kW single-phase seven-level flying capacitor multilevel inverter with an active energy buffer," *IEEE Trans. Power Electron*, vol. 32, no. 11, pp. 8570–8581. Fig 7e–© 2016 IEEE. Reprinted, with permission from B. Van Tassell et al., "Metacapacitors: Printed thin film, flexible capacitors for power conversion applications," *IEEE Trans. Power Electron*, vol. 31, no. 4, pp. 2695–2708.)

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Architecture and Control

Power electronics systems located at the grid edge are performing increasingly sophisticated energy management functions. The maximum power point tracking systems in solar farms manage power for thousands of solar panels. The energy routers in smart homes interface with energy sources (such as rooftop solar), batteries, EVs, and the electric grid. The power delivery systems in data centers manage power for millions of servers and data communication devices. Power converters in EVs interface with motors, onboard chargers, and dc fast chargers. These are emerging and important applications for society. There are opportunities to integrate the many functions in a grid-edge power electronics system and create mutual advantages. Modular, multicell, distributed, and reconfigurable power architectures will play important roles at the grid edge. Power electronics systems need to be "smarter" to form the grid and manage the sophisticated power flow. One way to improve the intelligence of power converters is to push toward granular power conversion with highly modular and distributed sensing and control.

Granular power conversion architectures comprise many modular building blocks that interface with multiple sources and loads and manage the multiway power flow. They fit particularly well to build energy routers at the grid edge. Switched-inductor, switched-capacitor, and bridge structures as well as magnetic couplers (or in general, matching networks) can be considered as the four basic building blocks. Figure 8 displays four canonical energy conversion cells as building blocks of granular power electronics, including a bridge structure, a switched-inductor cell, a switched-capacitor cell, and a magnetic coupler. A switched-inductor circuit creates multiple current paths, a switched-capacitor circuit creates multiple voltage domains, a bridge structure can efficiently perform dc-ac conversion, and a magnetic coupler utilizes magnetic fields to transfer energy from one voltage/current domain to another and provides voltage and current scaling. These building blocks can be merged together to synthesize

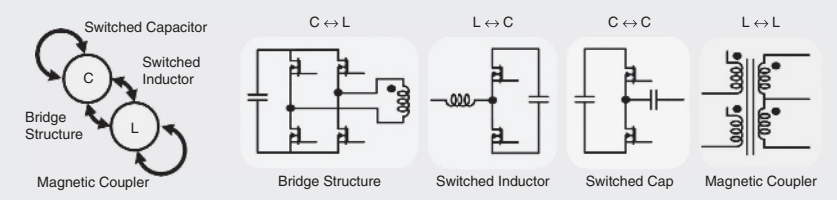

Figure 8 The canonical cells for granular power conversion, including a bridge-structure cell, a switched-inductor cell, a switched-capacitor cell, and a magnetic-coupler cell

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power electronics topologies that can maintain high performance across a wide operation range. Unlike designing and optimizing one specific power converter topology, granular power electronics need coordinated control of a large group of modular building blocks. Systematic approaches to designing and optimizing these building blocks are still unknown.

Many emerging energy systems at the grid edge comprise a large number of modular units: solar farms have thousands of solar panels, gridscale energy storage comprises tens of thousands of batteries or fuel cells, and data centers comprise millions of servers. Novel power delivery architectures that can reliably manage the sophisticated power flow in these highly intricate systems are needed. Differential power processing, and partial power processing techniques fit particularly well to these applications. Figure 9 presents an example multifunctional grid-interface architecture with many granular building blocks that can be linearly extended to higher power levels. Figure 10 depicts the principle, schematic, and prototype of a multicell reconfigurable multiple-input multiple-output (MIMO) energy router architecture, which can maintain high performance across a wide operation range.

High-Frequency Power Electronics at the Grid Edge: Advanced Functions

Grid-Forming Power Electronics

With the increasing deployment of distributed energy resources (DERs), conventional "firm" grid interconnection requirements and static hosting capacity planning do not facilitate the integration of a high percentage of DERs. Efficient, compact, responsive, and intelligent grid-interface converters are required in future energy systems. Power electronics for data centers, solar farms, EV charging stations, and grid-scale energy storage can function as distributed inverters and loads at the grid edge, which may help to stabilize the grid voltage and frequency by contributing to volt-var control. Deploying smart power electronics at the grid edge can substantially improve the DER hosting capacity. Each smart inverter or smart rectifier is a multiport energy system with energy storage capacity. They can provide the

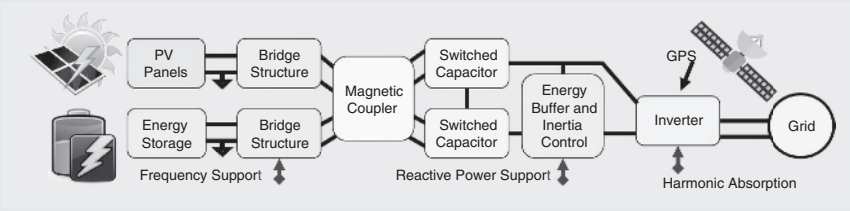

Figure 9 A multifunctional grid-interface architecture with granular building blocks

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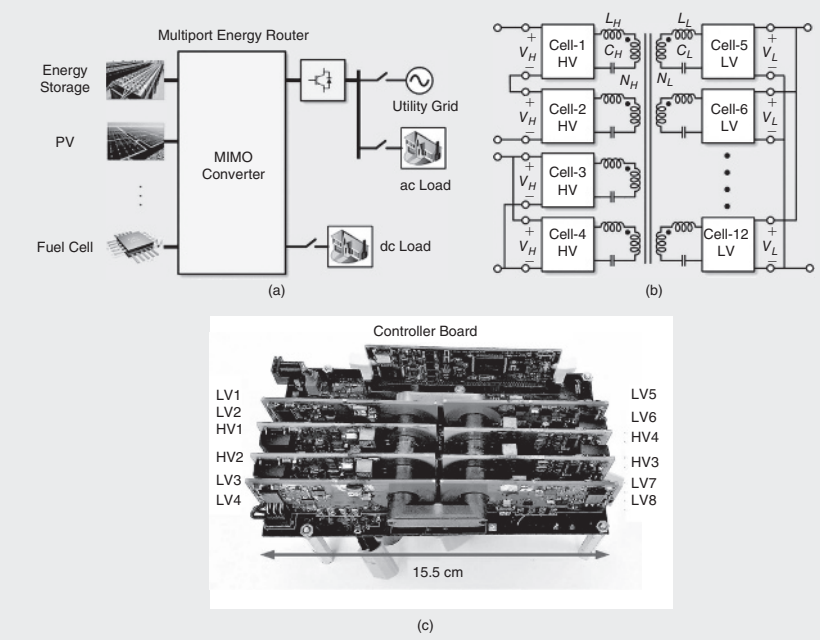

Figure 10 The (a) principle, (b) schematic, and (c) prototype of a multicell reconfigurable MIMO energy router. HV: high voltage; LV, low voltage.

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needed "inertia" to stabilize the grid dynamics, and they can autonomously cluster into nanogrids when facing cyberattacks or natural hazards as well as provide black-start capability when the system restores from blackouts. Smart inverters and smart rectifiers can actively sense the grid impedance and damp the inverter-grid oscillation. Innovations are sorely needed to enhance the capabilities of these grid-forming inverters and rectifiers.

Recently, there has been important progress in impedance-based grid stability analysis, high-frequency grid-interface power electronics, miniaturized phasor measurement unit, medium-voltage modular multilevel converters, and ultrahigh-density ac-dc inverters. There are also very active discussions on the concept of "grid-forming inverters," including grid-edge energy systems, virtual synchronous machines, virtual oscillators, virtual springs, and smart loads. Grid-interface power converters need to be able to function as current sources in a "stiff" grid and function as voltage sources in a "weak" grid. Conventionally, innovations in the "smart-inverter" domain often occur independently from the "topology" and "control" level. Innovations to address the challenges of coordinating a cluster of inverters and rectifiers at the grid

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edge securely and autonomously are needed.

The rapid development of microprocessors and digital control enables fast computation and quick response while performing complex tasks. The merging of the power grid and other existing and emerging large-scale information systems, such as 5G networks, optical communications, and global positioning systems, open exciting opportunities.

Distributed Observability and Controllability

Grid impedance affects the stability and dynamic performance of grid-interfaced power electronic systems, such as inverters and rectifiers on the load side. Adaptive control of such inverters, to guarantee stability under rapidly changing grid conditions (including both the hardware status, and the power flow), requires rapid online estimation of the grid impedance performed in real time. Such online measurement can be performed by injecting a current perturbation from the inverter into the grid and by observing the grid voltage responses. To minimize the impact on grid operation, the injection must be kept as small as possible while producing enough voltage perturbation that can be reliably measured and processed to extract its relevant information. High-frequency power electronics offers the unprecedented capability of performing grid impedance spectroscopy with higher accuracy and shorter time intervals (Figure 11).

The stable and efficient operation of the electric grid requires intricate control of the real and reactive power flow. Ideally, each inverter and rectifier should be controlled as pure resistors to maximize the efficiency and avoid other issues, such as converter-to-converter oscillations. However, due to the existence of line impedances and nonresistive loads as well as the reactance coming from the line-frequency filters in each power electronics device, the system may contain significant reactive power components at multiple harmonic frequencies and require

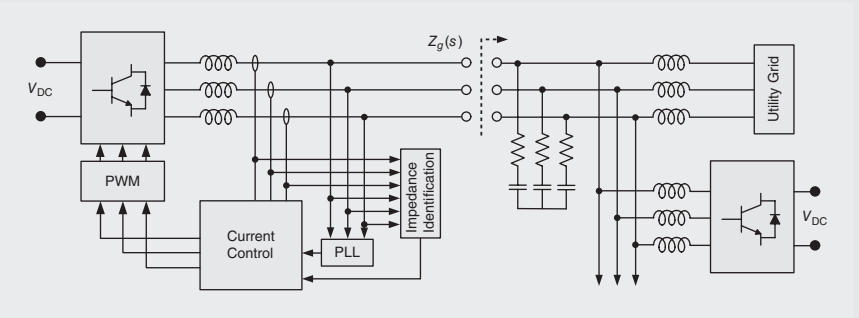

Figure 11 A grid impedance measurement setup. High-frequency power electronics enable fast and broadband impedance measurement. PLL: phase-locked loop; PWM: pulsewidth modulation (see Roinila et al.).

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active impedance compensation. High-frequency power electronics, if controlled appropriately, can inject or absorb the reactive power in these harmonic components and improve the stability of the future grid.

With granular observability and controllability at the edge, the recent advances in artificial intelligence and machine learning can enhance the supervisory control and data acquisition system at the distribution level. Machine learning algorithms can rapidly identify the grid status with reduced numbers of observations, and reinforcement learning can be used to explore optimal inverter control strategies.

Take the emerging "electric spring" concept as an example (Figure 12). Analogous to a mechanical spring, an electric spring is a power electronics device that can 1) provide electric voltage support, 2) store electric energy, and 3) damp low-frequency oscillations. It can compensate for the voltage drop in the distribution network and stabilize the frequency. To achieve these goals, electric springs need to accurately infer the grid status and respond rapidly to power flow dynamics: both ultrafast grid impedance estimation and ultrafast control are needed. Increasing the switching frequency of the electric spring devices can improve the control bandwidth, reduce the electromagnetic interference filter size, accelerate the overall control loop, and enable other grid response functions that have not been demonstrated before. Many similar active compensation devices have been proposed. Their performance and functionality can all benefit from higher switching frequency.

Power Quality, Low-Frequency Oscillation, and High-Frequency Harmonics

Power quality measures the electrical interaction between the power grid and the sources and the loads.

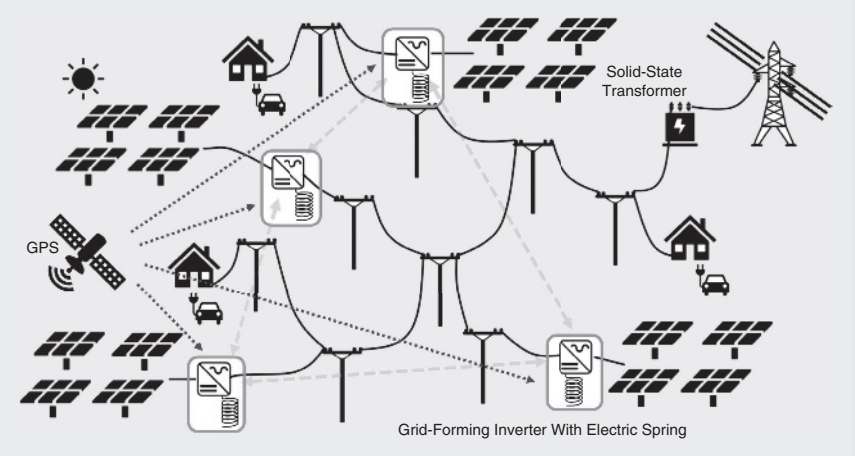

Figure 12 Grid-forming inverters and energy routers at the edge (see Hui, et al.)

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Conventionally, power quality comprises two parts: 1) voltage quality, the way in which the supply voltage impacts equipment operation, and 2) current quality, the way in which the equipment current impacts the system operation. With a large amount of power electronics at the grid edge, the voltage quality and current quality are cross-coupled and cannot be separated. High-frequency power electronics produces both low-frequency harmonics (below 1 kHz) and highfrequency harmonics (above 1 kHz). The low-frequency harmonics can be mitigated by advanced control, but the high-frequency harmonics appear in the distribution grid and cannot be easily eliminated, especially in the distributed grid without isolation transformers. Fixed frequency operation usually leads to groups of harmonics around the integer multiples of the switching frequencies. Hysteresis control, used in smaller converters, leads to a highly distorted frequency spectrum. If the switching frequency is close to a system resonant frequency, it causes a large high-frequency ripple on the voltage. An increasing penetration of DERs with power-electronic interfaces will lead to an increasing level of high-frequency harmonics. The physical principles and system consequences of high-frequency harmonics remain unclear. Standardized methods for measuring, islanding, and characterizing high-frequency current and voltage harmonics are not yet available.

Packetized Energy and Peer-to-Peer Energy Exchange

As the deployment of DERs expands, packetized power microgrids are emerging as promising solutions to effectively incorporate DERs (Figure 13). The full benefits of a microgrid can only be realized when energy can be freely exchanged and traded between sources and loads with economic incentives. Dynamic peer-to-peer energy trading requires

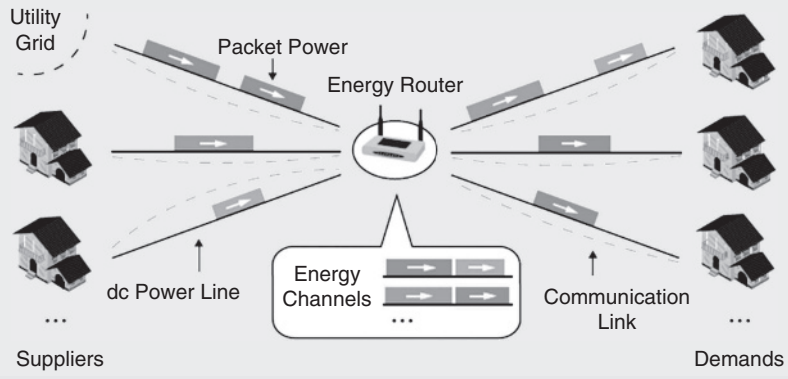

Figure 13 A dc packetized power microgrid

(Source: © 2020 IEEE. Reprinted, with permission from H. Zhang, H. Zhang, L. Song, Y. Li, Z. Han, and H. V. Poor, "Peer-to-peer energy trading in DC packetized power microgrids," *IEEE J. Sel. Areas Commun.*, vol. 38, no. 1, pp. 17–30.)

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rapid and precise balance of the sources and loads within a distribution network (Figure 14).

To enable packetized power delivery in microgrids, the sources and loads interact with energy routers that manage multiway power flow. Fast, stable, and efficient power flow management is needed. Higher switching frequencies lead to faster control of the multiway power flow and will enable the hardware implementation of more advanced trading and packetizing functions with improved energy efficiency, faster control bandwidth, and higher power density. High switching frequencies enable time-division multiple access, frequency-division multiple ac cess, code-division multiple access, and many other MIMO modulation schemes to be implemented in energy routers.

With independent control of real and reactive power in each converter, and with much simplified and improved utility operation and control, it is possible to achieve peer-to-peer energy trading among neighboring communities (in the same secondary feeder) or even among neighbors with trading

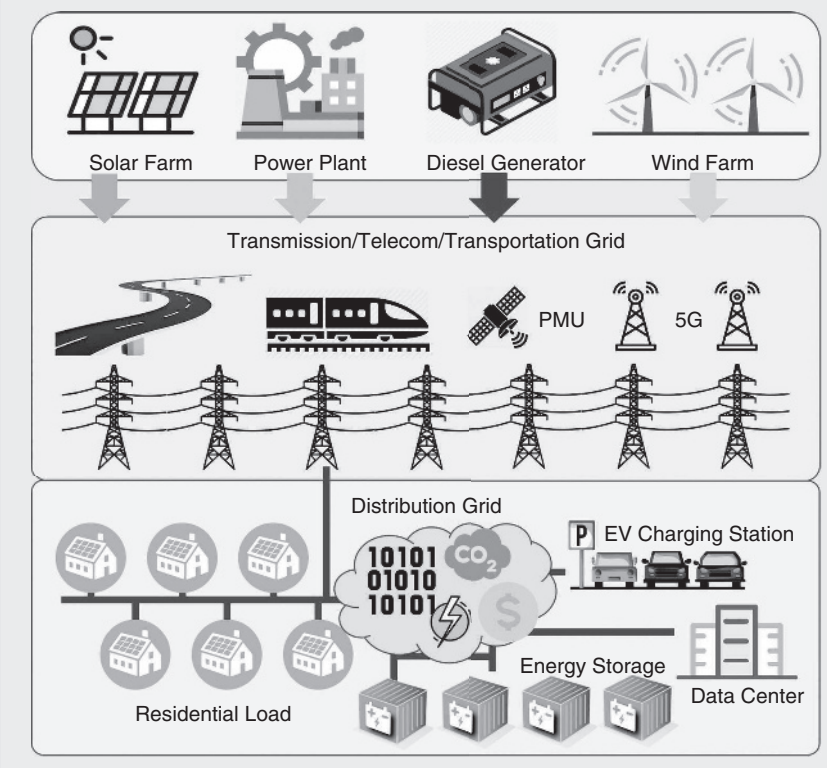

Figure 14 Peer-to-Peer energy trading at the grid edge. PMU: phasor measurement unit (see Tushar et al.).

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protocols across multiple timescales. With such peer-to-peer energy trading options, end users can sell their surplus energy either to the utility or directly to other end users, and they can provide the needed frequency support functions at the grid edge, making the best use of the distributed electricity generation and energy storage capability.

Peer-to-peer energy trading requires energy routing with system stability. High-frequency power electronics are needed to perform fast state estimation of the power flow (real and reactive) and ensure rapid and stable energy routing within the stability margin. High-frequency power electronics enable high-speed power-line communication, which is an emerging data transfer mechanism that can ensure transaction privacy and enable point-to-point, inverter-to-inverter communication on the distribution level.

Summary: A Bottom-Up Approach Toward the Future Grid

The future grid needs to supply more than 50% renewable energy and support a massive number of intelligent loads. Transitioning toward such a future grid requires bottom-up technical innovations including new semiconductor devices, new circuits and topologies, new system architectures, and advanced control. These fundamental technologies will improve the performance of power electronics systems and enable new functions that were not possible before. Continuous installation of intelligent EV chargers, solar inverters, and energy routers at the grid edge as well as the deployment of high-speed communication networks will make the future grid much more observable and controllable. The grid will be able to host intermittent renewable energy and become more robust against natural disasters and disturbances. Energy will be locally generated and consumed within a distribution grid, yielding significantly reduced stress on the transmission level. Innovations in grid operation and novel business models, such as demand response and peer-to-peer energy trading, will create value chains and foster an ecosystem to close the cycle for technical innovation.

With the electric grid delivering more than 50% of the societal energy consumption, the energy flow carried by the electric grid is mixing with the data flow on the information network, the pricing of electricity, and the geographical footprint of carbon emission. The security, privacy, and functionality of the electric grid; the transportation network; and the telecommunication network are coupling together. As more and more power electronics devices are installed at the grid edge, the one-way unidirectional power and information flow in the traditional electric grid will be replaced by the multiway bidirectional power and information flow in the future grid, and the optimal way of operating such an electric grid is still unknown. A wide range of economic, societal, and technical challenges need solutions coming from multidisciplinary systems thinking.

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Conclusion

The opportunities of investigating high-frequency power electronics at the grid edge are driven by fundamental technologies and motivated by high-impact applications. The performance benefits of high-frequency power electronics will reduce the initial cost barrier for technical innovation. The new functions and capabilities of high-frequency power electronics will make them attractive for large-scale adoption, further reducing the cost. The challenges for large-scale deployment of high-frequency power electronics exist in the "transition." How do we seamlessly switch from the current "grid-following" mode to the future "grid-forming" mode, without sacrificing the reliability, resiliency, and low cost of providing electricity?

Recent progress in other industries (e.g., transportation electrification, 5G communication, and IoT) are accelerating this transition. From small-scale solar-powered data centers to "net-zero" distribution substations that support large numbers of megawatt-level EV charging stations, a "bottom-up" approach, rooted in technologies and thriving in applications, can maximize the economic incentive of embedding distributed intelligence in power electronics, while mitigating the political, technical, economic, and societal barriers associated with the transition.

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15.1 INTRODUCTION TO DISTRIBUTION

Figure 15.1 shows the basic components of an electric power system [1–9]. Power plants convert energy from fuel (coal, gas, nuclear, oil, etc.) and from water, wind, solar, or other forms into electric energy. Power plant generators that are driven by turbines, with typical ratings varying from 50 to 1300 MVA, are of three-phase construction, with three-phase armature windings embedded in the slots of stationary armatures. Generator terminal voltages, which are limited by material and insulation capabilities, range from a few kV for older and smaller units up to 20 kV for newer and larger units.

During the last ten years, there has been an ever-growing number of distributed energy resources (DERs), including distributed solar PV arrays, wind-turbine generators, and battery storage systems, each with their power electronic converters (e.g., see the case studies in Chapter 1, 2, and 11). Many of these DERs connect to the power grid at the distribution level. Large solar PV plants and wind farms connect to the transmission system.

To reduce transmission energy losses, generator step-up (GSU) transformers at power plant substations increase voltage and decrease current. Both the GSU transformers and the buses in these substations are protected by circuit breakers, surge arresters, and other protection equipment.

The transmission system serves three basic functions:

1. It delivers energy from generators to the system.
2. It provides for energy interchange among utilities.
3. It supplies energy to the subtransmission and distribution system.

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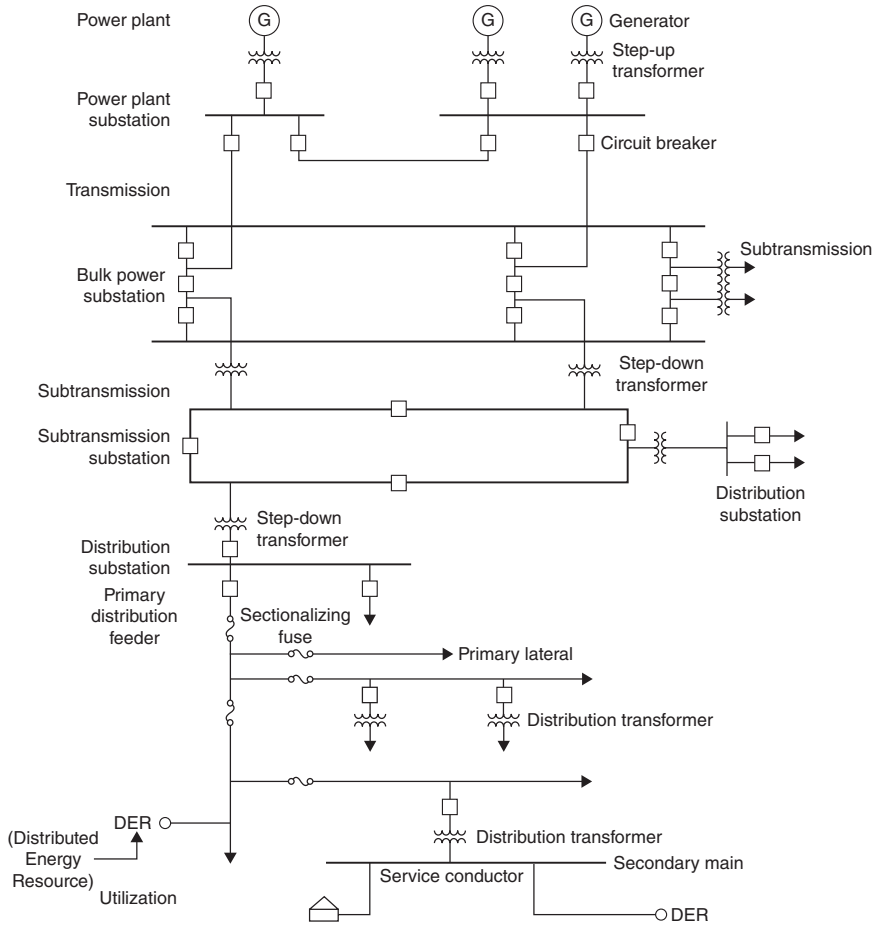

FIGURE 15.1

Basic components of an electric power system (J. D. Glover, "Electric Power Distribution," *Encyclopedia of Energy Technology and The Environment*, John Wiley & Sons, New York, NY, 1995. Copyright © 1995, John Wiley and Sons.)

The transmission system consists of a network of three-phase transmission lines and transmission substations, also called bulk power substations. Typical ac transmission voltages range from 230 up to 765 kV. Single-circuit three-phase ratings vary from 400 MVA at 230 kV up to 4000 MVA at 765 kV. In some cases, HVDC lines with solid-state converters are embedded in the transmission system, as well as back-to-back ac-dc links.

The subtransmission system consists of step-down transformers, substations, and subtransmission lines that connect bulk power substations to distribution substations. In some cases, a subtransmission line may be tapped, usually through a circuit breaker, to supply a single-customer distribution load, such as a large industrial plant. Typical subtransmission voltages range from 69 to 138 kV.

Distribution substations include step-down transformers (distribution substation transformers) that decrease subtransmission voltages to primary distribution voltages in the 2.2- to 46-kV range for local distribution. These transformers connect through associated circuit breaker and surge arrester protection

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to substation buses, which in turn connect through circuit breakers to three-phase primary distribution lines called distribution circuits or feeders. Each substation bus usually supplies several feeders. Typical distribution substation ratings vary from 15 MVA for older substations to 200 MVA or higher for newer installations. Distribution substations may also include equipment for regulating the primary voltage, such as load tap changers (LTCs) on the distribution substation transformers or separate voltage regulators.

Typical primary distribution feeder ratings include 4 MVA for 4.16-kV, 12 MVA for 13.8-kV, 20 MVA for 22.9-kV, and 30 MVA for 34.5-kV feeders. Feeders are usually segregated into several three-phase sections connected through sectionalizing fuses or switches. Each feeder section may have several single-phase laterals connected to it through fuses. Three-phase laterals may also be connected to the feeders through fuses or reclosers. Separate, dedicated primary feeders supply industrial or large commercial loads.

Feeders and laterals run along streets, as either overhead lines or underground cables, and supply distribution transformers that step the voltage down to the secondary distribution level (120 to 480 V). Distribution transformers, typically rated 5 to 5000 kVA, are installed on utility poles for overhead lines, on pads at ground level, or in vaults for underground cables. Distribution transformers are protected from overloads and faults by fuses or circuit breakers on the primary and/or the secondary side. From these transformers, energy flows through secondary mains and service conductors to supply single- or three-phase power directly to customer loads (residential, commercial, and light industrial).

Service conductors connect through meters, which determine kilowatt-hour consumption for customer billing purposes as well as other data for planning and operating purposes, to service panels located on customers' premises. Customers' service panels contain circuit breakers or fuses that connect to wiring that in turn supplies energy for utilization devices (lighting, appliances, motors, heating-ventilation-air conditioning, etc.).

Distribution of electric energy from distribution substations to meters at customers' premises has two parts:

1. Primary distribution, which distributes energy in the 2.2- to 46-kV range from distribution substations to distribution transformers, where the voltage is stepped down to customer utilization levels.
2. Secondary distribution, which distributes energy at customer utilization voltages of 120 to 480 V to meters at customers' premises.

15.2 PRIMARY DISTRIBUTION

Table 15.1 shows typical primary distribution voltages in the United States [1–9]. Primary voltages in the "15-kV class" predominate among U.S. utilities. The 2.5- and 5-kV classes are older primary voltages that are gradually being replaced by 15-kV class primaries. In some cases, higher 25- to 34.5-kV classes are used in new high-density load areas as well as in rural areas that have long feeders.

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Class, kV	Voltage, kV
2.5	2.4
5	4.16
8.66	7.2
15	12.47
	13.2
	13.8
25	22.9
	24.94
34.5	34.5

TABLE 15.1

Typical Primary Distribution Voltages in the United States

(Based on J. D. Glover, "Electric Power Distribution," *Encyclopedia of Energy Technology and The Environment,* John Wiley & Sons, New York, NY, 1995.)

The three-phase, four-wire multi-grounded primary system is the most widely used. Under balanced operating conditions, the voltage of each phase is equal in magnitude and 120° out of phase with each of the other two phases. The fourth wire in these Y-connected systems is used as a neutral for the primaries, or as a common neutral when both primaries and secondaries are present. Usually the windings of distribution substation transformers are Y-connected on the primary distribution side, with the neutral point grounded and connected to the common neutral wire. The neutral is also grounded at frequent intervals along the primary, at distribution transformers, and at customers' service entrances. Sometimes distribution substation transformers are grounded through an impedance (approximately one ohm) to limit short-circuit currents and improve coordination of protective devices.

The three-wire delta primary system is also popular, although not as widely used as the four-wire multi-grounded primary system. Three-wire delta primary systems are not being actively expanded. They are generally older and lower in voltage than the four-wire multi-grounded type. They are also popular in industrial systems.

Rural areas with low-density loads are usually served by overhead primary lines with distribution transformers, fuses, switches, and other equipment mounted on poles. Urban areas with high-density loads are served by underground cable systems with distribution transformers and switchgear installed in underground vaults or in ground-level cabinets. There is also an increasing trend toward underground residential distribution (URD), particularly single-phase primaries serving residential areas. Underground cable systems are highly reliable and usually unaffected by weather. But the installation costs of underground distribution are significantly higher than overhead.

Primary distribution includes three basic systems:

1. Radial
2. Loop
3. Primary network systems

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PRIMARY RADIAL SYSTEMS

The primary radial system, as illustrated in Figure 15.2, is a widely used, economical system often found in low-load-density areas [1, 3, 4, and 9]. It consists of separate three-phase feeder mains (or feeders) emanating from a distribution substation in a radial fashion, with each feeder serving a given geographical area. The photograph at the beginning of this chapter shows a distribution substation that supplies six radial feeders for a suburban residential area. A three-phase feeder main can be as short as a mile or two, or as long as 30 miles (48 km). Single-phase laterals (or branches) are usually connected to feeders through fuses so that a fault on a branch can be cleared without interrupting the feeder. Single-phase laterals are connected to different phases of the feeder so as to balance the loading on the three phases.

To reduce the duration of interruptions, overhead feeders can be protected by automatic reclosing devices located at the distribution substation, at the first overhead pole, or at other locations along the feeder [11]. As an example, Figure 15.3 shows a pole-mount recloser for a 22.9-kV circuit. Studies have shown that the large majority of faults on overhead primaries are temporary, caused by lightning flashover of line insulators, momentary contact of two conductors, momentary bird or animal contact, or momentary tree limb contact. The recloser or circuit breaker with reclosing relays opens the circuit either "instantaneously" or

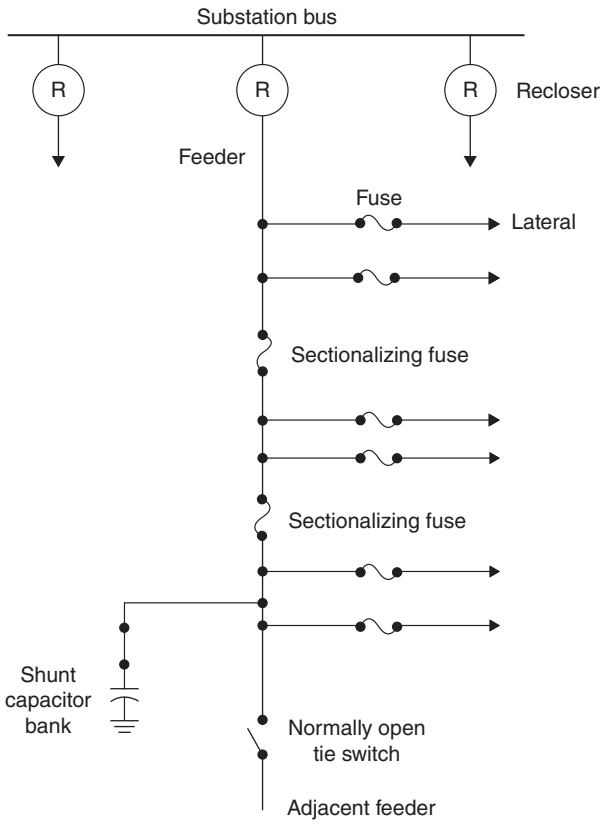

FIGURE 15.2

Primary radial system (J. D. Glover, "Electric Power Distribution," *Encyclopedia of Energy Technology and The Environment*, John Wiley & Sons, New York, NY, 1995. Copyright © 1995, John Wiley and Sons.)

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FIGURE 15.3

Pole-mount recloser for a three-phase 22.9-kV circuit. This recloser has an 800-A continuous current rating and a 16-kA interrupting rating. The 22.9-kV feeder is located near the top of the pole. There are two three-phase 4.16-kV circuits below the recloser. An antenna located below the 4.16-kV circuits is for remote control of the recloser from the dispatch center. A normally open bypass switch located on the top crossarm can be manually operated if the recloser fails to reclose. (Courtesy of Danvers Electric.)

with intentional time delay when a fault occurs, and then recloses after a short period of time. The recloser can repeat this open-and-reclose operation if the fault is still on the feeder. A popular reclosing sequence is one or two fast openings without intentional time delay (to clear temporary faults), followed by two delayed openings (allowing time for fuses to clear persistent downstream faults), followed by opening and lockout for persistent faults between the recloser and fuses. For safety purposes, the reclosing feature is bypassed during live line maintenance. Reclosing is not used on circuits that are primarily underground.

To further reduce the duration and extent of customer interruptions, sectionalizing fuses are installed at selected intervals along radial feeders. In the case of a fault, one or more fuses blow to isolate the fault, and the unfaulted section upstream remains energized. In addition, normally open tie switches to adjacent feeders are incorporated so that during emergencies, unfaulted sections of a feeder can be tied to the adjacent feeder. Spare capacity is often allocated to feeders to prevent overloads during such emergencies, or there may be enough diversity between loads on adjacent feeders to eliminate the need for spare capacity.

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Many utilities have also installed sectionalizers at intervals along radial lines (see also Section 11.5). A sectionalizer is a switching device that can automatically isolate faulted sections of a distribution circuit after an upstream circuit breaker or recloser has interrupted the fault current. Since sectionalizers have no capacity to interrupt fault currents, they must be used with a backup device, such as a recloser, that has the capacity to interrupt fault currents. Sectionalizers count the number of recloser operations during fault conditions. After a pre-set number of recloser openings, and while the recloser is open, the sectionalizer opens and isolates the faulted section of the line. This permits the recloser to close and reenergize the unfaulted section of the line before the recloser locks out. If a temporary fault is cleared during recloser operations, before the sectionalizer reaches its pre-set count, the operating mechanism of the sectionalizer is reset. There are also newer recloser technologies that enable reclosers to operate in a sectionalizer mode [32].

Figure 15.4 shows a primary radial feeder with a recloser and two sectionalizers. A fault results in the recloser operation and the start of the counters in both sectionalizers. The downstream sectionalizer is adjusted to open for a smaller count. For a permanent fault, the downstream sectionalizer with the smaller count trips first. If the fault is located between the sectionalizers, the upstream sectionalizer will trip next. If the fault is upstream of both sectionalizers, the recloser will trip on its final operation and lock out.

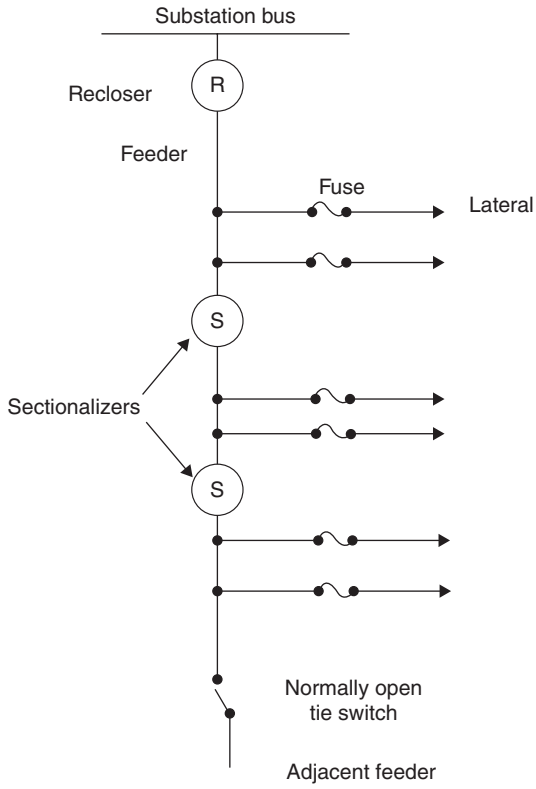

FIGURE 15.4

Primary radial feeder with recloser and two sectionalizers.

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In another scheme, sectionalizers with remote control can be used to isolate faulted sections of a feeder and reenergize unfaulted sections rapidly from a dispatch center before the repair crew is sent out. After a feeder is disconnected by a recloser or circuit breaker, fault indicators located along the feeder enable dispatchers to remotely determine the fault location, and then remotely operate sectionalizers to isolate the fault and rapidly restore power to many customers upstream of the fault. When equipped with remote control, the normally open switch at the end of the feeder (see Figure 15.4) can also be closed remotely by dispatchers to feed customers on unfaulted downstream sections of the feeder from an adjacent feeder. Figure 15.5 shows a radio-controlled sectionalizing switch on a 22.9-kV circuit.

Shunt capacitor banks, including fixed and switched banks, are used on primary feeders to reduce voltage drop, reduce power losses, and improve power factor. Capacitors are typically switched off during the night for light loads and

FIGURE 15.5

S&C normally open radio-controlled sectionalizing switch on a 22.9-kV circuit. This switch has a 1200-A load-break capability. (Courtesy of Danvers Electric.)

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FIGURE 15.6

Pole-mount three-phase 1800-kvar shunt capacitor bank for a 22.9-kV circuit. The capacitor bank, which is protected by 50-A type K fuses, is a switched bank. (Courtesy of Danvers Electric.)

switched on during the day for heavy loads. Figure 15.6 shows a pole-mount switched capacitor bank. Computer programs are available to determine the number, size, and location of capacitor banks that optimize voltage profile, power factor, and installation and operating costs. In some cases, voltage regulators are used on primary feeders.

One or more additional, independent feeders along separate routes may be provided for critical loads, such as hospitals that cannot tolerate long interruptions. Switching from the normal feeder to an alternate feeder can be done manually or automatically with circuit breakers and electrical interlocks to prevent the connection of a good feeder to a faulted feeder. Figure 15.7 shows a primary selective system, often used to supply concentrated loads over 300 kVA [3, 4, and 9].

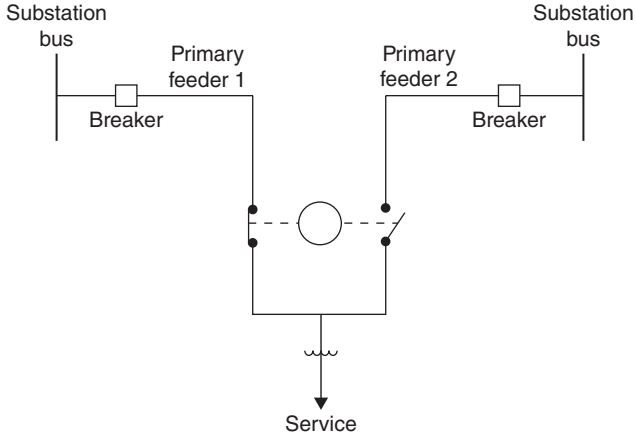

FIGURE 15.7

Primary selective system (J. D. Glover, "Electric Power Distribution," *Encyclopedia of Energy Technology and The Environment*, John Wiley & Sons, New York, NY, 1995. Copyright © 1995, John Wiley and Sons.)

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There are two primary feeders with automatic switching in front (upstream) of the distribution transformer. In case of feeder loss, automatic transfer to the other feeder is rapid and does not require fault locating before transfer.

PRIMARY LOOP SYSTEMS

The primary loop system, as illustrated in Figure 15.8 for overhead, is used where high service reliability is important [1, 3, 4 and 9]. The feeder loops around a load area and returns to the distribution substation, especially providing two-way feed from the substation. The size of the feeder conductors, which are kept the same throughout the loop, is usually selected to carry the entire load connected to the loop, including future load growth. Reclosers and tie switches (sectionalizers) are used to reduce customer interruptions and isolate faulted sections of the loop. The loop is normally operated with the tie switch (or tie recloser) open. Power to a customer at any one time is supplied through a single path from the distribution substation, depending on the open/close status of the reclosers/sectionalizers. Each of the circuit breakers at the distribution substation can be connected to separate bus sections and fed from separate distribution substation transformers.

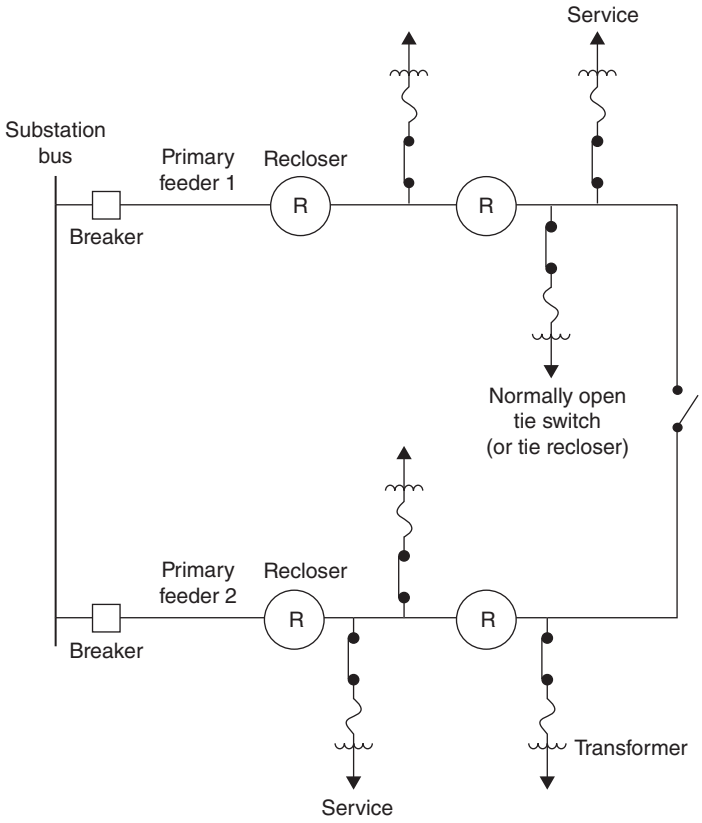

FIGURE 15.8

Overhead primary loop (J. D. Glover, "Electric Power Distribution," *Encyclopedia of Energy Technology and The Environment*, John Wiley & Sons, New York, NY, 1995. Copyright © 1995, John Wiley and Sons.)

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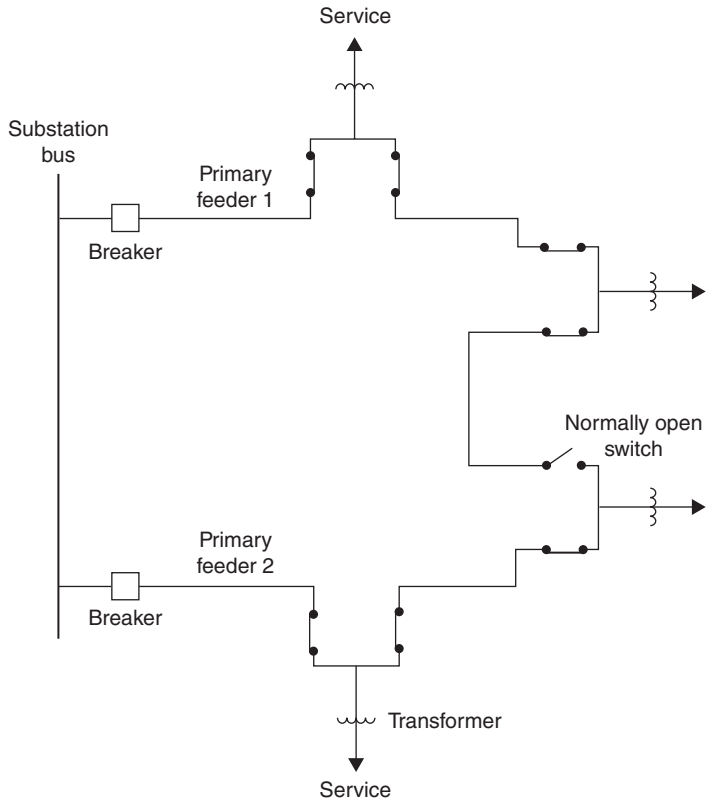

FIGURE 15.9

Underground primary loop (J. D. Glover, "Electric Power Distribution," *Encyclopedia of Energy Technology and The Environment*, John Wiley & Sons, New York, NY, 1995. Copyright © 1995, John Wiley and Sons.)

Figure 15.9 shows a typical primary loop for underground residential distribution (URD). The size of the cable, which is kept the same throughout the loop, is selected to carry the entire load, including future load growth. Underground primary feeder faults occur far less frequently than in overhead primaries but are generally permanent. The duration of outages caused by primary feeder faults is the time to locate the fault and perform switching to isolate the fault and restore service. Fault locators at each distribution substation transformer help to reduce fault locating times.

PRIMARY NETWORK SYSTEMS

Although the primary network system, as illustrated in Figure 15.10, provides higher service reliability and quality than a radial or loop system, only a few primary networks remain in operation in the United States today [1, 3, 4, and 9]. They are typically found in downtown areas of large cities with high load densities. The primary network consists of a grid of interconnected feeders supplied from a number of substations. Conventional distribution substations can be replaced by smaller, self-contained unit substations at selected network locations. Adequate voltage is maintained at utilization points by voltage regulators at distribution substations and by locating distribution transformers close to major load centers on the grid. However, it is difficult to maintain adequate voltage everywhere on the primary

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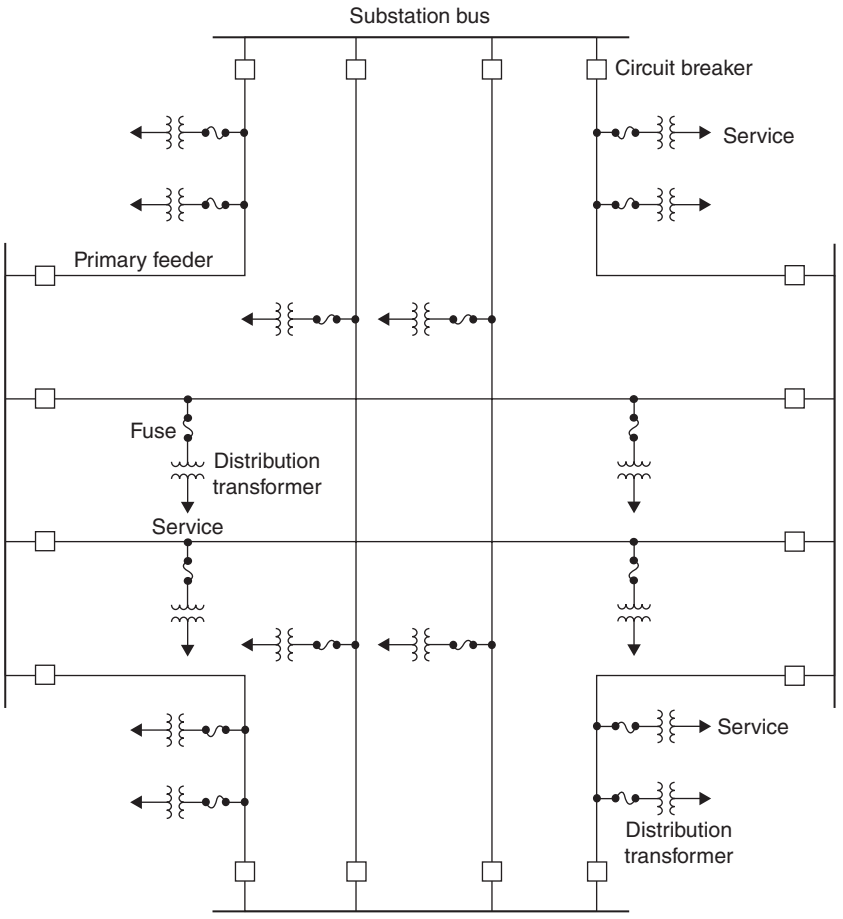

FIGURE 15.10

Primary network (J. D. Glover, "Electric Power Distribution," *Encyclopedia of Energy Technology and The Environment*, John Wiley & Sons, New York, NY, 1995. Copyright © 1995, John Wiley and Sons.)

grid under various operating conditions. Faults on interconnected grid feeders are cleared by circuit breakers at distribution substations, and in some cases, by fuses on the primary grid. Radial primary feeders protected by circuit breakers or fuses can be tapped off the primary grid or connected directly at distribution substations.

15.3 SECONDARY DISTRIBUTION

Secondary distribution distributes energy at customer utilization voltages from distribution transformers up to meters at customers' premises. Table 15.2 shows typical secondary voltages and applications in the United States [1–9]. In residential areas, 120/240-V, single-phase, three-wire service is the most common, where lighting loads and outlets are supplied by 120-V, single-phase connections, and large household appliances, such as electric ranges, clothes dryers, water heaters, and electric space heaters, are supplied by 240-V, single-phase connections. In urban areas serving high-density residential and commercial loads, 208 Y/120-V,

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Voltage	# Phases	# Wires	Application
120/240 V	Single-phase	Three	Residential
208 Y/120 V	Three-phase	Four	Residential/Commercial
480 Y/277 V	Three-phase	Four	Commercial/Industrial/High Rise

TABLE 15.2

Typical Secondary Distribution Voltages in the United States

(Based on J. D. Glover, "Electric Power Distribution," *Encyclopedia of Energy Technology and The Environment,* John Wiley & Sons, New York, NY, 1995.)

three-phase, four-wire service is common where lighting, outlets, and small motor loads are supplied by 120-V, single-phase connections, and larger motor loads are supplied by 208-V, three-phase connections. In areas with very high-density commercial and industrial loads as well as high-rise buildings, 480 Y/277-V, three-phase, four-wire service is common, with fluorescent lighting supplied by 277-V, single-phase connections and motor loads supplied by 480-V, three-phase connections. Separate 120-V radial systems fed by small transformers from the 480-V system are used to supply outlets in various offices, retail stores, or rooms.

Figure 15.11 shows a typical residential customer voltage profile along a radial feeder. In accordance with ANSI standards, during normal conditions, utilities in the United States are required to maintain customer voltage at the customer's service panel between 114 and 126 volts (±5%) based on a 120-V nominal secondary voltage. As shown in Figure 15.11, the first customer, closest to the substation,

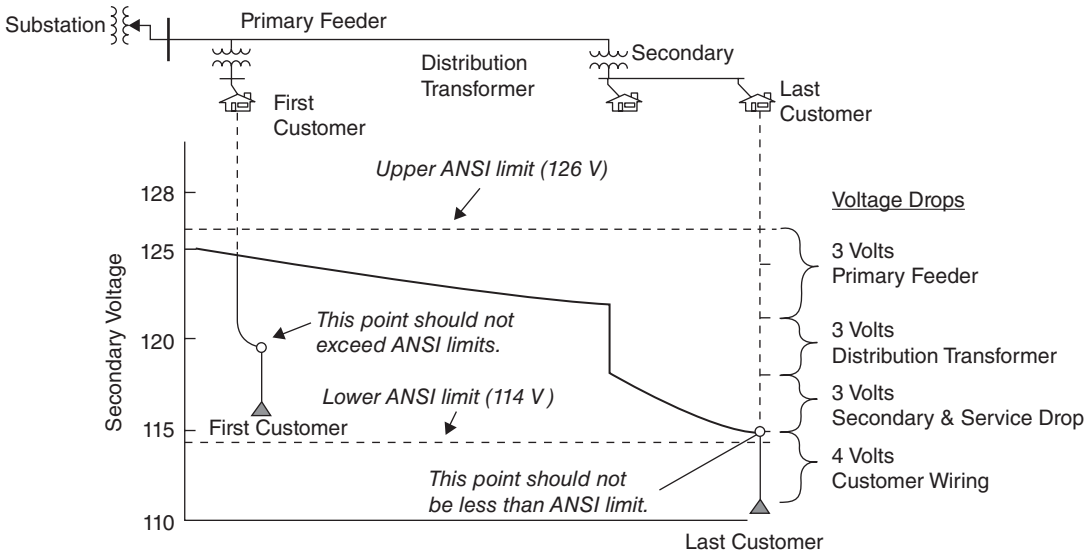

FIGURE 15.11

Typical residential customer voltage profile along a radial feeder, assuming no shunt capacitors or voltage regulators along the feeder

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has the highest voltage, and the last customer, furthest from the substation, has the lowest voltage. Proper distribution design dictates that the first customer's voltage is less than 126 V during light loads and the last customer's voltage is greater than 114 V during peak loads, so that all customers remain within 120 V ±5% under all normal loading conditions. Load-tap-changing distribution substation transformers and voltage regulators (see Section 15.4) as well as shunt capacitors (see Section 15.5) are used to maintain customer voltages within ANSI limits.

There are four general types of secondary systems:

1. Individual distribution transformer per customer
2. Common secondary main
3. Secondary network
4. Spot network

INDIVIDUAL DISTRIBUTION TRANSFORMER PER CUSTOMER

Figure.15.12 shows an individual distribution transformer with a single service supplying one customer, which is common in rural areas where distances between customers are large and long secondary mains are impractical [3 and 4]. This type of system may also be used for a customer that has an unusually large load or for a customer that would otherwise have a low-voltage problem with a common secondary main. Although transformer installation costs and operating costs due to no-load losses are higher than those of other types of secondary systems, the installation costs of secondary mains are avoided.

COMMON SECONDARY MAIN

Figure.15.13 shows a primary feeder connected through one or more distribution transformers to a common secondary main with multiple services to a group of customers [3 and 4]. This type of secondary system takes advantage of diversity among customer demands that allows a smaller capacity of the transformer supplying a group compared to the sum of the capacities of individual transformers for each customer in the group. Also, the large transformer supplying a group can handle motor starting currents and other abrupt load changes without severe voltage drops.

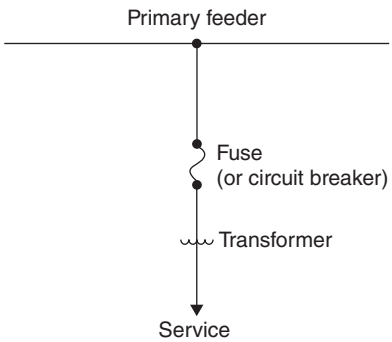

FIGURE 15.12

Individual distribution transformer supplying single-service secondary (J. D. Glover, "Electric Power Distribution," *Encyclopedia of Energy Technology and The Environment*, John Wiley & Sons, New York, NY, 1995. Copyright © 1995, John Wiley and Sons.)

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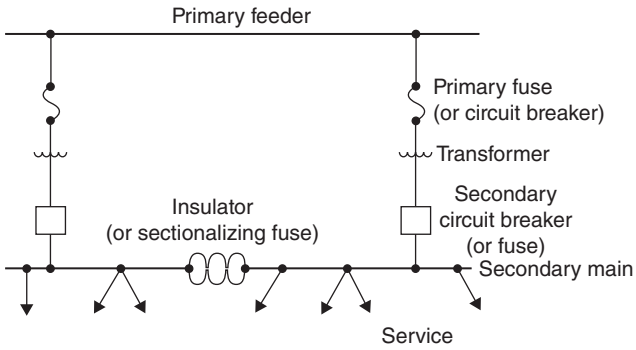

FIGURE 15.13

Common secondary main (J. D. Glover, "Electric Power Distribution," *Encyclopedia of Energy Technology and The Environment*, John Wiley & Sons, New York, NY, 1995. Copyright © 1995, John Wiley and Sons.)

In most cases, the common secondary main is divided into sections, where each section is fed by one distribution transformer and is also isolated from adjacent sections by insulators. In some cases, fuses are installed along a continuous secondary main, which results in banking of distribution transformers, also called banked secondaries.

SECONDARY NETWORK

Figure 15.14 shows a secondary network or secondary grid, which may be used to supply high-density load areas in downtown sections of cities, where the highest degree of reliability is required and revenues justify grid costs [1, 3, 4, and 9]. The

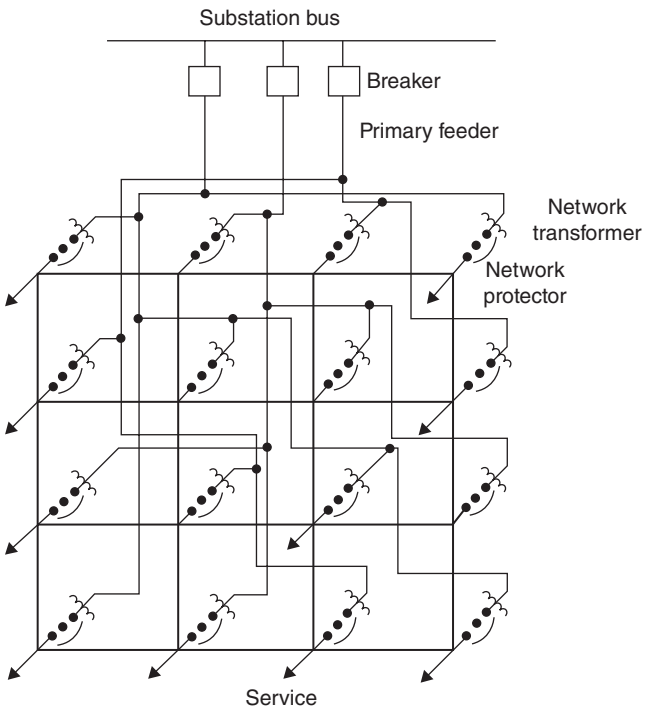

FIGURE 15.14

Secondary network (J. D. Glover, "Electric Power Distribution," *Encyclopedia of Energy Technology and The Environment*, John Wiley & Sons, New York, NY, 1995. Copyright © 1995, John Wiley and Sons.)

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underground secondary network is supplied simultaneously by two or more primary feeders through network transformers. Most networks are supplied by three or more primary feeders with transformers that have spare capacity so that the network can operate with two feeders out of service.

Secondary grids operate at either 208 Y/120 V or 480 Y/277 V in the United States. Commonly used secondary cable sizes range from 4/0 to 500 kcmil AWG [5].

More than 260 cities in the United States have secondary networks [5]. New York City has the largest secondary network system in the United States with approximately 23,000 network transformers feeding various secondary networks and an online monitoring system that continuously monitors transformer loadings. Some of the secondary networks in New York City are fed by as many as 24 primary feeders operating in parallel [9].

Network transformers are protected by network protectors between the transformers and secondary mains. A network protector is an electrically operated low-voltage air circuit breaker with relays and auxiliary devices that automatically opens to disconnect the transformer from the network when the transformer or the primary feeder is faulted, or when there is a power flow reversal. The network protector also has the ability to close automatically when a feeder is energized [5]. Fuses may also be used for backup of network protectors.

In many cases, especially on 208 Y/120-V secondary networks, the main protection of secondary cables has come from the ability of the cable system to "burn clear" with no fuse or other protective device. However, in many instances for 480 Y/277-V secondary networks, this practice was not able to successfully burn clear, resulting in fires and considerable damage. As a solution, special fuses called cable limiters are commonly used at tie points in the secondary network to isolate faulted secondary cables. Cable limiters, which are designed with restricted sections of copper that act like a fuse, do not limit the magnitude of fault current like current-limiting fuses. In high short-circuit locations on the secondary network, current-limiting fuses may be used instead of cable limiters.

In secondary network systems, a forced or scheduled outage of a primary feeder does not result in customer outages. Because the secondary mains provide parallel paths to customer loads, secondary cable failures usually do not result in customer outages, either. Also, each network is designed to share the load equally among transformers and to handle large motor starting and other abrupt load changes without severe voltage drops.

SPOT NETWORK

Figure 15.15 shows a spot network consisting of a secondary network supplying a single, concentrated load, such as a high-rise building or shopping center, where a high degree of reliability is required [1, 3, 4, and 9]. The secondary spot network bus is supplied simultaneously by two or more primary feeders through network transformers. In some cases, a spot network load as large as 25 MVA may be fed by up to six primary feeders. Most spot networks in the United States operate at a 480 Y/277-V secondary voltage [5]. Separate 120-V radial systems fed by small

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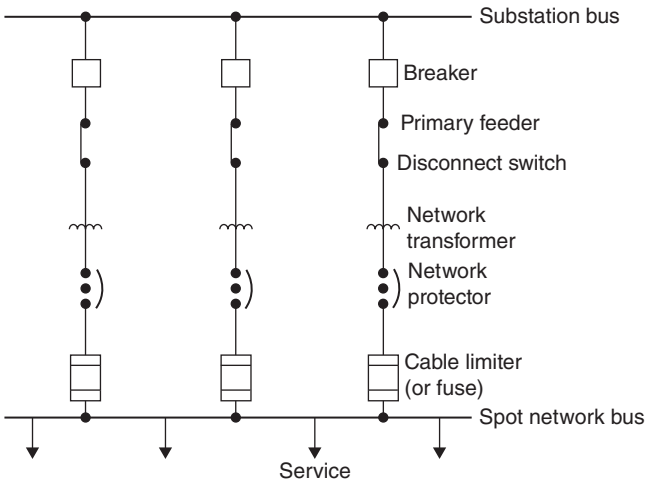

FIGURE 15.15

Secondary spot network (J. D. Glover, "Electric Power Distribution," *Encyclopedia of Energy Technology and The Environment*, John Wiley & Sons, New York, NY, 1995. Copyright © 1995, John Wiley and Sons.)

transformers from the 480-V system are used to supply outlets in various offices, retail stores, or rooms.

High service reliability and operating flexibility are achieved with a spot network fed by two or more primary feeders through network transformers. The secondary bus is continuously energized by all network transformers. Network protectors are used to automatically disconnect transformers from the spot network bus for transformer/feeder faults or for power-flow reversal, and cable limiters or fuses are used to protect against overloads and faults on secondary cables. Scheduled or forced outages of primary feeders occur without customer interruption or involvement. Spot networks also provide a very compact and reliable arrangement of components [5].

15.4 TRANSFORMERS IN DISTRIBUTION SYSTEMS

Transformers in distribution systems include distribution substation transformers and distribution transformers.

DISTRIBUTION SUBSTATION TRANSFORMERS

Distribution substation transformers come in a wide variety of ratings. Some of the typical characteristics of distribution substation transformers are given in Table 15.3.

Distribution substation transformers usually contain mineral oil for insulating and cooling purposes (older transformers manufactured prior to 1978 originally contained askarels with high PCB content, but many of these have either been retired or reclassified as non-PCB transformers using per-chloroethylene). In some units, an inert gas, such as nitrogen, fills the space above the oil, in order to keep moisture and air out of the oil, and the transformer tank is sealed. Some

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Class, kV	Voltage, kV
Rating of High Voltage Winding	34.5 to 230 kV
Rating of Low Voltage Winding	2.4 to 46 kV
MVA Rating (OA)	2.5 to 75 MVA
Transformer Impedance	5 to 12 %
Number of Transformers in Substation	1 to 4
Loading	OA, OA/FA, OA/FA/FOA, OA/FA/FA
High Side Protection	Circuit Switches, Circuit Breakers, Fuses
Relay Protection	Overcurrent, Differential, Under-Frequency
Feeder Protection	Circuit Breakers, Reclosers

TABLE 15.3

Typical Characteristics of Distribution Substation Transformers [5]

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sealed transformers have a pressure-relief diaphragm that is designed to rupture when the internal pressure exceeds a specified value, indicating possible deterioration of the insulation. Sealed transformers may also have a sudden pressure relay to either provide an alarm or deenergize the transformer when the internal pressure suddenly increases above a specified threshold [4].

Many distribution substation transformers have load tap changers (LTCs) that automatically regulate voltage levels based on loading conditions. Figure 15.16 shows a distribution substation transformer that has an internal LTC on the low-voltage side.

FIGURE 15.16

Three-phase 22.9-kVΔ/4.16-kVY distribution substation transformer rated 12 MVA OA/16 MVA FA1/20 MVA FA2. The transformer has fixed taps on the high-voltage side and an LTC on the low-voltage side. (Courtesy of Danvers Electric.)

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Some distribution substations have distribution substation transformers with fixed taps and separate voltage regulators. A voltage regulator is basically an autotransformer with taps that automatically raise or lower voltage, operating in a similar way as LTCs on distribution substation transformers. Figure 15.17 shows a voltage regulator at a distribution substation. In addition to voltage regulators for distribution substations, there are also pole-mount voltage regulators that can be placed on feeders.

Some outdoor distribution substation transformers are equipped with a tank on the top of the transformer called a "conservator," in which expansion and contraction of the oil takes place. Condensation of moisture and formation of sludge occur within the conservator, which is also provided with a sump pump to draw off the moisture and sludge [4].

Distribution substation transformers have MVA ratings that indicate the continuous load that the transformers carry without exceeding a specified temperature rise of either 55°C (for older transformers) or 65°C (for newer transformers) above a specified ambient temperature (typically 40°C). Also, distribution substation transformers are typically equipped with external radiators with fans and/or oil circulating pumps, in order to dissipate heat generated by copper and core losses. These transformers have multiple MVA ratings that include the following:

1. OA rating (passive convection with oil circulating pumps and fans off).
2. FA rating (with fans on but oil circulating pumps off).
3. FOA rating (with both fans and oil circulating pumps on). Some units, such as the one shown in Figure 15.16, may have two FA ratings, a lower FA rating with one of two sets of fans on, and a higher FA rating with both sets of fans on. Also, some units have water-cooled heat exchangers. The nameplate transformer impedance is usually given in percent using the OA rating as the base MVA [5].

FIGURE 15.17

Voltage regulators at the 69/13.8-kV Lunenburg distribution substation, Lunenburg MA. The regulators are General Electric Type VR1 with ratings of 7.96 kV (line-to neutral), 437 A. (Courtesy of Unitil Corporation.)

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Most utilities have a planning and operating policy of loading distribution substation transformers within their nameplate OA/FA/FOA ratings during normal conditions, but possibly above their nameplate ratings during short-term emergency conditions. If one transformer has a scheduled or forced outage, the remaining transformer or transformers can continuously carry the entire substation load.

EXAMPLE 15.1

Distribution substation transformer rated current and short-circuit current

A three-phase 230-kV Δ/34.5-kV Y distribution substation transformer rated 75 MVA OA/100 MVA FA/133 MVA FOA has a 7% impedance. (a) Determine the rated current on the low-voltage side of the transformer at its OA, FA, and FOA ratings. (b) Determine the per-unit transformer impedance using a system base of 100 MVA and 34.5 kV on the low-voltage side of the transformer. (c) Calculate the short-circuit current on the low-voltage side of the transformer for a three-phase bolted fault on the low-voltage side. Assume that the prefault voltage is 34.5 kV.

SOLUTION

a. At the OA rating of 75 MVA,

$I_{OA,\,L} = 75/(\sqrt{3} \times 34.5) = 7.372$ kA per phase

Similarly,

$I_{FA,\,L} = 100/(\sqrt{3} \times 34.5) = 9.829$ kA per phase

$I_{FOA,\,L} = 133/(\sqrt{3} \times 34.5) = 13.07$ kA per phase

b. The transformer impedance is 7% or 0.07 per unit based on the OA rating of 75 MVA. Using (3.3.11), the transformer per unit impedance on a 100-MVA system base is:

$Z_{puSystem\ Base} = 0.07(100/75) = 0.09333$ per unit

c. For a three-phase bolted fault, using the transformer ratings as the base quantities,

$$\begin{aligned} Isc3\phi &= 1.0/(0.07) = 14.286 \text{ per unit} \\ &= (14.286)(7.372) \\ &= 105.31 \text{ kA/phase} \end{aligned}$$

Note that in (c) above, the OA rating is used to calculate the short-circuit current, because the transformer manufacturer gives the per-unit transformer impedance using the OA rating as the base quantity.

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Typically there are two emergency loading criteria for distribution substation transformers:

1. A two-hour emergency rating, which gives time to perform switching operations and reduce loadings.
2. A longer-duration emergency rating (10 to 30 days), which gives time to replace a failed transformer with a spare that is in stock.

As one example, the distribution substation shown in the photograph at the beginning of this chapter has two transformers rated 9 MVA OA/12 MVA FA1/15 MVA FA2. The practice of the utility that owns the substation is normally to operate the substation at or below 15 MVA. As such, if there is a forced or scheduled outage of one transformer, the other transformer can supply all six 4.16-kV feeders without being loaded above its FA2 nameplate of 15 MVA. For this conservative operating practice, emergency transformer ratings above nameplate ratings are not used.

Some utilities operate their distribution substation transformers above nameplate ratings during normal operating conditions, as well as during emergency conditions. ANSI/IEEE C-57.91-2011, entitled *IEEE Guide for Loading Mineral-Oil-Immersed Transformers*, identifies the risks of transformer loads in excess of the nameplate rating and establishes limitations and guidelines, the application of which are intended minimize the risks to an acceptable limit [21, 22].

EXAMPLE 15.2

Distribution substation normal, emergency, and allowable ratings

As shown in Figure 15.18, a distribution substation is served by two 138-kV subtransmission lines, each connected to a 40-MVA (FOA nameplate rating) 138-kV Δ/12.5-kV Y distribution substation transformer, denoted TR1 and TR2. Both TR1 and TR2 are relatively new transformers with insulation systems designed for 65°C temperature rises under continuous loading conditions. Shunt capacitor banks are also installed at 12.5-kV bus 1 and bus 2. The utility that owns this substation has the following transformer loading criteria based on a percentage of the nameplate rating:

1. 128% for normal summer loading;
2. 170% during a two-hour summer emergency; AND
3. 155% during a 30-day summer emergency.

(a) Assuming a 5% reduction for unequal transformer loadings, determine the summer "normal" rating of the substation. (b) Determine the "allowable" summer rating of the substation under the single-contingency loss of one transformer. (c) Determine the 30-day summer emergency rating of the substation under the single-contingency loss of one transformer.

(*Continued*)

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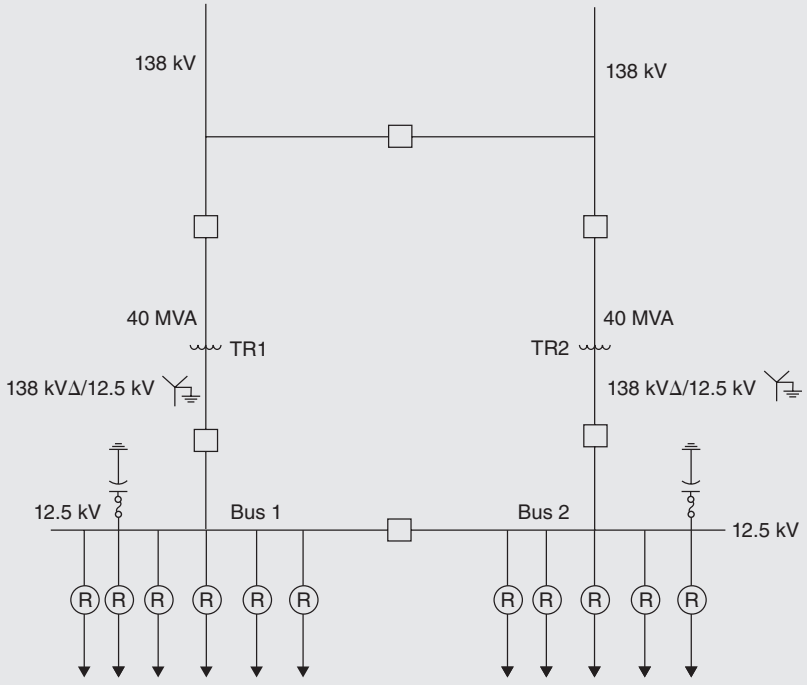
FIGURE 15.18

Distribution substation for Example 15.2



SOLUTION

a. During normal operations, both transformers are in service. Using a 5% reduction to account for unequal transformer loadings, the summer normal substation rating is $1.28 \times (40 + 40) \times 0.95 = 97$ MVA. With both transformers in service, the substation can operate as high as 97 MVA without exceeding the summer normal rating of 128% or 51.2 MVA for each transformer.

b. The summer allowable substation rating, based on the single-contingency loss of one transformer, is $1.7 \times 40 = 68$ MVA. The transformer that remains in service is allowed to operate at 170% of its nameplate rating (68 MVA) for two hours, which gives time to perform switching operations to reduce the transformer loading to its 30-day summer emergency rating. Note that, even though the normal summer substation rating is 97 MVA, it is only allowed to operate up to 68 MVA so that a transformer will not exceed its two-hour emergency rating in case the other transformer has an outage.

c. The 30-day summer emergency rating of the substation is $1.55 \times 40 = 62$ MVA. When one transformer has a permanent failure, the other can operate at 62 MVA for 30 days, which gives time to replace the failed transformer with a spare that is in stock.

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DISTRIBUTION TRANSFORMERS

Distribution transformers connect the primary system (2.4 to 34.5 kV) to the secondary system (480 V and lower). Distribution transformers may be installed outdoors on overhead poles (pole-mount), outdoors at ground level on pads (padmount transformers), indoors within buildings, or underground in manholes and vaults.

Pole-mount transformers for overhead distribution are liquid-filled transformers that can be either single-phase or three-phase, depending on the load requirements and the primary supply configuration. Pole-mount distribution transformers may be manufactured as conventional transformers with no integral surge protection, overload protection, or short-circuit protection, or alternatively, as completely self-protected (CSP) transformers.

For conventional pole-mount transformers, the protective devices are mounted external to the transformer. Typically a fuse cutout, which is a combination of a fuse and a switch, is installed adjacent to the conventional distribution transformer to disconnect it from the primary under overload conditions or an internal transformer failure. Similarly, a surge arrester is installed adjacent to the conventional transformer primary to protect it against transient overvoltages due to switching and lightning surges. Figure 15.19 shows three conventional single-phase pole-mount distribution transformers that are wired to form a three-phase bank rated 75 kVA supplying 120/208 V overhead secondary service for commercial customers.

For CSP transformers, a primary fuse is located within the transformer tank. The surge arrester is mounted outside the tank, but connected to the primary bushing. Circuit breakers on the secondary side of CSP transformers provide protection from overloads and are coordinated with primary fuses.

FIGURE 15.19

Three conventional single-phase pole-mount 25-kVA transformers. The transformers are wired to form a three-phase bank rated 75 kVA, 4160 VΔ-208/120 V grounded Y, which supplies secondary service for commercial customers. The transformers are supplied from a 4160-V primary through fused cutouts, with surge arresters mounted vertically on the sides of the transformer tanks. (Courtesy of Danvers Electric.)

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Padmount transformers for underground distribution are liquid-filled or dry-type transformers that can be either single-phase or three-phase, outdoors or indoors. Single-phase padmount distribution transformers are typically designed for underground residential and commercial distribution systems where safety, reliability, and aesthetics are especially important. Three-phase padmount distribution transformers are compact power centers usually for large commercial or industrial applications. Figure 15.20 shows a three-phase, liquid-filled padmount transformer that supplies 480/277 V underground secondary service to an industrial plant. Dry-type padmount distribution transformers whose insulation is solid (for example, glass, silica, epoxy, or polyester resins) are primarily used where safety is a major concern, in close proximity to people, such as at schools, hospitals, commercial buildings, and industrial plants, both indoors and outdoors.

Network transformers are large (300 to 2500 kVA), liquid-filled, three-phase distribution transformers that are designed for use in underground vaults or in specially designed rooms within buildings to supply power to either secondary networks or spot networks. Their voltage ratings vary from 4.16 to 34.5 kV Δ or grounded Y for the high-voltage windings, and either 216 grounded Y/125 V or 480 grounded Y/277 V for the low-voltage windings. Network transformers are designed to be connected through network protectors that are integrally

(a)

FIGURE 15.20

Three-phase oil-filled padmount transformer shown with doors closed (a) and open (b). This padmount, rated 1,500 kVA OA, kVΔ-480/277 V grounded Y with internal fuses on the high-voltage side, supplies secondary service to an industrial plant. (Courtesy of Danvers Electric.)

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(b)

FIGURE 15.20

(*Continued*)

mounted on the transformer. Figure 15.21 shows a network transformer from utility stock. Network transformers are built as either "vault type" (suitable for occasional submerged operation) or "subway type" (suitable for continuous submerged operation).

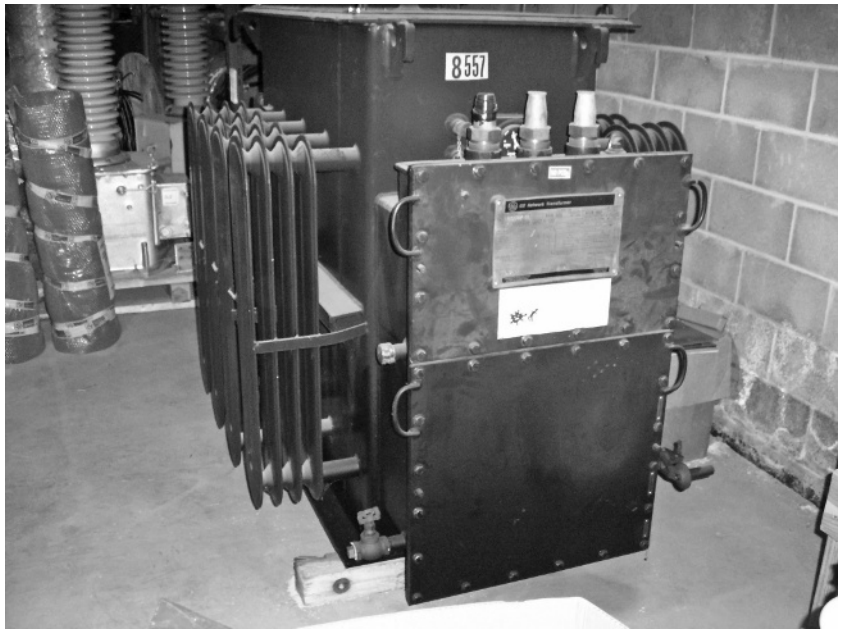

FIGURE 15.21

General Electric 500 kVA, 13.8 kVΔ-120/208 V grounded Y network transformer from utility stock (Courtesy of Unitil Corporation.)

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Table 15.4 shows typical kVA ratings of distribution transformers. The kVA ratings of distribution transformers are based on the continuous load the transformers can carry without exceeding a specified temperature rise of either 55°C (for older transformers) or 65°C (for newer transformers) above a specified ambient temperature (usually 40°C). When in service, distribution transformers are rarely loaded continuously at their rated kVA as they go through a daily load cycle. Oil-filled distribution transformers have a relatively long thermal time constant; that is, the load temperature rises slowly during load increases. As such, it is possible to load these transformers above their kVA ratings without compromising the life expectancy of the transformer. ANSI/EEE Std. C57.92-1981 is entitled *IEEE Guide for Loading Mineral-Oil-Immersed Overhead and Pad-Mounted Distribution Transformers Rated 500 kVA and Less with 65°C or 55°C Average Winding Rise* [23]. Table 15.5 shows a typical loading guide, based on this standard.

Note that in accordance with Table 15.5, short-time loadings can be as high as 89% above the nameplate kVA rating for short durations. Also note that dry-type distribution transformers, which are not considered as rugged as liquid-filled units of the same rating, are not normally loaded above their kVA ratings.

Single-Phase	Three-Phase
kVA	kVA
5	30
10	45
15	75
25	112.5
38	150
50	225
75	300
100	500
167	750
250	1000
333	1500
500	2500
	3000
	3750
	5000

TABLE 15.4

Standard Distribution Transformer kVA Ratings

J. D. Glover, "Electric Power Distribution," *Encyclopedia of Energy Technology and The Environment,* John Wiley & Sons, New York, NY, 1995. Reprinted with permission of John Wiley & Sons, Inc. (Based on ANSI/IEEE C.57 Standard)

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Period of Increased Loading, Hours	Average Initial Load in Per Unit of Transformer Rating		
	0.9	0.7	0.5
	Maximum Load in Per Unit of Transformer Rating		
0.5	1.59	1.77	1.89
1.0	1.40	1.54	1.60
2.0	1.24	1.33	1.37
4.0	1.12	1.17	1.19
8.0	1.06	1.08	1.08

TABLE 15.5

Permissible Daily Short-Time Loading of Liquid-Filled Distribution Transformers Based on Normal Life Expectancy [5]

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15.5 SHUNT CAPACITORS IN DISTRIBUTION SYSTEMS

Loads in electric power systems consume real power (MW) and reactive power (Mvar). At power plants, many of which are located at long distances from load centers, real power is generated and reactive power may either be generated, such as during heavy load periods, or absorbed as during light load periods. Unlike real power (MW), the generation of reactive power (Mvar) at power plants and transmission of the reactive power over long distances to loads is not economically feasible. Shunt capacitors, however, are widely used in primary distribution to supply reactive power to loads. They draw leading currents that offset the lagging component of currents in inductive loads. Shunt capacitors provide an economical supply of reactive power to meet reactive power requirements of loads as well as transmission and distribution lines operating at lagging power factor. They can also reduce line losses and improve voltage regulation.

EXAMPLE 15.3

Shunt capacitor bank at end of primary feeder

Figure 15.22 shows a single-line diagram of a 13.8-kV primary feeder supplying power to a load at the end of the feeder. A shunt capacitor bank is located at the load bus. Assume that the voltage at the sending end of the feeder is 5% above rated and that the load is Y-connected with $R_{Load} = 20\ \Omega$/phase in parallel with load $jX_{Load} = j40\ \Omega$/phase. (a) With the shunt capacitor bank out of service, calculate the following: (1) line current; (2) voltage drop across

(*Continued*)

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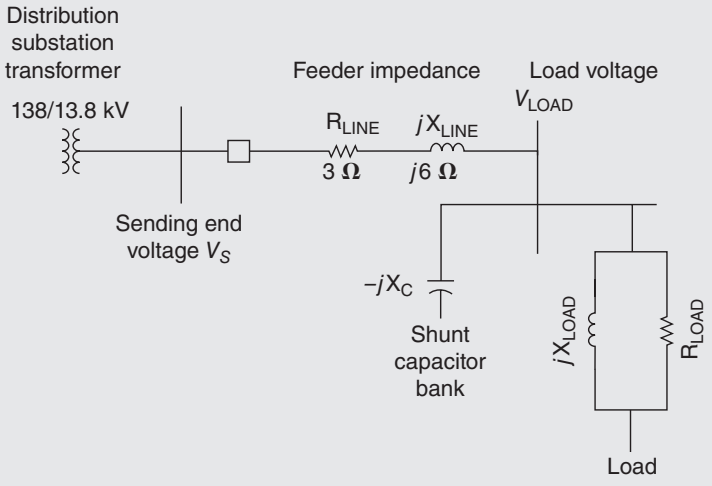
FIGURE 15.22
Single-line diagram of a primary feeder for Example 15.3



the line; (3) load voltage; (4) real and reactive power delivered to the load; (5) load power factor; (6) real and reactive line losses; and (7) real power, reactive power, and apparent power delivered by the distribution substation. (b) The capacitor bank is Y-connected with a reactance $X_C = 40\ \Omega$/phase. With the shunt capacitor bank in service, redo the calculations. Also, calculate the reactive power supplied by the capacitor bank. (c) Compare the results of (a) and (b).

SOLUTION

a. Without the capacitor bank, the total impedance seen by the source is

$$Z_{TOTAL} = R_{LINE} + jX_{LINE} + \frac{1}{\frac{1}{R_{LOAD}} + \frac{1}{jX_{LOAD}}}$$

$$Z_{TOTAL} = 3 + j6 + \frac{1}{\frac{1}{20} + \frac{1}{j40}}$$

$$Z_{TOTAL} = 3 + j6 + \frac{1}{0.0559\angle -26.57^\circ}$$

$$= 3 + j6 + 17.89\angle 26.56^\circ$$

$$Z_{TOTAL} = 3 + j6 + 16 + j8 = 19 + j14$$

$$= 23.60\angle 36.38^\circ\ \Omega/\text{phase}$$

1. The line current is

$$I_{LINE} = V_{SLN}/Z_{TOTAL} = \frac{1.05(13.8/\sqrt{3})\angle 0^\circ}{23.60\angle 36.38^\circ}$$

$$= 0.3545\angle -36.38^\circ\ \text{kA/phase}$$

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2. The voltage drop across the line is

$$V_{\text{DROP}} = Z_{\text{LINE}} I_{\text{LINE}} = (3 + j6)(0.3545\underline{/-36.38°})$$
$$= (6.708\underline{/63.43°})(0.3545\underline{/-36.38°})$$
$$= 2.378\underline{/27.05°}\ \text{kV}$$
$$|V_{\text{DROP}}| = 2.378\ \text{kV}$$

3. The load voltage is

$$V_{\text{LOAD}} = V_{\text{SLN}} - Z_{\text{LINE}} I_{\text{LINE}} = 1.05(13.8/\sqrt{3})\underline{/0°} - 2.378\underline{/27.05°}$$
$$= 8.366 - (2.117 + j1.081) = 6.249 - j1.081$$
$$= 6.342\underline{/-9.814°}\ \text{kV}_{\text{LN}}$$
$$|V_{\text{LOAD}}| = 6.342\sqrt{3} = 10.98\ \text{kV}_{\text{LL}}$$

4. The real and reactive power delivered to the three-phase load is

$$\text{P}_{\text{LOAD3}\phi} = 3(\text{V}_{\text{LOADLN}})^2/\text{R}_{\text{LOAD}} = 3(6.342)^2/20 = 6.033\ \text{MW}$$
$$\text{Q}_{\text{LOAD3}\phi} = 3(\text{V}_{\text{LOADLN}})^2/\text{X}_{\text{LOAD}} = 3(6.342)^2/40 = 3.017\ \text{Mvar}$$

5. The load power factor is

$$\text{p.f.} = \cos[\tan^{-1}(\text{Q/P})]$$
$$= \cos[\tan^{-1}(3.017/6.033)]$$
$$= 0.89\ \text{lagging}$$

6. The real and reactive line losses are

$$\text{P}_{\text{LINELOSS3}\phi} = 3\ \text{I}_{\text{LINE}}{}^2\text{R}_{\text{LINE}} = 3(0.3545)^2(3) = 1.131\ \text{MW}$$
$$\text{Q}_{\text{LINELOSS3}\phi} = 3\ \text{I}_{\text{LINE}}{}^2\text{X}_{\text{LINE}} = 3(0.3545)^2(6) = 2.262\ \text{Mvar}$$

7. The real power, reactive power, and apparent power delivered by the distribution substation are

$$\text{P}_{\text{SOURCE3}\phi} = \text{P}_{\text{LOAD3}\phi} + \text{P}_{\text{LINELOSS3}\phi} = 6.033 + 1.131 = 7.164\ \text{MW}$$
$$\text{Q}_{\text{SOURCE3}\phi} = \text{Q}_{\text{LOAD3}\phi} + \text{Q}_{\text{LINELOSS3}\phi} = 3.017 + 2.262 = 5.279\ \text{Mvar}$$
$$\text{S}_{\text{SOURCE3}\phi} = \sqrt{(7.164^2 + 5.279^2)} = 8.899\ \text{MVA}$$

(*Continued*)

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b. With the capacitor bank in service, the total impedance seen by the source is

$$Z_{TOTAL} = R_{LINE} + jX_{LINE} + \frac{1}{\frac{1}{R_{LOAD}} + \frac{1}{jX_{LOAD}} - \frac{1}{jX_C}}$$

$$Z_{TOTAL} = 3 + j6 + \frac{1}{\frac{1}{20} + \frac{1}{j40} - \frac{1}{j40}}$$

$$Z_{TOTAL} = 3 + j6 + \frac{1}{0.05} = 23 + j6 = 23.77\angle 14.62^\circ\ \Omega/\text{phase}$$

1. The line current is

$$I_{LINE} = V_{SLN}/Z_{TOTAL} = \frac{1.05(13.8/\sqrt{3})/0^\circ}{23.77\angle 14.62^\circ}$$

$$= 0.3520\angle -14.62^\circ\ \text{kA/phase}$$

2. The voltage drop across the line is

$$V_{DROP} = Z_{LINE}\, I_{LINE} = (6.708\angle 63.43^\circ)(0.3520\angle -14.62^\circ)$$

$$= 2.361\angle 48.81^\circ\ \text{kV}$$

$$|V_{DROP}| = 2.361\ \text{kV}$$

3. The load voltage is

$$V_{LOAD} = V_{SLN} - Z_{LINE}\, I_{LINE}$$

$$= 1.05(13.8/\sqrt{3})\angle 0^\circ - 2.361\angle 48.81^\circ$$

$$= 8.366 - (1.555 + j1.778)$$

$$= 6.81 - j1.778$$

$$= 7.038\angle -14.62^\circ\ \text{kV}_{LN}$$

$$|V_{LOAD}| = 7.038\sqrt{3} = 12.19\ \text{kV}_{LL}$$

4. The real and reactive power delivered to the three-phase load is

$$P_{LOAD3\phi} = 3(V_{LOADLN})^2/R_{LOAD} = 3(7.038)^2/20 = 7.430\ \text{MW}$$

$$Q_{LOAD3\phi} = 3(V_{LOADLN})^2/X_{LOAD} = 3(7.038)^2/40 = 3.715\ \text{Mvar}$$

5. The load power factor is

$$\text{p.f.} = \cos[\tan^{-1}(Q/P)]$$

$$= \cos[\tan^{-1}(3.715/7.430)]$$

$$= 0.89\ \text{lagging}$$

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6. The real and reactive line losses are

$$P_{LINELOSS3\phi} = 3\, I_{LINE}^2 R_{LINE} = 3(0.3520)^2(3) = 1.115 \text{ MW}$$

$$Q_{LINELOSS3\phi} = 3\, I_{LINE}^2 X_{LINE} = 3(0.3520)^2(6) = 2.230 \text{ Mvar}$$

7. The reactive power delivered by the shunt capacitor bank is

$$Q_C = 3(V_{LOADLN})^2/X_C = 3(7.038)^2/40 = 3.715 \text{ Mvar}$$

8. The real power, reactive power, and apparent power delivered by the distribution substation are

$$P_{SOURCE3\phi} = P_{LOAD3\phi} + P_{LINELOSS3\phi} = 7.430 + 1.115 = 8.545 \text{ MW}$$

$$\begin{aligned} Q_{SOURCE3\phi} &= Q_{LOAD3\phi} + Q_{LINELOSS3\phi} - Q_C \\ &= 3.715 + 2.230 - 3.715 \\ &= 2.230 \text{ Mvar} \end{aligned}$$

$$S_{SOURCE3\phi} = \sqrt{(8.545^2 + 2.230^2)} = 8.675 \text{ MVA}$$

c. Comparing the results of (a) and (b), with the shunt capacitor bank in service, the real power delivered to the load increases by 23% (from 6.033 to 7.430 MW) while at the same time

- The line current decreases
- The real and reactive line losses decrease
- The voltage drop across the line decreases
- The reactive power delivered by the source decreases
- The load voltage increases

The above benefits are achieved by having the shunt capacitor bank (instead of the distribution substation) deliver reactive power to the load.

The location of a shunt capacitor bank along a primary feeder is important. If there were only one load on the feeder, the best location for the capacitor bank would be directly at the load, so as to minimize I^2R losses and voltage drops on the feeder. Note that if shunt capacitors were placed at the distribution substation, I^2R feeder losses and feeder voltage drops would not be reduced because the total power including MW and Mvar would still have to be sent from the substation all the way to the load. Shunt capacitors at distribution substations, however, can be effective in reducing I^2R losses and voltage drops on the transmission or subtransmission lines that feed the distribution substations.

For a primary feeder that has a uniformly distributed load along the feeder, a common application is the "two-thirds" rule; that is, place 2/3 of the required

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reactive power 2/3 of the way down the feeder. Locating shunt capacitors 2/3 of the way down the feeder allows for good coordination between LTC distribution substation transformers or voltage regulators at the distribution substation. For other load distributions, computer software is available for optimal placement of shunt capacitor banks. Note that capacitors are rarely applied to secondary distribution systems due to their small economic advantage [3, 5].

During the daily load cycle, reactive power requirements change as a function of time. To meet the changing reactive power requirements, many utilities use a combination of fixed and switched capacitor banks. Fixed capacitor banks can be used to compensate for reactive power requirements at light loads, and switched capacitor banks can be added during heavy load conditions. The goal is to obtain a close-to-unity power factor throughout the day by switching capacitor banks on when needed and off when not needed. Methods of controlling switched capacitor banks include the following:

1. Voltage control
2. Current control
3. Var control
4. Time control
5. Temperature control
6. Radio dispatch/SCADA control

15.6 DISTRIBUTION SOFTWARE

Computer programs are available for the planning, design, and operation of electric power distribution systems. Program functions include:

1. Arc flash hazard and fault analysis
2. Capacitor placement optimization
3. Circuit breaker duty
4. Conductor and conduit sizing—ampacity and temperature calculations
5. Database management
6. Demand management
7. Distribution reliability evaluation
8. Distribution short circuit calculations
9. Fault detection and location
10. Graphics for single-line diagrams and mapping systems
11. Harmonic analysis
12. Motor starting

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13. Outage management
14. Power factor correction
15. Power flow/voltage drop computations
16. Power loss computations and costs of losses
17. Power quality and reliability
18. Relay and protective device coordination
19. Switching optimization
20. Tie capacity optimization
21. T & D modeling and analysis
22. Transformer sizing—load profile and life expectancy
23. Voltage/var optimization

Some of the vendors that offer distribution software packages are given as follows:

- ASPEN, San Mateo, CA
- Cyme International T&D, Saint Bruno, Quebec, CA (now part of Eaton)
- Electrocon International Inc., Ann Arbor, MI (now part of Siemens)
- ETAP, Irvine, CA
- Milsoft Utility Solutions Inc., Abilene, TX
- RTDS Technologies, Winnipeg, Manitoba, Canada

15.7 DISTRIBUTION RELIABILITY

Reliability in engineering applications, as defined in the *The Authoritative Dictionary of IEEE Standard Terms* (IEEE 100), is the probability that a device will function without failure over a specified time period or amount of usage. In the case of electric power distribution, reliability concerns have come from customers who want uninterrupted continuous power supplied to their facilities at minimum cost [11–17].

A typical goal for an electric utility is to have an overall average of one interruption of no more than two hours' duration per customer year. Given 8760 hours in a non-leap year, this goal corresponds to an Average Service Availability Index (ASAI) greater than or equal to 8758 service hours/8760 hours = 0.999772 = 99.9772%.

IEEE Standard 1366-2012, entitled *IEEE Guide for Electric Power Distribution Reliability Indices*, defines the following distribution reliability indices [24]:

System Average Interruption Frequency Index (SAIFI):

$$\text{SAIFI} = \frac{\Sigma \text{ Total Number of Customers Interrupted}}{\text{Total Number of Customers Served}} \tag{15.7.1}$$

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System Average Interruption Duration Index (SAIDI):

$$\text{SAIDI} = \frac{\Sigma \text{ Customer Minutes of Interruption}}{\text{Total Number of Customers Served}} \tag{15.7.2}$$

Customer Average Interruption Duration Index (CAIDI):

$$\text{CAIDI} = \frac{\Sigma \text{ Customer Minutes of Interruption}}{\text{Total Number of Customers Interrupted}} = \frac{\text{SAIDI}}{\text{SAIFI}} \tag{15.7.3}$$

Average Service Availability Index (ASAI):

$$\text{ASAI} = \frac{\text{Customer Hours Service Availability}}{\text{Customer Hours Service Demands}} \tag{15.7.4}$$

In accordance with IEEE Std. 1366-2012, when calculating the above reliability indices, momentary interruption events are not included. A momentary interruption event has an interruption duration that is limited to the time required to restore service by an interrupting device (including multiple re-closures of reclosers or circuit breakers). Switching operations must be completed within five minutes for a momentary interruption event. As such, customer interruption durations less than five minutes are excluded when calculating the reliability indices. IEEE Std. 1366-2012 also includes a method, when calculating reliability indices, for excluding major events, such as severe storms, for which the daily SAIDI exceeds a specified threshold.

The above formulas for reliability indices use customers' out-of-service and customer-minutes out-of-service data. Electric utilities with outage management systems including geographical information systems (GIS) and customer information systems (CIS) are able to keep track of this data very accurately. Some utilities in the United States are required to report distribution reliability indices to state public service commissions, while other utilities may voluntarily report these indices to regional power associations. Typical values for these indices are given in Table 15.6 [24].

SAIDI	SAIFI	CAIDI	ASAI
90 minutes/year	1.1 Interruptions/year	76 minutes/year	99.982%

TABLE 15.6

Typical Values of Reliability Indices [24] (Based on IEEE Std. 1366-2003, *IEEE Guide for Electric Power Distribution Reliability Indices*, 2004.)

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The following example uses outage data given in IEEE Std. 1366 [24].

EXAMPLE 15.4

Distribution reliability indices

Table 15.7 gives 2010 annual outage data (sustained interruptions) from a utility's CIS database for one feeder. This feeder (denoted circuit 7075) serves 2000 customers with a total load of 4 MW. Excluding momentary interruption events (less than five minutes duration) and major events, which are omitted from Table 15.7, calculate the SAIFI, SAIDI, CAIDI, and ASAI for this feeder.

Outage Date	Time at Beginning of Outage	Outage Duration (minutes)	Circuit	Number of Customers Interrupted
3/17/2010	12:12:20	8.17	7075	200
5/5/2010	00:23:10	71.3	7075	600
6/12/2010	09:30:10	30.3	7075	25
8/20/2010	15:45:39	267.2	7075	90
8/31/2010	08:20:00	120	7075	700
9/03/2010	17:10:00	10	7075	1500
10/07/2010	10:15:00	40	7075	100

TABLE 15.7

Customer Outage Data for Example 15.4 [24] (Based on IEEE Std. 1366-2012, *IEEE Guide for Electric Power Distribution Reliability Indices*.)

SOLUTION

Using the outage data from Table 15.7 in (15.7.1) through (15.7.4):

$$\text{SAIFI} = \frac{200 + 600 + 25 + 90 + 700 + 1500 + 100}{2000}$$

$$= 1.6075 \text{ interruptions/year}$$

$$\text{SAIDI} = \frac{(8.17 \times 200) + (71.3 \times 600) + (30.3 \times 25) + (267.2 \times 90) + (120 \times 700) + (10 \times 1500) + (40 \times 100)}{2000}$$

$$\text{SAIDI} = 86.110 \text{ minutes/year}$$

$$\text{CAIDI} = \frac{\text{SAIDI}}{\text{SAIFI}} = \frac{86.110}{1.6075} = 53.57 \text{ minutes/year}$$

$$\text{ASAI} = \frac{8760 \times 2000 - [(8.17 \times 200) + (71.3 \times 600) + (30.3 \times 25) + (267.2 \times 90) + (120 \times 700) + (10 \times 1500) + (40 \times 100)]/60}{8760 \times 2000}$$

$$\text{ASAI} = 0.999836 = 99.9836\%$$

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Table 15.8 lists basic outage reporting information recommended by an IEEE committee [15]. Table 15.9 lists generic and specific causes of outages, based on a U.S. Department of Energy study [16]. Many electric utilities routinely prepare distribution outage reports monthly, quarterly, and annually by town (municipality) or by district. The purposes of the reports are to monitor and evaluate distribution reliability, uncover weaknesses and potential problems, and make recommendations for improving reliability. These reports may include:

1. Frequency and duration reports, which provide data for the number of interruptions on distribution circuits together with power interrupted, average interruption duration, and causes.
2. Annual reports that sort outages according to cause of failure and according to circuit classification (for example, sort for each primary voltage, or sort for each conductor type, including overhead open wire, overhead spacer cable, underground direct-burial, and cable in conduit).
3. Five- or ten-year trends for reliability indices, and outage trends for specific causes such as tree-contact outages for overhead distribution or dig-in outages for underground distribution.
4. Lists of problem circuits such as the 20 "worst" (lowest ASAI) circuits in a district, or all circuits with repeated outages during the reporting period.

Methods for improving distribution reliability include replacement of older distribution equipment, upgrades of problem circuits, crew staffing and training for fast responses to outages and rapid restoration of service, formal maintenance programs, and public awareness programs to reduce hazards in the vicinity of distribution equipment, such as contractor dig-ins. Reliability evaluation has also become an important component of bid selections to procure new distribution equipment. Also, great strides in distribution reliability have come through distribution automation.

1. Type, design, manufacturer, and other descriptions for classification purposes
2. Date of installation, location on system, length in case of a line
3. Mode of failure (short circuit, false operation, etc.)
4. Cause of failure (lightning, tree, etc.)
5. Times (both out of service and back in service, rather than outage duration alone), date, meteorological conditions when the failure occurred
6. Type of outage, forced or scheduled, transient or permanent (momentary or sustained)

TABLE 15.8

Basic Outage Reporting Information [15]

© 1977 IEEE. Reprinted, with permission, from D. O. Koval and R. Billinton, "Evaluation of Distribution Circuit Reliability," Paper F77 067-2, IEEE Winter Power Meeting, New York, NY (January/February 1977).

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Weather	Miscellaneous	System Components	System Operation
Blizzard/ snow	Airplane/helicopter	Electrical & mechanical:	System conditions:
Cold	Animal/bird/snake	Fuel supply	Stability
Flood	Vehicle:	Generating unit failure	High/low voltage
Heat	Automobile/truck	Transformer failure	High/low frequency
Hurricane	Crane	Switchgear failure	Line overload
Ice	Dig-in	Conductor failure	Transformer overload
Lightning	Fire/explosion	Tower, pole attachment	Unbalanced load
Rain	Sabotage/vandalism	Insulation failure:	Neighboring power system
		Transmission line	
Tornado	Tree	Substation	Public appeal:
Wind	Unknown	Surge arrester	Commercial & industrial
Other	Other	Cable failure	All customers
		Voltage control equipment:	Voltage Reduction:
		Voltage regulator	0–2% voltage reduction
		Automatic tap changer	Greater than 2–8% voltage reduction
		Capacitor	
		Reactor	Rotating Blackout
		Protection and control:	Utility personnel:
		Relay failure	System operator error
		Communication signal error	Power plant operator error
		Supervisory control error	Field operator error
			Maintenance error
			Other

TABLE 15.9

Generic and Specific Causes of Outages [16]

© 1976 IEEE. Reprinted, with permission, from IEEE Committee Report, "List of Transmission and Distribution Components for Use in Outage Reporting and Reliability Calculations," *IEEE Transactions on Power Apparatus and Systems*, PAS 95(4) (July/August 1976), pp. 1210–1215.

15.8 DISTRIBUTION AUTOMATION

Throughout its existence, the electric power industry has been a leader in the application of electric, electronic, and later computer technology for monitoring, control, and automation. Initially, this technology consisted of simple meters to show voltages and flows, and telephones to call the manned substations to do control operations. Yet as early as the 1950s, supervisory control and the associated telemetering equipment was in widespread use, with a 1955 AIEE (American Institute of Electrical Engineers) report [25] indicating 31% of transmission-level stations (switching stations) had such control, and that the U.S. electric industry had more than 48,000 channel kilometers for communication, and that continuous monitoring of watts, vars, and voltage was widespread. By the late 1960s, increasingly sophisticated *energy management systems* (EMSs) were beginning to be deployed

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in electric utility control centers with applications that included automatic generation control, alarming, state estimation, on-line power flows, and contingency analysis. Initially, all of this monitoring, control, and automation was confined to the generators and the transmission-level substations. Because of its larger number of devices and more diffuse nature, the costs of monitoring and automating the distribution system could not be justified. However, this has been rapidly changing.

The monitoring, control, and automation of the distribution system is known under the general rubric of *distribution automation* (DA). While prototype DA systems date back to the 1970s, it has only been in the last 20 years or so, as communication and computer costs have continued to decrease, that they have become increasingly widespread.

A primary reason for DA is to reduce the duration of customer outages. As was mentioned in Section 15.2, the primary radial system (e.g., Figure 15.2) is widely used, with sectionalizing fuses and circuit breakers or reclosers used to clear faults, thereby avoiding prolonged outages for most customers upstream from the fault location. Sectionalizers (Figure 15.4) can also be used to isolate faulted sections of a distribution circuit after an upstream circuit breaker or recloser has interrupted the fault current. Furthermore, by closing normally open switches, the unfaulted downstream sections of a feeder can be fed from adjacent feeders. Prior to DA, these switches were manually controlled, requiring delays of potentially more than an hour for a line crew to travel to the switch location. DA can greatly reduce the time necessary to complete this process by either

1. providing the distribution system operators at dispatch centers with the ability to remotely control the various sectionalizers, or
2. automating the entire process.

Real-time monitoring of the voltages and power flows is used to ensure there is sufficient capacity to pick up the unfaulted load on the adjacent feeders.

An example of this situation is illustrated in Figure 15.23, which can also be seen in PowerWorld Simulator by loading case Figure 15.23. This case represents a feeder system modeled using the primary loop approach from Section 15.2 with a nominal 13.8-kV feeder voltage. A total of 10 loads are represented, with each load classified as either primarily residential (res), primarily commercial (com), or industrial (ind); the bus name suffix indicates the type. The left side of the loop goes from the substation distribution bus 2 to bus 8ind; the right side of the loop goes from the substation distribution bus 3 to bus 13ind. Substation buses 2 and 3 are connected by a normally open bus-tie breaker, while buses 8ind and 13ind are joined together by a normally open feeder segment to complete the loop. Each feeder line segment is assumed to use 336,400 26/7 ACSR line conductors and to be 0.96 kilometers in length, giving a per-unit impedance (on a 100-MVA base) of $0.0964 + j0.1995$. The two 12-MVA, 138/13.8-kV transformers have an impedance of $0.1 + j0.8$ per unit.

Assume a persistent fault occurs immediately downstream from the bus 3 breaker. This fault would be cleared by the bus 3 breaker, subjecting all of the customers of the right branch feeder to an outage. Without DA, a line crew

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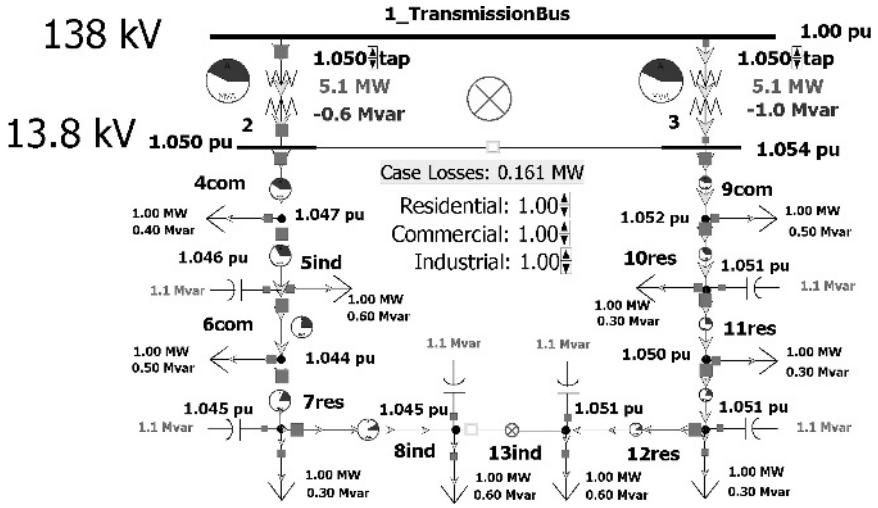

FIGURE 15.23

Primary loop feeder example, before fault

would need to be dispatched to locate the fault and then manually change the status of the appropriate sectionalizers to restore service to most customers on this feeder.

In contrast, even with a simple DA, which just consisted of having the ability to remotely control the sectionalizers and monitor flow values/voltages, service could be more quickly restored to most customers on the feeder. This could be accomplished by first opening the sectionalizer downstream from bus 9com, then closing the sectionalizer between buses 8ind and 13ind, then closing the distribution substation bus-tie breaker between buses 2 and 3 (after first balancing their taps to prevent circulating reactive power). This new configuration is shown in Figure 15.24.

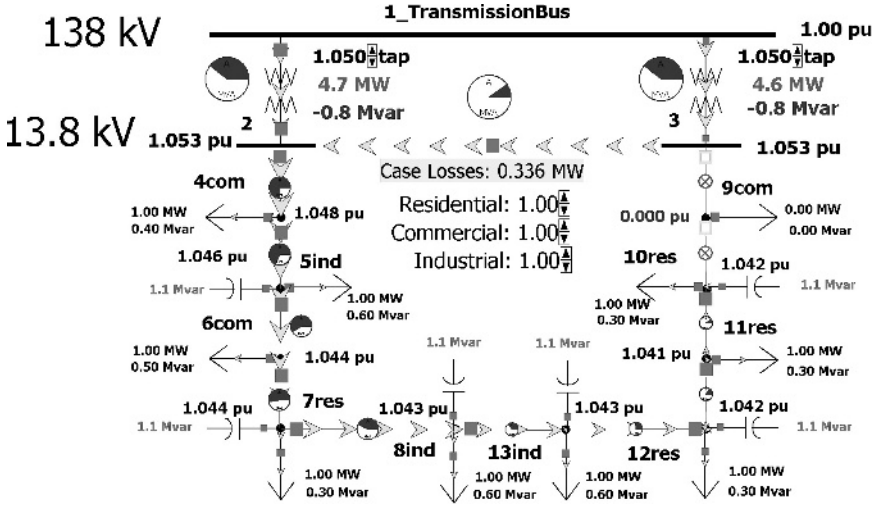

FIGURE 15.24

Primary loop feeder example, after fault and switching

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Another important application of DA is the use of voltage control devices, such as switched capacitor banks, load tap changers (LTCs) on the distribution substation transformers, or separate voltage regulators, to minimize distribution losses and to better manage the customer voltage. This is known as volt-var optimization. Since the feeder load is continually varying, these devices often need to be changed in order to maintain the desired feeder voltages and minimize feeder losses. Without DA, various techniques using only local information for local control have been employed. These included changing LTC transformer taps using the voltage sensed at the transformer, switching capacitors using local voltage, current, temperature, reactive power sensors, or just simple timers. While these approaches are better than nothing, they all have limitations. By providing a more global view of the entire feeder, DA can greatly improve the feeder voltage profile while minimizing losses.

To illustrate, again consider the Figure 15.23 case, which includes six 1.0-Mvar (nominal voltage) switched capacitors. The one-line display also shows the total system losses, and allows variation in the load multiplier for each of the three customer classes (residential, commercial, industrial) with a typical value for each ranging between 0.5 and 1.25. Initially, all of the capacitors are in-service with total losses of 0.161 MW. However, by manually opening the capacitors at buses 7res and 12res, the losses can be decreased modestly to 0.153 MW. Under lighter loading situations, the loss reduction from capacitor optimization can be even more significant.

DA can also help to reduce peak electric demands, potentially saving energy, by what is known as conservation voltage reduction (CVR). Because the power used by many loads is voltage dependent, the idea of CVR is to reduce the feeder voltage profile, while maintaining all customer voltages within the ANSI limits. A drop in the feeder voltage profile can result in an immediate decrease in power consumption. However, because some loads have their own closed-loop control systems, such as thermostats for electric heating, over the long-term, these loads tend to behave more as constant energy resources. Overall energy savings from CVR are feeder-specific and are estimated to be between 1% and 4% savings from initial deployments.

While the benefits of DA can be substantial, what had been holding back more widespread adoption of DA was the costs, both for the initial updated equipment installations and the ongoing costs associated with maintaining the monitoring and control infrastructure, such as communication costs. However, many of these costs have continued to decrease, resulting in much more widespread adoption of DA technology.

15.9 SMART GRIDS

Around 2008, the term "smart grid" entered the electric power industry lexicon, with its use being further cemented in the industry through the launch of the *IEEE Transactions on Smart Grid Journal* in 2010. While a relatively new word, the smart grid actually represents an evolutionary advancement on the technological innovation that has been present in the power industry since its inception in the 1880s. Such pervasive innovation over more than a century resulted in

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electrification being named the top engineering technology of the 20th century by the U.S. National Academy of Engineering in 2000. The smart grid represents a continuation of this application of advanced technology into this century to take advantage of near ubiquitous computing and communication.

As defined in the United States Department of Energy's *Smart Grid System Report* [29], "A smart grid uses digital technology to improve reliability, flexibility, security and efficiency of the electric system." This covers from the large generators, through the transmission and distribution systems to electricity consumers and a growing number of distributed-generation and storage resources. Probably the best definition of the attributes of the smart grid is also given in [29] with its listing of ten key characteristics:

1. Increased use of digital information and controls technology to improve reliability, security and efficiency of the electric grid;
2. Dynamic optimization of grid operations and resources, with full cybersecurity;
3. Deployment and integration of distributed resources and generation, including renewable resources;
4. Development and incorporation of demand response, demand-side resources, and energy-efficient resources;
5. Deployment of "smart" technologies for metering, communications concerning grid operations and status, and distribution automation;
6. Integration of "smart" appliances and consumer devices;
7. Deployment and integration of advanced electricity storage and peak-shaving technologies, including plug-in electric and hybrid electric vehicles, and thermal-storage air conditioning;
8. Provision to consumers of timely information and control options;
9. Development of standards for communication and interoperability of appliances and equipment connected to the electric grid, including the infrastructure serving the grid; and
10. Identification and lowering of unreasonable or unnecessary barriers to adoption of smart grid technologies, practices, and services.

While the smart grid covers large generation and high-voltage transmission, it is most germane to the distribution system and the end-use customer. The distribution system that, quoting from W.H. Kersting [28], "has traditionally been characterized as the most unglamorous component" of the power grid has now come front and center. Rather than just being a passive, radial conduit for power to flow from the networked transmission system, it is the means for supporting a bidirectional flow of information and energy to customers who are no longer content to just receive a monthly electric bill. The large, new load potential of electric vehicles requires that the home electric meter and the distribution system become smarter, since the grid cannot reasonably accommodate charging a large

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number of car batteries as people return to their garages in the early evening when the remainder of the electric load is at peak demand. Providing real-time prices to some loads can provide additional control flexibility. In many locations, distributed energy resources such as solar photovoltaics are resulting in situations in which distribution feeders have reversible power flows, looking like net generation when the sun is shining and load when it is not.

As noted in this chapter's case study, a newer term that is being used to describe the changes that are occurring at the interface between the distribution system and what is behind the electric meters is *grid edge*. While there is currently no single formal definition, the term grid edge often encompasses the physical hardware at the end user locations, the software that controls this hardware and is used to analyze its data, and the business models that are helping to create a more decentralized grid. Grid edge examples include smart electric meters, building controls, more controllable solar PV inverters, microgrids, price responsive load, peer-to-peer communication between grid edge devices, and functional aggregation of DERs to allow participation in electricity markets.

Underlying the entire grid is the need for a trustworthy and resilient cyber infrastructure. As more smart grid technologies are deployed, the result will be a power grid increasingly dependent on communication and computing from its center to its edge. Disruptions in this cyber infrastructure, either due to accidents, bugs, or deliberate attacks could result in wide scale blackouts.

PROBLEMS

SECTION 15.2

15.1 Are laterals on primary radial systems typically protected from short circuits? If so, how (by fuses, circuit breakers, or reclosers)?

15.2 What is the most common type of grounding on primary distribution systems?

15.3 What is the most common primary distribution voltage class in the United States?

15.4 Why are reclosers used on overhead primary radial systems and overhead primary loop systems? Why are they not typically used on underground primary radial systems and underground primary loop systems?

SECTION 15.3

15.5 What are the typical secondary distribution voltages in the United States?

15.6 What are the advantages of secondary networks? Name two disadvantages.

15.7 Using the Internet, name three cities in the Western Interconnection of the United States that have secondary network systems.

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SECTION 15.4

15.8 A three-phase 138-kV Δ/13.8-kV Y distribution substation transformer rated 40 MVA OA/50 MVA FA/65 MVA FOA has an 9% impedance. (a) Determine the rated current on the primary distribution side of the transformer at its OA, FA, and FOA ratings. (b) Determine the per-unit transformer impedance using a system base of 100 MVA and 13.8 kV on the primary distribution side of the transformer. (c) Calculate the short-circuit current on the primary distribution side of the transformer for a three-phase bolted fault on the primary distribution side. Assume that the prefault voltage is 13.8 kV.

15.9 As shown in Figure 15.25, an urban distribution substation has one 30-MVA (FOA) and three 33.3-MVA (FOA), 138-kV Δ/12.5-kV Y transformers denoted TR1-TR4, which feed through circuit breakers to a ring bus. The transformers are older transformers designed for a 55°C temperature rise. The ring bus contains eight bus-tie circuit breakers, two of which are normally open (NO), so as to separate the ring bus into two sections. TR1 and TR2 feed one section, and TR3 and TR4 feed the other section. Also, four capacitor banks, three banks rated at 6 Mvar and one at 9 Mvar, are connected to the ring

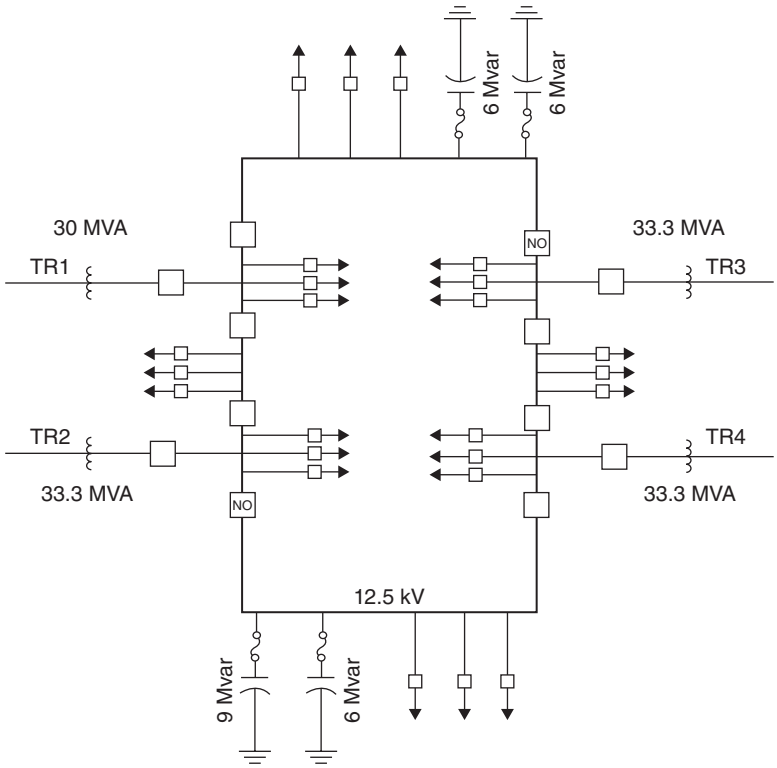

FIGURE 15.25
Distribution substation for Problems 15.9 and 15.10

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bus. Twenty-four 12.5-kV underground primary feeders are served from the substation, 12 from each section. The utility that owns this substation has the following transformer summer loading criteria based on a percentage of nameplate rating:

a. 120% for normal summer loading.

b. 150% during a two-hour emergency.

c. 130% 30-day emergency loading.

Determine the following summer ratings of this substation: (a) the normal summer rating with all four transformers in service; (b) the allowable substation rating, assuming the single-contingency loss of one transformer; and (c) the 30-day emergency rating under the single-contingency loss of one transformer. Assume that during a two-hour emergency, switching can be performed to reduce the total substation load by 10% and to approximately balance the loadings of the three transformers remaining in service. Assume a 5% reduction for unequal transformer loadings.

15.10 For the distribution substation given in Problem 15.9, assume that each of the four circuit breakers on the 12.5-kV side of the distribution substation transformers has a maximum continuous current of 2000 A/phase during both normal and emergency conditions. Determine the summer allowable substation rating under the single-contingency loss of one transformer, based on not exceeding the maximum continuous current of these circuit breakers at 12.5-kV operating voltage. Assume a 5% reduction for unequal transformer loadings. Comparing the results of this problem with Problem 15.9, what limits the substation allowable rating, the circuit breakers or the transformers?

SECTION 15.5

15.11 (a) How many Mvars of shunt capacitors are required to increase the power factor on a 10-MVA load from 0.85 to 0.9 lagging? (b) How many Mvars of shunt capacitors are required to increase the power factor on a 10-MVA load from 0.90 to 0.95 lagging? (c) Which requires more reactive power, improving a low power-factor load or a high power-factor load?

15.12 Rework Example 15.3 with $R_{Load} = 40\ \Omega$/phase, $X_{Load} = 60\ \Omega$/phase, and $X_C = 60\ \Omega$/phase.

SECTION 15.7

15.13 Table 15.10 gives 2010 annual outage data (sustained interruptions) from a utility's CIS database for feeder 8050. This feeder serves 4500 customers with a total load of 9 MW. Table 15.10 includes a major event that began on 11/04/2010 with 4000 customers out of service

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Outage Date	Time at Beginning of Outage	Outage Duration (minutes)	Circuit	Number of Customers Interrupted
1/15/2010	01:24:20	14.4	8050	342
4/4/2010	08:14:20	151.2	8050	950
7/08/2010	07:15:46	89.8	8050	125
9/10/2010	15:45:39	654.6	8050	15
10/11/2010	07:40:59	32.7	8050	2200
11/04/2010	22:30:00	10,053*	8050	4000*
12/01/2010	14:18:07	40	8050	370

TABLE 15.10

Customer Outage Data for Problem 15.13

* Major event due to ice storm

for approximately seven days (10,053 minutes) due to an ice storm. Momentary interruption events (less than five minutes' duration) are excluded from Table 15.10. Calculate the SAIFI, SAIDI, CAIDI, and ASAI for this feeder: (a) including the major event; and (b) excluding the major event.

15.14 Assume that a utility's system consists of two feeders: feeder 7075, serving 2000 customers, and feeder 8050, serving 4000 customers. Annual outage data during 2010 is given in Table 15.7 and 15.10 for these feeders. Calculate the SAIFI, SAIDI, CAIDI, and ASAI for the system, excluding the major event.

SECTION 15.8

PW **15.15** Open PowerWorld Simulator case Problem 15_15, which represents a lower load scenario for the Figure 15.23 case. Determine the optimal status of the six switched shunts to minimize the system losses.

PW **15.16** Open PowerWorld Simulator case Problem 15_16, which represents a lower load scenario for the Figure 15.23 case and has the LTC transformer taps each changed to 1.025. Determine the optimal status of the six switched shunts to minimize the system losses for this case.

PW **15.17** Open PowerWorld Simulator case Problem 15_17 and note the case losses. Then close the bus tie breaker between buses 2 and 3. How do the losses change? How can the case be modified to reduce the system losses?

PW **15.18** Usually in power flow studies, the load is treated as being independent of the bus voltage. That is, a constant power model is used. However, in reality, the load usually has some voltage dependence, so

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that decreasing the feeder voltage (such as for conversation voltage reduction) will result in less power consumption, at least temporarily. Open PowerWorld Simulator case Problem 15_18, which contains the Figure 15.23 system except the load model is set so 30% of the load is modeled as constant power, 40% of the load is modeled as constant current (the load varies linearly with the bus voltage magnitude), and 30% of the load is modeled as constant impedance (the load varies with the square of the bus voltage magnitude). By adjusting the tap positions for the two substation transformers and the capacitor banks, determine the operating configuration that minimizes the total load plus losses (shown on the display), with the constraint that all bus voltage must be at least 0.97 per unit.

PW **15.19** Repeat Problem 15.18, except using PowerWorld Simulator case Problem 15_19 (which has a different load level from the Problem 15.18 case).

SECTION 15.9

15.20 Select one of the smart grid characteristics from the list given in this section. Write a one-page (or other instructor-selected length) summary and analysis paper on a current news story that relates to this characteristic.

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APPENDIX

TABLE A.1

Typical average values of synchronous-machine constants

Constant (units)	Type	Symbol	Turbo-Generator (solid rotor)	Water-Wheel Generator (with dampers)	Synchronous Condenser	Synchronous Motor
Reactances (per unit)	Synchronous	X_d	1.1	1.15	1.80	1.20
		X_q	1.08	0.75	1.15	0.90
	Transient	X'_d	0.23	0.37	0.40	0.35
		X'_q	0.23	0.75	1.15	0.90
	Subtransient	X''_d	0.12	0.24	0.25	0.30
		X''_q	0.15	0.34	0.30	0.40
	Negative-sequence	X_2	0.13	0.29	0.27	0.35
	Zero-sequence	X_0	0.05	0.11	0.09	0.16
Resistances (per unit)	Positive-sequence	R (dc)	0.003	0.012	0.008	0.01
		R (ac)	0.005	0.012	0.008	0.01
	Negative-sequence	R_2	0.035	0.10	0.05	0.06
Time constants (seconds)	Transient	T'_{d0}	5.6	5.6	9.0	6.0
		T'_d	1.1	1.8	2.0	1.4
	Subtransient	$T''_d = T''_q$	0.035	0.035	0.035	0.036
	Armature	T_a	0.16	0.15	0.17	0.15

(Adapted from E. W. Kimbark, *Power System Stability: Synchronous Machines* (New York: Dover Publications, 1956/1968), Chap. 12)

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TABLE A.2

Typical transformer leakage reactances

Rating of Highest Voltage Winding kV	BIL of Highest Voltage Winding kV	Leakage Reactance per unit*	
Distribution Transformers			
2.4	30	0.023–0.049	
4.8	60	0.023–0.049	
7.2	75	0.026–0.051	
12	95	0.026–0.051	
23	150	0.052–0.055	
34.5	200	0.052–0.055	
46	250	0.057–0.063	
69	350	0.065–0.067	
Power Transformers 10 MVA and Below			
8.7	110	0.050–0.058	
25	150	0.055–0.058	
34.5	200	0.060–0.065	
46	250	0.060–0.070	
69	350	0.070–0.075	
92	450	0.070–0.085	
115	550	0.075–0.100	
138	650	0.080–0.105	
161	750	0.085–0.011	
Power Transformers Above 10 MVA			
		Self-Cooled or Forced-Air-Cooled	Forced-Oil-Cooled
8.7	110	0.050–0.063	0.082–0.105
34.5	200	0.055–0.075	0.090–0.128
46	250	0.057–0.085	0.095–0.143
69	350	0.063–0.095	0.103–0.158
92	450	0.060–0.118	0.105–0.180
115	550	0.065–0.135	0.107–0.195
138	650	0.070–0.140	0.117–0.245
161	750	0.075–0.150	0.125–0.250
230	900	0.070–0.160	0.120–0.270
345	1300	0.080–0.170	0.130–0.280
500	1550	0.100–0.200	0.160–0.340
765		0.110–0.210	0.190–0.350

*Per-unit reactances are based on the transformer rating

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TABLE A.3 Characteristics of copper conductors, hard drawn, 97.3% conductivity

Size of Conductor		Number of Strands	Diameter of Individual Strands (inches)	Outside Diameter (inches)	Breaking Strength (pounds)	Weight (pounds per mile)	Approx. Current Carrying Capacity* (amps)	Geometric Mean Radius at 60 Hz (feet)	r_a Resistance (Ohms per Conductor per Mile)								x_a Inductive Reactance (ohms per conductor per mile at 1 ft spacing)			x'_a Shunt Capacitive Reactance (megohms per conductor per mile at 1 ft spacing)		
									25°C (77°F)				50°C (122°F)									
Circular Mils	A.W.G. or B. & S.								dc	25 Hz	50 Hz	60 Hz	dc	25 Hz	50 Hz	60 Hz	25 Hz	50 Hz	60 Hz	25 Hz	50 Hz	60 Hz
1 000 000		37	0.1644	1.151	43 830	16 300	1 300	0.0368	0.0585	0.0594	0.0620	0.0634	0.0640	0.0648	0.0672	0.0685	0.1666	0.333	0.400	0.216	0.1081	0.0901
900 000		37	0.1560	1.092	39 510	14 670	1 220	0.0349	0.0650	0.0658	0.0682	0.0695	0.0711	0.0718	0.0740	0.0752	0.1693	0.339	0.406	0.220	0.1100	0.0916
800 000		37	0.1470	1.029	35 120	13 040	1 130	0.0329	0.0731	0.0739	0.0760	0.0772	0.0800	0.0806	0.0826	0.0837	0.1722	0.344	0.413	0.224	0.1121	0.0934
750 000		37	0.1424	0.997	33 400	12 230	1 090	0.0319	0.0780	0.0787	0.0807	0.0818	0.0853	0.0859	0.0878	0.0888	0.1739	0.348	0.417	0.226	0.1132	0.0943
700 000		37	0.1375	0.963	31 170	11 410	1 040	0.0308	0.0836	0.0842	0.0861	0.0871	0.0914	0.0920	0.0937	0.0947	0.1759	0.352	0.422	0.229	0.1145	0.0954
600 000		37	0.1273	0.891	27 020	9 781	940	0.0285	0.0975	0.0981	0.0997	0.1008	0.1068	0.1071	0.1088	0.1095	0.1799	0.360	0.432	0.235	0.1173	0.0977
500 000		37	0.1162	0.814	22 510	8 151	840	0.0260	0.1170	0.1175	0.1188	0.1196	0.1280	0.1283	0.1296	0.1303	0.1845	0.369	0.443	0.241	0.1205	0.1004
500 000		19	0.1622	0.811	21 590	8 151	840	0.0256	0.1170	0.1175	0.1188	0.1196	0.1280	0.1283	0.1296	0.1303	0.1853	0.371	0.445	0.241	0.1206	0.1005
450 000		19	0.1539	0.770	19 750	7336	780	0.0243	0.1300	0.1304	0.1316	0.1323	0.1422	0.1426	0.1437	0.1443	0.1879	0.376	0.451	0.245	0.1224	0.1020
400 000		19	0.1451	0.726	17 560	6521	730	0.0229	0.1462	0.1466	0.1477	0.1484	0.1600	0.1603	0.1613	0.1619	0.1909	0.382	0.458	0.249	0.1245	0.1038
350 000		19	0.1357	0.679	15 590	5706	670	0.0214	0.1671	0.1675	0.1684	0.1690	0.1828	0.1831	0.1840	0.1845	0.1943	0.389	0.466	0.254	0.1269	0.1058
350 000		12	0.1708	0.710	15 140	5706	670	0.0225	0.1671	0.1675	0.1684	0.1690	0.1828	0.1831	0.1840	0.1845	0.1918	0.384	0.460	0.251	0.1253	0.1044
300 000		19	0.1257	0.629	13 510	4891	610	0.01987	0.1950	0.1953	0.1961	0.1966	0.213	0.214	0.214	0.215	0.1982	0.396	0.476	0.259	0.1296	0.1080
300 000		12	0.1581	0.657	13 170	4891	610	0.0208	0.1950	0.1953	0.1961	0.1966	0.213	0.214	0.214	0.215	0.1957	0.392	0.470	0.258	0.1281	0.1068
250 000		19	0.1147	0.574	11 360	4076	540	0.01813	0.234	0.234	0.235	0.235	0.256	0.256	0.257	0.257	0.203	0.406	0.487	0.266	0.1329	0.1108
250 000		12	0.1443	0.600	11 130	4075	540	0.01902	0.234	0.234	0.235	0.235	0.256	0.256	0.257	0.257	0.200	0.401	0.481	0.263	0.1313	0.1094
211 600	4/0	19	0.1055	0.528	9617	3450	480	0.01668	0.276	0.277	0.277	0.278	0.302	0.303	0.303	0.303	0.207	0.414	0.497	0.272	0.1359	0.1132
211 600	4/0	12	0.1328	0.552	9483	3450	490	0.01750	0.276	0.277	0.277	0.278	0.302	0.303	0.303	0.303	0.205	0.409	0.491	0.269	0.1343	0.1119
211 600	4/0	7	0.1739	0.522	9154	3450	480	0.01579	0.276	0.277	0.277	0.278	0.302	0.303	0.303	0.303	0.210	0.420	0.503	0.273	0.1363	0.1136
167 800	3/0	12	0.1183	0.492	7556	2736	420	0.01559	0.349	0.349	0.349	0.350	0.381	0.381	0.382	0.382	0.210	0.421	0.505	0.277	0.1384	0.1153
167 800	3/0	7	0.1548	0.464	7366	2736	420	0.01404	0.349	0.349	0.349	0.350	0.381	0.381	0.382	0.382	0.216	0.431	0.518	0.281	0.1405	0.1171
133 100	2/0	7	0.1379	0.414	5926	2170	360	0.01252	0.440	0.440	0.440	0.440	0.481	0.481	0.481	0.481	0.222	0.443	0.532	0.289	0.1445	0.1205
105 500	1/0	7	0.1228	0.368	4752	1720	310	0.01113	0.555	0.555	0.555	0.555	0.606	0.607	0.607	0.607	0.227	0.455	0.546	0.298	0.1488	0.1240
83 690	1	7	0.1093	0.328	3804	1364	270	0.00992	0.699	0.699	0.699	0.699	0.765				0.233	0.467	0.560	0.306	0.1528	0.1274
83 690	1	3	0.1670	0.360	3620	1351	270	0.01016	0.692	0.692	0.692	0.692	0.757				0.232	0.464	0.557	0.299	0.1495	0.1246
66 370	2	7	0.0974	0.292	3045	1082	230	0.00883	0.881	0.882	0.887	0.887	0.964				0.239	0.478	0.574	0.314	0.1570	0.1308
66 370	2	3	0.1487	0.320	2913	1071	240	0.00903	0.873				0.955				0.238	0.476	0.571	0.307	0.1537	0.1281
66 370	2	1	. . . .	0.258	3003	1061	220	0.00836	0.864				0.945				0.242	0.484	0.581	0.323	0.1614	0.1345
52 630	3	7	0.0867	0.260	2433	858	200	0.00787	1.112				1.216				0.245	0.490	0.588	0.322	0.1611	0.1343
52 630	3	3	0.1325	0.285	2359	850	200	0.00805	1.101				1.204				0.244	0.488	0.585	0.316	0.1578	0.1315
52 630	3	1		0.229	2439	841	190	0.00745	1.090	Same as dc			1.192	Same as dc			0.248	0.496	0.595	0.331	0.1656	0.1380
41 740	4	3	0.1180	0.254	1879	674	180	0.00717	1.388				1.518				0.250	0.499	0.599	0.324	0.1619	0.1349
41 740	4	1	. . . .	0.204	1970	667	170	0.00663	1.374				1.503				0.254	0.507	0.609	0.339	0.1697	0.1415
33 100	5	3	0.1050	0.226	1505	534	150	0.00638	1.750				1.914				0.256	0.511	0.613	0.332	0.1661	0.1384
33 100	5	1	. . . .	0.1819	1591	529	140	0.00590	1.733				1.895				0.260	0.519	0.623	0.348	0.1738	0.1449
26 250	6	3	0.0935	0.201	1205	424	130	0.00568	2.21				2.41				0.262	0.523	0.628	0.341	0.1703	0.1419
26 250	6	1		0.1620	1280	420	120	0.00526	2.18				2.39				0.265	0.531	0.637	0.356	0.1779	0.1483
20 820	7	1	. . . .	0.1443	1030	333	110	0.00468	2.75				3.01				0.271	0.542	0.651	0.364	0.1821	0.1517
16 510	8	1	. . . .	0.1285	826	264	90	0.00417	3.47				3.80				0.277	0.554	0.665	0.372	0.1862	0.1552

*For conductor at 75°C, air at 25°C, wind 1.4 miles per hour (2 ft/sec), frequency = 60 Hz.

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TABLE A.4 Characteristics of aluminum cable, steel, reinforced (Aluminum Company of America)—ACSR

Code Word	Circular Mils Aluminum	Aluminum		Strand Diameter (inches)	Steel	Strand Diameter (inches)	Outside Diameter (inches)	Copper Equivalent* Circular Mils or A.W.G.	Ultimate Strength (pounds)	Weight (pounds per mile)	Geometric Mean Radius at 60 Hz (feet)	Approx. Current Carrying Capacity† (amps)	r_a Resistance (Ohms per Conductor per Mile)								x_a Inductive Reactance (ohms per conductor per mile at 1 ft spacing all currents)	x'_d Shunt Capacitive Reactance (megohms per conductor per mile at 1 ft spacing)
													25°C (77°F) Small Currents				50°C (122°F) Current Approx 75% Capacity‡					
													dc	25 Hz	50 Hz	60 Hz	dc	25 Hz	50 Hz	60 Hz	60 Hz	60 Hz
Coree	2 515 000	76	..	0.1819	19	0.0849	1.880		61 700		0.0621									0.0450	0.337	0.0755
Thrasher	2 312 000	76	..	0.1744	19	0.0814	1.802		57 300		0.0595									0.0482	0.342	0.0767
Kiwi	2 167 000	72	4	0.1735	7	0.1157	1.735		49 800		0.0570									0.0511	0.348	0.0778
Bluebird	2 156 000	84	4	0.1602	19	0.0961	1.762		60 300		0.0588									0.0505	0.344	0.0774
Chukar	1 781 000	84	4	0.1456	19	0.0874	1.602		51 000		0.0534									0.0598	0.355	0.0802
Falcon	1 590 000	54	3	0.1716	19	0.1030	1.545	1 000 000	56 000	10 777	0.0520	1 380	0.0587	0.0588	0.0590	0.0591	0.0646	0.0656	0.0675	0.0684	0.359	0.0814
Parrot	1 510 500	54	3	0.1673	19	0.1004	1.506	950 000	53 200	10 237	0.0507	1 340	0.0618	0.0619	0.0621	0.0622	0.0680	0.0690	0.0710	0.0720	0.362	0.0821
Plover	1 431 000	54	3	0.1628	19	0.0977	1.465	900 000	50 400	9 699	0.0493	1 300	0.0652	0.0653	0.0655	0.0656	0.0718	0.0729	0.0749	0.0760	0.365	0.0830
Martin	1 351 000	54	3	0.1582	19	0.0949	1.424	850 000	47 600	9 160	0.0479	1 250	0.0691	0.0692	0.0694	0.0695	0.0761	0.0771	0.0792	0.0803	0.369	0.0838
Pheasant	1 272 000	54	3	0.1535	19	0.0921	1.382	800 000	44 800	8 621	0.0465	1 200	0.0734	0.0735	0.0737	0.0738	0.0808	0.0819	0.0840	0.0851	0.372	0.0847
Grackle	1 192 500	54	3	0.1486	19	0.0892	1.338	750 000	43 100	8 082	0.0450	1 160	0.0783	0.0784	0.0786	0.0788	0.0862	0.0872	0.0894	0.0906	0.376	0.0857
Finch	1 113 000	54	3	0.1436	19	0.0862	1.293	700 000	40 200	7 544	0.0435	1 110	0.0839	0.0840	0.0842	0.0844	0.0924	0.0935	0.0957	0.0969	0.380	0.0867
Curlew	1 033 500	54	3	0.1384	7	0.1384	1.246	650 000	37 100	7 019	0.0420	1 060	0.0903	0.0905	0.0907	0.0909	0.0994	0.1005	0.1025	0.1035	0.385	0.0878
Cardinal	954 000	54	3	0.1329	7	0.1329	1.196	600 000	34 200	6 479	0.0403	1 010	0.0979	0.0980	0.0981	0.0982	0.1078	0.1088	0.1118	0.1128	0.390	0.0890
Canary	900 000	54	3	0.1291	7	0.1291	1.162	566 000	32 300	6 112	0.0391	970	0.104	0.104	0.104	0.104	0.1145	0.1155	0.1175	0.1185	0.393	0.0898
Crane	874 500	54	3	0.1273	7	0.1273	1.146	550 000	31 400	5 940	0.0386	950	0.107	0.107	0.107	0.108	0.1178	0.1188	0.1218	0.1228	0.395	0.0903
Condor	795 000	54	3	0.1214	7	0.1214	1.093	500 000	28 500	5 399	0.0368	900	0.117	0.118	0.118	0.119	0.1288	0.1308	0.1358	0.1378	0.401	0.0917
Drake	795 000	26	2	0.1749	7	0.1360	1.108	500 000	31 200	5 770	0.0375	900	0.117	0.117	0.117	0.117	0.1288	0.1288	0.1288	0.1288	0.399	0.0912
Mallard	795 000	30	2	0.1678	19	0.0977	1.140	500 000	38 400	6 517	0.0393	910	0.117	0.117	0.117	0.117	0.1288	0.1288	0.1288	0.1288	0.393	0.0904
Crow	715 500	54	3	0.1151	7	0.1151	1.036	450 000	26 300	4 859	0.0349	830	0.131	0.131	0.131	0.132	0.1442	0.1452	0.1472	0.1482	0.407	0.0932
Starling	715 500	26	2	0.1659	7	0.1290	1.051	450 000	28 100	5 193	0.0355	840	0.131	0.131	0.131	0.131	0.1442	0.1442	0.1442	0.1442	0.405	0.0928
Redwing	715 500	30	2	0.1544	19	0.0926	1.081	450 000	34 600	5 865	0.0372	840	0.131	0.131	0.131	0.131	0.1442	0.1442	0.1442	0.1442	0.399	0.0920
Flamingo	666 600	54	3	0.1111	7	0.1111	1.000	419 000	24 500	4 527	0.0337	800	0.140	0.140	0.141	0.141	0.1541	0.1571	0.1591	0.1601	0.412	0.0943
Rook	636 000	54	3	0.1085	7	0.1085	0.977	400 000	23 600	4 319	0.0329	770	0.147	0.147	0.148	0.148	0.1618	0.1638	0.1678	0.1688	0.414	0.0950
Grosbeak	636 000	26	2	0.1564	7	0.1216	0.990	400 000	25 000	4 616	0.0335	780	0.147	0.147	0.147	0.147	0.1618	0.1618	0.1618	0.1618	0.412	0.0946
Egret	636 000	30	2	0.1456	19	0.0874	1.019	400 000	31 500	5 213	0.0351	780	0.147	0.147	0.147	0.147	0.1618	0.1618	0.1618	0.1618	0.406	0.0937
Peacock	605 000	54	3	0.1059	7	0.1059	0.953	380 500	22 500	4 109	0.0321	750	0.154	0.155	0.155	0.155	0.1695	0.1715	0.1755	0.1775	0.417	0.0957
Squab	605 000	26	2	0.1525	7	0.1186	0.966	380 500	24 100	4 391	0.0327	760	0.154	0.154	0.154	0.154	0.1700	0.1720	0.1720	0.1720	0.415	0.0953
Dove	556 500	26	2	0.1463	7	0.1138	0.927	350 000	22 400	4 039	0.0313	730	0.168	0.168	0.168	0.168	0.1849	0.1859	0.1859	0.1859	0.420	0.0965
Eagle	556 500	30	2	0.1362	7	0.1362	0.953	350 000	27 700	4 588	0.0328	730	0.168	0.168	0.168	0.168	0.1849	0.1859	0.1859	0.1859	0.415	0.0957
Hawk	477 000	26	2	0.1355	7	0.1054	0.858	300 000	19 430	3 462	0.0290	670	0.196	0.196	0.196	0.196	0.216				0.430	0.0988
Hen	477 000	30	2	0.1261	7	0.1261	0.883	300 000	23 300	3 933	0.0304	670	0.196	0.196	0.196	0.196	0.216				0.424	0.0980
Ibis	397 500	26	2	0.1236	7	0.0981	0.783	250 000	16 190	2 885	0.0265	590	0.235				0.259				0.441	0.1015
Lark	397 500	30	2	0.1151	7	0.1151	0.806	250 000	19 980	3 277	0.0270	600	0.235		Same as dc		0.259		Same as dc		0.435	0.1006
Linnet	336 400	26	2	0.1138	7	0.0855	0.721	4/0	14 050	2 442	0.0244	530	0.278				0.306				0.451	0.1039
Oriole	336 400	30	2	0.1059	7	0.1059	0.741	4/0	17 040	2 774	0.0255	530	0.278				0.306				0.445	0.1032
Ostrich	300 000	26	2	0.1074	7	0.0835	0.680	188 700	12 650	2 178	0.0230	490	0.311				0.342				0.458	0.1057
Piper	300 000	30	2	0.1000	7	0.1000	0.700	188 700	15 430	2 473	0.0241	500	0.311				0.342				0.462	0.1049
Partridge	266 800	26	2	0.1013	7	0.0788	0.642	3/0	11 250	1 936	0.0217	460	0.350				0.385				0.465	0.1074

*Based on copper 97% aluminum 61% conductivity.
†For conductor at 75°C, air at 25°C, wind 1.4 miles per hour (2 ft/sec), frequency = 60 Hz
‡ Current Approx 75% Capacity" is 75% of the "Approx Current Carrying Capacity in Amps" and is approximately the current which will produce 50°C conductor temp (25°C rise) with 25°C air temp., wind 1.4 miles per hour

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